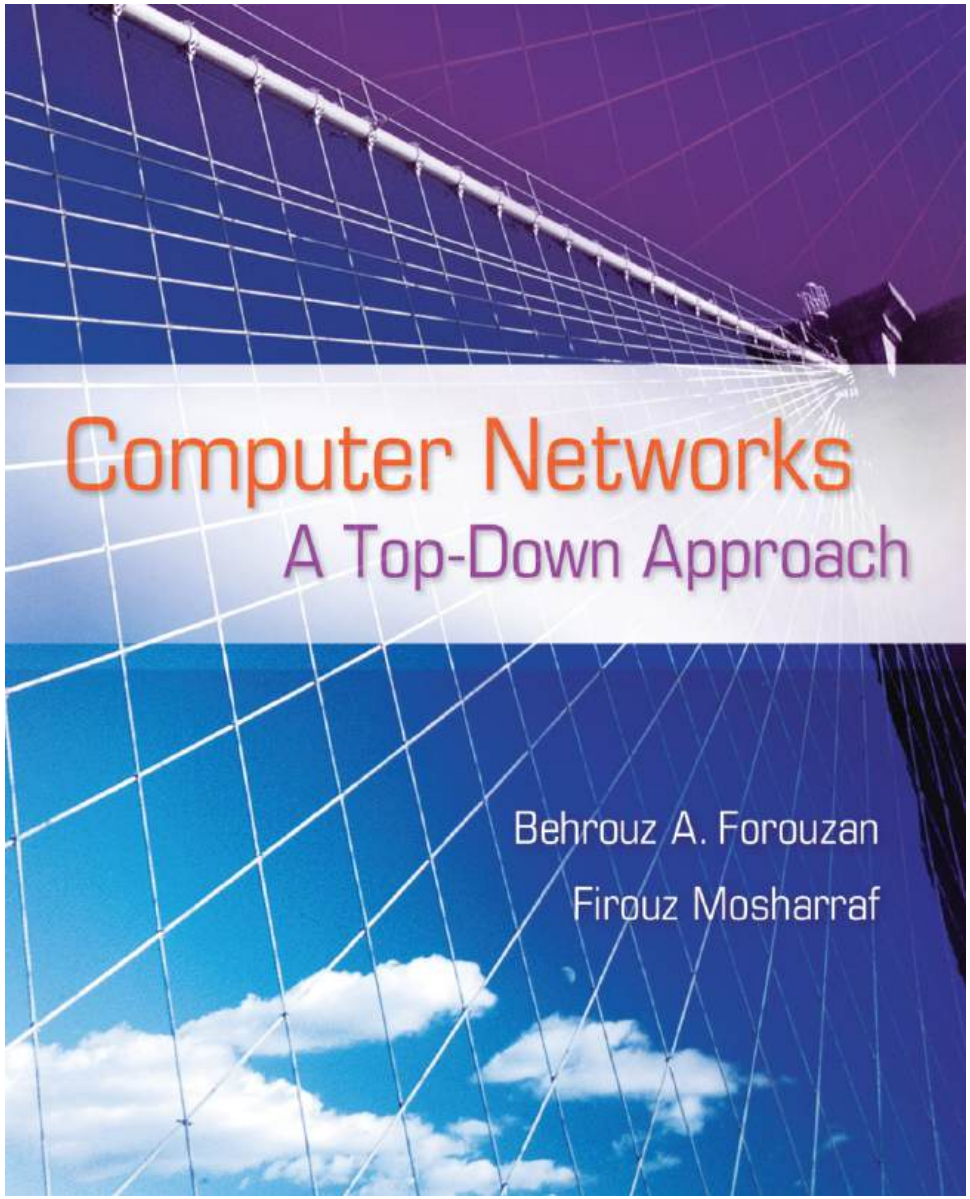
Chapter 1

Introduction



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Chapter 1: Outline

1.1 Overview of the Internet

1.2 Protocol Layering

1.3 Internet History

1.4 Standards and Administration

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Chapter 1: Objective

- ***We introduce local area networks (LANs) and wide area networks (WANs) and show that an internet or the Internet is a combination of these networks.***

- ***We introduce the concept of protocol layering to show how the task to be done by the Internet is divided into smaller tasks. We also discuss TCP/IP protocol suite and show the duty of each layer.***

- ***We give a brief history of the Internet.***

- ***We introduce the administration of the Internet and define the standards and their lifetime.***

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1-1 OVERVIEW OF THE INTERNET

We start our journey by first defining a network. We then show how we can connect networks to create small internetworks. Finally, we show the structure of the Internet and open the gate to study the Internet in the next ten chapters.

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1.1.1 Networks

A network is the interconnection of a set of devices capable of communication. In this definition, a device can be a host such as a large computer, desktop, laptop, workstation, cellular phone, or security system. A device in this definition can also be a connecting device such as a router a switch, a modem that changes the form of data, and so on.

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1.1.1 Networks (Continued)

- **Local Area Networks**
- **Wide Area Networks**
  - Point-to-Point WANs
  - Switched WANs
- **Internetwork**

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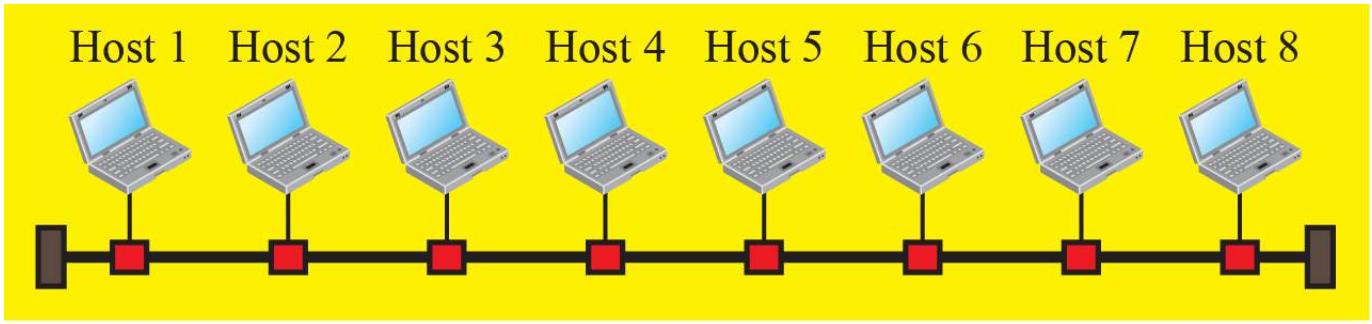
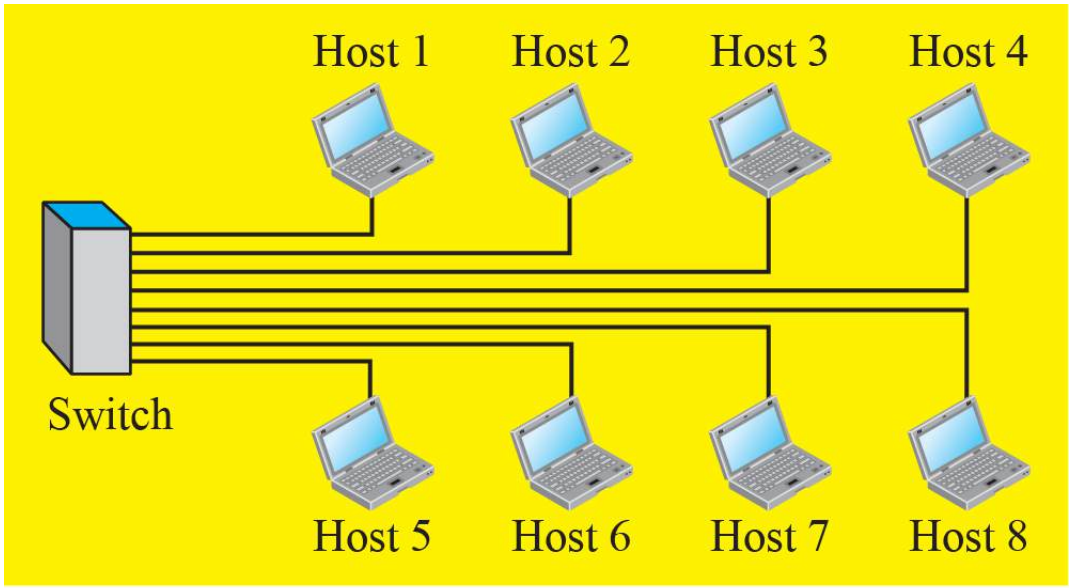
Figure 1.1: An Isolated LAN in the past and today



a. LAN with a common cable (past)



b. LAN with a switch (today)

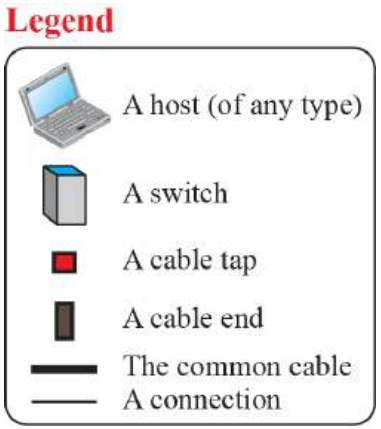


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Figure 1.2: A Point-to-Point WAN

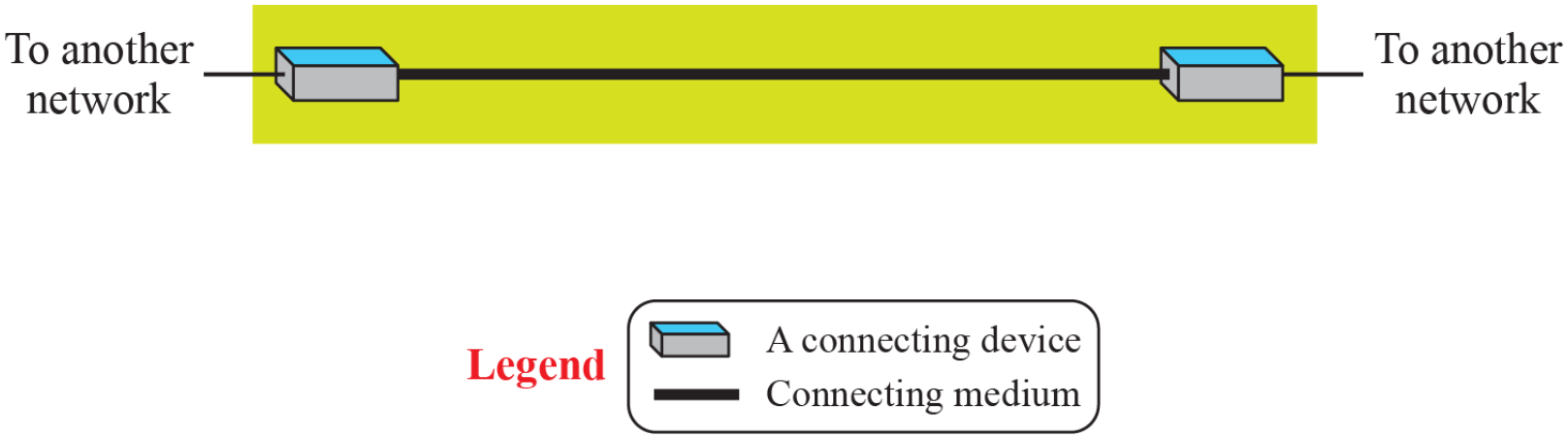

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Figure 1.3: A Switched WAN

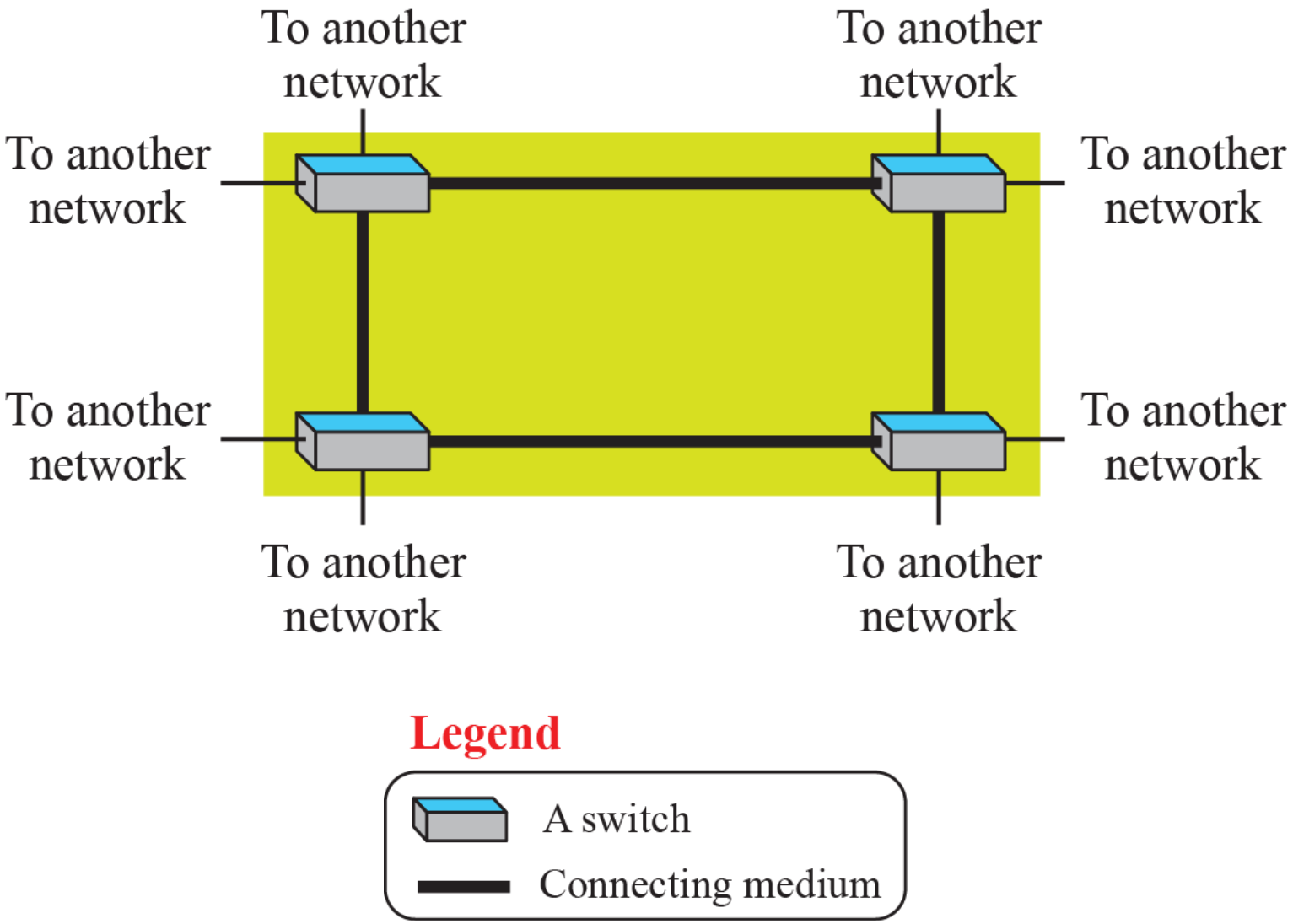

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Figure 1.4: An internetwork made of two LANs and one WAN

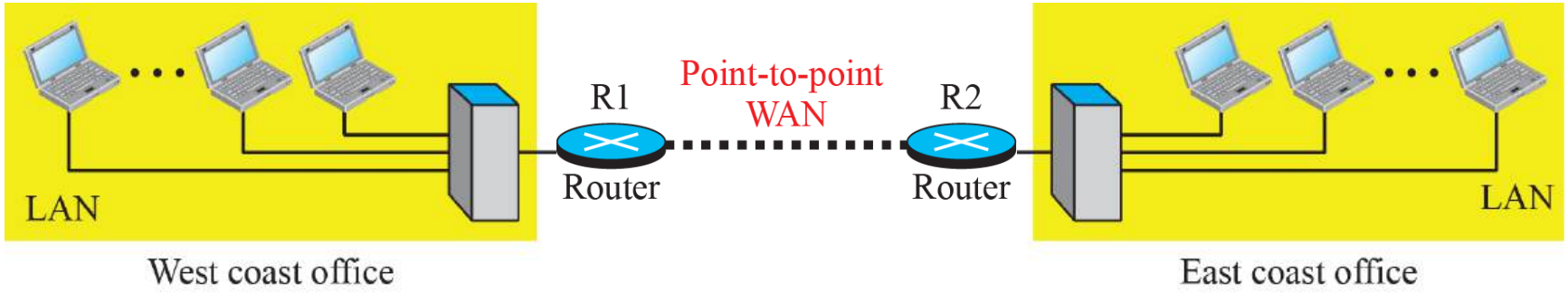

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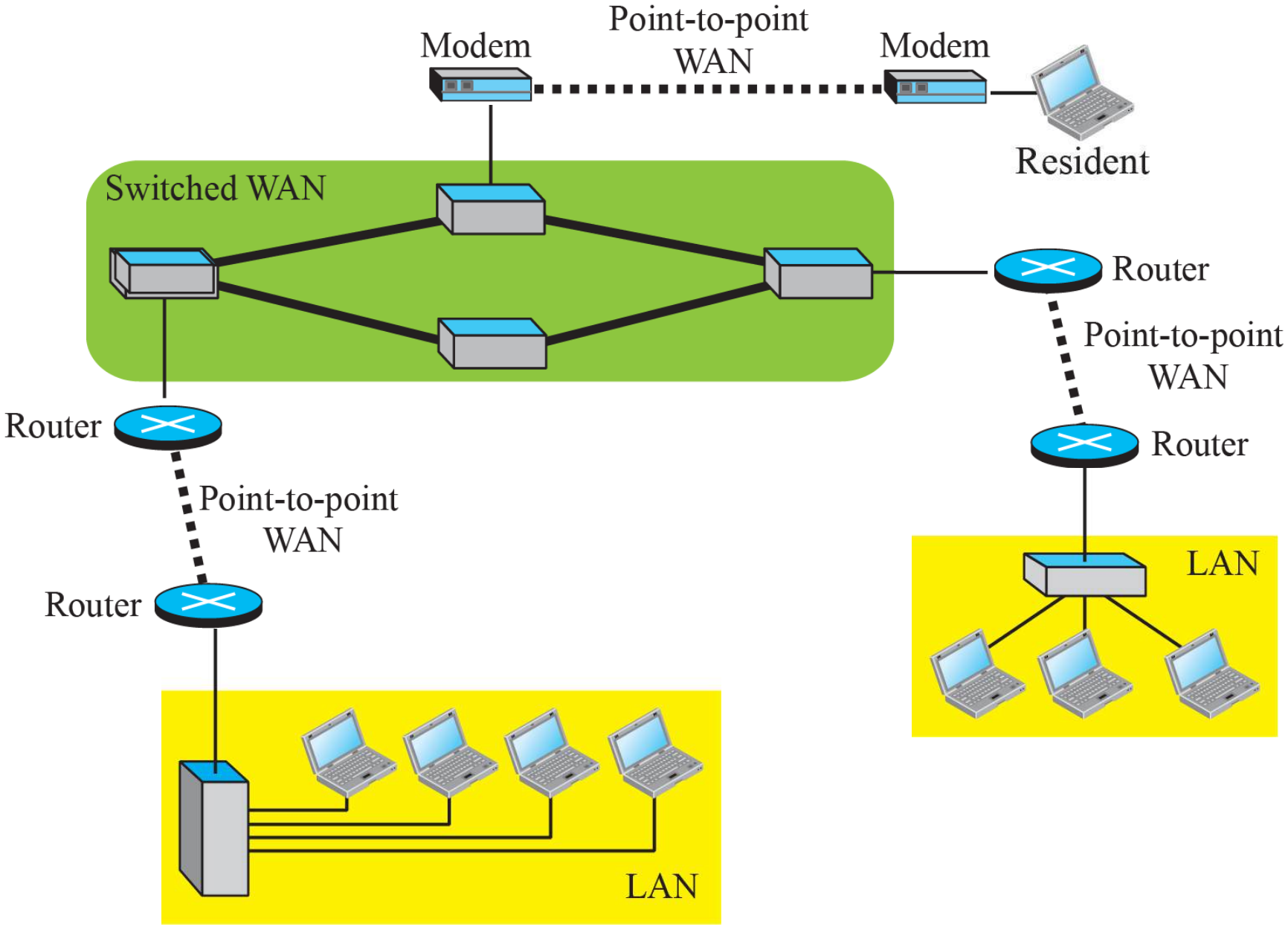
Figure 1.5: A heterogeneous network made of WANs and LANs



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1.1.2 Switching

An internet is a switched network in which a switch connects at least two links together. A switch needs to forward data from a link to another link when required.

- **Circuit-Switched Network**
- **Packet-Switched Network**

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Figure 1.6: A circuit-switched network

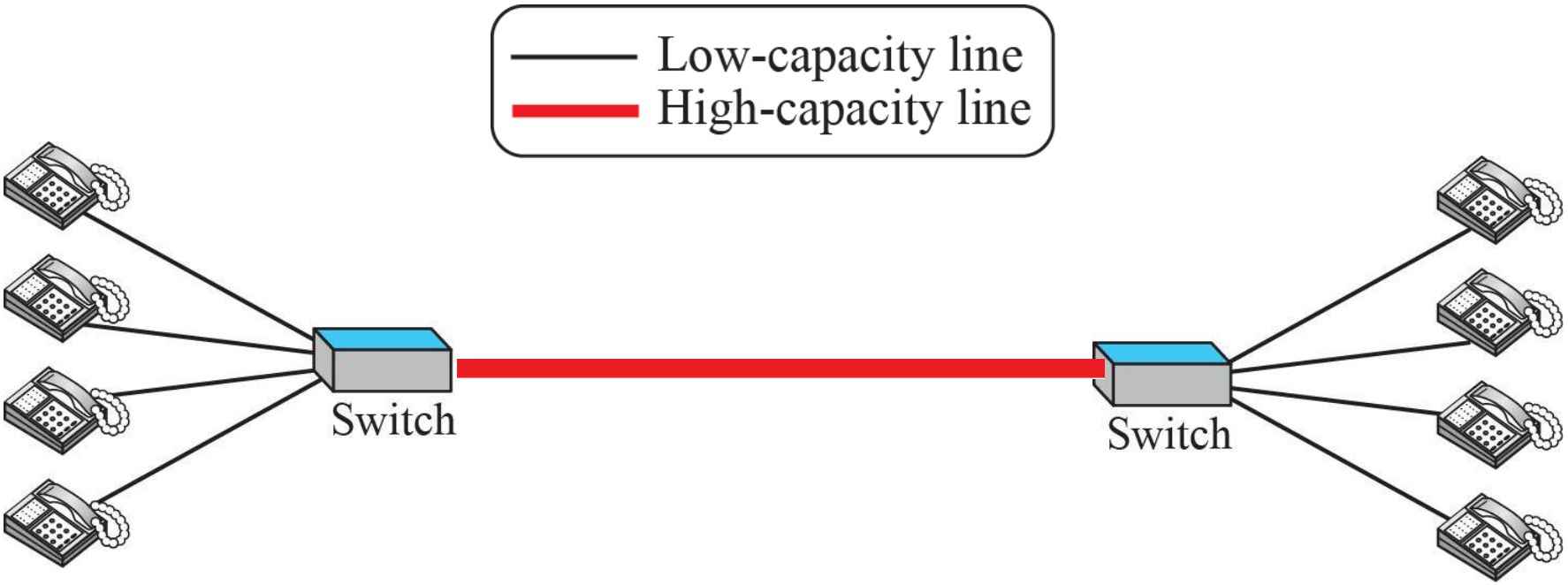

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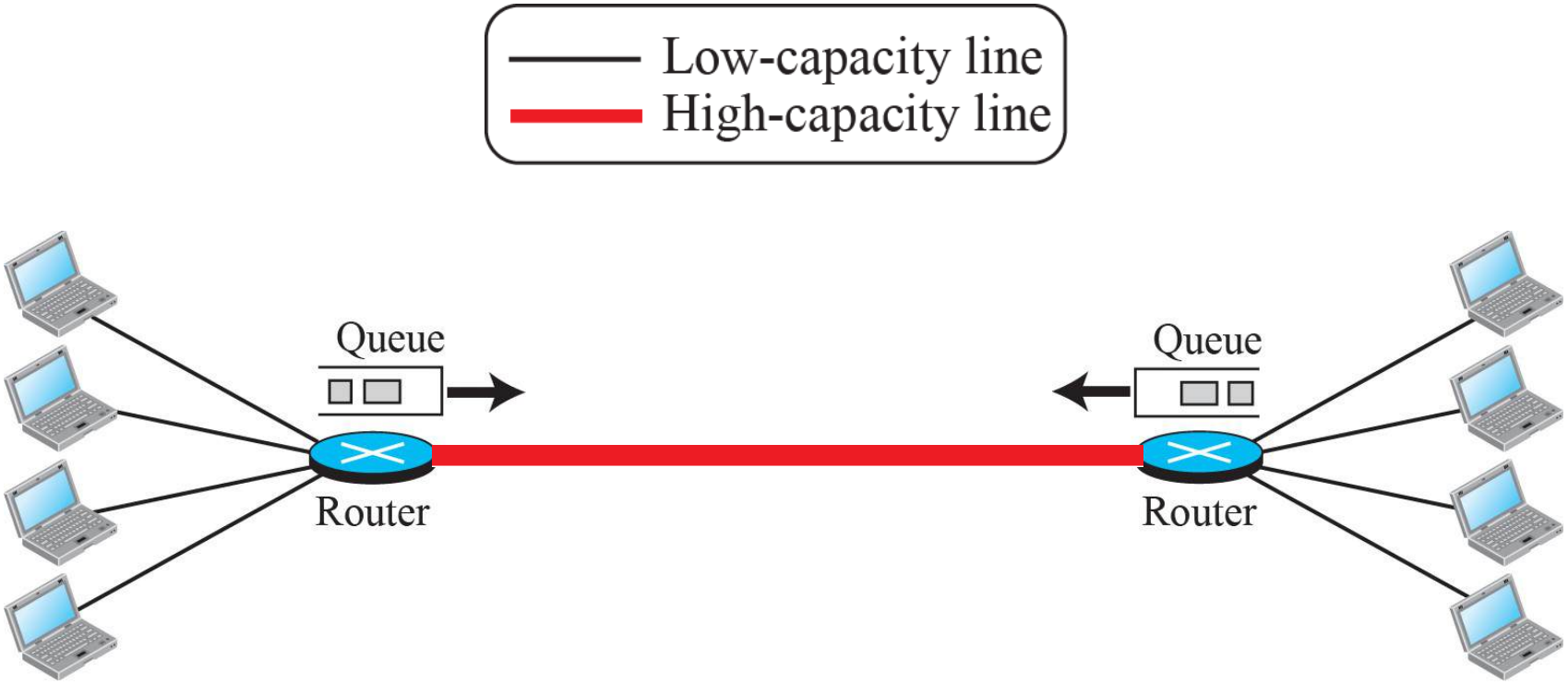
Figure 1.7: A packet-switched network



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1.1.3 The Internet

The most notable internet is called the Internet and is composed of thousands of inter-connected networks. Figure1.8 shows a conceptual (not geographical) view of the Internet.

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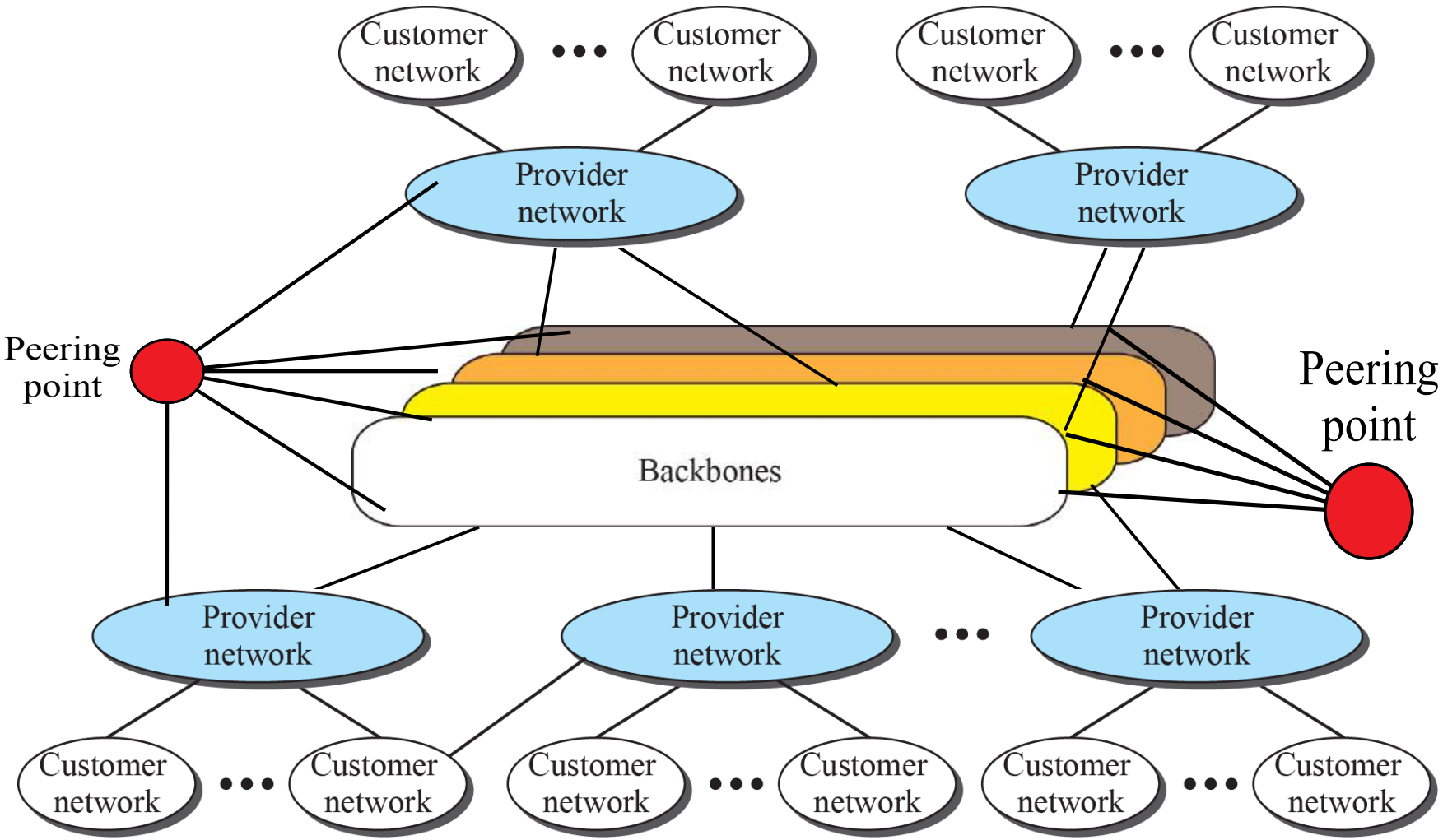
Figure 1.8: The Internet today



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1.1.4 Accessing the Internet

The Internet today is an internetwork that allows any user to become part of it. The user, however, needs to be physically connected to an ISP. The physical connection is normally done through a point-to-point WAN. In this section, we briefly describe how this can happen, but we postpone the technical details of the connection until Chapters 6 and 7.

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1.1.4 Accessing the Internet (continued)

- **Using Telephone Networks**
  - Dial-up Service
  - DSL
- **Using Cable Networks**
- **Using Wireless Networks**
- **Direct Connection**

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1.1.5 Hardware and Software

We have given the overview of the Internet structure. For communication to happen, we need both hardware and software. This is similar to a complex computation in which we need both a computer and a program. In the next section, we show how these combinations of hardware and software are coordinated with each other using protocol layering.

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1-2 PROTOCOL LAYERING

A word we hear all the time when we talk about the Internet is protocol. A protocol defines the rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively. When communication is simple, we may need only one simple protocol; when the communication is complex, we need a protocol at each layer, or protocol layering.

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1.2.1 Scenarios

Let us develop two simple scenarios to better understand the need for protocol layering.

- *First Scenario (Figure 1.9)*
- *Second Scenario (Figure 1.10)*
- *Principle of Protocol Layering*
- *Logical Connections*

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Figure 1.9: A single-layer protocol

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Figure 1.10: A three-layer protocol

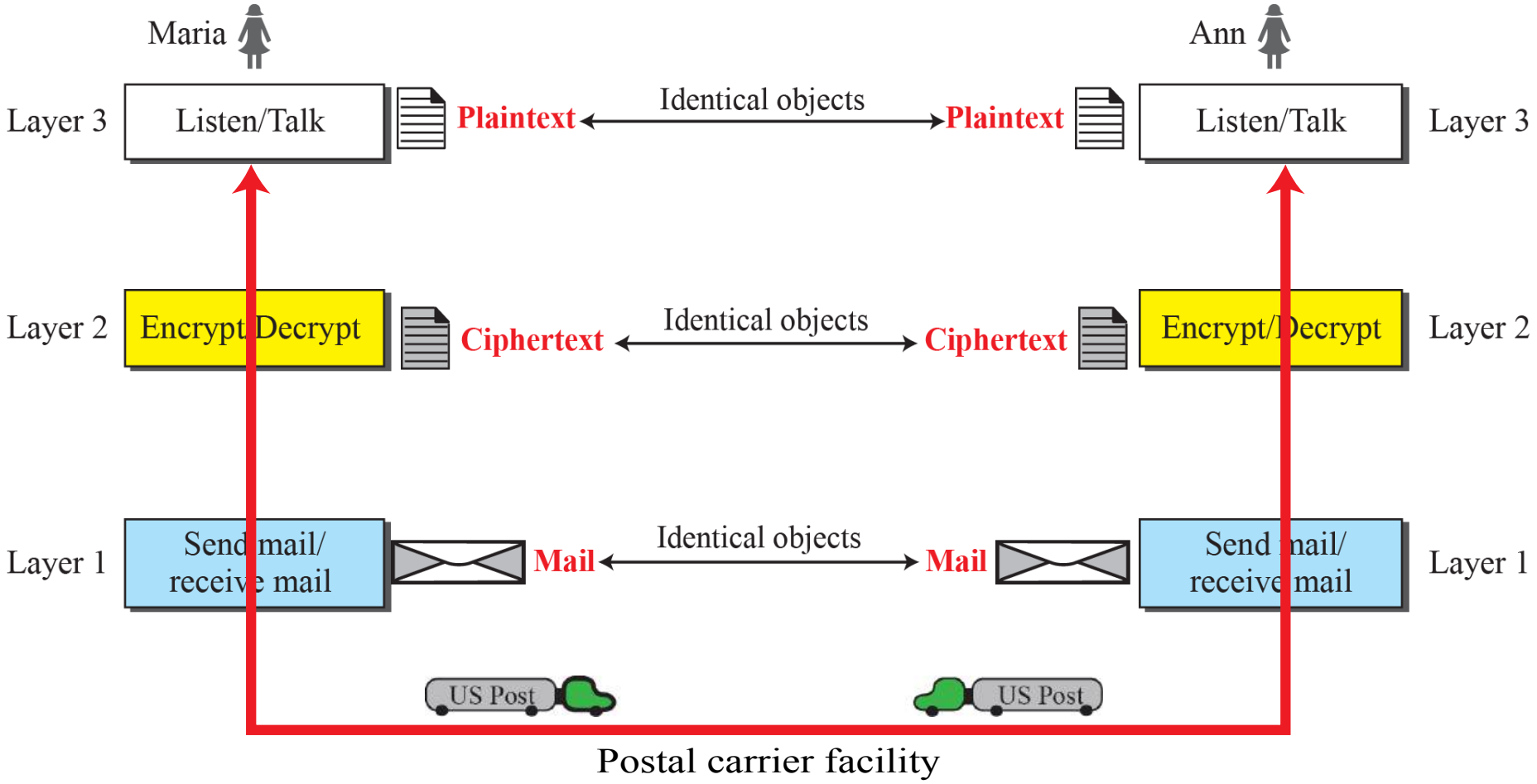

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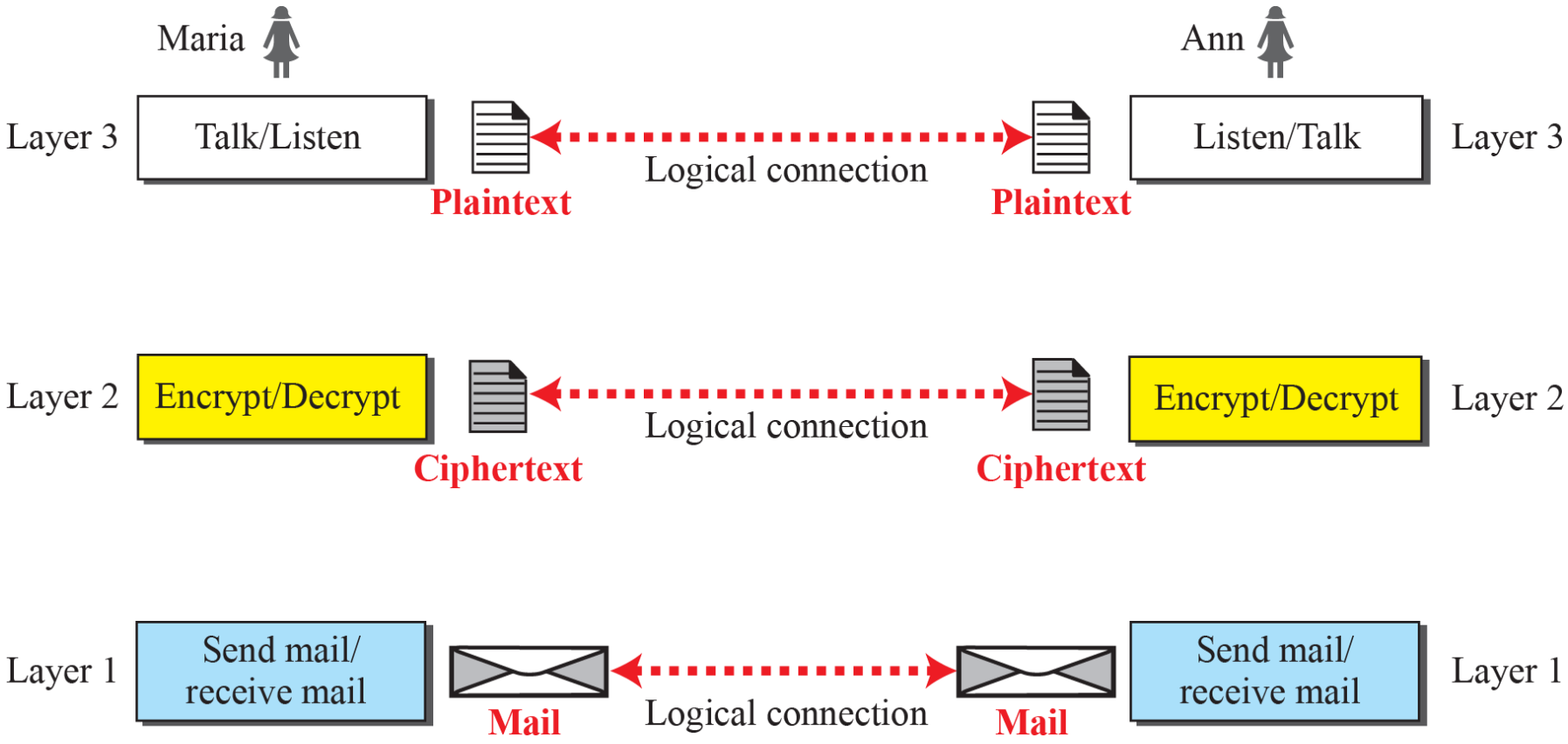
Figure 1.11: Logical connection between peer layers



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1.2.2 TCP/IP Protocol Suite

TCP/IP is a protocol suite used in the Internet today. It is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality. The term hierarchical means that each upper level protocol is supported by the services provided by one or more lower level protocols. The original TCP/IP protocol suite was defined as four software layers built upon the hardware. Today, however, TCP/IP is thought of as a five-layer model.

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1.2.2 TCP/IP Protocol Suite (continued)

- ***Layered Architecture***
- ***Layered in the Suite***
- ***Description of Each Layer***
  - *Application Layer*
  - *Transport Layer*
  - *Network Layer*
  - *Data-link Layer*
  - *Physical Layer*

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1.2.2 TCP/IP Protocol Suite (continued)

- ***Encapsulation and Decapsulation***
  - *Encapsulation at the Source Host*
  - *Decapsulation and Encapsulation at Router*
  - *Decapsulation at the Destination Host*
- ***Addressing***
- ***Multiplexing and Demultiplexing***

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Figure 1.12: Layers in the TCP/IP protocol suite

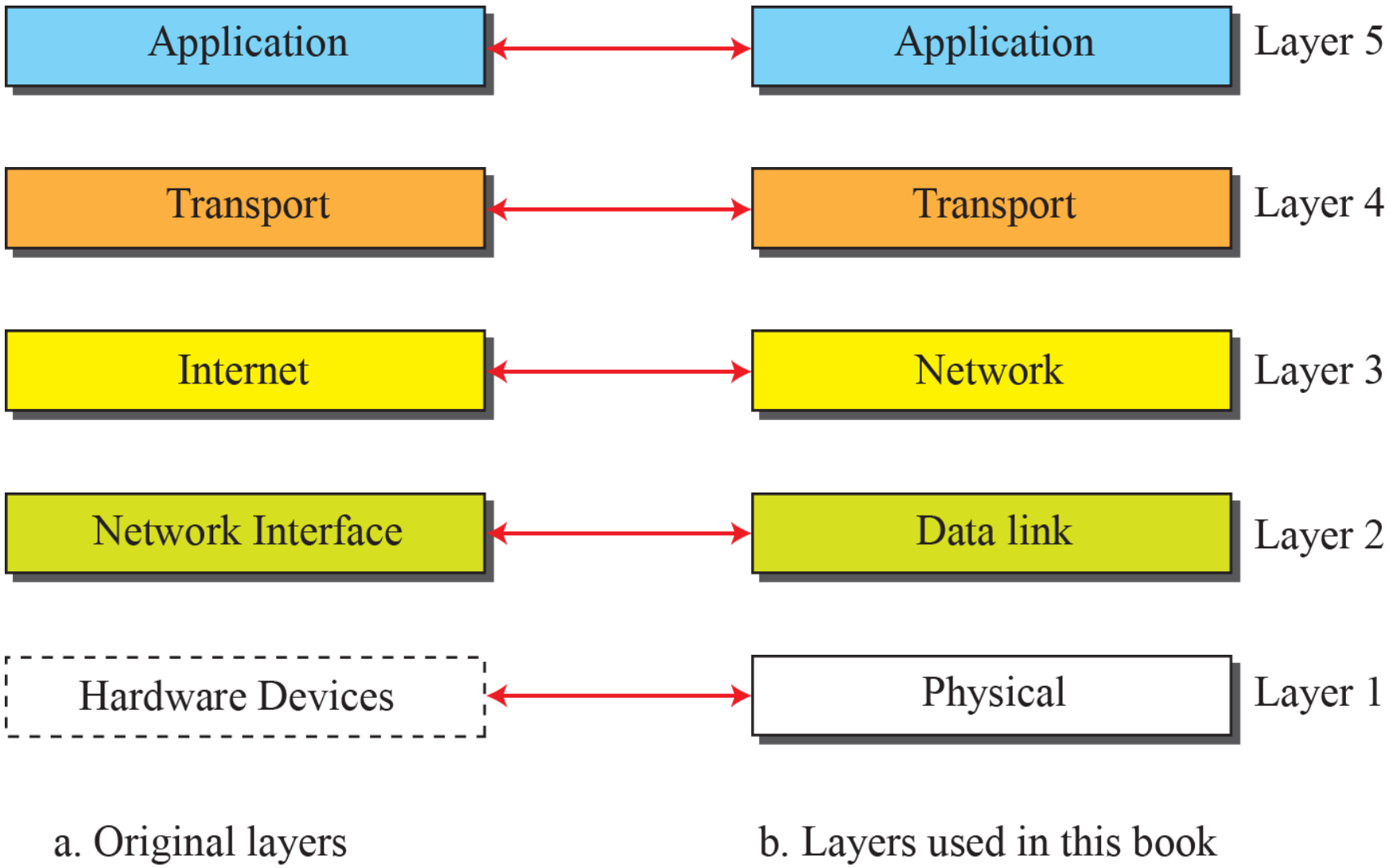

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Figure 1.13: Communication through an internet

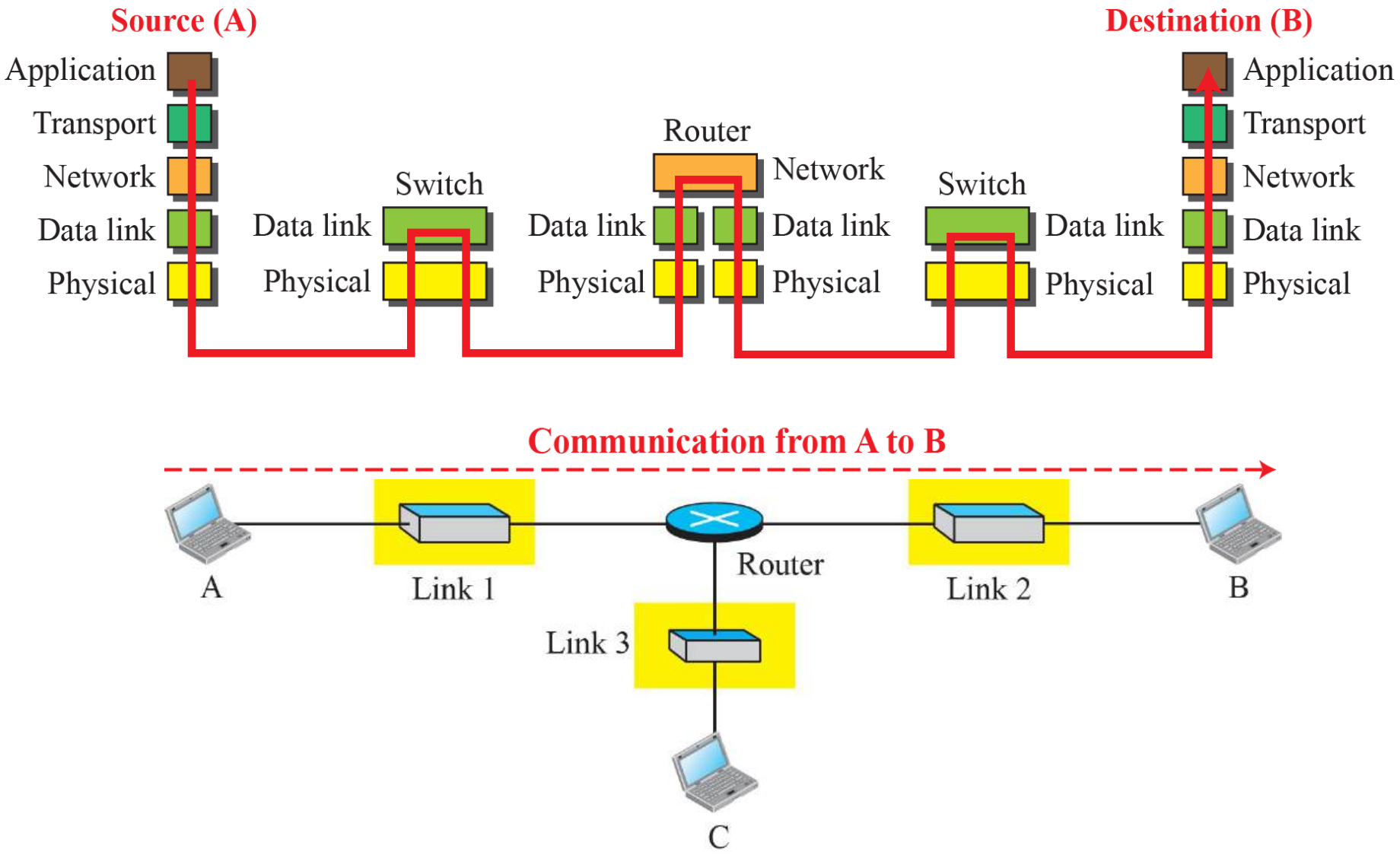

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Figure 1.14: Logical connections between layers in TCP/IP

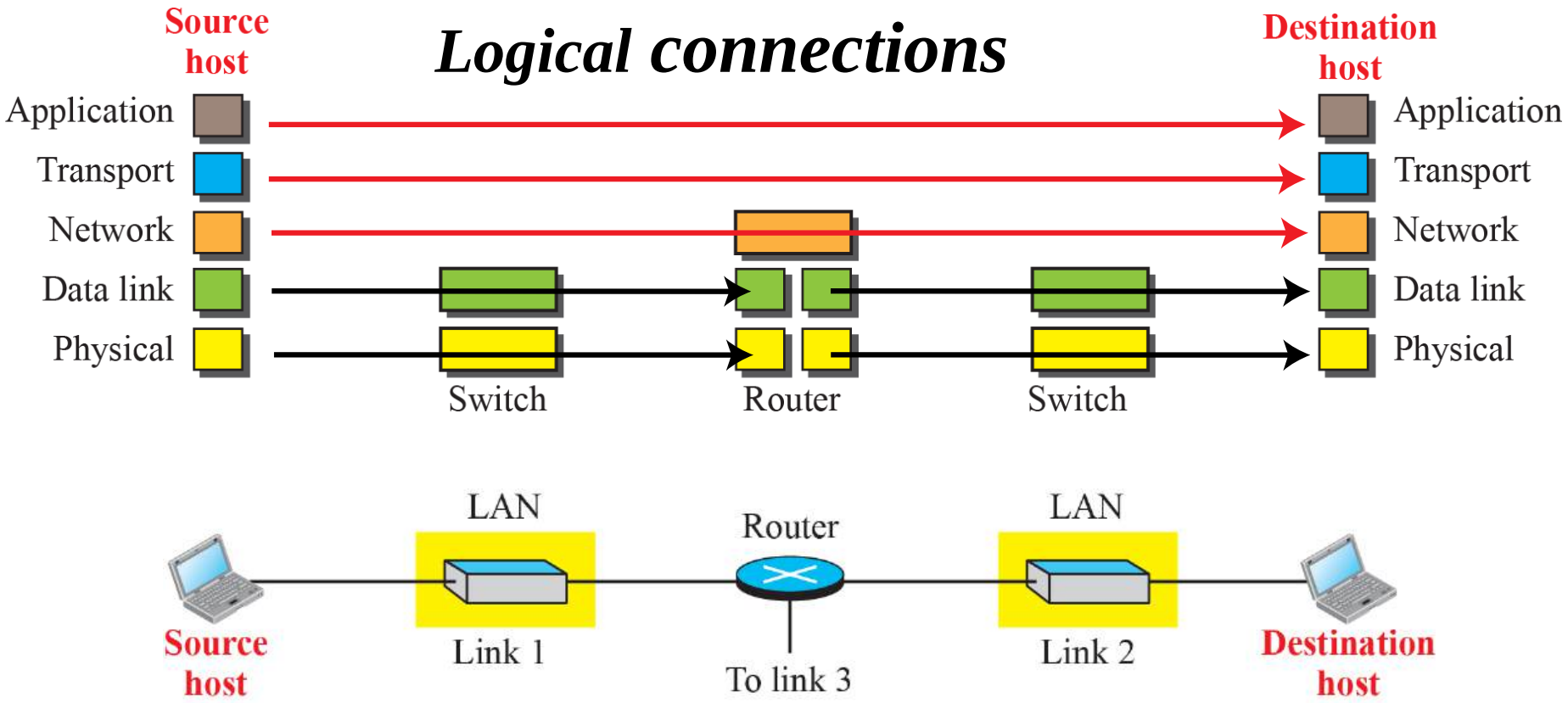

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Figure 1.15: Identical objects in the TCP/IP protocol suite

Notes: We have not shown switches because they don't change objects.

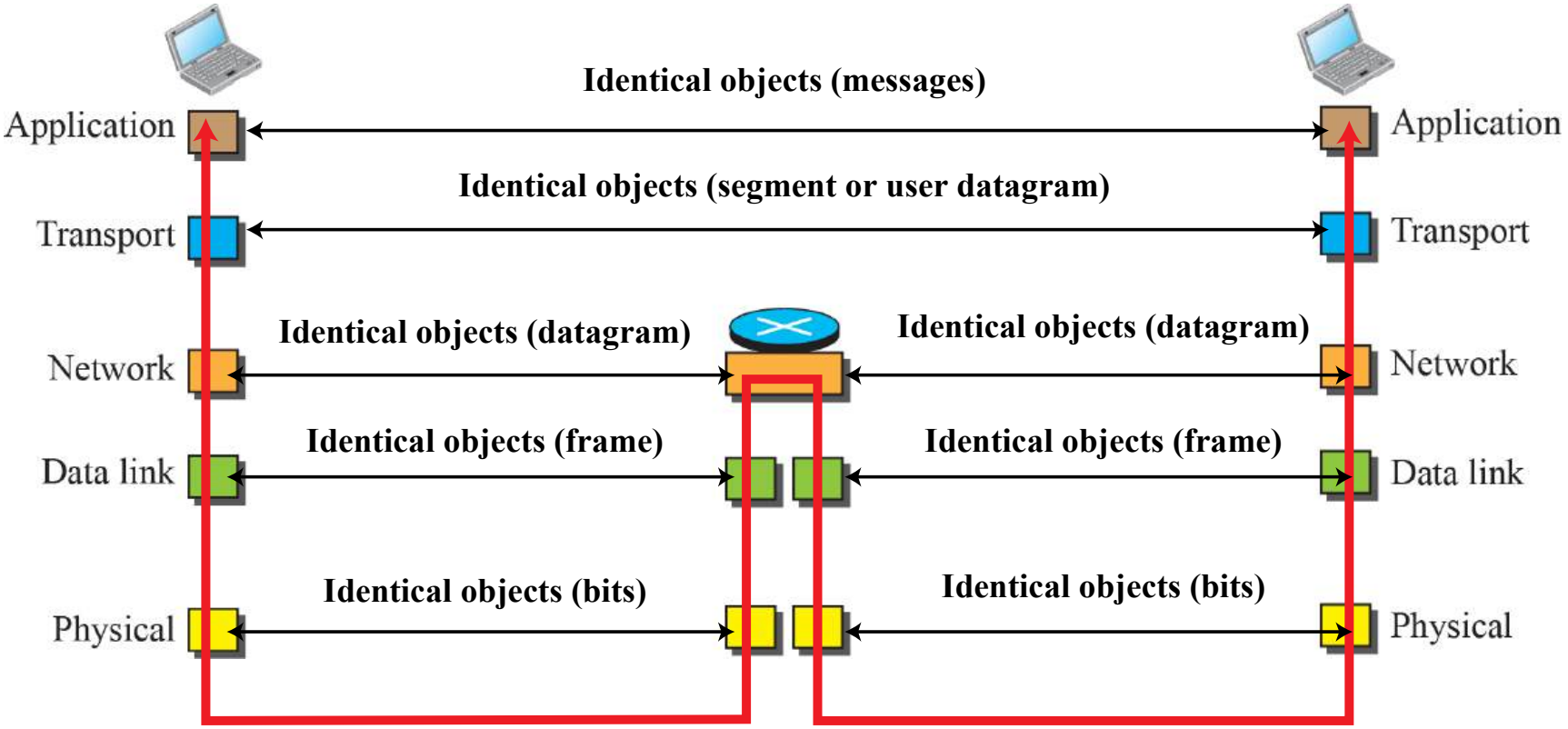

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Figure 1.16: Encapsulation / Decapsulation

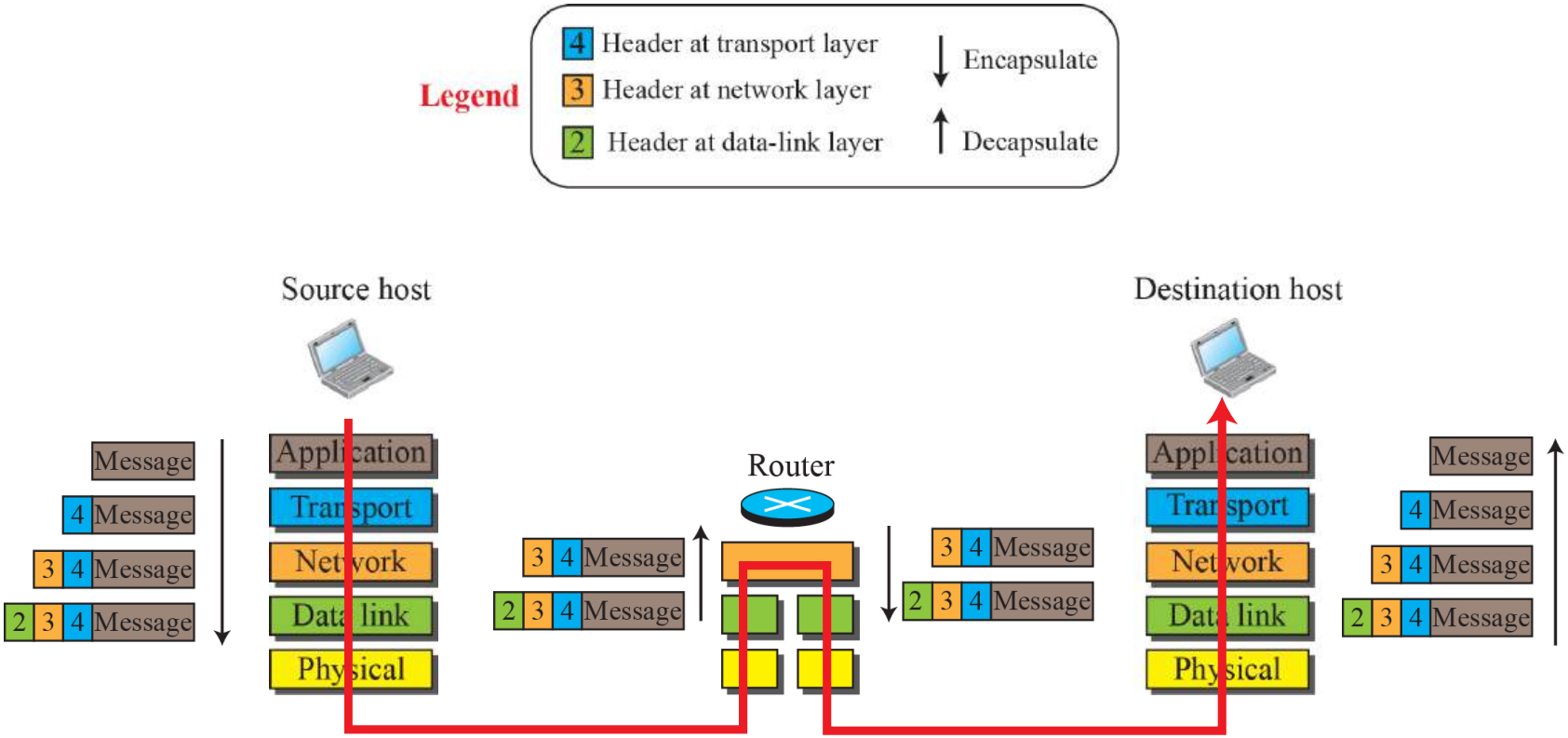

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Figure 1.17: Addressing in the TCP/IP protocol suite

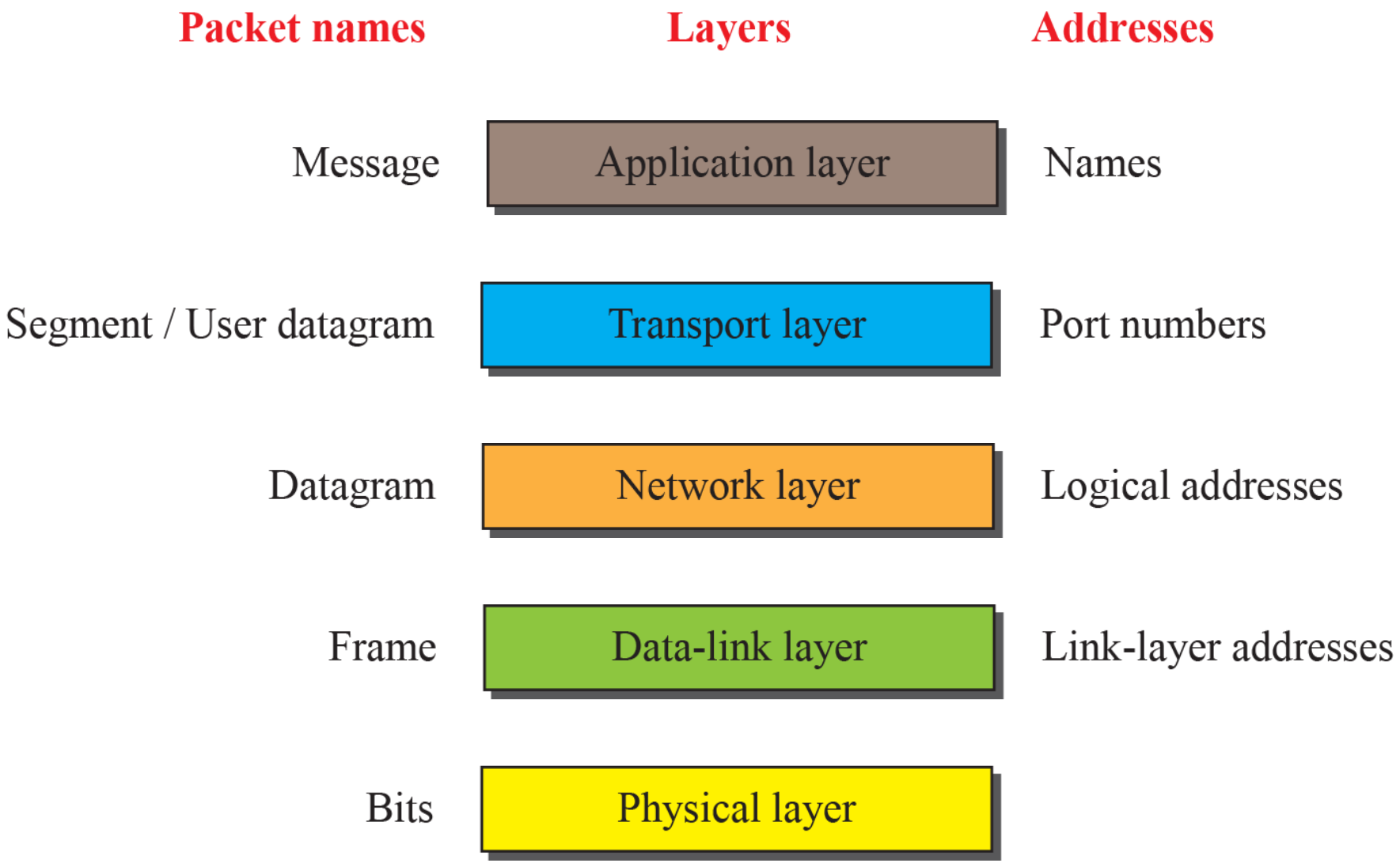

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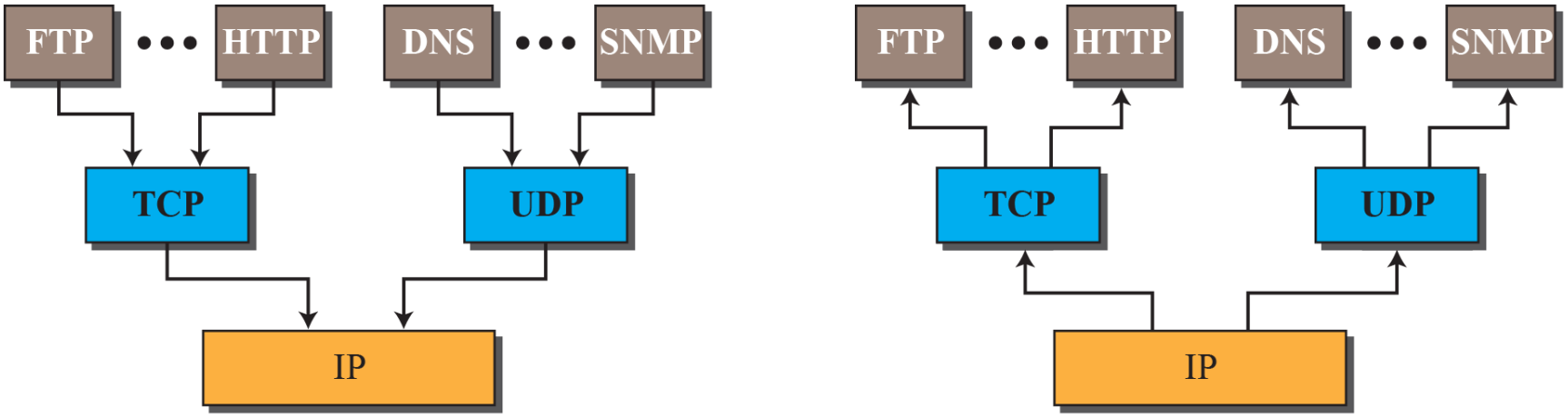
Figure 1.18: Multiplexing and demultiplexing



a. Multiplexing at source

b. Demultiplexing at destination

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1.2.2 The OSI Model

Established in 1947, ISO is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late 1970s.

- ***OSI versus TCP/IP***
- ***Lack of OSI Model's Success***

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Figure 1.19: The OSI model

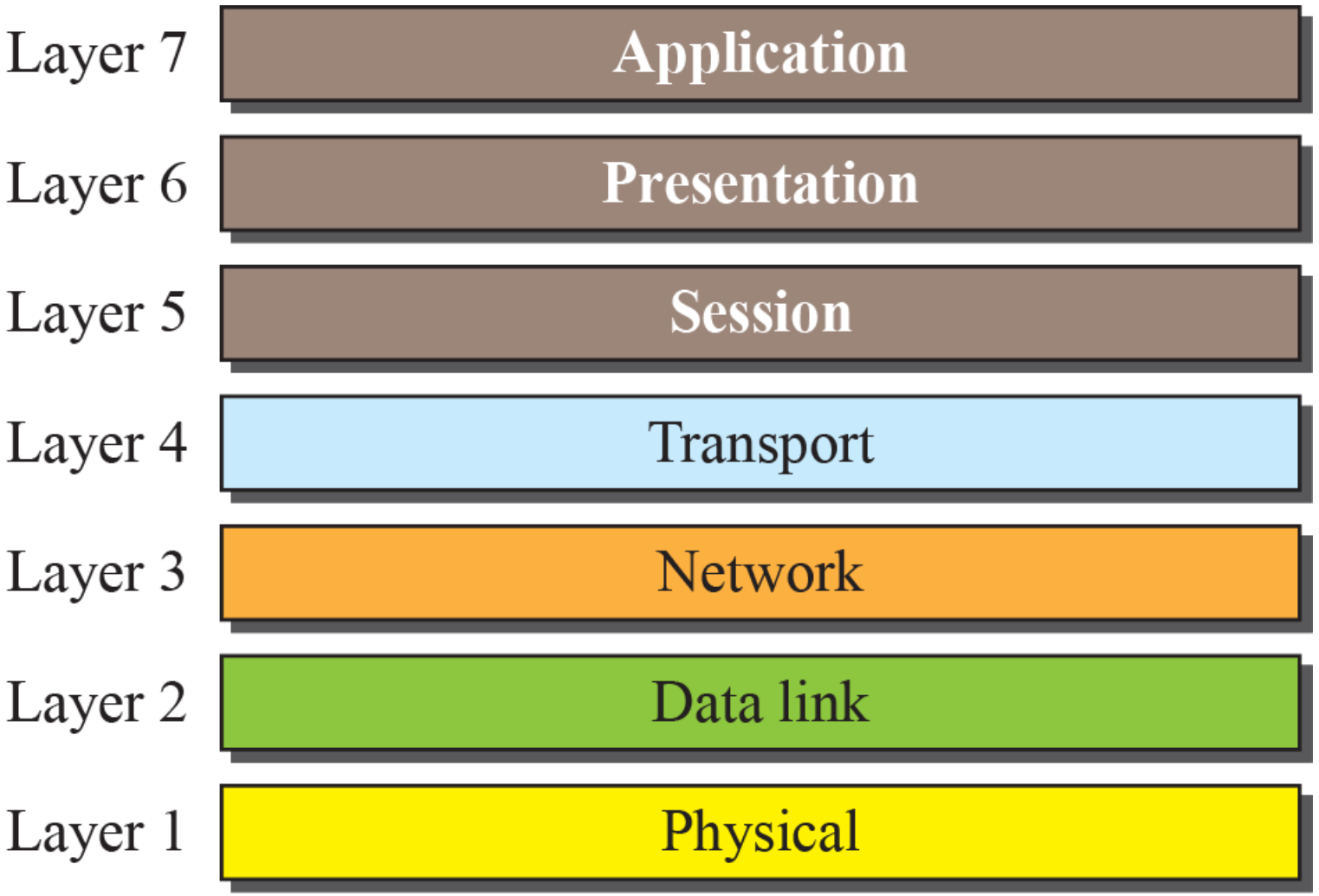

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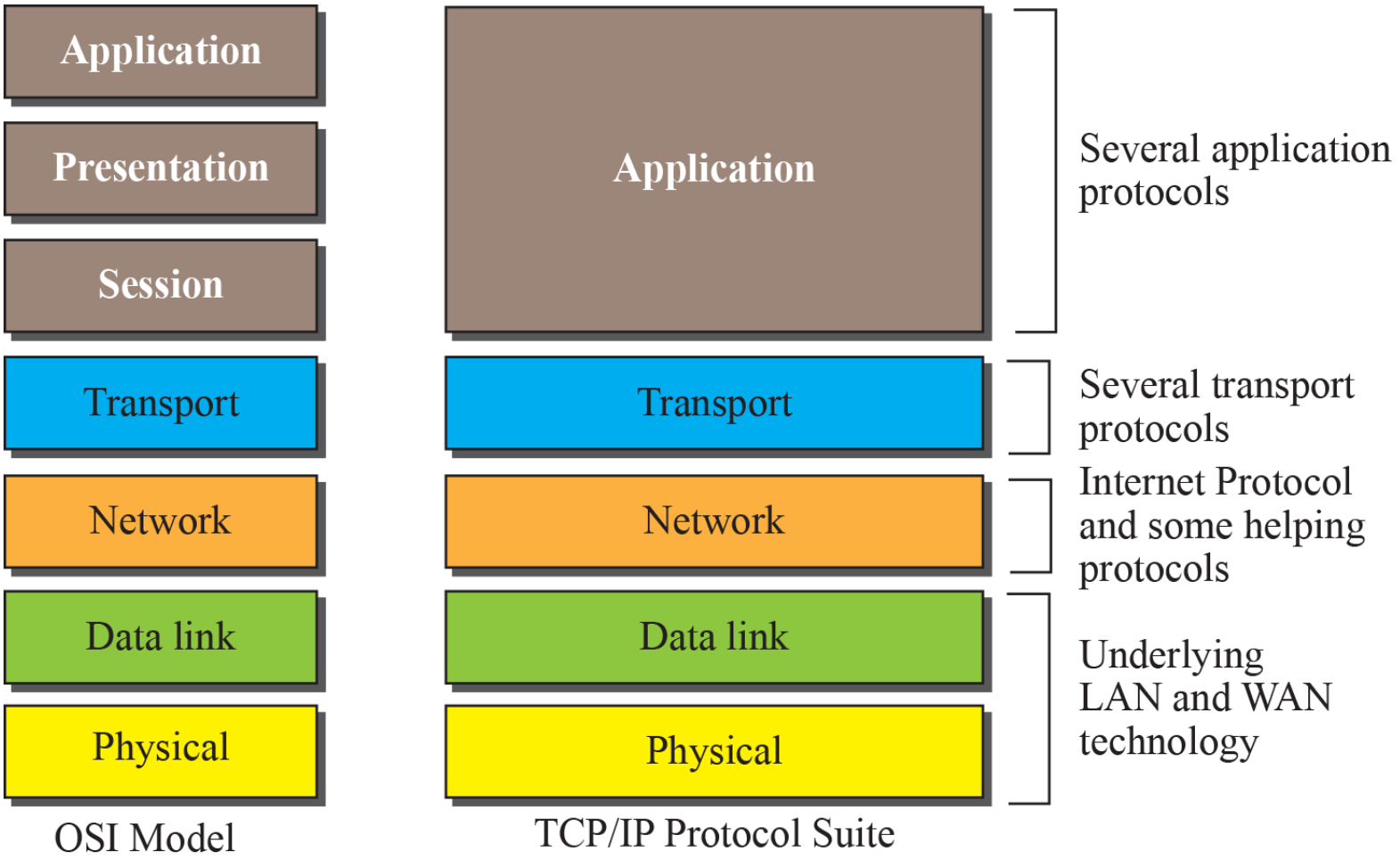
Figure 1.20: TCP/IP and OSI model



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1-3 INTERNET HISTORY

Now that we have given an overview of the Internet and its protocol, let us give a brief history of the Internet. This brief history makes it clear how the Internet has evolved from a private network to a global one in less than forty years.

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1.3.2 Early History

There were some communication networks, such as telegraph and telephone networks, before 1960. These networks were suitable for constant-rate communication at that time, which means that after a connection was made between two users, the encoded message (telegraphy) or voice (telephony) could be exchanged.

- ***Birth of Packet-Switched Networks***
- ***ARPANET***

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1.3.3 Birth of the Internet

In 1972, Vint Cerf and Bob Kahn, both of whom were part of the core ARPANET group, collaborated on what they called the Internetting Project. They wanted to link dissimilar networks so that a host on one network could communicate with a host on another. There were many problems to overcome: diverse packet sizes, diverse interfaces, and diverse transmission rates, as well as differing reliability requirements.

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1.3.3 Birth of the Internet (continued)

- TCP/IP
- MILNET
- CSNET
- NSFNET
- ANSNET

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1.3.3 Internet Today

Today, we witness a rapid growth both in the infrastructure and new applications. The Internet today is a set of pier networks that provide services to the whole world. What has made the Internet so popular is the invention of new applications.

- ***World Wide Web***
- ***Multimedia***
- ***Peer-to-Peer Applications***

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1-4 STANDARDS AND ADMINISTRATION

In the discussion of the Internet and its protocol, we often see a reference to a standard or an administration entity. In this section, we introduce these standards and administration entities for those readers that are not familiar with them; the section can be skipped if the reader is familiar with them.

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1.4.1 Internet Standards

An Internet standard is a thoroughly tested specification that is useful to and adhered to by those who work with the Internet. It is a formalized regulation that must be followed. There is a strict procedure by which a specification attains Internet standard status. A specification begins as an Internet draft.

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1.4.1 Internet Standards (Continued)

- **Maturity Levels**
  - Proposed Standard
  - Draft Standard
  - Internet Standard
  - Historic
  - Experimental
  - Informational

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1.4.1 Internet Standards (Continued)

- **Requirement Levels**
  - Required
  - Recommended
  - Elective
  - Limited Use
  - Not Recommended

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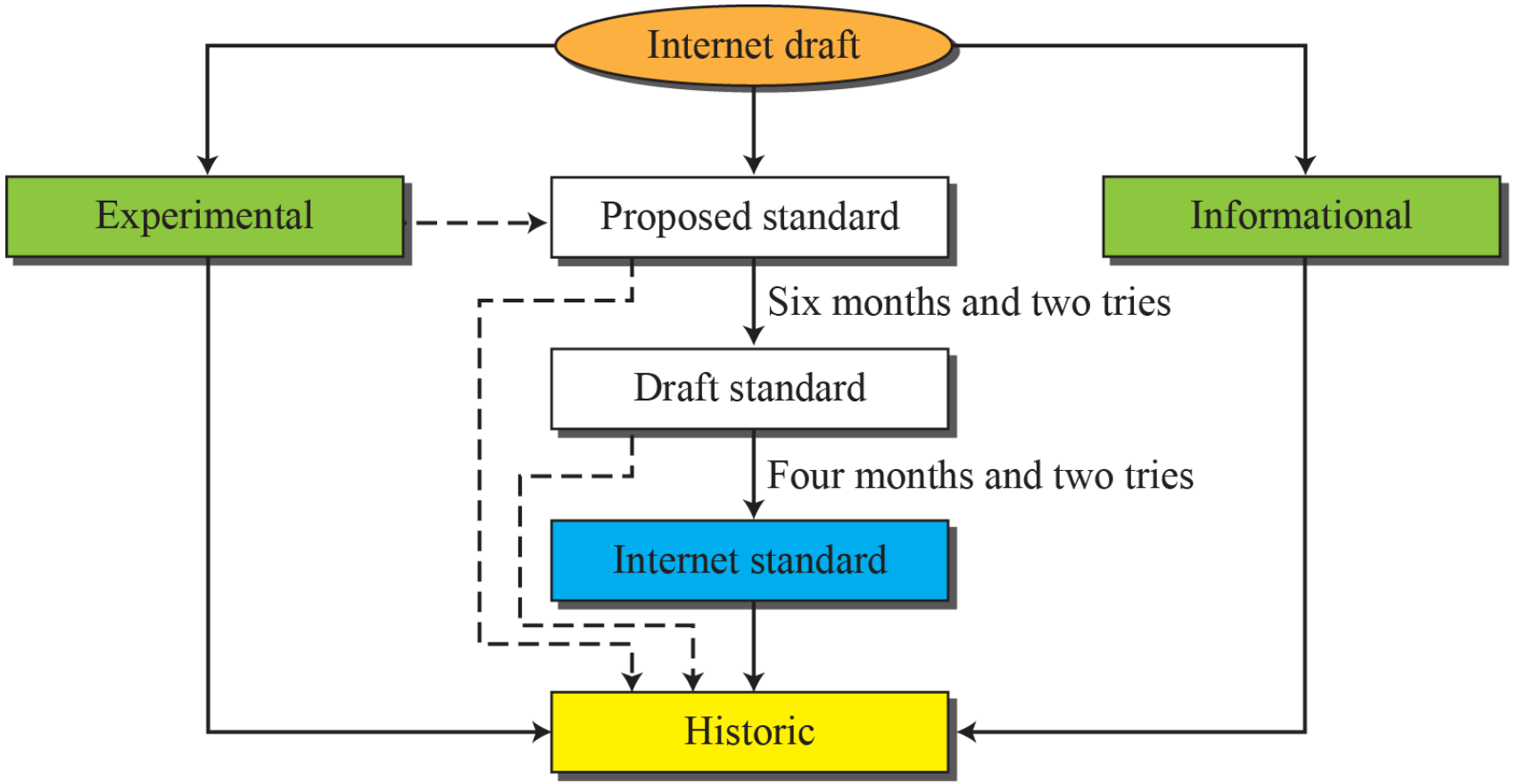
Figure 1.21: Maturity levels of an RFC



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1.4.2 Internet Administration

The Internet, with its roots primarily in the research domain, has evolved and gained a broader user base with significant commercial activity. Various groups that coordinate Internet issues have guided this growth and development. Appendix D gives the addresses, e-mail addresses, and telephone numbers for some of these groups.

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1.4.2 Internet Administration (continued)

- ISOC
- IAB
- IETF
- IANA and ICANN
- Network Information Center (NIC)

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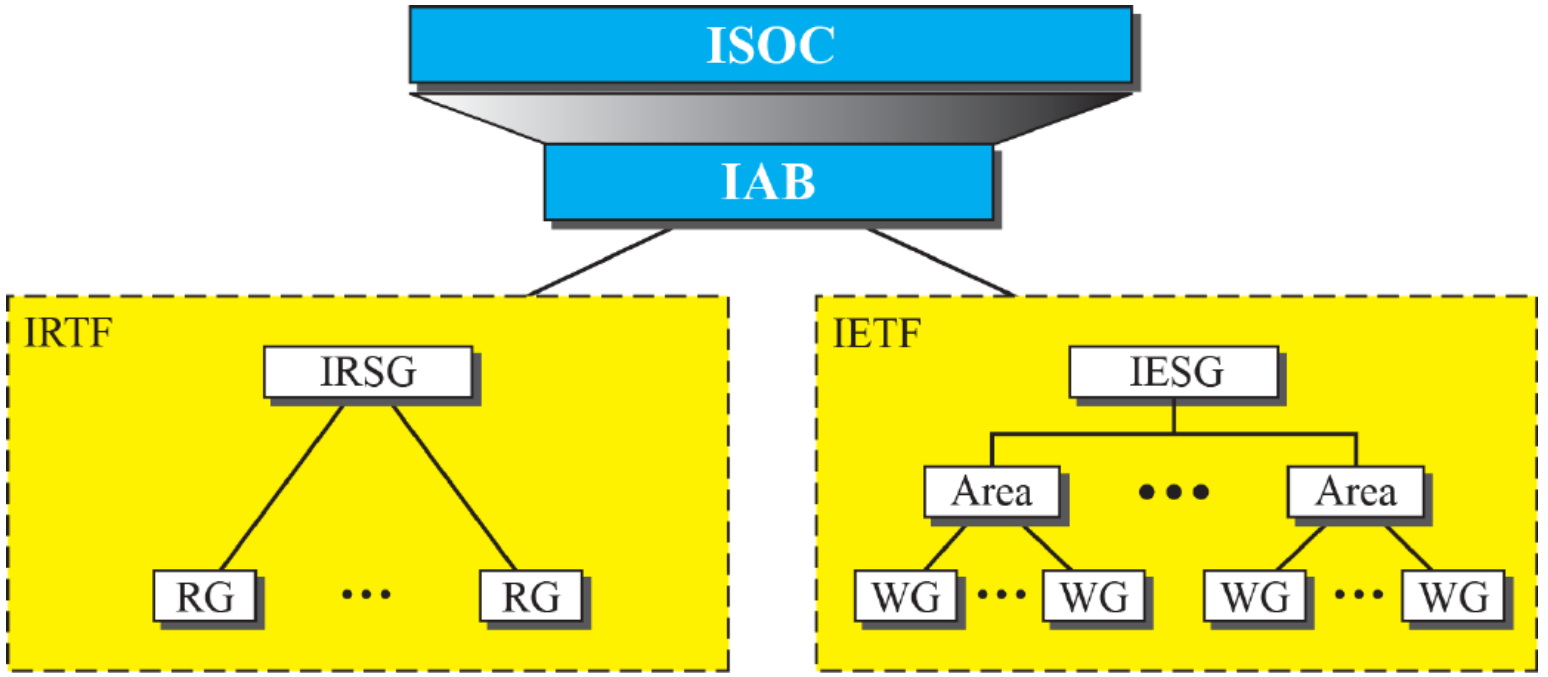
Figure 1.22: Internet administration



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Chapter 1: Summary

- *A network is a set of devices connected by communication links. The Internet today is made up of many wide and local area networks joined by connecting devices and switching stations. Most end users who want Internet connection use the services of Internet service providers (ISPs). There are backbone ISPs, regional ISPs, and local ISPs.*

- *A protocol is a set of rules that governs communication. In protocol layering, we need to follow two principles to provide bidirectional communication. First, each layer needs to perform two opposite tasks. Second, two objects under each layer at both sides should be identical. TCP/IP is a hierarchical protocol suite made of five layers: application, transport, network, data-link, and physical.*

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Chapter 1: Summary (continued)

- *The history of internetworking started with ARPA in the mid-1960s. The birth of the Internet can be associated with the work of Cerf and Kahn and the invention of a gateway to connect networks. The Internet administration has evolved with the Internet. We discussed ISOC, IAB, IETF, IRTF, ICANN, and NIC.*

- *An Internet standard is a thoroughly tested specification. An Internet draft is a working document with no official status and a six-month lifetime. A draft may be published as a Request for Comment (RFC). RFCs go through maturity levels and are categorized according to their requirement level.*

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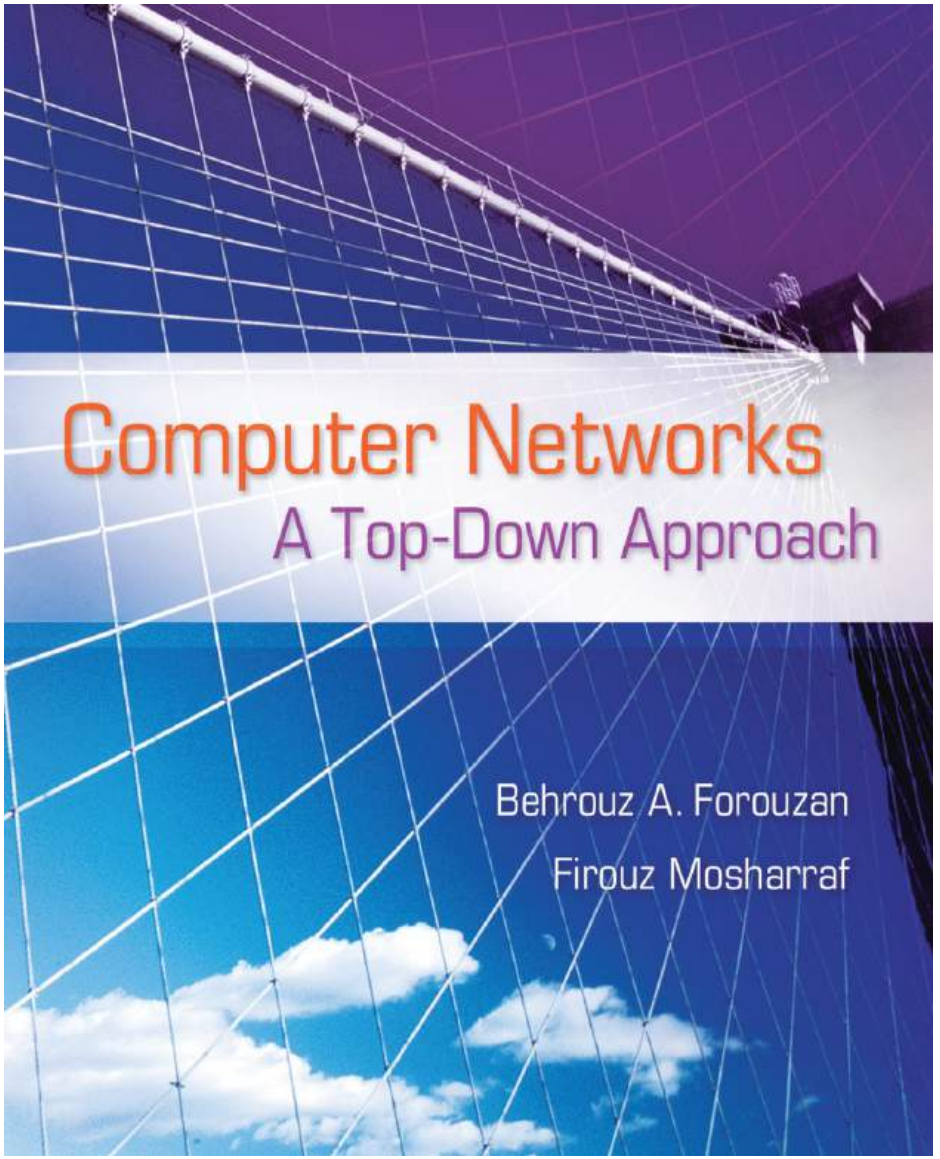
Chapter 2

Application Layer



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Chapter 2: Outline

2.1 INTRODUCTION

2.2 CLIENT-SERVER PARADIGM

2.3 STANDARD APPLICATIONS

2.4 PEER-TO-PEER PARADIGM

2.5 SOCKET-INTERFACE PROGRAMMING

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Chapter 2: Objective

- *We introduce the nature of services provided by the Internet: the client-server paradigm and the peer-to-peer paradigm.*
- *We discuss the concept of the client-server paradigm.*
- *We discuss some predefined or standard applications based on the client-server paradigm such as surfing the Web, file transfer, e-mail, and so on.*
- *We discuss the concept and protocols in the peer-to-peer paradigm such as Chord, Pastry, and Kademlia.*
- *We show how a new application can be created in the client-server paradigm by writing two programs in the C language.*

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2-1 INTRODUCTION

The application layer provides services to the user. Communication is provided using a logical connection, which means that the two application layers assume that there is an imaginary direct connection through which they can send and receive messages. Figure 2.1 shows the idea behind this logical connection.

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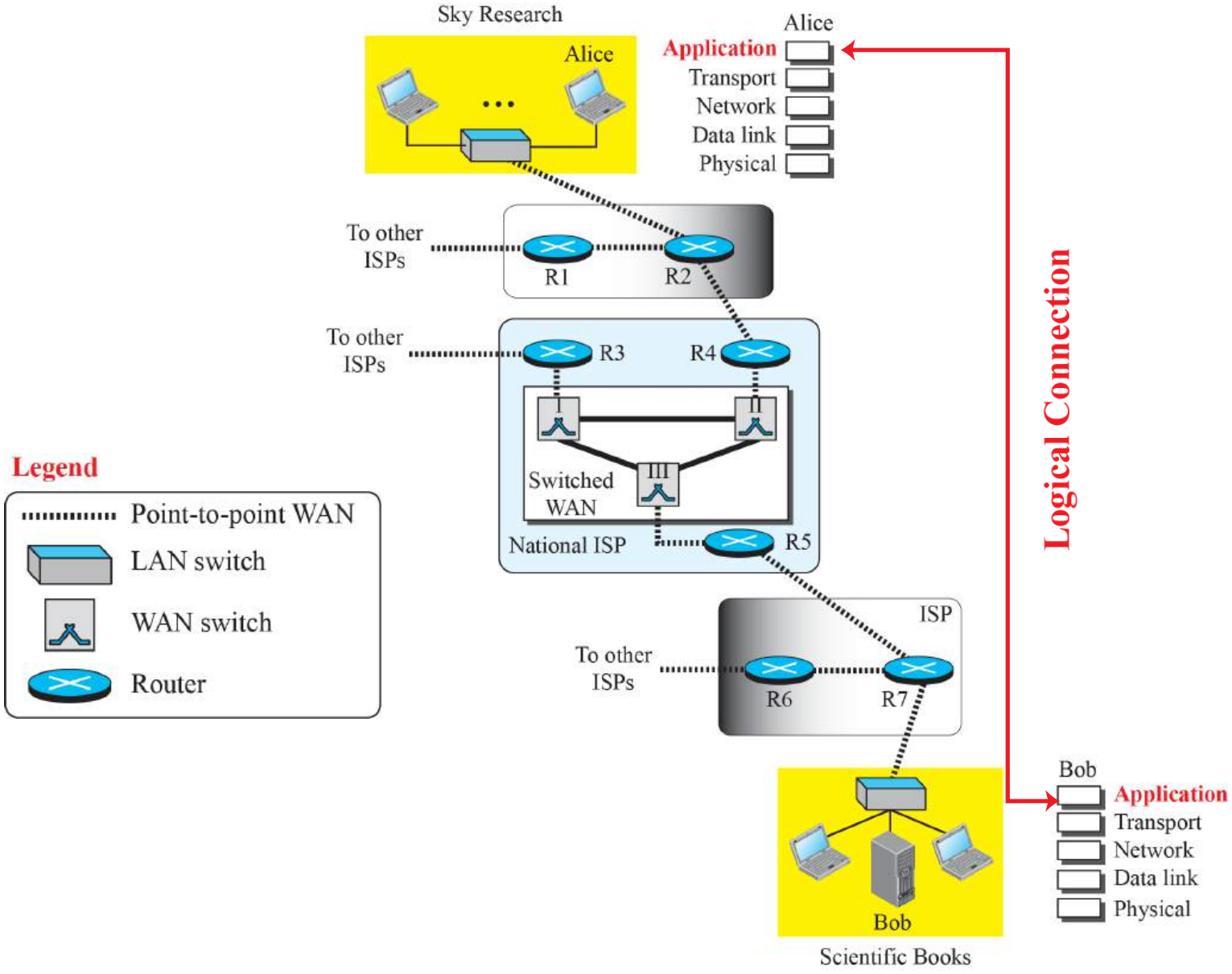
Figure 2.1: Logical connection at the application layer



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2.1.1 Providing Services

The Internet was originally designed to provide service to users around the world. Since the application layer is the only layer that provides services to the Internet user, it allows new application protocols to be easily added to the Internet, which has been occurring during the lifetime of the Internet. When the Internet was created, only a few application protocols were available to the users; today we cannot give a number for these protocols because new ones are being added constantly.

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2.1.1 Providing Services (Cont.)

- **Standard and Nonstandard Protocols**
  - Standard Application-Layer Protocols
  - Nonstandard Application-Layer Protocols

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2.1.2 Application-Layer Paradigm

It should be clear that to use the Internet we need two application programs to interact with each other: one running on a computer somewhere in the world, the other running on another computer somewhere else in the world. The two programs need to send messages to each other through the Internet infrastructure. However, we have not discussed what the relationship should be between these programs. Should both application programs be able to request services and provide services, or should the application programs just do one or the other?

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2.1.2 Application-Layer Paradigm (cont)

- ***Traditional Paradigm: Client-Server***
- ***New Paradigm: Peer-to-Peer***
- ***Mixed Paradigm***

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Figure 2.2: Example of a client-server paradigm

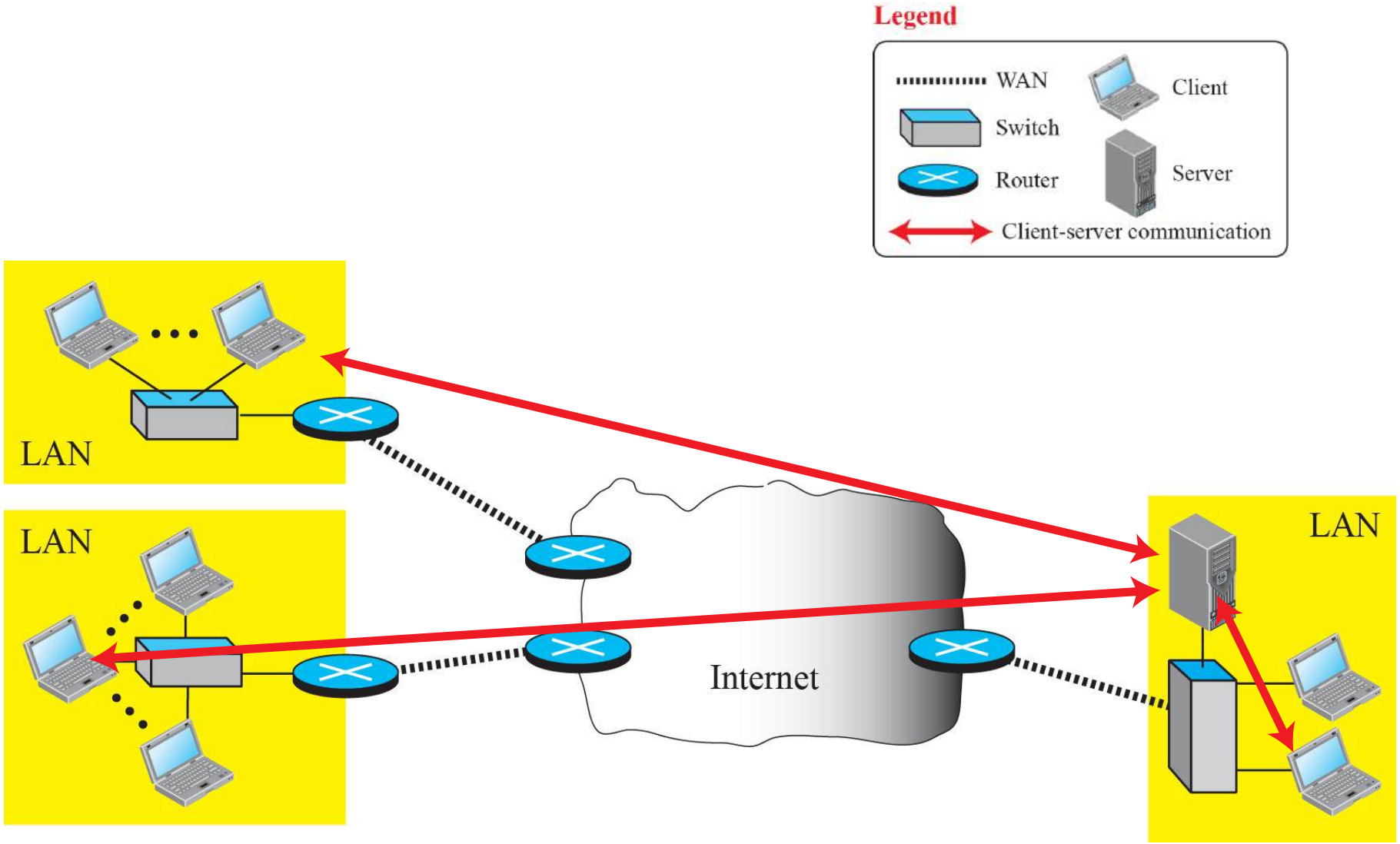

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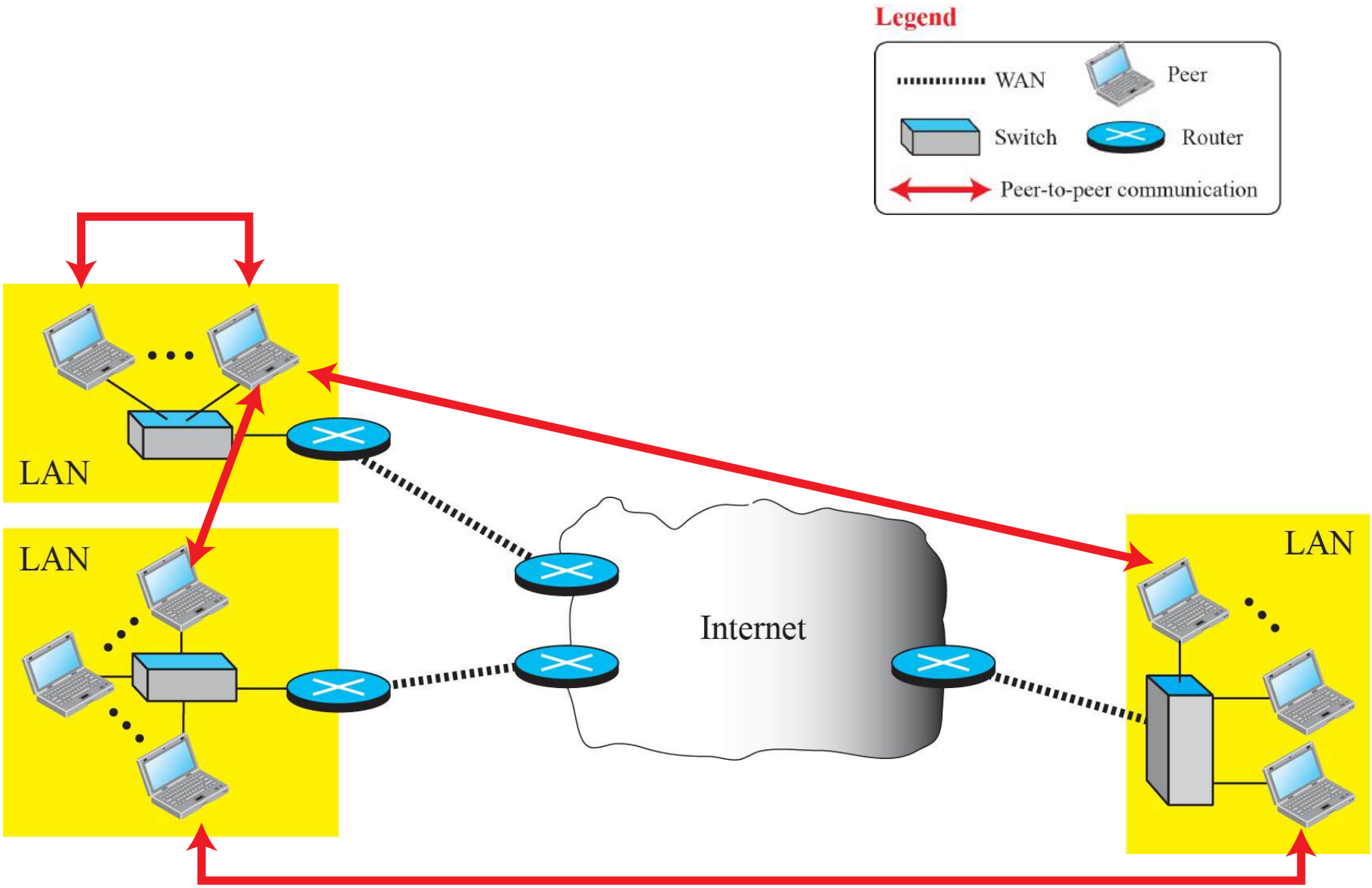
Figure 2.3: Example of a peer-to-peer paradigm



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2-2 CLIENT-SERVER PARADIGM

In this paradigm, communication at the application layer is between two running application programs called processes: a client and a server. A client is a running program that initializes the communication by sending a request; a server is another application program that waits for a request from a client.

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2.2.1 Application Programming Interface

A computer language has a set of instructions for mathematical operations, a set of instructions for string manipulation, a set of instructions for input/output access, and so on. If we need a process to be able to communicate with another process, we need a new set of instructions to tell the lowest four layers of the TCP/IP suite to open the connection, send and receive data from the other end, and close the connection. A set of instructions of this kind is normally referred to as Application Programming Interface (API).

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2.2.1 (continued)

- **Sockets**
- **Socket Addresses**
- **Finding Socket Addresses**
  - Server Site
  - Client Site

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Figure 2.4: Position of the socket interface

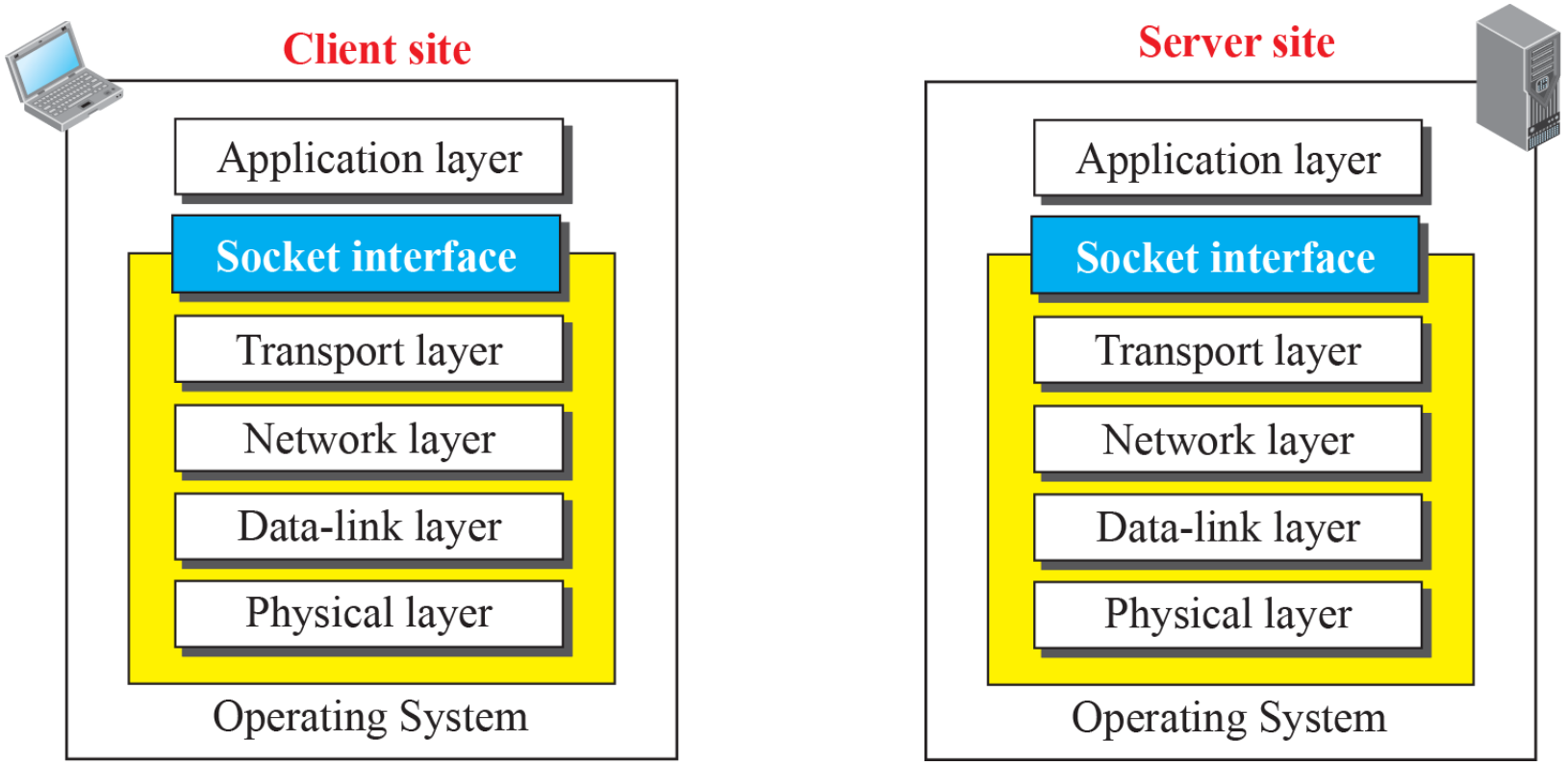

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Figure 2.5: ***A Sockets used like other sources and sinks***

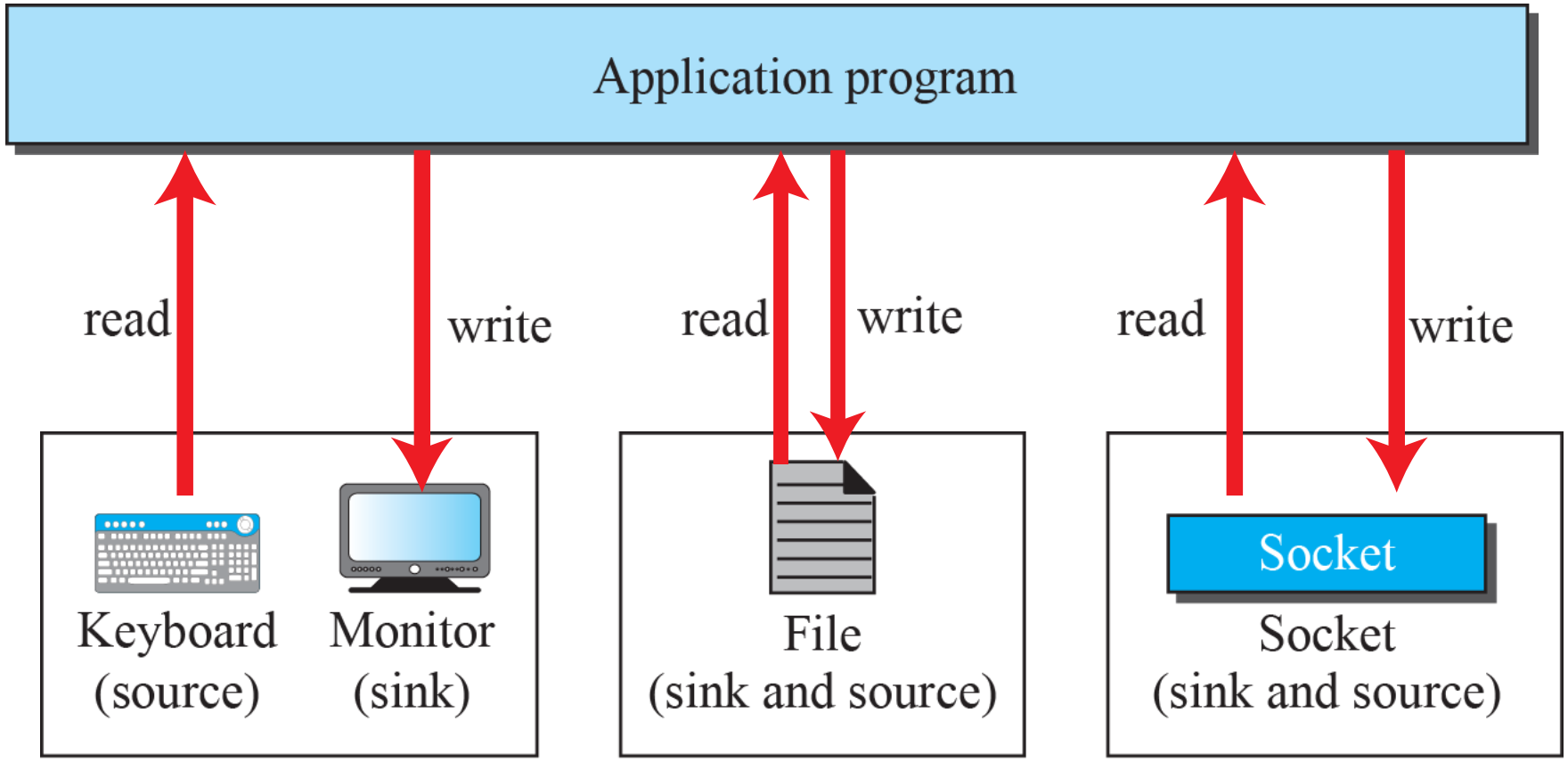

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Figure 2.6: Use of sockets in process-to-process communication

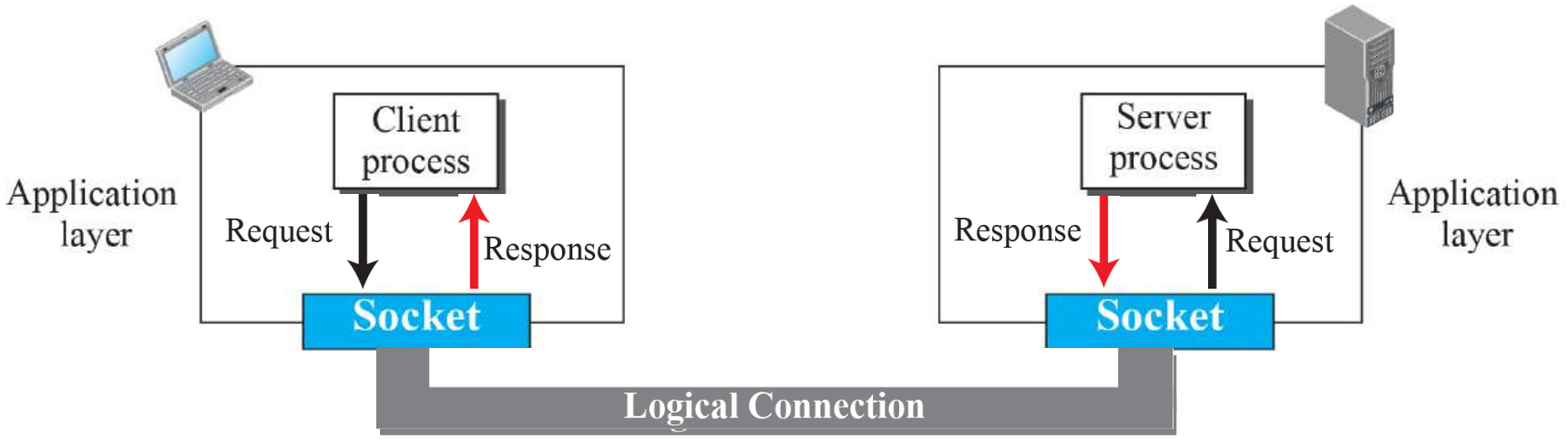

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Figure 2.7: A socket address

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Example 2.1

We can find a two-level address in telephone communication. A telephone number can define an organization, and an extension can define a specific connection in that organization. The telephone number in this case is similar to the IP address, which defines the whole organization; the extension is similar to the port number, which defines the particular connection.

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2.2.2 Using Services of Transport Layer

A pair of processes provide services to the users of the Internet, human or programs. A pair of processes, however, need to use the services provided by the transport layer for communication because there is no physical communication at the application layer. There are three common transport layer protocols in the TCP/IP suite: UDP, TCP, and SCTP.

- **UDP Protocol**
- **TCP Protocol**
- **SCTP Protocol**

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2-3 STANDARD CLIENT-SERVER APPLICATIONS

During the lifetime of the Internet, several application programs have been developed. We do not have to redefine them, but we need to understand what they do. For each application, we also need to know the options available to us. The study of these applications can help us to create customized applications in the future.

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2.3.1 World Wide Web and HTTP

In this section, we first introduce the World Wide Web (abbreviated WWW or Web). We then discuss the Hyper Text Transfer Protocol (HTTP), the most common client-server application program used in relation to the Web.

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2.3.1 (continued)

- **World Wide Web**
  - Architecture
  - Uniform Resource Locator (URL)
  - Web Documents
- **HyperText Transfer Protocol (HTTP)**
  - Nonpersistent versus Persistent Connections
  - Message Formats
  - Conditional Request
  - Cookies

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2.3.1 (continued)

- **Web Caching: Proxy Server**
  - Proxy Server Location
  - Cache Update
- **HTTP Security**

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Example 2.2

Assume we need to retrieve a scientific document that contains one reference to another text file and one reference to a large image. Figure 2.8 shows the situation.

The main document and the image are stored in two separate files in the same site (file A and file B); the referenced text file is stored in another site (file C). Since we are dealing with three different files, we need three transactions if we want to see the whole document.

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Figure 2.8: Example 2.2 (Retrieving two files and one image

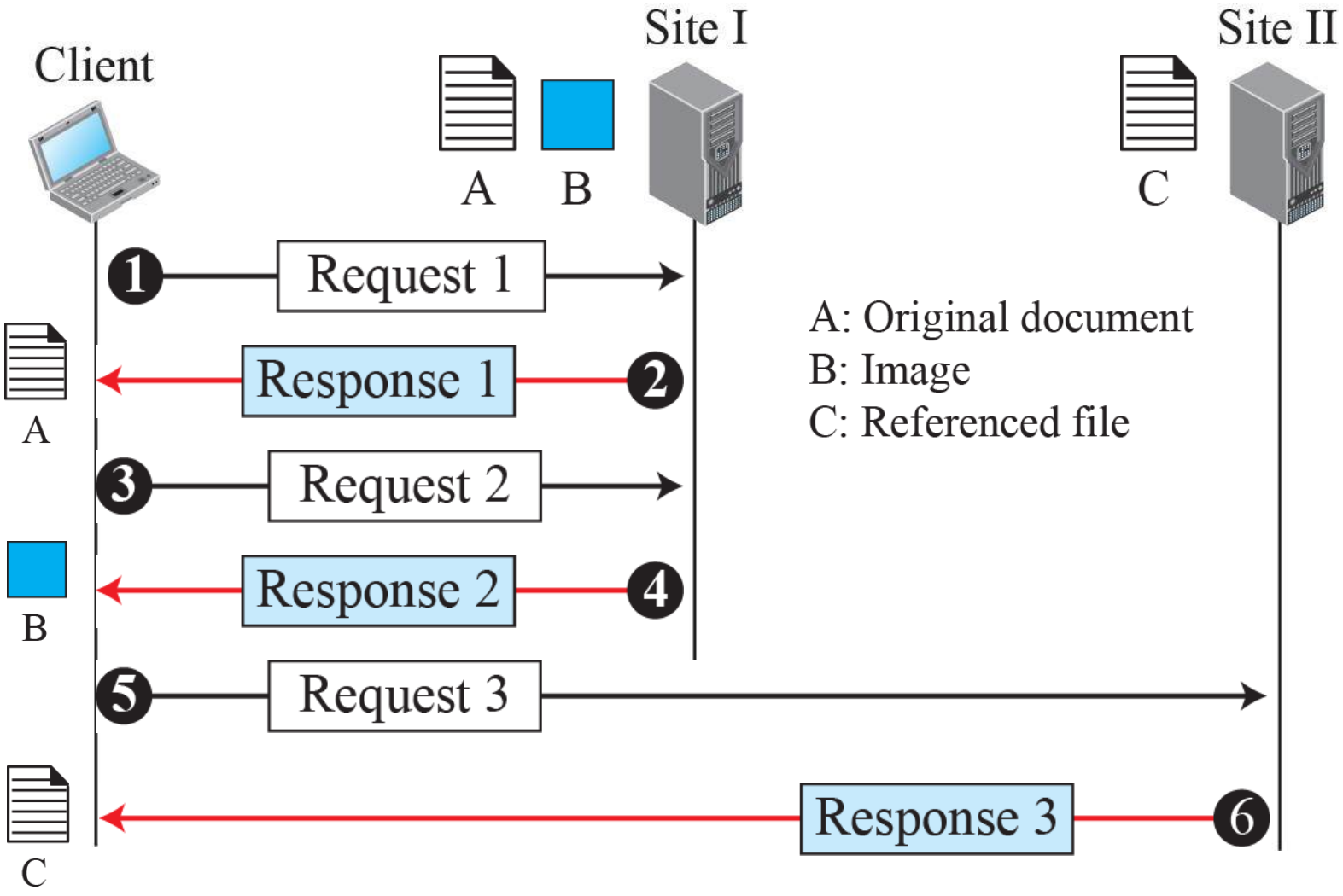

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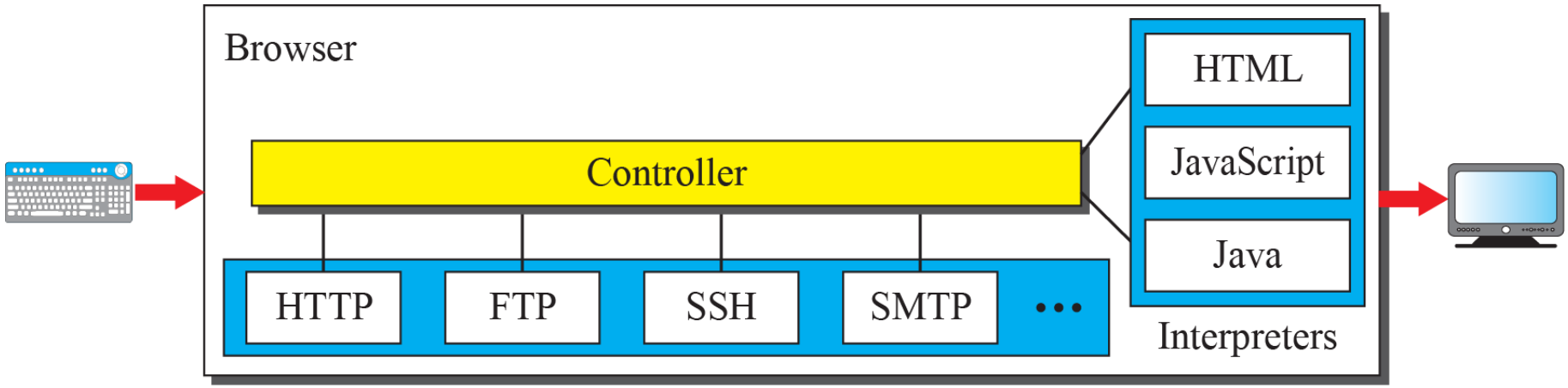
Figure 2.9: Browser



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Example 2.3

The URL **http://www.mhhe.com/compsci/forouzan/** defines the web page related to one of the of the computer in the McGraw-Hill company (the three letters www are part of the host name and are added to the commercial host). The path is *compsci/forouzan/*, which defines Forouzan's web page under the directory *compsci* (computer science).

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Example 2.4

Figure 2.10 shows an example of a *nonpersistent* connection. The client needs to access a file that contains one link to an image. The text file and image are located on the same server. Here we need two connections. For each connection, TCP requires at least three handshake messages to establish the connection, but the request can be sent with the third one. After the connection is established, the object can be transferred. After receiving an object, another three handshake messages are needed to terminate the connection, as we will see in Chapter 3.

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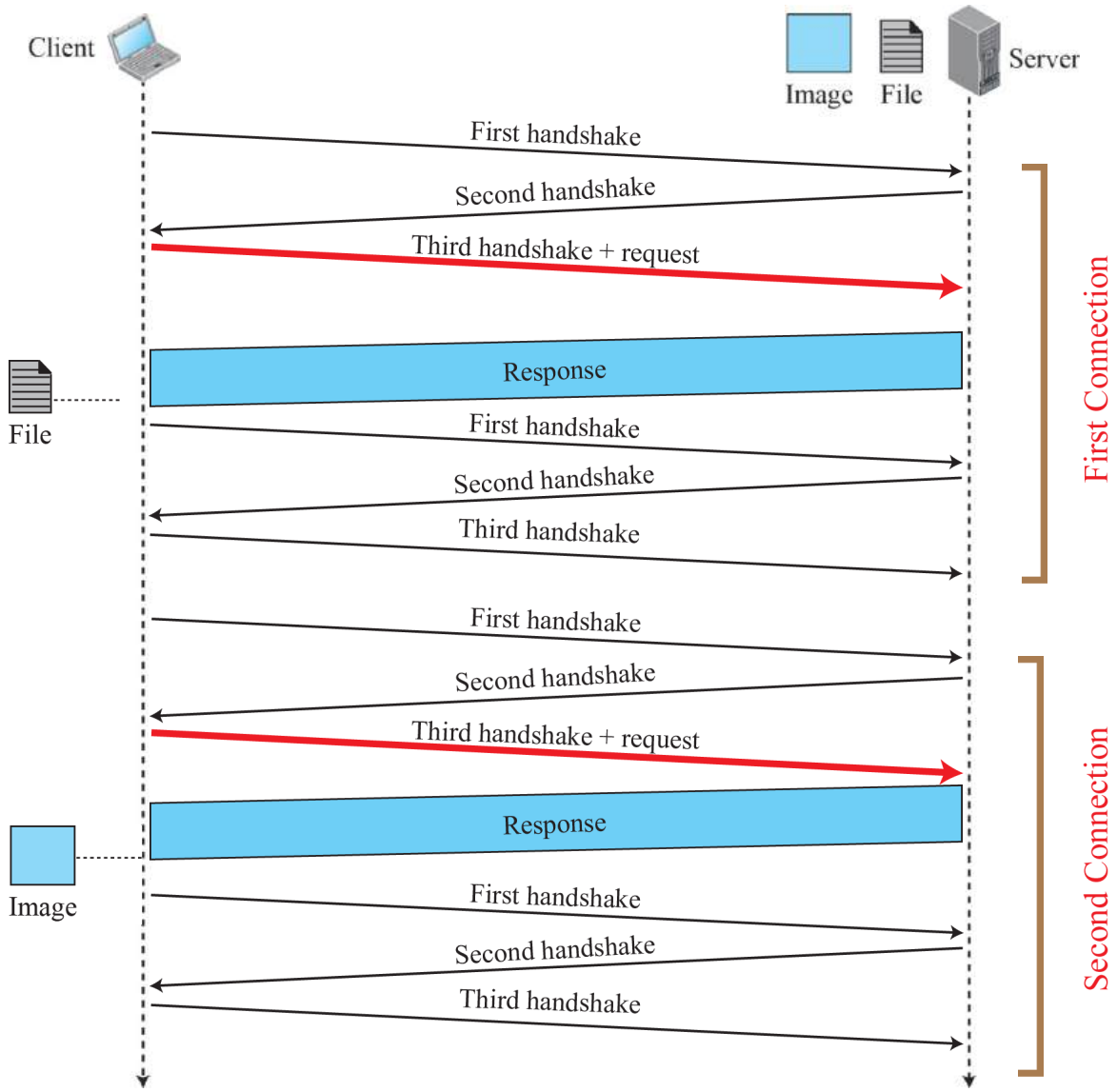
Figure 2.10: Example 2.4



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Example 2.5

Figure 2.11 shows the same scenario as in Example 2.4, but using a persistent connection. Only one connection establishment and connection termination is used, but the request for the image is sent separately.

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Figure 2.11: Example 2.5

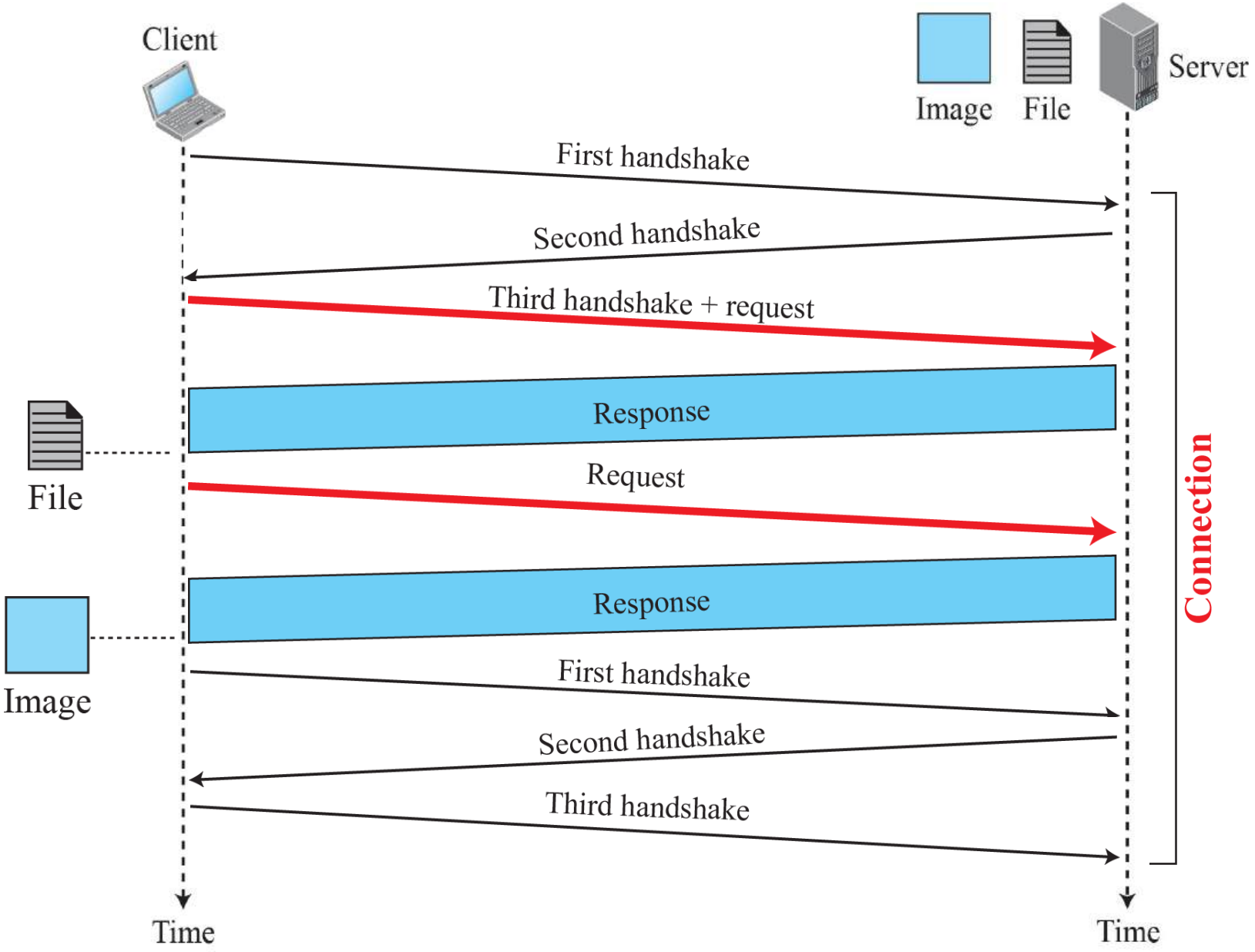

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Figure 2.12: Formats of the request and response messages

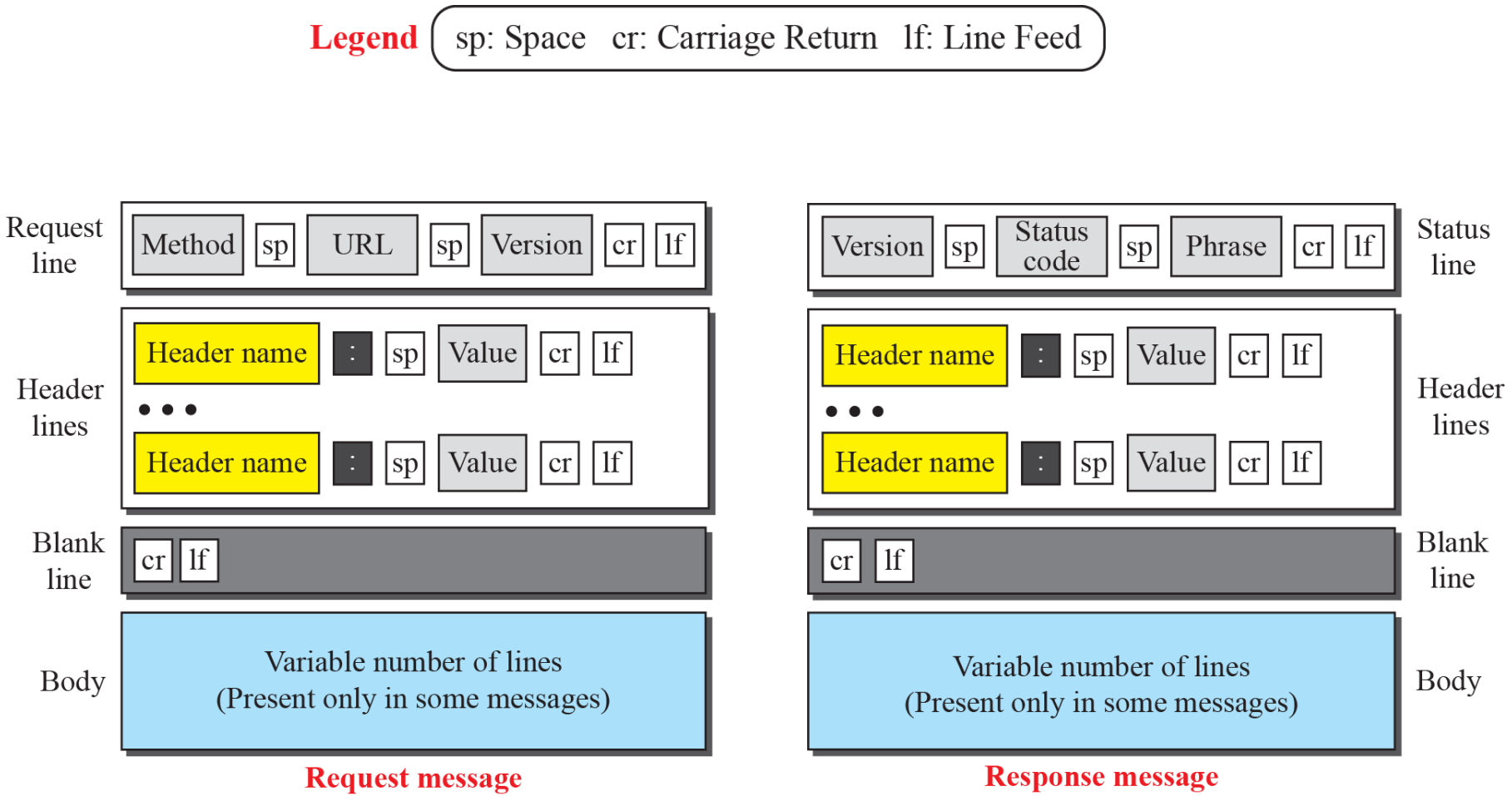

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Table 2.1: Methods

Method	*Action*
GET	Requests a document from the server
HEAD	Requests information about a document but not the document itself
PUT	Sends a document from the client to the server
POST	Sends some information from the client to the server
TRACE	Echoes the incoming request
DELETE	Removes the web page
CONNECT	Reserved
OPTIONS	Inquires about available options

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Table 2.2: Request Header Names

Header	*Description*
User-agent	Identifies the client program
Accept	Shows the media format the client can accept
Accept-charset	Shows the character set the client can handle
Accept-encoding	Shows the encoding scheme the client can handle
Accept-language	Shows the language the client can accept
Authorization	Shows what permissions the client has
Host	Shows the host and port number of the client
Date	Shows the current date
Upgrade	Specifies the preferred communication protocol
Cookie	Returns the cookie to the server (explained later)
If-Modified-Since	If the file is modified since a specific date

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Table 2.3: Response Header Names

Header	*Description*
Date	Shows the current date
Upgrade	Specifies the preferred communication protocol
Server	Gives information about the server
Set-Cookie	The server asks the client to save a cookie
Content-Encoding	Specifies the encoding scheme
Content-Language	Specifies the language
Content-Length	Shows the length of the document
Content-Type	Specifies the media type
Location	To ask the client to send the request to another site
Accept-Ranges	The server will accept the requested byte-ranges
Last-modified	Gives the date and time of the last change

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Example 2.6

This example retrieves a document (see Figure 2.13). We use the GET method to retrieve an image with the path **/usr/bin/image1**. The request line shows the method (GET), the URL, and the HTTP version (1.1). The header has two lines that show that the client can accept images in the GIF or JPEG format. The request does not have a body. The response message contains the status line and four lines of header. The header lines define the date, server, content encoding (MIME version, which will be described in electronic mail), and length of the document. The body of the document follows the header..

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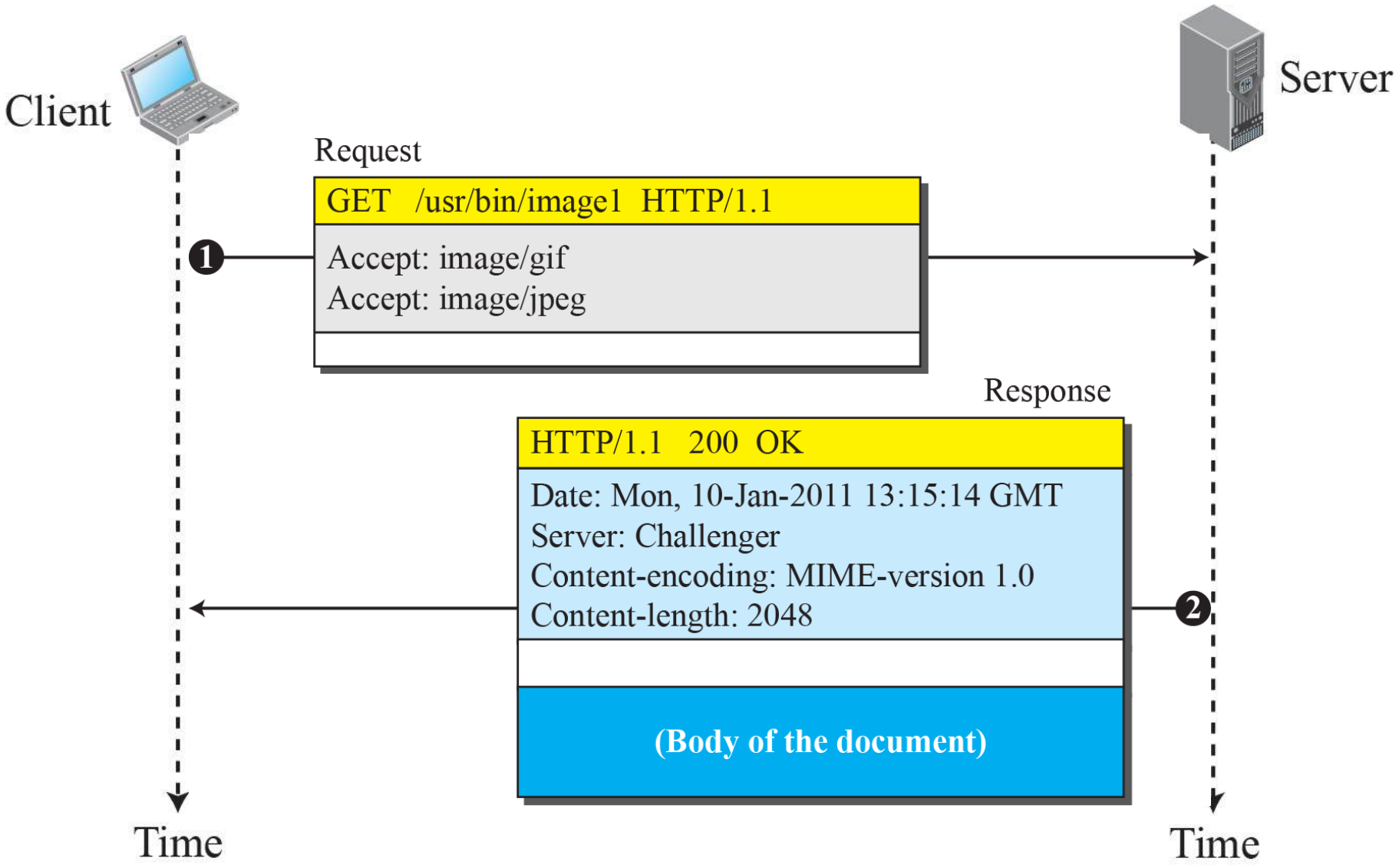
Figure 2.13: Example 2.6



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Example 2.7

In this example, the client wants to send a web page to be posted on the server. We use the PUT method. The request line shows the method (PUT), URL, and HTTP version (1.1). There are four lines of headers. The request body contains the web page to be posted. The response message contains the status line and four lines of headers. The created document, which is a CGI document, is included as the body (see Figure 2.14).

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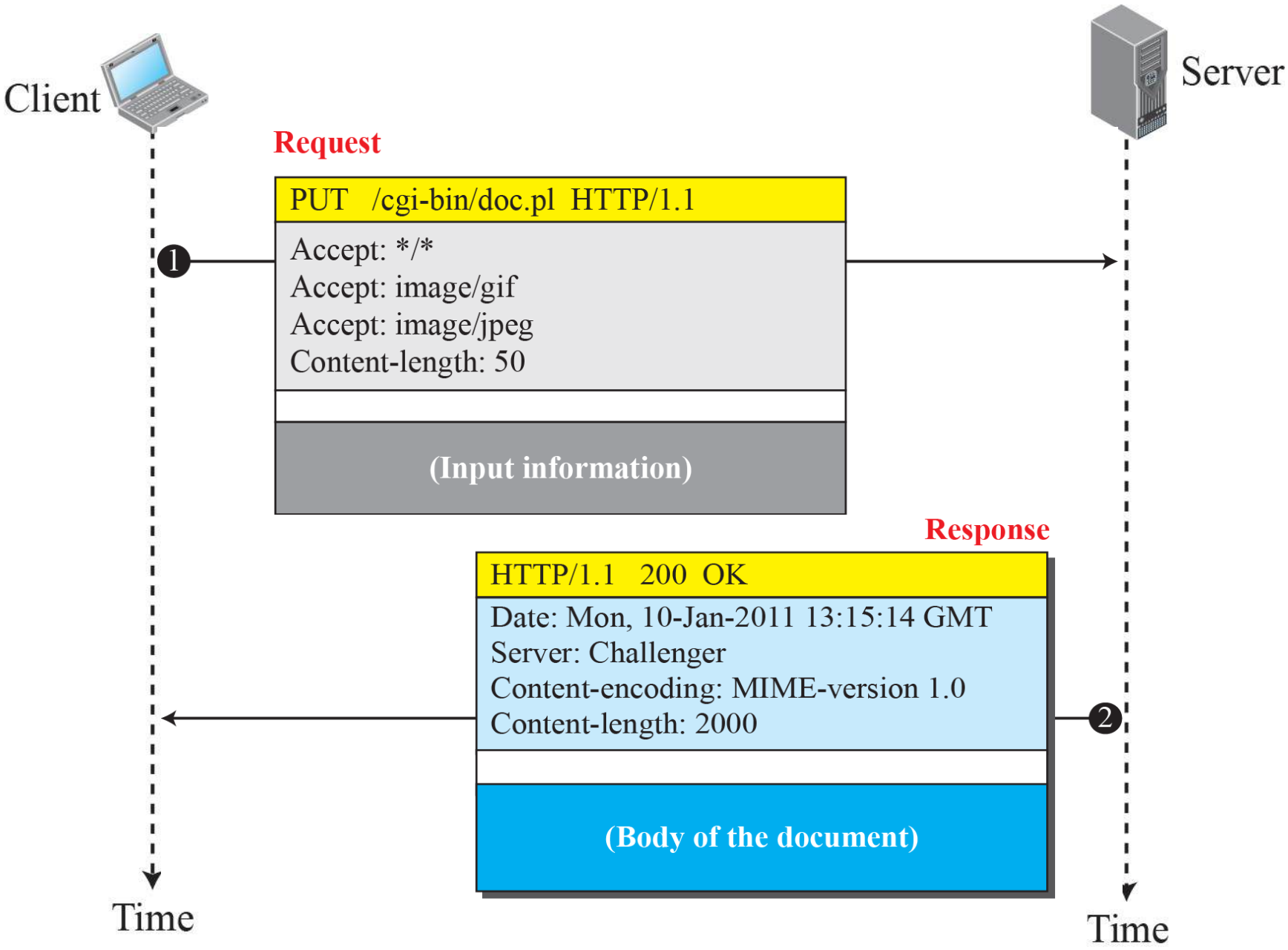
Figure 2.14: Example 2.7



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Example 2.8

The following shows how a client imposes the modification data and time condition on a request.

GET http://www.commonServer.com/information/file1 HTTP/1.1	Request line
If-Modified-Since: Thu, Sept 04 00:00:00 GMT	Header line
	Blank line

The status line in the response shows the file was not modified after the defined point in time. The body of the response message is also empty.

HTTP/1.1 304 Not Modified	Status line
Date: Sat, Sept 06 08 16:22:46 GMT	First header line
Server: commonServer.com	Second header line
	Blank line
(Empty Body)	Empty body

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Example 2.9

Figure 2.15 shows a scenario in which an electronic store can benefit from the use of cookies. Assume a shopper wants to buy a toy from an electronic store named BestToys. The shopper browser (client) sends a request to the BestToys server. The server creates an empty shopping cart (a list) for the client and assigns an ID to the cart (for example, 12343). The server then sends a response message, which contains the images of all toys available, with a link under each toy that selects the toy if it is being clicked. This response message also includes the Set-Cookie header line whose value is 12343. The client displays the images and stores the cookie value in a file named BestToys.

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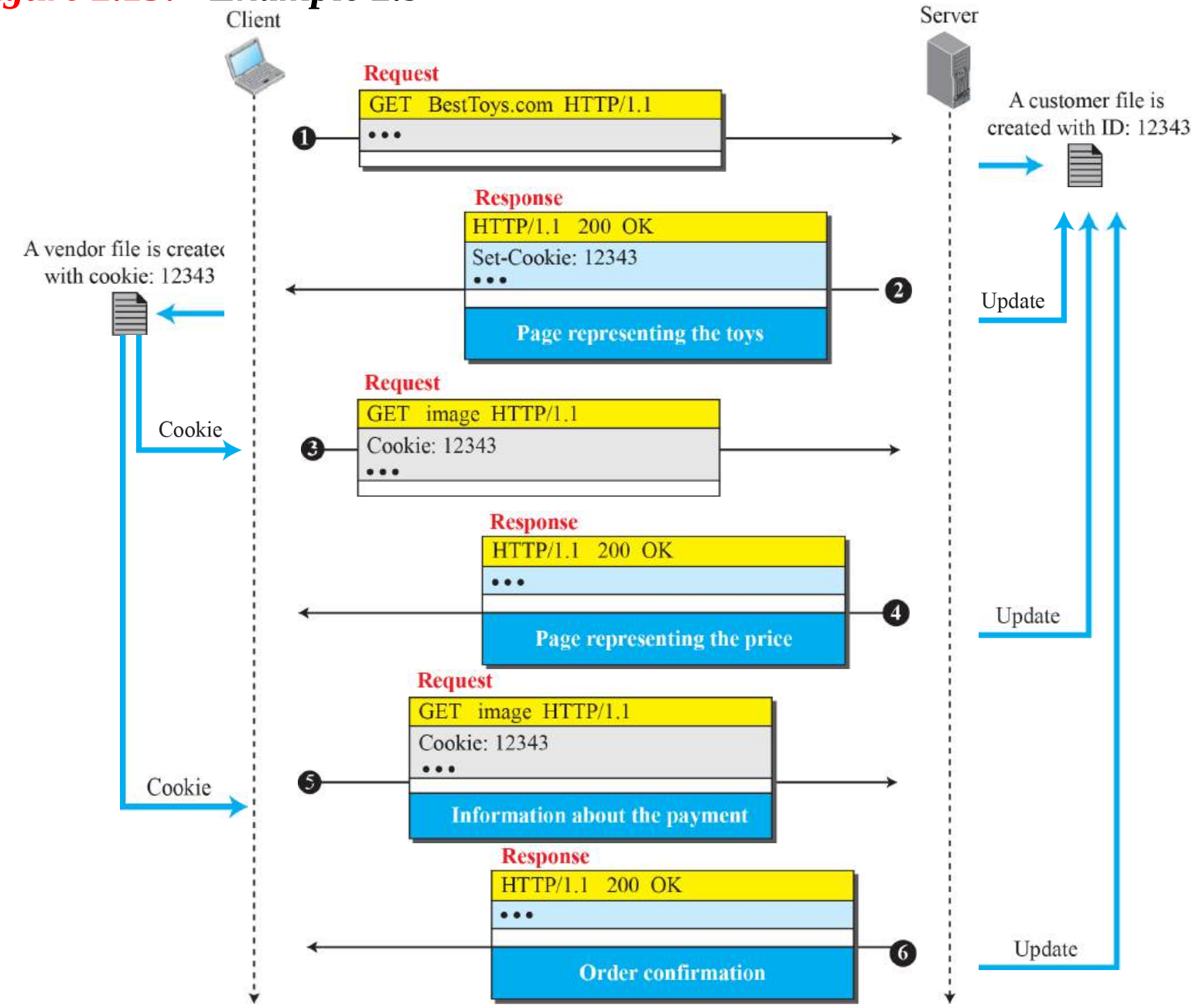
Figure 2.15: Example 2.9



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Example 2.10

Figure 2.16 shows an example of a use of a proxy server in a local network, such as the network on a campus or in a company. The proxy server is installed in the local network. When an HTTP request is created by any of the clients (browsers), the request is first directed to the proxy server If the proxy server already has the corresponding web page, it sends the response to the client. Otherwise, the proxy server acts as a client and sends the request to the web server in the Internet. When the response is returned, the proxy server makes a copy and stores it in its cache before sending it to the requesting client.

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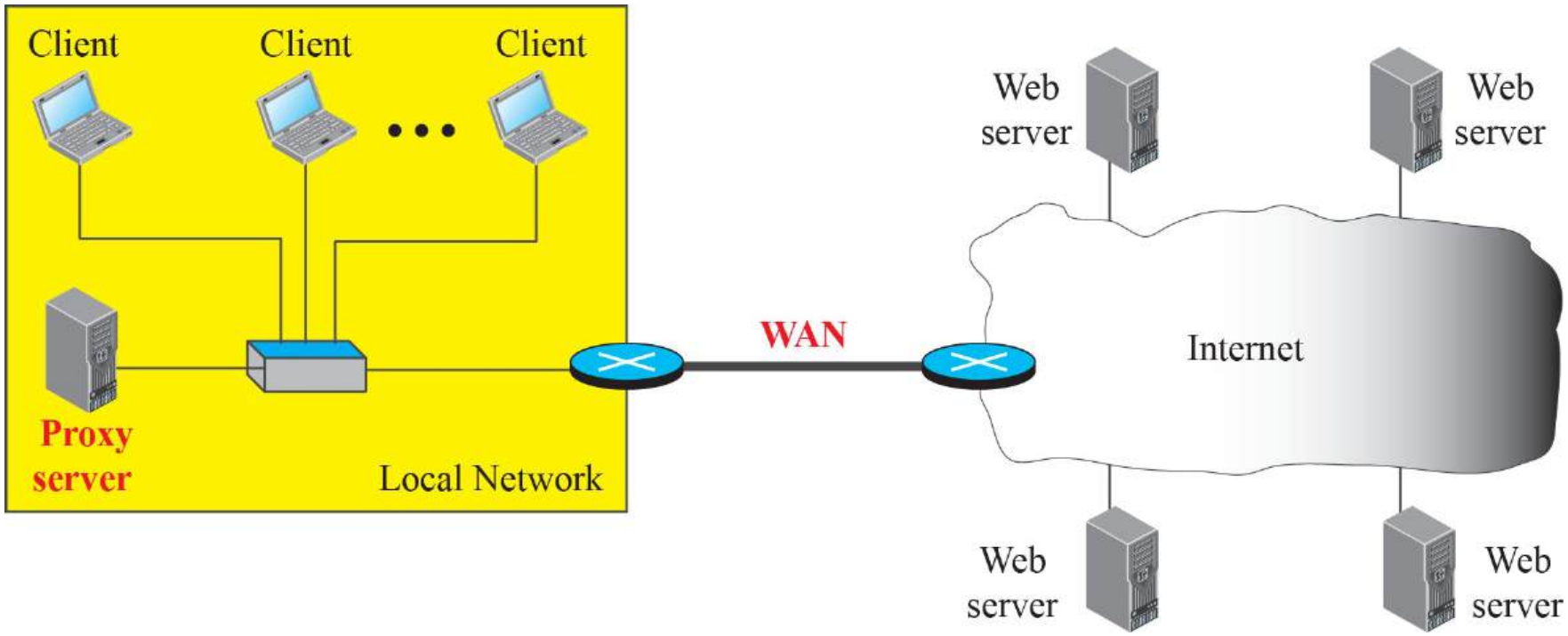
Figure 2.16: Example of a proxy server



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2.3.2 FTP

File Transfer Protocol (FTP) is the standard protocol provided by TCP/IP for copying a file from one host to another. Although transferring files from one system to another seems simple and straightforward, some problems must be dealt with first. For example, two systems may use different file name conventions. Two systems may have different ways to represent data. All of these problems have been solved by FTP in a very simple and elegant approach.

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2.3.2 (continued)

- **Lifetimes of Two Connections**
- **Control Connection**
- **Data Connection**
  - Communication over Data Connection
  - File Transfer
- **Security for FTP**

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Figure 2.17: FTP

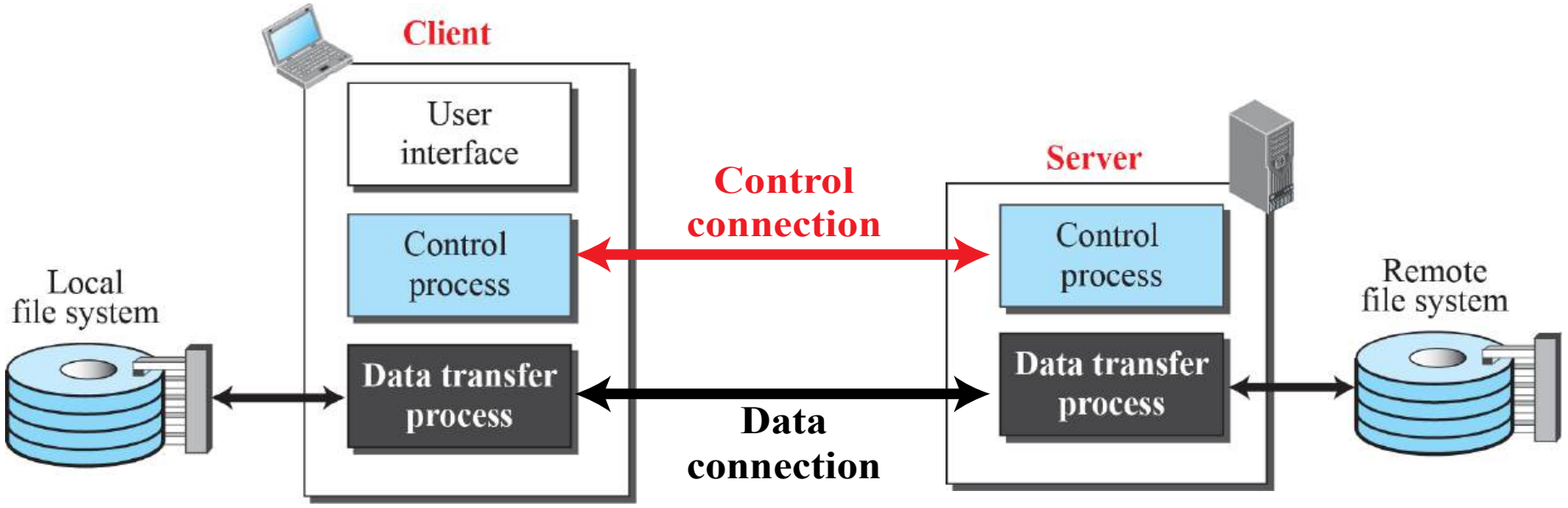

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Table 2.4: Some FTP commands

Command	*Argument(s)*	*Description*
ABOR		Abort the previous command
CDUP		Change to parent directory
CWD	Directory name	Change to another directory
DELE	File name	Delete a file
LIST	Directory name	List subdirectories or files
MKD	Directory name	Create a new directory
PASS	User password	Password
PASV		Server chooses a port
PORT	port identifier	Client chooses a port
PWD		Display name of current directory
QUIT		Log out of the system
RETR	File name(s)	Retrieve files; files are transferred from server to client
RMD	Directory name	Delete a directory
RNFR	File name (old)	Identify a file to be renamed

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Table 2.4: Some FTP commands (continued)

Command	*Argument(s)*	*Description*
RNTO	File name (new)	Rename the file
STOR	File name(s)	Store files; file(s) are transferred from client to server
STRU	**F**, **R**, or **P**	Define data organization (**F**: file, **R**: record, or **P**: page)
TYPE	**A**, **E**, **I**	Default file type (**A**: ASCII, **E**: EBCDIC, **I**: image)
USER	User ID	User information
MODE	**S**, **B**, or **C**	Define transmission mode (**S**: stream, **B**: block, or **C**: compressed

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Table 2.5: Some responses in FTP

Code	*Description*	*Code*	*Description*
125	Data connection open	**250**	Request file action OK
150	File status OK	**331**	User name OK; password is needed
200	Command OK	**425**	Cannot open data connection
220	Service ready	**450**	File action not taken; file not available
221	Service closing	**452**	Action aborted; insufficient storage
225	Data connection open	**500**	Syntax error; unrecognized command
226	Closing data connection	**501**	Syntax error in parameters or arguments
230	User login OK	**530**	User not logged in

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Example 2.11

Figure 2.18 shows an example of using FTP for retrieving a file. The figure shows only one file to be transferred. The control connection remains open all the time, but the data connection is opened and closed repeatedly. We assume the file is transferred in six sections. After all records have been transferred, the server control process announces that the file transfer is done. Since the client control process has no file to retrieve, it issues the QUIT command, which causes the service connection to be closed.

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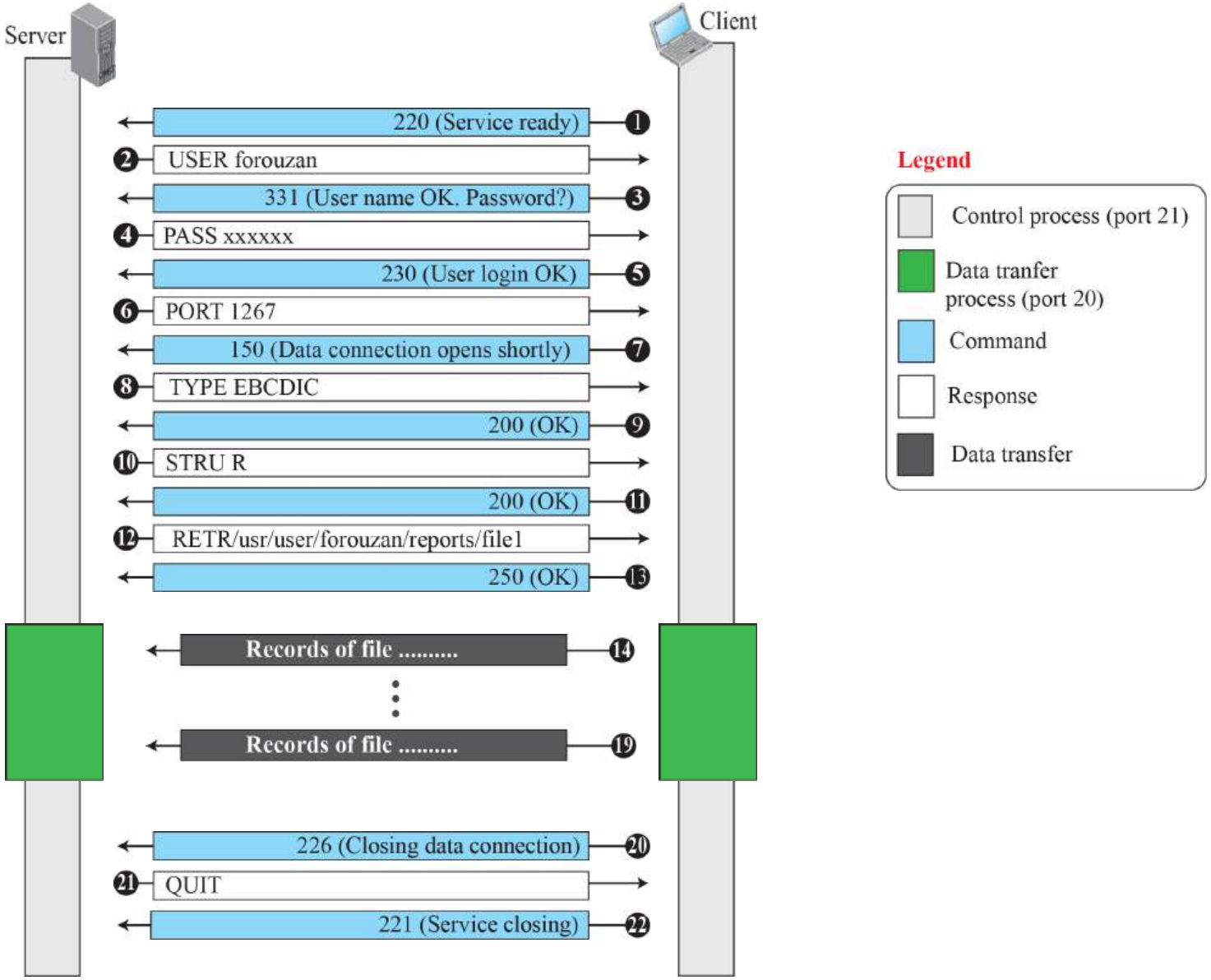
Figure 2.18: Example 2.11



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Example 2.12

The following shows an actual FTP session that lists the directories.

```
$ ftp voyager.deanza.fhda.edu
Connected to voyager.deanza.fhda.edu.
220 (vsFTPd 1.2.1)
530 Please login with USER and PASS.
Name (voyager.deanza.fhda.edu:forouzan): forouzan
331 Please specify the password.
Password:**********
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
227 Entering Passive Mode (153,18,17,11,238,169)
150 Here comes the directory listing.
drwxr-xr-x    2    3027    411    4096    Sep 24    2002    business
drwxr-xr-x    2    3027    411    4096    Sep 24    2002    personal
drwxr-xr-x    2    3027    411    4096    Sep 24    2002    school
226 Directory send OK.
ftp> quit
221 Goodbye.
```


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2.3.3 Electronic Mail

Electronic mail (or e-mail) allows users to exchange messages. The nature of this application, however, is different from other applications discussed so far. In an application such as HTTP or FTP, the server program is running all the time, waiting for a request from a client. When the request arrives, the server provides the service. In the case of electronic mail, the situation is different.

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2.3.3 Continued

First, e-mail is considered a one-way transaction. When Alice sends an e-mail to Bob, she may expect a response, but this is not a mandate. Bob may or may not respond. If he does respond, it is another one-way transaction. Second, it is neither feasible nor logical for Bob to run a server program and wait until someone sends an e-mail to him. Bob may turn off his computer when he is not using it. This means that the idea of client/ server programming should be implemented in another way: using some intermediate computers (servers).

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2.3.3 (continued)

- **Architecture**

- **User Agent**
  - Sending Mail
  - Receiving Mail
  - Addresses
  - Mailing List or Group List

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2.3.1 (continued)

- **MIME**
  - MIME Headers
- **Web-Based Mail**
  - Case I
  - Case II
- **E-Mail Security**

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2.3.1 (continued)

- **Message Transfer Agent: SMTP**
  - Commands and Responses
  - Mail Transfer Phases
- **Message Access Agent: POP and IMAP**
  - POP3
  - IMAP4

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Figure 2.19: Common scenario

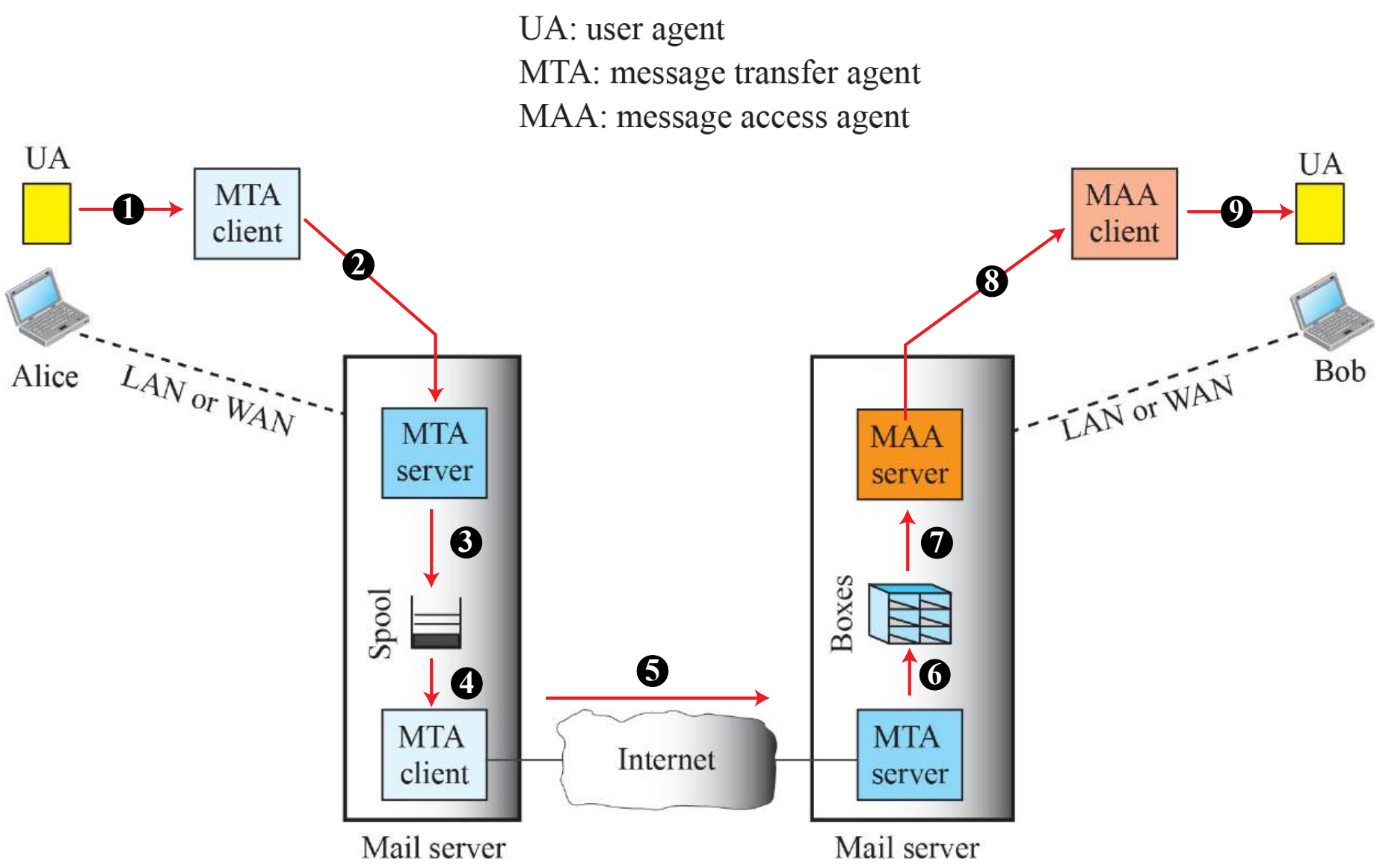

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Figure 2.20: Format of an e-mail

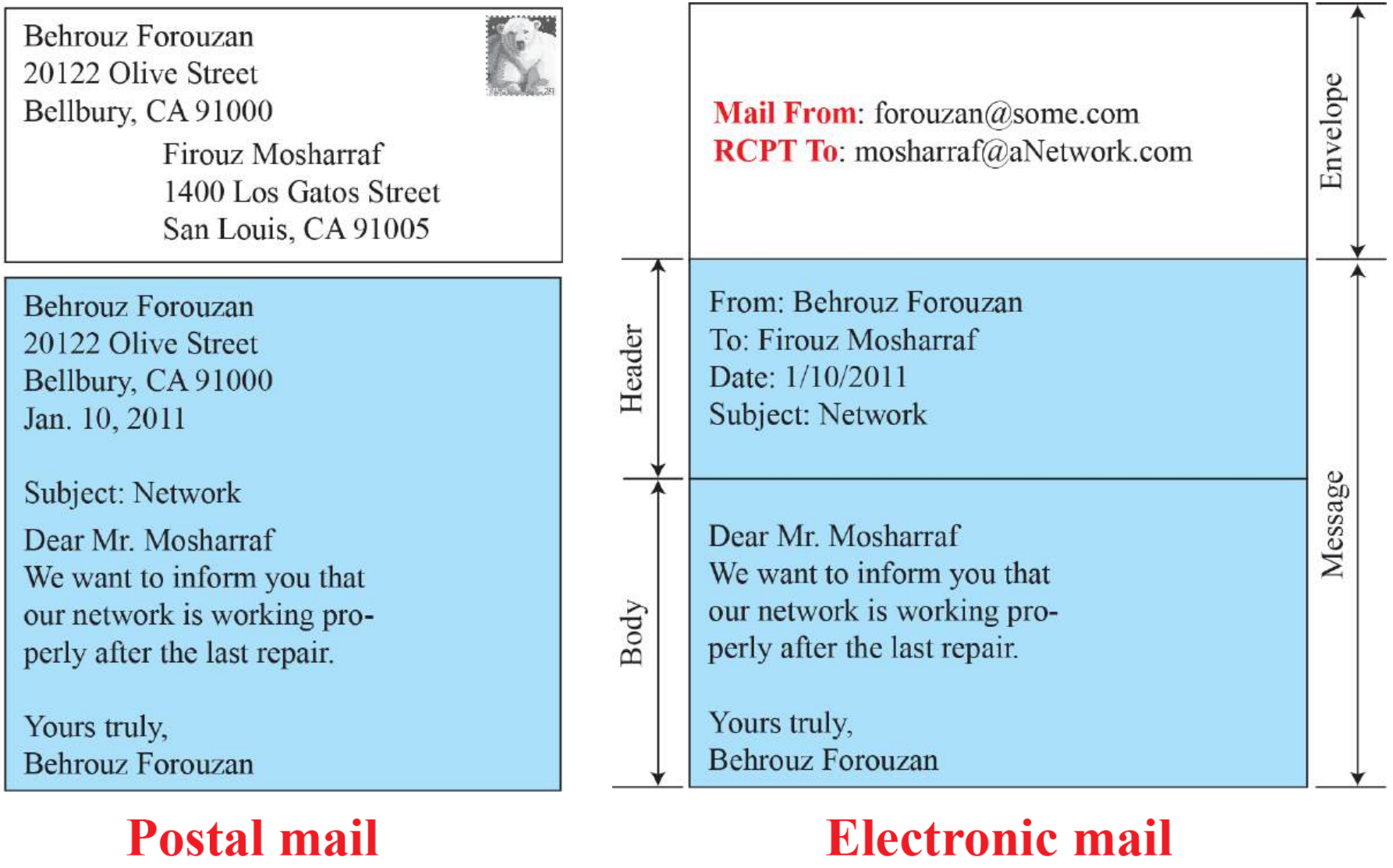

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Figure 2.21: E-mail address

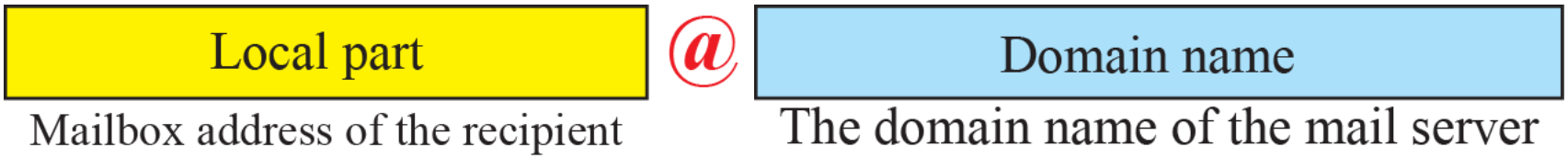

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Figure 2.22: Protocols used in electronic mail

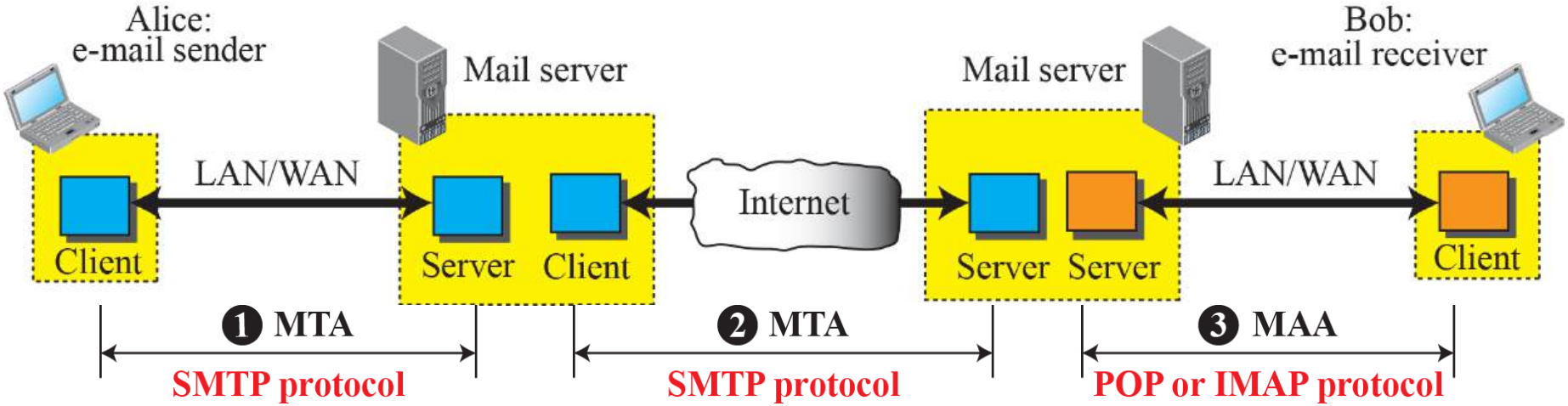

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Table 2.6: SMTP Commands

Keyword	*Argument(s)*	*Description*
HELO	Sender's host name	Identifies itself
MAIL FROM	Sender of the message	Identifies the sender of the message
RCPT TO	Intended recipient	Identifies the recipient of the message
DATA	Body of the mail	Sends the actual message
QUIT		Terminates the message
RSET		Aborts the current mail transaction
VRFY	Name of recipient	Verifies the address of the recipient
NOOP		Checks the status of the recipient
TURN		Switches the sender and the recipient
EXPN	Mailing list	Asks the recipient to expand the mailing list.
HELP	Command name	Asks the recipient to send information about the command sent as the argument
SEND FROM	Intended recipient	Specifies that the mail be delivered only to the terminal of the recipient, and not to the mailbox
SMOL FROM	Intended recipient	Specifies that the mail be delivered to the terminal *or* the mailbox of the recipient
SMAL FROM	Intended recipient	Specifies that the mail be delivered to the terminal *and* the mailbox of the recipient

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Table 2.7: SMTP responses

Code	*Description*
	Positive Completion Reply
211	System status or help reply
214	Help message
220	Service ready
221	Service closing transmission channel
250	Request command completed
251	User not local; the message will be forwarded
	Positive Intermediate Reply
354	Start mail input
	Transient Negative Completion Reply
421	Service not available
450	Mailbox not available
451	Command aborted: local error
452	Command aborted; insufficient storage

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Table 2.7: SMTP responses (continued)

Permanent Negative Completion Reply	
500	Syntax error; unrecognized command
501	Syntax error in parameters or arguments
502	Command not implemented
503	Bad sequence of commands
504	Command temporarily not implemented
550	Command is not executed; mailbox unavailable
551	User not local
552	Requested action aborted; exceeded storage location
553	Requested action not taken; mailbox name not allowed
554	Transaction failed

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Example 2.13

To show the three mail transfer phases, we show all of the steps described above using the information depicted in Figure 2.23. In the figure, we have separated the messages related to the envelope, header, and body in the data transfer section. Note that the steps in this figure are repeated two times in each e-mail transfer: once from the e-mail sender to the local mail server and once from the local mail server to the remote mail server. The local mail server, after receiving the whole e-mail message, may spool it and send it to the remote mail server at another time.

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Figure 2.23: Example 2.13

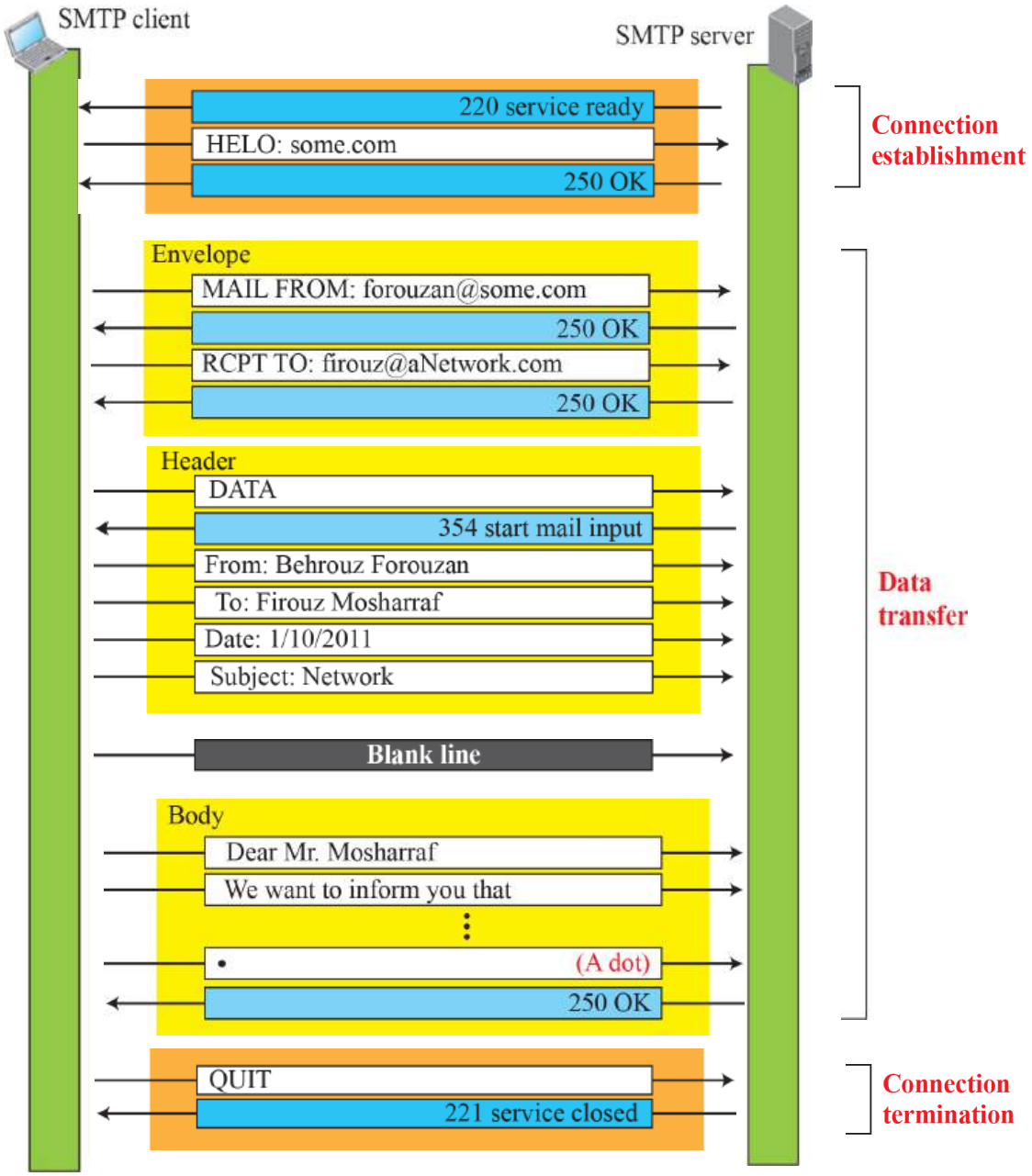

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Figure 2.24: POP3

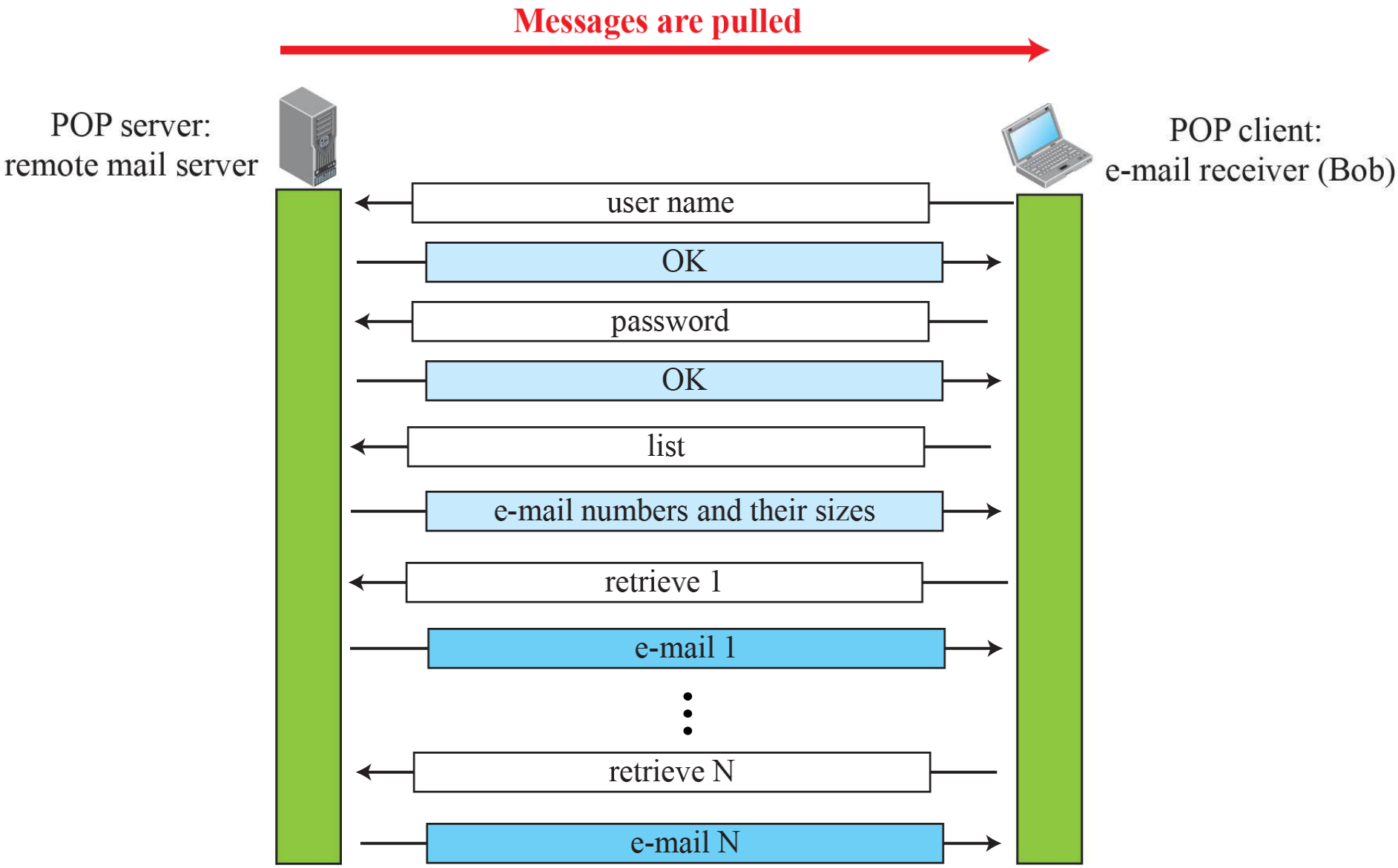

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Figure 2.25: MIME

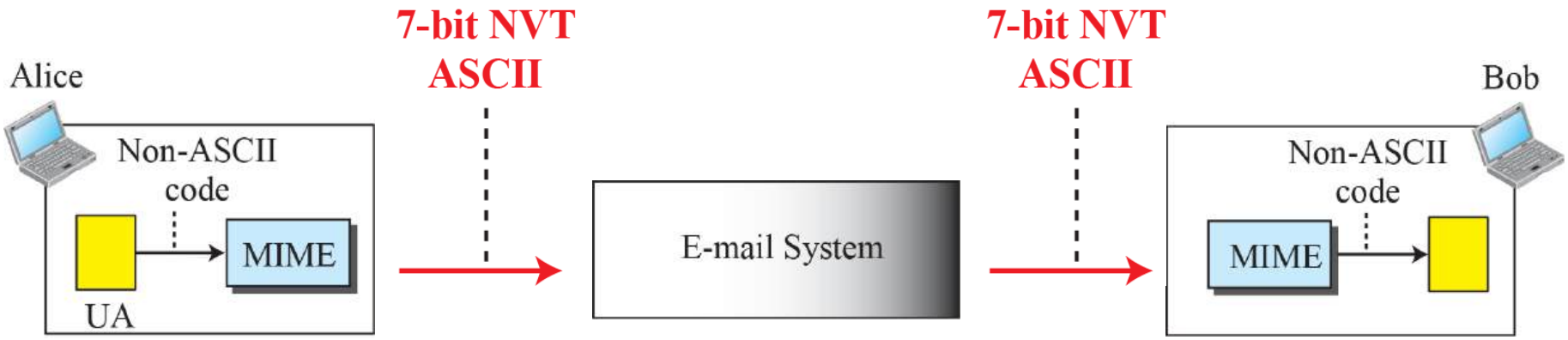

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Figure 2.26: MIME header

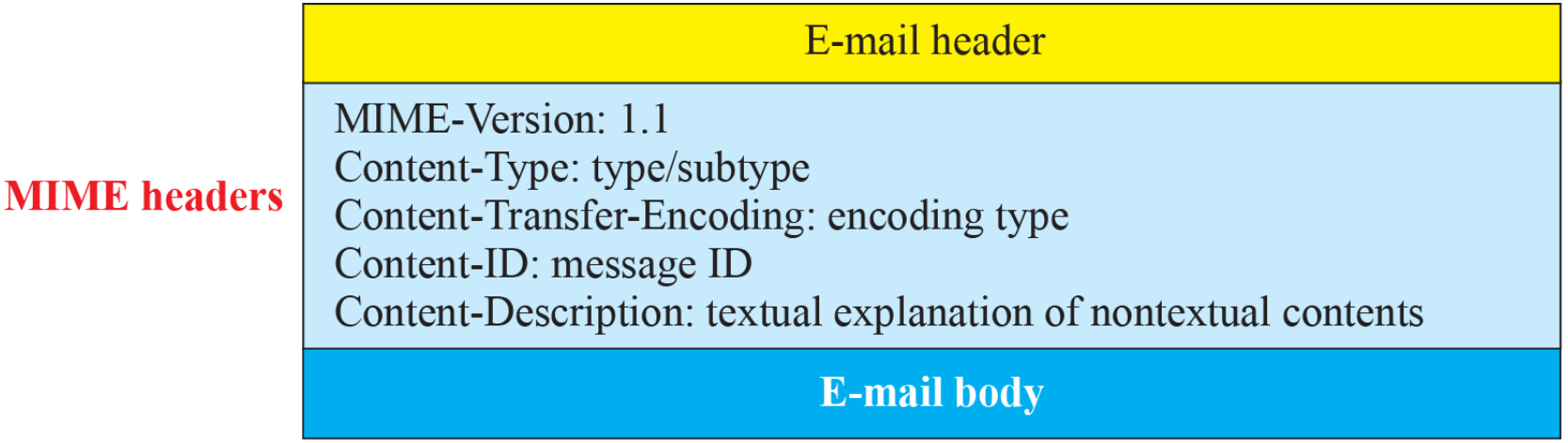

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Table 2.8: Data Types and Subtypes in MIME

Type	*Subtype*	*Description*
Text	Plain	Unformatted
	HTML	HTML format (see Appendix C)
Multipart	Mixed	Body contains ordered parts of different data types
	Parallel	Same as above, but no order
	Digest	Similar to Mixed, but the default is message/RFC822
	Alternative	Parts are different versions of the same message
Message	RFC822	Body is an encapsulated message
	Partial	Body is a fragment of a bigger message
	External-Body	Body is a reference to another message
Image	JPEG	Image is in JPEG format
	GIF	Image is in GIF format
Video	MPEG	Video is in MPEG format
Audio	Basic	Single channel encoding of voice at 8 KHz
Application	PostScript	Adobe PostScript
	Octet-stream	General binary data (eight-bit bytes)

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Table 2.9: Methods for Content-Transfer-Encoding

Type	*Description*
7-bit	NVT ASCII characters with each line less than 1000 characters
8-bit	Non-ASCII characters with each line less than 1000 characters
Binary	Non-ASCII characters with unlimited-length lines
Base64	6-bit blocks of data encoded into 8-bit ASCII characters
Quoted-printable	Non-ASCII characters encoded as an equal sign plus an ASCII code

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Figure 2.27: *Base64 conversion*

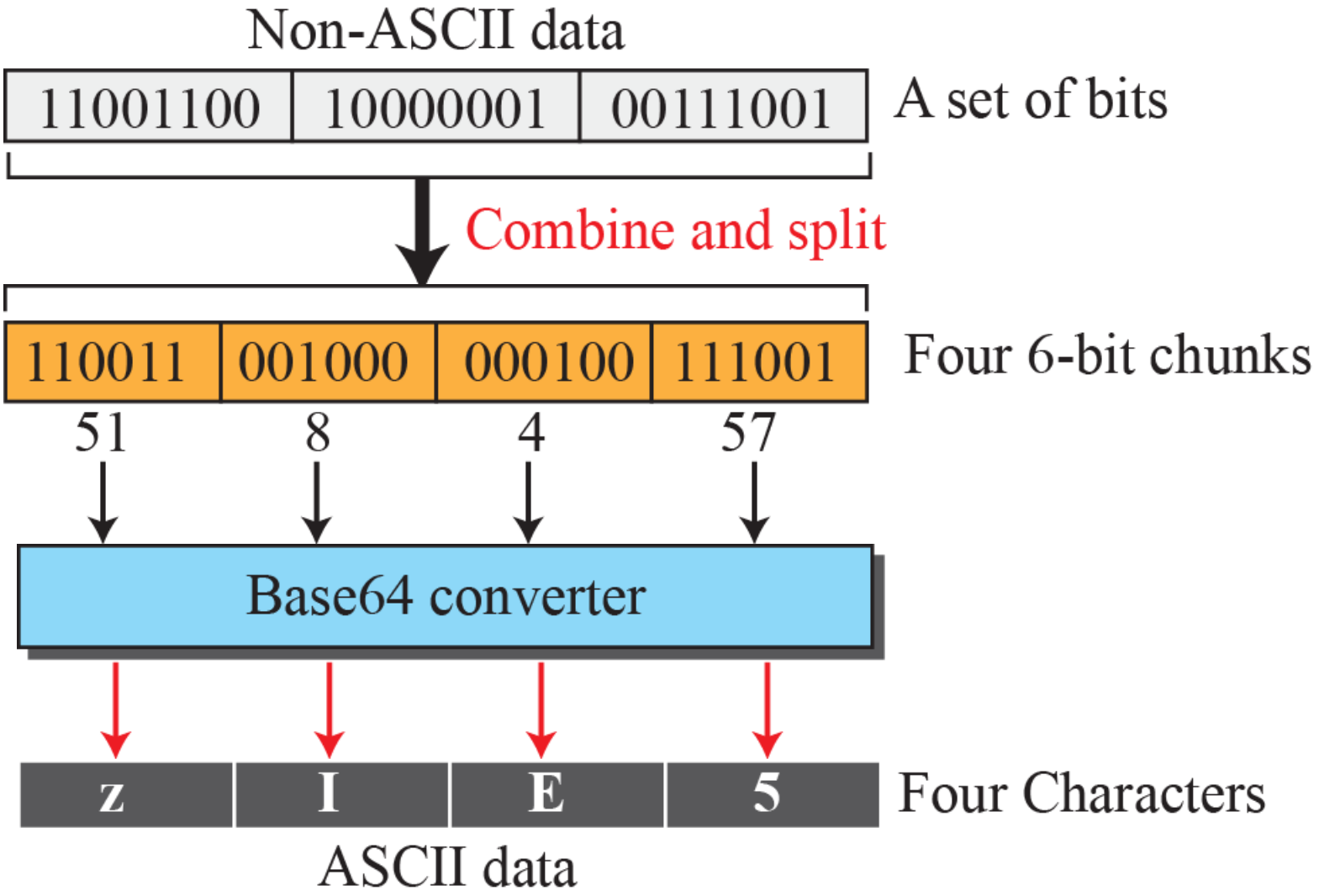

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Table 2.10: Base64 Converting Table

Value	Code	Value	Code	Value	Code	Value	Code	Value	Code	Value	Code
0	**A**	11	**L**	22	**W**	33	**h**	44	**s**	55	**3**
1	**B**	12	**M**	23	**X**	34	**i**	45	**t**	56	**4**
2	**C**	13	**N**	24	**Y**	35	**j**	46	**u**	57	**5**
3	**D**	14	**O**	25	**Z**	36	**k**	47	**v**	58	**6**
4	**E**	15	**P**	26	**a**	37	**l**	48	**w**	59	**7**
5	**F**	16	**Q**	27	**b**	38	**m**	49	**x**	60	**8**
6	**G**	17	**R**	28	**c**	39	**n**	50	**y**	61	**9**
7	**H**	18	**S**	29	**d**	40	**o**	51	**z**	62	+
8	**I**	19	**T**	30	**e**	41	**p**	52	**0**	63	/
9	**J**	20	**U**	31	**f**	42	**q**	53	**1**		
10	**K**	21	**V**	32	**g**	43	**r**	54	**2**		

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Figure 2.28: Quoted-printable

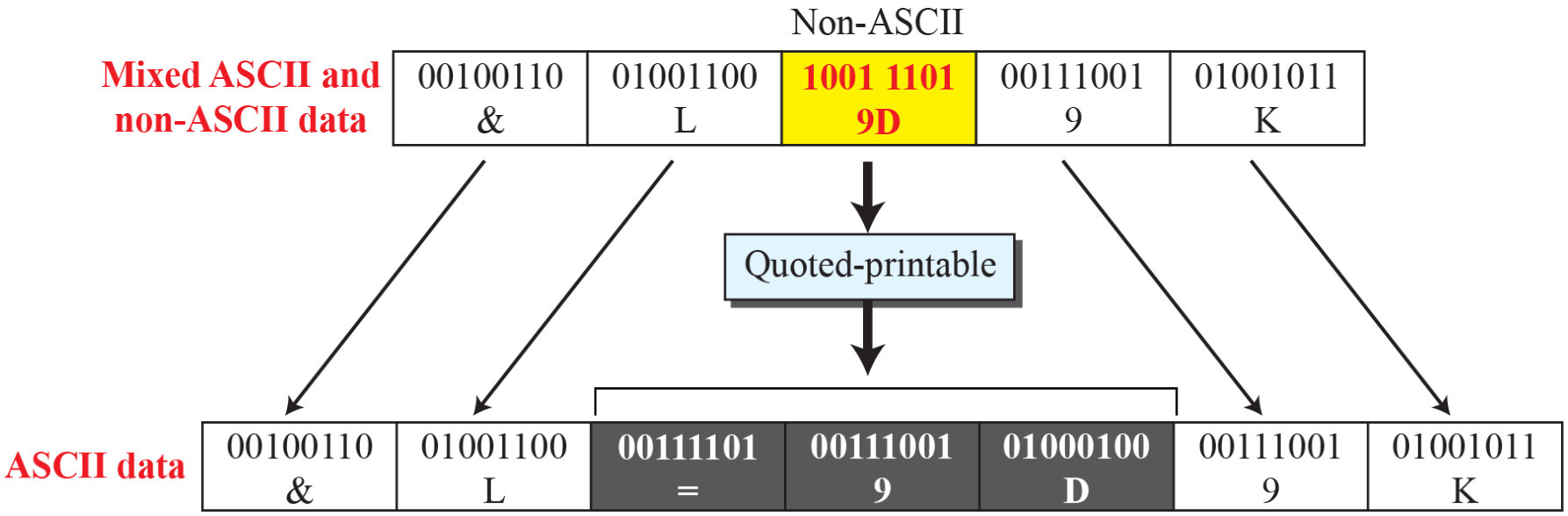

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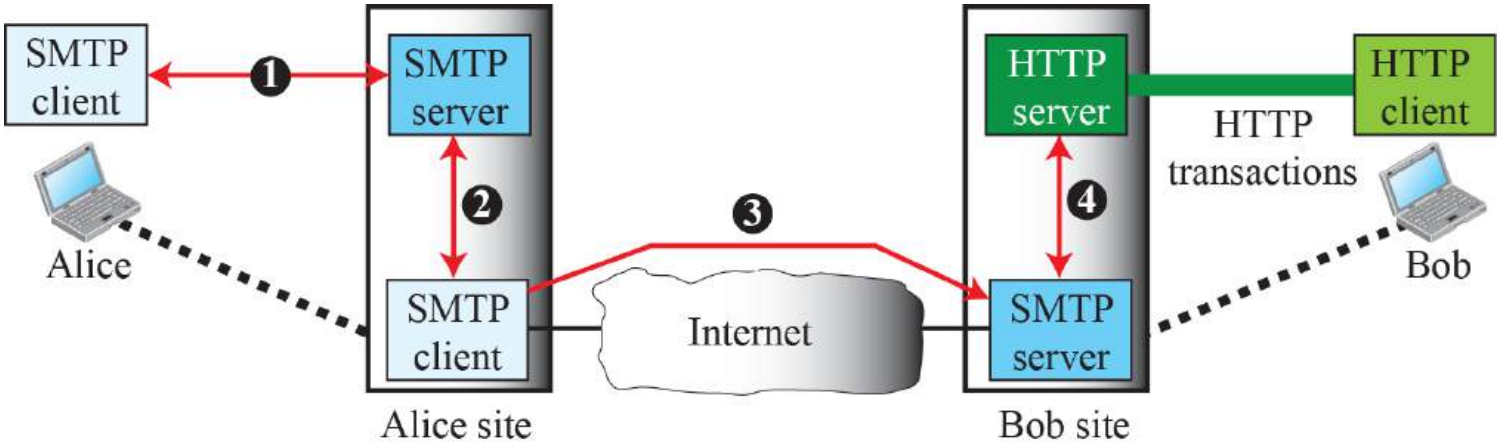
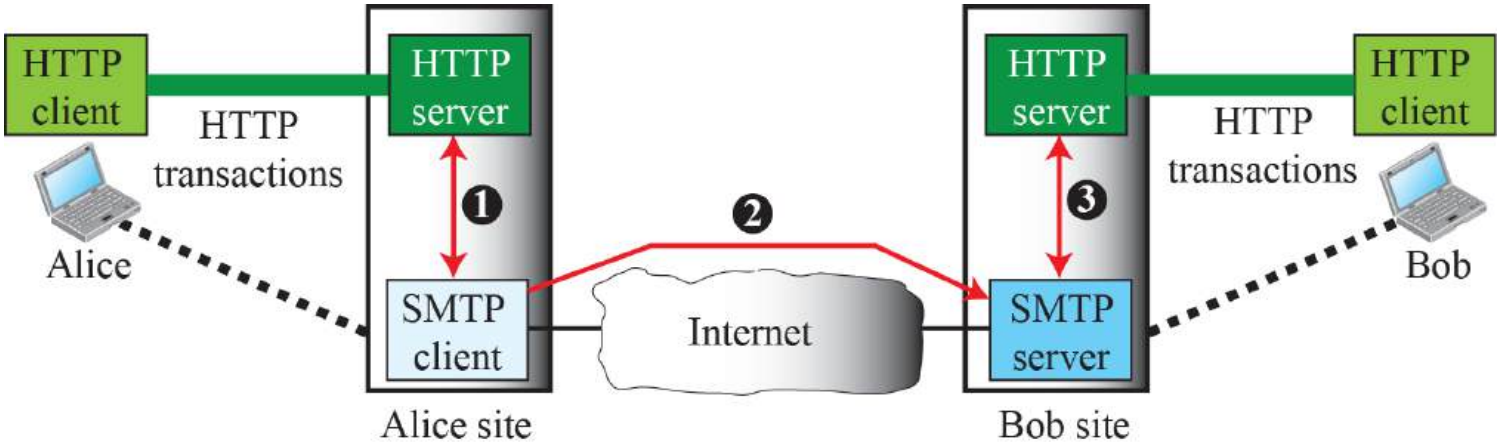
Figure 2.29: Web-based e-mail, cases I and II



Case 1: Only receiver uses HTTP



Case 2: Both sender and receiver use HTTP

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2.3.4 TELNET

A server program can provide a specific service to its corresponding client program. However, it is impossible to have a client/server pair for each type of service we need. Another solution is to have a specific client/server program for a set of common scenarios, but to have some generic client/server programs that allow a user on the client site to log into the computer at the server site and use the services available there. We refer to these generic client/server pairs as remote logging applications. One of the original remote logging protocols is ***TELNET****.*

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2.3.4 (continued)

- ***Local versus Remote Logging***
- ***Network Virtual Terminal (NVT)***
- ***Options***
- ***User Interface***

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Figure 2.30: Local versus remote logging

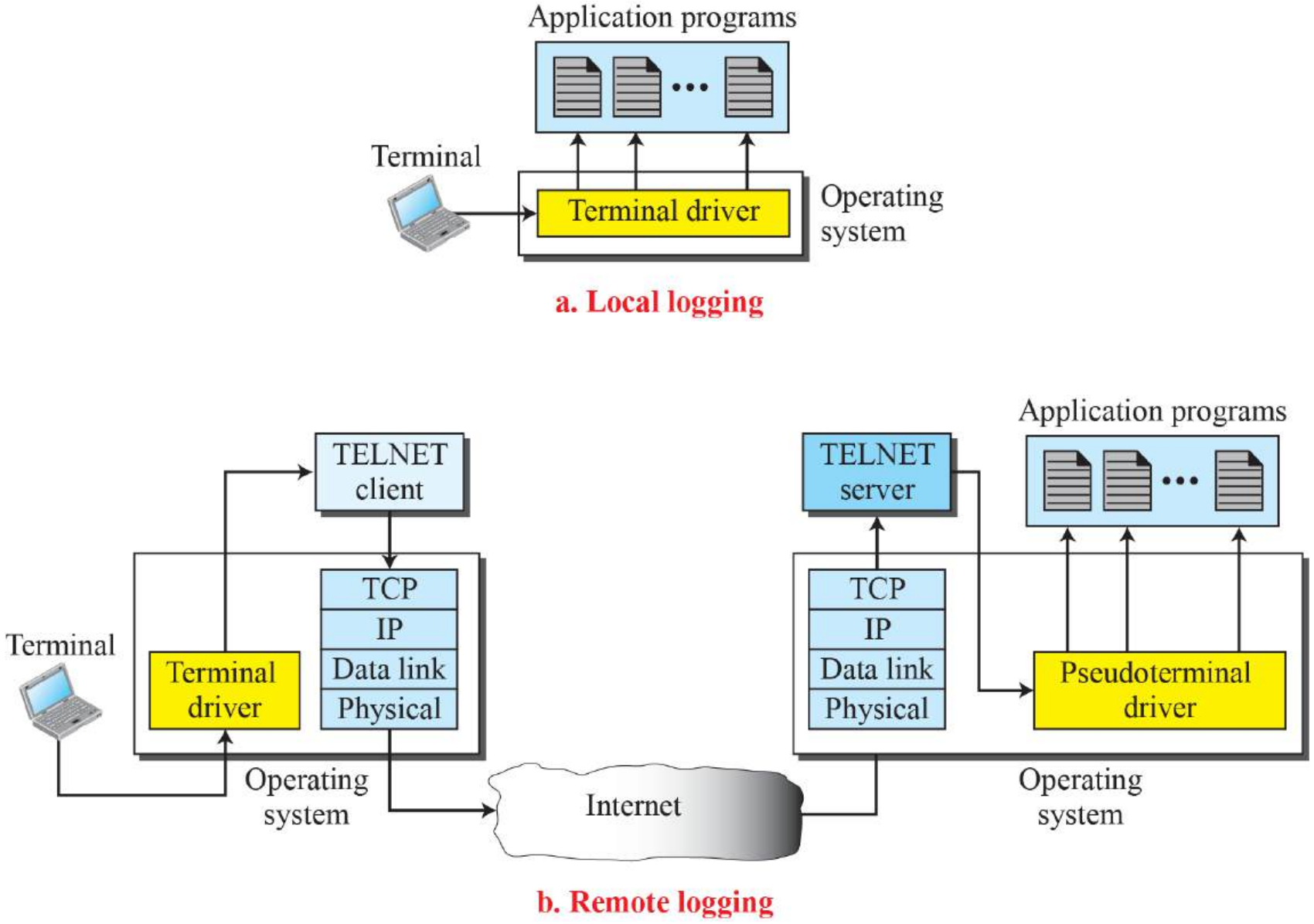

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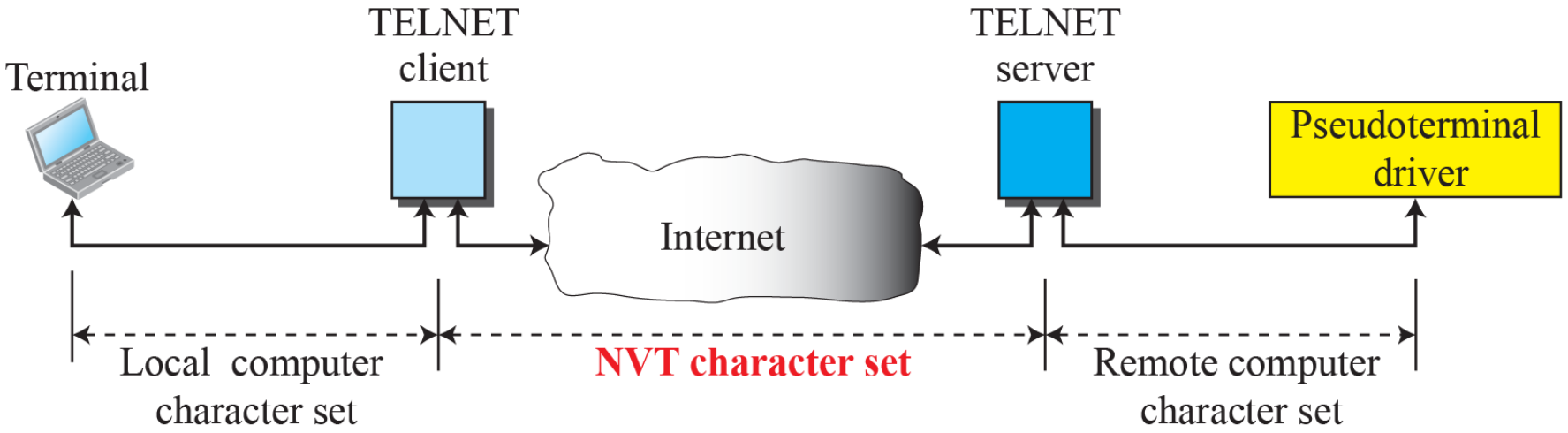
Figure 2.31: Concept of NVT



NVT character format

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Table 2.11: Examples of interface commands

Command	*Meaning*	*Command*	*Meaning*
open	Connect to a remote computer	**set**	Set the operating parameters
close	Close the connection	**status**	Display the status information
display	Show the operating parameters	**send**	Send special characters
mode	Change to line or character mode	**quit**	Exit TELNET

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2.3.5 Secure Shell (SSH)

Although Secure Shell (SSH) is a secure application program that can be used today for several purposes such as remote logging and file transfer, it was originally designed to replace TELNET. There are two versions of SSH: SSH-1 and SSH-2, which are totally incompatible. The first version, SSH-1, is now deprecated because of security flaws in it. In this section, we discuss only SSH-2.

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2.3.5 (continued)

- **Components**
  - SSH Transport-Layer Protocol (SSH-TRANS)
  - SSH Authentication Protocol (SSH-AUTH)
  - SSH Connection Protocol (SSH-CONN)
- **Applications**
  - SSH for Remote Logging
  - SSH for File Transfer
- **Port Forwarding**
- **Format of the SSH Packets**

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Figure 2.32: Components of SSH

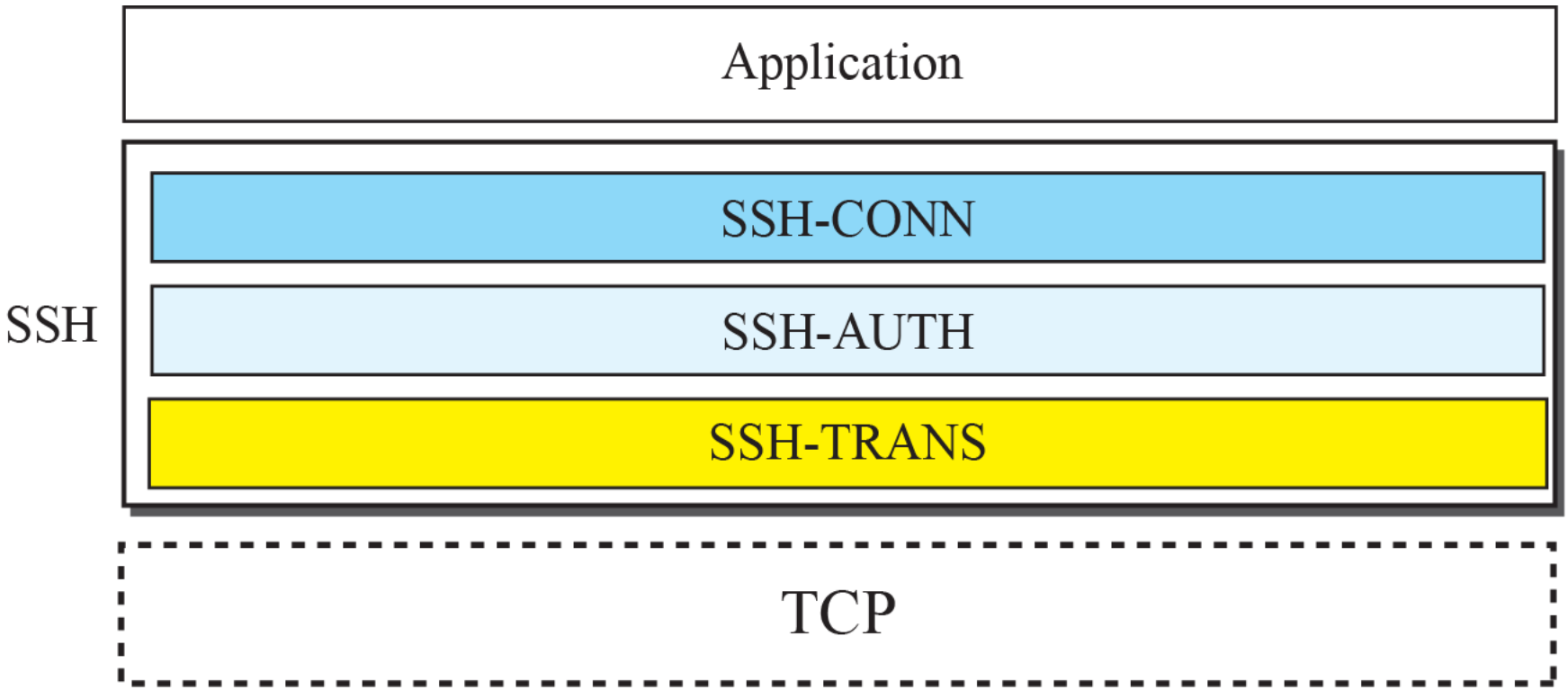

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Figure 2.33: Port Forwarding

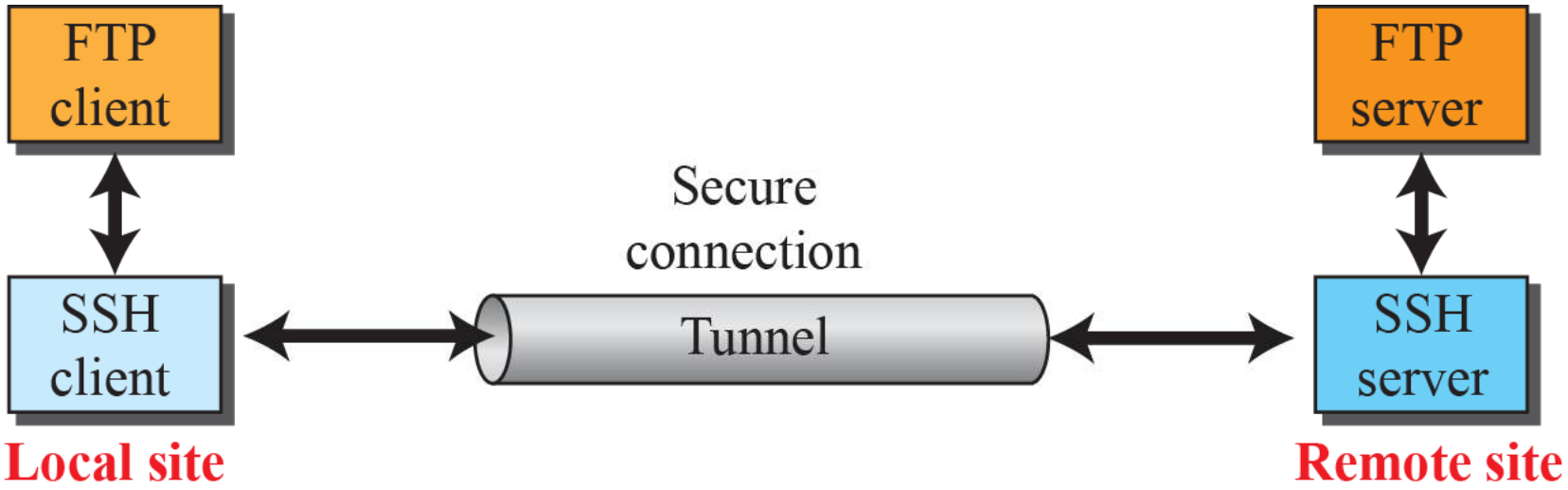

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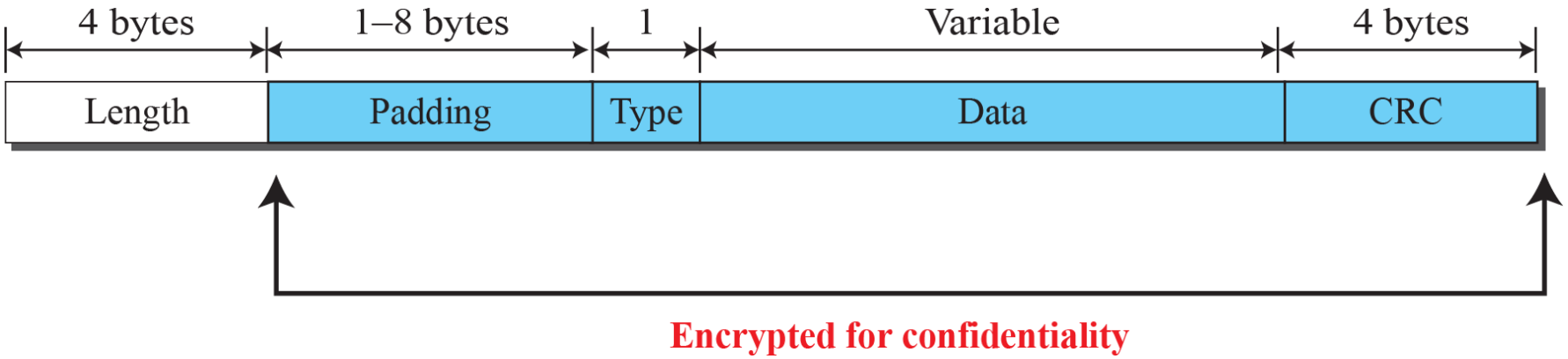
Figure 2.34: SSH Packet Format



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2.3.6 Domain Name System (DNS)

To identify an entity, TCP/IP protocols use the IP address, which uniquely identifies the connection of a host to the Internet. However, people prefer to use names instead of numeric addresses. Therefore, the Internet needs to have a directory system that can map a name to an address. This is analogous to the telephone network. A telephone network is designed to use telephone numbers, not names. People can either keep a private file to map a name to the corresponding telephone number or can call the telephone directory to do so.

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2.3.6 (continued)

- **Name Space**
  - Domain Name Space
  - Domain
  - Distribution of Name Space
  - Zone
  - Root Server
- **DNS in the Internet**
  - Generic Domains
  - Country Domains

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2.3.1 (continued)

- **Resolution**
  - Recursive Resolution
  - Iterative Resolution
  - Caching
- **Resource Records**
- **DNS Messages**
- **Encapsulation**
- **Registrars**
- **DDNS**
- **Security of DNS**

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Figure 2.35: Purpose of DNS

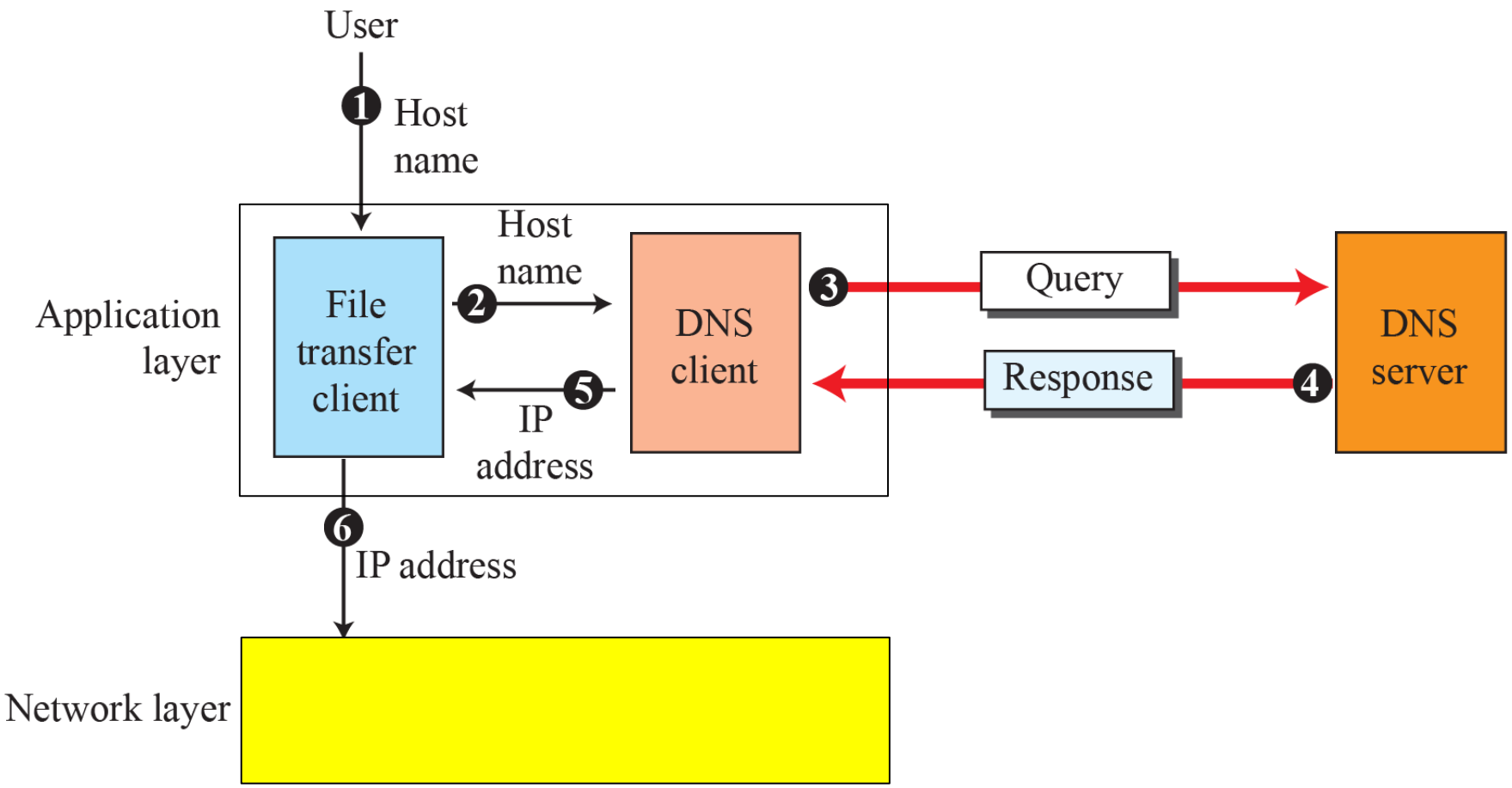

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Figure 2.36: Domain name space

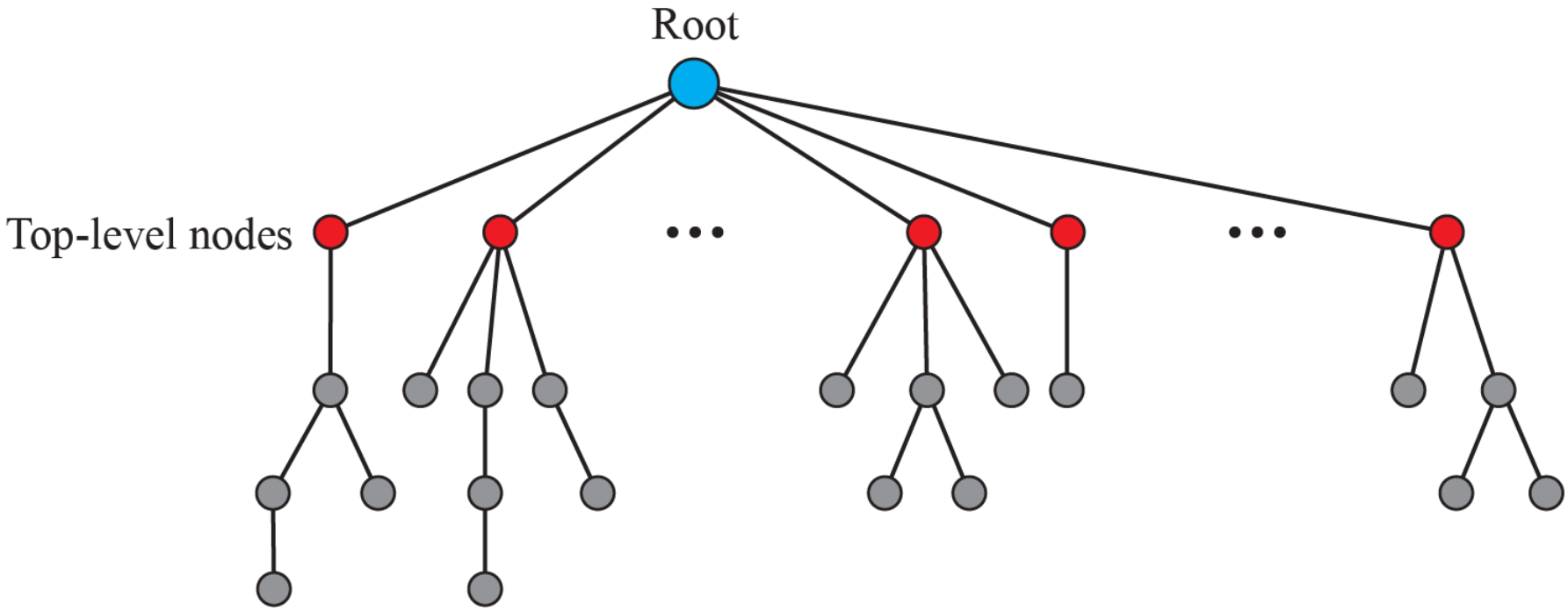

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Figure 2.37: Domain names and labels

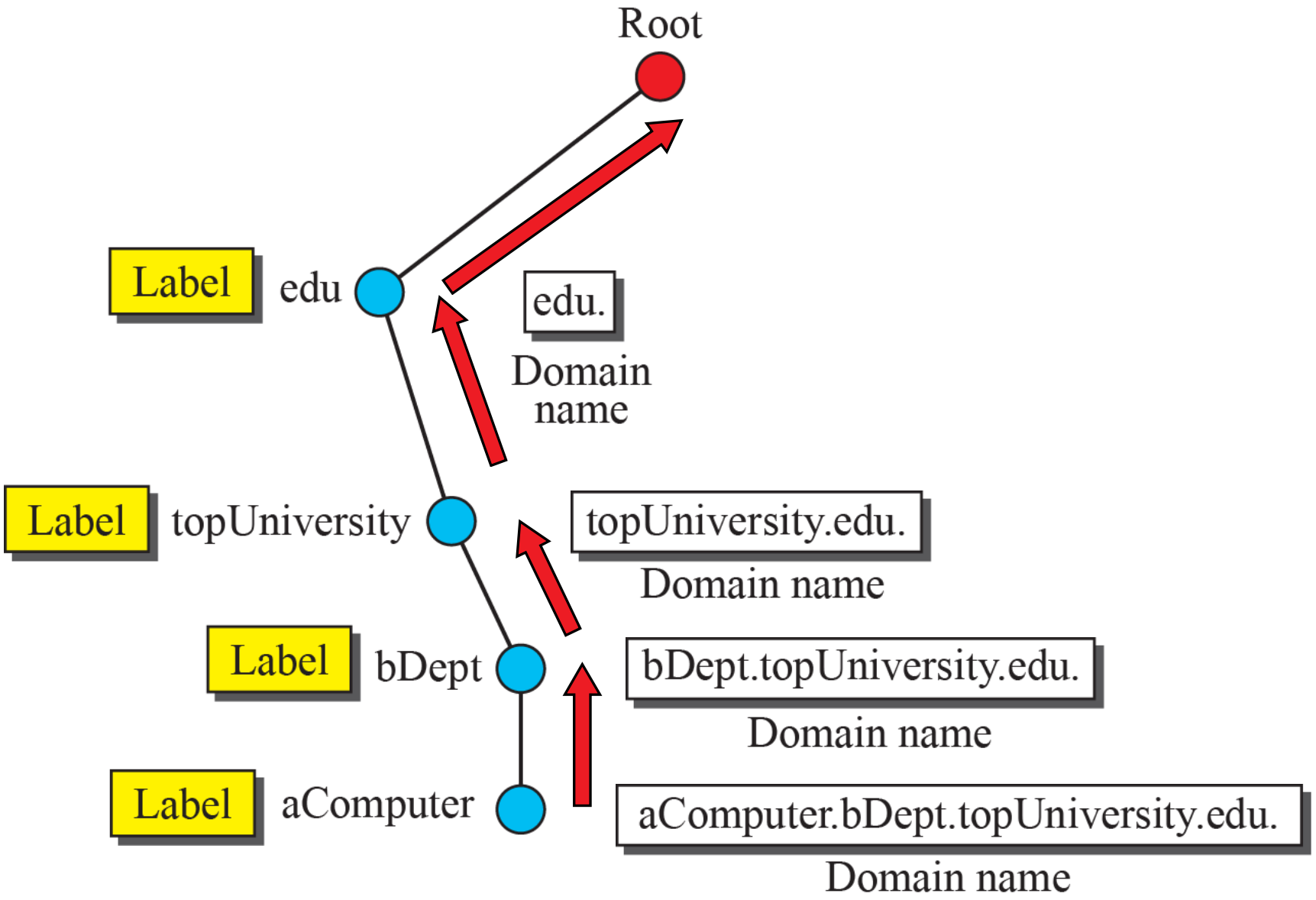

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Figure 2.38: *Domains*

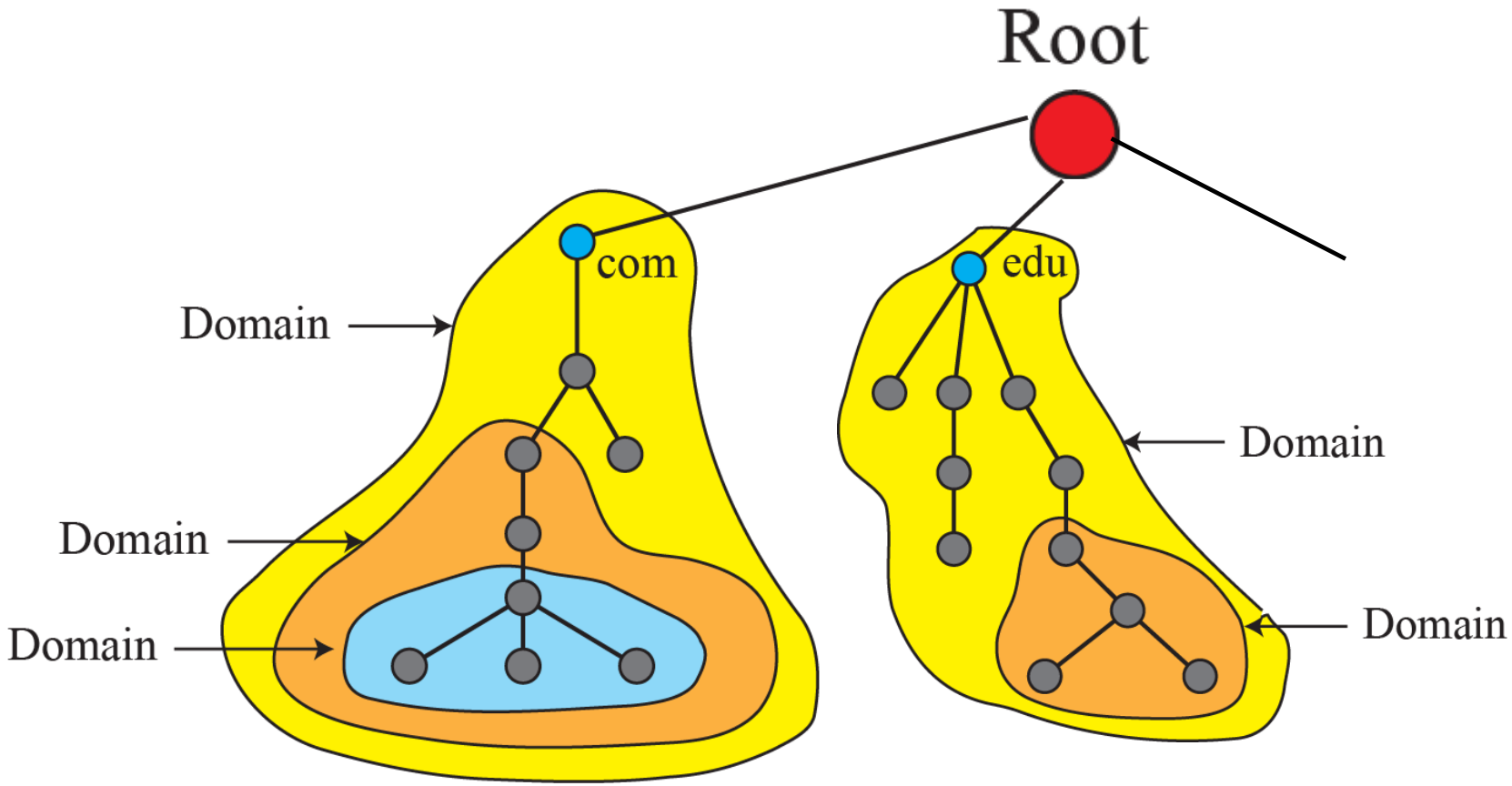

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Figure 2.39: Hierarchy of name servers

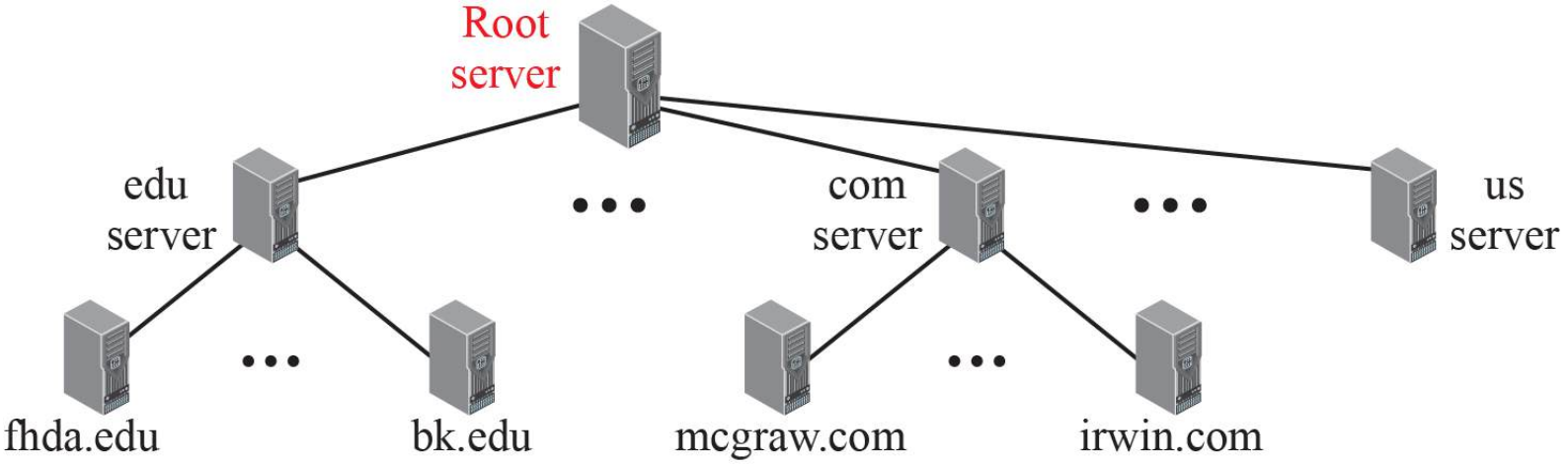

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Figure 2.40: Zone

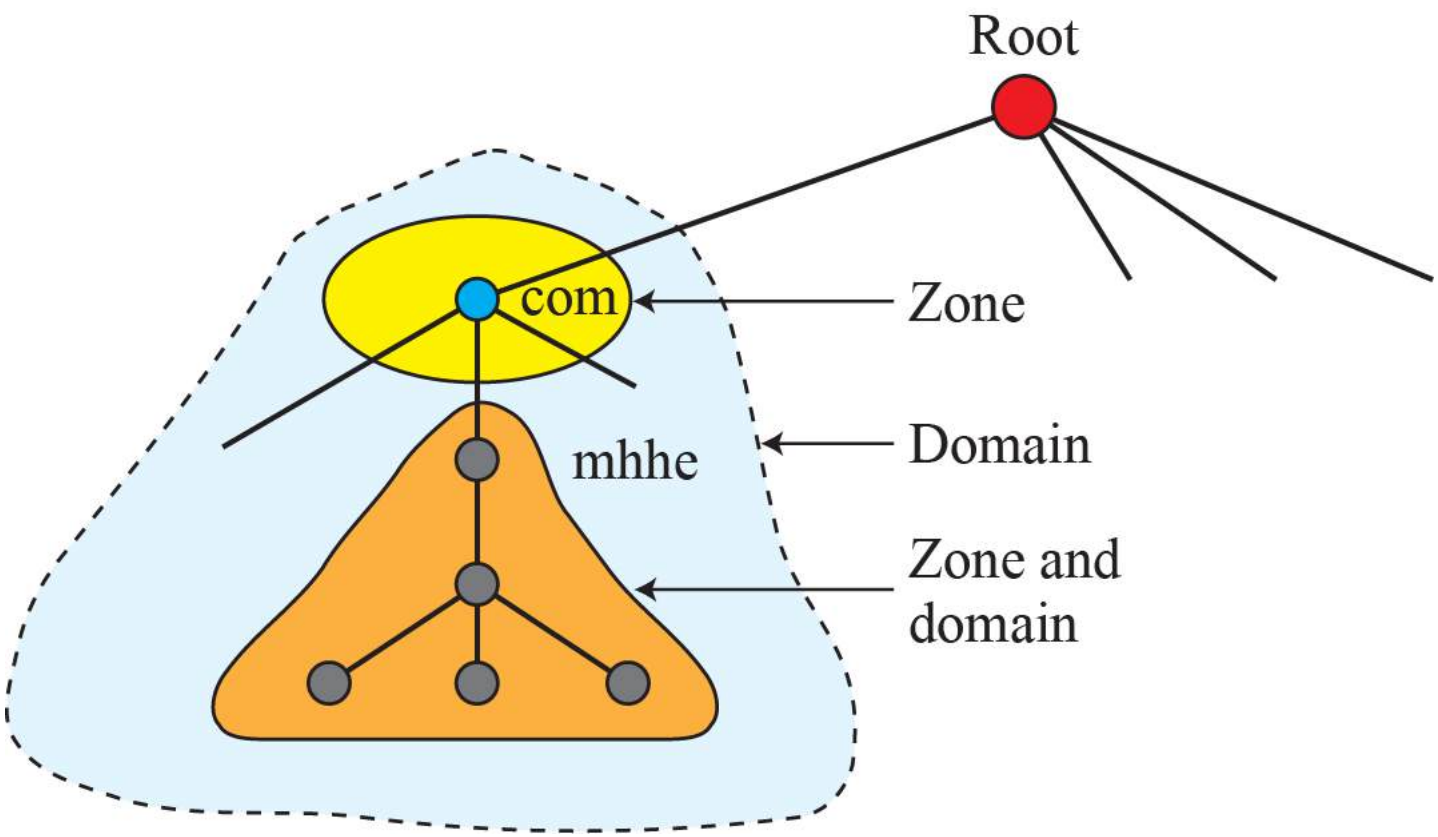

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Figure 2.41: Generic domains

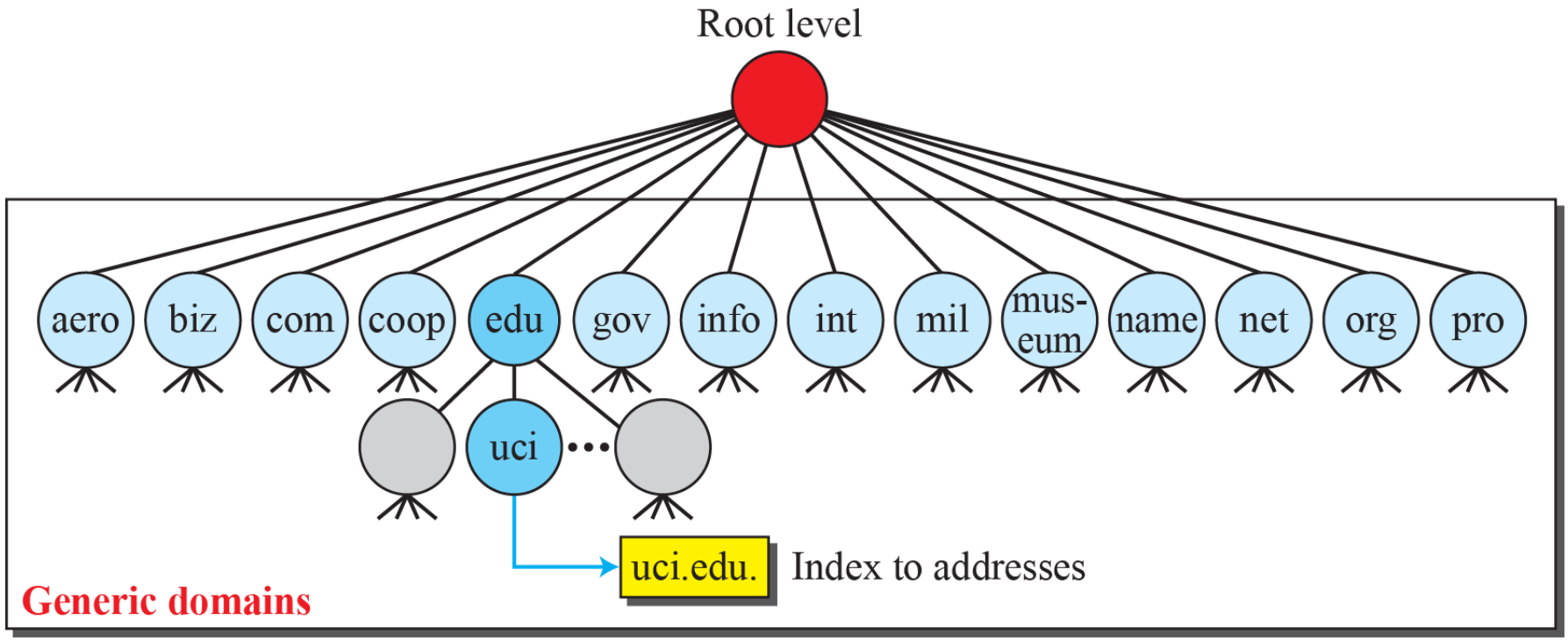

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Table 2.12: Generic domain labels

Label	*Description*	*Label*	*Description*
aero	Airlines and aerospace	**int**	International organizations
biz	Businesses or firms	**mil**	Military groups
com	Commercial organizations	**museum**	Museums
coop	Cooperative organizations	**name**	Personal names (individuals)
edu	Educational institutions	**net**	Network support centers
gov	Government institutions	**org**	Nonprofit organizations
info	Information service providers	**pro**	Professional organizations

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Figure 2.42: Country domains

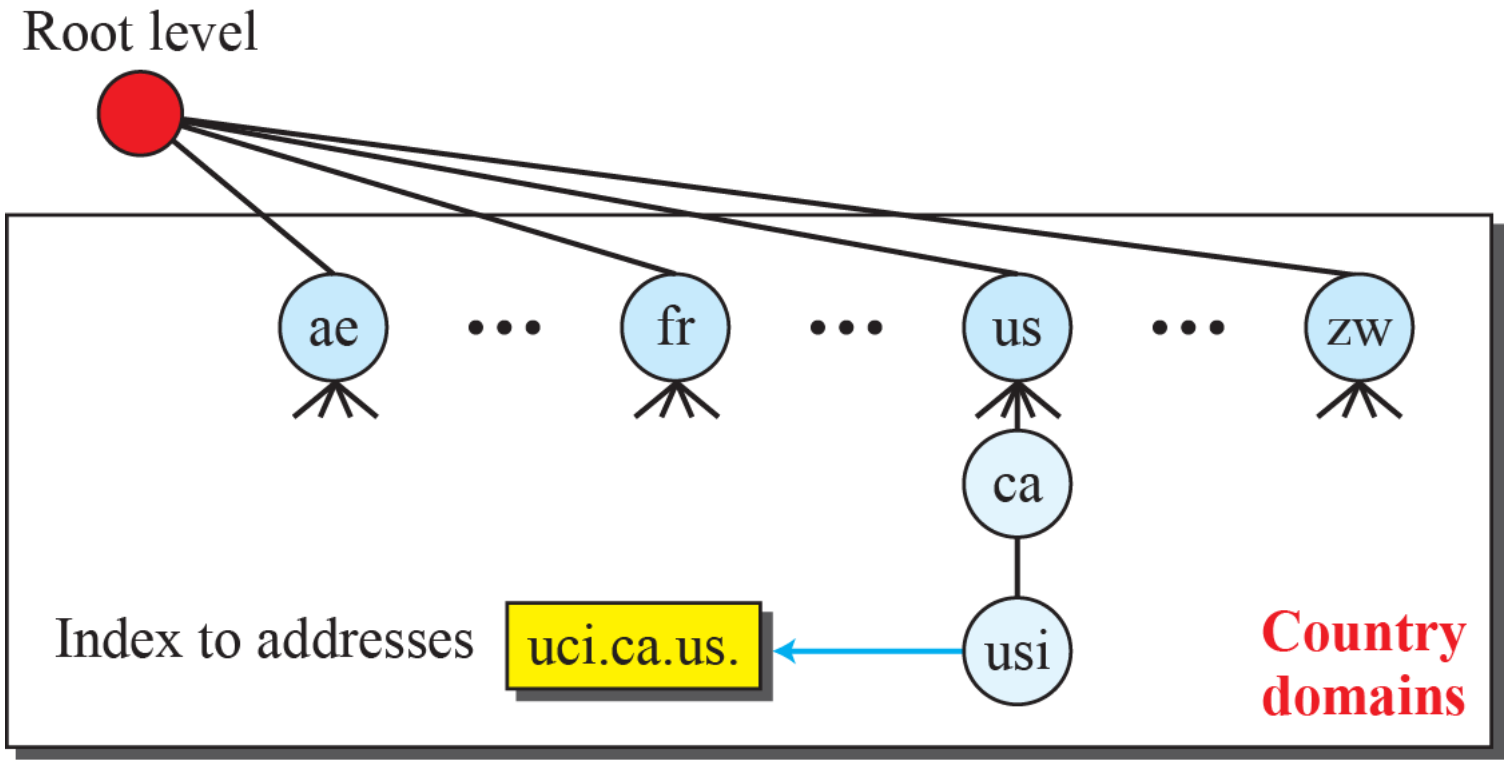

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Figure 2.43: Recursive resolution

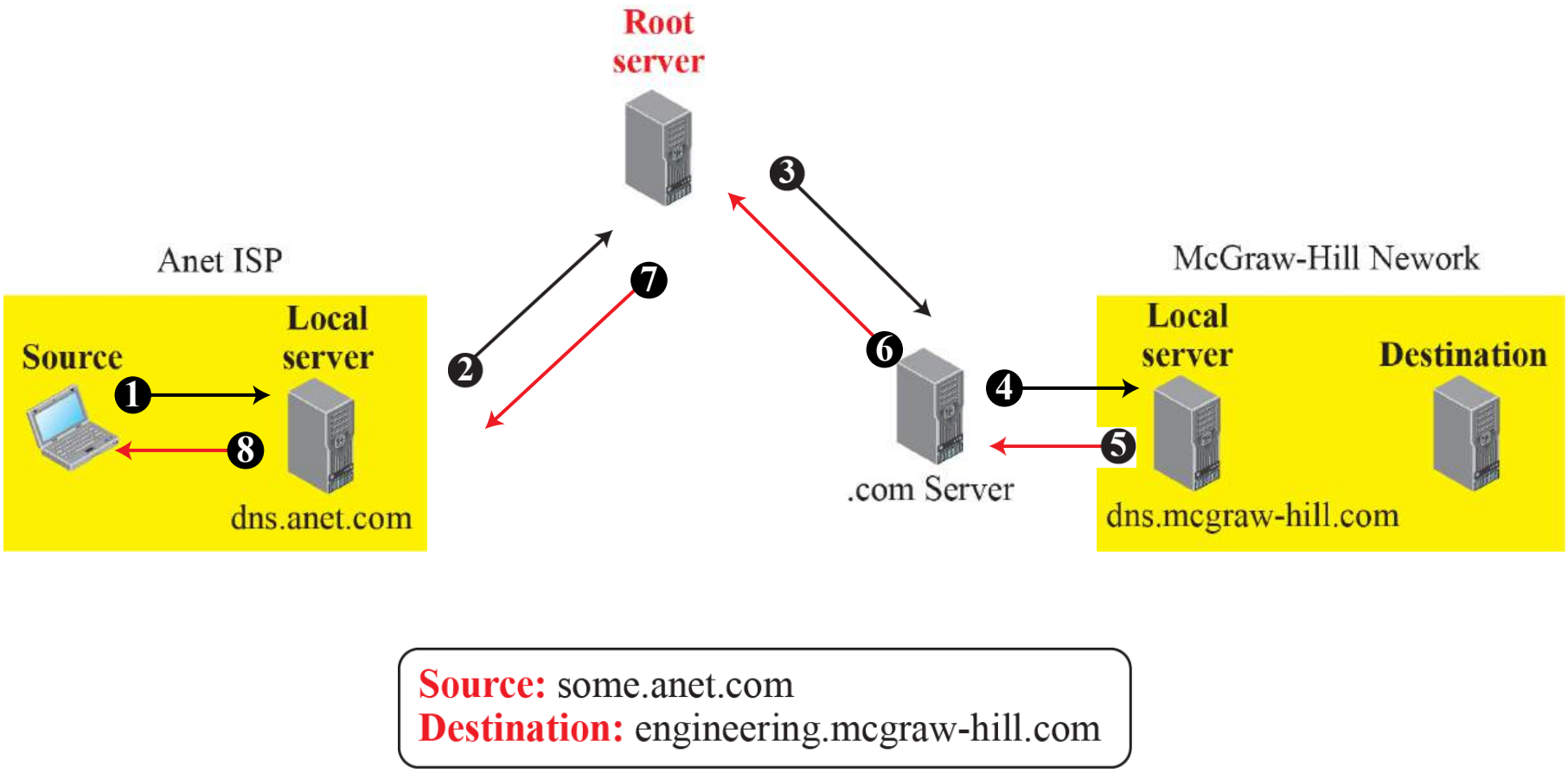

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Figure 2.44: Iterative resolution

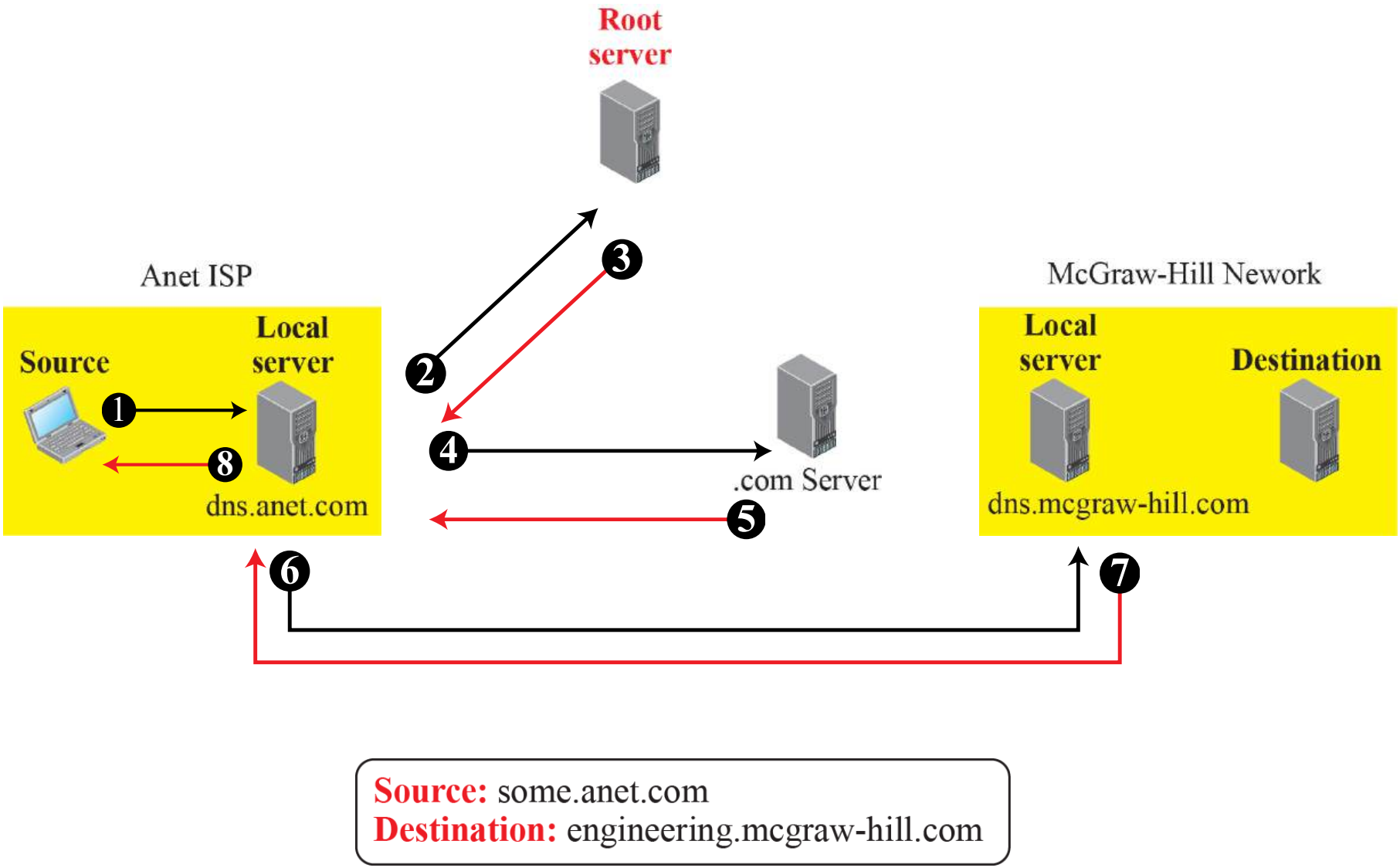

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Table 2.13: DNS types

Type	*Interpretation of value*
A	A 32-bit IPv4 address (see Chapter 4)
NS	Identifies the authoritative servers for a zone
CNAME	Defines an alias for the official name of a host
SOA	Marks the beginning of a zone
MX	Redirects mail to a mail server
AAAA	An IPv6 address (see Chapter 4)

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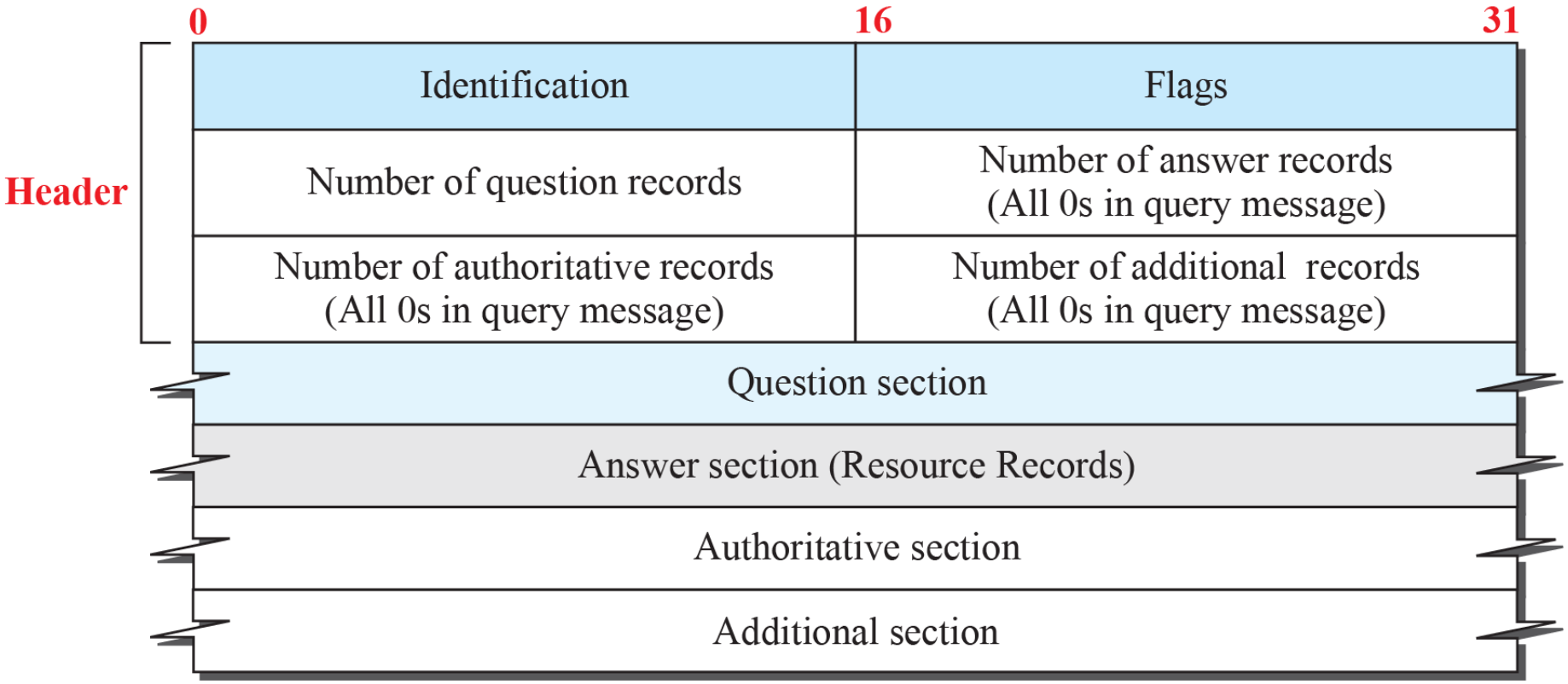
Figure 2.45: DNS message



Note:

The query message contains only the question section. The response message includes the question section, the answer section, and possibly two other sections.

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Example 2.14

In UNIX and Windows, the nslookup utility can be used to retrieve address/name mapping. The following shows how we can retrieve an address when the domain name is given.

```
$nslookup www.forouzan.biz
Name:   www.forouzan.biz
Address: 198.170.240.179
```


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2-4 PEER-TO-PERR PARADIGM

In this section, we discuss the peer-to peer paradigm. Peer-to-peer gained popularity with Napster, an online music file. Napster paved the way for peer-to-peer file-distribution models that came later. Gnutella was followed by Fast-
Track, BitTorrent, WinMX, and GNUnet.

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2.4.1 P2P Networks

Internet users that are ready to share their resources become peers and form a network. When a peer in the network has a file) to share, it makes it available to the rest of the peers. An interested peer can connect itself to the computer where the file is stored and download it. After a peer downloads a file, it can make it available for other peers to download. As more peers join and download that file, more copies of the file become available to the group.

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2.4.1 (continued)

- **Centralized Networks**
- **Decentralized Network**
  - *Unstructured Networks*
  - *Structured Networks*

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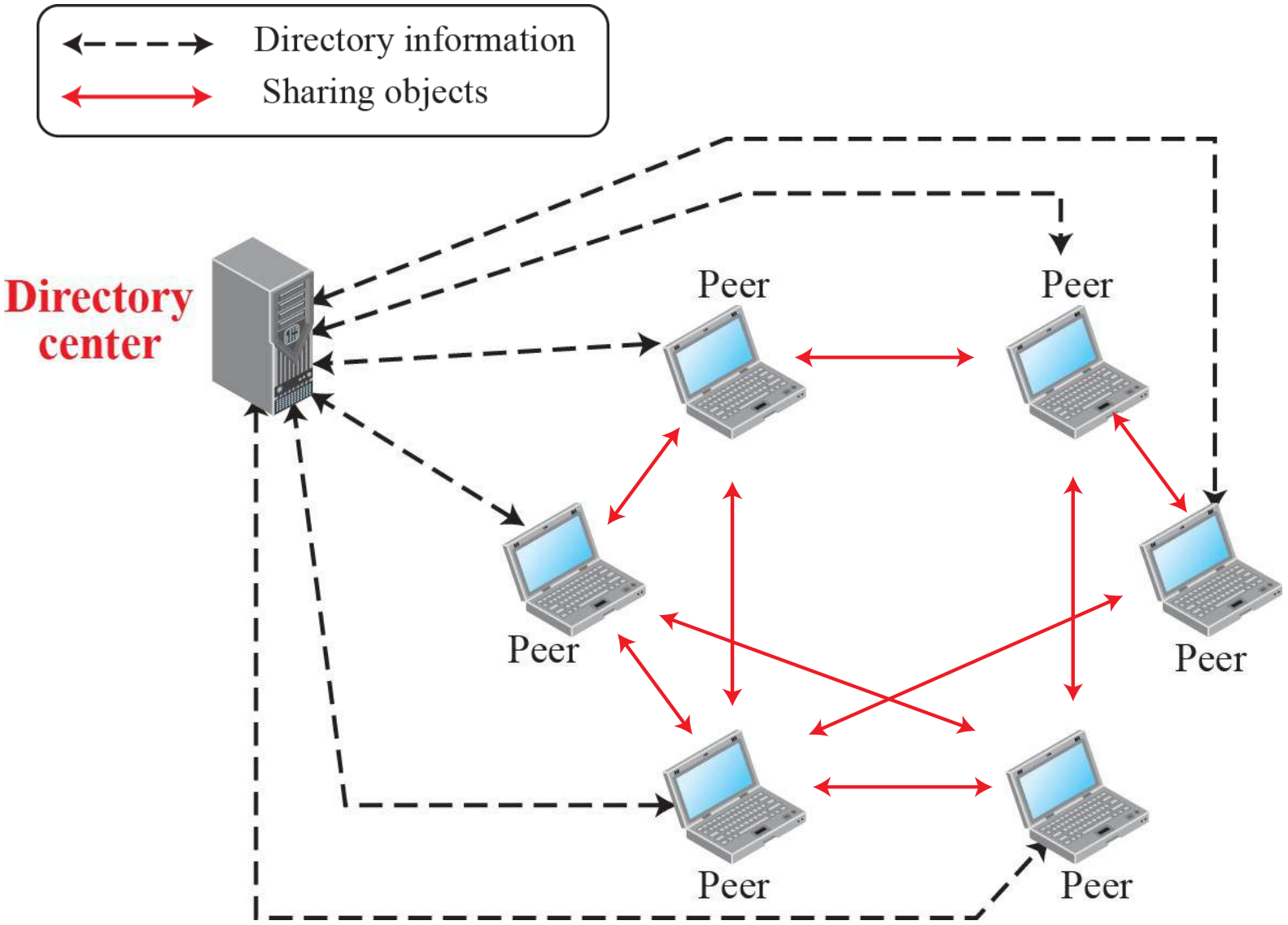
Figure 2.46: Centralized network



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2.4.2 Distributed Hash Function

A Distributed Hash Table (DHT) distributes data among a set of nodes according to some predefined rules. Each peer in a DHT-based network becomes responsible for a range of data items. To avoid the flooding overhead that we discussed for unstructured P2P networks, DHT-based networks allow each peer to have a partial knowledge about the whole network. This knowledge can be used to route the queries about the data items to the responsible nodes using effective and scalable procedures.

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2.4.2 (continued)

- **Address Space**
  - Hashing Peer Identifier
  - Hashing Object Identifier
  - Storing the Object
  - Routing
  - Arrival and Departure of Nodes

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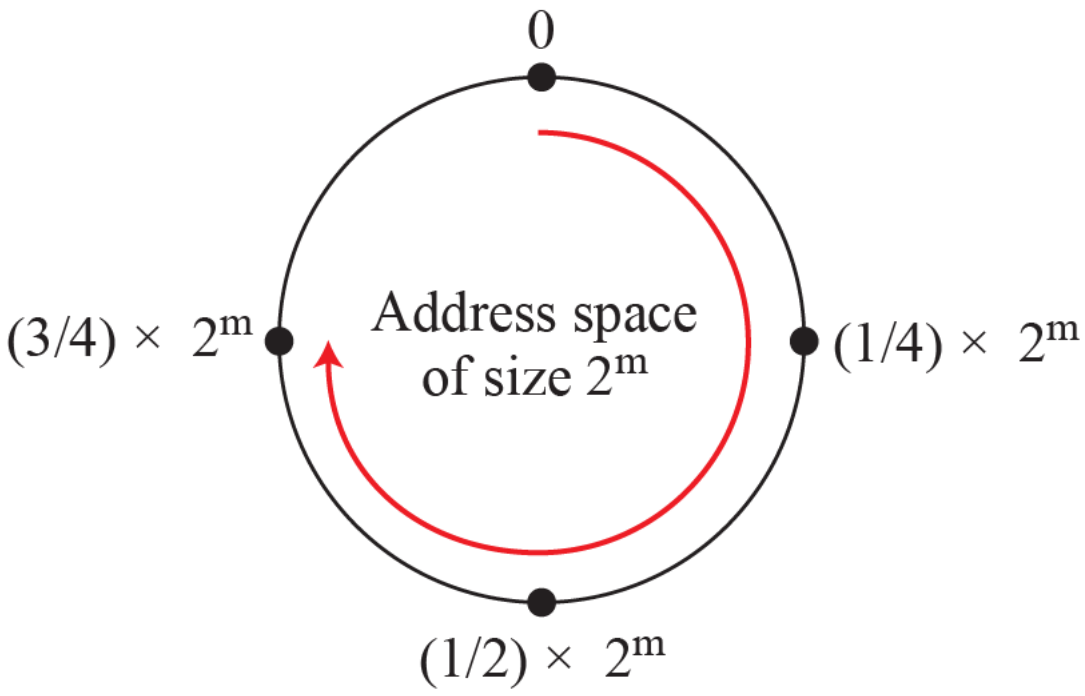
Figure 2.47: Address space

Note:

1. Space range is 0 to $2^m - 1$.
2. Calculation is done modulo 2^m.



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Example 2.15

Although the normal value of m is 160, for the purpose of demonstration, we use $m = 5$ to make our examples tractable. In Figure 2.48, we assume that several peers have already joined the group. The node N5 with IP address 110.34.56.20 has a file named Liberty that wants to share with its peers. The node makes a hash of the file name, "Liberty," to get the key = 14. Since the closest node to key 14 is node N17, N5 creates a reference to file name (key), its IP address, and the port number (and possibly some other information about the file) and sends this reference to be stored in node N17. In other words, the file is stored in N5, the key of the file is k14 (a point in the DHT ring), but the reference to the file is stored in node N17.

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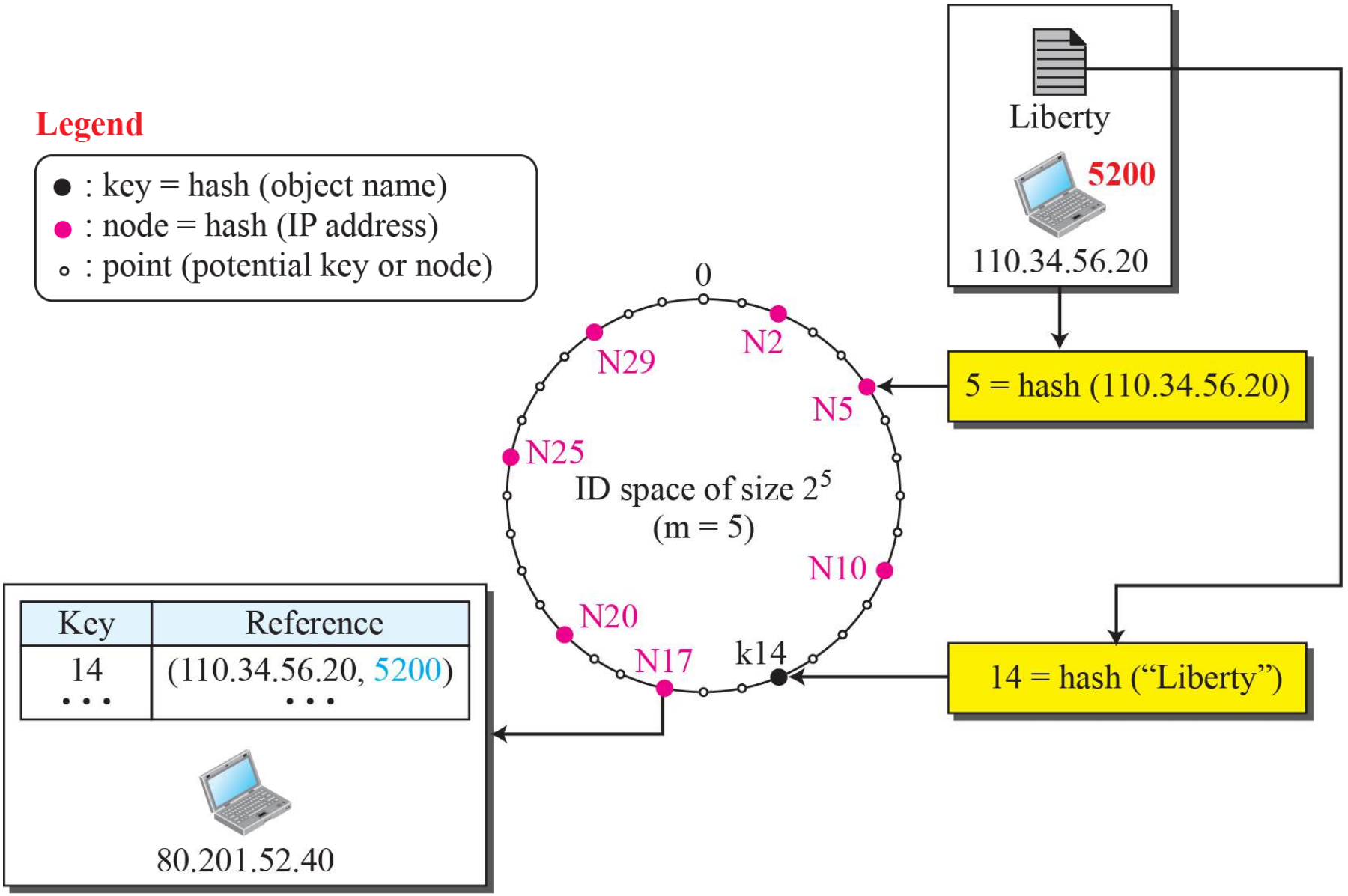
Figure 2.48: Example 2.15



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2.4.3 Chord

There are several protocols that implement DHT systems. In this section, we introduce the Chord protocol for its simplicity and elegant approach to routing queries. Chord was published by Stoica et al in 2001. We briefly discuss the main feature of this algorithm here.

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2.4.3 (continued)

- ***Identifier Space***
- ***Finger Table***
- ***Interface***
  - *Lookup*
  - *Stabilize*
  - *Fix_Finger*
  - *Join*
  - *Leave or Fail*

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Table 2.14: Finger table

i	*Target Key*	*Successor of Target Key*	*Information about Successor*
1	N + 1	Successor of N + 1	IP address and port of successor
2	N + 2	Successor of N + 2	IP address and port of successor
⋮	⋮	⋮	⋮
m	$N + 2^{m-1}$	Successor of $N + 2^{m-1}$	IP address and port of successor

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Figure 2.49: An example of a ring in Chord

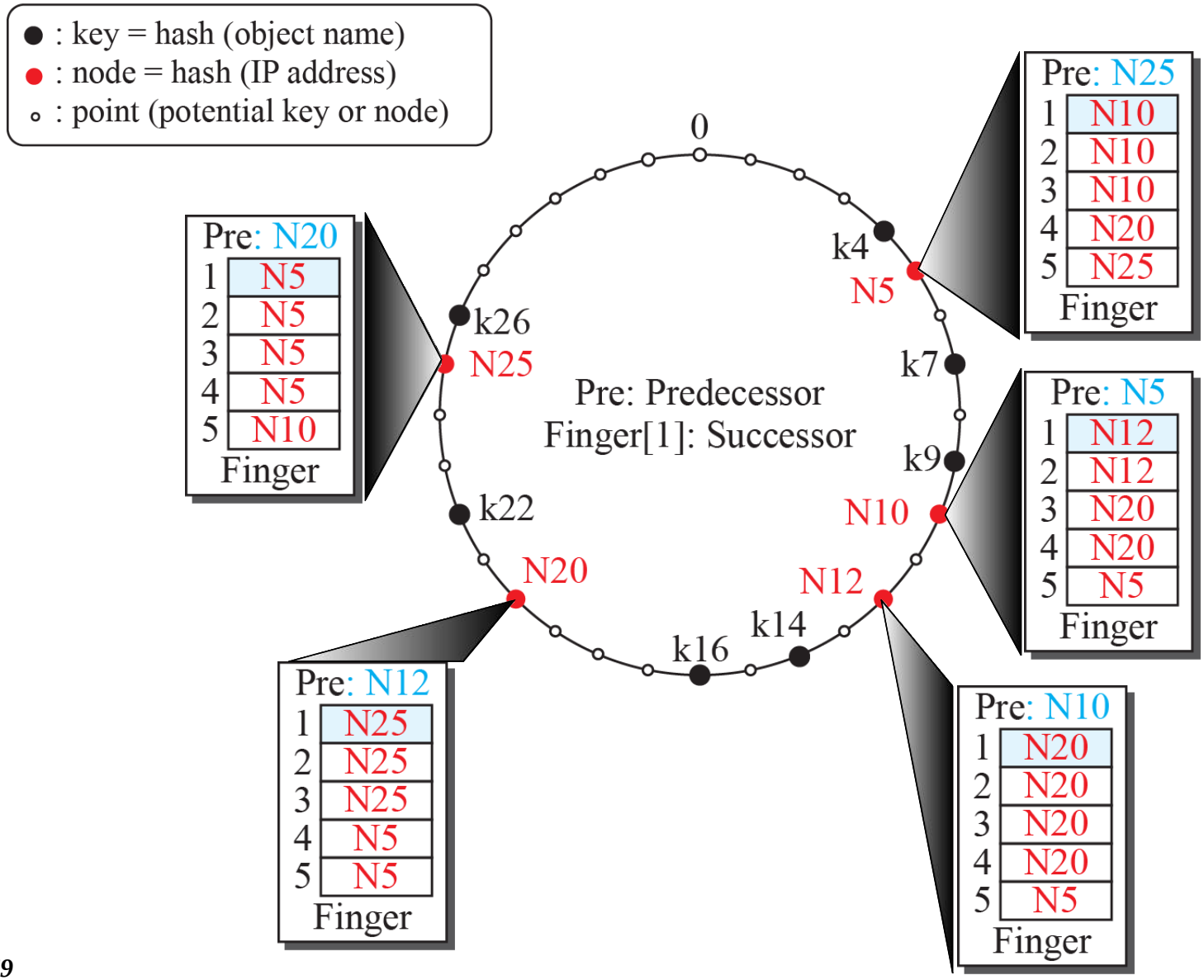

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Table 2.15: Lookup

```
Lookup (key)
{
        if (node is responsible for the key)
                return (node's ID)
        else
                 return find_succesor (key)
}

find_successor (id)
{
        x = find_ predecessor (id)
        return x.finger[1]
}
```


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Table 2.15: Lookup (continued)

```
find_predecessor (id)
{
        x = N                                          // N is the current node
        while (id ∉ (x, x.finger[1]]
        {
                x = x.find_closest_predecessor (id)    // Let x find it
        }
        return x
}

find_closest_predecessor (id)
{
        for (i = m downto 1)
        {
                if (finger [i] ∈ (N, id))              //N is the current node
                        return (finger [i])
        }
        return N                                       //The node itself is closest predecessor
}
```


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Example 2.16

Assume node N5 in Figure 2.49 needs to find the responsible node for key k14. Figure 2.50 shows the sequence of 8 eventsto do so.

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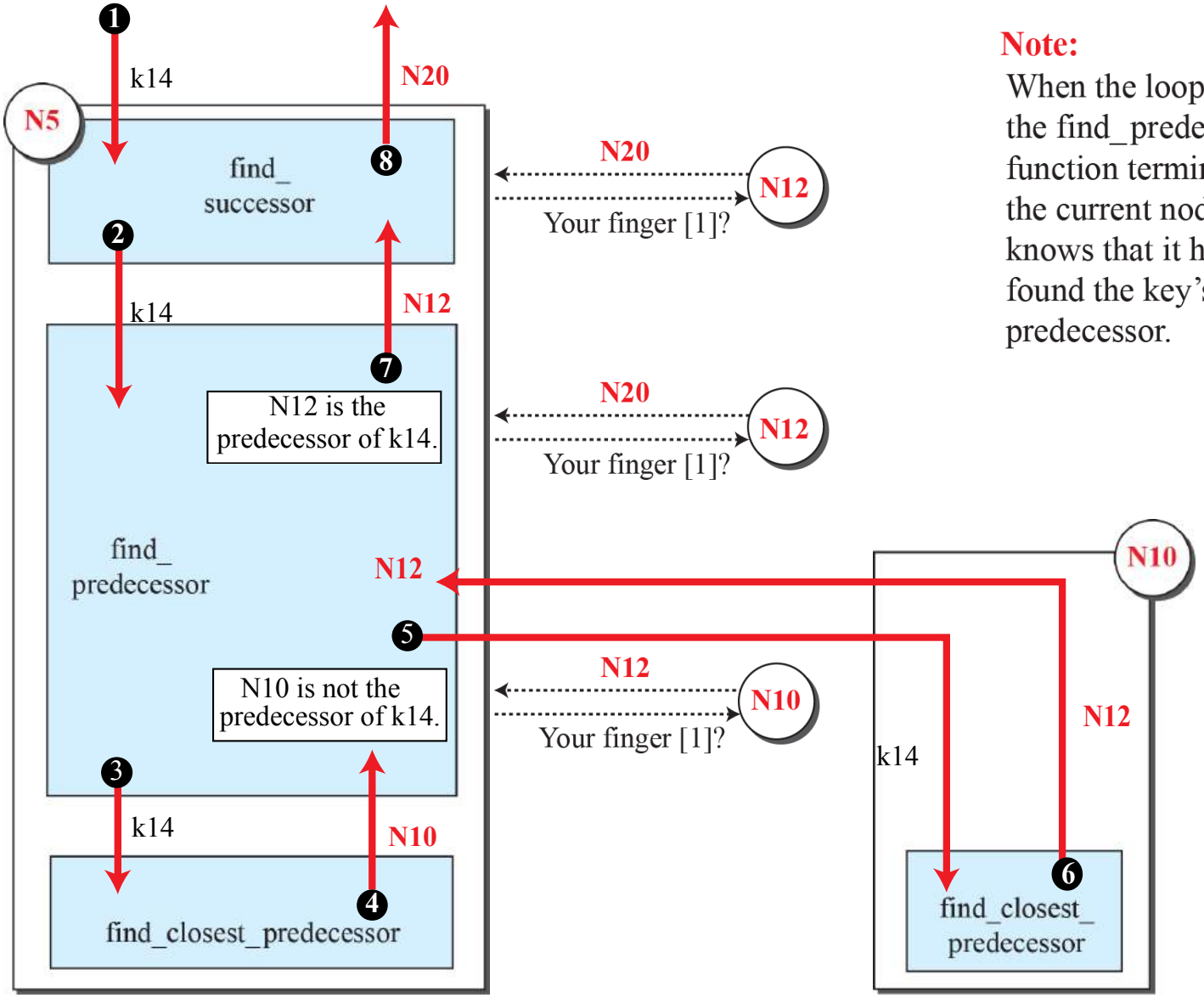
Figure 2.50: Example 2.16



Note:
When the loop in the find_predecessor function terminates, the current node knows that it has found the key's predecessor.

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Table 2.16: Stabilize

```
Stabilize ()
{
        P = finger[1].Pre                    //Ask the successor to return its predecessor
        if(P∈ (N, finger[1]))  finger[1] = P      // P is the possible successor of N
        finger[1].notify (N)                      // Notify P to change its predecessor
}
Notify (x)
{
            if (Pre = null or x ∈ (Pre, N))      Pre = x
}
```


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Table 2.17: Fix_Finger

```
Fix_Finger ()
{
        Generate (i ∈ (1, m])                    //Randomly generate i such as 1< i ≤ m
        finger[i] = find_successor (N + 2^(i − 1))        // Find value of finger[i]
}
```


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Table 2.18: Join

```
Join (x)
{
        Initialize (x)
        finger[1].Move_Keys (N)
}

Initialize (x)
{
        Pre = null
        if (x = null) finger[1] = N
        else finger[1] = x. Find_Successor (N)
}

Move_Keys (x)
{
        for (each key k)
        {
                if (x ∈ [k, N)) move (k to node x)          // N is the current node
        }
}
```


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Example 2.17

We assume that node N17 joins the ring in Figure 2.49 with the help of N5. Figure 2.51 shows the ring after the ring has been stabilized. The following five steps shows the process:

1. N17 set its predecessor to null and its successor to N20.
2. N17 then asks N20 to send k14 and k16 to N17.
3. N17 validates its own successor and asks N20 to change its predecessor to N17
4. The predecessor of N17 is updated to N12.
5. The finger table of nodes N17, N10, N5, and N12 is changed.

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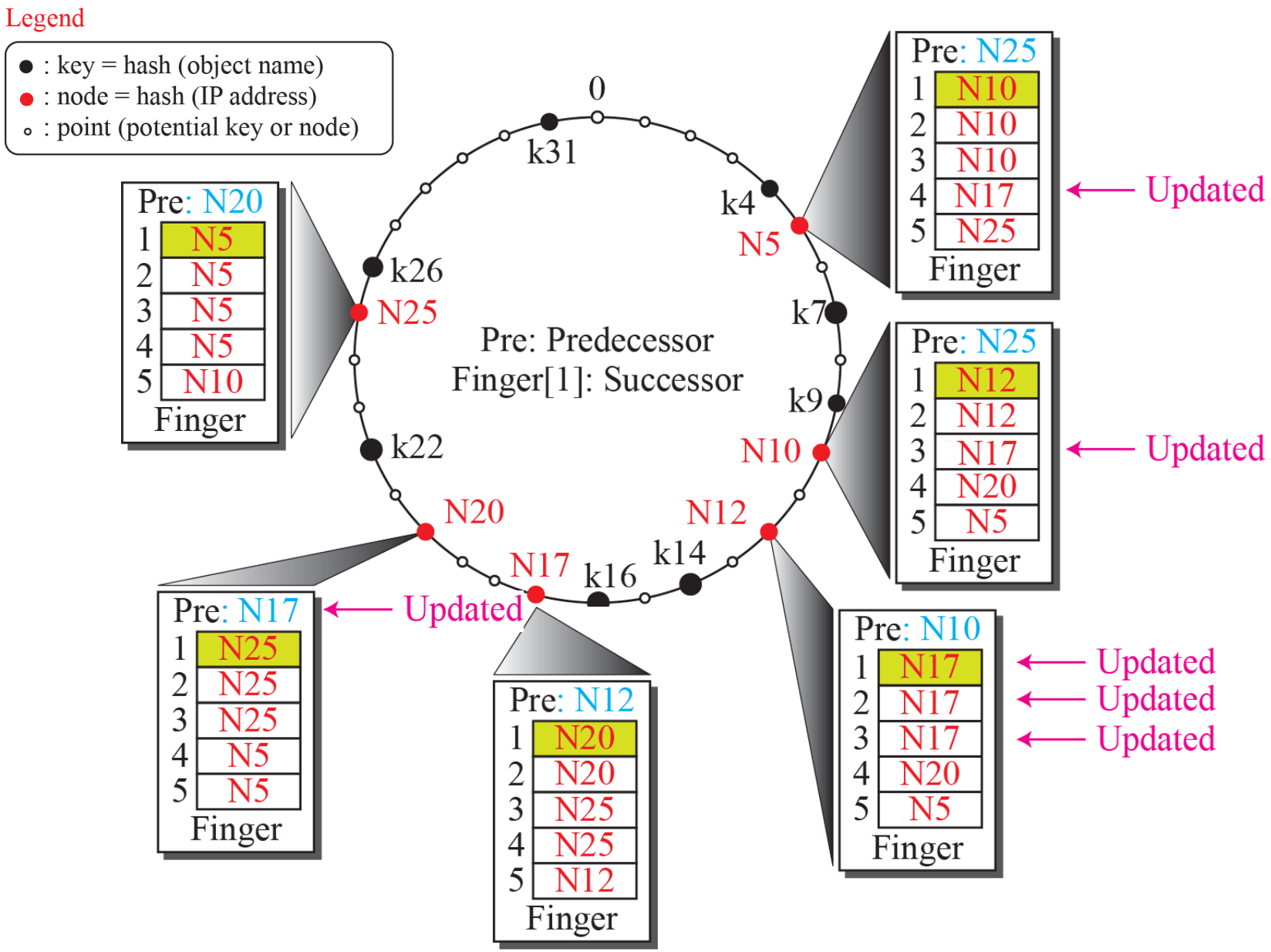
Figure 2.51: Example 2.17



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Example 2.18

We assume that a node, N10, leaves the ring in Figure 2.51. Figure 2.52 shows the ring after it has been stabilized. The following shows the process:

1. Node N5 finds out about N10's departure when it does not receive a pong message to its ping message. Node N5 changes its successor to N12 in the list of successors.
2. Node N5 immediately launches the *stabilize* function and asks N12 to change its predecessor to N5.
3. Hopefully, k7 and k9, which were under the responsibility of N10, have been duplicated in N12 before the departure of N10.

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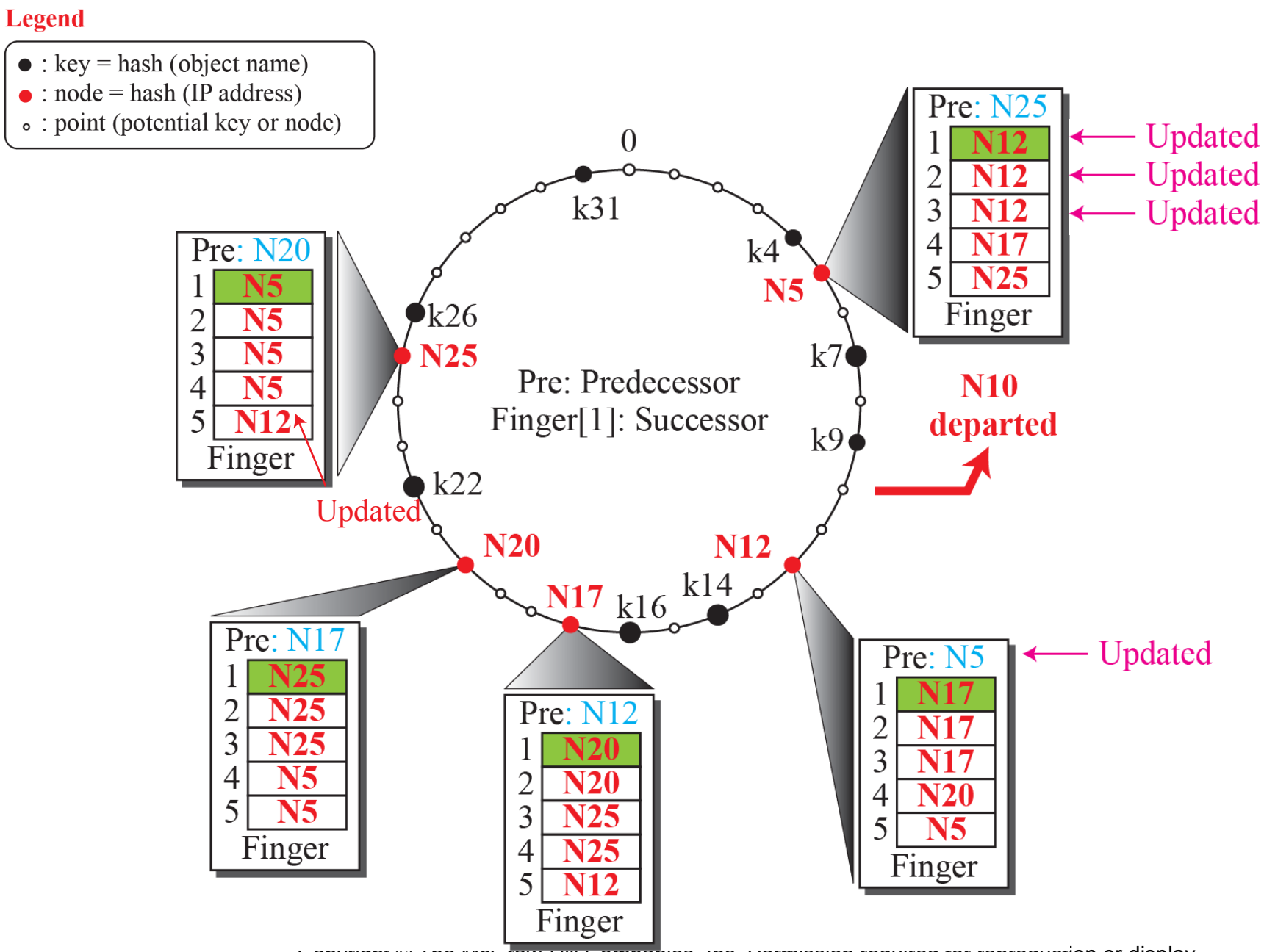
Figure 2.52: Example 2.18



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2.4.4 Pastry

Another popular protocol in the P2P paradigm is Pastry, designed by Rowstron and Druschel. Pastry uses DHT, as described before, but there are some fundamental differences between Pastry and Chord in the identifier space and routing process that we describe next.

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2.4.4 (continued)

- **Identifier Space**
- **Routing**
  - Routing Table
  - Leaf Set
- **Lookup**
- **Join**
- **Leave or Fail**
- **Application**

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Table 2.19: Routing table for a node in Pastry

Common prefix length	*0*	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*A*	*B*	*C*	*D*	*E*	*F*
0																
1																
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
31																

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Example 2.19

Let us assume that $m = 8$ bits and $b = 2$. This means that we have up to $2^m = 256$ identifiers, and each identifier has m/b = 4 digits in base $2^b = 4$. Figure 2.53 shows the situation in which there are some live nodes and some keys mapped to these nodes. The key k1213 is stored in two nodes because it is equidistant from them.

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Figure 2.53: An example of a Pastry ring

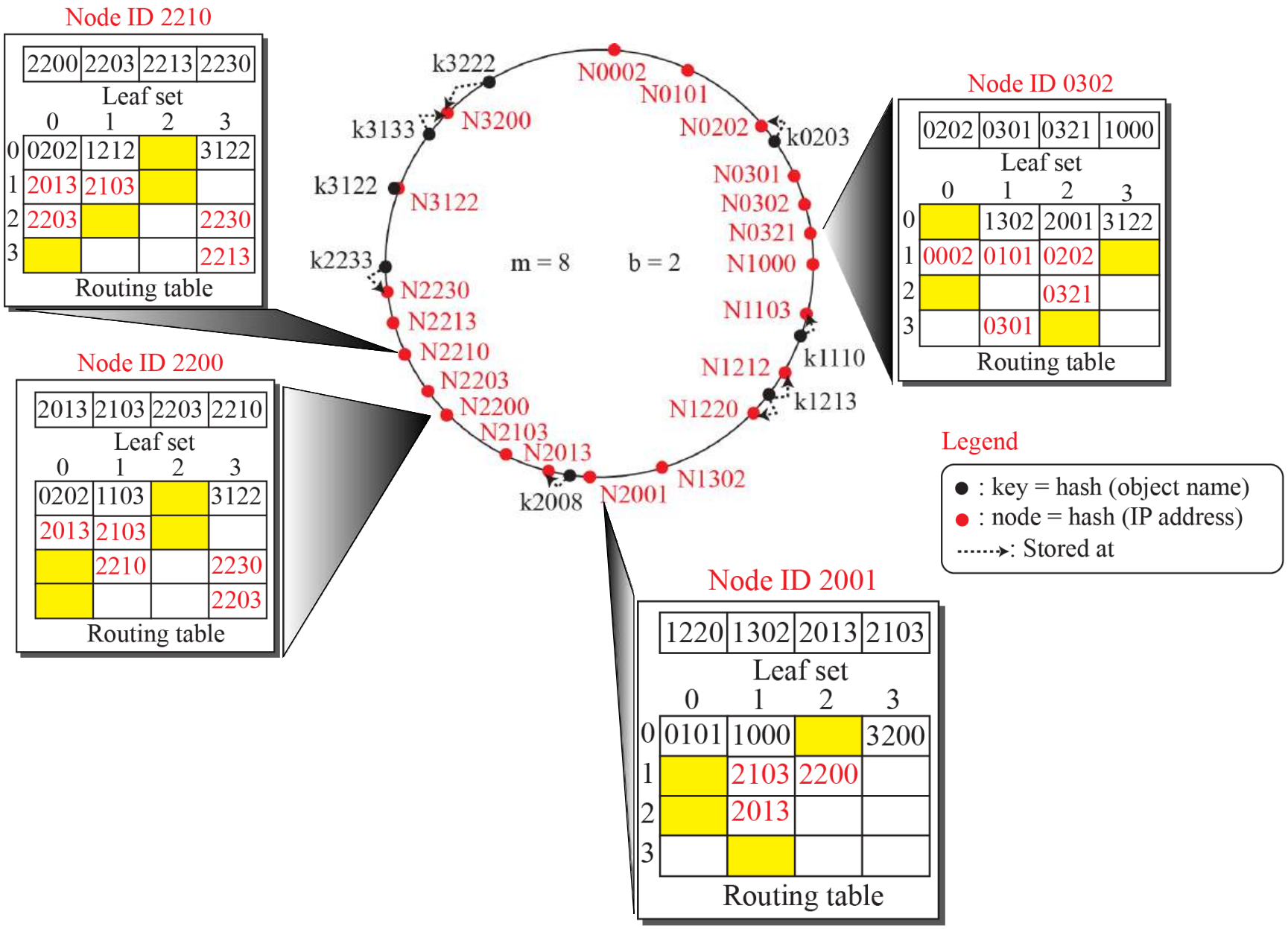

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Table 2.20: Lookup (Pastry)

```
Lookup (key)
{
        if (key is in the range of N's leaf set)
                forward the message to the closest node in the leaf set
        else
                route (key, Table)
}
route (key, Table)
{
        p = length of shared prefix between key and N
        v = value of the digit at position p of the key          // Position starts from 0
        if (Table [p, v] exists)
                forward the message to the node in Table [p, v]
        else
                forward the message to a node sharing a prefix as long as the current node, but
                numerically closer to the key.
}
```


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Example 2.20

In Figure 2.53, we assume that node N2210 receives a query to find the node responsible for key 2008. Since this node is not responsible for this key, it first checks its leaf set. The key 2008 is not in the range of the leaf set, so the node needs to use its routing table. Since the length of the common prefix is 1, p = 1. The value of the digit at position 1 in the key is v = 0. The node checks the identifier in Table [1, 0], which is 2013. The query is forwarded to node 2013, which is actually responsible for the key. This node sends its information to the requesting node.

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Example 2.21

In Figure 2.53, we assume that node N0302 receives a query to find the node responsible for the key 0203. This node is not responsible for this key, but the key is in the range of its leaf set. The closest node in this set is the node N0202. The query is sent to this node, which is actually responsible for this node. Node N0202 sends its information to the requesting node.

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Example 2.22

Figure 2.54 shows how a new node X with node identifier N2212 uses the information in four nodes in Figure 2.53 to create its initial routing table and leaf set for joining the ring. Note that the contents of these two tables will become closer to what they should be in the updating process. In this example, we assume that node 0302 is a nearby node to node 2212 based on the proximity metric.

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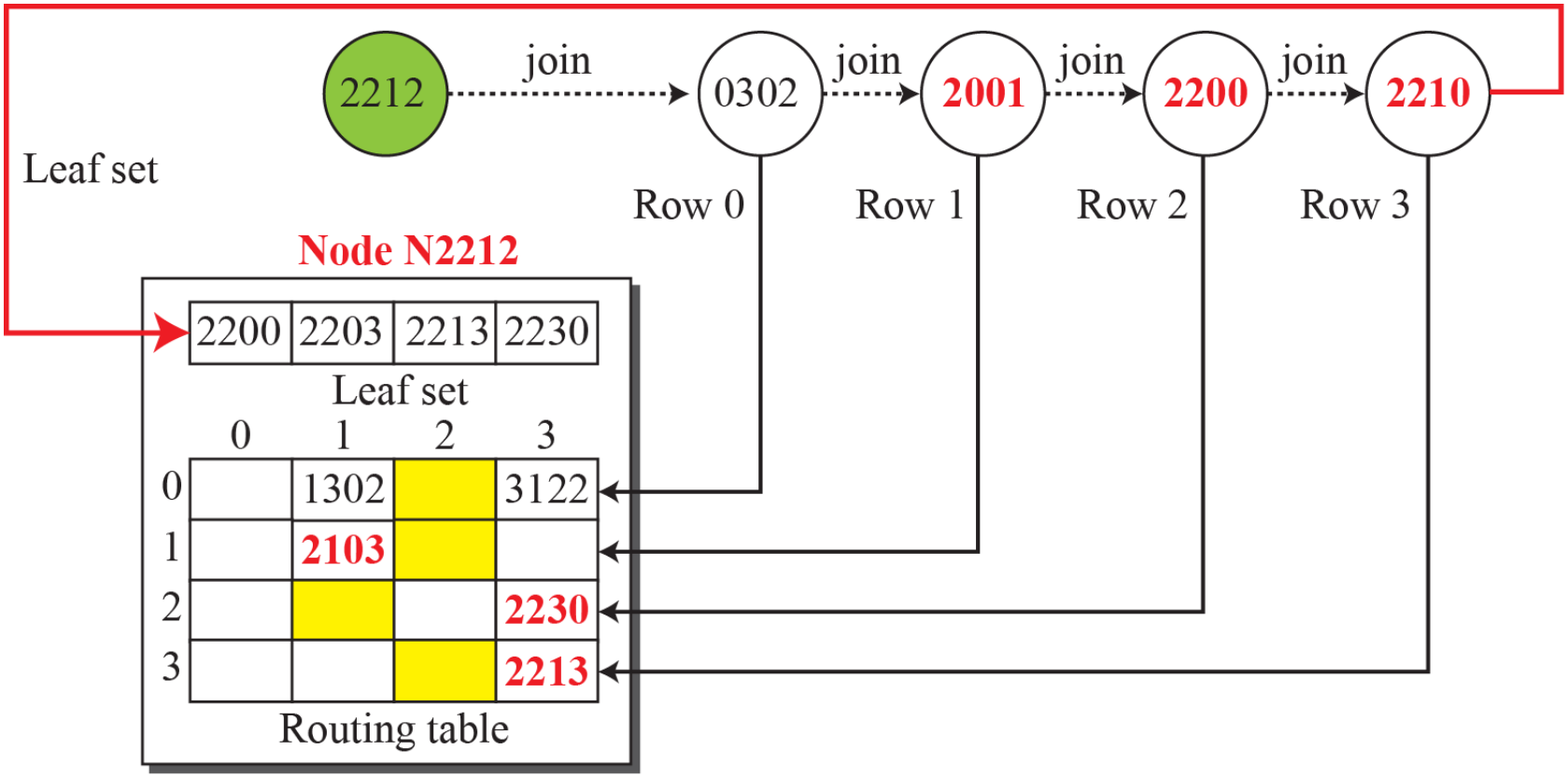
Figure 2.54: Example 2.22



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2.4.5Kademlia

Another DHT peer-to-peer network is Kademlia, designed by Maymounkov and Mazières. Kademlia, like Pastry, routes messages based on the distance between nodes, but the interpretation of the distance metric in Kademlia is different from the one in Pastry, as we describe below. In this network, the distance between the two identifiers (nodes or keys) is measured as the bitwise exclusive-or (XOR), between them.

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2.4.5 (continued)

- ***Identifier Space***
- ***Routing Table***
- ***K-Buckets***
  - *Parallel Query*
  - *Concurrent Updating*
- ***Join***
- ***Leave or Fail***

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Example 2.23

For simplicity, let us assume that $m = 4$. In this space, we can have 16 identifiers distributed on the leaves of a binary tree. Figure 2.55 shows the case with only eight live nodes and five keys.

As the figure shows, the key k3 is stored in N3 because $3 \oplus 3 = 0$. Although the key k7 looks numerically equidistant from N6 and N8, it is stored only in N6 because $6 \oplus 7 = 1$ but $6 \oplus 8 = 14$. Another interesting point is that the key k12 is numerically closer to N11, but it is stored in N15 because $11 \oplus 12 = 7$, but $15 \oplus 12 = 3$.

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Figure 2.55: *Example 2.23*

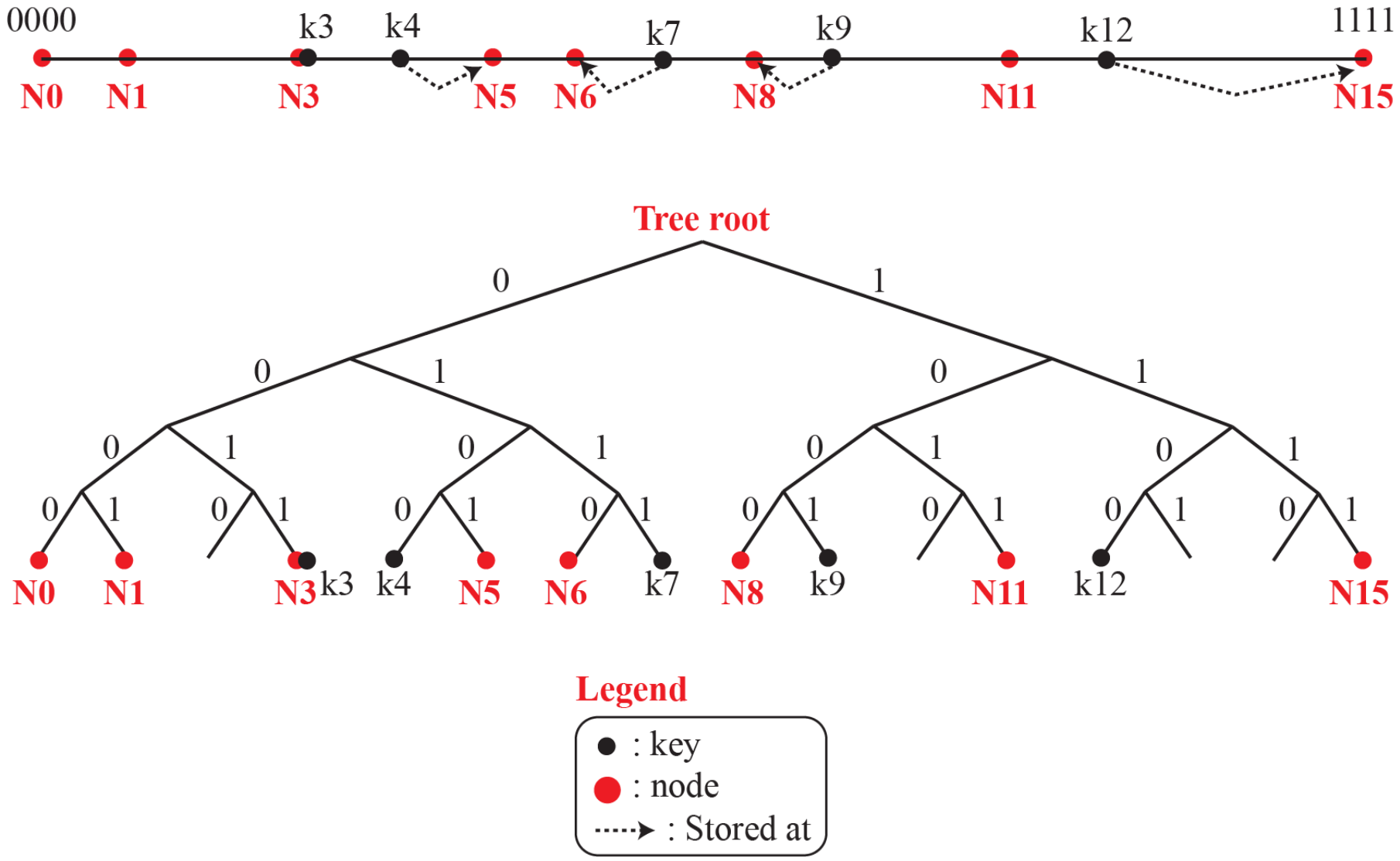

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Table 2.21: Routing table for a node in Kademlia

Common prefix length	*Identifiers*
0	Closest node(s) in subtree with common prefix of length 0
1	Closest node(s) in subtree with common prefix of length 1
⋮	⋮
$m-1$	Closest node(s) in subtree with common prefix of length $\boldsymbol{m}-1$

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Example 2.24

Let us find the routing table for Example 2.23. To make the example simple, we assume that each row uses only one identifier. Since m = 4, each node has four subtrees corresponding to four rows in the routing table. The identifier in each row represents the node that is closest to the current node in the corresponding subtree. Figure 2.56 shows all routing tables, but only three of the subtrees. We have chosen these three, out of eight, to make the figure smaller.

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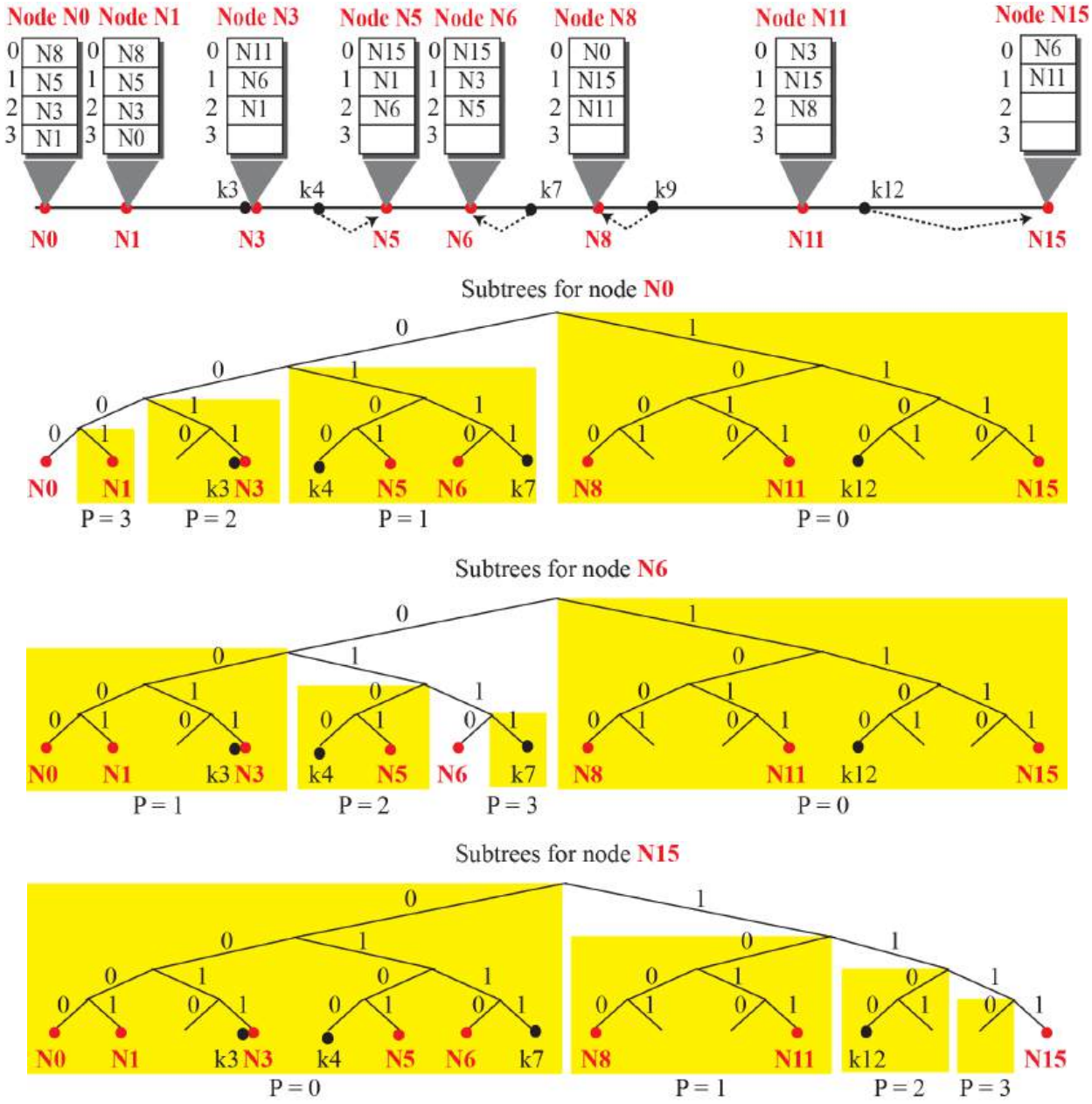
Figure 2.56: Example 2.24



P : Prefix lengh

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Example 2.25

In Figure 2.56, we assume node N0 $(0000)_2$ receives a lookup message to find the node responsible for k12 $(1100)_2$. The length of the common prefix between the two identifiers is 0. Node N0 sends the message to the node in row 0 of its routing table, node N8. Now node N8 $(1000)_2$needs to look for the node closest to k12 $(1100)_2$. The length of the common prefix between the two identifiers is 1. Node N8 sends the message to the node in row 1 of its routing table, node N15, which is responsible for k12. The routing process is terminated. The route is N0 → N8 → N15. It is interesting to note that node N15, $(1111)_2$, and k12, $(1100)_2$, have a common prefix of length 2, but row 2 of N15 is empty, which means that N15 itself is responsible for k12.

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Example 2.26

In Figure 2.56, we assume node N5 $(0101)_2$ receives a lookup message to find the node responsible for k7 $(0111)_2$. The length of the common prefix between the two identifiers is 2. Node N5 sends the message to the node in row 2 of its routing table, node N6, which is responsible for k7. The routing process is terminated. The route is N5 → N6.

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Example 2.27

In Figure 2.56, we assume node N11 $(1011)_2$ receives a lookup message to find the node responsible for k4 $(0100)_2$. The length of the common prefix between the two identifiers is 0. Node N11 sends the message to the node in row 0 of its routing table, node N3. Now node N3 $(0011)_2$needs to look for the node closest to k4 $(0100)_2$. The length of the common prefix between the two identifiers is 1. Node N3 sends the message to the node in row 1 of its routing table, node N6. And so on. Theroute is N11 → N3→ N6 → N5.

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2.4.6 BitTorrent

BitTorrent is a P2P protocol, designed by Bram Cohen, for sharing a large file among a set of peers. However, the term sharing in this context is different from other file sharing protocols. Instead of one peer allowing another peer to download the whole file, a group of peers takes part in the process to give all peers in the group a copy of the file. File sharing is done in a collaborating process called a ***torrent****.*

- ***BitTorrent with A Tracker***

- ***Trackerless BitTorrent***

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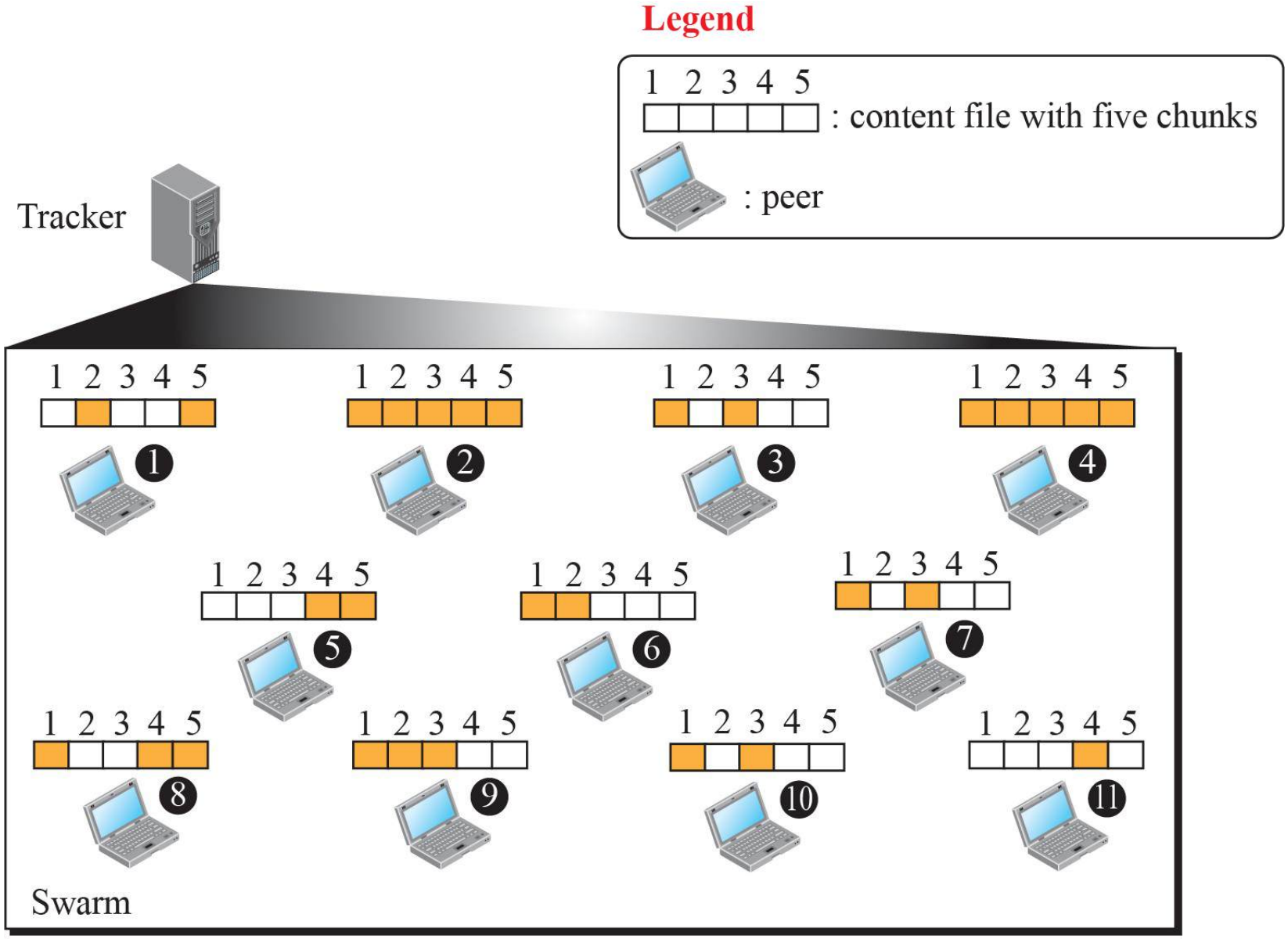
Figure 2.57: Example of a torrent



Note: Peers 2 and 4 are seeds; others are leeches.

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2-5 SOCKET-INTERFACE PROGRAMMING

In this section, we show how to write some simple client-server programs using C, a procedural programming language. We chose the C language in this section; In Chapter 11, we expand this idea in Java, which provides a more compact version.

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2.5.1 Socket Interface in C

In this section, we show how this interface is implemented in the C language. The important issue in socket interface is to understand the role of a socket in communication. The socket has no buffer to store data to be sent or received. It is capable of neither sending nor receiving data. The socket just acts as a reference or a label. The buffers and necessary variables are created inside the operating system.

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2.5.1 (continued)

- **Data Structure for Socket**
- **Header Files**
- **Communication Using UDP**
  - Sockets Used for UDP
  - Communication Flow Diagram
  - Programming Examples
- **Communication Using TCP**
  - Sockets Used in TCP
  - Communication Flow Diagram

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Figure 2.58: Socket data structure

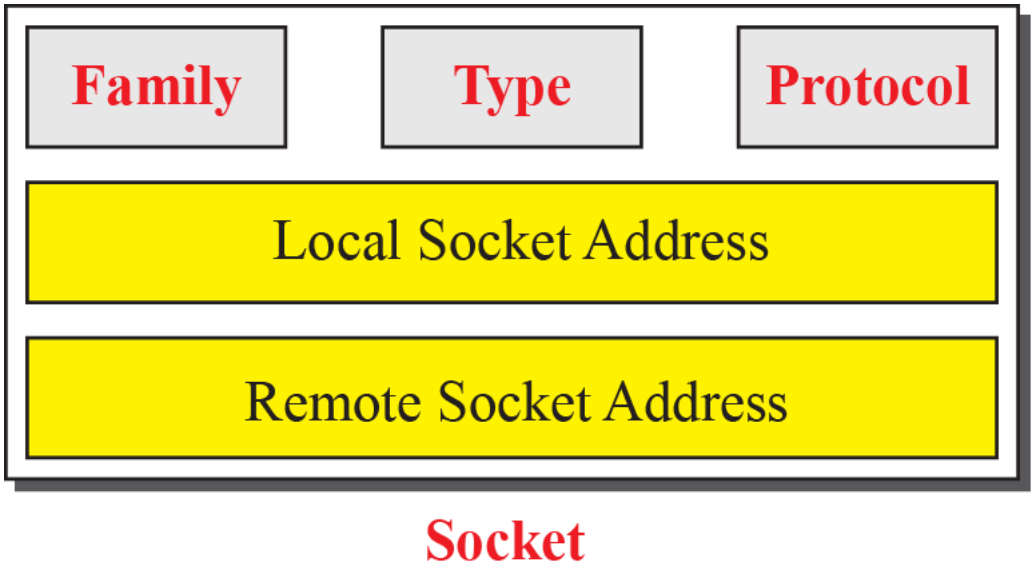

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Figure 2.59: Sockets for UDP communication

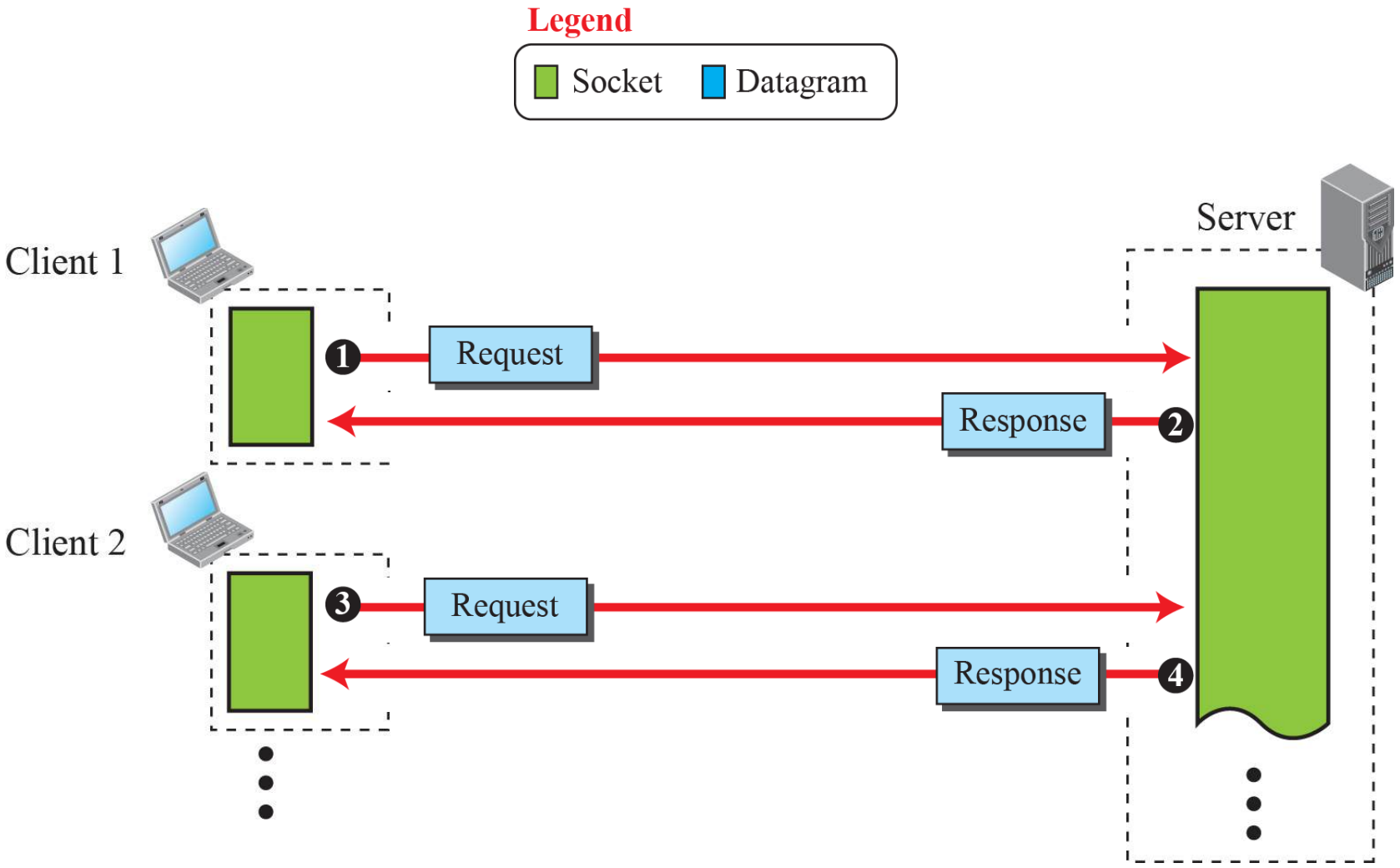

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Figure 2.60: Flow diagram for iterative UDP communication

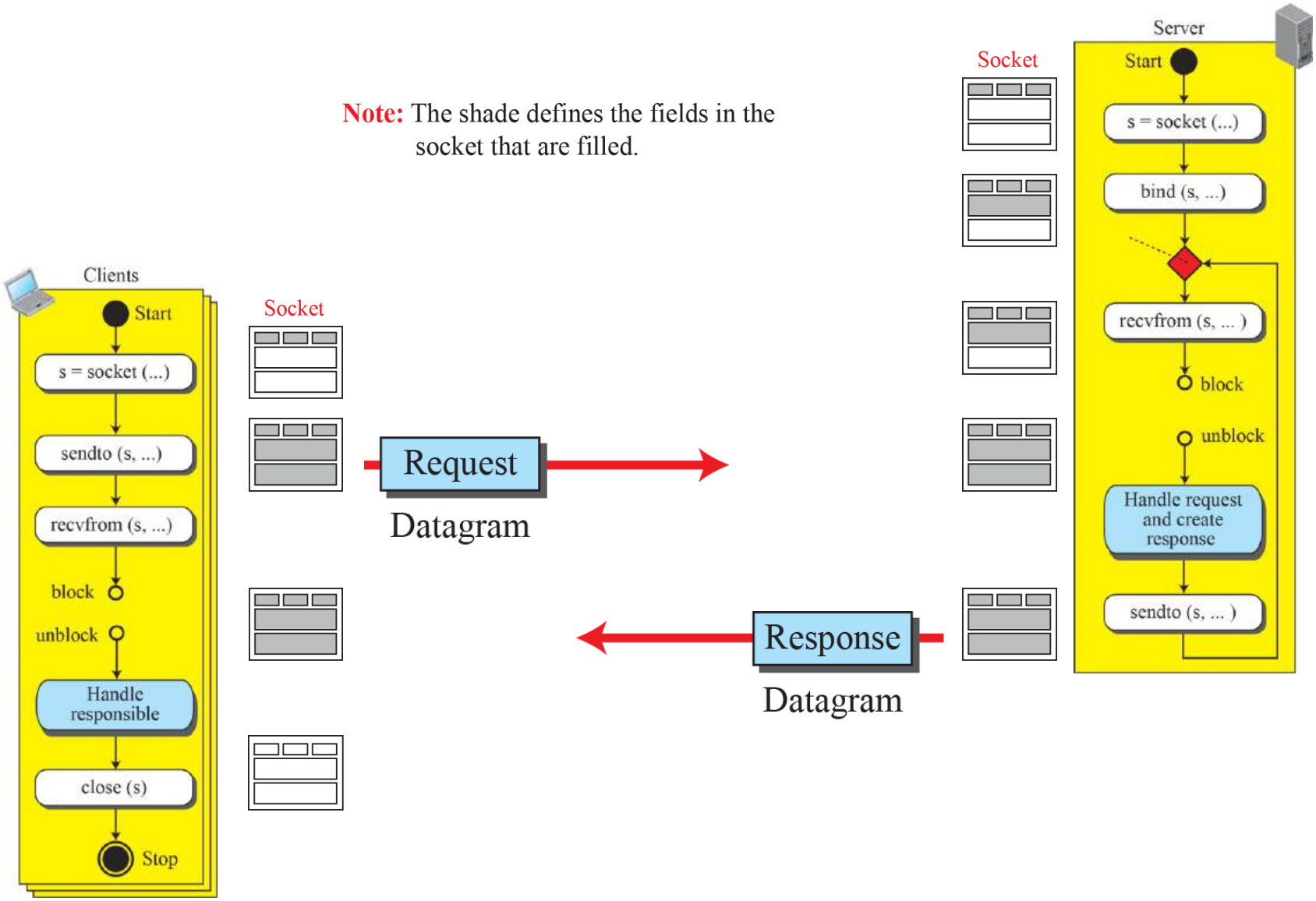

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Table 2.22: Echo server program using UDP

```
1  // UDP echo server program
2  #include "headerFiles.h"
3  int main (void)
4  {
5     // Declare and define variables
6     int s;                                  // Socket descriptor (reference)
7     int len;                                // Length of string to be echoed
8     char   buffer [256];                    // Data buffer
9     struct sockaddr_in servAddr;            // Server (local) socket address
10    struct sockaddr_in clntAddr;            // Client (remote) socket address
11    int  clntAddrLen;                       // Length of client socket address
12    // Build local (server) socket address
13    memset (&servAddr, 0, sizeof (servAddr));          // Allocate memory
14    servAddr.sin_family =   AF_INET;                   // Family field
15    servAddr.sin_port = htons (SERVER_PORT)            // Default port number
16    servAddr.sin_addr.s_addr =  htonl (INADDR_ANY);    // Default IP address
17    // Create socket
18    if ((s = socket (PF_INET, SOCK_DGRAM, 0) < 0);
```


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Table 2.22: Echo server program using UDP (continued)

```
19      {
20              perror ("Error: socket failed!");
21              exit (1);
22      }
23      // Bind socket to local address and port
24      if ((bind (s, (struct sockaddr*) &servAddr, sizeof (servAddr)) < 0);
25      {
26              perror ("Error: bind failed!");
27              exit (1);
28      }
29      for ( ; ; )         // Run forever
30      {
31              // Receive String
32              len = recvfrom (s, buffer, sizeof (buffer), 0,
33                      (struct sockaddr*)&clntAddr, &clntAddrLen);
34              // Send String
35              sendto (s, buffer, len, 0, (struct sockaddr*)&clntAddr, sizeof(clntAddr));
36      } // End of for loop
37  } // End of echo server program
```


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Table 2.23: Echo client program using UDP

```
1  // UDP echo client program
2  #include "headerFiles.h"
3  int main (int argc, char* argv[ ])          // Three arguments to be checked later
4  {
5       // Declare and define variables
6       int   s;                               // Socket descriptor
7       int   len;                             // Length of string to be echoed
8       char* servName;                        // Server name
9       int   servPort;                        // Server port
10      char* string;                          // String to be echoed
11      char buffer[256 + 1];                  // Data buffer
12      struct sockaddr_in servAddr;           // Server socket address
13      // Check and set program arguments
14      if (argc != 3)
15      {
16           printf ("Error: three arguments are needed!");
17           exit(1);
18      }
```


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Table 2.23: Echo client program using UDP (continued)

```
19      servName = argv[1];
20      servPort = atoi (argv[2]);
21      string =  argv[3];
22      // Build server socket address
23      memset (&servAddr, 0, sizeof (servAddr));
24      servAddr.sin_family = AF_INET;
25      inet_pton (AF_INET, servName, &servAddr.sin_addr);
26      servAddr.sin_port = htons (servPort);
27      // Create socket
28      if ((s = socket (PF_INET, SOCK_DGRAM, 0) < 0);
29      {
30          perror ("Error: Socket failed!");
31          exit (1);
32      }
33      // Send echo string
34      len =  sendto (s, string, strlen (string), 0, (struct sockaddr)&servAddr, sizeof (servAddr));
35      // Receive echo string
36      recvfrom (s, buffer, len, 0, NULL, NULL);
37      // Print and verify echoed string
```


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Table 2.23: Echo client program using UDP (continued)

```
38        buffer [len] = '\0';
39        printf ("Echo string received: ";
40        fputs (buffer, stdout);
41        // Close the socket
42        close (s);
43        // Stop the program
44        exit (0);
45  } // End of echo client program
```


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Figure 2.61: Sockets used in TCP communication

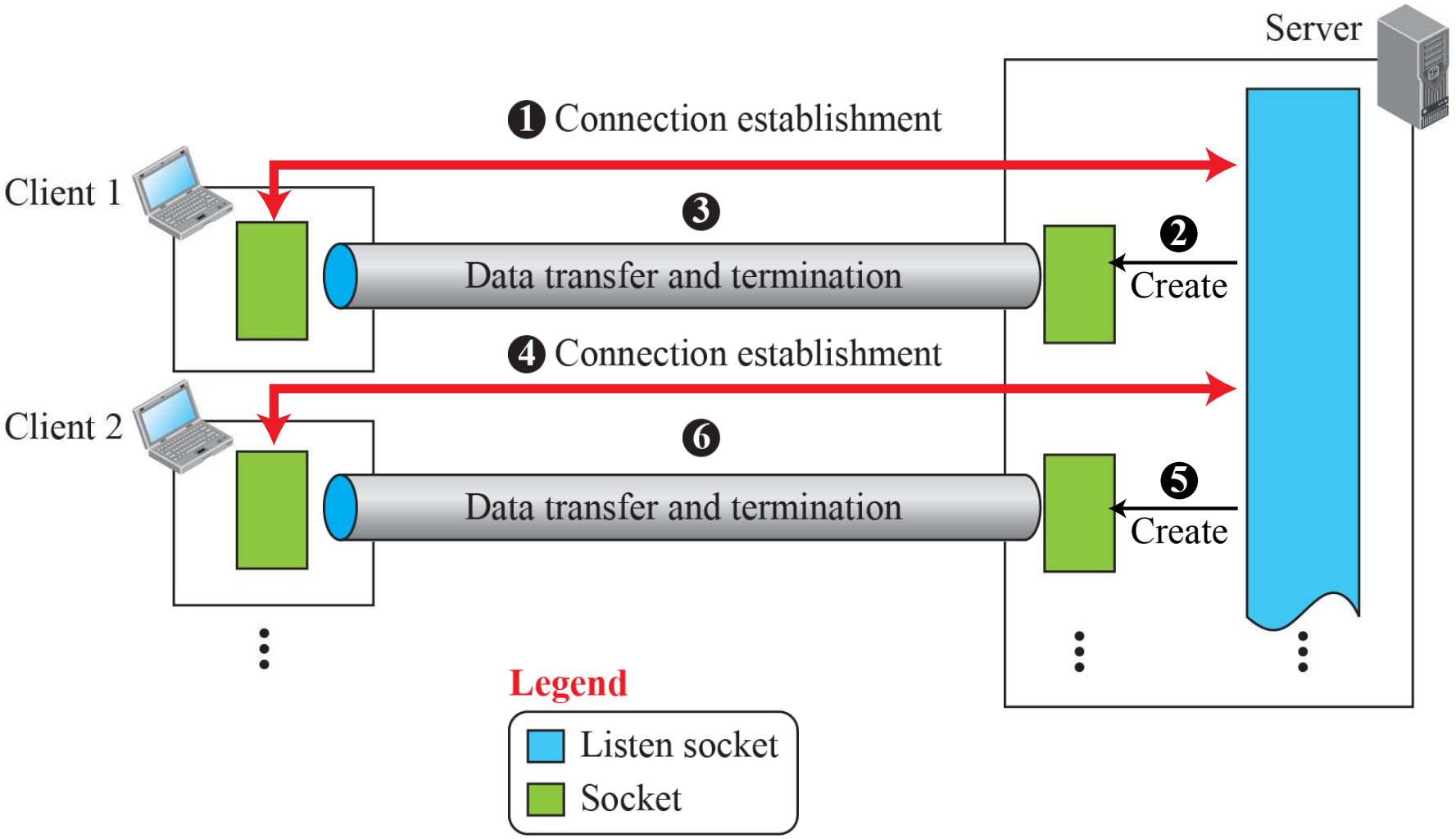

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Figure 2.62: Flow diagram for iterative TCP communication

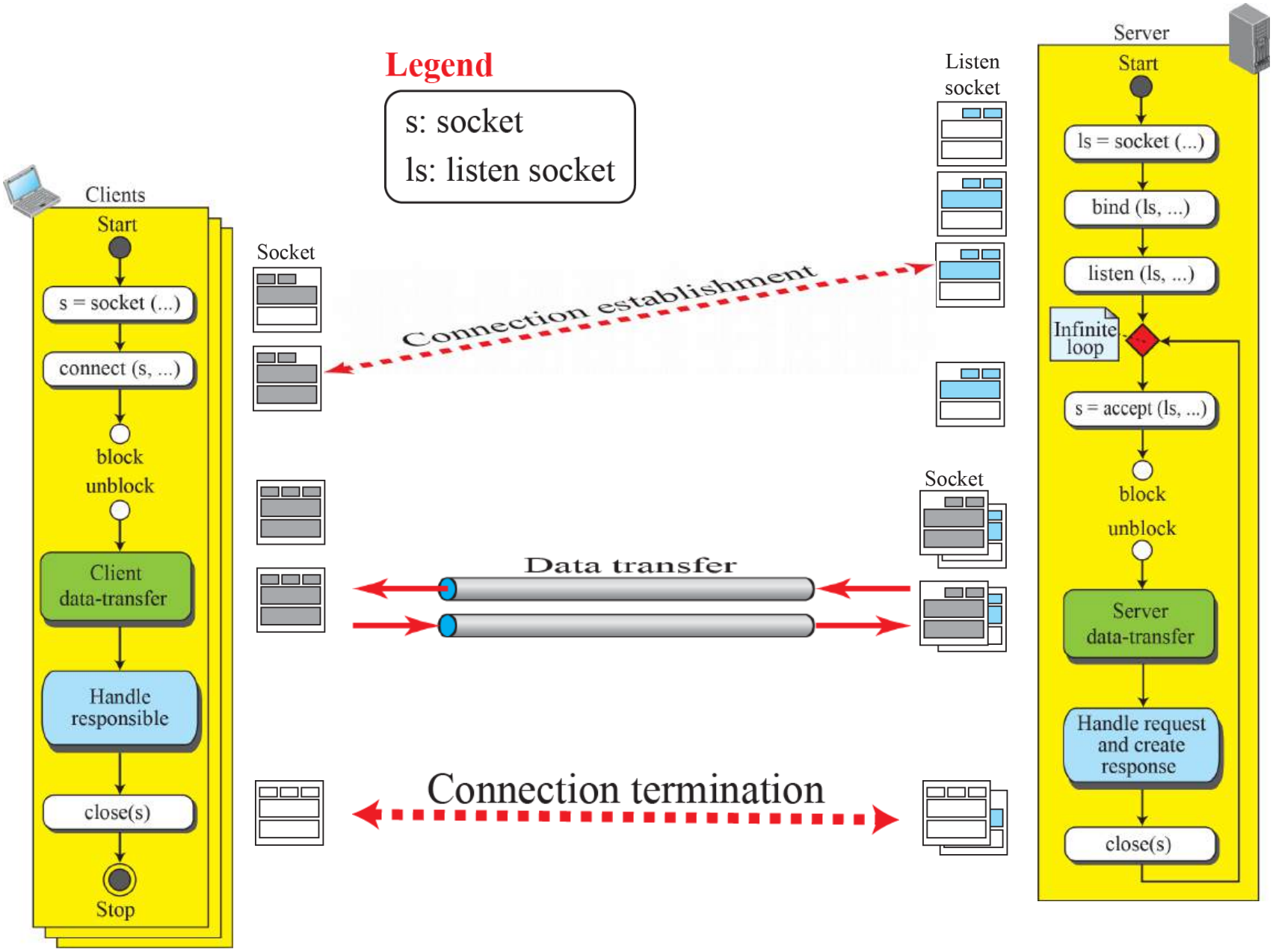

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Figure 2.63: Flow diagram for data-transfer boxes

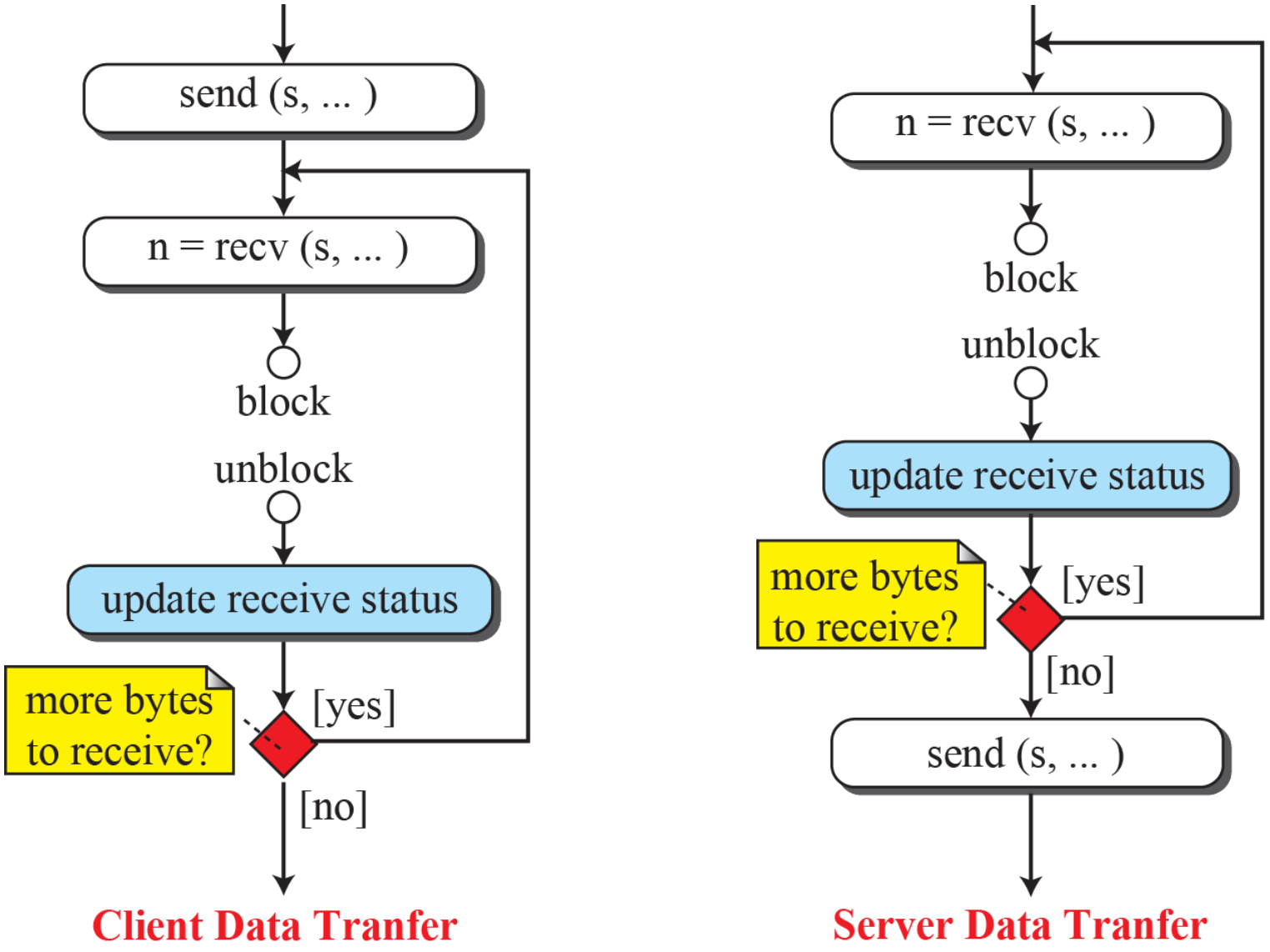

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Figure 2.64: Buffer used for receiving

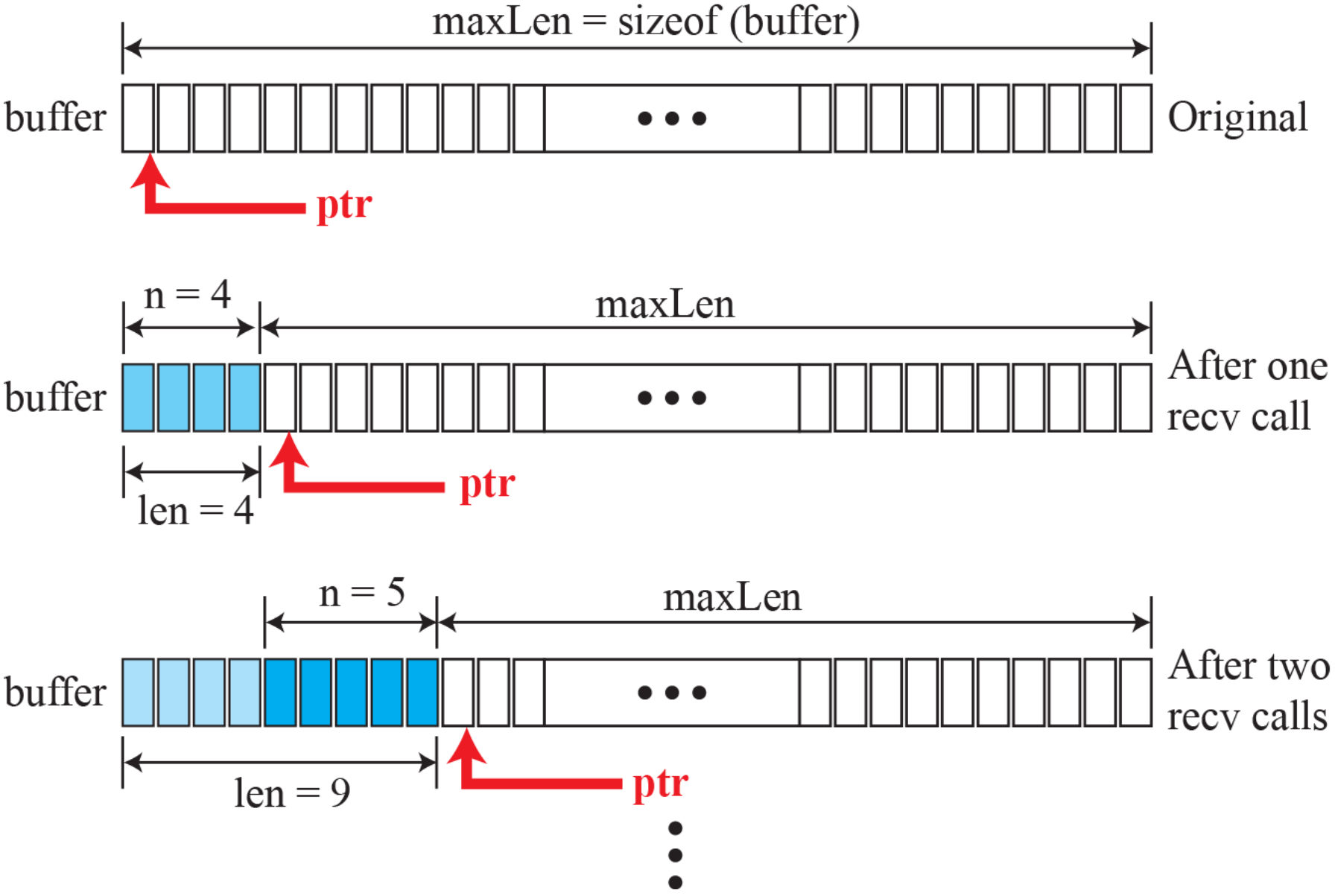

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Table 2.24: Echo server program using the services of TCP

```
1  // Echo server program
2  #include "headerFiles.h"
3  int main (void)
4  {
5      // Declare and define
6      int ls;                          // Listen socket descriptor (reference)
7      int s;                           // socket descriptor (reference)
8      char buffer [256];               // Data buffer
9      char* ptr = buffer;              // Data buffer
10     int len = 0;                     // Number of bytes to send or receive
11     int maxLen = sizeof (buffer);    // Maximum number of bytes to receive
12     int n = 0;                       // Number of bytes for each recv call
13     int waitSize = 16;               // Size of waiting clients
14     struct sockaddr_in serverAddr;   // Server address
15     struct sockaddr_in clientAddr;   // Client address
16     int clntAddrLen;                 // Length of client address
17     // Create local (server) socket address
18     memset (&servAddr, 0, sizeof (servAddr));
19     servAddr.sin_family = AF_INET;
20     servAddr.sin_addr.s_addr = htonl (INADDR_ANY);     // Default IP address
```


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Table 2.24: TCP Echo server program(continued)

```
21      servAddr.sin_port = htons (SERV_PORT);          // Default port
22      // Create listen socket
23      if (ls = socket (PF_INET, SOCK_STREAM, 0) < 0);
24      {
25          perror ("Error: Listen socket failed!");
26          exit (1);
27      }
28      // Bind listen socket to the local socket address
29      if (bind (ls, &servAddr, sizeof (servAddr)) < 0);
30      {
31          perror ("Error: binding failed!");
32          exit (1);
33      }
34      // Listen to connection requests
35      if (listen (ls, waitSize) < 0);
36      {
37          perror ("Error: listening failed!");
38          exit (1);
39      }
40      // Handle the connection
```


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Table 2.24: TCP Echo server program (continued)

```
41      for ( ; ; )                                    // Run forever
42      {
43          // Accept connections from client
44          if (s = accept (ls, &clntAddr, &clntAddrLen) < 0);
45          {
46              perror ("Error: accepting failed!);
47              exit (1);
48          }
49          // Data transfer section
50          while ((n = recv (s, ptr, maxLen, 0)) > 0)
51          {
52              ptr + = n;                  // Move pointer along the buffer
53              maxLen - = n;               // Adjust maximum number of bytes to receive
54              len + = n;                  // Update number of bytes received
55          }
56          send (s, buffer, len, 0);       // Send back (echo) all bytes received
57          // Close the socket
58          close (s);
59      } // End of for loop
60  } // End of echo server program
```


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Table 2.25: Echo client program using the services of TCP

```
1   // TCP echo client program
2   #include "headerFiles.h"
3   int main (int argc, char* argv[ ])          // Three arguments to be checked later
4   {
5       // Declare and define
6       int  s;                                 // Socket descriptor
7       int  n;                                 // Number of bytes in each recv call
8       char* servName;                         // Server name
9       int servPort;                           // Server port number
10      char* string;                           // String to be echoed
11      int  len;                               // Length of string to be echoed
12      char  buffer [256 + 1];                 // Buffer
13      char* ptr = buffer;                     // Pointer to move along the buffer
14      struct sockaddr_in serverAddr;          // Server socket address
15      // Check and set arguments
16      if (argc != 3)
17      {
18          printf ("Error: three arguments are needed!");
19          exit (1);
20      }
```


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Table 2.25: TCP Echo client program (continued)

```
21      servName = arg [1];
22      servPort = atoi (arg [2]);
23      string = arg [3];
24      // Create remote (server) socket address
25      memset (&servAddr, 0, sizeof(servAddr);
26      serverAddr.sin_family = AF_INET;
27      inet_pton (AF_INET, servName, &serverAddr.sin_addr); // Server IP address
28      serverAddr.sin_port = htons (servPort);              // Server port number
29      // Create socket
30      if ((s = socket (PF_INET, SOCK_STREAM, 0) < 0);
31      {
32          perror ("Error: socket creation failed!");
33          exit (1);
34      }
35      // Connect  to the server
36      if (connect (sd, (struct sockaddr*)&servAddr, sizeof(servAddr)) < 0);
37      {
38          perror ("Error: connection failed!");
```


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Table 2.24: TCP Echo client program(continued)

```
39              exit (1);
40          }
41          // Data transfer section
42          send (s, string, strlen(string), 0);
43          while ((n = recv (s, ptr, maxLen, 0)) > 0)
44          {
45              ptr + = n;                  // Move pointer along the buffer
46              maxLen - = n;               // Adjust the maximum number of bytes
47              len += n;                   // Update the length of string received
48          } // End of while loop
49          // Print and verify the echoed string
50          buffer [len] = '\0';
51          printf ("Echoed string received: ");
52          fputs (buffer, stdout);
53          // Close socket
54          close (s);
55          // Stop program
56          exit (0);
57      } // End of echo client program
```


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Chapter 2: Summary

- *Applications in the Internet are designed using either a client-server paradigm or a peer-to-peer paradigm. In a client-server paradigm, an application program, called a server, provides services and another application program, called a client, receives services. A server program is an infinite program; a client program is finite. In a peer-to-peer paradigm, a peer can be both a client and a server.*

- *The World Wide Web (WWW) is a repository of information linked together from points all over the world. Hypertext and hypermedia documents are linked to one another through pointers. The HyperText Transfer Protocol (HTTP) is the main protocol used to access data on the World Wide Web (WWW).*

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Chapter 2: Summary (continued)

- *File Transfer Protocol (FTP) is a TCP/IP client-server application for copying files from one host to another. FTP requires two connections for data transfer: a control connection and a data connection. FTP employs NVT ASCII for communication between dissimilar systems.*

- *Electronic mail is one of the most common applications on the Internet. The e-mail architecture consists of several components such as user agent (UA), main transfer agent (MTA), and main access agent (MAA). The protocol that implements MTA is called Simple Main Transfer Protocol (SMTP). Two protocols are used to implement MAA: Post Office Protocol, version 3 (POP3) and Internet Mail Access Protocol, version 4 (IMAP4).*

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Chapter 2: Summary (continued)

- *File Transfer Protocol (FTP) is a TCP/IP client-server application for copying files from one host to another. FTP requires two connections for data transfer: a control connection and a data connection. FTP employs NVT ASCII for communication between dissimilar systems.*

- *TELNET is a client-server application that allows a user to log into a remote machine, giving the user access to the remote system. When a user accesses a remote system via the TELNET process, this is comparable to a time-sharing environment.*

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Chapter 2: Summary (continued)

- *The Domain Name System (DNS) is a client-server application that identifies each host on the Internet with a unique name. DNS organizes the name space in a hierarchical structure to decentralize the responsibilities involved in naming. TELNET is a client-server application that allows a user to log into a remote machine, giving the user access to the remote system.*

- *In a peer-to-peer network, Internet users that are ready to share their resources become peers and form a network. Peer-to-peer networks are divided into centralized and decentralized.*

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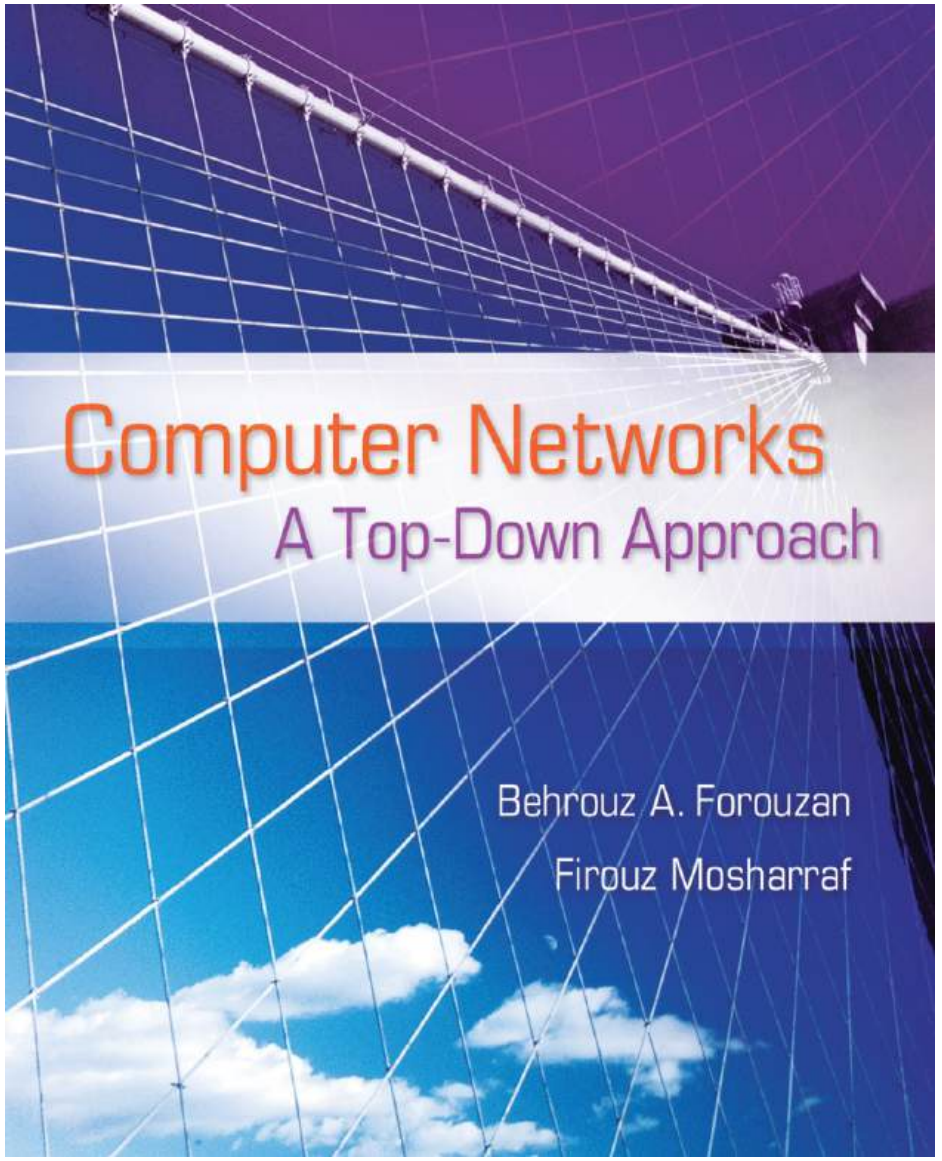
Chapter 3

Transport Layer



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Chapter 3: Outline

3.1 INTRODUCTION

3.2 TRANSPORT-LAYER PROTOCOLS

3.3 USER DATAGRAM PROTOCOL

3.4 TRANSMISSION CONTROL PROTOCOL

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Chapter 3: Objective

- *We introduce general services we normally require from the transport layer, such as process-to-process communication, addressing, multiplexing, error, flow, and congestion control.*
- *We discuss general transport-layer protocols such as Stop-and-Wait, Go-Back-N, and Selective-Repeat.*
- *We discuss UDP, which is the simpler of the two protocols we discuss in this chapter.*
- *We discuss TCP services and features. We then show how TCP provides a connection-oriented service using a transition diagram. Finally, we discuss flow and error control, and congestion control in TCP.*

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3-1 INTRODUCTION

The transport layer provides a process-to-process communication between two application layers. Communication is provided using a logical connection, which means that the two application layers assume that there is an imaginary direct connection through which they can send and receive messages.

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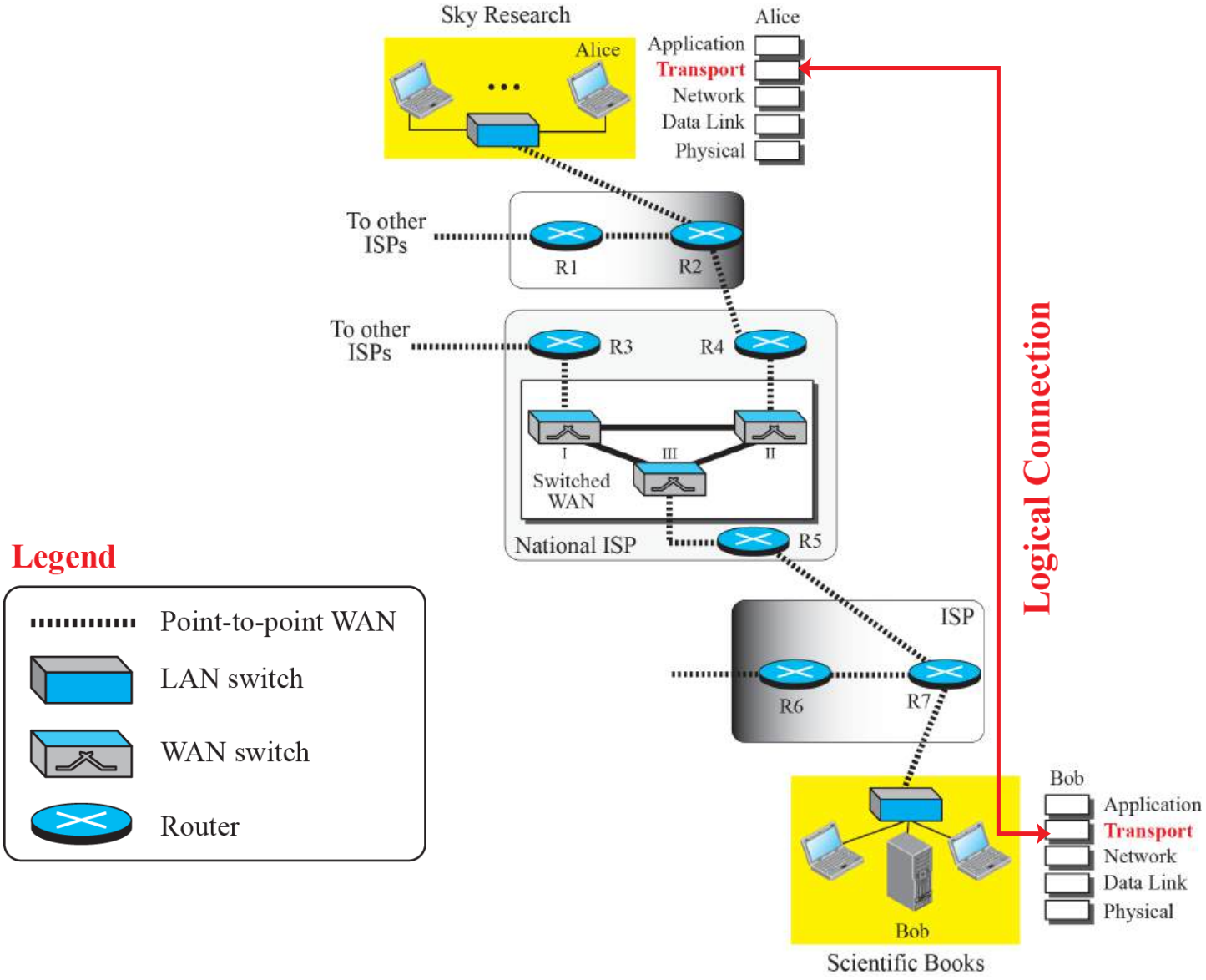
Figure 3.1: Logical connection at the transport layer



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3.1.1 Transport-Layer Services

As we discussed in Chapter 1, the transport layer is located between the network layer and the application layer. The transport layer is responsible for providing services to the application layer; it receives services from the network layer. In this section, we discuss the services that can be provided by the transport layer; in the next section, we discuss several transport-layer protocols.

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3.1.1 (continued)

- **Process-to-Process Communication**
- **Addressing: Port Numbers**
- **ICANN Ranges**
  - Well-known ports
  - Registered ports
  - Dynamic ports
- **Encapsulation and Decapsulation**
- **Multiplexing and Demultiplexing**

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3.1.1 (continued)

- **Flow Control**
  - Pushing or Pulling
  - Flow Control at Transport Layer
  - Buffers
- **Error Control**
  - Sequence Numbers
  - Acknowledgment
- **Combination of Flow and Error Control**
  - Sliding Window

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3.1.1 (continued)

- **Congestion Control**
- **Connectionless and Connection-Oriented**
  - Connectionless Service
  - Connection-Oriented Service
  - Finite State Machine
- **Multiplexing and Demultiplexing**

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Figure 3.2: Network layer versus transport layer

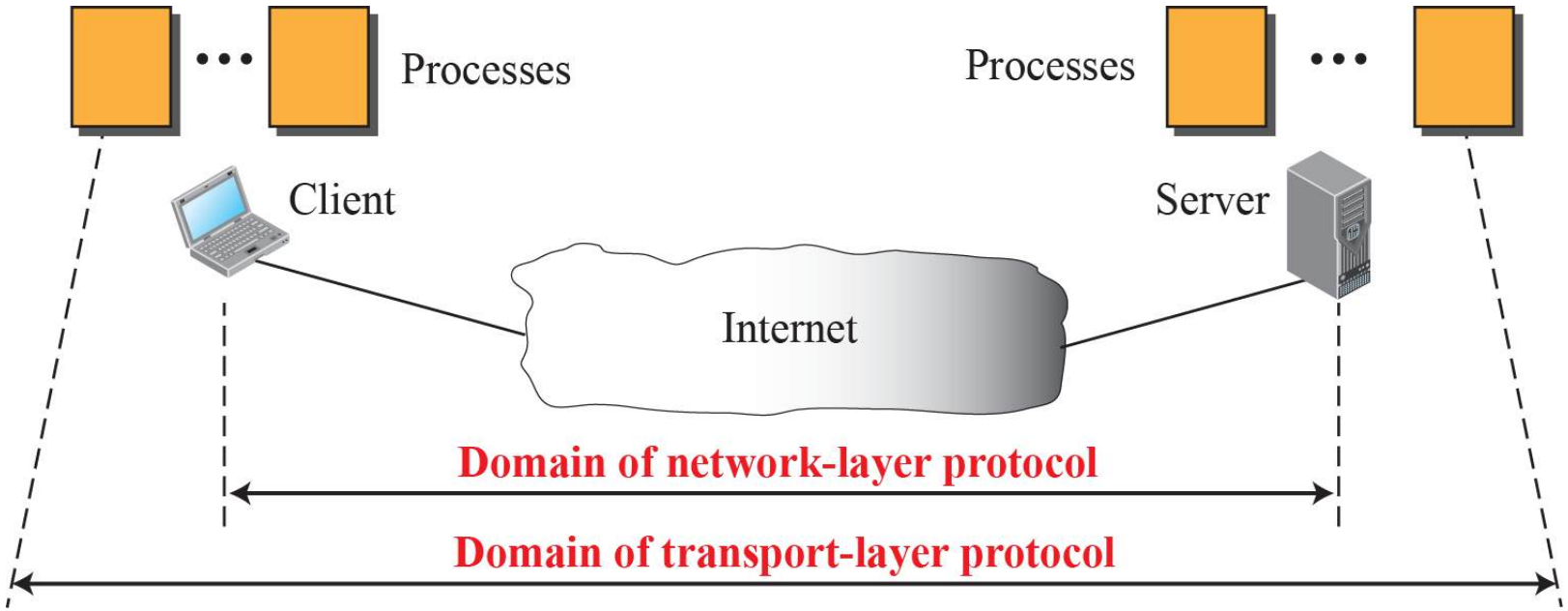

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Figure 3.3: Port numbers

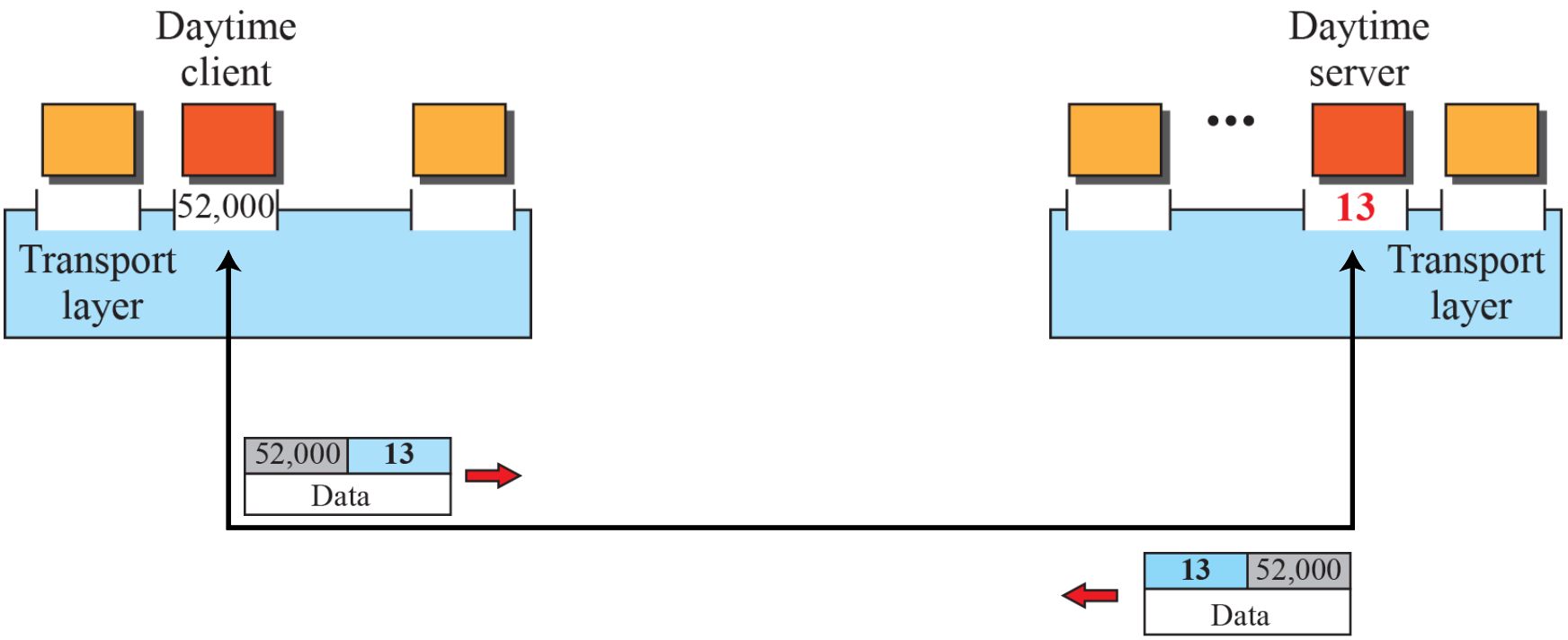

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Figure 3.4: IP addresses versus port numbers

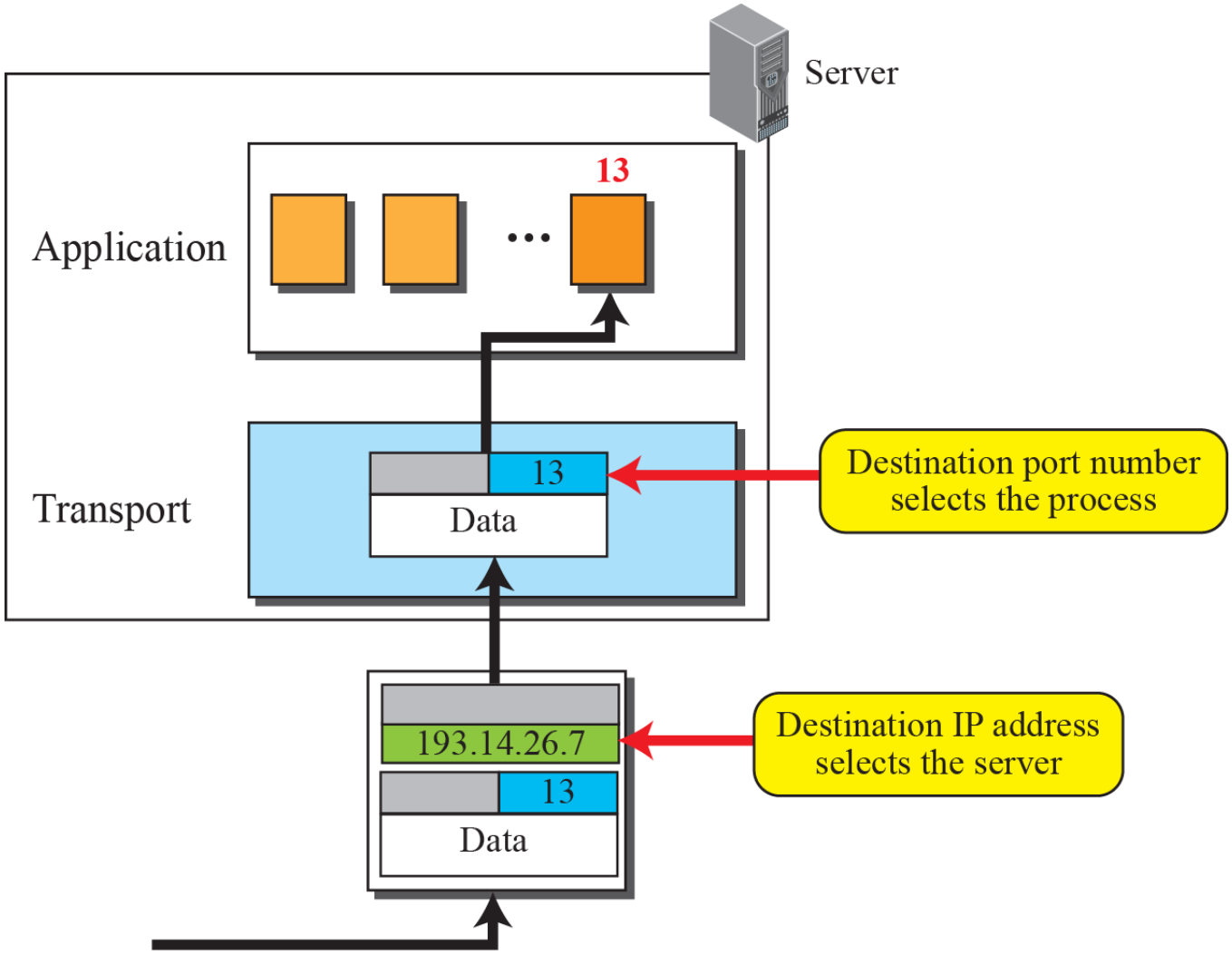

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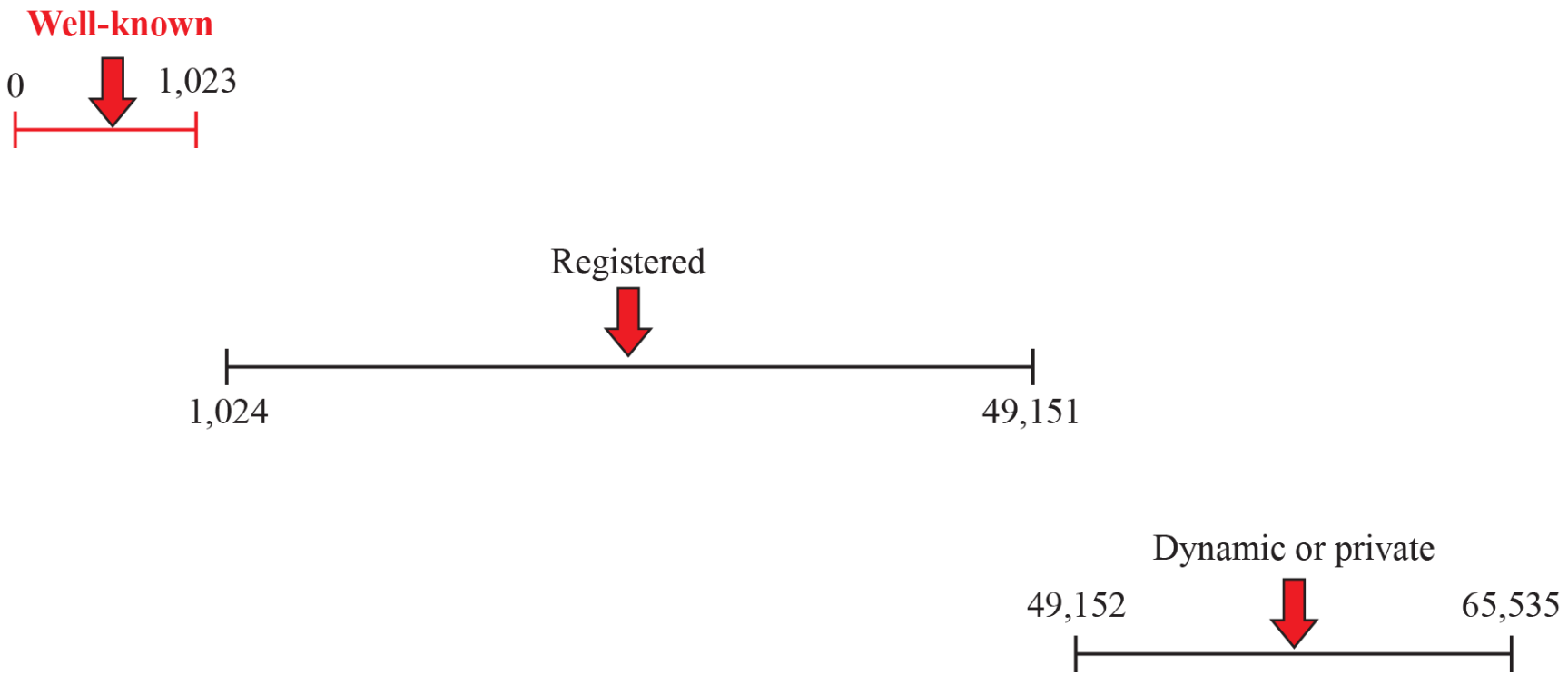
Figure 3.5: ICANN ranges



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Example 3.1

In UNIX, the well-known ports are stored in a file called /etc/services. We can use the *grep* utility to extract the line corresponding to the desired application.

```
$grep    tftp/etc/services
tftp 69/tcp
tftp 69/udp
```

SNMP (see Chapter 9) uses two port numbers (161 and 162), each for a different purpose.

```
$grep    snmp/etc/services
snmp161/tcp#Simple Net Mgmt Proto
snmp161/udp#Simple Net Mgmt Proto
snmptrap162/udp#Traps for SNMP
```


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Figure 3.6: ***Socket address***

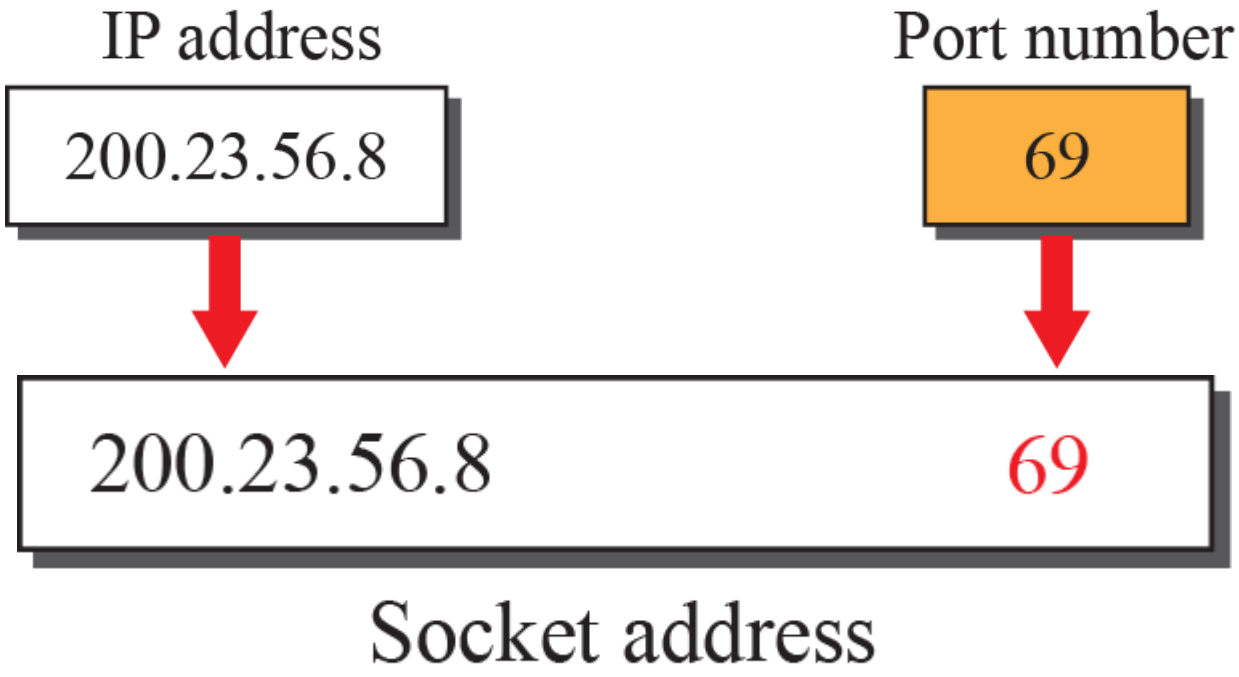

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Figure 3.7: Encapsulation and decapsulation

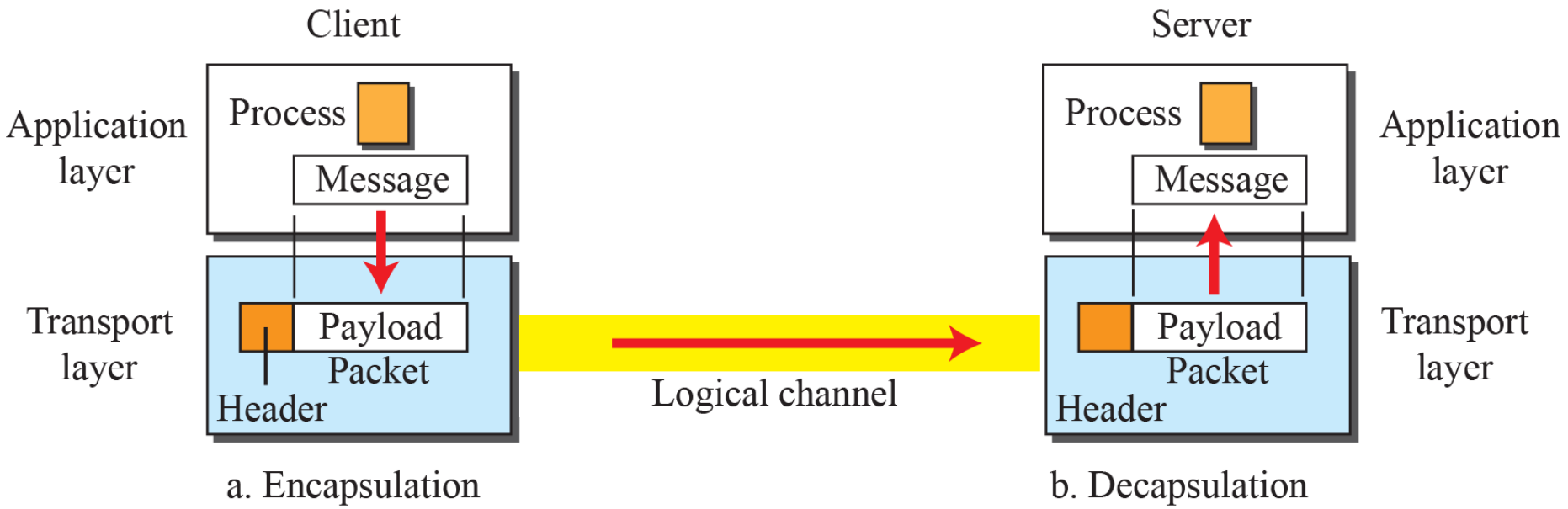

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Figure 3.8: Multiplexing and demultiplexing

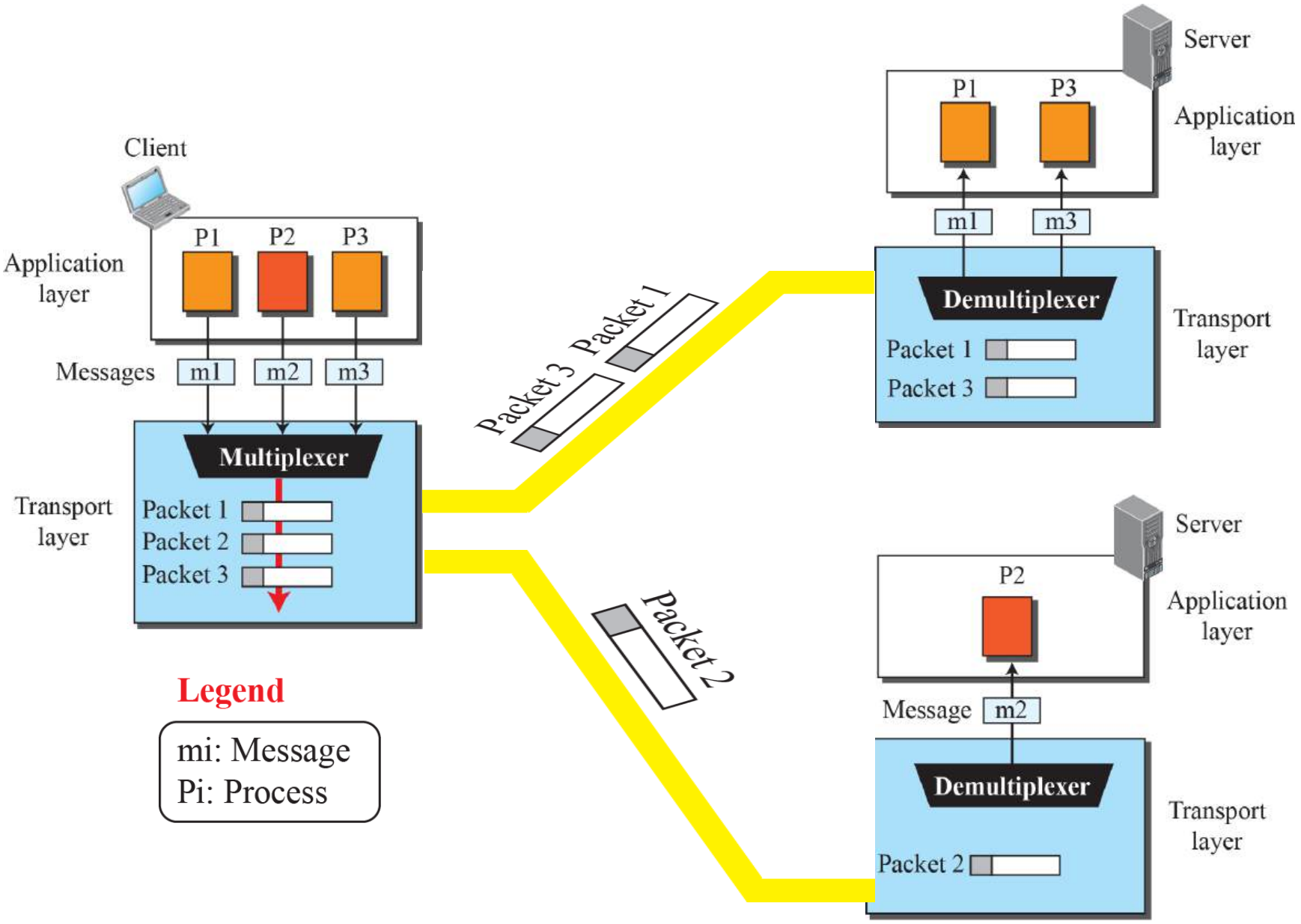

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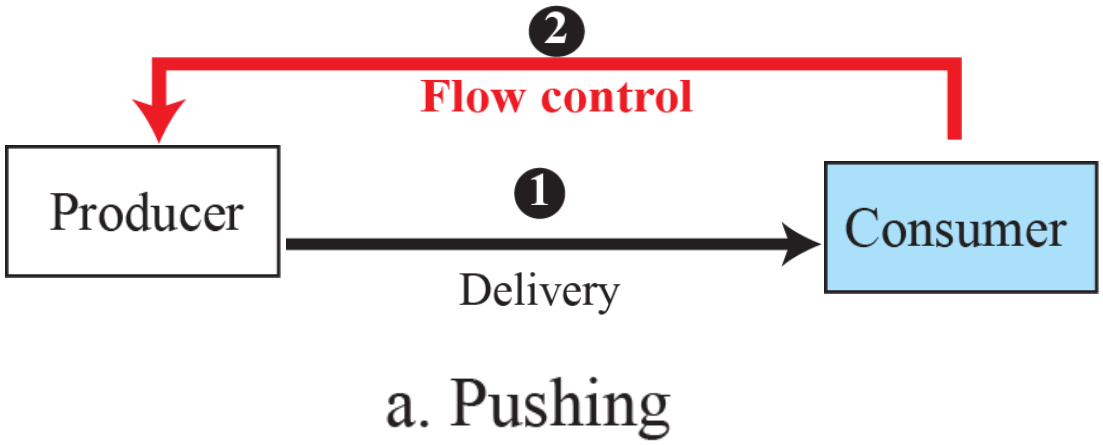
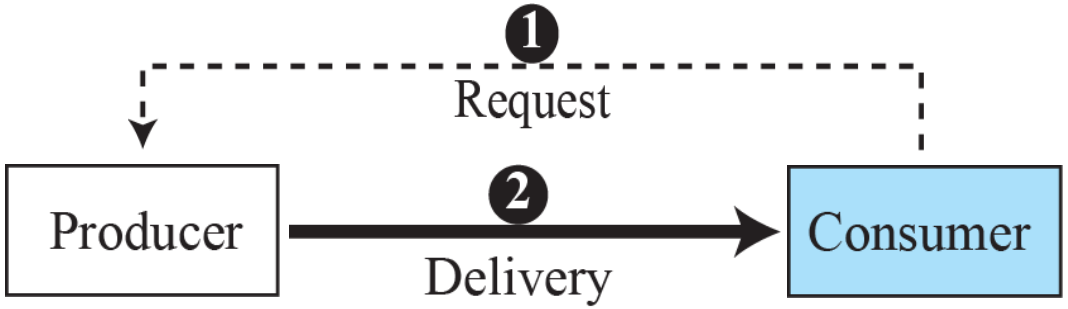
Figure 3.9: Pushing or pulling



a. Pushing



b. Pulling

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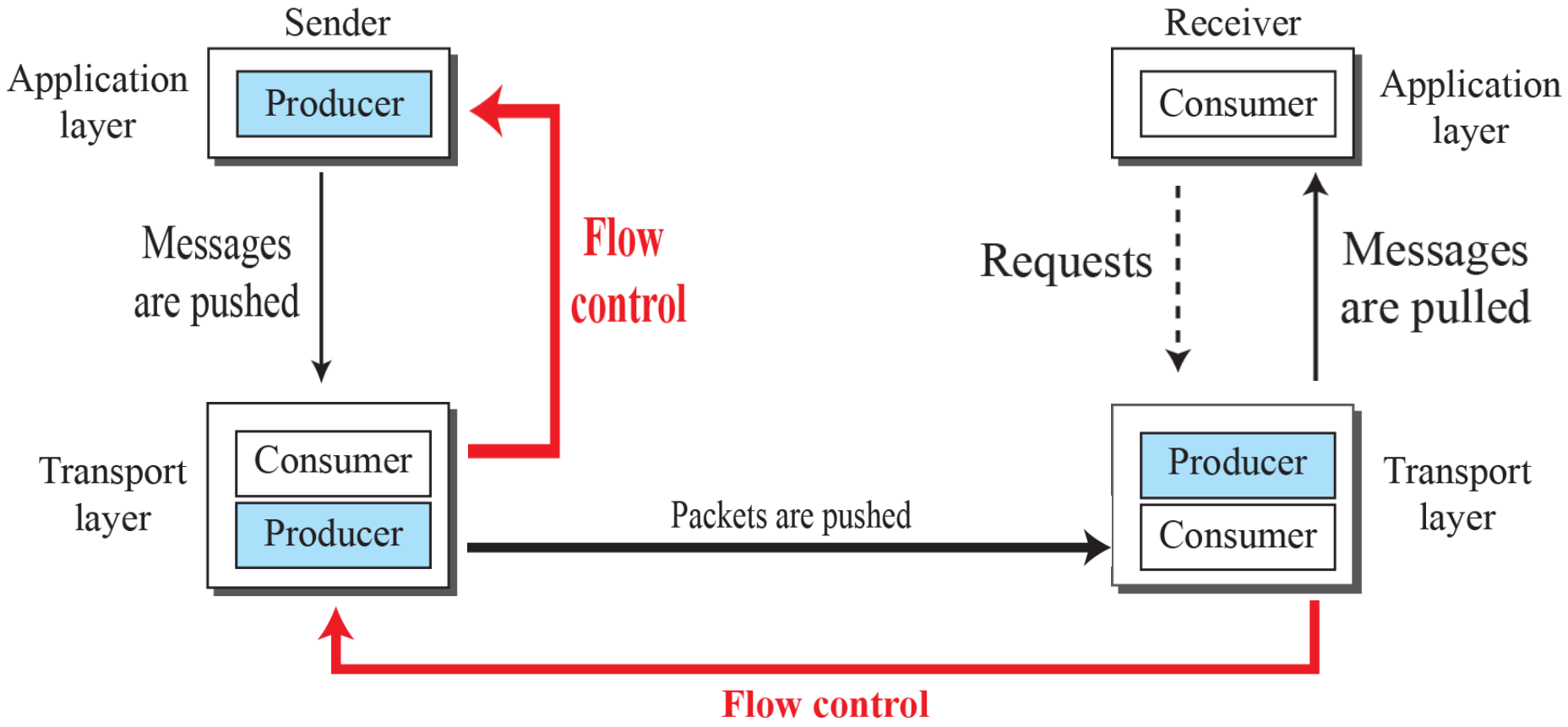
Figure 3.10: Flow control at the transport layer



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Example 3.2

The above discussion requires that the consumers communicate with the producers on two occasions: when the buffer is full and when there are vacancies. If the two parties use a buffer with only one slot, the communication can be easier. Assume that each transport layer uses one single memory location to hold a packet. When this single slot in the sending transport layer is empty, the sending transport layer sends a note to the application layer to send its next chunk; when this single slot in the receiving transport layer is empty, it sends an acknowledgment to the sending transport layer to send its next packet. As we will see later, however, this type of flow control, using a single-slot buffer at the sender and the receiver, is inefficient.

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Figure 3.11: ***Error control at the transport layer***

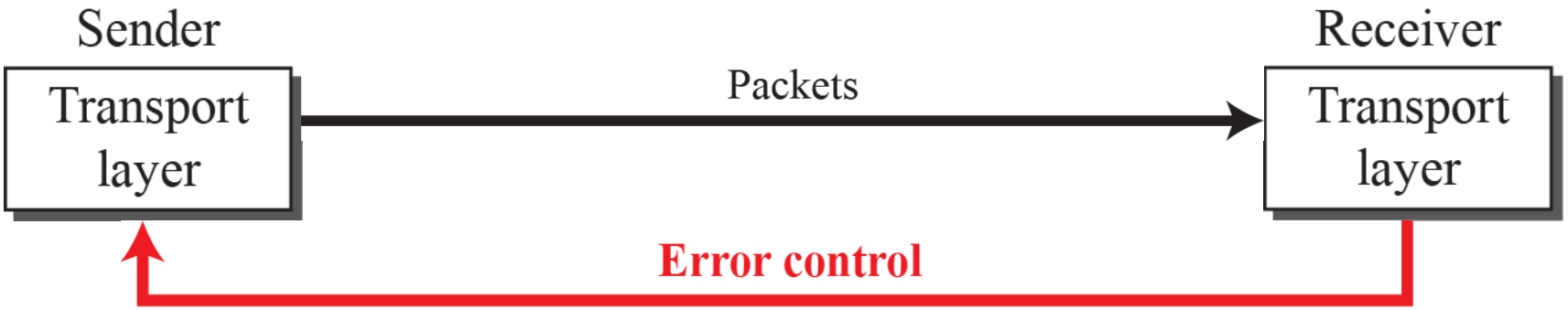

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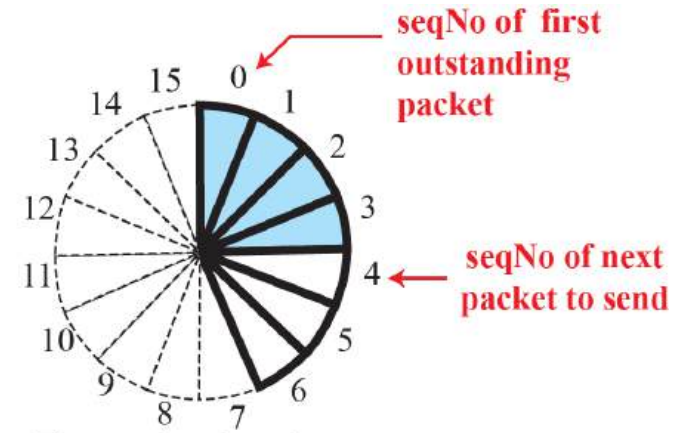
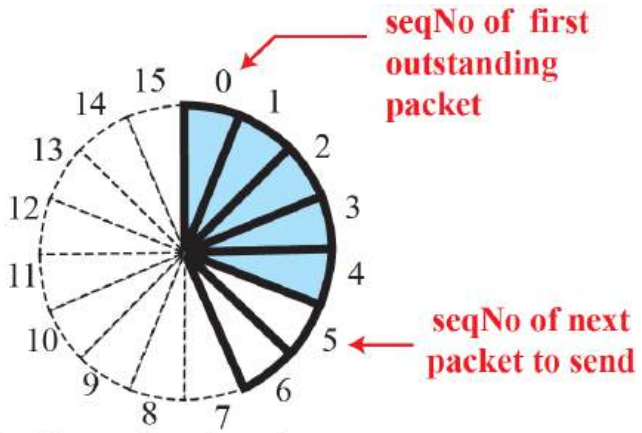
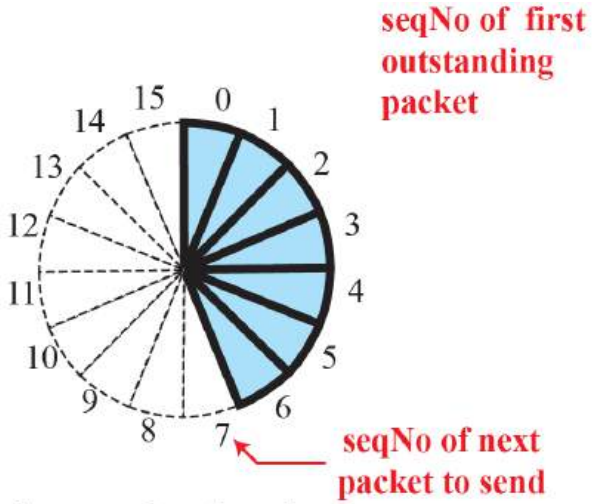
Figure 3.12: Sliding window in circular format



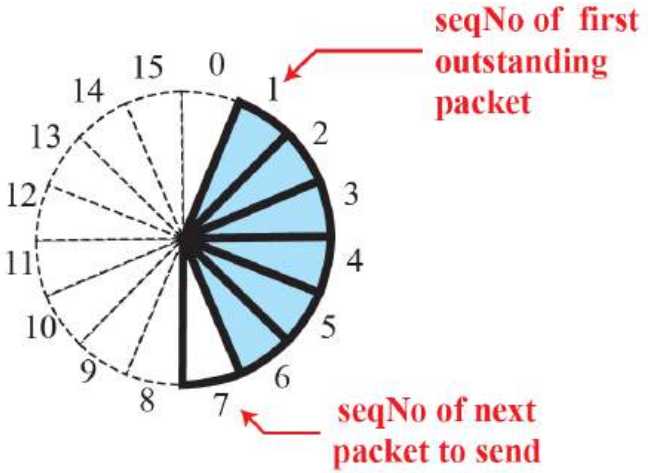
a. Four packets have been sent.



b. Five packets have been sent.



c. Seven packets have been sent; window is full.



d. Packet 0 has been acknowledged; window slides.

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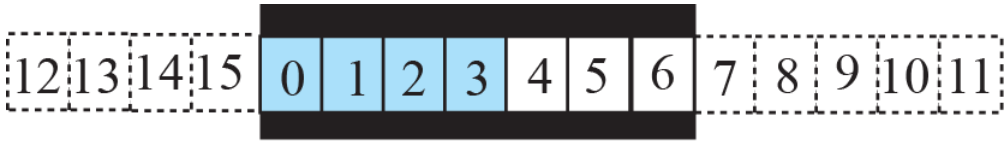
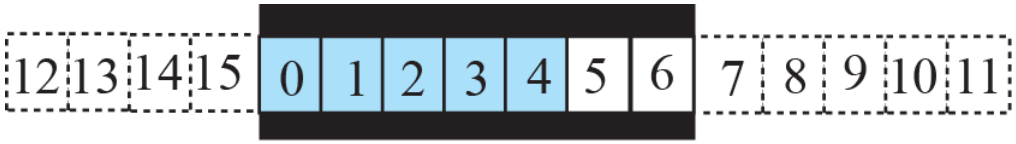
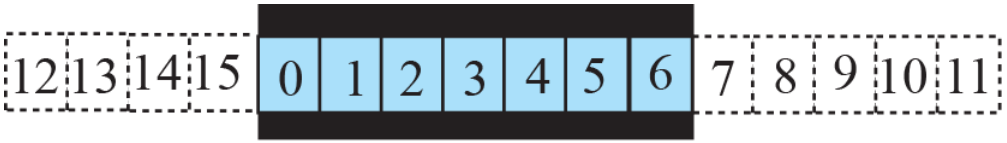
Figure 3.13: Sliding window in linear format



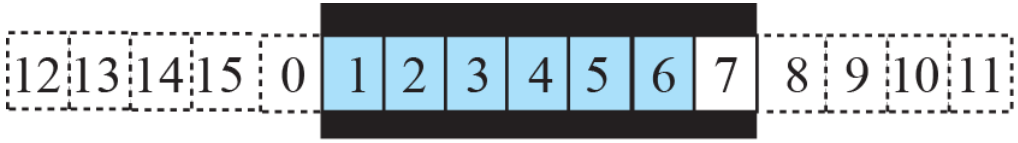
a. Four packets have been sent.



b. Five packets have been sent.



c. Seven packets have been sent;
window is full.



d. Packet 0 has been acknowledged;
window slides.

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Figure 3.14: Connectionless service

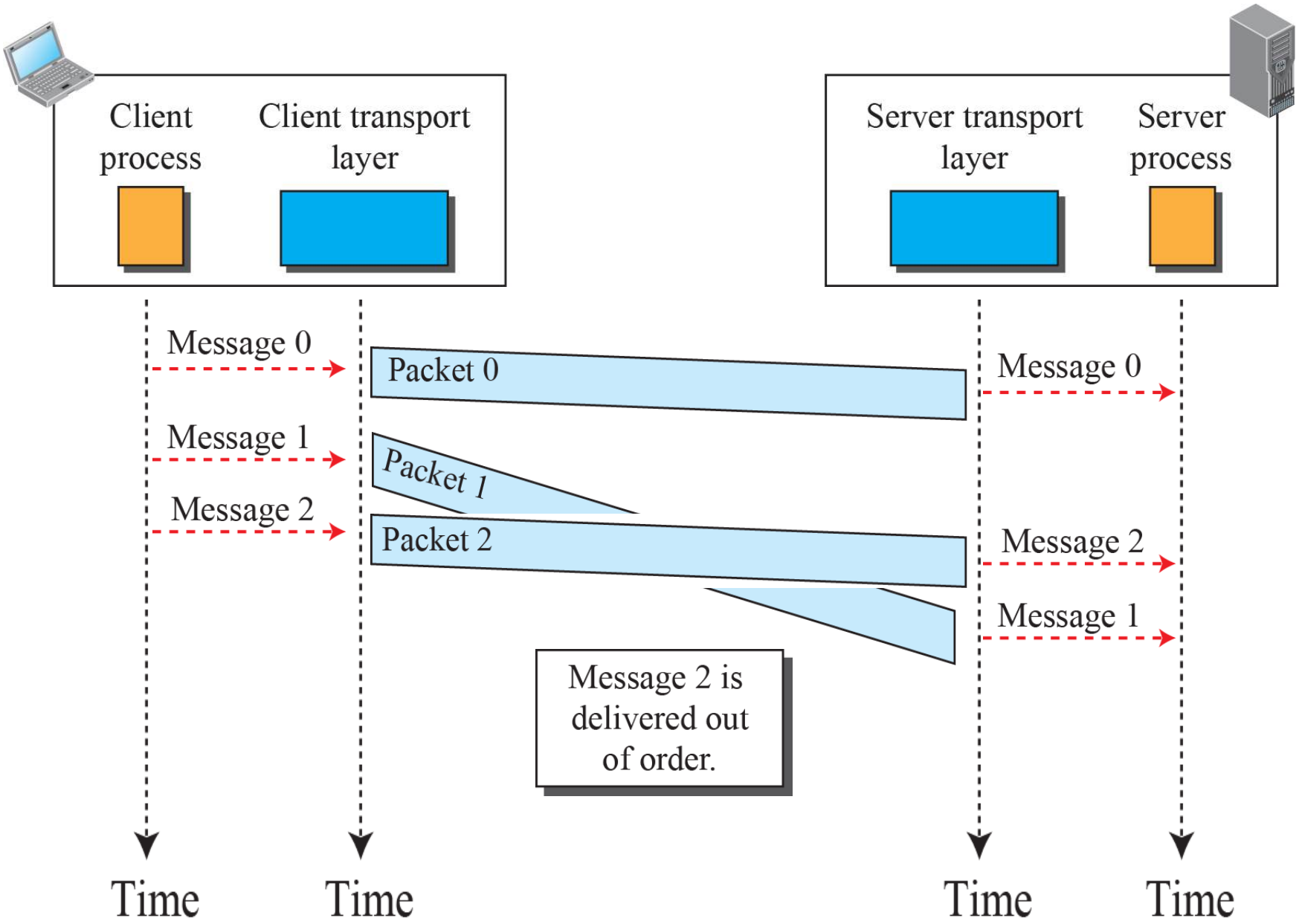

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Figure 3.15: Connection-oriented service

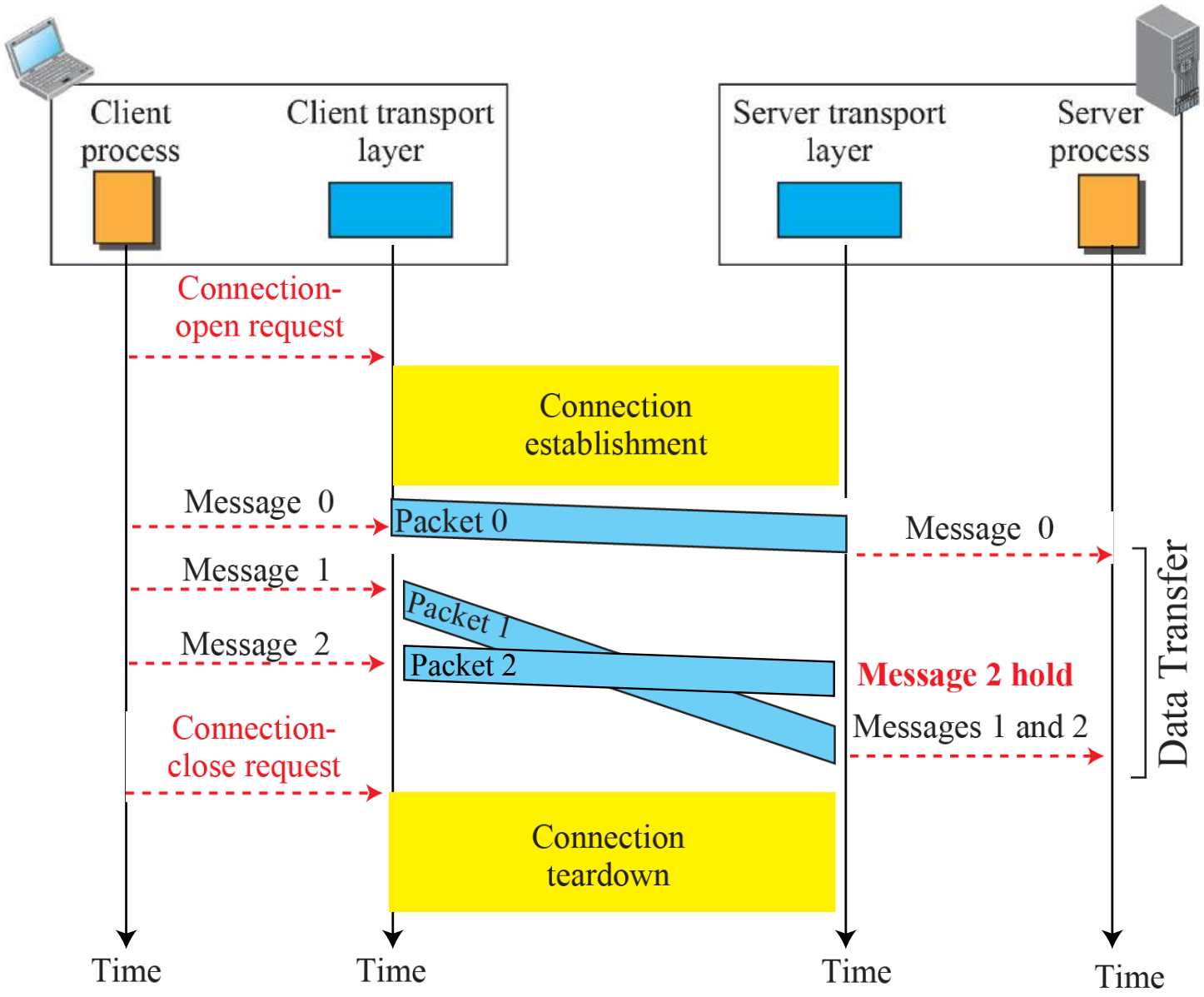

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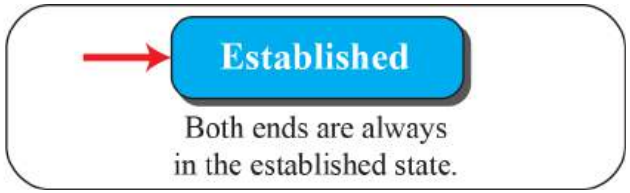
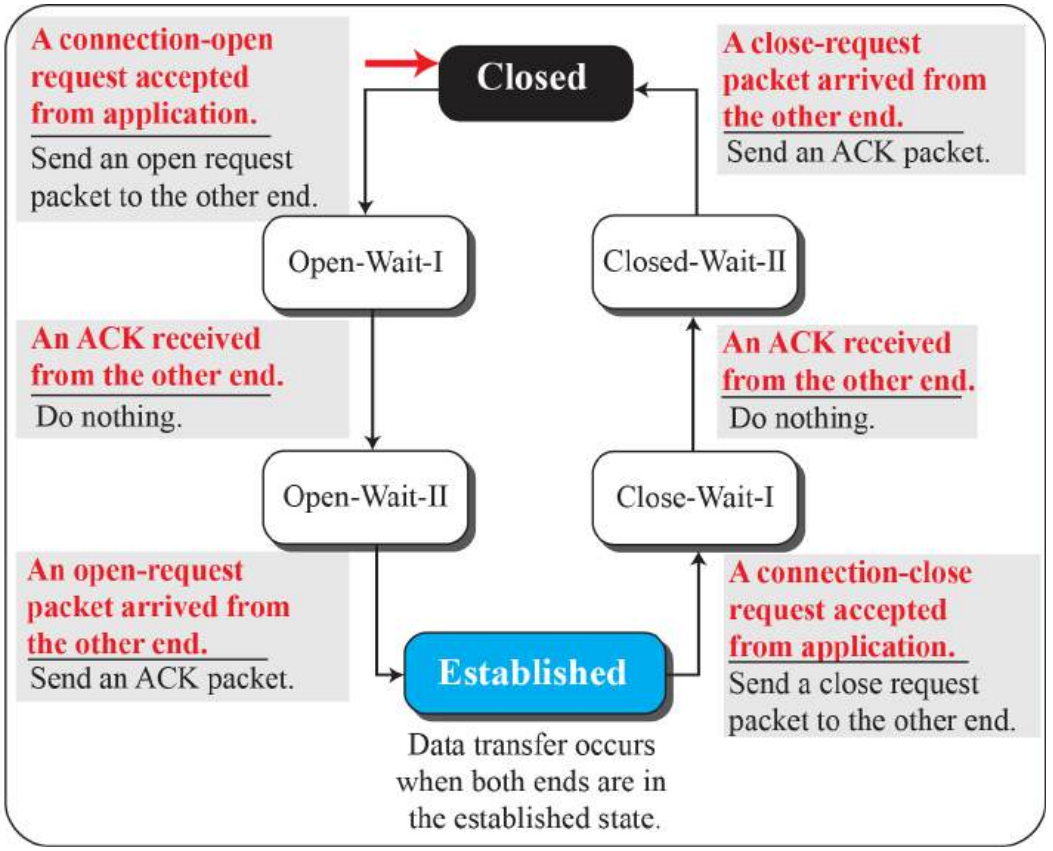
Figure 3.16: Connectionless and connection-oriented service represented as FSMs

FSM for connectionless transport layer



Note:
The colored arrow shows the starting state.

FSM for connection-oriented transport layer



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3-2 TRANSPORT-LAYER PROTOCOLS

We can create a transport-layer protocol by combining a set of services described in the previous sections. To better understand the behavior of these protocols, we start with the simplest one and gradually add more complexity.

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3.2.1 Simple Protocol

Our first protocol is a simple connectionless protocol with neither flow nor error control. We assume that the receiver can immediately handle any packet it receives. In other words, the receiver can never be overwhelmed with incoming packets. Figure 3.17 shows the layout for this protocol.

- ***FSMs***

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Figure 3.17: Simple protocol

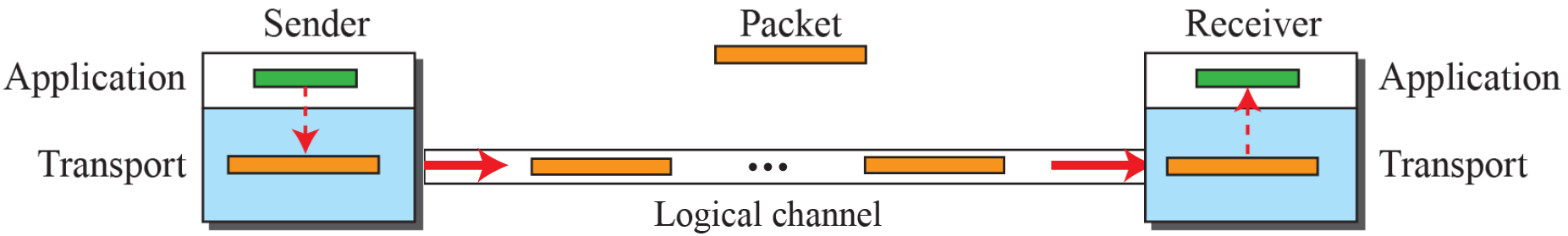

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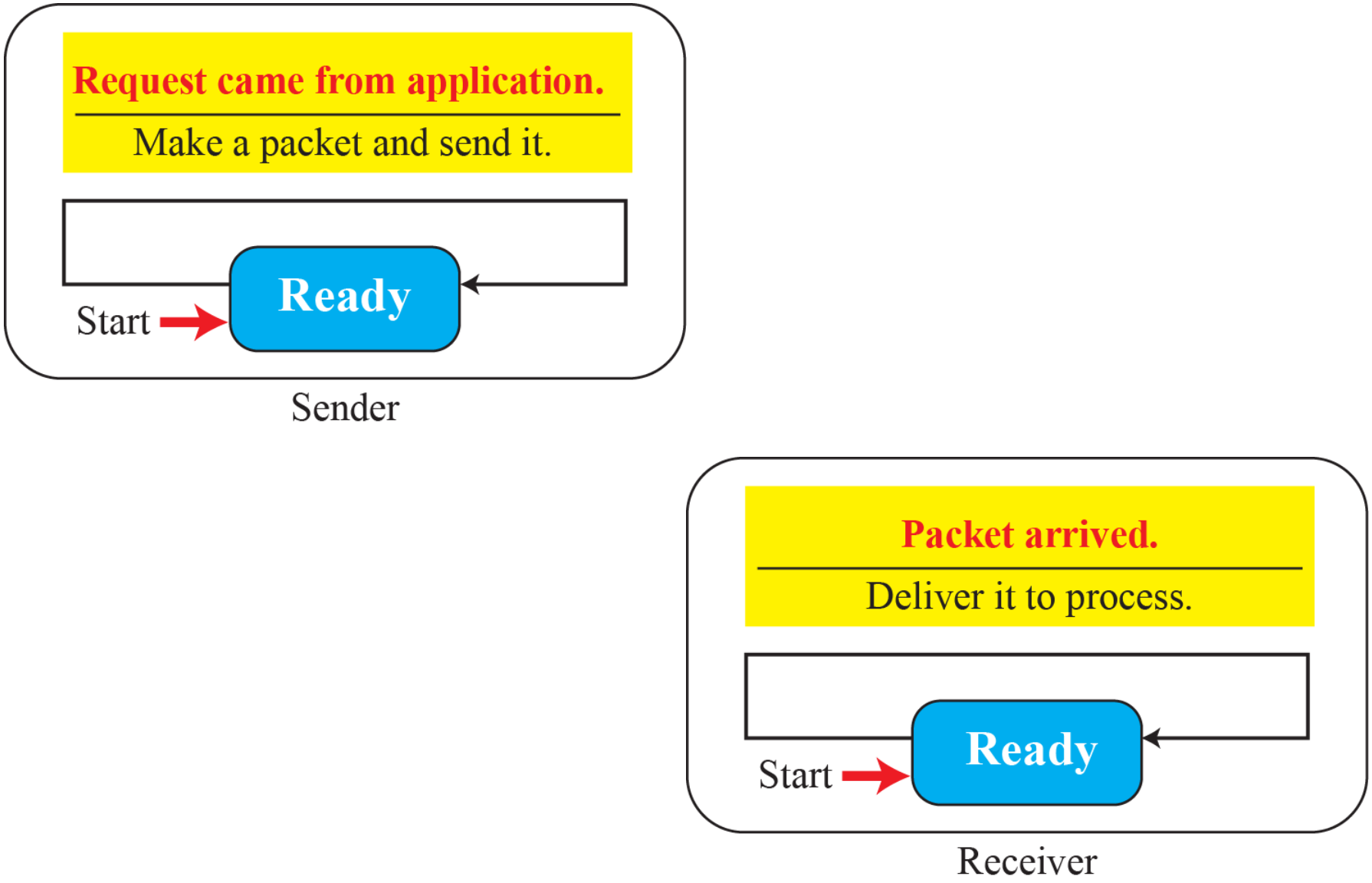
Figure 3.18: FSMs for the simple protocol



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Example 3.3

Figure 3.19 shows an example of communication using this protocol. It is very simple. The sender sends packets one after another without even thinking about the receiver.

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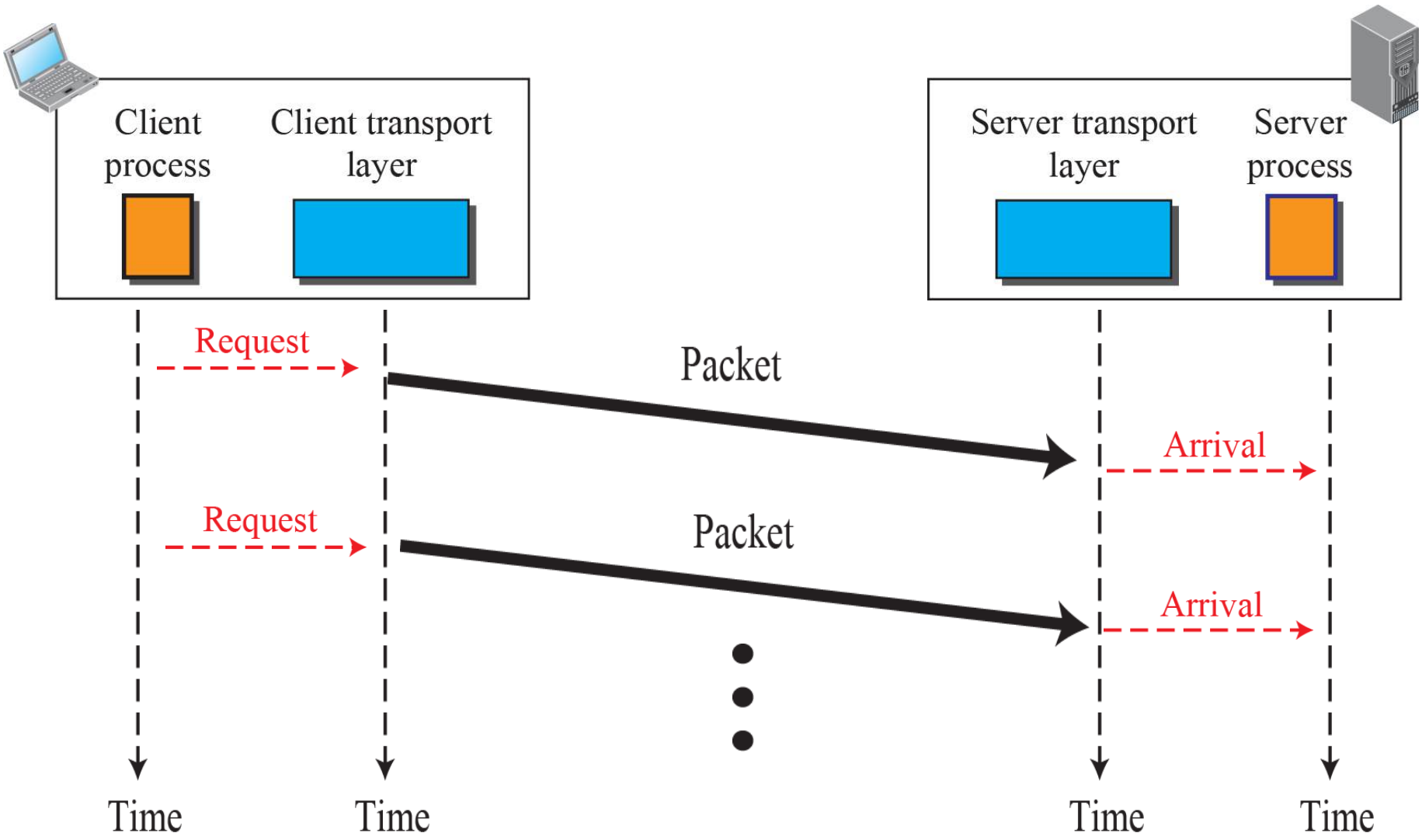
Figure 3.19: Flow diagram for Example 3.3



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3.2.2 Stop-and-Wait Protocol

Our second protocol is a connection-oriented protocol called the Stop-and-Wait protocol, which uses both flow and error control. Both the sender and the receiver use a sliding window of size 1. The sender sends one packet at a time and waits for an acknowledgment before sending the next one. To detect corrupted packets, we need to add a checksum to each data packet. When a packet arrives at the receiver site,

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3.2.2 (continued)

- Sequence Numbers
- Acknowledgment Numbers
- FSMs
  - Sender
  - Receiver
- Efficiency
- Pipelining

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Figure 3.20: Stop-and-Wait protocol

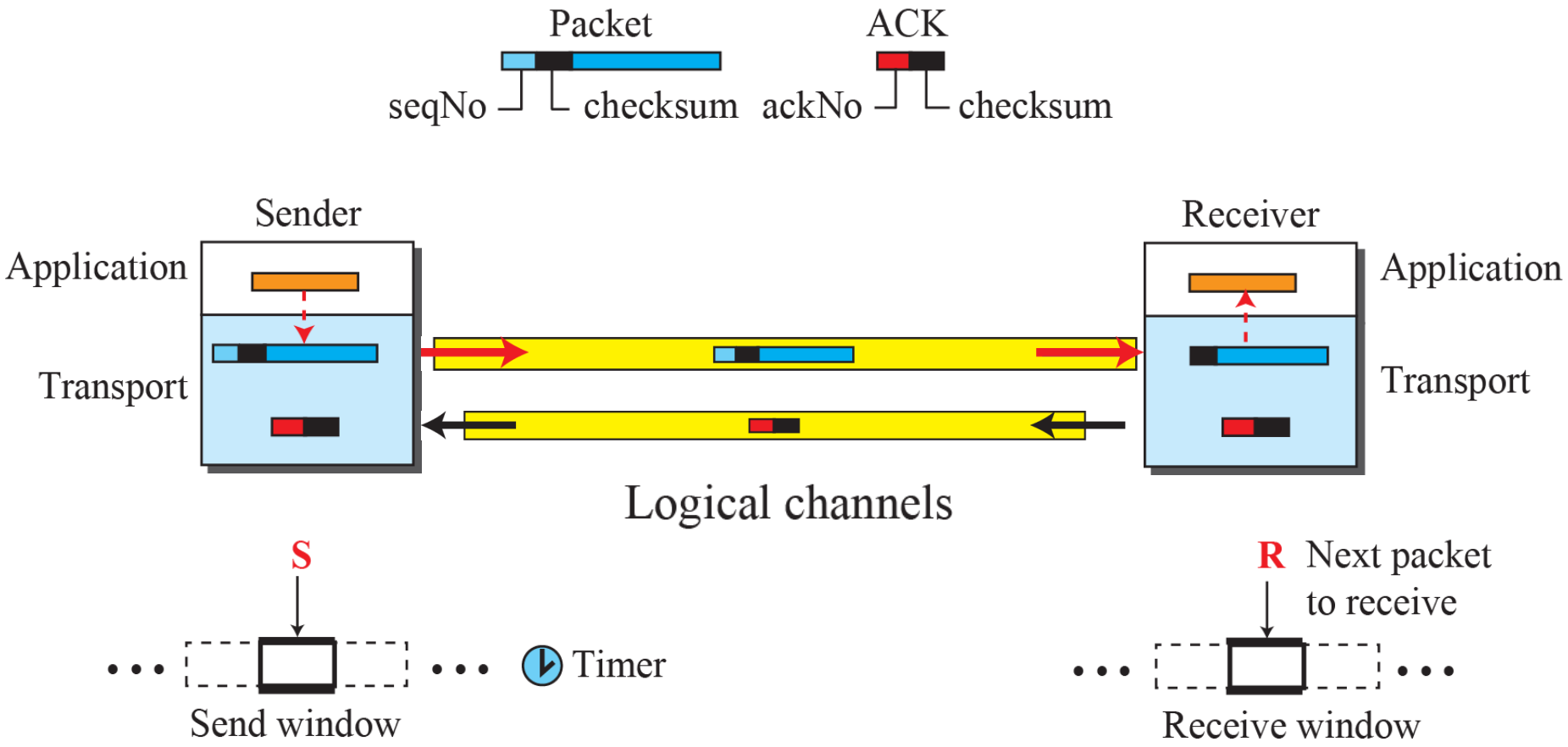

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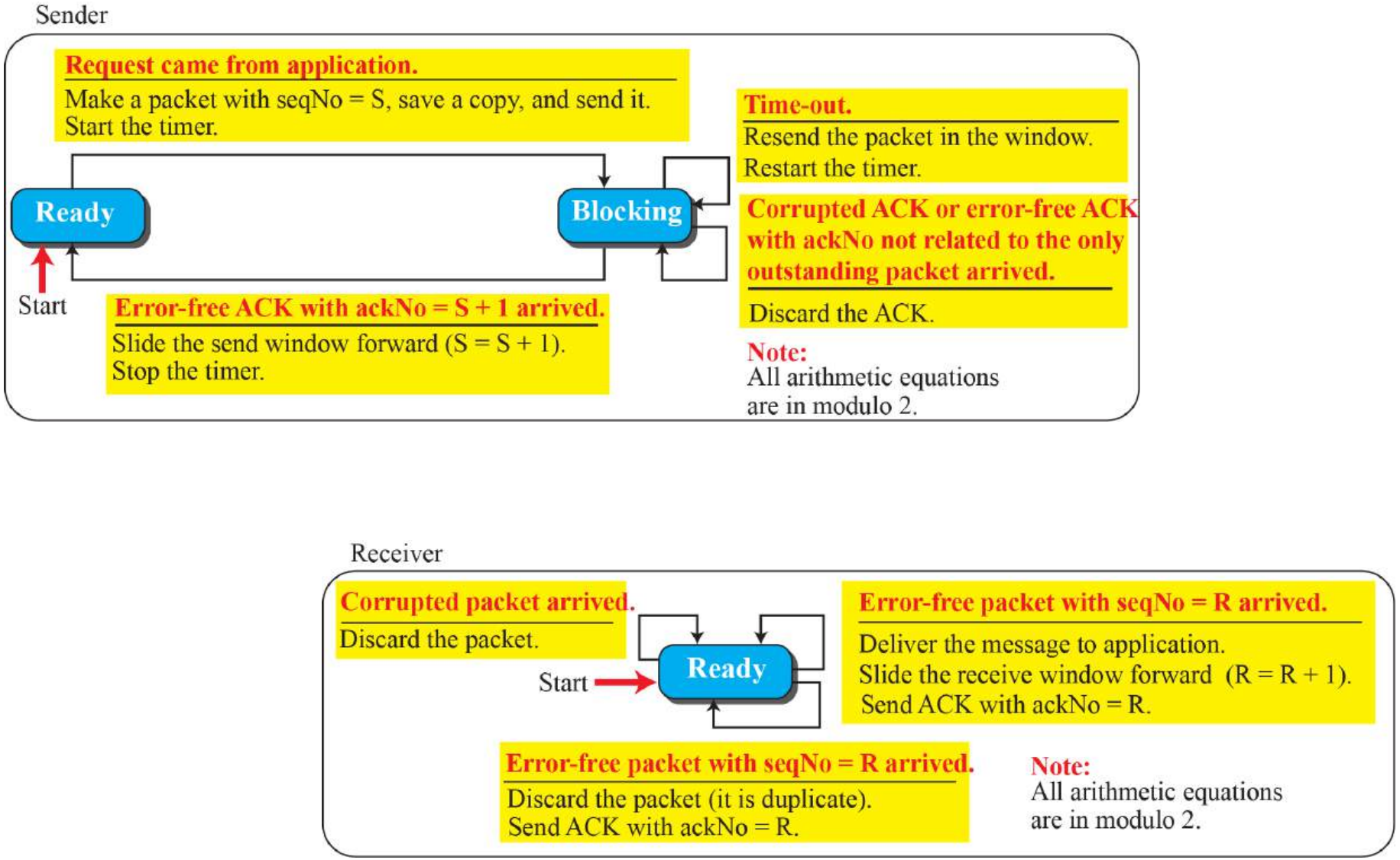
Figure 3.21: FSM for the Stop-and-Wait protocol



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Example 3.4

Figure 3.22 shows an example of the Stop-and-Wait protocol. Packet 0 is sent and acknowledged. Packet 1 is lost and resent after the time-out. The resent packet 1 is acknowledged and the timer stops. Packet 0 is sent and acknowledged, but the acknowledgment is lost. The sender has no idea if the packet or the acknowledgment is lost, so after the time-out, it resends packet 0, which is acknowledged.

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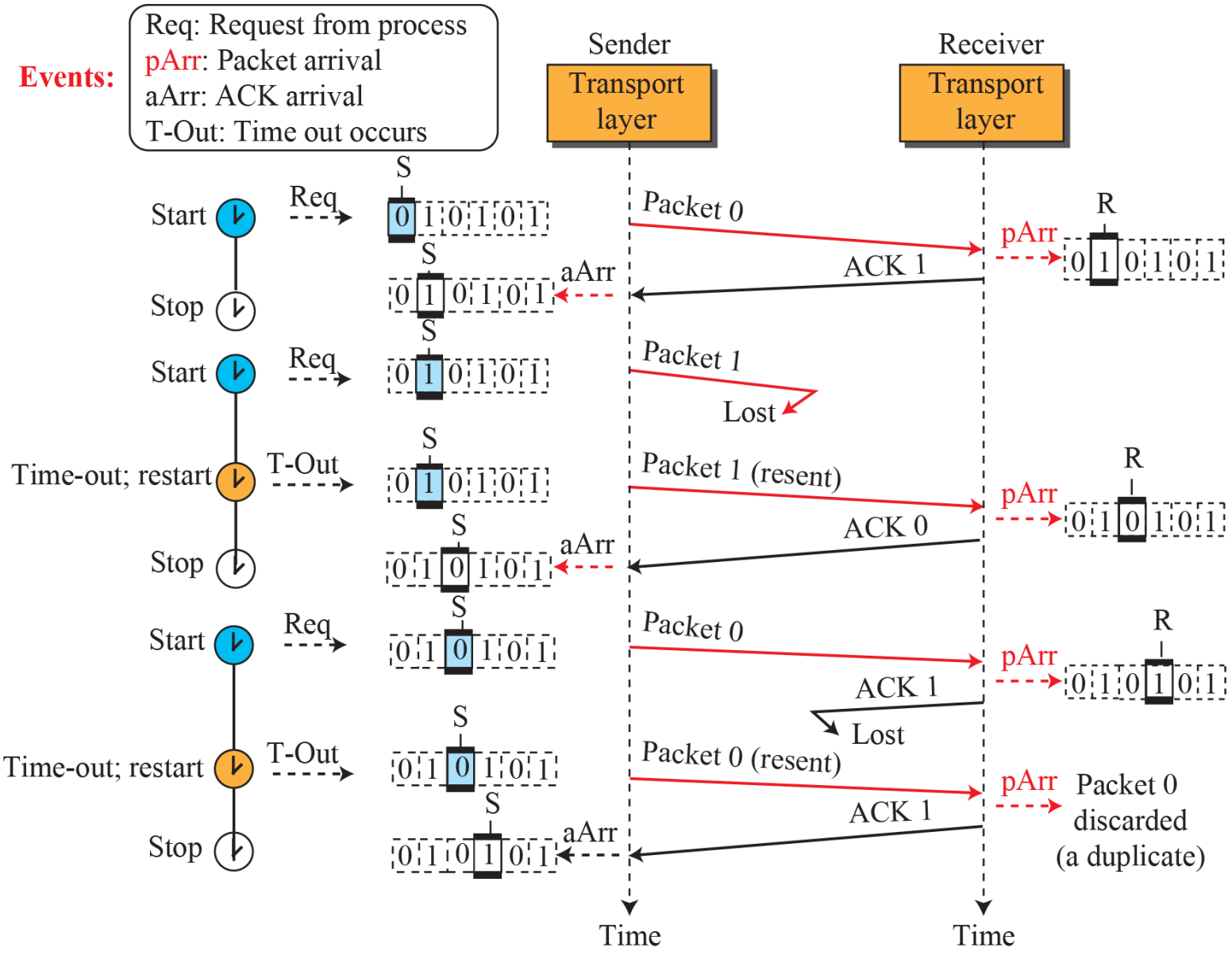
Figure 3.22: Flow diagram for Example 3.4



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Example 3.5

Assume that, in a Stop-and-Wait system, the bandwidth of the line is 1 Mbps, and 1 bit takes 20 milliseconds to make a round trip. What is the bandwidth-delay product? If the system data packets are 1,000 bits in length, what is the utilization percentage of the link?

Solution

The bandwidth-delay product is $(1 \times 10^6) \times (20 \times 10^{-3}) = 20{,}000$ bits. The system can send 20,000 bits during the time it takes for the data to go from the sender to the receiver and the acknowledgment to come back. However, the system sends only 1,000 bits. The link utilization is only 1,000/20,000, or 5 percent.

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Example 3.6

What is the utilization percentage of the link in Example 3.5 if we have a protocol that can send up to 15 packets before stopping and worrying about the acknowledgments?

Solution

The bandwidth-delay product is still 20,000 bits. The system can send up to 15 packets or 15,000 bits during a round trip. This means the utilization is 15,000/20,000, or 75 percent. Of course, if there are damaged packets, the utilization percentage is much less because packets have to be resent.

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3.2.3 Go-Back-N Protocol

To improve the efficiency of transmission multiple packets must be in transition while the sender is waiting for acknowledgment. In this section, we discuss one protocol that can achieve this goal; in the next section, we discuss a second. The first is called Go-Back-N (GBN) (the rationale for the name will become clear later).

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3.2.3 (continued)

- Sequence Numbers
- Acknowledgment Numbers
- Send Window
- Receive Window
- Timers
- Resending packets

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3.2.3 (continued)

- **FSMs**
  - Sender
  - Receiver
- **Send Window Size**
- **Go-Back-N versus Stop-and-Wait**

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Figure 3.23: Go-Back-N protocol

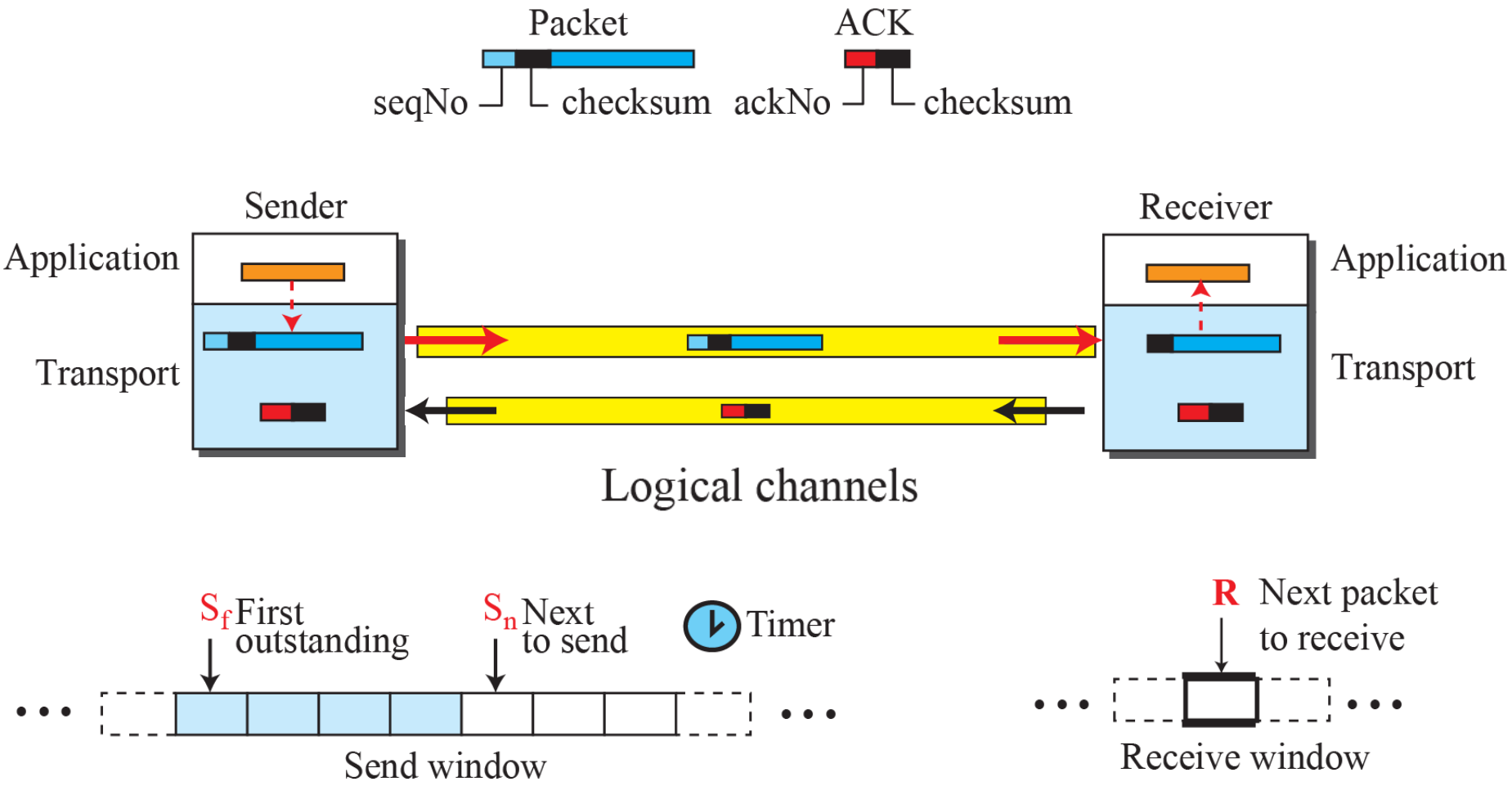

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Figure 3.24: Send window for Go-Back-N

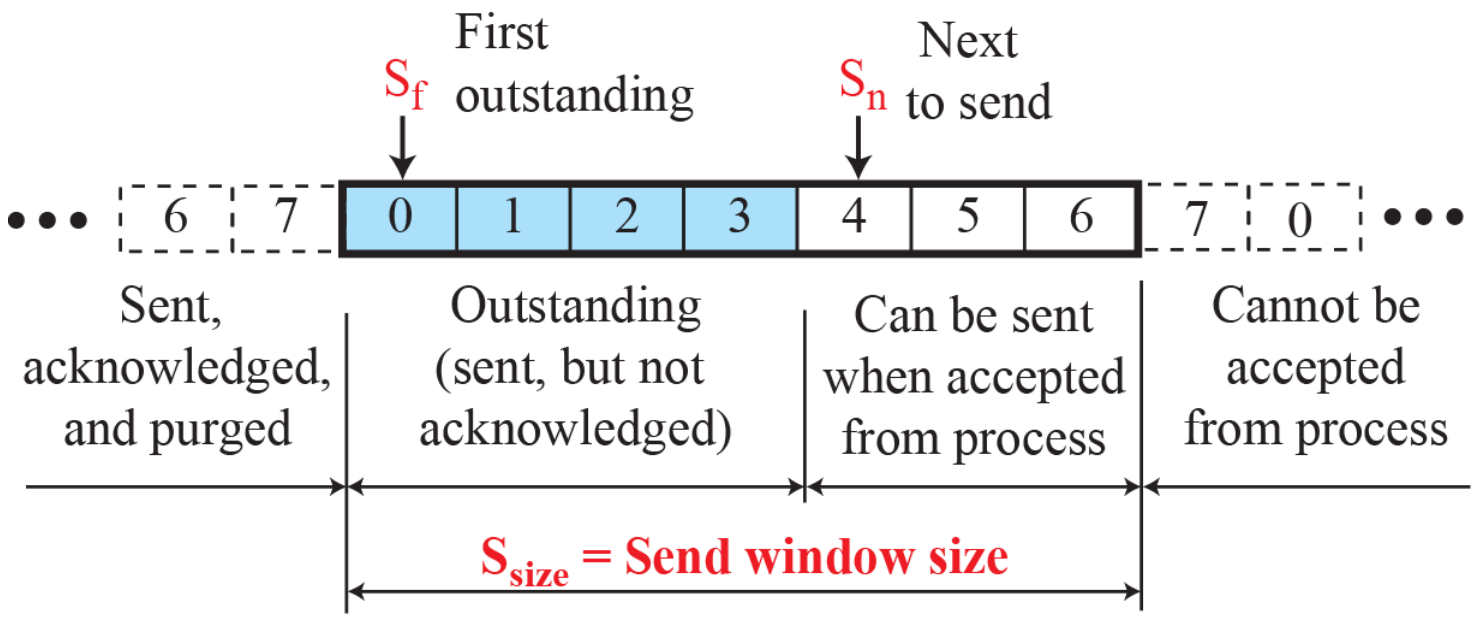

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Figure 3.25: ***Sliding the send window***

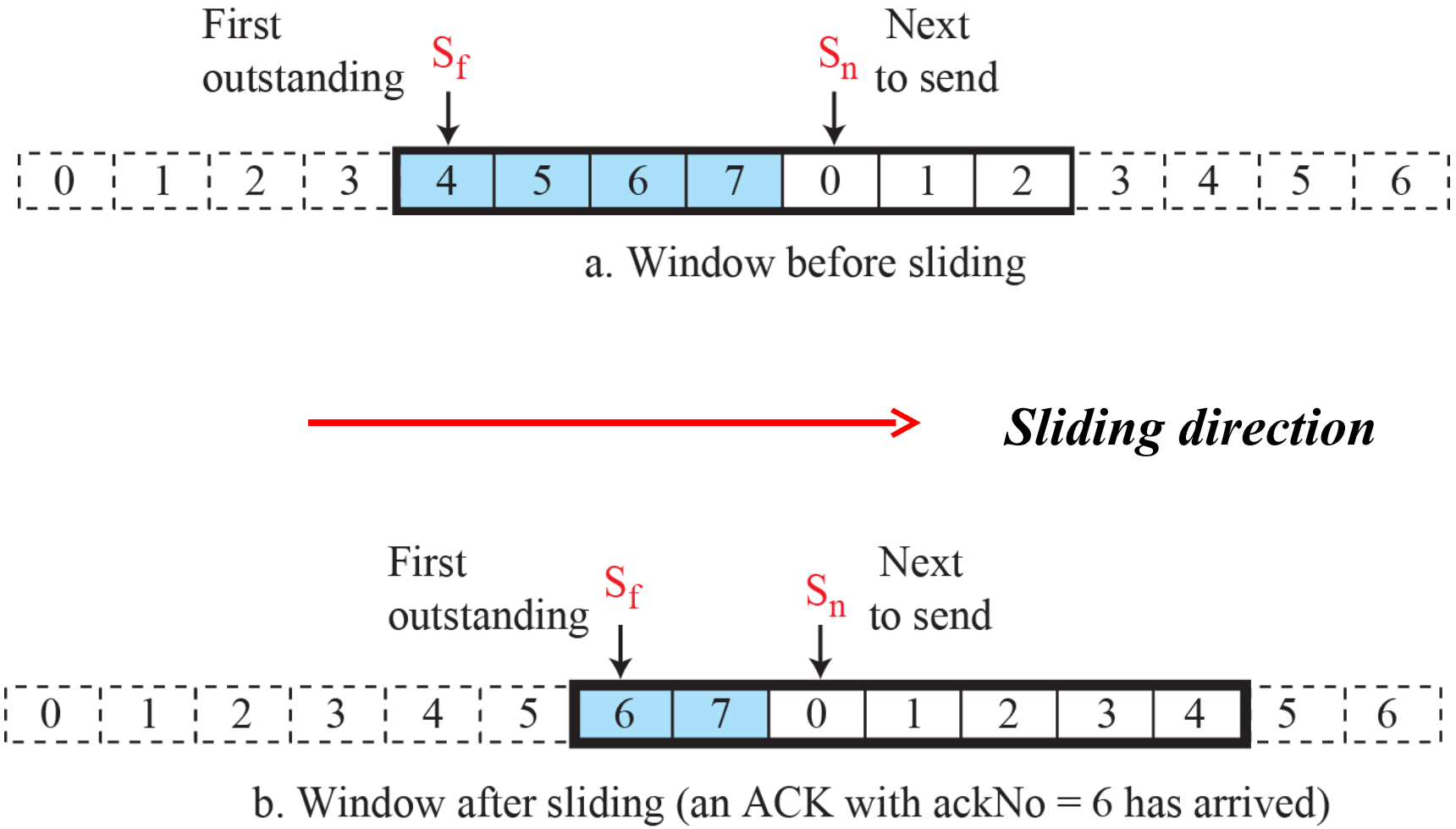

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Figure 3.26: Receive window for Go-Back-N

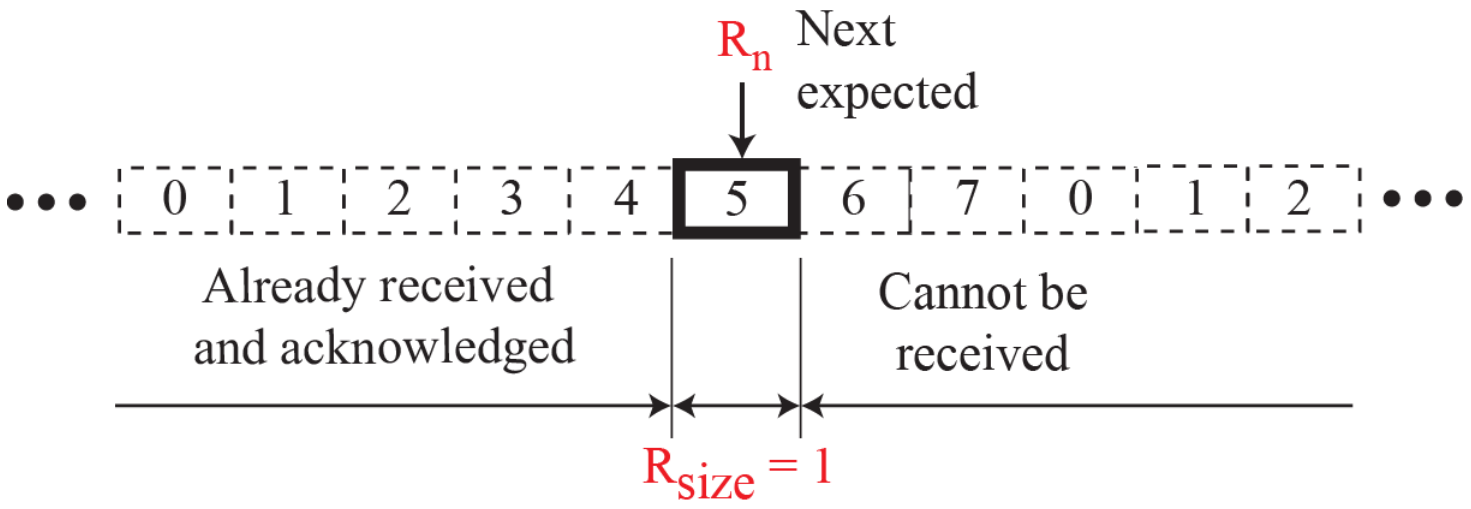

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Figure 3.27: FSMs for the Go-Back-N protocol

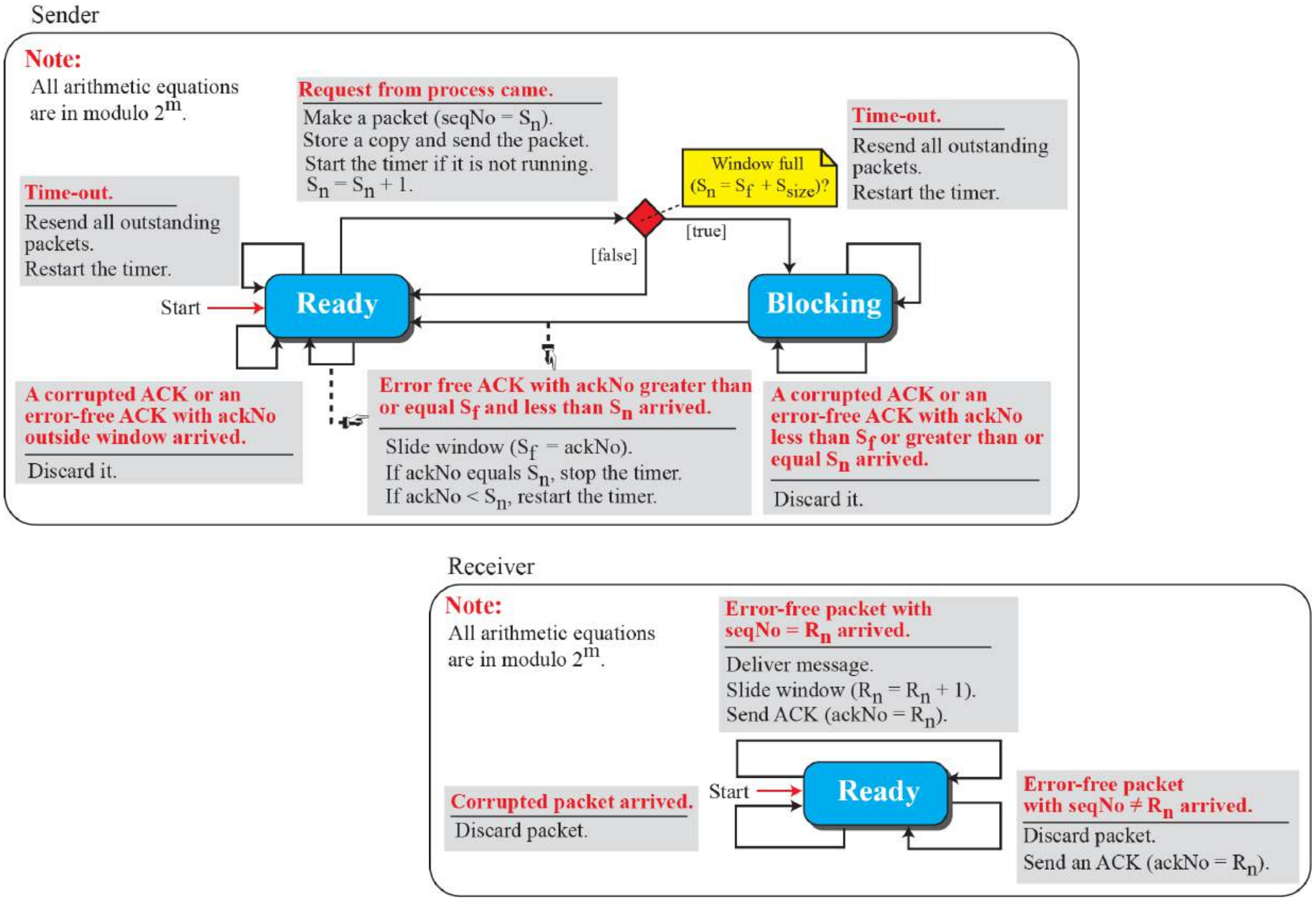

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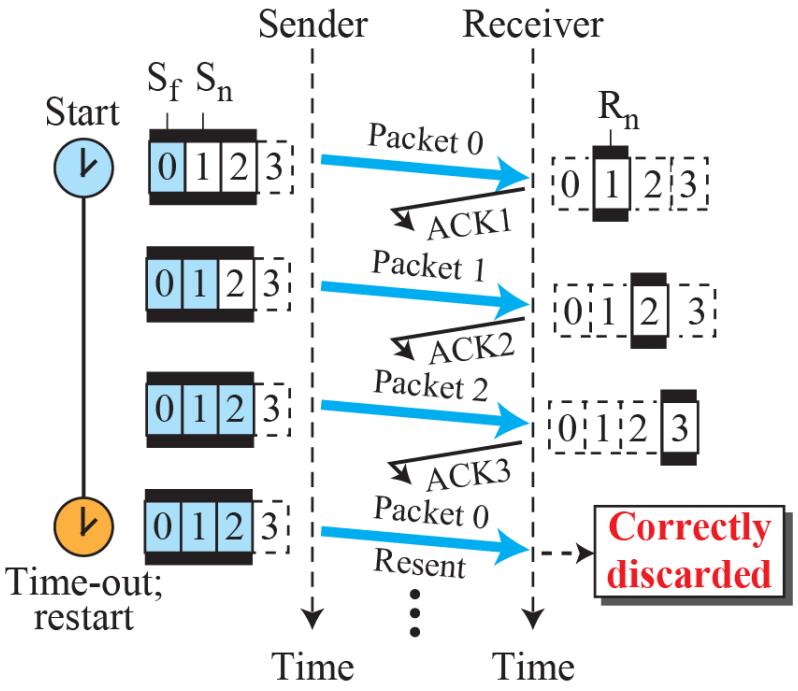
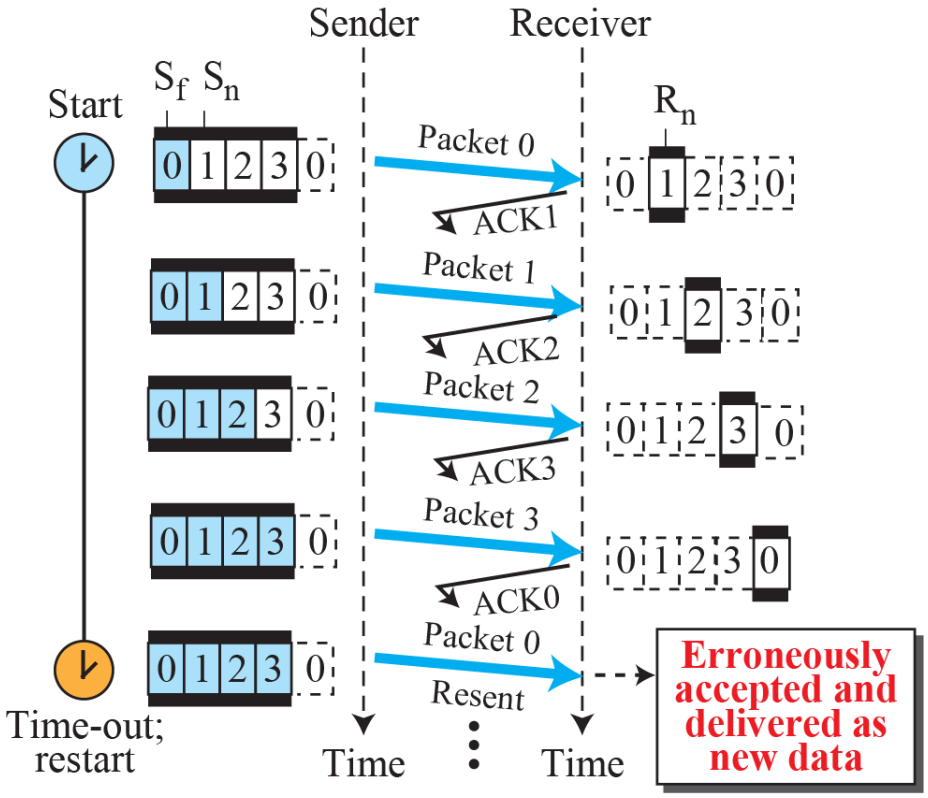
Figure 3.28: Send window size for Go-Back-N



a. Send window of size < 2^m



b. Send window of size = 2^m

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Example 3.7

Figure 3.29 shows an example of Go-Back-*N*. This is an example of a case where the forward channel is reliable, but the reverse is not. No data packets are lost, but some ACKs are delayed and one is lost. The example also shows how cumulative acknowledgments can help if acknowledgments are delayed or lost.

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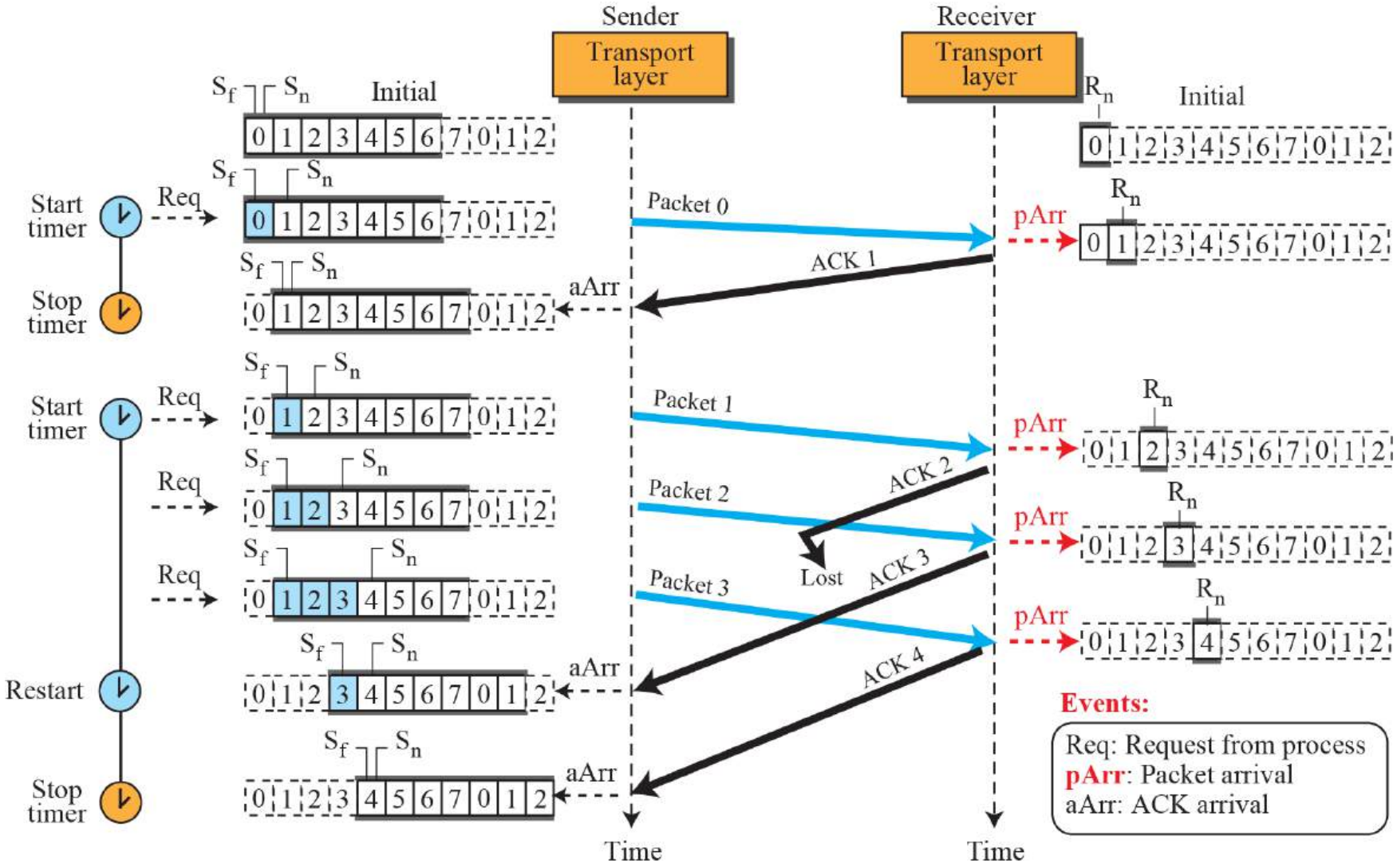
Figure 3.29: Flow diagram for Example 3.7



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Example 3.8

Figure 3.30 shows what happens when a packet is lost. Packets 0, 1, 2, and 3 are sent. However, packet 1 is lost. The receiver receives packets 2 and 3, but they are discarded because they are received out of order (packet 1 is expected). When the receiver receives packets 2 and 3, it sends ACK1 to show that it expects to receive packet 1. However, these ACKs are not useful for the sender because the ackNo is equal to S_f, not greater that S_f . So the sender discards them. When the time-out occurs, the sender resends packets 1, 2, and 3, which are acknowledged.

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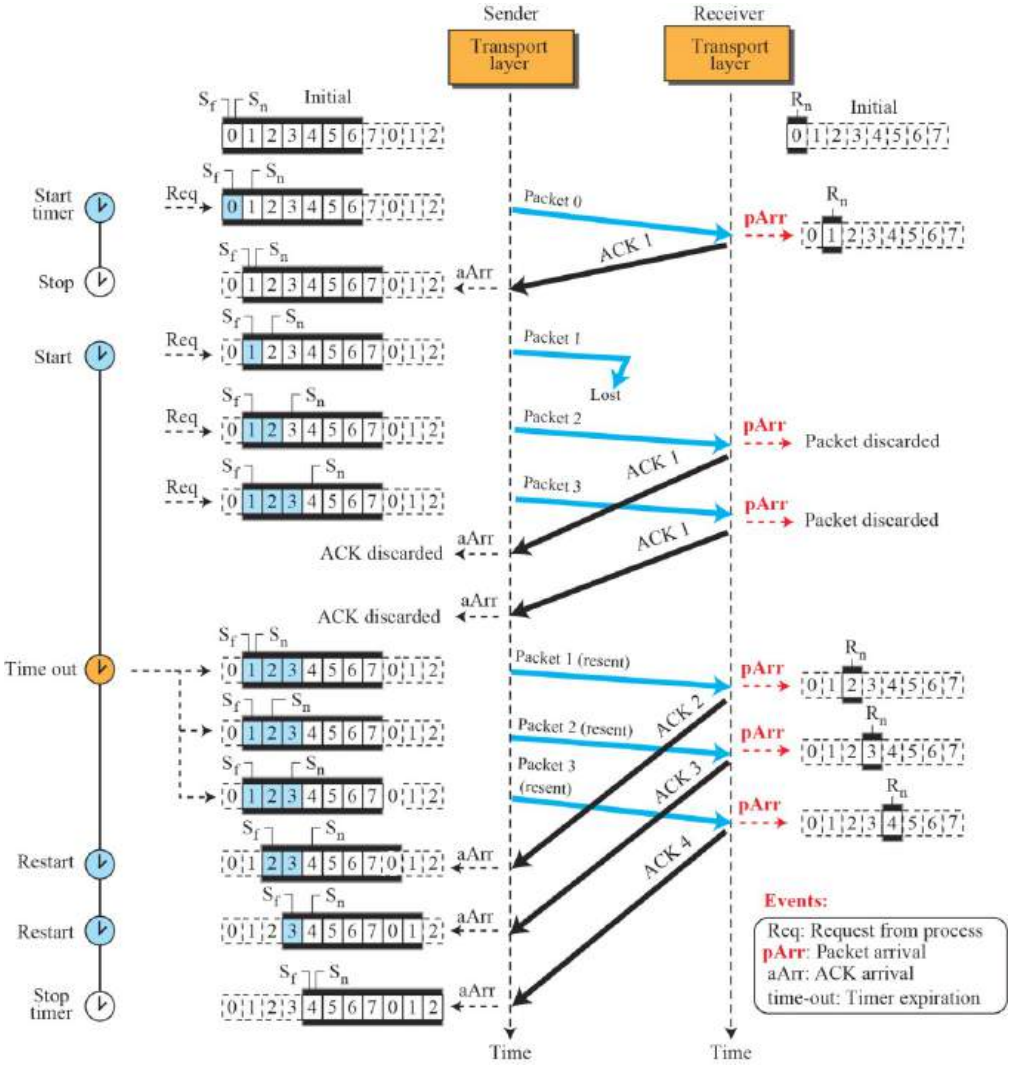
Figure 3.30: Flow diagram for Example 3.8



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3.2.4 Selective-Repeat Protocol

The Go-Back-N protocol simplifies the process at the receiver. The receiver keeps track of only one variable, and there is no need to buffer out-of-order packets; they are simply discarded. Another protocol, called the Selective-Repeat (SR) protocol, has been devised, which, as the name implies, resends only selective packets, those that are actually lost. The outline of this protocol is shown in Figure 3.31.

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3.2.4 (continued)

- ***Windows***
- ***Timer***
- ***Acknowledgments***
- ***FSMs***
  - *Sender*
  - *Receiver*

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Figure 3.31: Outline of Selective-Repeat

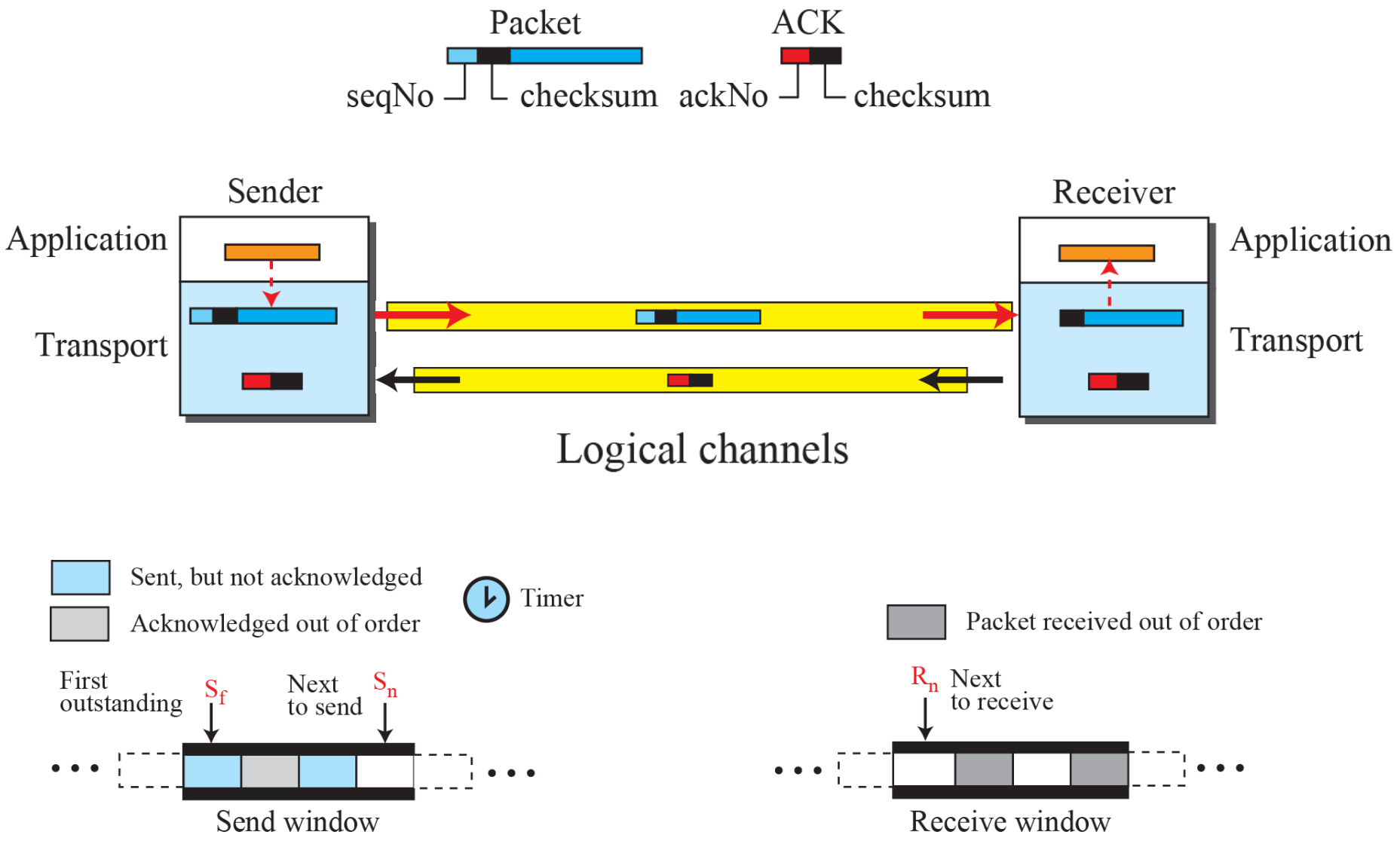

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Figure 3.32: Send window for Selective-Repeat protocol

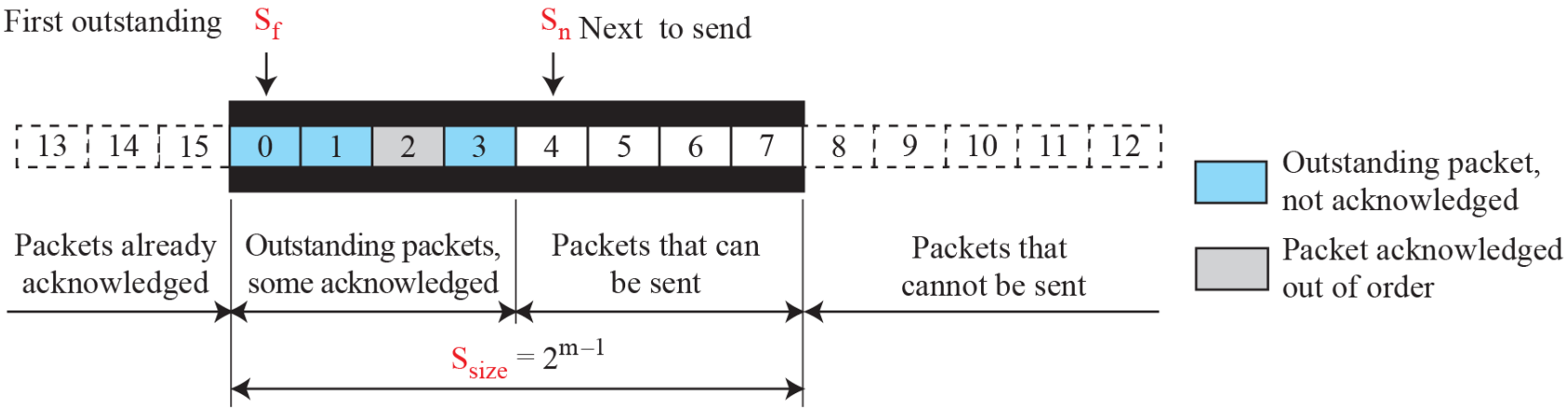

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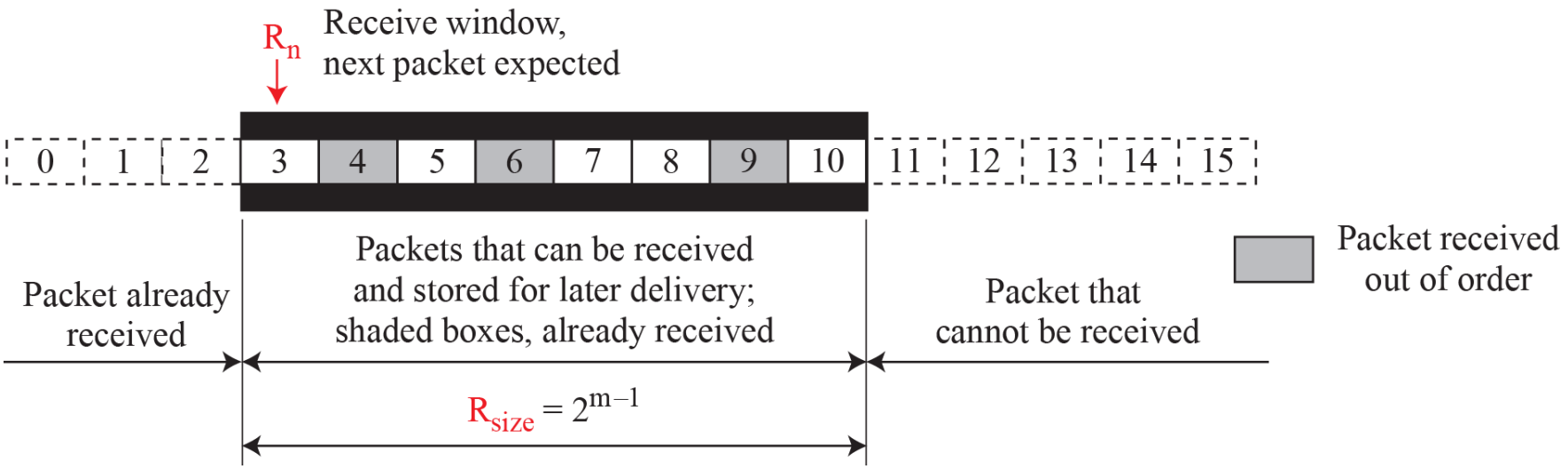
Figure 3.33: Receive window for Selective-Repeat protocol



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Example 3.9

Assume a sender sends 6 packets: packets 0, 1, 2, 3, 4, and 5. The sender receives an ACK with ackNo = 3. What is the interpretation if the system is using GBN or SR?

Solution

If the system is using GBN, it means that packets 0, 1, and 2 have been received uncorrupted and the receiver is expecting packet 3. If the system is using SR, it means that packet 3 has been received uncorrupted; the ACK does not say anything about other packets.

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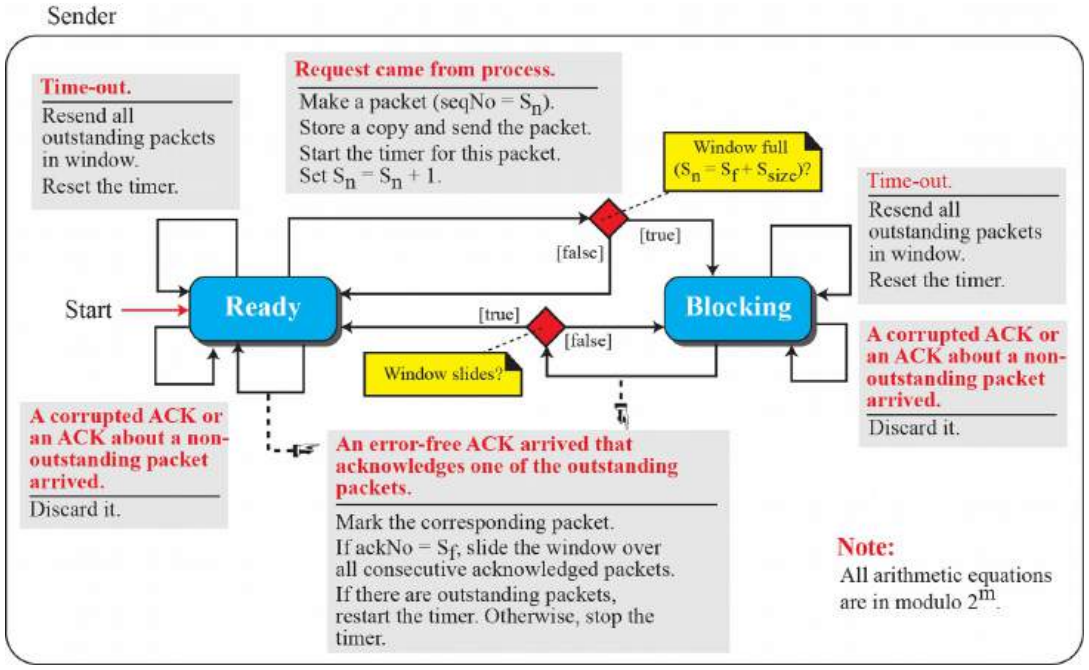
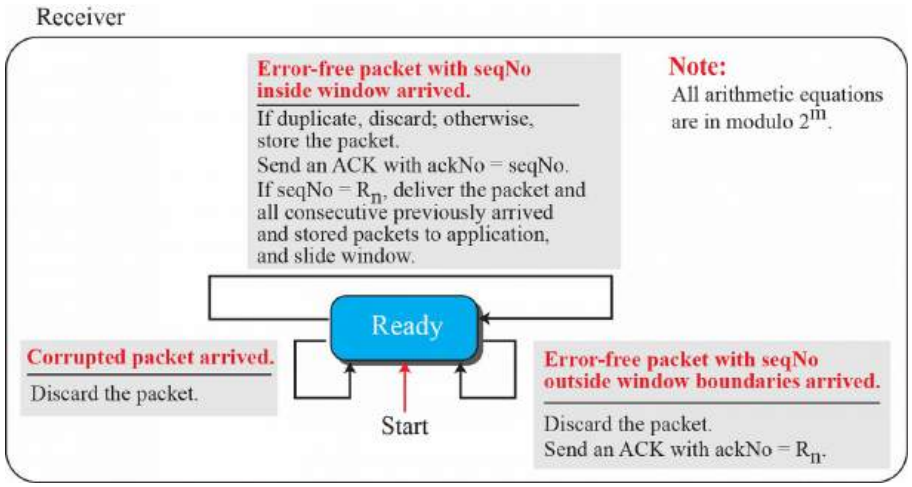
Figure 3.34: *FSMs for SR protocol*





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Example 3.10

This example is similar to Example 3.8 (Figure 3.30) in which packet 1 is lost. We show how Selective-Repeat behaves in this case. Figure 3.35 shows the situation.

At the sender, packet 0 is transmitted and acknowledged. Packet 1 is lost. Packets 2 and 3 arrive out of order and are acknowledged. When the timer times out, packet 1 (the only unacknowledged packet) is resent and is acknowledged. The send window then slides.

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Example 3.10 (continued)

At the receiver site we need to distinguish between the acceptance of a packet and its delivery to the application layer. At the second arrival, packet 2 arrives and is stored and marked (shaded slot), but it cannot be delivered because packet 1 is missing. At the next arrival, packet 3 arrives and is marked and stored, but still none of the packets can be delivered. Only at the last arrival, when finally a copy of packet 1 arrives, can packets 1, 2, and 3 be delivered to the application layer. There are two conditions for the delivery of packets to the application layer: First, a set of consecutive packets must have arrived. Second, the set starts from the beginning of the window.

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Figure 3.35: Flow diagram for Example 3.10

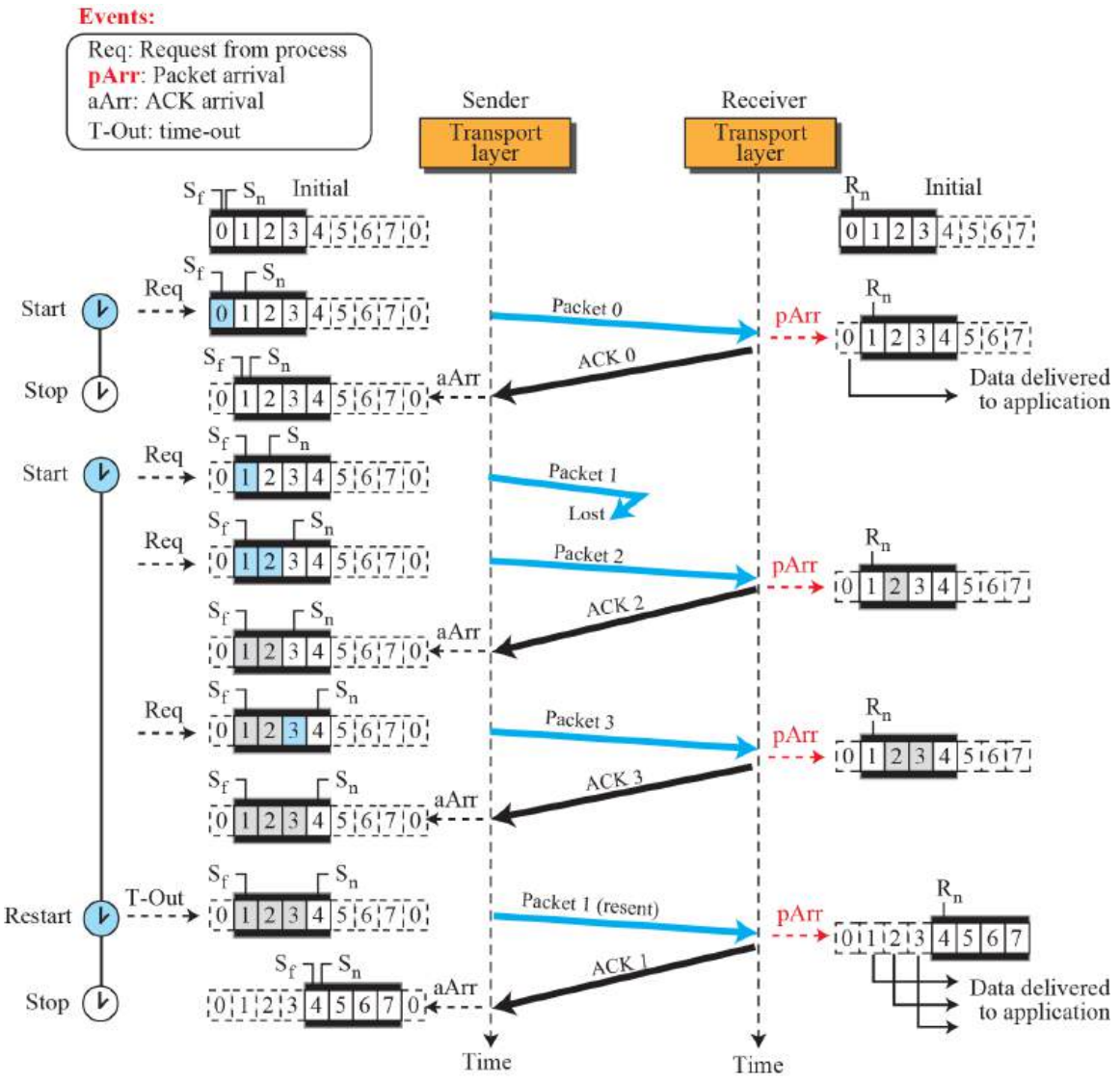

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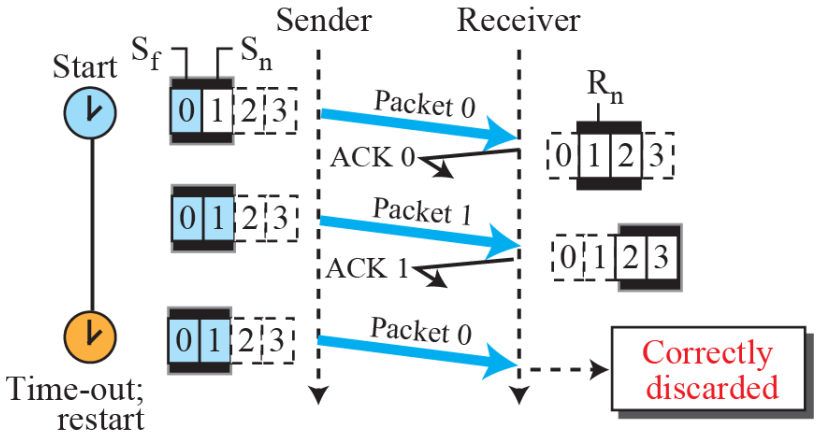
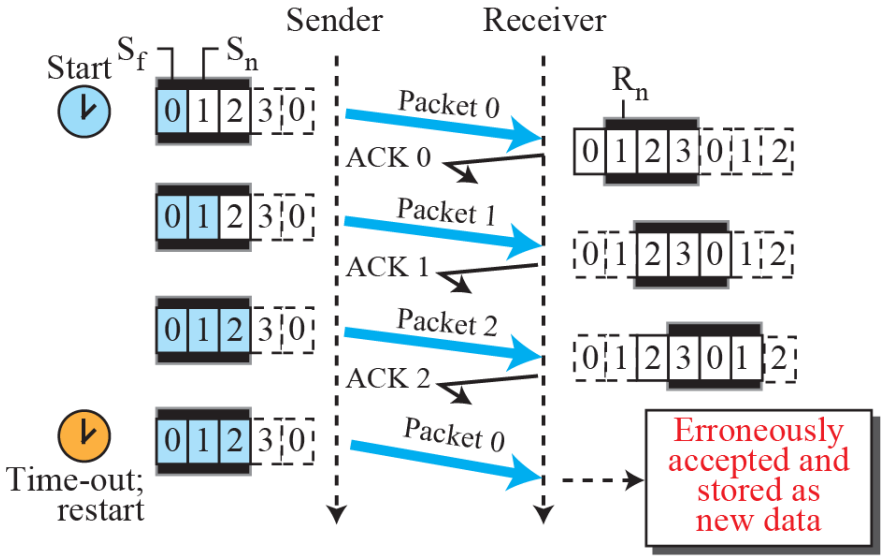
Figure 3.36: Selective-Repeat, window size



a. Send and receive windows of size = 2^{m-1}



b. Send and receive windows of size > 2^{m-1}

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3.2.5 Bidirectional Protocols

The four protocols we discussed earlier in this section are all unidirectional: data packets flow in only one direction and acknowledgments travel in the other direction. In real life, data packets are normally flowing in both directions: from client to server and from server to client. This means that acknowledgments also need to flow in both directions. A technique called piggybacking is used to improve the efficiency of the bidirectional protocols.

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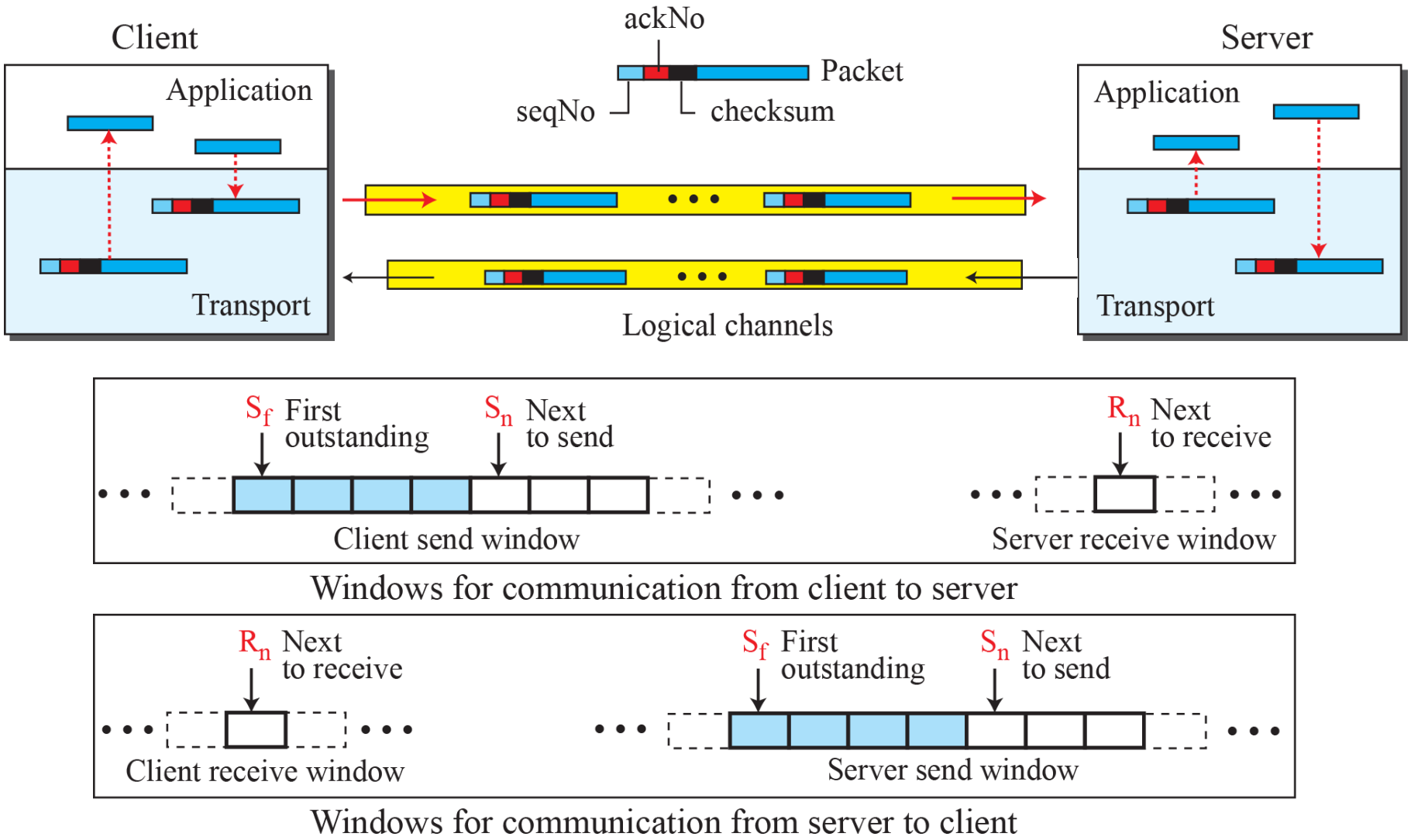
Figure 3.37: Design of piggybacking in Go-Back-N



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3.2.6 Internet Transport-Layer Protocols

A network is the interconnection of a set of devices capable of communication. In this definition, a device can be a host such as a large computer, desktop, laptop, workstation, cellular phone, or security system. A device in this definition can also be a connecting device such as a router a switch, a modem that changes the form of data, and so on.

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Figure 3.38: ***Position of transport-layer protocols in the TCP/IP protocol suite***

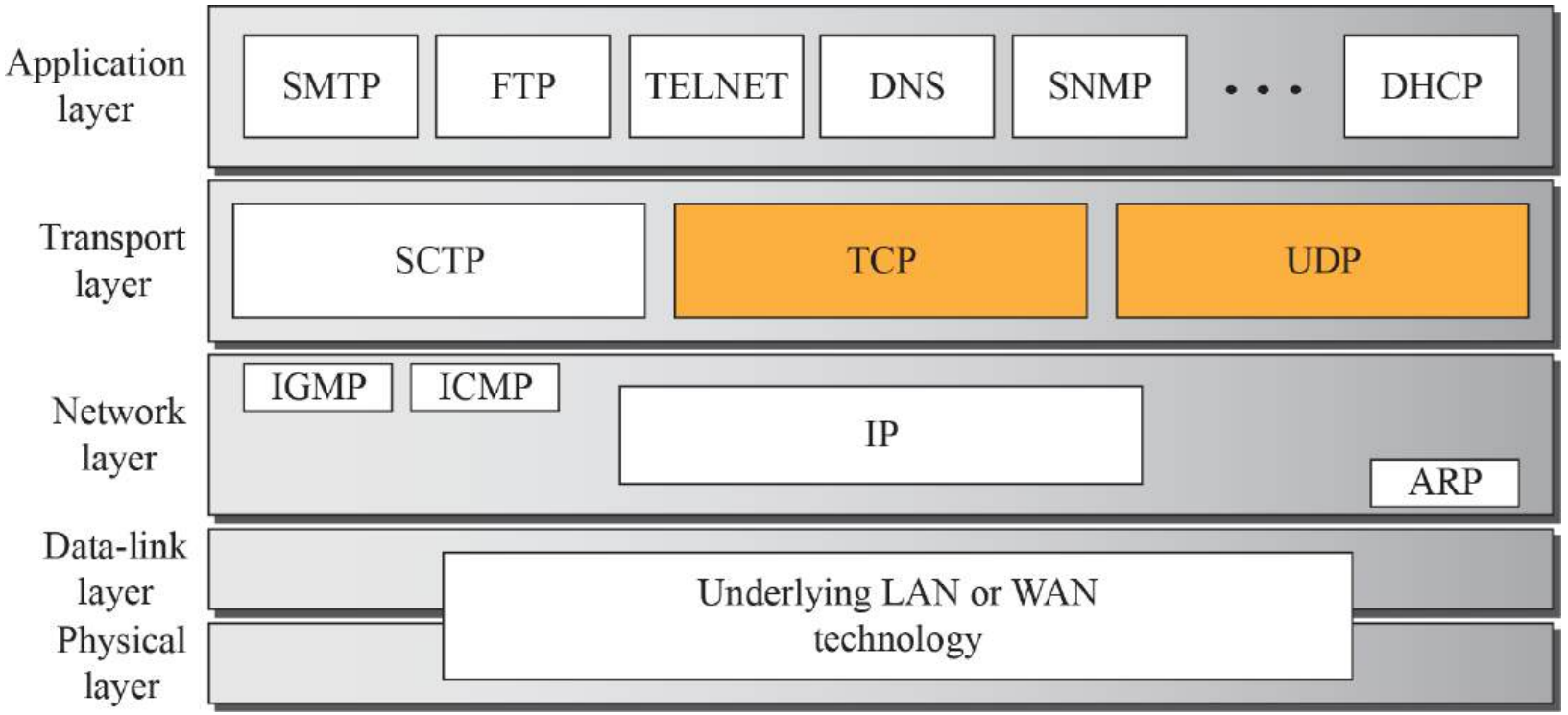

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Table 3.1: Some well-known ports used with UDP and TCP

Port	*Protocol*	*UDP*	*TCP*	*Description*
7	Echo	√		Echoes back a received datagram
9	Discard	√		Discards any datagram that is received
11	Users	√	√	Active users
13	Daytime	√	√	Returns the date and the time
17	Quote	√	√	Returns a quote of the day
19	Chargen	√	√	Returns a string of characters
20, 21	FTP		√	File Transfer Protocol
23	TELNET		√	Terminal Network
25	SMTP		√	Simple Mail Transfer Protocol
53	DNS	√	√	Domain Name Service
67	DHCP	√	√	Dynamic Host Configuration Protocol
69	TFTP	√		Trivial File Transfer Protocol
80	HTTP		√	Hypertext Transfer Protocol
111	RPC	√	√	Remote Procedure Call
123	NTP	√	√	Network Time Protocol
161, 162	SNMP		√	Simple Network Management Protocol

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3-3 USER DATAGRAM PROTOCOL (UDP)

The User Datagram Protocol (UDP) is a connectionless, unreliable transport protocol. It does not add anything to the services of IP except for providing process-to-process instead of host-to-host communication. UDP is a very simple protocol using a minimum of overhead.

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3.3.1 User Datagram

UDP packets, called user datagrams, have a fixed size header of 8 bytes made of four fields, each of 2 bytes (16 bits). Figure 3.39 shows the format of a user datagram. The first two fields define the source and destination port numbers. The third field defines the total length of the user datagram, header plus data. The 16 bits can define a total length of 0 to 65,535 bytes.

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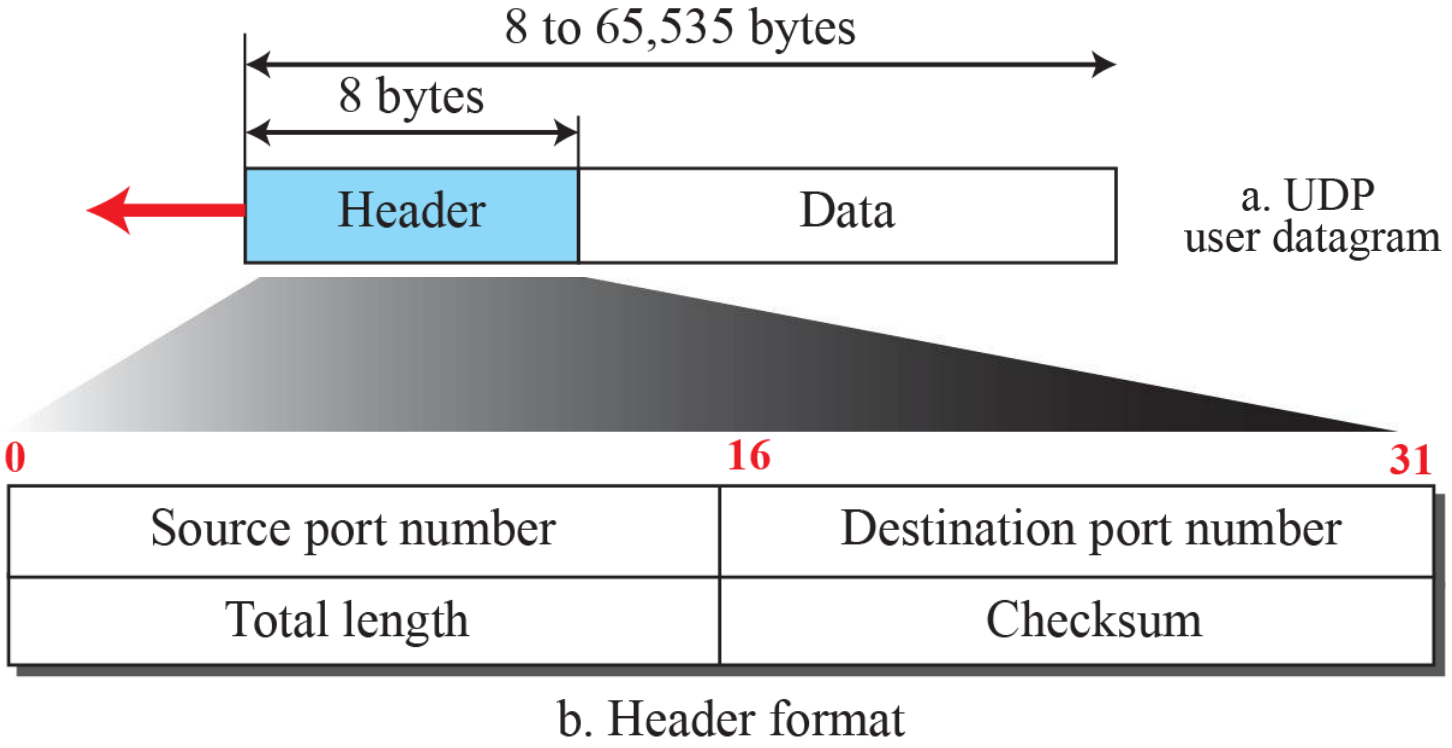
Figure 3.39: User datagram packet format



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Example 3.11

The following is the contents of a UDP header in hexadecimal format.

CB84000D001C001C

a. What is the source port number?
b. What is the destination port number?
c. What is the total length of the user datagram?
d. What is the length of the data?
e. Is the packet directed from a client to a server or vice versa?
f. What is the client process?

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Example 3.11 (continue)

Solution

a. The source port number is the first four hexadecimal digits $(CB84)_{16}$ or 52100

b. The destination port number is the second four hexadecimal digits $(000D)_{16}$ or 13.

c. The third four hexadecimal digits $(001C)_{16}$ define the length of the whole UDP packet as 28 bytes.

d. The length of the data is the length of the whole packet minus the length of the header, or $28 - 8 = 20$ bytes.

e. Since the destination port number is 13 (well-known port), the packet is from the client to the server.

f. The client process is the Daytime (see Table 3.1).

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3.3.2 UDP Services

Earlier we discussed the general services provided by a transport-layer protocol. In this section, we discuss what portions of those general services are provided by UDP.

- ***Process-to-Process Communication***
- ***Connectionless Services***
- ***Flow Control***
- ***Error Control***

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3.3.2 (continued)

- **Checksum**
  - *Optional Inclusion of Checksum*
- **Congestion Control**
- **Encapsulation and Decapsulation**
- **Queuing**
- **Multiplexing and Demultiplexing**
- **Comparison : UDP and Simple Protocol**

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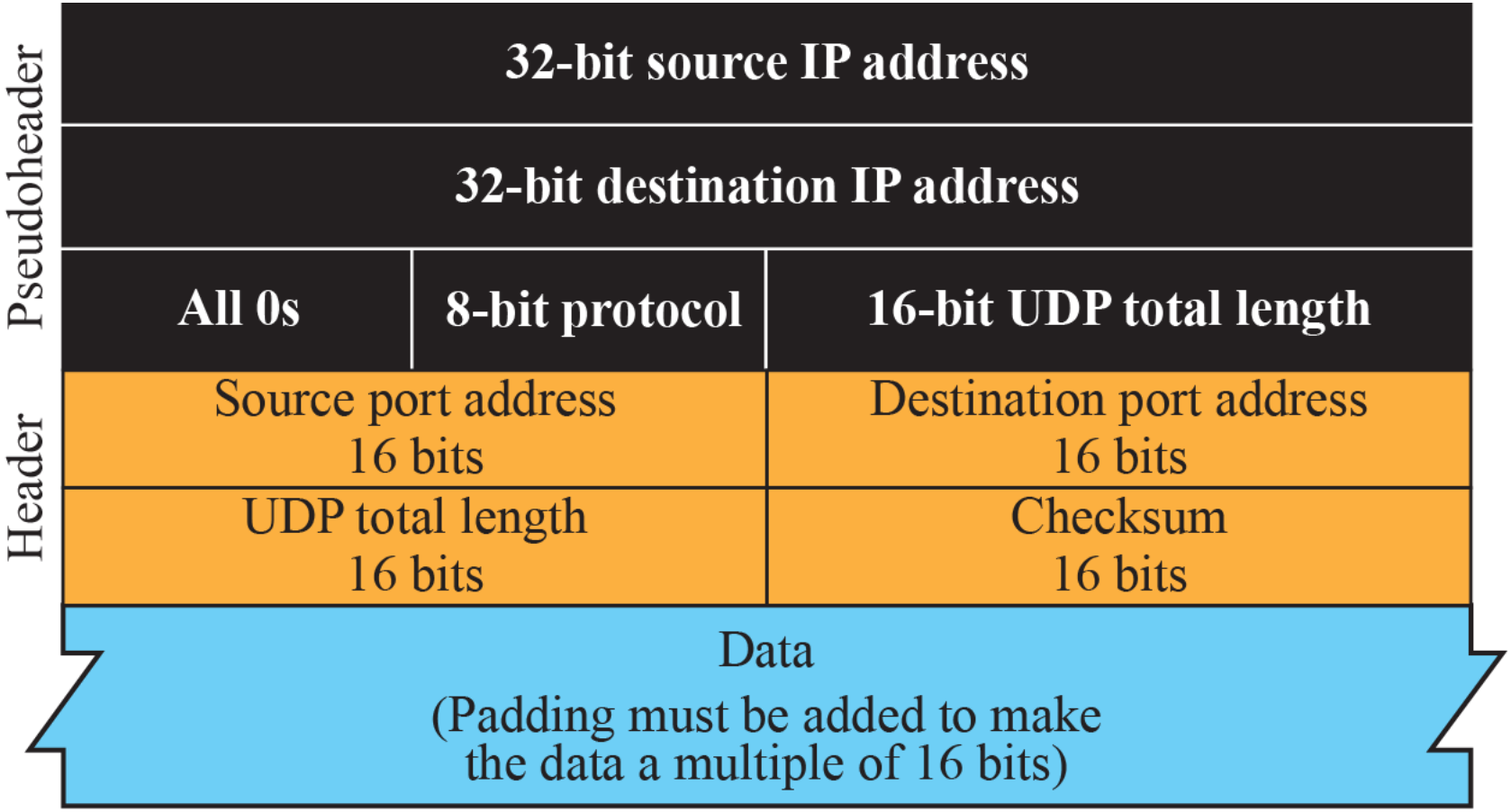
Figure 3.40: Pseudoheader for checksum calculation



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Example 3.12

What value is sent for the checksum in one of the following hypothetical situations?

a. The sender decides not to include the checksum.

b. The sender decides to include the checksum, but the value of the sum is all 1s.

c. The sender decides to include the checksum, but the value of the sum is all 0s.

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Example 3.12 (continued)

Solution

a. The value sent for the checksum field is all 0s to show that the checksum is not calculated.

b. When the sender complements the sum, the result is all 0s; the sender complements the result again before sending. The value sent for the checksum is all 1s. The second complement operation is needed to avoid confusion with the case in part a.

c. This situation never happens because it implies that the value of every term included in the calculation of the sum is all 0s, which is impossible; some fields in the pseudoheader have nonzero values.

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3.3.3 UDP Applications

Although UDP meets almost none of the criteria we mentioned earlier for a reliable transport-layer protocol, UDP is preferable for some applications. The reason is that some services may have some side effects that are either unacceptable or not preferable. An application designer sometimes needs to compromise to get the optimum.

In this section, we first discuss some features of UDP that may need to be considered when one designs an application program and then show some typical applications.

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3.3.3 (continued)

- **UDP Features**
  - Connectionless Service
  - Lack of Error Control
  - Lack of Congestion Control
- **Typical Applications**

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Example 3.13

A client-server application such as DNS (see Chapter 2) uses the services of UDP because a client needs to send a short request to a server and to receive a quick response from it. The request and response can each fit in one user datagram. Since only one message is exchanged in each direction, the connectionless feature is not an issue; the client or server does not worry that messages are delivered out of order.

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Example 3.14

A client-server application such as SMTP (see Chapter 2), which is used in electronic mail, cannot use the services of UDP because a user can send a long e-mail message, which may include multimedia (images, audio, or video). If the application uses UDP and the message does not fit in one single user datagram, the message must be split by the application into different user datagrams. Here the connectionless service may create problems. The user datagrams may arrive and be delivered to the receiver application out of order. The receiver application may not be able to reorder the pieces. This means the connectionless service has a disadvantage for an application program that sends long messages.

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Example 3.15

Assume we are downloading a very large text file from the Internet. We definitely need to use a transport layer that provides reliable service. We don't want part of the file to be missing or corrupted when we open the file. The delay created between the deliveries of the parts is not an overriding concern for us; we wait until the whole file is composed before looking at it. In this case, UDP is not a suitable transport layer.

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Example 3.16

Assume we are using a real-time interactive application, such as Skype. Audio and video are divided into frames and sent one after another. If the transport layer is supposed to resend a corrupted or lost frame, the synchronizing of the whole transmission may be lost. The viewer suddenly sees a blank screen and needs to wait until the second transmission arrives. This is not tolerable. However, if each small part of the screen is sent using one single user datagram, the receiving UDP can easily ignore the corrupted or lost packet and deliver the rest to the application program. That part of the screen is blank for a very short period of time, which most viewers do not even notice.

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3-4 TRANSMISSION CONTROL PROTOCOL

Transmission Control Protocol (TCP) is a connection-oriented, reliable protocol. TCP explicitly defines connection establishment, data transfer, and connection teardown phases to provide a connection-oriented service. TCP uses a combination of GBN and SR protocols to provide reliability.

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3.4.1 TCP Services

A Before discussing TCP in detail, let us explain the services offered by TCP to the processes at the application layer.

- **Process-to-Process Communication**
- **Stream Delivery Service**
  - Sending and Receiving Buffers
  - Segments

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3.4.1 (continued)

- ***Full-Duplex Communication***
- ***Multiplexing and Demultiplexing***
- ***Connection-Oriented Service***
- ***Reliable Service***

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Figure 3.41: *Stream delivery*

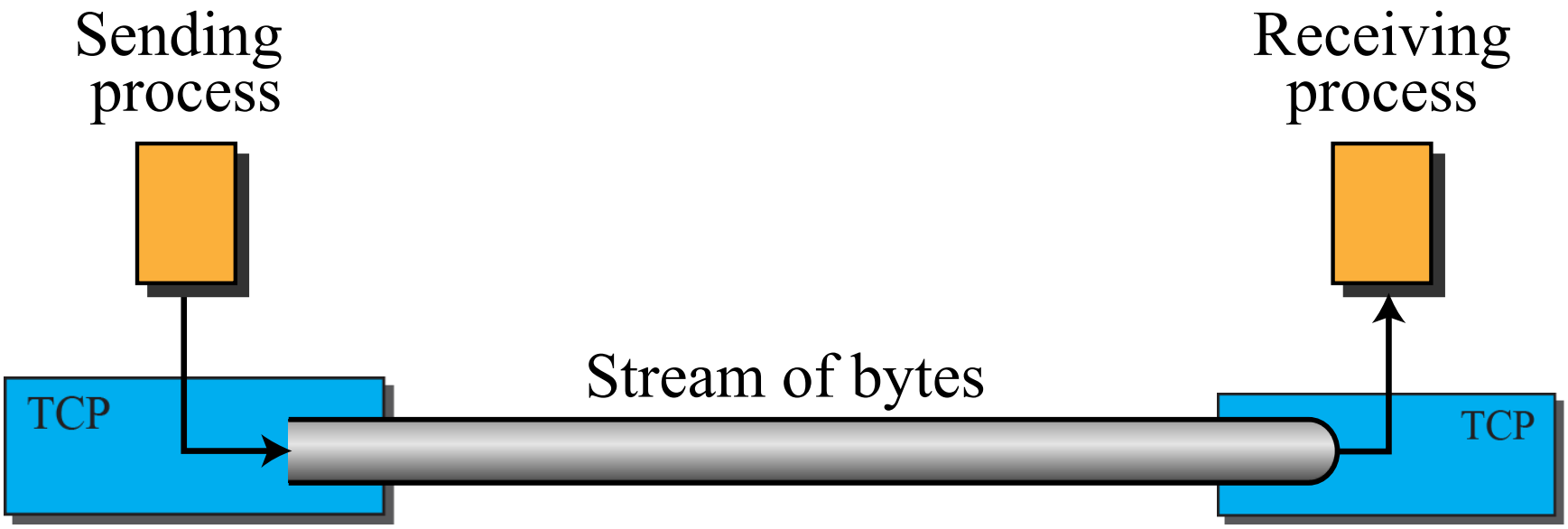

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Figure 3.42: Sending and receiving buffers

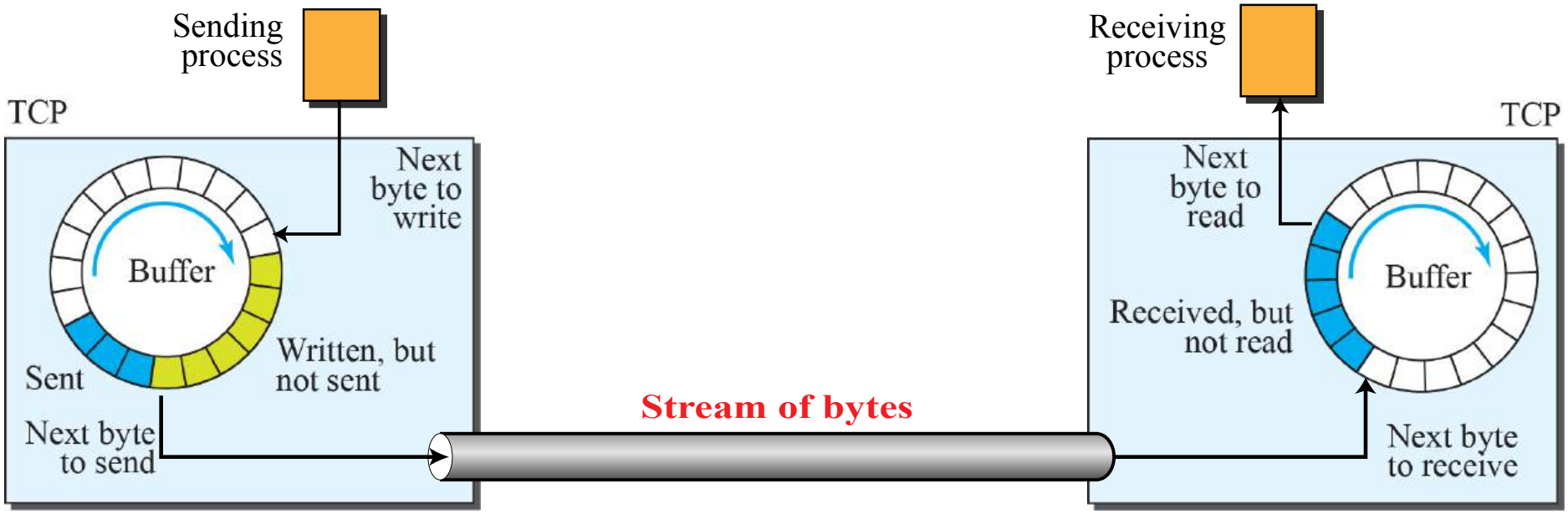

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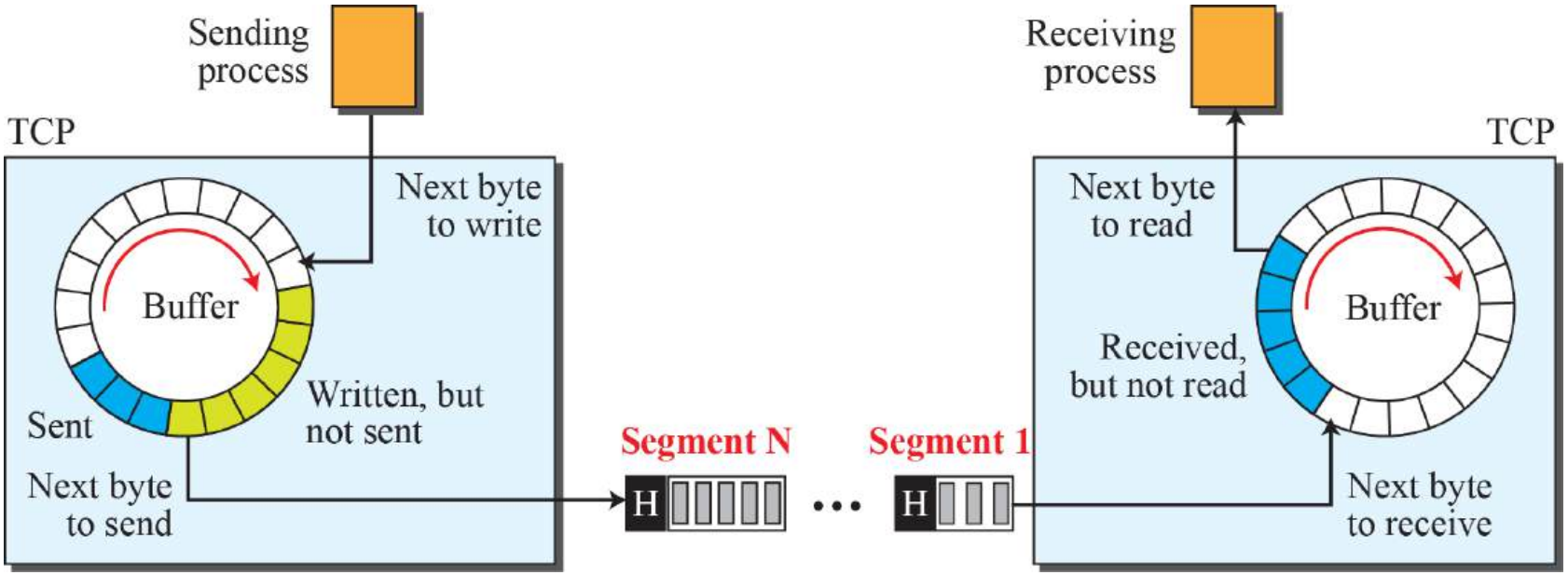
Figure 3.43: TCP segments



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3.4.2 TCP Features

To provide the services mentioned in the previous section, TCP has several features that are briefly summarized in this section and discussed later in detail.

- **Numbering System**
  - Byte Number
  - Sequence Number
  - Acknowledgment Number

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Example 3.17

Suppose a TCP connection is transferring a file of 5,000 bytes. The first byte is numbered 10,001. What are the sequence numbers for each segment if data are sent in five segments, each carrying 1,000 bytes?

Solution

The following shows the sequence number for each segment:

Segment 1	→	Sequence Number:	10,001	**Range:**	10,001	to	11,000
Segment 2	→	Sequence Number:	11,001	**Range:**	11,001	to	12,000
Segment 3	→	Sequence Number:	12,001	**Range:**	12,001	to	13,000
Segment 4	→	Sequence Number:	13,001	**Range:**	13,001	to	14,000
Segment 5	→	Sequence Number:	14,001	**Range:**	14,001	to	15,000

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3.4.3 Segment

Before discussing TCP in more detail, let us discuss the TCP packets themselves. A packet in TCP is called a segment.

- *Format*
- *Encapsulation*

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Figure 3.44: TCP segment format

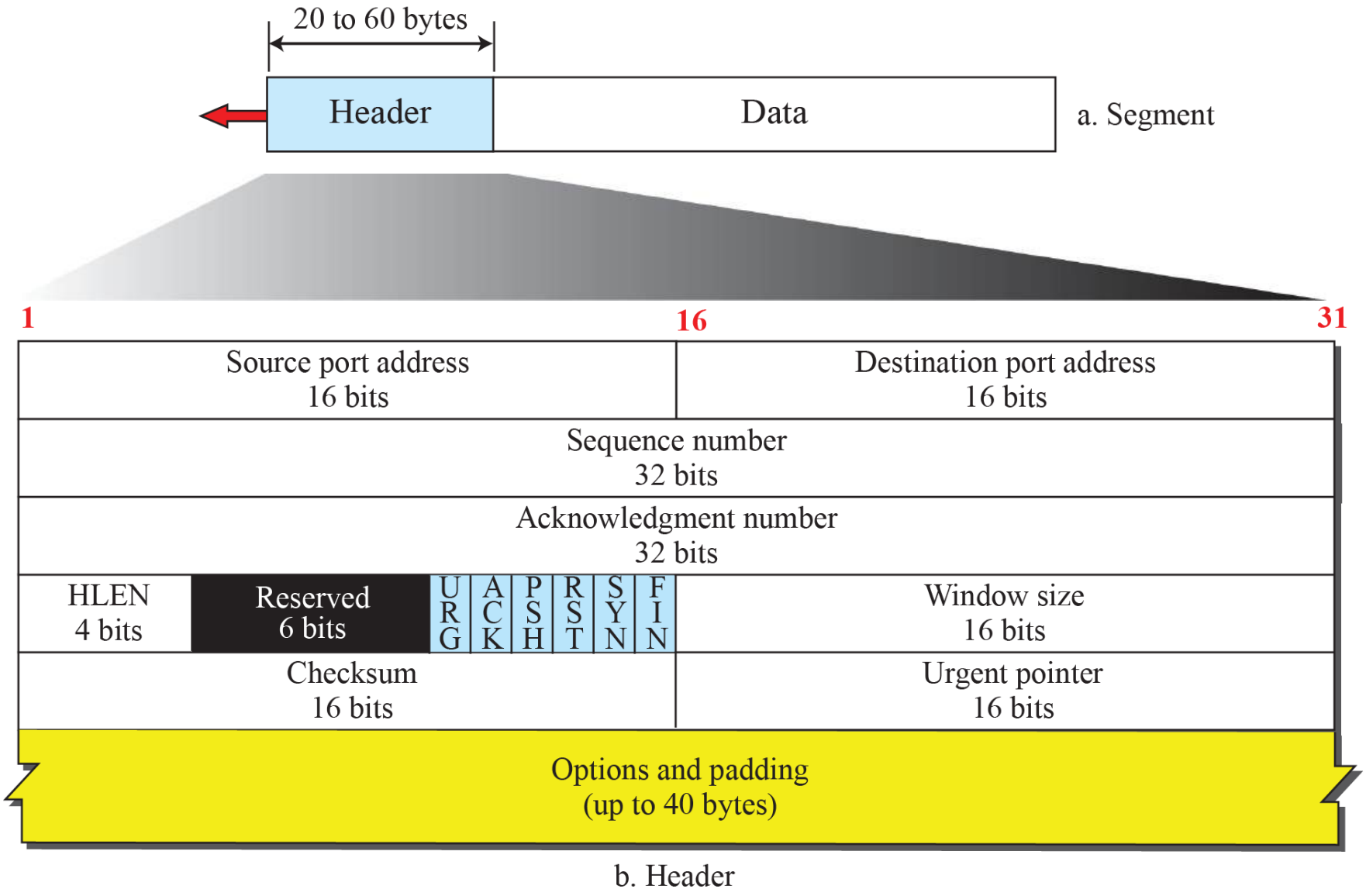

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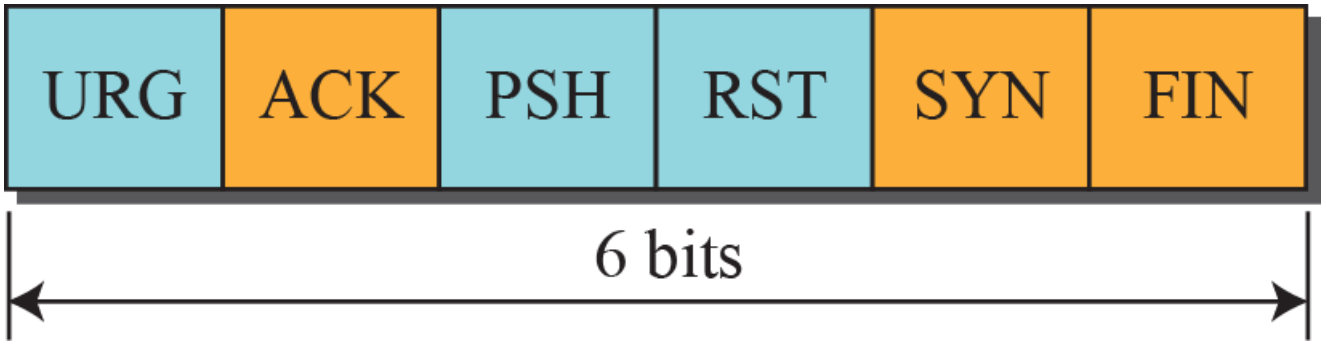
Figure 3.45: Control field



URG: Urgent pointer is valid
ACK: Acknowledgment is valid
PSH: Request for push
RST: Reset the connection
SYN: Synchronize sequence numbers
FIN: Terminate the connection

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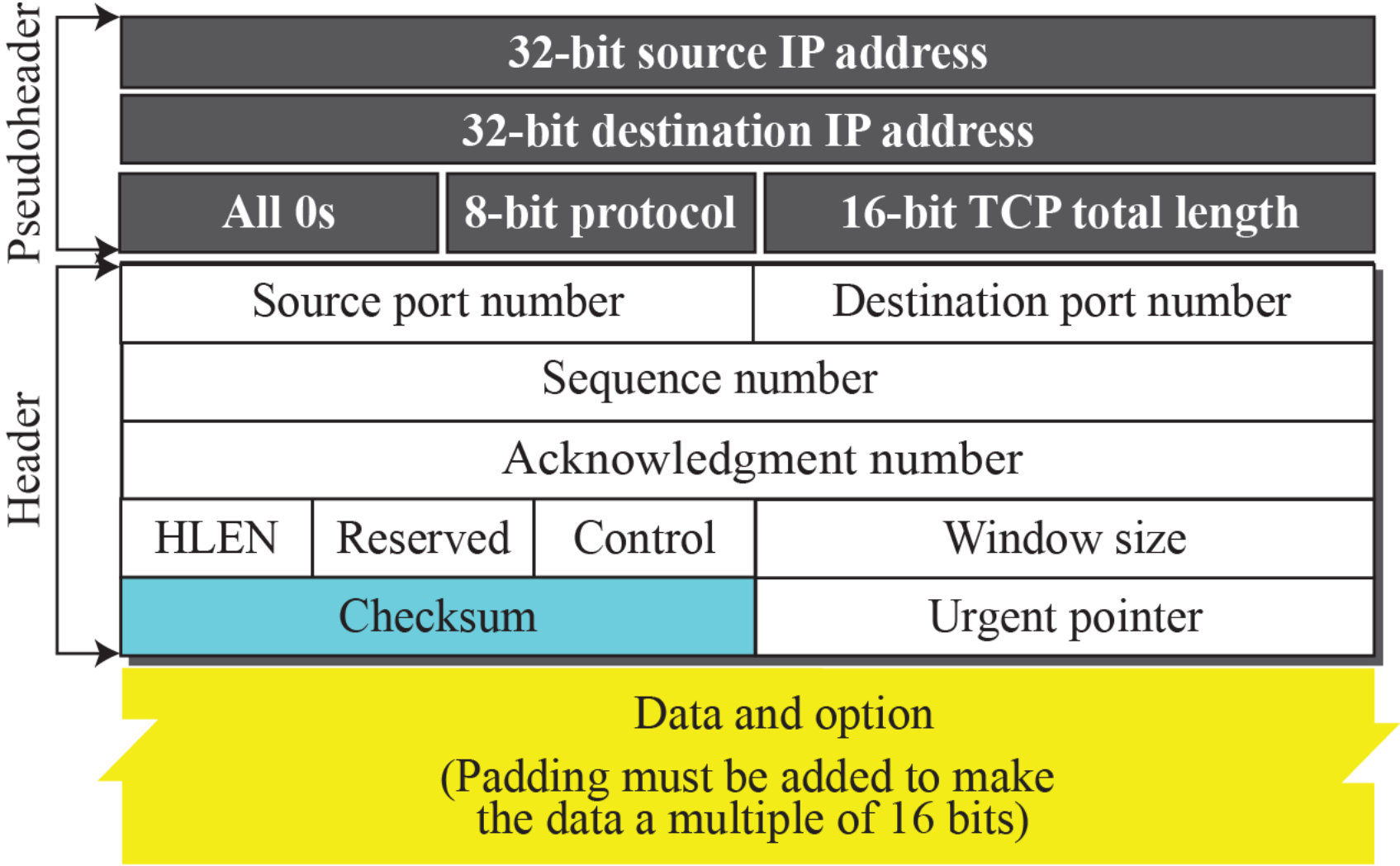
Figure 3.46: *Pseudoheader added to the TCP datagram*



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3.4.4 A TCP Connection

TCP is connection-oriented. As discussed before, a connection-oriented transport protocol establishes a logical path between the source and destination. All of the segments belonging to a message are then sent over this logical path. Using a single logical pathway for the entire message facilitates the acknowledgment process as well as retransmission of damaged or lost frames.

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3.4.4 (continued)

- **Connection Establishment**
  - Three-Way Handshaking
  - SYN Flooding Attack
- **Data Transfer**
  - Pushing Data
  - Urgent Data
- **Connection Termination**
  - Three-Way Handshaking
  - Half-Close
- **Connection Reset**

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Figure 3.47: Connection establishment using three-way handshaking

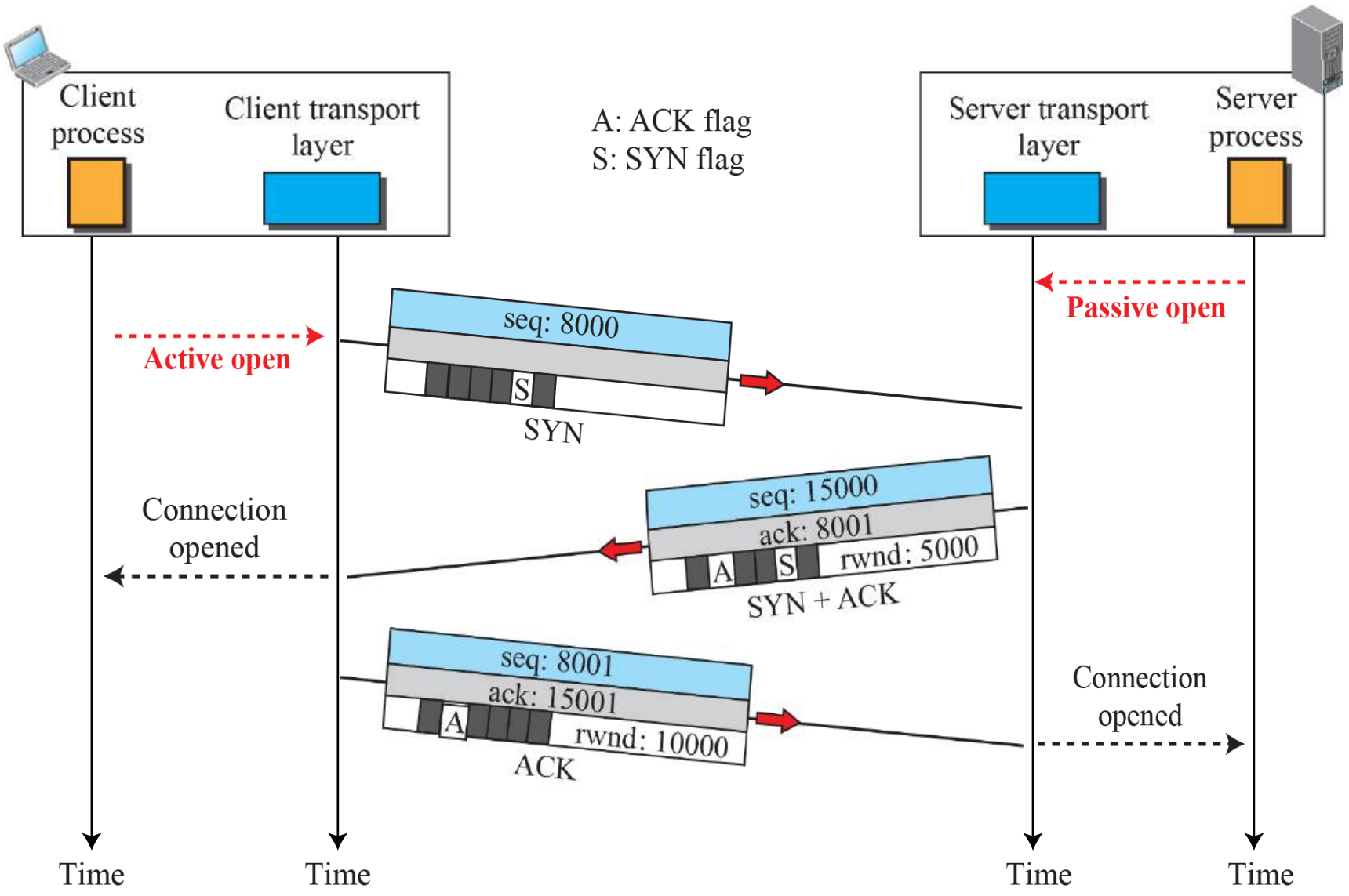

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Figure 3.48: Data transfer

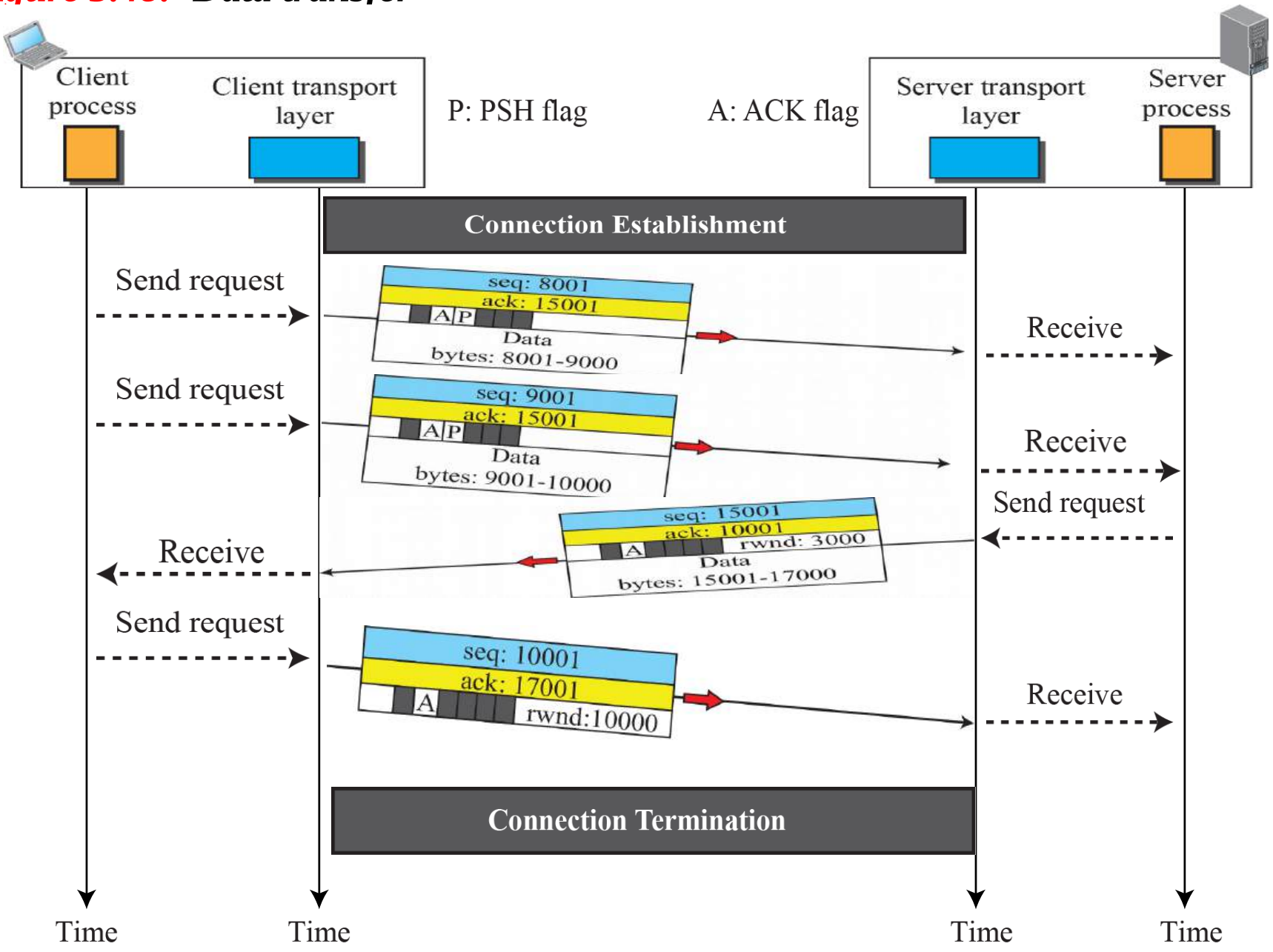

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Figure 3.49: Connection termination using three-way handshaking

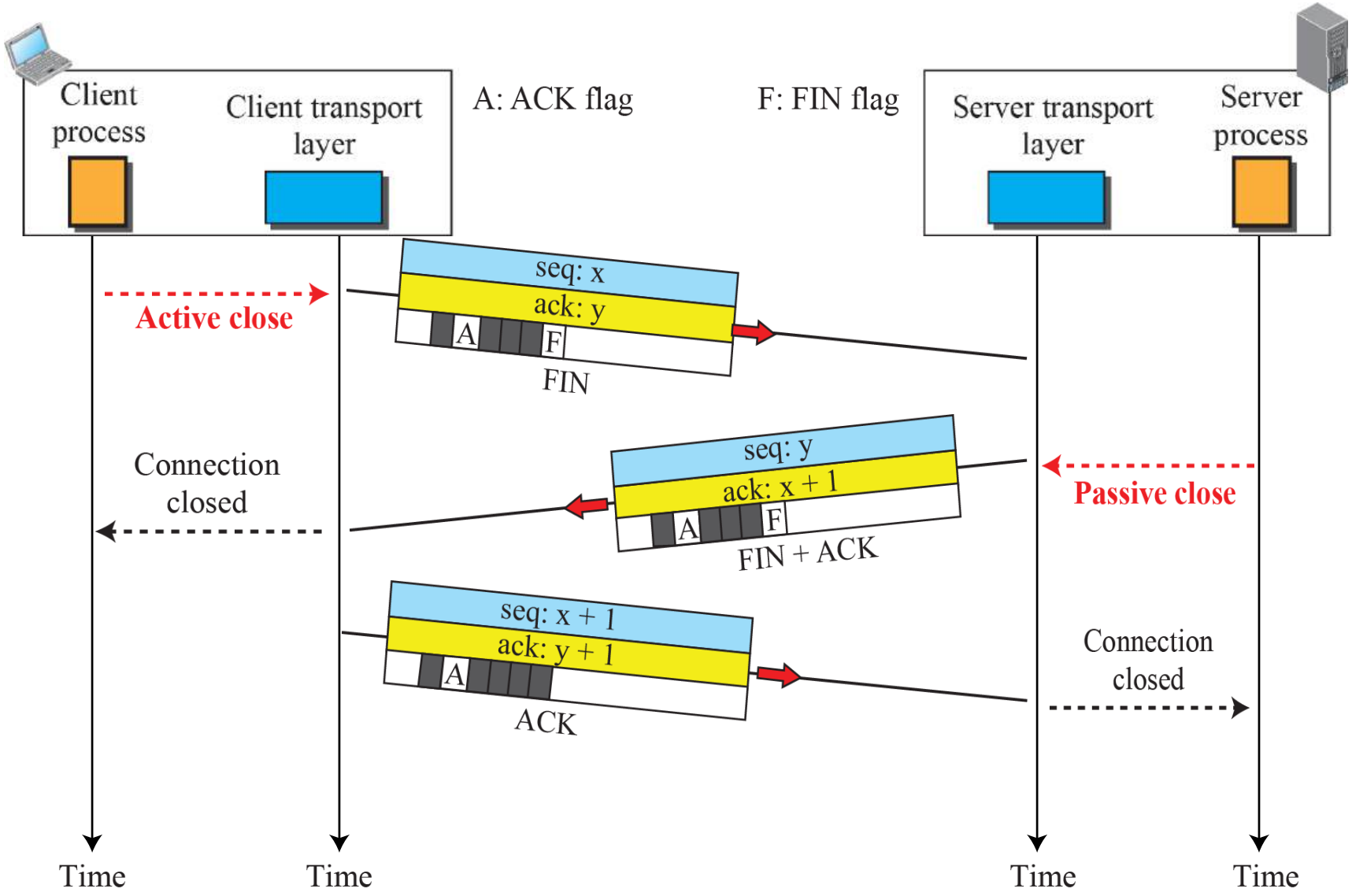

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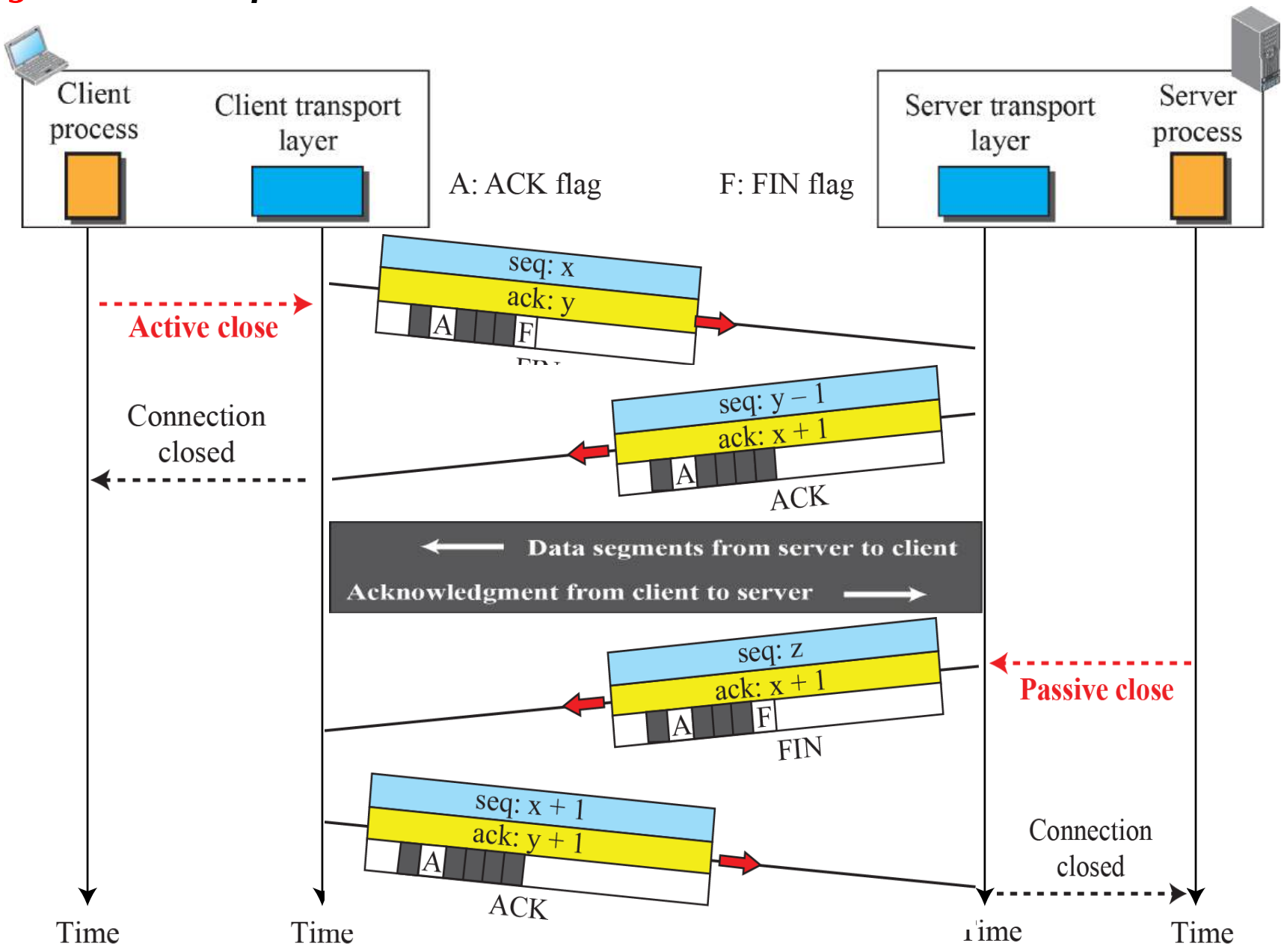
Figure 3.50: *Half-close*



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3.4.5 State Transmission Diagram

To keep track of all the different events happening during connection establishment, connection termination, and data transfer, TCP is specified as the finite state machine (FSM) as shown in Figure 3.51.

- **Scenarios**
  - *A Half-Close Scenario*

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Figure 3.51: *State transition diagram*

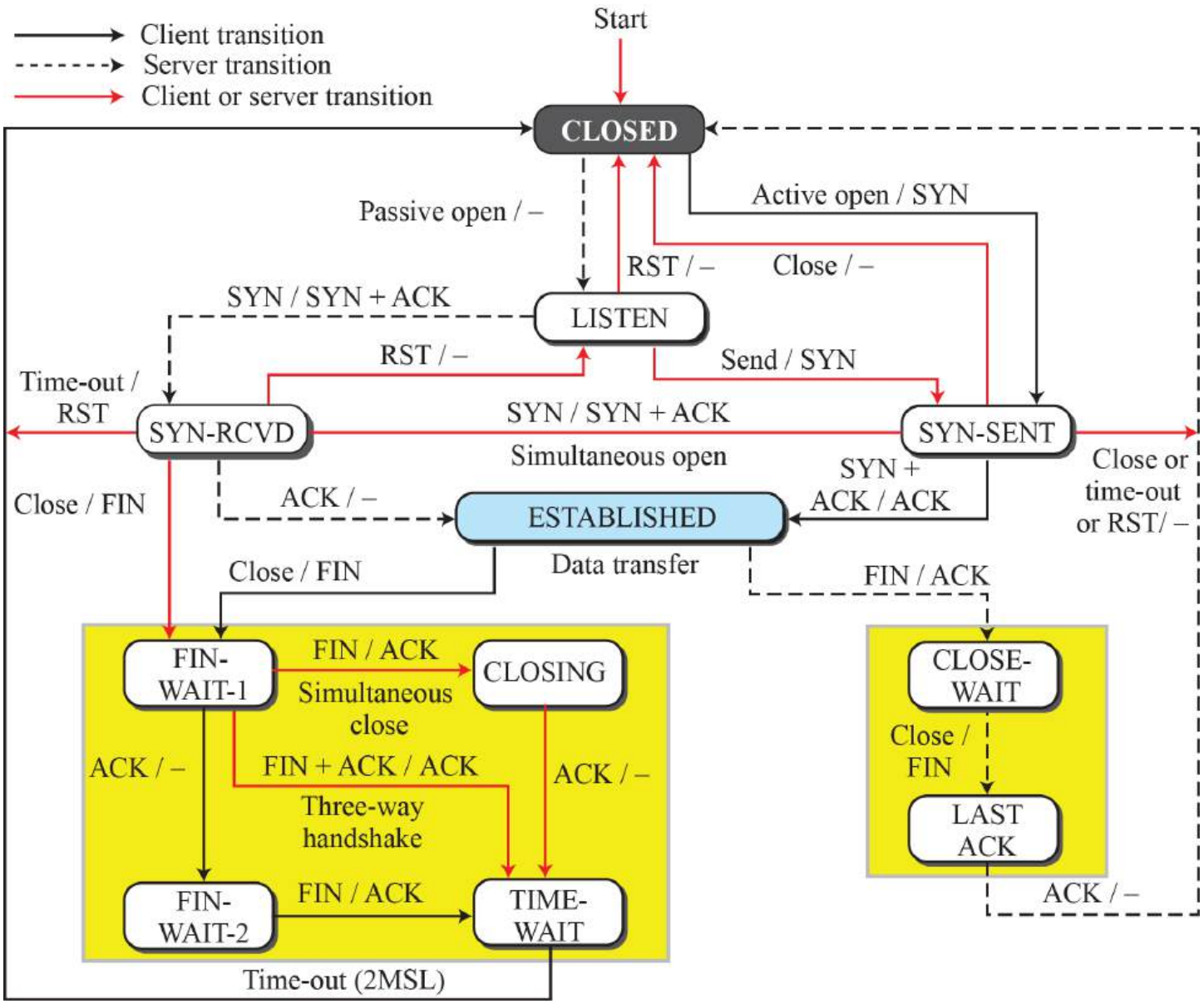

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Table 3.2: States for TCP

State	*Description*
CLOSED	No connection exists
LISTEN	Passive open received; waiting for SYN
SYN-SENT	SYN sent; waiting for ACK
SYN-RCVD	SYN+ACK sent; waiting for ACK
ESTABLISHED	Connection established; data transfer in progress
FIN-WAIT-1	First FIN sent; waiting for ACK
FIN-WAIT-2	ACK to first FIN received; waiting for second FIN
CLOSE-WAIT	First FIN received, ACK sent; waiting for application to close
TIME-WAIT	Second FIN received, ACK sent; waiting for 2MSL time-out
LAST-ACK	Second FIN sent; waiting for ACK
CLOSING	Both sides decided to close simultaneously

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Figure 3.52: Transition diagram with half-close connection termination

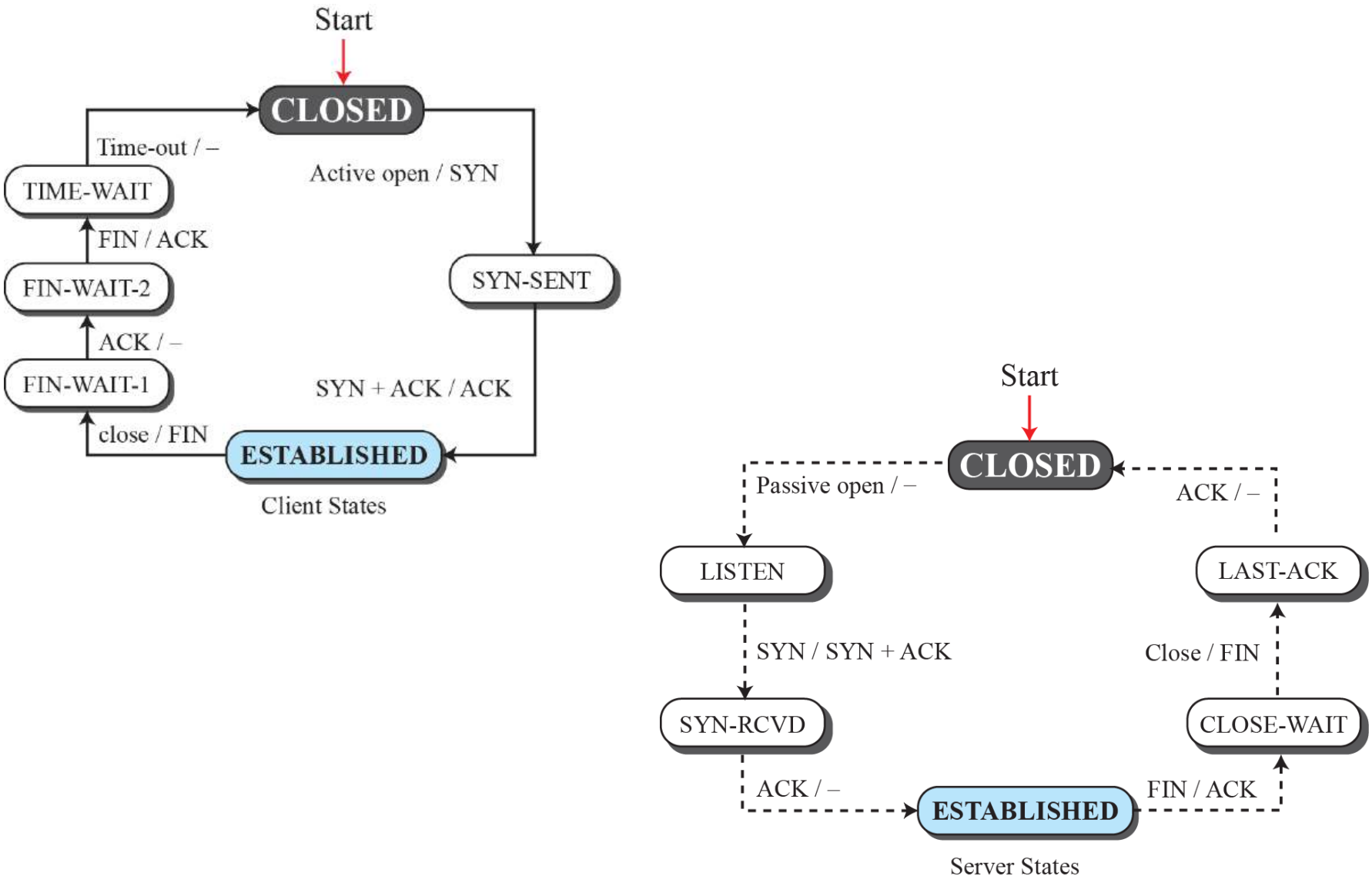

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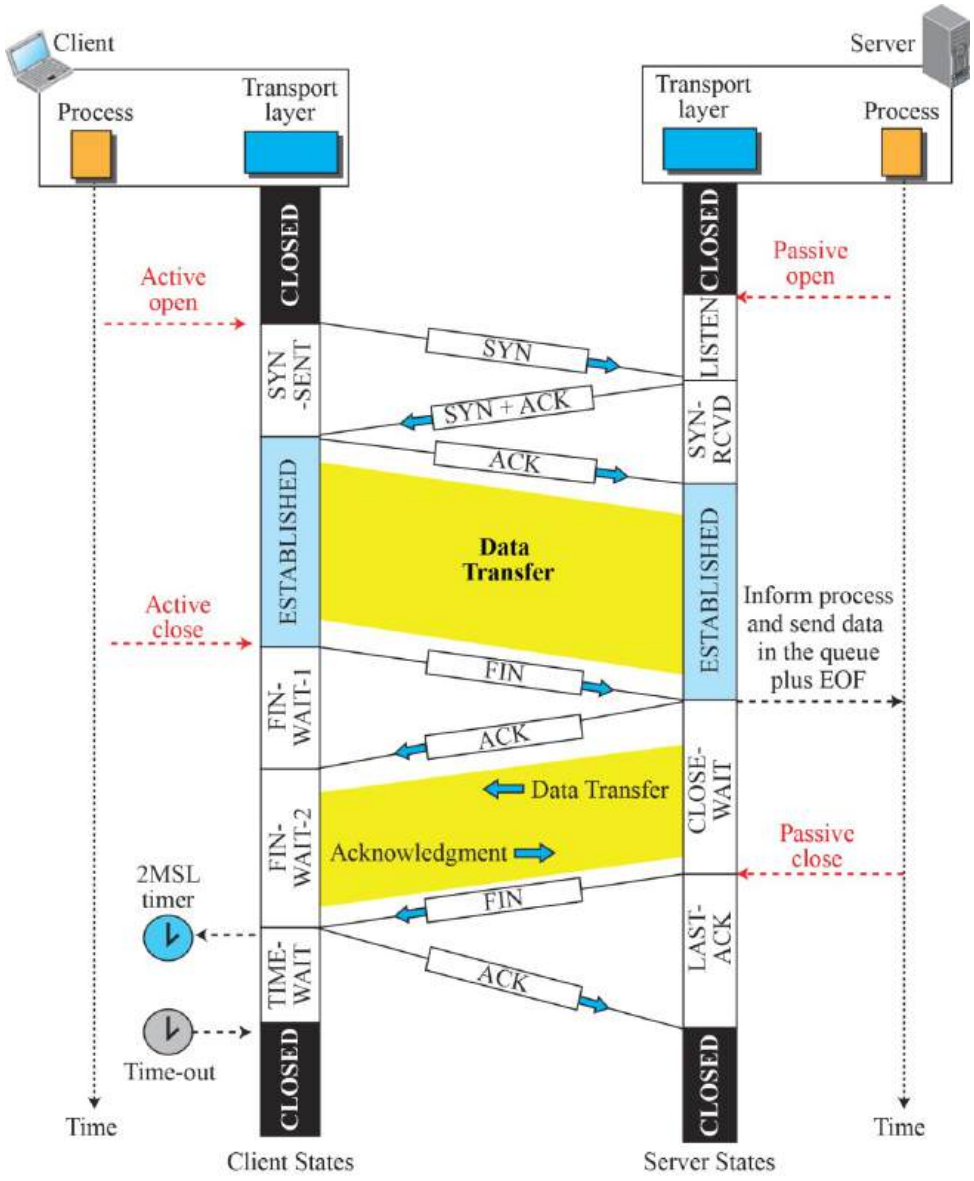
Figure 3.53: Time-line diagram for a common scenario



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3.4.6 Windows in TCP

TCP uses two windows (send window and receive window) for each direction of data transfer, which means four windows for a bidirectional communication. To make the discussion simple, we make an unrealistic unidirectional; the bidirectional communication can be inferred using two unidirectional communications with piggybacking.

- ***Send Window***
- ***Receive Window***

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Figure 3.54: Send window in TCP

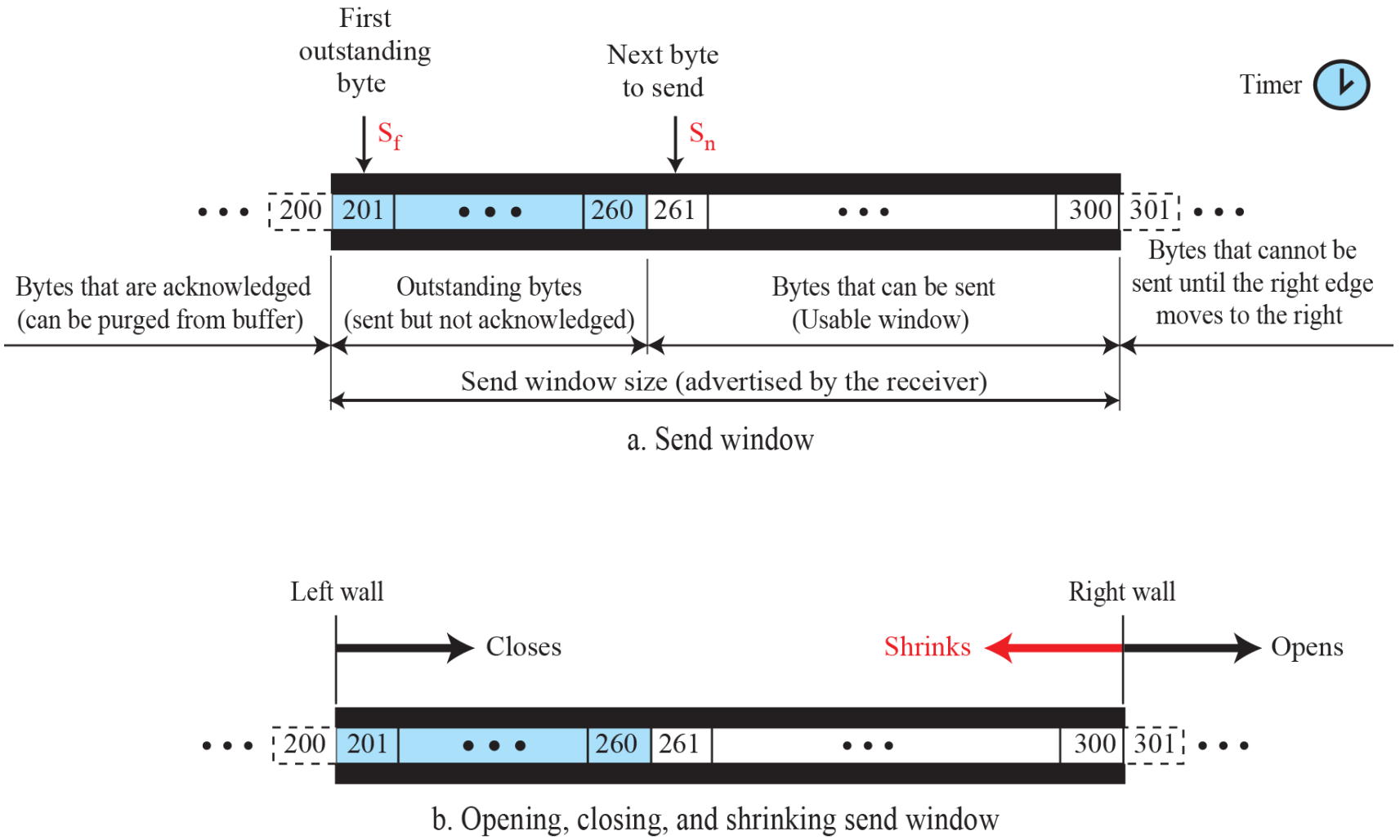

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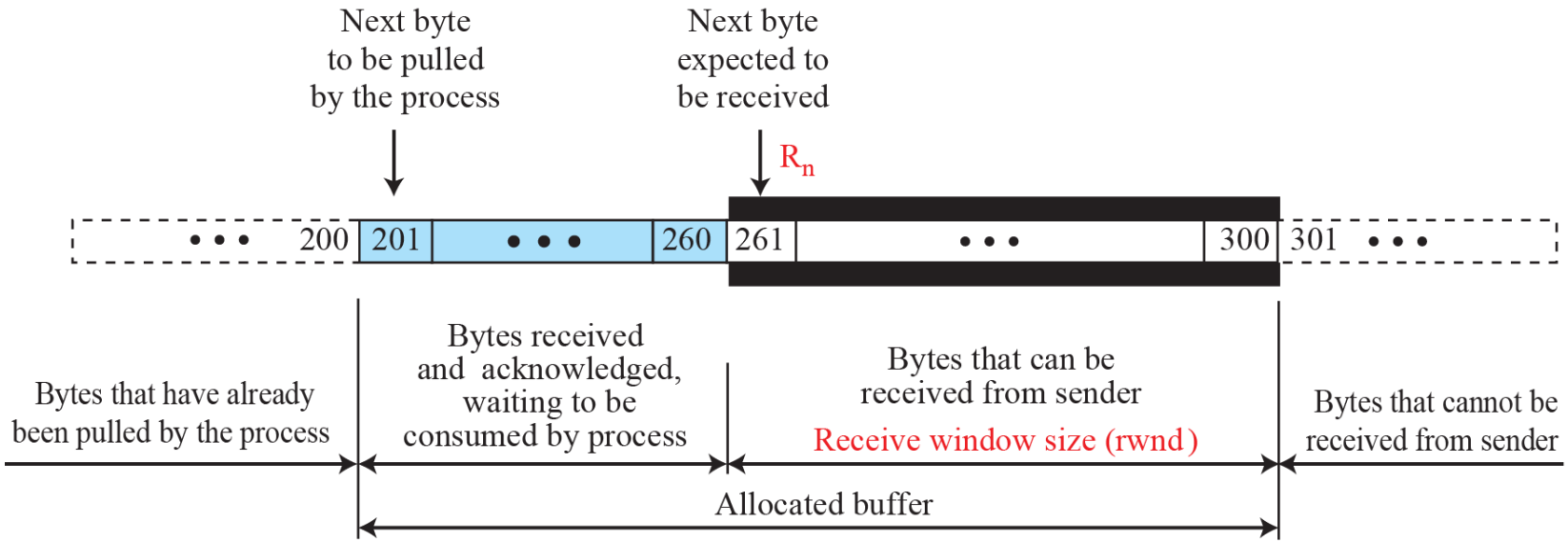
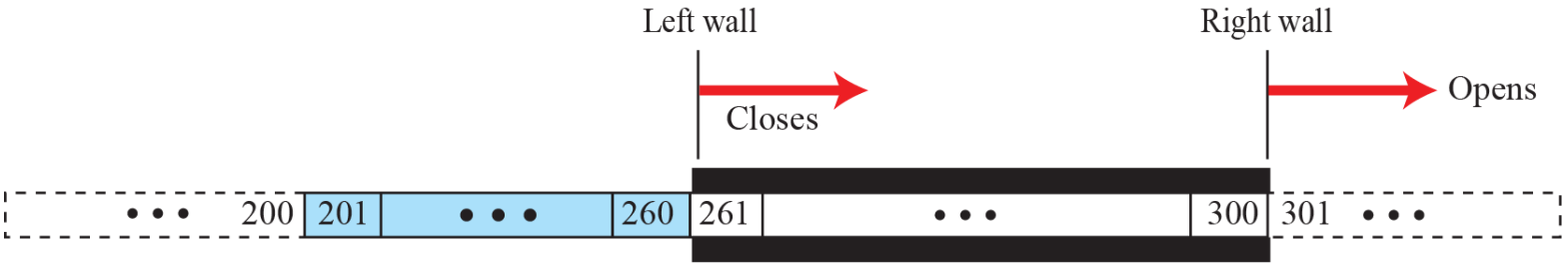
Figure 3.55: Receive window in TCP



a. Receive window and allocated buffer



b. Opening and closing of receive window

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3.4.7 Flow Control

As discussed before, flow control balances the rate a producer creates data with the rate a consumer can use the data. TCP separates flow control from error control. In this section we discuss flow control, ignoring error control. We assume that the logical channel between the sending and receiving TCP is error-free.

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3.4.7 (continued)

- **Opening and Closing Windows**
  - A Scenario
- **Shrinking of Windows**
  - Window Shutdown
- **Silly Window Syndrome**
  - Syndrome Created by the Sender
  - Syndrome Created by the Receiver

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Figure 3.56: Data flow and flow control feedbacks in TCP

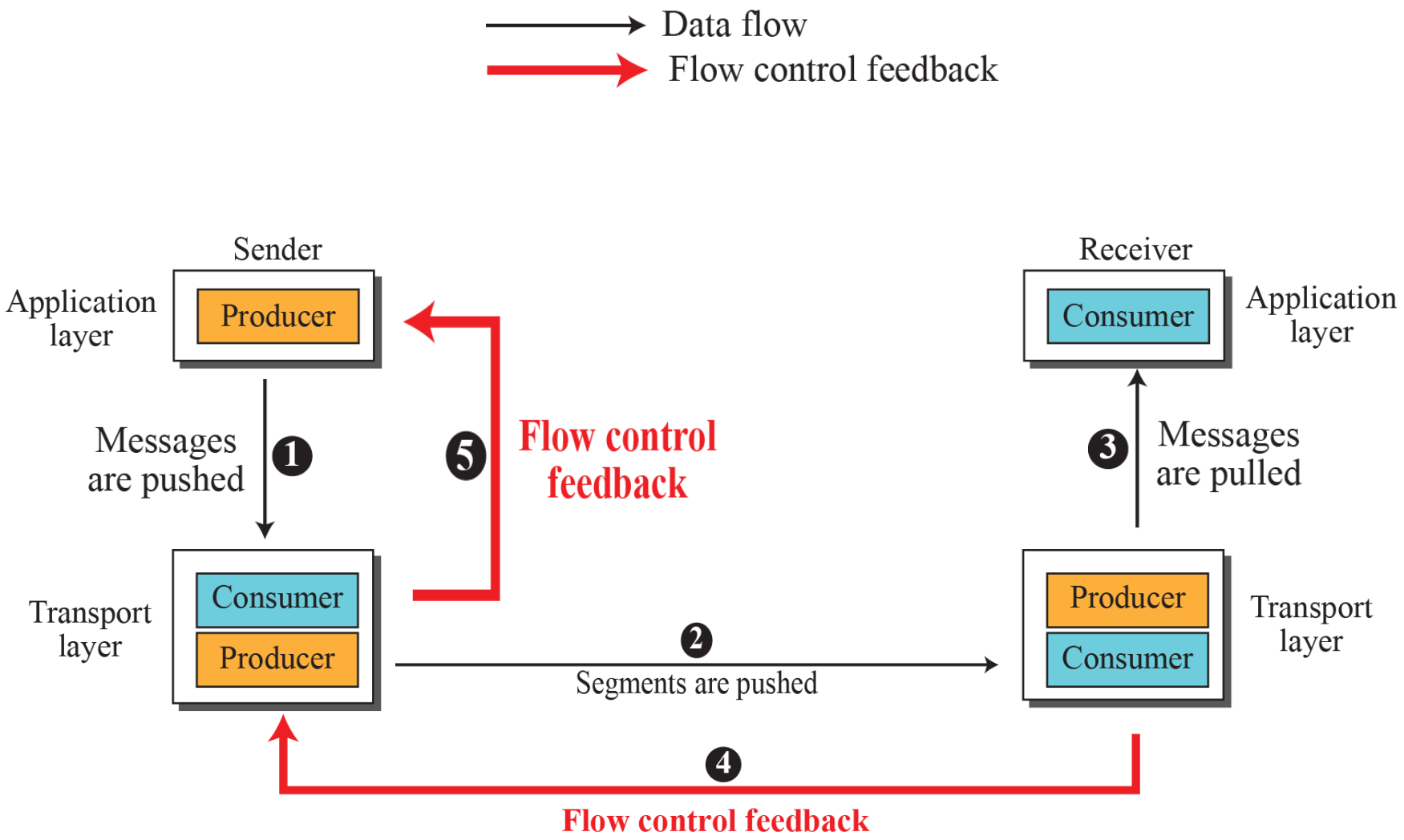

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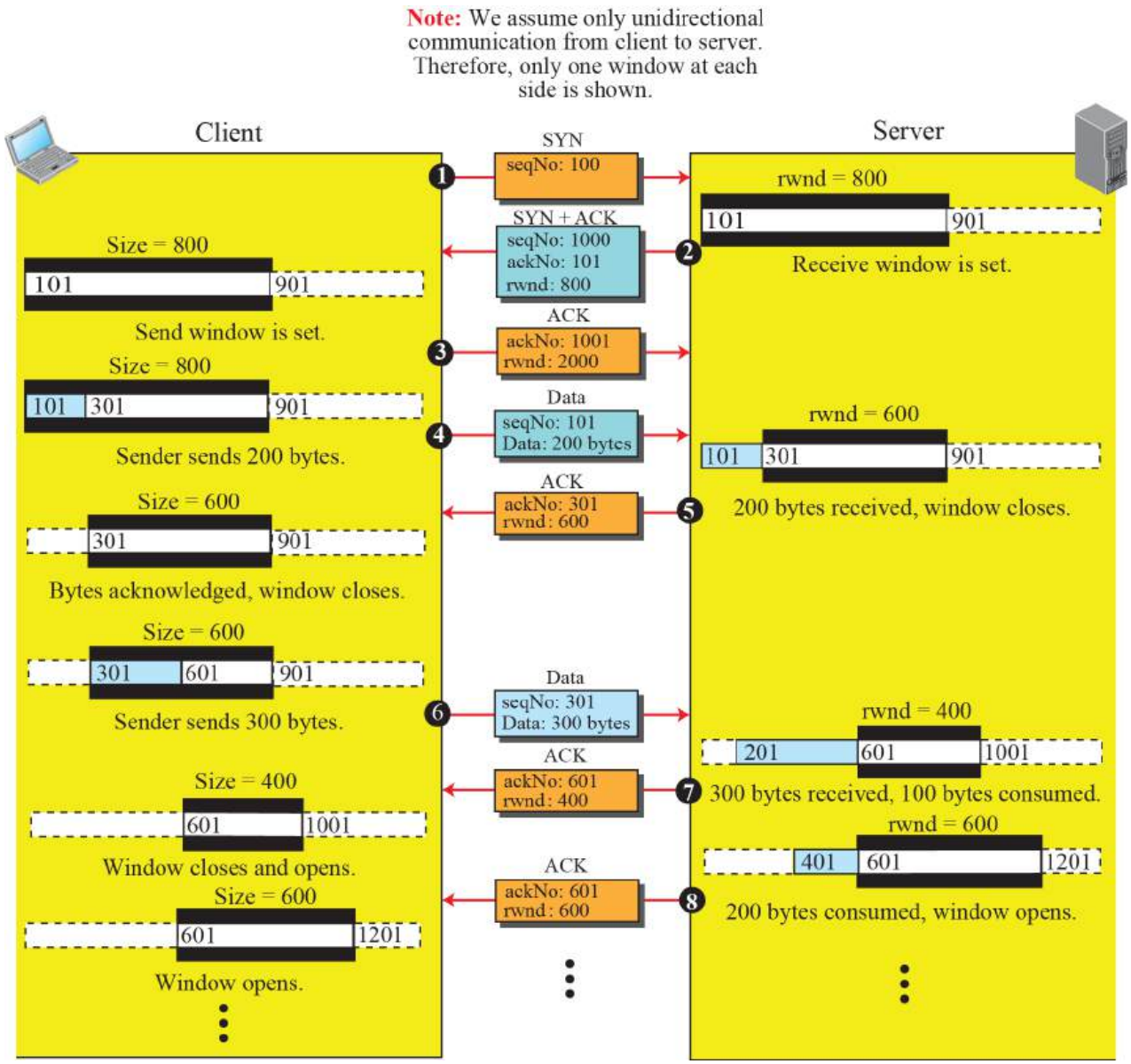
Figure 3.57: An example of flow control



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Example 3.18

Figure 3.58 shows the reason for this mandate.

Part a of the figure shows the values of the last acknowledgment and *rwnd*. Part b shows the situation in which the sender has sent bytes 206 to 214. Bytes 206 to 209 are acknowledged and purged. The new advertisement, however, defines the new value of *rwnd* as 4, in which $210 + 4 < 206 + 12$. When the send window shrinks, it creates a problem: byte 214, which has already been sent, is outside the window. The relation discussed before forces the receiver to maintain the right-hand wall of the window to be as shown in part a, because the receiver does not know which of the bytes 210 to 217 has already been sent. described above.

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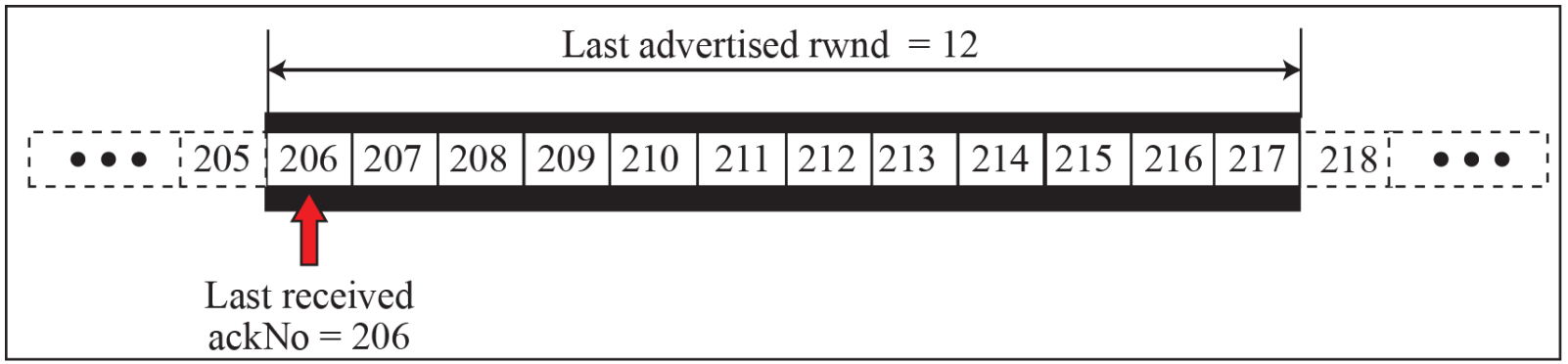
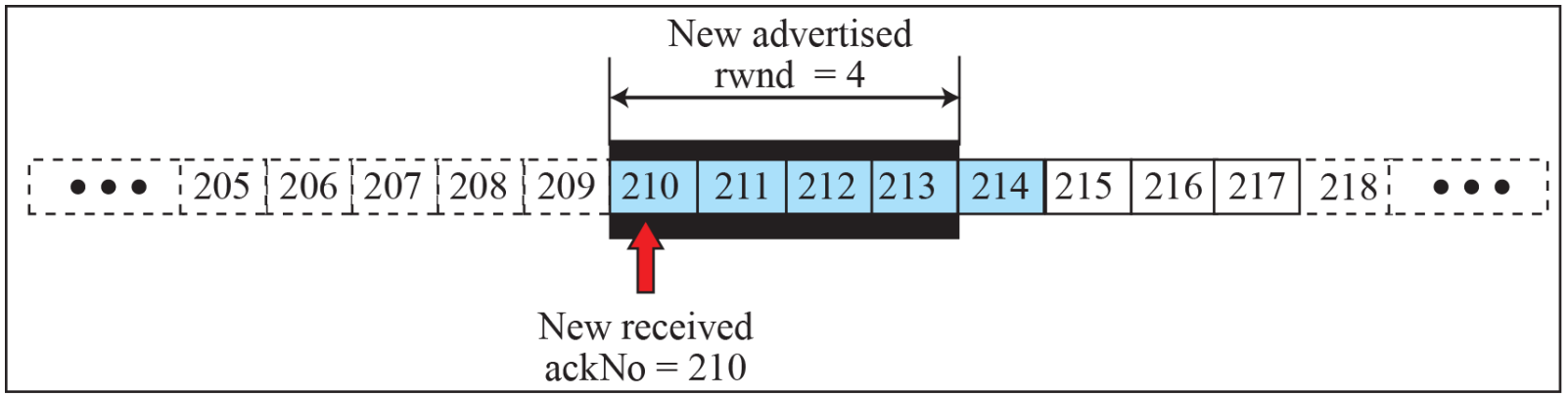
Figure 3.58: *Example 3.18*



a. The window after the last advertisement



b. The window after the new advertisement; window has shrunk

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3.4.8 Error Control

TCP is a reliable transport-layer protocol. This means that an application program that delivers a stream of data to TCP relies on TCP to deliver the entire stream to the application program on the other end in order, without error, and without any part lost or duplicated.

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3.4.8 (continued)

- **Checksum**
- **Acknowledgment**
  - Cumulative Acknowledgment (ACK)
  - Selective Acknowledgment (SACK)
- **Generating Acknowledgments**
- **Retransmission**
  - Retransmission after RTO
  - Retransmission after Three Duplicate ACK
- **Out-of-Order Segments**

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3.4.8 (continued)

- **FSMs for Data Transfer in TCP**
  - Sender-Side FSM
  - Receiver-Side FSM
- **Some Scenarios**
  - Normal Operation
  - Lost Segment
  - Fast Retransmission
  - Delayed Segment
  - Duplicate Segment
  - Automatically Corrected Lost ACK
  - Correction by Resending a Segment
  - Deadlock Created by Lost Acknowledgment

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Figure 3.59: *Simplified FSM for the TCP sender side*

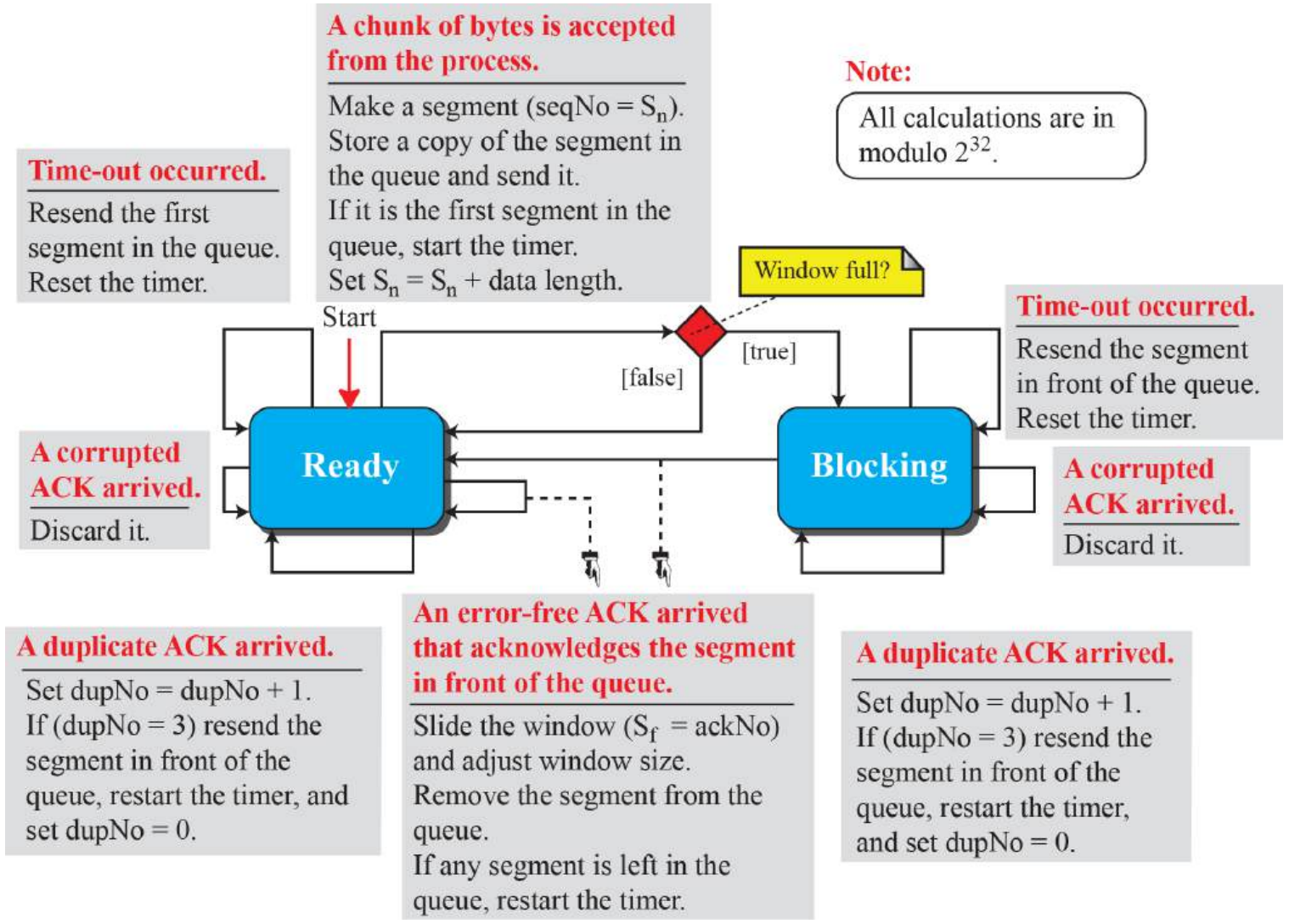

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Figure 3.60: Simplified FSM for the TCP receiver side

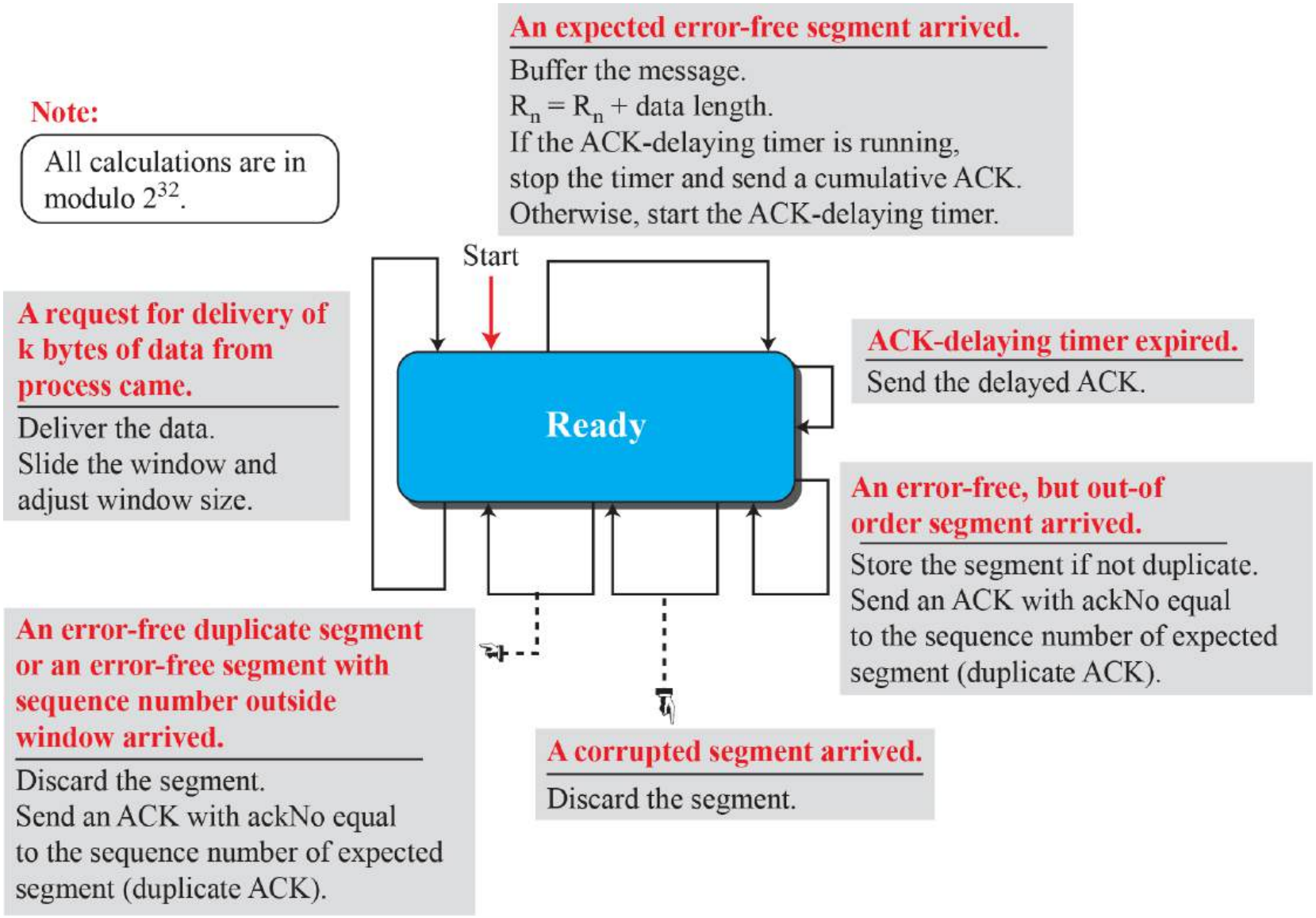

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Figure 3.61: Normal operation

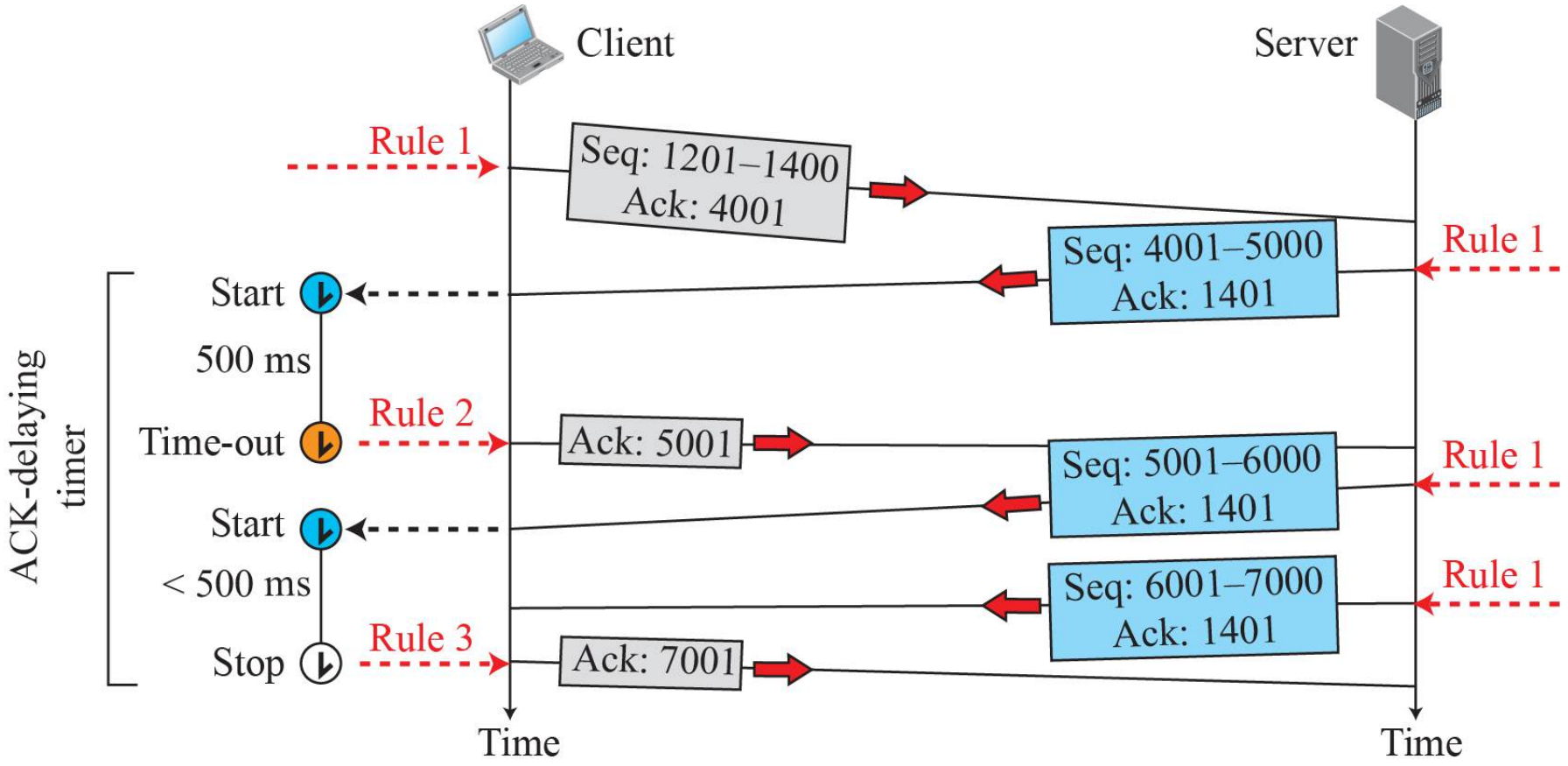

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Figure 3.62: Lost segment

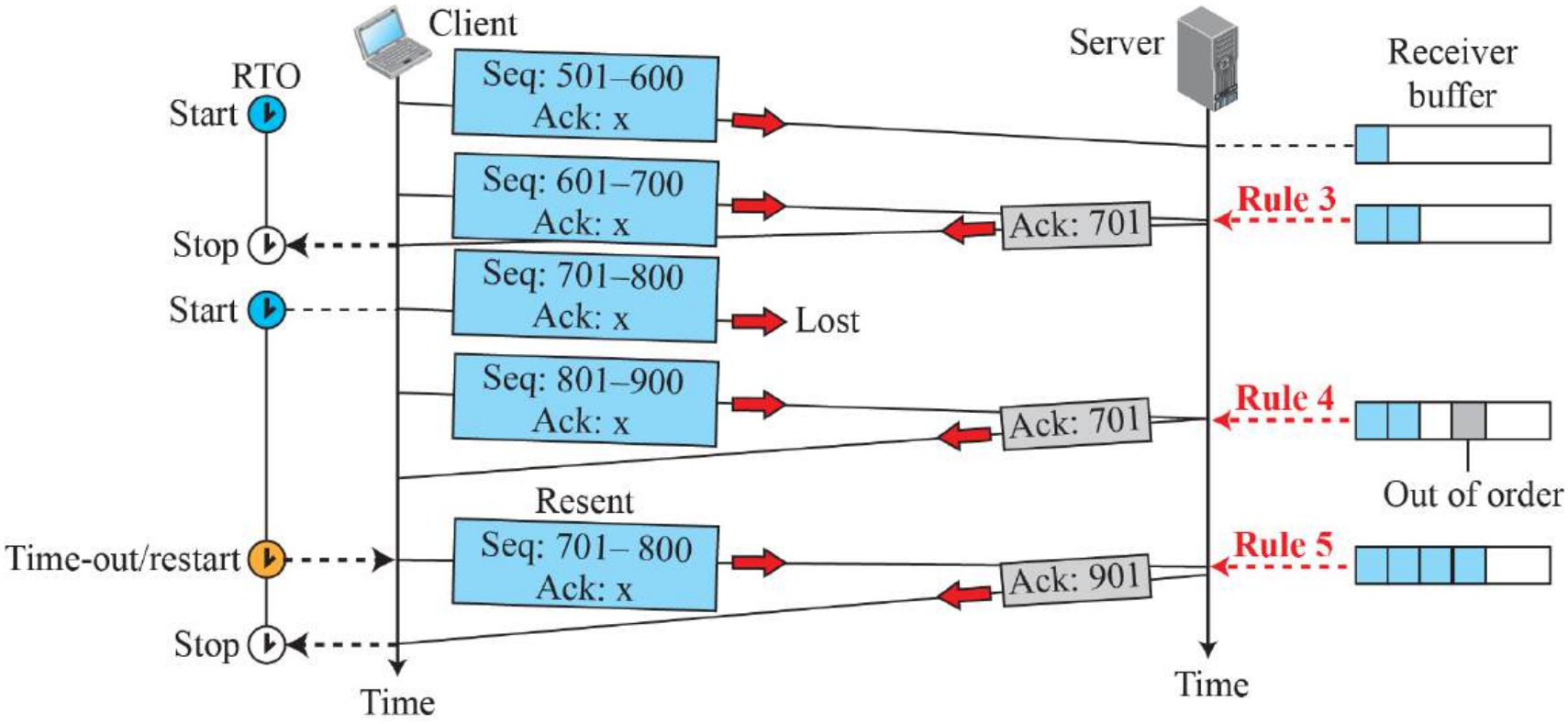

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Figure 3.63: Fast retransmission

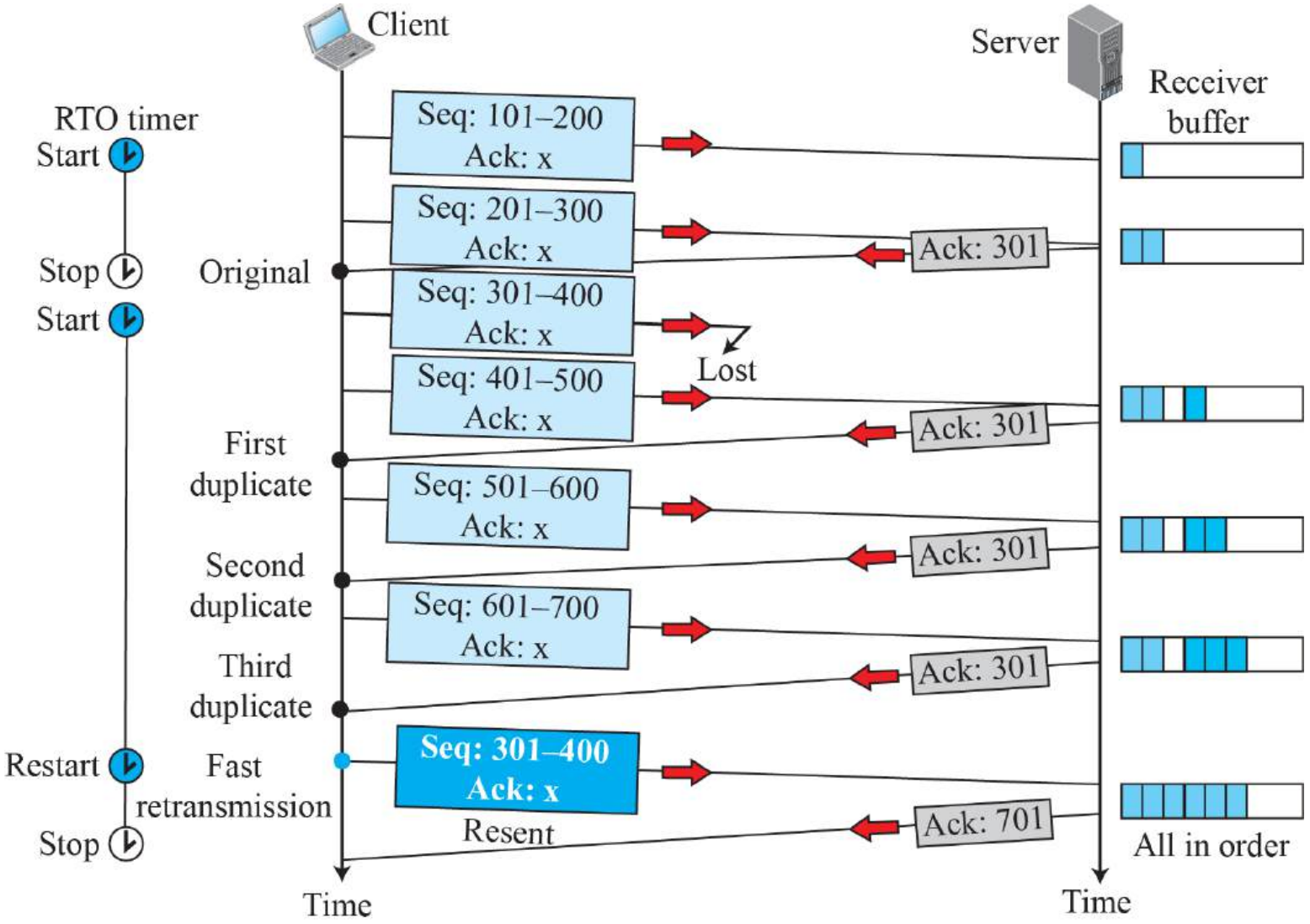

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Figure 3.64: Lost acknowledgment

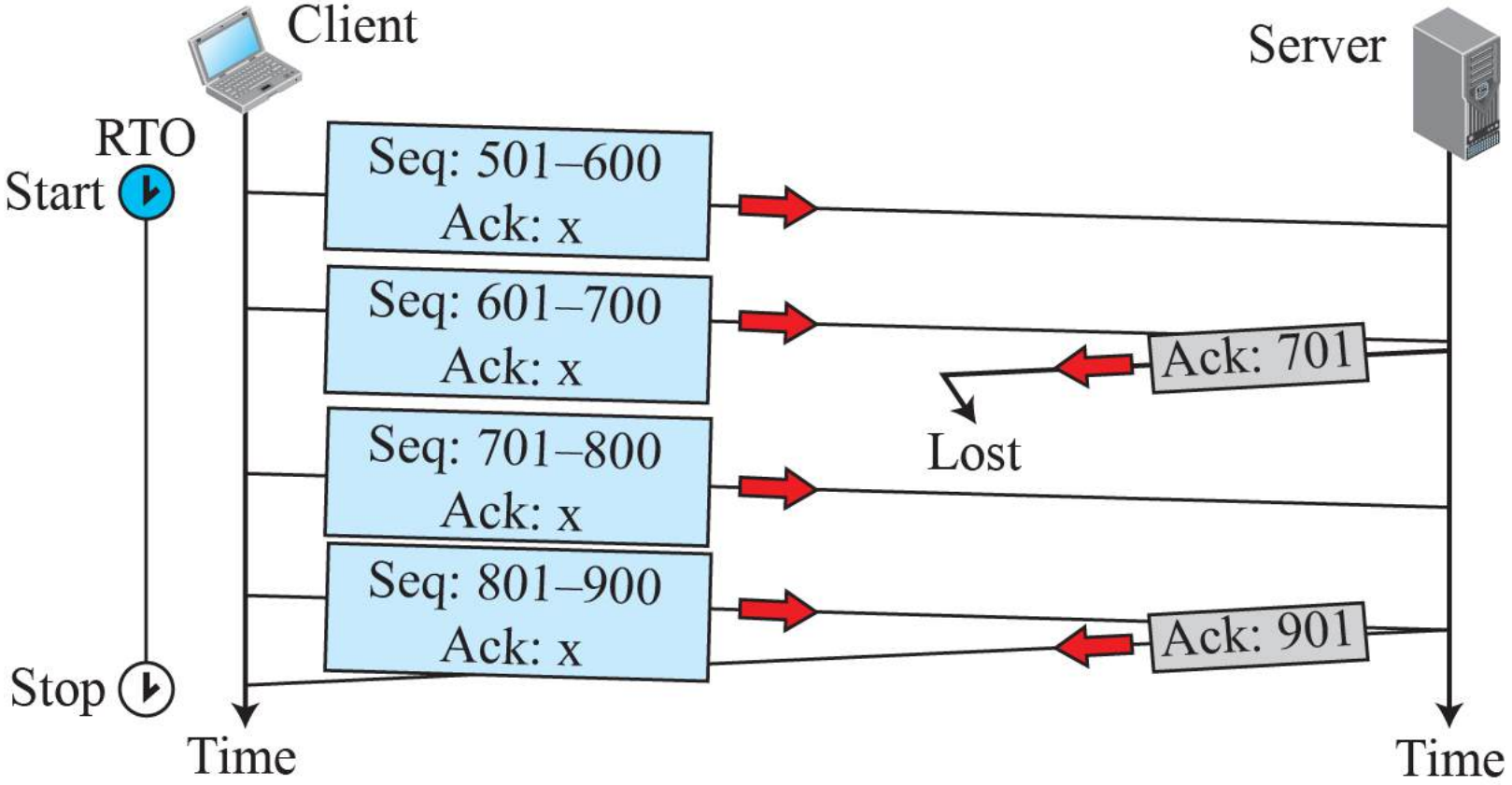

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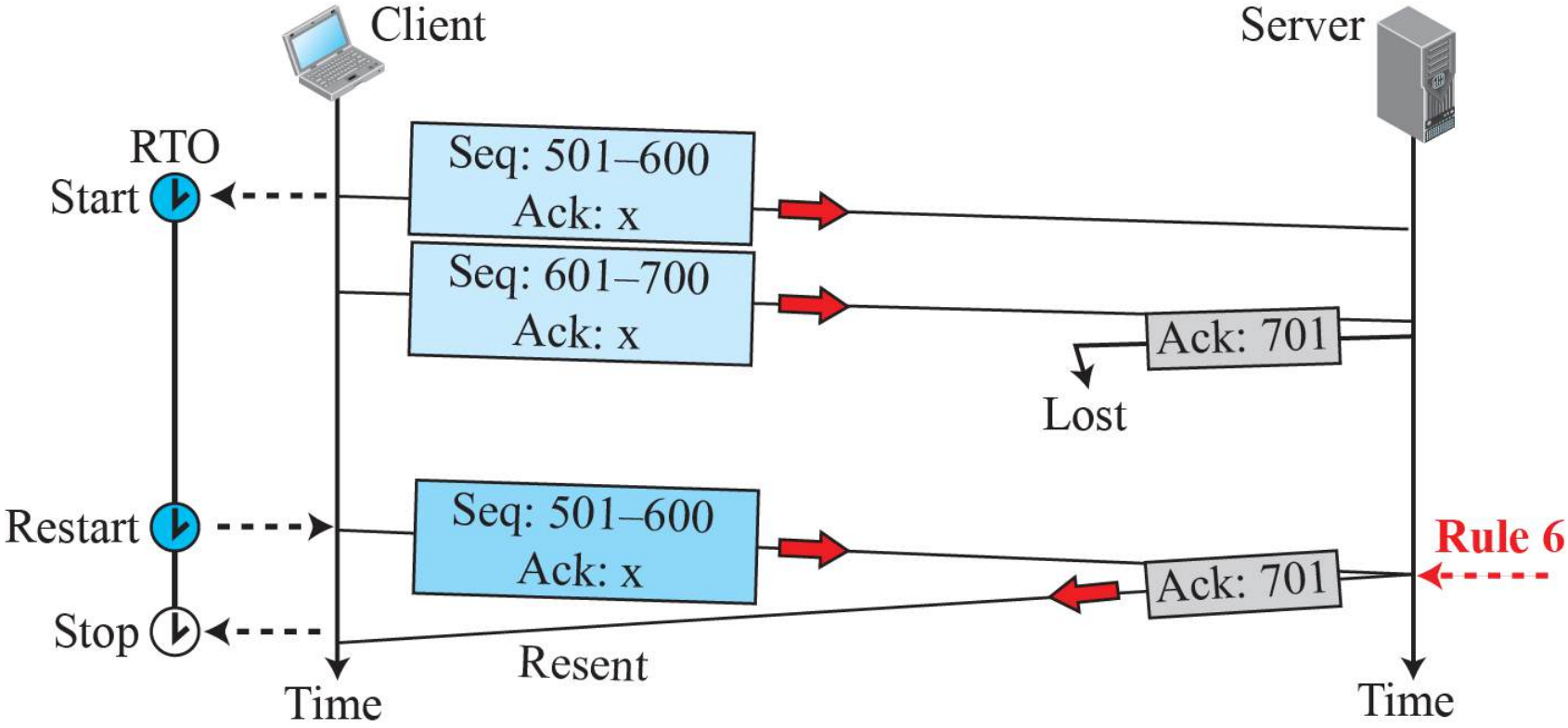
Figure 3.65: Lost acknowledgment corrected by resending a segment



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3.4.9 TCP Congestion Control

TCP uses different policies to handle the congestion in the network. We describe these policies in this section.

- **Congestion Window**
- **Congestion Detection**
- **Congestion Policies**
  - Slow Start: Exponential Increase
  - Congestion Avoidance: Additive Increase

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3.4.9 (continued)

- **Policy Transition**
  - Taho TCP
  - Reno TCP
  - NewReno TCP
- **Additive Increase, Multiplicative Decrease**
- **TCP Throughput**

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Figure 3.66: *Slow start, exponential increase*

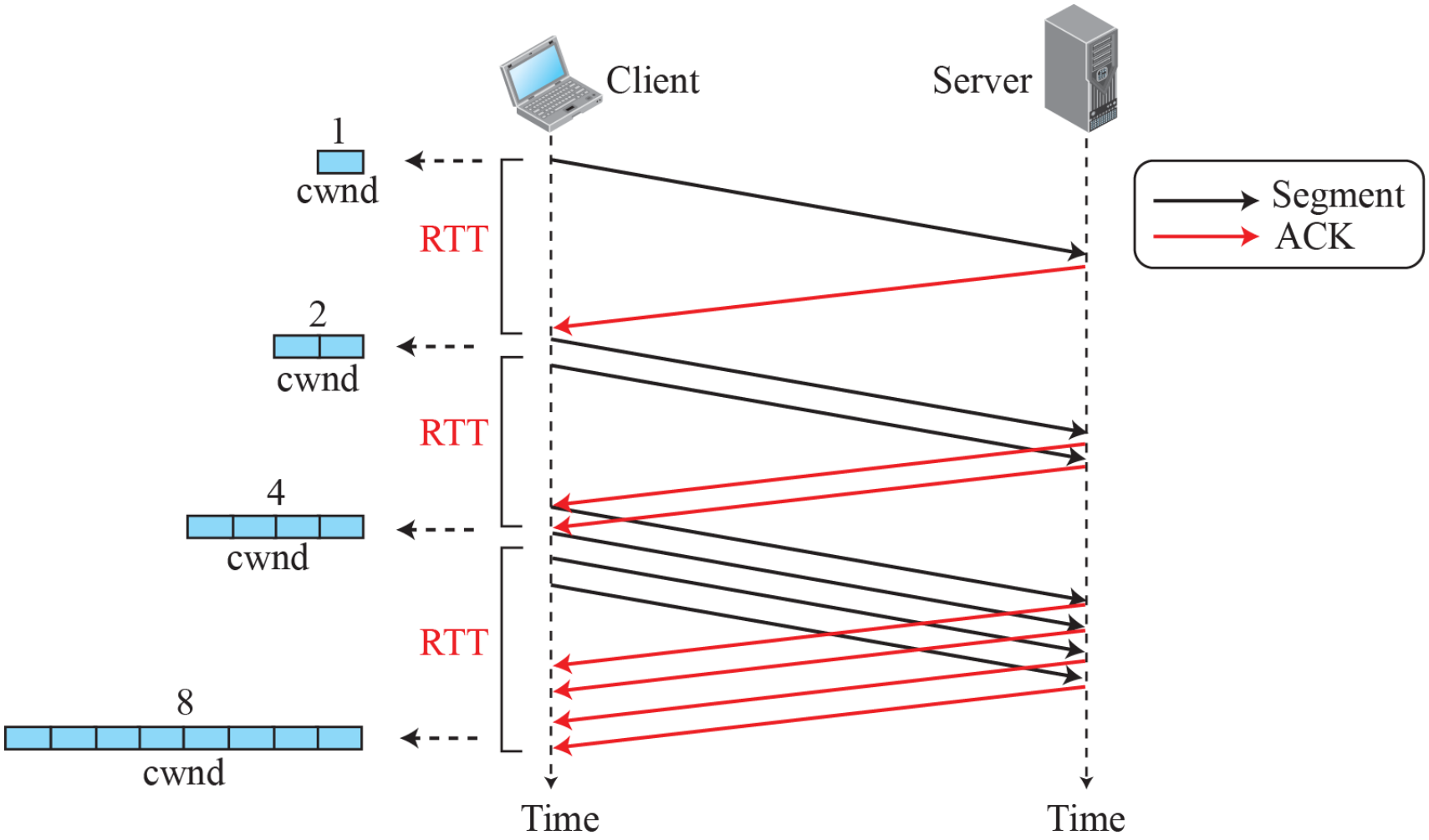

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Figure 3.67: Congestion avoidance, additive increase

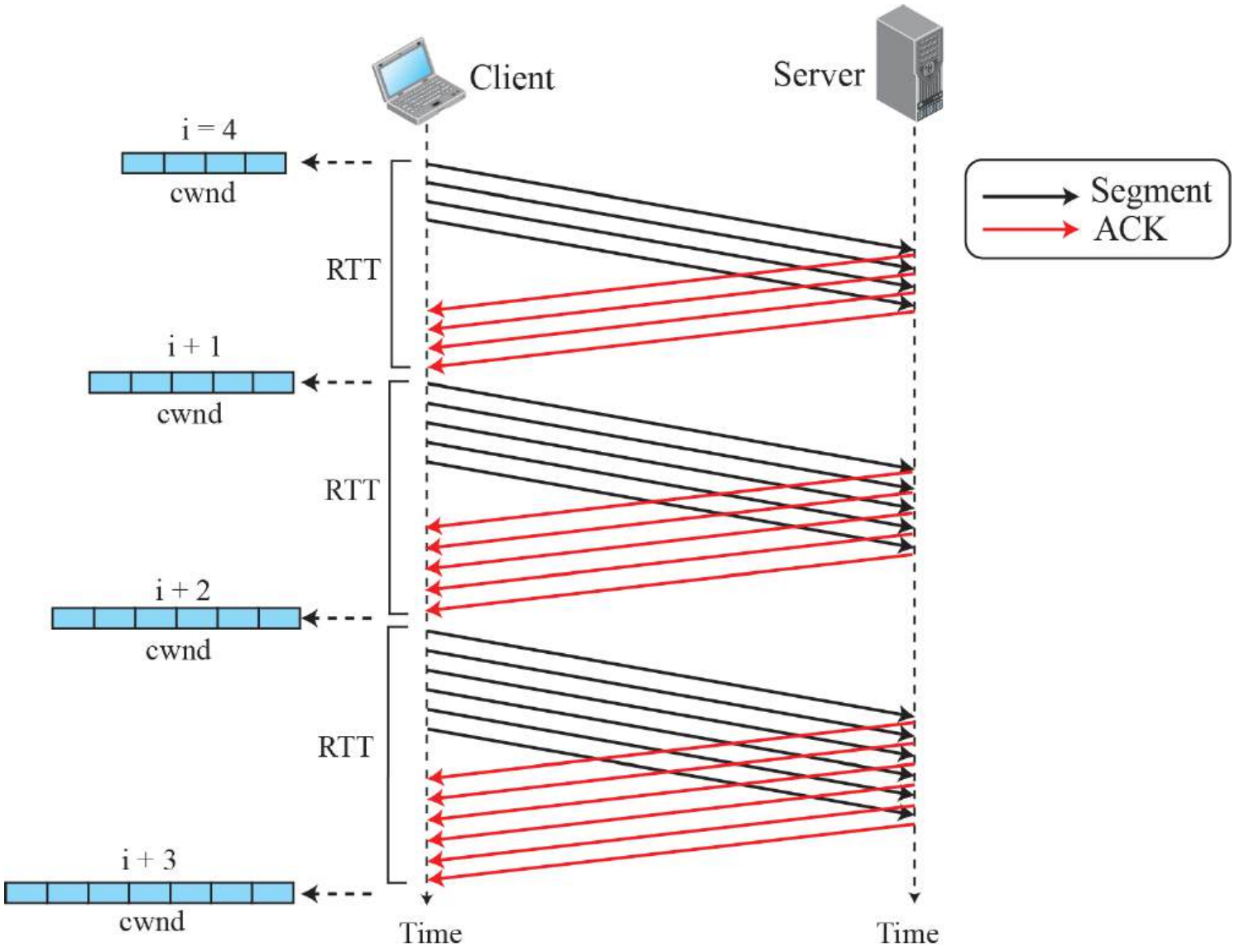

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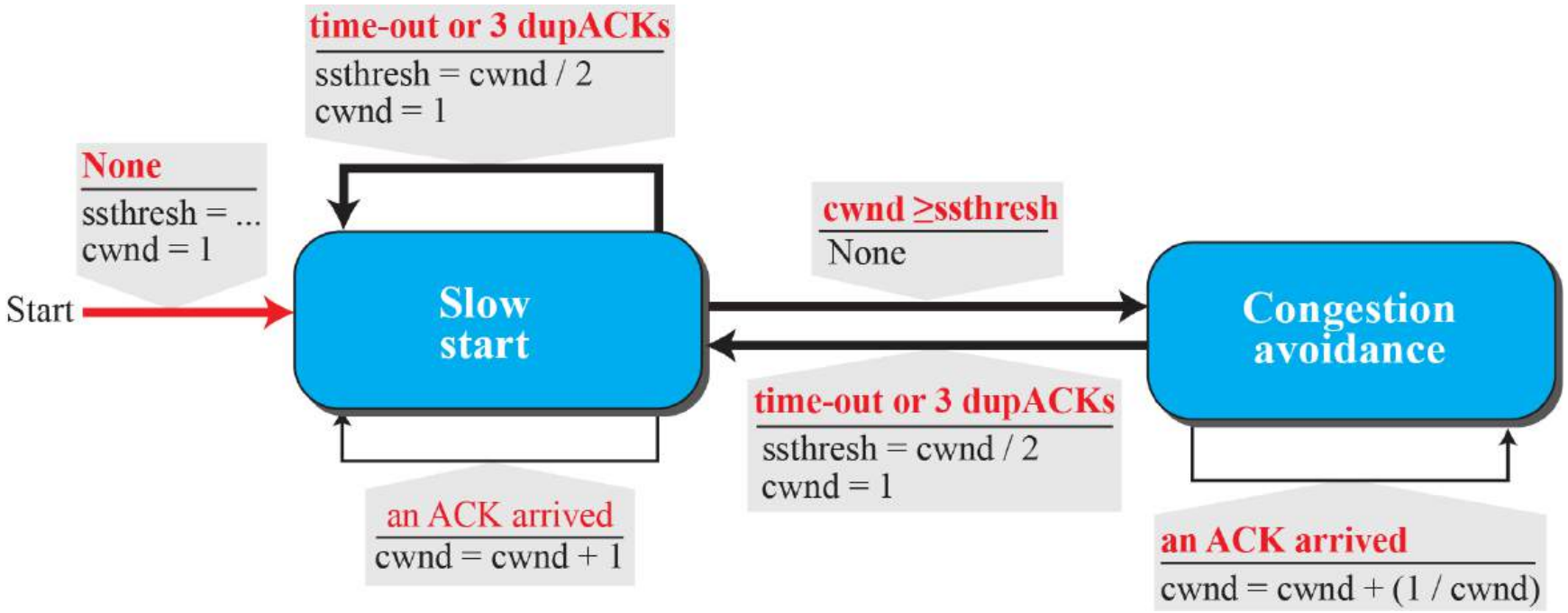
Figure 3.68: FSM for Taho TCP



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Example 3.19

Figure 3.69 shows an example of congestion control in a Taho TCP. TCP starts data transfer and sets the *ssthresh* variable to an ambitious value of 16 MSS. TCP begins at the slow-start (SS) state with the *cwnd* = 1. The congestion window grows exponentially, but a time-out occurs after the third RTT (before reaching the threshold). TCP assumes that there is congestion in the network. It immediately sets the new *ssthresh* = 4 MSS (half of the current cwnd, which is 8) and begins a new slow start (SA) state with *cwnd* = 1 MSS. The congestion grows exponentially until it reaches the newly set threshold. TCP now moves to the congestion avoidance (CA) state and the congestion window grows additively until it reaches *cwnd* = 12 MSS.

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Example 3.19 (continued)

At this moment, three duplicate ACKs arrive, another indication of the congestion in the network. TCP again halves the value of ssthresh to 6 MSS and begins a new slow-start (SS) state. The exponential growth of the cwnd continues. After RTT 15, the size of cwnd is 4 MSS. After sending four segments and receiving only two ACKs, the size of the window reaches the *ssthresh* (6) and the TCP moves to the congestion avoidance state. The data transfer now continues in the congestion avoidance (CA) state until the connection is terminated after RTT 20.

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Figure 3.69: Example of Taho TCP

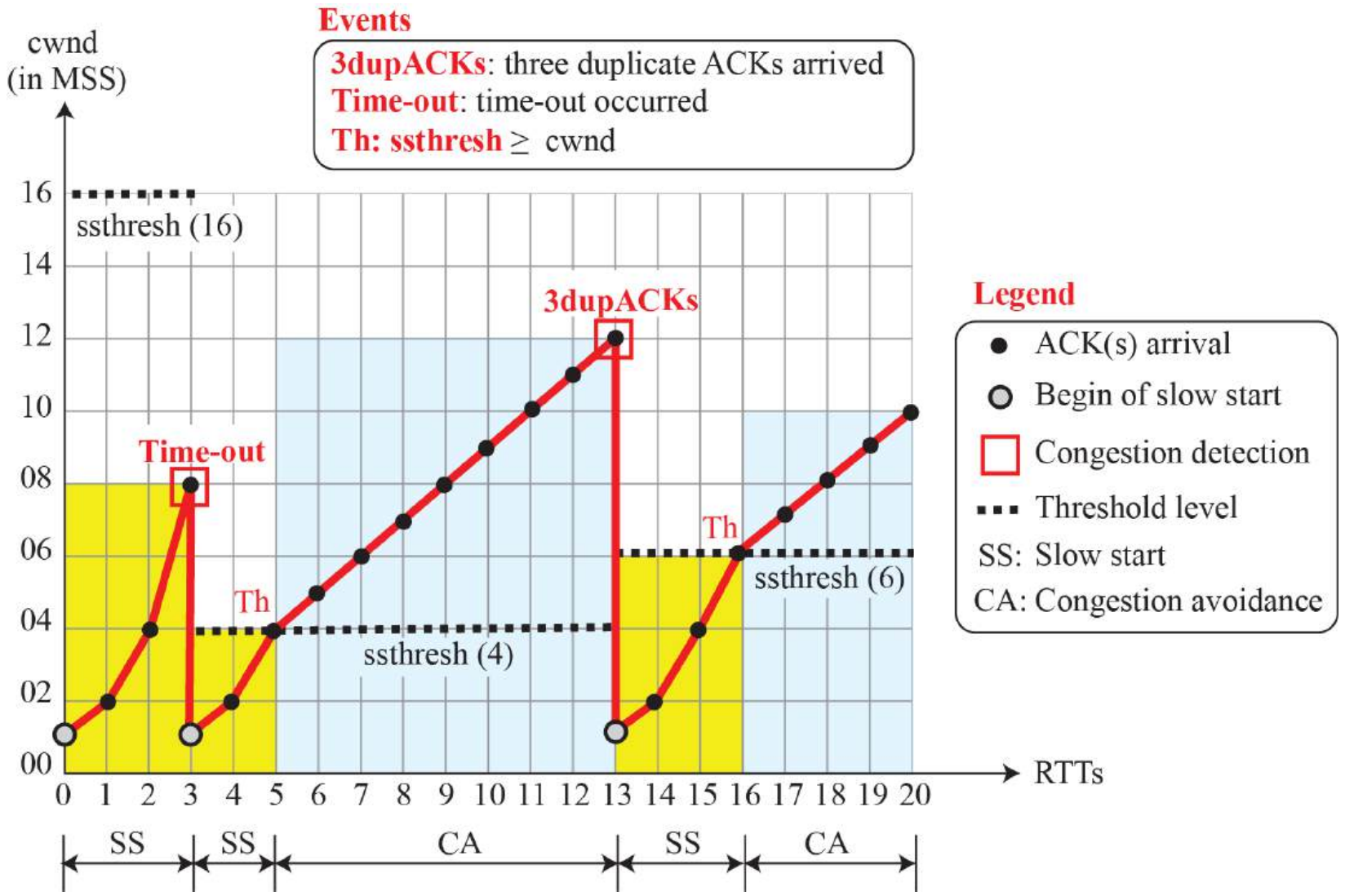

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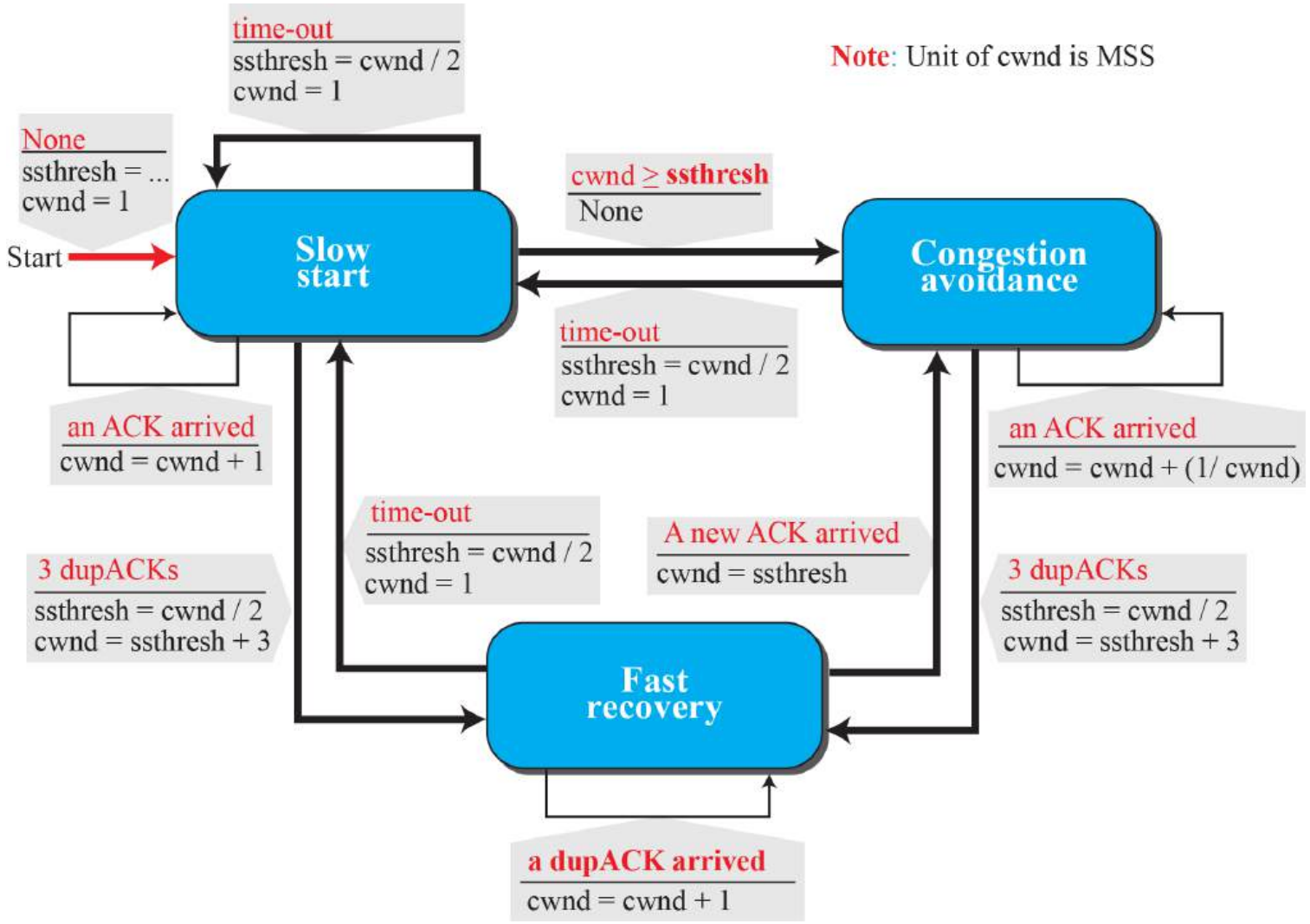
Figure 3.70: FSM for Reno TCP



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Example 3.20

Figure 3.71 shows the same situation as Figure 3.69, but in Reno TCP. The changes in the congestion window are the same until RTT 13 when three duplicate ACKs arrive. At this moment, Reno TCP drops the ssthresh to 6 MSS, but it sets the cwnd to a much higher value (*ssthresh* + 3 = 9 MSS) instead of 1 MSS. It now moves to the fast recovery state. We assume that two more duplicate ACKs arrive until RTT 15, where *cwnd*grows exponentially. In this moment, a new ACK (not duplicate) arrives that announces the receipt of the lost segment. It now moves to the congestion avoidance state, but first deflates the congestion window to 6 MSS as though ignoring the whole fast-recovery state and moving back to the previous track.

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Figure 3.71: Example of a Reno TCP

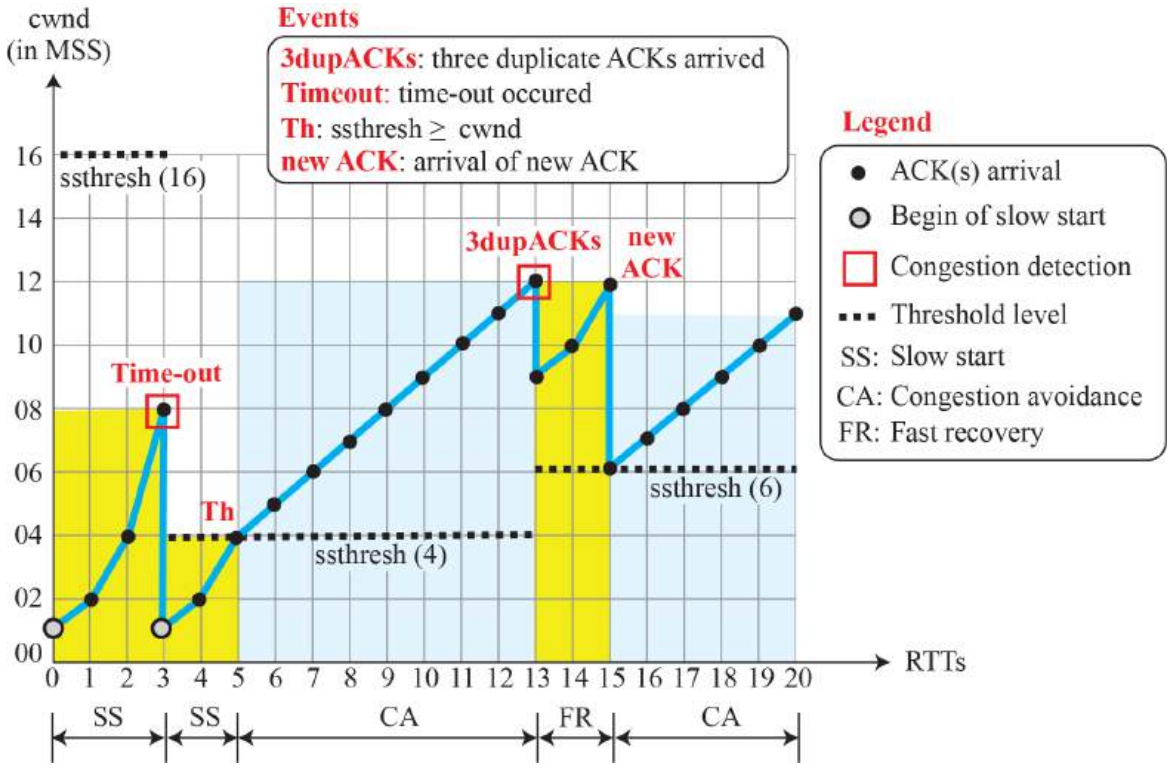

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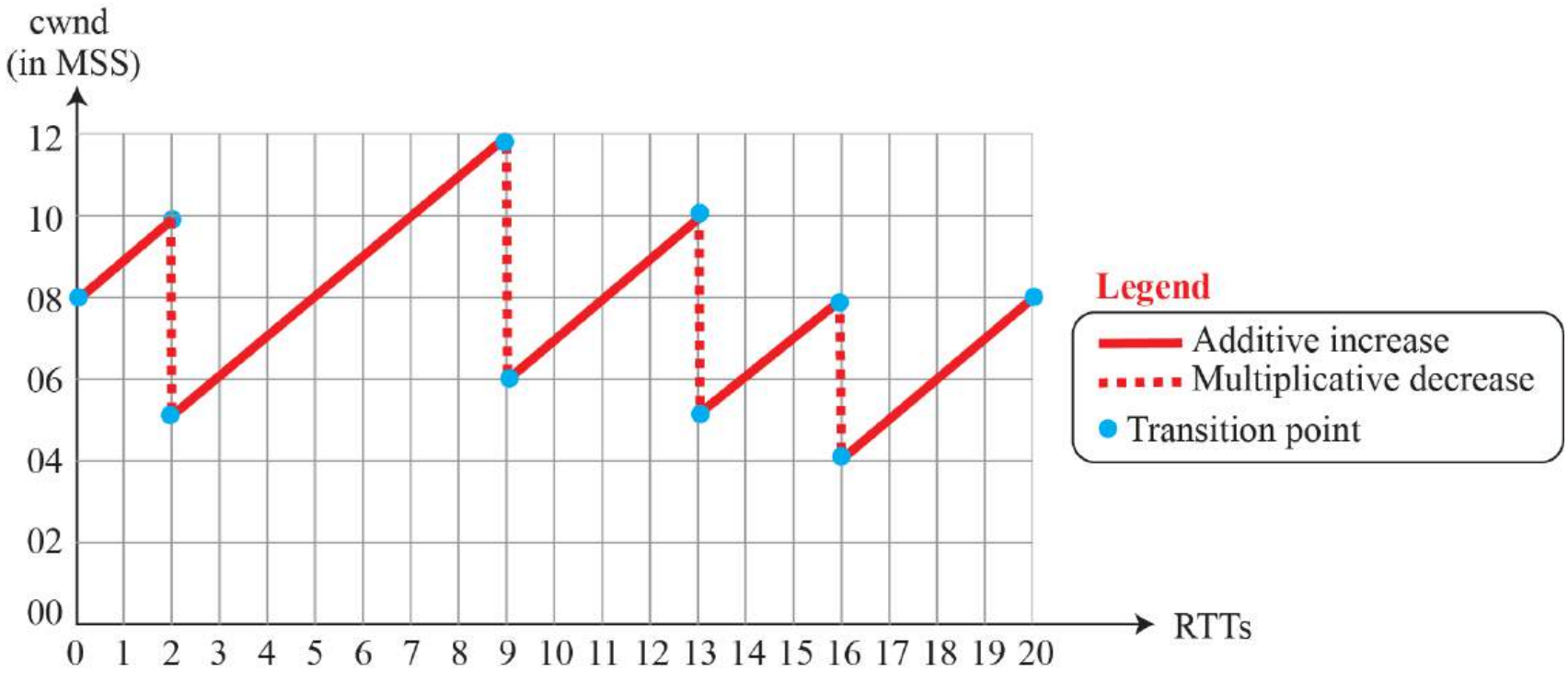
Figure 3.72: Additive increase, multiplicative decrease (AIMD)



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Example 3.21

If MSS = 10 KB (kilobytes) and RTT = 100 ms in Figure 3.72, we can calculate the throughput as shown below.

$$W_{max} = (10 + 12 + 10 + 8 + 8) / 5 = 9.6 \text{ MSS}$$

$$\text{Throughput} = (0.75\ W_{max} / \text{RTT}) = 0.75 \times 960 \text{ kbps} / 100 \text{ ms} = 7.2 \text{ Mbps}$$

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3.4.10 TCP Timers

To perform their operations smoothly, most TCP implementations use at least four timers.

- **Retransmission Timer**
  - Round-Trip Time (RTT)
  - Karn's Algorithm
  - Exponential Backoff
- **Persistence Timer**
- **Keepalive Timer**
- **TIME-WAIT Timer**

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Example 3.22

Let us give a hypothetical example. Figure 3.73 shows part of a connection. The figure shows the connection establishment and part of the data transfer phases.

1. When the SYN segment is sent, there is no value for RTTM, RTTS, or RTTD. The value of RTO is set to 6.00 seconds. The following shows the value of these variables at this moment:

RTO = 6

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Example 3.22 (continued)

2. When the SYN+ACK segment arrives, RTTM is measured and is equal to 1.5 seconds. The following

shows the values of these variables:

$$RTT_M = 1.5$$
$$RTT_S = 1.5$$
$$RTT_D = (1.5)/2 = 0.75$$
$$RTO = 1.5 + 4 \times 0.75 = 4.5$$

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Example 3.22 (continued)

3. When the first data segment is sent, a new RTT measurement starts. Note that the sender does not start an RTT measurement when it sends the ACK segment, because it does not consume a sequence number and there is no time-out. No RTT measurement starts for the second data segment because a measurement is already in progress.

$$RTT_M = 2.5$$

$$RTT_S = (7/8) \times (1.5) + (1/8) \times (2.5) = 1.625$$

$$RTT_D = (3/4) \times (0.75) + (1/4) \times |1.625 - 2.5| = 0.78$$

$$RTO = 1.625 + 4 \times (0.78) = 4.74$$

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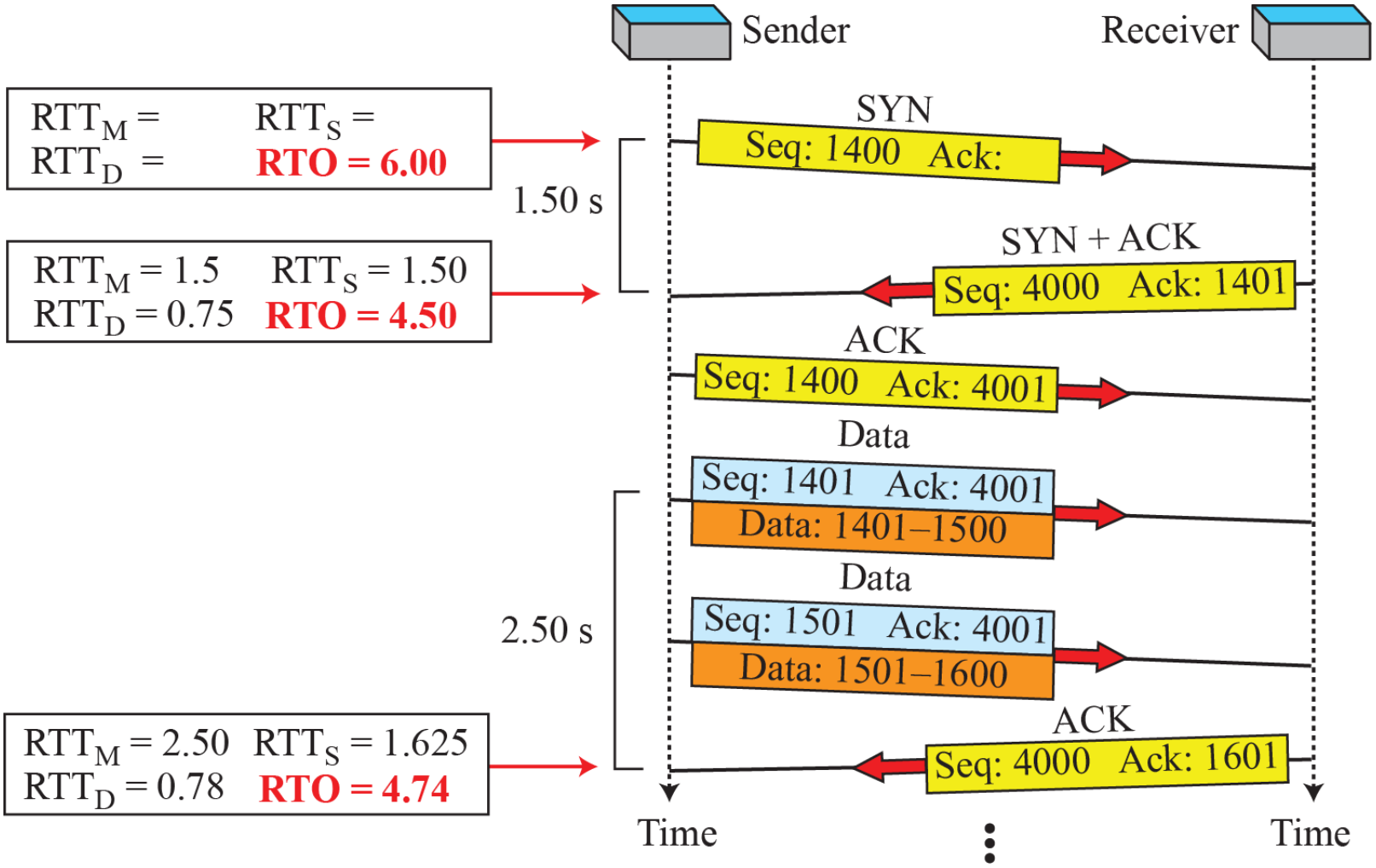
Figure 3.73: Example 3.22



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Example 3.23

Figure 3.74 is a continuation of the previous example. There is retransmission and Karn's algorithm is applied. The first segment in the figure is sent, but lost. The RTO timer expires after 4.74 seconds. The segment is retransmitted and the timer is set to 9.48, twice the previous value of RTO. This time an ACK is received before the time-out. We wait until we send a new segment and receive the ACK for it before recalculating the RTO (Karn's algorithm).

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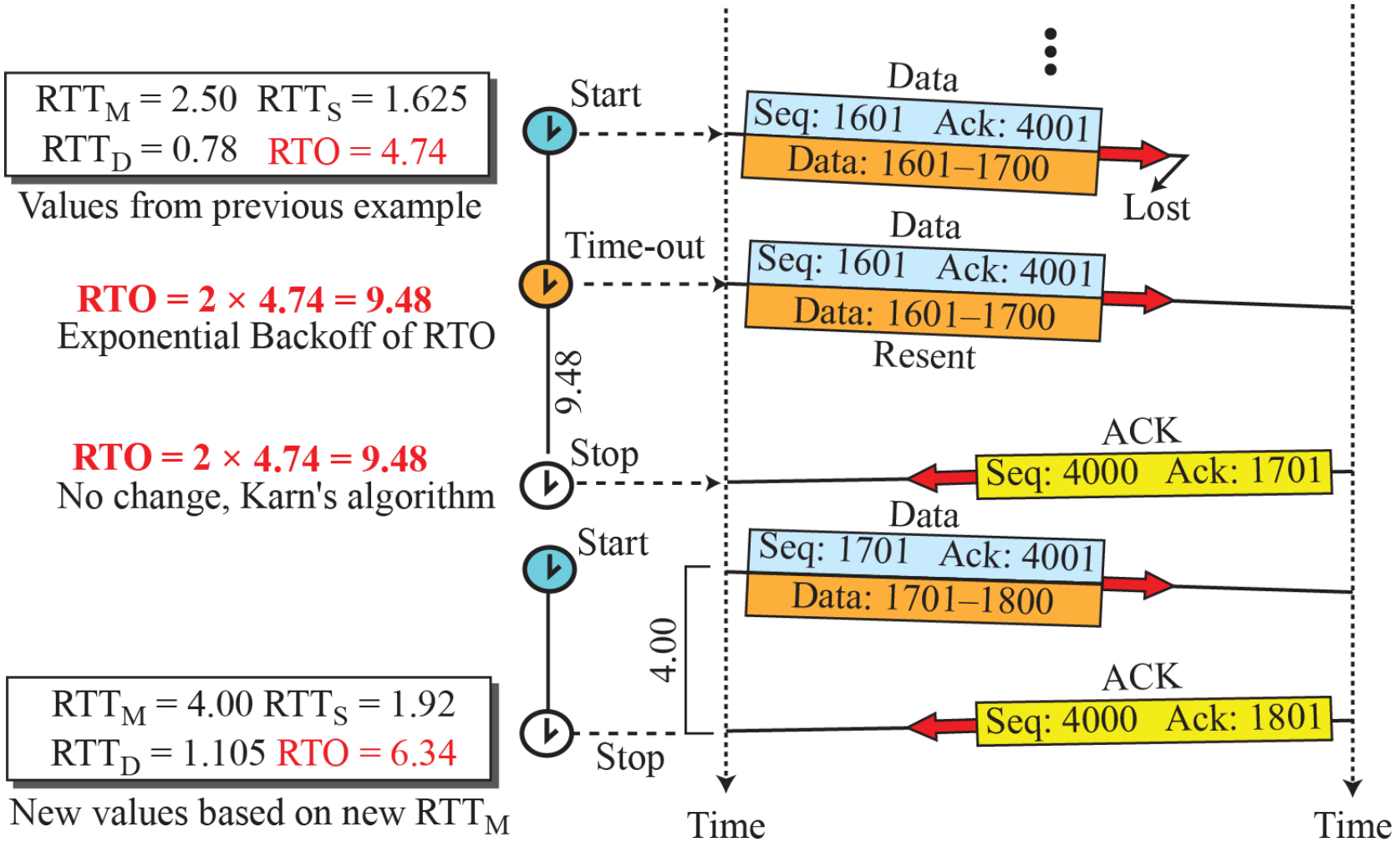
Figure 3.74: Example 3.23



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3.4.11 Options

The TCP header can have up to 40 bytes of optional information. Options convey additional information to the destination or align other options. These option are included on the book website for further reference.

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Chapter 3: Summary

- *The main duty of a transport-layer protocol is to provide process-to-process communication. To define the processes, we need port numbers. The client program defines itself with an ephemeral port number. The server defines itself with a well-known port number. To send a message from one process to another, the transport-layer protocol encapsulates and decapsulates messages. Flow control balances the exchange of data items between a producer and a consumer. A transport-layer protocol can provide two types of services: connectionless and connection-oriented. In a connectionless service, the sender sends packets to the receiver without any connection establishment. In a connection-oriented service, the client and the server first need to establish a connection between themselves.*

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Chapter 3: Summary (continued)

- *We have discussed several common transport-layer protocols in this chapter. The Stop-and-Wait protocol provides both flow and error control, but is inefficient. The Go-Back-N protocol is the more efficient version of the Stop-and-Wait protocol and takes advantage of pipelining. The Selective-Repeat protocol, a modification of the Go-Back-N protocol, is better suited to handle packet loss. All of these protocols can be implemented bidirectionally using piggybacking.*

- *UDP is a transport protocol that creates a process-to-process communication. UDP is a (mostly) unreliable and connectionless protocol that requires little overhead and offers fast delivery. The UDP packet is called a user datagram.*

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Chapter 3: Summary (continued)

- *Transmission Control Protocol (TCP) is another transport-layer protocol in the TCP/IP protocol suite. TCP provides process-to-process, full-duplex, and connection-oriented service. The unit of data transfer between two devices using TCP software is called a segment. A TCP connection consists of three phases: connection establishment, data transfer, and connection termination. TCP software is normally implemented as a finite state machine (FSM).*

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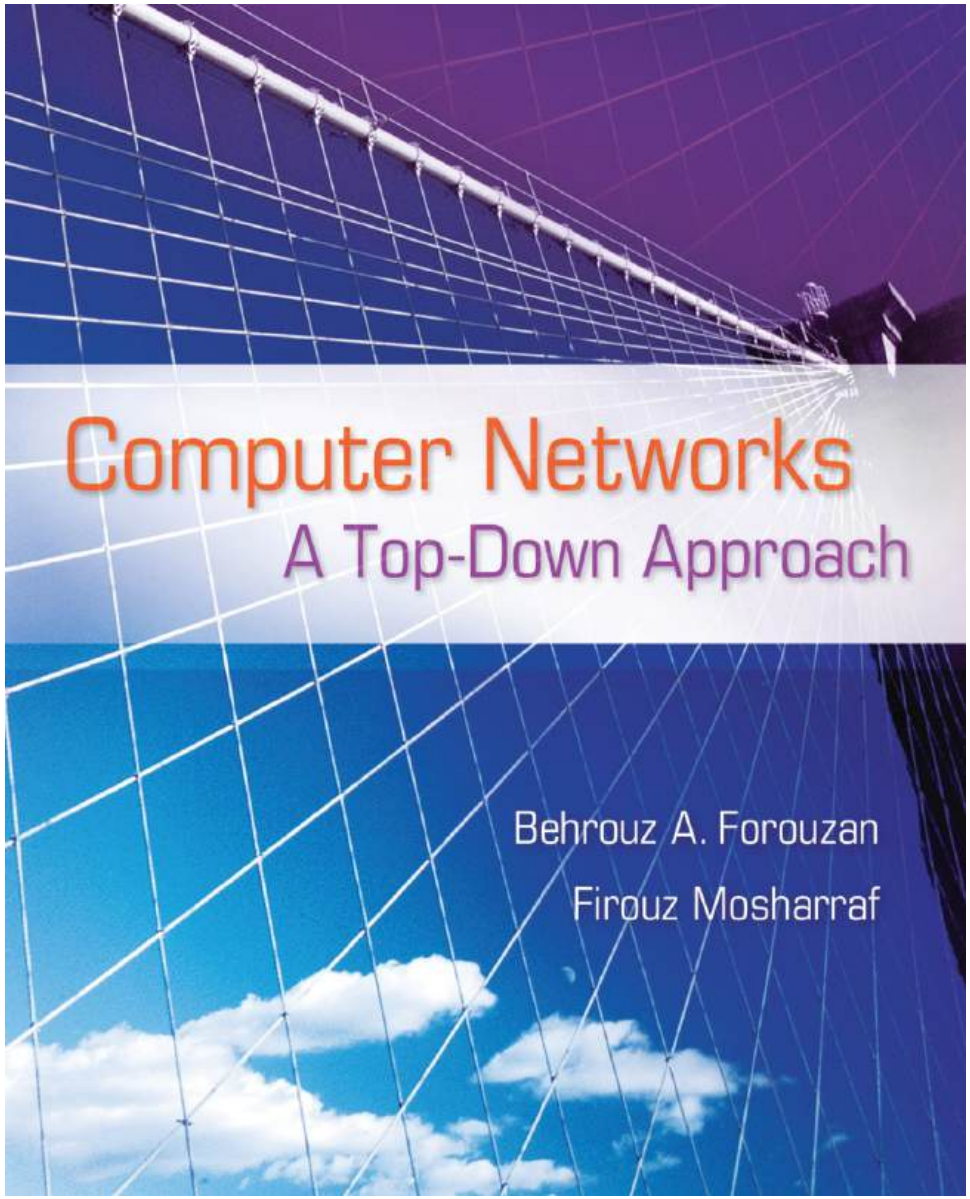
Chapter 4

Network Layer



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Chapter 4: Outline

4.1 INTRODUCTION

4.2 NETWORK-LAYER PROTOCOLS

4.3 UNICAST ROUTING

4.4 MULTICAST ROUTING

4.5 NEXT GENERATION IP

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Chapter 4: Objective

- *We first discuss services that can be provided at the network layer: packetizing, routing, and forwarding.*
- *We then discuss the network layer at the TCP/IP suite: IPv4 and ICMPv4. We also discuss IPv4 addressing and related issues.*
- *We then concentrate on the unicast routing and unicast routing protocols.*
- *We then move to multicasting and multicast routing protocols protocol.*
- *We finally discuss the new generation of network-layer protocols, IPv6 and ICMPv6.*

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4-1 INTRODUCTION

Figure 4.1 shows the communication between Alice and Bob at the network layer. This is the same scenario we used in Chapters 2 and 3 to show the communication at the application and the transport layers, respectively.

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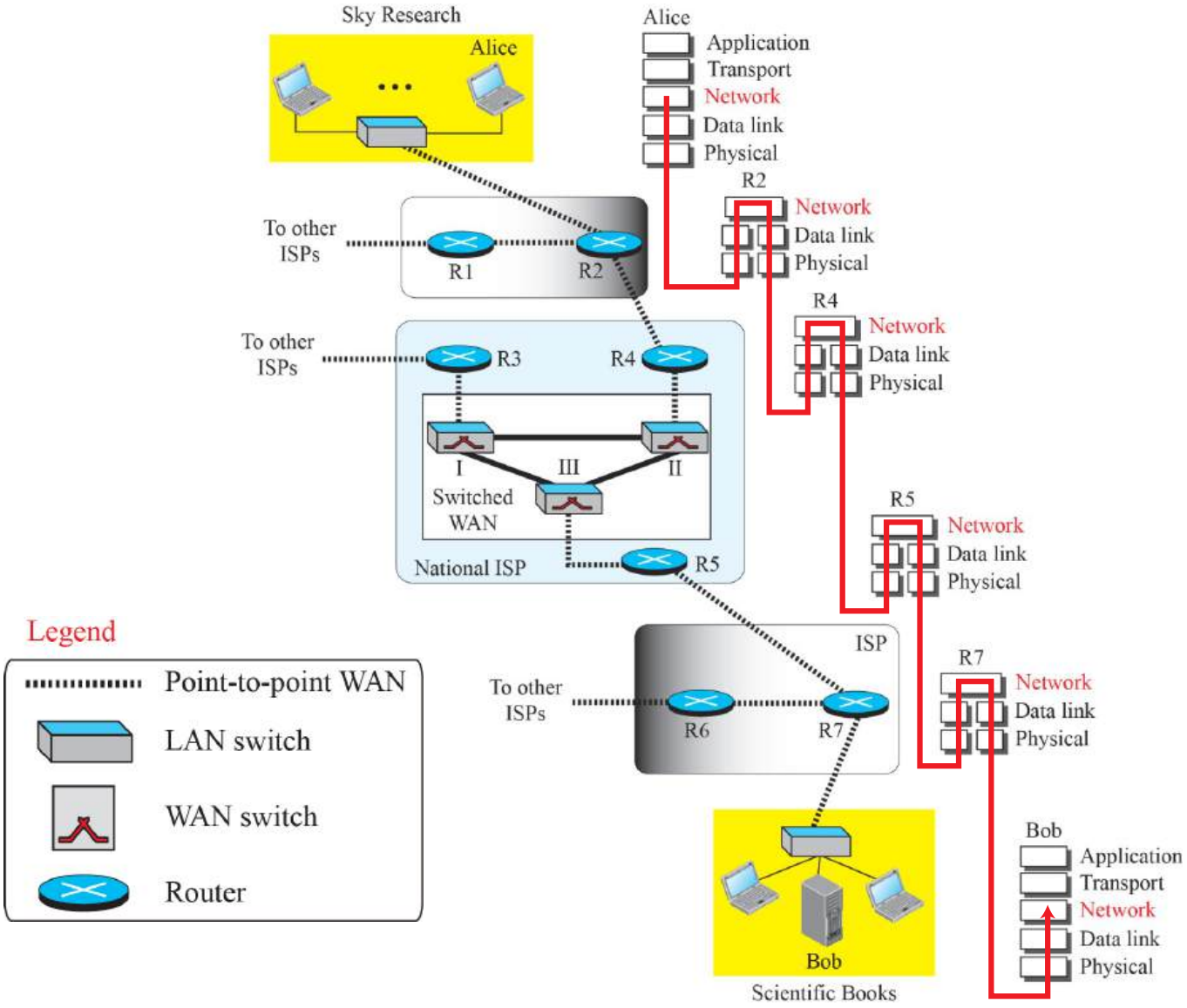
Figure 4.1: Communication at the network layer



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4.1.1 Network-Layer Services

Before discussing the network layer in the Internet today, let's briefly discuss the network-layer services that, in general, are expected from a network-layer protocol.

- ***Packetizing***
- ***Routing***
- ***Forwarding***

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4.1.1 (continued)

- *Error Control*
- *Flow Control*
- *Congestion Control*
- *Quality of Service*
- *Security*

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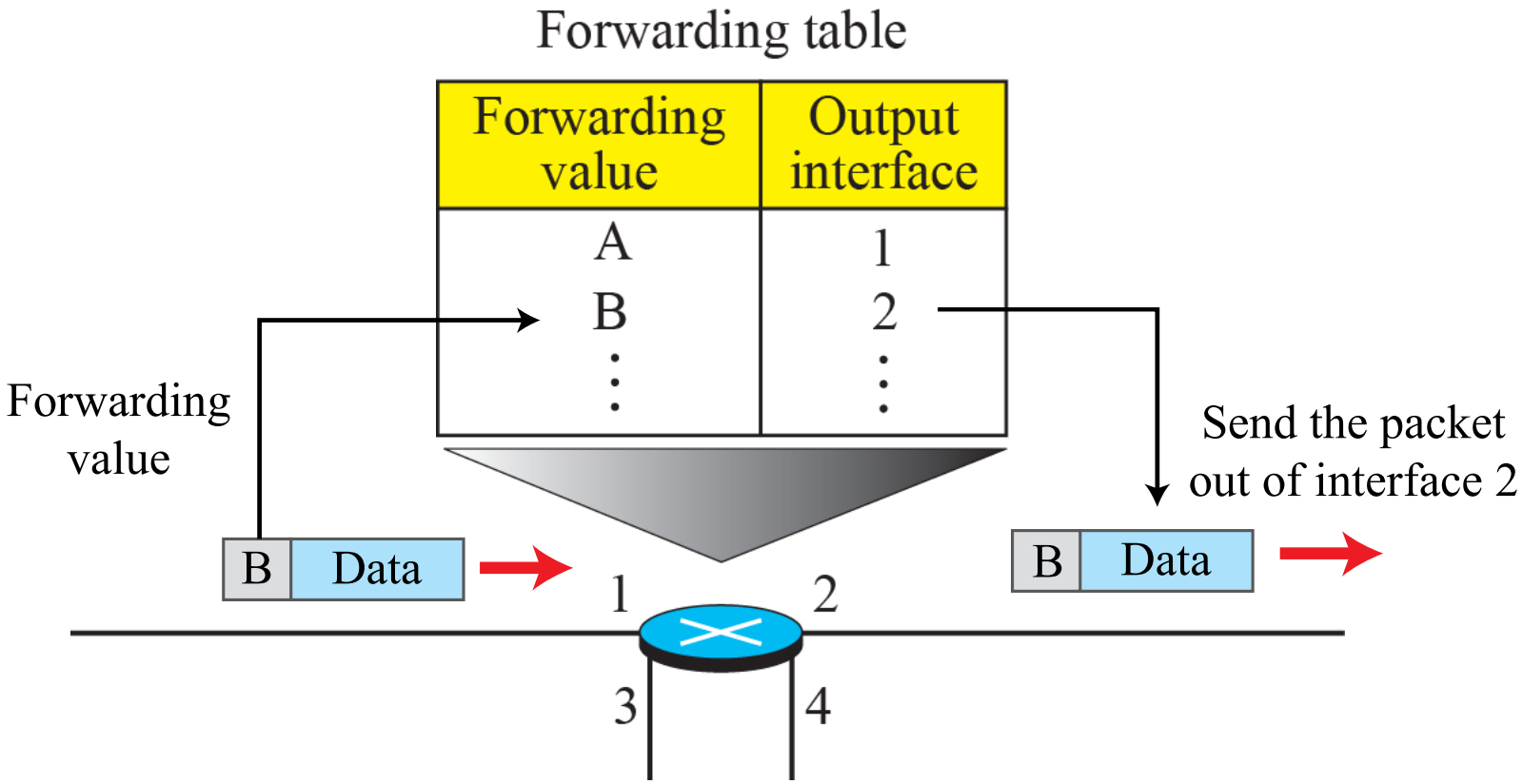
Figure 4.2: Forwarding process



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4.1.2 Packet Switching

From the discussion of routing and forwarding in the previous section, we infer that a kind of switching occurs at the network layer. A router, in fact, is a switch that creates a connection between an input port and an output port (or a set of output ports), just as an electrical switch connects the input to the output to let electricity flow.

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4.1.2 (continued)

- **Datagram Approach**
- **Virtual-Circuit Approach**
  - Setup Phase
  - Data-Transfer Phase
  - Teardown Phase

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Figure 4.3: A connectionless packet-switched network

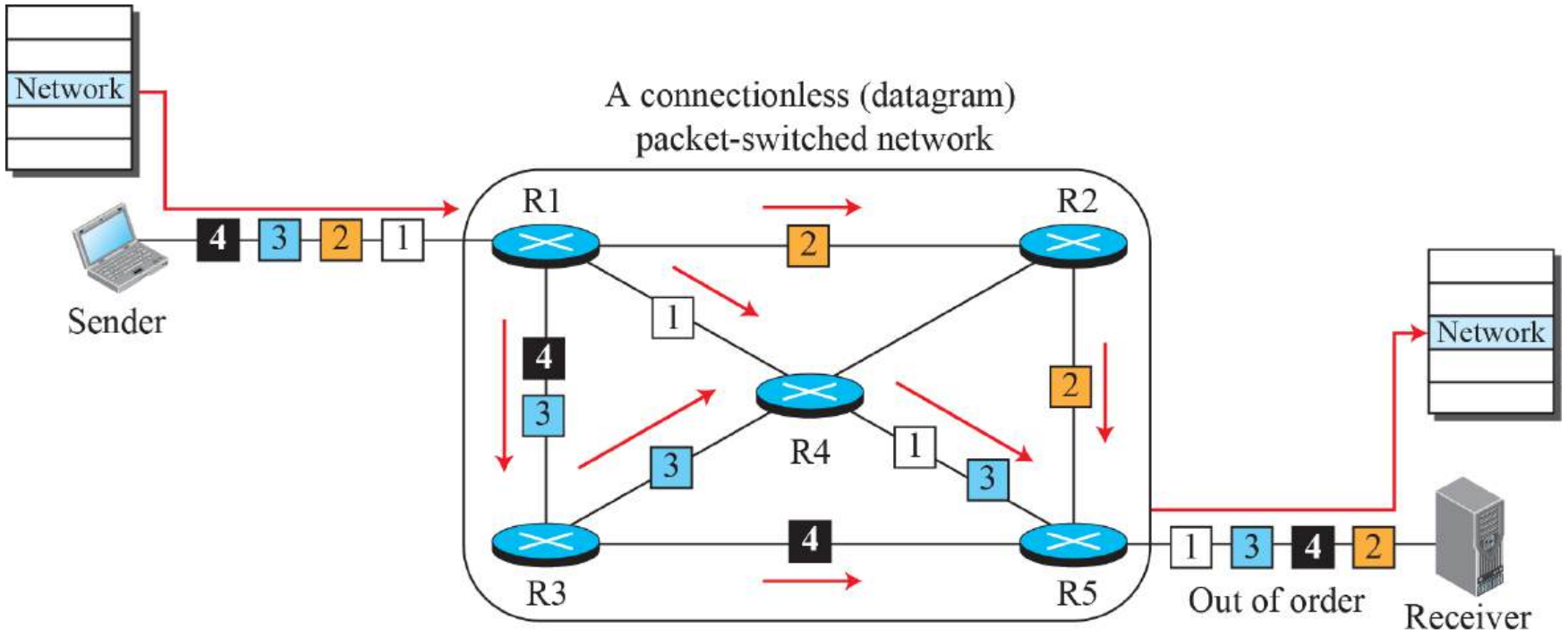

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Figure 4.4: ***Forwarding process in a router when used in a connectionless network***

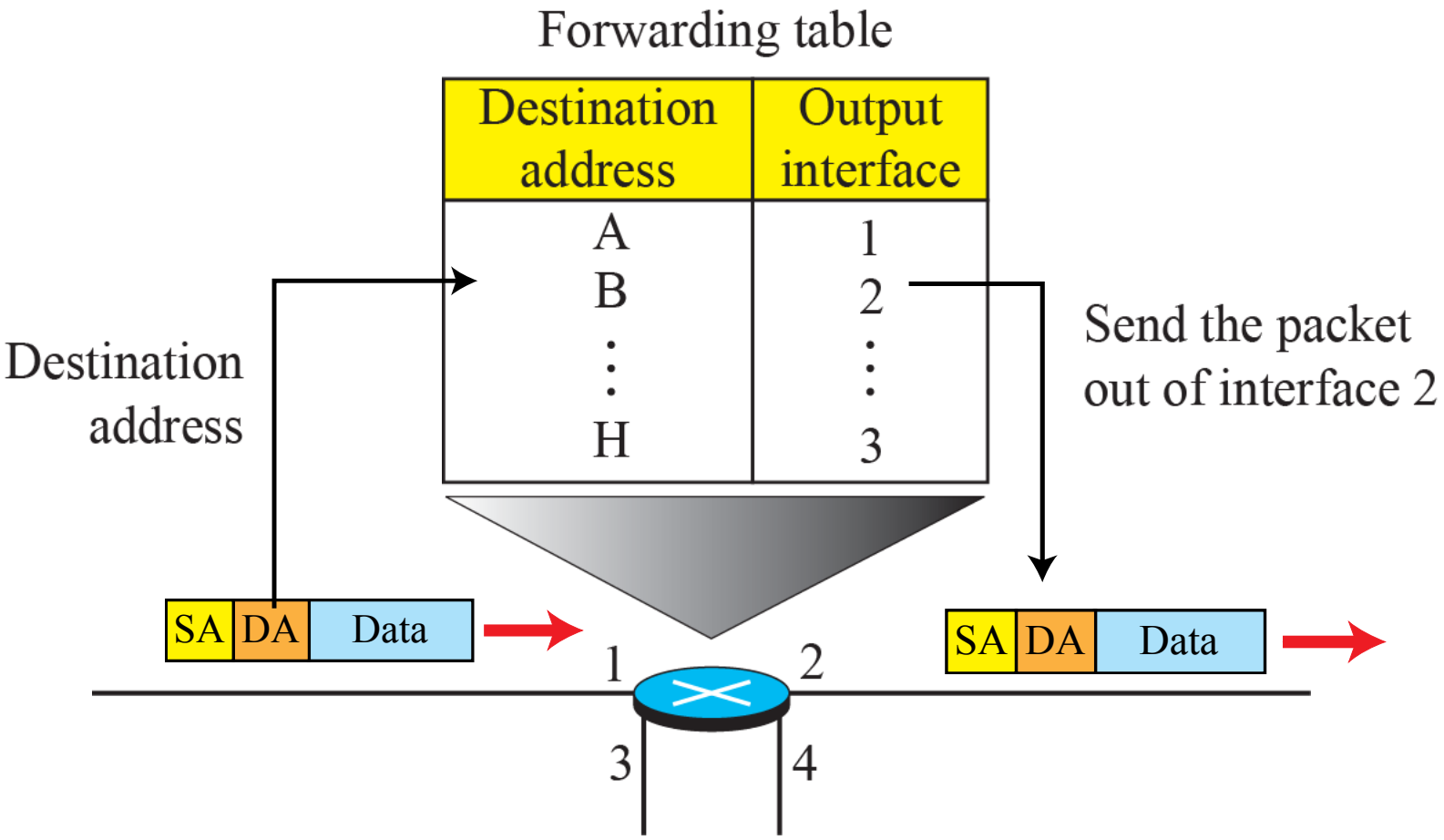

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Figure 4.5: A virtual-circuit packet-switched network

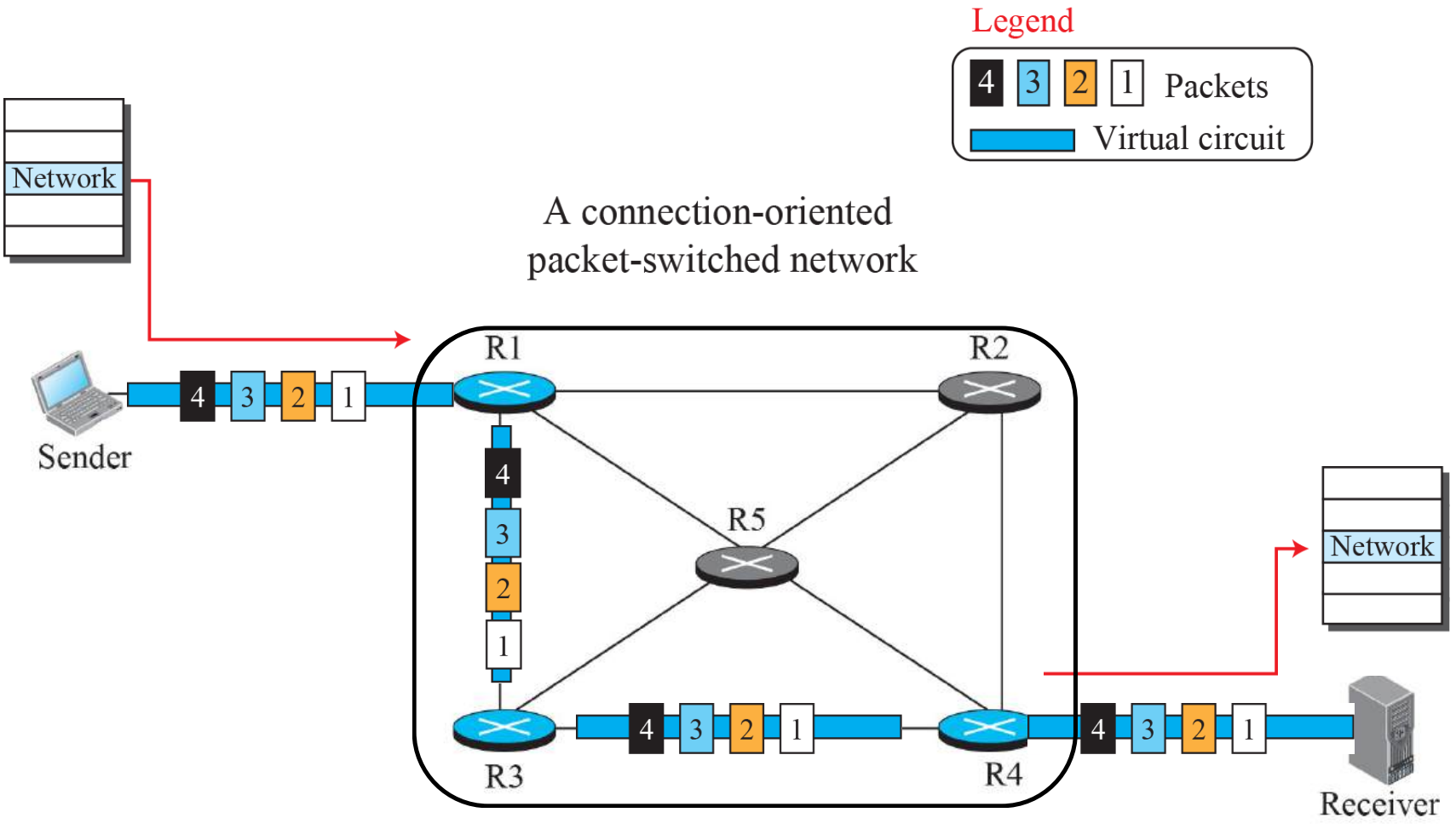

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Figure 4.6: ***Forwarding process in a router when used in a virtual circuit network***

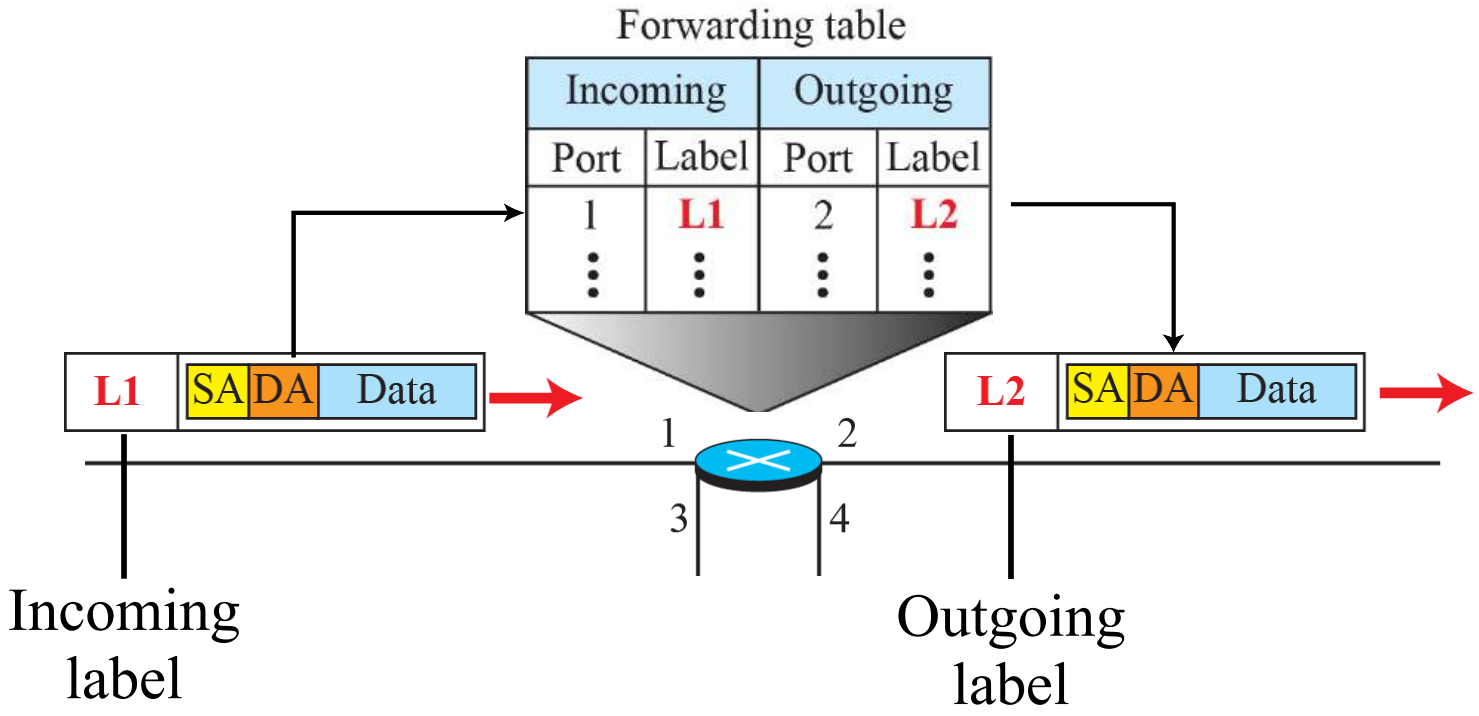

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Figure 4.7: Sending request packet in a virtual-circuit network

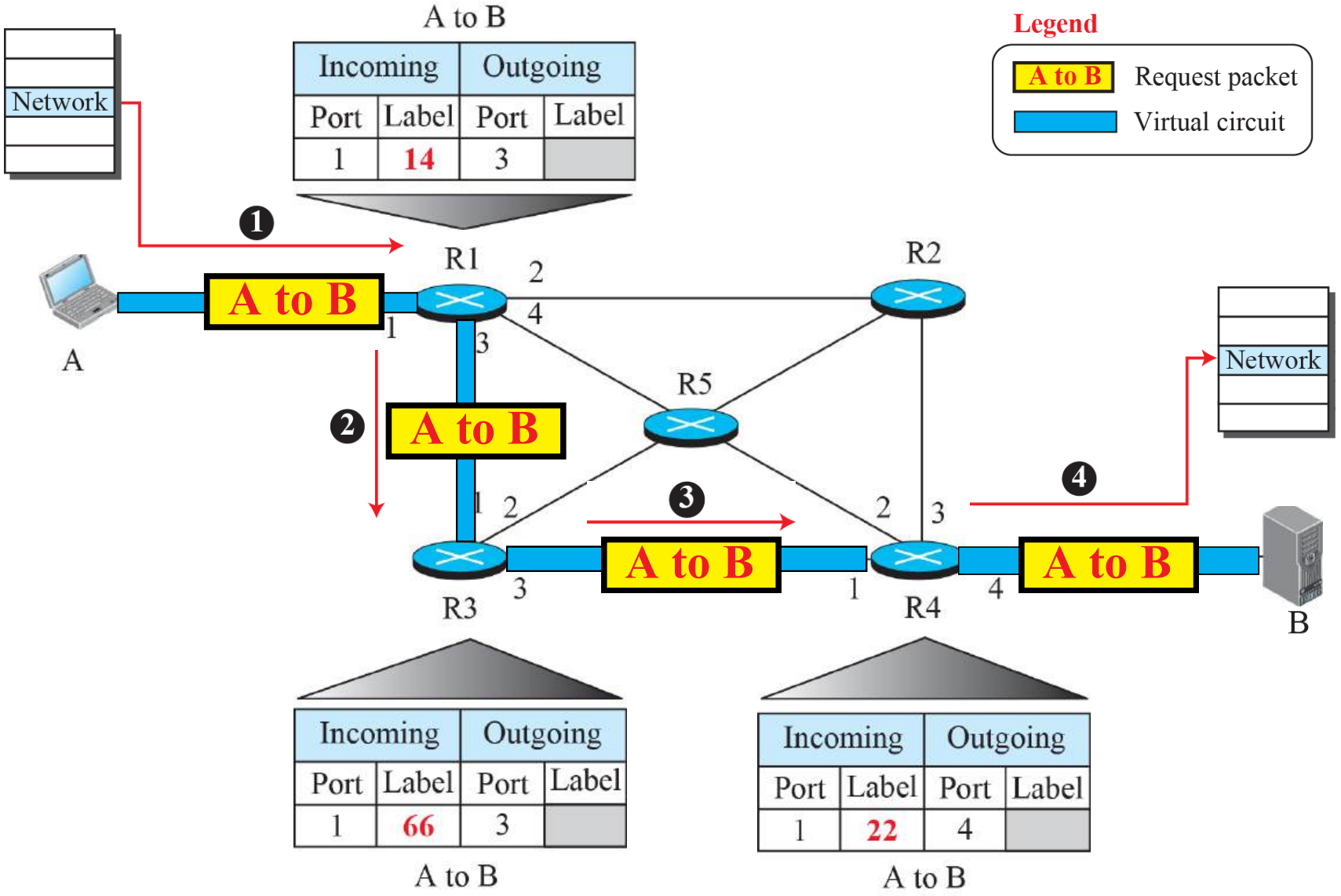

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Figure 4.8: Sending acknowledgments in a virtual-circuit network

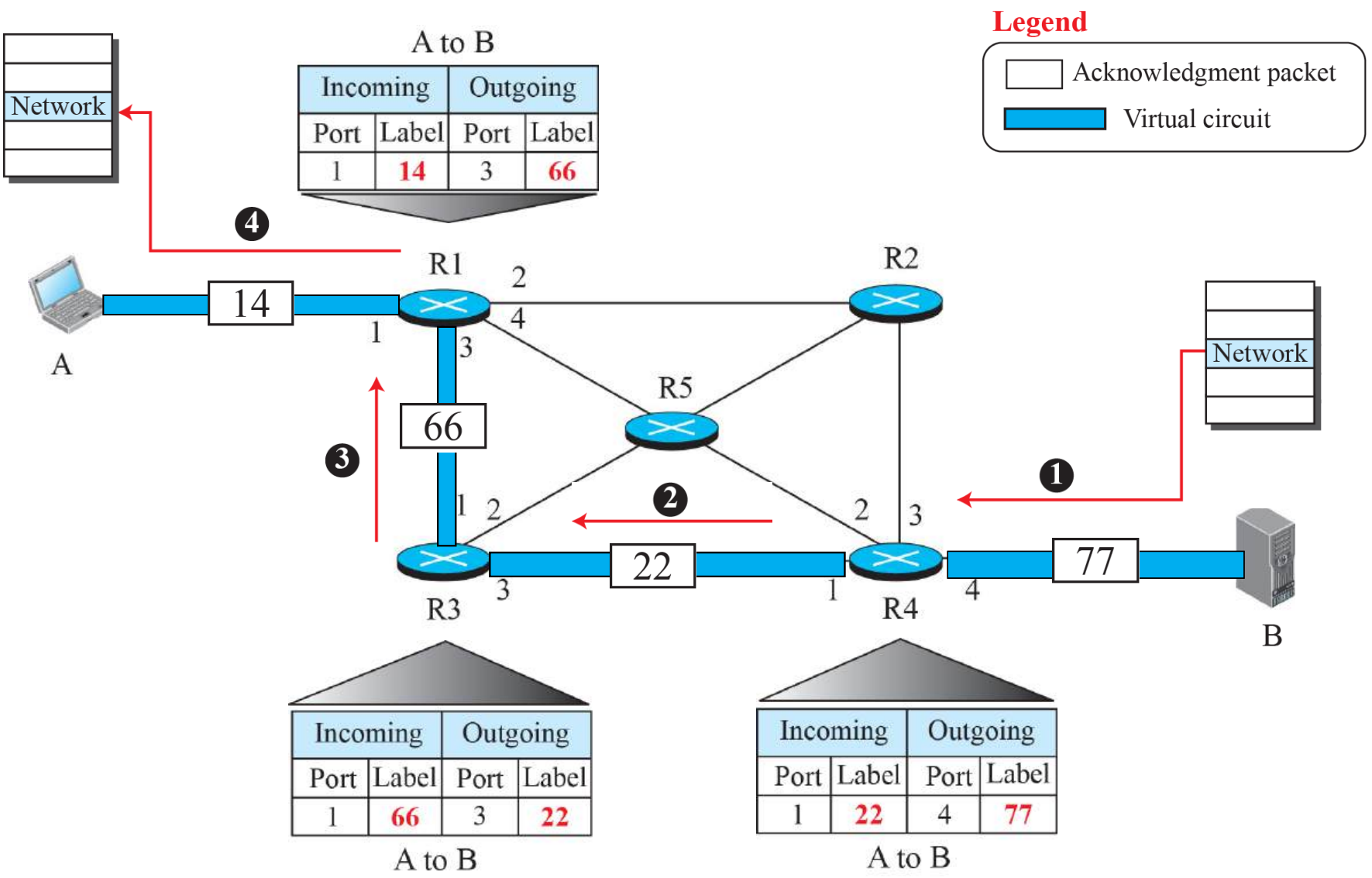

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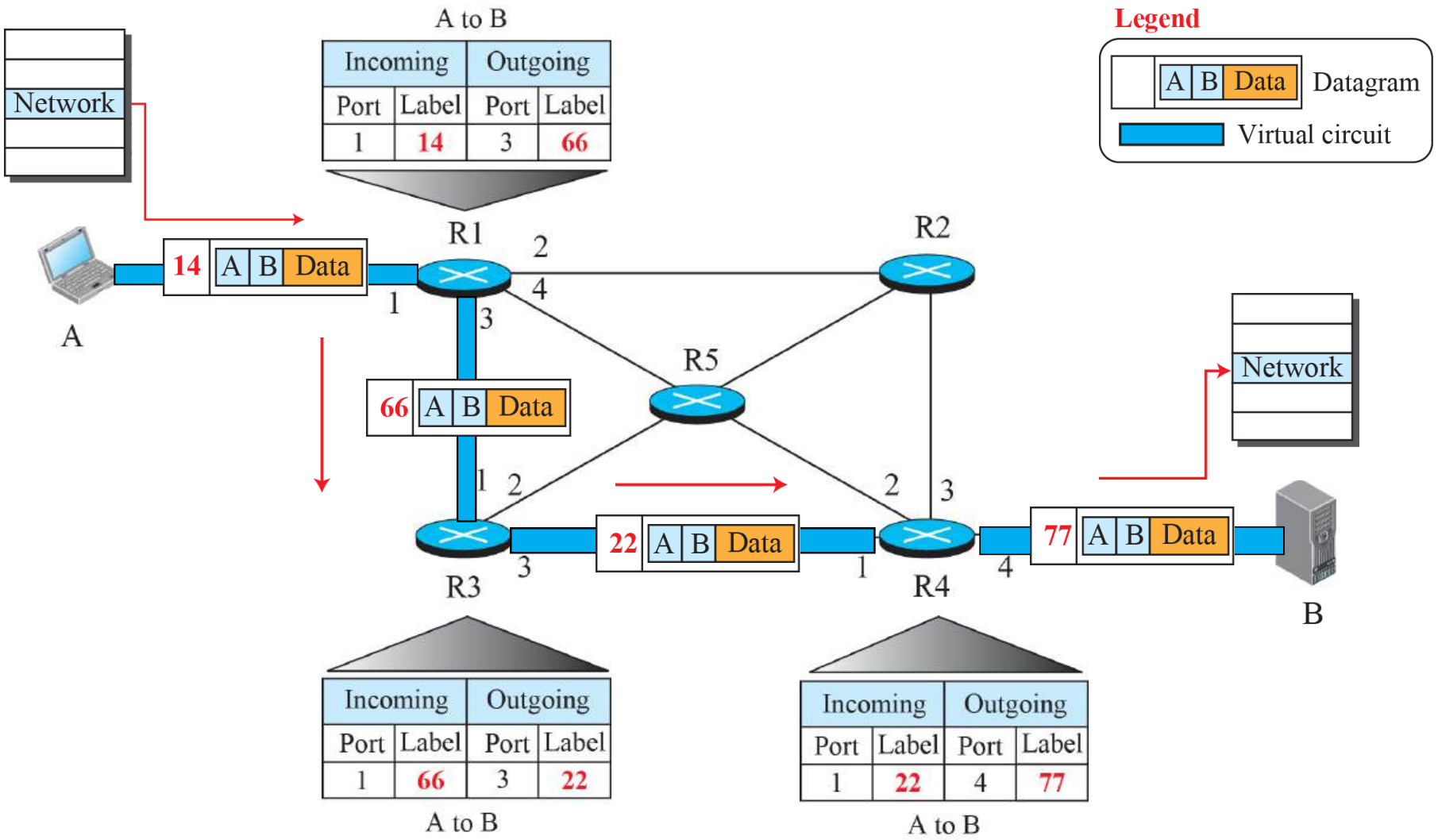
Figure 4.8: *Sending acknowledgments in a virtual-circuit network*



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4.1.3 Network-Layer Performance

The upper-layer protocols that use the service of the network layer expect to receive an ideal service, but the network layer is not perfect. The performance of a network can be measured in terms of delay, throughput, and packet loss. We first define these three terms in a packet-switched network before we discuss their effects on performance.

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4.1.3 (continued)

- **Delay**
  - Transmission Delay
  - Propagation Delay
  - Processing Delay
  - Queuing Delay
  - Total Delay
- **Throughput**
- **Packet Loss**

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Figure 4.10: Throughput in a path with three links in a series

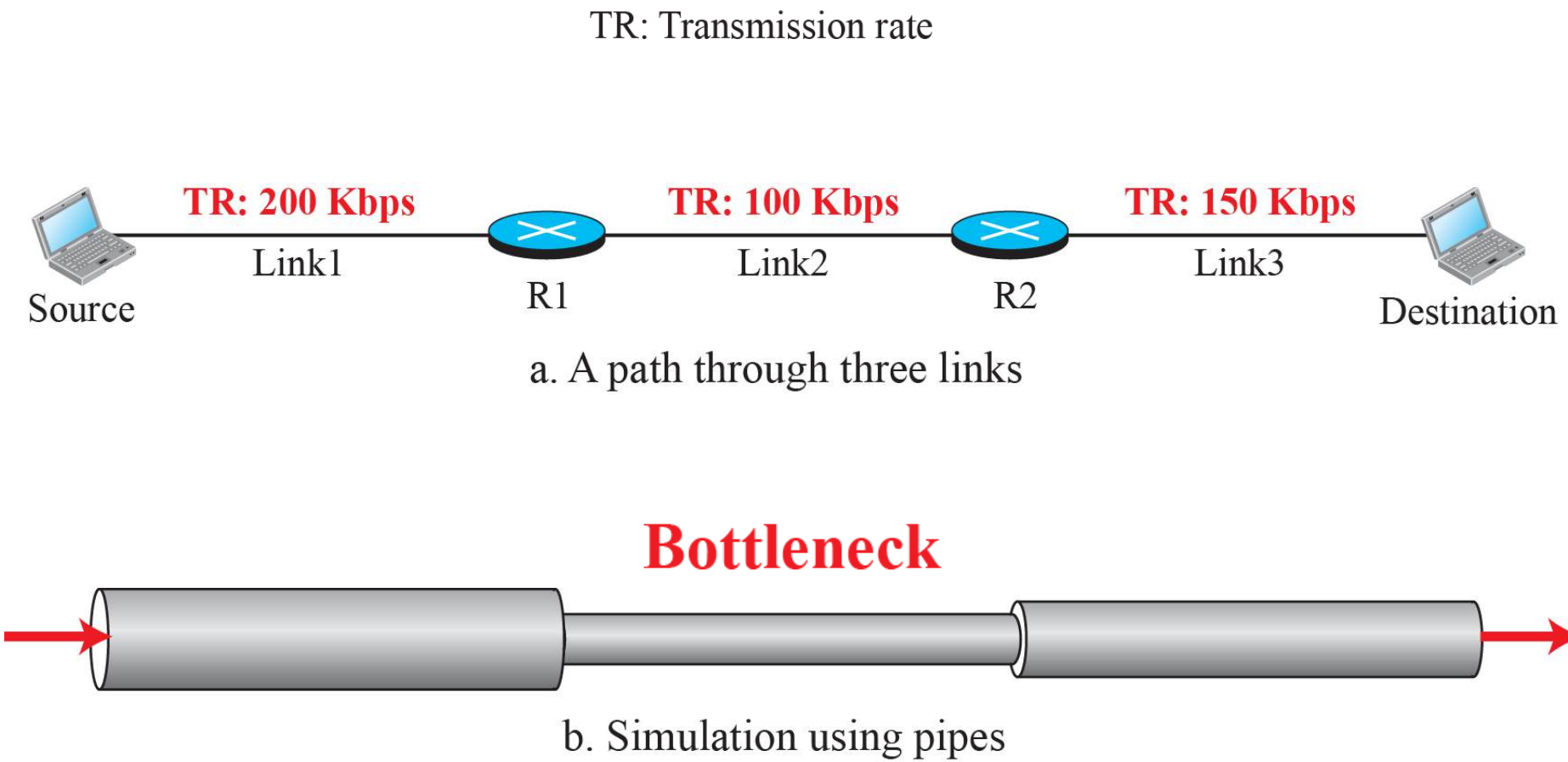

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Figure 4.11: A path through the Internet backbone

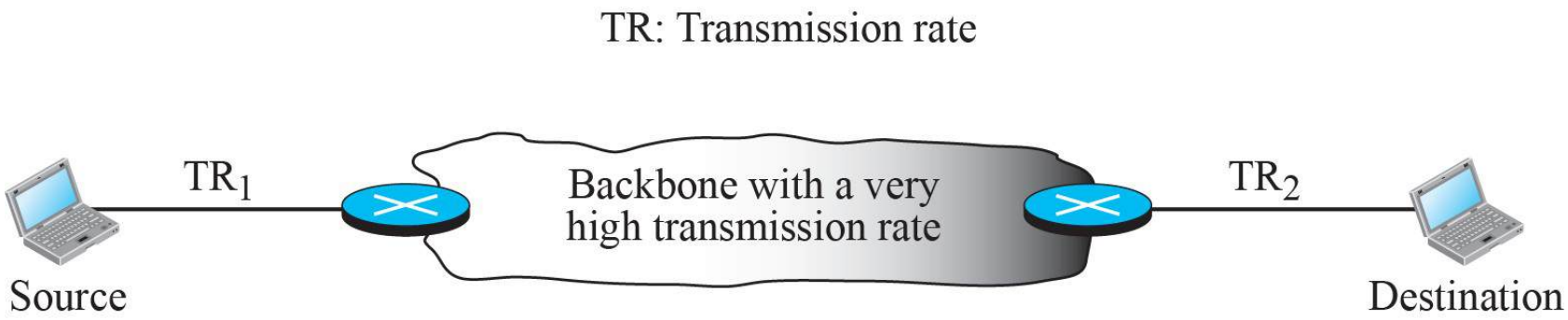

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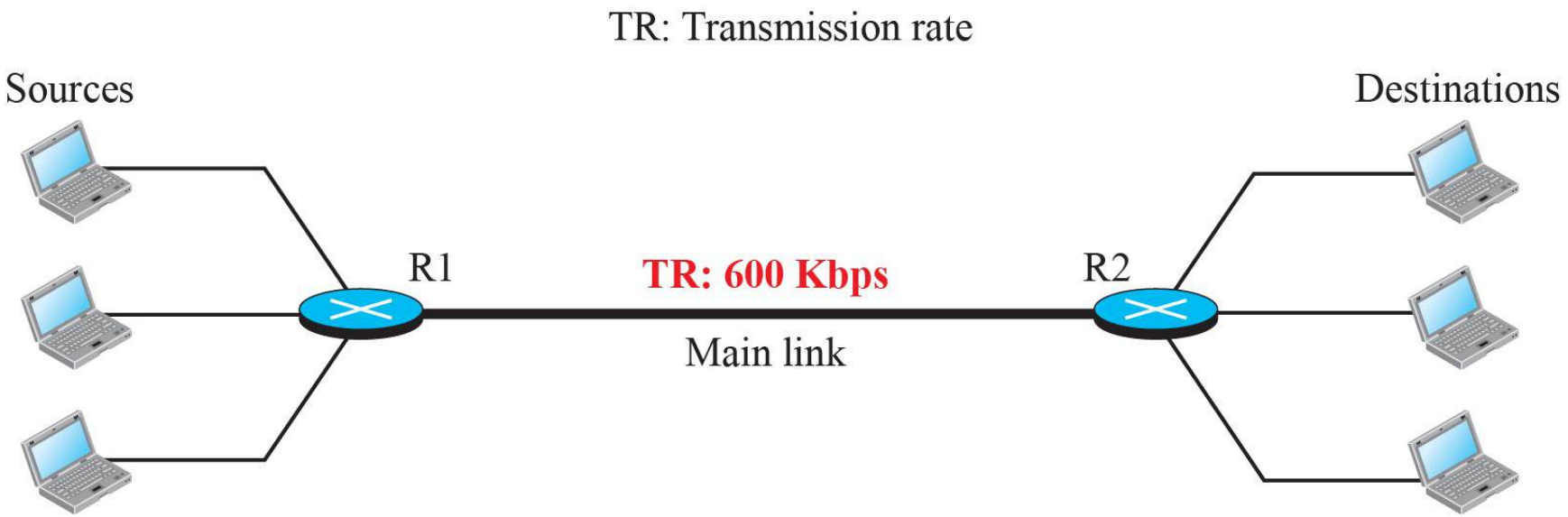
Figure 4.12: Effect of throughput in shared links



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4.1.4 Network-Layer Congestions

In Chapter 3, we discussed congestion at the transport layer. Although congestion at the network layer is not explicitly addressed in the Internet model, the study of congestion at this layer may help us to better understand the cause of congestion at the transport layer and find possible remedies to be used at the network layer. Congestion at the network layer is related to two issues, throughput and delay, which we discussed in the previous section.

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4.1.4 (continued)

Congestion Control

Open-Loop Congestion Control

- Retransmission Policy
- Window Policy
- Acknowledgment Policy
- Discarding Policy
- Admission Policy

Closed-Loop Congestion Control

- Backpressure
- Choke Packet
- Implicit Signaling
- Explicit Signaling

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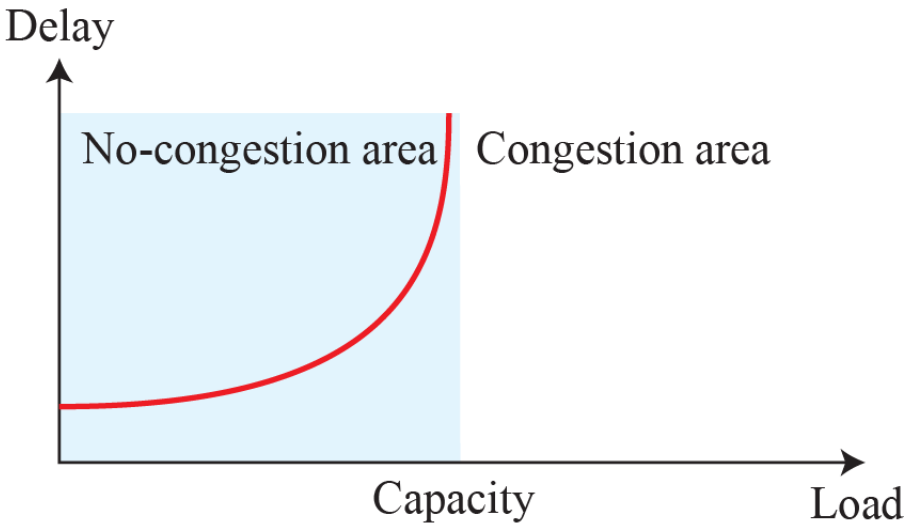
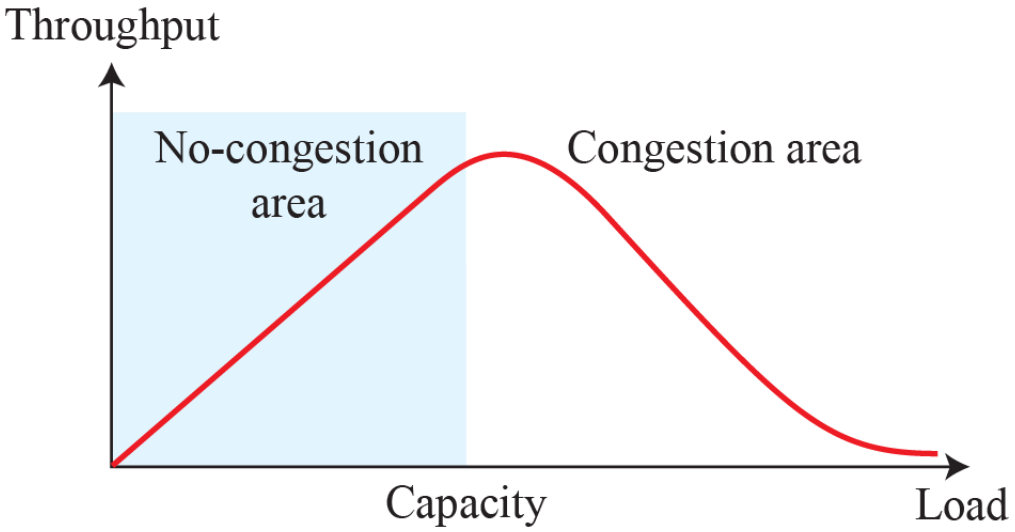
Figure 4.13: Packet delay and throughput as functions of load



a. Delay as a function of load



b. Throughput as a function of load

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Figure 4.14: Backpressure method for alleviating congestion

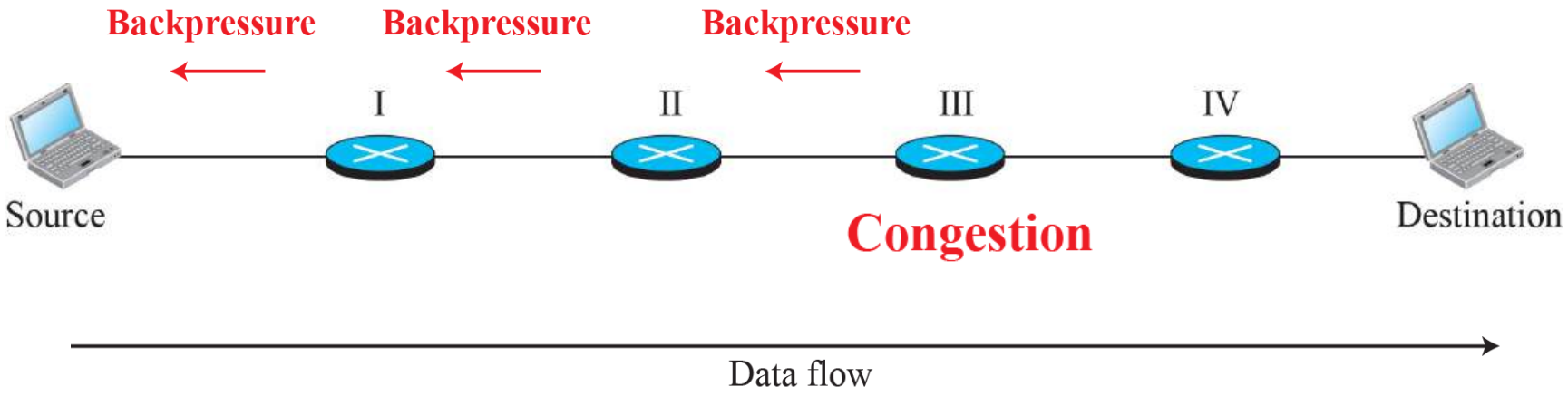

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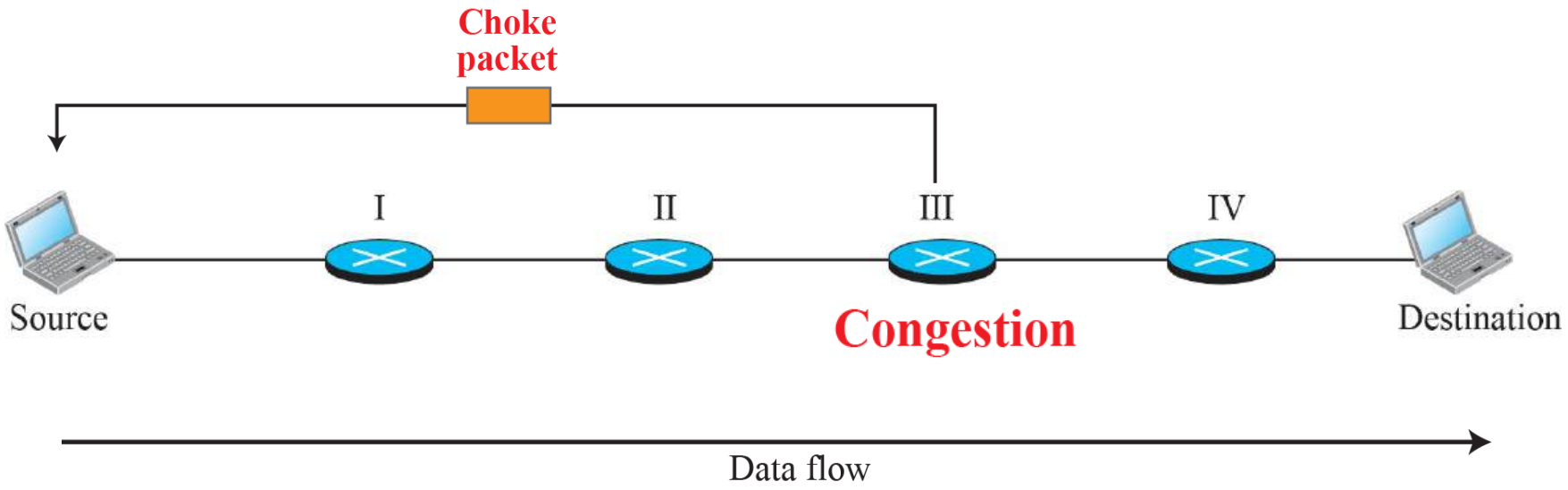
Figure 4.15: Choke packet



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4.1.5 Structure of A Router

In our discussion of forwarding and routing, we represented a router as a black box that accepts incoming packets from one of the input ports (interfaces), uses a forwarding table to find the output port from which the packet departs, and sends the packet from this output port. In this section we open the black box and look inside. However, our discussion won't be very detailed; entire books have been written about routers. We just give an overview to the reader.

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4.1.5 (continued)

- **Components**
  - Input Ports
  - Output Ports
  - Routing Processor
  - Switching Fabrics
    - Crossbar Switch
    - Banyan Switch
    - Batcher-Banyan Switch

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Figure 4.16: Router components

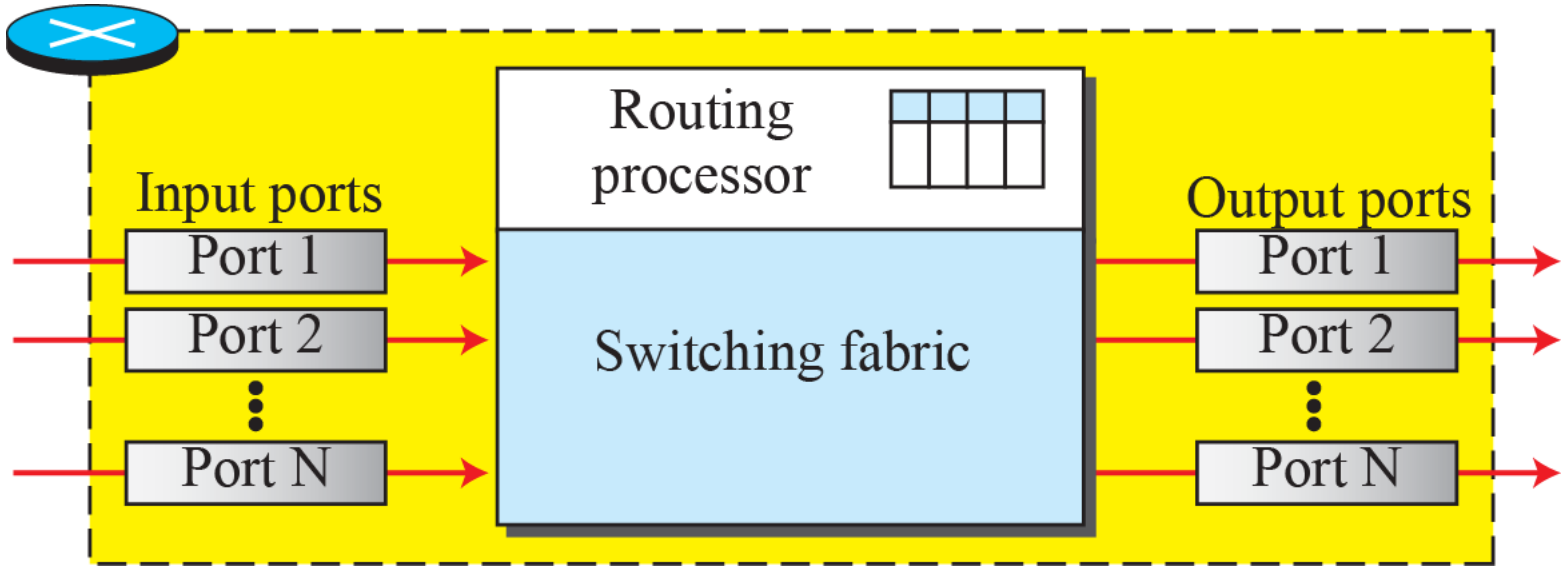

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Figure 4.17: *Input port*

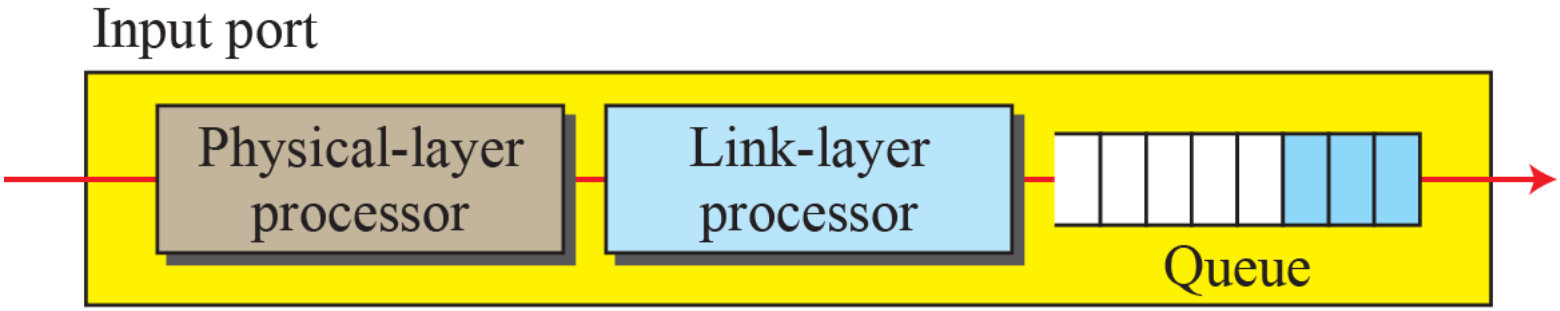

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Figure 4.18: *Output port*

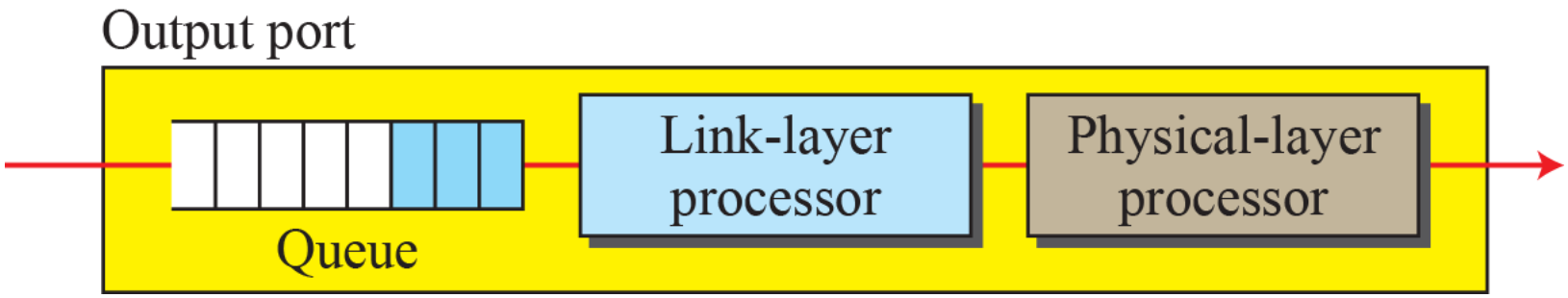

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Figure 4.19: Crossbar switch

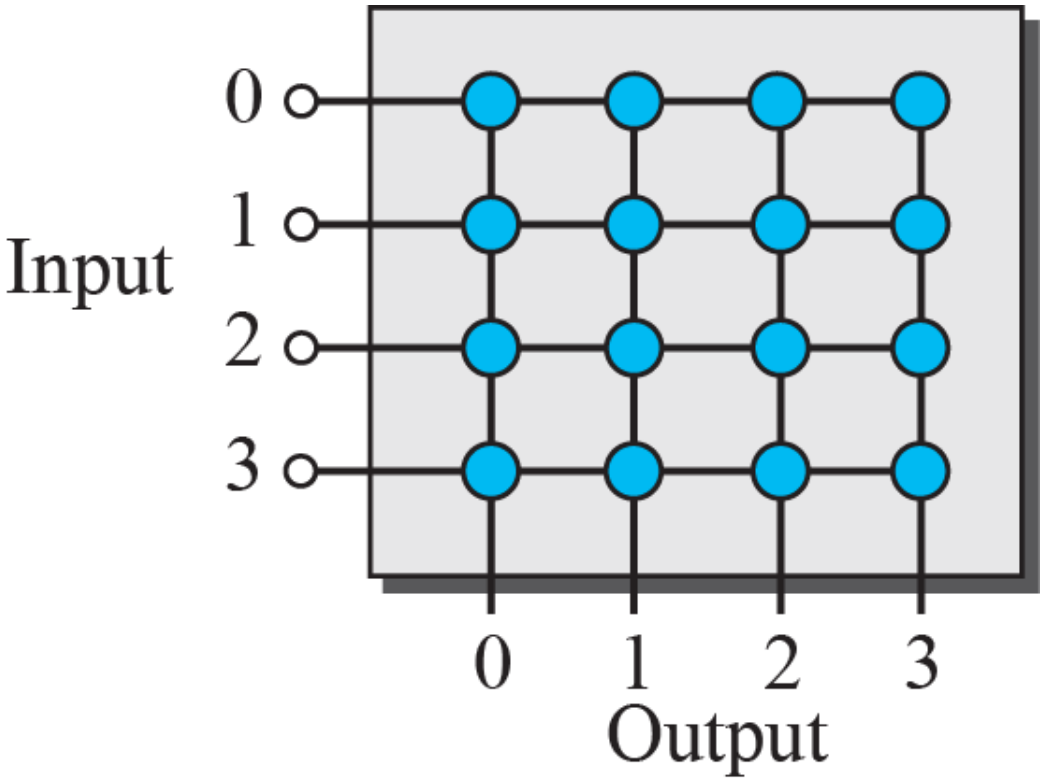

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Figure 4.20: ***Banyan switch***

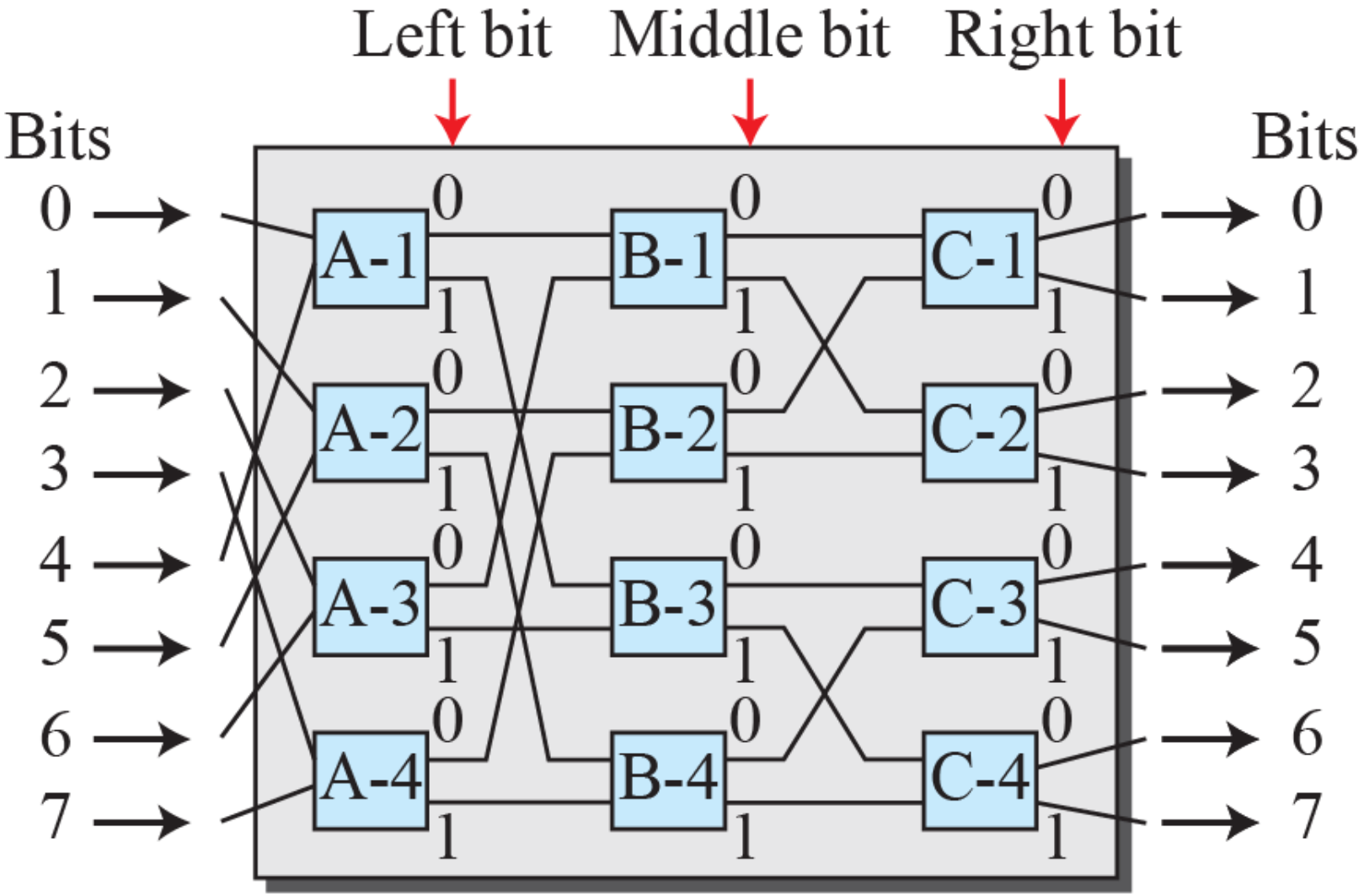

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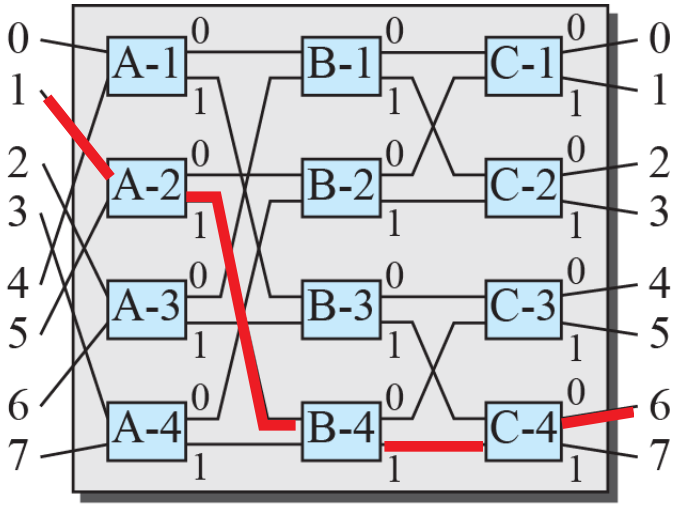
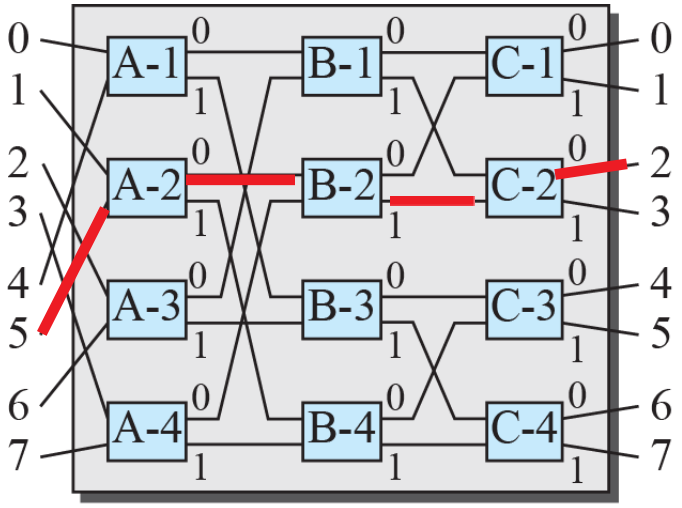
Figure 4.21: Examples of routing in a banyan switch



a. Input 1 sending to output 6 (110)



b. Input 5 sending to output 2 (010)

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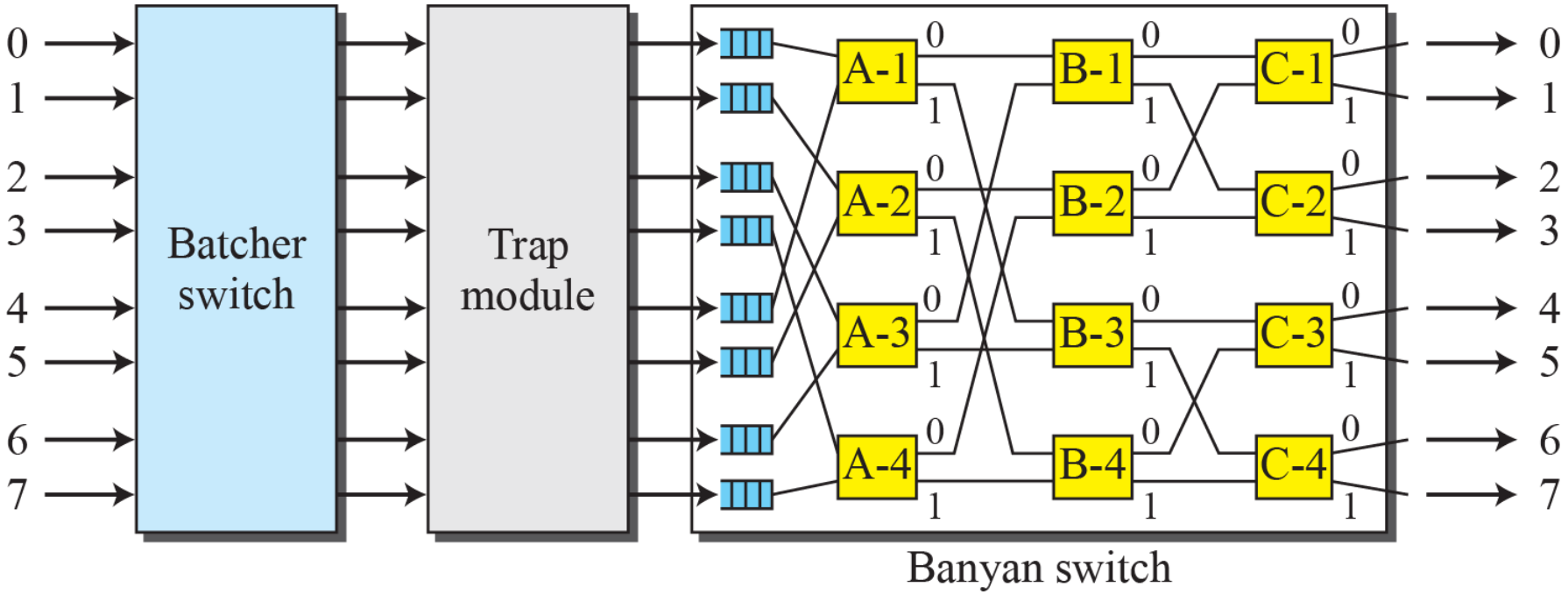
Figure 4.22: Batcher-banyan switch



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4-2 NETWORK-LAYER PROTOCOLS

In this section, we show how the network layer is implemented in the TCP/IP protocol suite. The protocols in the network layer have gone through several versions; in this section, we concentrate on the current version (4), in the last section of this chapter, we briefly discuss version 6, which is on the horizon.

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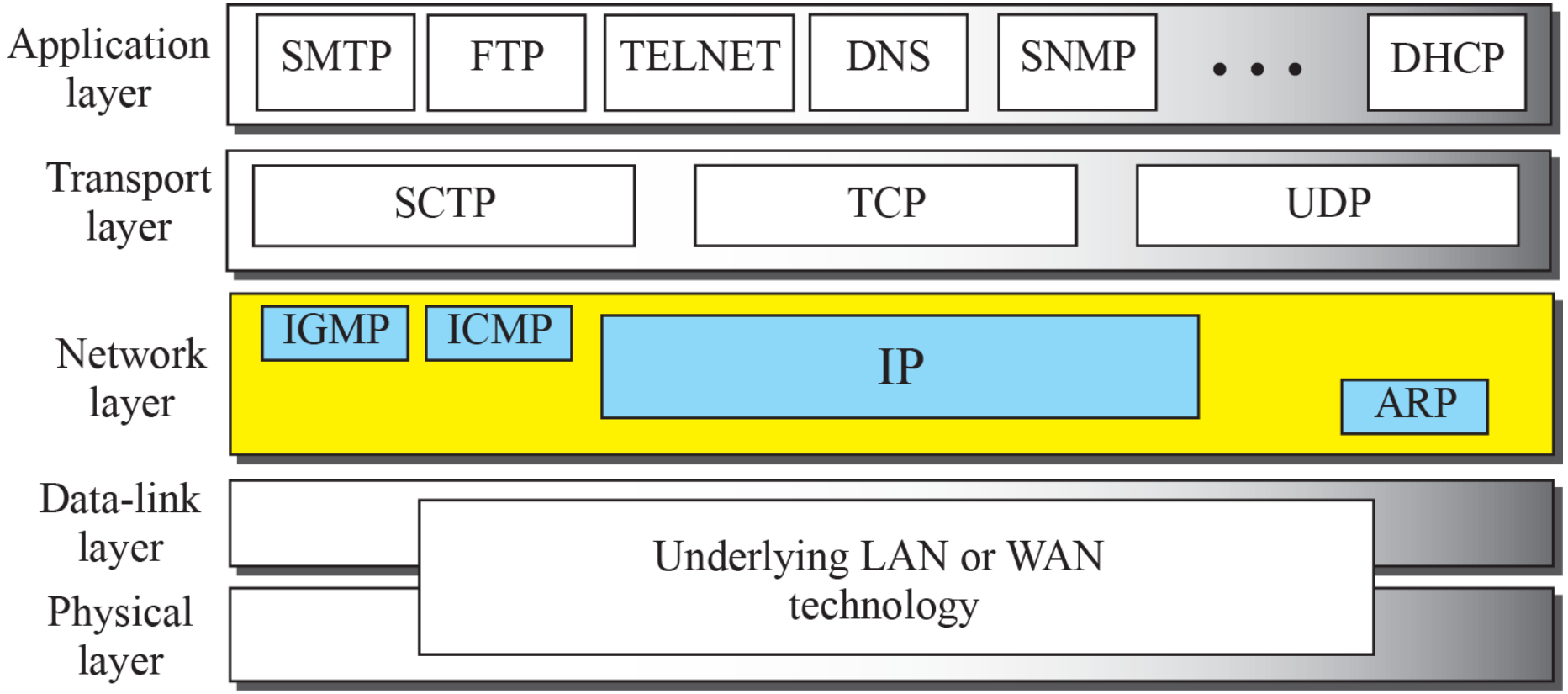
Figure 4.23: ***Position of IP and other network-layer protocols in TCP/IP protocol suite***



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4.2.1 IPv4 Datagram Format

Packets used by the IP are called datagrams. Figure 4.24 shows the IPv4 datagram format. A datagram is a variable-length packet consisting of two parts: header and payload (data). The header is 20 to 60 bytes in length and contains information essential to routing and delivery. It is customary in TCP/IP to show the header in 4-byte sections.

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4.2.1 (continued)

- **Fragmentation**
  - Maximum Transfer Unit (MTU)
  - Fields Related to Fragmentation

- **Security of IPv4 Datagrams**
  - Packet Sniffing
  - Packet Modification
  - IP Spoofing
  - IPSec

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Figure 4.24: IP datagram

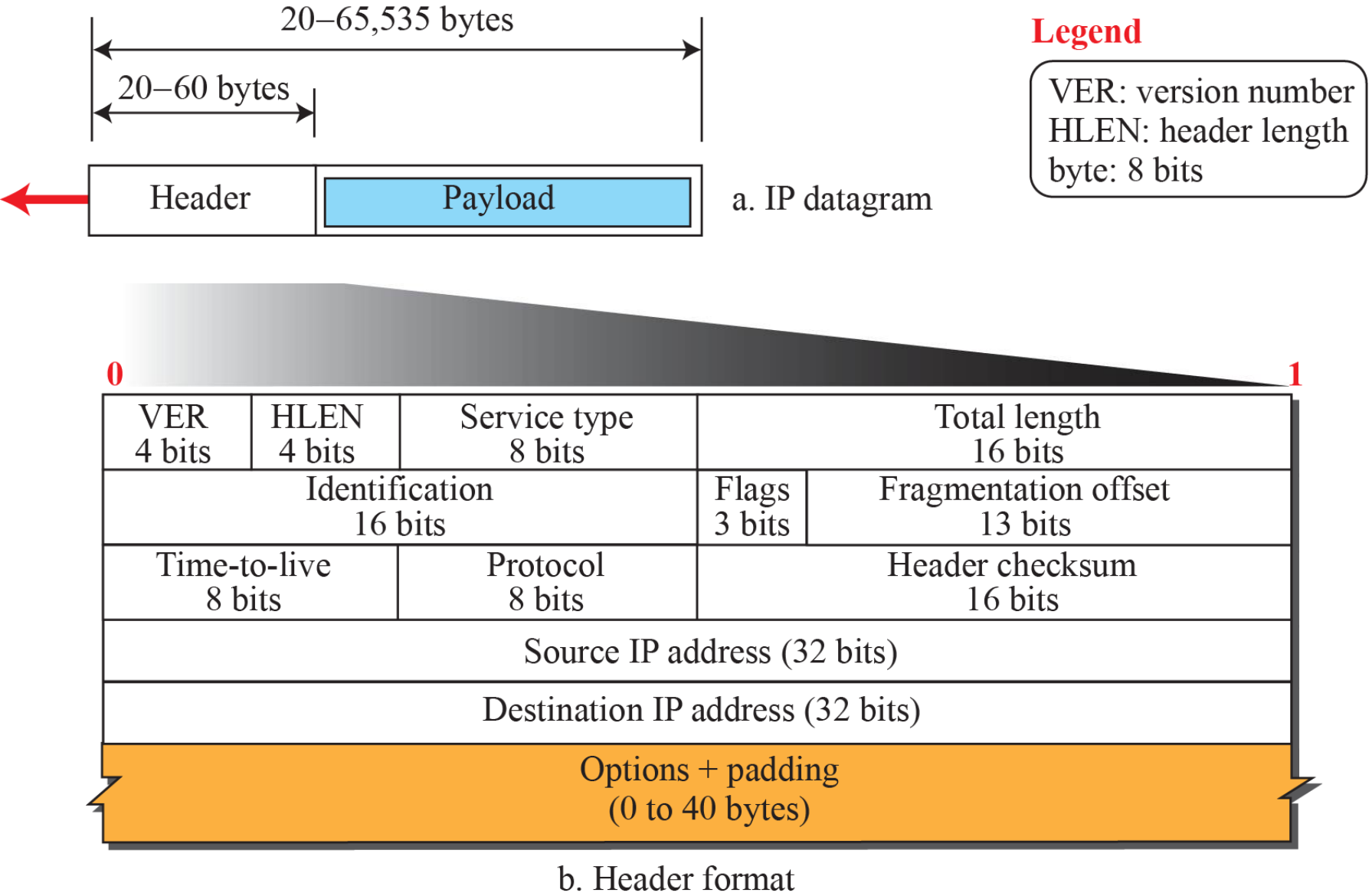

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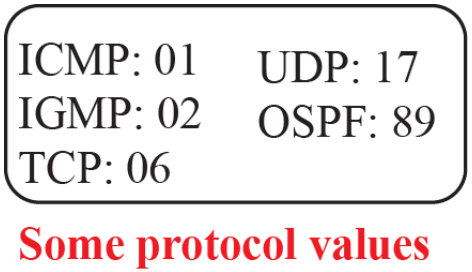
Figure 4.25: ***Multiplexing and demultiplexing using the value of the protocol field***



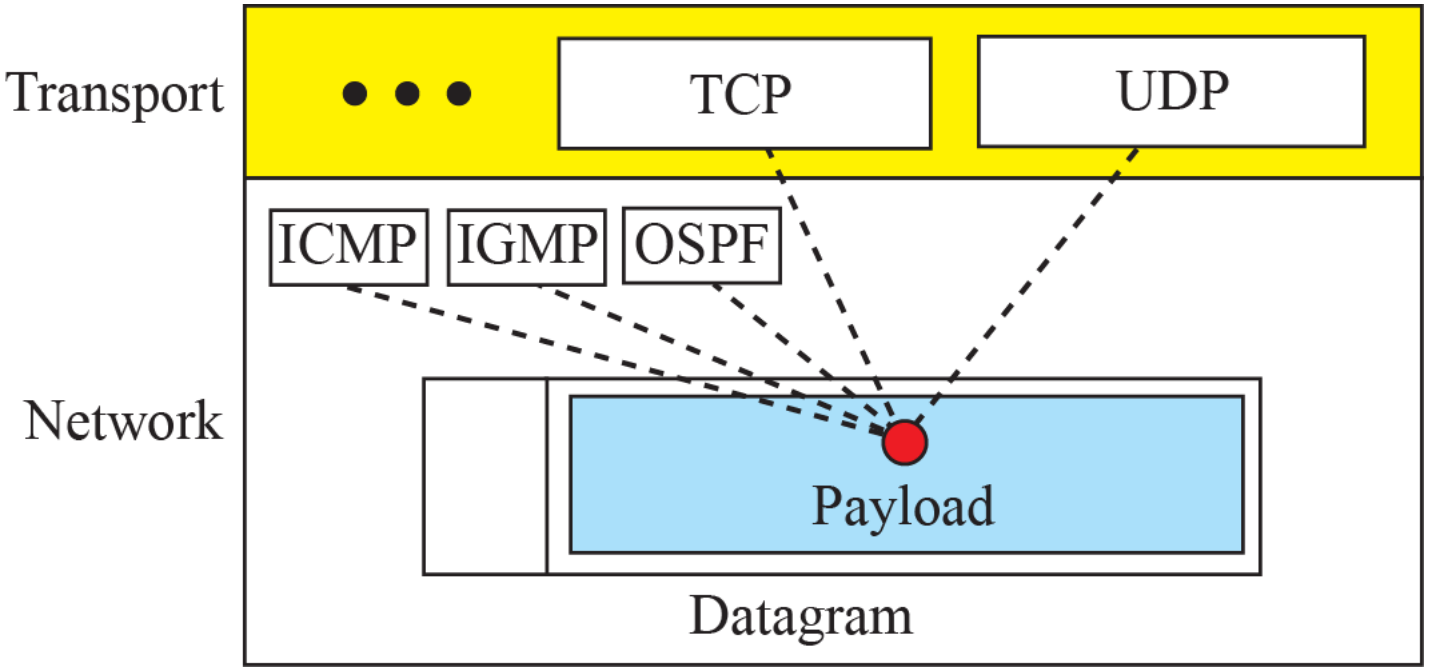


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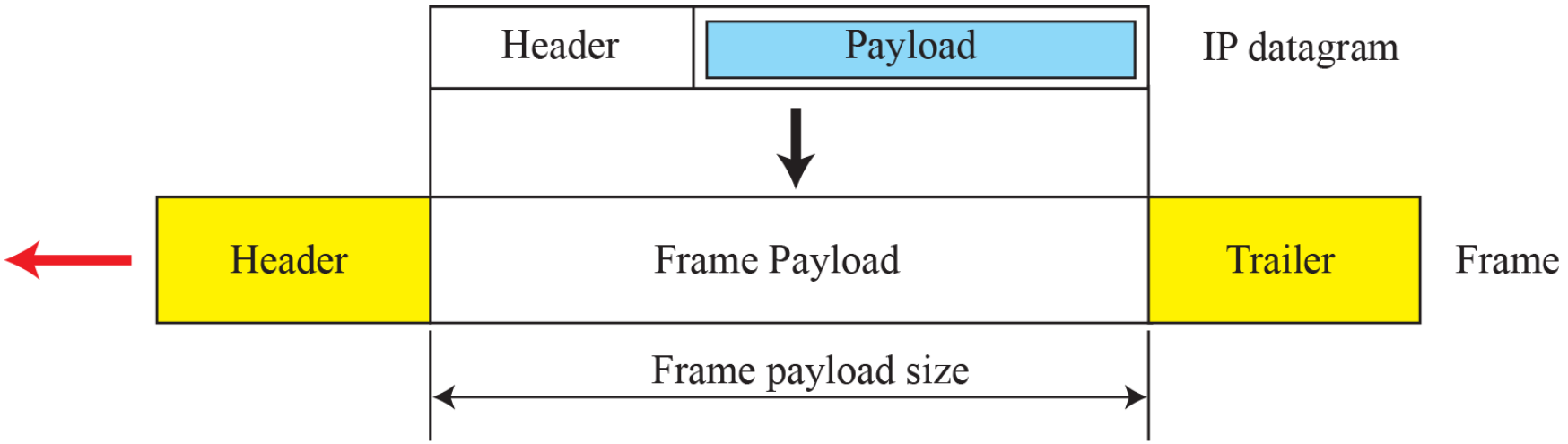
Figure 4.26: Maximum transfer unit (MTU)



MTU: Maximum size of frame payload

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Figure 4.27: Fragmentation example

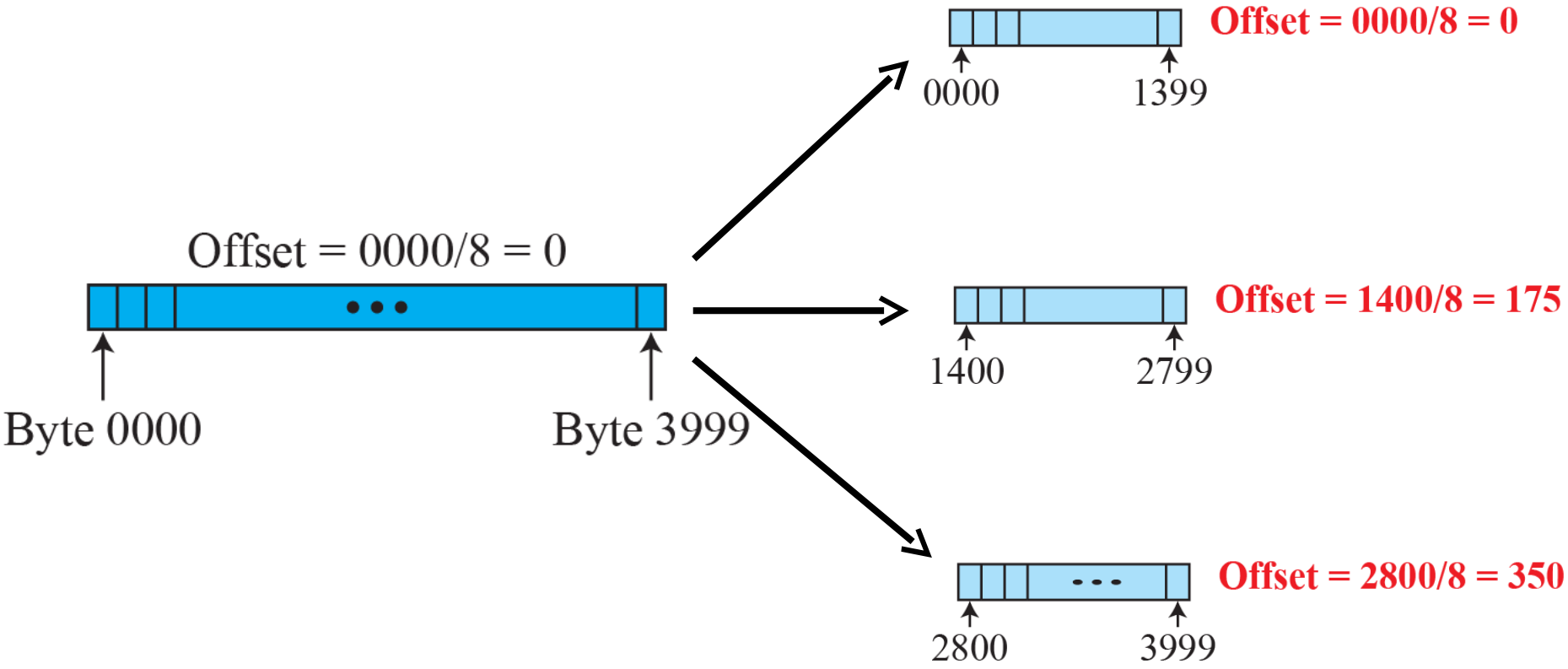

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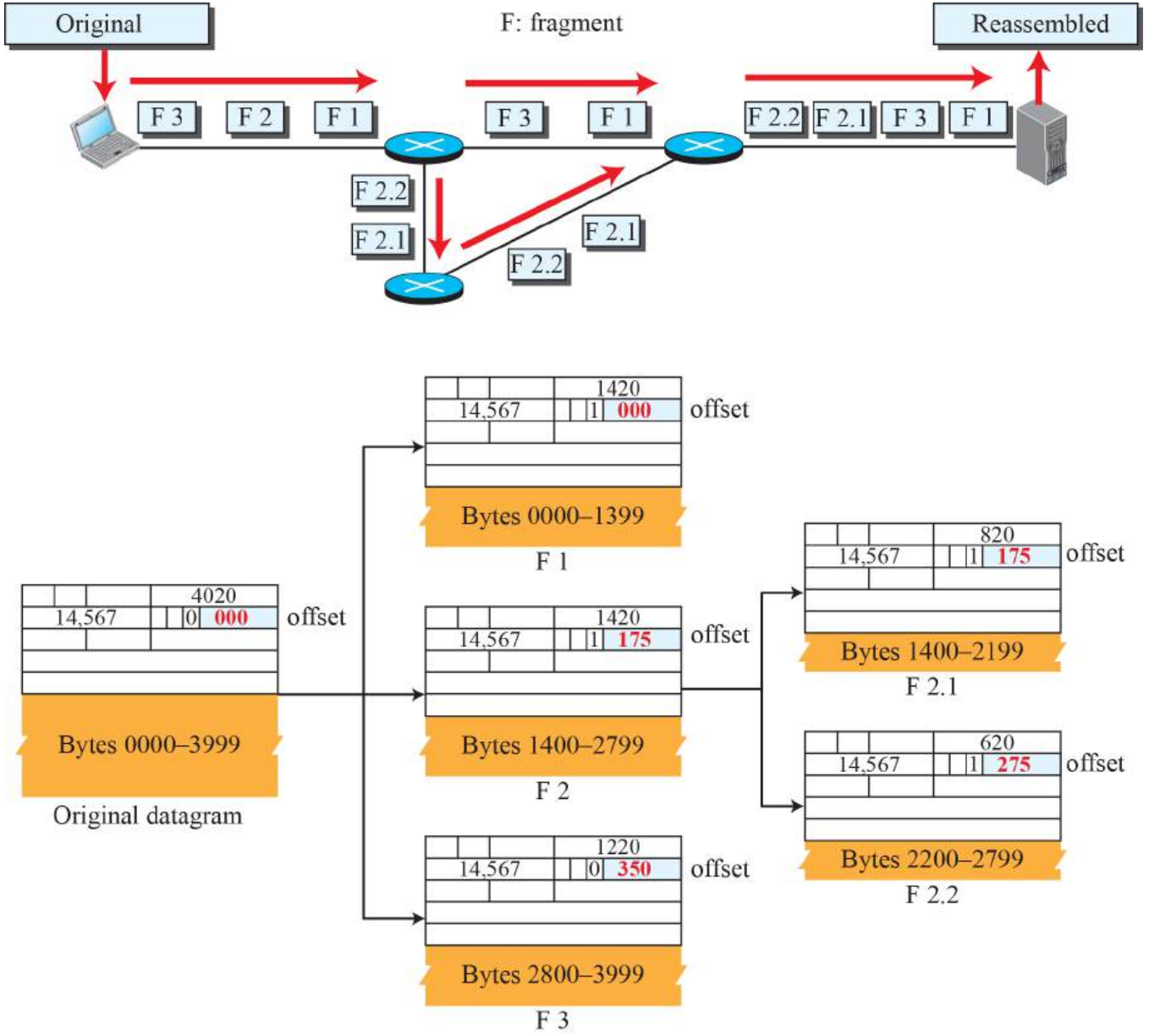
Figure 4.28: Detailed fragmentation example



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4.2.2 IPv4 Addresses

The identifier used in the IP layer of the TCP/IP protocol suite to identify the connection of each device to the Internet is called the Internet address or IP address. An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a host or a router to the Internet. The IP address is the address of the connection, not the host or the router, because if the device is moved to another network, the IP address may be changed.

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4.2.2 (continued)

- **Address Space**
- **Notation**
- **Hierarchy in Addressing**
- **Classful Addressing**
  - Address Depletion
  - Subnetting and Supernetting
  - Advantage of Classful Addressing

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4.2.2 (continued)

- **Classless Addressing**
  - Prefix Length: Slash Notation
  - Extracting information from an address
  - Address Mask
  - Network Address
  - Block Allocation
  - Subnetting
  - Address Aggregation
  - Special Addresses

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4.2.2 (continued)

- **Dynamic Host Configuration Protocol (DHCP)**
  - DHCP Message Format
  - DHCP Operation
  - Two Well-Known Ports
  - Using FTP
  - Error Control
  - Transition States
- **NAT**
  - Address Translation
  - Translation Table

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Figure 4.29: ***Three different notations in IPv4 addressing***

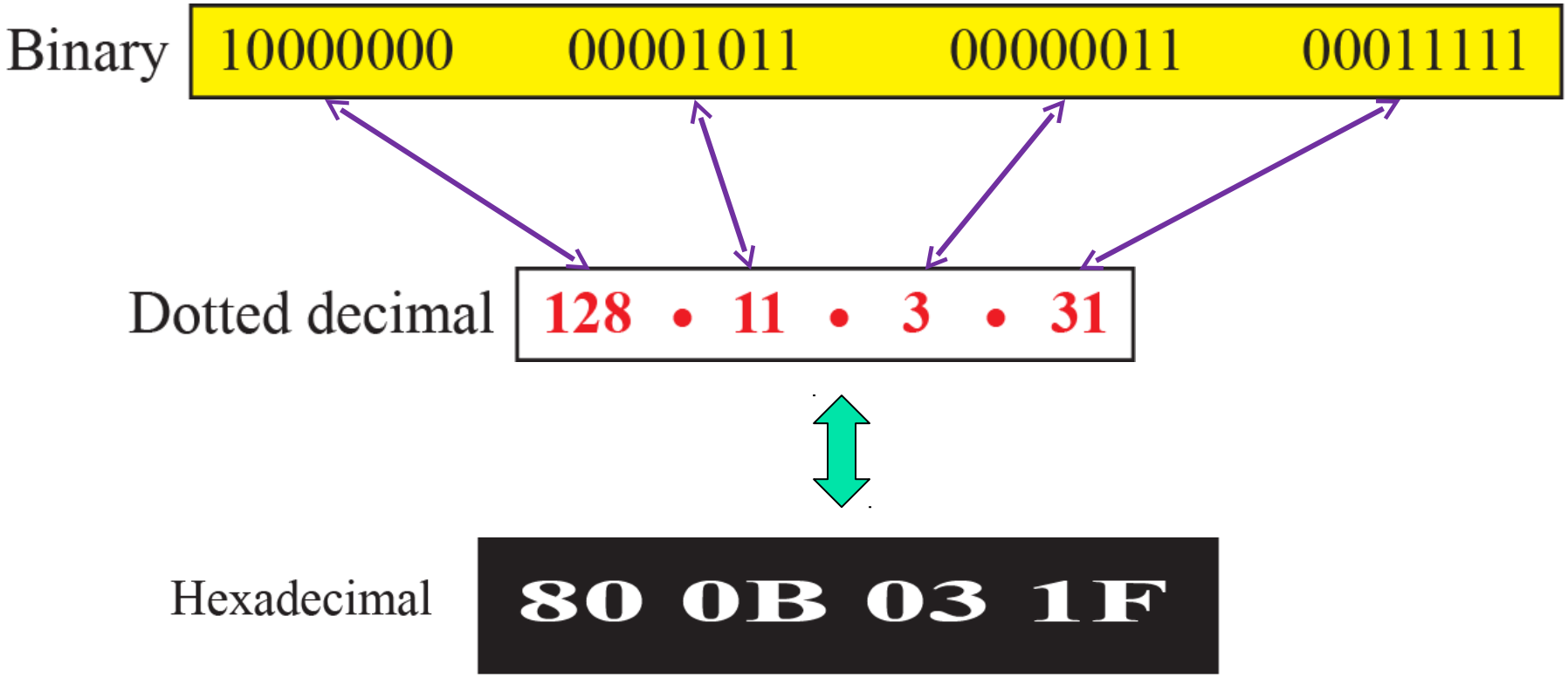

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Figure 4.30: Hierarchy in addressing

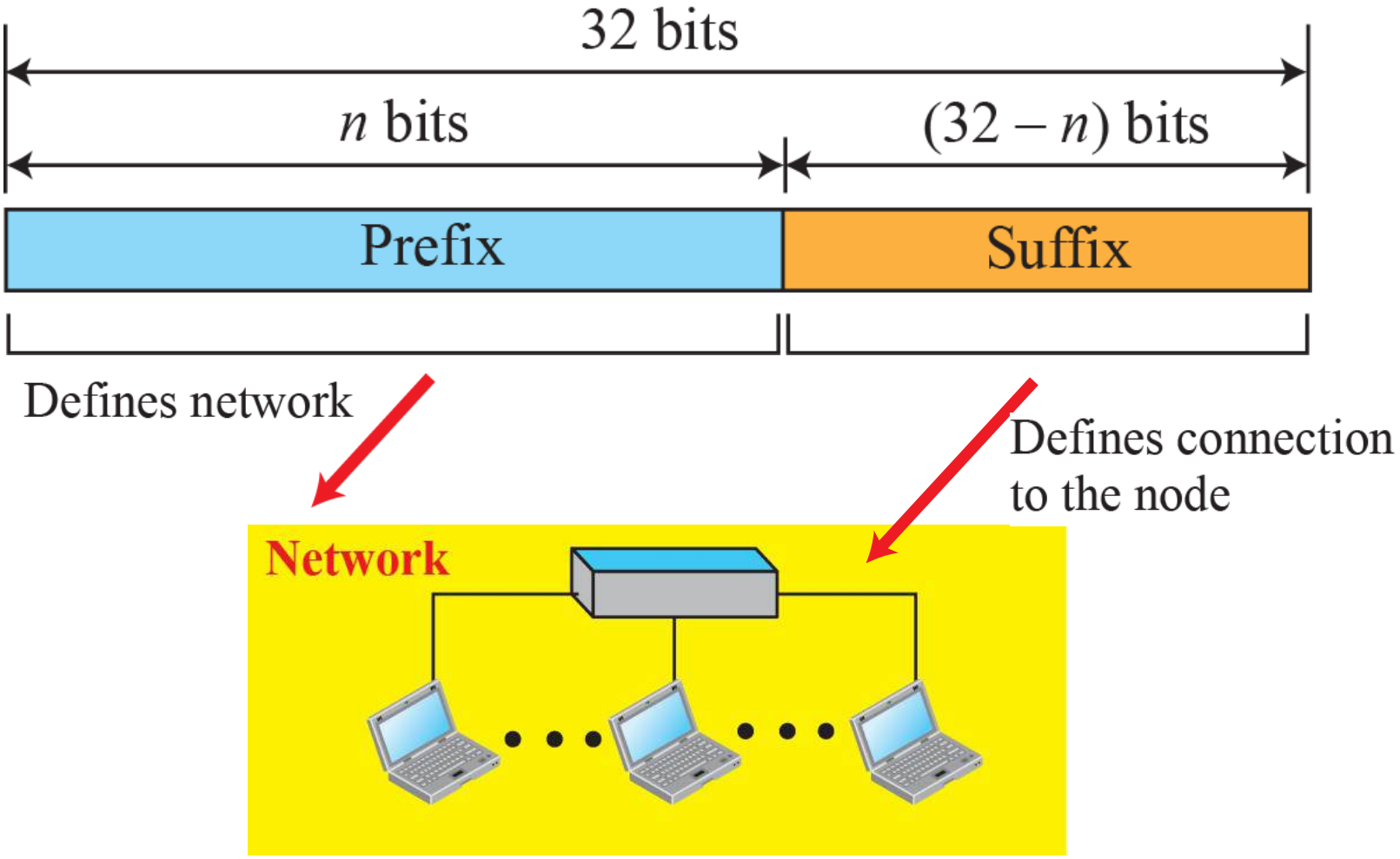

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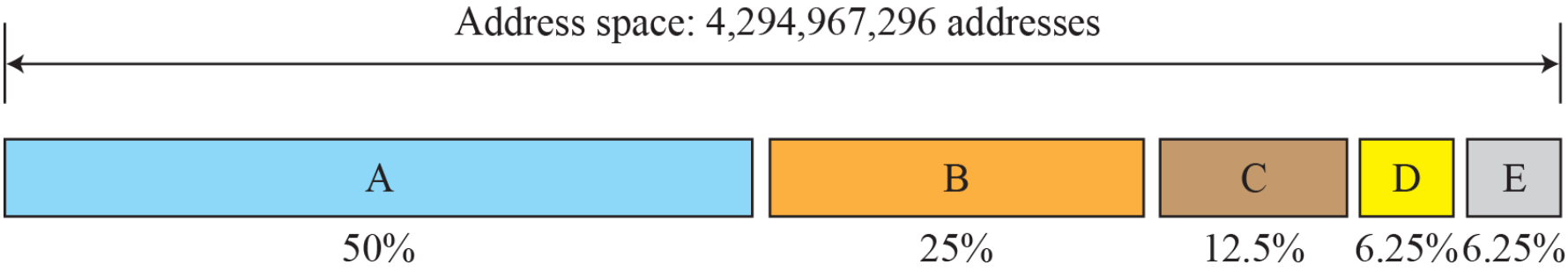
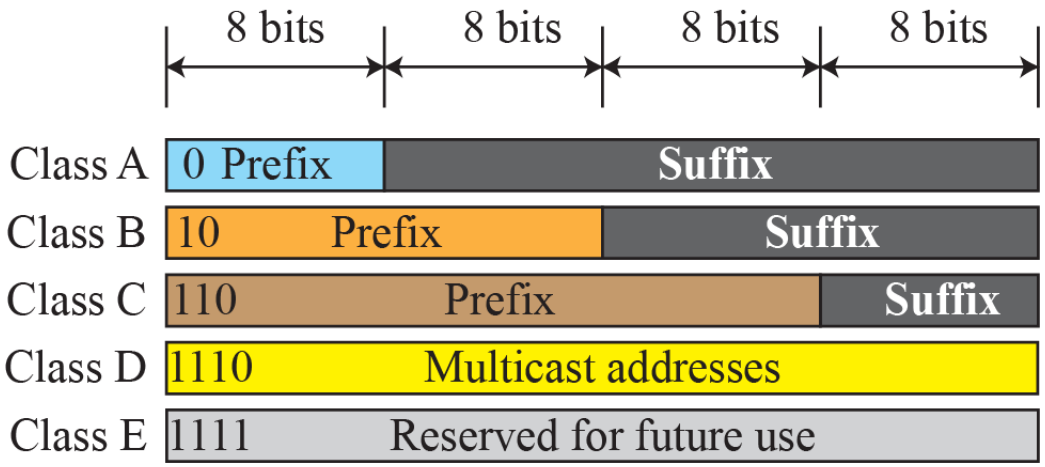
Figure 4.31: Occupation of the address space in classful addressing





Class	Prefixes	First byte
A	n = 8 bits	0 to 127
B	n = 16 bits	128 to 191
C	n = 24 bits	192 to 223
D	Not applicable	224 to 239
E	Not applicable	240 to 255

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Figure 4.32: Variable-length blocks in classless addressing

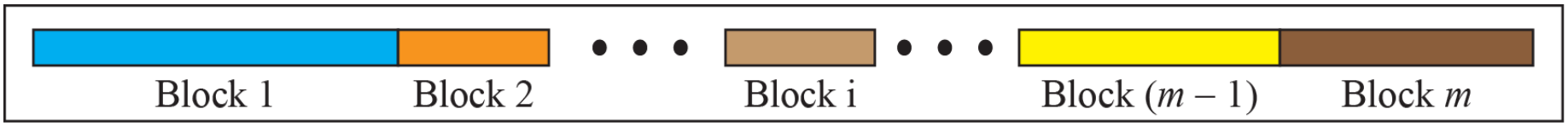

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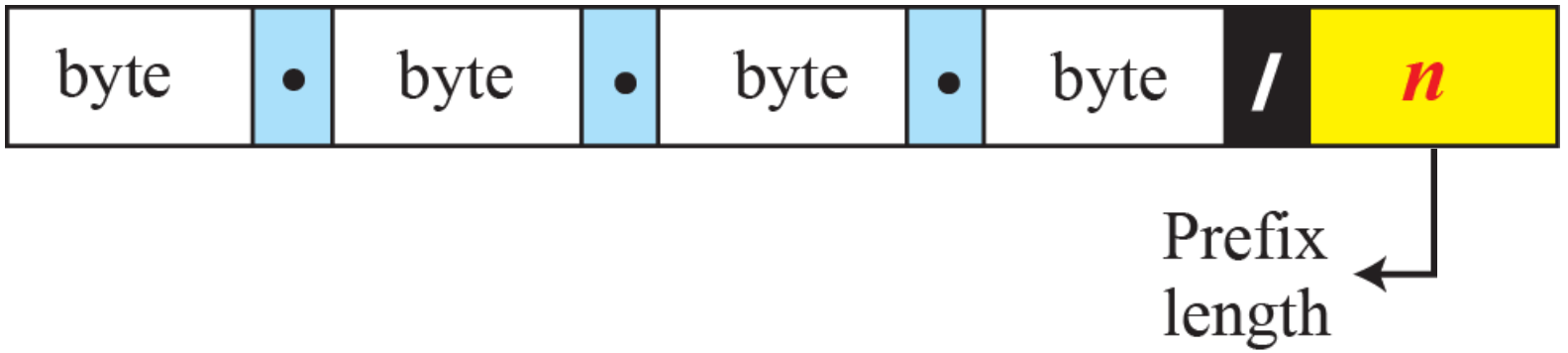
Figure 4.33: *Slash notation (CIDR)*



Examples:

12.24.76.8/**8**

23.14.67.92/**12**

220.8.24.255/**25**

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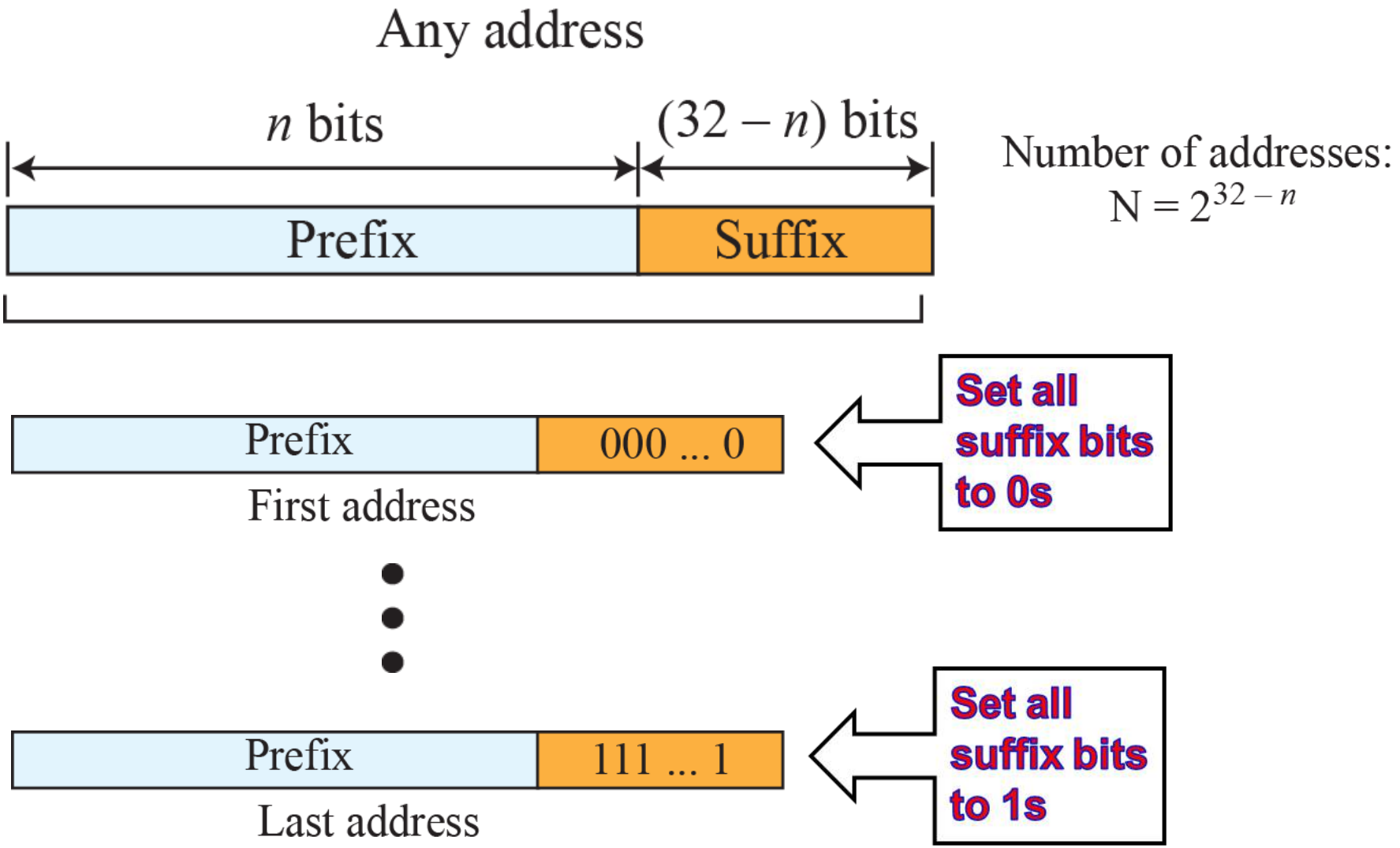
Figure 4.34: Information extraction in classless addressing



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Example 4.1

A classless address is given as 167.199.170.82/27. We can find the above three pieces of information as follows. The number of addresses in the network is $2^{32-n} = 2^5 = 32$ addresses. The first address can be found by keeping the first 27 bits and changing the rest of the bits to 0s.

Address: 167.199.170.82/27	10100111	11000111	10101010	01010010
First address: 167.199.170.64/27	10100111	11000111	10101010	01000000

The last address can be found by keeping the first 27 bits and changing the rest of the bits to 1s.

Address: 167.199.170.82/27	10100111	11000111	10101010	01011111
Last address: 167.199.170.95/27	10100111	11000111	10101010	01011111

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Example 4.2

We repeat Example 4.1 using the mask. The mask in dotted-decimal notation is 256.256.256.224 The AND, OR, and NOT operations can be applied to individual bytes using calculators and applets at the book website.

Number of addresses in the block:	N = **NOT** (mask) + 1 = 0.0.0.31 + 1 = 32 addresses
First address:	First = (address) **AND** (mask) = 167.199.170. 82
Last address:	Last = (address) **OR** (**NOT** mask) = 167.199.170. 255

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Example 4.3

In classless addressing, an address cannot per se define the block the address belongs to. For example, the address 230.8.24.56 can belong to many blocks. Some of them are shown below with the value of the prefix associated with that block.

Prefix length:16	→	Block:	230.8.0.0	to	230.8.255.255
Prefix length:20	→	Block:	230.8.16.0	to	230.8.31.255
Prefix length:26	→	Block:	230.8.24.0	to	230.8.24.63
Prefix length:27	→	Block:	230.8.24.32	to	230.8.24.63
Prefix length:29	→	Block:	230.8.24.56	to	230.8.24.63
Prefix length:31	→	Block:	230.8.24.56	to	230.8.24.57

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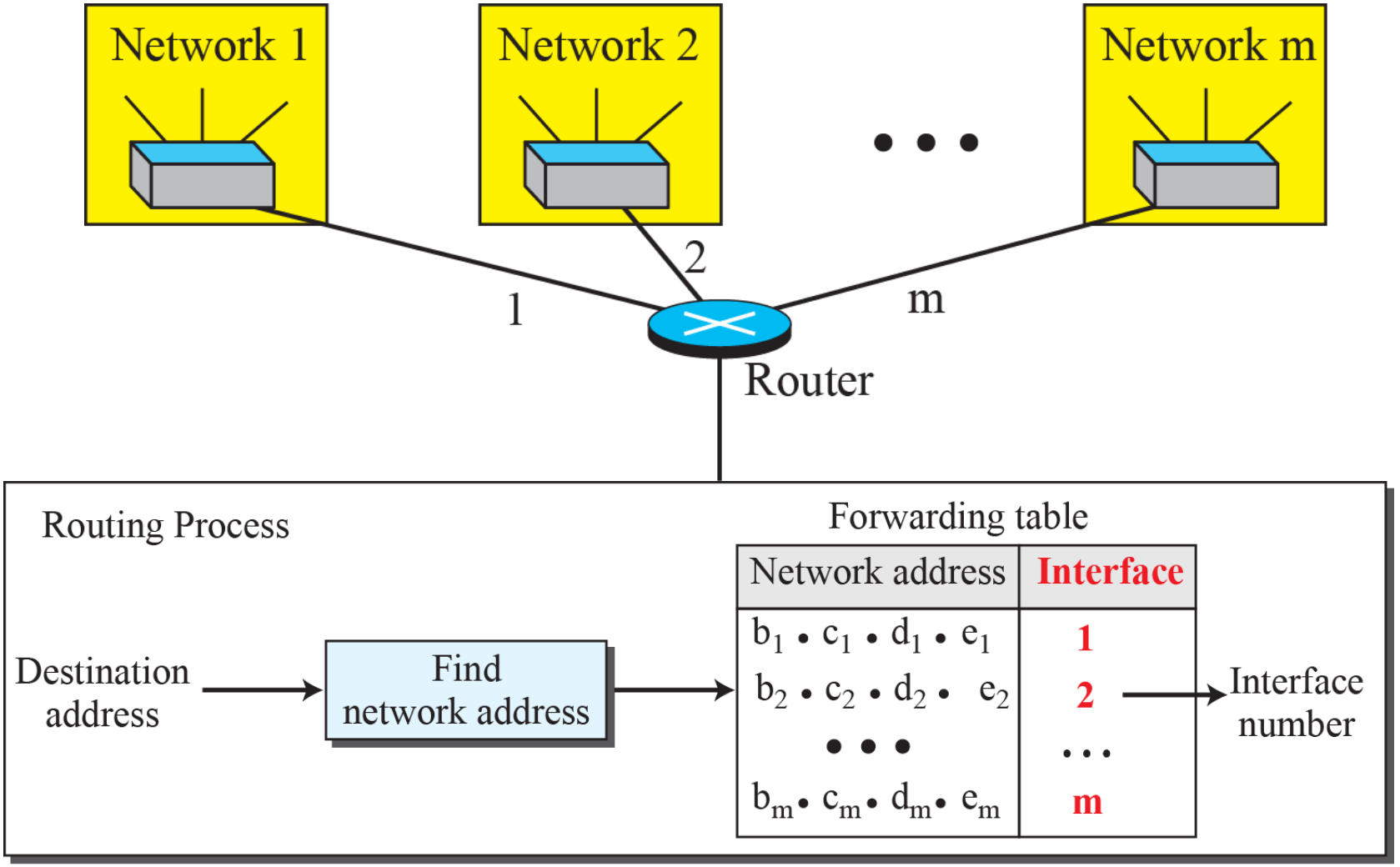
Figure 4.35: Network address



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Example 4.4

An ISP has requested a block of 1000 addresses. Since 1000 is not a power of 2, 1024 addresses are granted. The prefix length is calculated as $n = 32 - \log_2 1024 = 22$. An available block, 18.14.12.0/**22**, is granted to the ISP. It can be seen that the first address in decimal is 302,910,464, which is divisible by 1024.

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Example 4.5

An organization is granted a block of addresses with the beginning address 14.24.74.0/**24**. The organization needs to have 3 subblocks of addresses to use in its three subnets: one subblock of 10 addresses, one subblock of 60 addresses, and one subblock of 120 addresses. Design the subblocks.

Solution

There are $2^{32-24} = 256$ addresses in this block. The first address is 14.24.74.0/**24**; the last address is 14.24.74.255/**24**. To satisfy the third requirement, we assign addresses to subblocks, starting with the largest and ending with the smallest one.

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Example 4.5 (continued)

a. The number of addresses in the largest subblock, which requires 120 addresses, is not a power of 2. We allocate 128 addresses. The subnet mask for this subnet can be found as $n_1 = 32 - \log_2 128 = 25$. The first address in this block is 14.24.74.0/**25**; the last address is 14.24.74.127/**25**.

b. The number of addresses in the second largest subblock, which requires 60 addresses, is not a power of 2 either. We allocate 64 addresses. The subnet mask for this subnet can be found as $n_2 = 32 - \log_2 64 = 26$. The first address in this block is 14.24.74.128/**26**; the last address is 14.24.74.191/**26**.

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Example 4.5 (continued)

c. The number of addresses in the largest subblock, which requires 120 addresses, is not a power of 2. We allocate 128 addresses. The subnet mask for this subnet can be found as $n_1 = 32 - \log_2 128 = 25$. The first address in this block is 14.24.74.0**/25**; the last address is 14.24.74.127**/25**.

If we add all addresses in the previous subblocks, the result is 208 addresses, which means 48 addresses are left in reserve. The first address in this range is 14.24.74.208. The last address is 14.24.74.255. We don't know about the prefix length yet. Figure 4.36 shows the configuration of blocks. We have shown the first address in each block.

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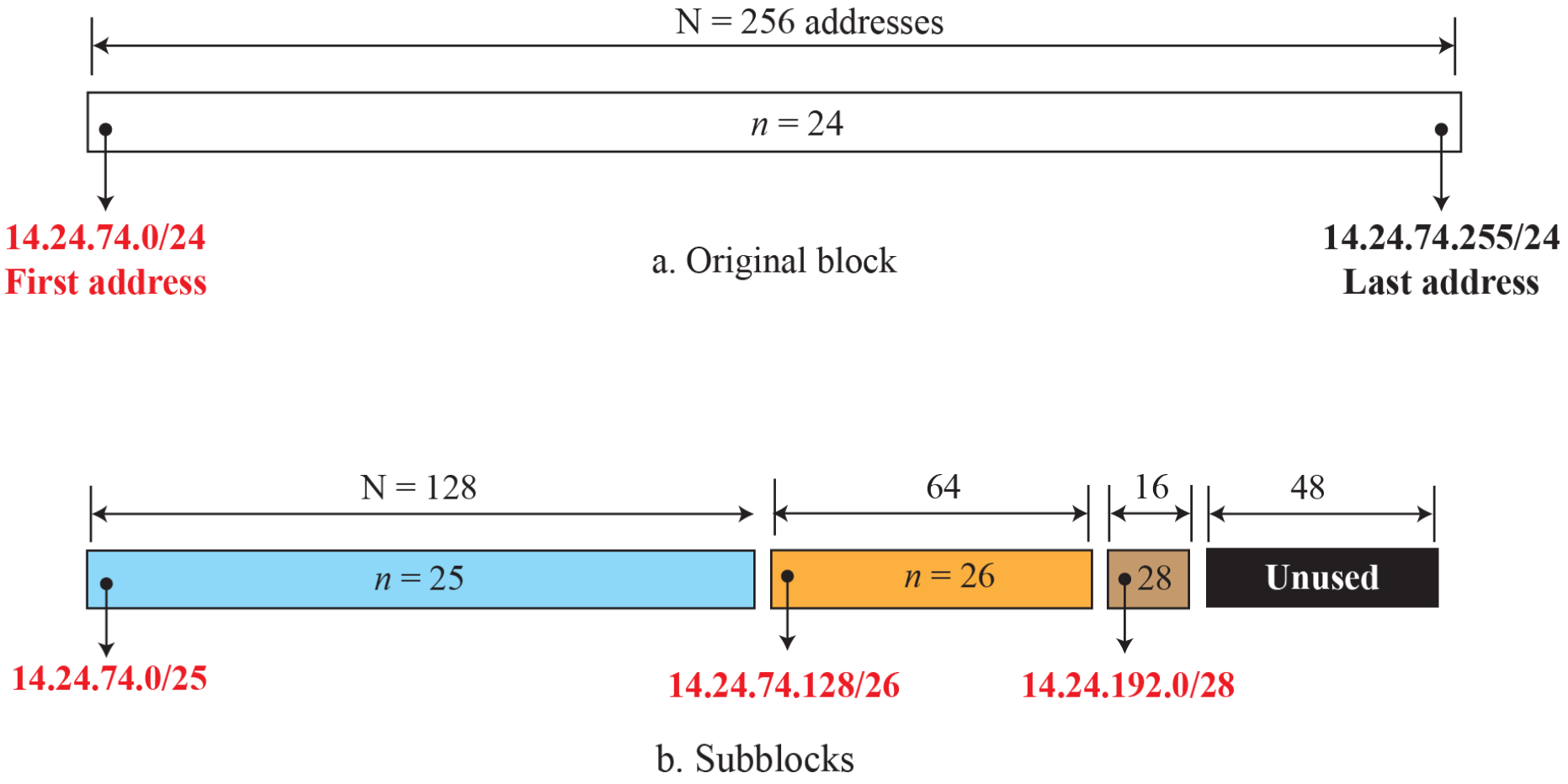
Figure 4.36: Solution to Example 4.5



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Example 4.6

Figure 4.37 shows how four small blocks of addresses are assigned to four organizations by an ISP. The ISP combines these four blocks into one single block and advertises the larger block to the rest of the world. Any packet destined for this larger block should be sent to this ISP. It is the responsibility of the ISP to forward the packet to the appropriate organization. This is similar to routing we can find in a postal network. All packages coming from outside a country are sent first to the capital and then distributed to the corresponding destination.

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Figure 4.37: Example of address aggregation

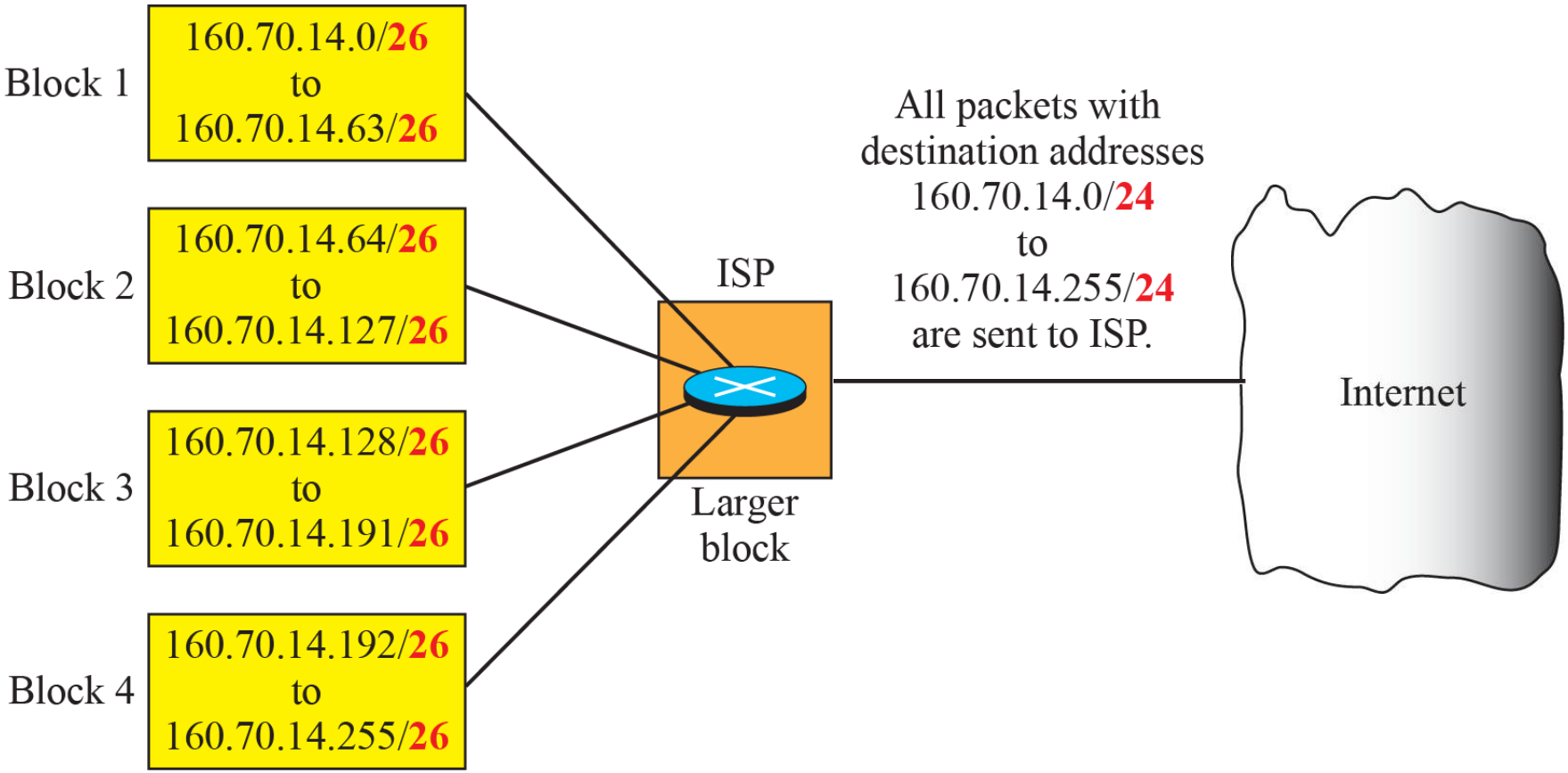

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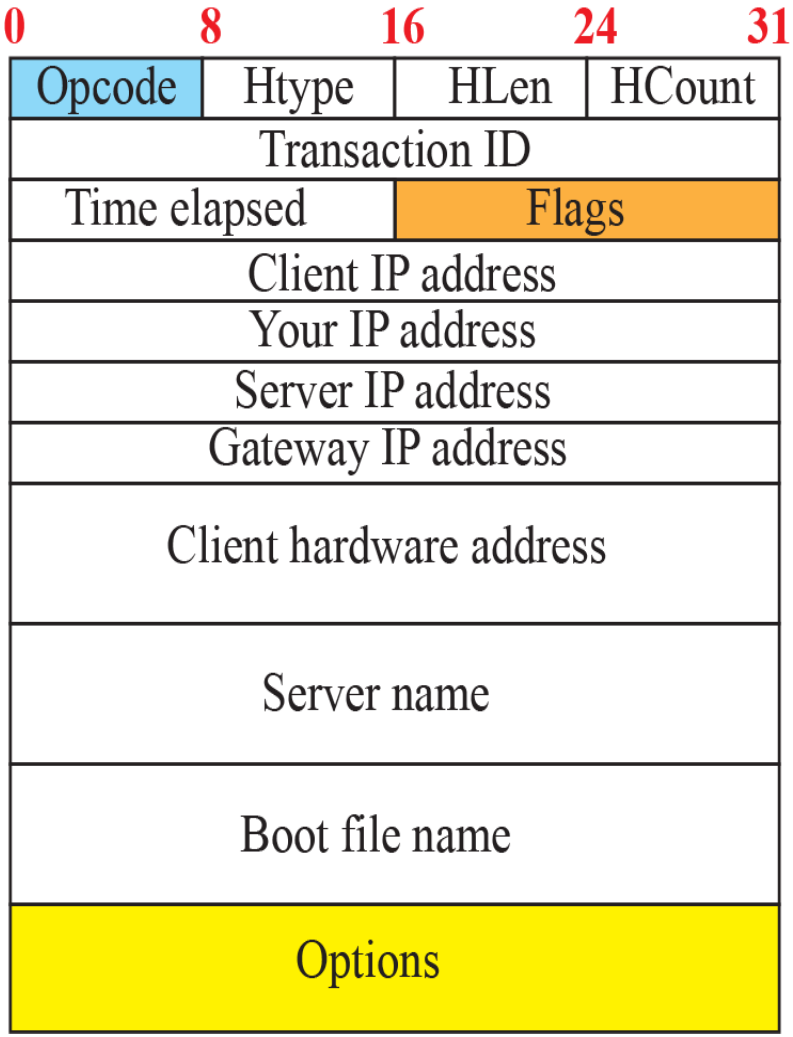
Figure 4.38: DHCP message format



Fields:

Opcode: Operation code, request (1) or reply (2)
Htype: Hardware type (Ethernet, ...)
HLen: Lengh of hardware address
HCount: Maximum number of hops the packet can travel
Transaction ID: An integer set by client and repeated by the server
Time elapsed: The number of seconds since the client started to boot
Flags: First bit defines unicast (0) or multicast (1); other 15 bits not used
Client IP address: Set to 0 if the client does not know it
Your IP address: The client IP address sent by the server
Server IP address: A broadcast IP address if client does not know it
Gateway IP address: The address of default router
Server name: A 64-byte domain name of the server
Boot file name: A 128-byte file name holding extra information
Options: A 64-byte field with dual purpose described in text

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Figure 4.39: ***Option format***

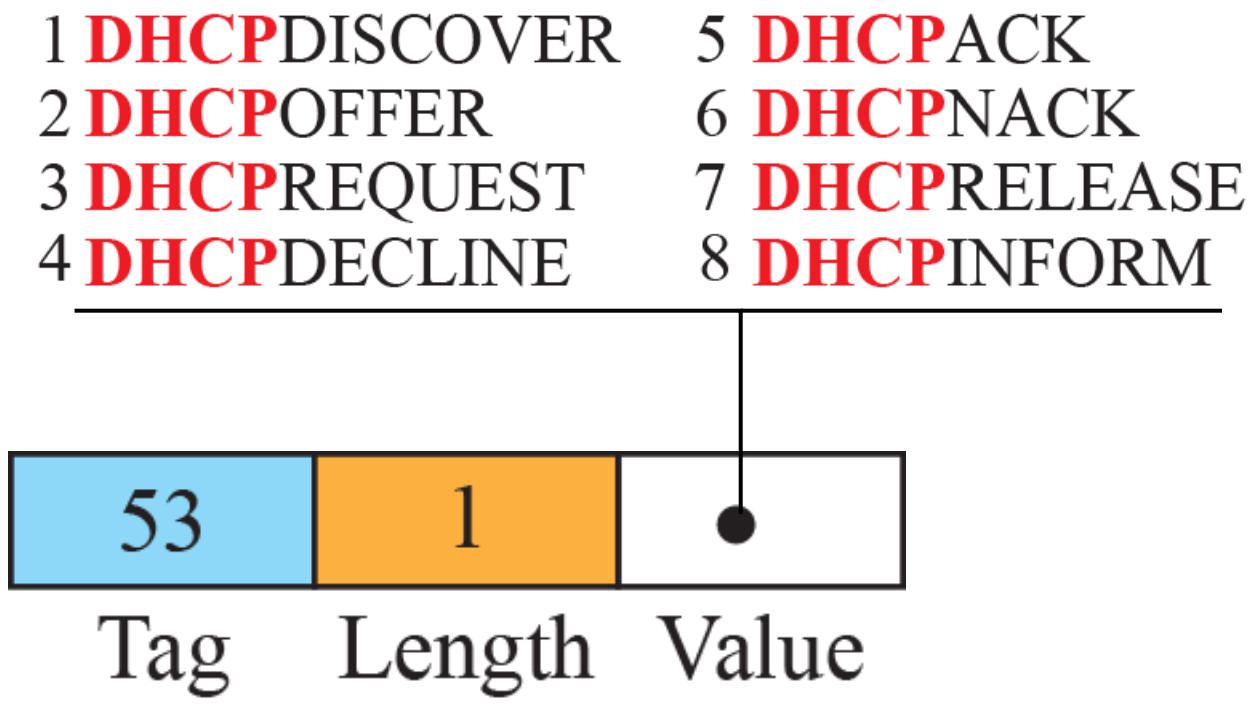

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Figure 4.40: Operation of DHCP

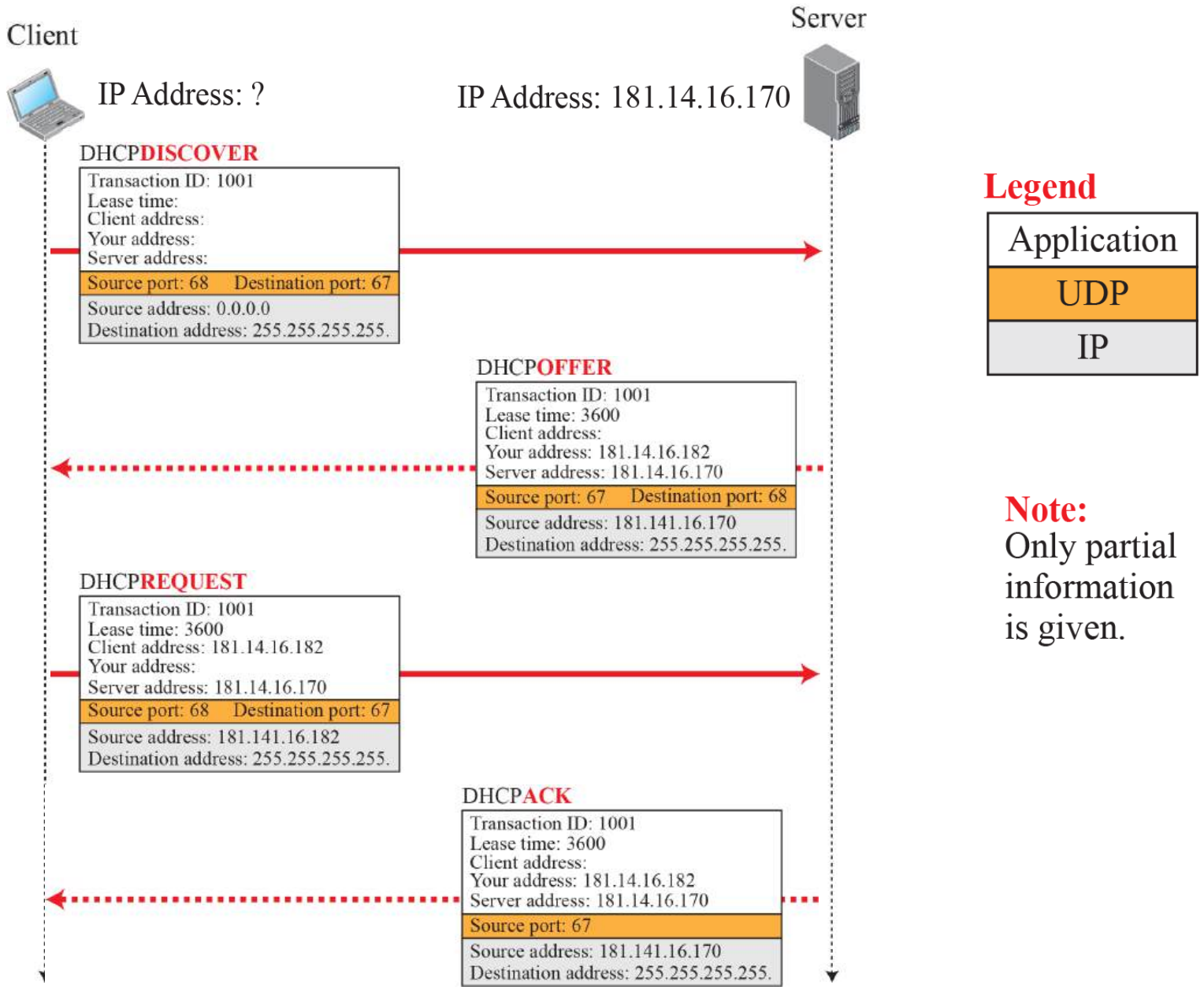

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Figure 4.41: FSM for the DHCP client

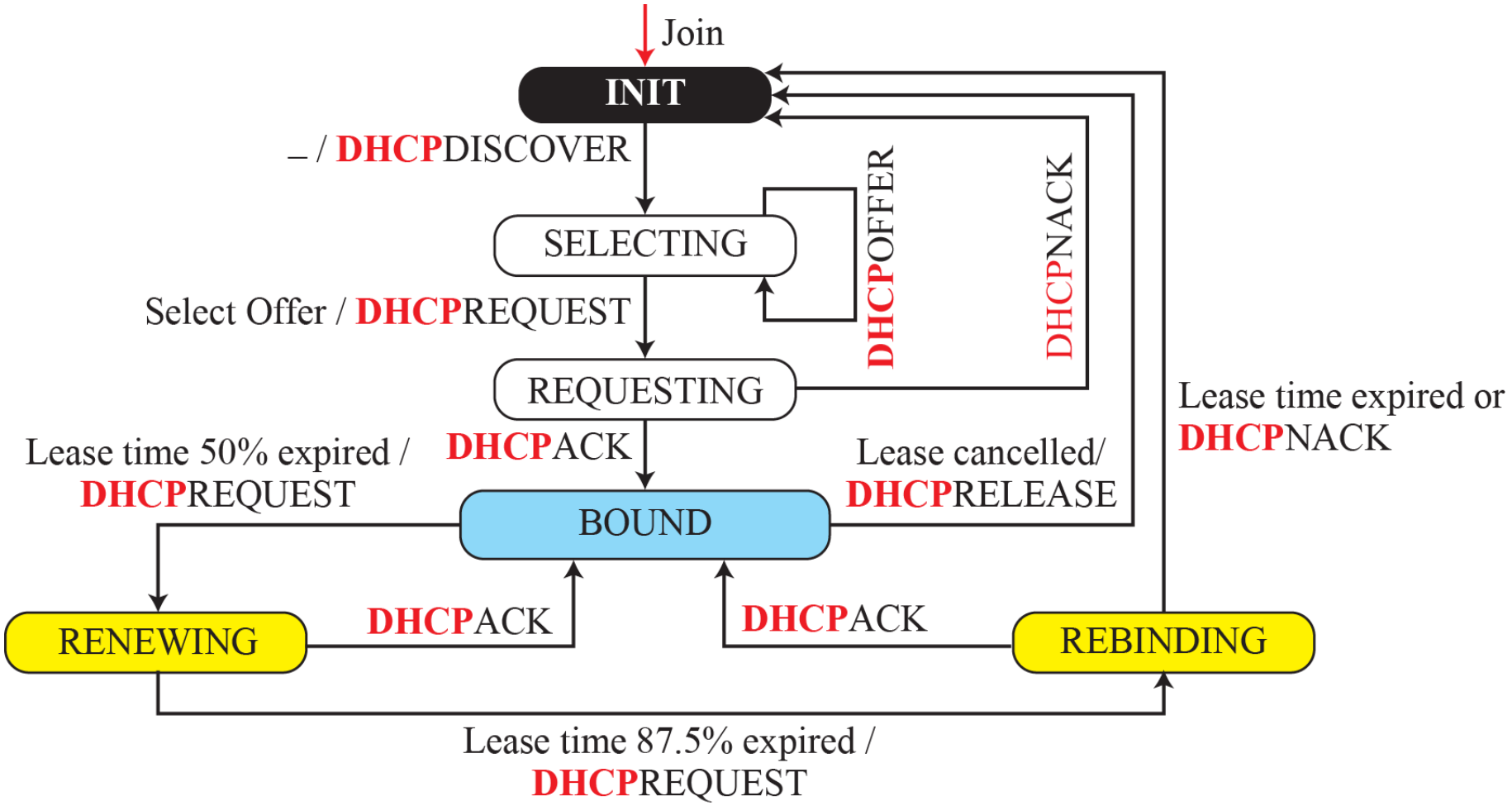

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Figure 4.42: NAT

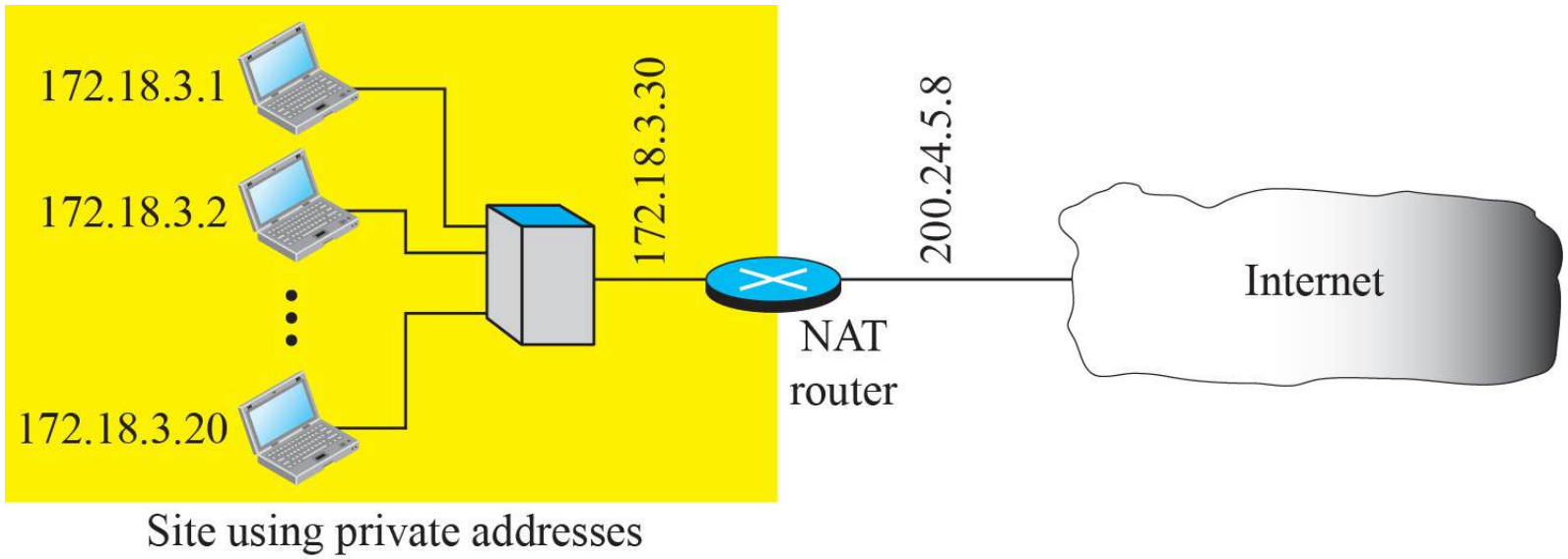

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Figure 4.43: Address translation

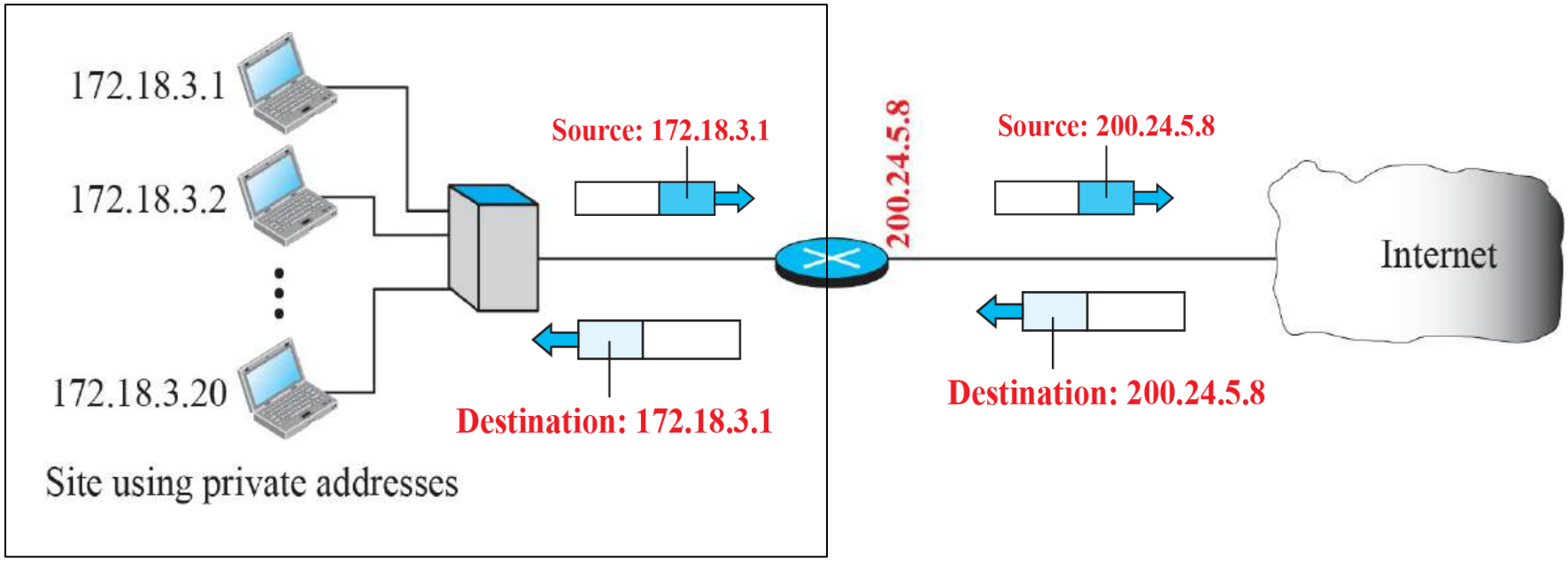

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Figure 4.44: Translation

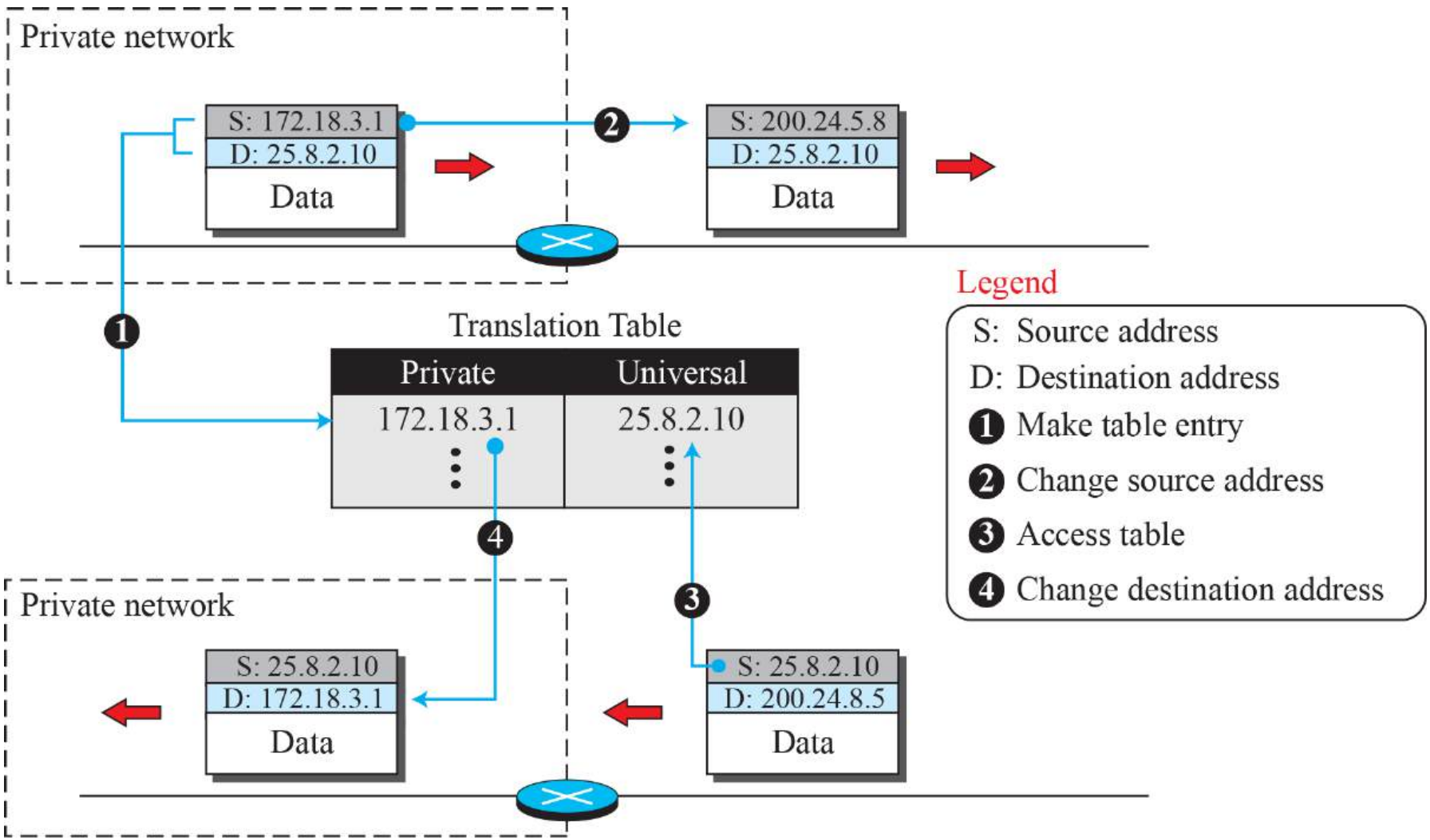

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Table 4.1: Five-column translation table

Private address	*Private port*	*External address*	*External port*	*Transport protocol*
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
⋮	⋮	⋮	⋮	⋮

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4.2.3 Forwarding of IP Packets

We discussed the concept of forwarding at the network layer earlier in this chapter. In this section, we extend the concept to include the role of IP addresses in forwarding. As we discussed before, forwarding means to place the packet in its route to its destination. Since the Internet today is made of a combination of links (networks), forwarding means to deliver the packet to the next hop (which can be the final destination or the intermediate connecting device).

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4.2.3 (continued)

- **Forwarding Based On Destination Address**
  - Address Aggregation
  - Longest Mask Matching
  - Hierarchical Routing
  - Geographical Routing
  - Forwarding Table Search Algorithms
- **Forwarding Based on Label**
  - Multi-Protocol Label Switching (MPLS)
  - A New Header
  - Hierarchical Routing

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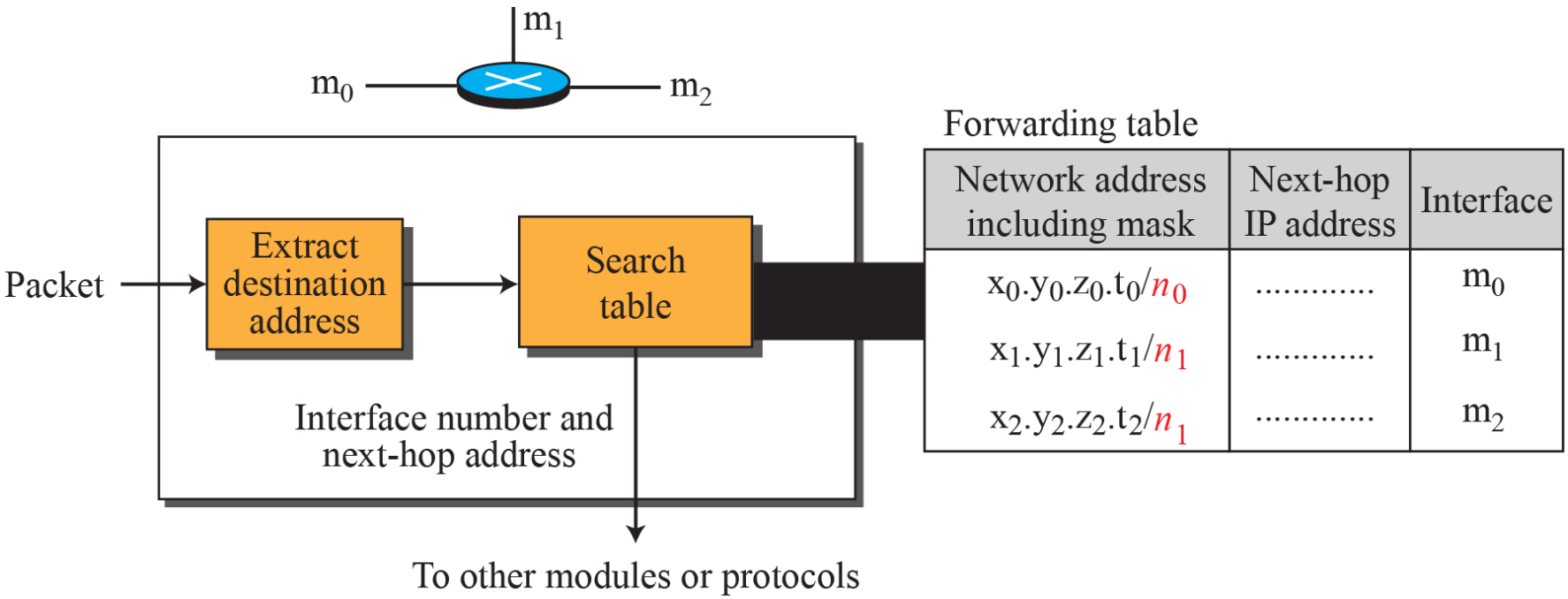
Figure 4.45: Simplified forwarding module in classless address



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Example 4.7

Make a forwarding table for router R1 using the configuration in Figure 4.46.

Solution

Table 4.2 shows the corresponding table.

Table 4.2: Forwarding table for router R1 in Figure 4.46

Network address/mask	*Next hop*	*Interface*
180.70.65.192/**26**	—	m2
180.70.65.128/**25**	—	m0
201.4.22.0/**24**	—	m3
201.4.16.0/**22**	—	m1
Default	180.70.65.200	m2

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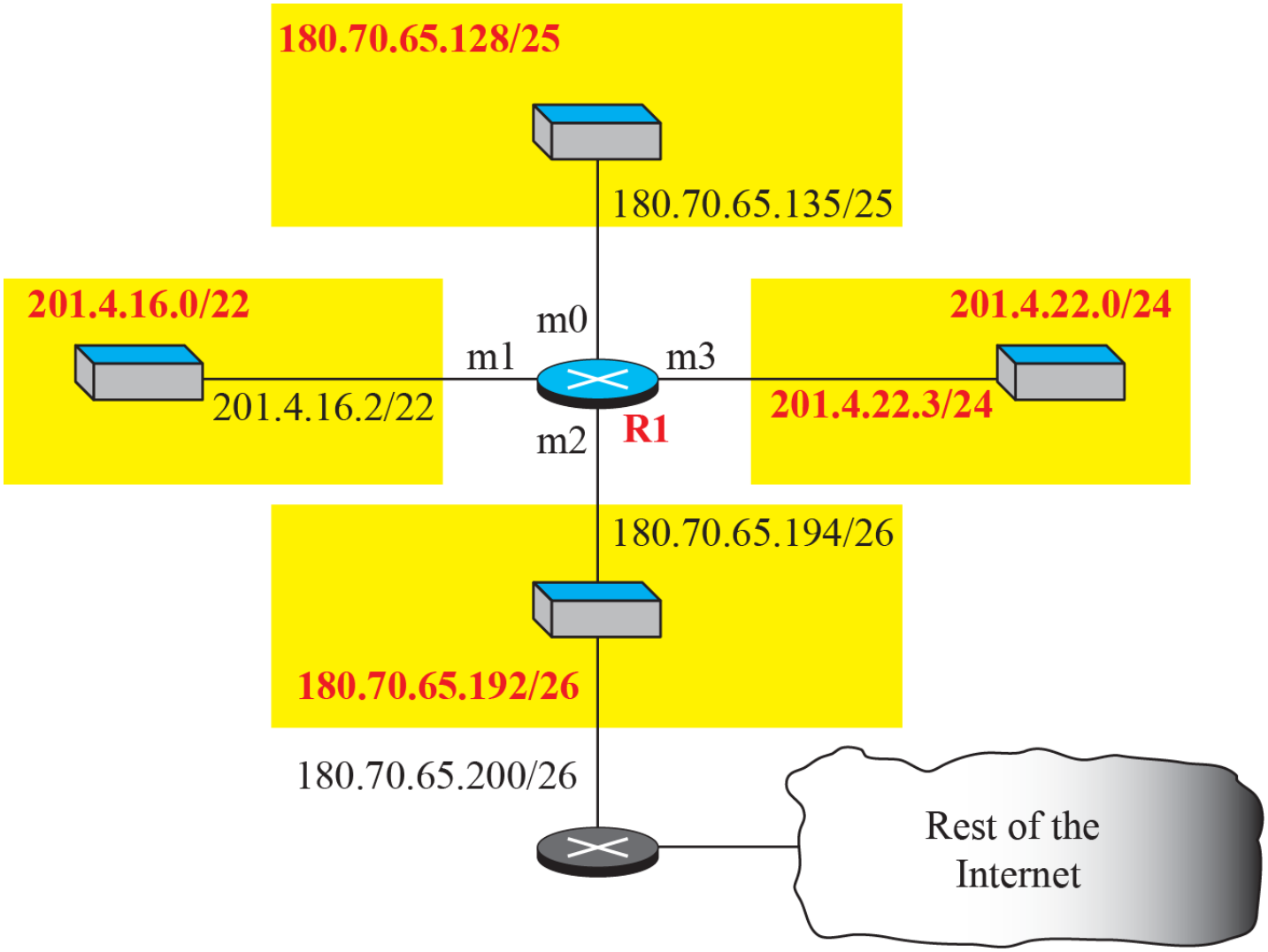
Figure 4.46: Configuration for Example 4.7



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Example 4.8

Instead of Table 4.2, we can use Table 4.3, in which the network address/mask is given in bits.

Table 4.3: Forwarding table for router R1 using prefix bits

Leftmost bits in the destination address	*Next hop*	*Interface*
10110100 01000110 01000001 11	—	m2
10110100 01000110 01000001 1	—	m0
11001001 00000100 00011100	—	m3
11001001 00000100 000100	—	m1
Default	180.70.65.200	m2

When a packet arrives whose leftmost 26 bits in the destination address match the bits in the first row, the packet is sent out from interface m2. And so on.

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Example 4.9

Show the forwarding process if a packet arrives at R1 in Figure 4.46 with the destination address 180.70.65.140.

Solution

The router performs the following steps:

1. The first mask (/26) is applied to the destination address. The result is 180.70.65.128, which does not match the corresponding network address.
2. The second mask (/25) is applied to the destination address. The result is 180.70.65.128, which matches the corresponding network address. The next-hop address and the interface number m0 are extracted for forwarding the packet (see Chapter 5).

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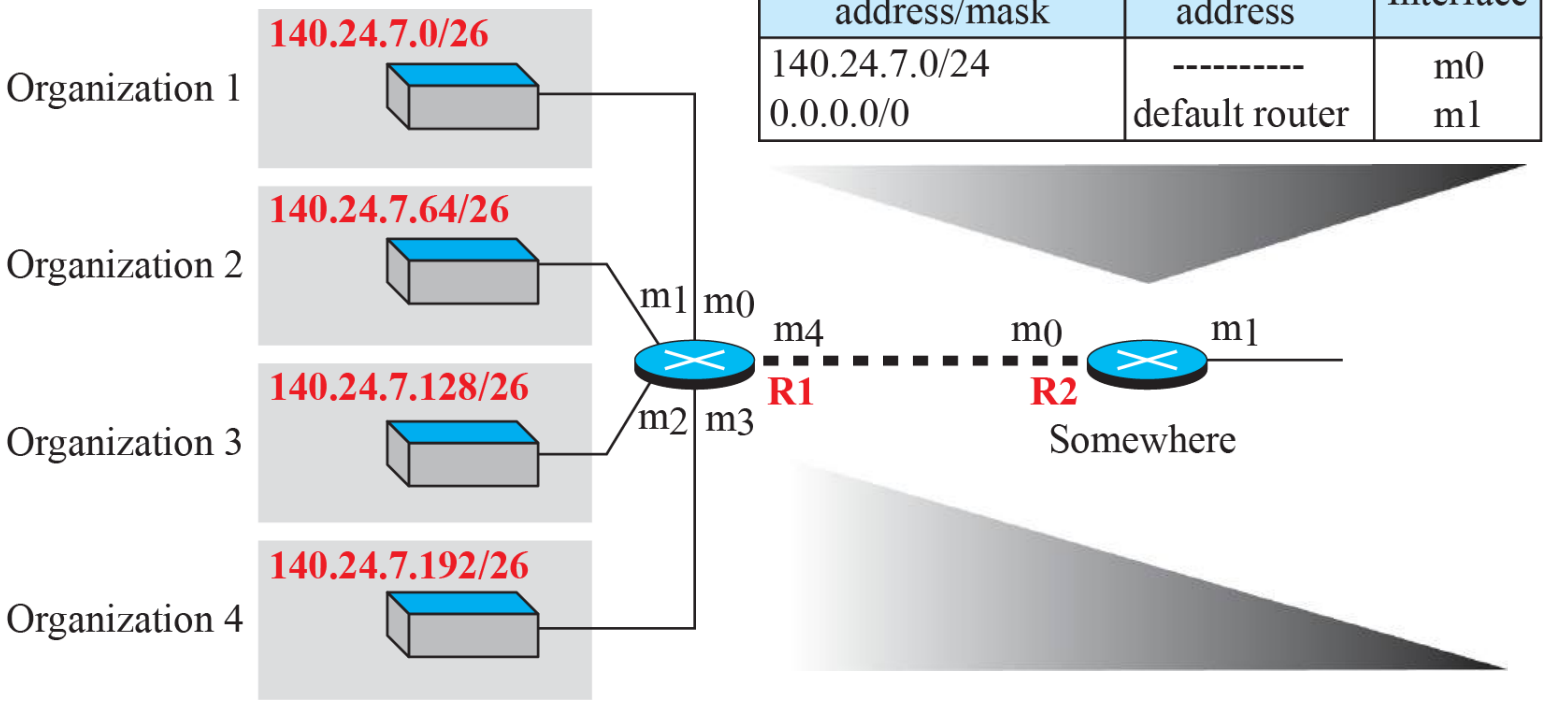
Figure 4.47: Address aggregation



Forwarding table for R2

Network address/mask	Next-hop address	Interface
140.24.7.0/24	----------	m0
0.0.0.0/0	default router	m1

Forwarding table for R1

Network address/mask	Next-hop address	Interface
140.24.7.0/26	----------	m0
140.24.7.64/26	----------	m1
140.24.7.128/26	----------	m2
140.24.7.192/26	----------	m3
0.0.0.0/0	address of R2	m4

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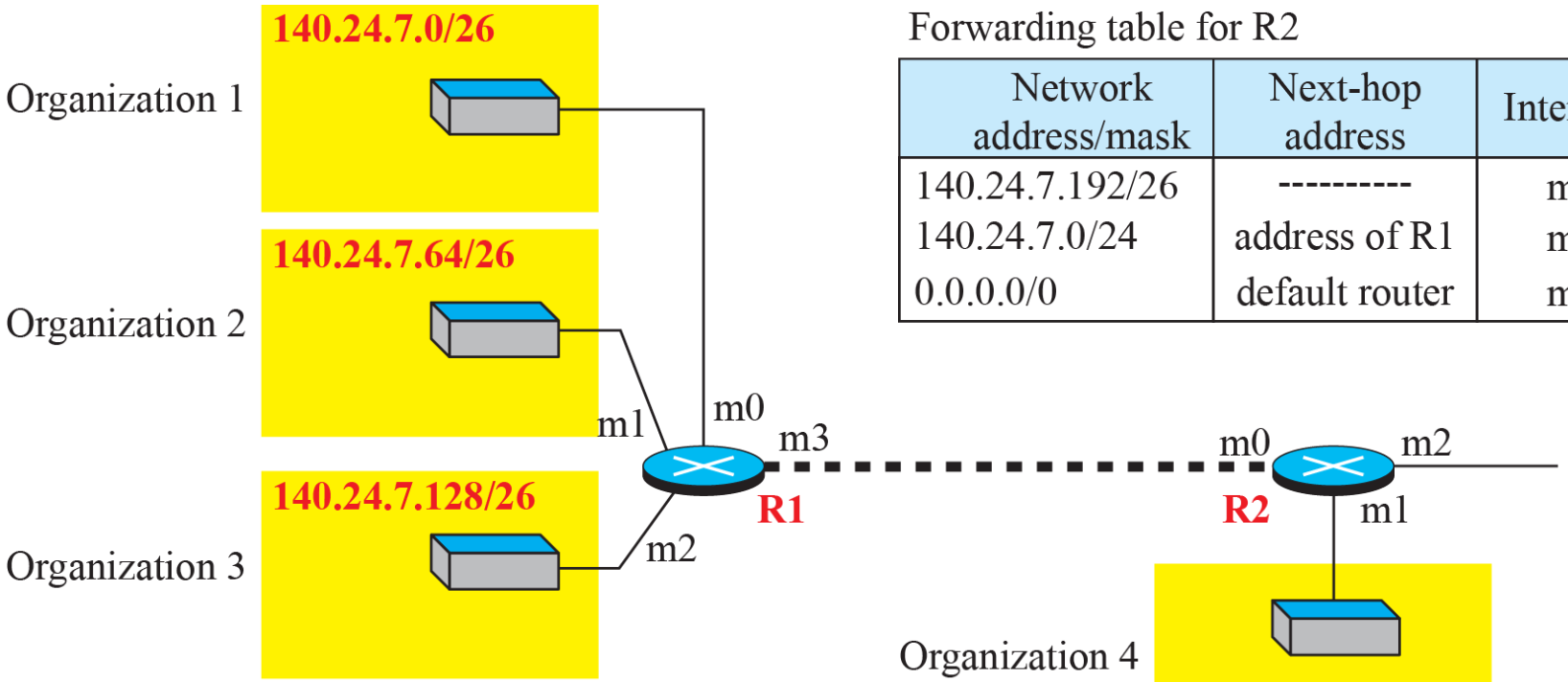
Figure 4.48: Longest mask matching



Forwarding table for R2

Network address/mask	Next-hop address	Interface
140.24.7.192/26	----------	m1
140.24.7.0/24	address of R1	m0
0.0.0.0/0	default router	m2

Forwarding table for R1

Network address/mask	Next-hop address	Interface
140.24.7.0/26	----------	m0
140.24.7.64/26	----------	m1
140.24.7.128/26	----------	m2
0.0.0.0/0	default router	m3

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Example 4.10

As an example of hierarchical routing, let us consider Figure 4.49. A regional ISP is granted 16,384 addresses starting from 120.14.64.0. The regional ISP has decided to divide this block into 4 subblocks, each with 4096 addresses. Three of these subblocks are assigned to three local ISPs, the second subblock is reserved for future use. Note that the mask for each block is **/20** because the original block with mask /18 is divided into 4 blocks.

The figure also shows how local and small ISPs have assigned addresses.

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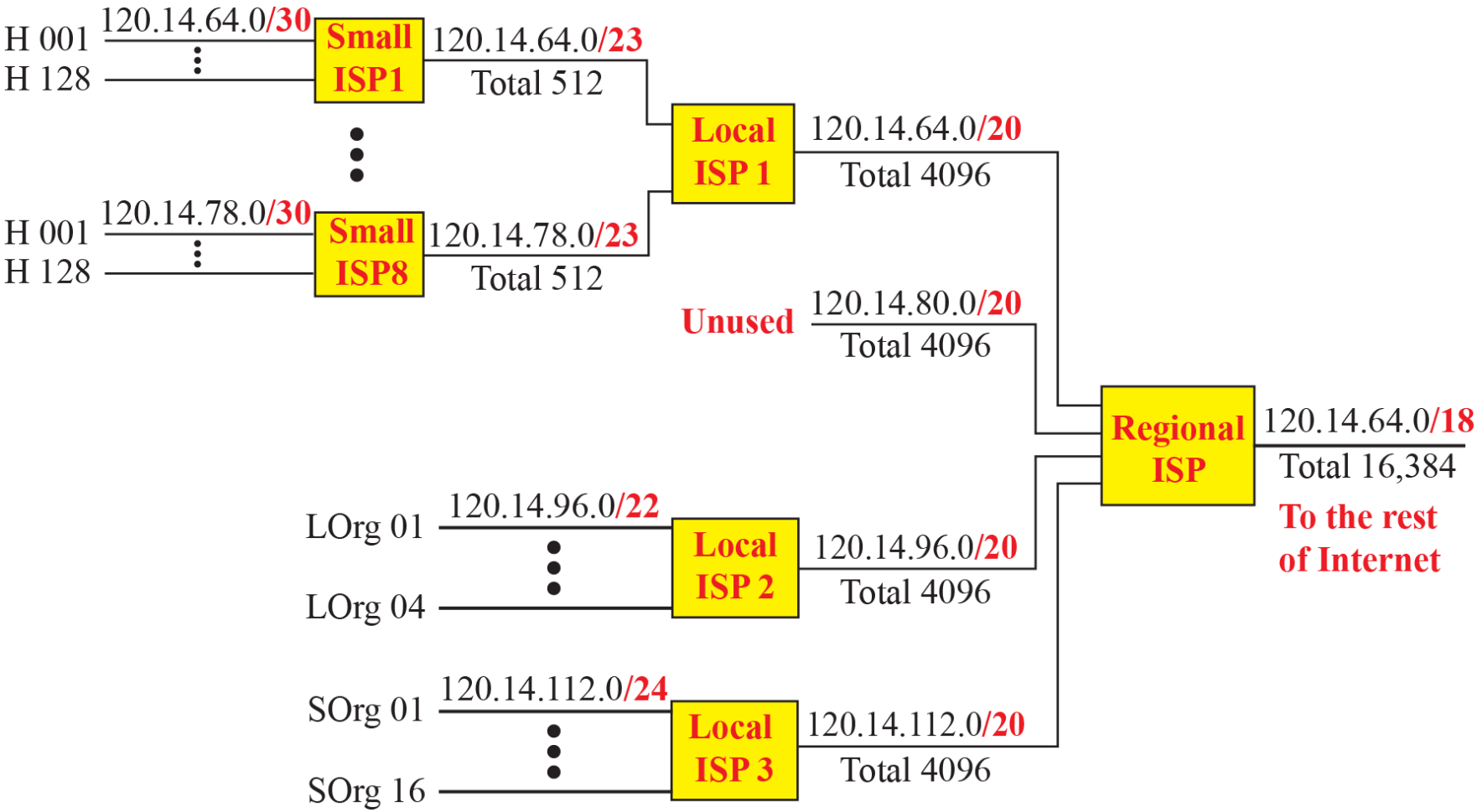
Figure 4.49: Hierarchical routing with ISPs



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Example 4.11

Figure 4.50 shows a simple example of searching in a forwarding table using the longest mask algorithm. Although there are some more efficient algorithms today, the principle is the same.

When the forwarding algorithm gets the destination address of the packet, it needs to delve into the mask column. For each entry, it needs to apply the mask to find the destination network address. It then needs to check the network addresses in the table until it finds the match. The router then extracts the next-hop address and the interface number to be delivered to the data-link layer.

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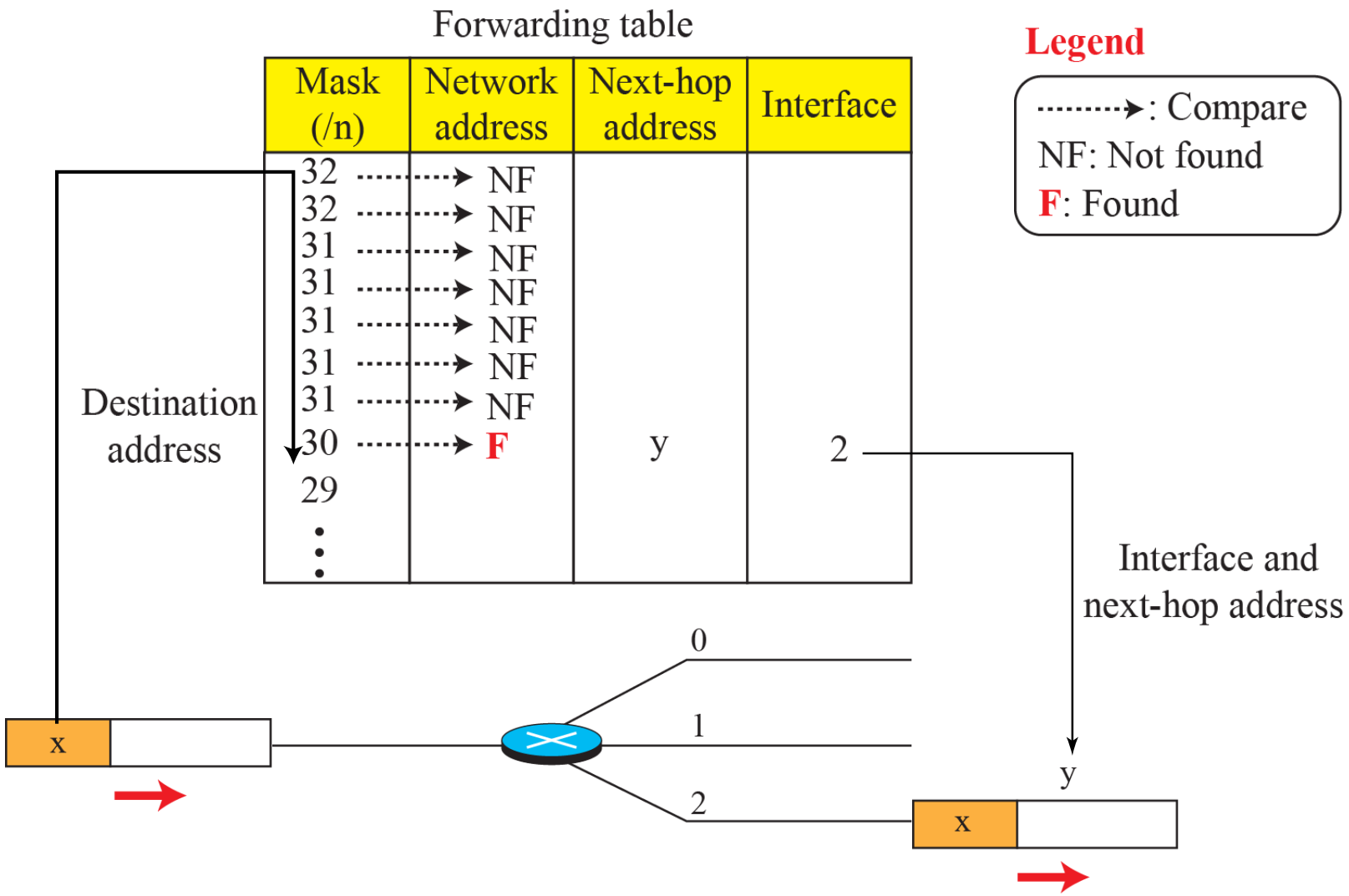
Figure 4.50: Example 4.11: Forwarding based on destination address



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Example 4.12

Figure 4.51 shows a simple example of using a label to access a switching table. Since the labels are used as the index to the table, finding the information in the table is immediate.

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Figure 4.51: Example 4.12: Forwarding based on label

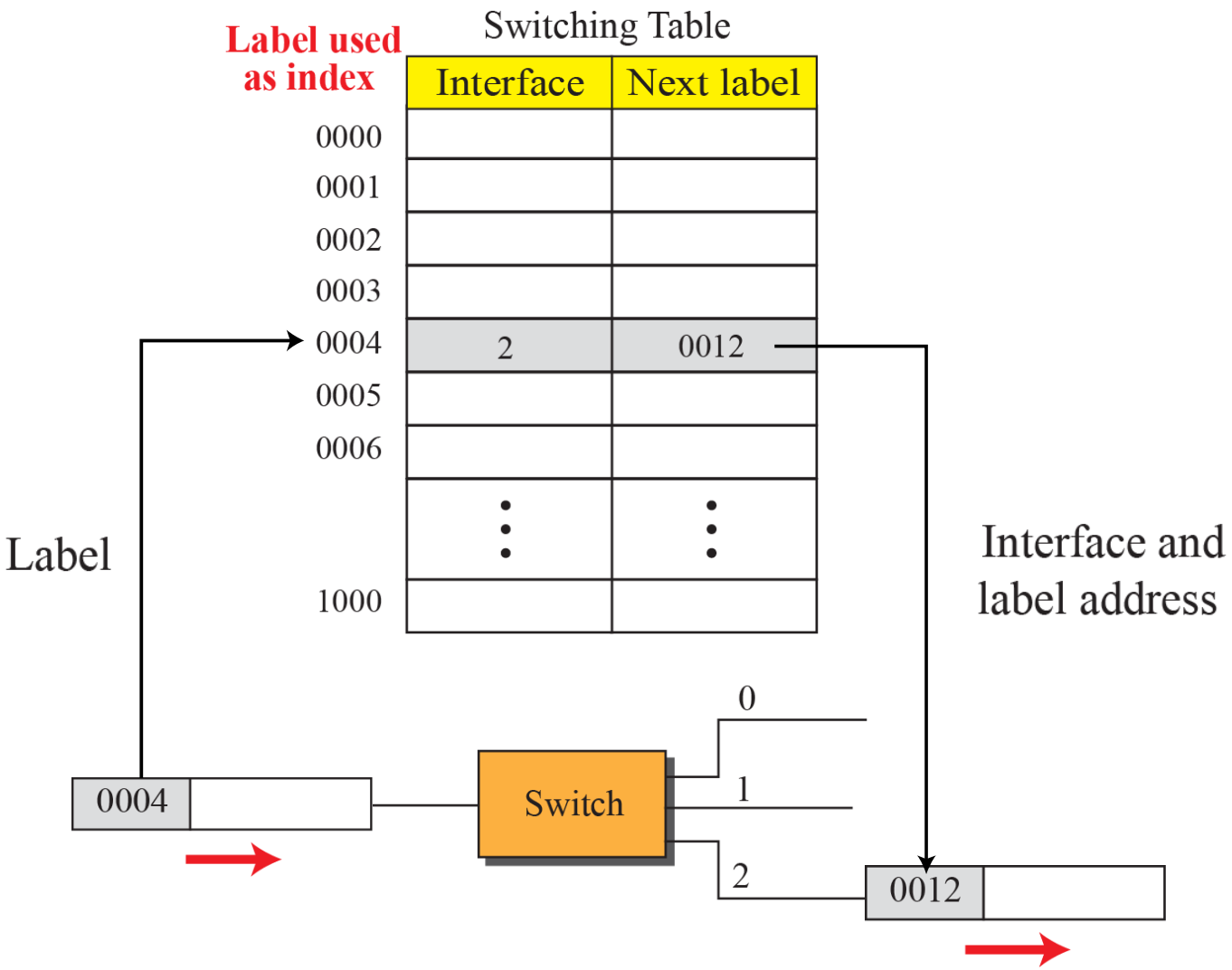

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Figure 4.52: MPLS header added to an IP packet

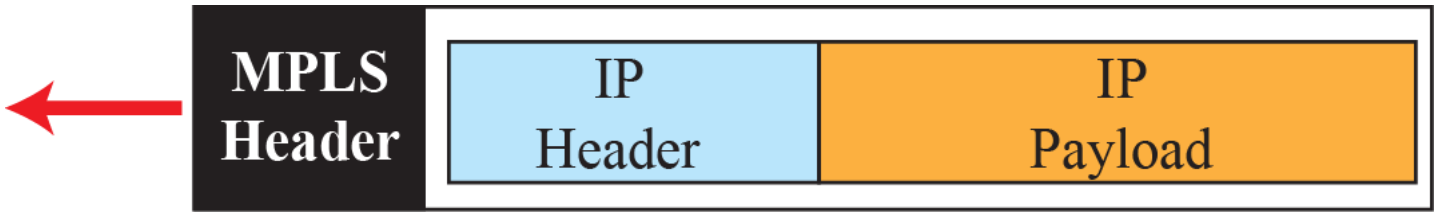

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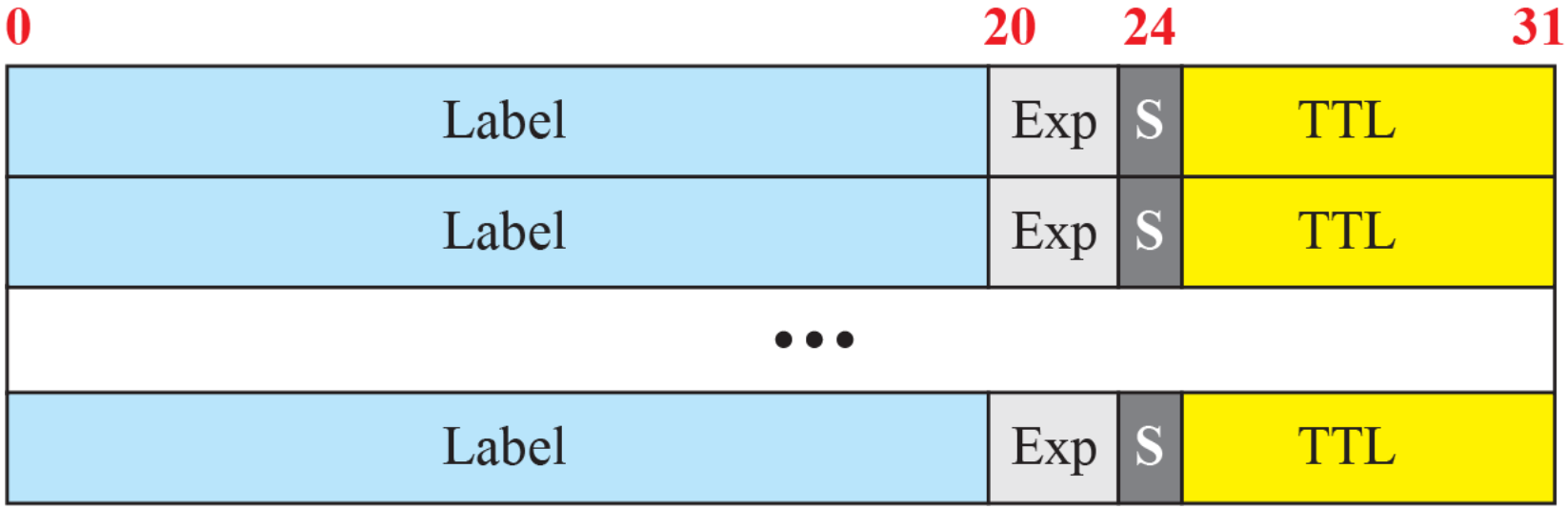
Figure 4.53: MPLS header made of a stack of labels



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4.2.4 ICMPv4

The IPv4 has no error-reporting or error-correcting mechanism. What happens if something goes wrong? There are examples of situations where an error has occurred and the IP protocol has no built-in mechanism to notify the original host.

The IP protocol also lacks a mechanism for host and management queries.

The Internet Control Message Protocol version 4 (ICMPv4) has been designed to compensate for the above two deficiencies.

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4.2.4 (continued)

- **Messages**
  - Message Format
  - Error Reporting Messages
  - Query Messages

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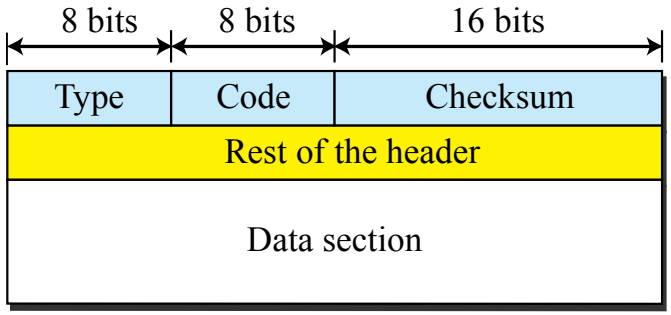
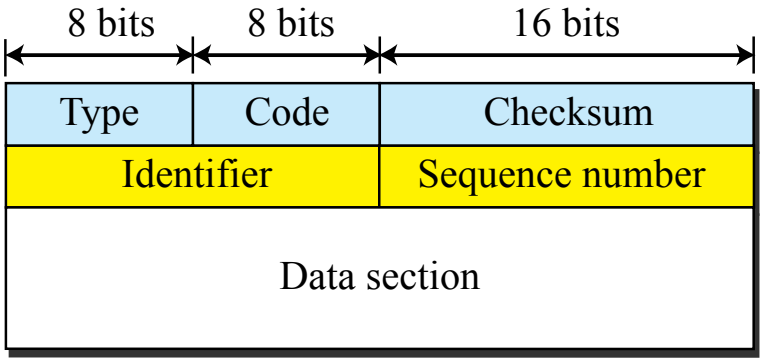
Figure 4.54: General format of ICMP messages



Error-reporting messages



Query messages

Type and code values

Error-reporting messages

03: Destination unreachable (codes 0 to 15)
04: Source quench (only code 0)
05: Redirection (codes 0 to 3)
11: Time exceeded (codes 0 and 1)
12: Parameter problem (codes 0 and 1)

Query messages

08 and 00: Echo request and reply (only code 0)
13 and 14: Timestamp request and reply (only code 0)

Note: See the book website for more explanation about the code values.

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Example 4.13

One of the tools that a host can use to test the liveliness of another host is the ping program. The ping program takes advantage of the ICMP echo request and echo reply messages. A host can send an echo request (type 8, code 0) message to another host, which, if alive, can send back an echo reply (type 0, code 0) message. To some extent, the ping program can also measure the reliability and congestion of the router between the two hosts by sending a set of request-reply messages.

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Example 4.13 (continued)

The following shows how we send a ping message to the *auniversity.edu site*.

```
$ ping auniversity.edu
PING auniversity.edu (152.181.8.3)  56 (84)  bytes of data.
64 bytes from auniversity.edu (152.181.8.3): icmp_seq=0   ttl=62   time=1.91 ms
64 bytes from auniversity.edu (152.181.8.3): icmp_seq=1   ttl=62   time=2.04 ms
64 bytes from auniversity.edu (152.181.8.3): icmp_seq=2   ttl=62   time=1.90 ms
64 bytes from auniversity.edu (152.181.8.3): icmp_seq=3   ttl=62   time=1.97 ms
64 bytes from auniversity.edu (152.181.8.3): icmp_seq=4   ttl=62   time=1.93 ms
64 bytes from auniversity.edu (152.181.8.3): icmp_seq=5   ttl=62   time=2.00 ms
--- auniversity.edu statistics ---
6 packets transmitted, 6 received, 0% packet loss
rtt min/avg/max = 1.90/1.95/2.04 ms
```


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Example 4.14

The *traceroute* program in UNIX or *tracert* in Windows can be used to trace the path of a packet from a source to the destination. It can find the IP addresses of all the routers that are visited along the path. The program is usually set to check for the maximum of 30 hops (routers) to be visited. The number of hops in the Internet is normally less than this. The traceroute program is different from the ping program. The ping program gets help from two query messages; the traceroute program gets help from two error-reporting messages: time-exceeded and destination-unreachable

Figure 4.55 shows an example in which $n = 3$.

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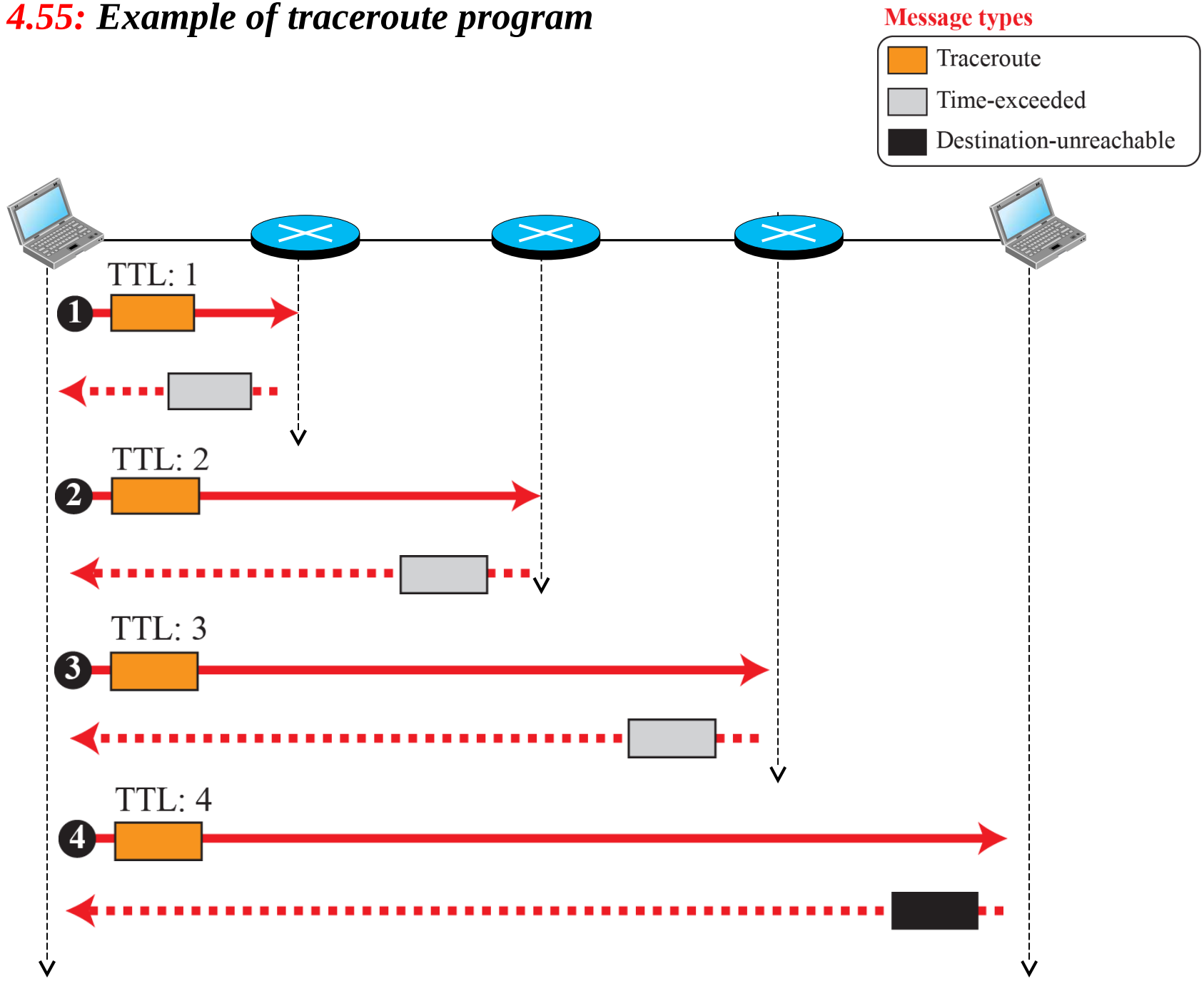
Figure 4.55: *Example of traceroute program*



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Example 4.14 (continued)

The *traceroute* program also sets a timer to find the round-trip time for each router and the destination. Most *traceroute* programs send three messages to each device, with the same TTL value, to be able to find a better estimate for the round-trip time. The following shows an example of a *traceroute* program, which uses three probes for each device and gets three RTTs.

```
$ traceroute printers.com
traceroute to printers.com (13.1.69.93), 30 hops max, 38 byte packets
1  route.front.edu      (153.18.31.254)   0.622 ms   0.891 ms   0.875 ms
2  ceneric.net          (137.164.32.140)  3.069 ms   2.875 ms   2.930 ms
3  satire.net           (132.16.132.20)   3.071 ms   2.876 ms   2.929 ms
4  alpha.printers.com   (13.1.69.93)      5.922 ms   5.048 ms   4.922 ms
```


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4-3 UNICAST ROUTING

In an internet, the goal of the network layer is to deliver a datagram from its source to its destination or destinations. If a datagram is destined for only one destination (one-to-one delivery), we have unicast routing. In this section and the next, we discuss only unicast routing; multicast and broadcast routing will be discussed later in the chapter.

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4.3.1 General Idea

In unicast routing, a packet is routed, hop by hop, from its source to its destination by the help of forwarding tables. The source host needs no forwarding table because it delivers its packet to the default router in its local network. The destination host needs no forwarding table either because it receives the packet from its default router in its local network. This means that only the routers that glue together the networks in the internet need forwarding tables.

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4.3.1 (continued)

- An Internet as a Graph
- Least-Cost Routing
- Least-Cost Trees

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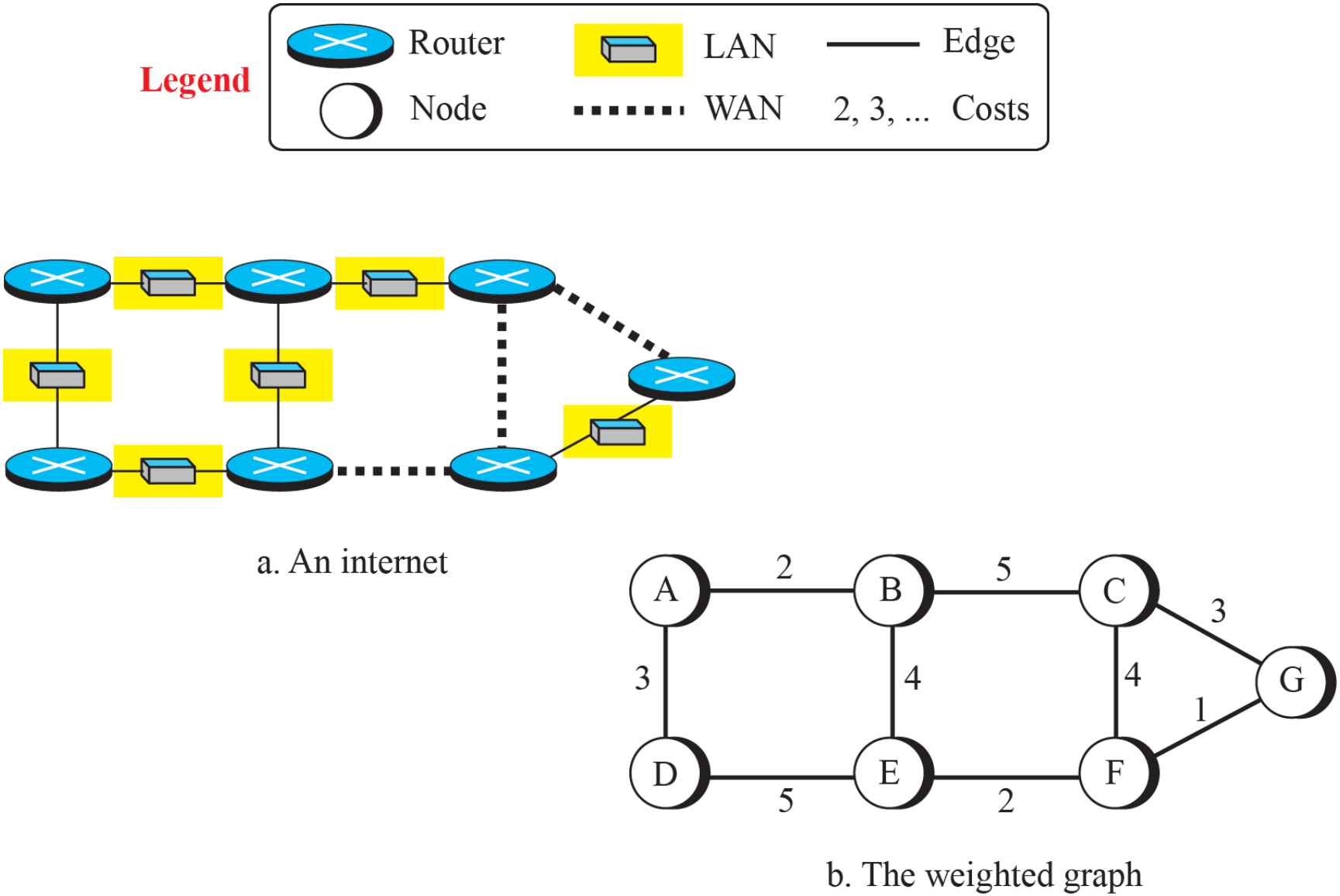
Figure 4.56: An internet and its graphical representation



a. An internet

b. The weighted graph

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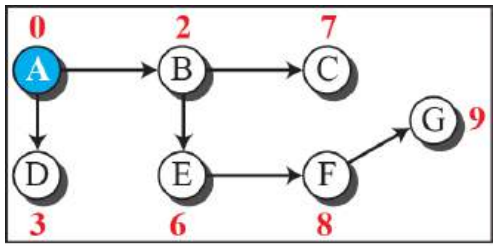
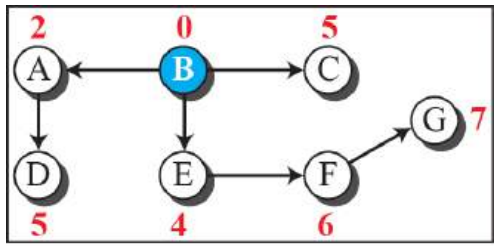
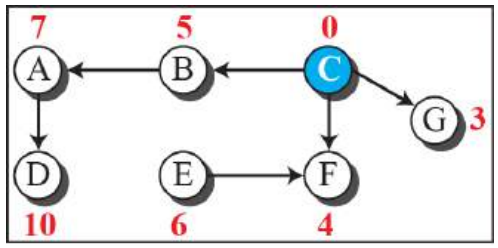
Figure 4.57: Least-cost trees for nodes in the internet of Figure 4.56

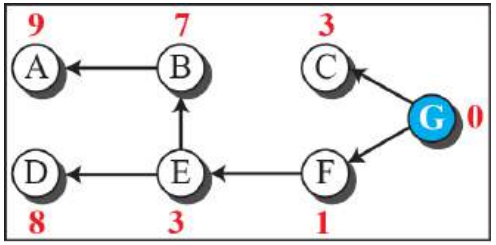


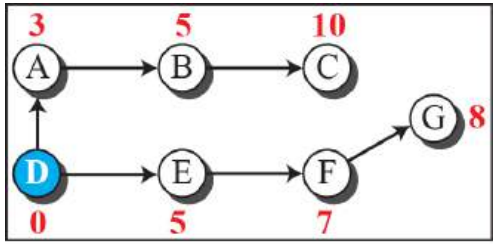


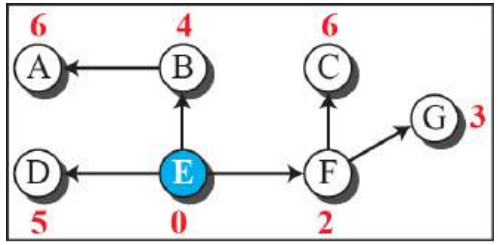


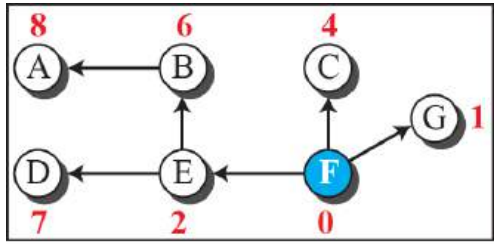
Legend

●	Root of the tree
○	Intermediate or end node
1, 2, ...	Total cost from the root









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4.3.2 Routing Algorithm

After discussing the general idea behind least-cost trees and the forwarding tables that can be made from them, now we concentrate on the routing algorithms. Several routing algorithms have been designed in the past. The differences between these methods are in the way they interpret the least cost and the way they create the least-cost tree for each node. In this section, we discuss the common algorithm; later we show how a routing protocol in the Internet implements one of these algorithms.

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4.3.2 (continued)

Distance-Vector Routing

- Bellman-Ford Equation
- Distance Vectors
- Distance-Vector Routing Algorithm
- Count to Infinity
- Two-Node Loop
  - Split Horizon
  - Poisoned Reverse
- Three-Node Instability

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4.3.2 (continued)

- **Link-State Routing**
  - Link-State Database (LSDB)
  - Least-Cost Trees (Dijkstra's Algorithm)
- **Path-Vector Routing**
  - Spanning Trees
  - Creation of Spanning Trees
  - Path-Vector Algorithm

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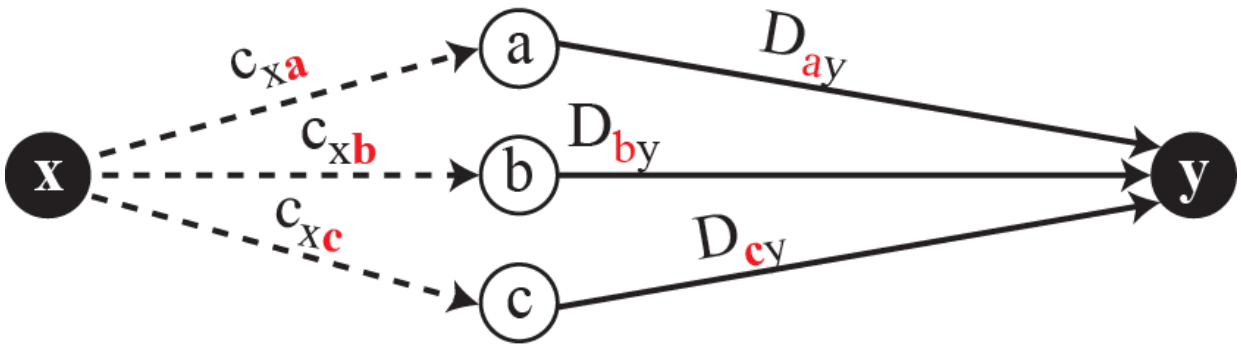
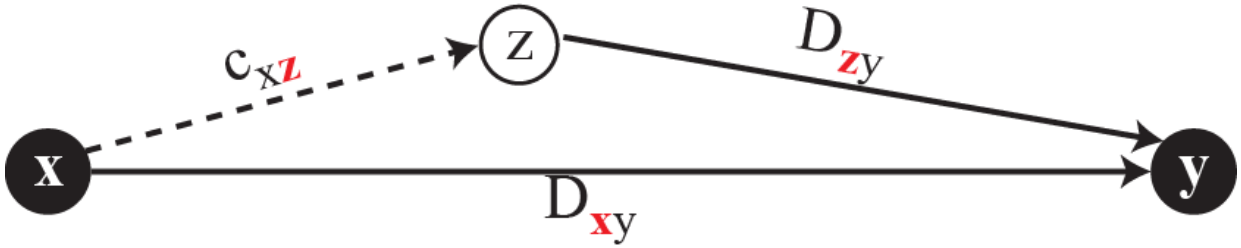
Figure 4.58: Graphical idea behind Bellman-Ford equation



a. General case with three intermediate nodes



b. Updating a path with a new route

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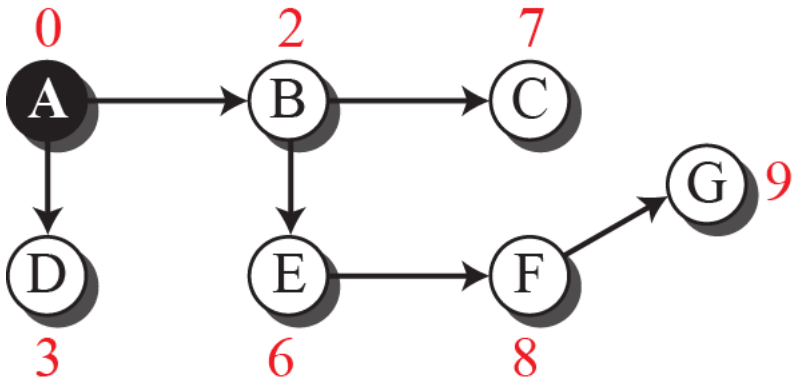
Figure 4.59: The distance vector corresponding to a tree



a. Tree for node A

	A
A	0
B	2
C	7
D	3
E	6
F	8
G	9

b. Distance vector for node A

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Figure 4.60: The first distance vector for an internet

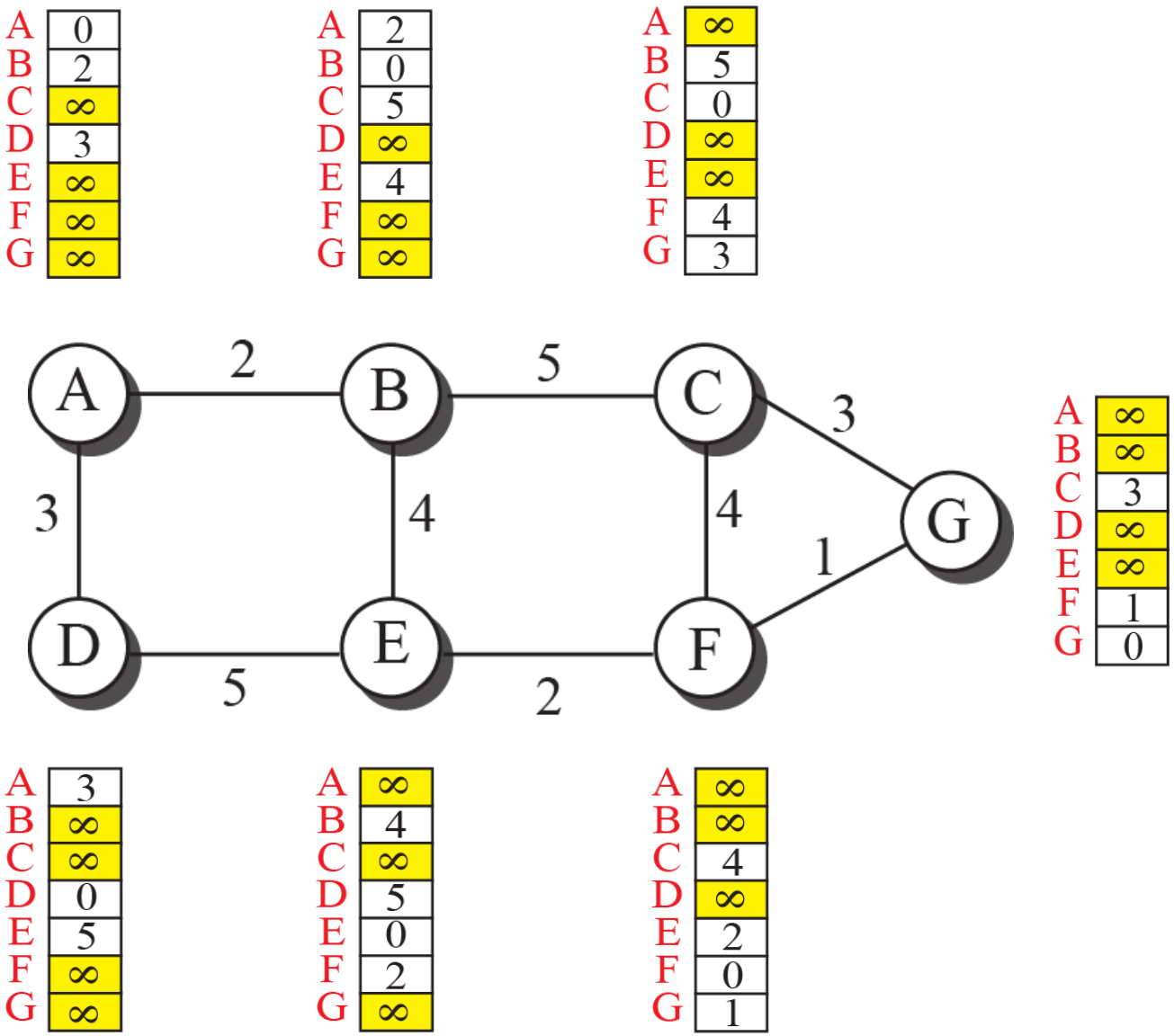

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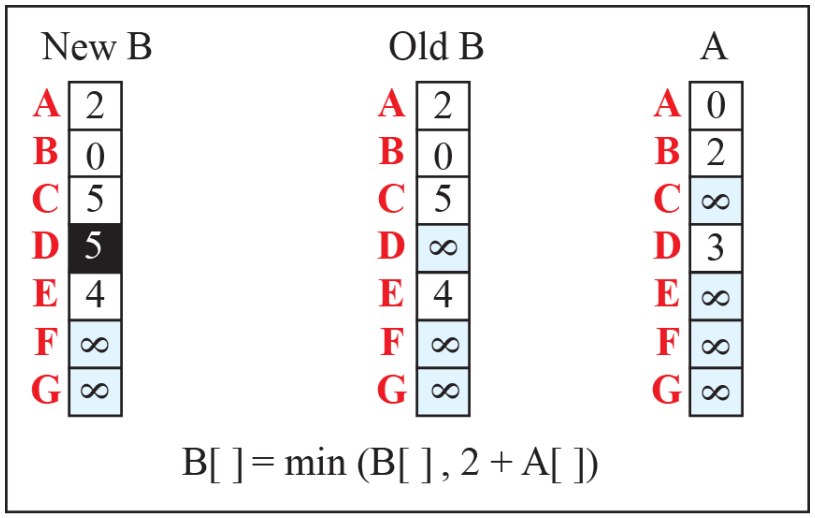
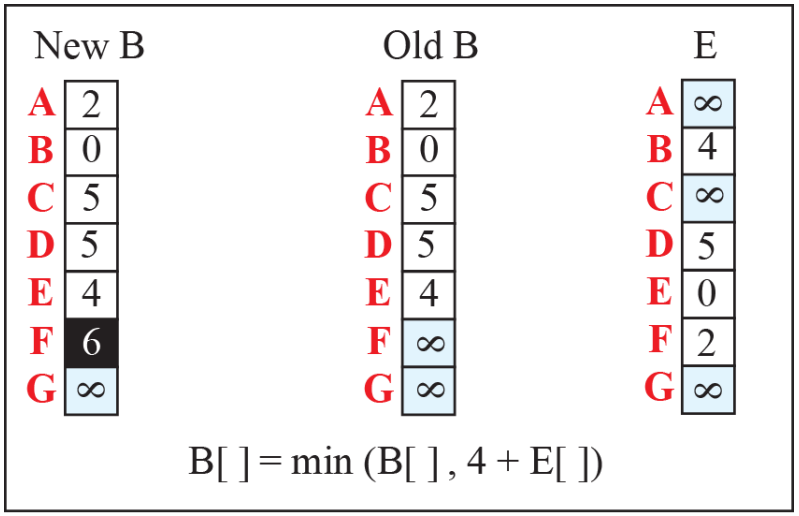
Figure 4.61: Updating distance vectors



a. First event: B receives a copy of A's vector.

Note:
X[ ]: the whole vector



b. Second event: B receives a copy of E's vector.

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Table 4.4: Distance-Vector Routing Algorithm for A Node

```
1  Distance_Vector_Routing ( )
2  {
3      // Initialize (create initial vectors for the node)
4      D[myself] = 0
5      for (y = 1 to N)
6      {
7          if (y is a neighbor)
8              D[y] = c[myself][y]
9          else
10             D[y] = ∞
11     }
12     send vector {D[1], D[2], ..., D[N]} to all neighbors
13     // Update (improve the vector with the vector received from a neighbor)
14     repeat (forever)
15     {
16         wait (for a vector D_w from a neighbor w or any change in the link)
17         for (y = 1 to N)
18         {
19             D[y] = min [D[y], (c[myself][w] + D_w[y])]        // Bellman-Ford equation
20         }
21         if (any change in the vector)
22             send vector {D[1], D[2], ..., D[N]} to all neighbors
23     }
24 } // End of Distance Vector
```


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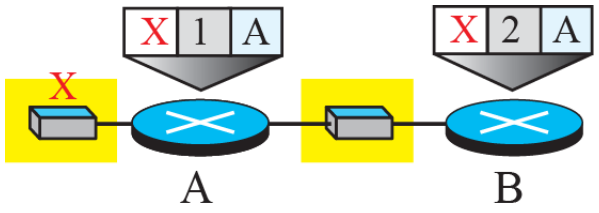
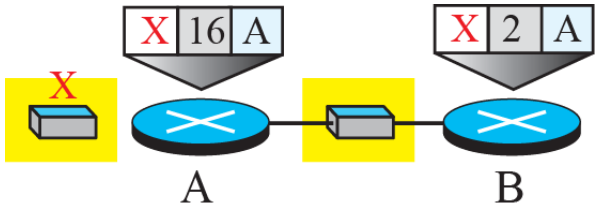
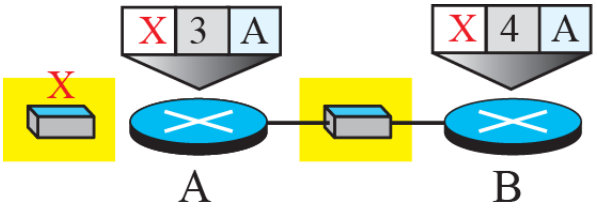
Figure 4.62: Two-node instability



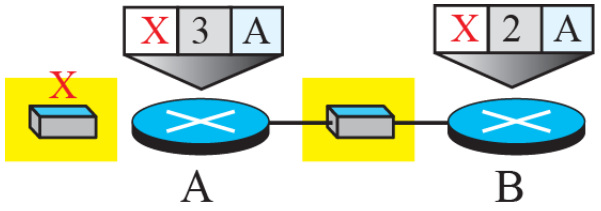
a. Before failure



b. After link failure

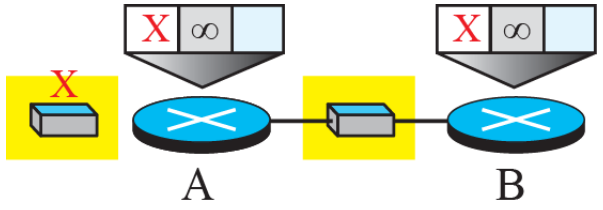


d. After B is updated by A



c. After A is updated by B

• • •



e. Finally

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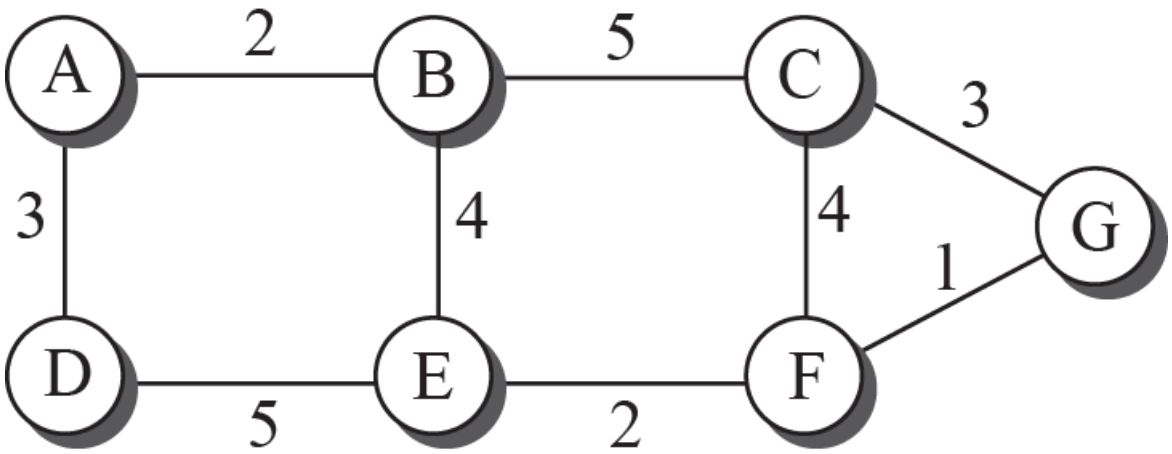
Figure 4.63: Example of a link-state database



a. The weighted graph

	A	B	C	D	E	F	G
A	0	2	∞	3	∞	∞	∞
B	2	0	5	∞	4	∞	∞
C	∞	5	0	∞	∞	4	3
D	3	∞	∞	0	5	∞	∞
E	∞	4	∞	5	0	2	∞
F	∞	∞	4	∞	2	0	1
G	∞	∞	3	∞	∞	1	0

b. Link state database

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Figure 4.64: LSPs created and sent out by each node to build LSDB

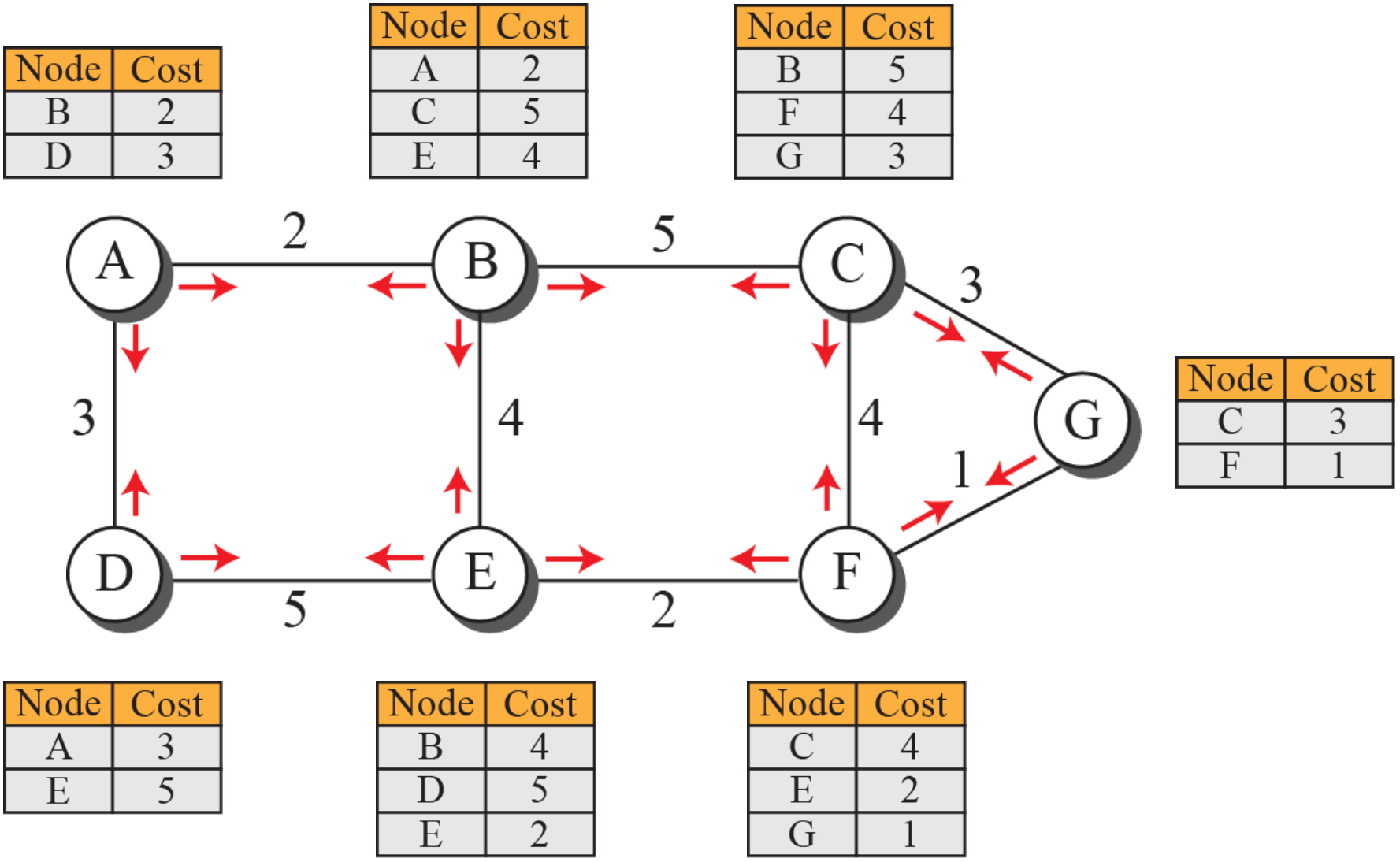

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Table 4.5: Dijkstra's Algorithm

```
1  Dijkstra's Algorithm ( )
2  {
3      // Initialization
4      Tree = {root}                    // Tree is made only of the root
5      for (y = 1 to N)                 // N is the number of nodes
6      {
7          if (y is the root)
8              D[y] = 0                 // D[y] is shortest distance from root to node y
9          else if (y is a neighbor)
10             D[y] = c[root][y]        // c[x][y] is cost between nodes x and y in LSDB
11         else
12             D[y] = ∞
13     }
14     // Calculation
15     repeat
16     {
17         find a node w, with D[w] minimum among all nodes not in the Tree
18         Tree = Tree ∪ {w}            // Add w to tree
19         // Update distances for all neighbor of w
20         for (every node x, which is neighbor of w and not in the Tree)
21         {
22             D[x] = min{D[x], (D[w] + c[w][x])}
23         }
24     } until (all nodes included in the Tree)
25 } // End of Dijkstra
```


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Figure 4.65: Least-cost tree

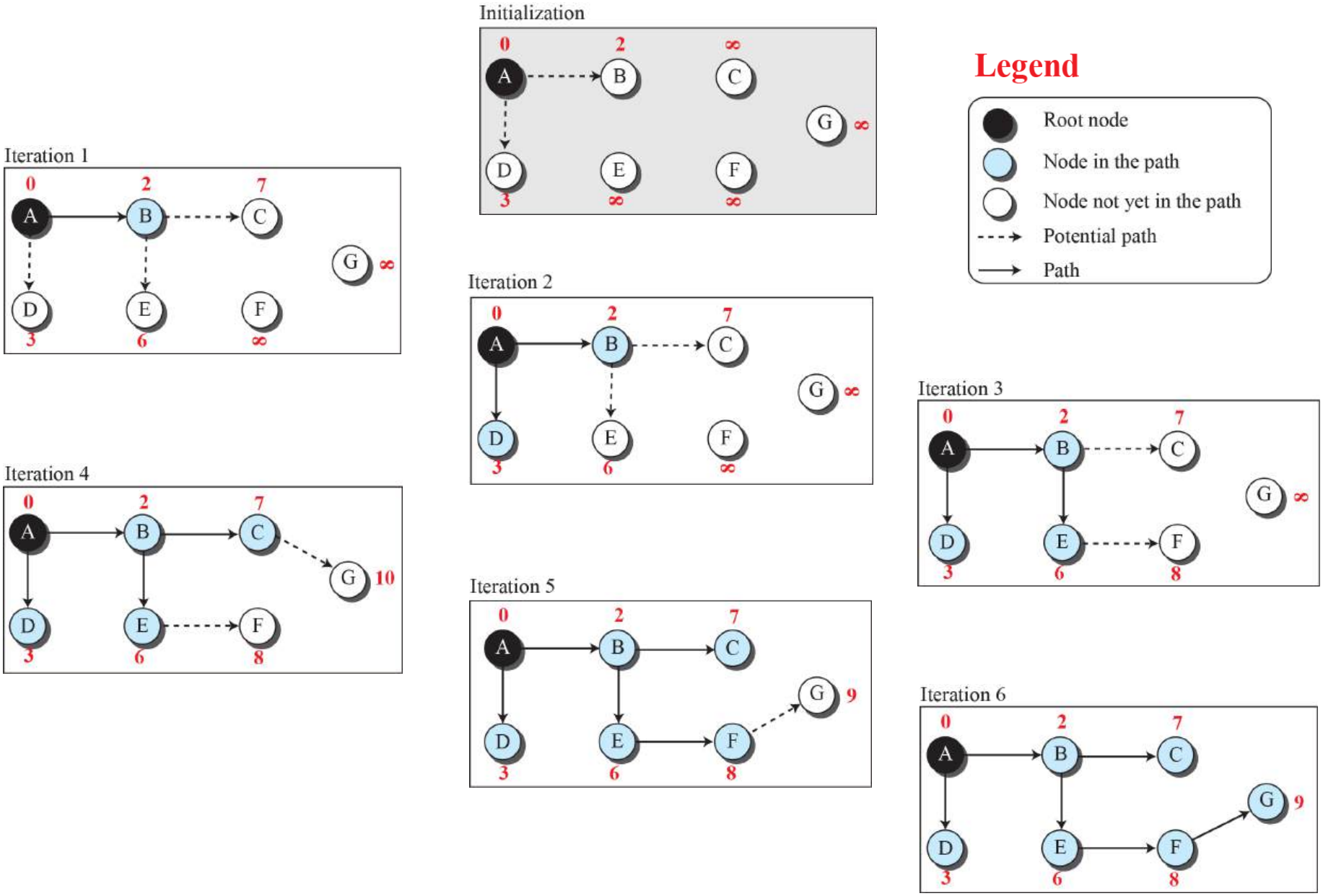

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Figure 4.66: Spanning trees in path-vector routing

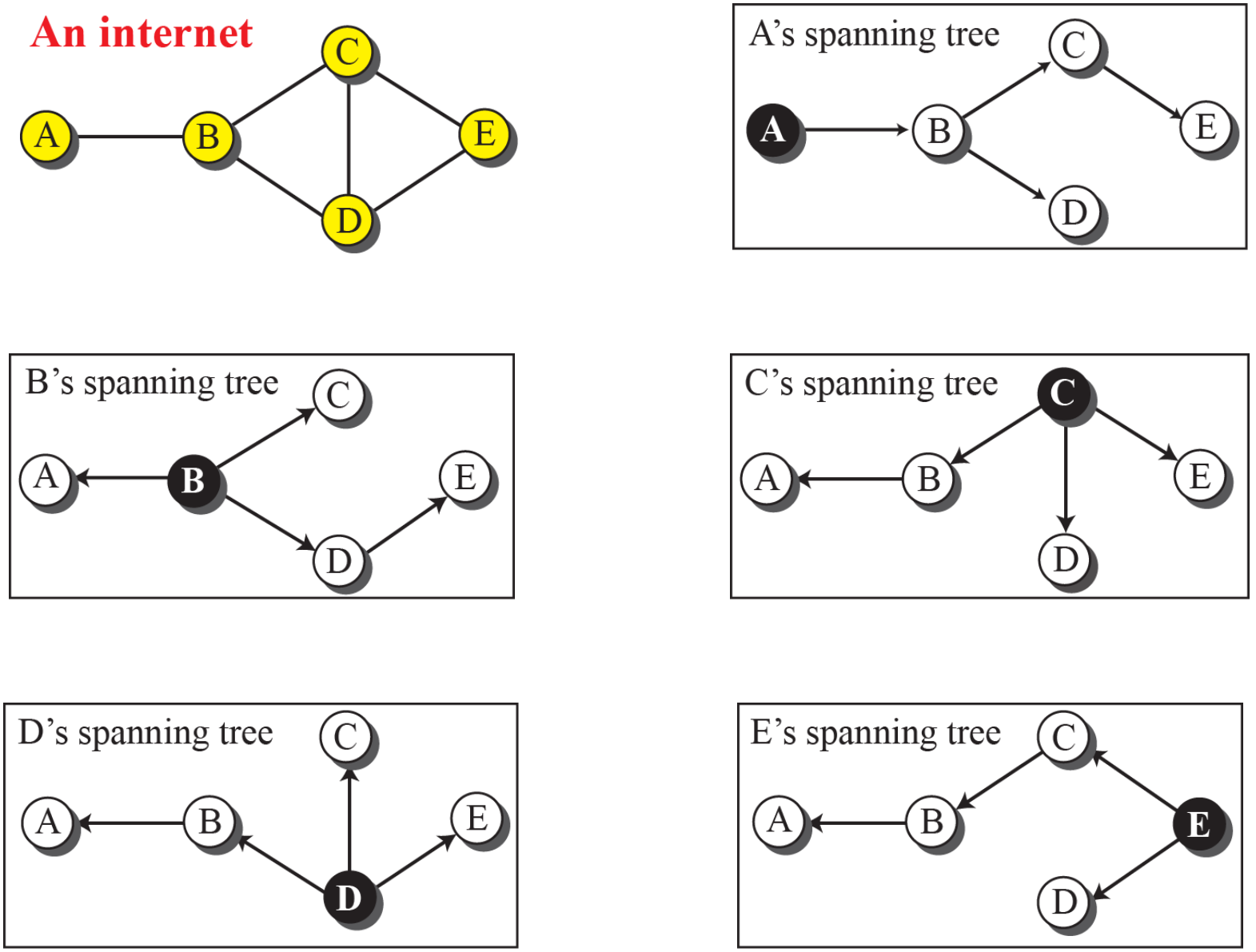

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Figure 4.67: Path vectors made at booting time

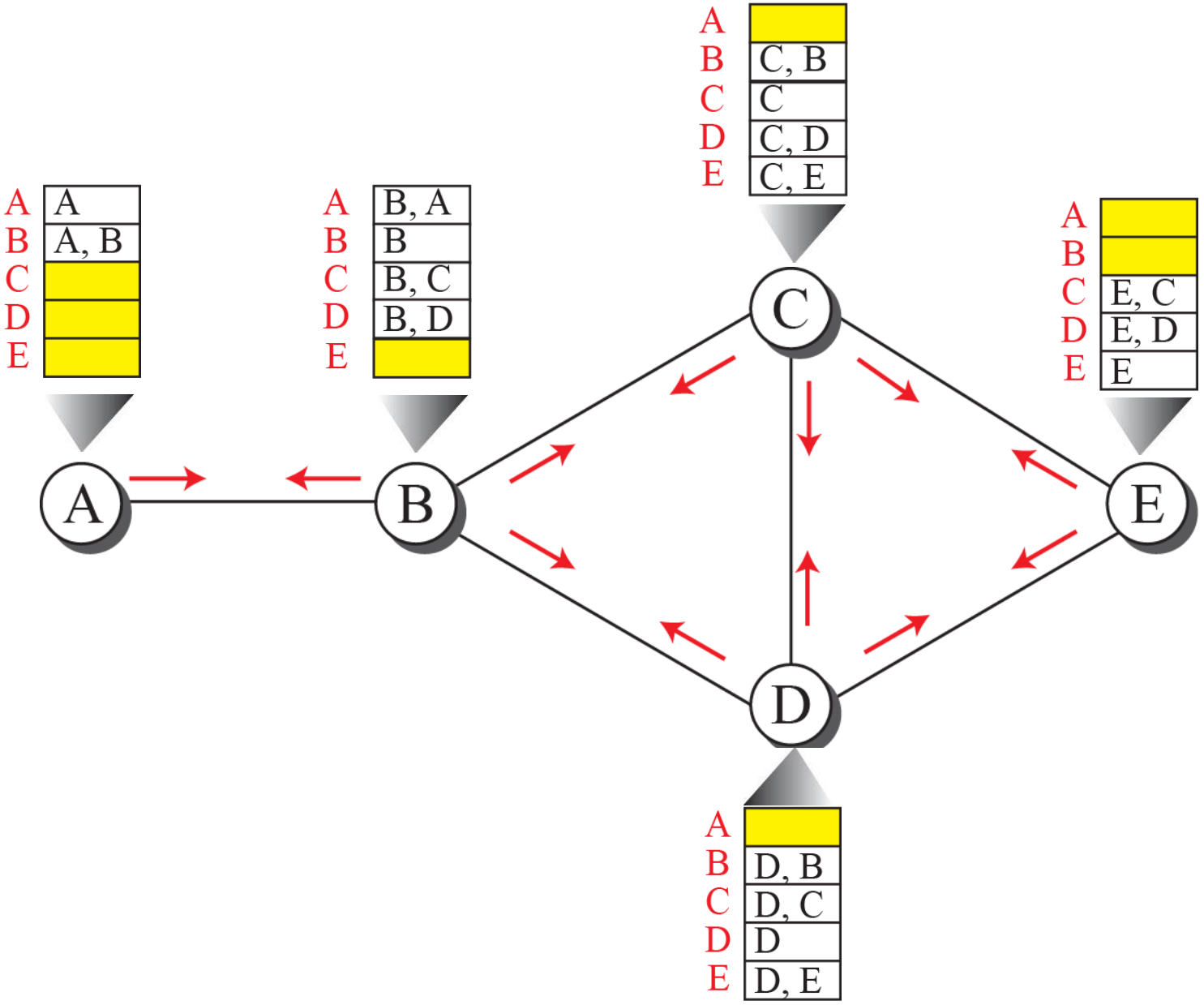

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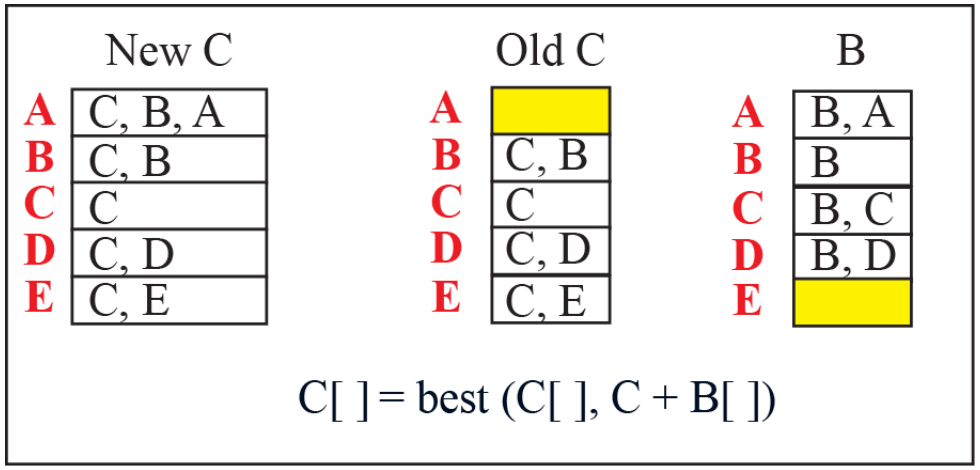
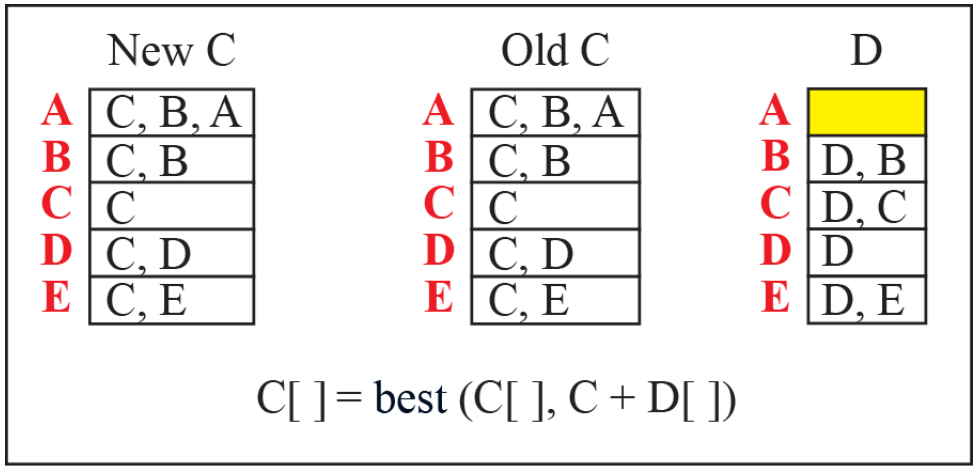
Figure 4.68: *Updating path vectors*



Event 1: C receives a copy of B's vector

Note:
X [ ]: vector X
Y: node Y



Event 2: C receives a copy of D's vector

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Table 4.6: Path-vector algorithm for a node

```
1   Path_Vector_Routing ( )
2   {
3       // Initialization
4       for (y = 1 to N)
5       {
6           if (y is myself)
7               Path[y] = myself
8           else if (y is a neighbor)
9               Path[y] = myself + neighbor node
10          else
11              Path[y] = empty
12      }
13      Send vector {Path[1], Path[2], ..., Path[y]} to all neighbors
14      // Update
15      repeat (forever)
16      {
17          wait (for a vector Path_w from a neighbor w)
18          for (y = 1 to N)
19          {
20              if (Path_w includes myself)
21                  discard the path                    // Avoid any loop
22              else
23                  Path[y] = best {Path[y], (myself + Path_w[y])}
24          }
25          If (there is a change in the vector)
26              Send vector {Path[1], Path[2], ..., Path[y]} to all neighbors
27      }
28  } // End of Path Vector
```


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4.3.3 Unicast Routing Protocols

In the previous section, we discussed unicast routing algorithms; in this section, we discuss unicast routing protocols used in the Internet. Although three protocols we discuss here are based on the corresponding algorithms we discussed before, a protocol is more than an algorithm. A protocol needs to define its domain of operation, the messages exchanged, communication between routers, and interaction with protocols in other domains. After an introduction, we discuss three common protocols used in the Internet: RIP, OSPF, and BGP.

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4.3.3 (continued)

- **Internet Structure**
  - Hierarchical Routing
  - Autonomous Systems
- **Routing Information Protocol (RIP)**
  - Hop Count
  - Forwarding Tables
  - RIP Implementation
  - Performance

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4.3.3 (continued)

- **Open Shortest Path First (OSPF)**
  - Metric
  - Forwarding Tables
  - Areas
  - Link-State Advertisement
  - OSPF Implementation
  - Performance
- **Border Gateway Protocol Version 4 (BGP4)**
  - Introduction
  - Path Attributes
  - Route Selection
  - Messages
  - Performance

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Figure 4.69: *Internet structure*

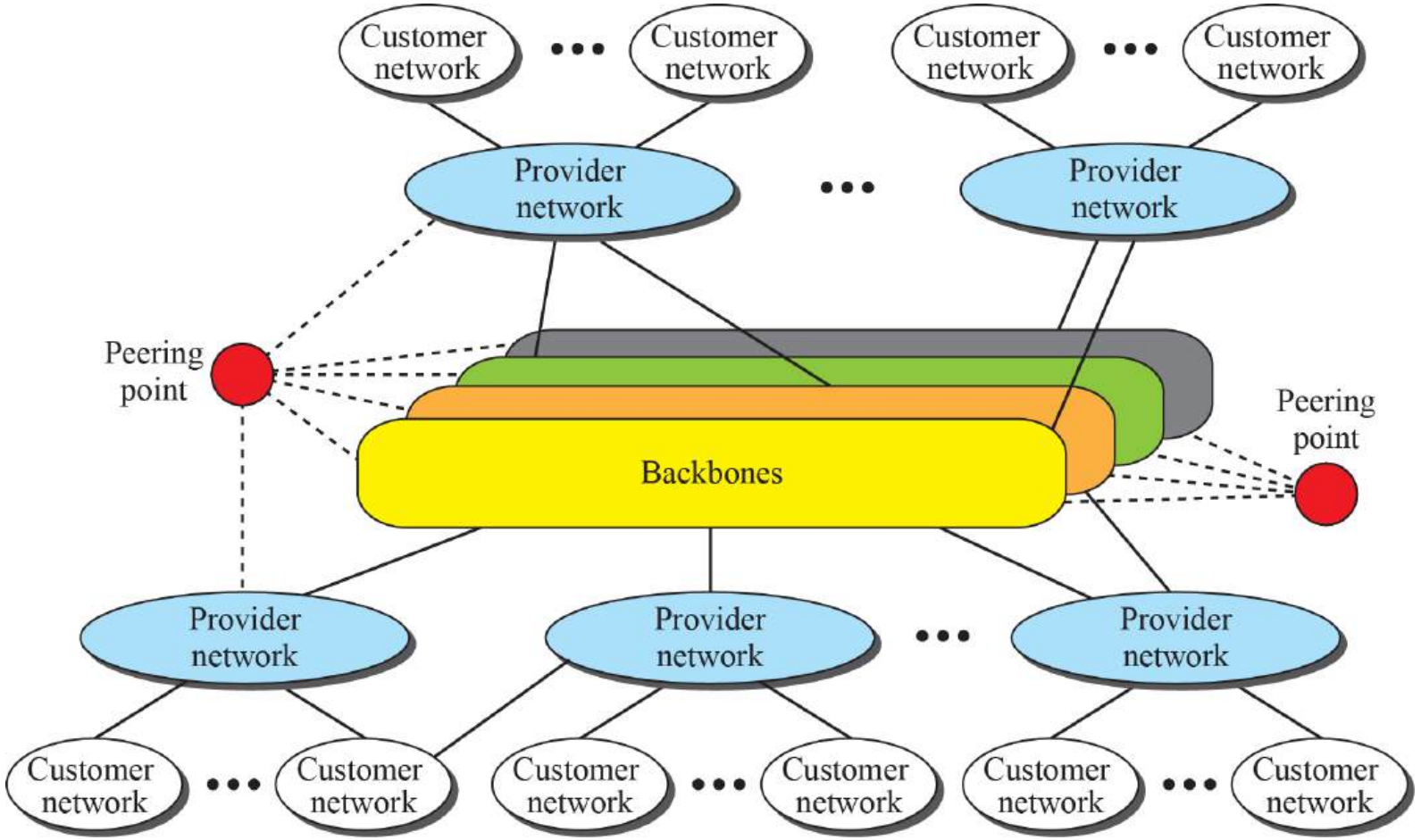

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Figure 4.70: *Hop counts in RIP*

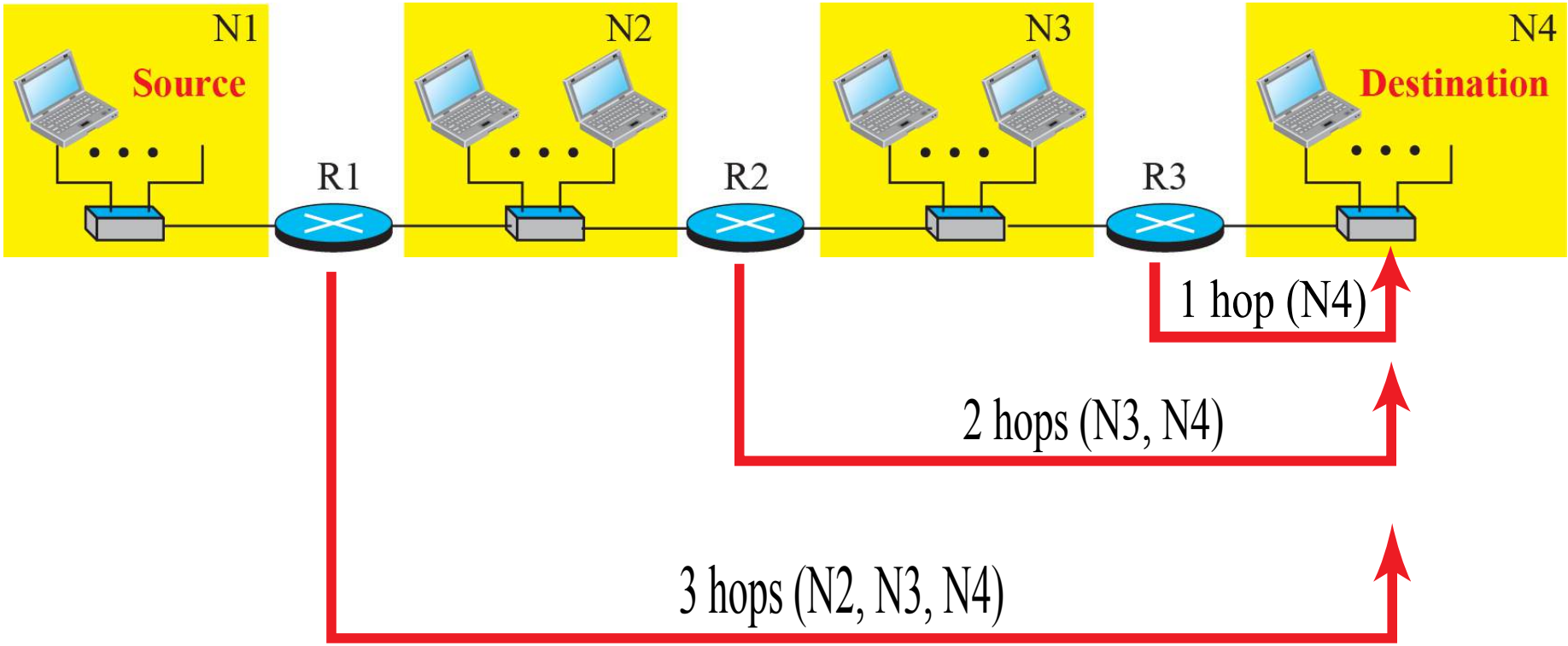

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Figure 4.71: Forwarding tables

Forwarding table for R1

Destination network	Next router	Cost in hops
N1	——	1
N2	——	1
N3	R2	2
N4	R2	3

Forwarding table for R2

Destination network	Next router	Cost in hops
N1	R1	2
N2	——	1
N3	——	1
N4	R3	2

Forwarding table for R3

Destination network	Next router	Cost in hops
N1	R2	3
N2	R2	2
N3	——	1
N4	——	1

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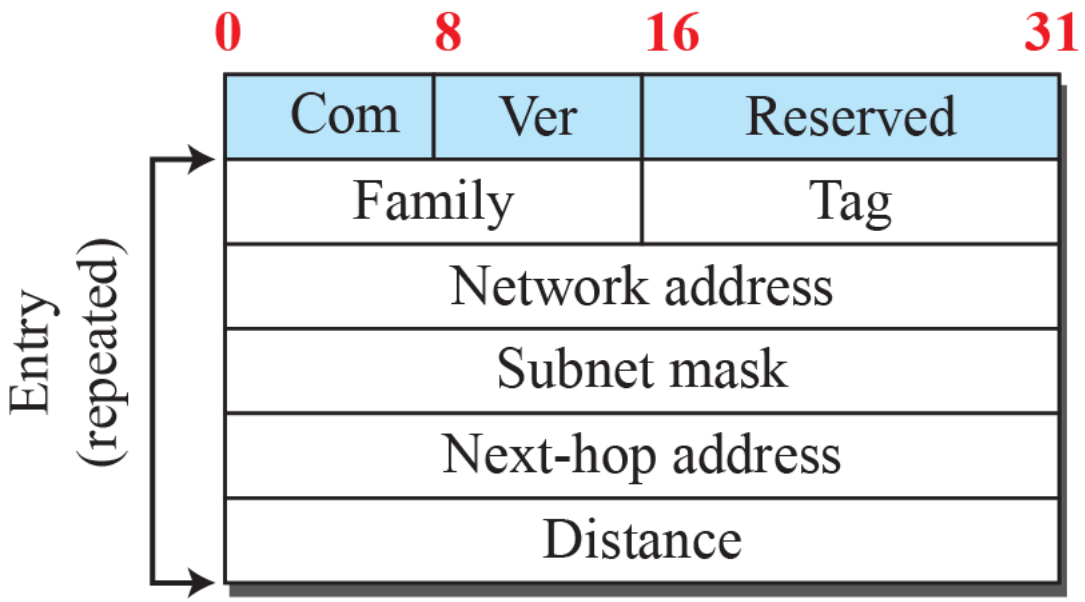
Figure 4.72: RIP message format



Fields

Com: Command, request (1), response (2)
Ver: Version, current version is 2
Family: Family of protocol, for TCP/IP value is 2
Tag: Information about autonomous system
Network address: Destination address
Subnet mask: Prefix length
Next-hop address: Address length
Distance: Number of hops to the destination

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Example 4.15

Figure 4.73 shows a more realistic example of the operation of RIP in an autonomous system. First, the figure shows all forwarding tables after all routers have been booted. Then we show changes in some tables when some update messages have been exchanged. Finally, we show the stabilized forwarding tables when there is no more change.

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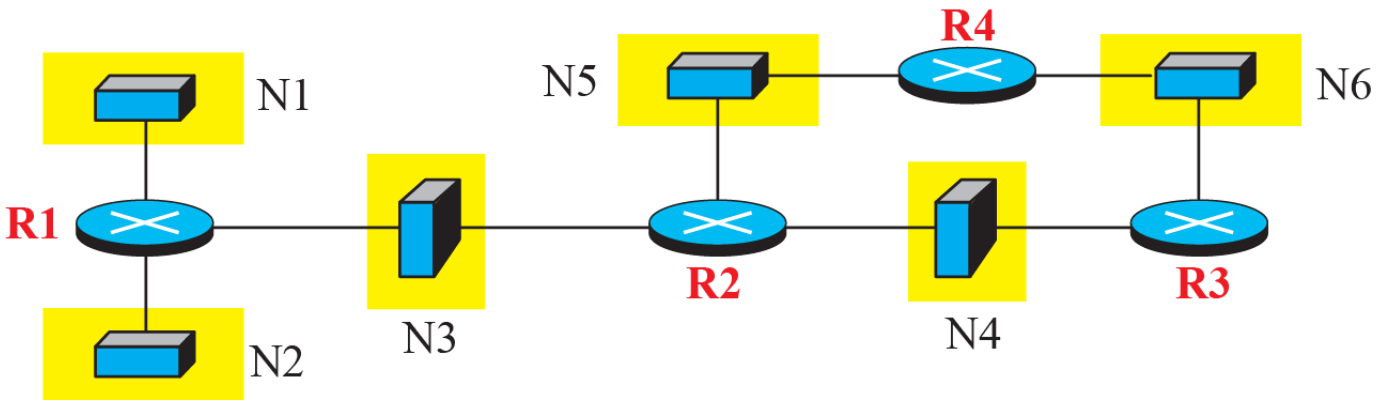
Figure 4.73: Example of an autonomous system using RIP (Part I)



Legend

Des.: Destination network
N. R.: Next router
Cost: Cost in hops

R1

Des.	N. R.	Cost
N1	———	1
N2	———	1
N3	———	1

R2

Des.	N. R.	Cost
N3	———	1
N4	———	1
N5	———	1

R3

Des.	N. R.	Cost
N4	———	1
N6	———	1

R4

Des.	N. R.	Cost
N5	———	1
N6	———	1

Forwarding tables after all routers booted

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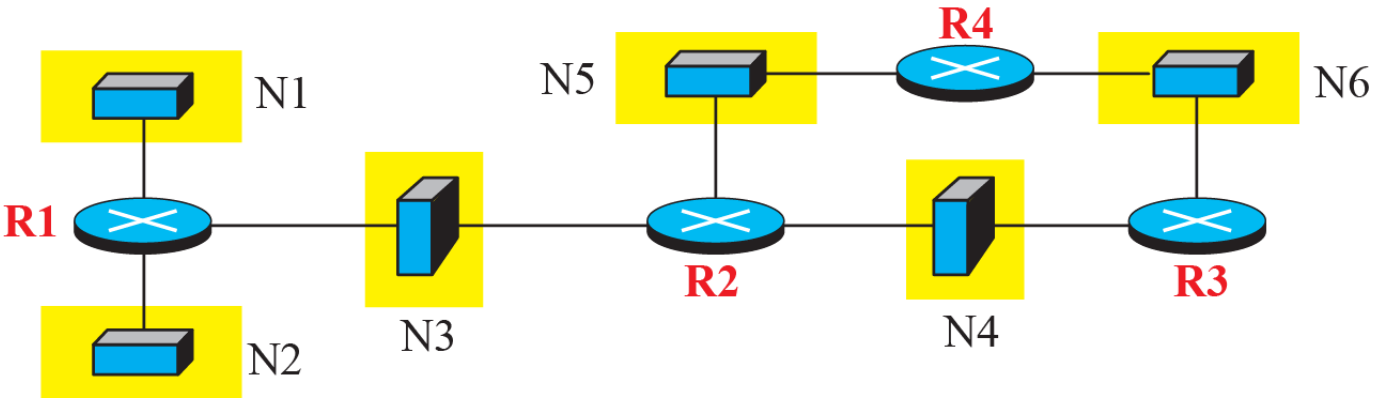
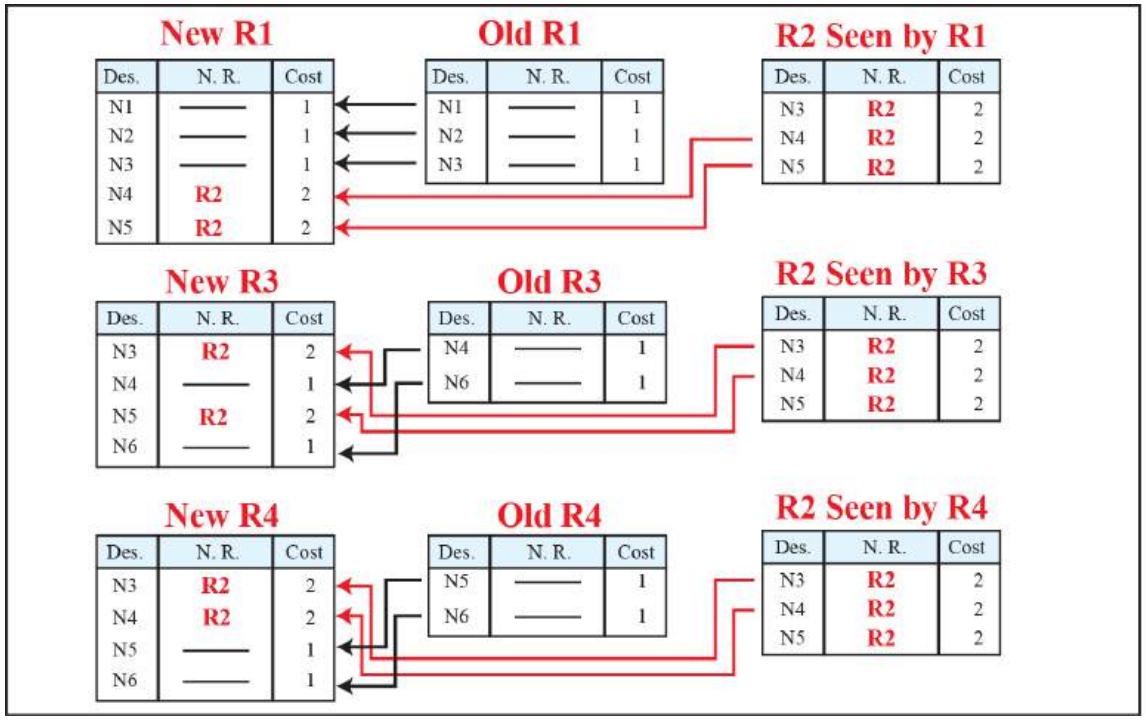
Figure 4.73: Example of an autonomous system using RIP (Part II)





Changes in the forwarding tables of R1, R3, and R4 after they receive a copy of R2's table

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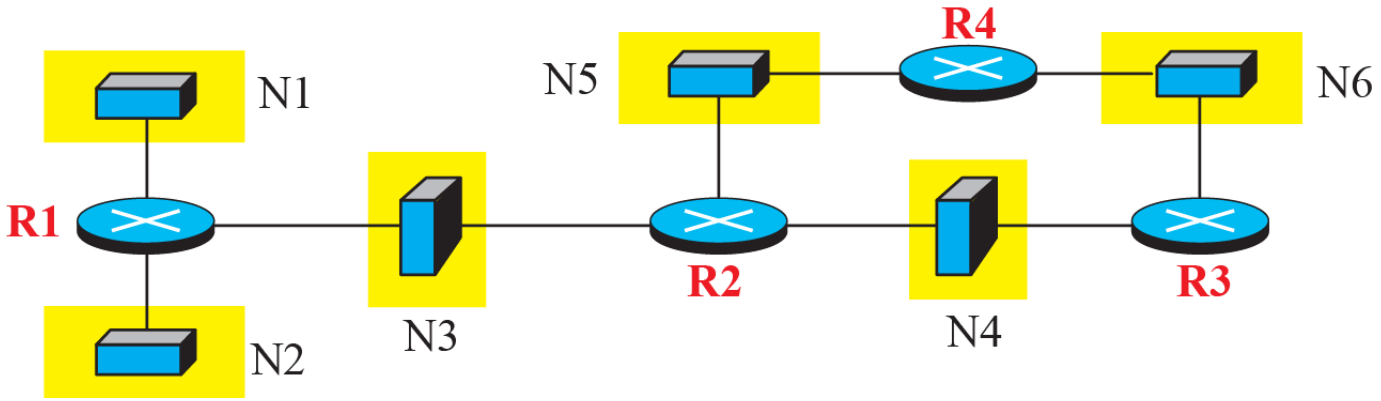
Figure 4.73: Example of an autonomous system using RIP (Part III)



Legend

Des.: Destination network
N. R.: Next router
Cost: Cost in hops

Forwarding tables for all routers after they have been stablized

Final R1

Des.	N. R.	Cost
N1	———	1
N2	———	1
N3	———	1
N4	R2	2
N5	R2	2
N6	R2	3

Final R2

Des.	N. R.	Cost
N1	R1	2
N2	R1	2
N3	———	1
N4	———	1
N5	———	1
N6	R3	2

Final R3

Des.	N. R.	Cost
N1	R2	3
N2	R2	3
N3	R2	2
N4	———	1
N5	R2	2
N6	———	1

Final R4

Des.	N. R.	Cost
N1	R2	3
N2	R2	3
N3	R2	2
N4	R2	2
N5	———	1
N6	———	1

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Figure 4.74: Metric in OSPF

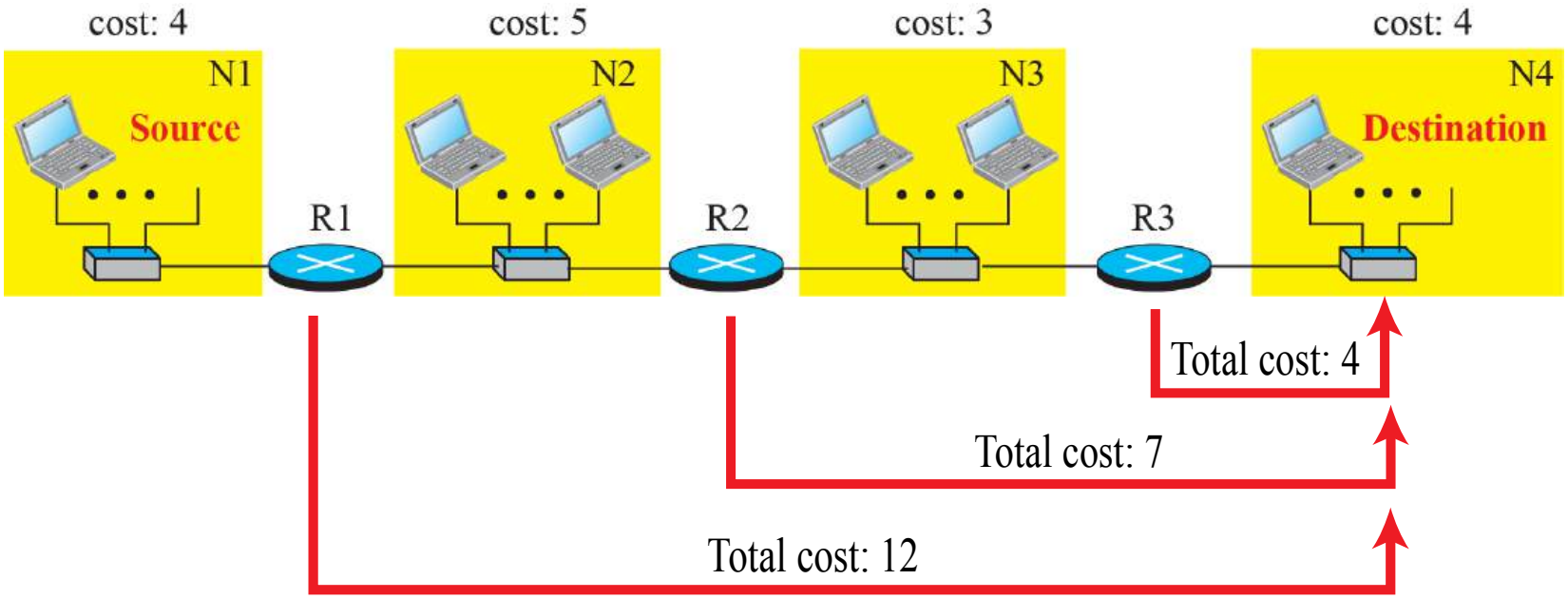

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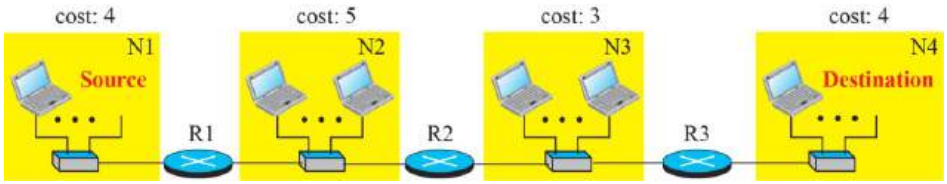
Figure 4.75: Forwarding tables in OSPF

Forwarding table for R1

Destination network	Next router	Cost
N1	——	
N2	——	
N3	R2	
N4	R2	12



The internet from previous figure

Forwarding table for R2

Destination network	Next router	Cost
N1	R1	9
N2	——	5
N3	——	3
N4	R3	7

Forwarding table for R3

Destination network	Next router	Cost
N1	R2	12
N2	R2	8
N3	——	3
N4	——	4

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Figure 4.76: Areas in an autonomous system

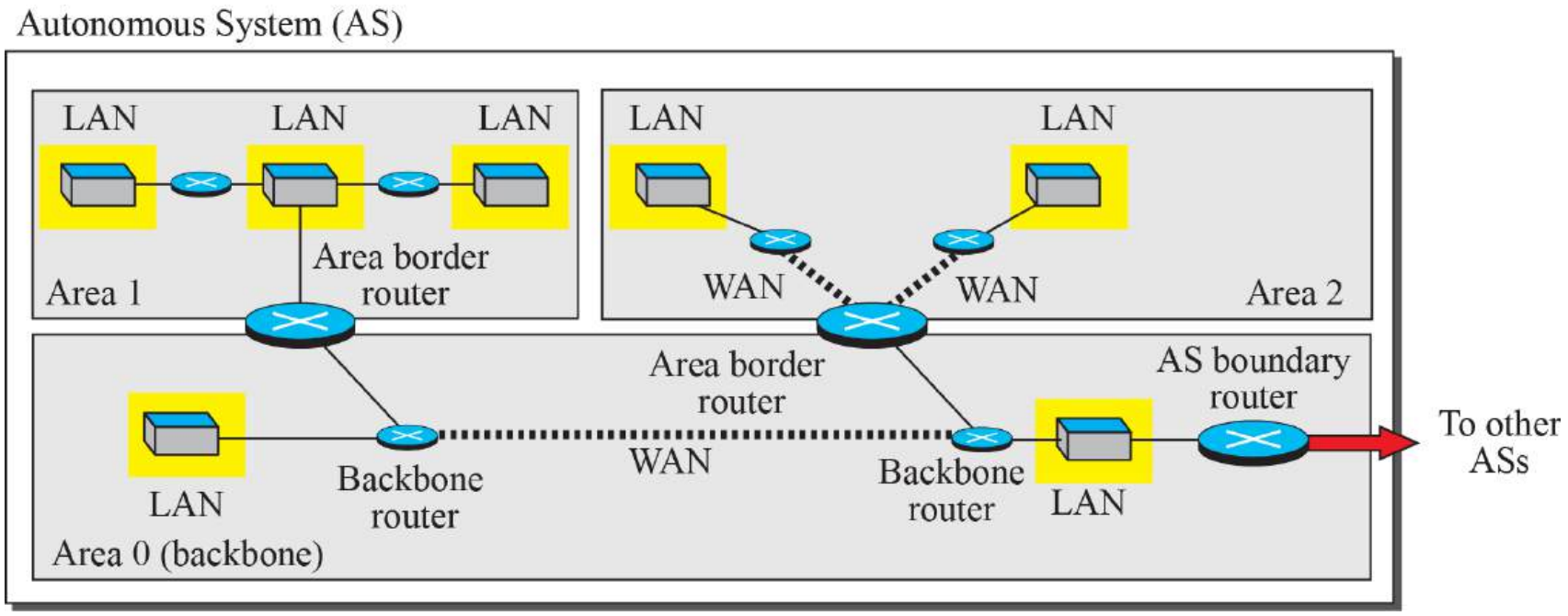

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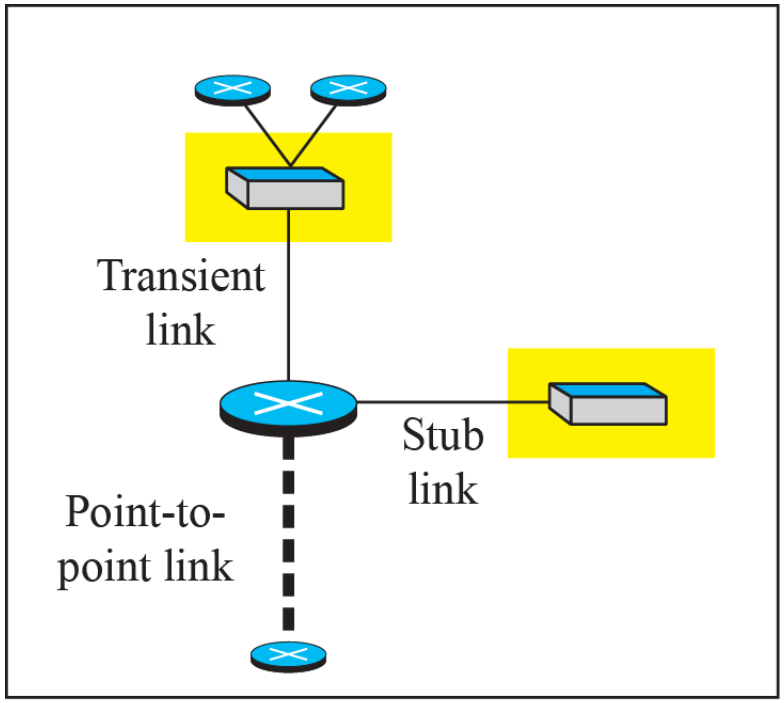
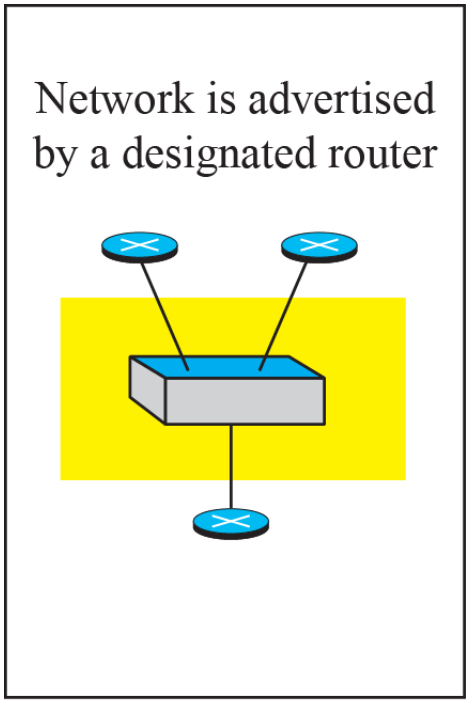
Figure 4.77: Five different LSPs (Part I)



a. Router link



b. Network link

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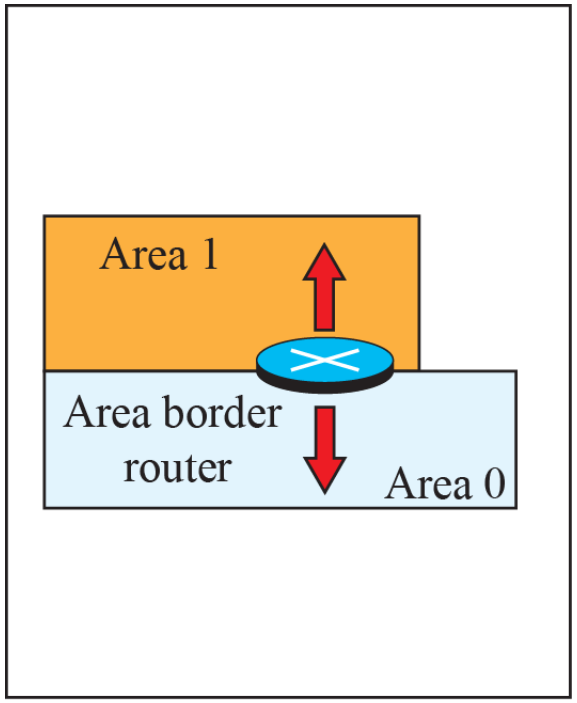
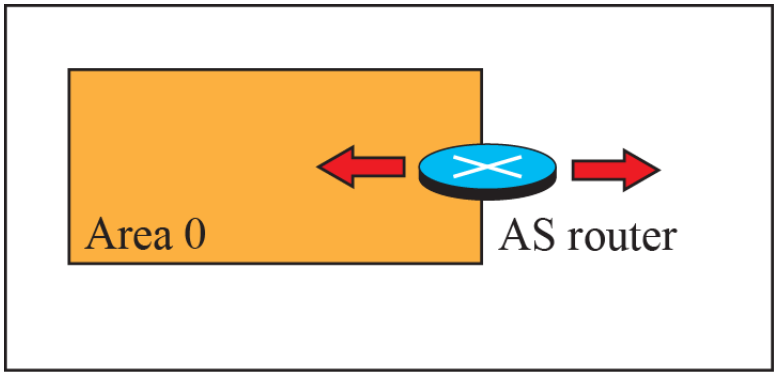
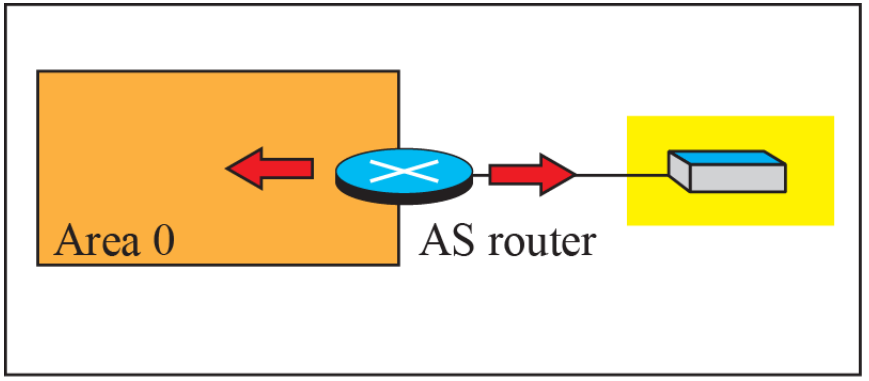
Figure 4.77: Five different LSPs (Part II)



c. Summary link to network



d. Summary link to AS



e. External link

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Figure 4.78: OSPF message formats (Part I)

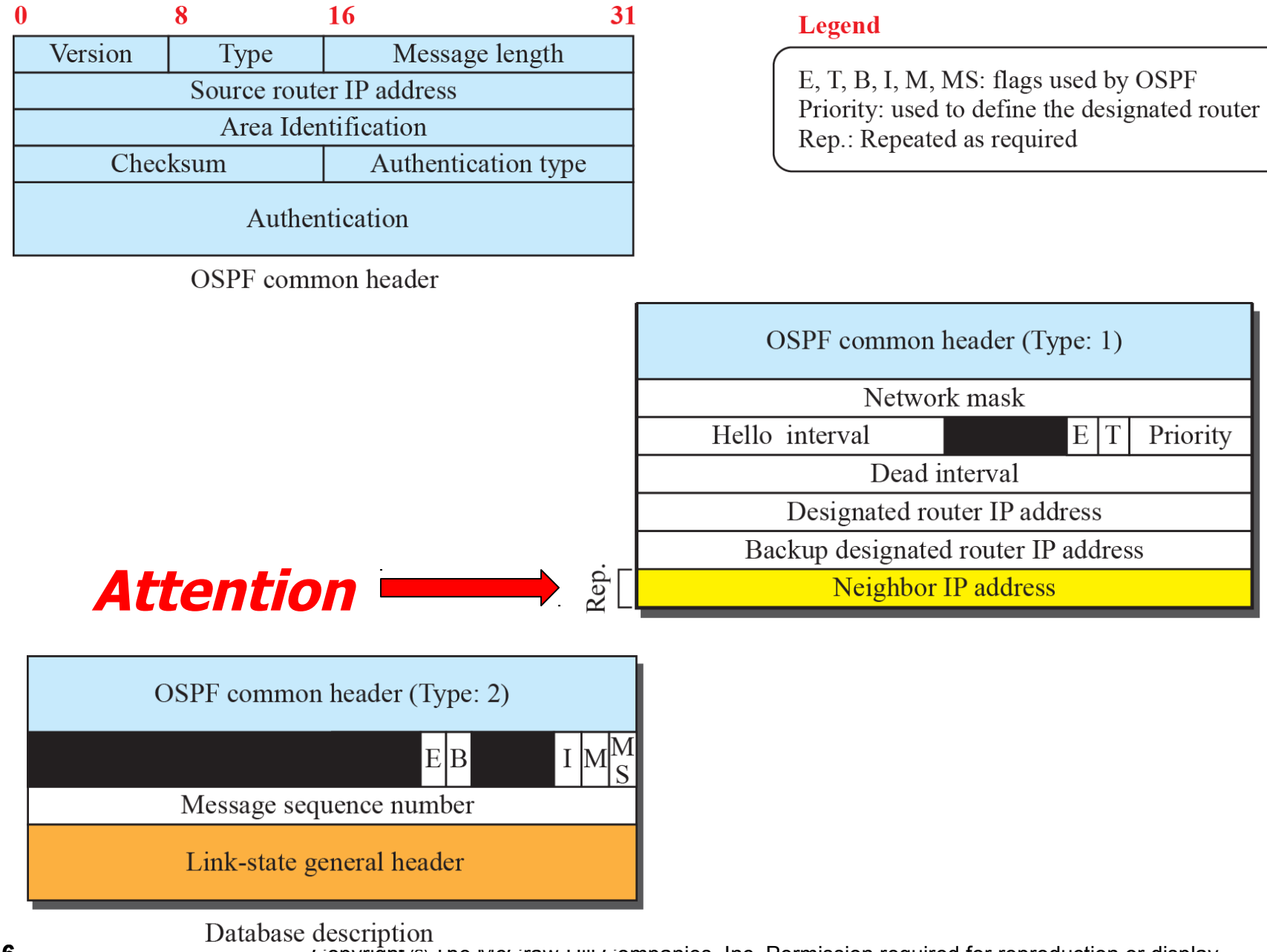

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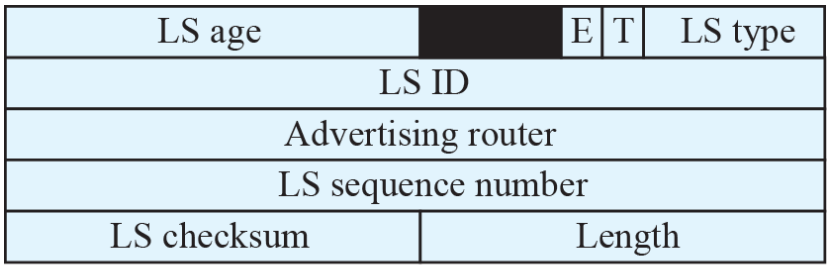
Figure 4.78: OSPF message formats (Part II)

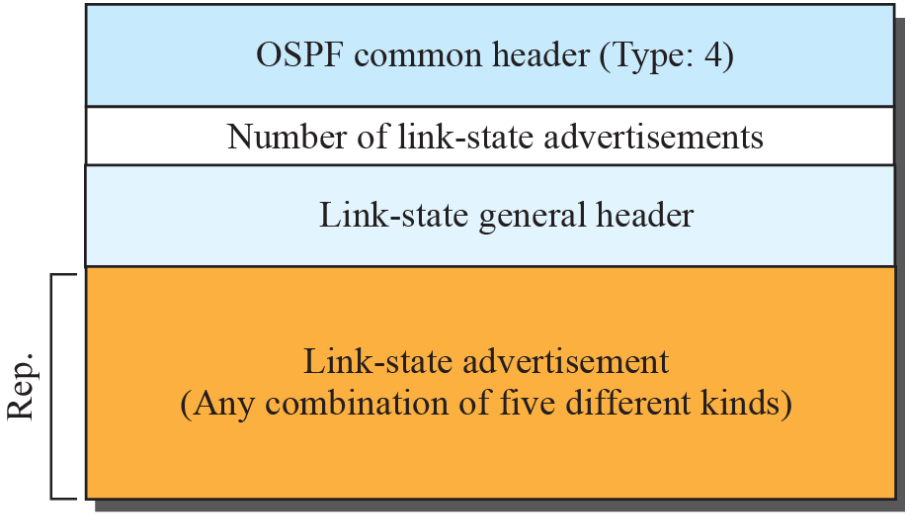


Link-state general header

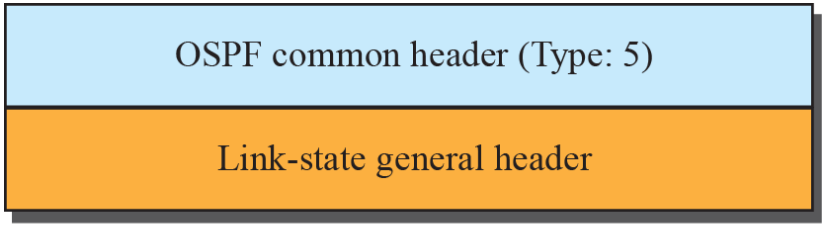
Legend

E, T, B, I, M, MS: flags used by OSPF
Priority: used to define the designated router
Rep.: Repeated as required

Attention



Link-state update



Link-state acknowledgment

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Figure 4.79: A sample internet with four ASs

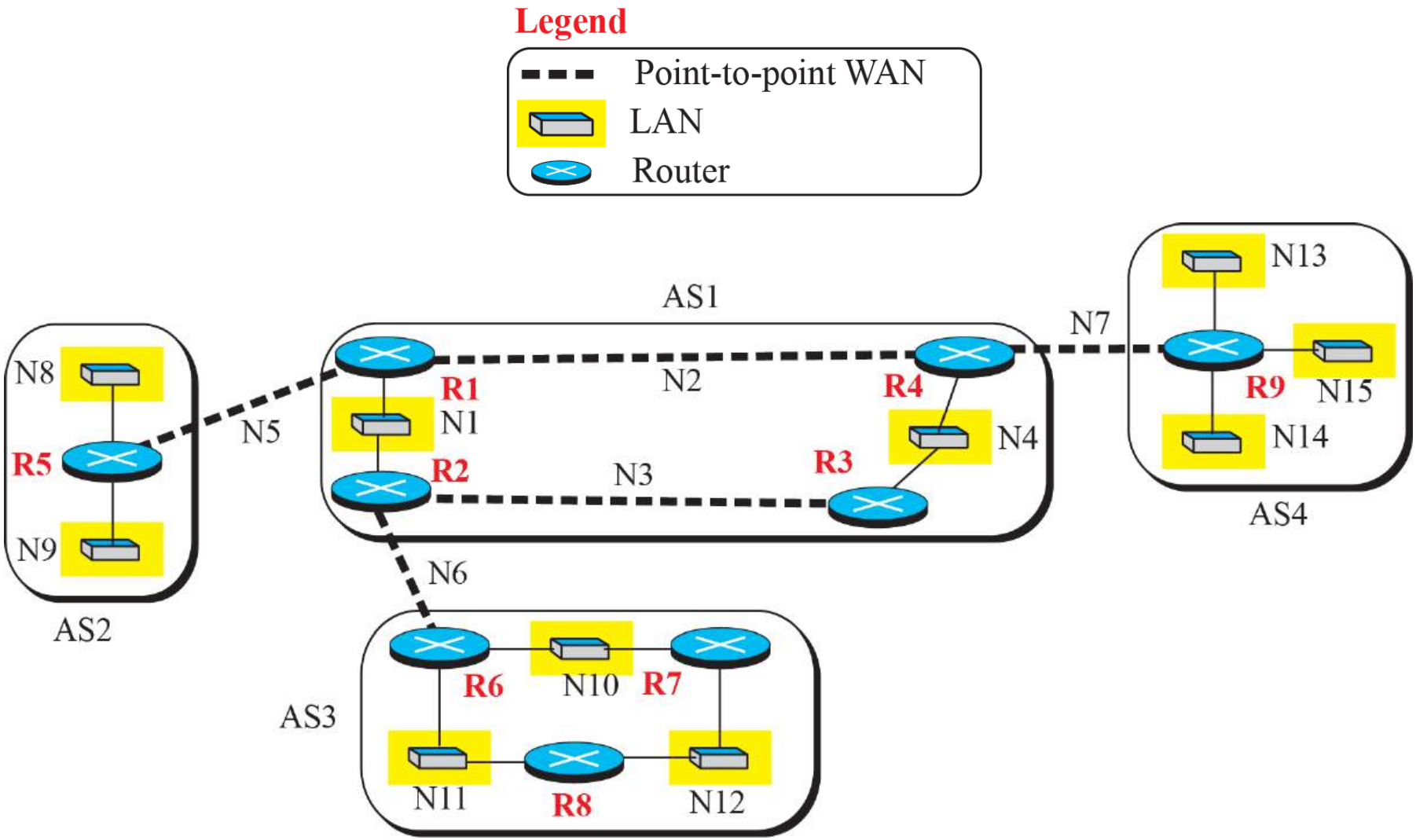

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Figure 4.80: eBGP operation

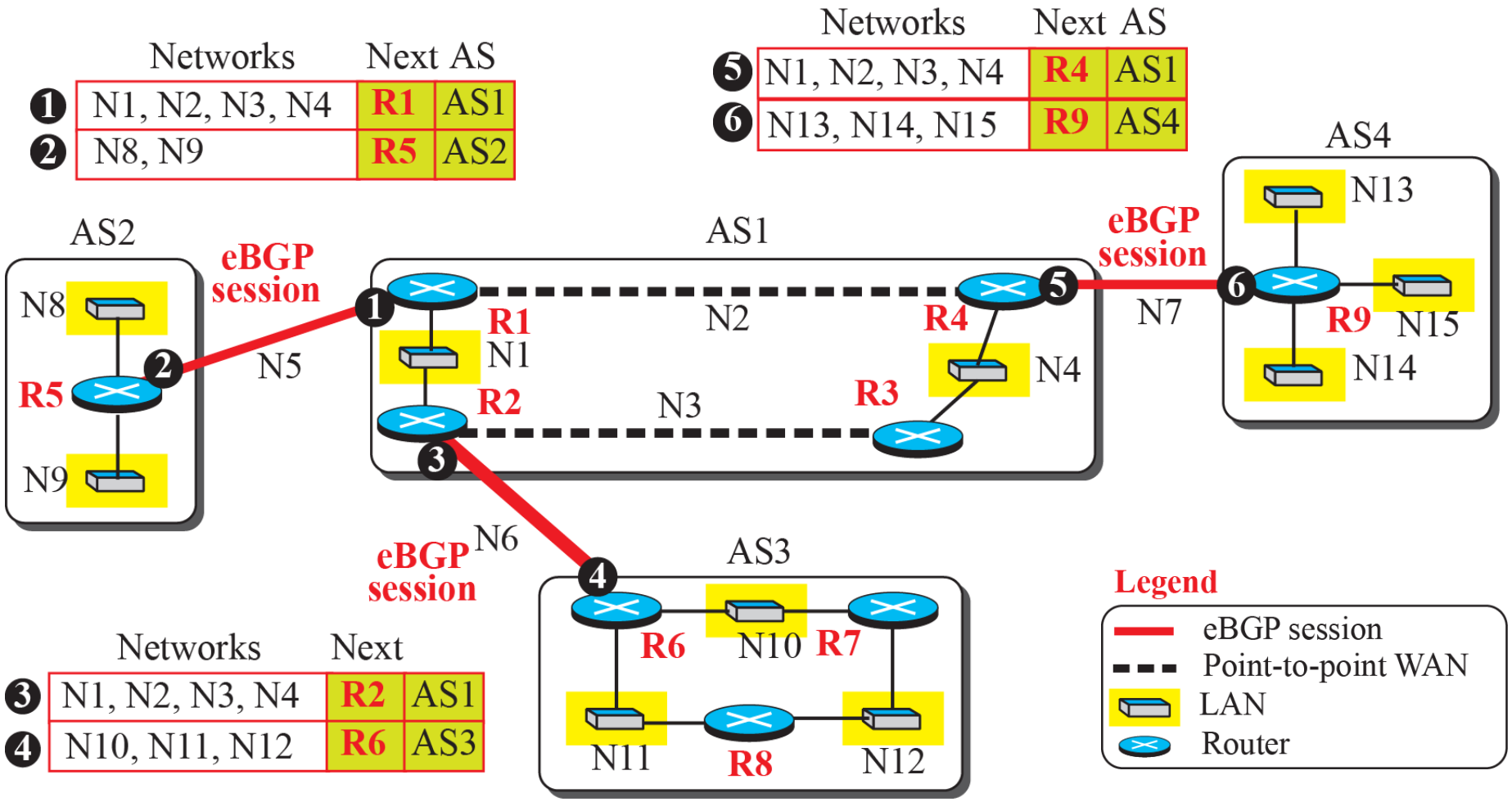

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Figure 4.81: Combination of eBGP and iBGP sessions in our internet

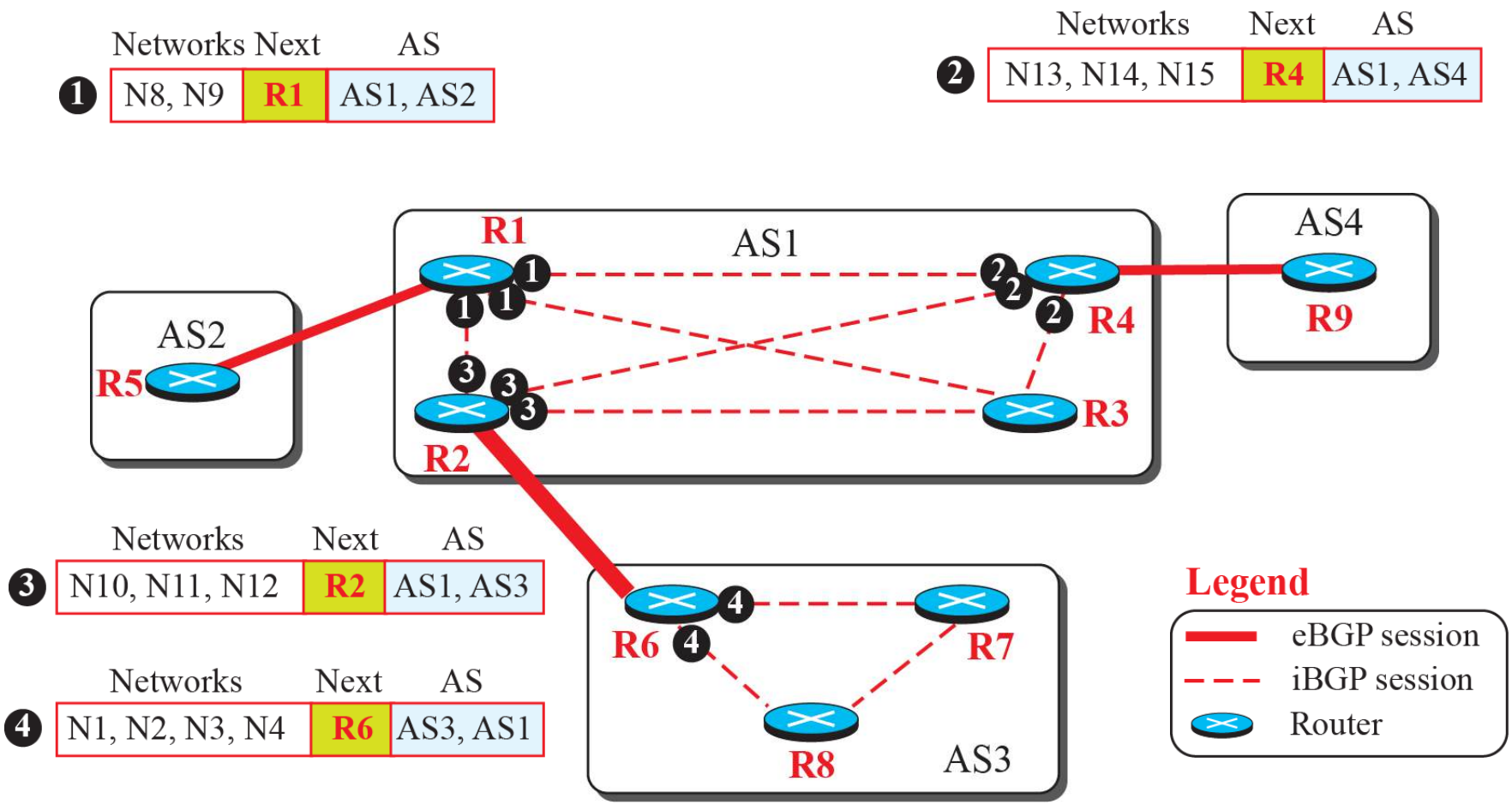

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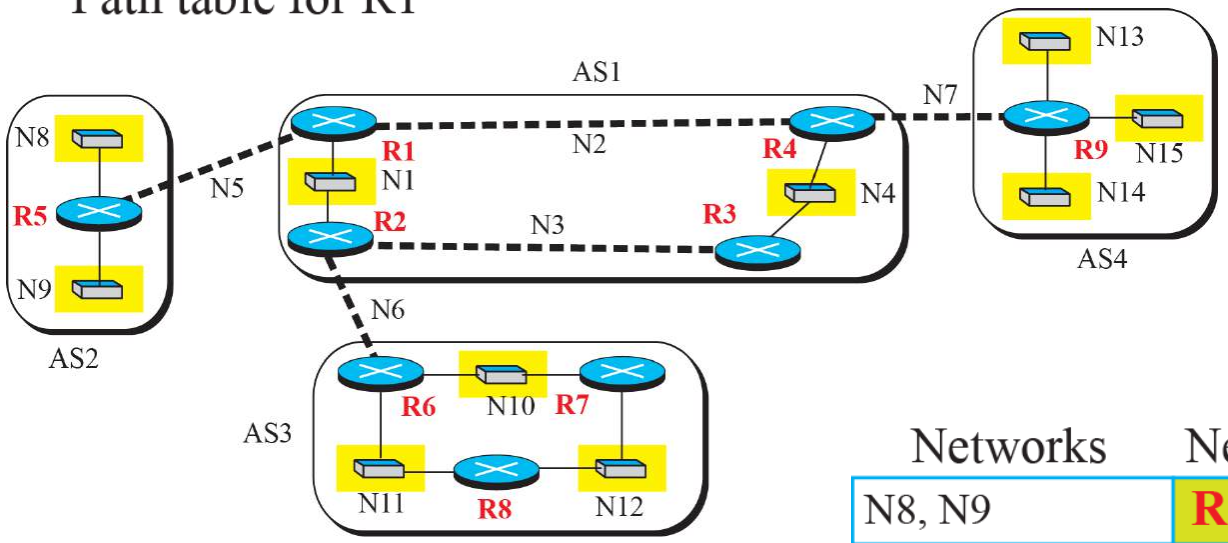
Figure 4.82: Finalized BGP path tables (Part I)

Networks	Next	Path
N8, N9	R5	AS1, AS2
N10, N11, N12	R2	AS1, AS3
N13, N14, N15	R4	AS1, AS4

Path table for R1



Networks	Next	Path
N8, N9	R1	AS1, AS2
N10, N11, N12	R6	AS1, AS3
N13, N14, N15	R1	AS1, AS4

Path table for R2

Networks	Next	Path
N8, N9	R2	AS1, AS2
N10, N11, N12	R2	AS1, AS3
N13, N14, N15	R4	AS1, AS4

Path table for R3

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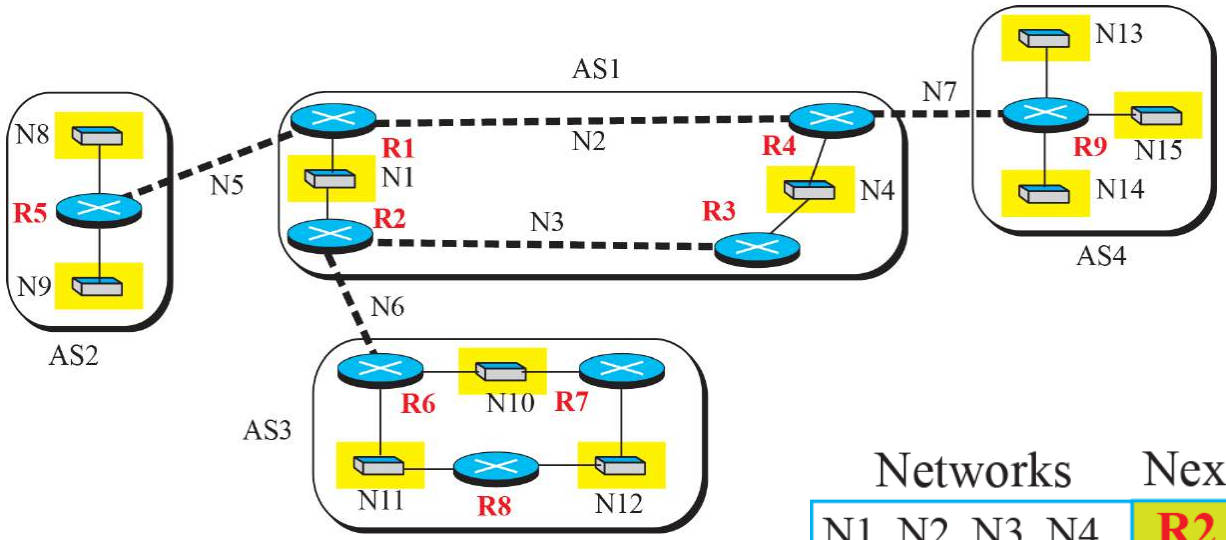
Figure 4.82: Finalized BGP path tables (Part II)

Networks	Next	Path
N1, N2, N3, N4	R1	AS2, AS1
N10, N11, N12	R1	AS2, AS1, AS3
N13, N14, N15	R1	AS2, AS1, AS4

Path table for R5

Networks	Next	Path
N8, N9	R1	AS1, AS2
N10, N11, N12	R1	AS1, AS3
N13, N14, N15	R9	AS1, AS4

Path table for R4



Networks	Next	Path
N1, N2, N3, N4	R2	AS3, AS1
N8, N9	R2	AS3, AS1, AS2
N13, N14, N15	R2	AS3, AS1, AS4

Path table for R6

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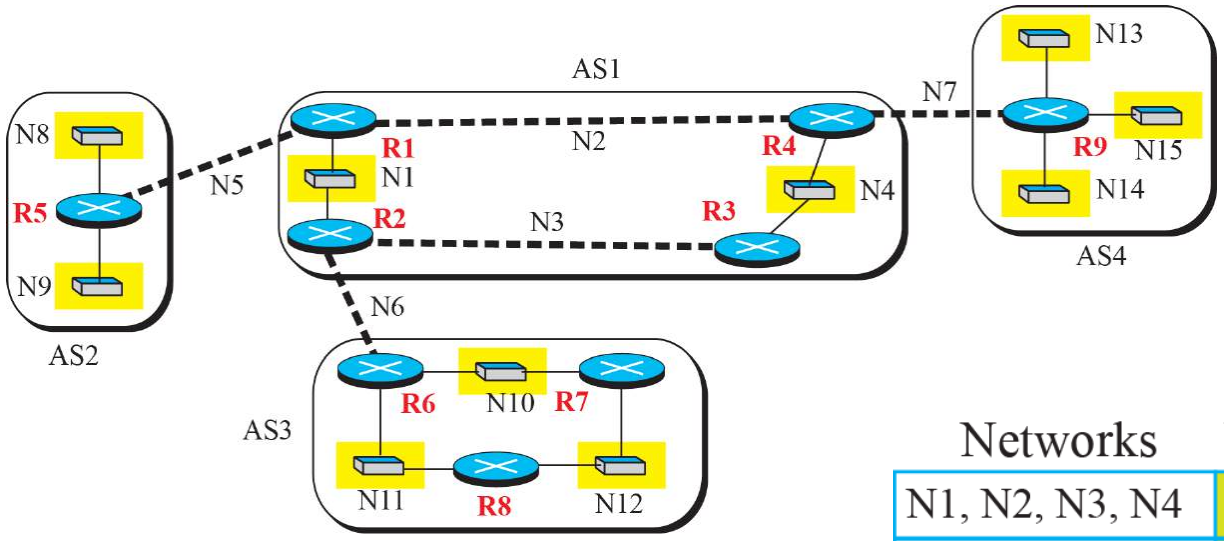
Figure 4.82: Finalized BGP path tables (Part III)

Networks	Next	Path
N1, N2, N3, N4	R4	AS4, AS1
N8, N9	R4	AS4, AS1, AS2
N10, N11, N12	R4	AS4, AS1, AS3

Path table for R9



Networks	Next	Path
N1, N2, N3, N4	R6	AS3, AS1
N8, N9	R6	AS3, AS1, AS2
N13, N14, N15	R6	AS3, AS1, AS4

Path table for R8

Networks	Next	Path
N1, N2, N3, N4	R6	AS3, AS1
N8, N9	R6	AS3, AS1, AS2
N13, N14, N15	R6	AS3, AS1, AS4

Path table for R7

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Figure 4.83: Forwarding tables after injection from BGP (Part I)

Des.	Next	Cost
N1	—	1
N4	R4	2
N8	R5	1
N9	R5	1
N10	R2	2
N11	R2	2
N12	R2	2
N13	R4	2
N14	R4	2
N15	R4	2

Table for R1

Des.	Next	Cost
N1	—	1
N4	R3	2
N8	R1	2
N9	R1	2
N10	R6	1
N11	R6	1
N12	R6	1
N13	R3	3
N14	R3	3
N15	R3	3

Table for R2

Des.	Next	Cost
N1	R2	2
N4	—	1
N8	R2	3
N9	R2	3
N10	R2	2
N11	R2	2
N12	R2	2
N13	R4	2
N14	R4	2
N15	R4	2

Table for R3

Des.	Next	Cost
N1	R1	2
N4	—	1
N8	R1	2
N9	R1	2
N10	R3	3
N11	R3	3
N12	R3	3
N13	R9	1
N14	R9	1
N15	R9	1

Table for R4

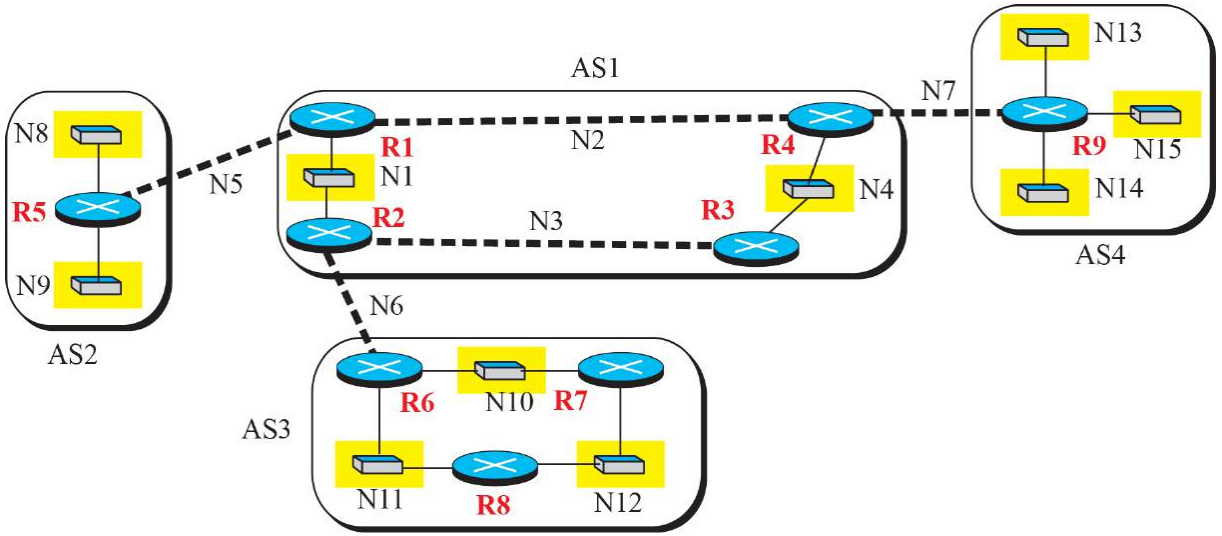

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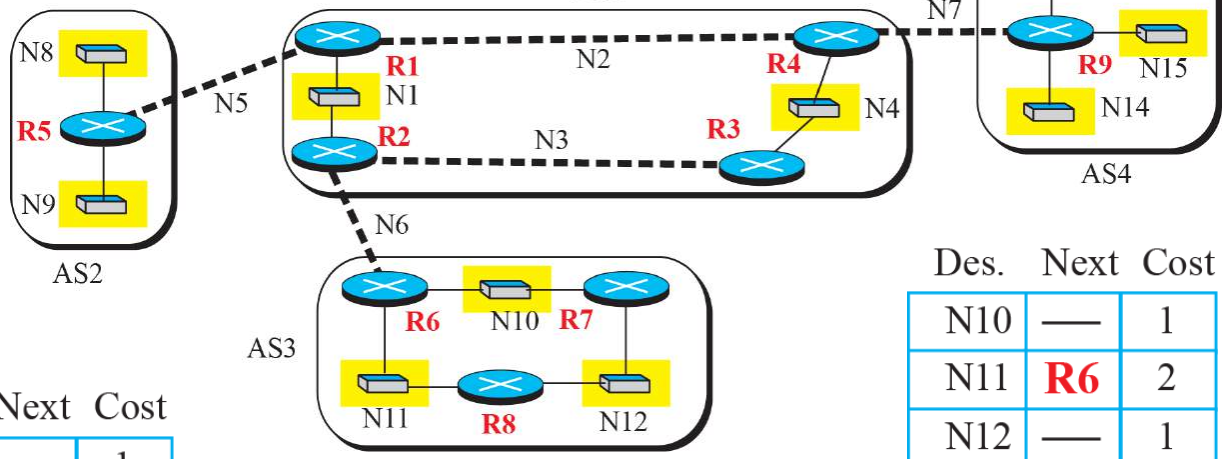
Figure 4.83: Forwarding tables after injection from BGP (Part II)



Des.	Next	Cost
N8	—	1
N9	—	1
0	**R1**	1

Table for R5

Des.	Next	Cost
N13	—	1
N14	—	1
N15	—	1
0	**R4**	1

Table for R9

Des.	Next	Cost
N10	—	1
N11	—	1
N12	**R7**	2
0	**R2**	1

Table for R6

Des.	Next	Cost
N10	**R6**	2
N11	—	1
N12	—	1
0	**R6**	2

Table for R8

Des.	Next	Cost
N10	—	1
N11	**R6**	2
N12	—	1
0	**R6**	2

Table for R7

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Figure 4.84: Format of path attribute

O: Optional bit (set if attribute is optional)
P: Partial bit (set if an optional attribute in lost in transit)
T: Transitive bit (set if attribute is transitive)
E: Extended bit (set if attribute length is two bytes)

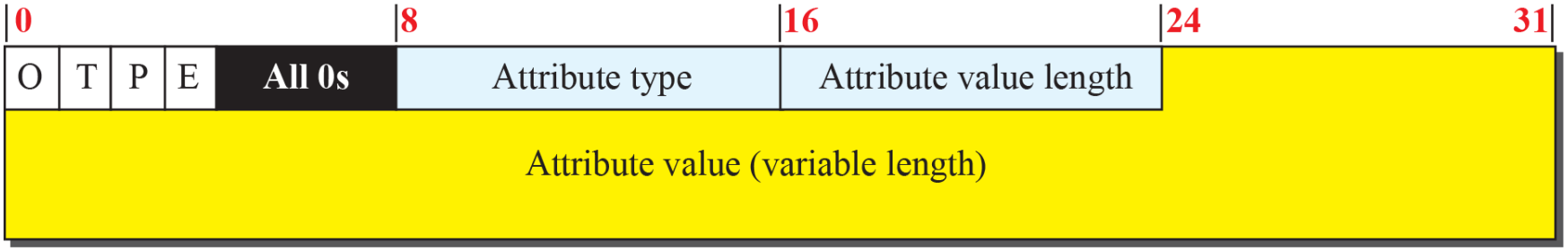

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Figure 4.85: Flow diagram for route selection

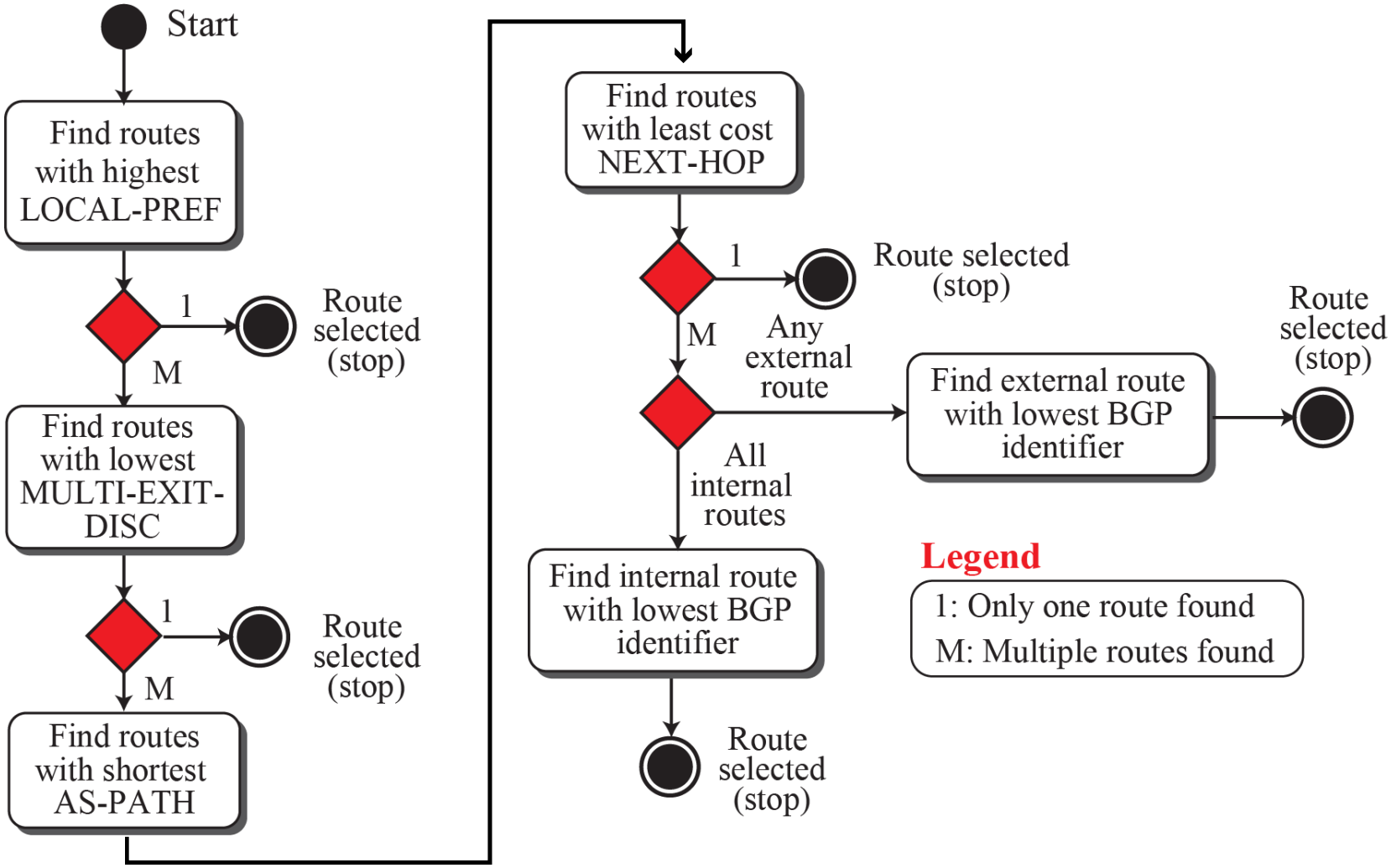

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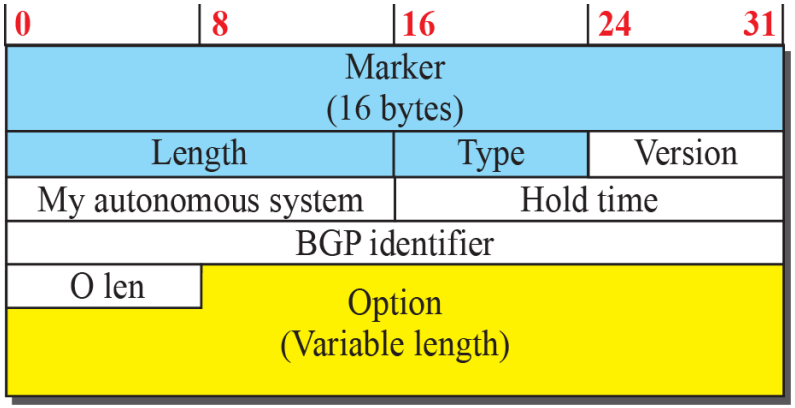
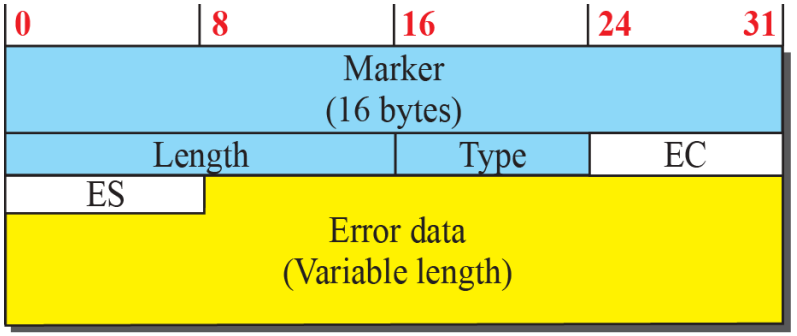
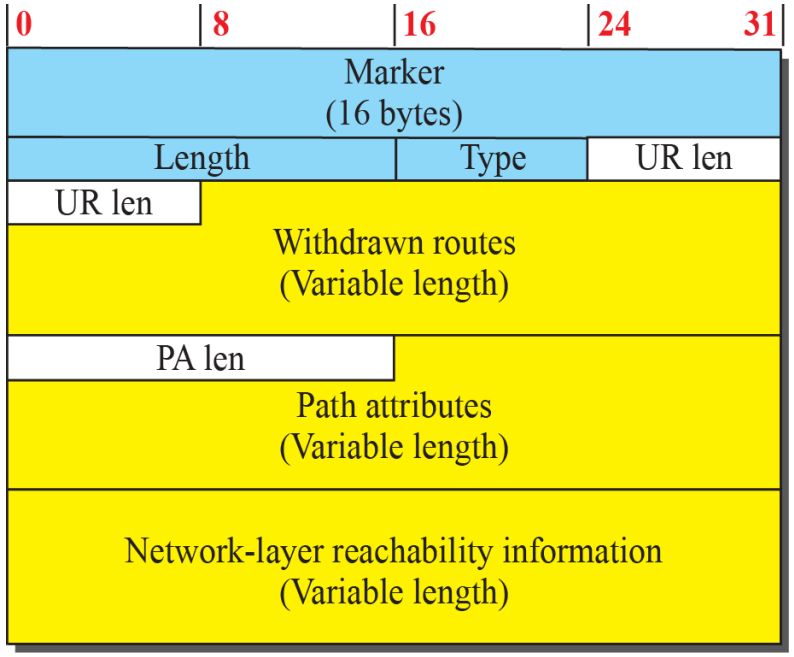
Figure 4.86: BGP messages



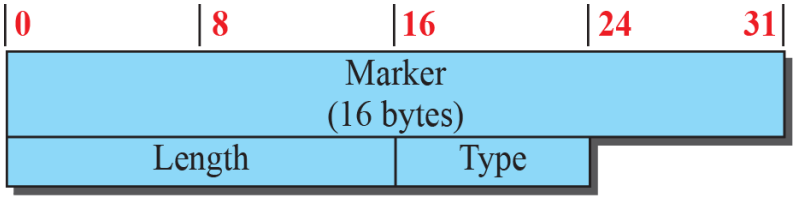
Open message (type 1)



Notification message (type 3)



Update message (type 2)



Keepalive message (type 4)

Fields in common header

Marker: Reserved for authentication
Length: Length of total message in bytes
Type: Type of message (1 to 4)

Abbreviations

O len: Option length
EC: Error code
ES: Error subcode
UR len: Unfeasible route length
PA len: Path attribute length

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4-4 MULTICAST ROUTING

Communication in the Internet today is not only unicasting; multicasting communication is growing fast. In this section, we first discuss the general ideas behind unicasting, multicasting, and broadcasting. We then talk about some basic issues in multicast routing. Finally, we discuss multicasting routing protocols in the Internet.

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4.4.1 Introduction

From the previous sections, we have learned that forwarding a datagram by a router is normally based on the prefix of the destination address in the datagram, which defines the network to which the destination host is connected. Understanding the above forwarding principle, we can now define unicasting, multicasting, and broadcasting. Let us clarify these terms as they relate to the Internet.

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4.4.1 (continued)

- **Unicasting**
- **Multicasting**
  - Multicasting versus Multiple Unicasting
  - Emulation of Multicasting with Unicasting
  - Multicast Applications
- **Broadcasting**

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Figure 4.87: Unicasting

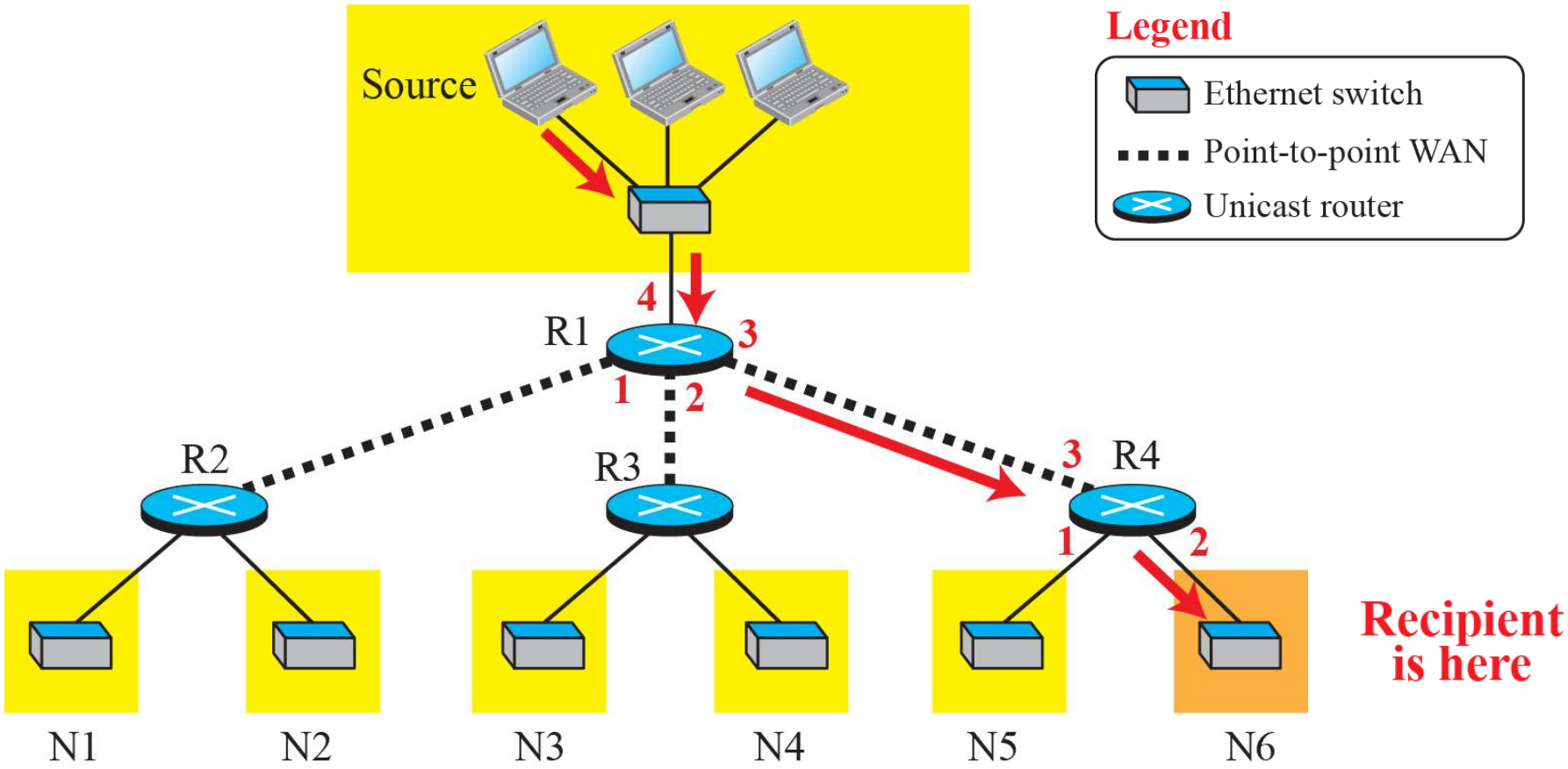

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Figure 4.88: Multicasting

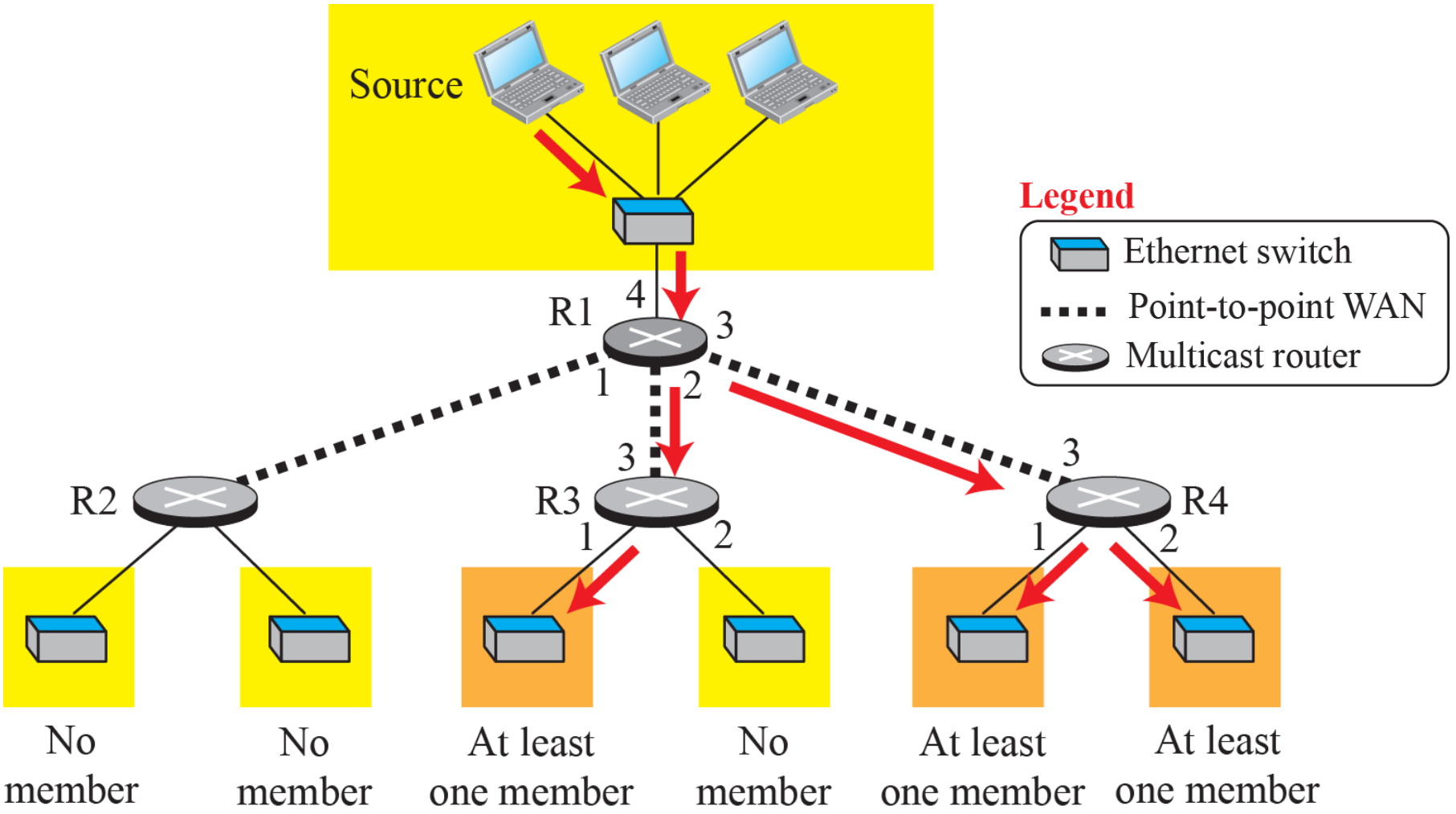

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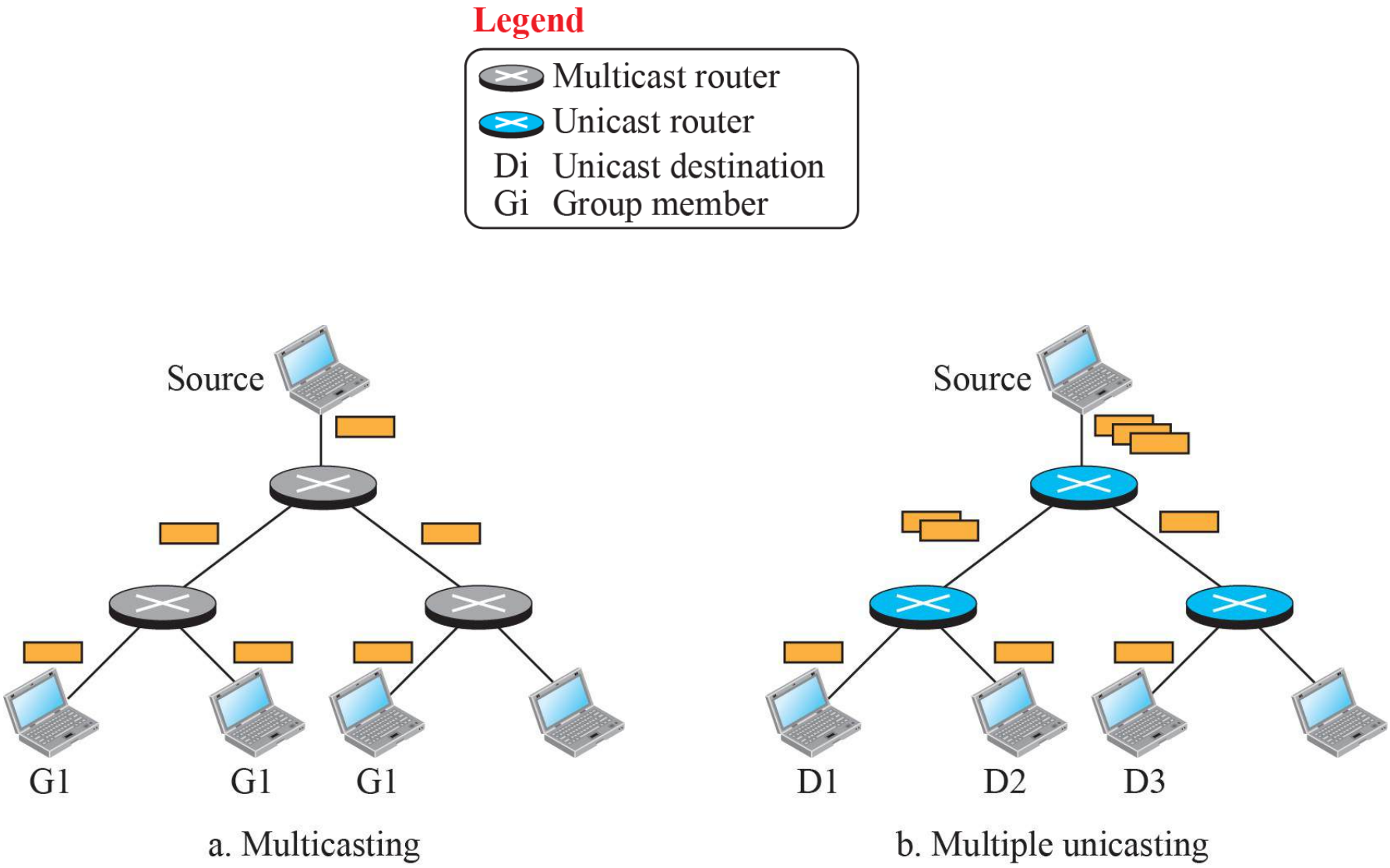
Figure 4.89: Multicasting versus multiple unicasting



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4.4.2 Multicasting Basics

Before discussing multicast routing protocols in the Internet, we need to discuss some multicasting basics: multicast addressing, collecting information about multicast groups, and multicast optimal trees.

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4.4.2 (continued)

Multicast Addresses

Multicast Addresses in IPv4

- Local Network Control Block
- Internetwork Control Block.
- Source-Specific Multicast (SSM) Block.
- GLOP Block.
- Administratively Scoped Block.

Selecting Multicast Address

- Limited Group
- Larger Group

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4.4.2 (continued)

- **Collecting Information about Groups**
  - Internet Group Management Protocol (IGMP)
- **Multicast Forwarding**
- **Two Approaches to Multicasting**
  - Source-Based Tree Approach
  - Group-Shared Tree Approach

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Figure 4.90: Needs for multicast addresses

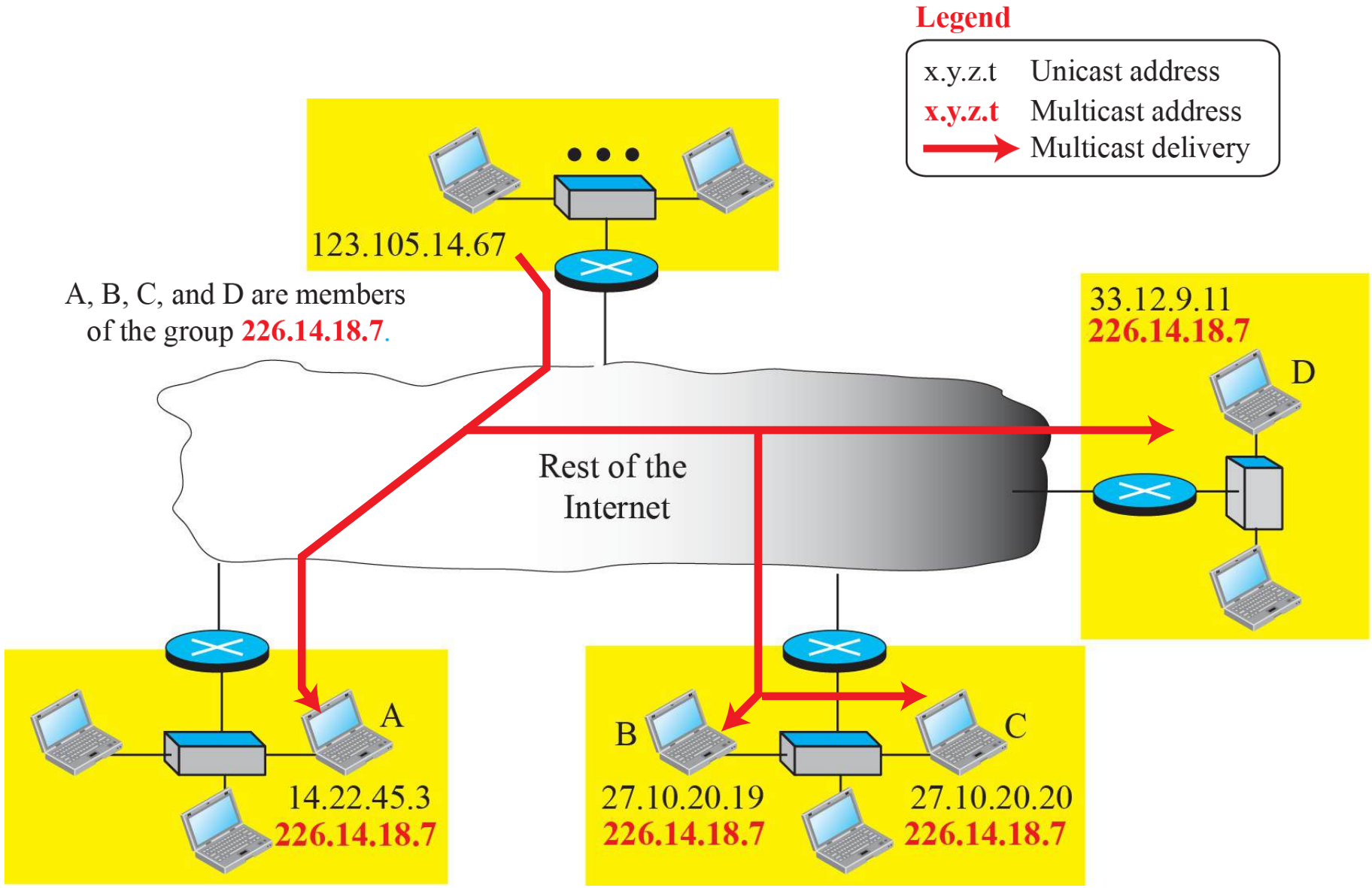

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Figure 4.91: A multicast address in binary

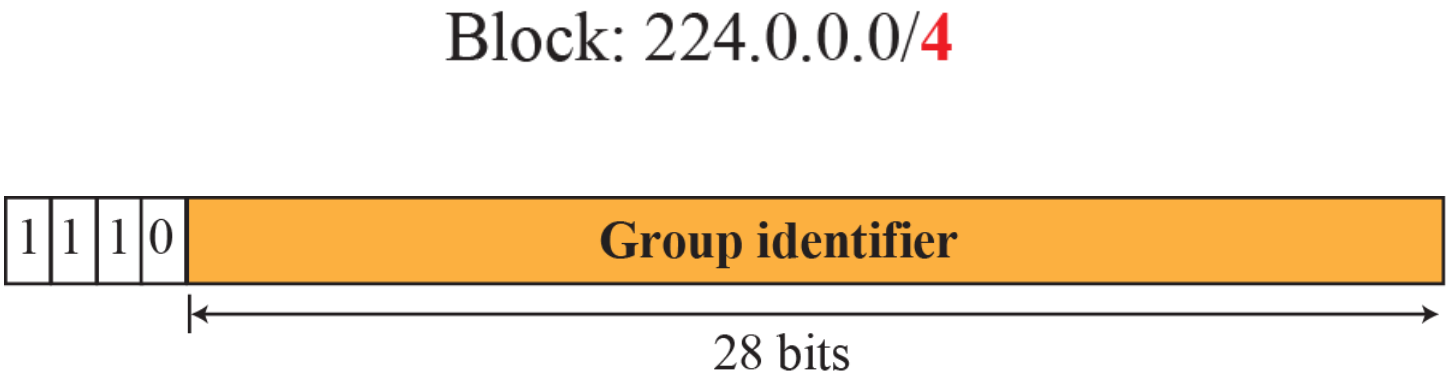

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Figure 4.92: Unicast versus multicast advertisement

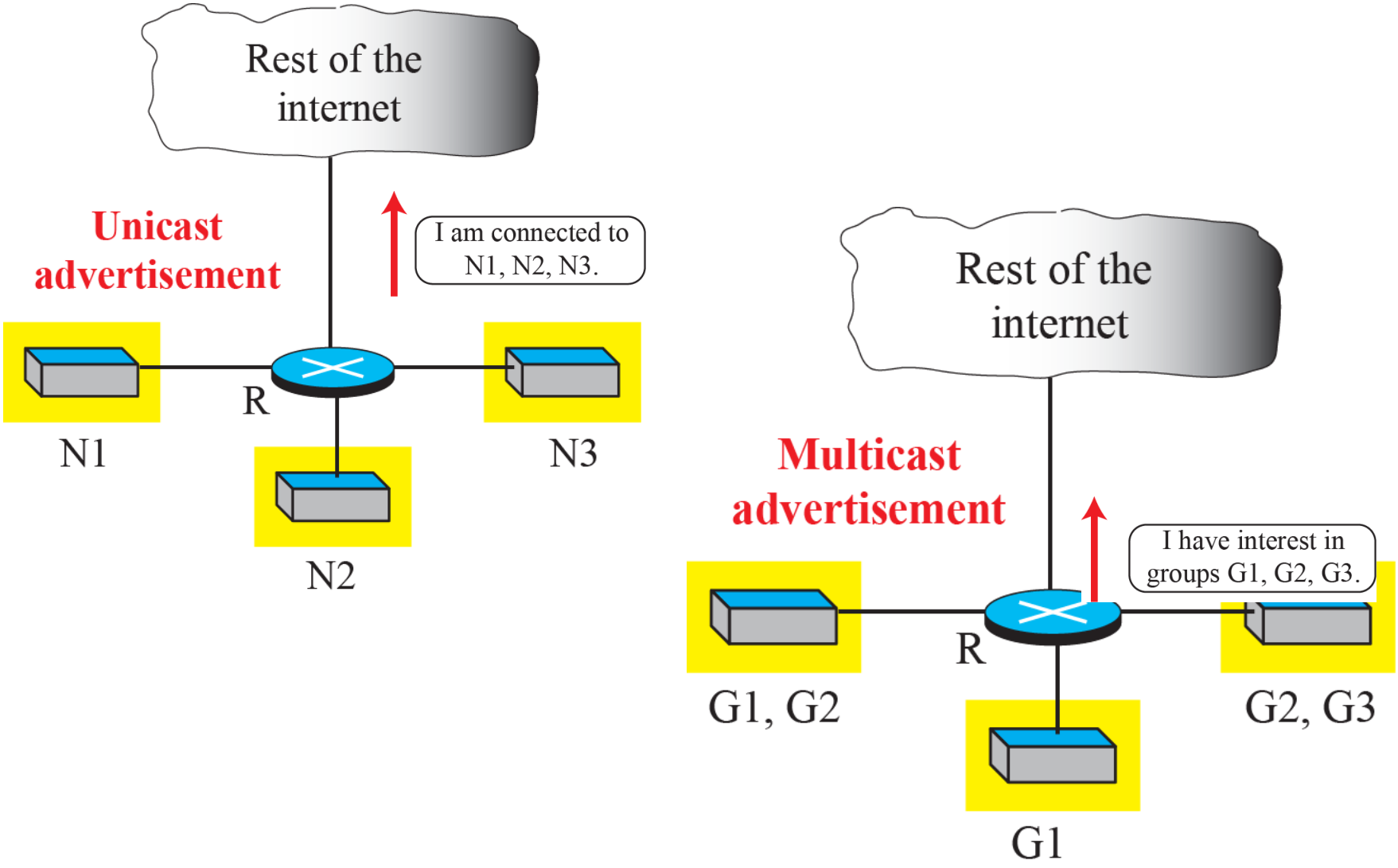

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Figure 4.93: IGMP operation

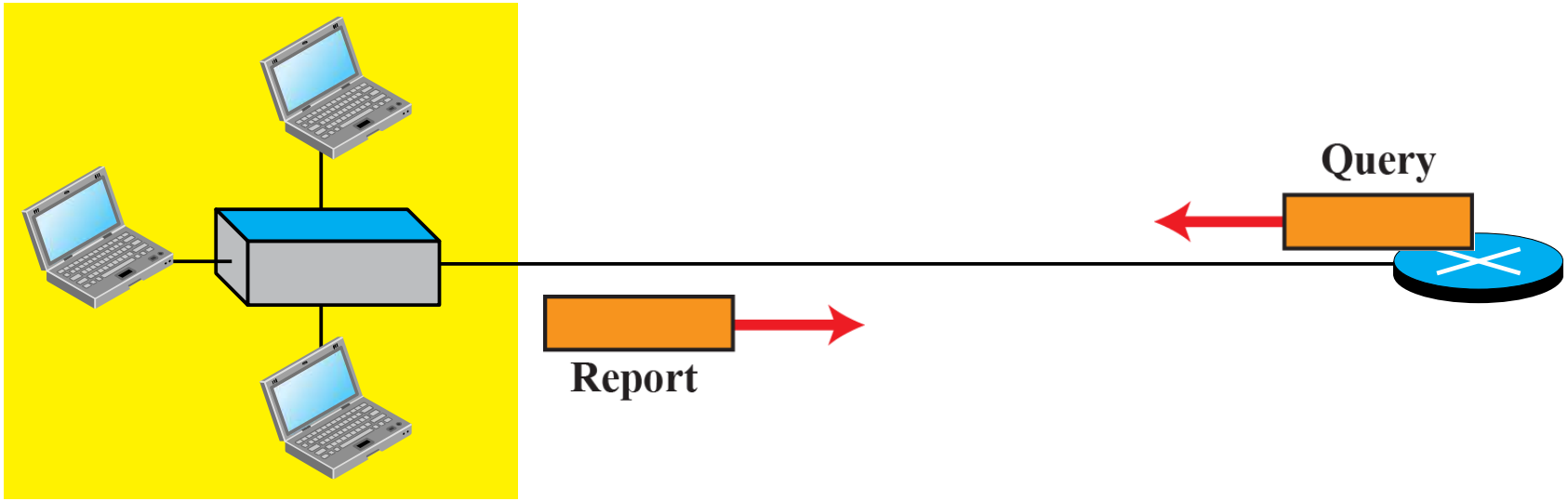

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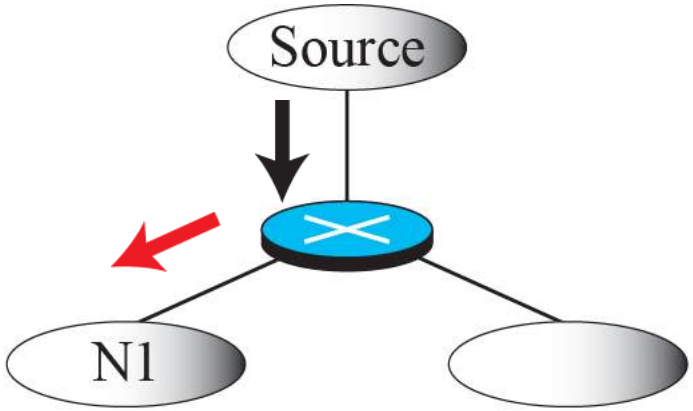
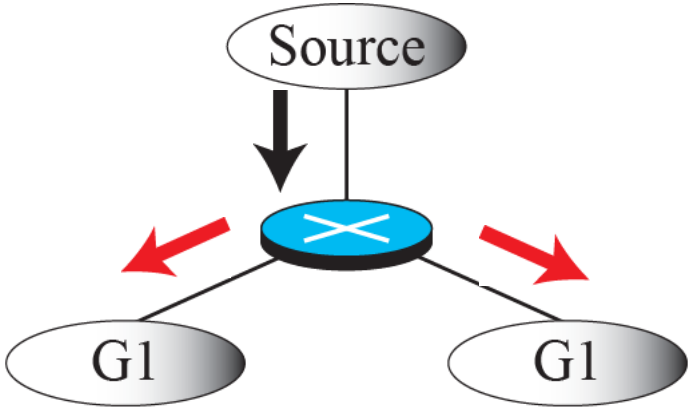
Figure 4.94: Destination in unicasting and multicasting



a. Destination in unicasting is one



b. Destination in mulicasting is more than one

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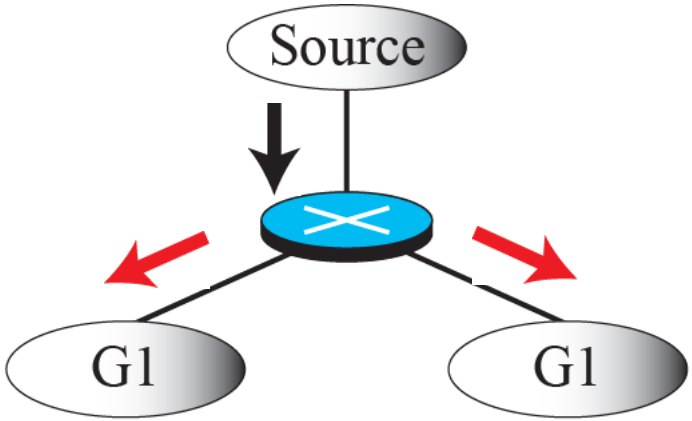
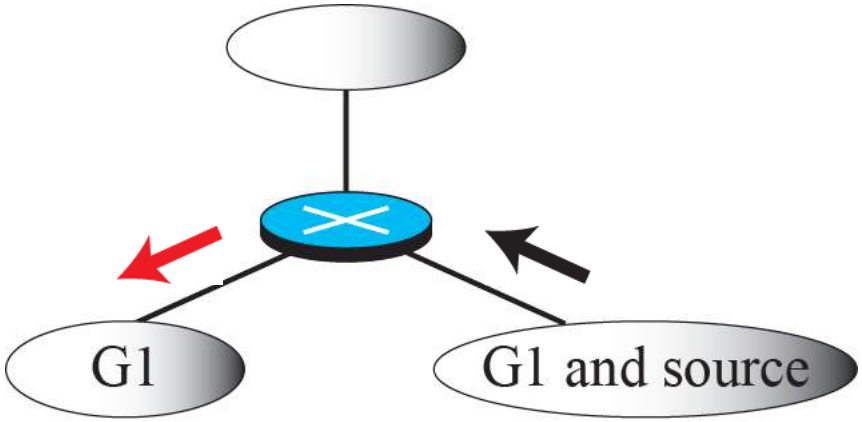
Figure 4.95: Forwarding depends on the destination and the source



a. Packet sent out of two interfaces



b. Packet sent out of one interface

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4.4.3 Intradomain Routing Protocols

During the last few decades, several intradomain multicast routing protocols have emerged. In this section, we discuss three of these protocols. Two are extensions of unicast routing protocols (RIP and OSPF), using the source-based tree approach; the third is an independent protocol which is becoming more and more popular. It can be used in two modes, employing either the source-based tree approach or the shared-group tree approach.

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4.4.3 (continued)

- **Multicast Distance Vector (DVMRP)**
  - Reverse Path Forwarding (RPF)
  - Reverse Path Broadcasting (RPB)
  - Reverse Path Multicasting (RPM)
- **Multicast Link State (MOSPF)**
- **Protocol Independent Multicast (PIM)**

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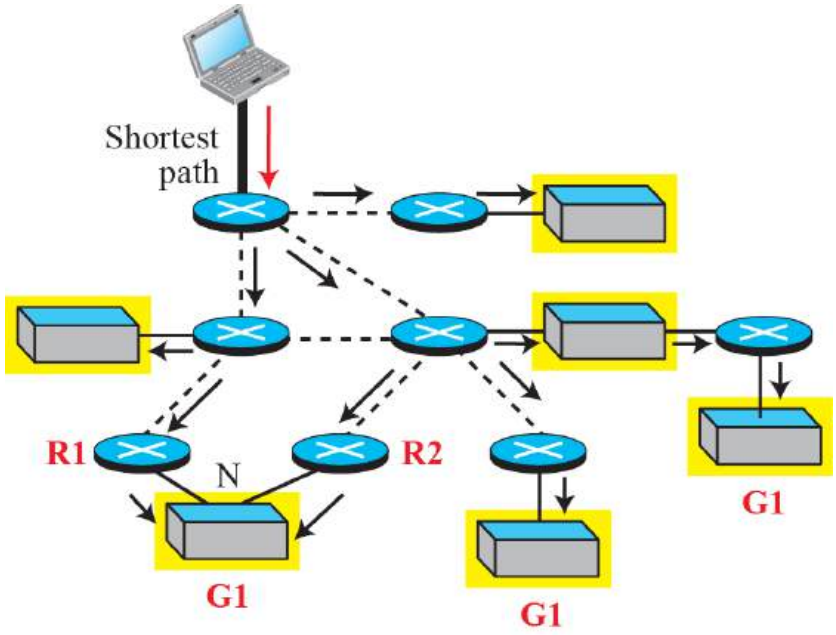
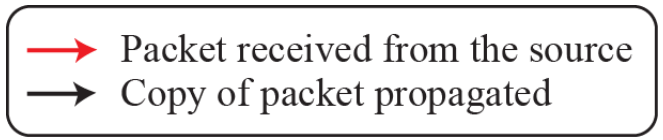
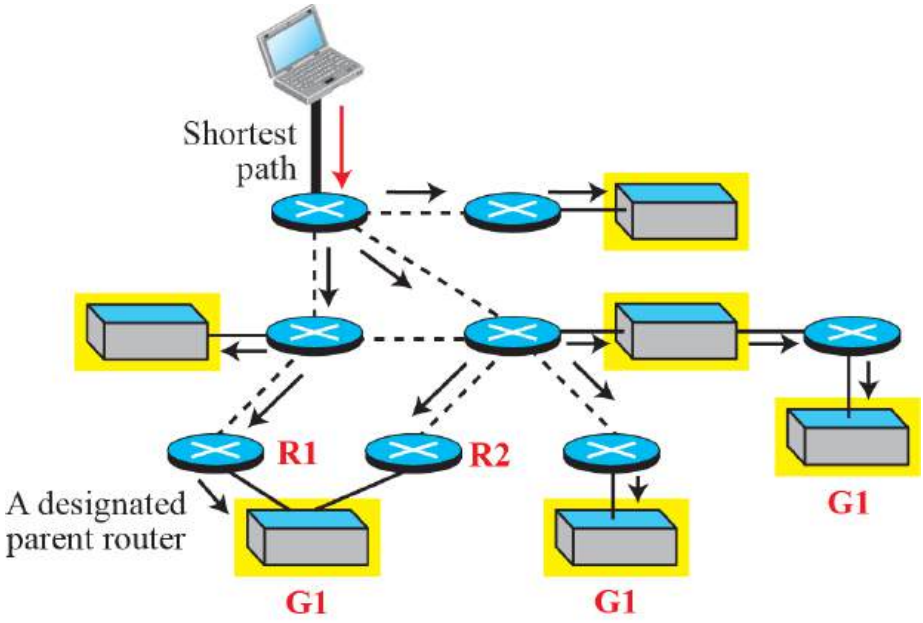
Figure 4.96: RPF versus RPB



a. Using RPF, N receives two copies.





b. Using RPB, N receives only one copy.

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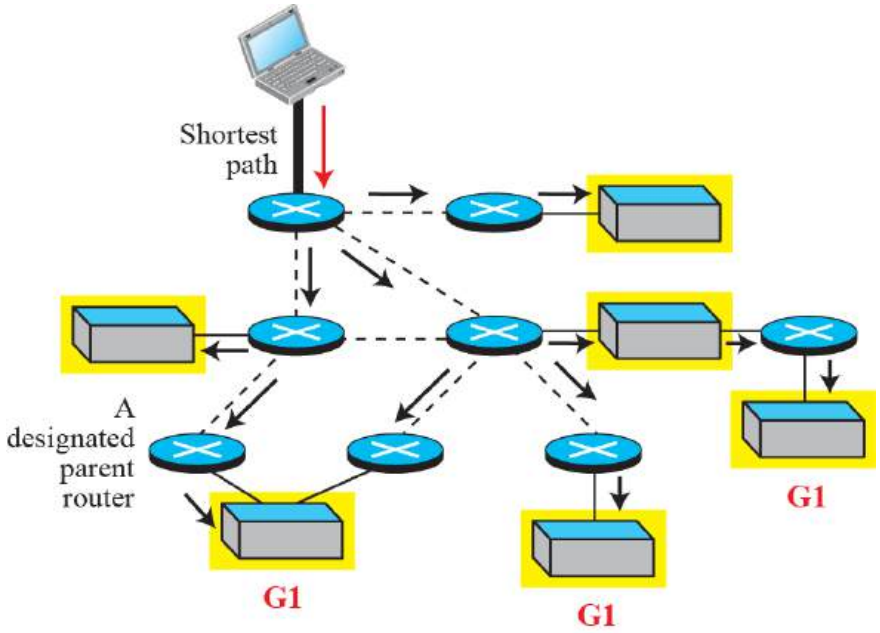
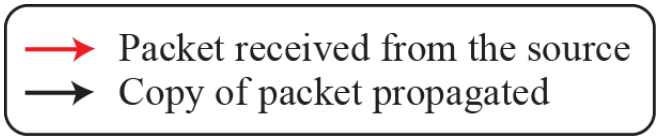
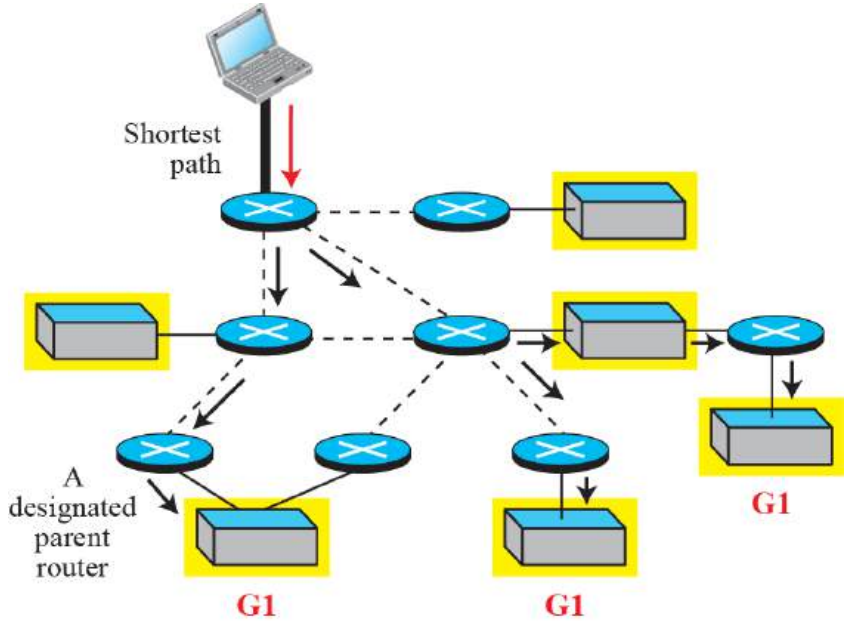
Figure 4.97: RPB versus RPM



a. Using RPB, all networks receive a copy.





b. Using RPM, only members receive a copy.

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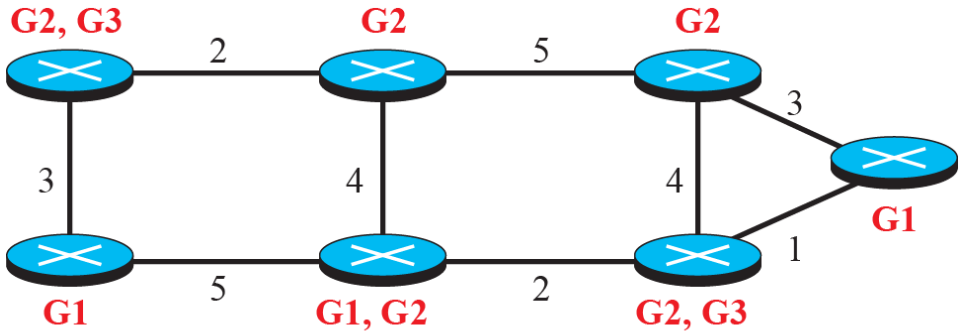
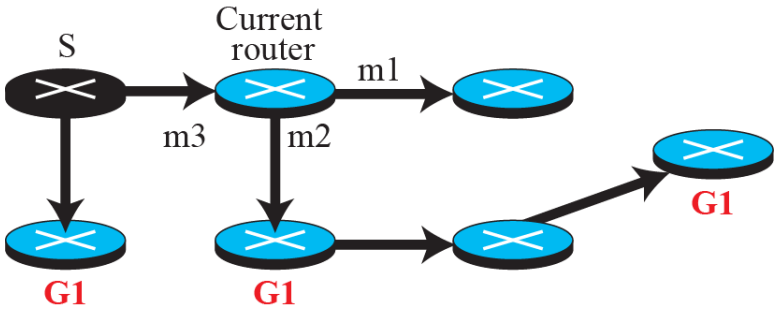
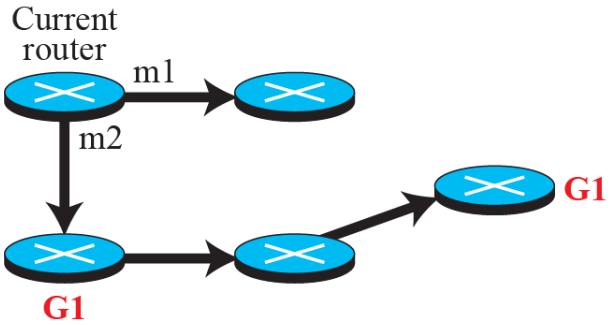
Figure 4.98: Example of tree formation in MOSPF



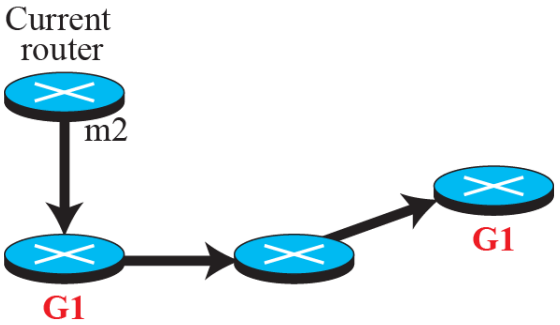
a. An internet with some active groups



b. S-G1 shortest-path tree



c. S-G1 subtree seen by current router



d. S-G1 pruned subtree

Forwarding table for current router

Group-Source	Interface
S, G1	m2
...	...

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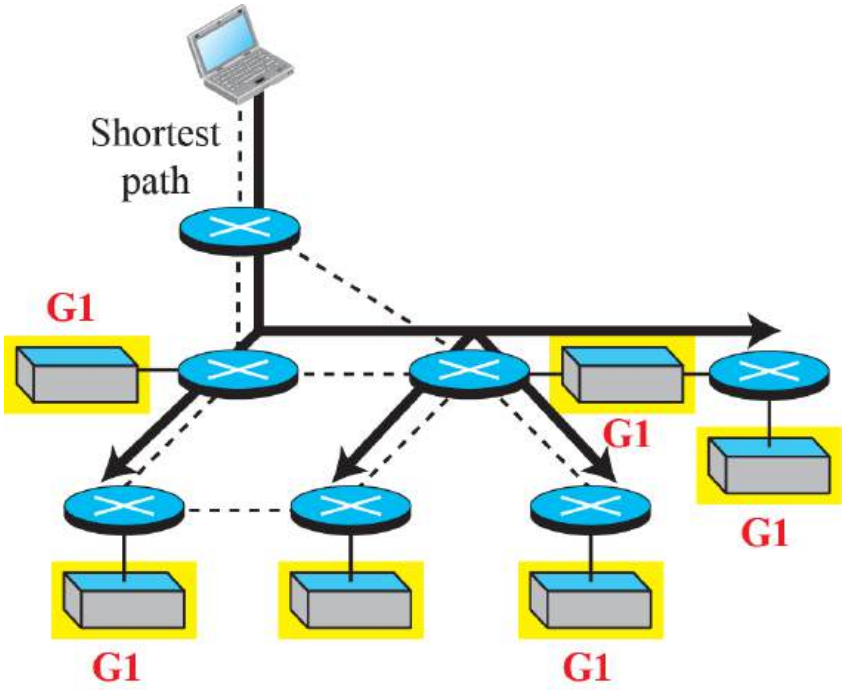
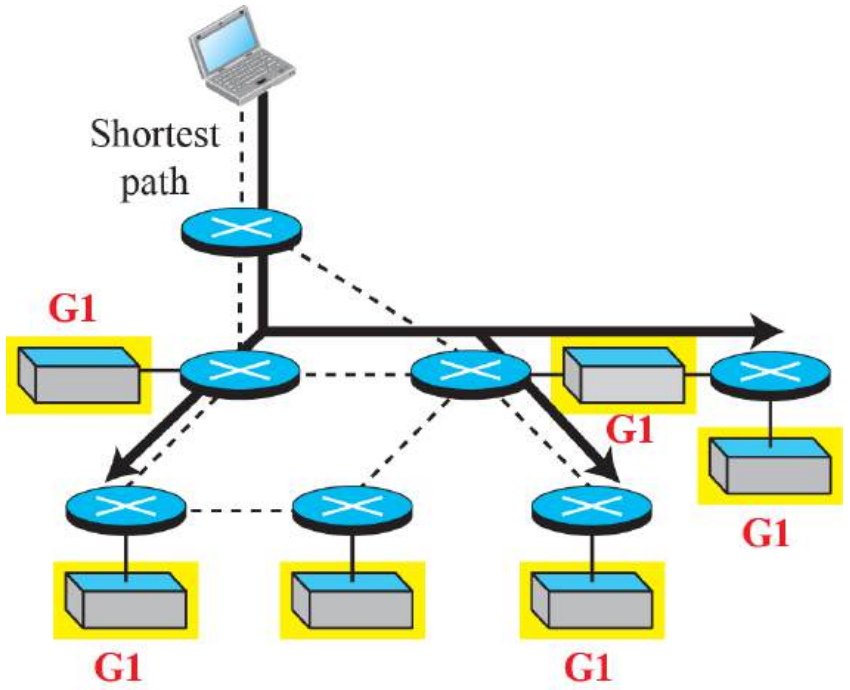
Figure 4.99: Idea behind PIM-DM



a. First packet is broadcast.



b. Second packet is multicast.

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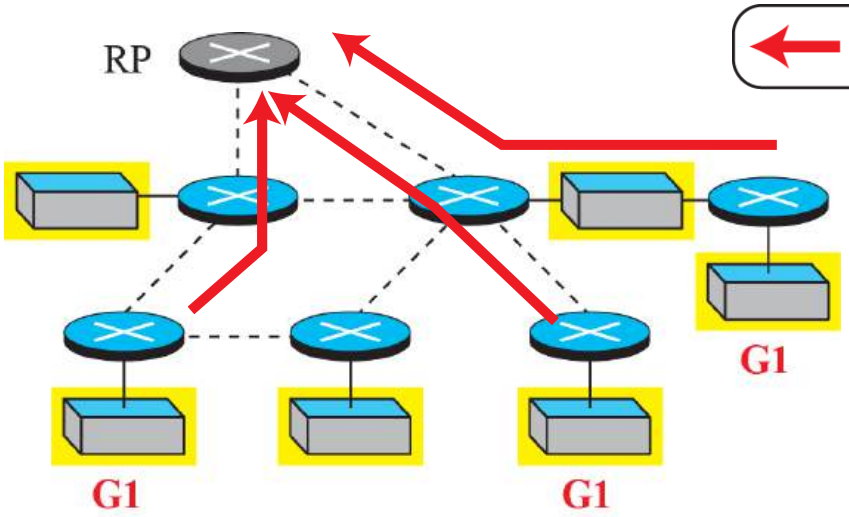
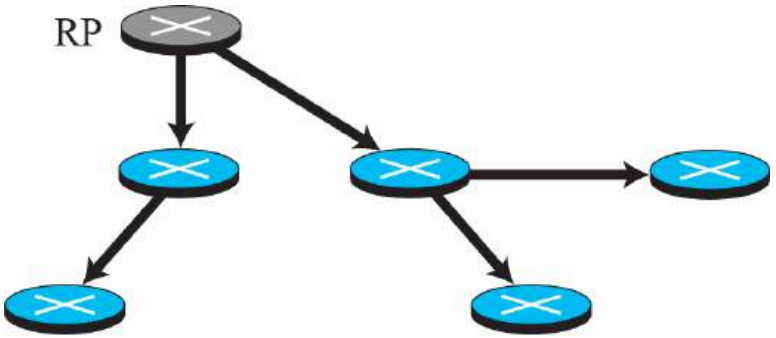
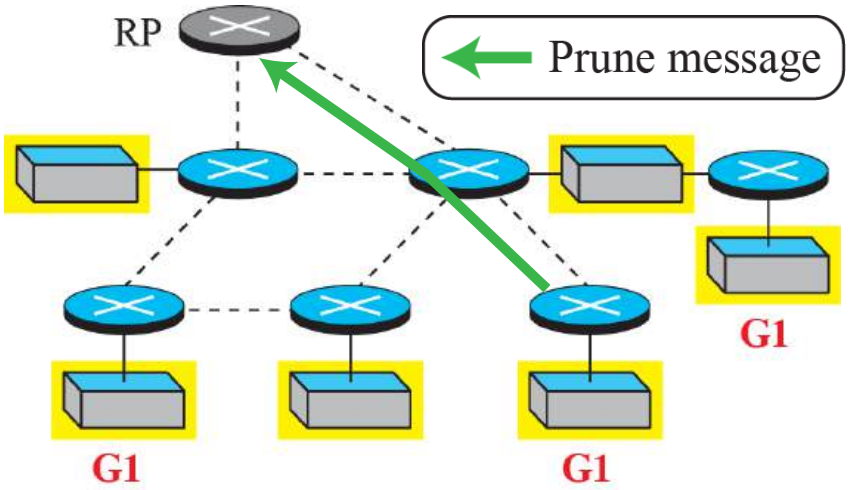
Figure 4.100: Idea behind PIM-SM



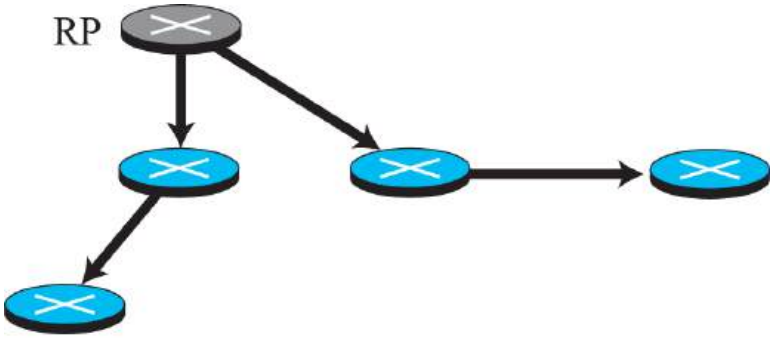
a. Three networks join group G1



b. Multicast tree after joins



c. One network leaves group G1



d. Multicast tree after pruning

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4.4.4 Interdomain Routing Protocols

The three protocols we discussed for multicast routing, DVMRP, MOSPF, and PIM, are designed to provide multicast communication inside an autonomous system. When the members of the groups are spread among different domains (ASs), we need an interdomain multicast routing protocol. One common protocol for interdomain multicast routing is called Multicast Border Gateway Protocol (MBGP), which is the extension of BGP protocol we discussed for interdomain unicast routing. MBGP provides two paths between ASs: one for unicasting and one for multicasting.

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4-5 NEXT GENERATION IP

The address depletion of IPv4 and other shortcomings of this protocol prompted a new version of IP protocol in the early 1990s. The new version, which is called Internet Protocol version 6 (IPv6) or IP new generation (IPng) was a proposal to augment the address space of IPv4 and at the same time redesign the format of the IP packet and revise some auxiliary protocols such as ICMP.

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4.5.1 Packet Format

The IPv6 packet is shown in Figure 4.101. Each packet is composed of a base header followed by the payload. The base header occupies 40 bytes, whereas payload can be up to 65,535 bytes of information.

- ***Concept of Flow and Priority in IPv6***
- ***Fragmentation and Reassembly***
- ***Extension Headers***

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Figure 4.101: IPv6 datagram

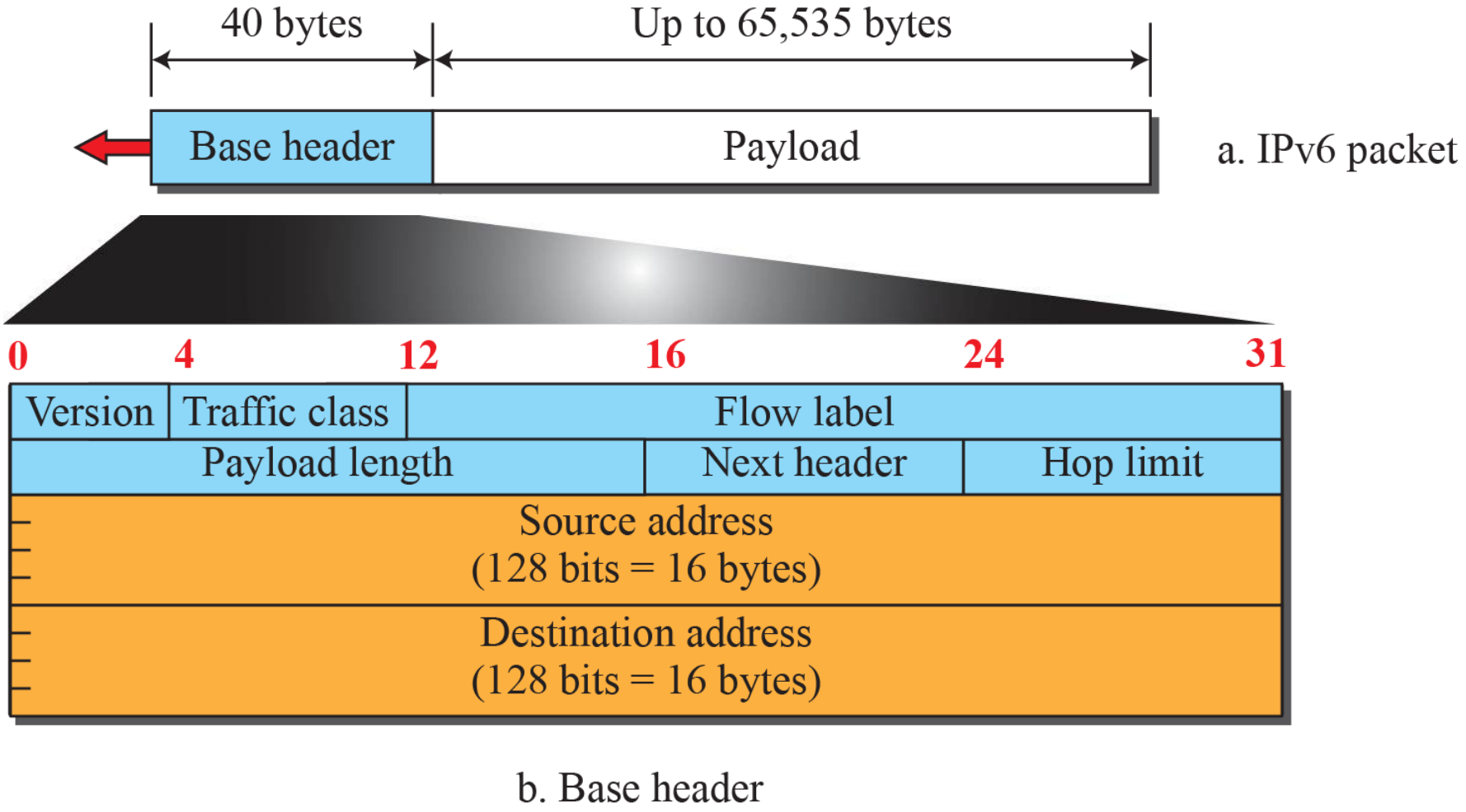

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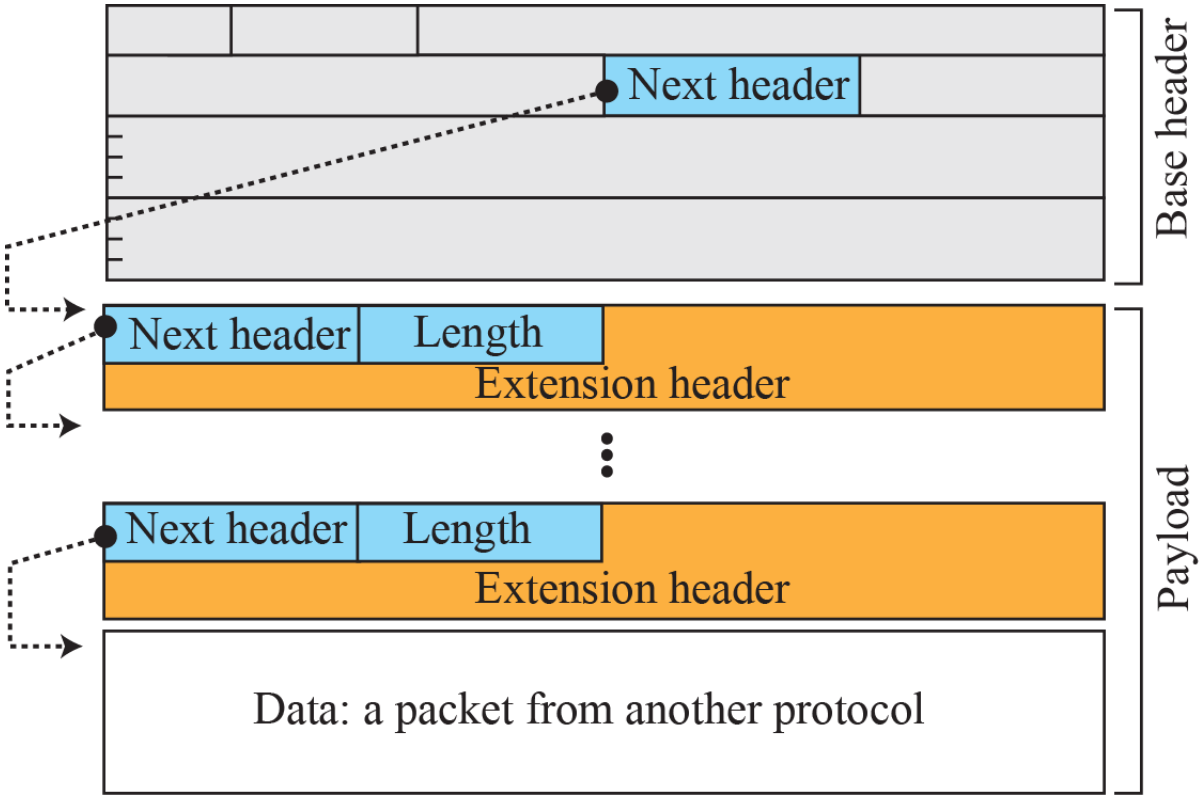
Figure 4.102: Payload in an IPv6 datagram



Some next-header codes

00: Hop-by-hop option
02: ICMPv6
06: TCP
17: UDP
43: Source-routing option
44: Fragmentation option
50: Encrypted security payload
51:Authentication header
59: Null (no next header)
60: Destination option

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4.5.2 IPv6 Addressing

The main reason for migration from IPv4 to IPv6 is the small size of the address space in IPv4. In this section, we show how the huge address space of IPv6 prevents address depletion in the future. We also discuss how the new addressing responds to some problems in the IPv4 addressing mechanism. An IPv6 address is 128 bits or 16 bytes (octets) long, four times the address length in IPv4.

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4.5.2 (continued)

- **Address Space**
- **Three Address Types**
- **Address Space Allocation**
  - Global Unicast Addresses
  - Special Addresses
  - Other Assigned Blocks

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Table 4.7: Prefixes for assigned IPv6 addresses

Block prefix	*CIDR*	*Block assignment*	*Fraction*
0000 0000	0000::/8	Special addresses	1/256
001	**2000::/3**	**Global unicast**	**1/8**
1111 110	FC00::/7	Unique local unicast	1/128
1111 1110 10	FE80::/10	Link local addresses	1/1024
1111 1111	FF00::/8	Multicast addresses	1/256

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Figure 4.103: *Global unicast address*

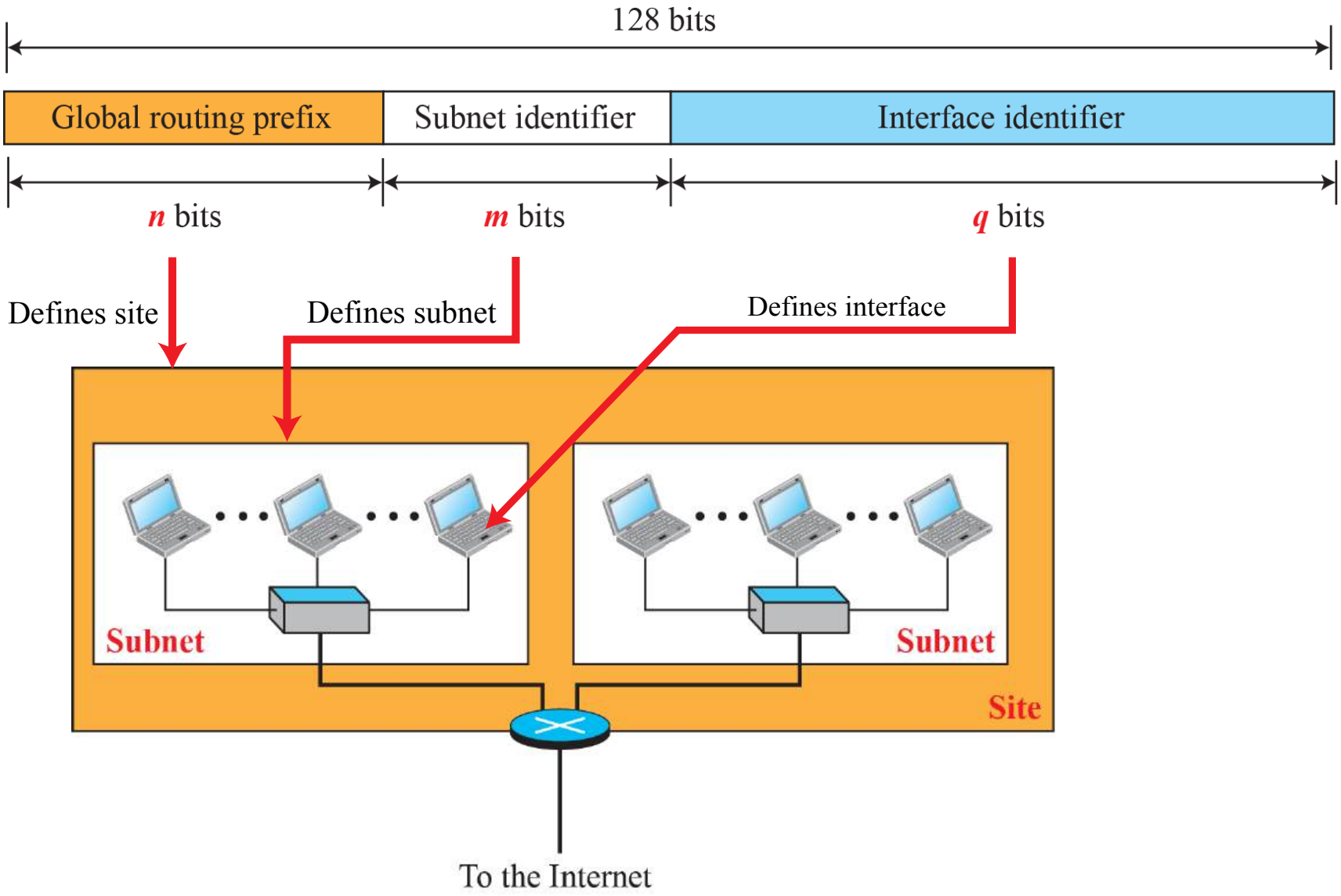

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Figure 4.104: *Special addresses*

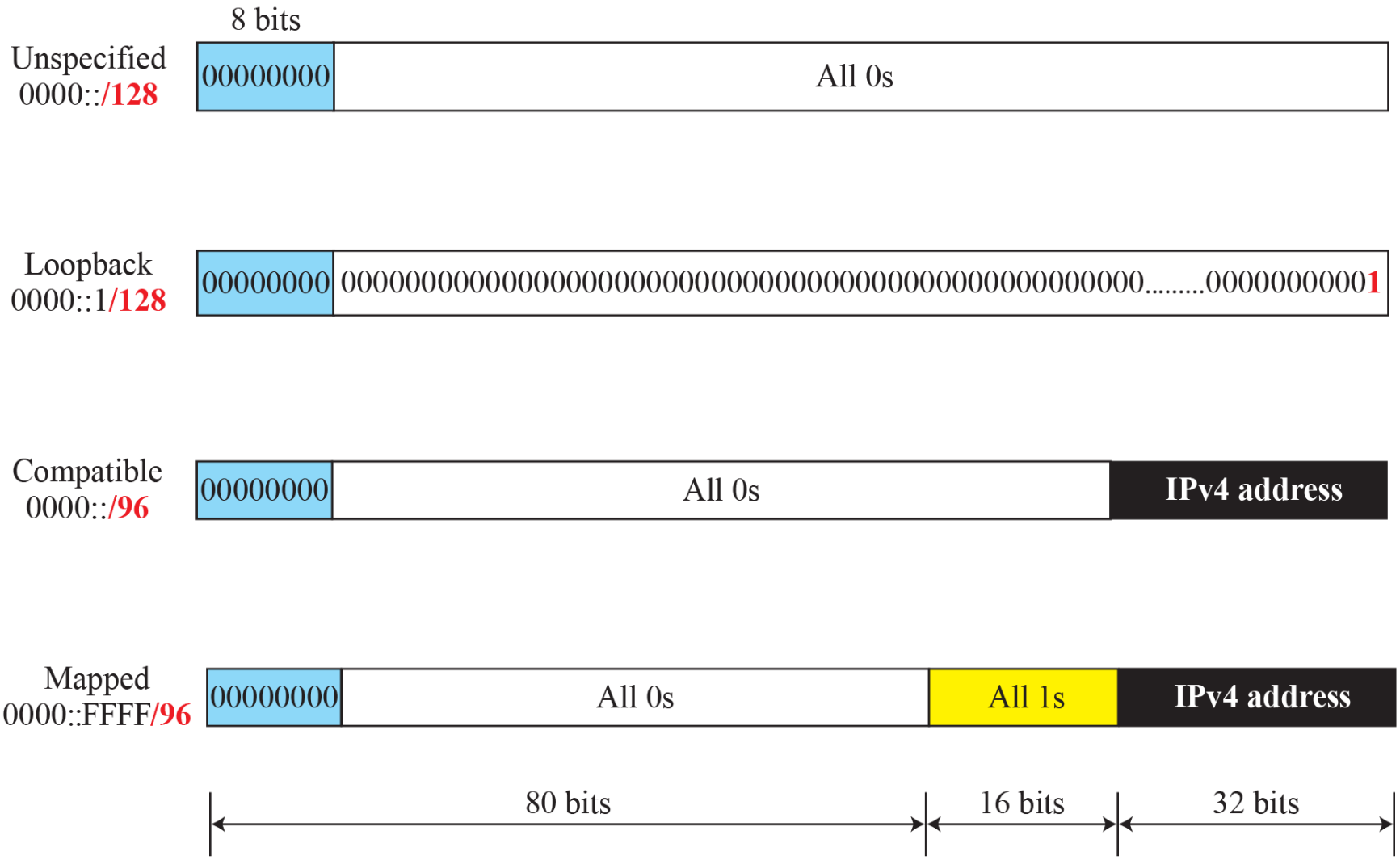

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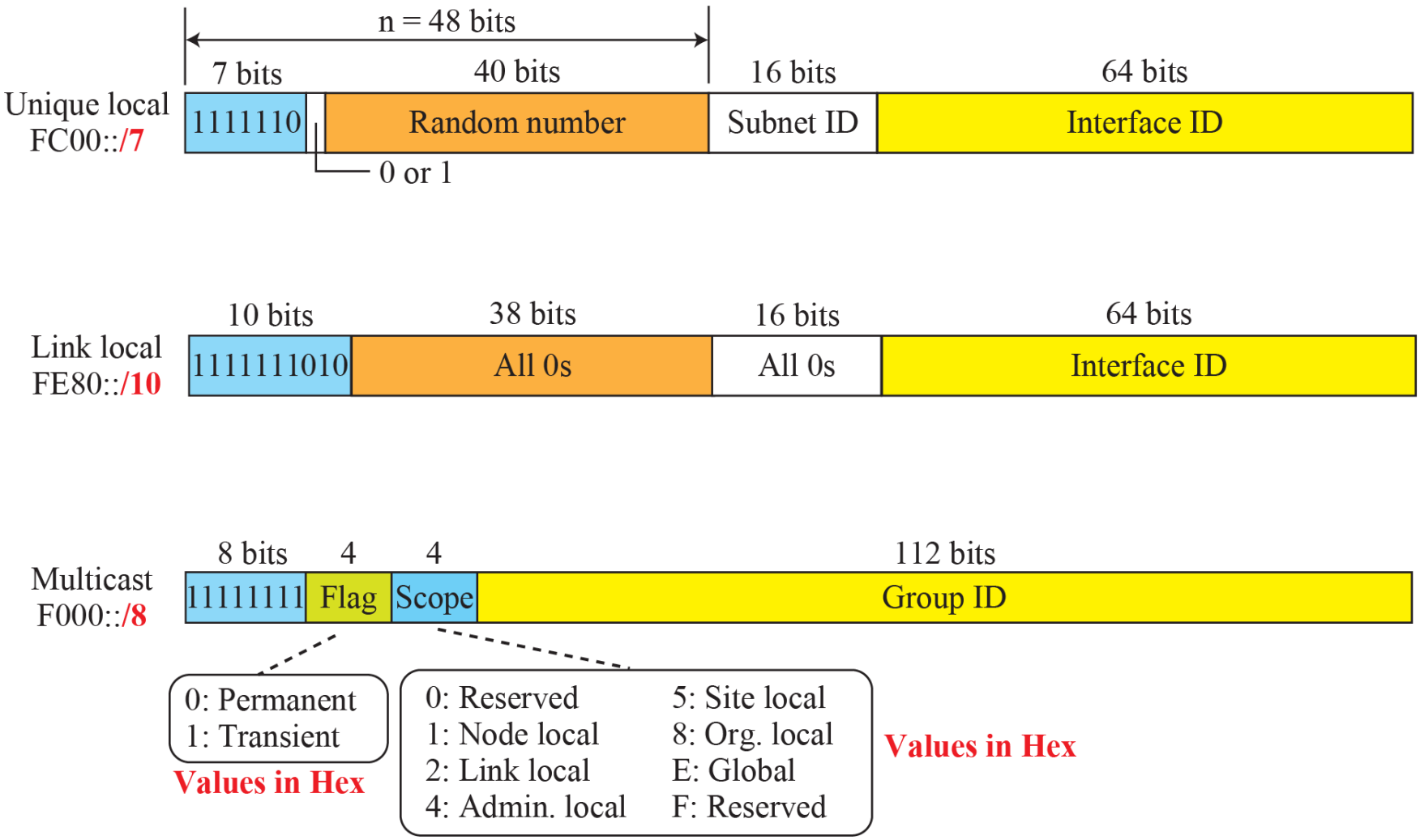
Figure 4.105: Unique local unicast block



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4.5.3 Transition from IPv4 to IPv6

Although we have a new version of the IP protocol, how can we make the transition to stop using IPv4 and start using IPv6? The first solution that comes to mind is to define a transition day on which every host or router should stop using the old version and start using the new version.

- **Dual Stack**
  - Tunneling
  - Header Translation

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Figure 4.106: ***Dual stack***

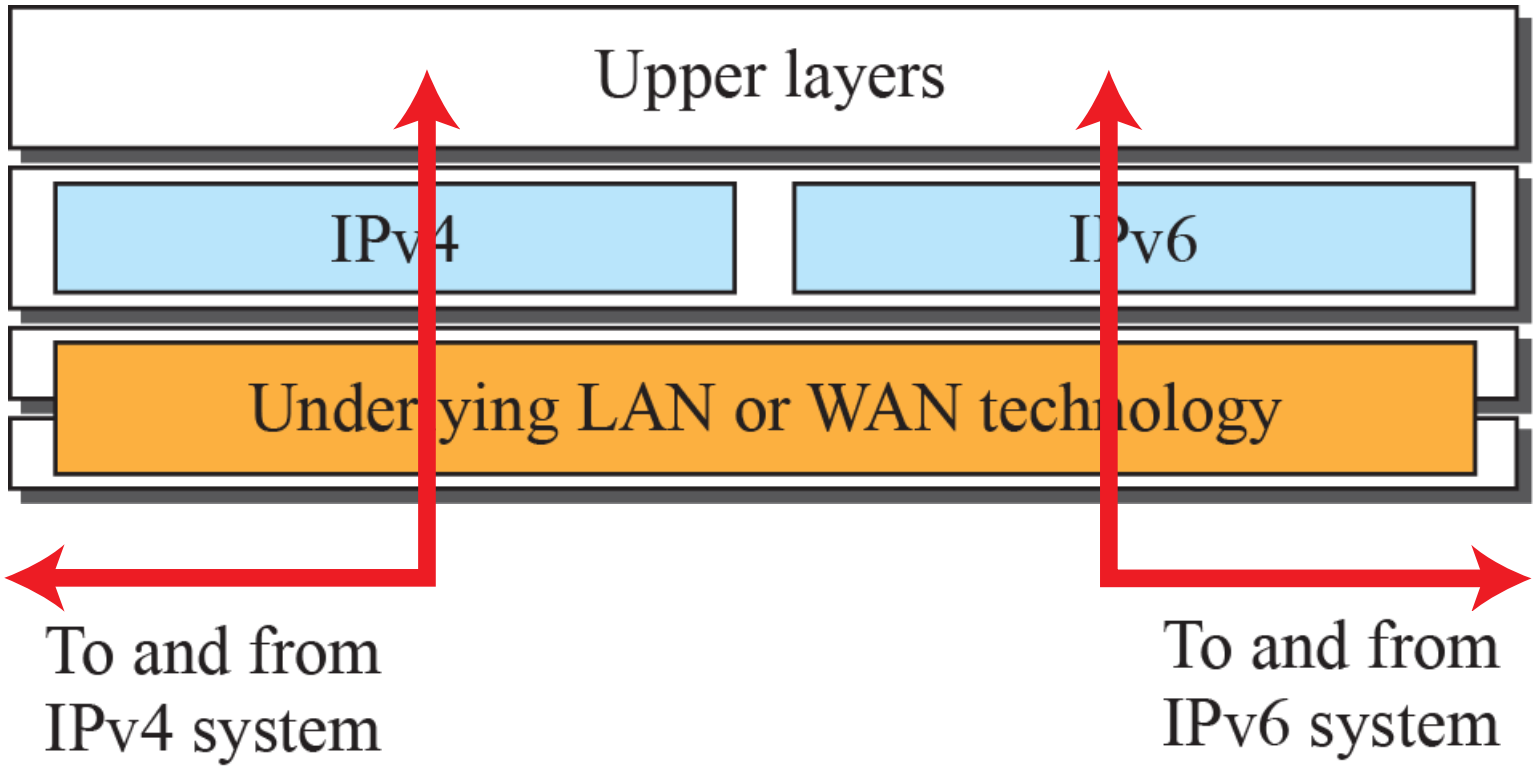

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Figure 4.107: Tunneling strategy

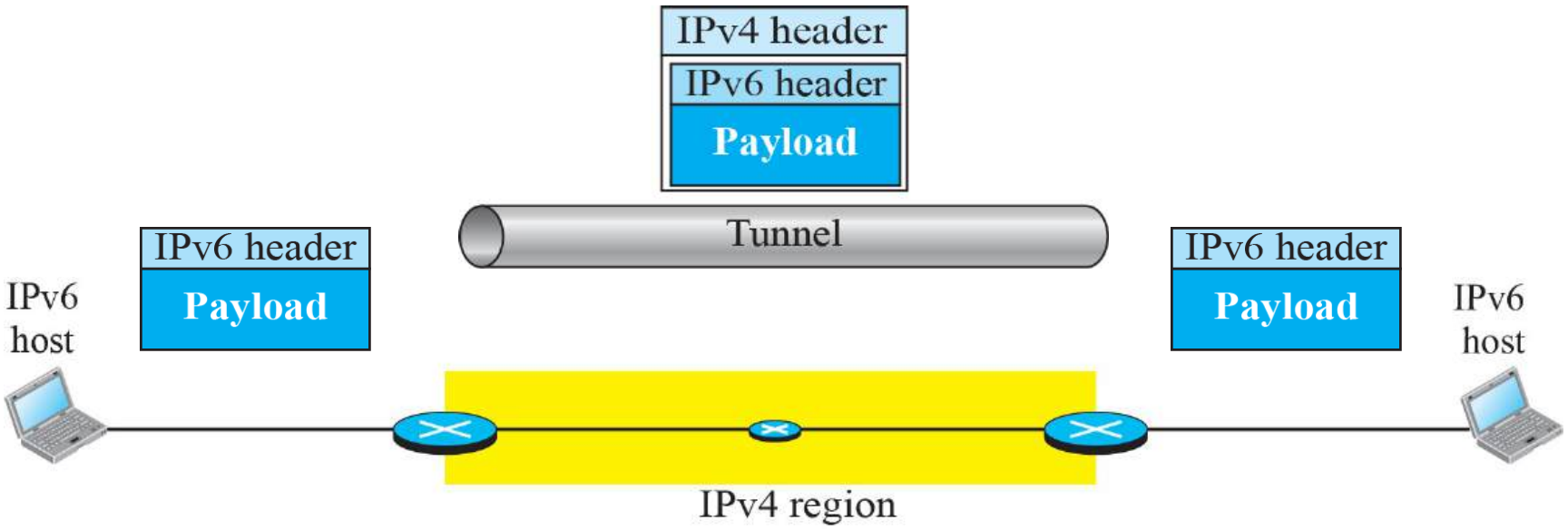

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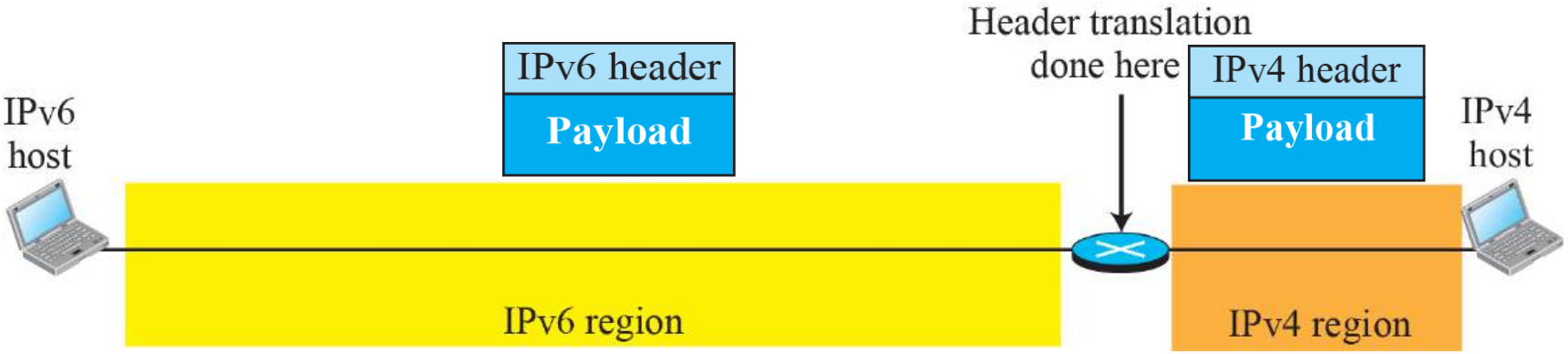
Figure 4.108: Header translation strategy



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4.5.4 ICMPv6

Another protocol that has been modified in version 6 of the TCP/IP protocol suite is ICMP. ICMPv6 is more complicated than ICMPv4: some protocols that were independent in version 4 are now part of ICMPv6 and some new messages have been added to make it more useful. Figure 4.109 compares the network layer of version 4 to that of version 6. The ICMP, ARP (discussed in Chapter 5), and IGMP protocols in version 4 are combined into one single protocol, ICMPv6.

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Figure 4.109: Comparison of network layer in version 4 and version 6

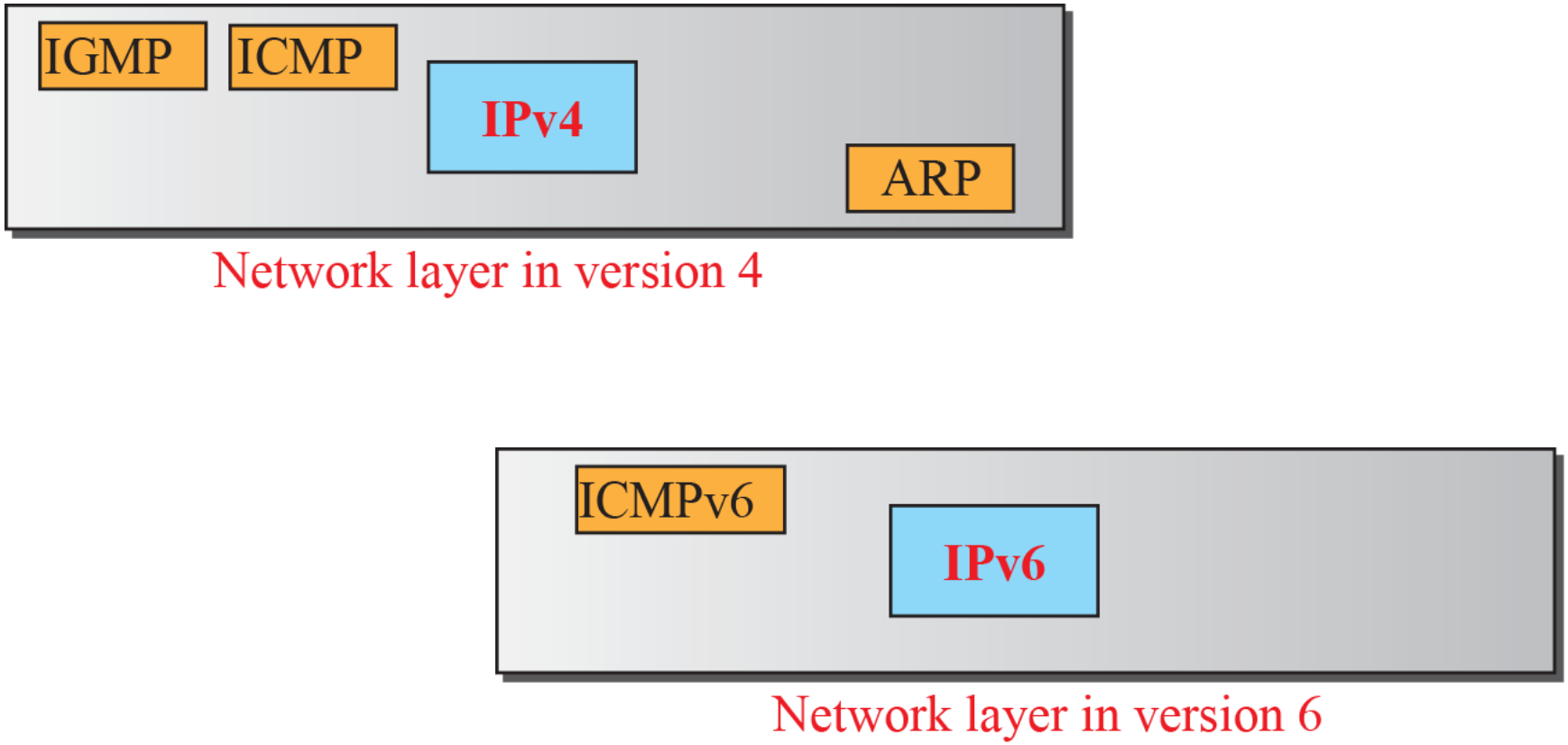

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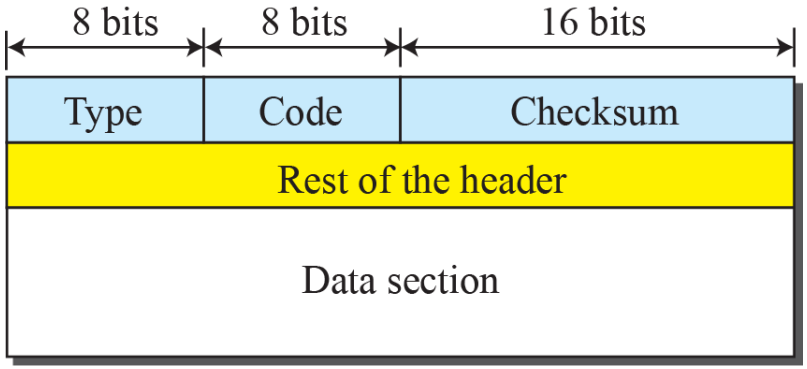
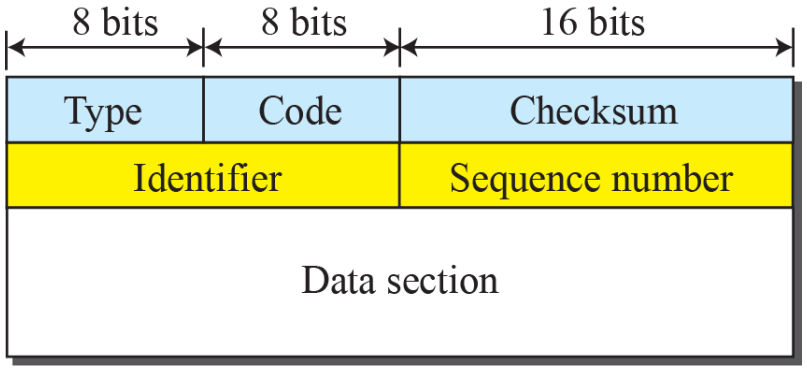
Figure 4.110: ICMPv6 messages (Part I)



Error-reporting messages



Informational messages

Type and code values

Error-reporting messages

01: Destination unreachable (codes 0 to 6)
02: Packet too big (code 0)
03: Time exceeded (codes 0 and 1)
04: Parameter problem (codes 0 to 2)

Informational messages

128 and 129: Echo request and reply (code 0)

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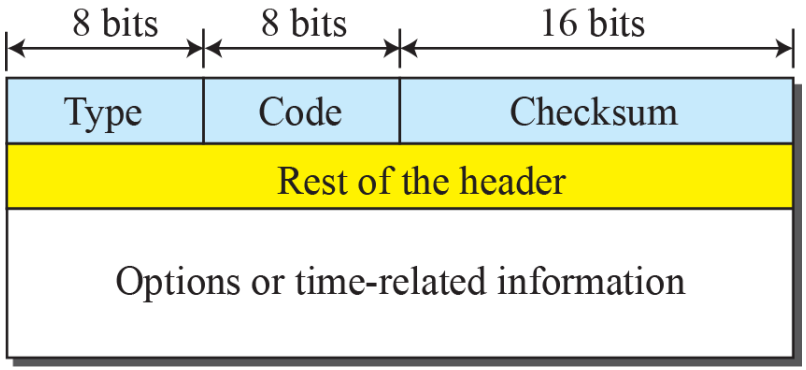
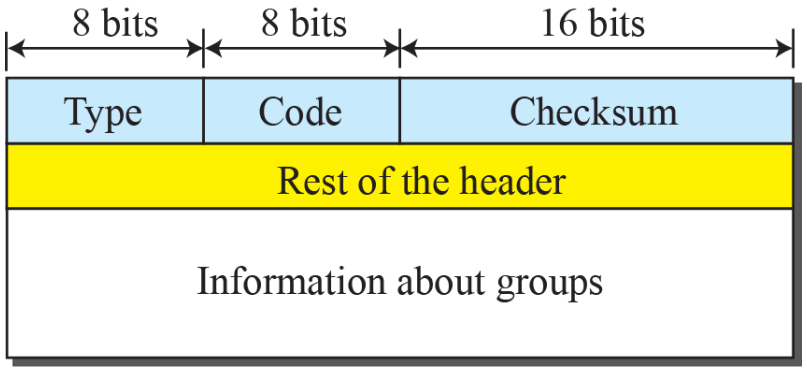
Figure 4.110: ICMPv6 messages (Part II)



Neighbor-discovery messages



Group-membership messages

Type and code values

Neighbor-discovery messages

133 and 134: Router solicitation and advertisement (code 0)
135 and 136: Neighbor solicitation and advertisement (code 0)
137: Redirection (codes 0 to 3)
141 and 142: Inverse neighbor solicitation and advertisement (code 0)

Group-membership messages

130: Membership query (code 0)
131: Membership report

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Chapter 4: Summary

- *The Internet is made of many networks (or links) connected through connecting devices, each of which acts as a router or as a switch. Two types of switching are traditionally used in networking: circuit switching and packet switching. The network layer is designed as a packet-switched network.*

- *The network layer supervises the handling of packets by the underlying physical networks. The delivery of a packet can be direct or indirect. Two categories of forwarding are defined: forwarding based on the destination address of the IP datagram and forwarding based on the label attached to an IP datagram.*

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Chapter 4: Summary (continued)

- *IPv4 is an unreliable connectionless protocol responsible for source-to-destination delivery. Packets in the IP layer are called datagrams. The identifiers used in the IP layer of the TCP/IP protocol suite are called the IP addresses. An IPv4 address is 32 bits long, divided into two parts: the prefix and the suffix. All addresses in the block have the same prefix; each address has a different suffix.*

- *The Internet Control Message Protocol (ICMP) supports the unreliable and connectionless Internet Protocol (IP).*

- *To be able to route a packet, a router needs a forwarding table. Routing protocols are specific application programs that have the duty of updating forwarding tables.*

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Chapter 4: Summary (continued)

- *Multicasting is the sending of the same message to more than one receiver simultaneously. The Internet Group Management Protocol (IGMP) is involved in collecting local membership group information.*

- *IPv6, the latest version of the Internet Protocol, has a 128-bit address space. IPv6 uses hexadecimal colon notation with abbreviation methods available.*

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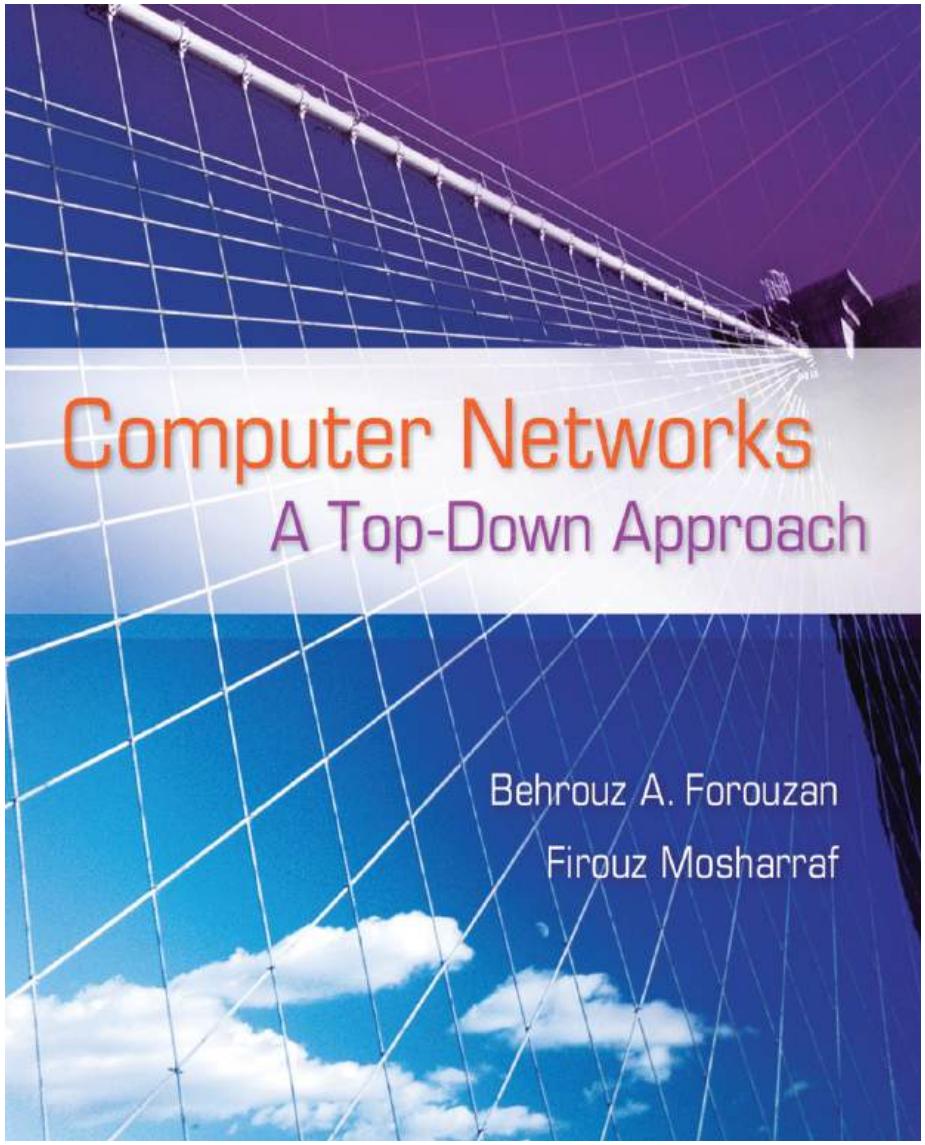
Chapter 5

Data-Link Layer: Wired Networks



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Chapter 5: Outline

5.1 INTRODUCTION

5.2 DATA LINK CONTROL (DLC)

5.3 MULTIPLE ACCESS PROTOCOLS

5.4 LINK-LAYER ADDRESSING

5.5 WIRED LANS: ETHERNET PROTOCOL

5.6 OTHER WIRED NETWORKS

5.7 CONNECTING DEVICES

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Chapter 5: Objective

- *We introduce the concept of nodes and links and the types of links, and show how the data-link layer is actually divided into two sublayers: data link control and media access control.*

- *We discuss data link control (DLC) of the data-link layer and explain services provided by this layer, such as framing, flow and error control, and error detection.*

- *We discuss the media access control (MAC) sublayer of the data-link layer. We explain different approaches such as random access, controlled access, and channelization.*

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Chapter 5: Objective (continued)

- *We discuss link-layer addressing and how the link-layer address of a node can be found using the Address Resolution Protocol (ARP).*
- *We introduce the wired LANs and in particular Ethernet, the dominant LAN protocol today. We move through different generations of Ethernet and show how it has evolved.*
- *We discuss other wired networks that we encounter in the Internet today, such as point-to-point networks and switched networks.*
- *We discuss connecting devices used in the lower three layers of the TCP/IP protocol such as hubs, link-layer switches, and routers.*

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5-1 INTRODUCTION

The Internet is a combination of networks glued together by connecting devices (routers or switches). If a datagram is to travel from a host to another host, it needs to pass through these networks. Figure 5.1 shows communication between Alice and Bob, using the same scenario we followed in the last three chapters.

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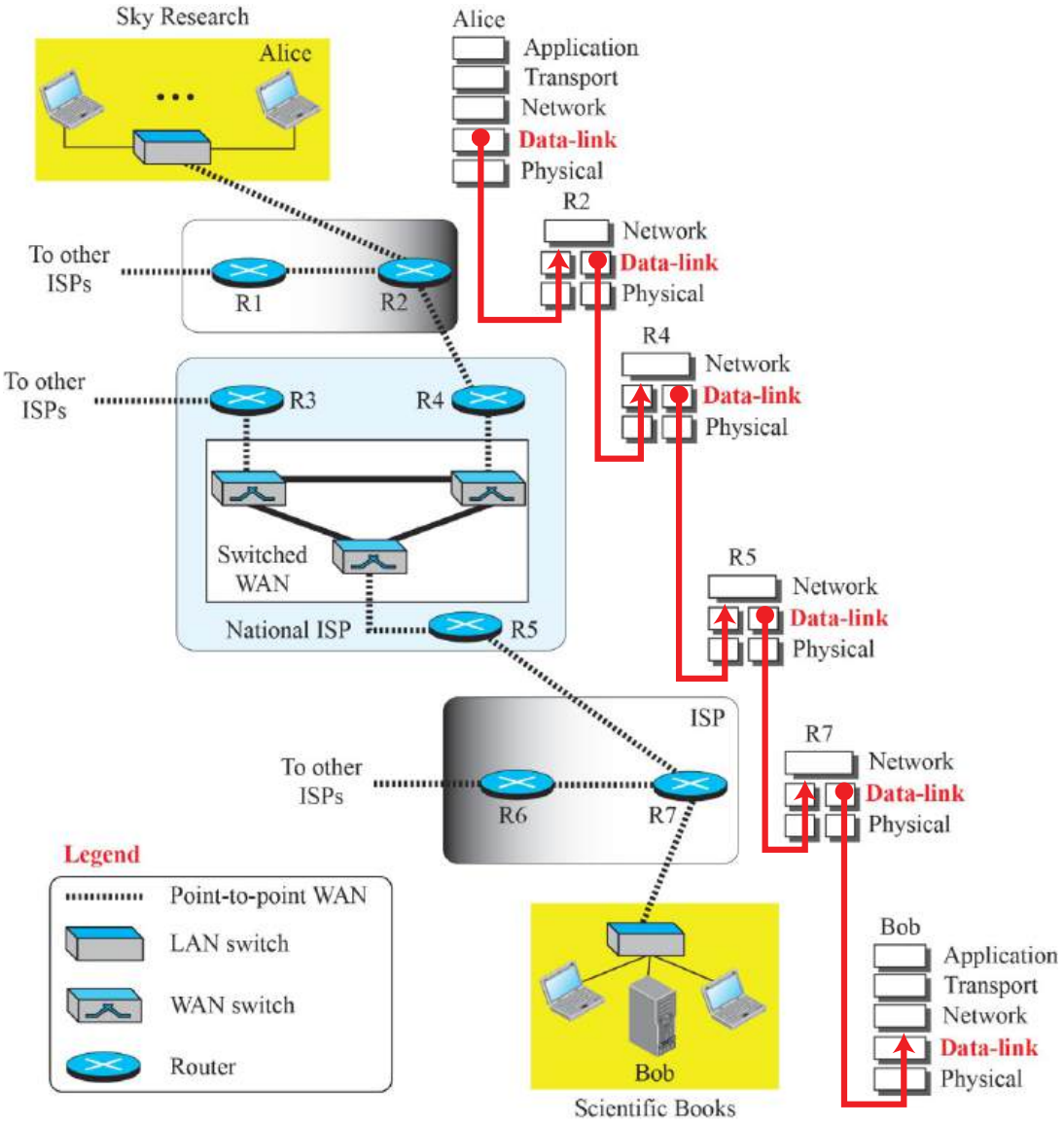
Figure 5.1: Communication at the data-link layer



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5.1.1 Nodes and Links

Although communication at the application, transport, and network layers is end-to-end, communication at the data-link layer is node-to node. As we have learned in the previous chapters, a data unit from one point in the Internet needs to pass through many networks (LANs and WANs) to reach another point. Theses LANs and WANs are connected by routers. It is customary to refer to the two end hosts and the routers as nodes and the networks in between as links.

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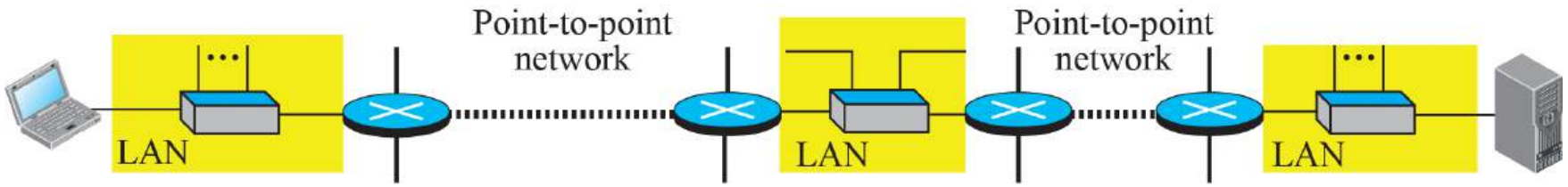
Figure 5.2: Nodes and Links



a. A small part of the Internet

b. Nodes and links

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5.1.2 Two Types of Links

Although two nodes are physically connected by a transmission medium such as cable or air, we need to remember that the data-link layer controls how the medium is used. We can have a data-link layer that uses the whole capacity of the medium; we can also have a data-link layer that uses only part of the capacity of the link. In other words, we can have a point-to-point link or a broadcast link.

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5.1.3 Two Sublayers

To better understand the functionality of and the services provided by the link layer, we can divide the data-link layer into two sublayers: data link control (DLC) and media access control (MAC). This is not unusual because, as we will see later in this chapter and in the next chapter, LAN protocols actually use the same strategy. The data link control sublayer deals with all issues common to both point-to-point and broadcast links; the media access control sublayer deals only with issues specific to broadcast links.

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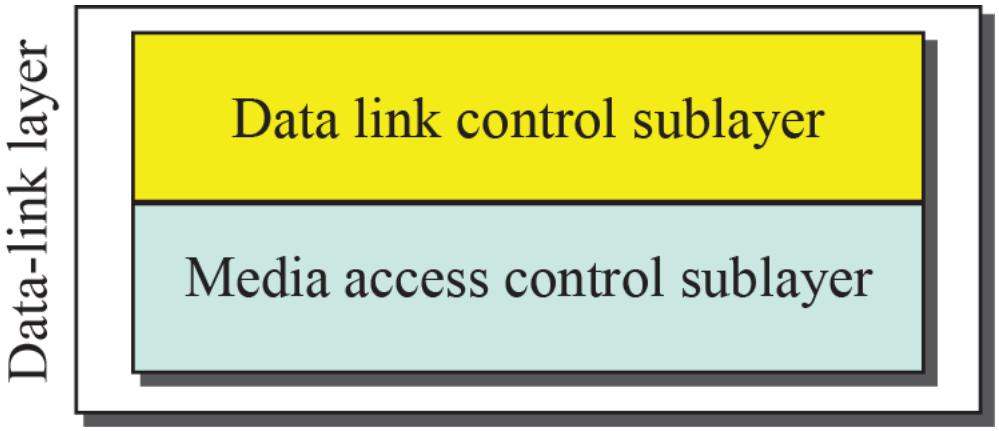
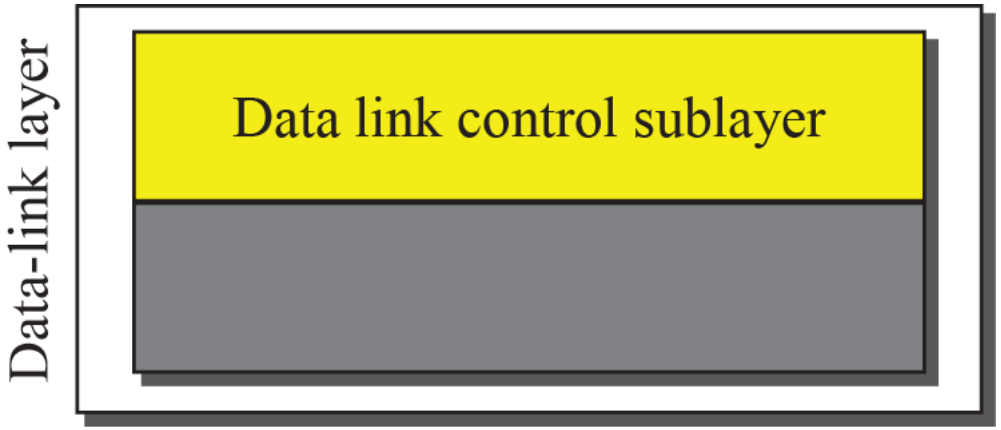
Figure 5.3: Dividing the data-link layer into two sublayers



a. Data-link layer of a broadcast link



b. Data-link layer of a point-to-point link

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5-2 DATA LINK CONTROL (DLC)

The data link control deals with procedures for communication between two adjacent nodes. Data link control (DLC) functions include framing, flow and error control, and error detection and correction. In this section, we first discuss framing, or how to organize the bits that are carried by the physical layer. We then discuss flow and error control. Techniques for error detection are discussed at the end of this section.

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5.2.1 Framing

Data transmission in the physical layer means moving bits in the form of a signal from the source to the destination. The data-link layer, on the other hand, needs to pack bits into frames, so that each frame is distinguishable from another.

- **Frame Size**
  - Character-Oriented Framing
  - Bit-Oriented Framing

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Figure 5.4: A frame in a character-oriented protocol

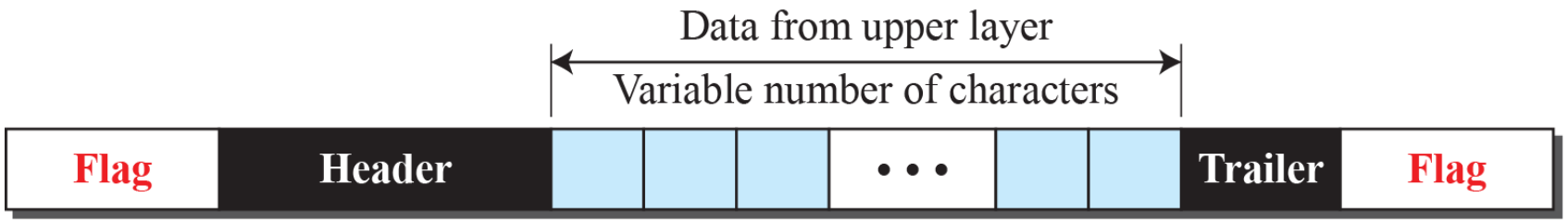

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Figure 5.5: Byte stuffing and unstuffing

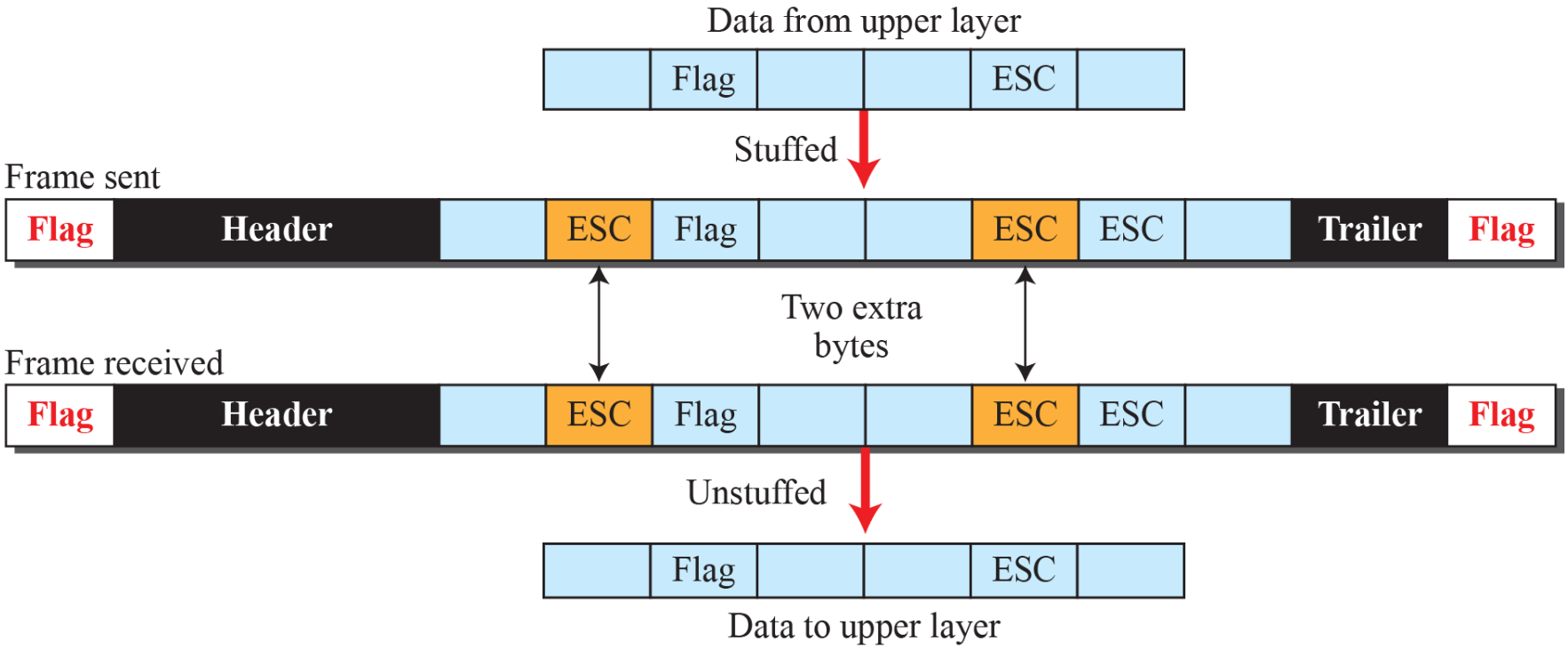

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Figure 5.6: A frame in a bit-oriented protocol

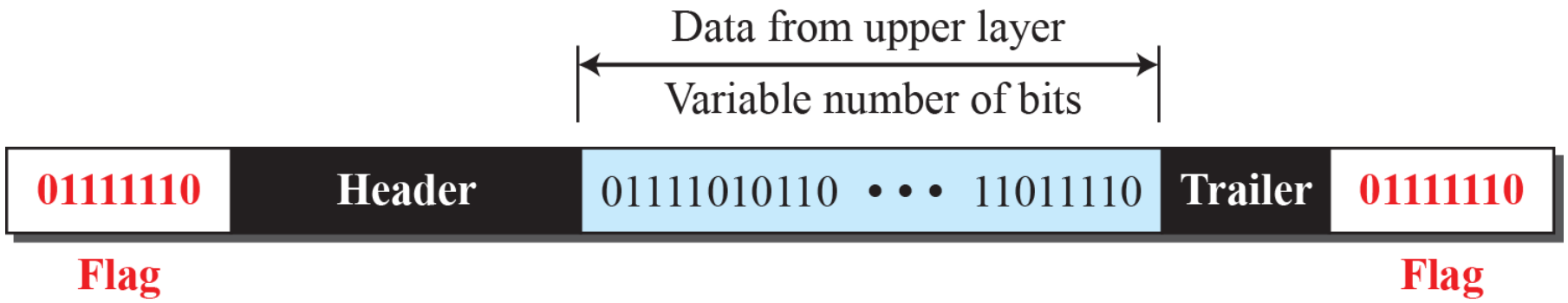

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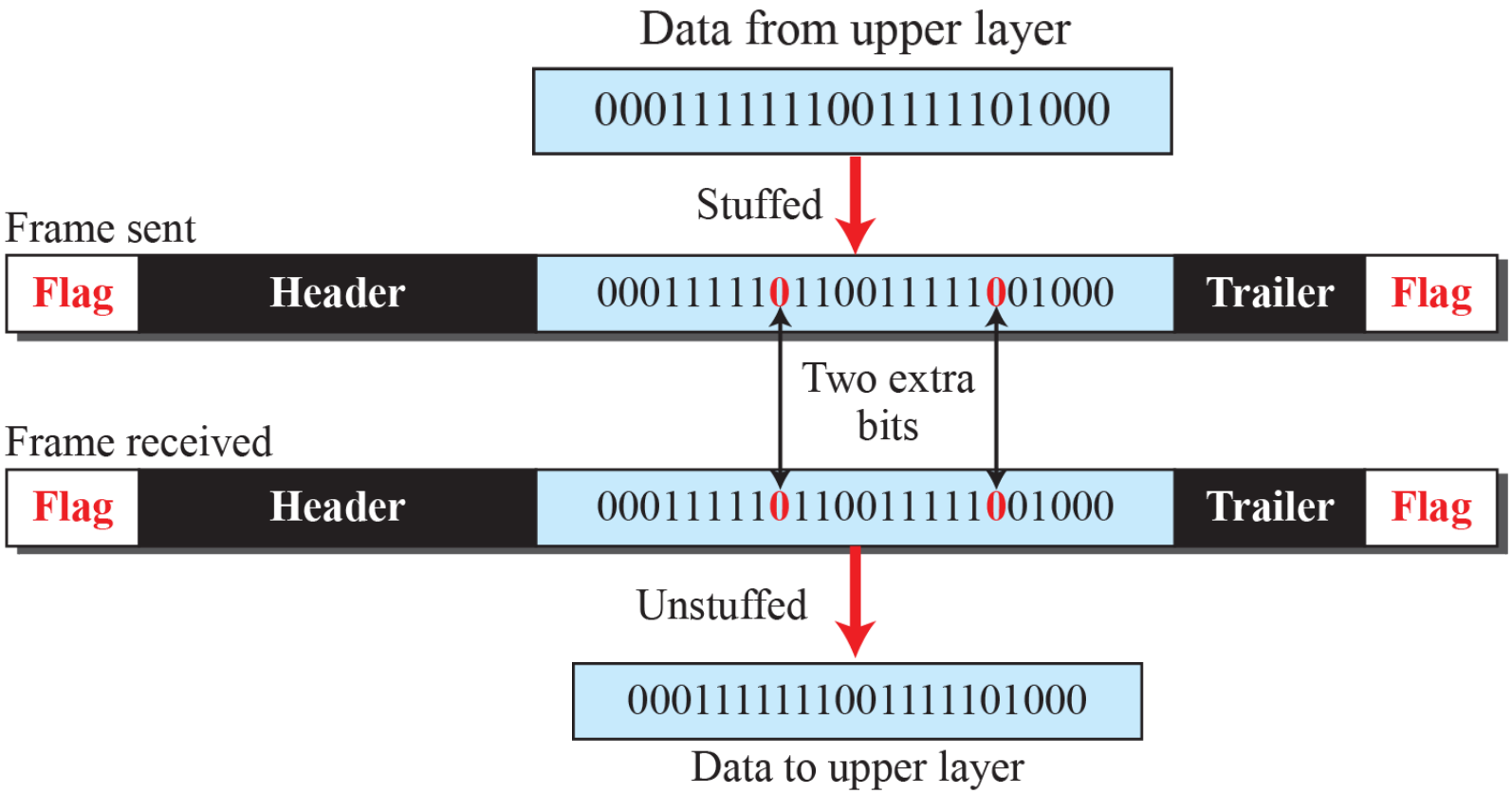
Figure 5.7: Bit stuffing and unstuffing



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5.2.2 Flow and Error Control

We defined flow and error control in Chapter 3. One of the responsibilities of the data-link control sublayer is flow and error control at the data-link layer.

- ***Flow Control***
- ***Error Control***

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5.2.3 Error Detection and Correction

At the data-link layer, if a frame is corrupted between the two nodes, it needs to be corrected before it continues its journey to other nodes. However, most link-layer protocols simply discard the frame and let the upper-layer protocols handle the retransmission of the frame. Some wireless protocols, however, try to correct the corrupted frame.

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5.2.3 (continued)

- **Introduction**
  - Types of Errors
  - Redundancy
  - Detection versus Correction
  - Coding
- **Block Coding**
  - Error Detection
  - Hamming Distance
  - Minimum Hamming Distance for Error Detection

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5.2.3 (continued)

- **Linear Block Codes**
  - Minimum Distance for Linear Block Codes
  - Parity-Check Code
- **Cyclic Codes**
  - Cyclic Redundancy Check
  - Polynomials
  - Requirement
  - Performance
  - Advantages of Cyclic Codes

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5.2.3 (continued)

- **Checksum**
  - Concept
  - Internet Checksum
  - Algorithm
  - Other Approaches to the Checksum

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Figure 5.8: Single-bit and burst error

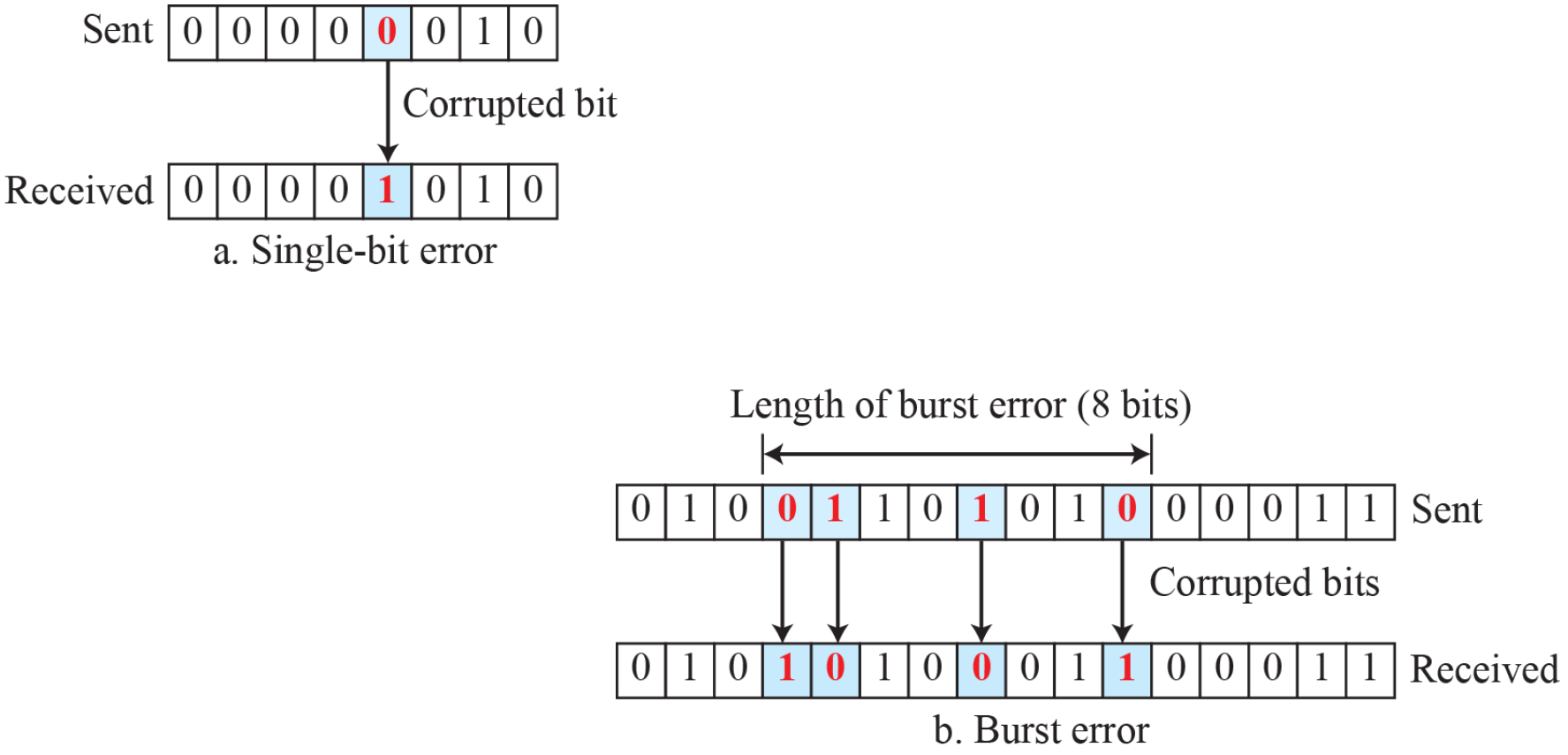

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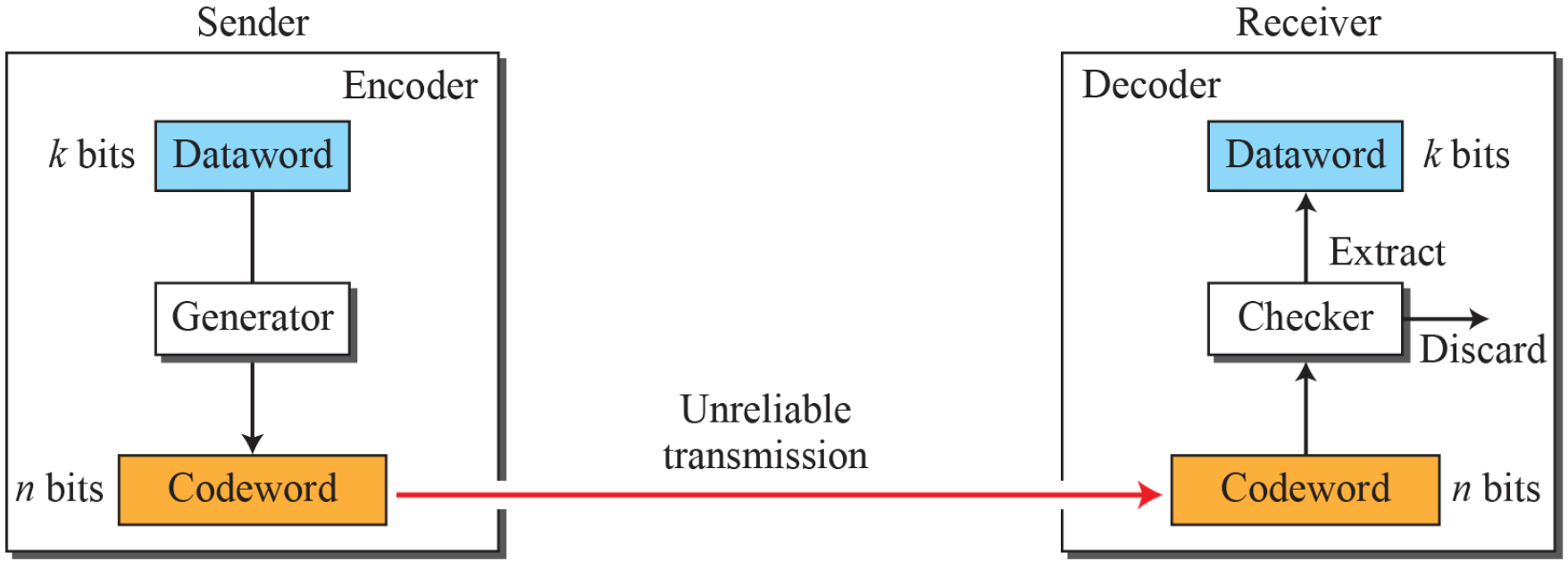
Figure 5.9: Process of error detection in block coding



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Example 5.1

Let us assume that k = 2 and n = 3. Table 5.1 shows the list of datawords and codewords. Later, we will see how to derive a codeword from a dataword.

Table 5.1: A code for error detection in Example 5.1

Datawords	*Codewords*	*Datawords*	*Codewords*
00	000	10	101
01	011	11	110

Assume the sender encodes the dataword 01 as 011 and sends it to the receiver. Consider the following cases:

1. The receiver receives 011. It is a valid codeword. The receiver extracts the dataword 01 from it.
2. The codeword is corrupted during transmission, and 111 is received (the leftmost bit is corrupted). This is not a valid codeword and is discarded.
3. The codeword is corrupted during transmission, and 000 is received (the right two bits are corrupted). This is a valid codeword. The receiver incorrectly extracts the dataword 00. Two corrupted bits have made the error undetectable.

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Example 5.2

Let us find the Hamming distance between two pairs of words.

1. The Hamming distance $d(000, 011)$ is 2 because $(000 \oplus 011)$ is 011 (two 1s).
2. The Hamming distance $d(10101, 11110)$ is 3 because $(10101 \oplus 11110)$ is 01011 (three 1s).

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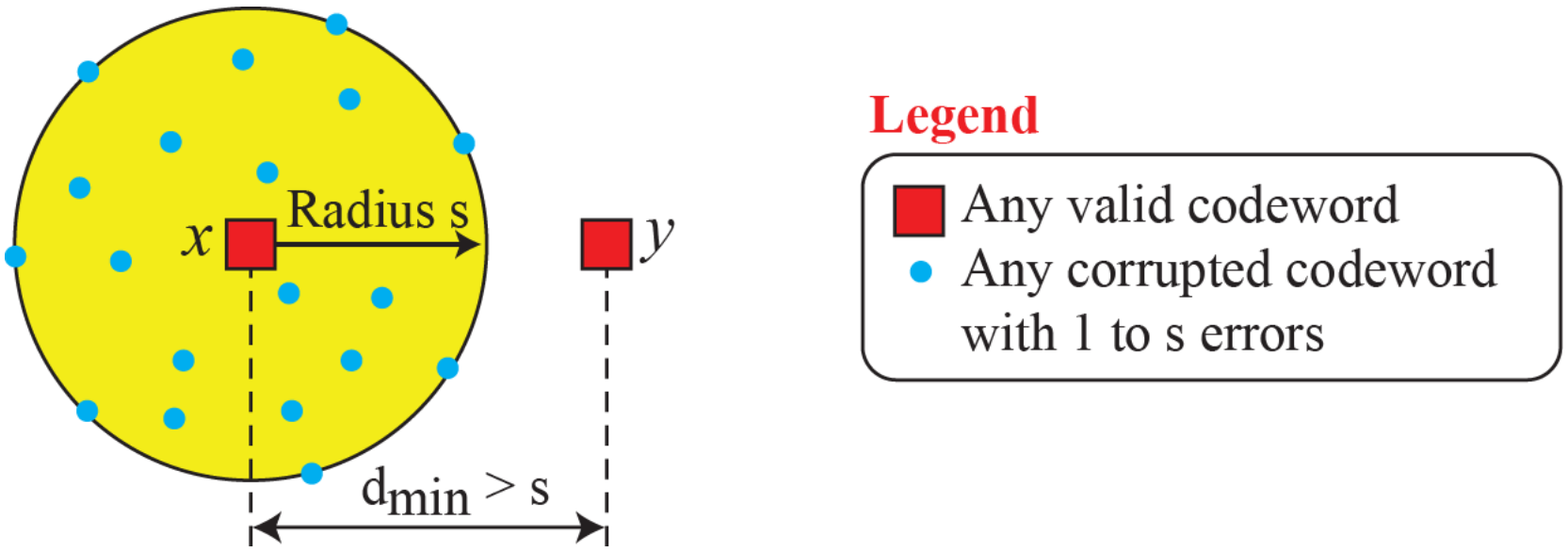
Figure 5.10: Geometric concept explaining d_{min} in error detection



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Example 5.3

The minimum Hamming distance for our first code scheme (Table 5.1) is 2. This code guarantees detection of only a single error. For example, if the third codeword (101) is sent and one error occurs, the received codeword does not match any valid codeword. If two errors occur, however, the received codeword may match a valid codeword and the errors are not detected.

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Example 5.4

A code scheme has a Hamming distance $d_{min} = 4$. This code guarantees the detection of up to three errors ($d = s + 1$ or $s = 3$).

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Example 5.5

The code in Table 5.1 is a linear block code because the result of XORing any codeword with any other codeword is a valid codeword. For example, the XORing of the second and third codewords creates the fourth one.

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Example 5.6

In our first code (Table 5.1), the numbers of 1s in the nonzero codewords are 2, 2, and 2. So the minimum Hamming distance is $\boldsymbol{d}_{\text{min}} = 2$.

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Table 5.2: Simple parity-check code C(5, 4)

Datawords	Codewords	*Datawords*	Codewords
0000	00000	1000	10001
0001	00011	1001	10010
0010	00101	1010	10100
0011	00110	1011	10111
0100	01001	1100	11000
0101	01010	1101	11011
0110	01100	1110	11101
0111	01111	1111	11110

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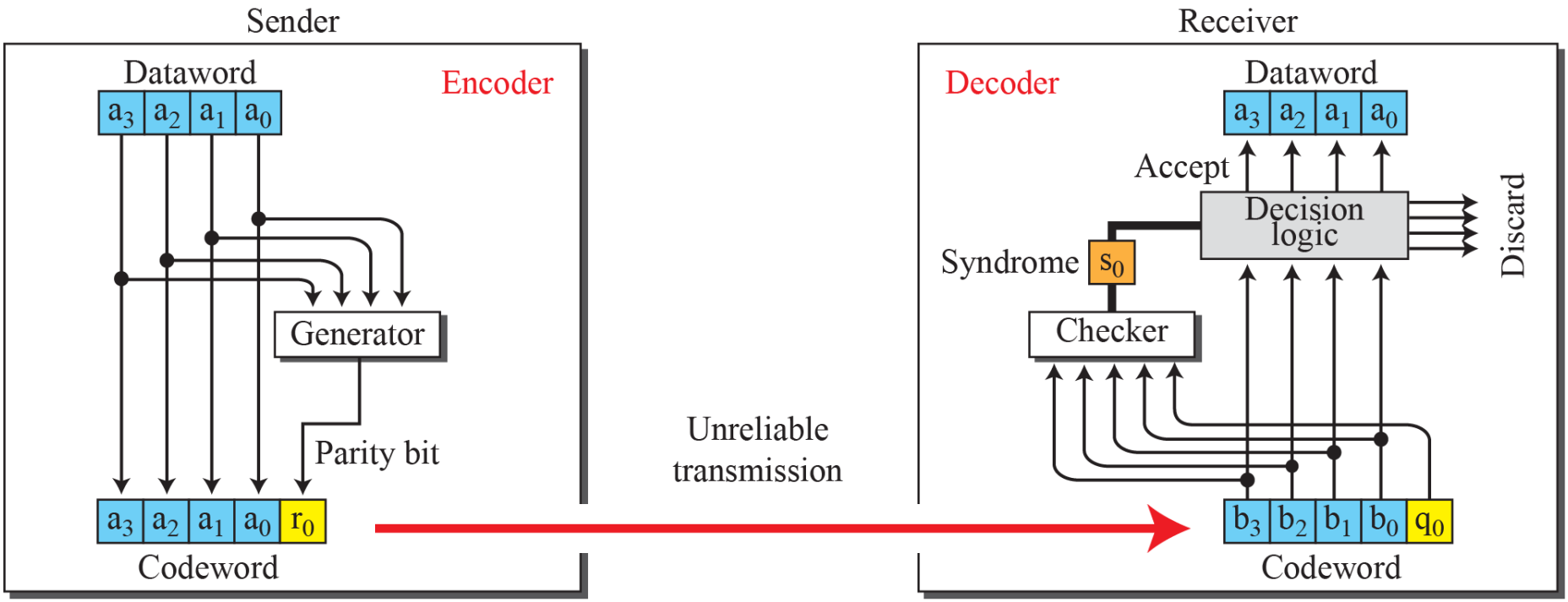
Figure 5.11: Encoder and decoder for simple parity-check code



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Example 5.7

Let us look at some transmission scenarios. Assume the sender sends the dataword 1011. The codeword created from this dataword is 10111, which is sent to the receiver. We examine five cases:

1. No error occurs; the received codeword is 10111. The syndrome is 0. The dataword 1011 is created.
2. One single-bit error changes a_1. The received codeword is 10**0**11. The syndrome is 1. No dataword is created.
3. One single-bit error changes r_0. The received codeword is 1011**0**. The syndrome is 1. No dataword is created. Note that although none of the dataword bits are corrupted, no dataword is created because the code is not sophisticated enough to show the position of the corrupted bit.
4. An error changes r_0 and a second error changes a_3. The received codeword is **0**011**0**. The syndrome is 0. The dataword 0011 is created at the receiver. Note that here the dataword is wrongly created due to the syndrome value. The simple parity-check decoder cannot detect an even number of errors. The errors cancel each other out and give the syndrome a value of 0.
5. Three bits—a_3, a_2, and a_1—are changed by errors. The received codeword is **010**11. The syndrome is 1. The dataword is not created. This shows that the simple parity check, guaranteed to detect one single error, can also find any odd number of errors.

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Table 5.3: A CRC code with C(7, 4)

Dataword	*Codeword*	*Dataword*	*Codeword*
0000	0000000	1000	1000101
0001	0001011	1001	1001110
0010	0010110	1010	1010011
0011	0011101	1011	1011000
0100	0100111	1100	1100010
0101	0101100	1101	1101001
0110	0110001	1110	1110100
0111	0111010	1111	1111111

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Figure 5.12: CRC encoder and decoder

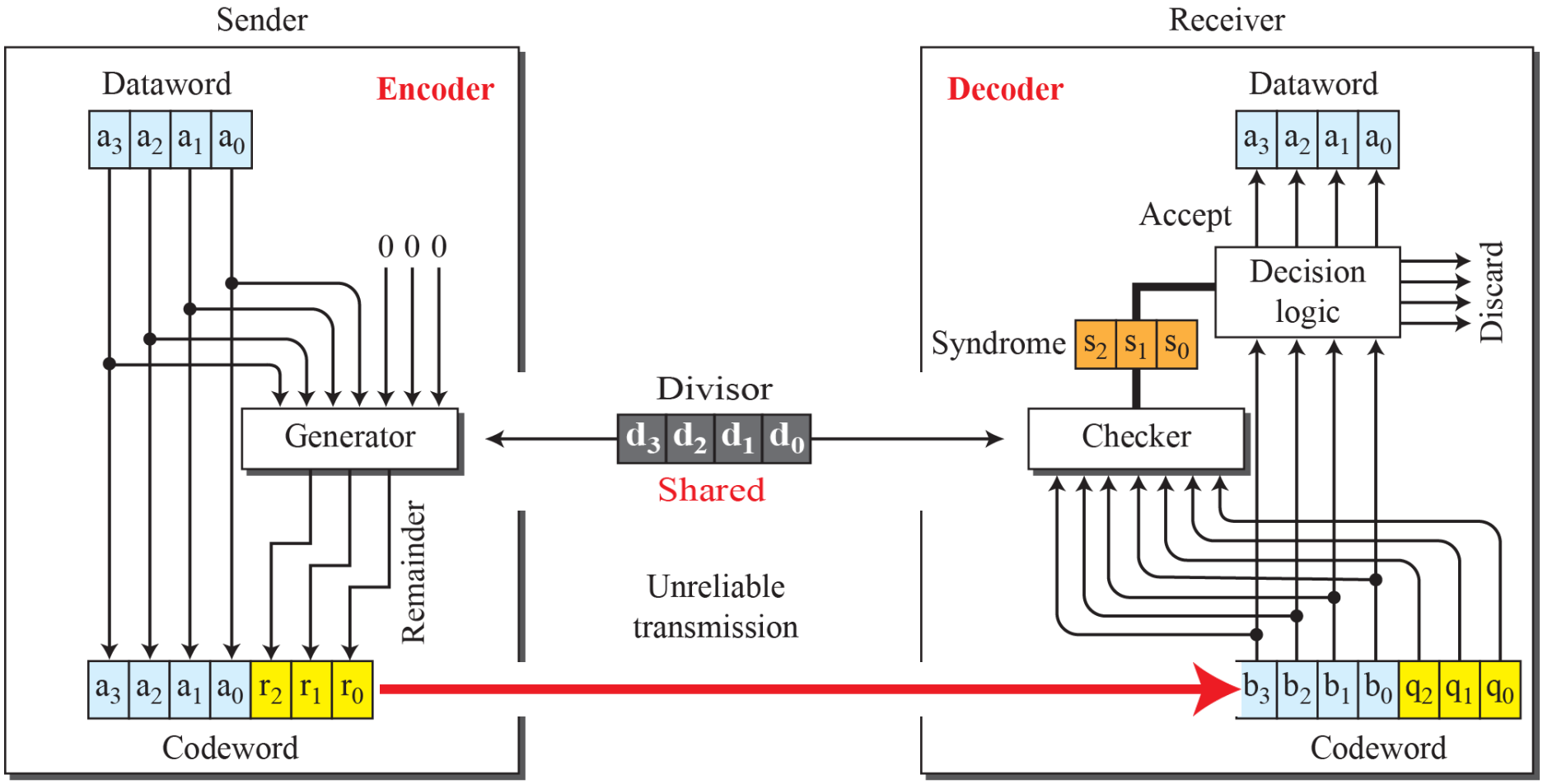

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Figure 5.13: Division in CRC encoder

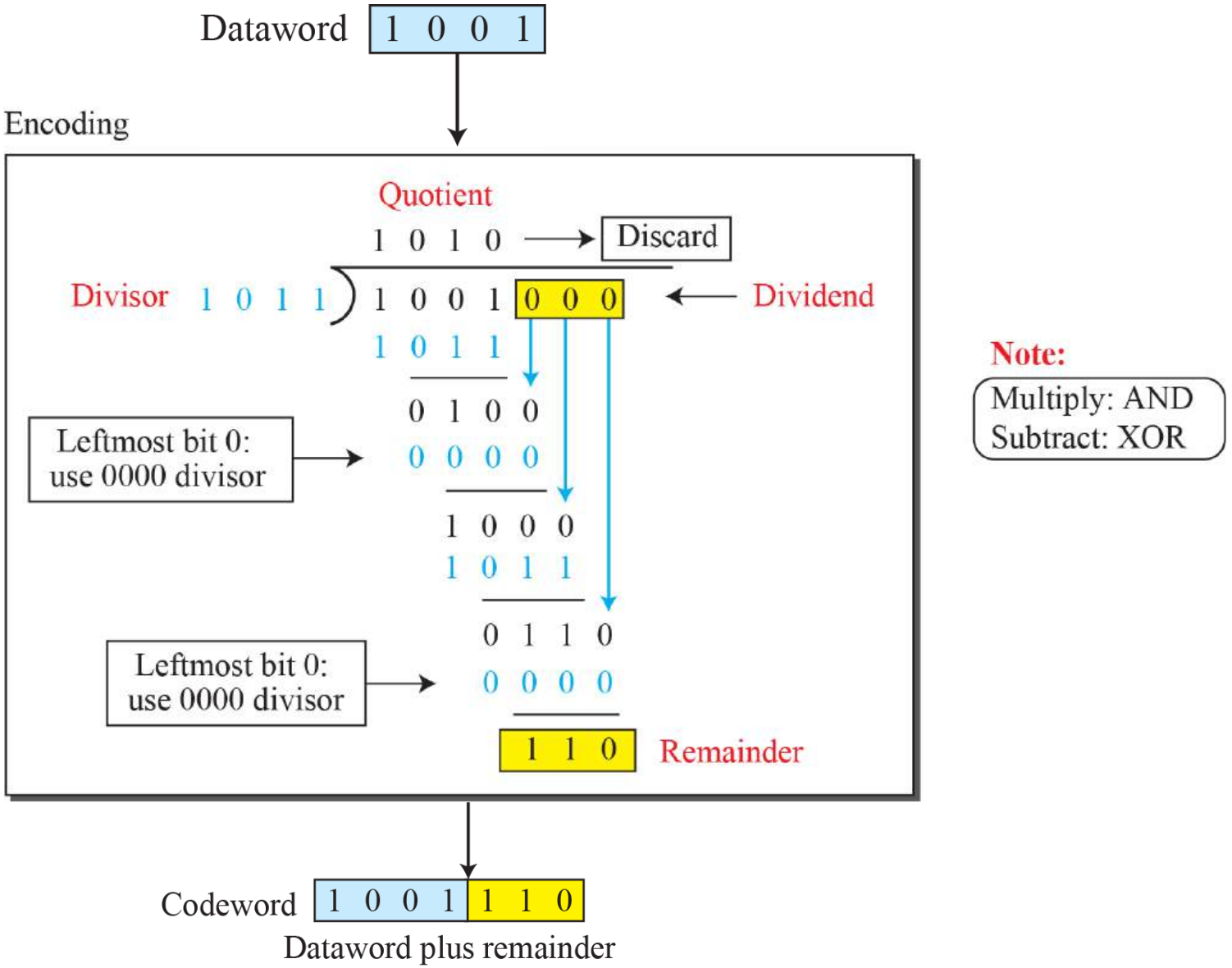

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Figure 5.14: Division in the CRC decoder for two cases

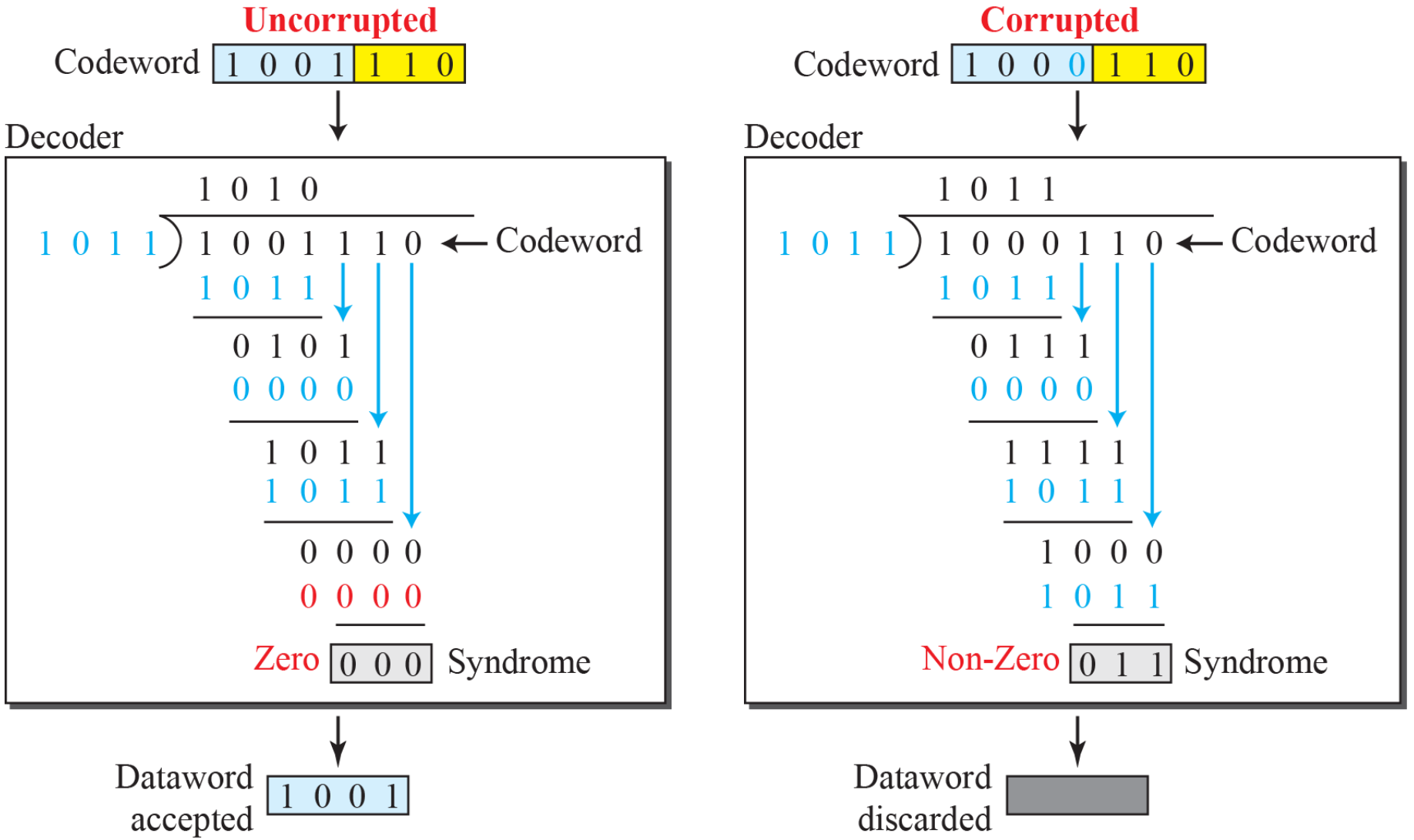

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Table 5.4: Standard polynomials

Name	*Binary*	*Application*
CRC-8	100000111	ATM header
CRC-10	11000110101	ATM AAL
CRC-16	1000100000100001	HDLC
CRC-32	100000100110000010001110110110111	LANs

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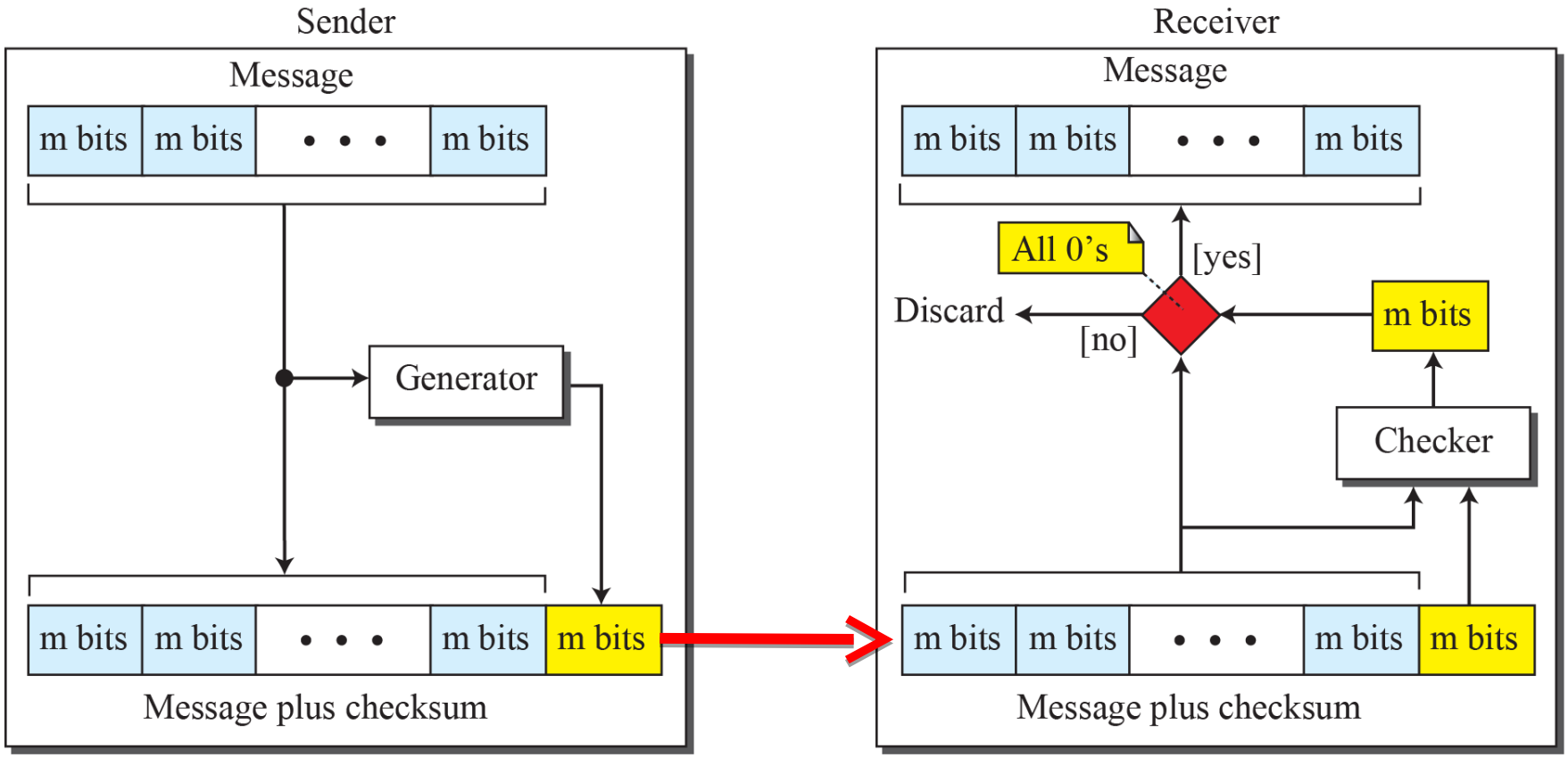
Figure 5.15: Checksum



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Example 5.8

Suppose the message is a list of five 4-bit numbers that we want to send to a destination. In addition to sending these numbers, we send the sum of the numbers. For example, if the set of numbers is (7, 11, 12, 0, 6), we send (7, 11, 12, 0, 6, **36**), where **36** is the sum of the original numbers. The receiver adds the five numbers and compares the result with the sum. If the two are the same, the receiver assumes no error, accepts the five numbers, and discards the sum. Otherwise, there is an error somewhere and the message not accepted.

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Example 5.9

In the previous example, the decimal number 36 in binary is (100100)2. To change it to a 4-bit number we add the extra leftmost bit to the right four bits as shown below.

$$(10)_2 + (0100)_2 = (0110)_2 \rightarrow (6)_{10}$$

Instead of sending 36 as the sum, we can send 6 as the sum (7, 11, 12, 0, 6, 6). The receiver can add the first five numbers in one's complement arithmetic. If the result is 6, the numbers are accepted; otherwise, they are rejected.

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Example 5.10

Let us use the idea of the checksum in Example 5.9. The sender adds all five numbers in one's complement to get the sum = 6. The sender then complements the result to get the checksum = **9**, which is 15 − 6. Note that 6 = (0110)2 and **9**= (1001)2; they are complements of each other. The sender sends the five data numbers and the checksum (7, 11, 12, 0, 6, **9**). If there is no corruption in transmission, the receiver receives (7, 11, 12, 0, 6, **9**) and adds them in one's complement to get 15.

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Figure 5.16: Example 5.10

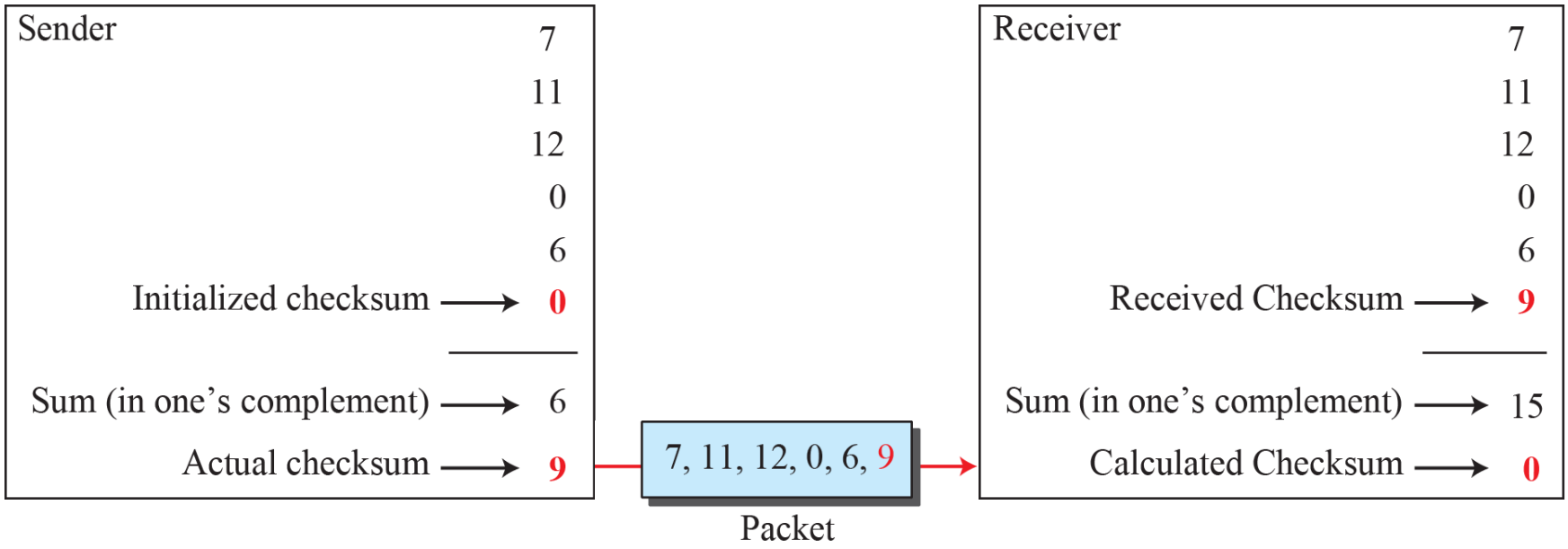

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Table 5.5: Procedure to calculate the traditional checksum

Sender	*Receiver*
1. The message is divided into 16-bit words.	1. The message and the checksum is received.
2. The value of the checksum word is initially set to zero.	2. The message is divided into 16-bit words.
3. All words including the checksum are added using one's complement addition.	3. All words are added using one's complement addition.
4. The sum is complemented and becomes the checksum.	4. The sum is complemented and becomes the new checksum.
5. The checksum is sent with the data.	5. If the value of the checksum is 0, the message is accepted; otherwise, it is rejected.

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Figure 5.17: Algorithm to calculate a traditional checksum

Notes:

a. Word and Checksum are each 16 bits, but Sum is 32 bits.
b. Left(Sum) can be found by shifting Sum 16 bits to the right.
c. Right(Sum) can be found by ANDing Sum with $(0000FFFF)_{16}$.
d. After Checksum is found, truncate it to 16 bits.

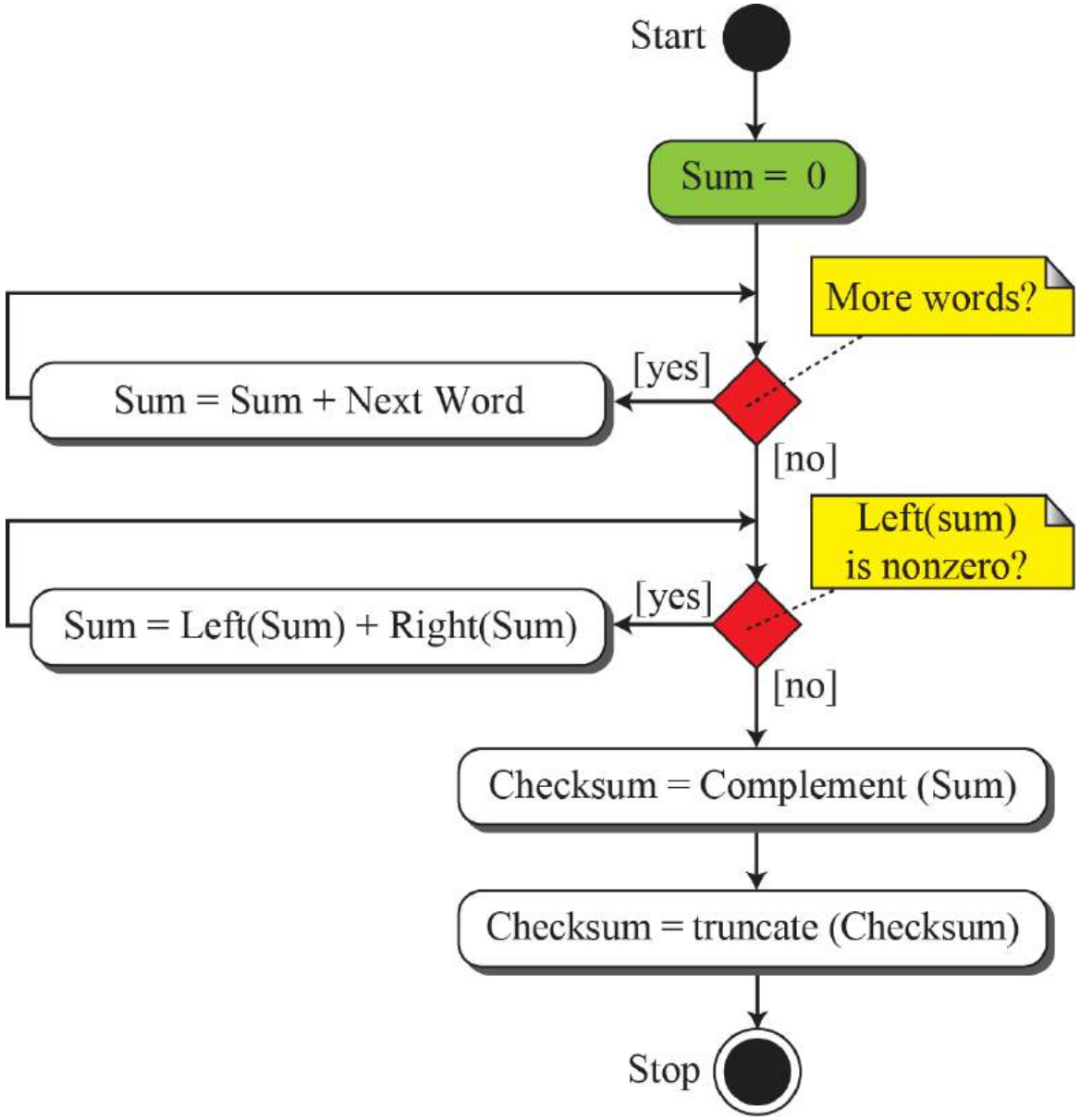

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Figure 5.18: Algorithm to calculate an 8-bit Fletcher checksum

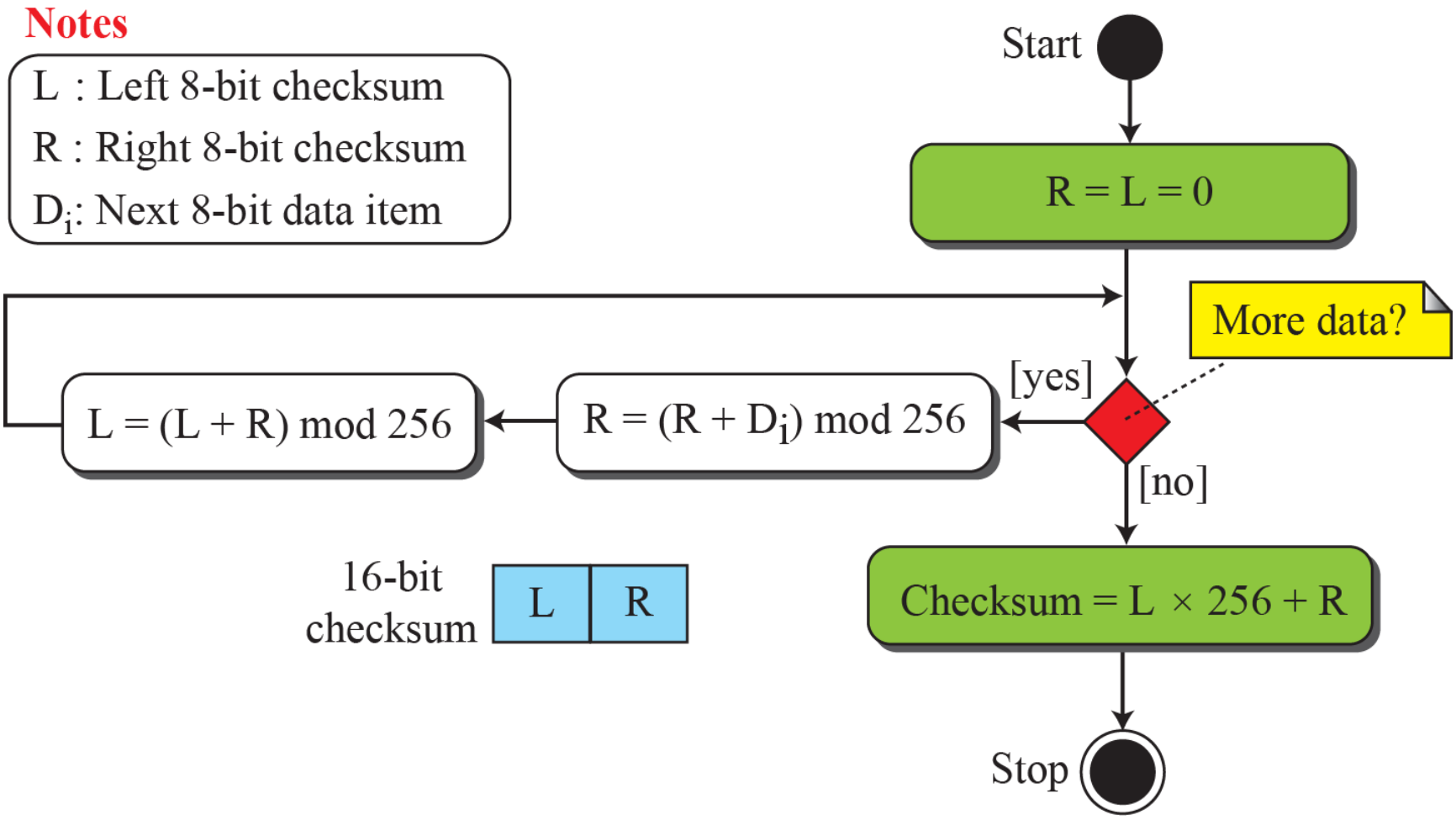

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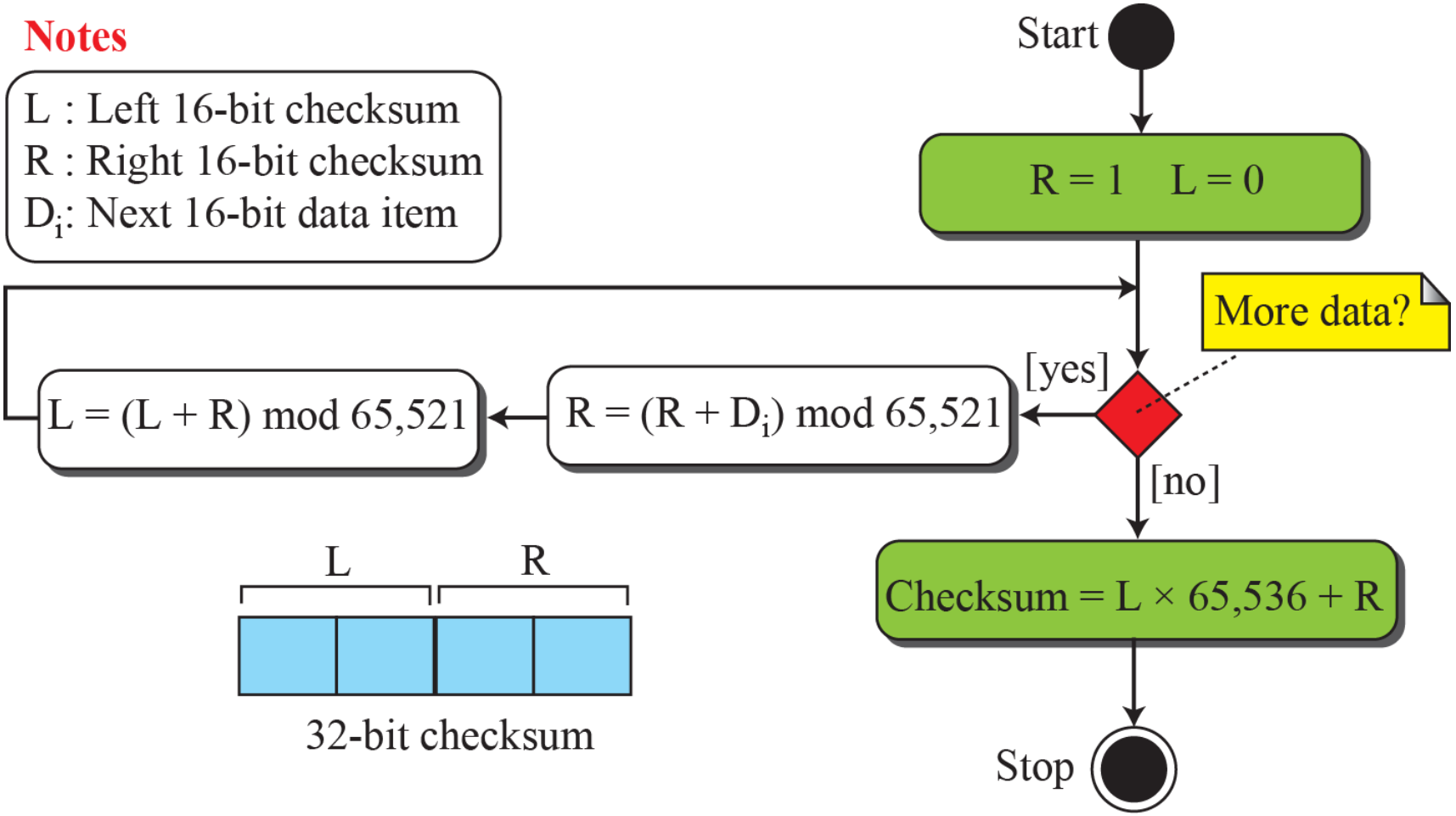
Figure 5.19: Algorithm to calculate an Adler checksum



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5.2.4 Two DLC Protocols

After finishing all issues related to the DLC sublayer, we discuss two DLC protocols that actually implemented those concepts. The first, HDLC, is the base of many protocols that have been designed for LANs. The second, Point to-Point, is a protocol derived from HDLC and is used for point-to-point links.

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5.2.4 (continued)

- **HDLC**
  - Configuration and Transfer Modes
  - Frames
- **Point-to-Point Protocol (PPP)**
  - Services
  - Framing
  - Transition Phases
  - Multiplexing
  - Multilink PPP

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Figure 5.20: Normal response mode

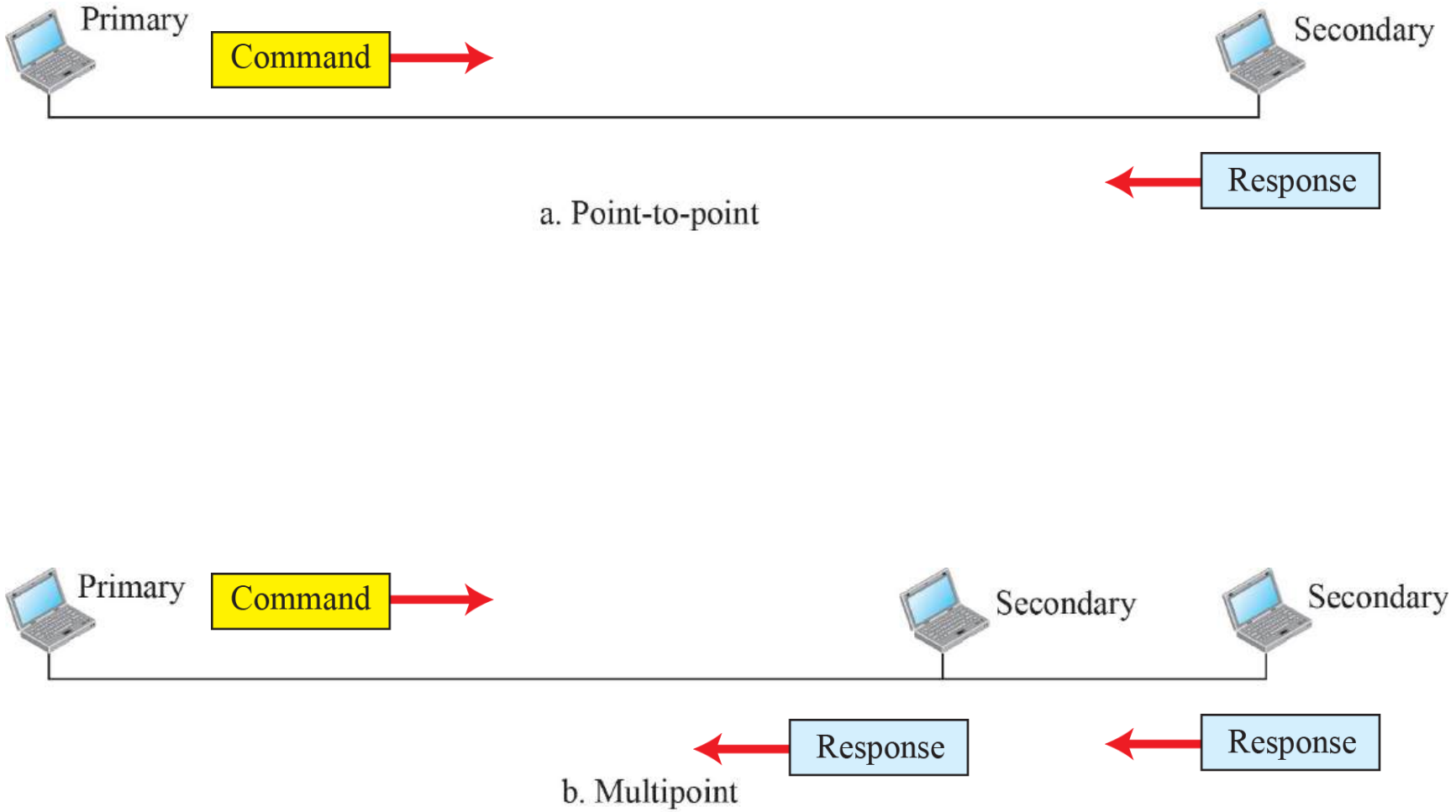

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Figure 5.21: Asynchronous balanced mode

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Figure 5.22: HDLC frames

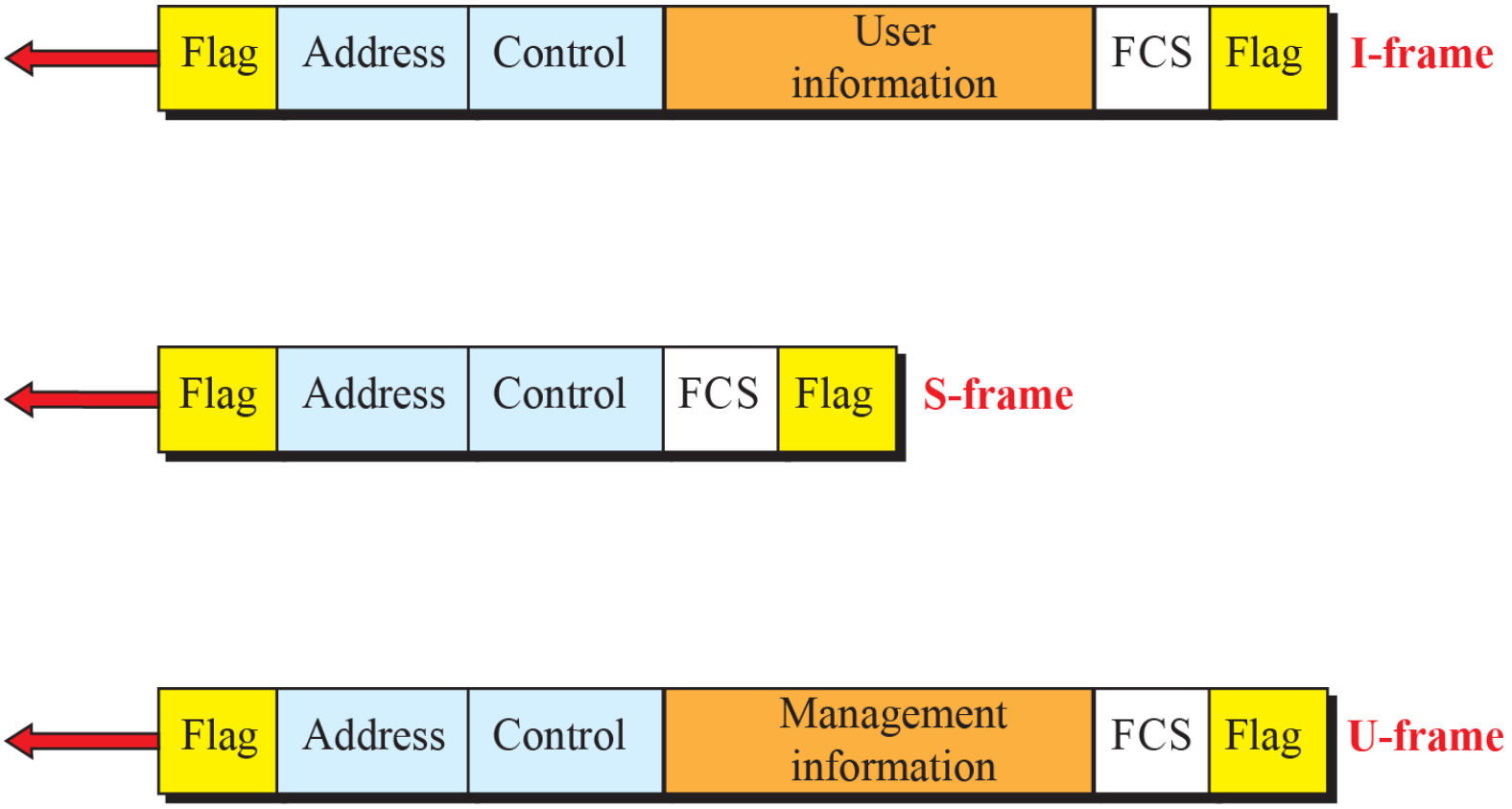

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Figure 5.23: ***Control field format for the different frame types***

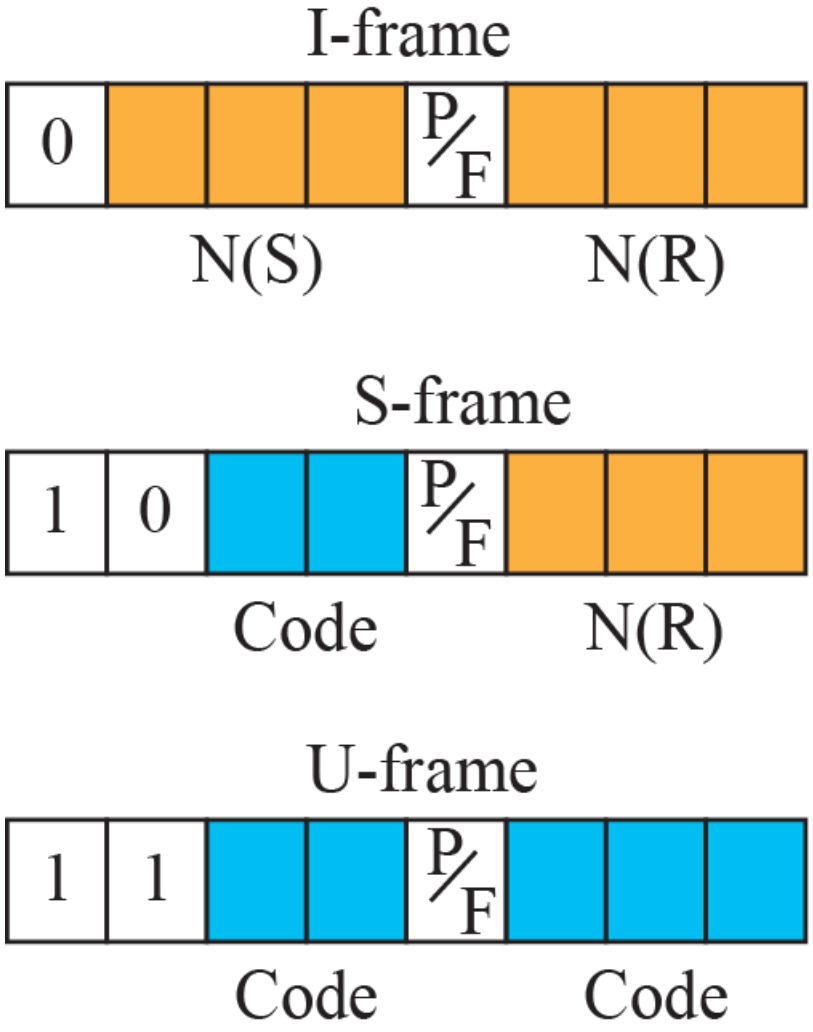

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Figure 5.24: PPP frame format

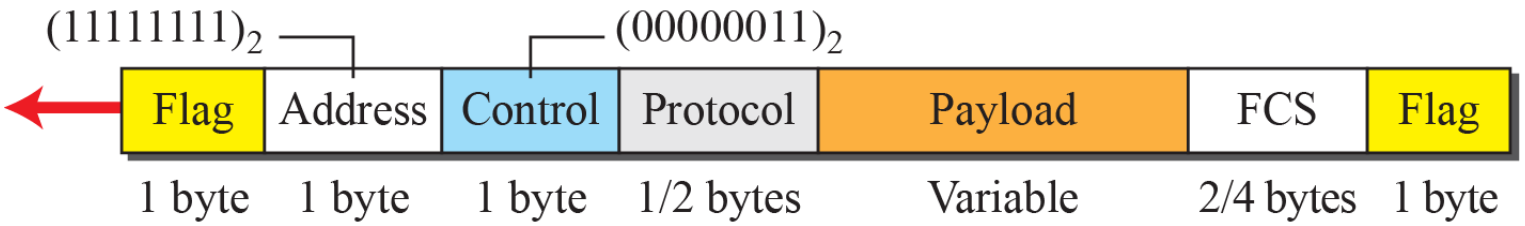

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Figure 5.25: Transition phases

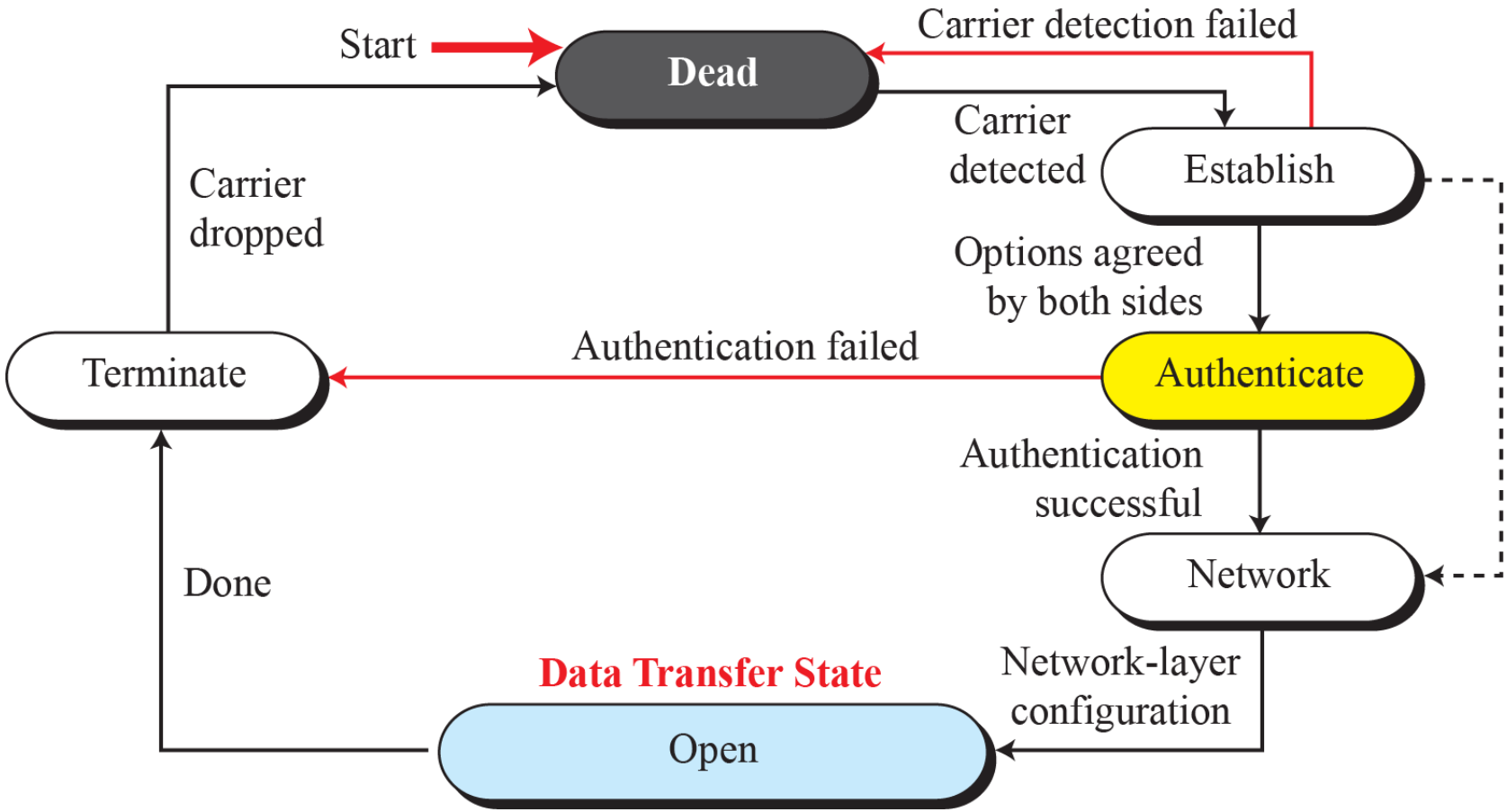

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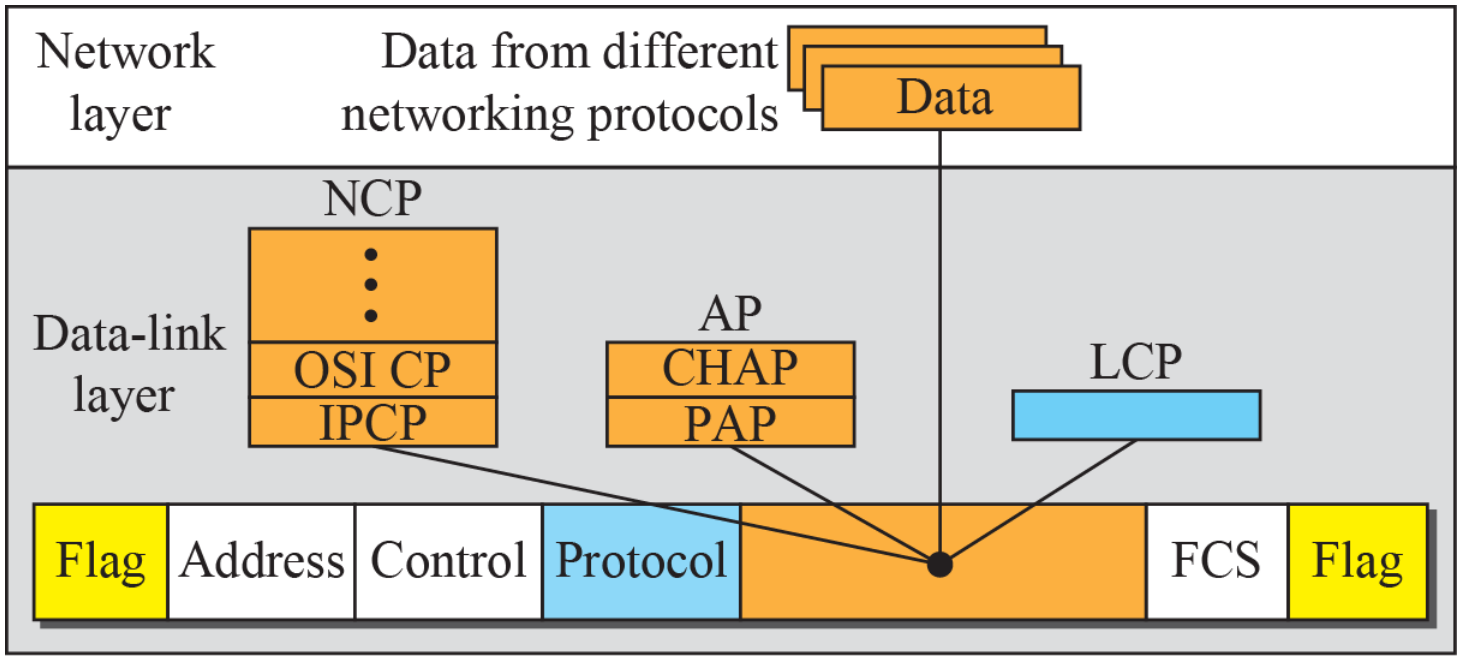
Figure 5.26: Multiplexing in PPP



Legend

LCP : Link control protocol
AP : Authentication protocol
NCP: Network control protocol

Protocol values:

LCP : 0xC021
AP : 0xC023 and 0xC223
NCP: 0x8021 and ....
Data: 0x0021 and ....

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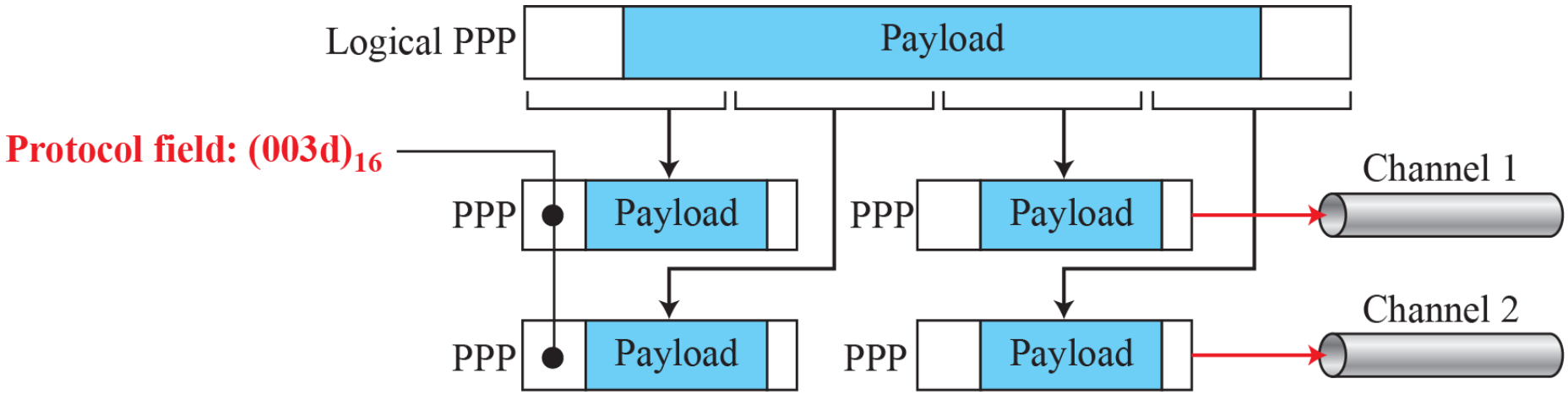
Figure 5.27: Multilink PPP



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5-3 MULTIPLE ACCESS PROTOCOLS

We said that the data-link layer is divided into two sublayers: data link control (DLC) and media access control (MAC). We discussed DLC in the previous section; we talk about MAC in this section.

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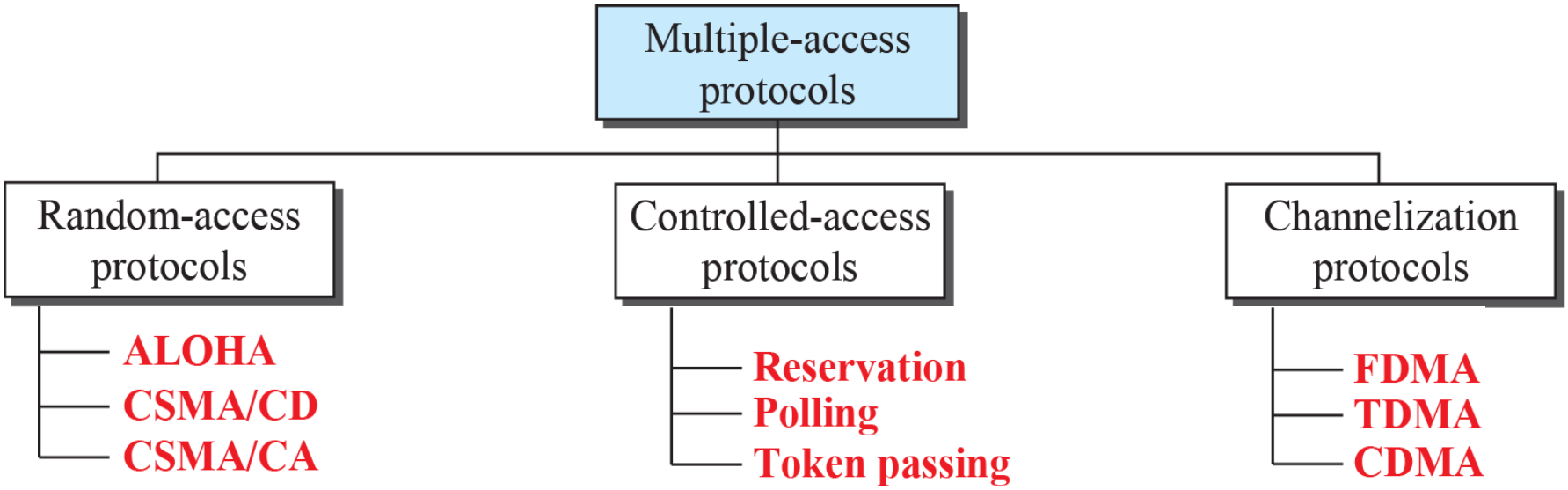
Figure 5.28: Taxonomy of multiple-access protocols



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5.3.1 Random Access

In random-access or contention methods, no station is superior to another station and none is assigned the control over another. At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send. This decision depends on the state of the medium (idle or busy). In other words, each station can transmit when it desires on the condition that it follows the predefined procedure, including the testing of the state of the medium.

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5.3.1 (continued)

- **ALOHA**
  - Pure ALOHA
  - Slotted ALOHA
- **CSMA**
  - Vulnerable Time
  - Persistence Methods
- **CSMA/CD**
  - Minimum Frame Size
  - Procedure
  - Energy Level
  - Throughput
  - Traditional Ethernet
- **CSMA/CA**

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Figure 5.29: Frames in a pure ALOHA network

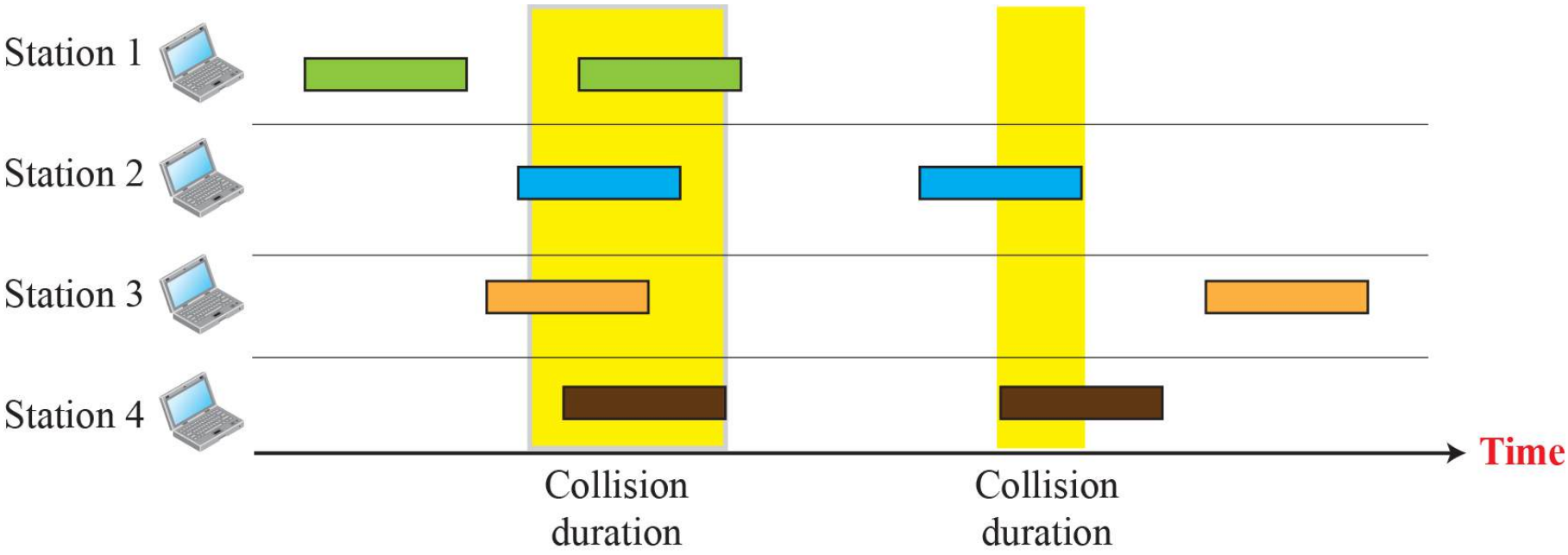

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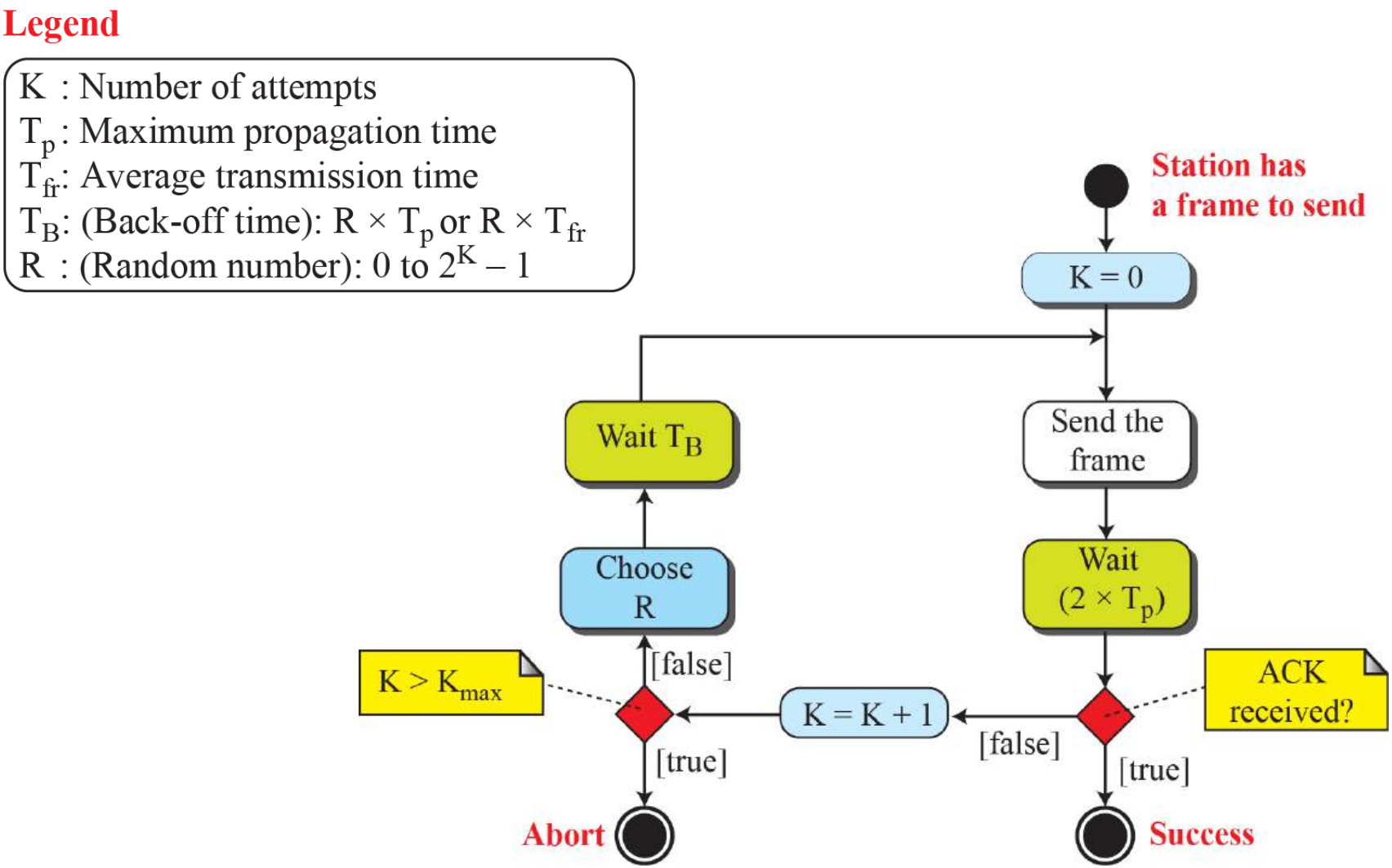
Figure 5.30: Procedure for pure ALOHA protocol



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Example 5.11

The stations on a wireless ALOHA network are a maximum of 600 km apart. If we assume that signals propagate at 3×10^8 m/s, we find $T_p = (600 \times 10^3) / (3 \times 10^8) = 2$ ms. For $K = 2$, the range of R is $\{0, 1, 2, 3\}$. This means that T_B can be 0, 2, 4, or 6 ms, based on the outcome of the random variable R.

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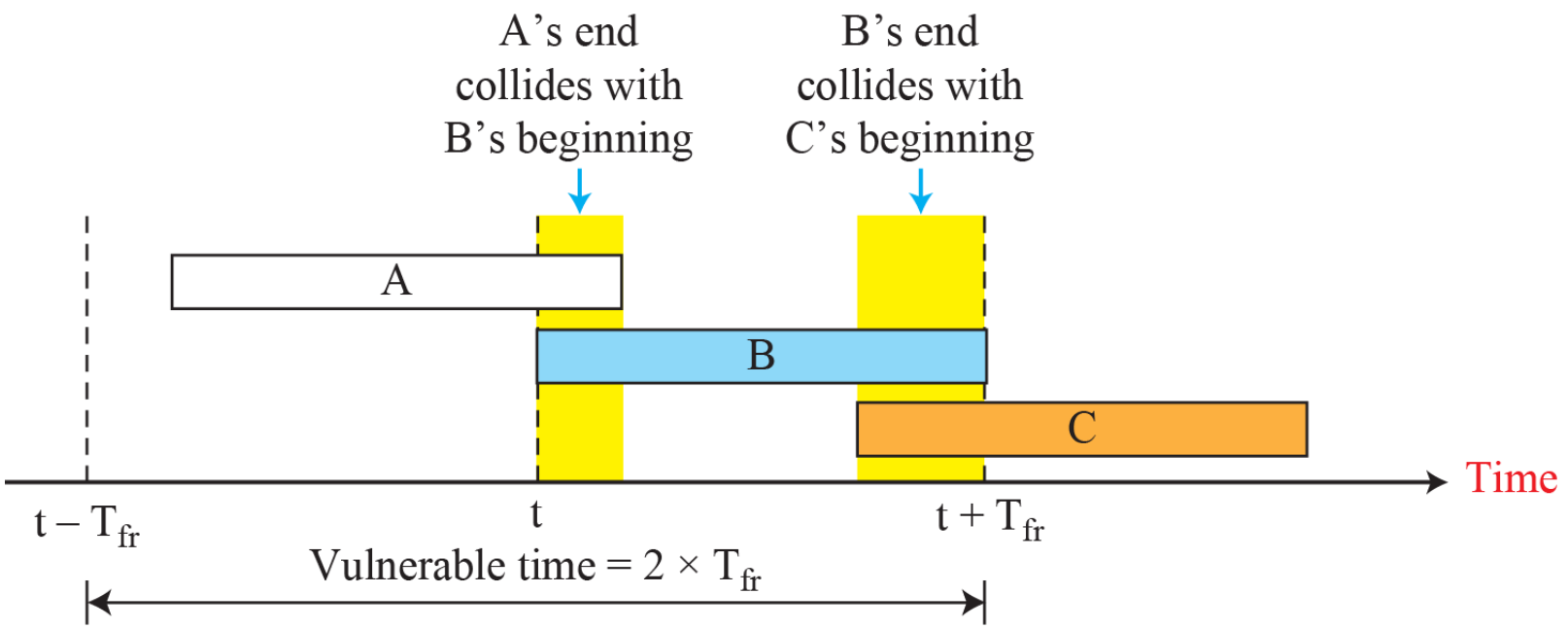
Figure 5.31: Vulnerable time for pure ALOHA protocol



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Example 5.12

A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?

Solution

Average frame transmission time T_{fr} is 200 bits/200 kbps or 1 ms. The vulnerable time is 2×1 ms $= 2$ ms. This means no station should send later than 1 ms before this station starts transmission and no station should start sending during the period (1 ms) that this station is sending.

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Example 5.13

A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces

a. 1000 frames per second?

b. 500 frames per second?

c. 250 frames per second?

Solution

The frame transmission time is 200/200 kbps or 1 ms.

a. If the system creates 1000 frames per second, or 1 frame per millisecond, then $G = 1$. In this case $S = G \times e^{-2G} = 0.135$ (13.5 percent). This means that the throughput is $1000 \times 0.135 = 135$ frames. Only 135 frames out of 1000 will probably survive.

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Example 5.13 (continued)

b. If the system creates 500 frames per second, or 1/2 frames per millisecond, then G = 1/2. In this case $S = G \times e^{-2G} = 0.184$ (18.4 percent). This means that the throughput is $500 \times 0.184 = 92$ and that only 92 frames out of 500 will probably survive. Note that this is the maximum throughput case, percentage-wise.

c. If the system creates 250 frames per second, or 1/4 frames per millisecond, then G = 1/4. In this case $S = G \times e^{-2G} = 0.152$ (15.2 percent). This means that the throughput is $250 \times 0.152 = 38$. Only 38 frames out of 250 will probably survive

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Figure 5.32: Frames in a slotted ALOHA network

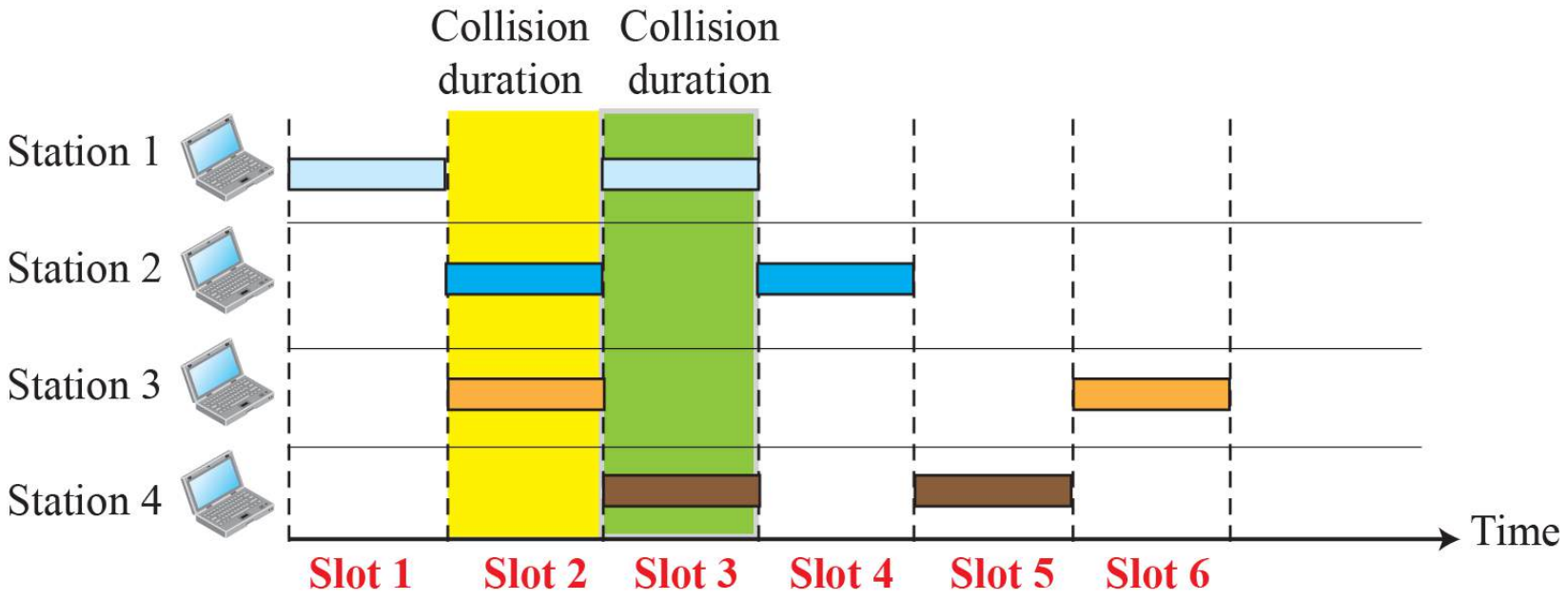

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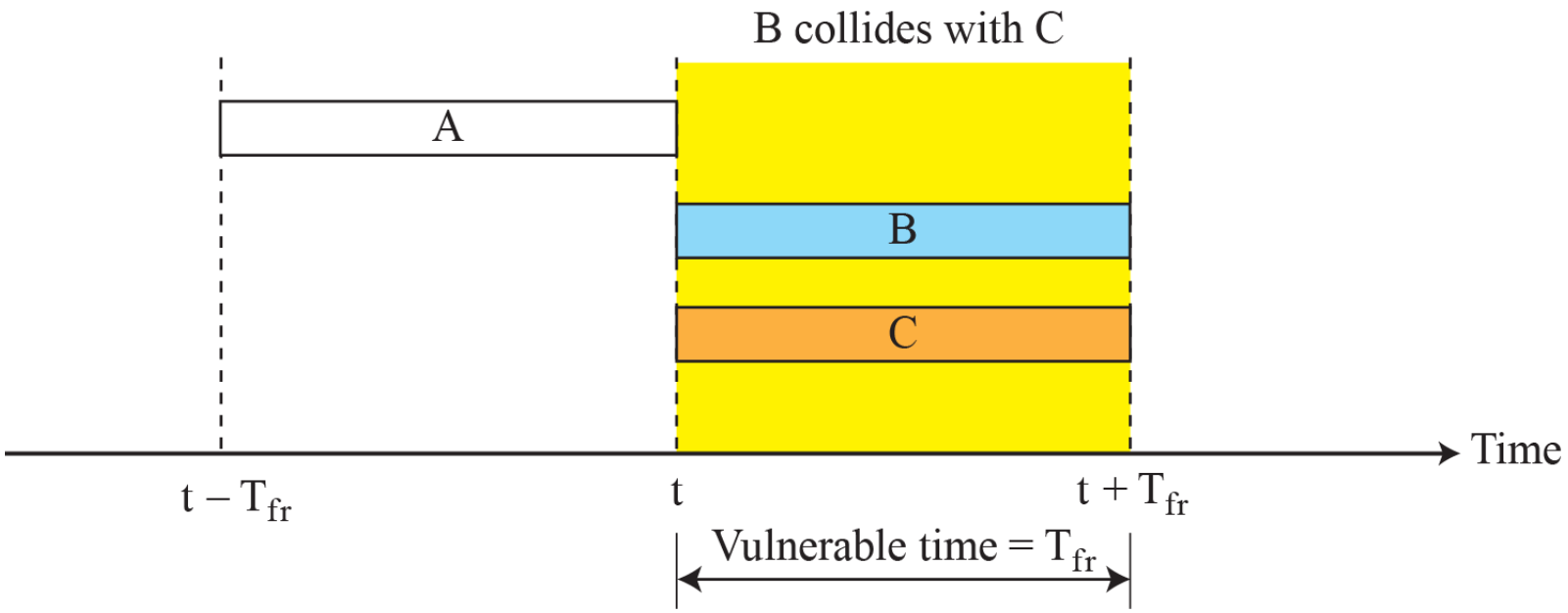
Figure 5.33: Vulnerable time for slotted ALOHA protocol



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Example 5.14

A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system (all stations together) produces

a. 1000 frames per second.
b. 500 frames per second.
c. 250 frames per second.

Solution

This situation is similar to the previous exercise except that the network is using slotted ALOHA instead of pure ALOHA. The frame transmission time is 200/200 kbps or 1 ms.

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Example 5.14 (continued)

a) In this case G is 1. So $S = G \times e^{-G} = 0.368$ (36.8 percent). This means that the throughput is $1000 \times 0.0368 = 368$ frames. Only 368 out of 1000 frames will probably survive. Note that this is the maximum throughput case, percentage-wise.

b) Here G is 1/2. In this case $S = G \times e^{-G} = 0.303$ (30.3 percent). This means that the throughput is $500 \times 0.0303 = 151$. Only 151 frames out of 500 will probably survive.

c) Now G is 1/4. In this case $S = G \times e^{-G} = 0.195$ (19.5 percent). This means that the throughput is $250 \times 0.195 = 49$. Only 49 frames out of 250 will probably survive.

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Figure 5.34: Space/time model of a collision in CSMA (Part I: model)

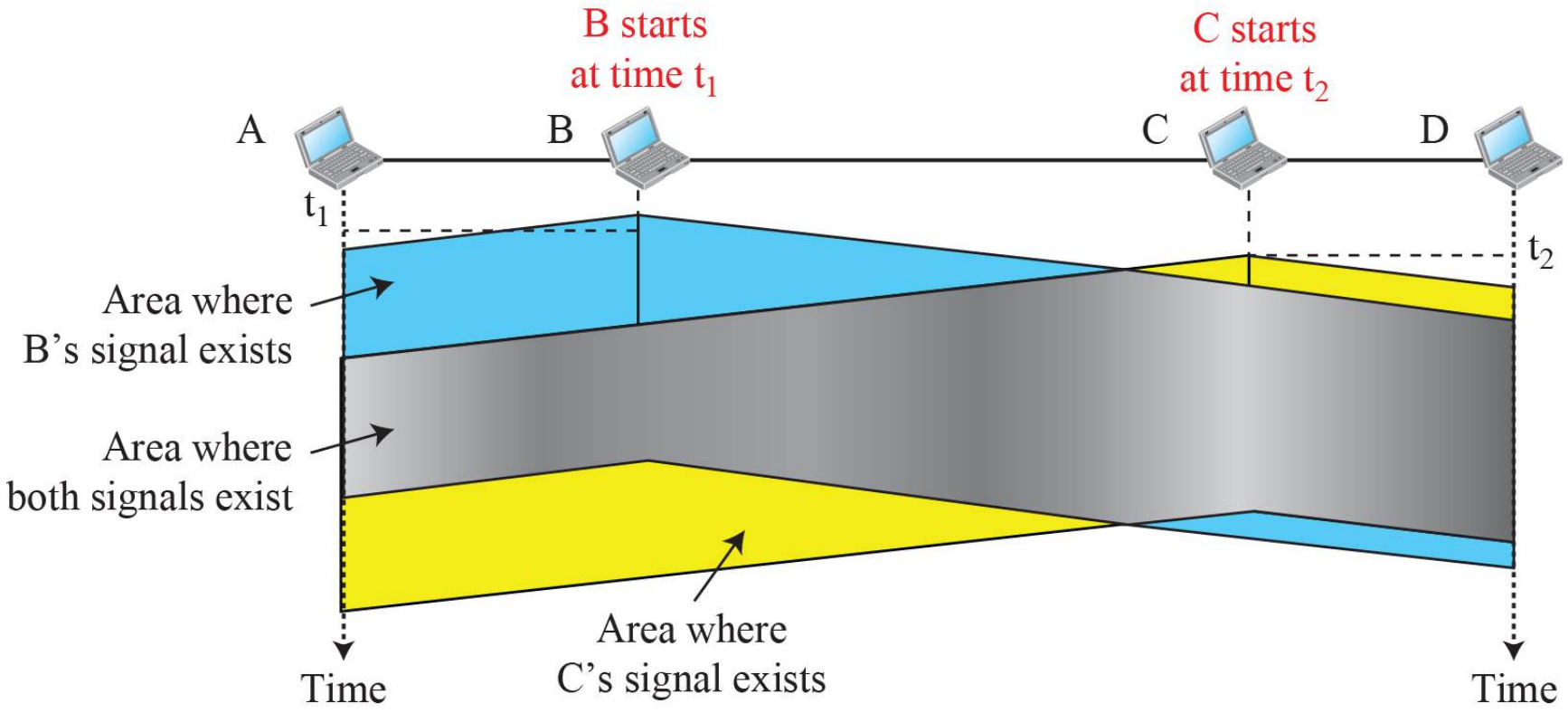

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Figure 5.34: Space/time model of a collision in CSMA Part II: timing)

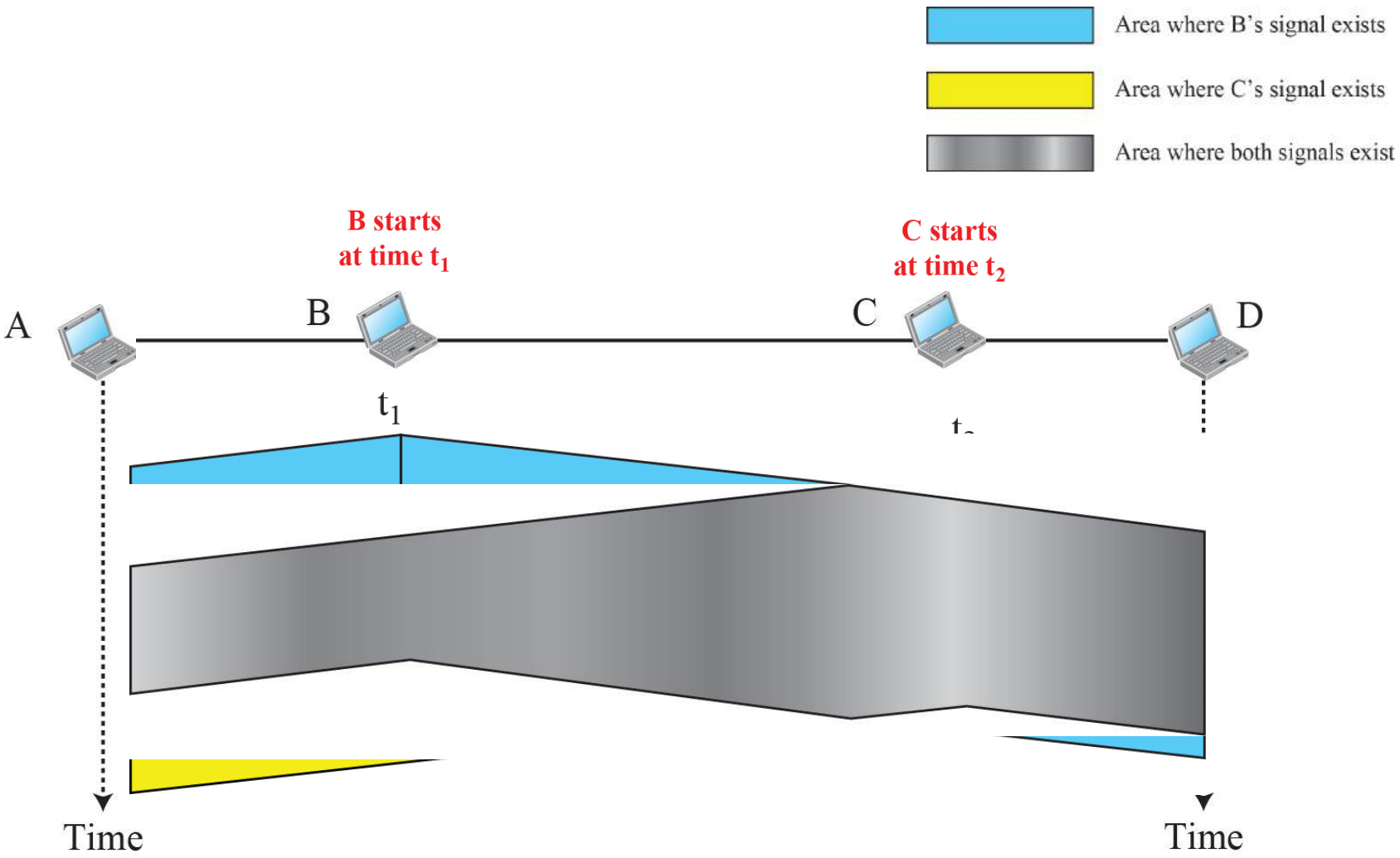

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Figure 5.35: Vulnerable time in CSMA

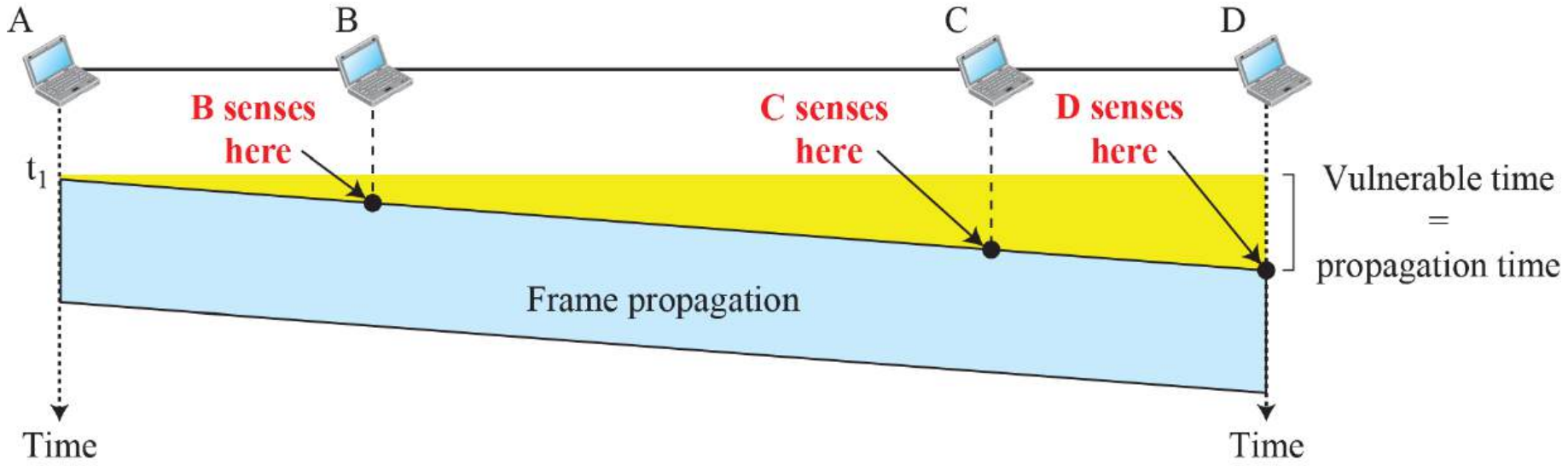

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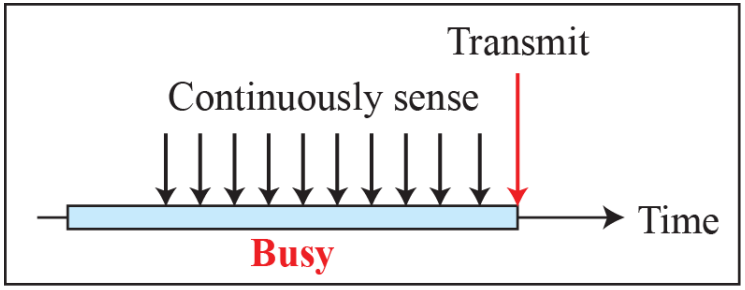
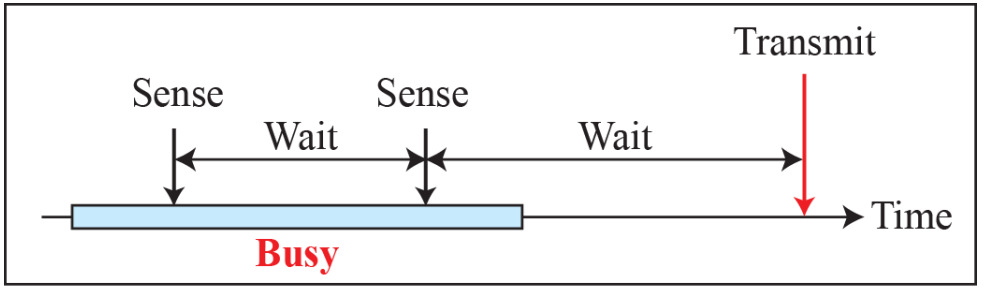
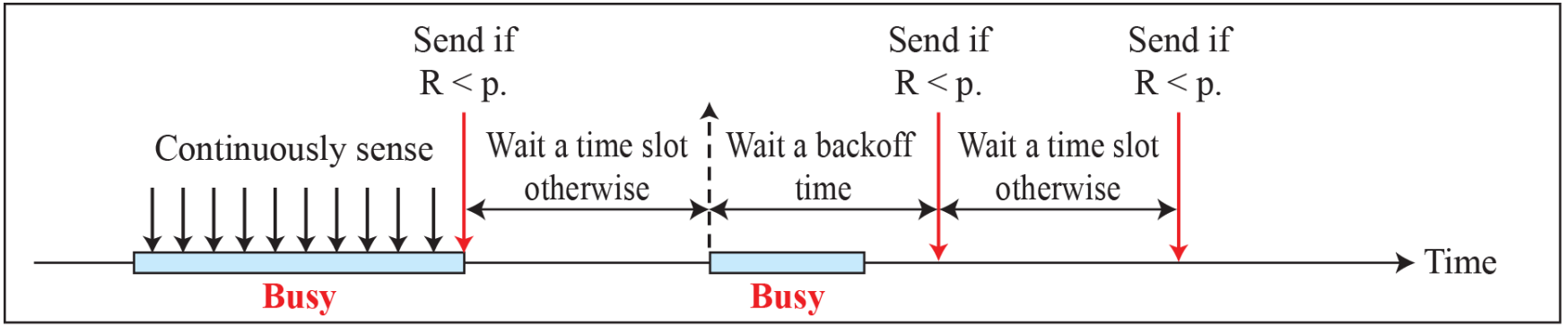
Figure 5.36: Behavior of three persistence methods



a. 1-persistent



b. Nonpersistent



c. *p*-persistent

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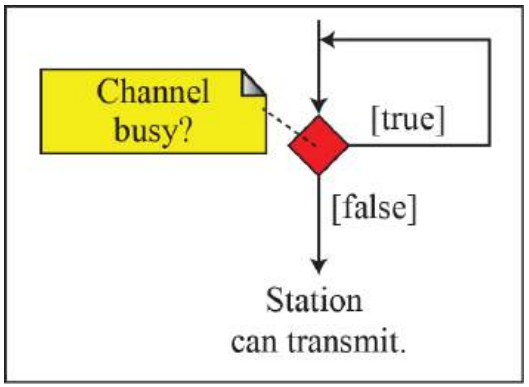
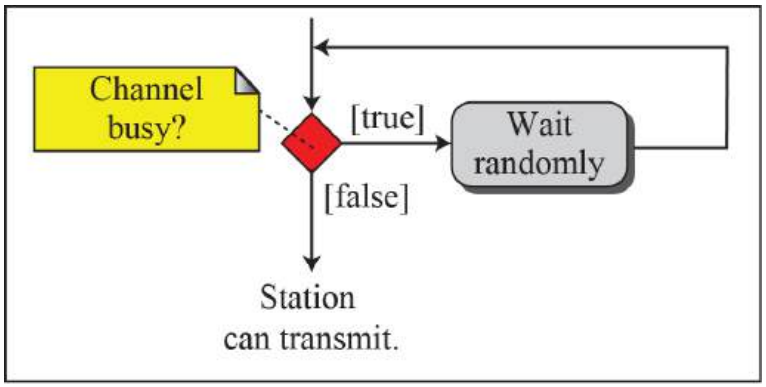
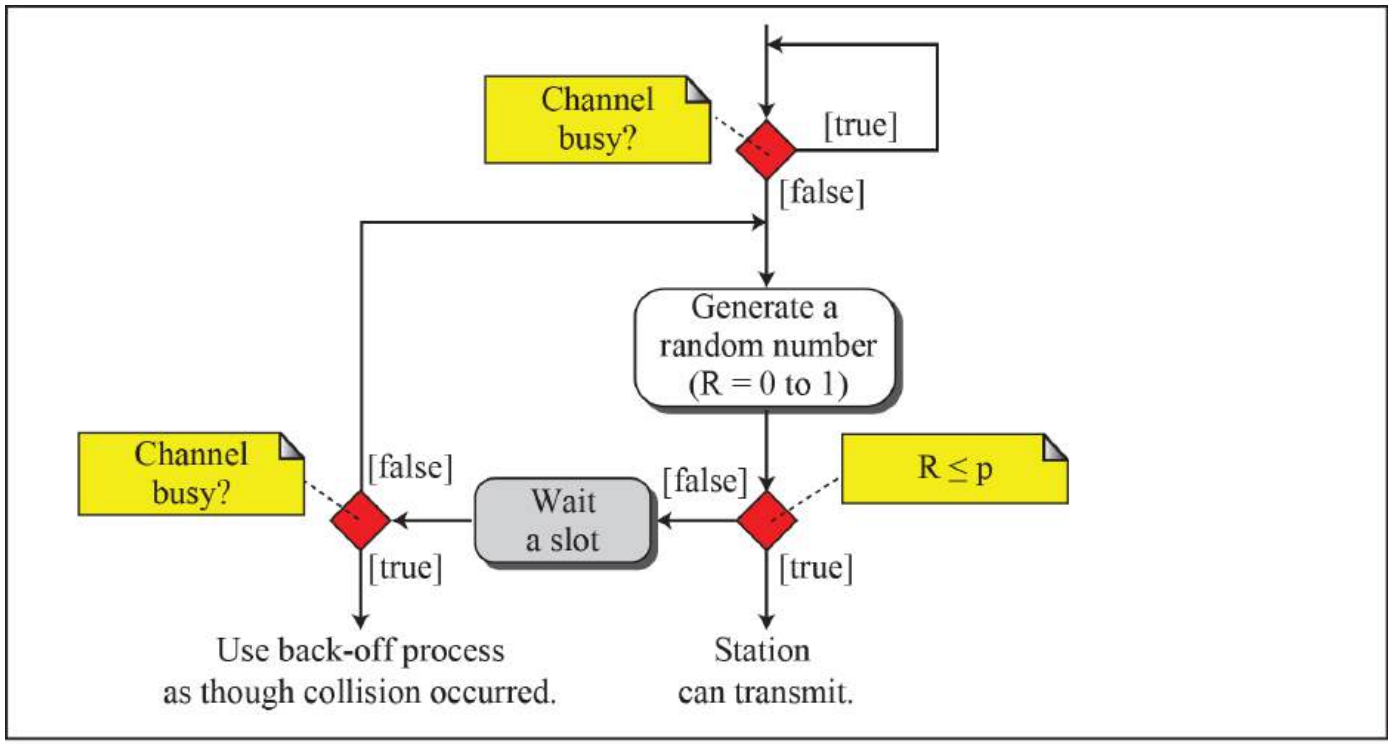
Figure 5.37: Flow diagram for three persistence methods



a. 1-persistent



b. Nonpersistent



c. *p*-persistent

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Figure 5.38: Collision of the first bits in CSMA/CD

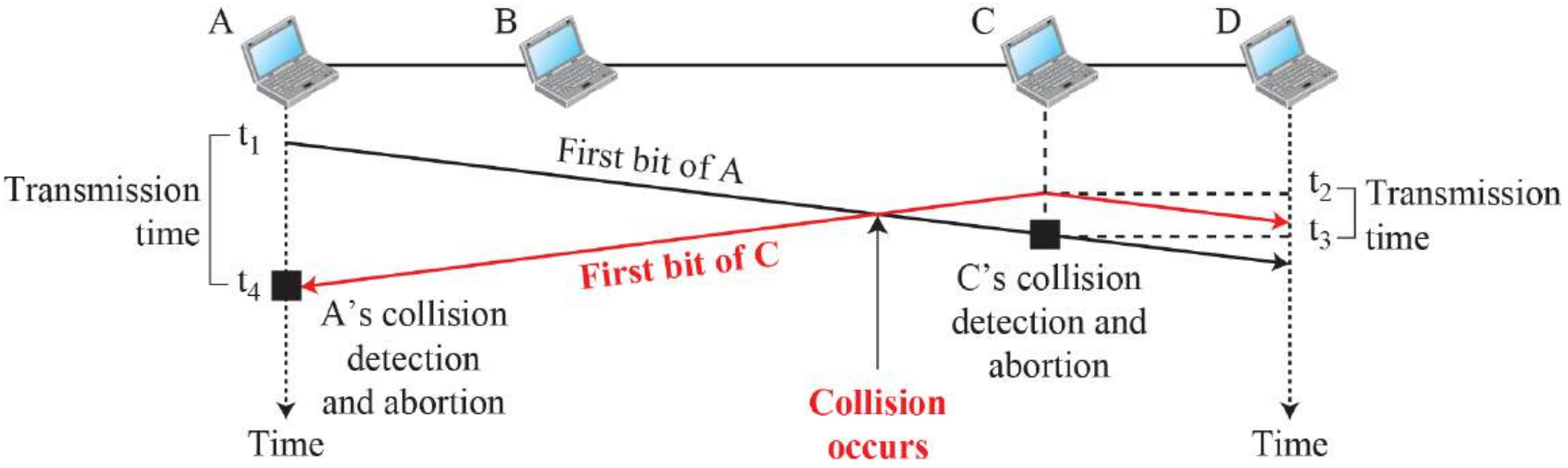

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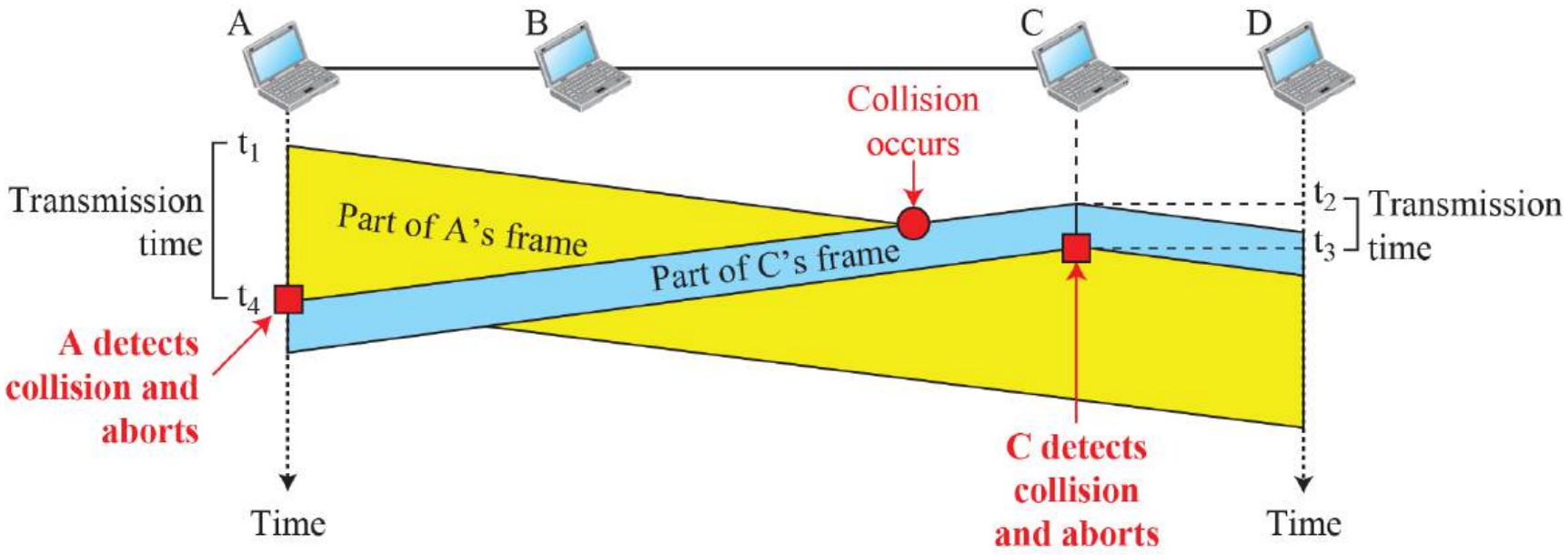
Figure 5.39: Collision and abortion in CSMA/CD



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Example 5.15

A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal, as we see later) is 25.6 μs, what is the minimum size of the frame?

Solution

The minimum frame transmission time is $T_{fr} = 2 \times T_p = 51.2$ μs. This means, in the worst case, a station needs to transmit for a period of 51.2 μs to detect the collision. The minimum size of the frame is 10 Mbps × 51.2 μs = 512 bits or 64 bytes. This is actually the minimum size of the frame for Standard Ethernet, as we will see later in the chapter.

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Figure 5.40: Flow diagram for the CSMA/CD

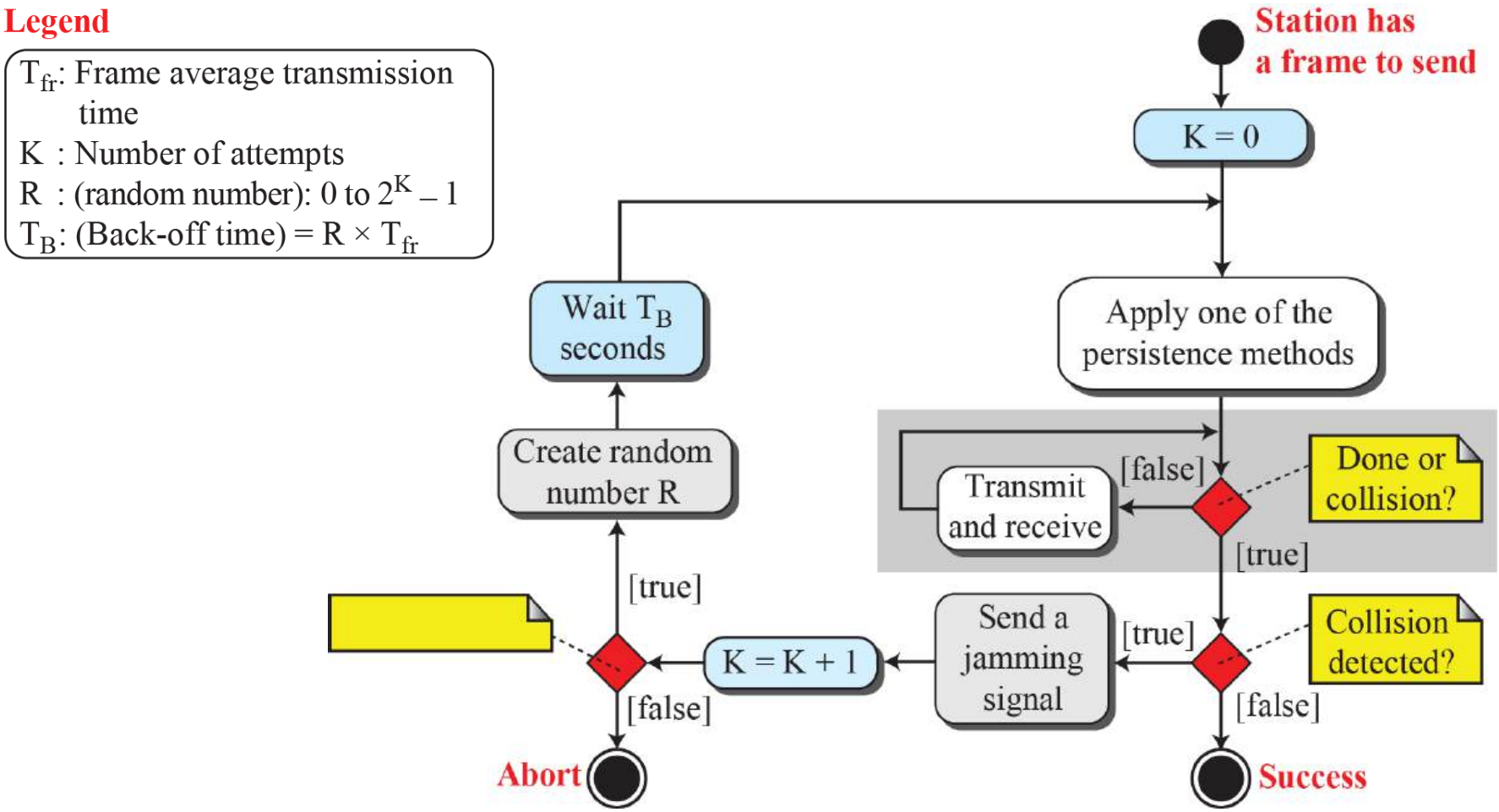

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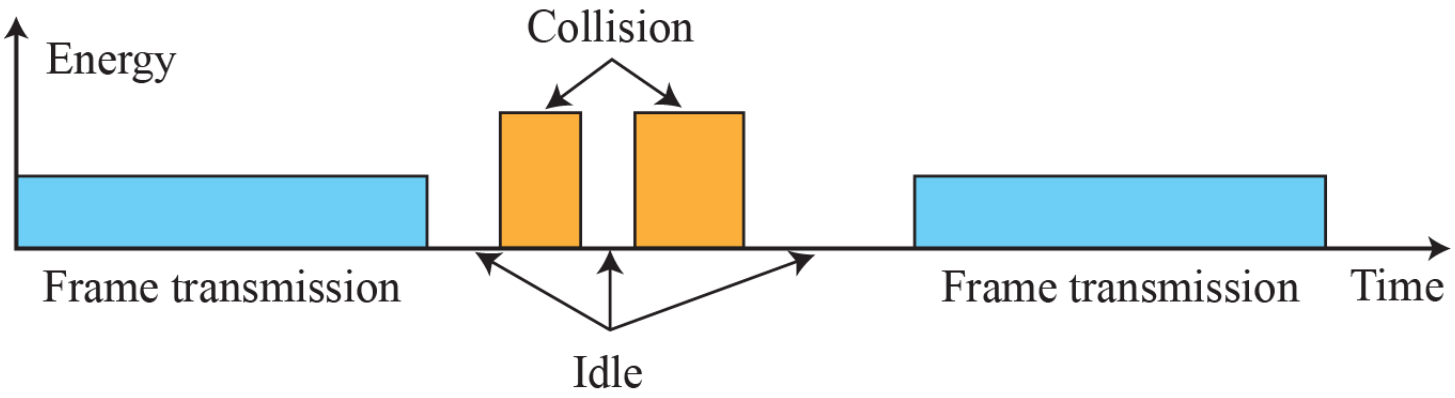
Figure 5.41: Energy level during transmission, idleness, or collision



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5.3.2 Controlled Access

In controlled access, the stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations. We discuss three controlled-access methods.

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5.3.2 (continued)

- **Reservation**
- **Polling**
  - Select
  - Poll
- **Token Passing**
  - Logical Ring

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Figure 5.42: Reservation access method

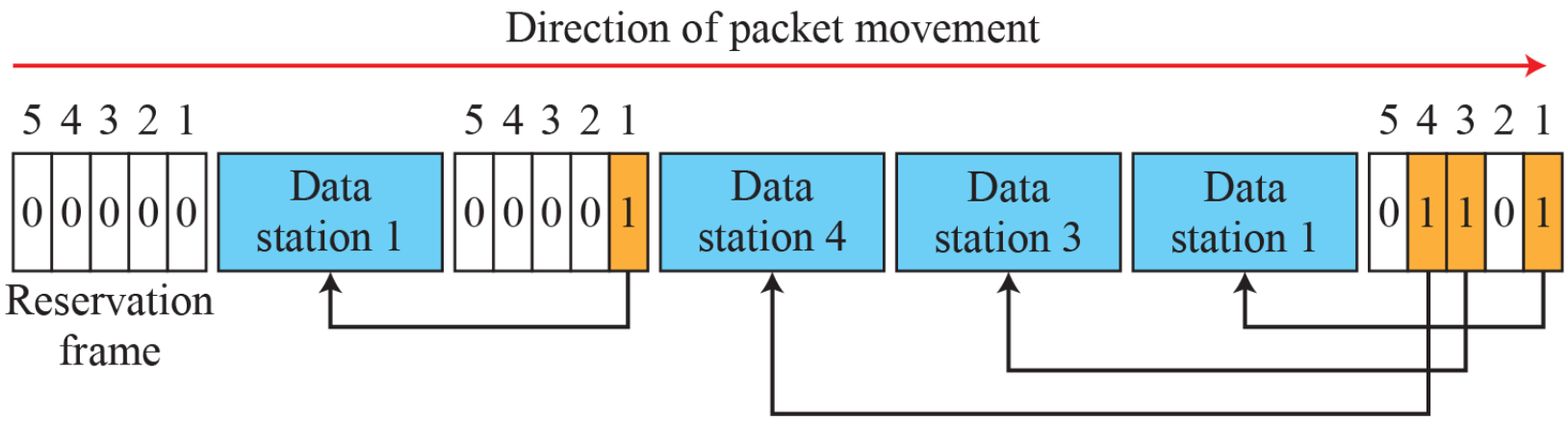

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Figure 5.43: ***Select and poll functions in polling-access method***

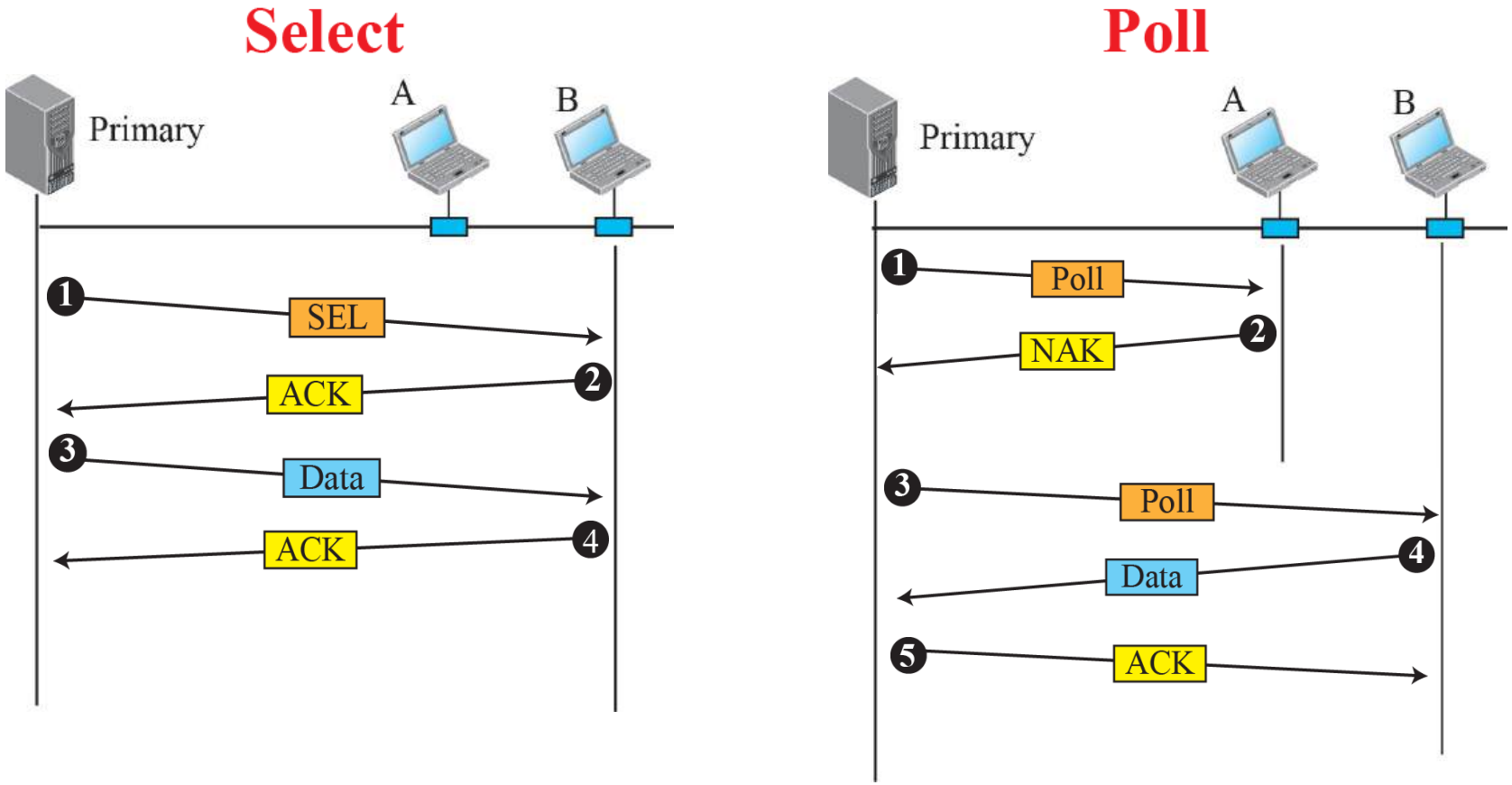

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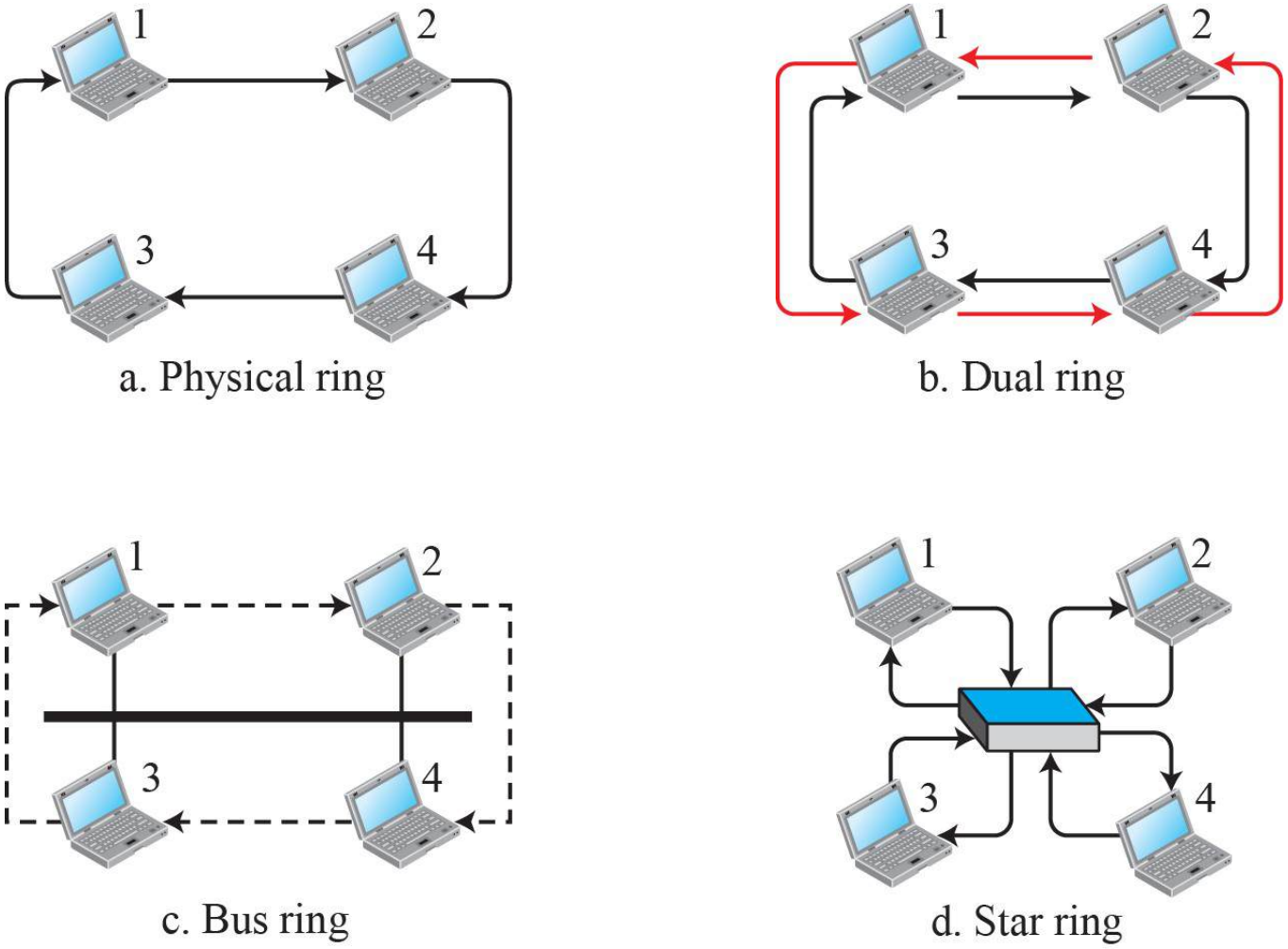
Figure 5.44: Logical ring and physical topology in token-passing access method



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5.3.3 Channelization

Channelization (or channel partition, as it is sometimes called) is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, among different stations. Since these methods are normally used in wireless networks, we postpone their discussion until the next chapter.

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5-4 LINK-LAYER ADDRESSING

In Chapter 4, we discussed IP addresses as the identifiers at the network layer that define the exact points in the Internet where the source and destination hosts are connected. However, in a connectionless internetwork such as the Internet we cannot make a datagram reach its destination using only IP addresses.

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5-4 Continued

- ***Address Resolution Protocol (ARP)***
  - *Packet Format*
- ***An Example***
  - *Activities at the Alice Site*
  - *Activities at Routers*
  - *Activities at Bob's Site*

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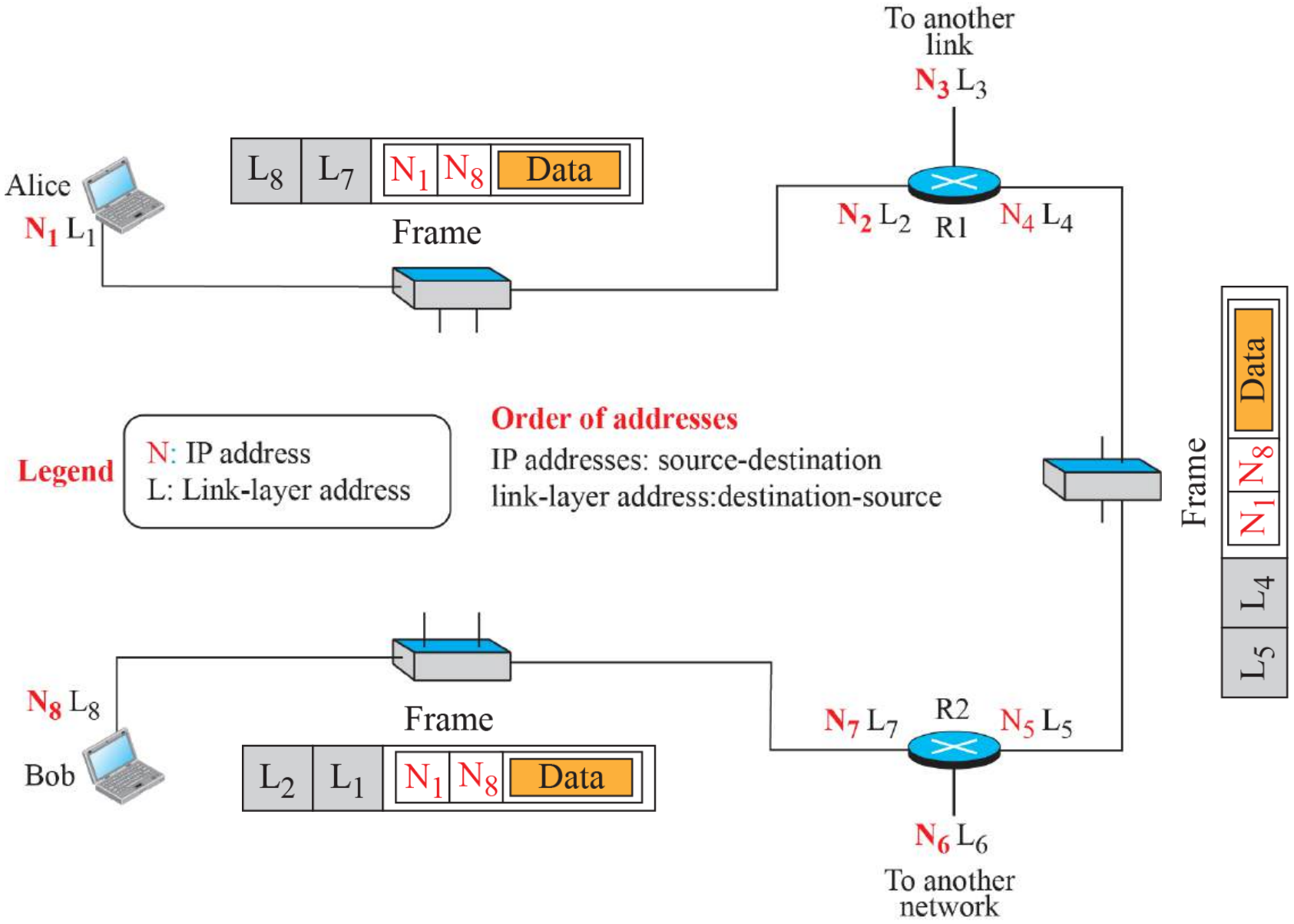
Figure 5.45: IP addresses and link-layer addresses in a small internet



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Example 5.16

As we discuss later in the chapter, the link-layer addresses in the most common LAN, Ethernet, are 48 bits (six bytes) that are presented as 12 hexadecimal digits separated by colons; for example, the following is a link-layer address of a computer.

A2:34:45:11:92:F1

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Figure 5.46: Position of ARP in TCP/IP protocol suite

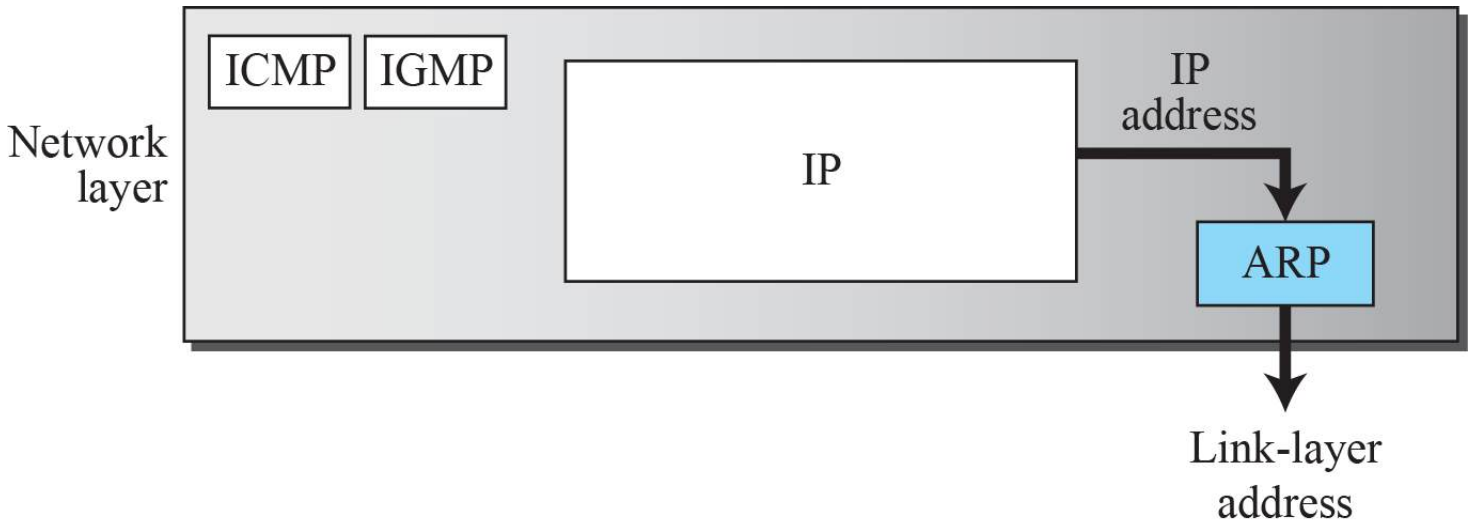

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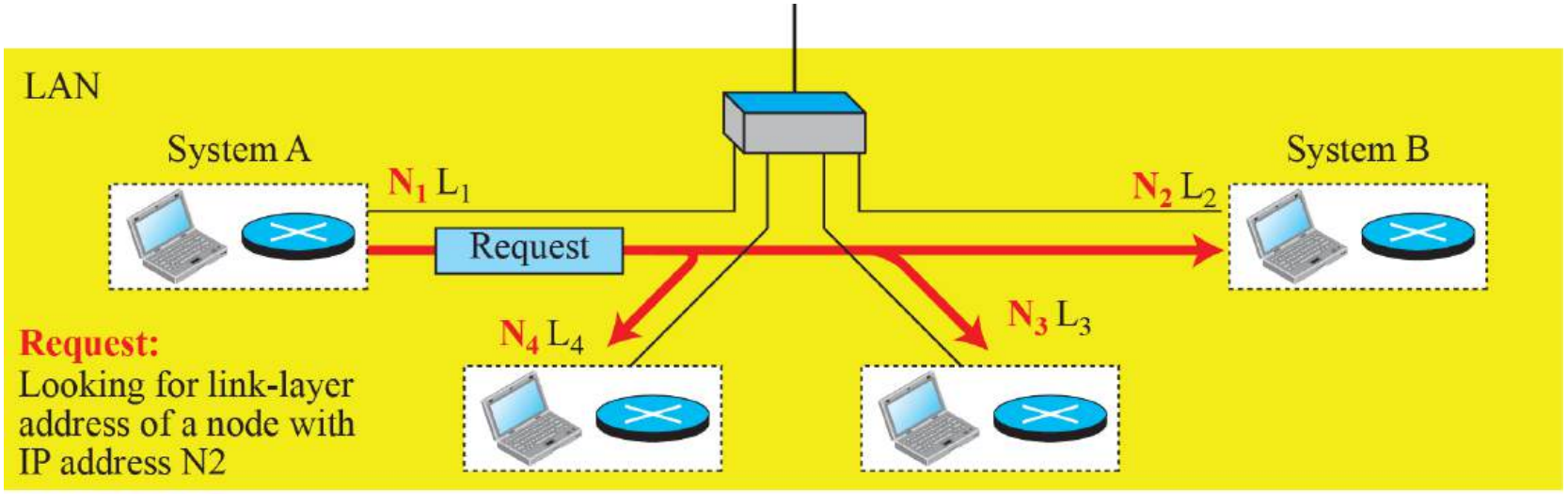
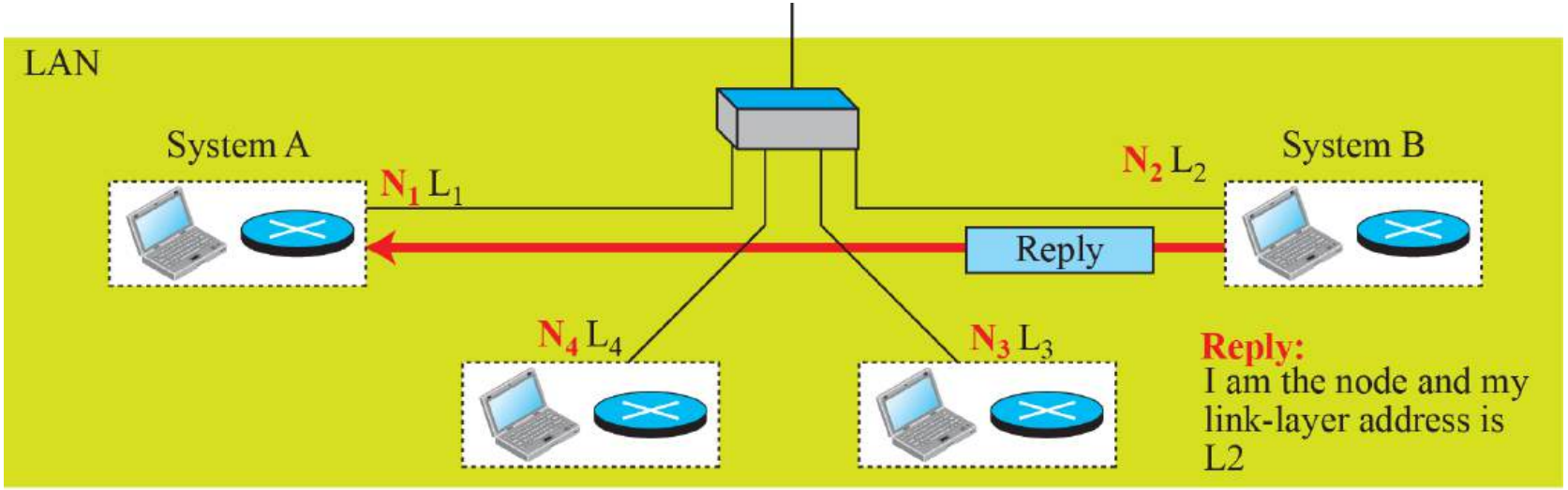
Figure 5.47: ARP operation



a. ARP request is broadcast



b. ARP reply is unicast

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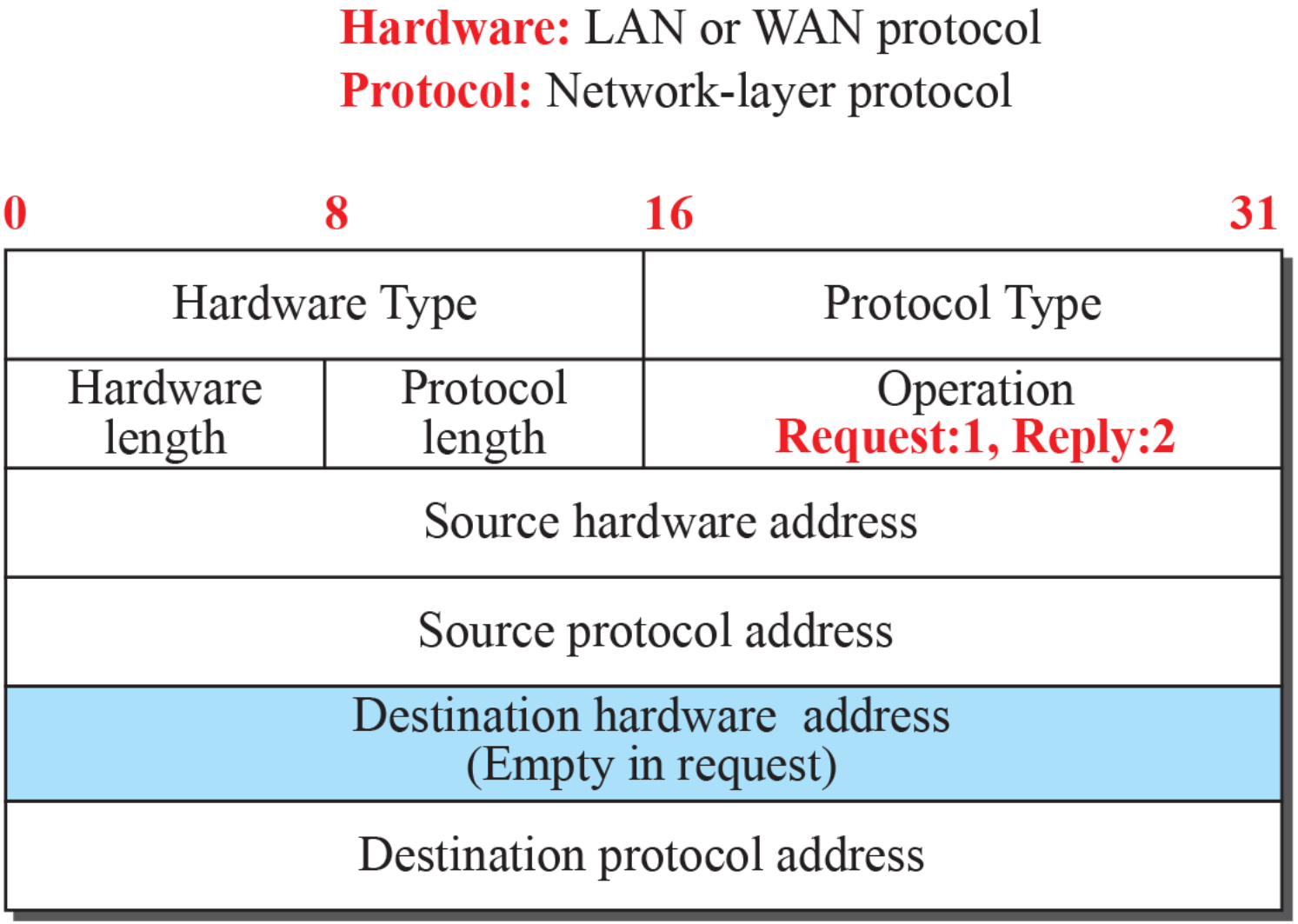
Figure 5.48: ***ARP packet***



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Example 5.17

A host with IP address N1 and MAC address L1 has a packet to send to another host with IP address N2 and physical address L2 (which is unknown to the first host). The two hosts are on the same network. Show the ARP request and reply packets encapsulated in Ethernet frames (see Figure 5.55).

Solution

Figure 5.49 shows the ARP request and reply packets. Note that the ARP data field in this case is 28 bytes, and that the individual addresses do not fit in the 4-byte boundary. That is why we do not show the regular 4-byte boundaries for these addresses. Also note that the IP addresses are shown in hexadecimal.

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Figure 5.49: Example 5.17

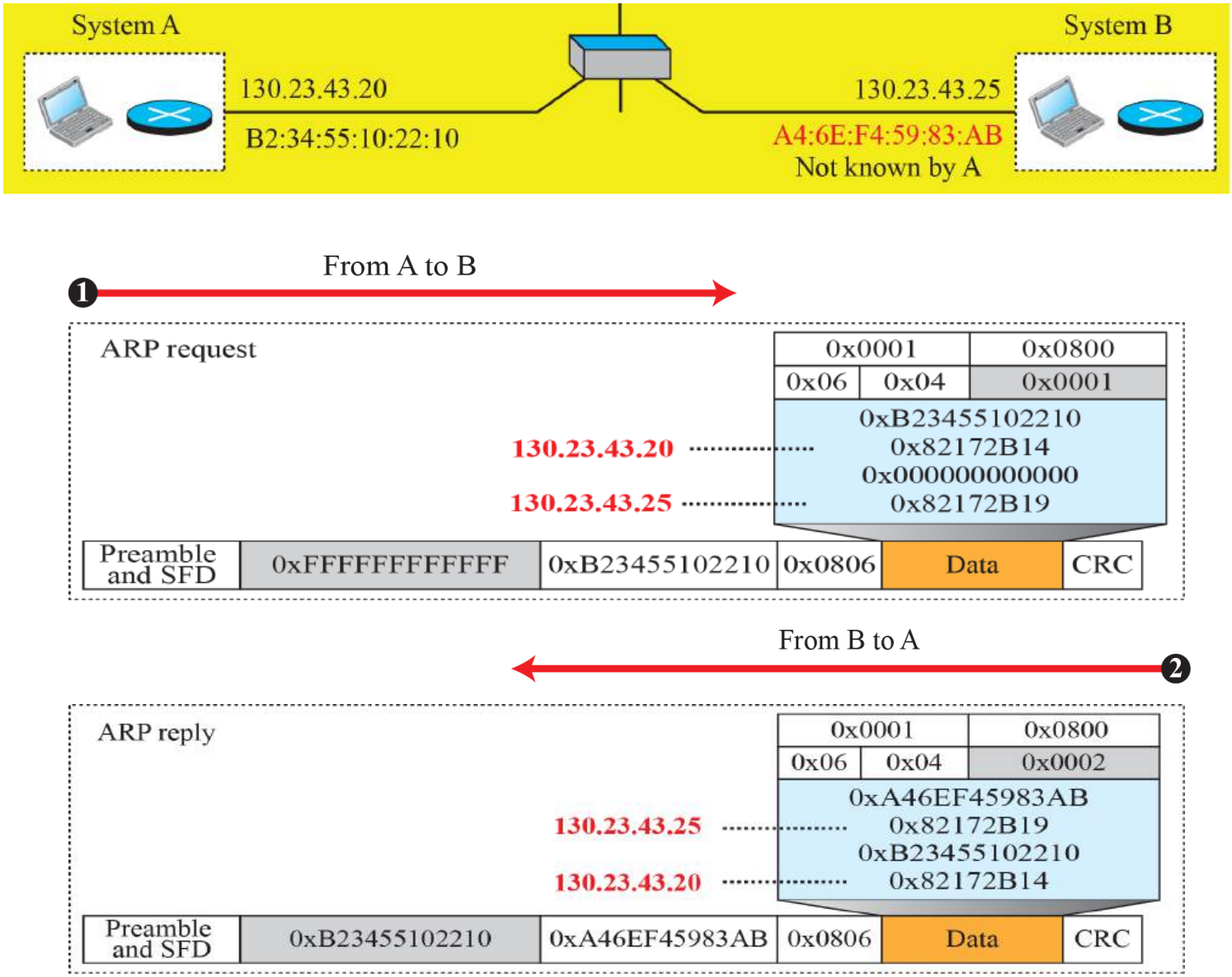

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Figure 5.50: The internet for our example

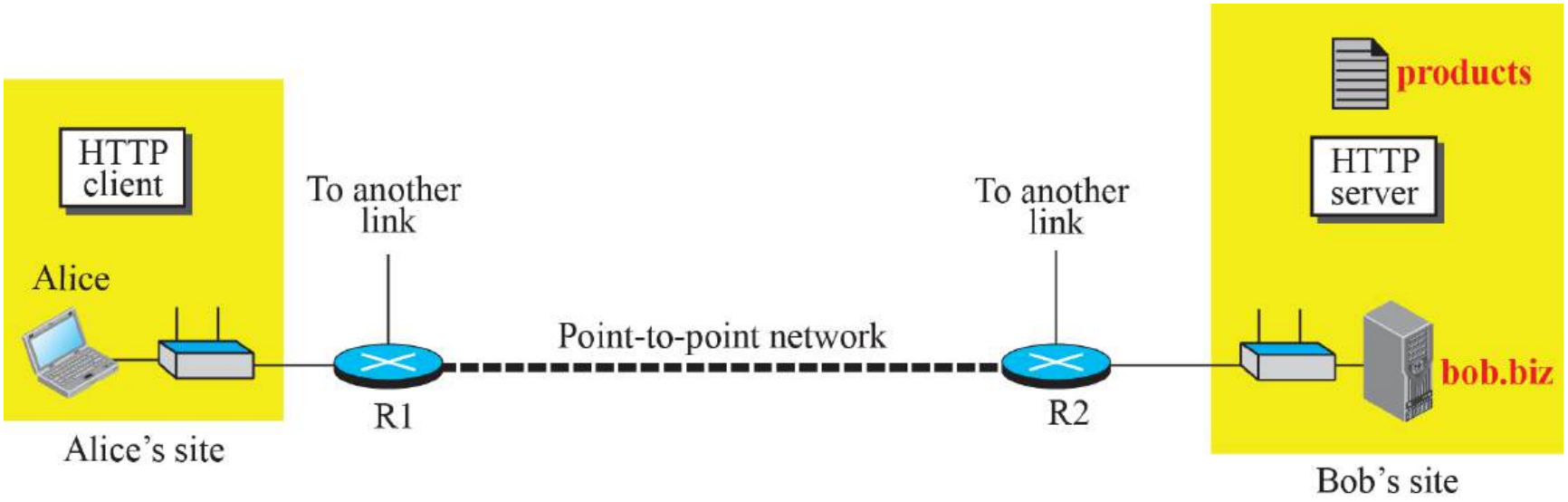

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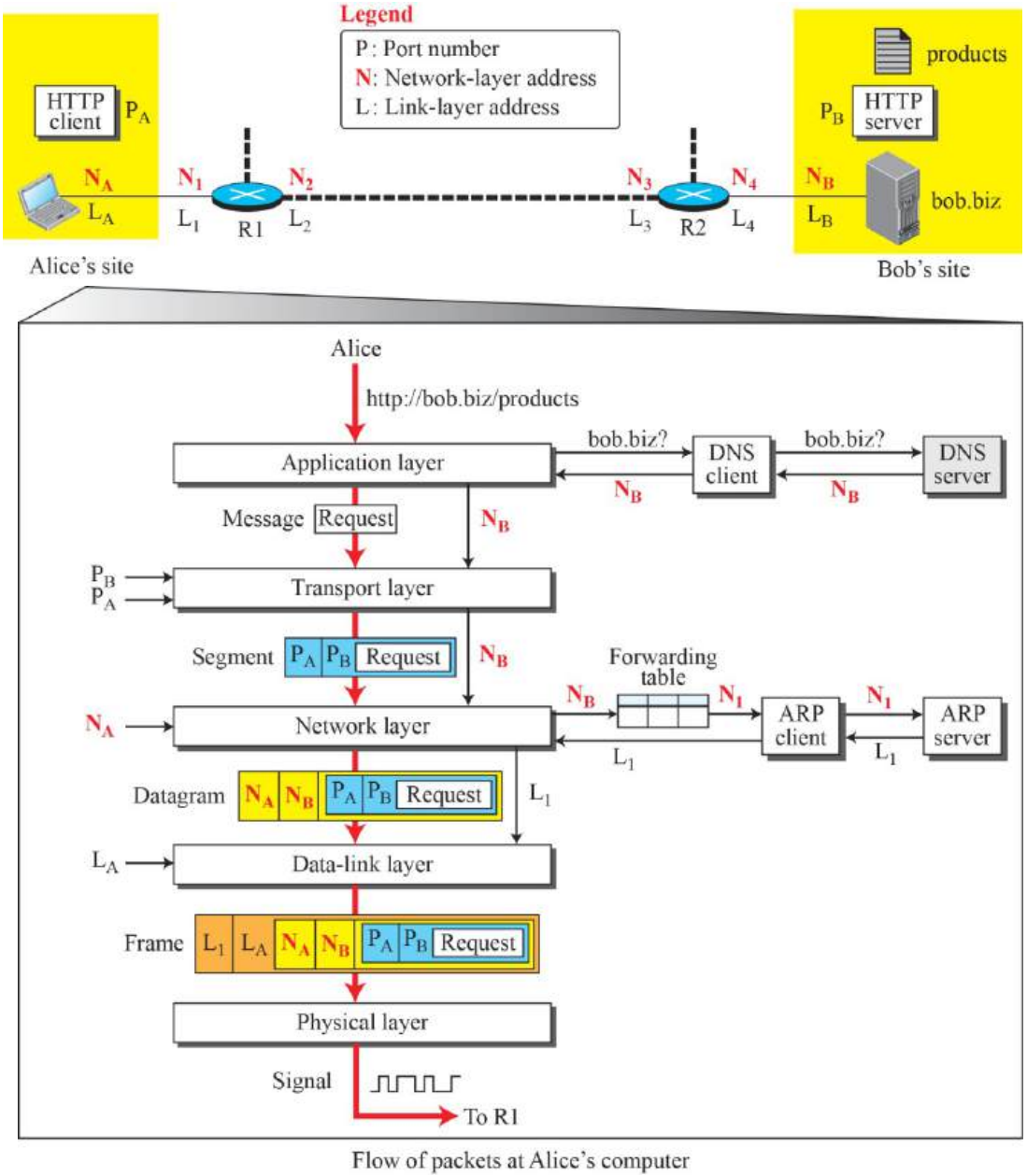
Figure 5.51: Flow of packets at Alice's computer



Flow of packets at Alice's computer

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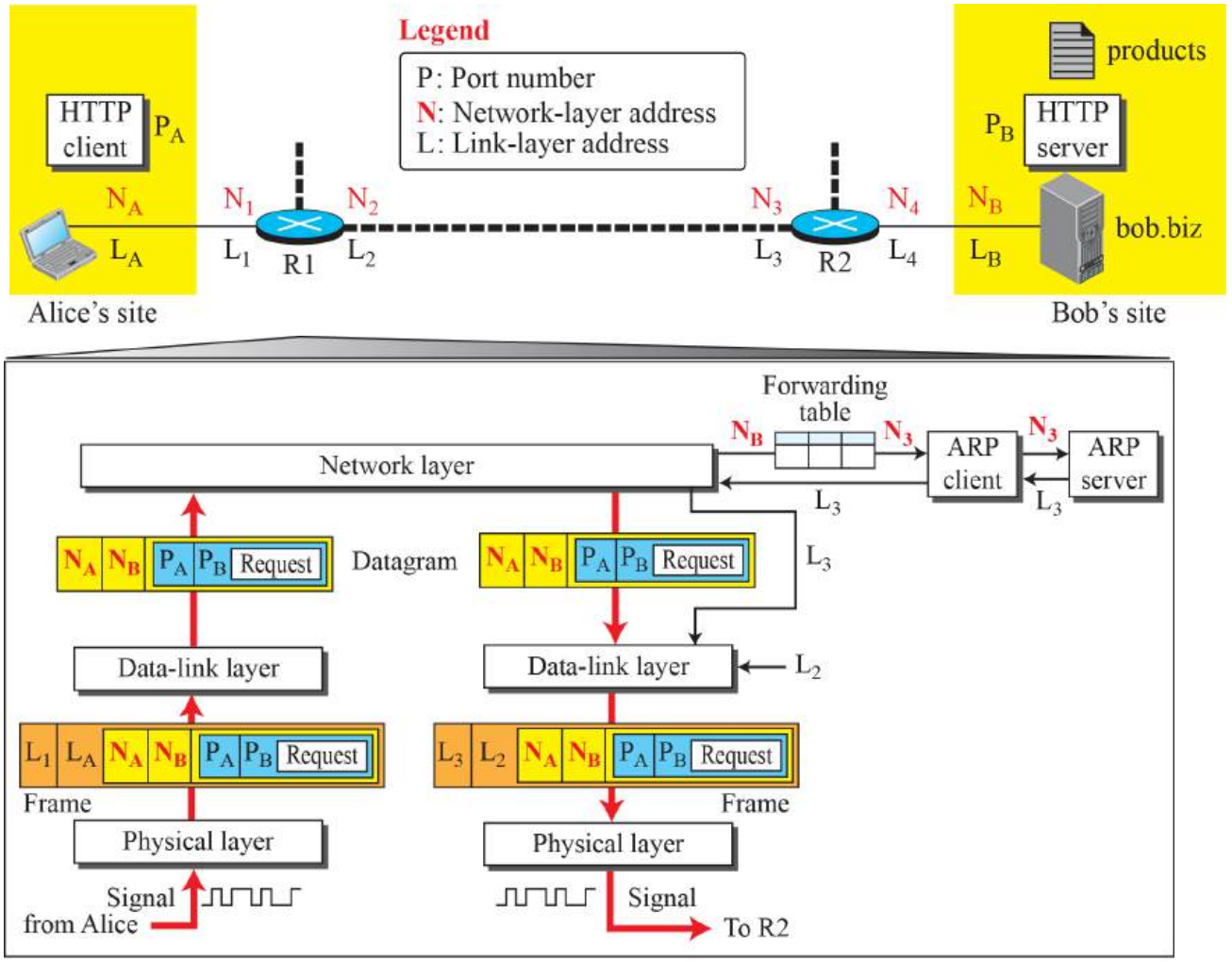
Figure 5.52: Flow of activities at router R1



Flow of packets at Router R1

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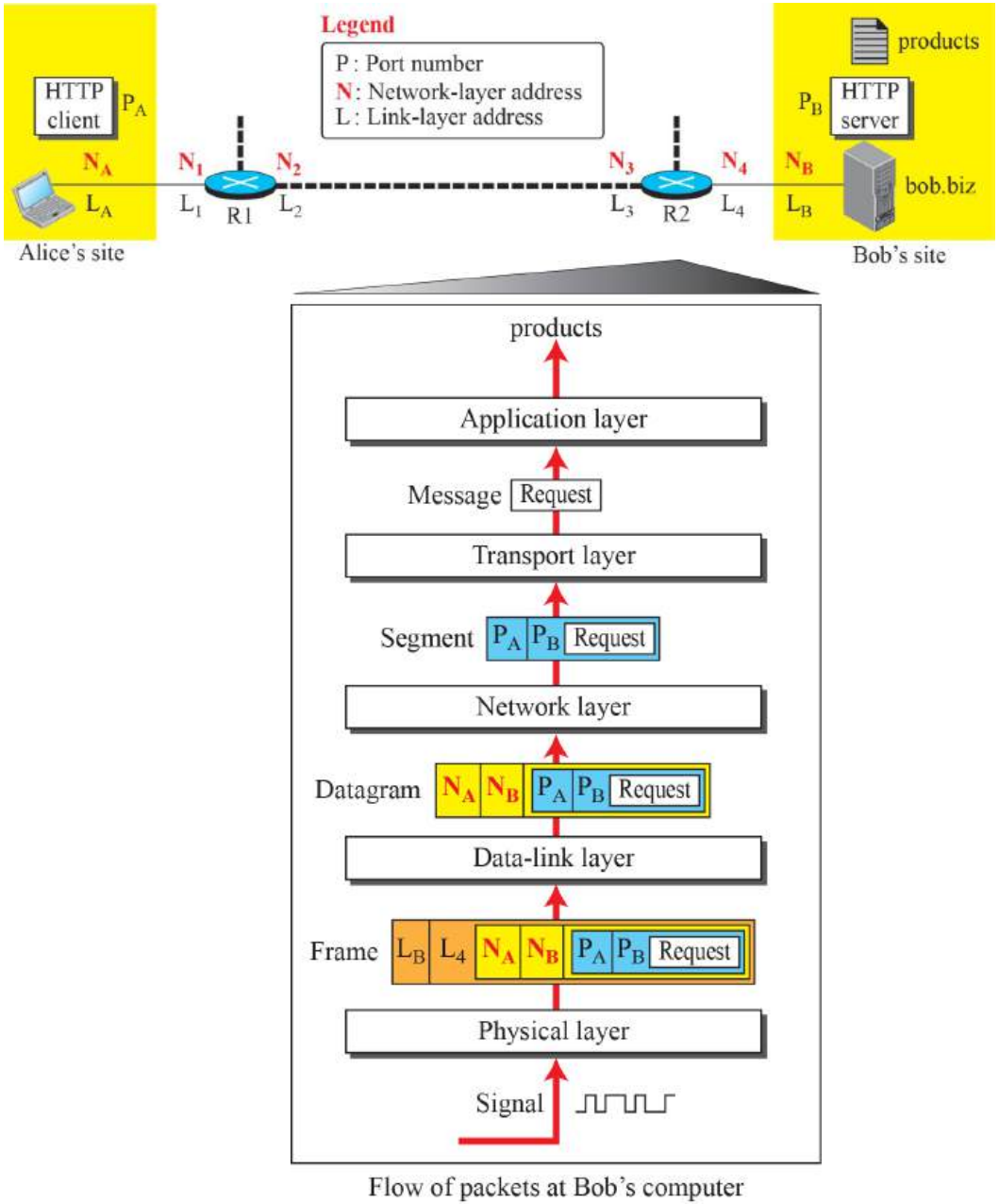
Figure 5.53: Activities at Bob's site



Flow of packets at Bob's computer

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5-5 WIRED LANS: ETHERNET PROTOCOL

TCP/IP accepts any protocol at the data-link and physical layers. These two layers are actually the territory of the local and wide area networks. This means that when we discuss these two layers, we are talking about networks that are using them. We can have wired or wireless networks. We discuss wired networks in this chapter and wireless networks in the next chapter.

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5.5.1 IEEE Project 802

In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers. Project 802 is a way of specifying functions of the physical layer and the data-link layer of major LAN protocols.

- *Logical Link Control (LLC)*
- *Media Access Control (MAC)*

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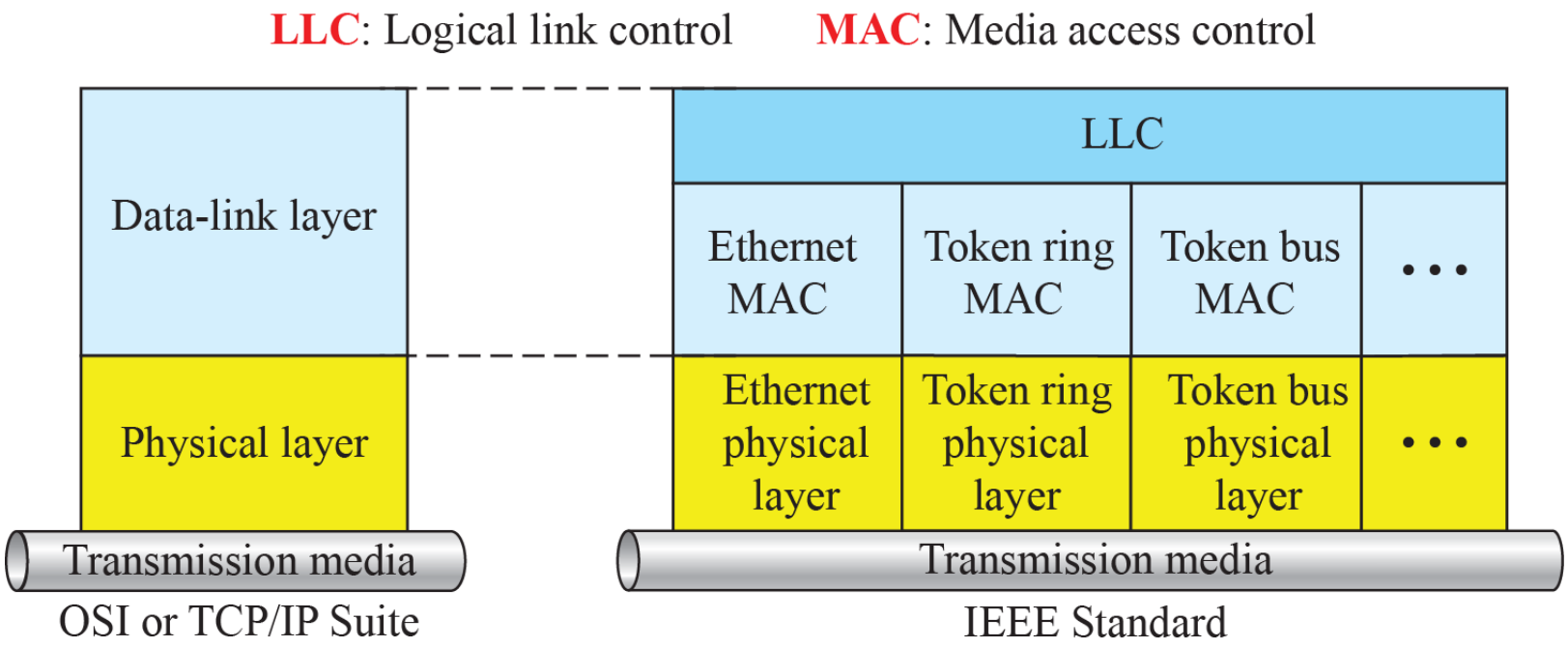
Figure 5.54: IEEE standard for LANs



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5.5.2 Standard Ethernet

We refer to the original Ethernet technology with he data rate of 10 Mbps as the Standard Ethernet. Although most implementations have moved to other technologies in the Ethernet evolution, there are some features of the Standard Ethernet that have not changed during the evolution. We discuss this standard version to pave the way for understanding the other three technologies.

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5.5.2 (continued)

- **Frame Format**
- **Connectionless and Unreliable Service**
- **Frame Length**
- **Addressing**
  - Transmission of Address Bits
  - Unicast, Multicast, and Broadcast Addresses
  - Distinguish between Unicast, Multicast, and Broadcast Transmission
- **Access Method**
- **Efficiency of Standard Ethernet**
- **Implementation**

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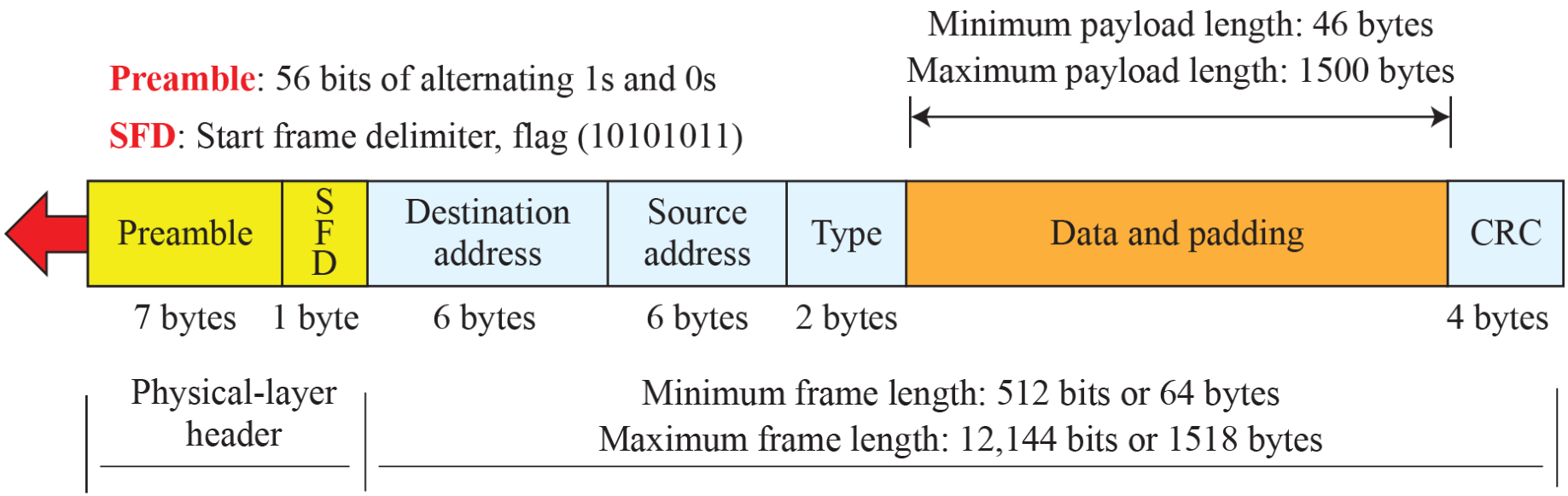
Figure 5.55: Ethernet frame



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Example 5.18

Show how the address 47:20:1B:2E:08:EE is sent out online.

Solution

The address is sent left to right, byte by byte; for each byte, it is sent right to left, bit by bit, as shown below:

Hexadecimal	47	20	1B	2E	08	EE
Binarys	01000111	00100000	00011011	00101110	00001000	11101110
Transmitted ←	11100010	00000100	11011000	01110100	00010000	01110111

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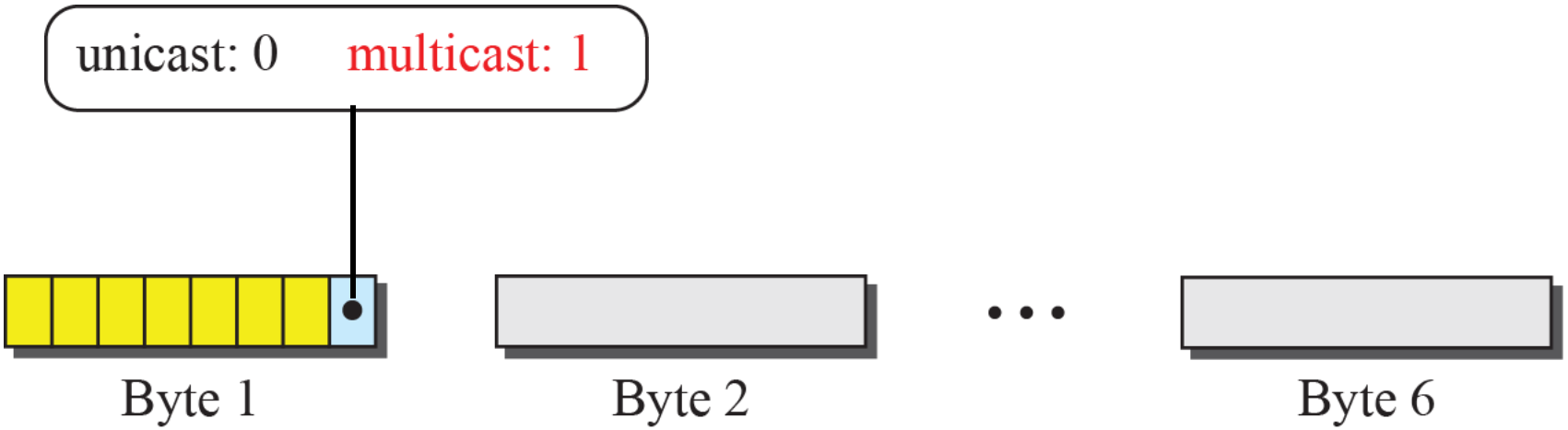
Figure 5.56: Unicast and multicast addresses



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Example 5.19

Define the type of the following destination addresses:

a. 4A:30:10:21:10:1A

b. 47:20:1B:2E:08:EE

c. FF:FF:FF:FF:FF:FF

Solution

To find the type of the address, we need to look at the second hexadecimal digit from the left. If it is even, the address is unicast. If it is odd, the address is multicast. If all digits are Fs, the address is broadcast. Therefore, we have the following:

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Example 5.19 (continued)

a. This is a unicast address because A in binary is 1010 (even).

b. This is a multicast address because 7 in binary is 0111 (odd).

c. This is a broadcast address because all digits are Fs in hexadecimal.

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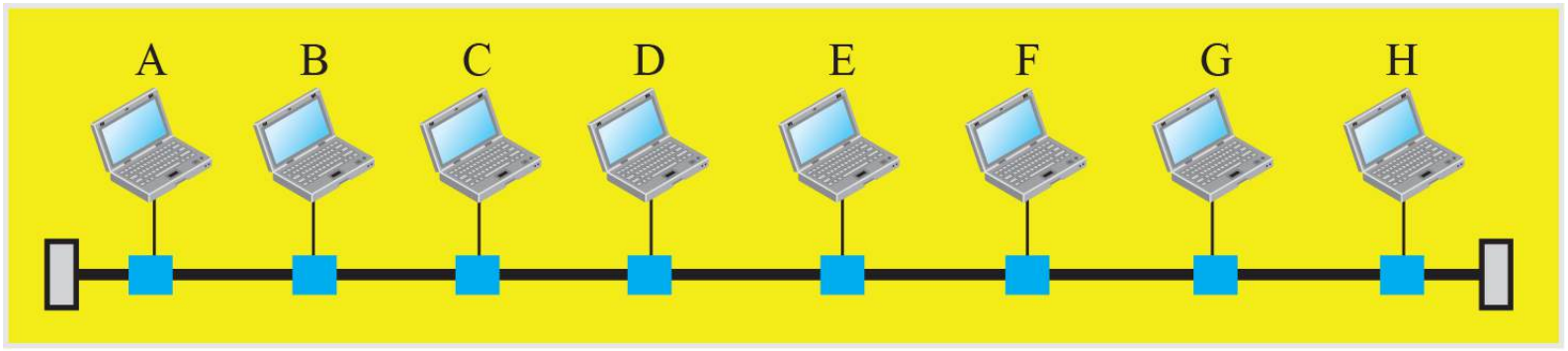
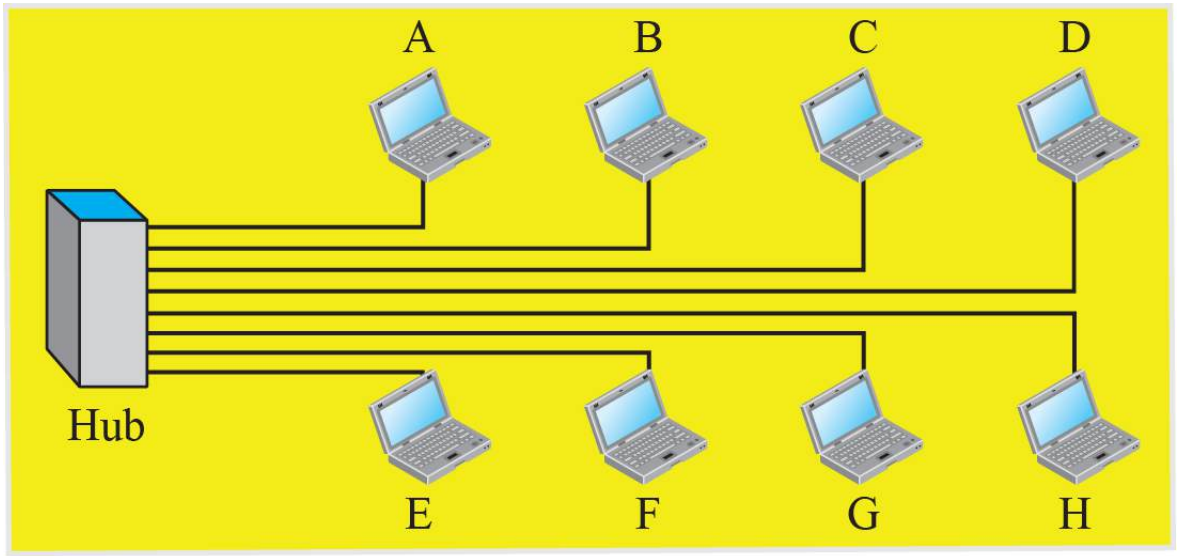
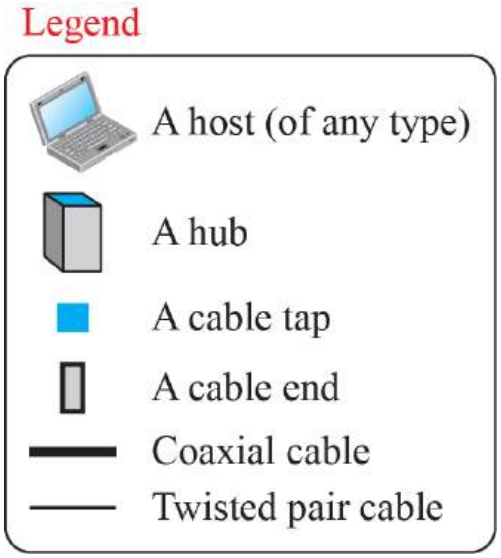
Figure 5.57: Implementation of standard Ethernet



a. A LAN with a bus topology using a coaxial cable





b. A LAN with a star topology using a hub

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Example 5.20

In the Standard Ethernet with the transmission rate of 10 Mbps, we assume that the length of the medium is 2500 m and the size of the frame is 512 bits. The propagation speed of a signal in a cable is normally 2×10^8 m/s.

Propagation delay $= 2500/(2 \times 10^8) = 12.5$ µs	Transmission delay $= 512/(10^7) = 51.2$ µs
$a = 12.5/51.2 = 0.24$	Efficiency = 39%

The example shows that a = 0.24, which means only 0.24 of a frame occupies the whole medium in this case. The efficiency is 39 percent, which is considered moderate; it means that only 61 percent of the time the medium is occupied but not used by a station.

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Table 5.6: Summary of Standard Ethernet implementations

Implementation	*Medium*	*Medium Length*	*Encoding*
10Base5	Thick coax	500 m	Manchester
10Base2	Thin coax	185 m	Manchester
10Base-T	2 UTP	100 m	Manchester
10Base-F	2 Fiber	2000	Manchester

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5.5.3 Fast Ethernet

Fast Ethernet was designed to operate at 100 Mbps. The designers of the Fast Ethernet needed to make it compatible with the Standard Ethernet. The MAC sublayer was left unchanged, which meant the frame format and the maximum and minimum size could also remain unchanged.

- **Access Method**
- **Autonegotiation**
- **Implementation**

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Table 5.7: Summary of Fast Ethernet implementations

Implementation	*Medium*	*Medium Length*	*Wires*	*Encoding*
100Base-TX	STP	100 m	2	4B5B + MLT-3
100Base-FX	Fiber	185 m	2	4B5B + NRZ-I
100Base-T4	UTP	100 m	4	Two 8B/6T

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5.5.4 Gigabit Ethernet

The need for an even higher data rate resulted in the design of the Gigabit Ethernet Protocol (1000 Mbps). The IEEE committee calls the Standard 802.3z. The goals of the Gigabit Ethernet were to upgrade the data rate to 1 Gbps, but keep the address length, the frame format, and the maximum and minimum frame length the same.

- ***MAC Sublayer***
- ***Implementation***

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Table 5.8: Summary of Gigabit Ethernet implementations

Implementation	*Medium*	*Medium Length*	*Wires*	*Encoding*
1000Base-SX	Fiber S-W	550 m	2	8B/10B + NRZ
1000Base-LX	Fiber L-W	5000 m	2	8B/10B + NRZ
1000Base-CX	STP	25 m	2	8B/10B + NRZ
1000Base-T4	UTP	100 m	4	4D-PAM5

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5.5.5 10-Gigabit Ethernet

In recent years, there has been another look into the Ethernet for use in metropolitan areas. The idea is to extend the technology, the data rate, and the coverage distance so that the Ethernet can be used as LAN and MAN (metropolitan area network). The IEEE committee created 10-Gigabit Ethernet and called it Standard 802.3ae.

- ***Implementation***

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Table 5.9: Summary of 10-Gigabit Ethernet implementations

Implementation	*Medium*	*Medium Length*	*Number of wires*	*Encoding*
10GBase-SR	Fiber 850 nm	300 m	2	64B66B
10GBase-LR	Fiber 1310 nm	10 Km	2	64B66B
10GBase-EW	Fiber 1350 nm	40 Km	2	SONET
10GBase-X4	Fiber 1310 nm	300 m to 10 Km	2	8B10B

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5.5.6 Virtual LANs

A station is considered part of a LAN if it physically belongs to that LAN. The criterion of membership is geographic. What happens if we need a virtual connection between two stations belonging to two different physical LANs? We can roughly define a virtual local area network (VLAN) as a local area network configured by software, not by physical wiring.

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5.5.6 (continued)

- **Membership**
  - Interface Numbers
  - MAC Addresses
  - IP Addresses
  - Multicast IP Addresses
  - Combination

- **Configuration**
  - Manual Configuration
  - Automatic Configuration
  - Semiautomatic Configuration

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5.5.6 (continued)

- **Communication between Switches**
  - Table Maintenance
  - Frame Tagging
  - Time-Division Multiplexing (TDM)
- **IEEE Standard**
- **Advantages**
  - Cost and Time Reduction
  - Creating Virtual Work Groups
  - Security

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Figure 5.58: A switch connecting three LANs

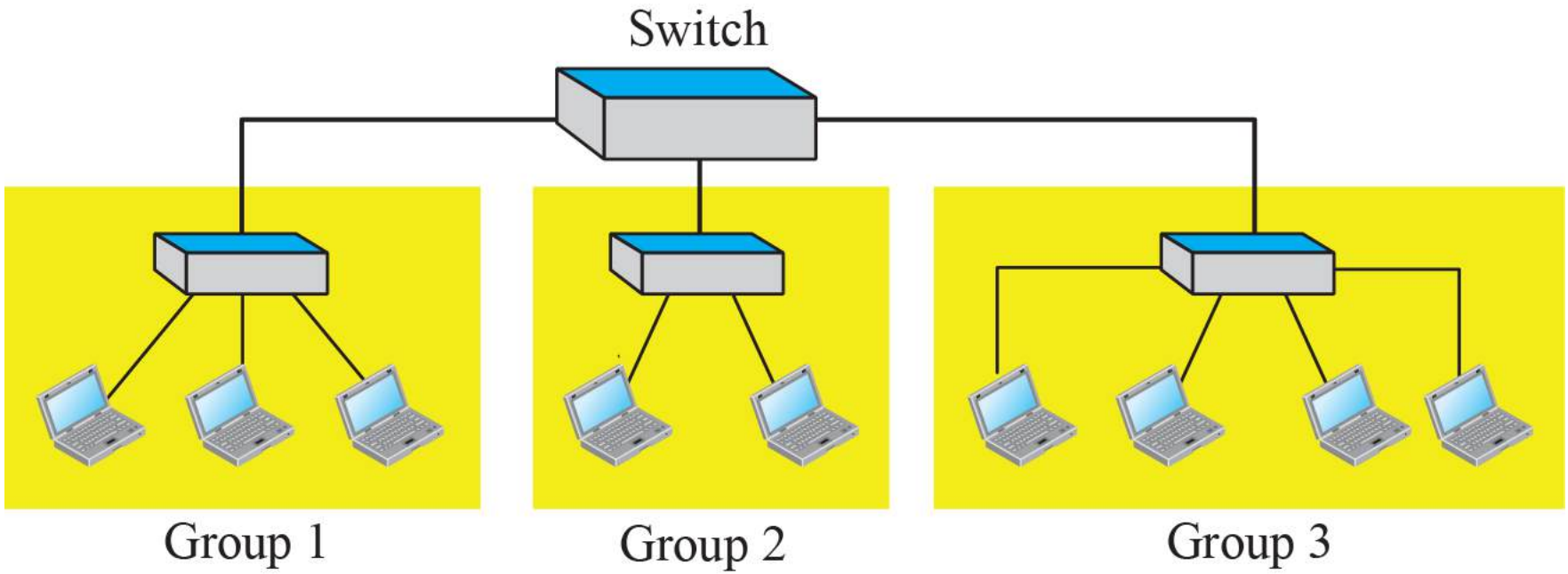

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Figure 5.59: A switch using VLAN software

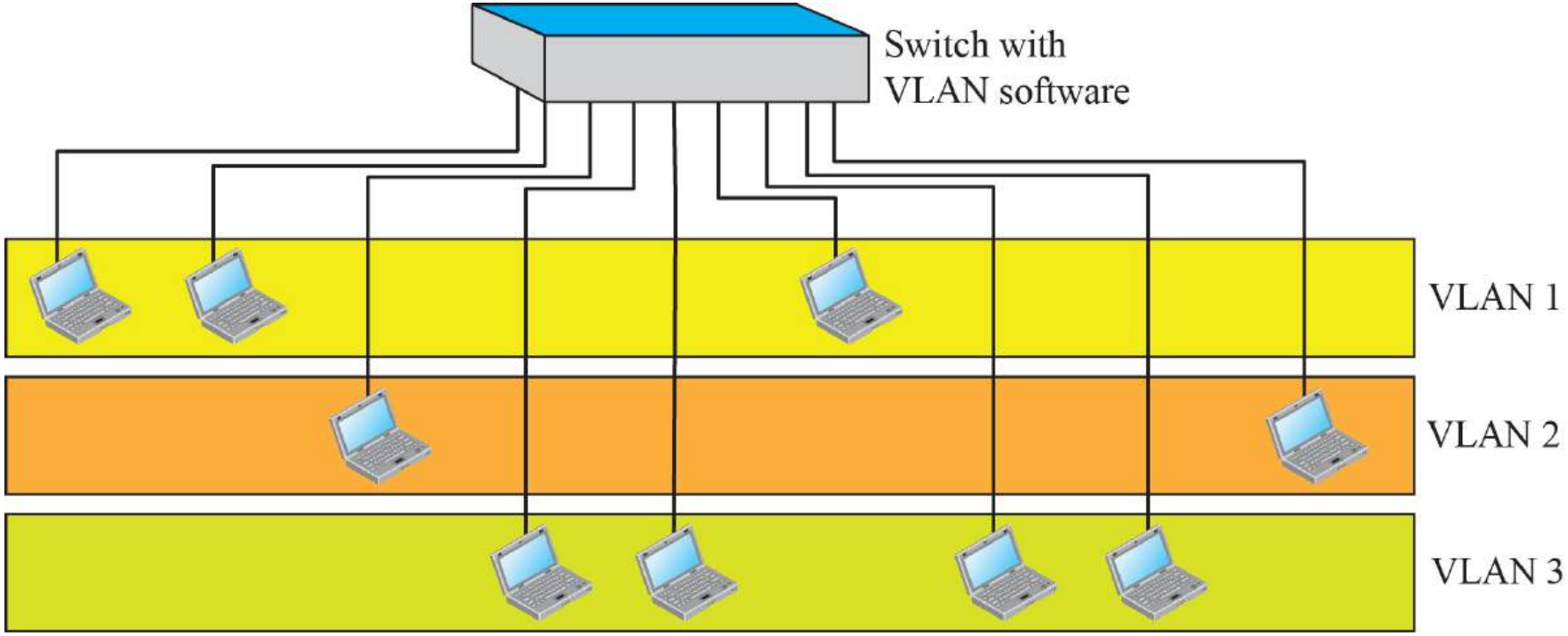

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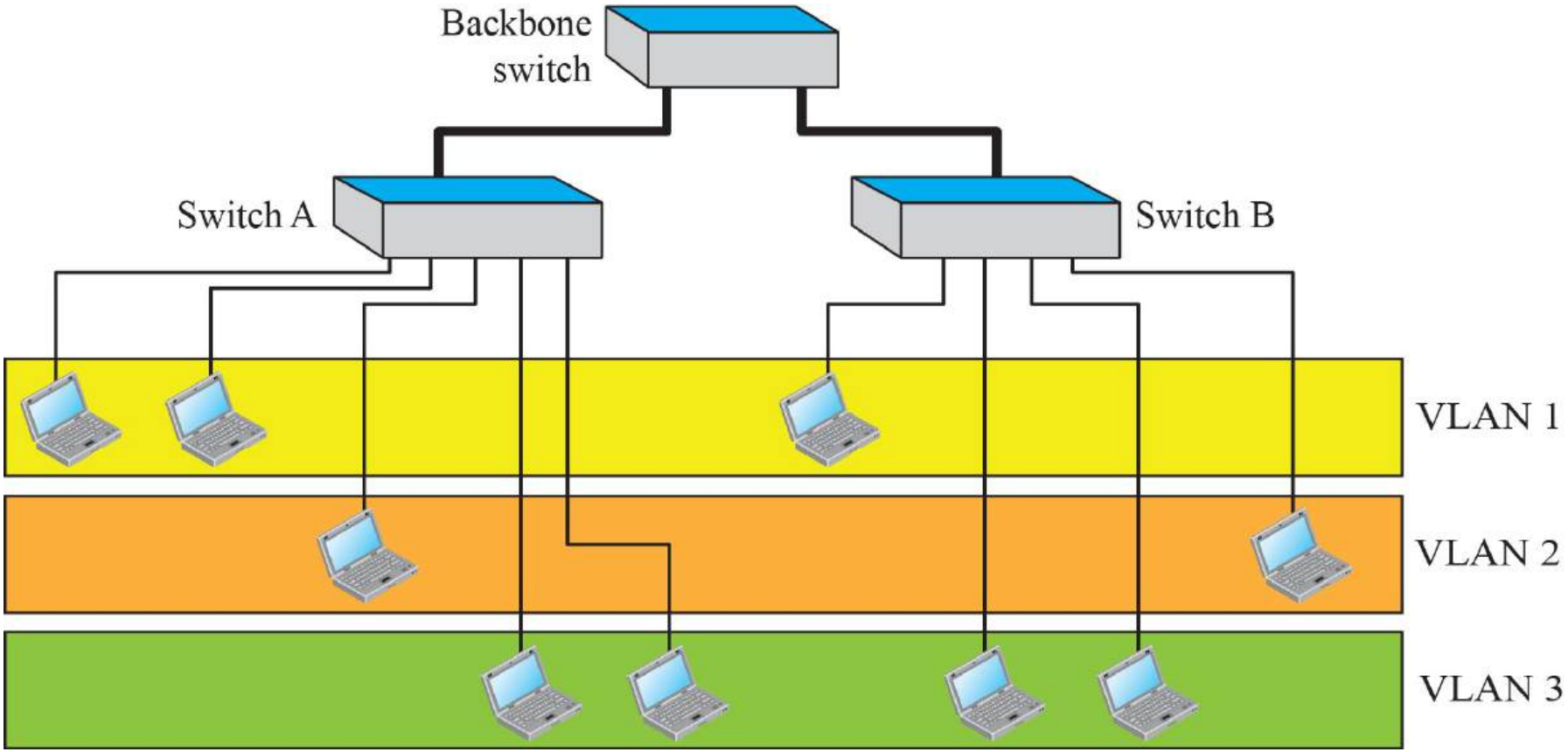
Figure 5.60: Two switches in a backbone using VLAN software



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5-6 OTHER WIRED NETWORKS

As we discussed in Chapter 1, the networks that we encounter in the Internet are either LANs or WANs. However, sometimes the terminology is under dispute. For example, some access networks such as dial-up connection or cable connection are called WANs by some people and MANs by others.

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5.6.1 Point-to-Point Networks

Some point-to-point networks, such as dial-up, DSL, and cable are used to provide internet access from Internet user premises. Since these networks use a dedicated connection between the two devices, they do not use media access control (MAC). The only protocol that is needed is PPP, as we discussed before.

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5.6.1 (continued)

- **Dial-up**
- **Digital Subscriber Line (DSL)**
  - Using Existing Local Loops
- **Cable**
  - Traditional Cable Networks
  - Hybrid Fiber-Coaxial (HFC) Network
  - Cable TV for Data Transfer
  - Sharing
  - CM and CMTS

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Figure 5.61: Dial-up network to provide Internet access

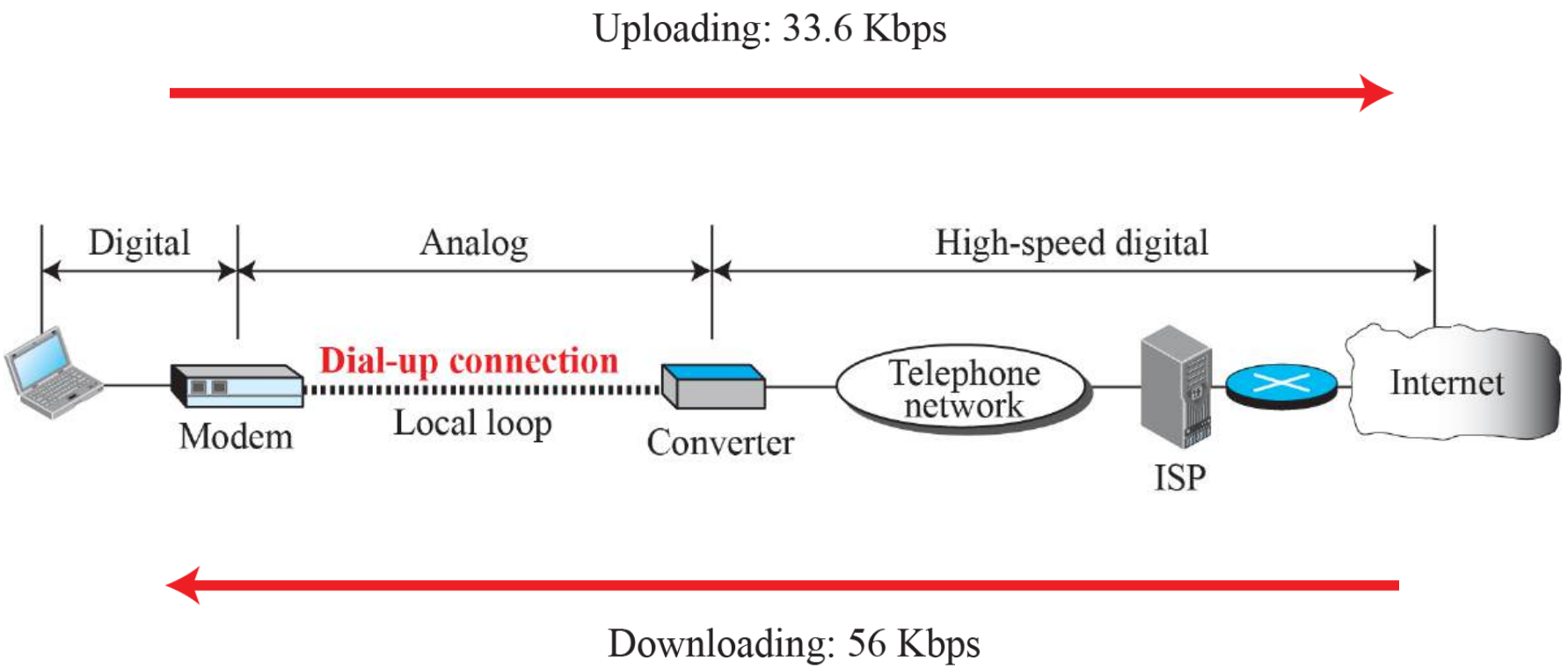

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Figure 5.62: ASDL point-to-point network

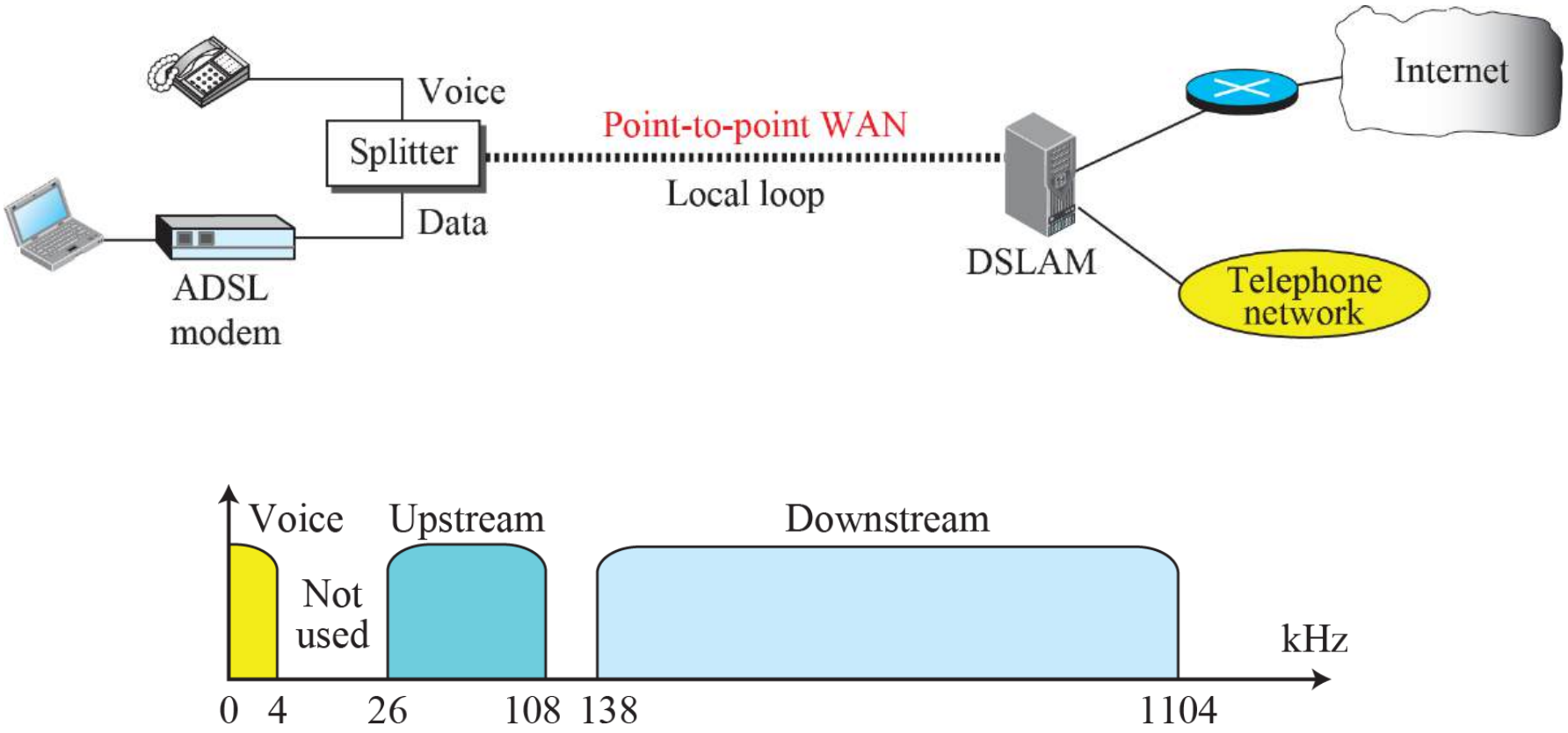

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Figure 5.63: Traditional cable TV network

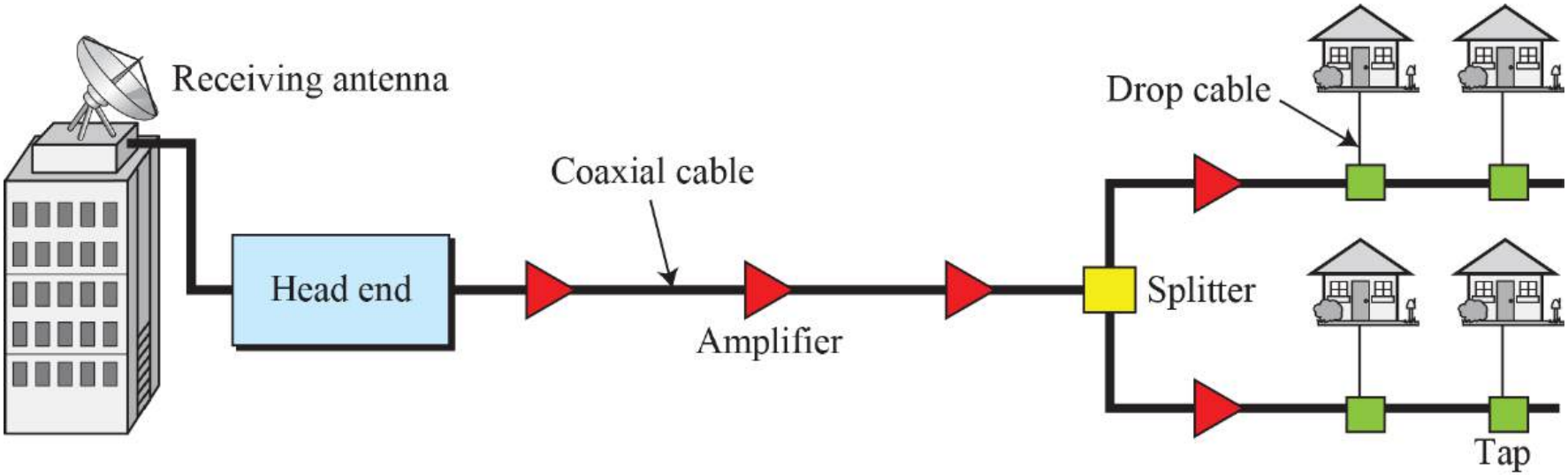

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Figure 5.64: Hybrid Fiber-Coaxial (HFC) Network

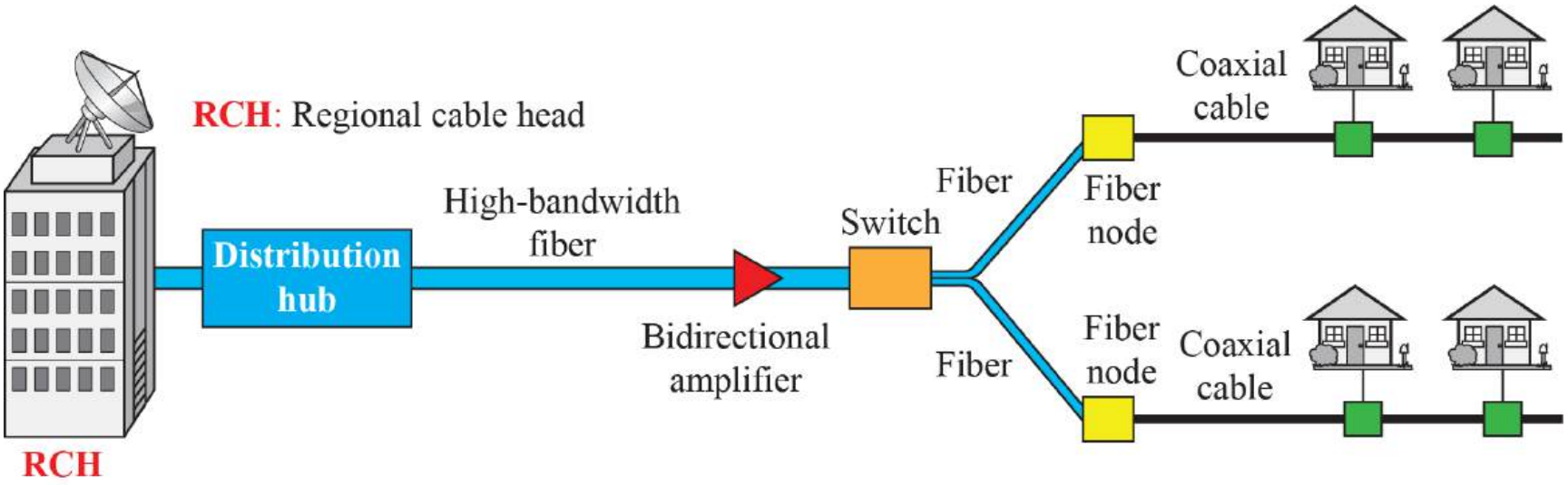

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Figure 5.65: Division of coaxial cable band by CATV

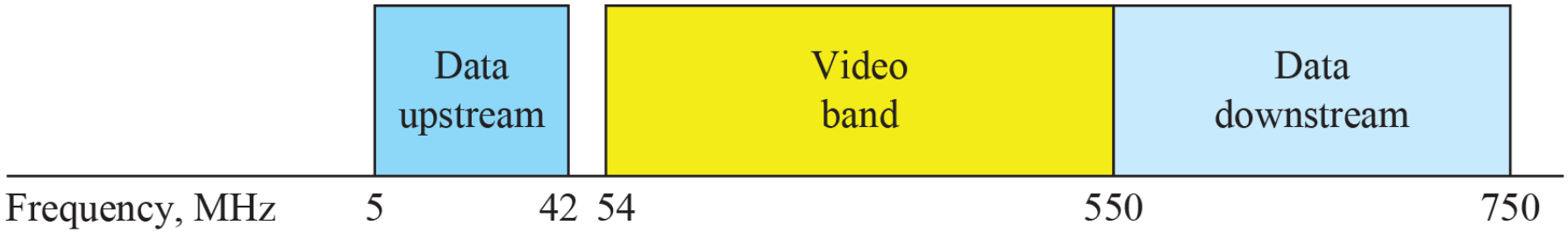

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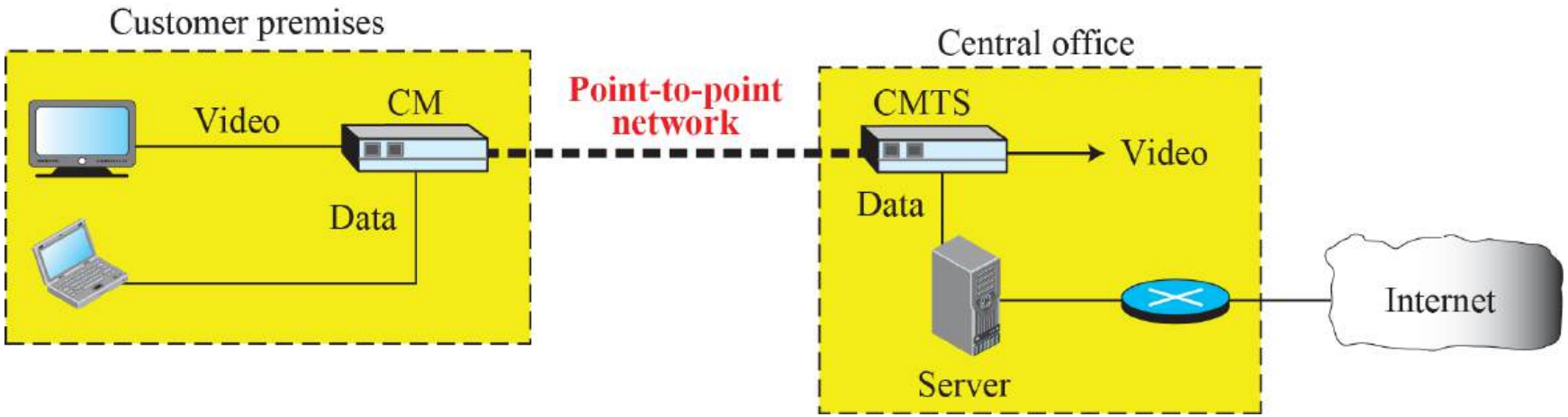
Figure 5.66: Cable modem transmission system (CMTS)



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5.6.2 SONET

In this section, we introduce a high-speed network, SONET, that is used as a transport network to carry loads from other networks. We first discuss SONET as a protocol, and we then show how SONET networks can be constructed from the standards defined in the protocol.

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5.6.2 (continued)

- **Architecture**
  - Signals
  - SONET Devices
- **Connections**
  - Sections
  - Lines
  - Paths
- **SONET Layers**
  - Path Layer
  - Line Layer
  - Section Layer
  - Photonic Layer

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5.6.2 (continued)

- **SONET Frames**
  - Frame, Byte, and Bit Transmission
  - STS-1 Frame Format
- **STS Multiplexing**
  - Add/Drop Multiplexer
- **SONET Networks**
  - Linear Network
  - Ring Networks
  - Mesh Networks
- **Virtual Tributaries**

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Table 5.10: SONET rates

STS	*OC*	*Rate (Mbps)*	*STS*	*OC*	*Rate (Mbps)*
STS-1	OC-1	51.840	STS-24	OC-24	1244.160
STS-3	OC-3	155.520	STS-36	OC-36	1866.230
STS-9	OC-9	466.560	STS-48	OC-48	2488.320
STS-12	OC-12	622.080	STS-96	OC-96	4976.640
STS-18	OC-18	933.120	STS-192	OC-192	9953.280

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Figure 5.67: A simple network using SONET equipment

ADM: Add/drop multiplexer

STS MUX: Synchronous transport signal multiplexer

STS DEMUX: Synchronous transport signal demultiplexer

R: Regenerator

T: Terminal

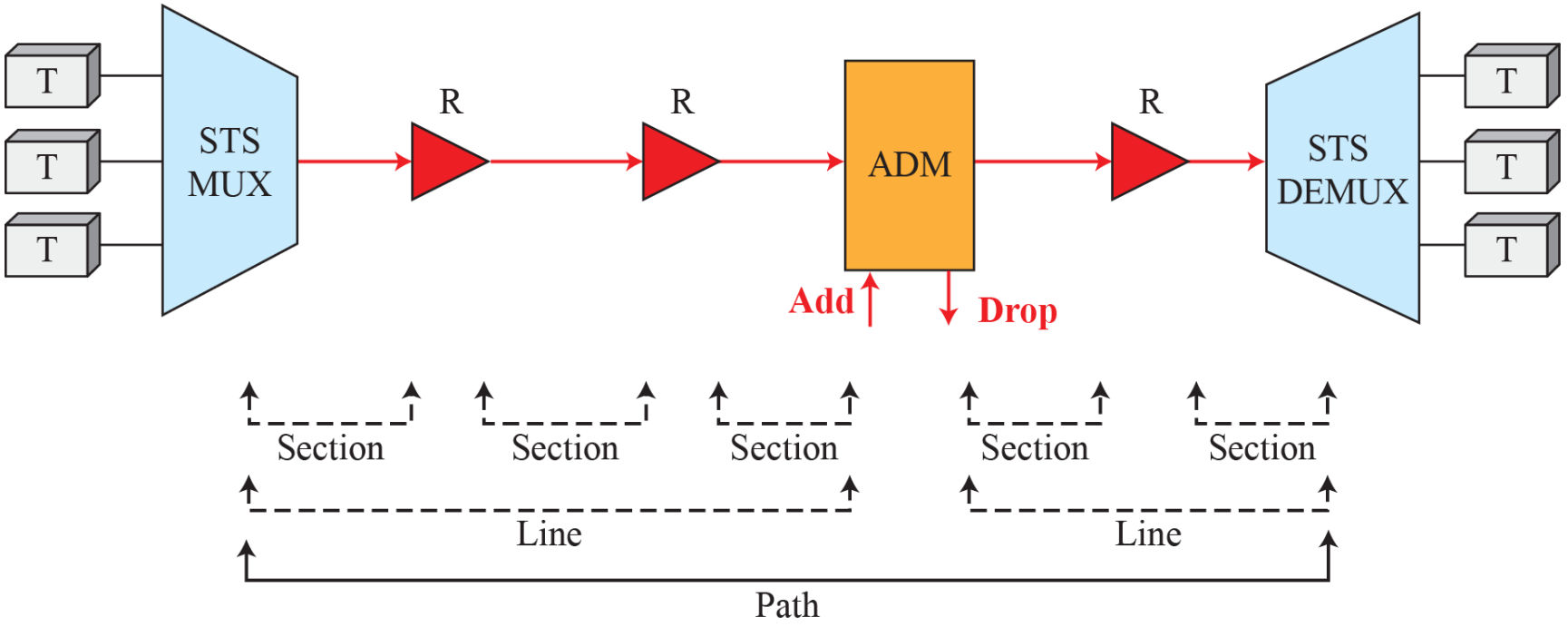

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Figure 5.68: ***SONET layers compared with OSI or the Internet layers***

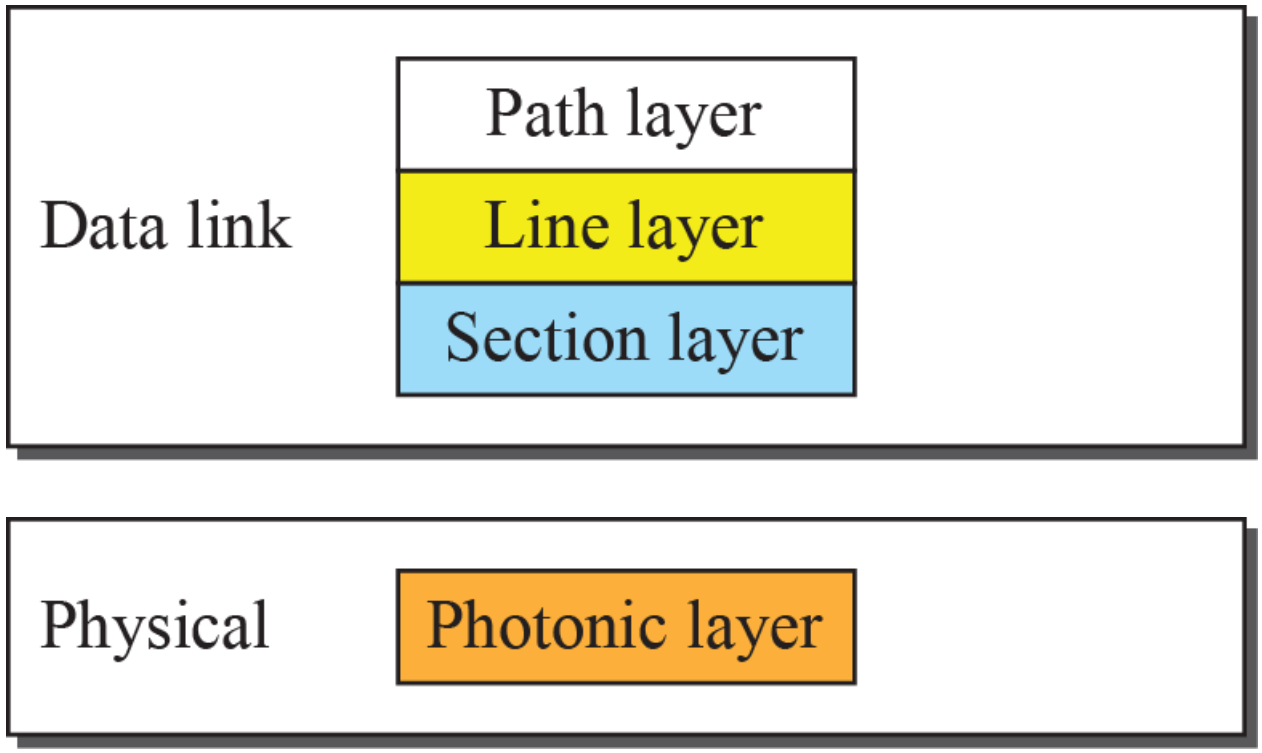

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Figure 5.69: An STS-1 and an STS-n frame

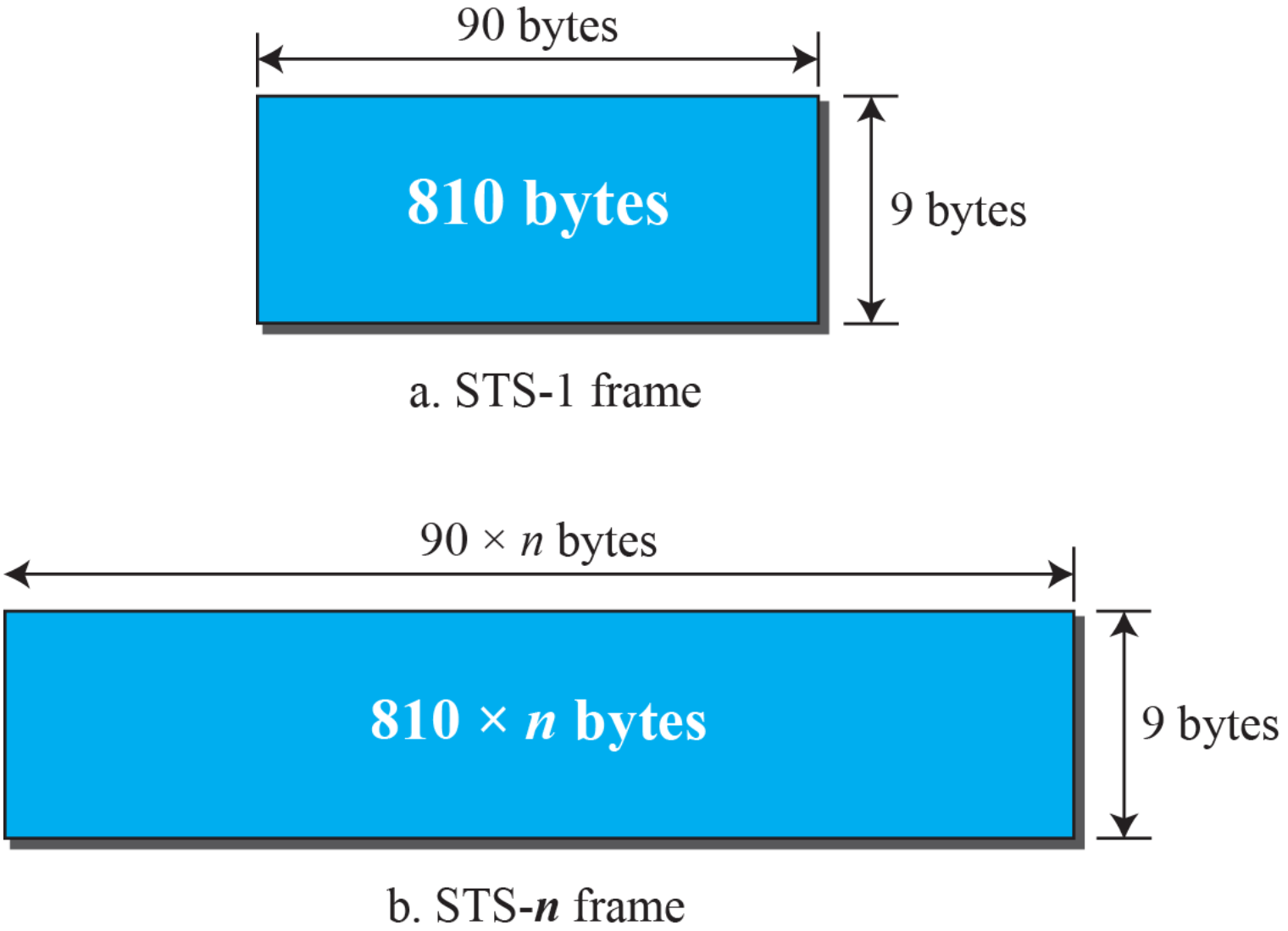

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Figure 5.70: STS-1 frames in transition

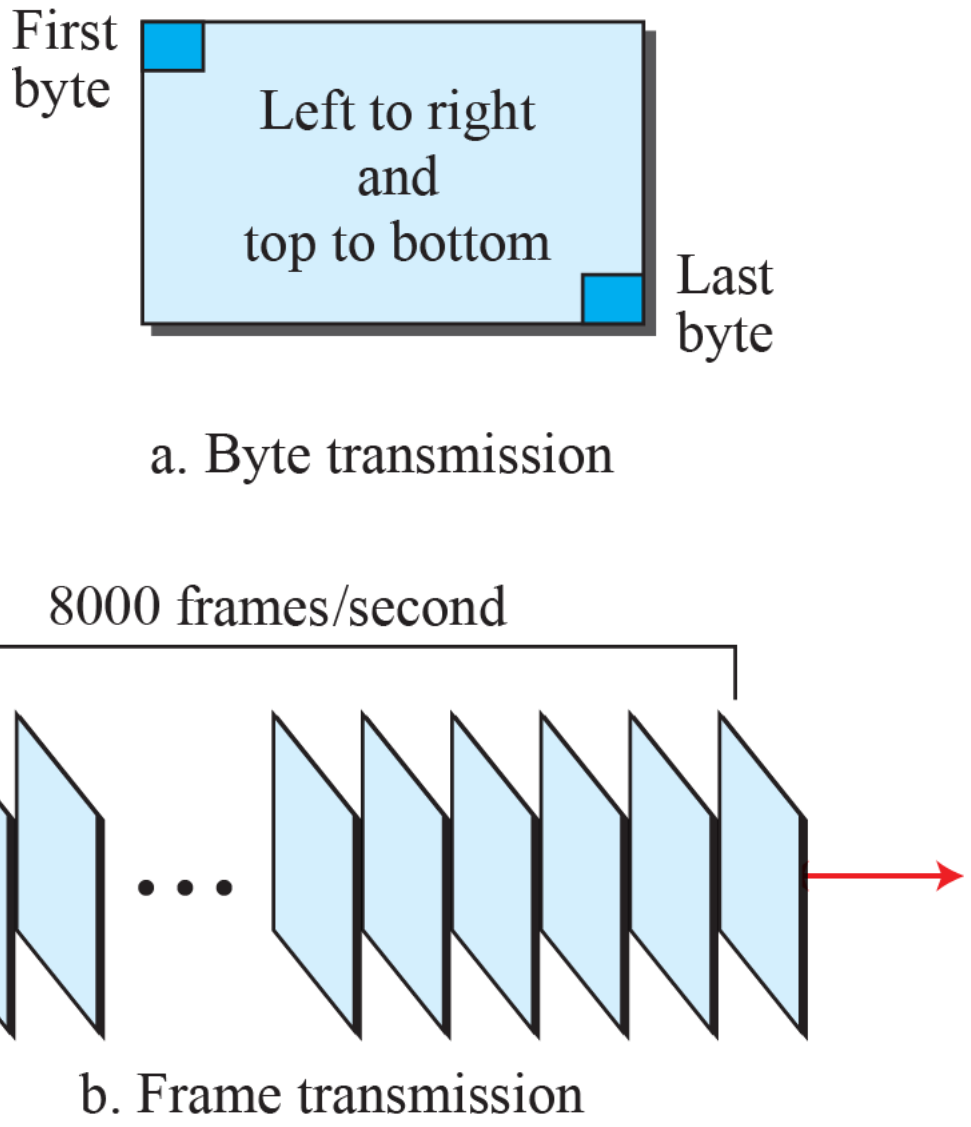

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Figure 5.71: ***STS-1 frame overheads***

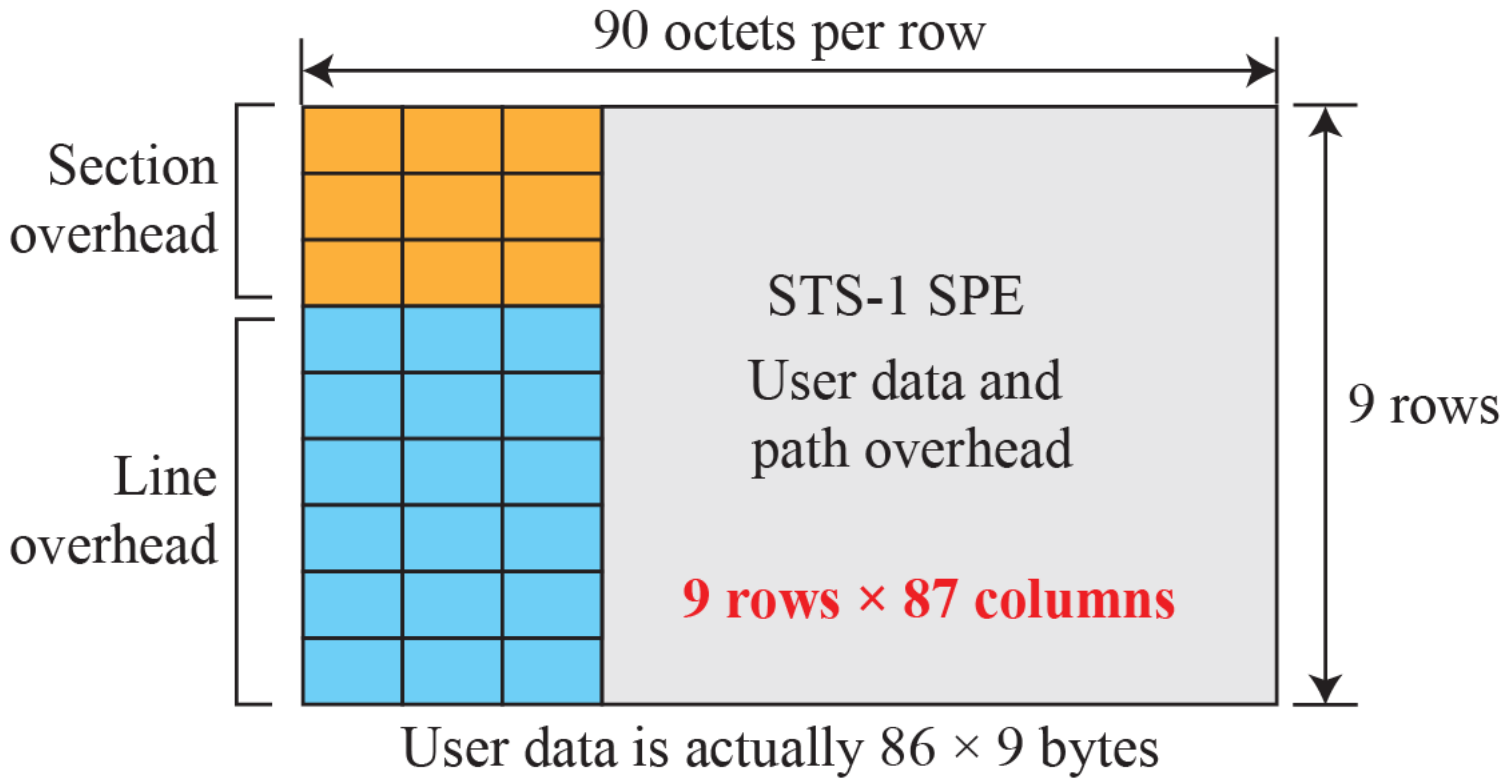

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Figure 5.72: A linear SONET network

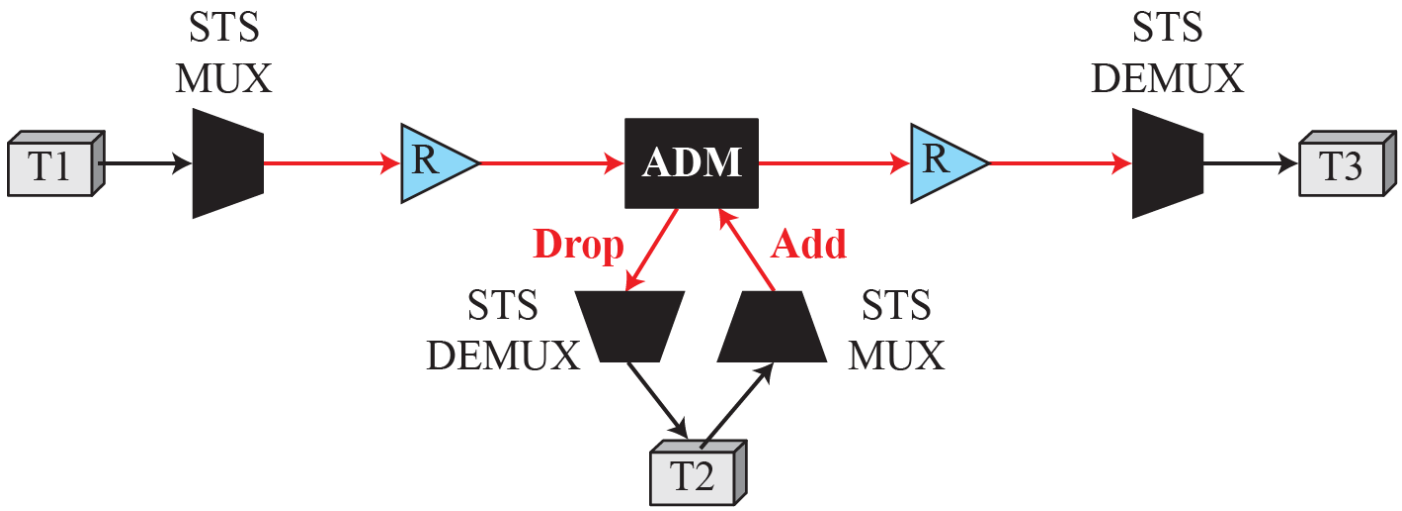

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Figure 5.73: A unidirectional path switching ring

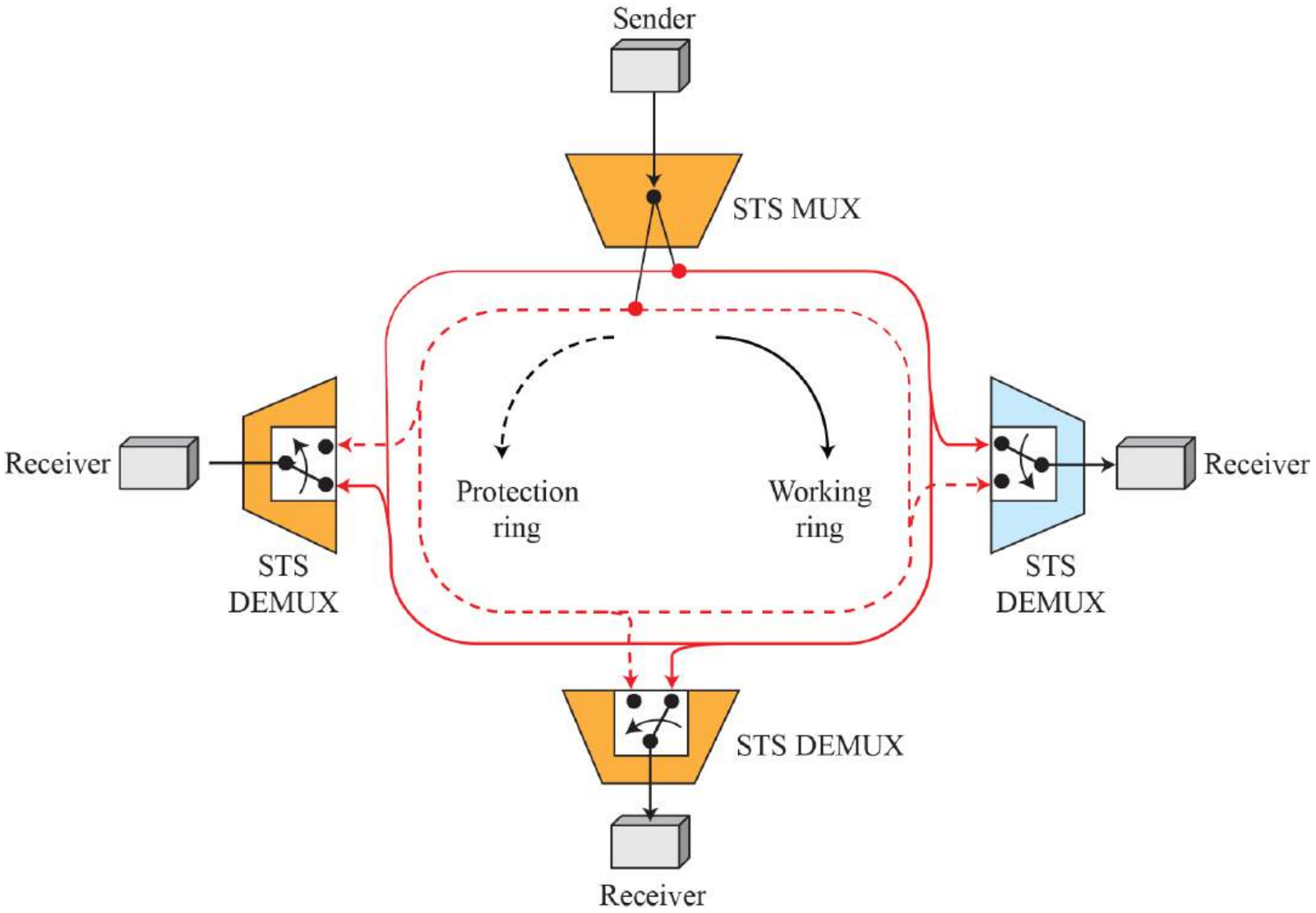

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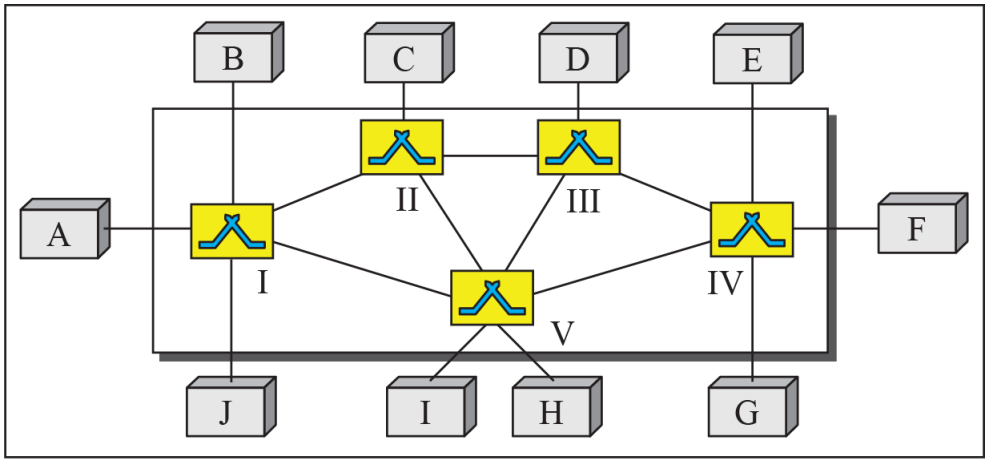
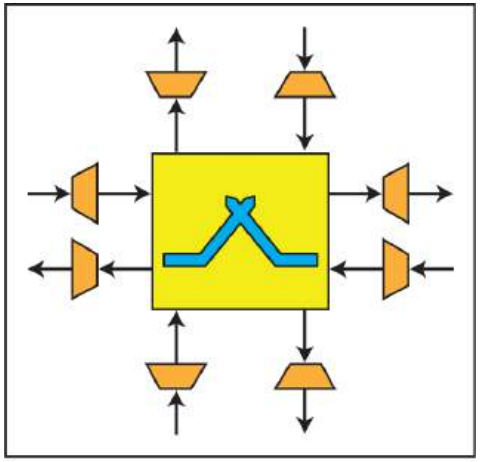
Figure 5.74: A mesh SONET network



a. SONET mesh network



b. Cross-connect switch

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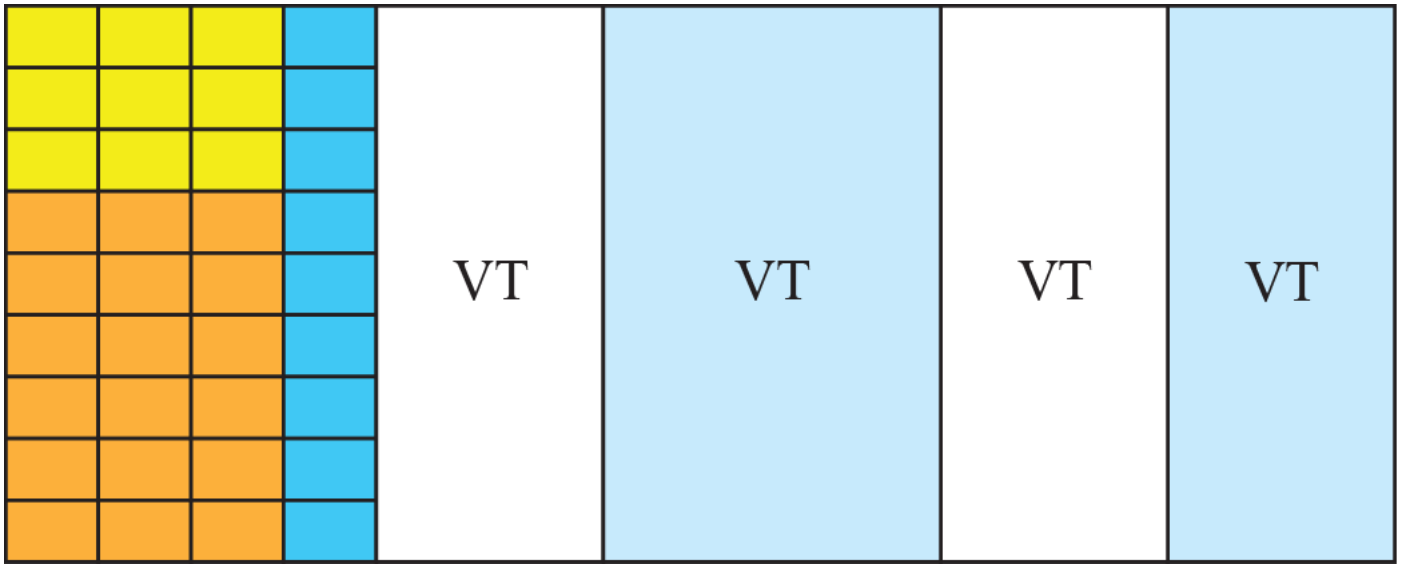
Figure 5.75: Virtual tributaries



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5.6.3 Switched Network: ATM

Asynchronous Transfer Mode (ATM) is a switched wide area network based on the cell relay protocol designed by the ATM forum and adopted by the ITU-T.

- ***Architecture***
  - *Virtual Connection*
  - *Connection Establishment and Release*
  - *Switching*
  - *ATM Layers*
  - *Congestion Control and Quality of Service*

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Figure 5.76: ATM multiplexing

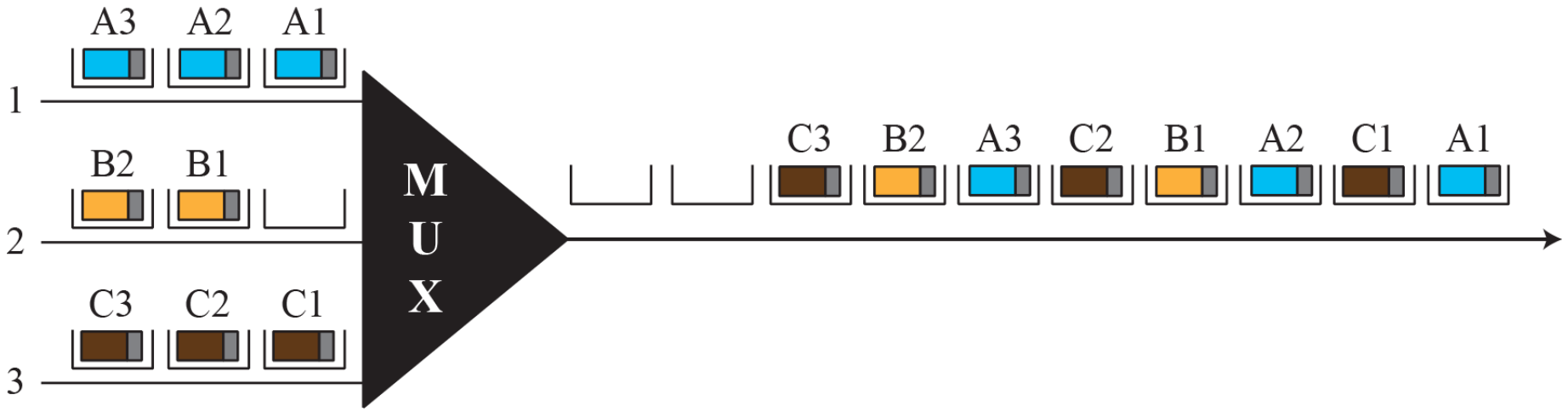

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Figure 5.77: Architecture of an ATM network

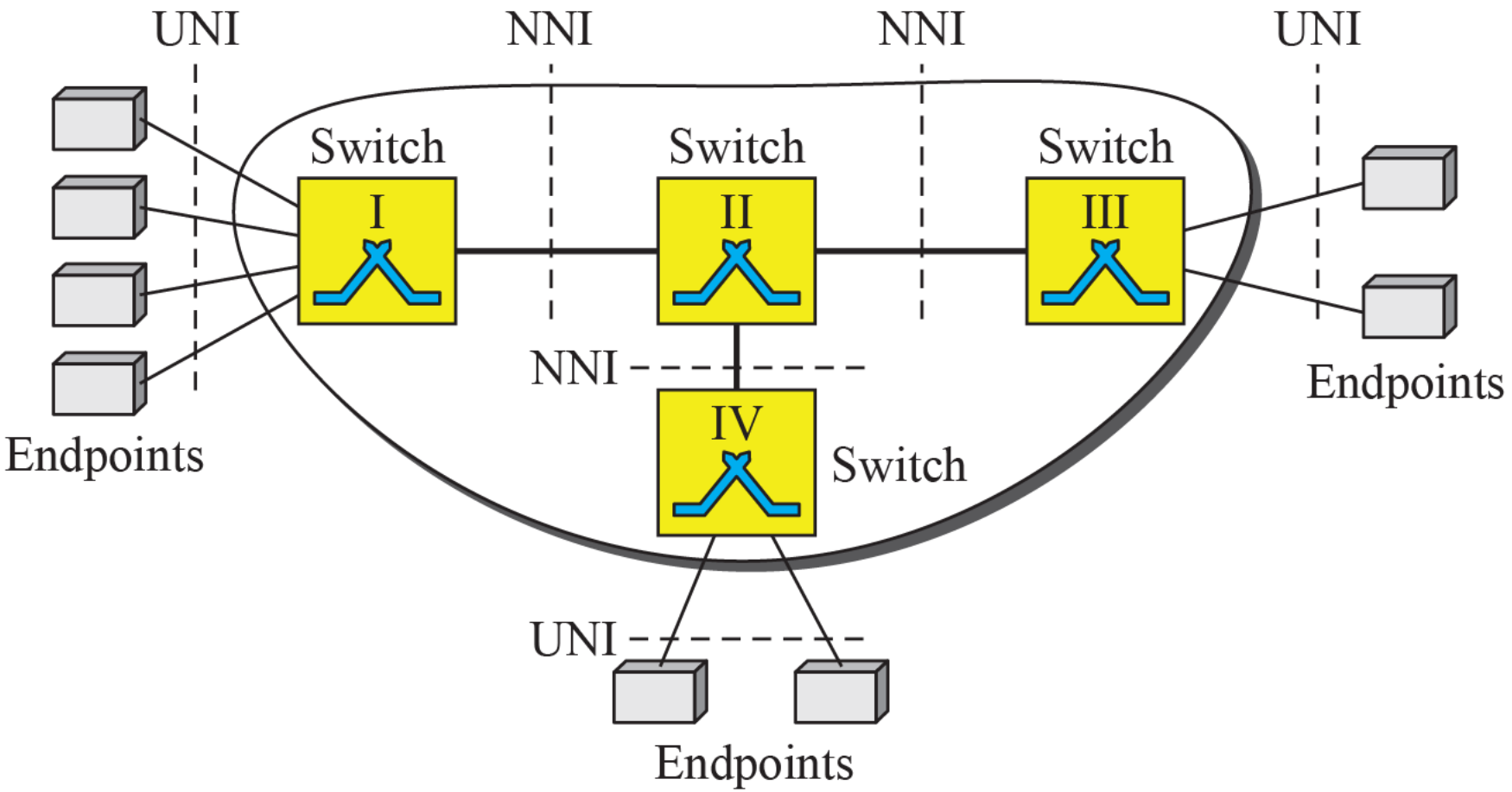

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Figure 5.78: TP, VPs, and VCs

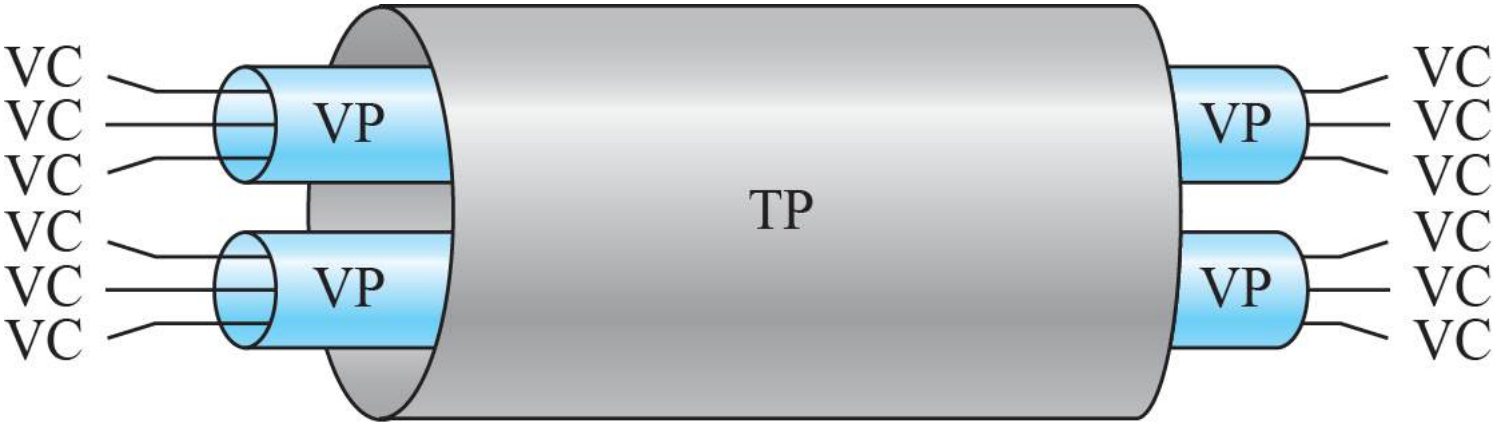

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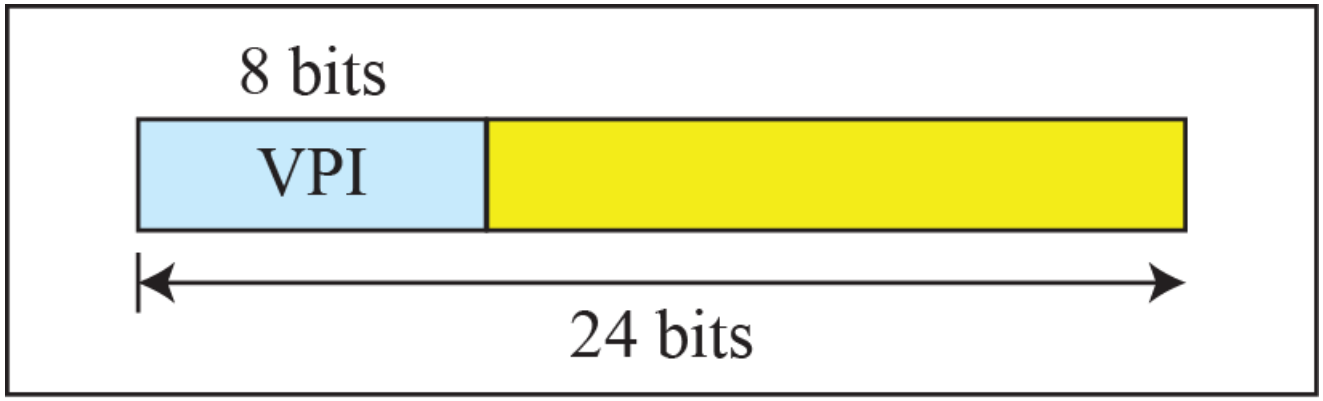
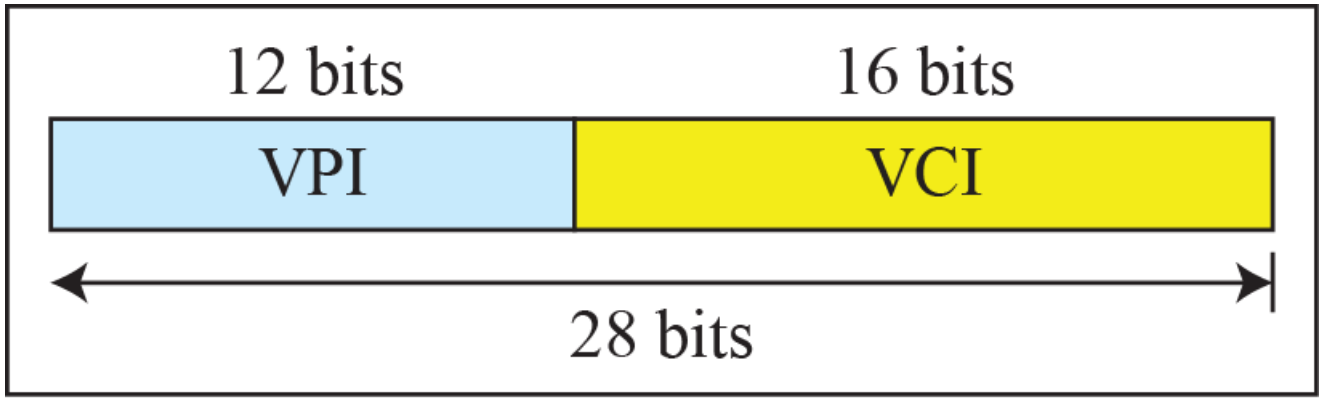
Figure 5.79: Virtual connection identifiers in UNIs and NNIs



a. VPI and VCI in a UNI



b. VPI and VCI in an NNI

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Figure 5.80: ***An ATM cell***

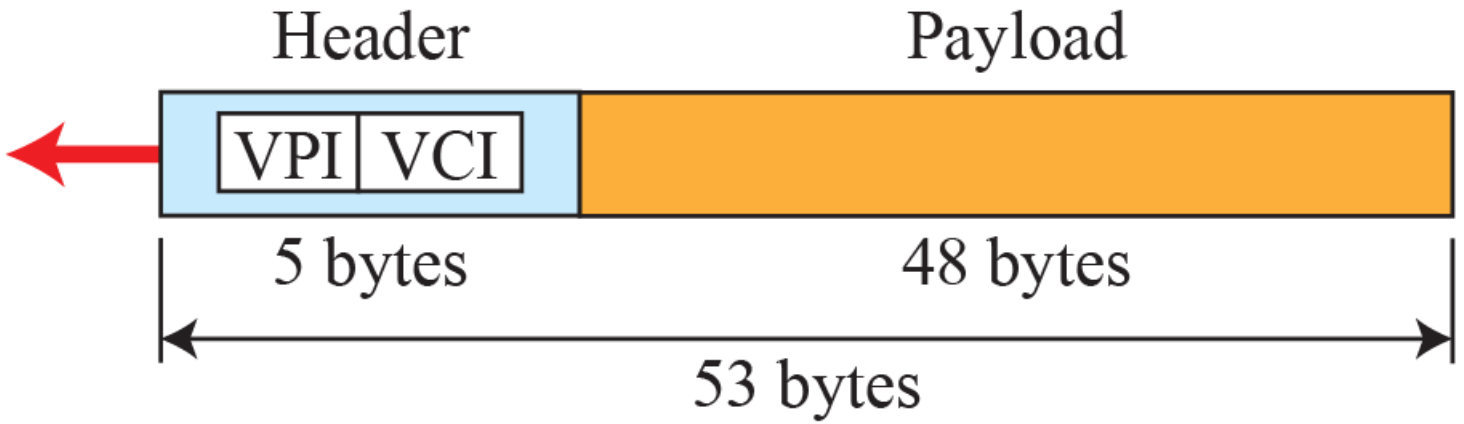

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Figure 5.81: *Routing with a switch*

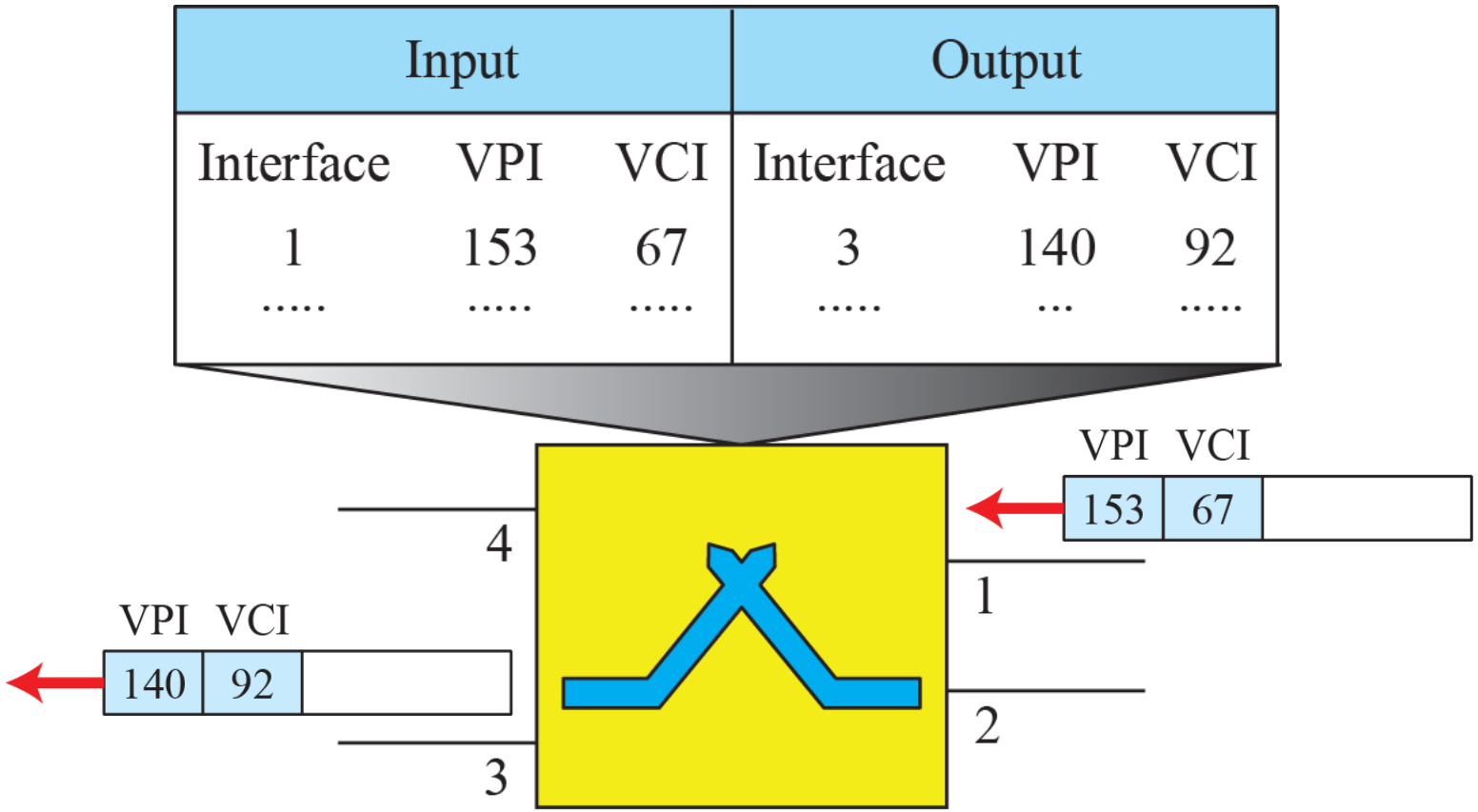

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Figure 5.82: ATM layers

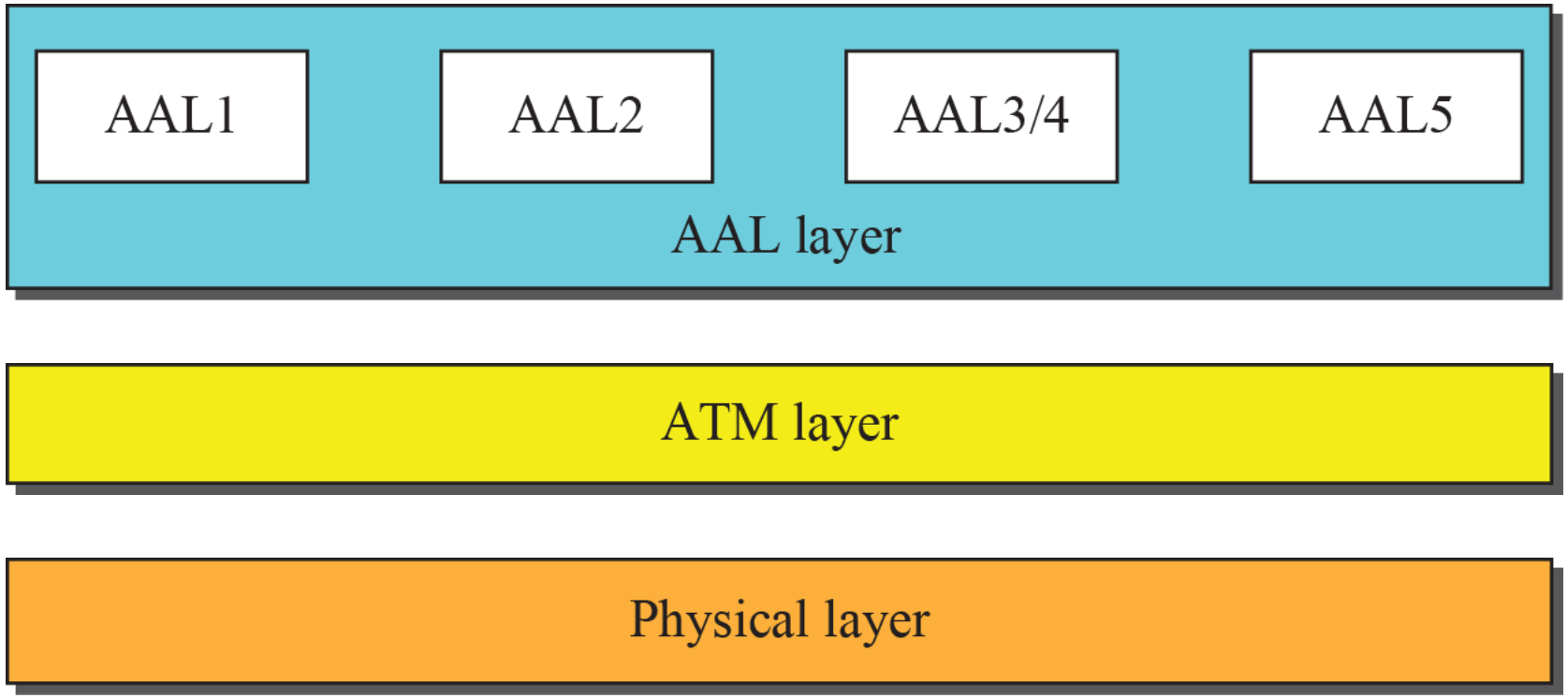

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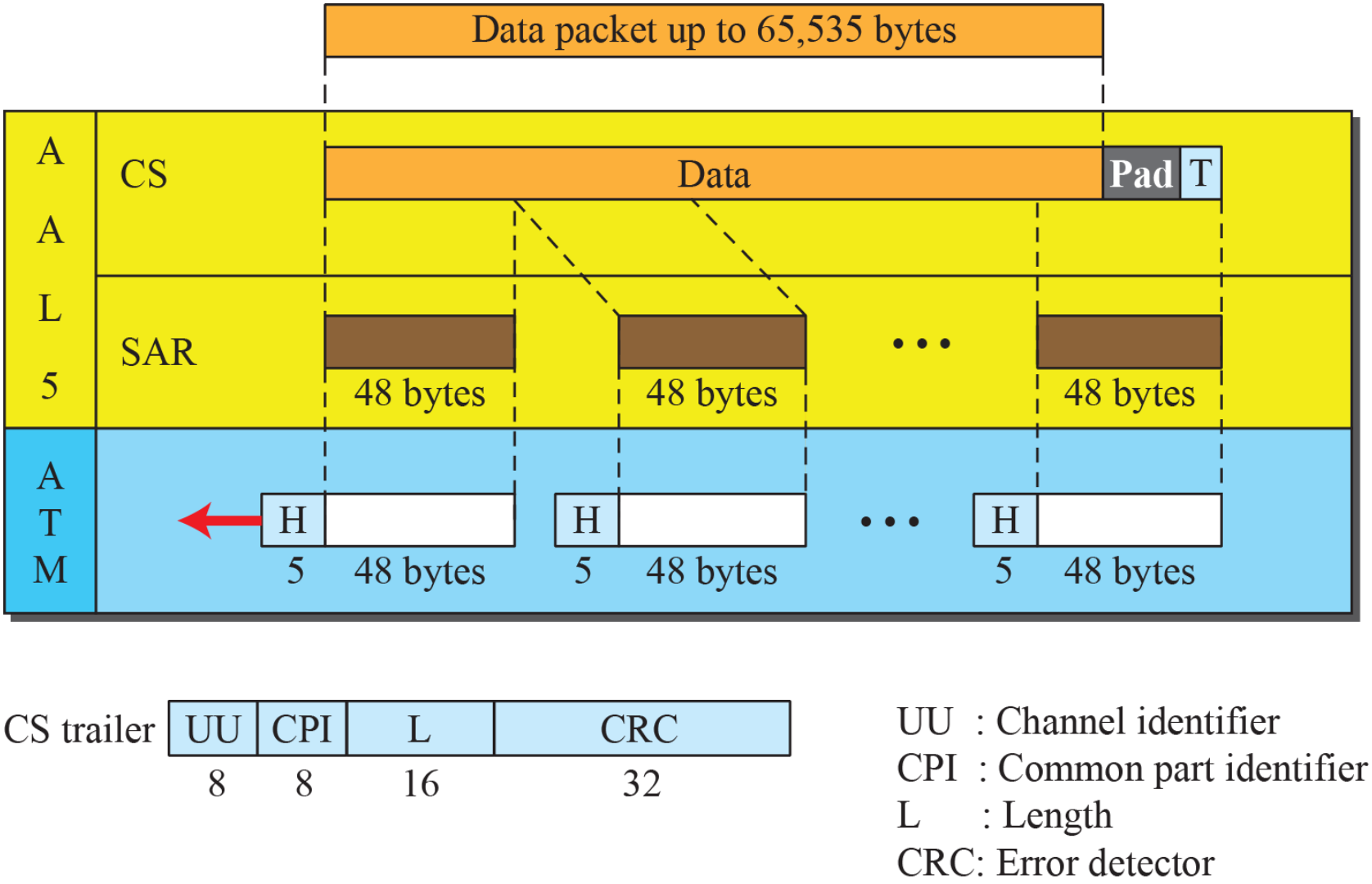
Figure 5.83: AAL5



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5-7 CONNECTING DEVICES

Hosts and networks do not normally operate in isolation. We use connecting devices to connect hosts together to make a network or to connect networks together to make an internet. Connecting devices can operate in different layers of the Internet model. We discuss three kinds of connecting devices: repeaters link-layer switches, and routers.

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5.7.1 Repeater or Hubs

A repeater is a device that operates only in the physical layer. Signals that carry information within a network can travel a fixed distance before attenuation endangers the integrity of the data. A repeater receives a signal and, before it becomes too weak or corrupted, regenerates and retimes the original bit pattern.

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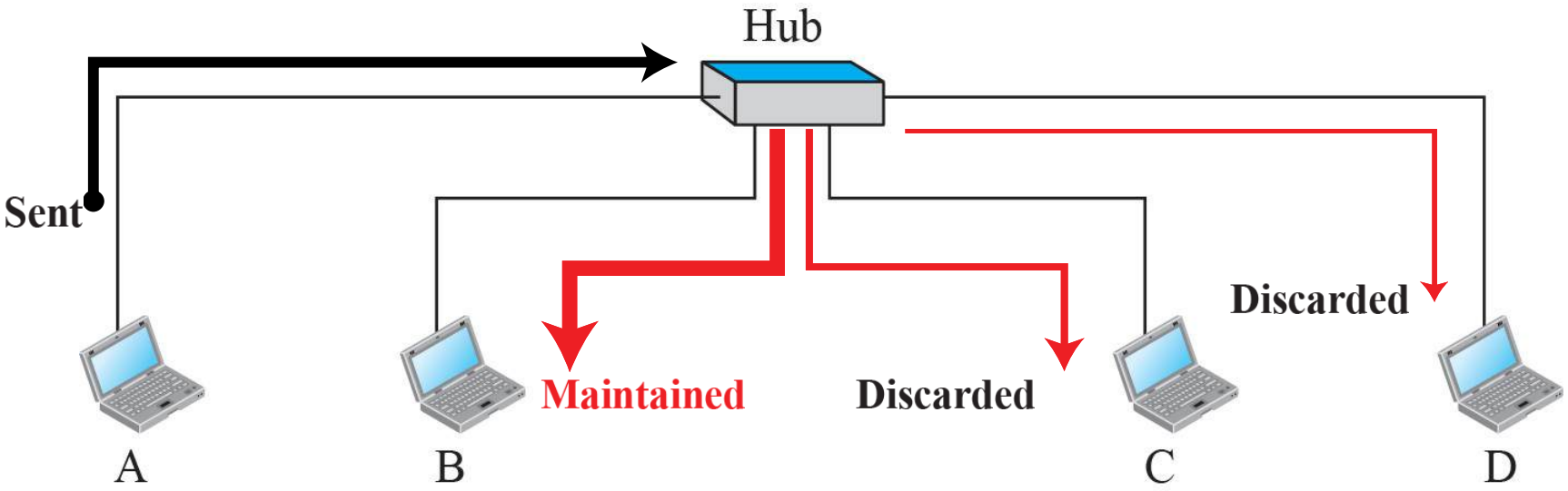
Figure 5.84: Repeater or hub



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5.7.2 Link-Layer Switches

A link-layer switch operates in both the physical and the data-link layers. As a physical layer device, it regenerates the signal it receives. As a link-layer device, the link-layer switch can check the MAC addresses (source and destination) contained in the frame.

- **Filtering**
- **Transparent Switches**
  - Forwarding
  - Learning

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Figure 5.85: Link-Layer Switch

Switching table

Address	Port
71:2B:13:45:61:41	1
71:2B:13:45:61:42	2
64:2B:13:45:61:12	3
64:2B:13:45:61:13	4

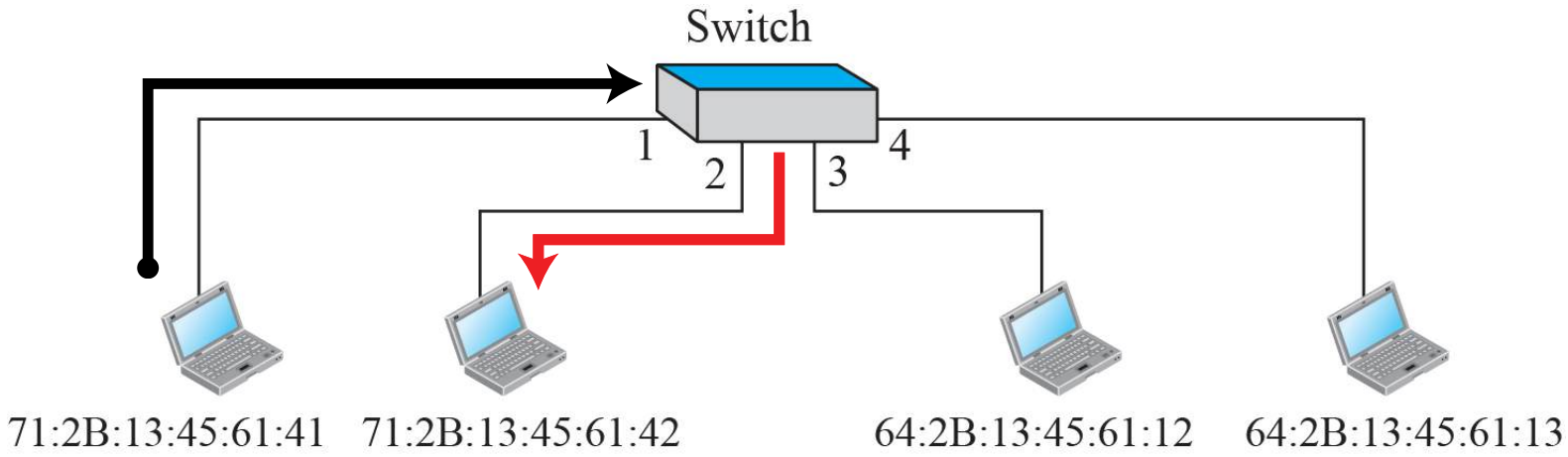

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Figure 5.86: Learning switch

Address	Port

a. Original

Address	Port
71:2B:13:45:61:41	1

b. After A sends a frame to D

Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4

c. After D sends a frame to B

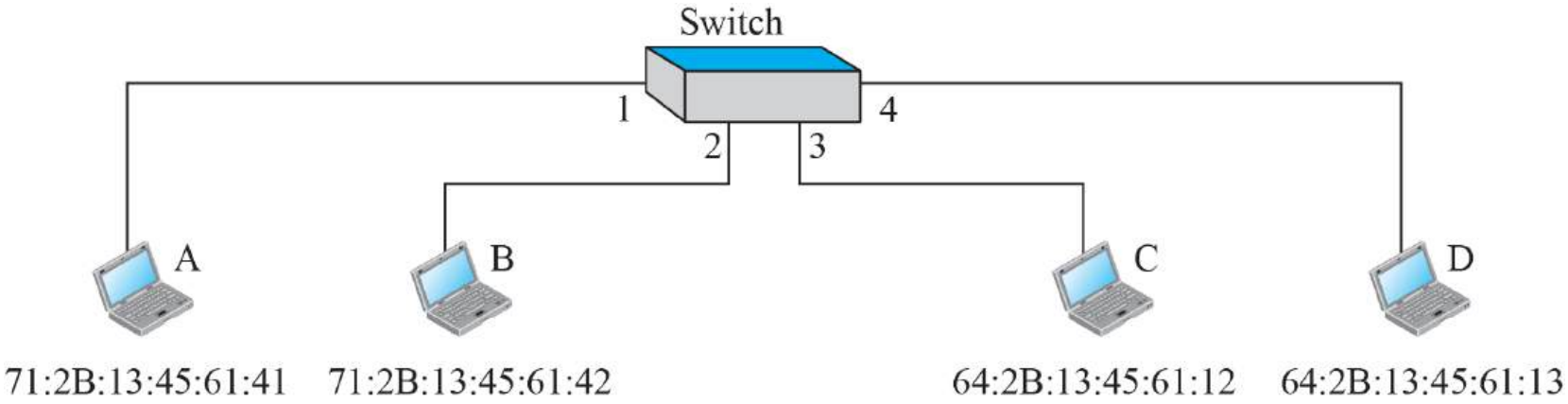
Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4
71:2B:13:45:61:42	2

d. After B sends a frame to A

Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4
71:2B:13:45:61:42	2
64:2B:13:45:61:12	3

e. After C sends a frame to D

Gradual building of Table



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5.7.3 Routers

We discussed routers in Chapter 4. In this chapter, we mention routers to compare them with a two-layer switch and a hub. A router is a three-layer device; it operates in the physical, data-link, and network layers.

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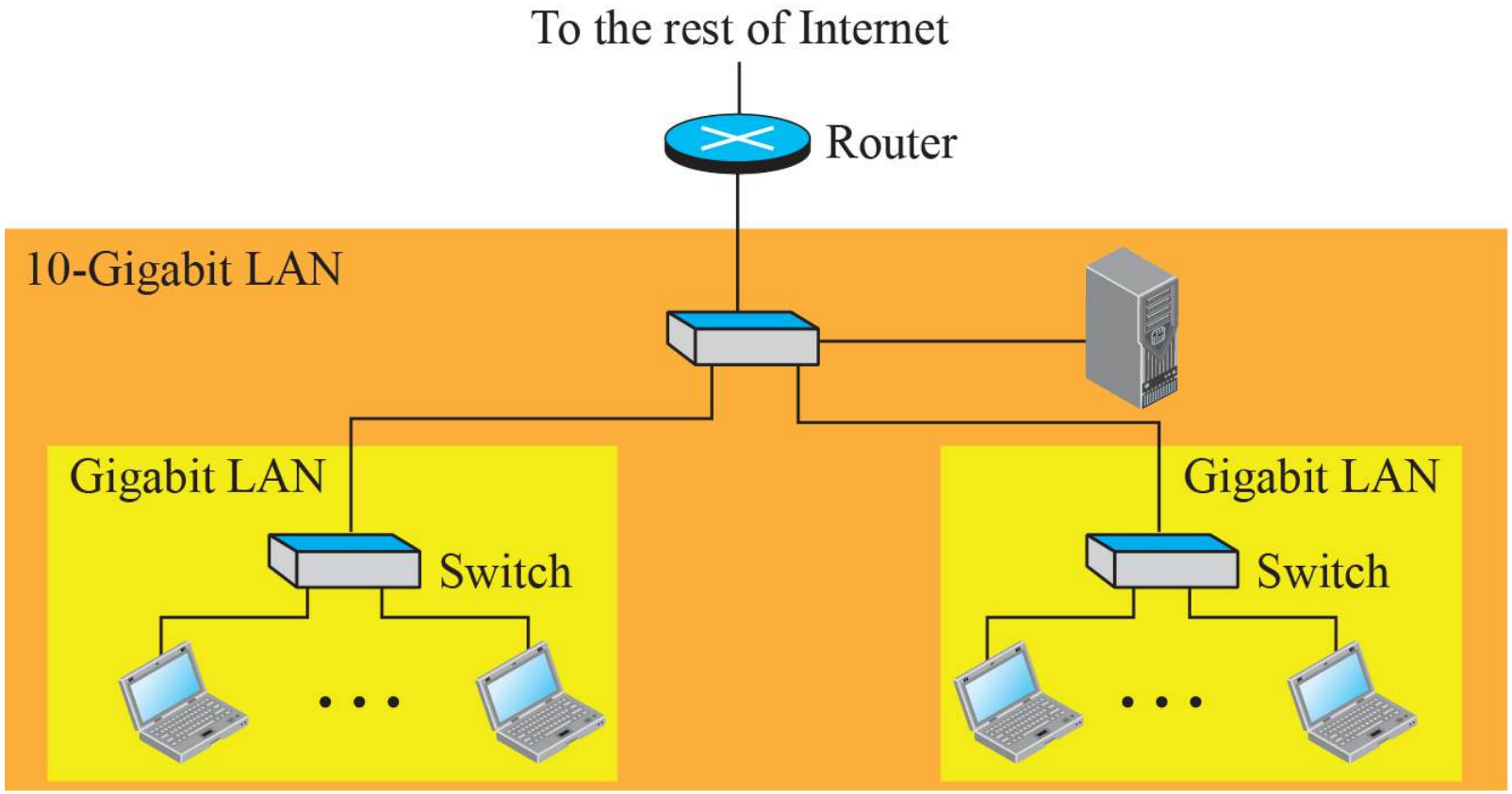
Figure 5.87: Routing example



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Chapter 5: Summary

- *We can consider the data-link layer as two sublayers. The upper sublayer is responsible for data link control, and the lower sublayer is responsible for resolving access to the shared media. Data link control (DLC) deals with the design and procedures for communication between two adjacent nodes: node-to-node communication. This sublayer is responsible for framing and error control. Error control deals with data corruption during transmission. We discussed two link-layer protocols in this chapter: HDLC and PPP.*

- *Many formal protocols have been devised to handle access to a shared link. We categorize them into three groups: random access protocols, controlled access protocols, and channelization protocols.*

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Chapter 5: Summary (continued)

- *At the data-link layer, we use link-layer addressing. The system normally finds the link-layer address of the next node using the Address Resolution Protocol.*

- *Ethernet is the most widely used local area network protocol. The data-link layer of Ethernet consists of the LLC sublayer and the MAC sublayer. The MAC sublayer is responsible for the operation of the CSMA/CD access method and framing. A virtual local area network (VLAN) is configured by software, not by physical wiring. Membership in a VLAN can be based on port numbers, MAC addresses, IP addresses, IP multicast addresses, or a combination of these features.*

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Chapter 5: Summary (continued)

- *We discussed two access networks: DSL and Cable. We also discussed two wide area networks: SONET and ATM.*

- *We also discussed connecting devices in this chapter. A repeater is a connecting device that operates in the physical layer of the Internet model. A switch is a connecting device that operates in the physical and data-link layers of the Internet model. A transparent switch can forward and filter frames and automatically build its forwarding table. A router is a connecting device that operates in the first three layers of the TCP/IP suite.*

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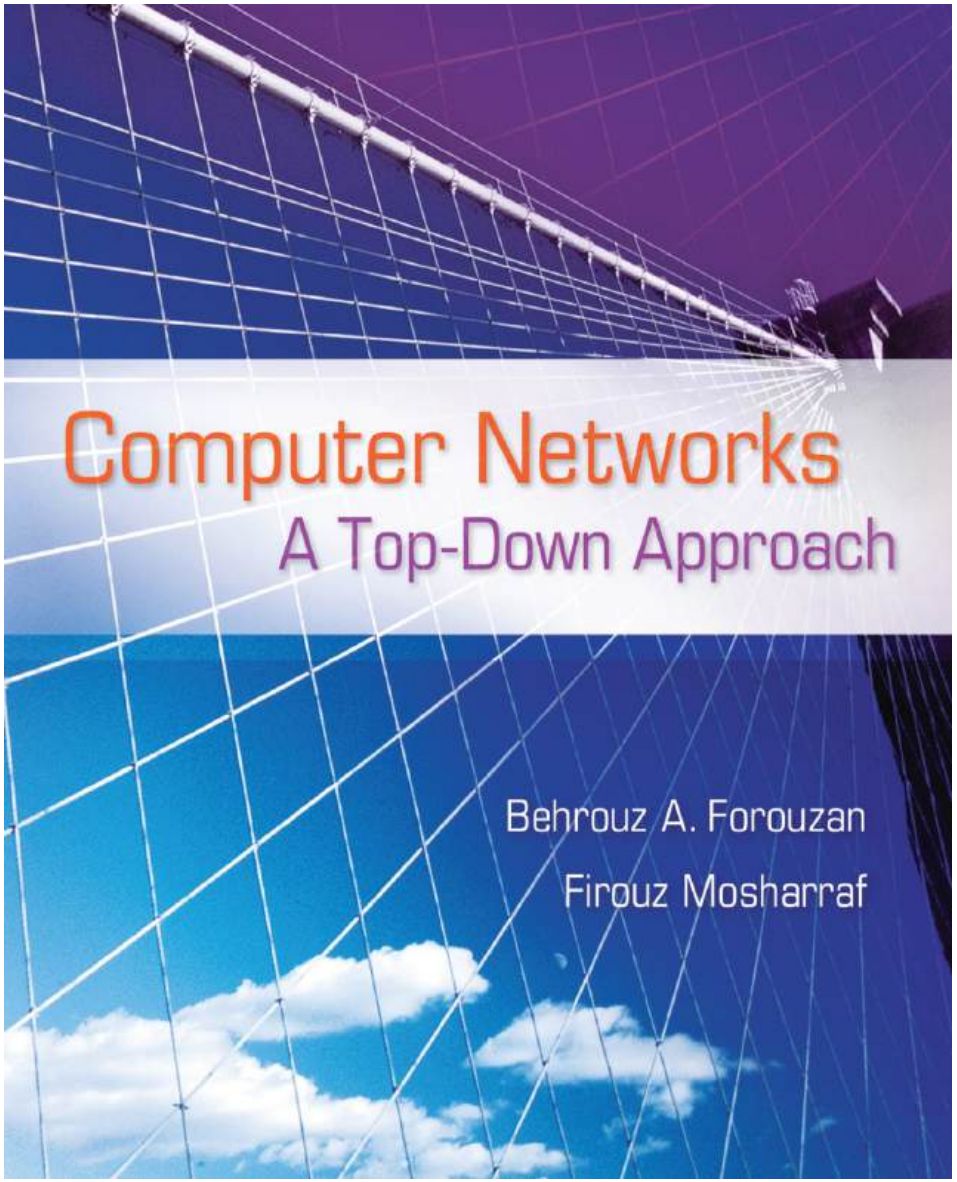
Chapter 6

Wireless Networks and Mobile IP



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Chapter6: Outline

6.1 WIRLESS LANS

6.2 OTHER WIRELESS NETWORKS

6.3 MOBILE IP

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Chapter 6: Objective

- *We introduce wired LANs, using IEEE project 802.11, the dominant standard. Next, we cover the Bluetooth LANs that are used as stand-alone LANs with many applications. We also discuss WiMAX technology, which is the counterpart of last-mile wired networks such as DSL or cable.*

- *We then discuss other wireless networks that can be categorized as wireless WANs or wireless broadband networks. For this purpose, we first discuss the channelization access method that is used in cellular telephones.*

- *We finally talk about mobile IP, which provides mobile access to the Internet. Our discussion include addressing, a big issue in mobile networking, and three phases of mobile access.*

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6-1 WIRELESS LANS

Wireless communication is one of the fastest-growing technologies. The demand for connecting devices without the use of cables is increasing everywhere. Wireless LANs can be found on college campuses, in office buildings, and in many public areas.

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6.1.1 Introduction

Before we discuss a specific protocol related to wireless LANs, let us talk about them in general.

- **Architectural Comparison**
  - Medium
  - Hosts
  - Isolated LANs
  - Connection to Other Networks
  - Moving between Environments

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6.1.1 (continued)

- **Characteristics**
  - Attenuation
  - Interference
  - Multipath Propagation
  - Error
- **Access Control**

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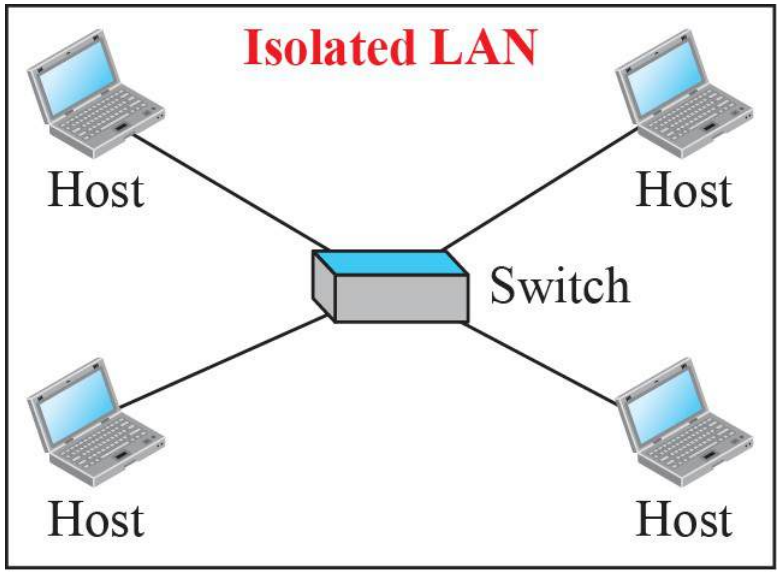
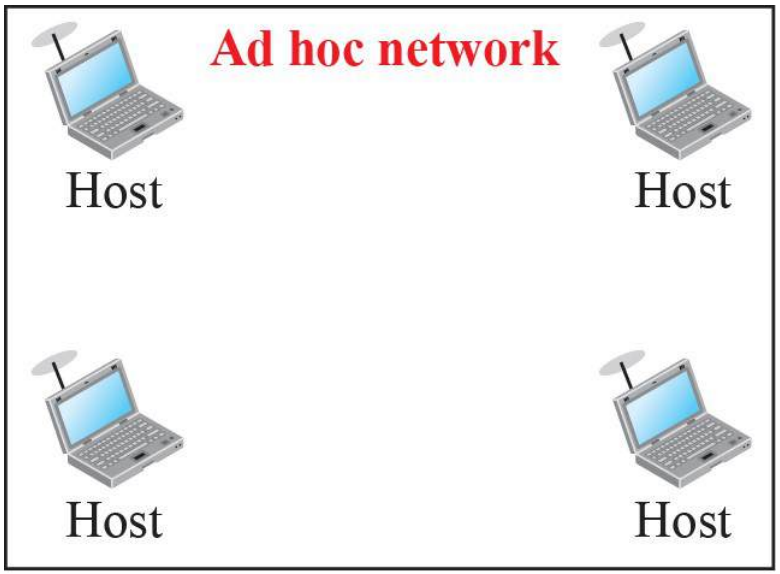
Figure 6.1: Isolated LANs: wired versus wireless



Wired



Wireless

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Figure 6.2: ***Connection of a wired LAN and a wireless LAN to other networks***

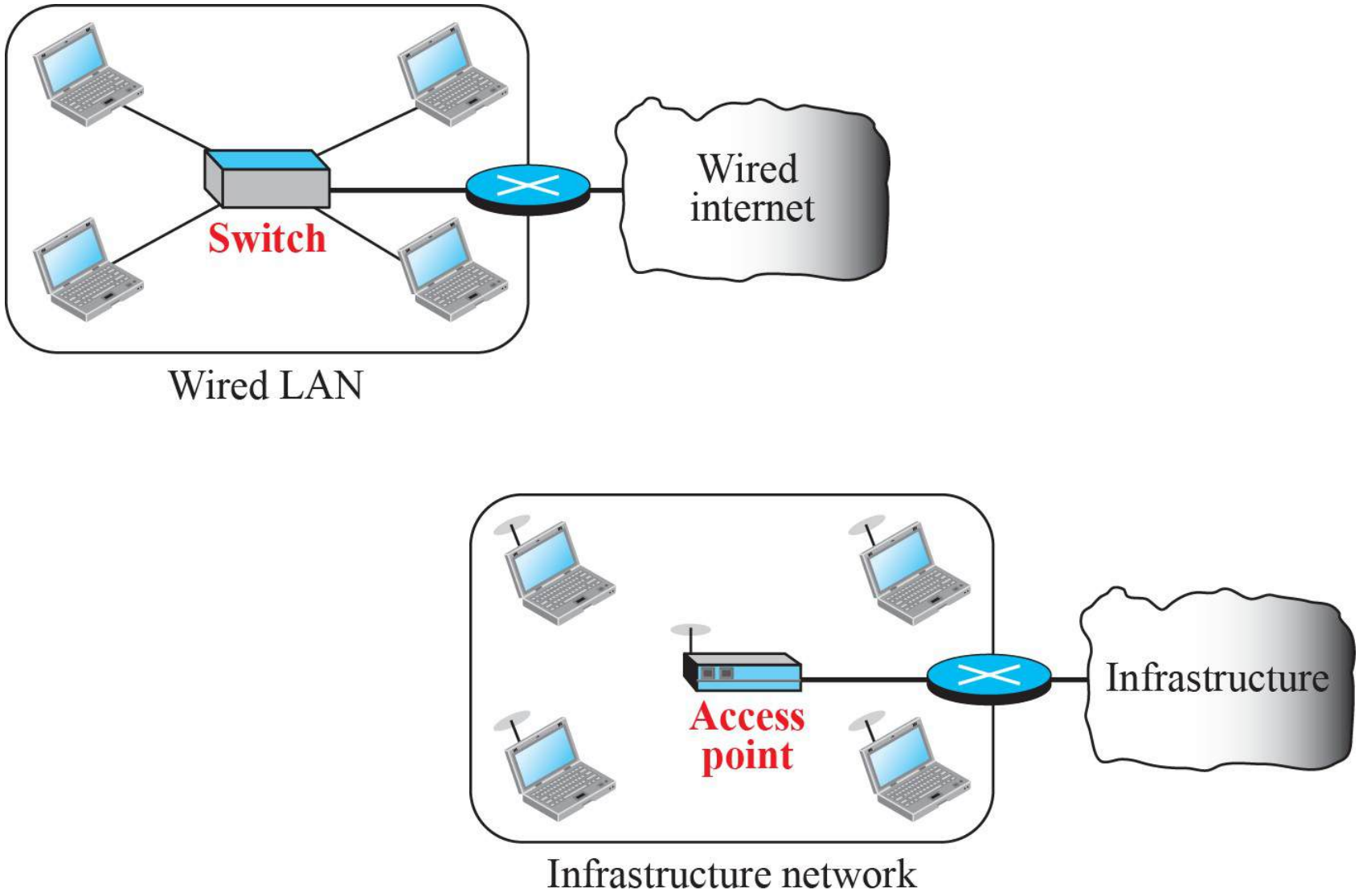

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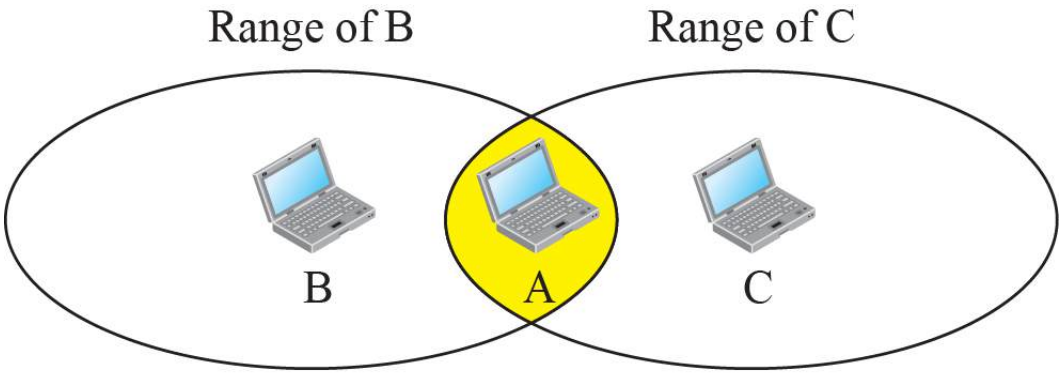
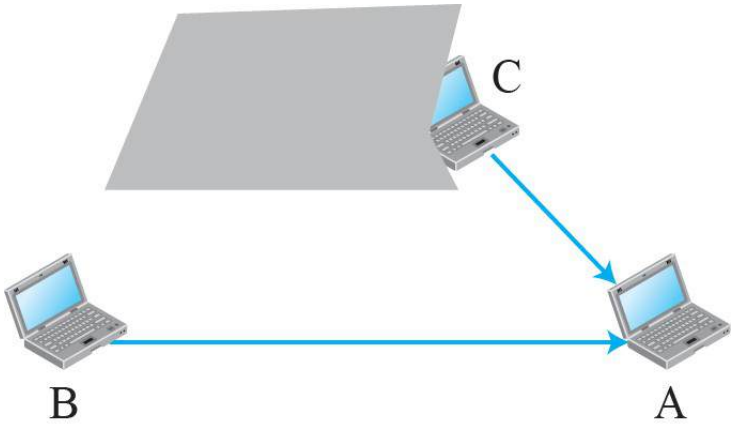
Figure 6.3: Hidden station problem



a. Stations B and C are not in each other's range.



b. Stations B and C are hidden from each other.

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6.1.2 IEEE 802.11 Project

IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data-link layers. In some countries, including the United States, the public uses the term WiFi (short for wireless fidelity) as a synonym for wireless LAN. WiFi, however, is a wireless LAN that is certified by the WiFi Alliance, a global, nonprofit industry association of more than 300 member companies devoted to promoting the growth of wireless LANs.

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6.1.2 (continued)

- **Architecture**
  - Basic Service Set
  - Extended Service Set
  - Station Types

- **MAC Sublayer**
  - Distributed Coordination Function (DCF)
  - Point Coordination Function (PCF)
  - Fragmentation
  - Frame Format
  - Frame Types

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6.1.2 (continued)

- ***Addressing Mechanism***
- ***Exposed Station Problem***
- ***Physical Layer***
  - *IEEE 802.11 FHSS*
  - *IEEE 802.11 DSSS*
  - *IEEE 802.11 Infrared*
  - *IEEE 802.11a OFDM*
  - *IEEE 802.11b DSSS*
  - *IEEE 802.11g*
  - *IEEE 802.11n*

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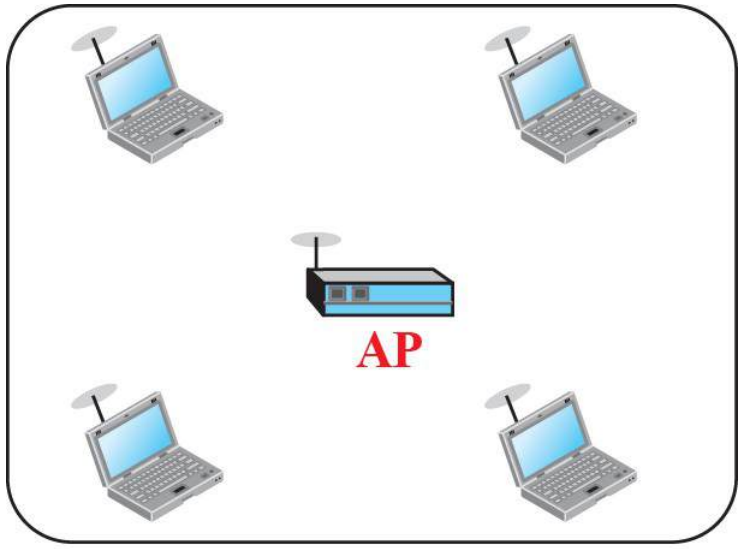
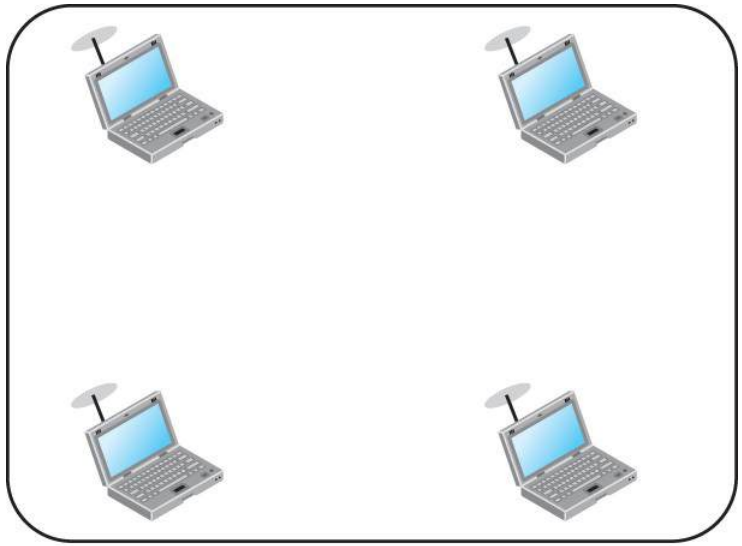
Figure 6.4: Basic service sets (BSSs)



Infrastructure BSS



Ad hoc BSS

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Figure 6.5: Extended service set (ESS)

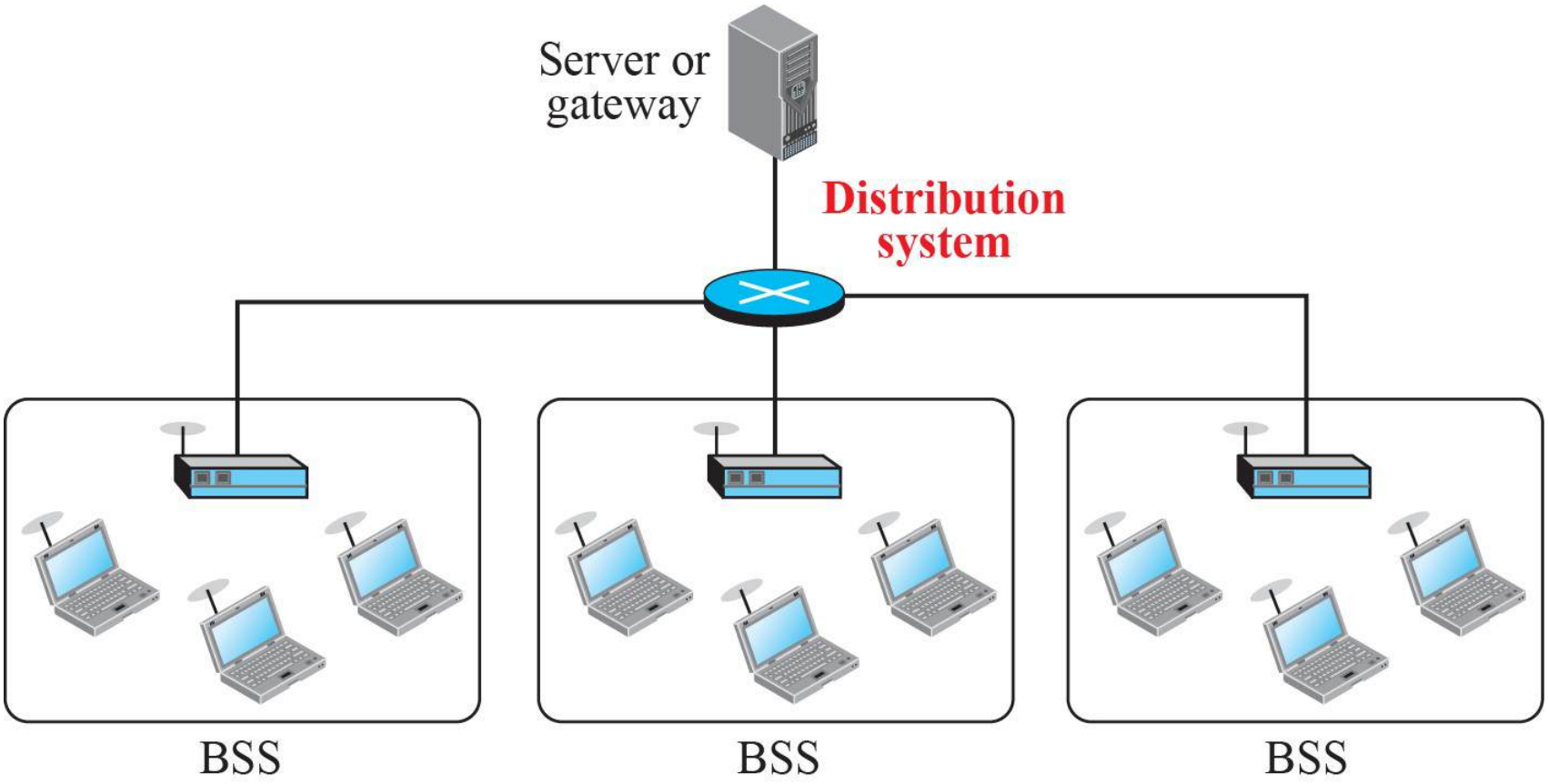

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Figure 6.6: MAC layers in IEEE 802.11 standard

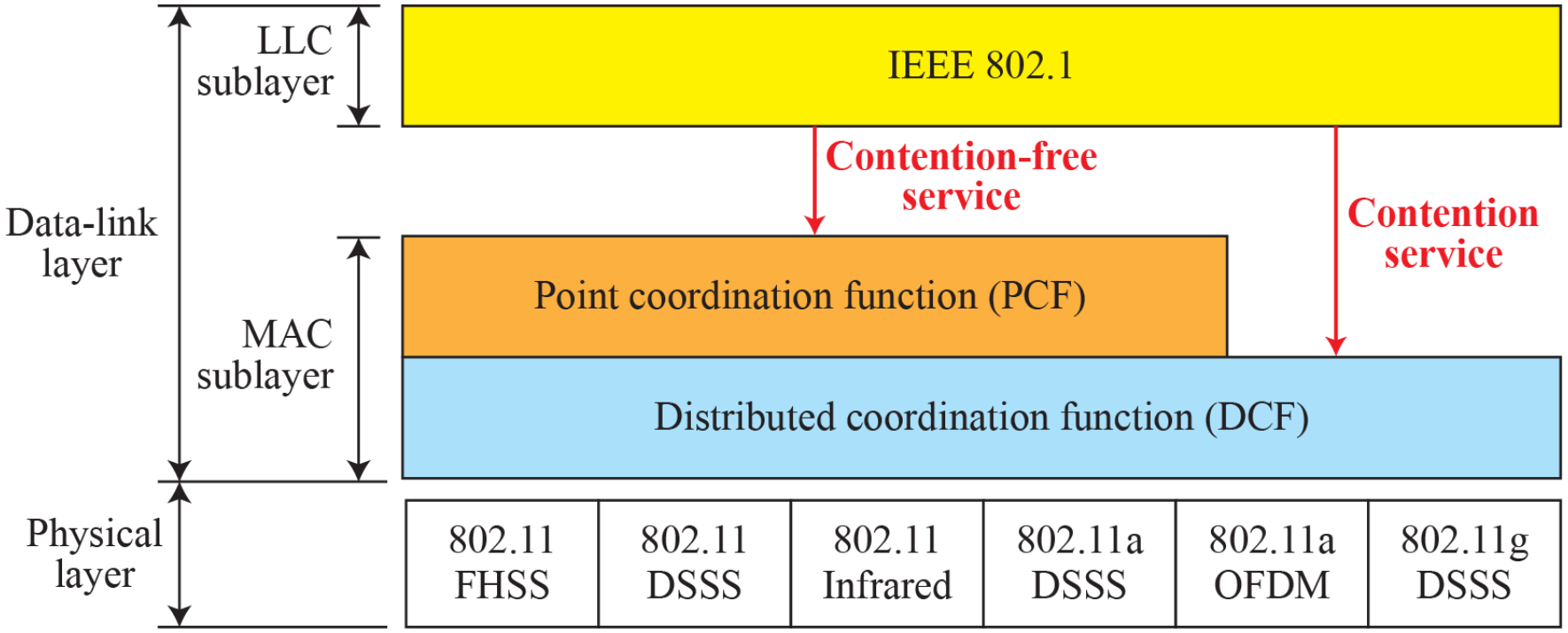

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Figure 6.7: Flow diagram of CSMA/CA

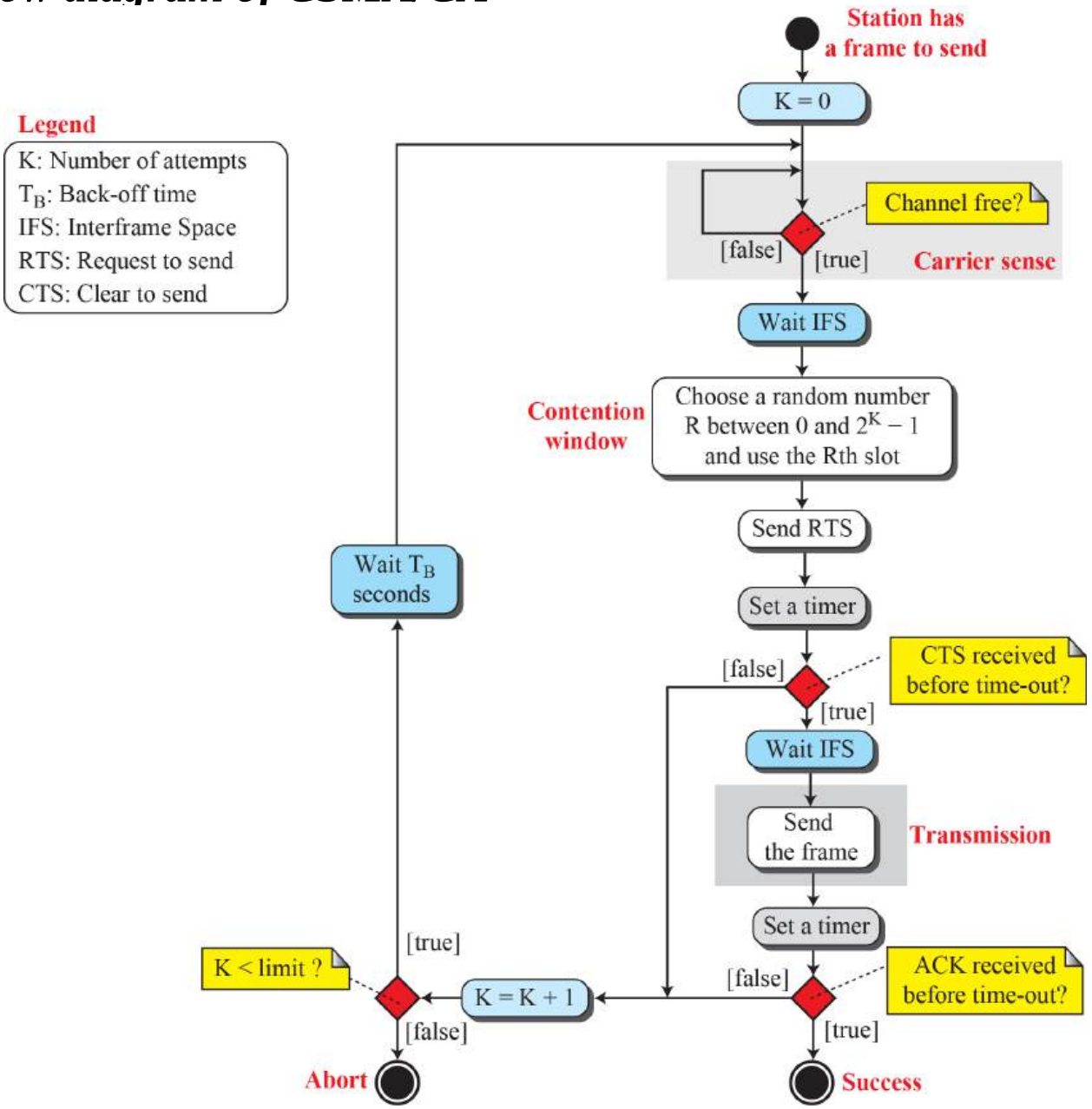

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Figure 6.8: Contention window

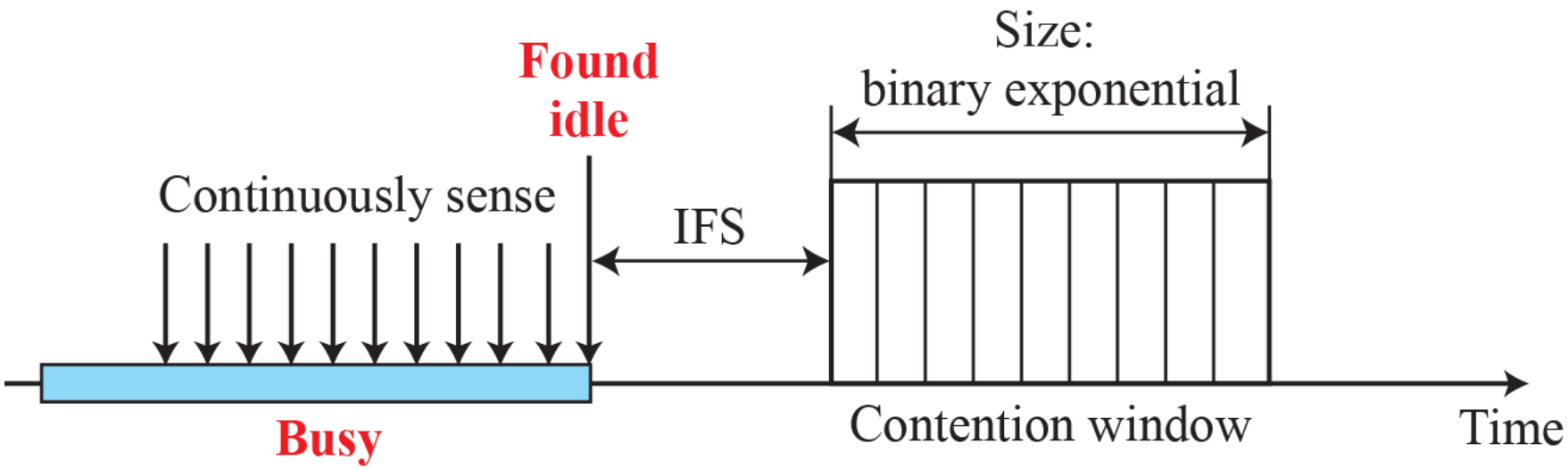

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Figure 6.9: CSMA/CA and NAV

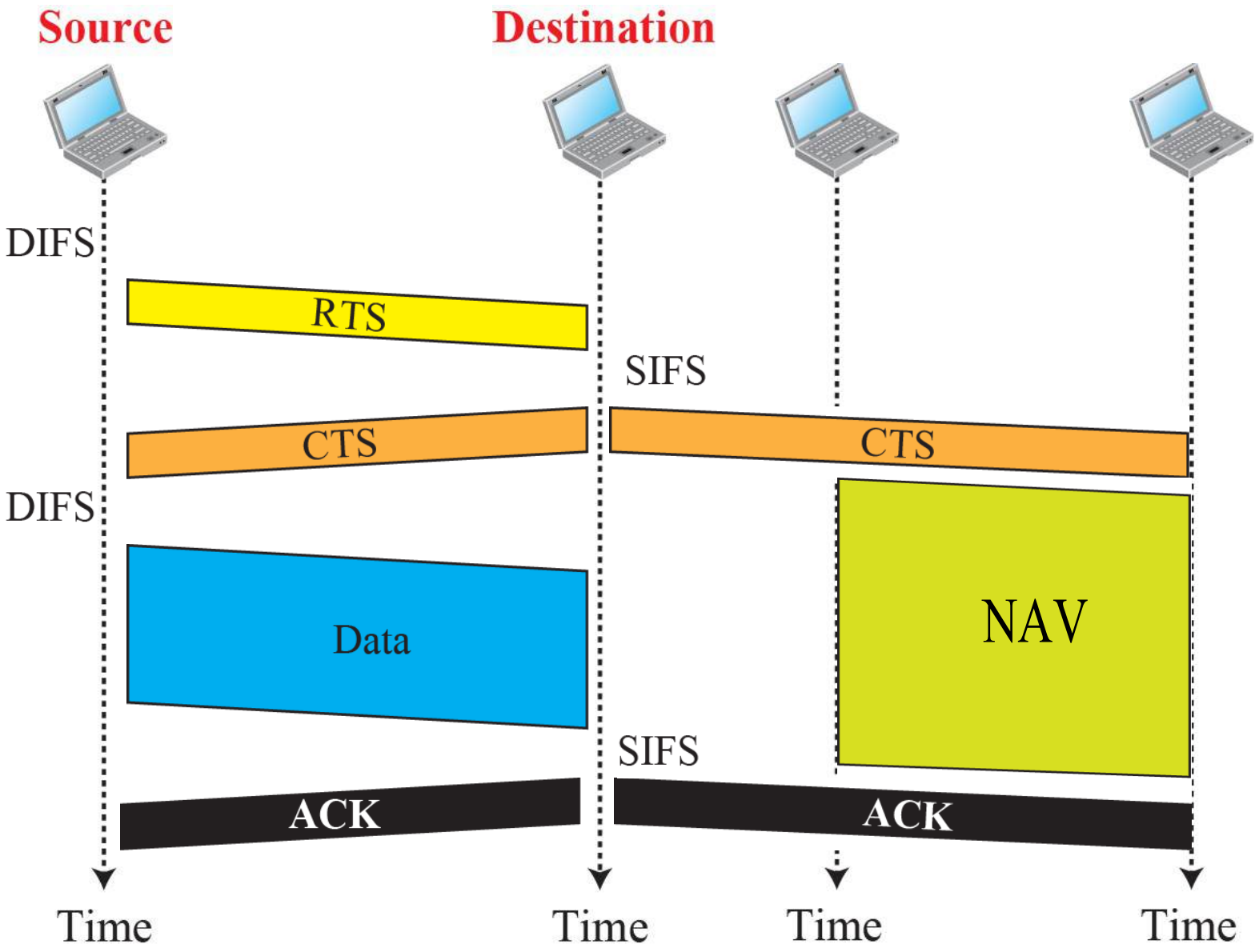

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Figure 6.10: Example of repetition interval

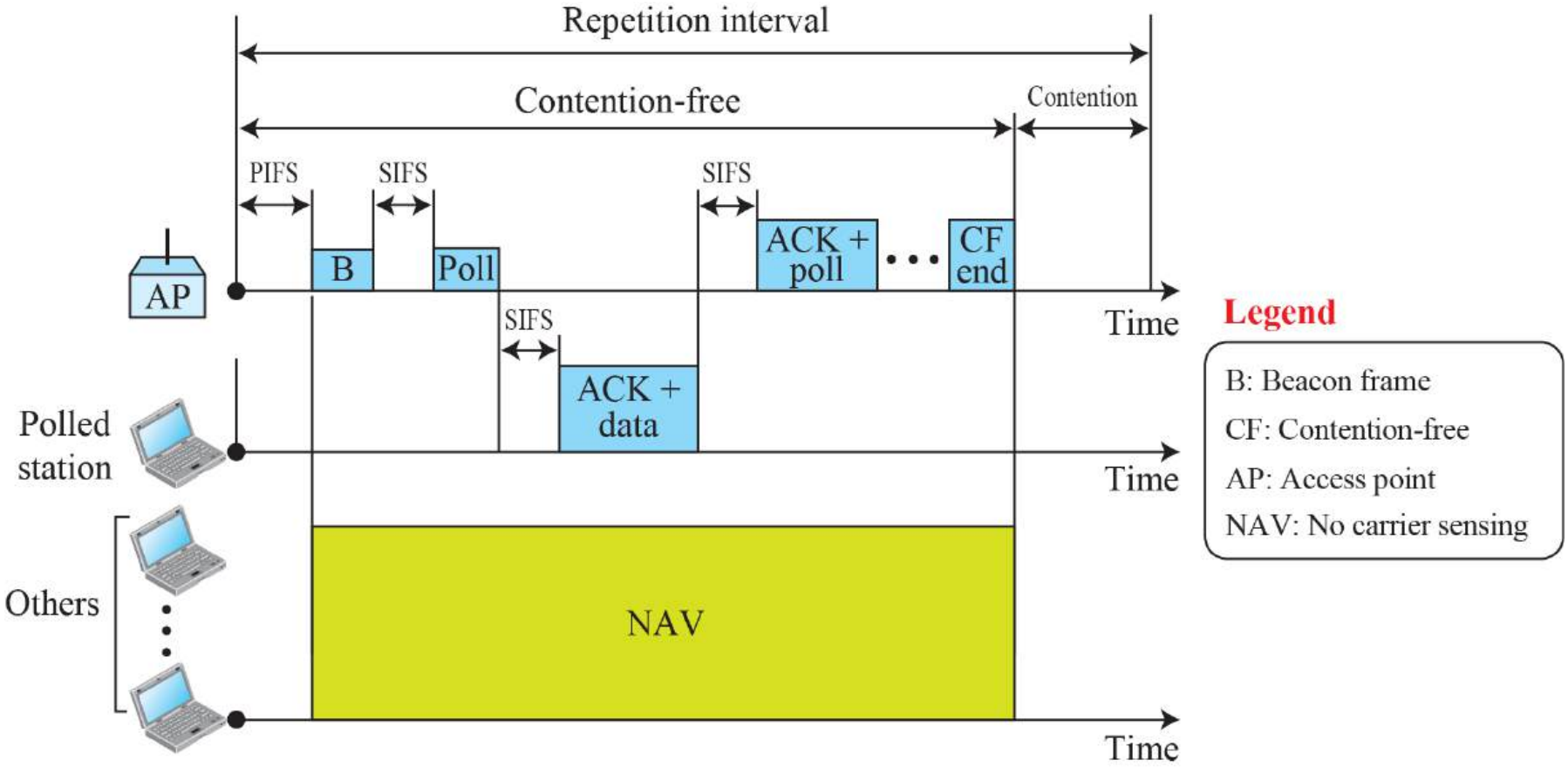

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Figure 6.11: Frame format

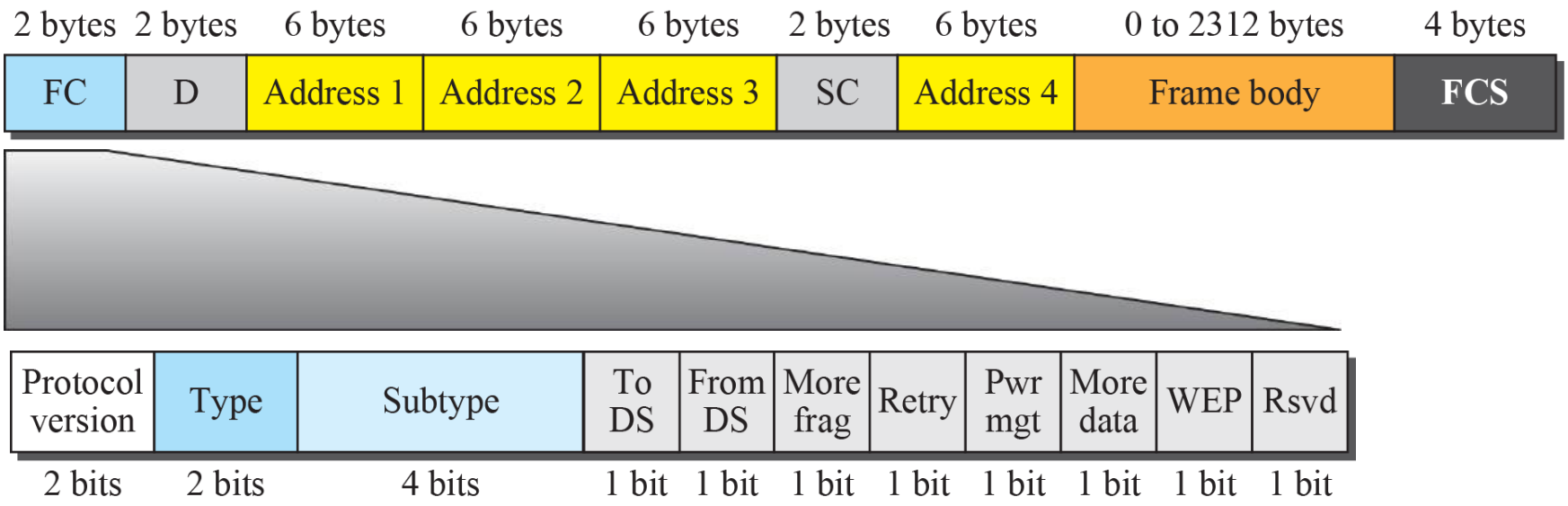

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Table 6.1: Subfields in FC field

Field	*Explanation*
Version	Current version is 0
Type	Type of information: management (00), control (01), or data (10)
Subtype	Subtype of each type (see Table 6.2)
To DS	Defined later
From DS	Defined later
More flag	When set to 1, means more fragments
Retry	When set to 1, means retransmitted frame
Pwr mgt	When set to 1, means station is in power management mode
More data	When set to 1, means station has more data to send
WEP	Wired equivalent privacy (encryption implemented)
Rsvd	Reserved

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Figure 6.12: Control frames

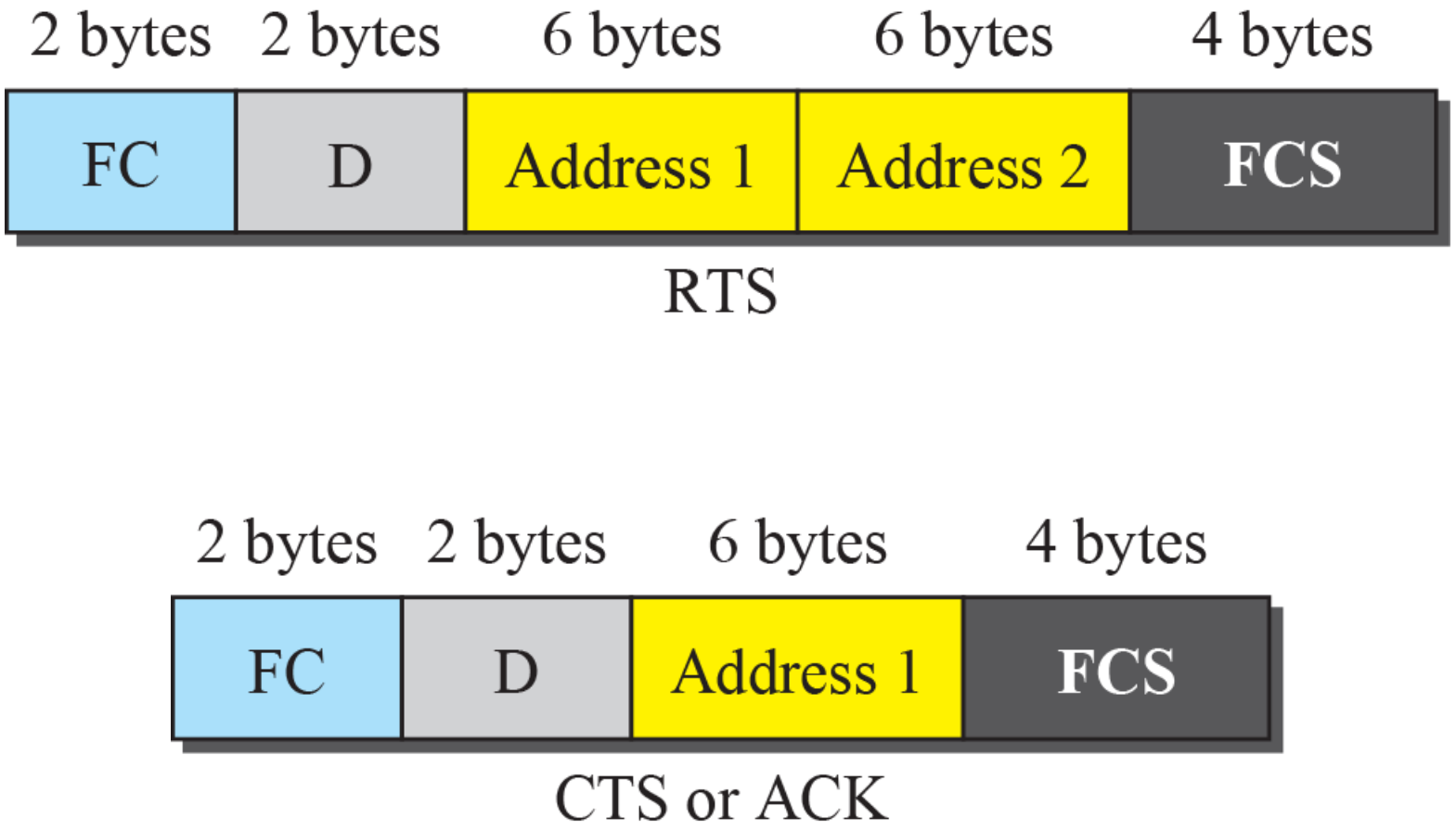

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Table 6.2: Values of subfields in control frames

Subtype	*Meaning*
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)

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Table 6.3: Addresses

To DS	*From DS*	*Address 1*	*Address 2*	*Address 3*	*Address 4*
0	0	Destination	Source	BSS ID	N/A
0	1	Destination	Sending AP	Source	N/A
1	0	Receiving AP	Source	Destination	N/A
1	1	Receiving AP	Sending AP	Destination	Source

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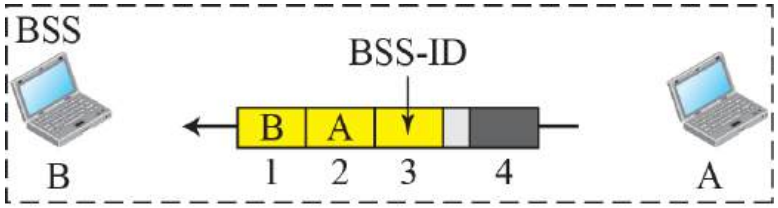
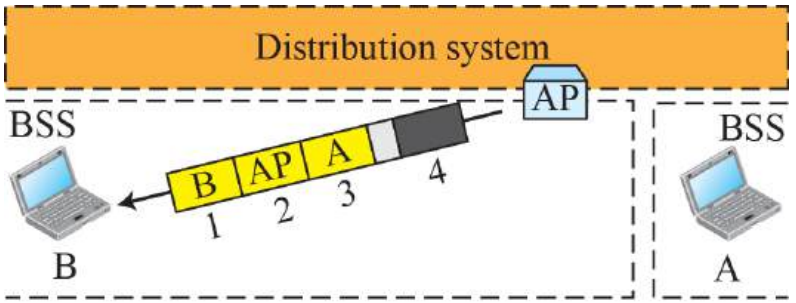
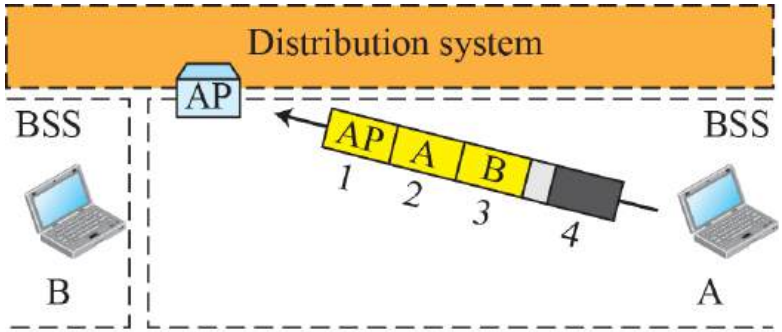
Figure 6.13: Addressing mechanisms



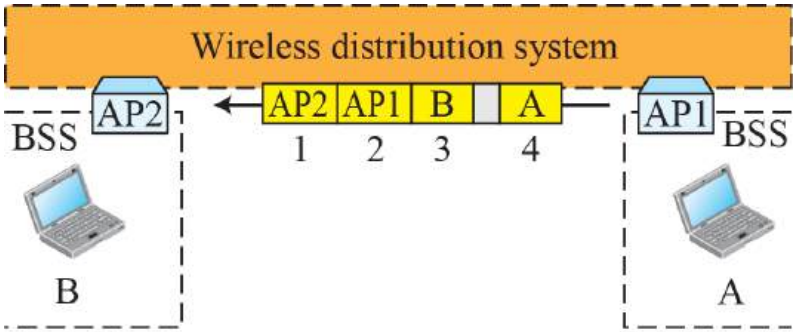
a. Case 1



b. Case 2



c. Case 3



d. Case 4

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Figure 6.14: Exposed station problem

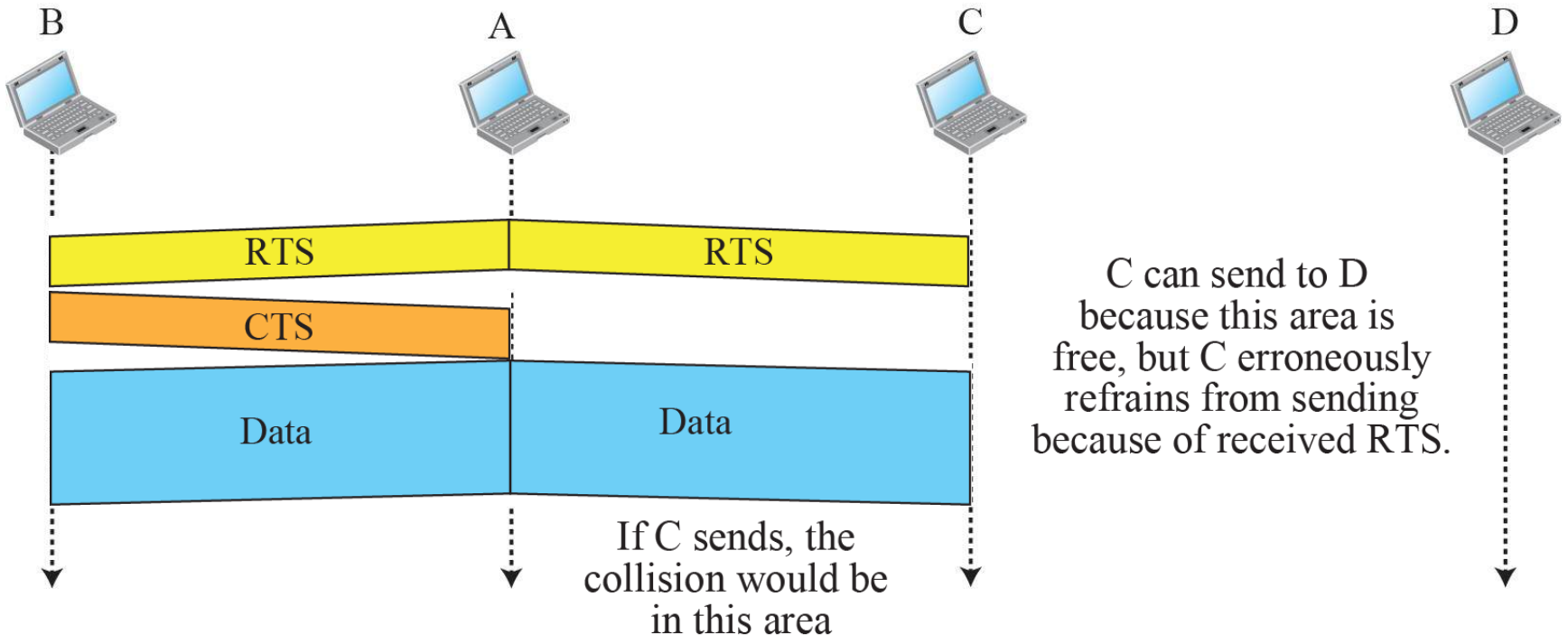

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Table 6.4: Specifications

IEEE	*Technique*	*Band*	*Modulation*	*Rate (Mbps)*
802.11	FHSS	2.400–4.835 GHz	FSK	1 and 2
	DSSS	2.400–4.835 GHz	PSK	1 and 2
	None	Infrared	PPM	1 and 2
802.11a	OFDM	5.725–5.850 GHz	PSK or QAM	6 to 54
802.11b	DSSS	2.400–4.835 GHz	PSK	5.5 and 11
802.11g	OFDM	2.400–4.835 GHz	Different	22 and 54
802.11n	OFDM	5.725–5.850 GHz	Different	600

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Figure 6.15: Physical layer of IEEE 802.11 FHSS

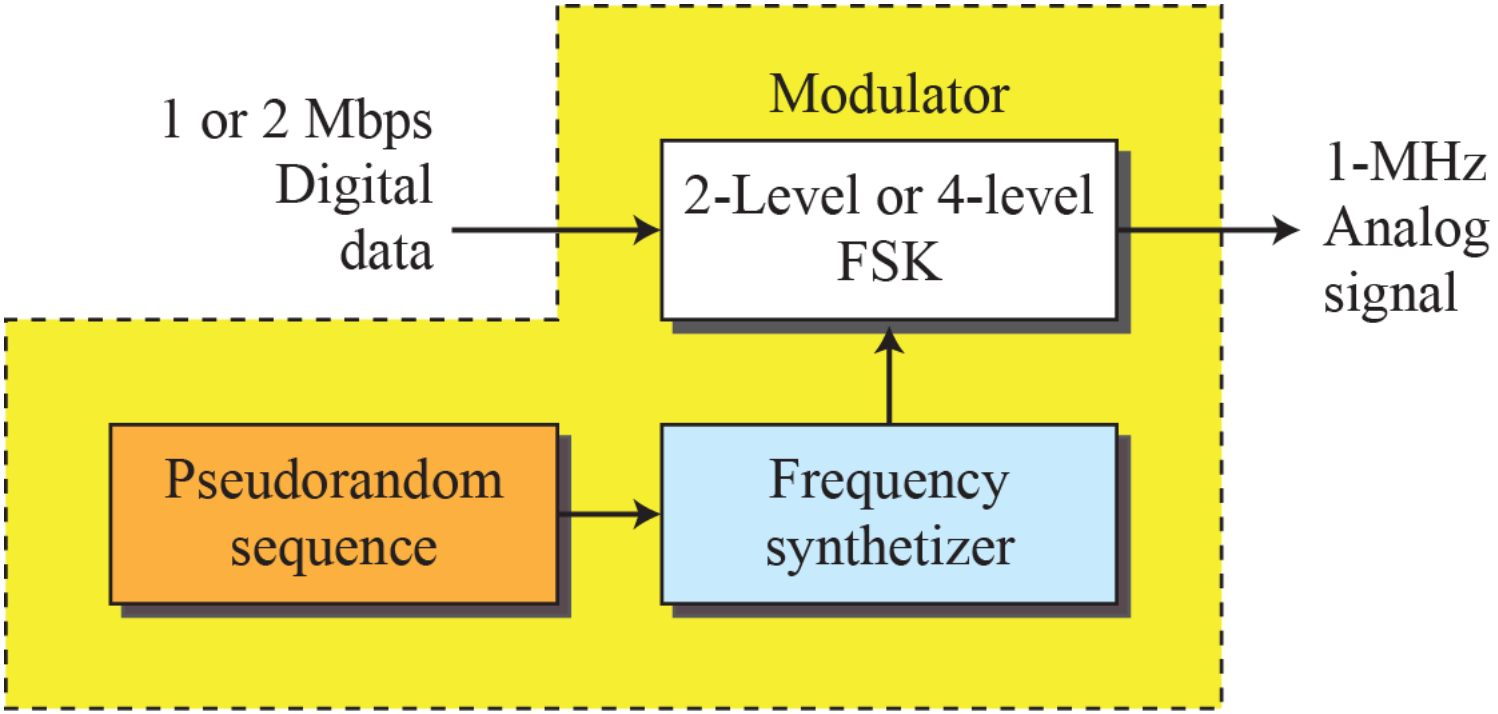

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Figure 6.16: Physical layer of IEEE 802.11 DSSS

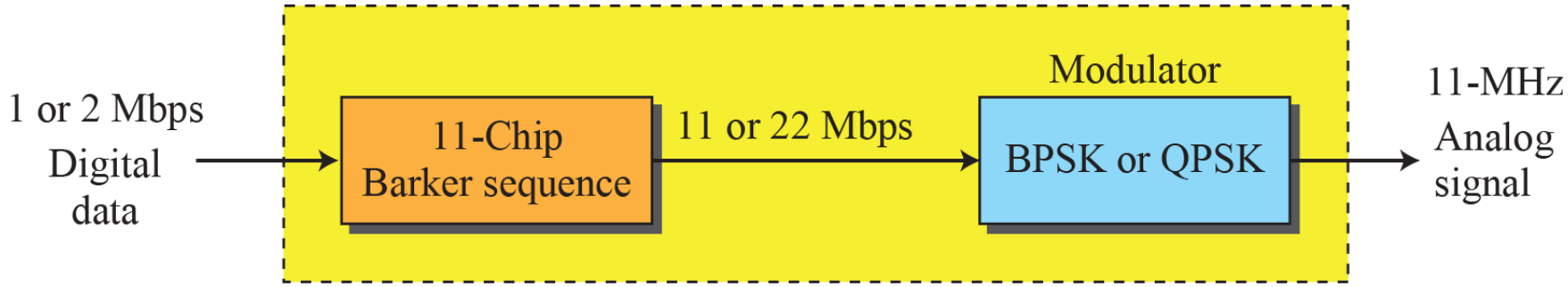

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Figure 6.17: Physical layer of IEEE 802.11 infrared

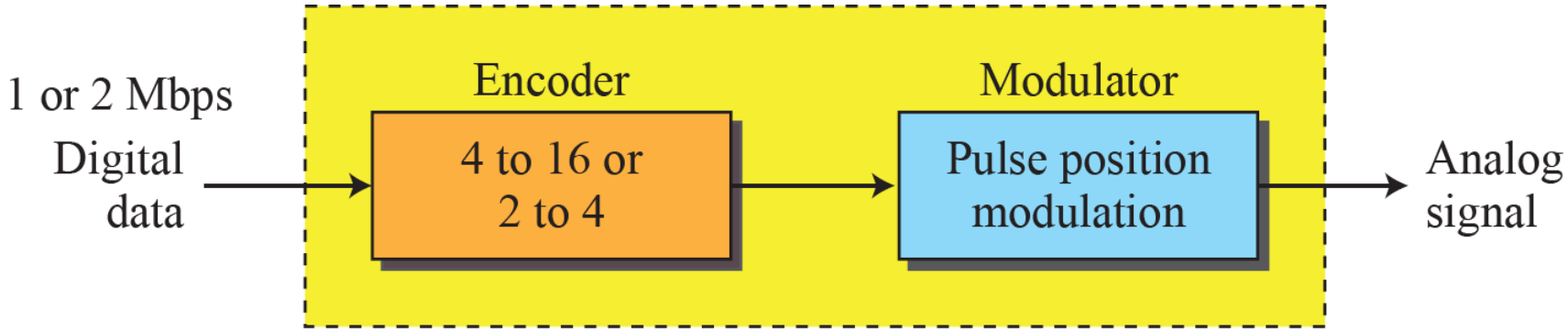

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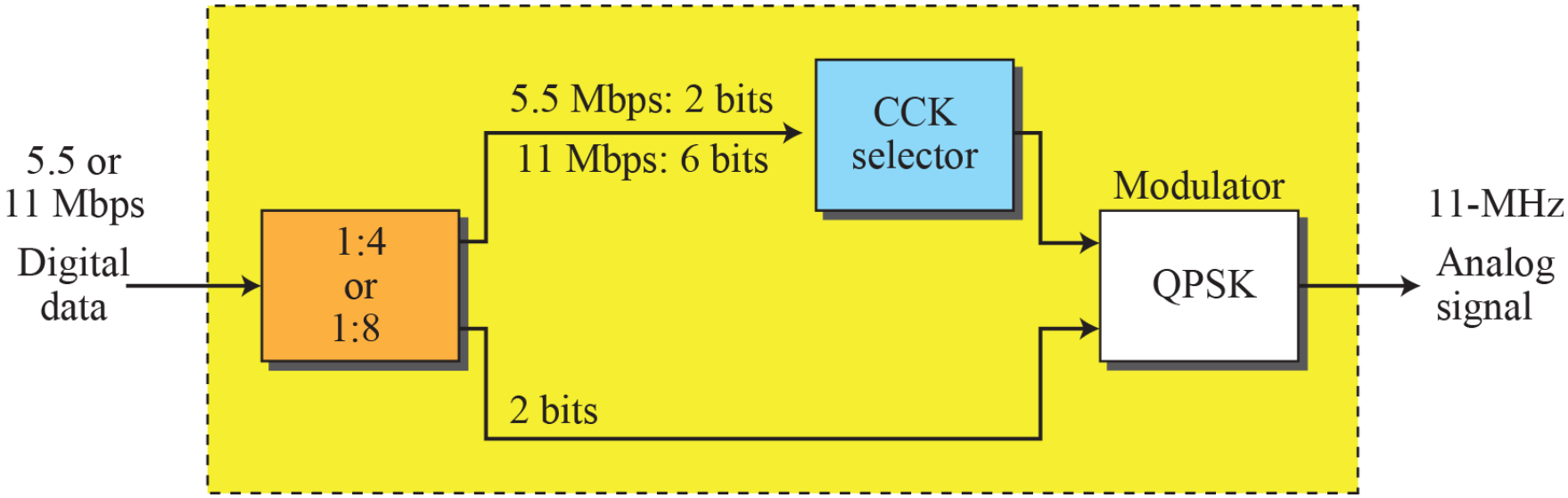
Figure 6.18: Physical layer of IEEE 802.11b



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6.1.3 Bluetooth

Bluetooth is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers (desktop and laptop), cameras, printers, and even coffee makers when they are at a short distance from each other. A Bluetooth LAN is an ad hoc network, which means that the network is formed spontaneously; the devices, sometimes called gadgets, find each other and make a network called a piconet.

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6.1.3 (continued)

- **Architecture**
  - Piconets
  - Scatternet
  - Bluetooth Devices

- **Bluetooth Layers**
  - L2CAP
  - Baseband Layer
  - Radio Layer

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Figure 6.19: ***Piconet***

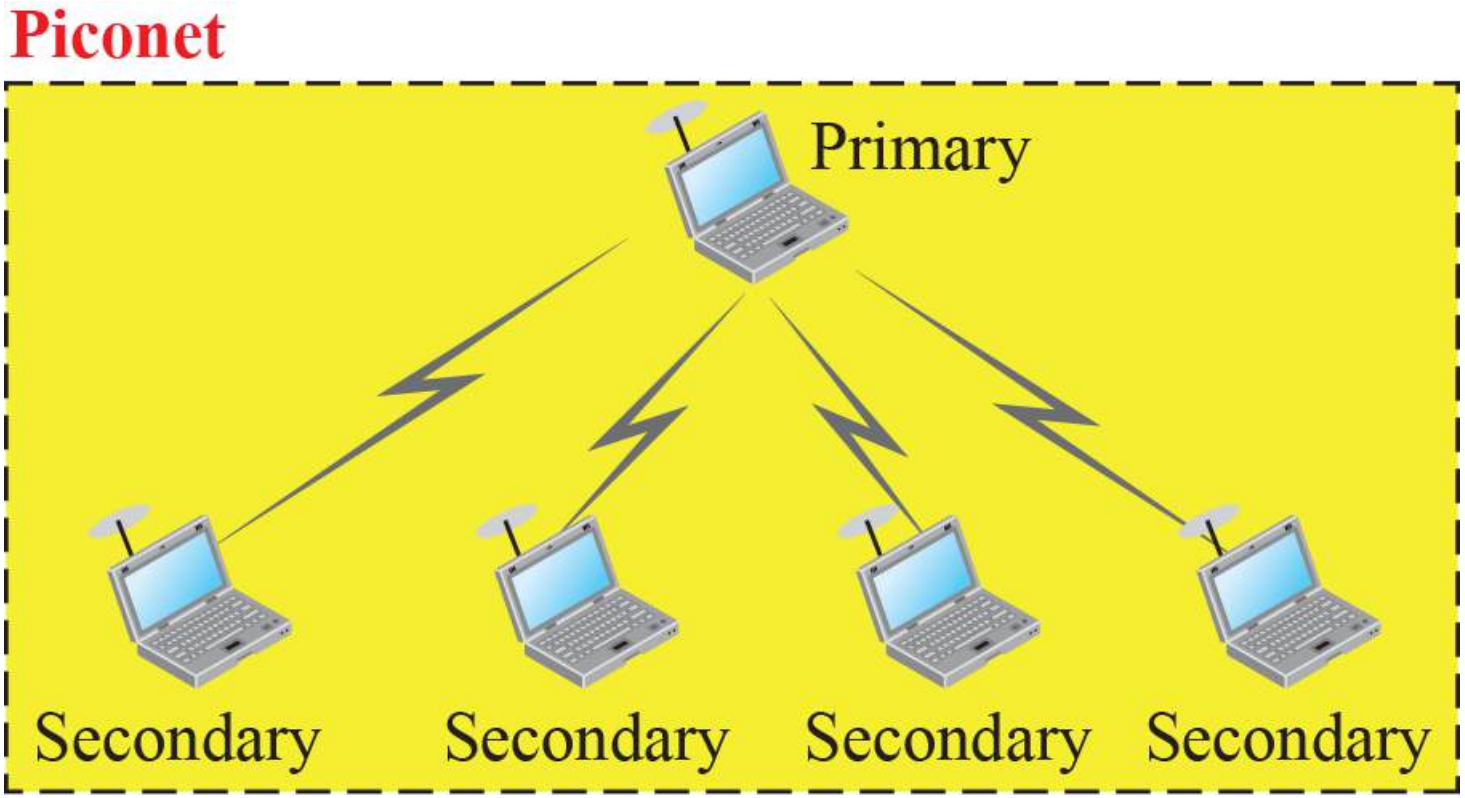

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Figure 6.20: ***Scatternet***

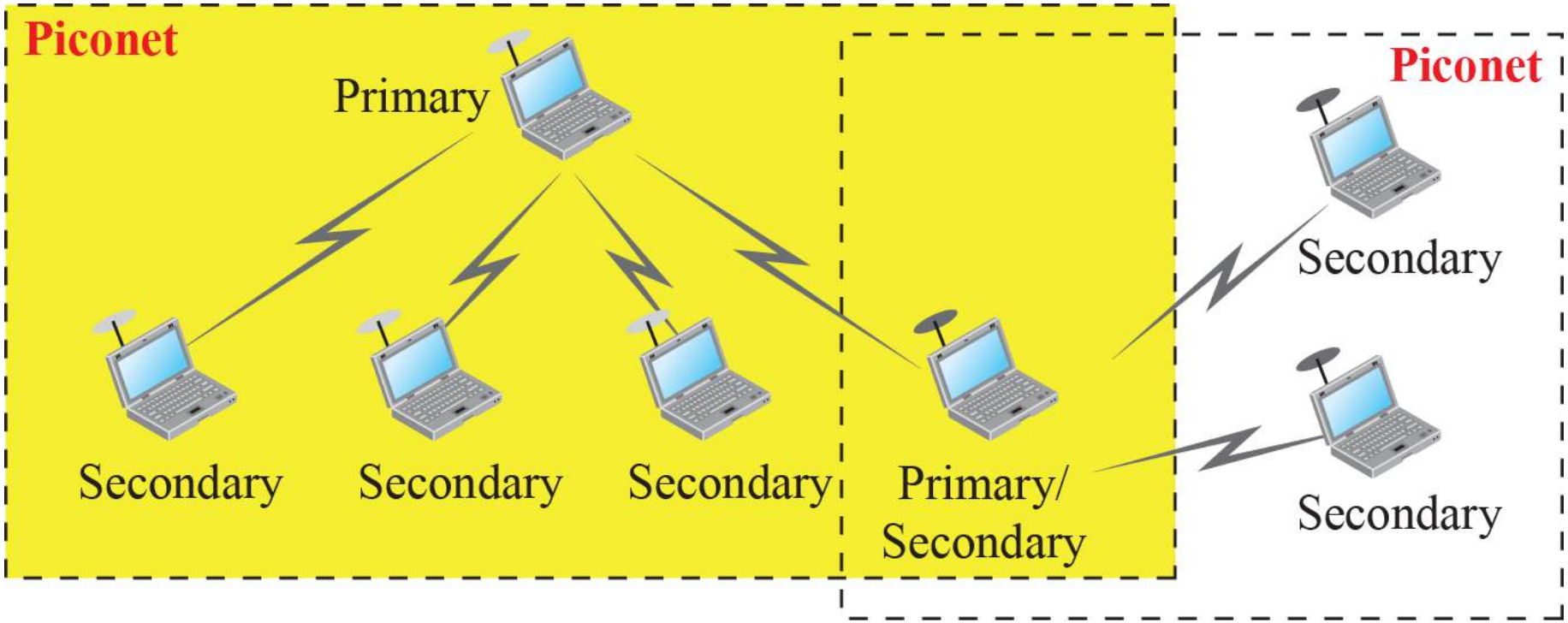

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Figure 6.21: Bluetooth layers

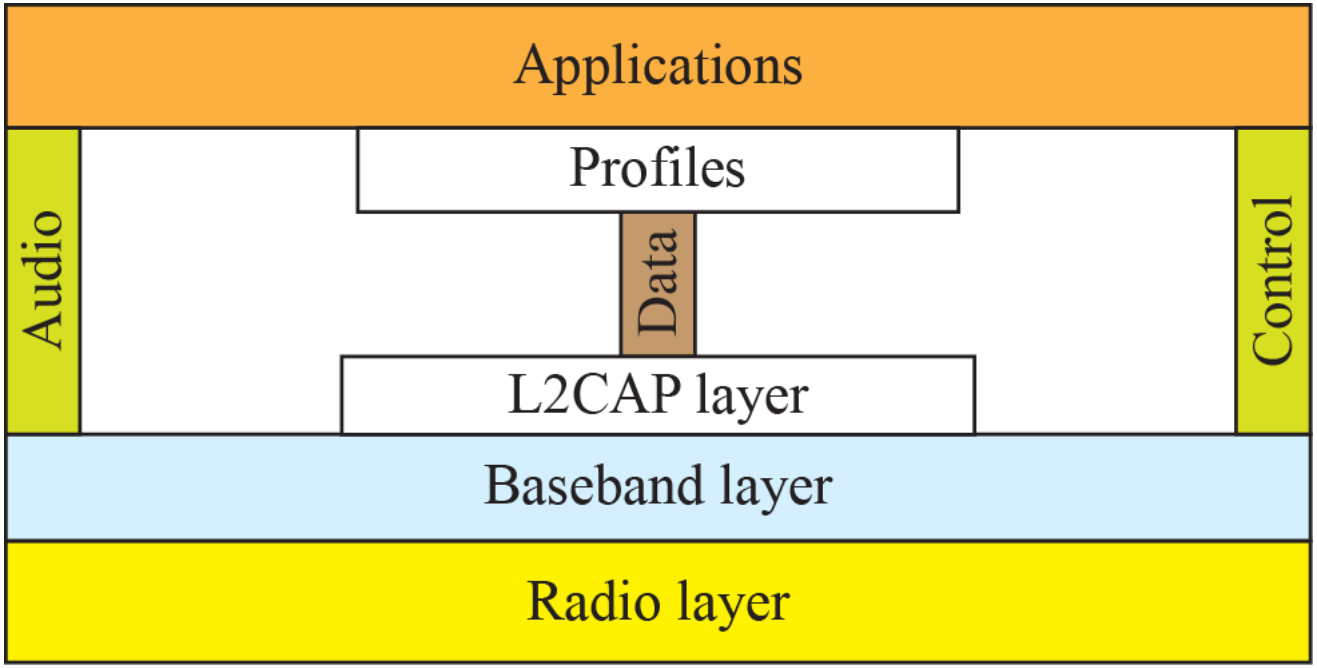

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Figure 6.22: L2CAP data packet format

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Figure 6.23: Single-secondary communication

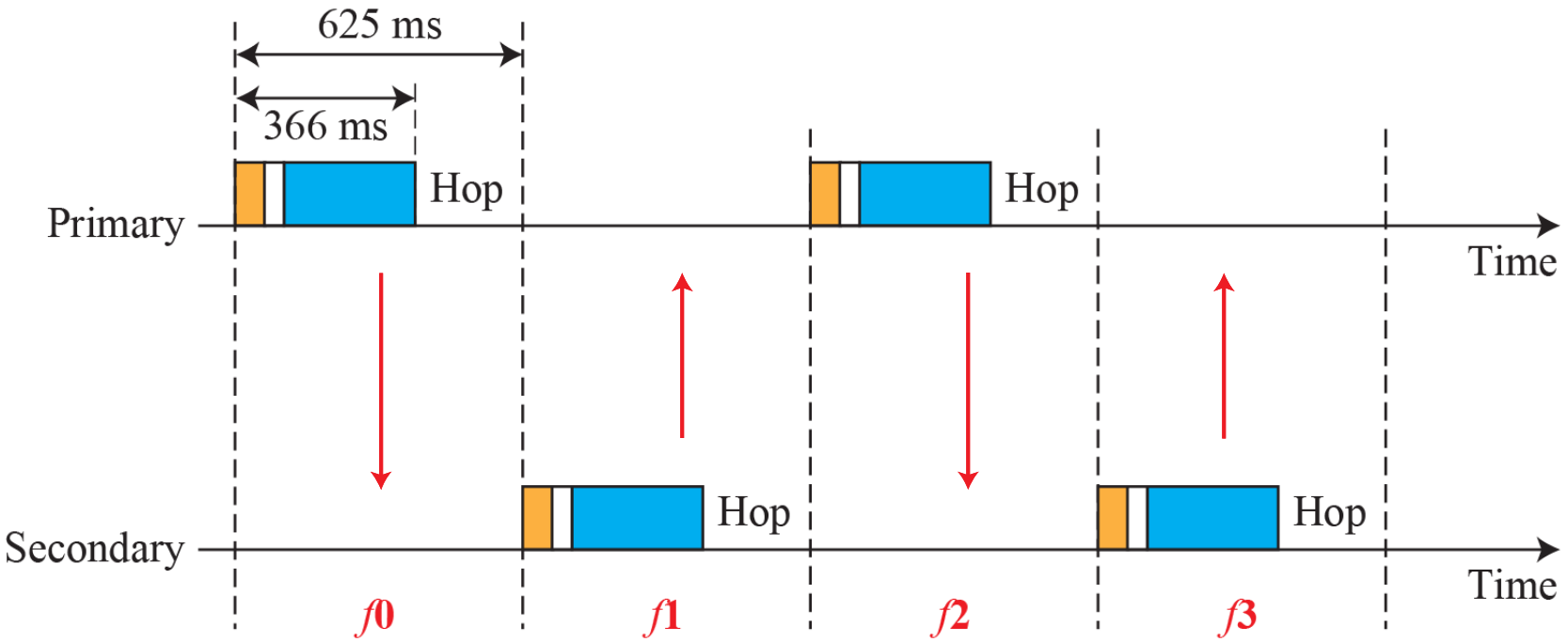

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Figure 6.24: Multiple-secondary communication

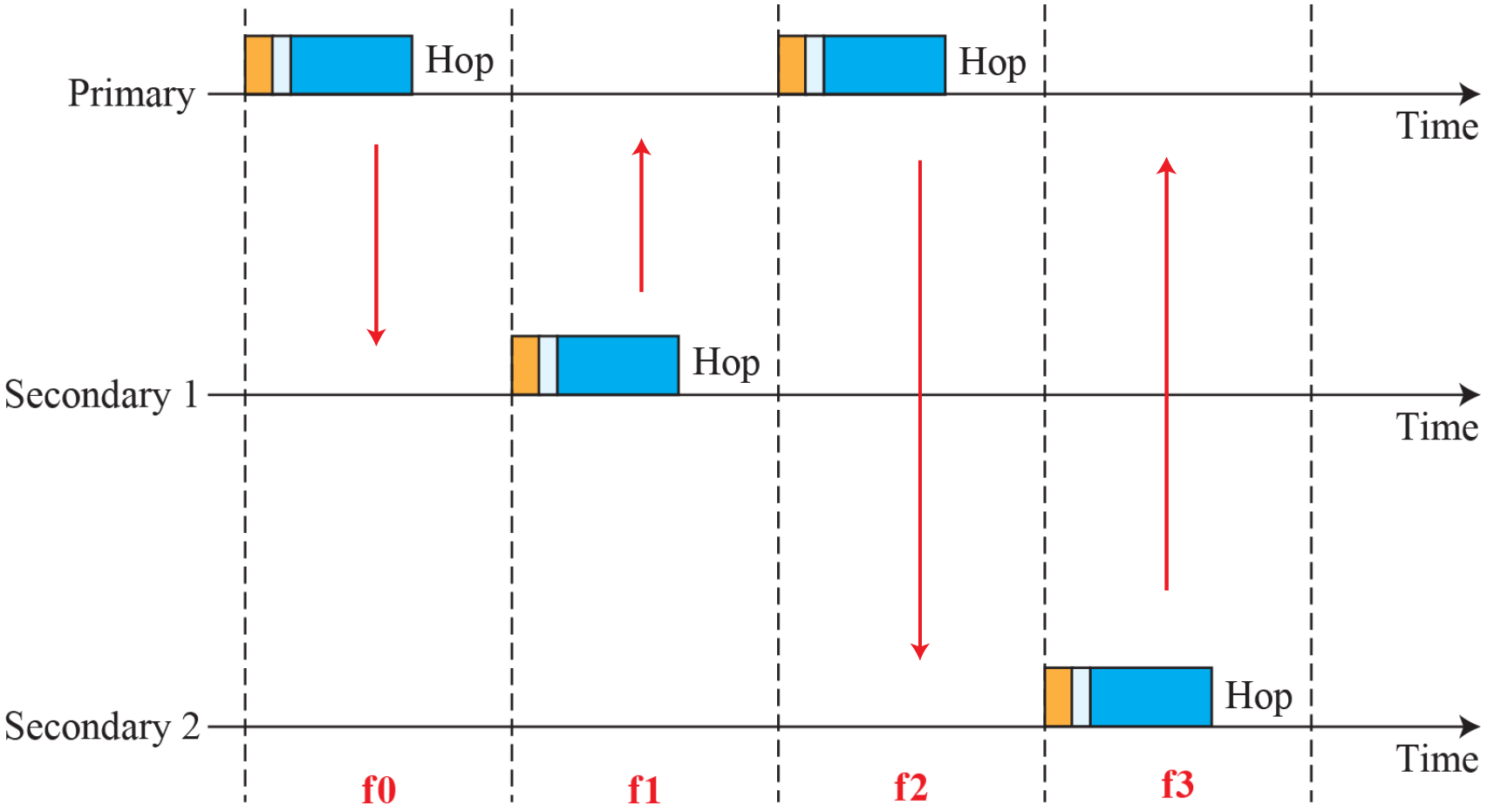

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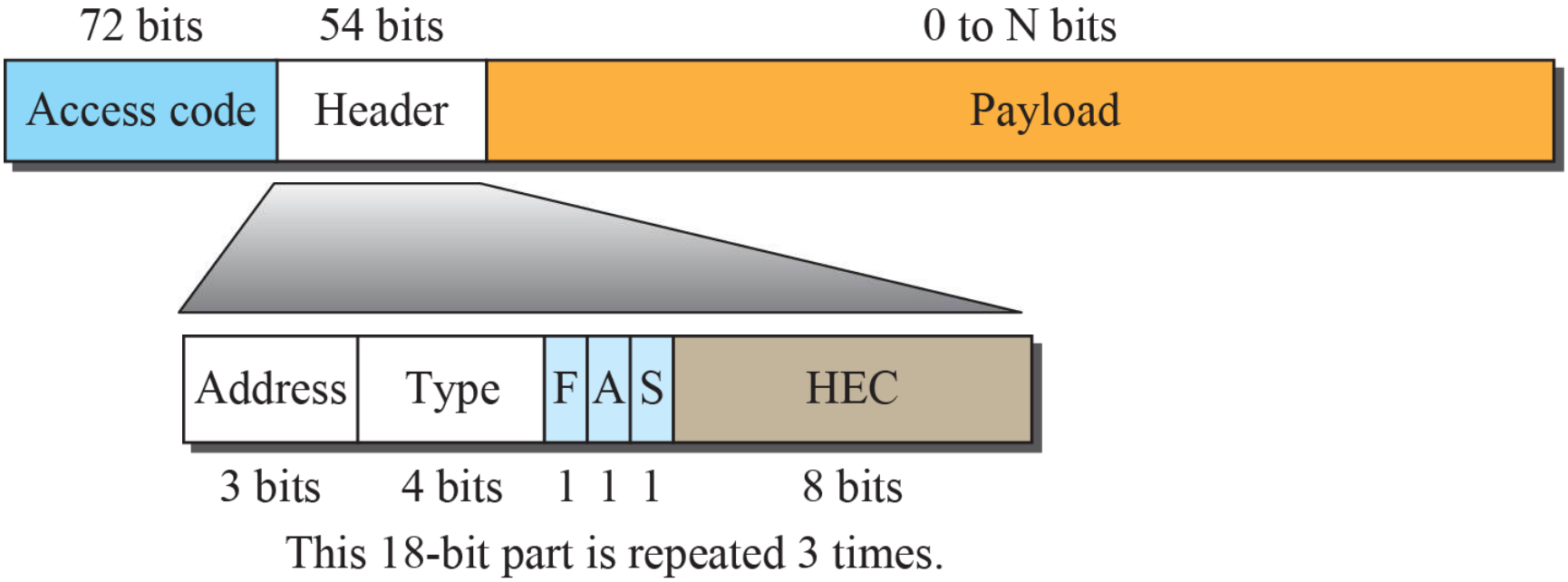
Figure 6.25: Frame format types



N = 240 for 1-slot frame

N = 1490 for 3-slot frame

N = 2740 for 5-slot frame

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6.1.4 WiMax

Worldwide Interoperability for Microwave Access (WiMAX) is an IEEE standard 802.16 (for fixed wireless) and 802.16e (for mobile wireless) that aims to provide the "last mile" broadband wireless access alternative to cable modem, telephone DSL service..

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6.1.4 (continued)

- **Architecture**
  - Base Station
  - Subscriber Stations
  - Portable Unit
- **Data-Link Layer**
- **Physical Layer**
- **Application**

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6-2 OTHER WIRELESS NETWORKS

In this section, we concentrate on other wireless networks. We first discuss cellular telephony, which is ubiquitous. We then talk about satellite networks. Before we discuss the above-mentioned wireless networks, let us discuss one access method that we postponed from Chapter 5: channelization, which is used in cellular and other wireless networks.

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6.2.1 Channelization

Channelization (or channel partition, as it is sometime called) is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations. In this section, we discuss three channelization protocols: FDMA, TDMA, and CDMA.

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6.2.1 (continued)

- ***Frequency-Division Multiple Access (FDMA)***
- ***Time-Division Multiple Access (TDMA)***
- ***Code-Division Multiple Access (CDMA))***
  - *Analogy*
  - *Idea*
  - *Chips*
  - *Data Representation*
  - *Encoding and Decoding*
  - *Signal Level*
  - *Sequence Generation*

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Figure 6.26: Frequency-division multiple access (FDMA)

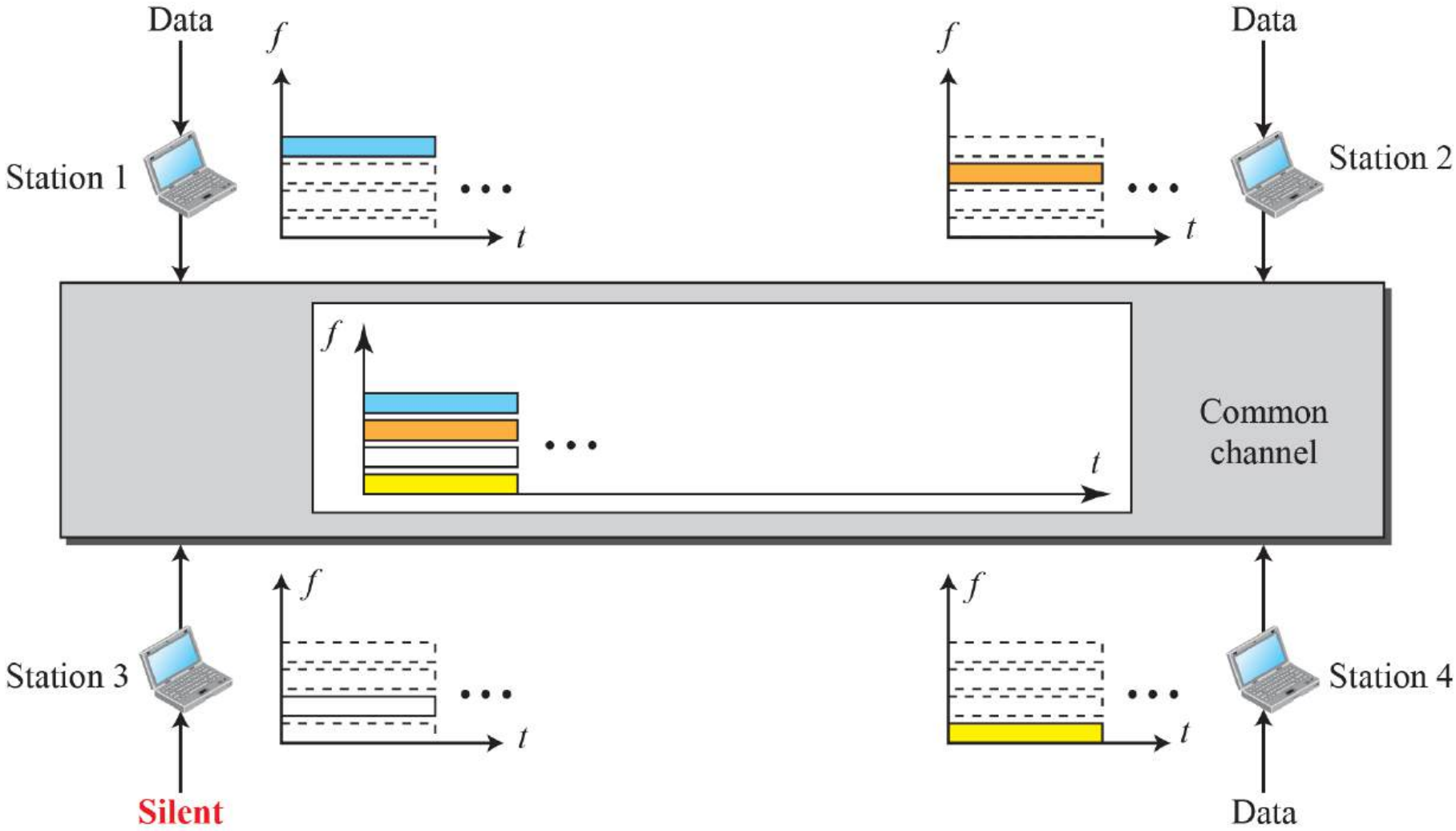

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Figure 6.27: Time-division multiple access (TDMA)

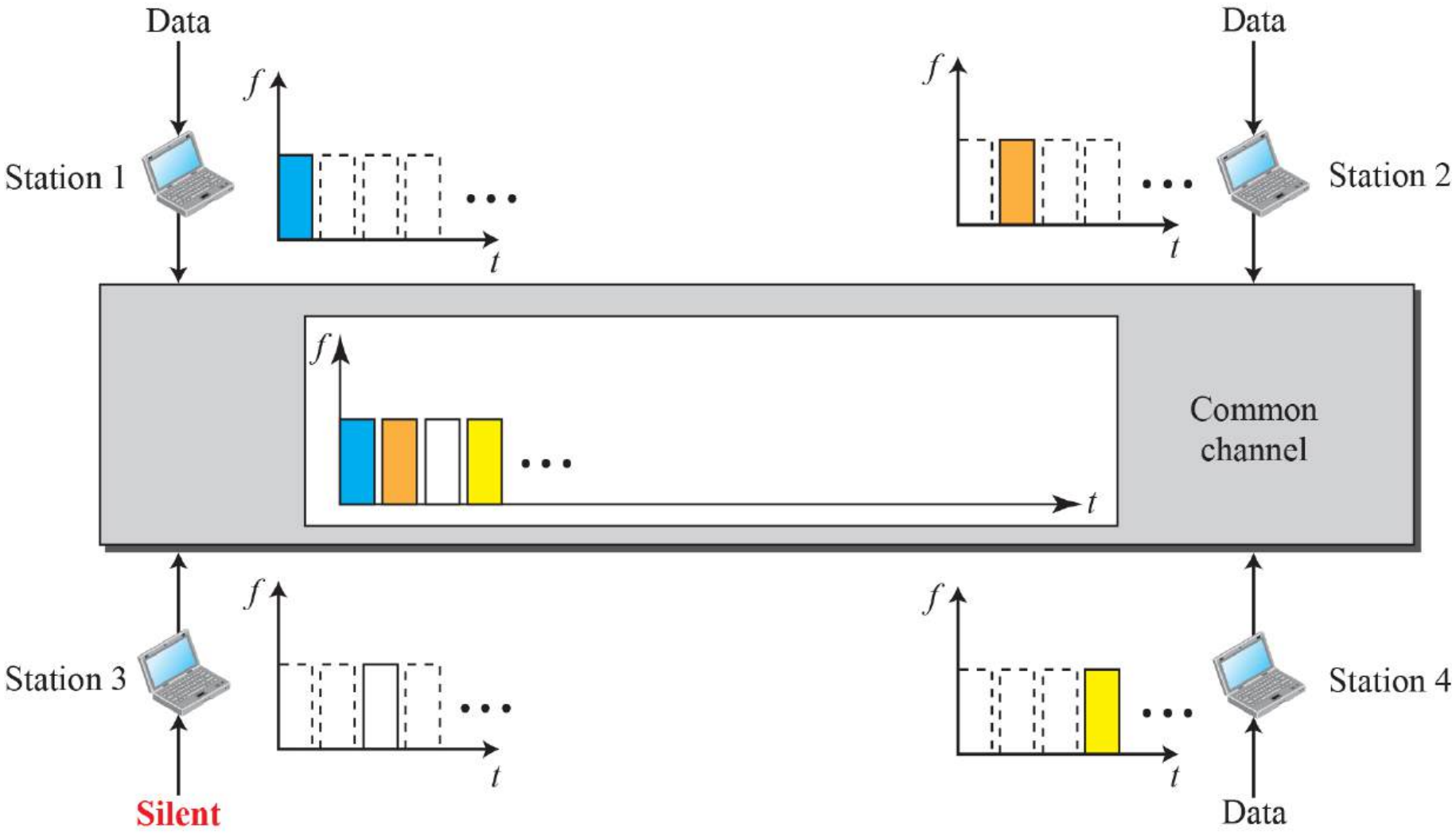

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Figure 6.28: Simple idea of communication with code

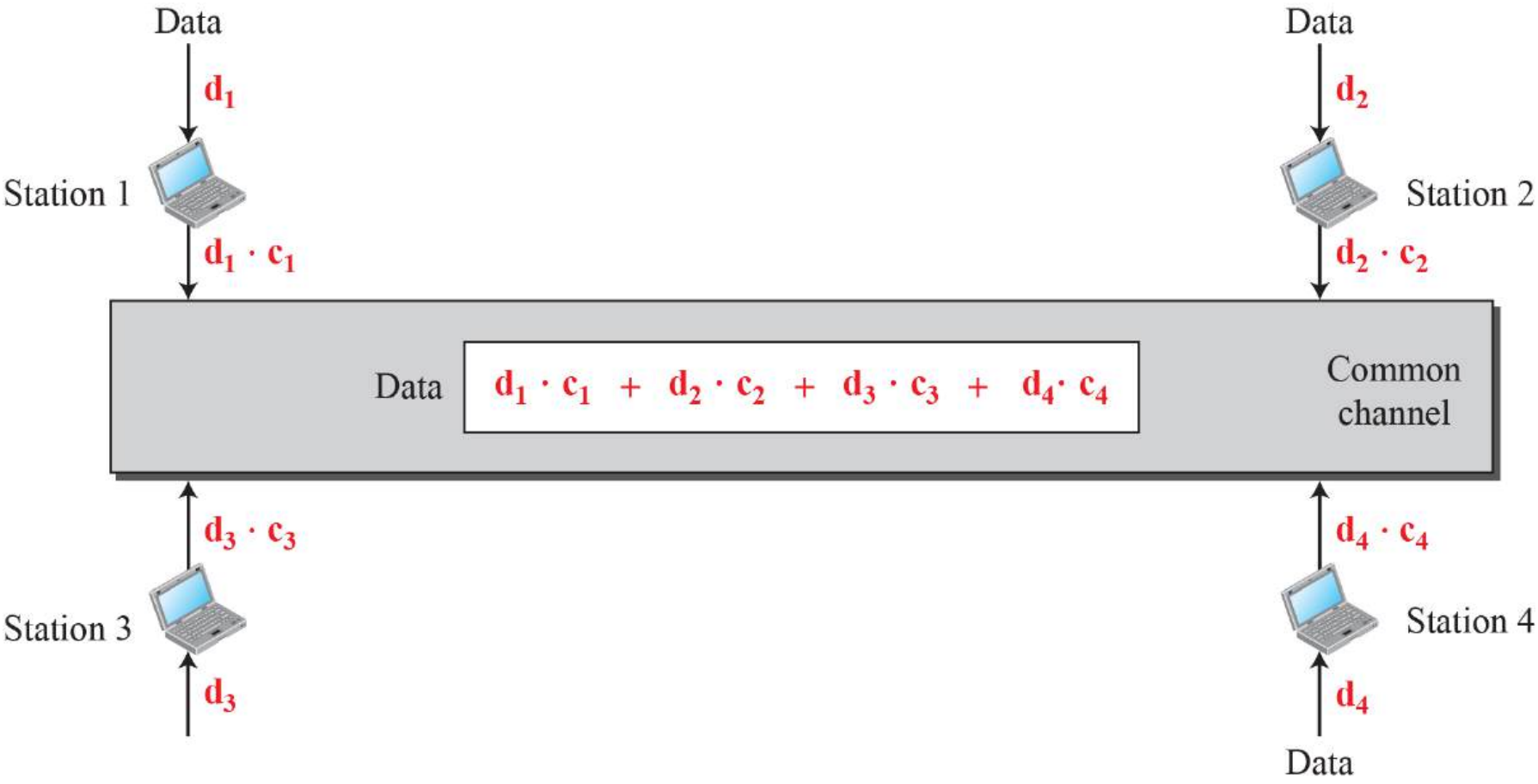

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Figure 6.29: Chip sequences

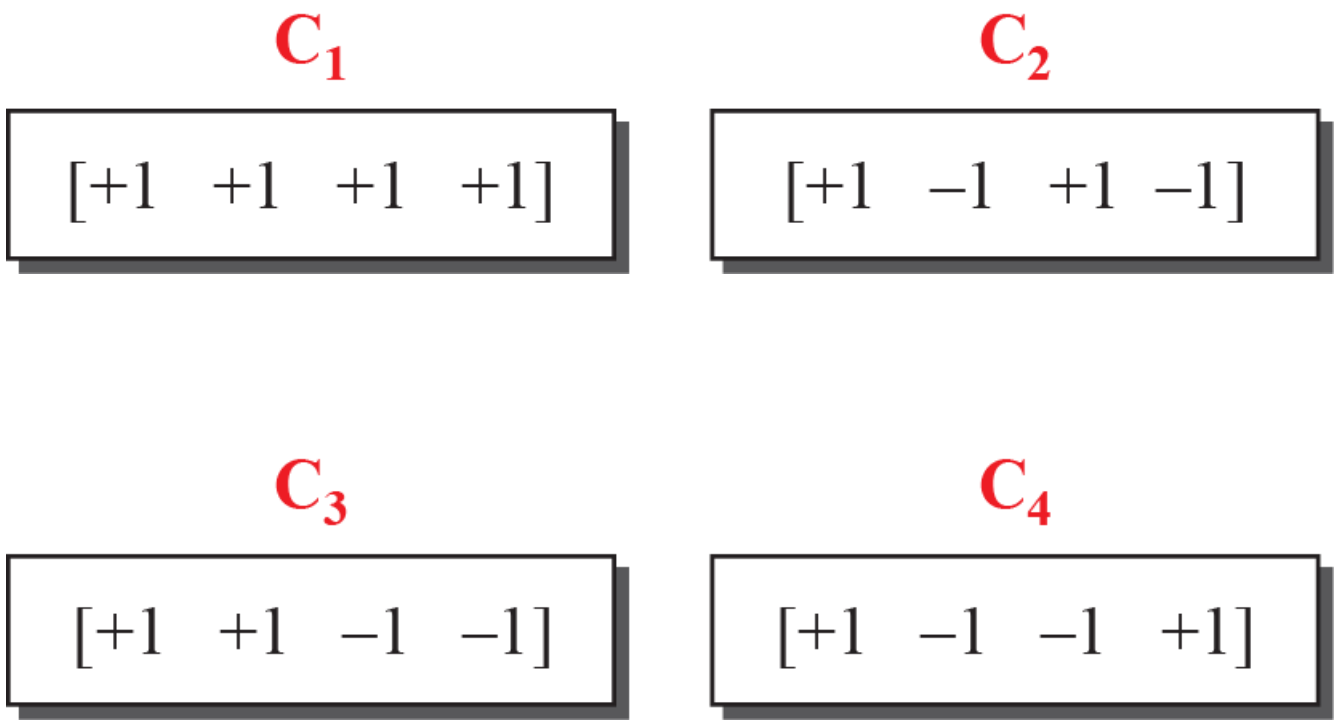

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Figure 6.30: Data representation in CDMA

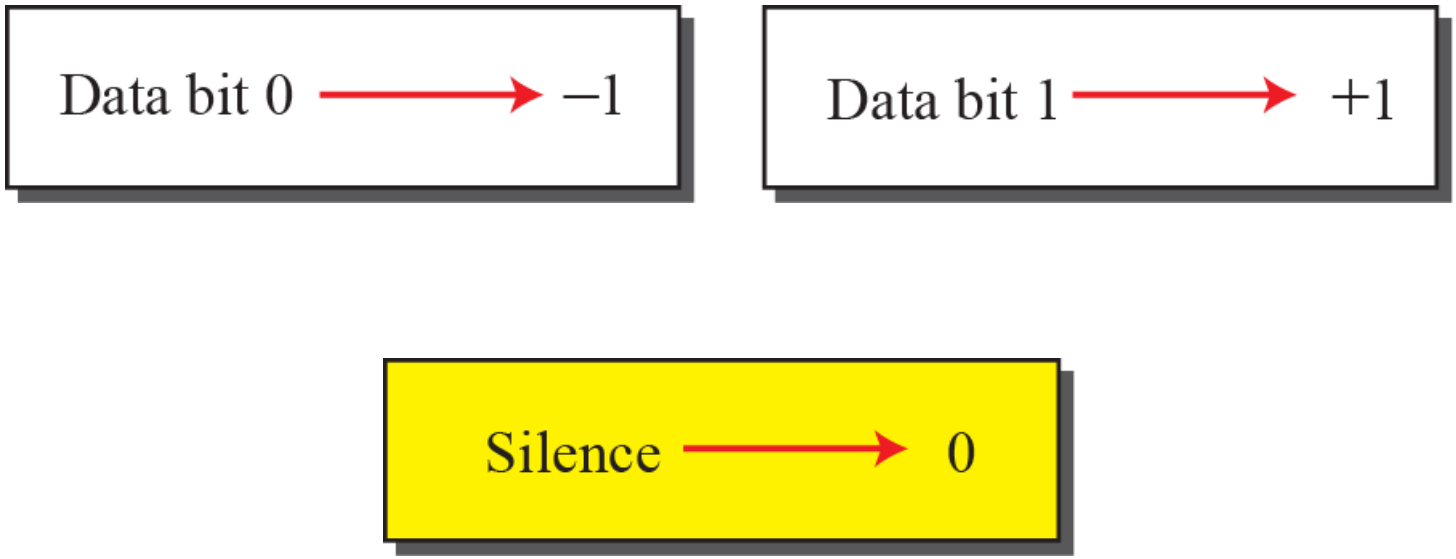

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Figure 6.31: Sharing channel in CDMA

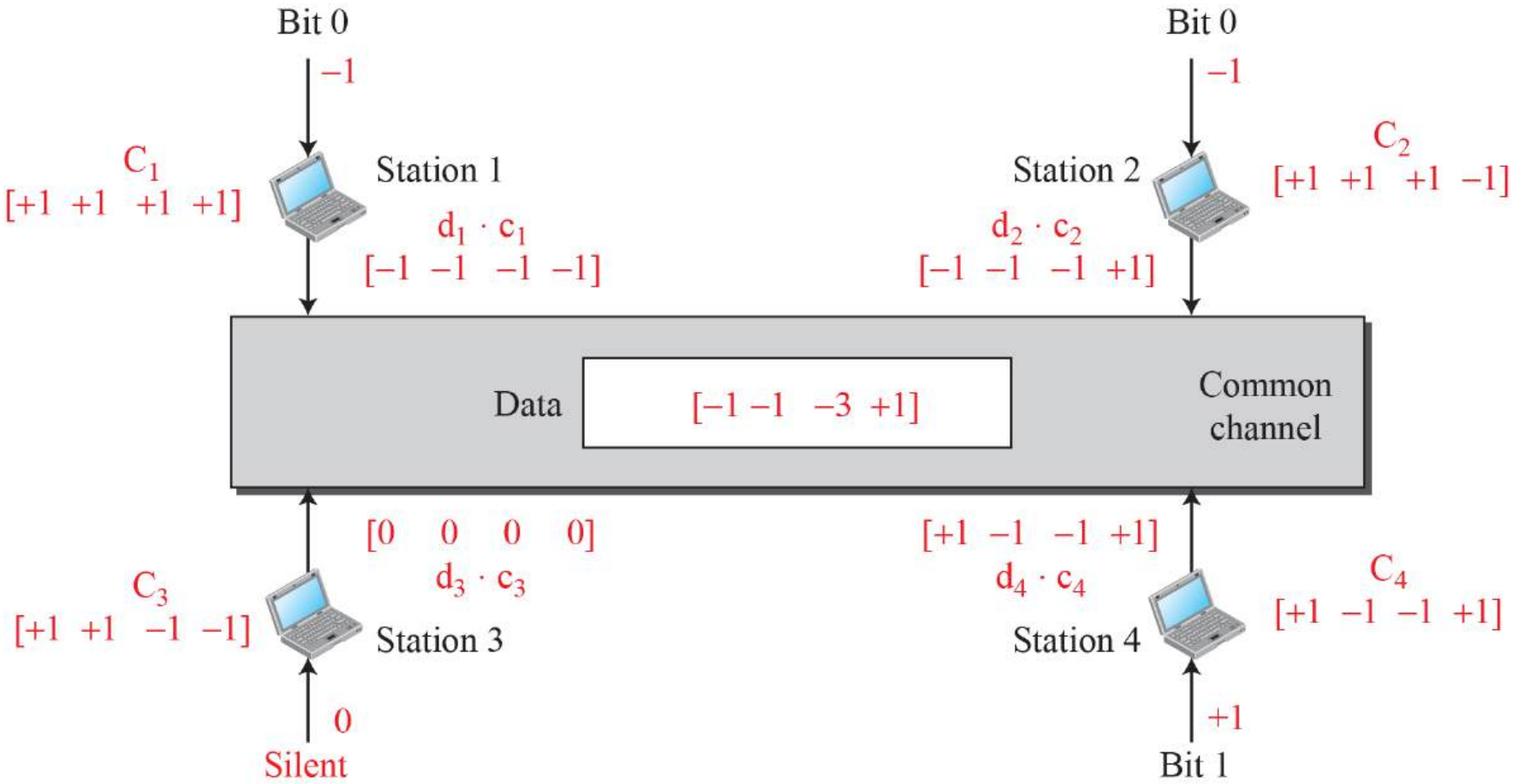

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Figure 6.32: Digital signal created by four stations in CDMA

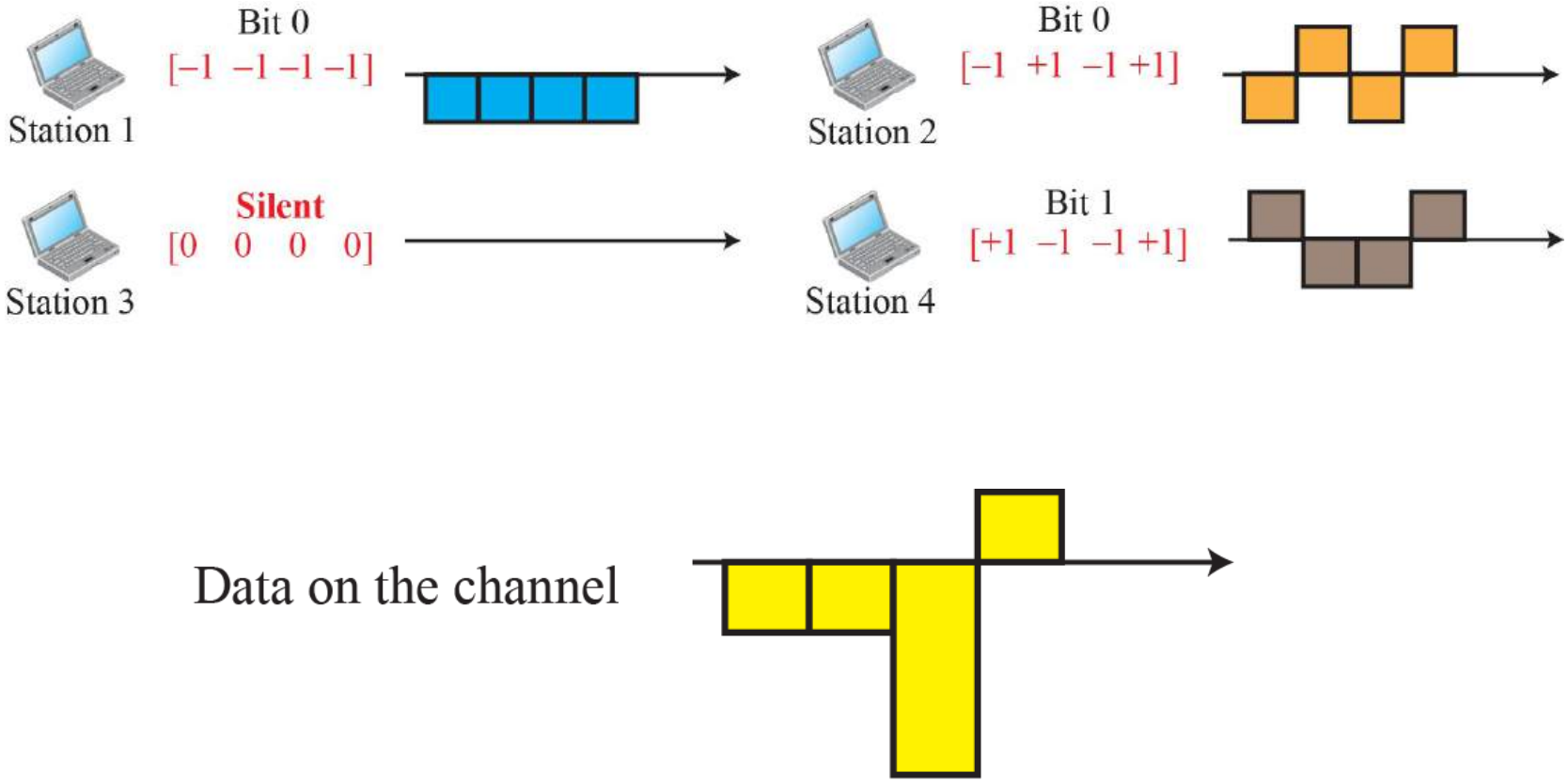

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Figure 6.33: Decoding of the composite signal for one in CDMA

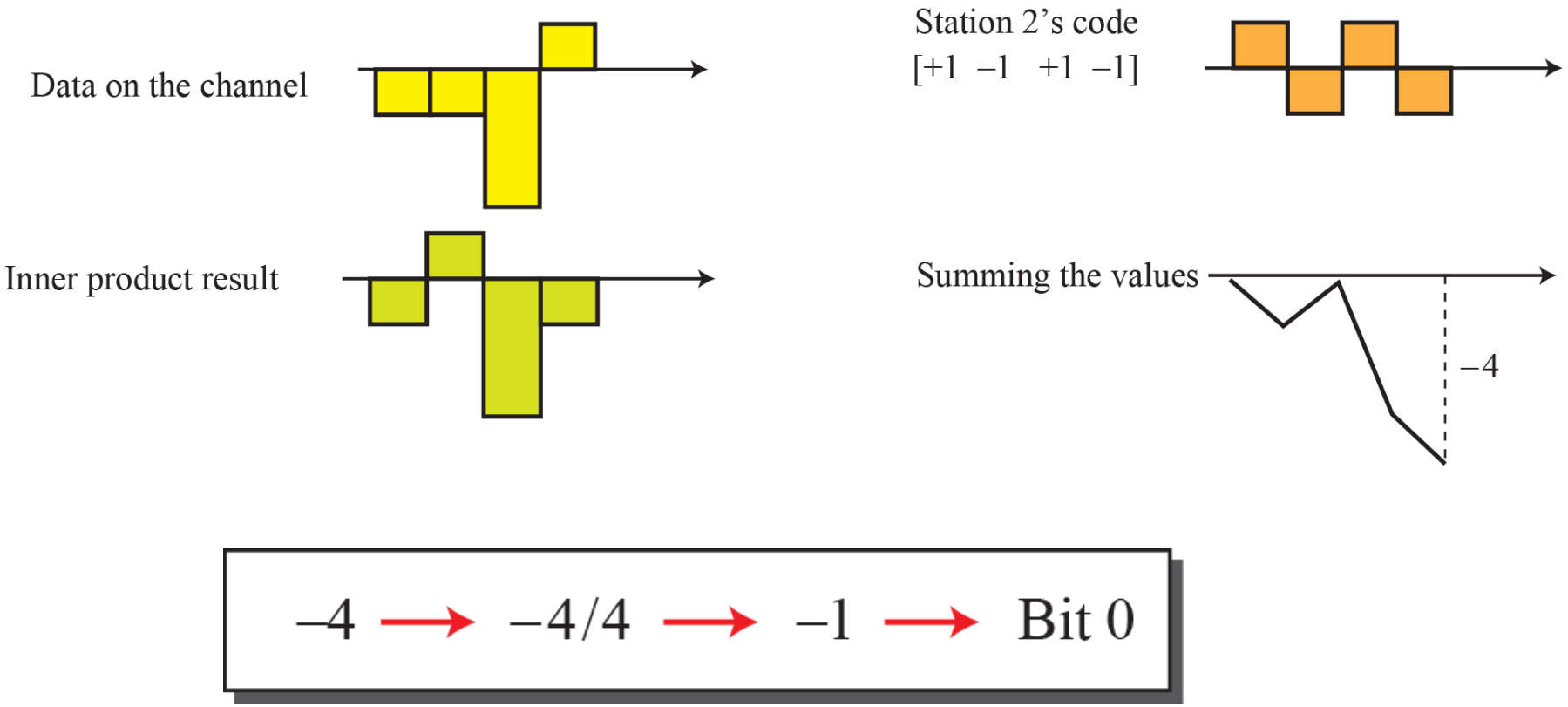

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Figure 6.34: General rules and examples of creating Walsh tables

$$W_1 = [+1] \qquad W_{2N} = \begin{bmatrix} W_N & W_N \\ W_N & \overline{W_N} \end{bmatrix}$$

a. Two basic rules

$$W_2 = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix} \qquad W_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$

b. Generation of W_1, W_2, and W_4

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Example 6.1

Find the chips for a network with

a. Two stations

b. Four stations

Solution

We can use the rows of W_2 and W_4 in Figure 6.34:

a. For a two-station network, we have [+1 +1] and [+1 −1].

b. For a four-station network we have [+1 +1 +1 +1], [+1 −1 +1 −1], [+1 +1 −1 −1], and [+1 −1 −1 +1].

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Example 6.2

What is the number of sequences if we have 90 stations in our network?

Solution

The number of sequences needs to be 2^m. We need to choose $m = 7$ and $N = 2^7$ or 128. We can then use 90 of the sequences as the chips.

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Example 6.3

Prove that a receiving station can get the data sent by a specific sender if it multiplies the entire data on the channel by the sender's chip code and then divides it by the number of stations.

Solution

Let us prove this for the first station, using our previous four-station example. We can say that the data on the channel $D = (d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4)$. The receiver, which wants to get the data sent by station 1, multiplies these data by c_1.

$$
\begin{aligned}
[D \cdot c_1] / 4 &= [(d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4) \cdot c_1] / 4 \\
&= [d_1 \cdot c_1 \cdot c_1 + d_2 \cdot c_2 \cdot c_1 + d_3 \cdot c_3 \cdot c_1 + d_4 \cdot c_4 \cdot c_1] / 4 \\
&= [d_1 \times 4 + d_2 \times 0 + d_3 \times 0 + d_4 \times 0] / 4 = [d_1 \times 4] / 4 = d_1
\end{aligned}
$$

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6.2.2 Cellular Telephony

Cellular telephony is designed to provide communications between two moving units, called mobile stations (MSs), or between one mobile unit and one stationary unit, often called a land unit. A service provider must be able to locate and track a caller, assign a channel to the call, and transfer the channel from base station to base station as the caller moves out of range.

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6.2.2 (continued)

- ***Frequency-Reuse Principle***
- ***Transmitting***
- ***Receiving***
  - *Handoff*
- ***Roaming***
- ***First Generation (1G)***
  - *AMPS*

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6.2.2 (continued)

- **Second Generation (2G)**
  - D-AMPS
  - GSM
  - IS-95
- **Third Generation (3G)**
  - IMT-2000 Radio Interface
- **Fourth Generation (4G)**
  - Access Scheme
  - Modulation
  - Radio System
  - Antenna
  - Applications

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Figure 6.35: Cellular system

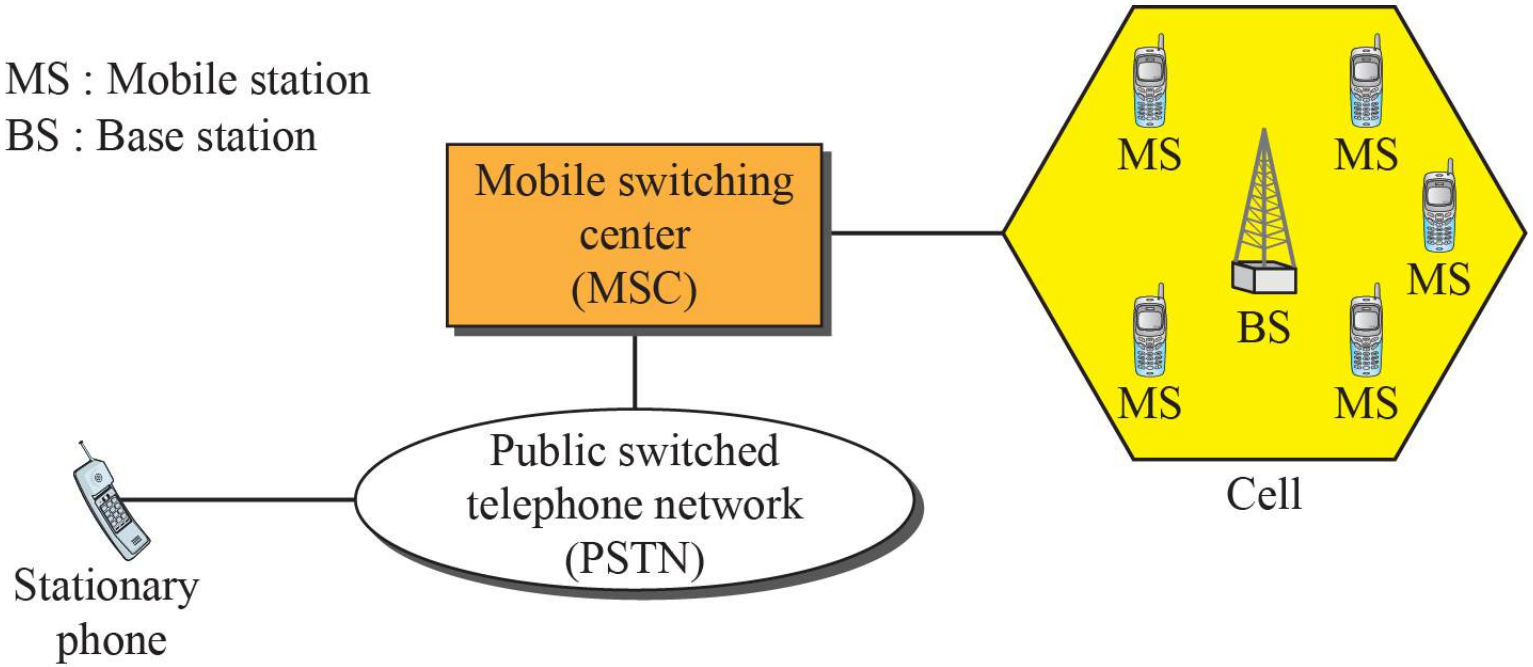

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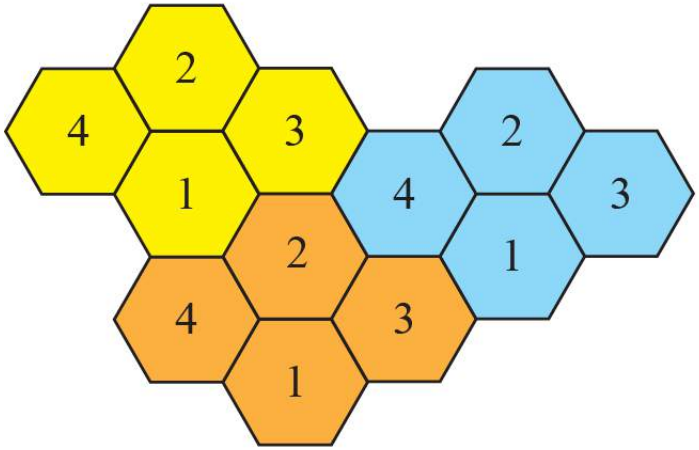
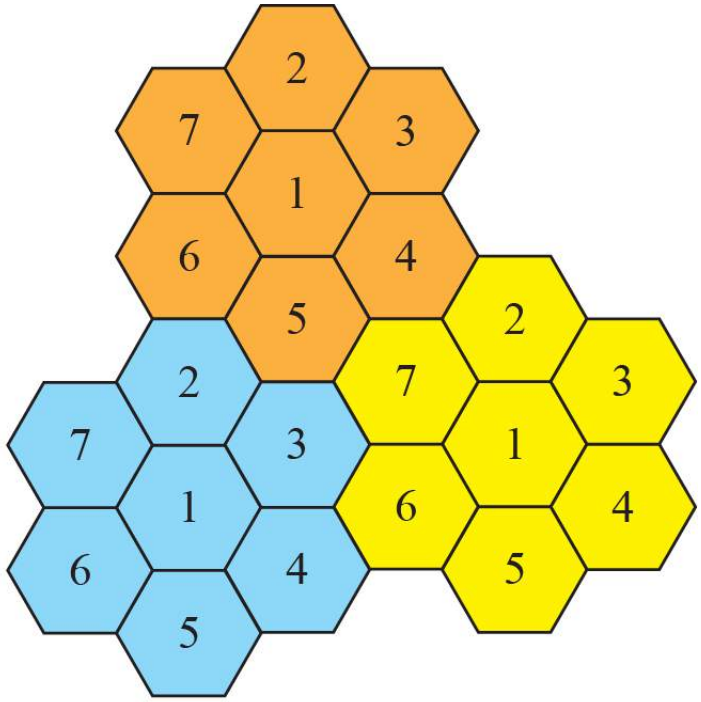
Figure 6.36: Frequency reuse patterns



a. Reuse factor of 4



b. Reuse factor of 7

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Figure 6.37: Cellular bands for AMPS

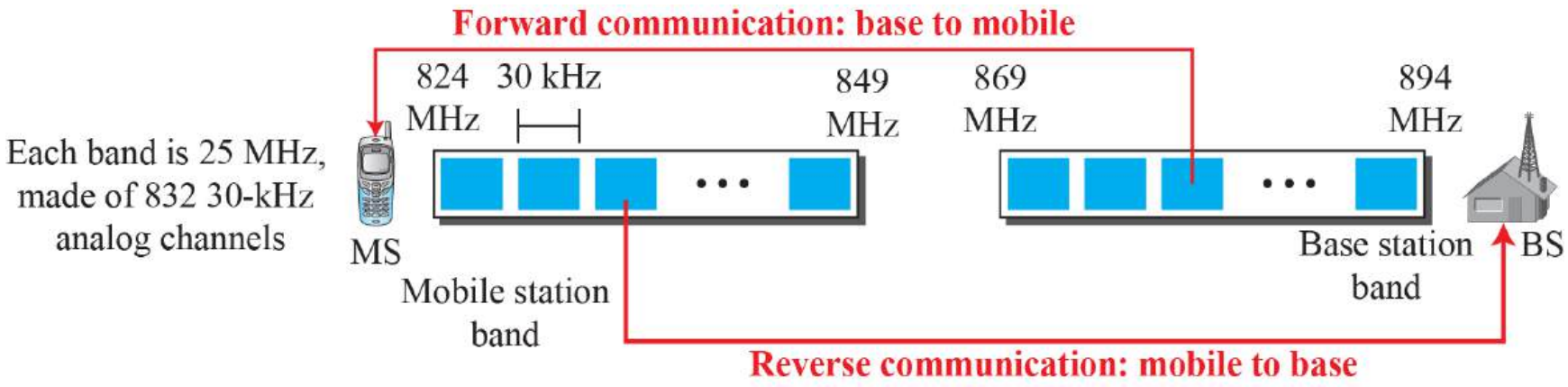

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Figure 6.38: AMPS reverse communication band

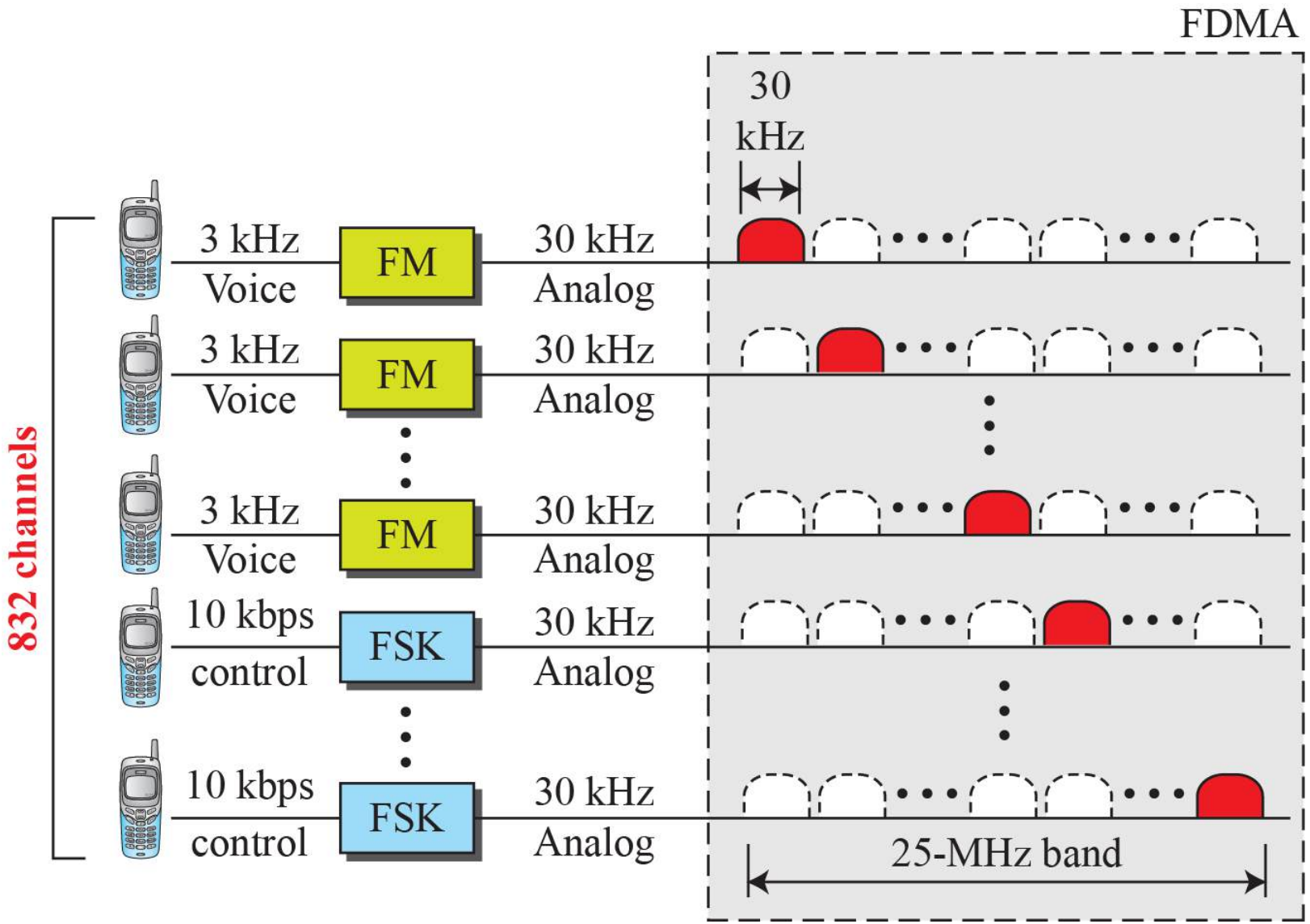

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Figure 6.39: D-AMPS

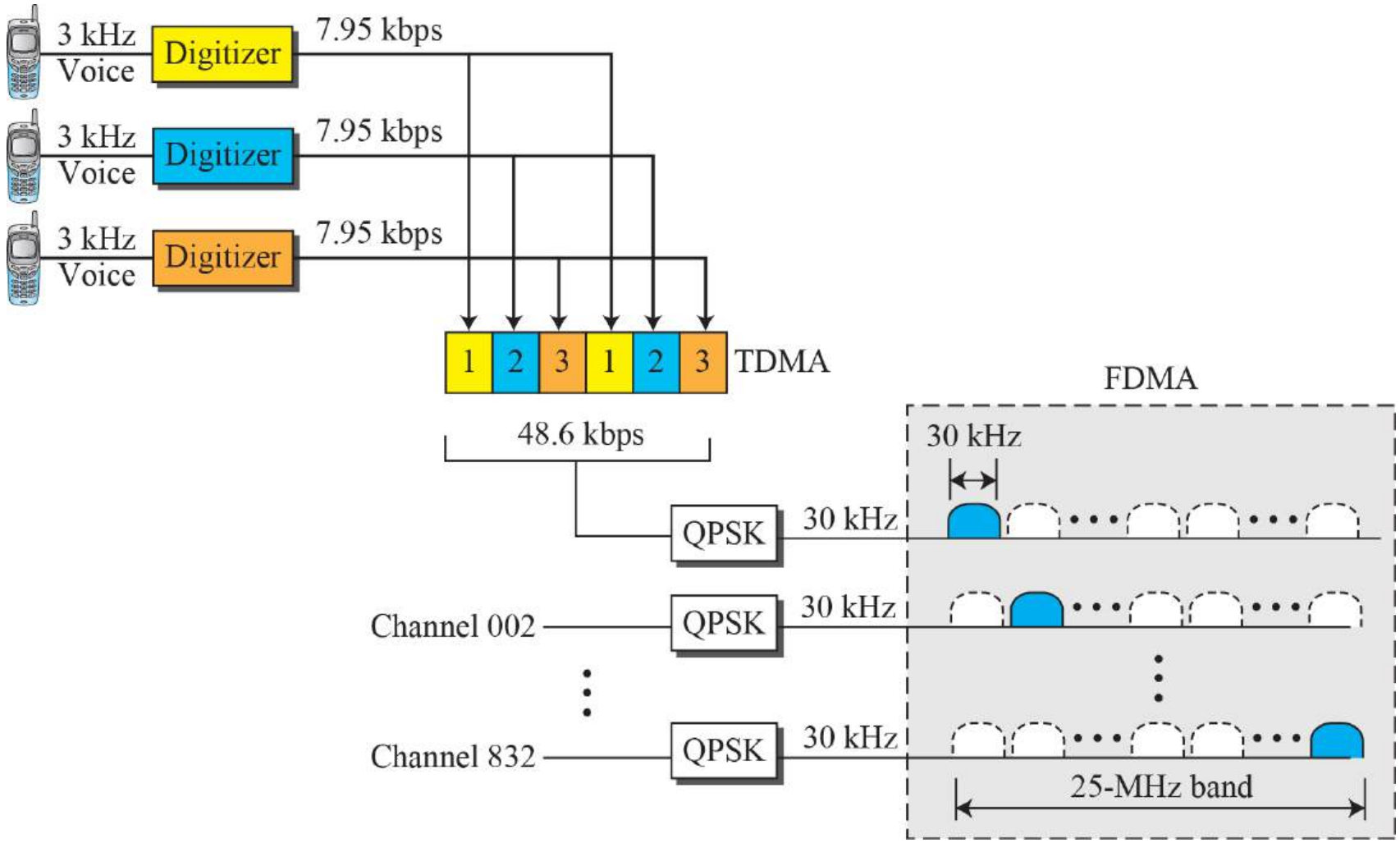

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Figure 6.40: ***GSM bands***

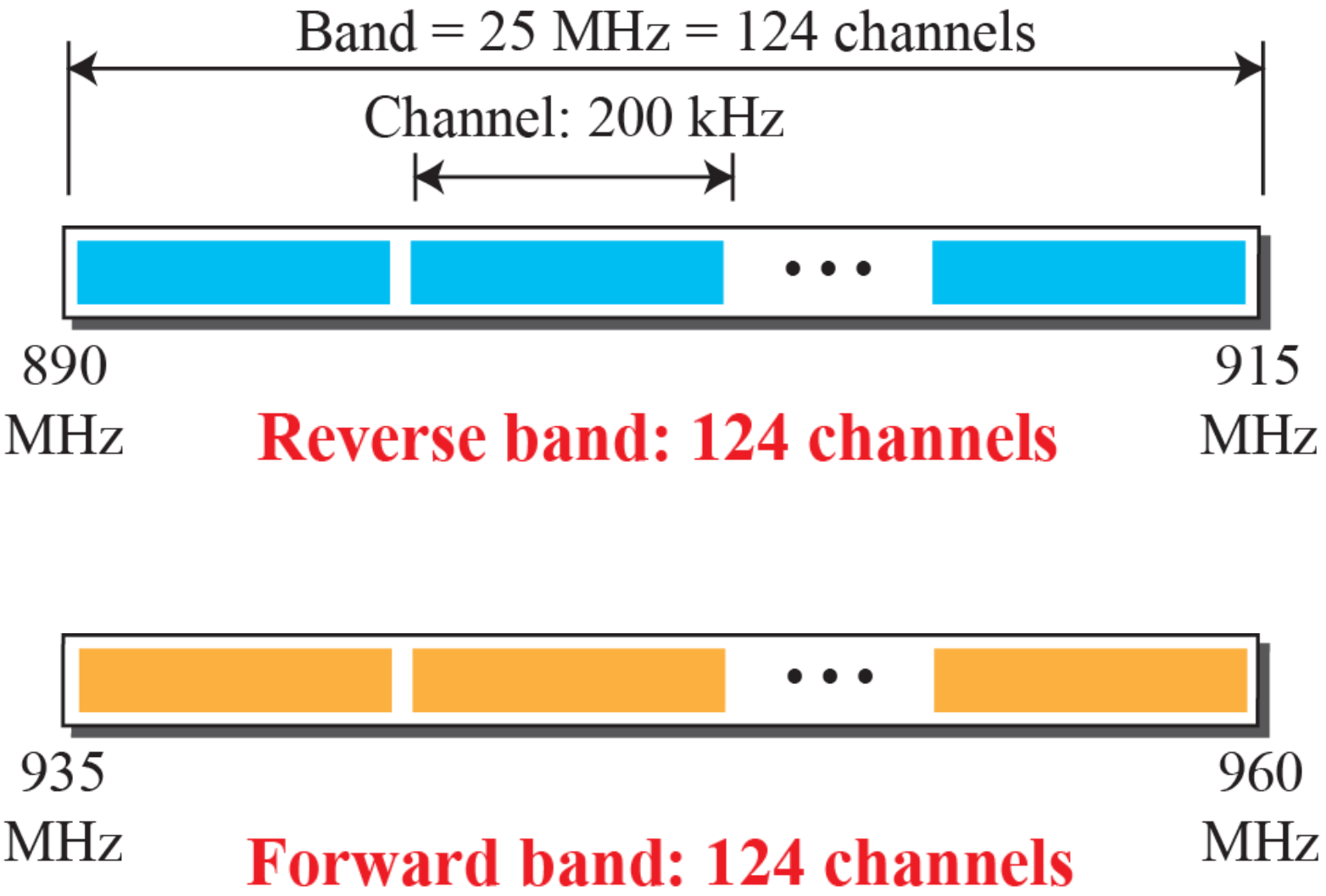

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Figure 6.41: GSM

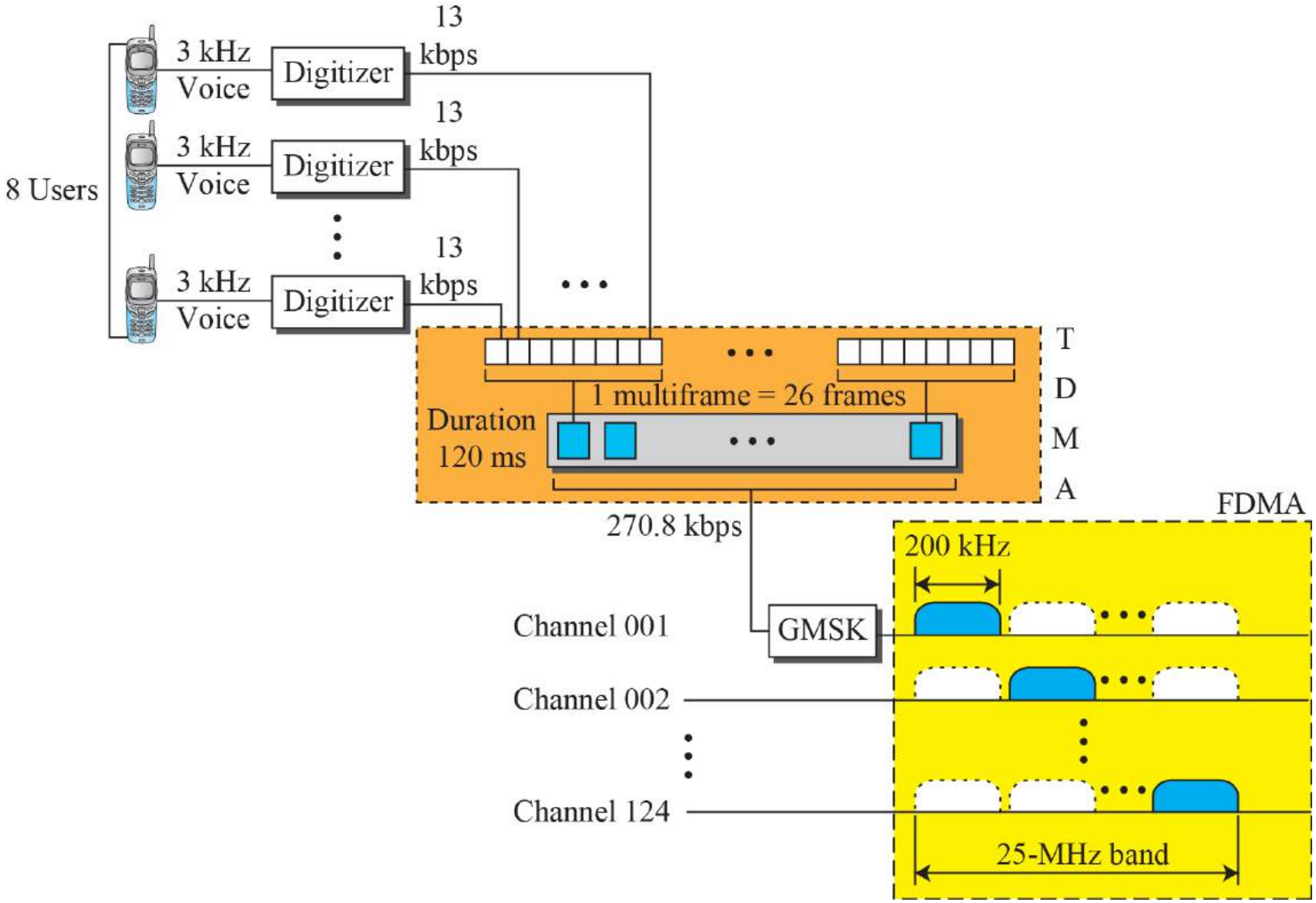

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Figure 6.42: Multiframe components

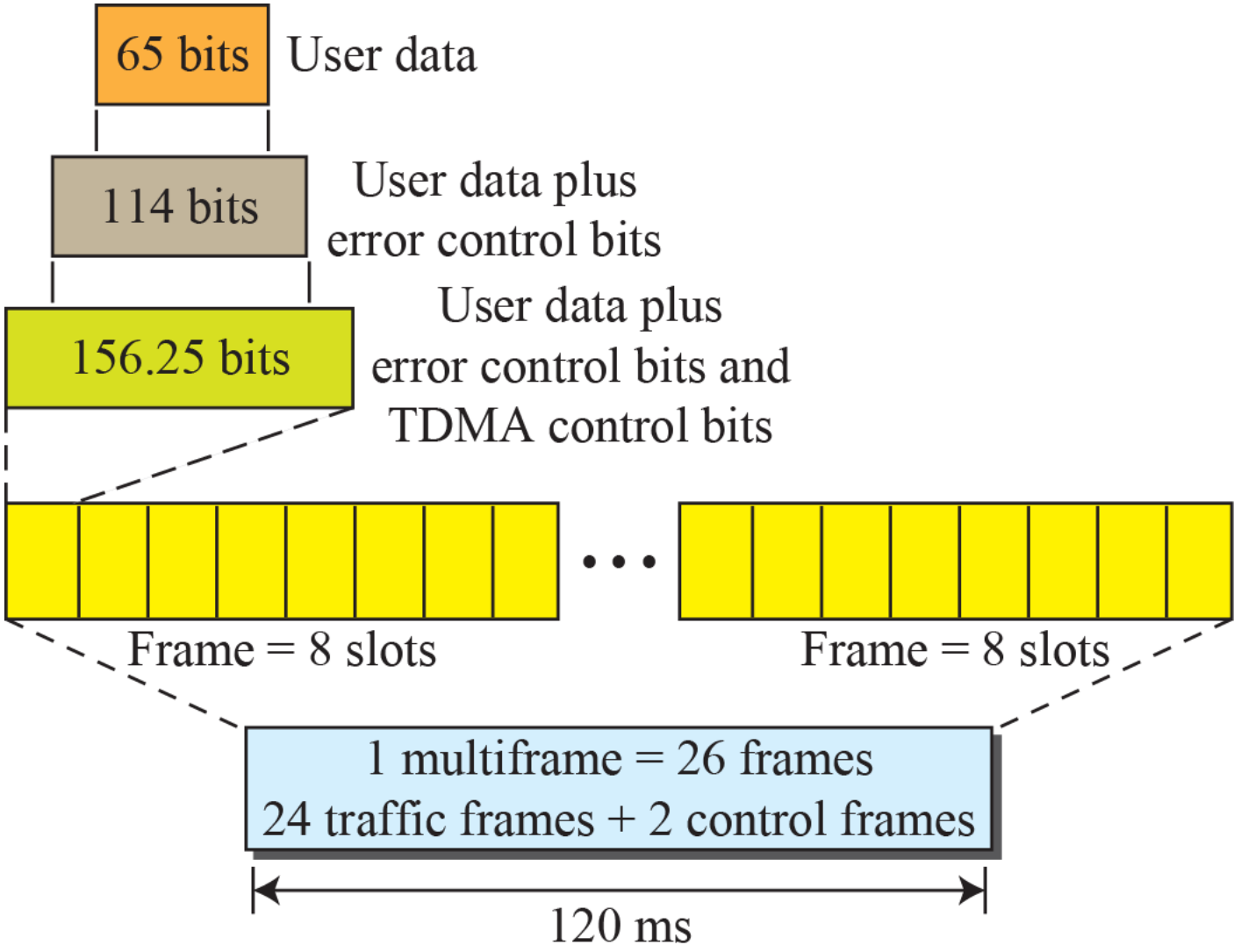

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Figure 6.43: IS-95 forward transmission

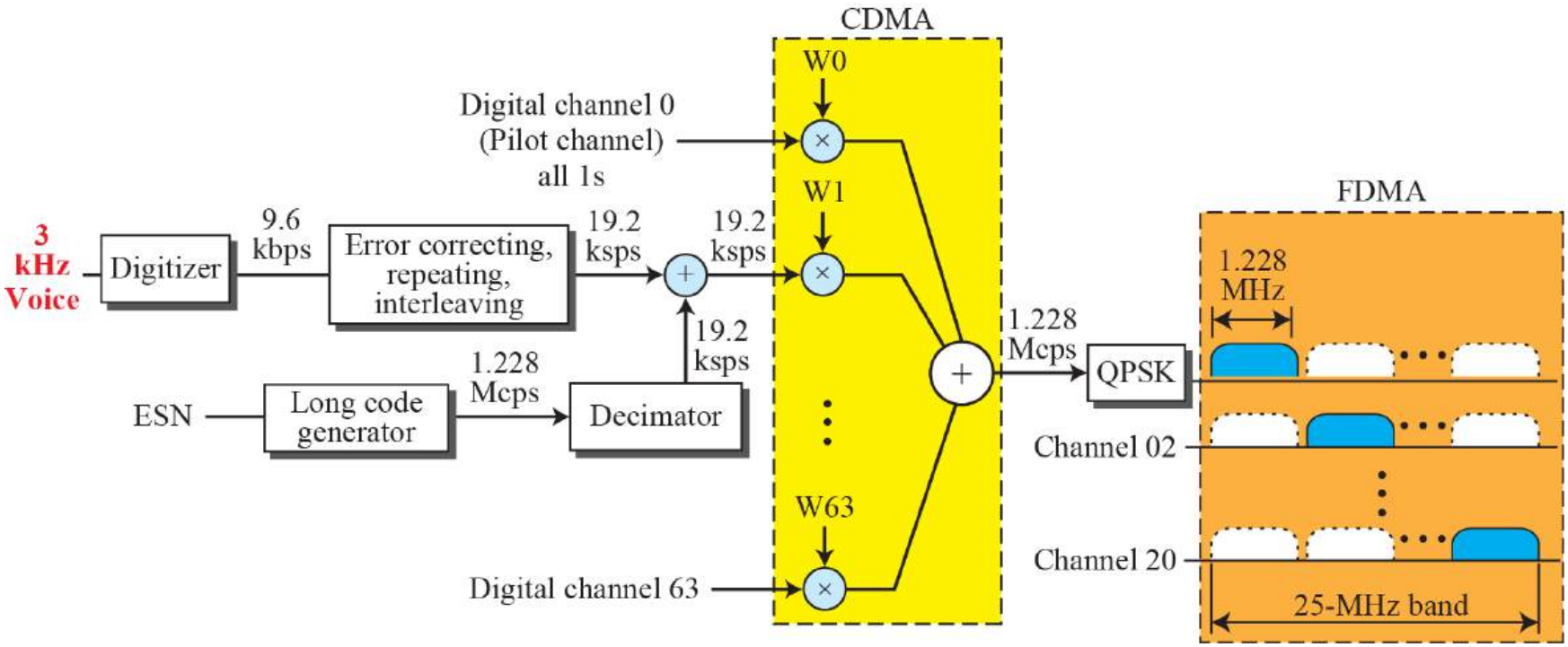

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Figure 6.44: S-95 reverse transmission

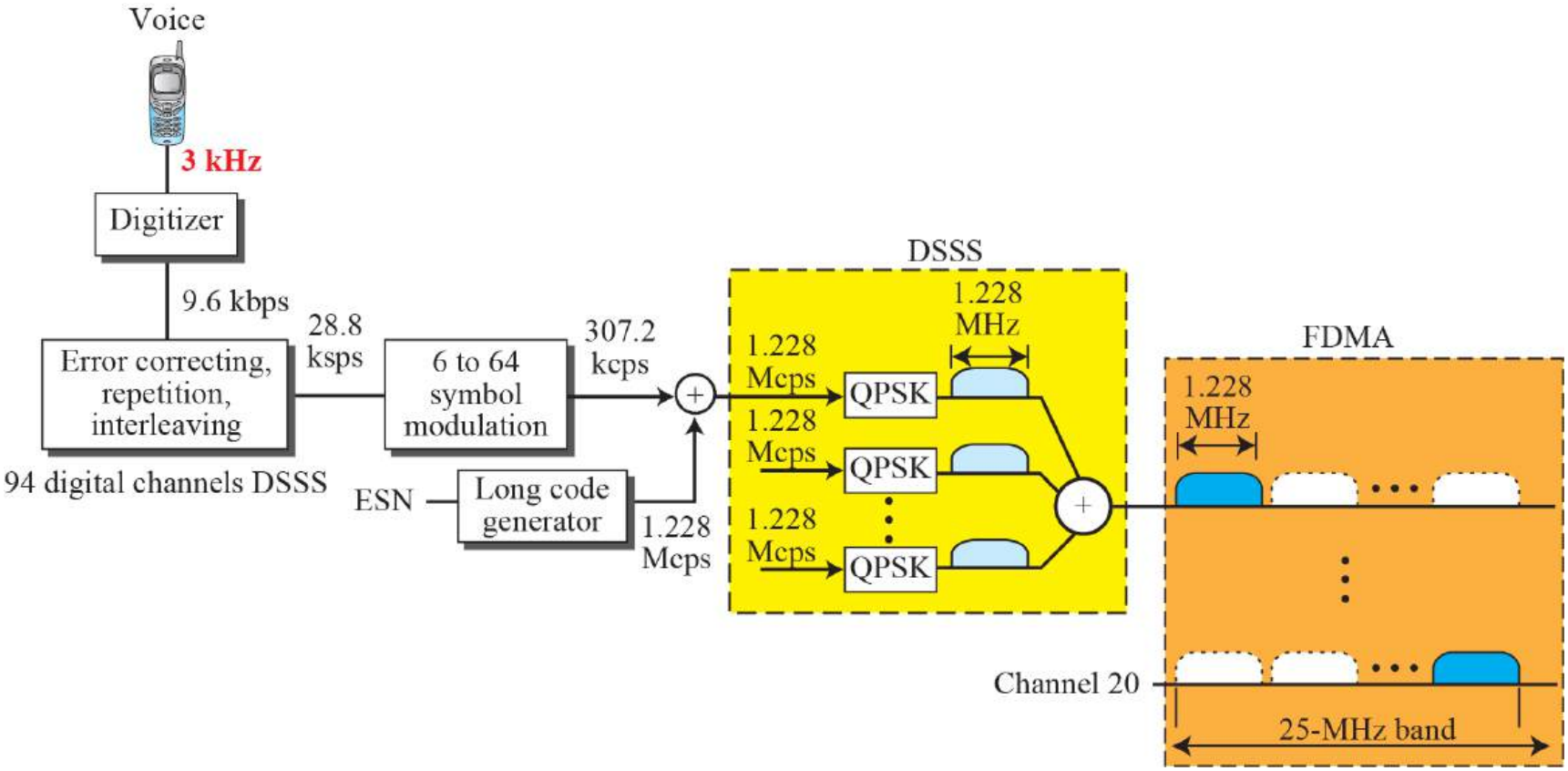

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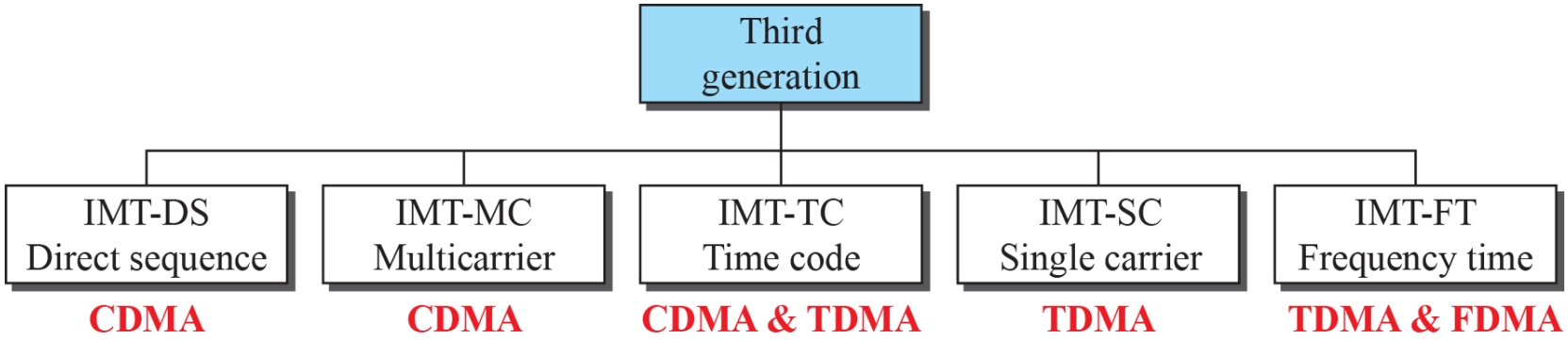
Figure 6.45: IMT-2000 radio interfaces



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6.2.3 Satellite Networks

A satellite network is a combination of nodes, some of which are satellites, that provides communication from one point on the Earth to another. A node in the network can be a satellite, an Earth station, or an end-user terminal or telephone.

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6.2.3 (continued)

- **Orbits**
- **Footprint**
- **Three Categories of Satellites**
  - Frequency Bands for Satellite Communication
- **GEO Satellites**
- **MEO Satellites**
  - Global Positioning System (GPS)
- **LEO Satellites**

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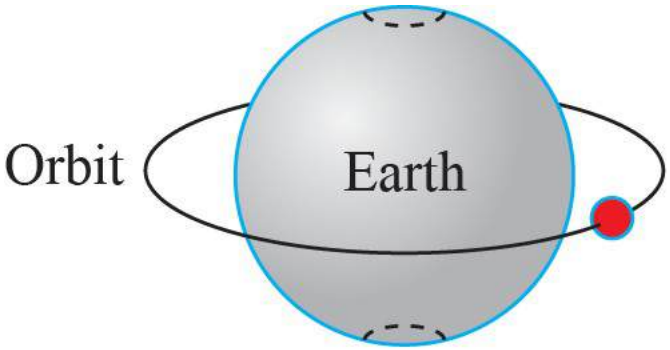
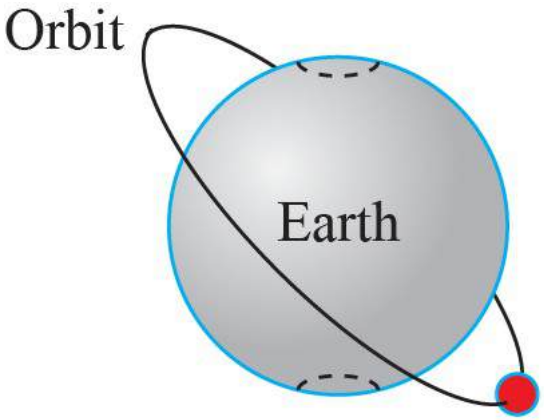
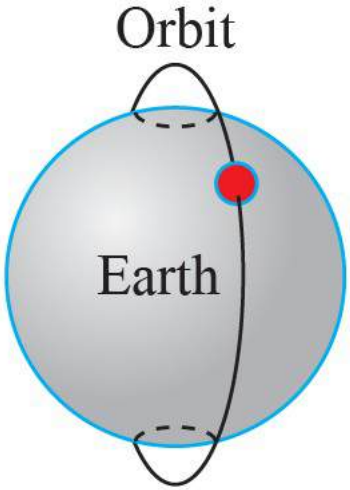
Figure 6.46: ***Satellite orbits***



a. Equatorial-orbit satellite



b. Inclined-orbit satellite



c. Polar-orbit satellite

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Example 6.4

What is the period of the moon, according to Kepler's law?

$$\text{Period} = C \times \text{distance}^{1.5}$$

Here C is a constant approximately equal to 1/100. The period is in seconds and the distance in kilometers.

Solution

The moon is located approximately 384,000 km above the Earth. The radius of the Earth is 6378 km. Applying the formula, we get the following.

$$\text{Period} = (1/100) \times (384{,}000 + 6378)^{1.5} = 2{,}439{,}090 \text{ s} = 1 \text{ month}$$

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Example 6.5

According to Kepler's law, what is the period of a satellite that is located at an orbit approximately 35,786 km above the Earth?

Solution

Applying the formula, we get the following.

$$\textbf{Period} = (1/100) \times (35{,}786 + 6378)^{1.5} = 86{,}579 \text{ s} = 24 \text{ h}$$

This means that a satellite located at 35,786 km has a period of 24 h, which is the same as the rotation period of the Earth. A satellite like this is said to be *stationary* to the Earth. The orbit, as we will see, is called a *geostationary orbit*.

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Figure 6.47: Satellite orbit altitudes

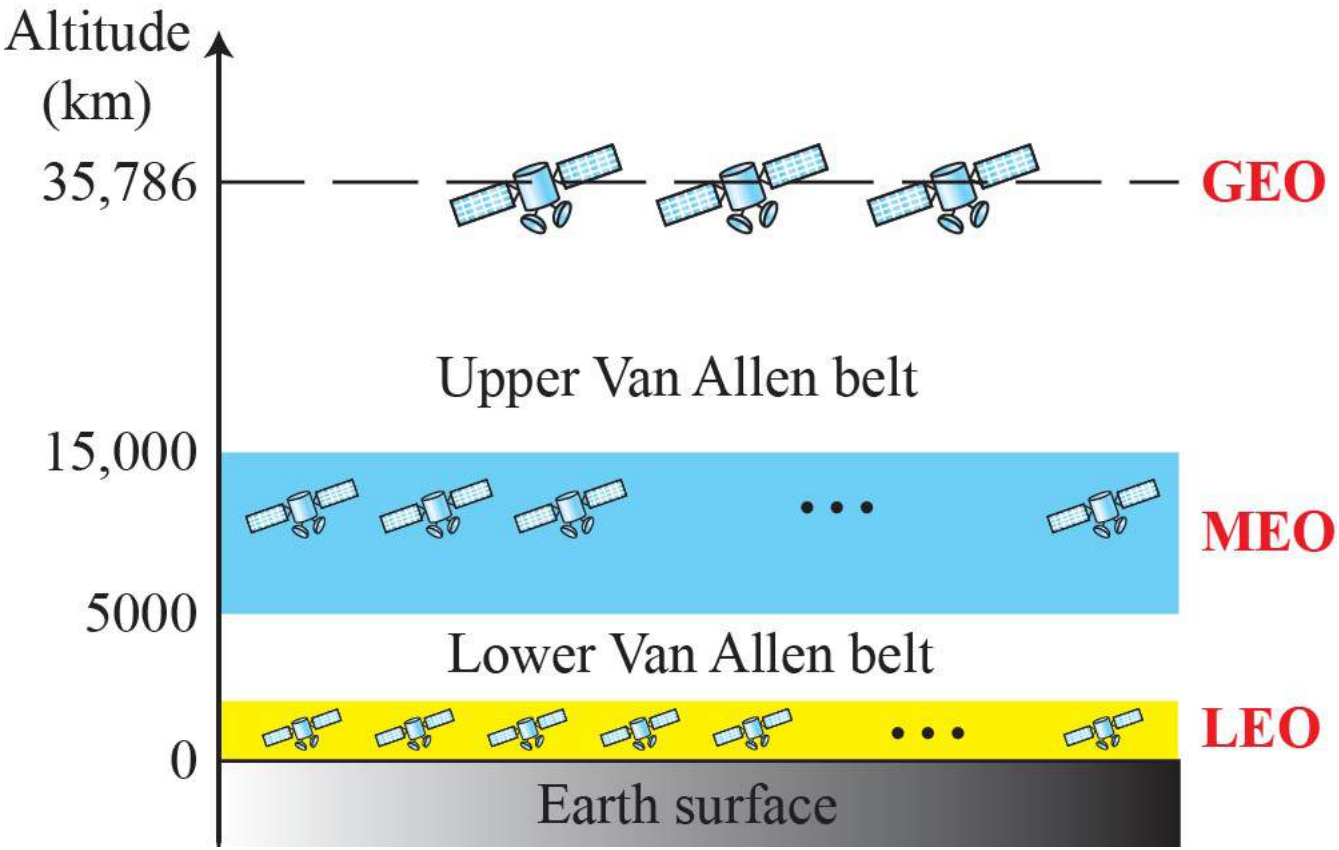

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Table 6.5: Satellite frequency bands

Band	*Downlink, GHz*	*Uplink, GHz*	*Bandwidth, MHz*
L	1.5	1.6	15
S	1.9	2.2	70
C	4.0	6.0	500
Ku	11.0	14.0	500
Ka	20.0	30.0	3500

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Figure 6.48: *Satellites in geostationary orbit*

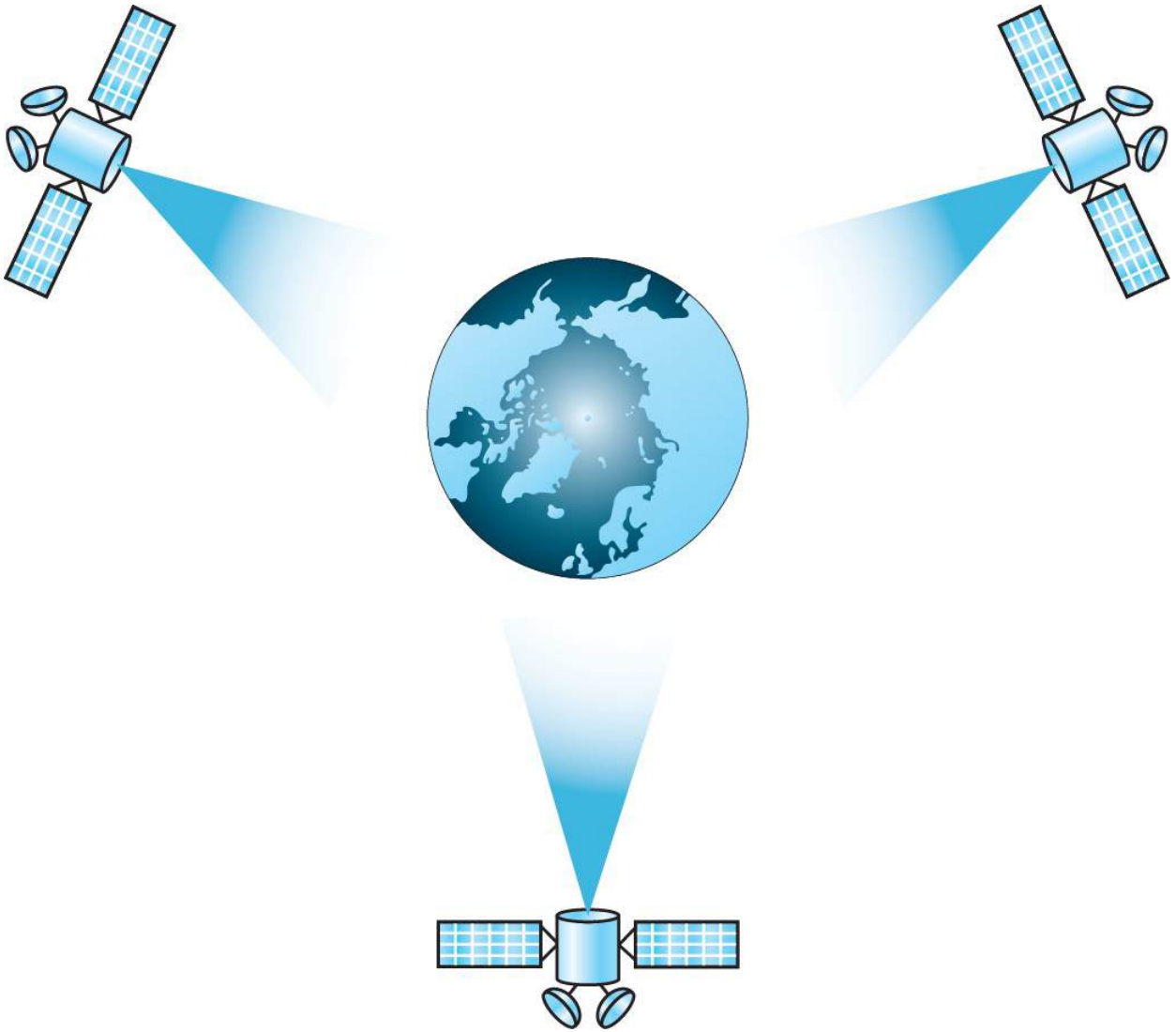

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Figure 6.49: ***Orbits for global positioning system (GPS) satellites***

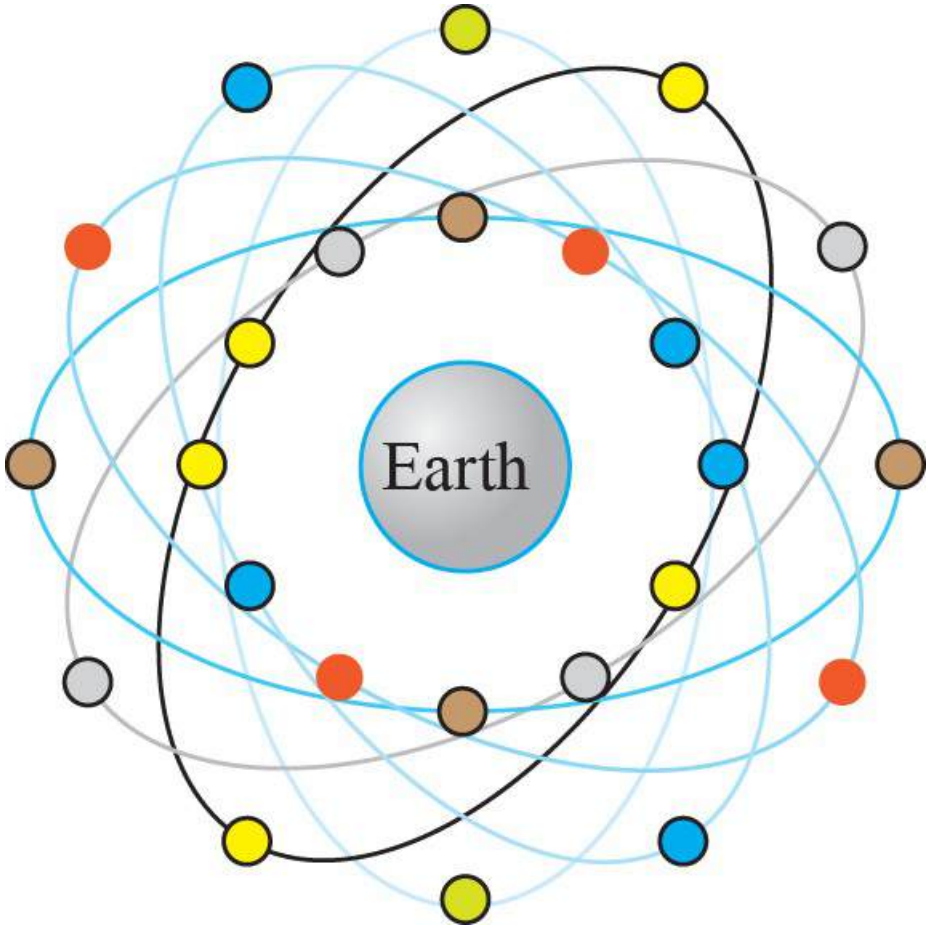

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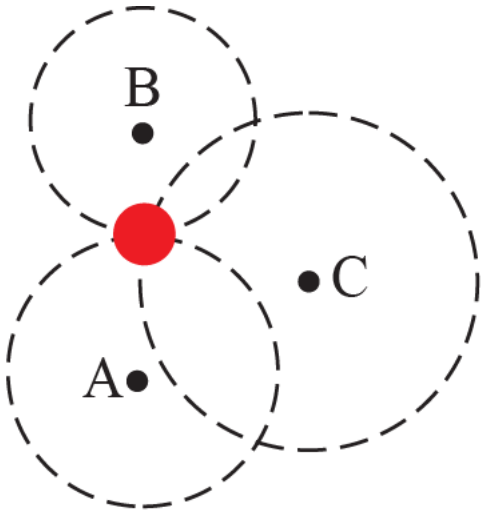
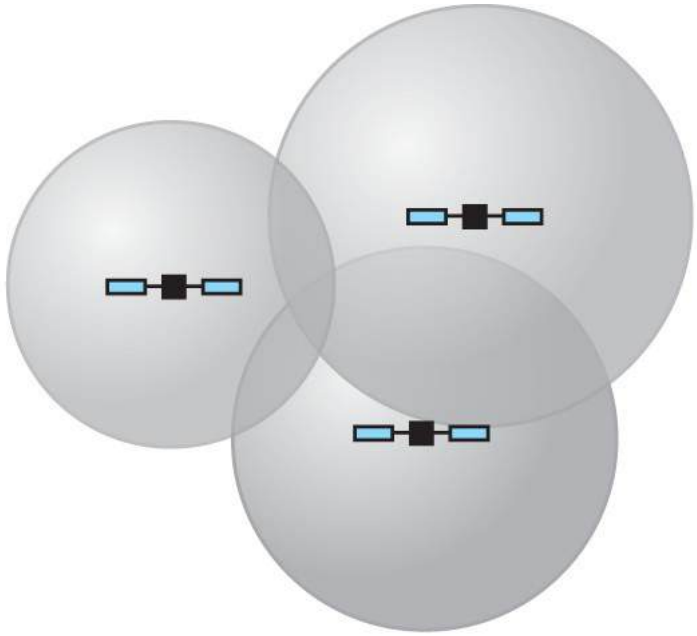
Figure 6.50: Trilateration on a plane



a. Two-dimensional trilateration



b. Three-dimensional trilateration

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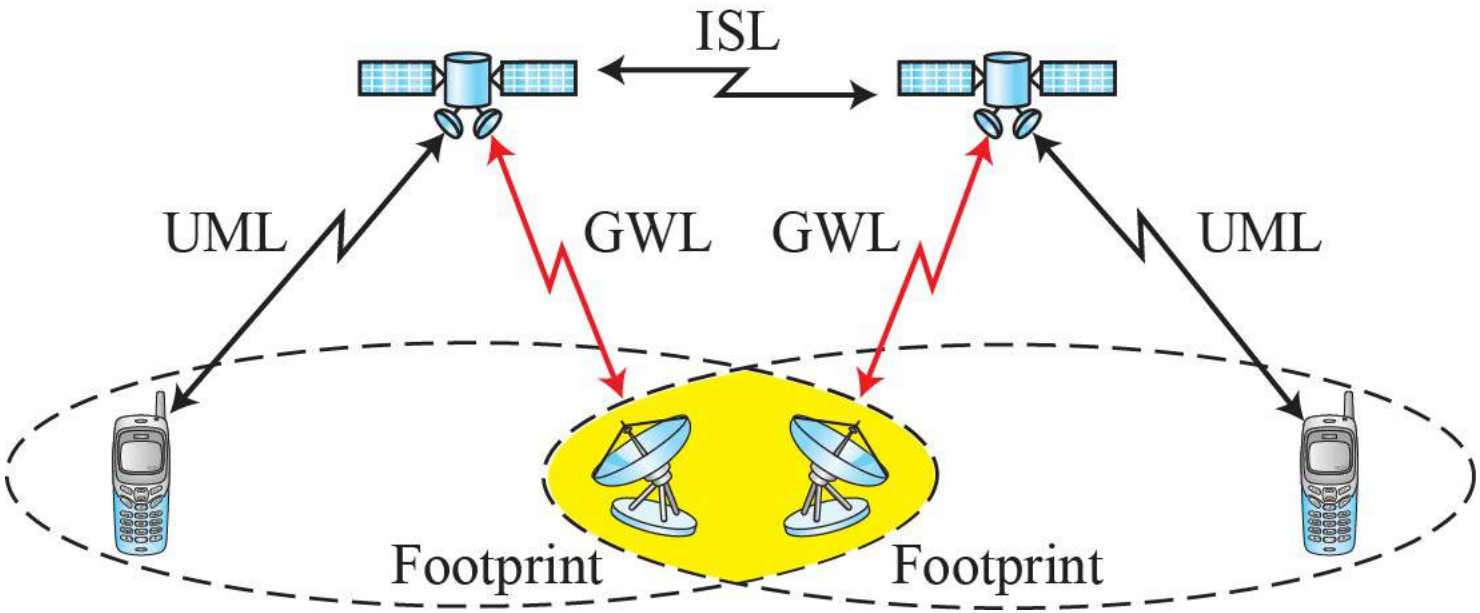
Figure 6.51: LEO satellite system



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6-3 MOBILE IP

As mobile and personal computers such as notebooks become increasingly popular, we need to think about mobile IP, the extension of IP protocol that allows mobile computers to be connected to the Internet at any location where the connection is possible. In this section, we discuss this issue.

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6.3.1 Addressing

The main problem that must be solved in providing mobile communication using the IP protocol is addressing.

- ***Stationary Hosts***
- ***Mobile Hosts***
  - *Changing the Address*
  - *Two Addresses*

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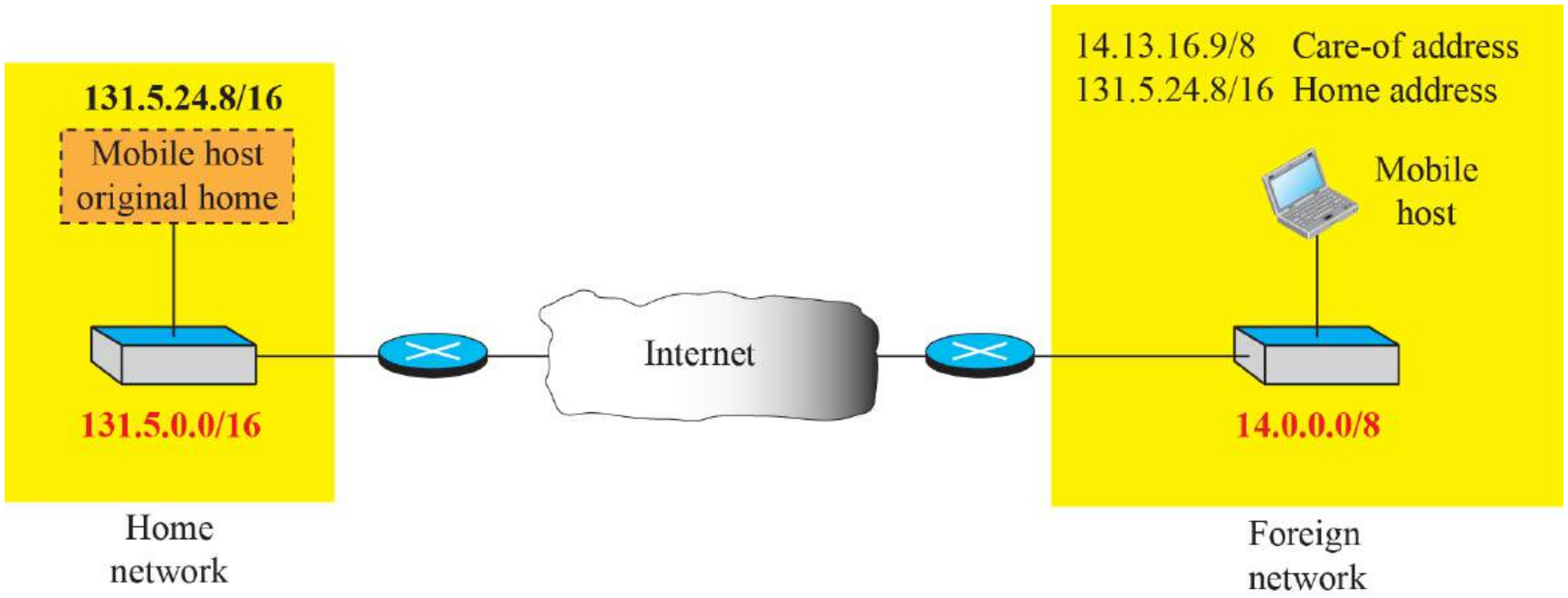
Figure 6.52: Home address and care-of address



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6.3.2 Agents

To make the change of address transparent to the rest of the Internet requires a home agent and a foreign agent. Figure 6.53 shows the position of a home agent relative to the home network and a foreign agent relative to the foreign network.

- **Home Agent**
- **Foreign Agent**

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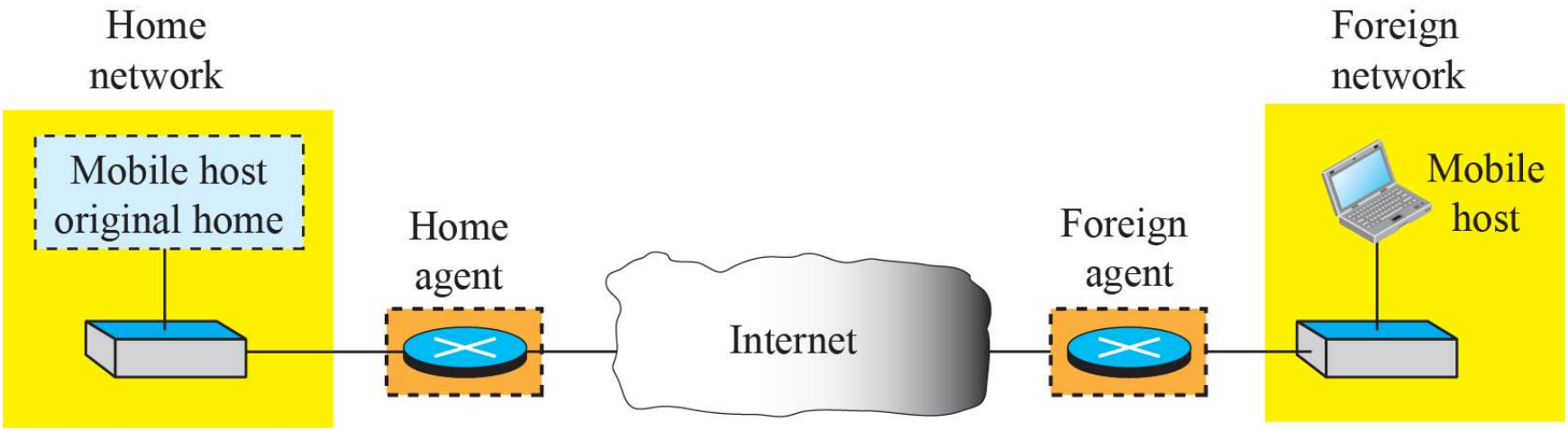
Figure 6.53: Home agent and foreign agent



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6.3.3 Three Phases

To communicate with a remote host, a mobile host goes through three phases: agent discovery, registration, and data transfer, as shown in Figure 6.54.

- **Agent Discovery**
  - Agent Advertisement
  - Agent Solicitation

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6.3.3 (continued)

- **Registration**
  - Request and Reply
  - Encapsulation
- **Data Transfer**
  - From Remote Host to Home Agent
  - From Home Agent to Foreign Agent
  - From Foreign Agent to Mobile Host
  - From Mobile Host to Remote Host
  - Transparency

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Figure 6.54: Remote host and mobile host communication

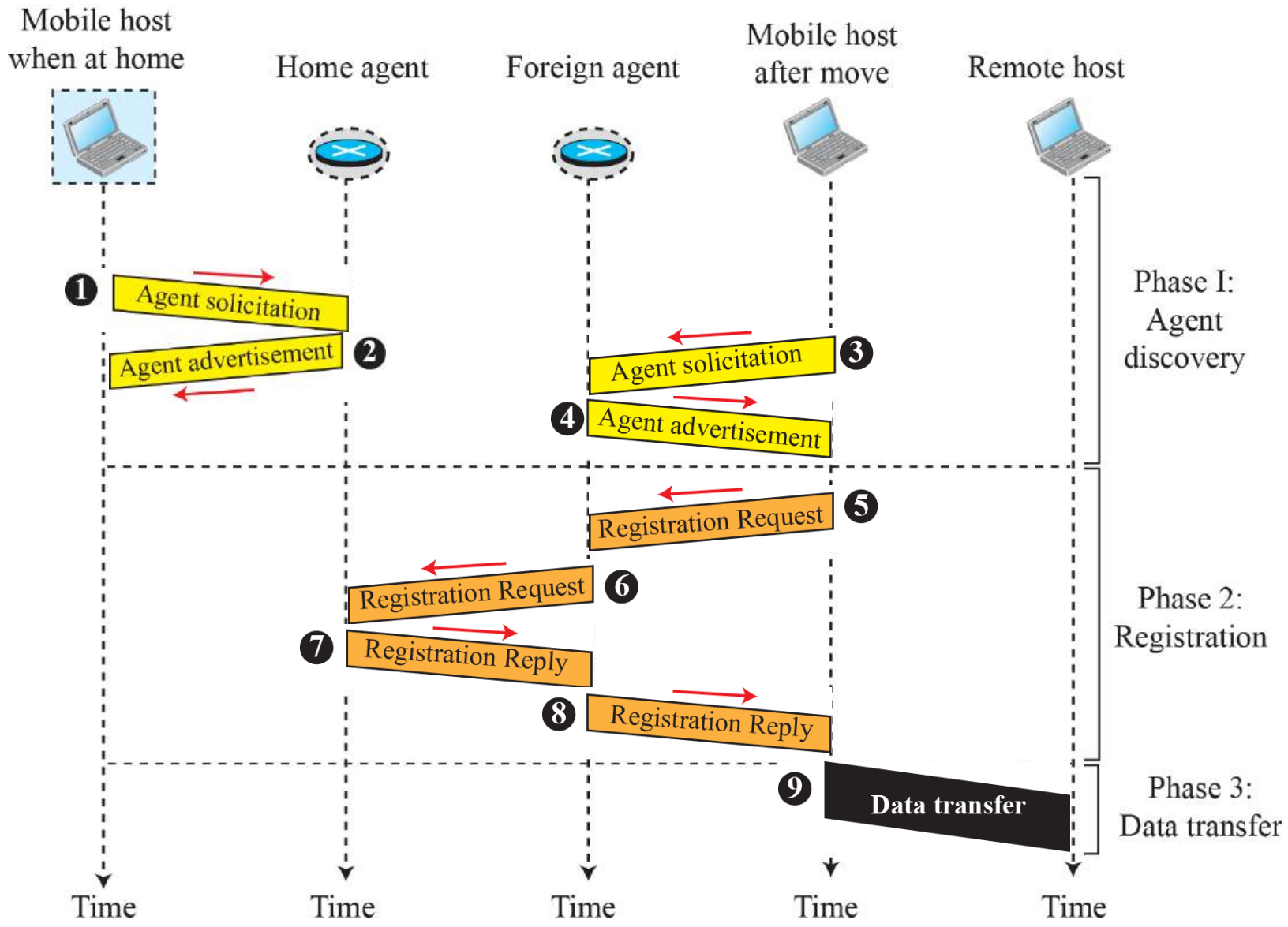

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Figure 6.55: Agent advertisement

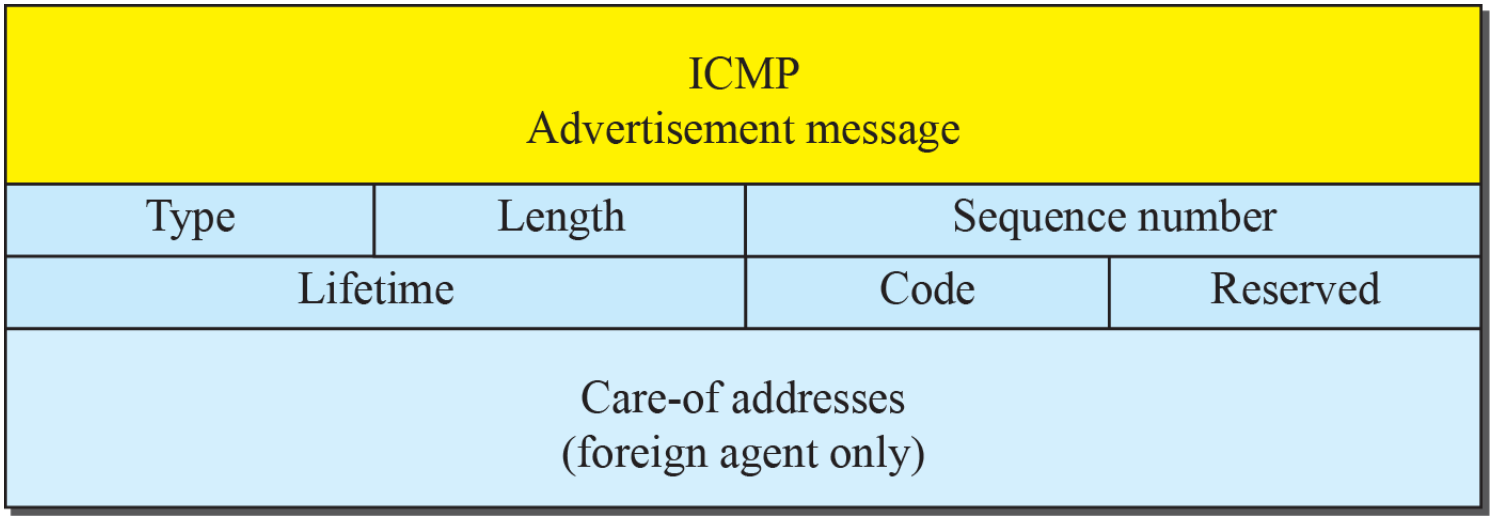

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Table 6.6: Code Bits

Bit	*Meaning*
0	Registration required. No collocated care-of address.
1	Agent is busy and does not accept registration at this moment.
2	Agent acts as a home agent.
3	Agent acts as a foreign agent.
4	Agent uses minimal encapsulation.
5	Agent uses generic routing encapsulation (GRE).
6	Agent supports header compression.
7	Unused (0).

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Figure 6.56: Registration request format

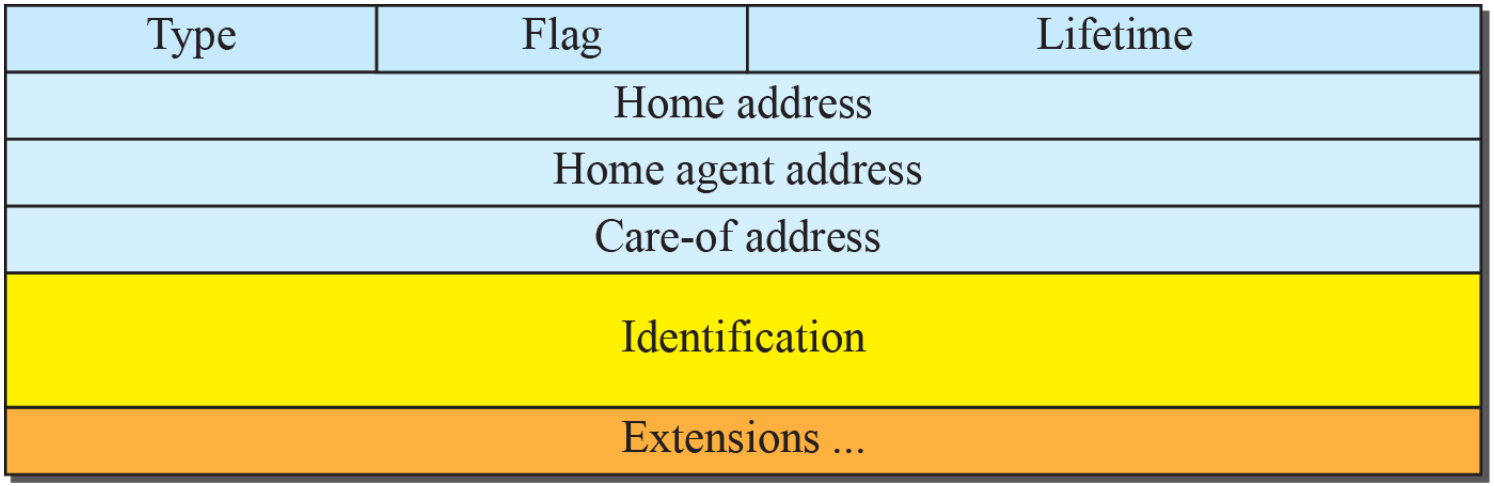

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Table 6.7: Registration request flag field bits

Bit	*Meaning*
0	Mobile host requests that home agent retain its prior care-of address.
1	Mobile host requests that home agent tunnel any broadcast message.
2	Mobile host is using collocated care-of address.
3	Mobile host requests that home agent use minimal encapsulation.
4	Mobile host requests generic routing encapsulation (GRE).
5	Mobile host requests header compression.
6–7	Reserved bits.

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Figure 6.57: Registration reply format

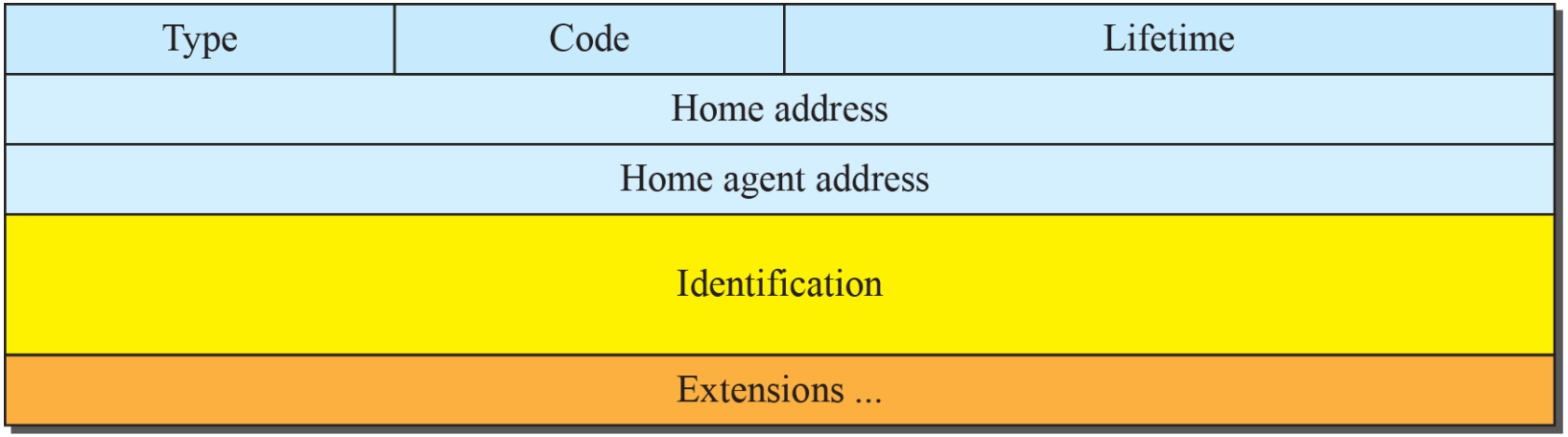

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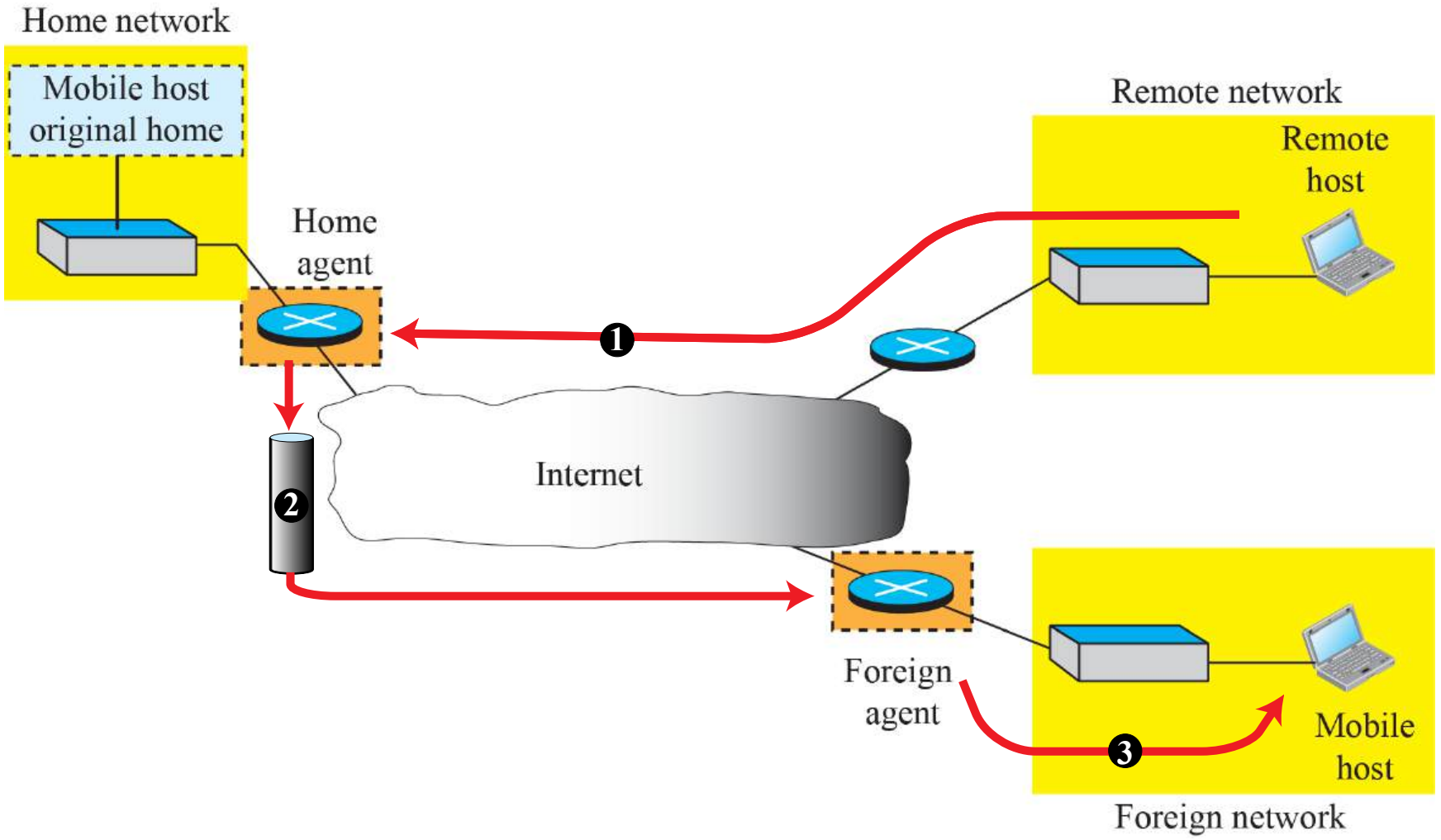
Figure 6.58: Data transfer



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6.3.4 Inefficiency in Mobile IP

Communication involving mobile IP can be inefficient. The inefficiency can be severe or moderate. The severe case is called double crossing or 2X. The moderate case is called triangle routing or dog-leg routing.

- ***Double Crossing***
- ***Triangle Routing***
- ***Solution***

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Figure 6.59: Double crossing

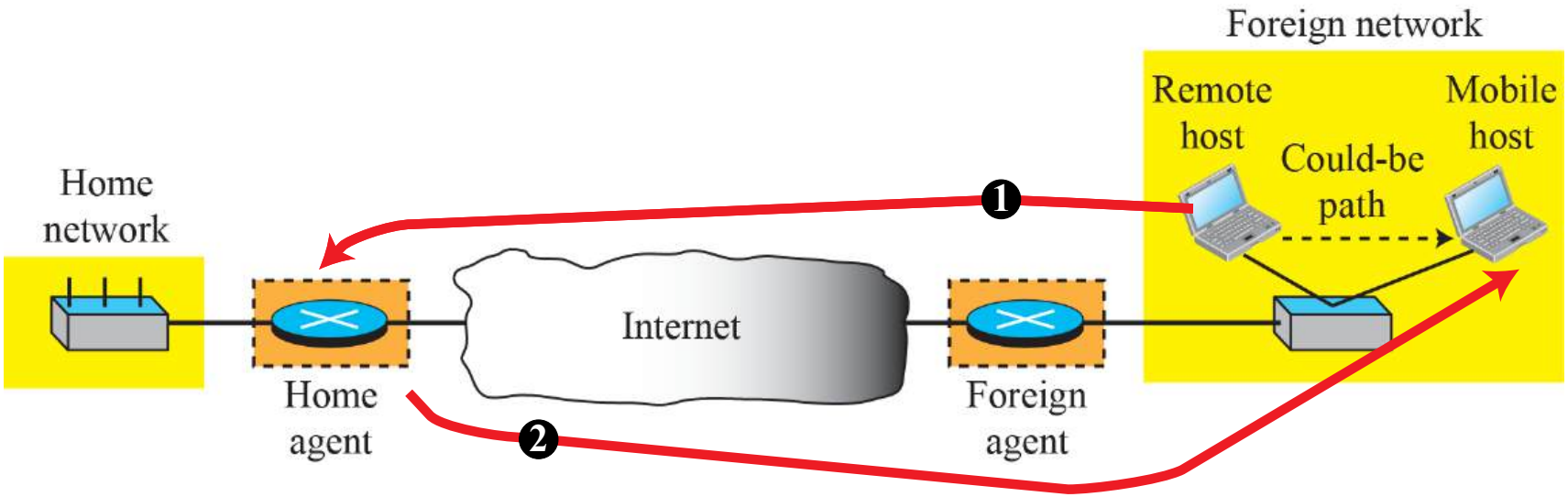

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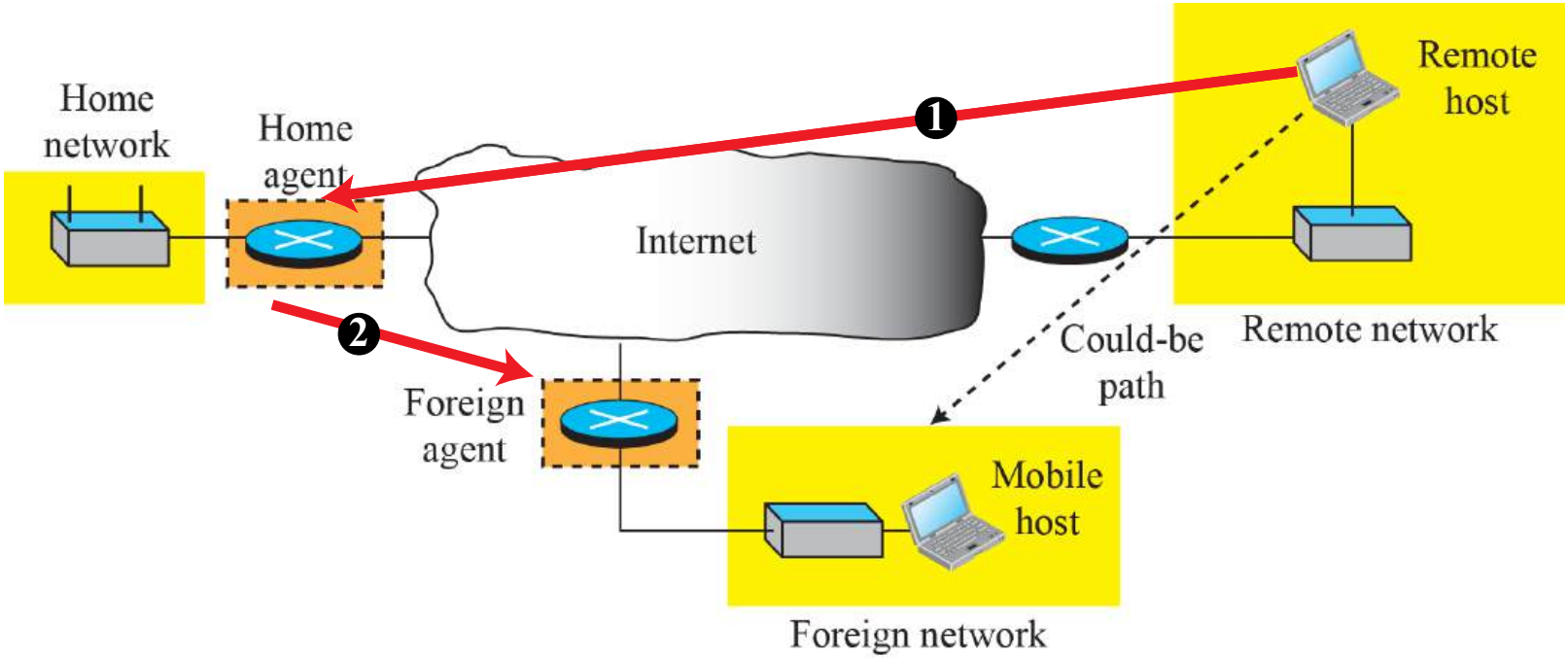
Figure 6.60: Triangle routing



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Chapter 6: Summary

- *Wireless LANs became formalized with the IEEE 802.11 standard, which defines two services: basic service set (BSS) and extended service set (ESS). The access method used in the distributed coordination function (DCF) MAC sublayer is CSMA/CA. The access method used in the point coordination function (PCF) MAC sublayer is polling.*

- *Bluetooth is a wireless LAN technology that connects devices (called gadgets) in a small area. A Bluetooth network is called a piconet.*

- *WiMAX is a wireless access network that may replace DSL and cable in the future.*

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Chapter 6: Summary (continued)

- *Cellular telephony provides communication between two devices. One or both may be mobile. A cellular service area is divided into cells. Cellular telephony has gone through four generations.*

- *A satellite network uses satellites to provide communication between any points on Earth. We have discussed several systems: including GEO, MEO, and LEO.*

- *Mobile IP is an enhanced version of the Internetworking Protocol (IP). A mobile host has a home address on its home network and a care-of address on its foreign network. When the mobile host is on a foreign network, a home agent relays messages (for the mobile host) to a foreign agent. A foreign agent sends relayed messages to a mobile host.*

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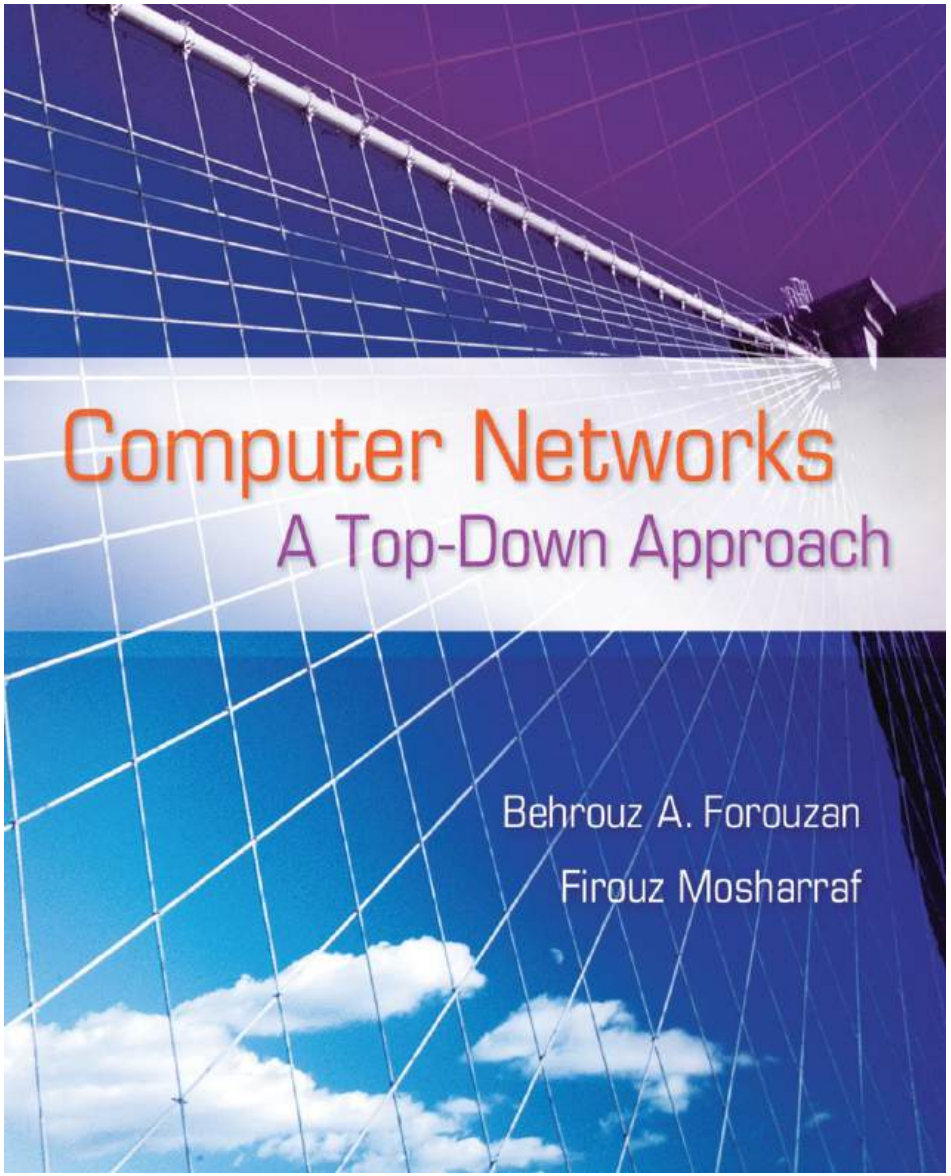
Chapter 7

Physical Layer and Transmission Media



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Chapter 7: Outline

7.1 DATA AND SIGNAL

7.2 DIGITAL TRANSMISSION

7.3 ANALOG TRANSMISSION

7.4 BANDWIDTH UTILIZATION

7.5 TRANSMISSION MEDIA

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Chapter 7: Objective

- *We first discuss the relationship between data and signals. We then show how data and signals can be both analog and digital.*
- *We then concentrate on digital transmission. We show how to convert digital and analog data to digital signals.*
- *Next, we concentrate on analog transmission. We show how to convert digital and analog data to analog signals.*
- *We then talk about multiplexing techniques and how they can combine several channels.*
- *Finally, we go below the physical layer and discuss the transmission media.*

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7-1 DATA AND SIGNALS

At the physical layer, the communication is node-to-node, but the nodes exchange electromagnetic signals. Figure 7.1 uses the same scenario we showed in four earlier chapters, but the communication is now at the physical layer.

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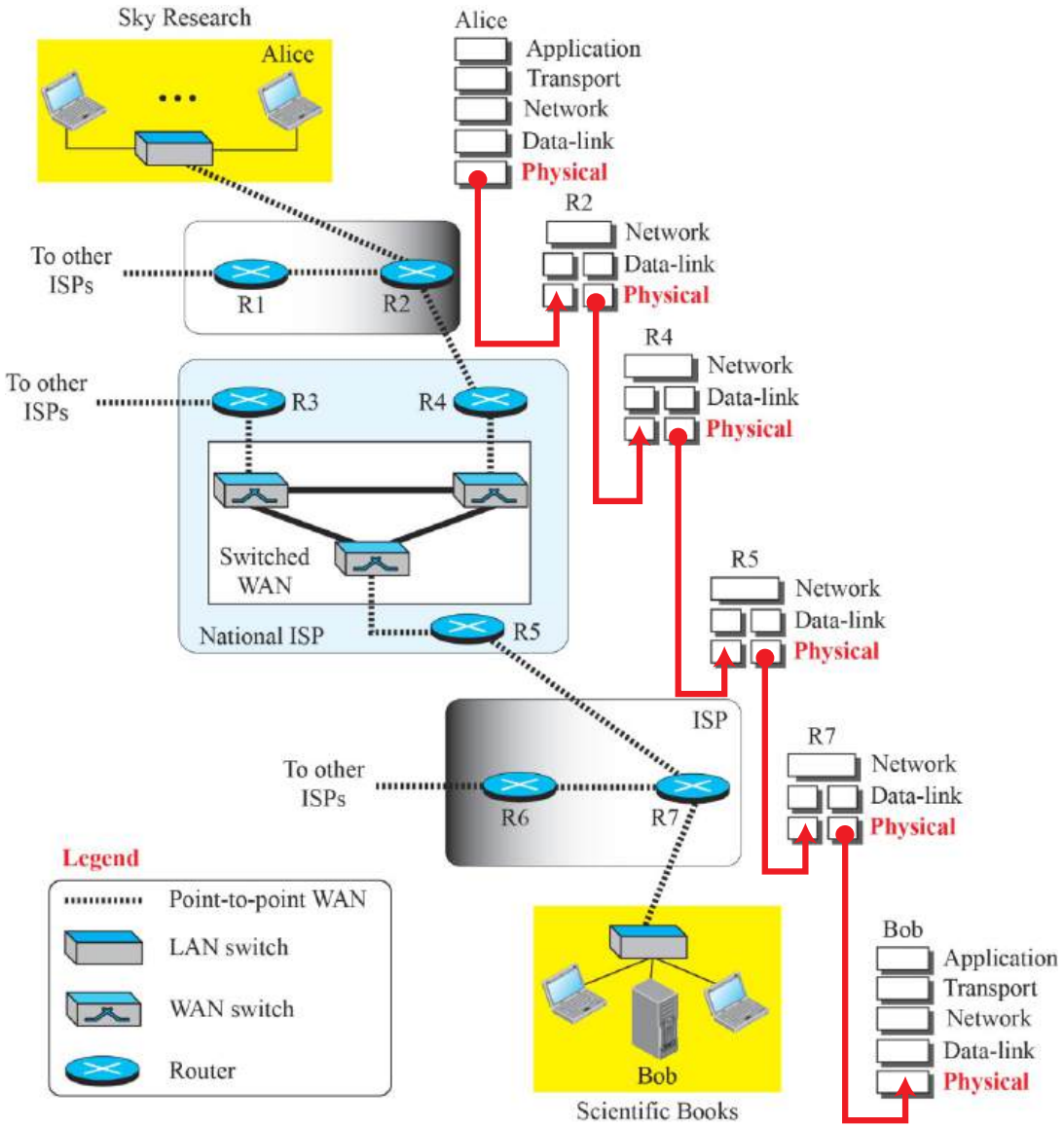
Figure 7.1: Communication at the physical layer



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7.1.1 Analog and Digital

Data can be analog or digital. The term analog data refers to information that is continuous. Digital data take on discrete values.

Like the data they represent, signals can be either analog or digital. An analog signal has infinitely many levels of intensity over a period of time. A digital signal, on the other hand, can have only a limited number of defined values. Although each value can be any number, it is often as simple as 1 and 0.

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7.1.1 (continued)

- **Analog Signals**
  - Time and Frequency Domains
  - Composite Signals
  - Bandwidth
- **Digital Signals**
  - Bit Rate
  - Bit Length
  - Digital Signal as a Composite Analog Signal
  - Transmission of Digital Signals
  - Baseband Transmission
  - Broadband Transmission

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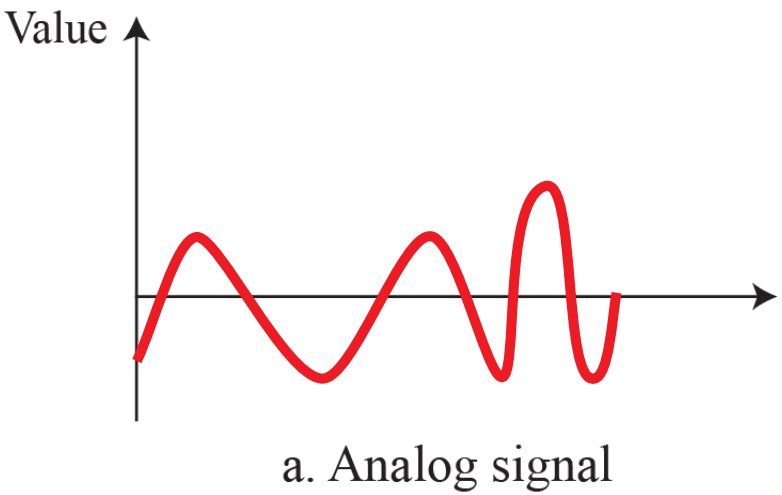
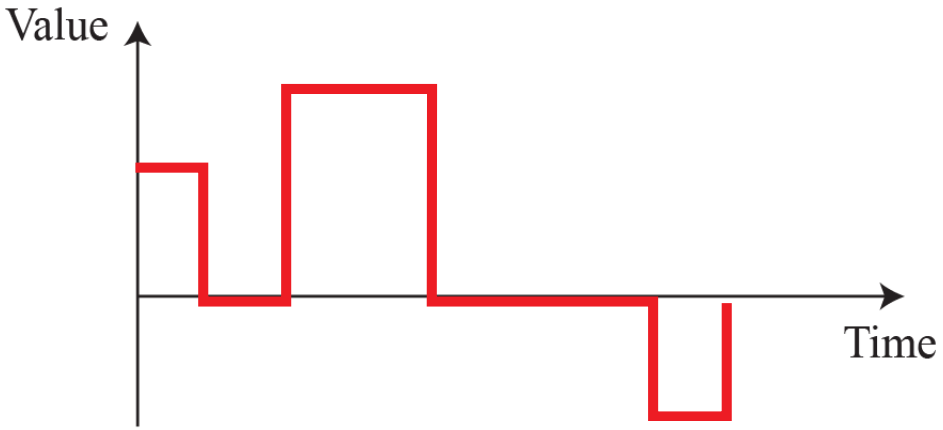
Figure 7.2: Comparison of analog and digital signals



a. Analog signal



b. Digital signal

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Figure 7.3: A sine wave

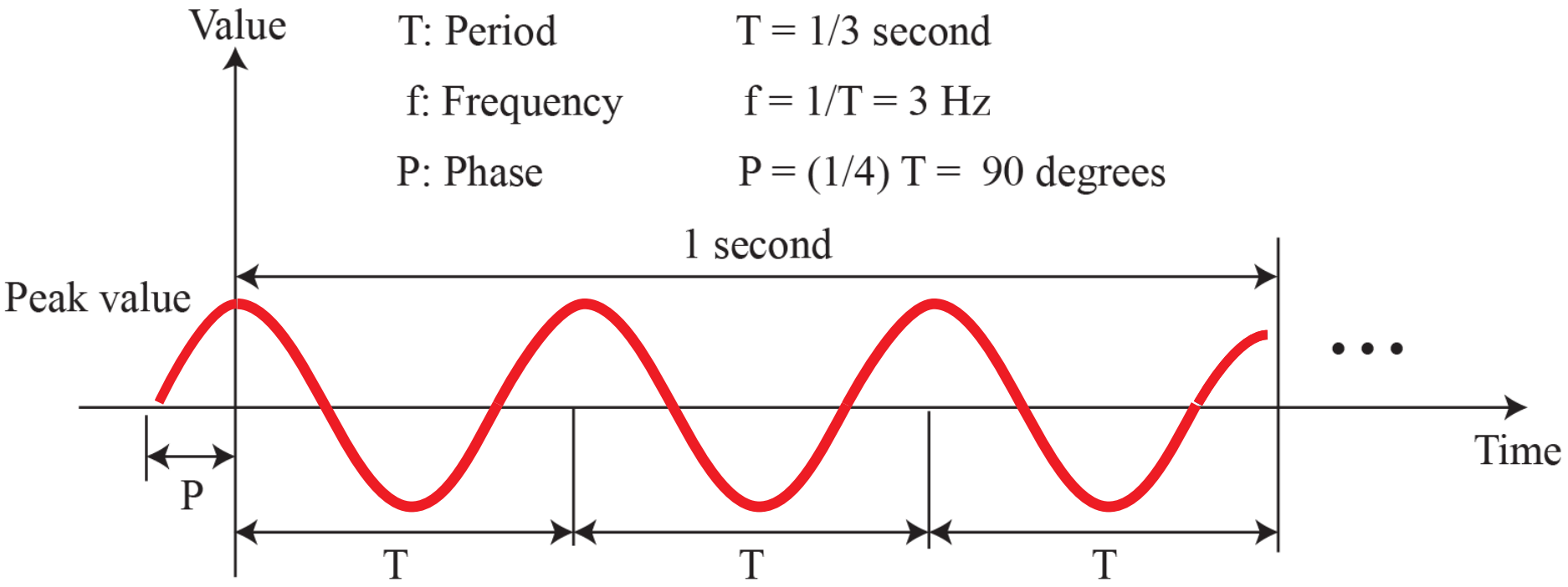

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Figure 7.4: Wavelength and period

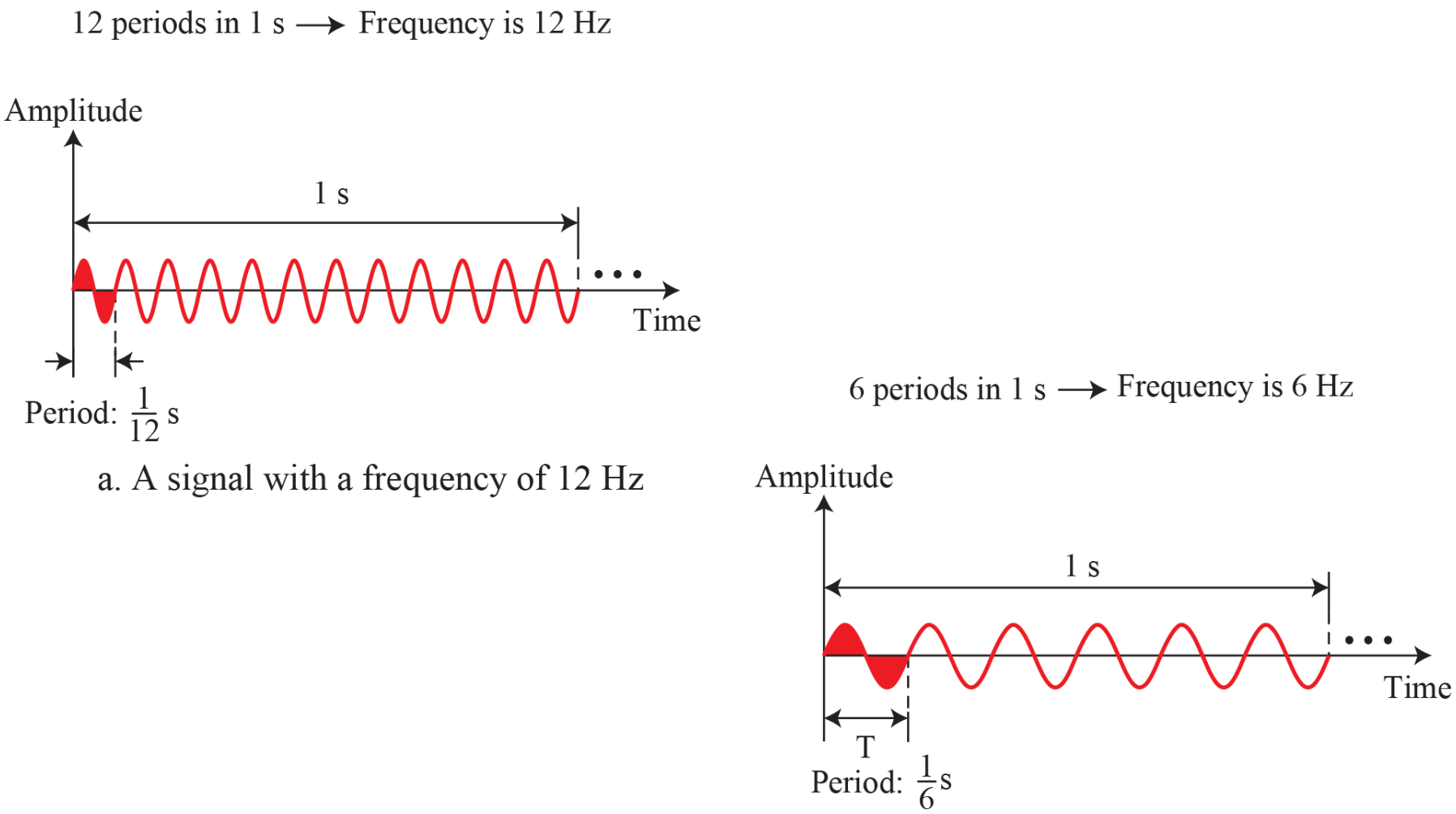

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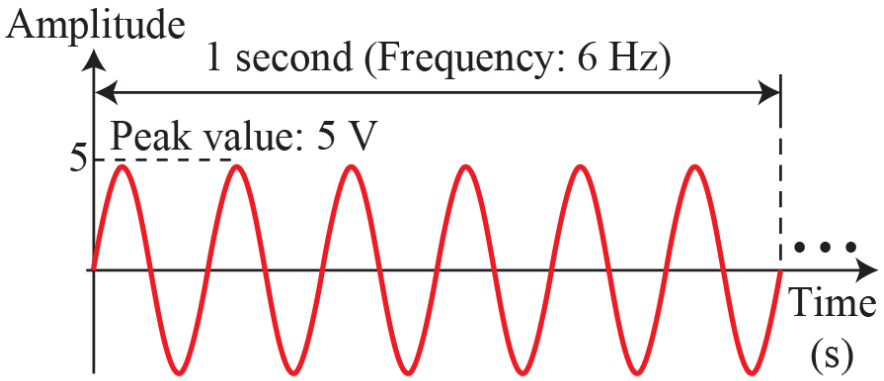
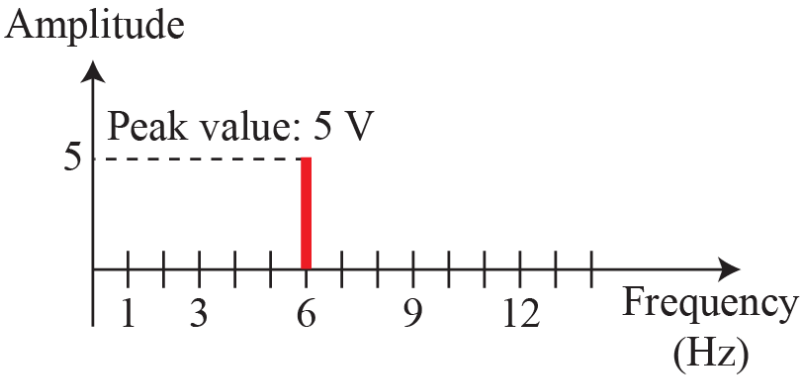
Figure 7.5: The time-domain and frequency-domain plots of a sine wave



a. A sine wave in the time domain



b. The same sine wave in the frequency domain

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Example 7.1

The frequency domain is more compact and useful when we are dealing with more than one sine wave. For example, Figure 7.6 shows three sine waves, each with different amplitude and frequency. All can be represented by three spikes in the frequency domain.

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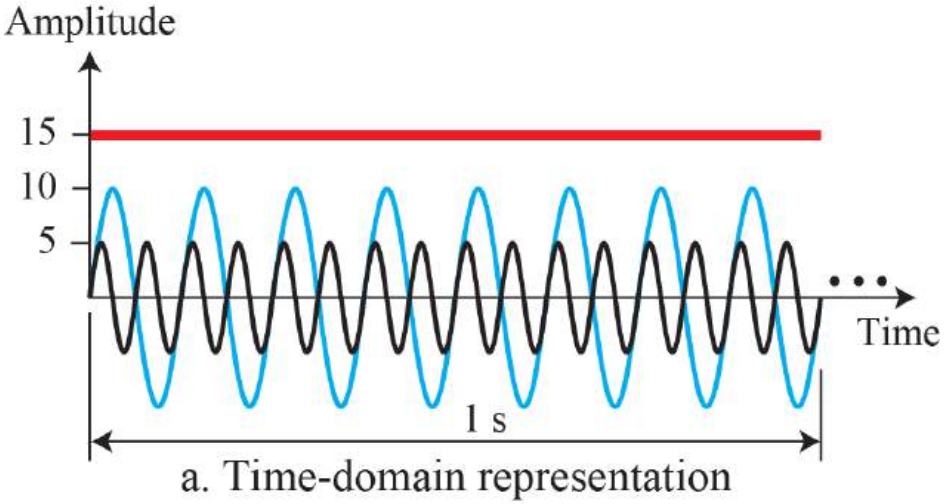
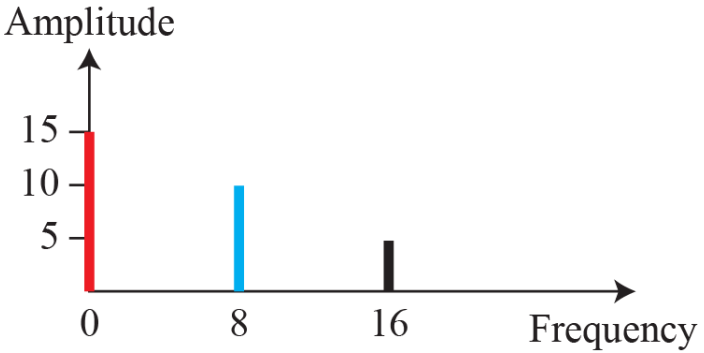
Figure 7.6: The time domain and frequency domain of three sine waves



a. Time-domain representation



b. Frequency-domain representation

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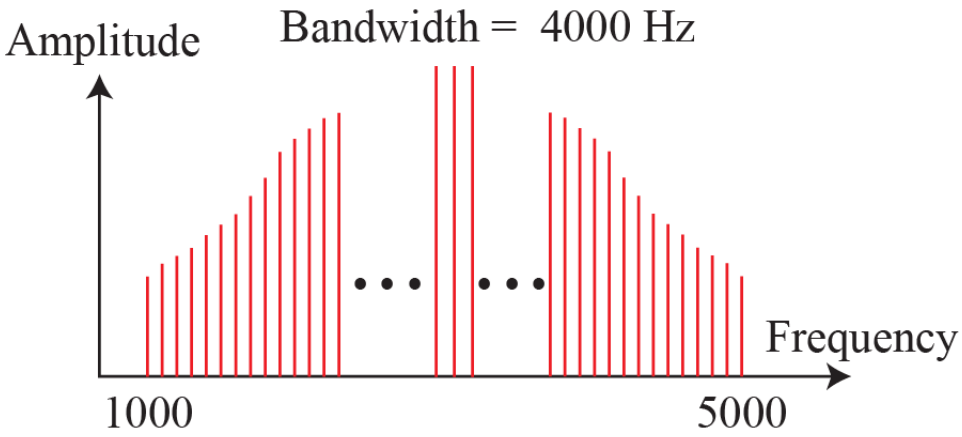
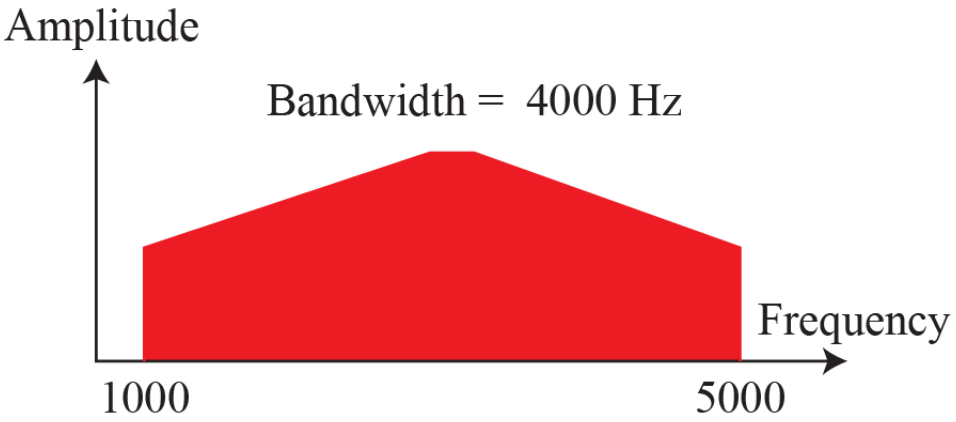
Figure 7.7: The bandwidth of periodic and nonperiodic composite signals



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

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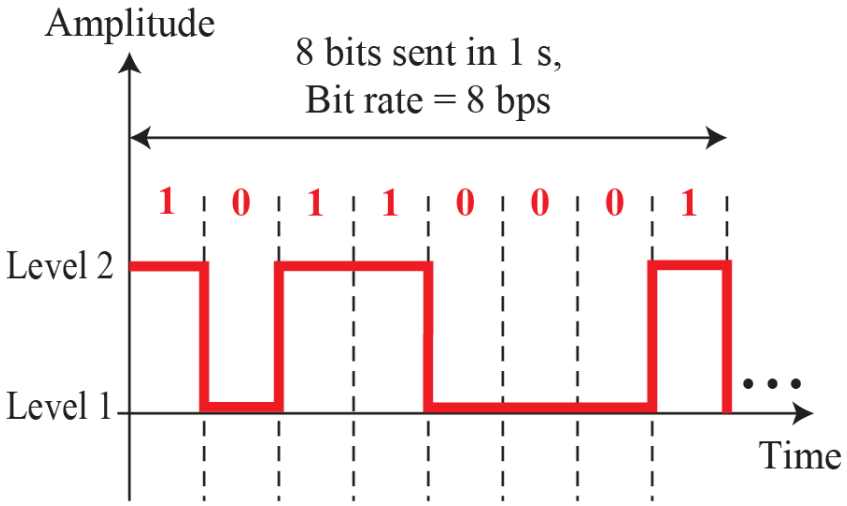
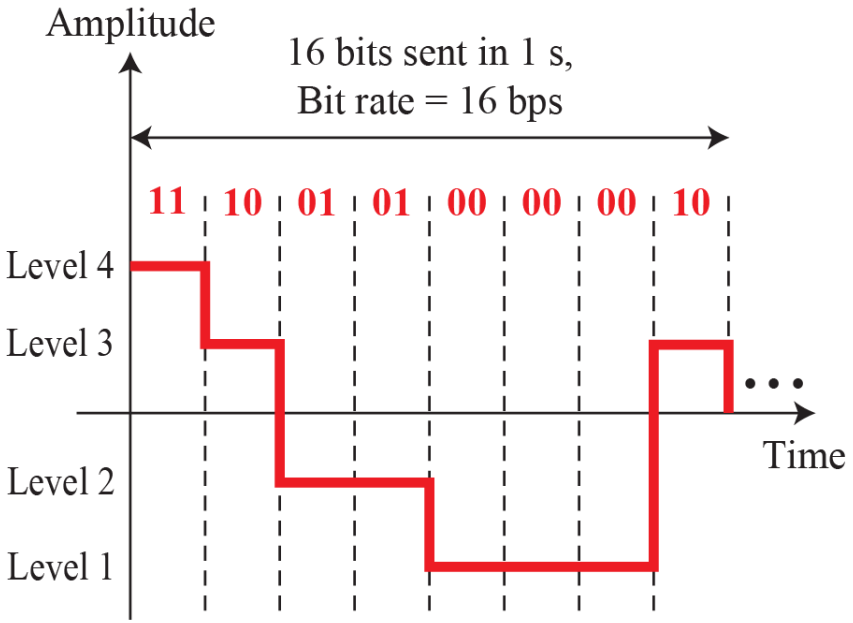
Figure 7.8: Two digital signals: one with two signal levels and the other with four signal levels



a. A digital signal with two levels



b. A digital signal with four levels

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Example 7.2

Assume we need to download text documents at the rate of 100 pages per minute. What is the required bit rate of the channel? A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits, the bit rate is

$$100 \times 24 \times 80 \times 8 = 1{,}536{,}000 \text{ bps} = 1.536 \text{ Mbps}.$$

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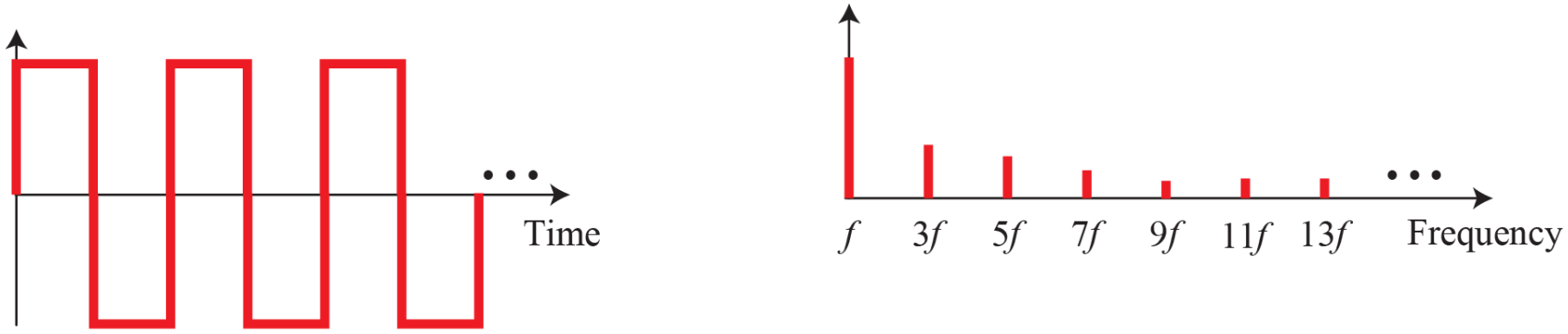
Figure 7.9: The time and frequency domains of periodic and nonperiodic digital signals



a. Time and frequency domains of periodic digital signal

b. Time and frequency domains of nonperiodic digital signal

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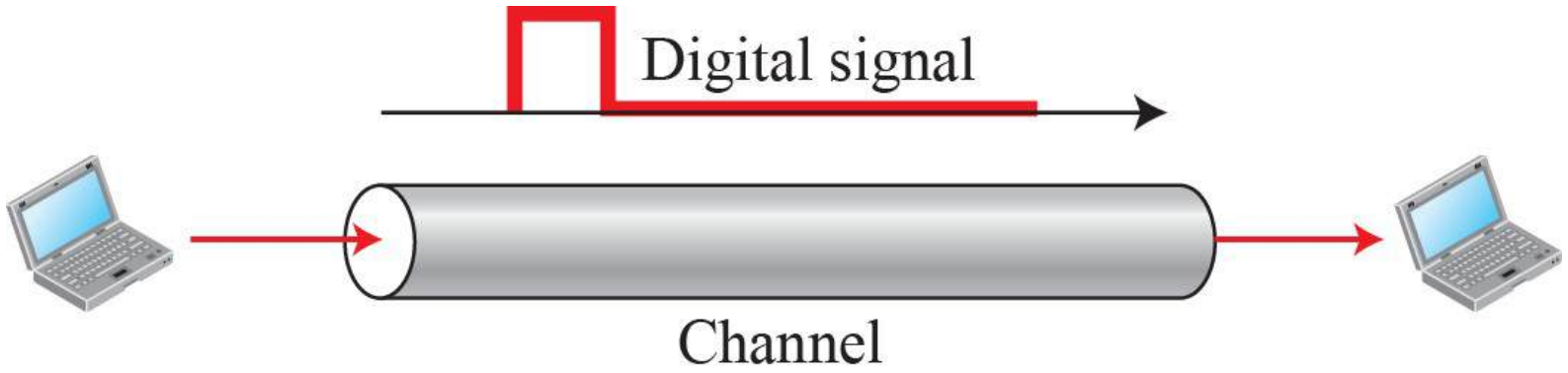
Figure 7.10: Baseband transmission



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Example 7.3

An example of a dedicated channel where the entire bandwidth of the medium is used as one single channel is a LAN. Almost every wired LAN today uses a dedicated channel for two stations communicating with each other.

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Figure 7.11: Bandwidth of a band-pass channel

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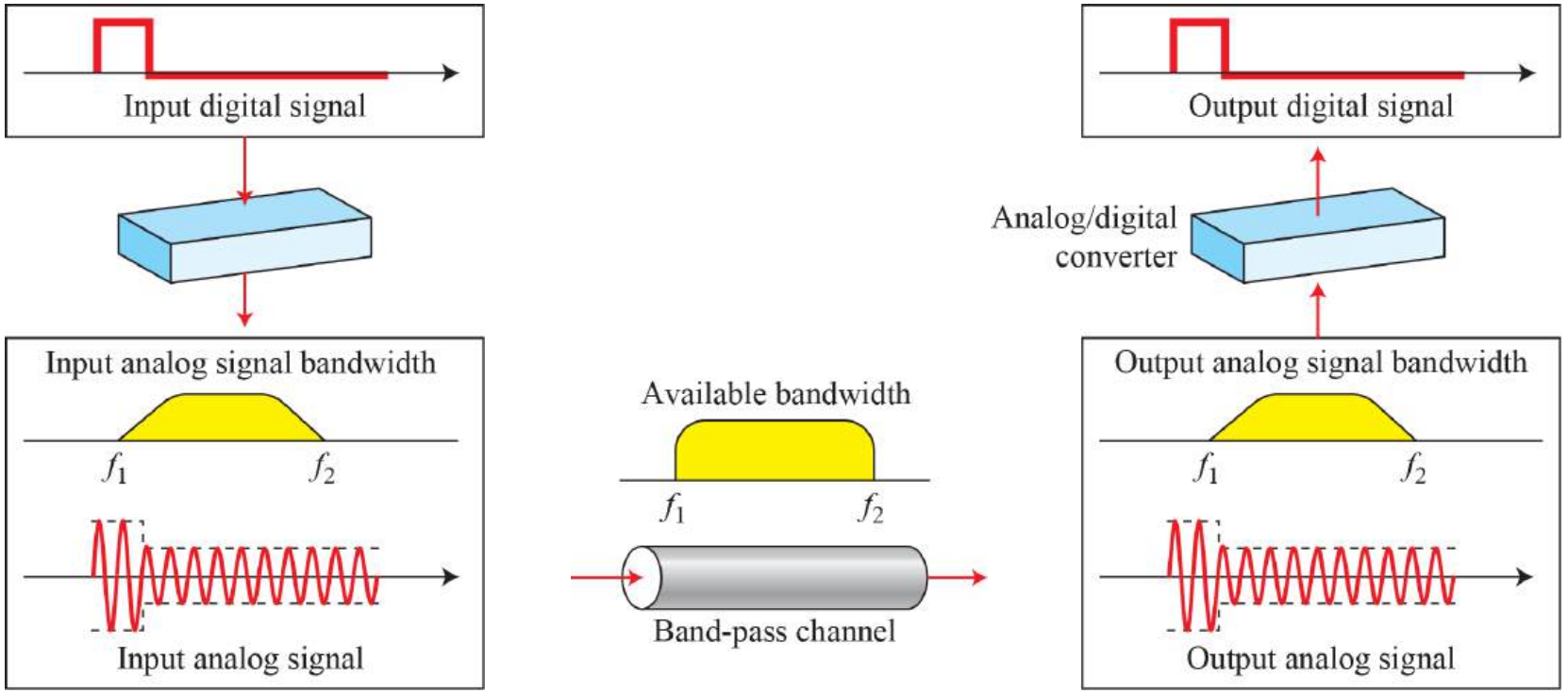
Figure 7.12: Modulation of a digital signal for transmission on a band-pass channel



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Example 7.4

An example of broadband transmission using modulation is the sending of computer data through a telephone subscriber line, the line connecting a resident to the central telephone office. Although this channel can be used as a low-pass channel, it is normally considered a band-pass channel. One reason is that the bandwidth is so narrow (4 kHz) that if we treat the channel as low-pass and use it for baseband transmission, the maximum bit rate can be only 8 kbps (explained later). The solution is to consider the channel a band-pass channel, convert the digital signal from the computer to an analog signal, and send the analog signal.

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Example 7.5

A second example is the digital cellular telephone. For better reception, digital cellular phones digitize analog voice. Although the bandwidth allocated to a company providing digital cellular phone service is very wide, we still cannot send the digitized signal without conversion. The reason is that we have only a band-pass channel available between caller and callee. For example, if the available bandwidth is W and we allow 1000 couples to talk simultaneously, this means the available channel is W/1000, just part of the entire bandwidth. We need to convert the digitized voice to a composite analog signal before transmission.

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7.1.2 Transmission Impairment

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are attenuation, distortion, and noise.

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7.1.2 (continued)

- **Attenuation**
- **Distortion**
- **Noise**
  - *Signal-to-Noise Ratio (SNR)*

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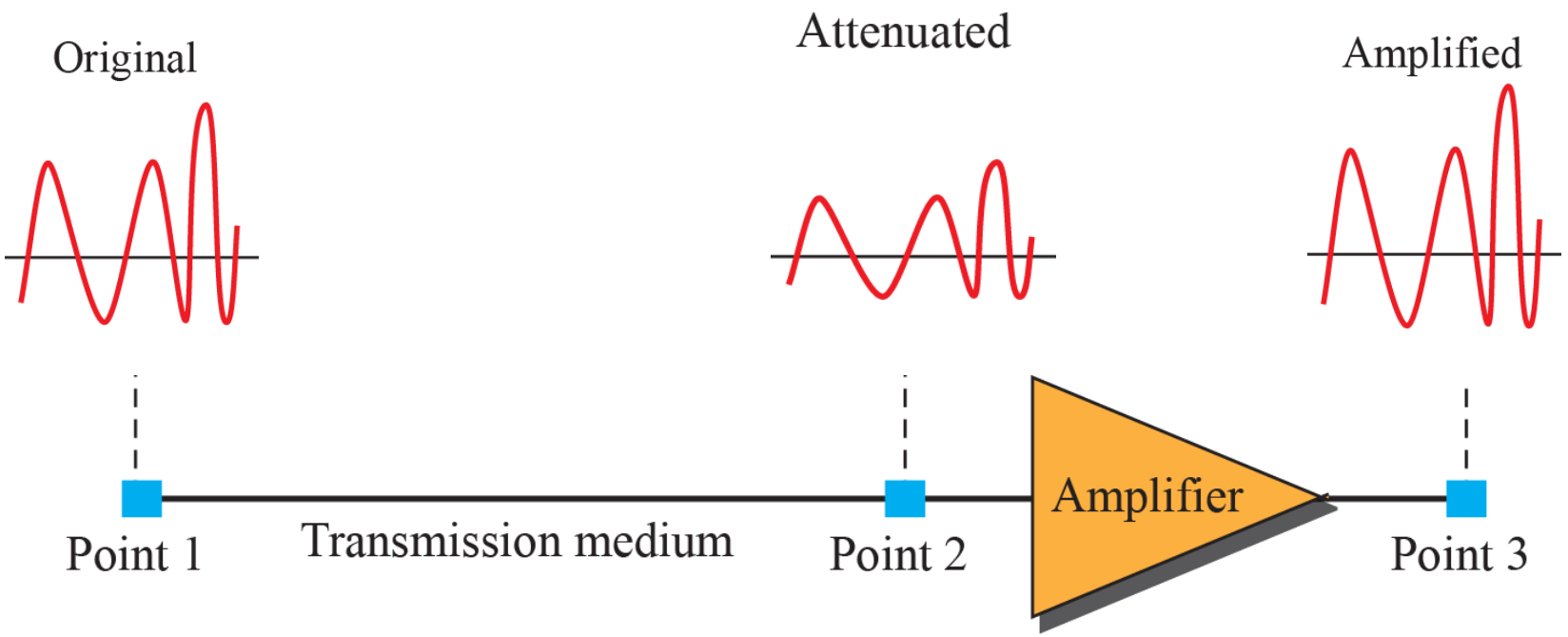
Figure 7.13: Attenuation and amplification



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Example 7.6

Suppose a signal travels through a transmission medium and its power is reduced to one half. This means that P2 = 0.5 P1. In this case, the attenuation (loss of power) can be calculated as

$$10 \log_{10} P_2/P_1 = 10 \log_{10} (0.5\, P_1)\, /P_1 = 10 \log_{10} 0.5 = 10 \times (-0.3) = -3 \text{ dB}.$$

A loss of 3 dB (−3 dB) is equivalent to losing one-half the power.

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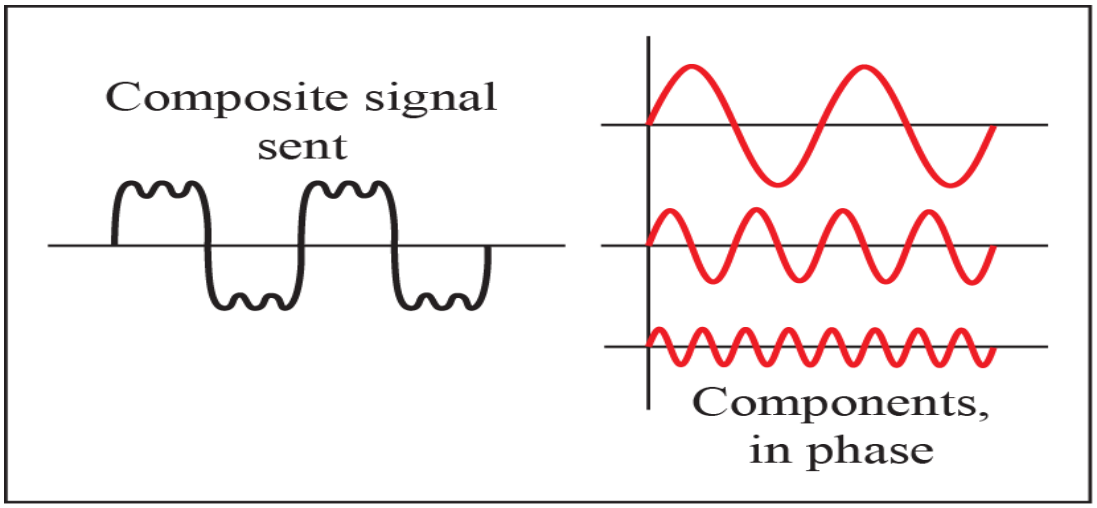
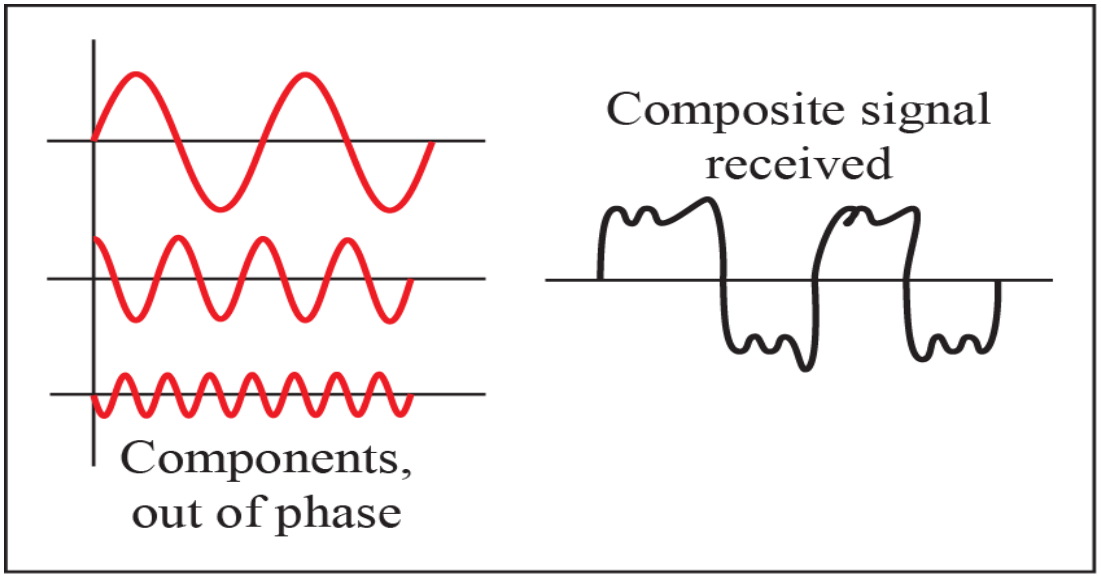
Figure 7.14: Distortion



At the sender



At the receiver

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Figure 7.15: Noise

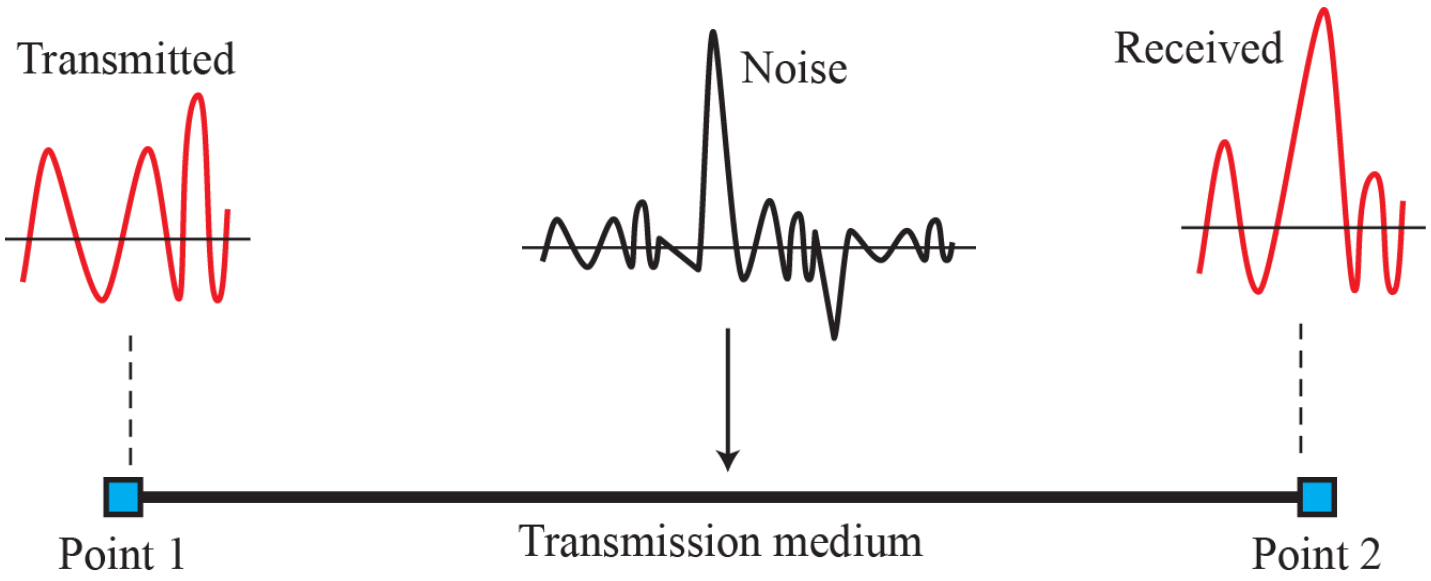

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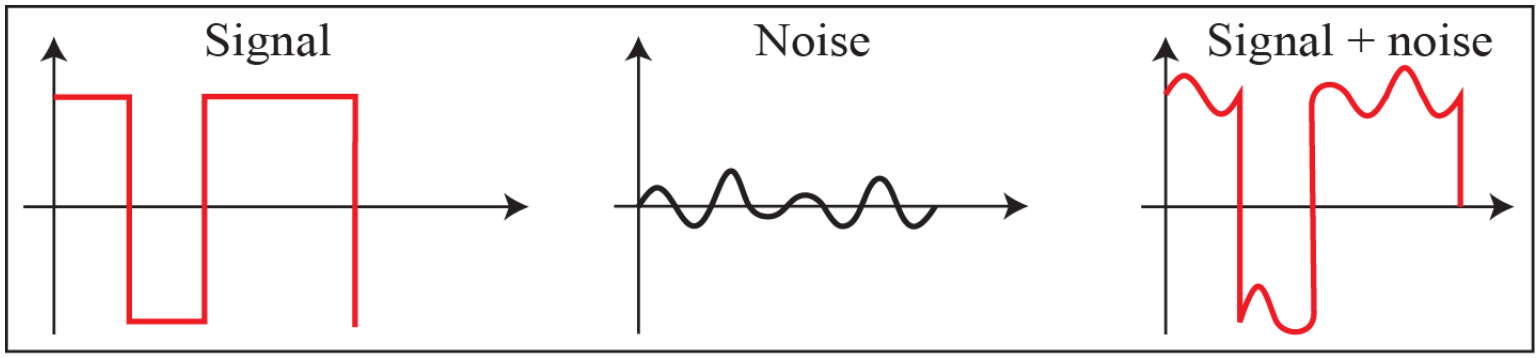
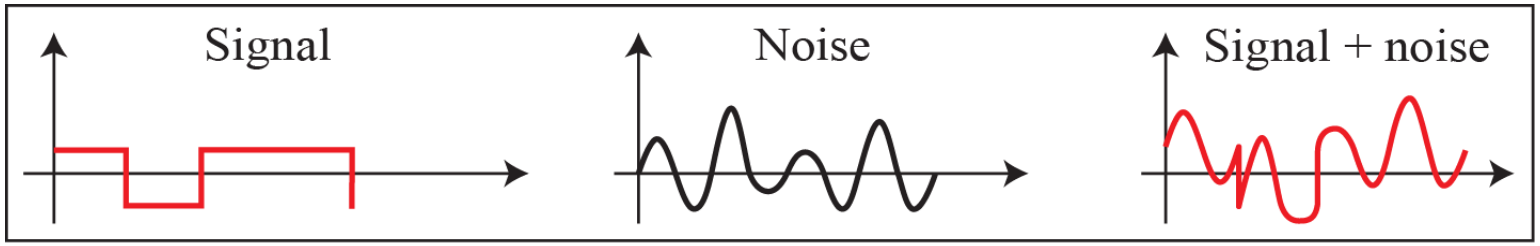
Figure 7.16: Two cases of SNR: a high SNR and a low SNR



a. High SNR



b. Low SNR

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7.1.3 Data Rate Limits

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel. Data rate depends on three factors:

1. *The bandwidth available*
2. *The level of the signals we use*
3. *The quality of the channel (the level of noise)*

Two theoretical formulas were developed to calculate the data rate: one by Nyquist for a noiseless channel, another by Shannon for a noisy channel.

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7.1.3 (continued)

- **Noiseless Channel: Nyquist Bit Rate**
- **Noisy Channel: Shannon Capacity**
- **Using Both Limits**

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Example 7.7

We need to send 265 kbps over a noiseless (ideal) channel with a bandwidth of 20 kHz. How many signal levels do we need? We can use the Nyquist formula as shown:

$$265{,}000 = 2 \times 20{,}000 \times \log_2 L \quad \rightarrow \quad \log_2 L = 6.625 \qquad L = 2^{6.625} = 98.7 \text{ levels}$$

Since this result is not a power of 2, we need to either increase the number of levels or reduce the bit rate. If we have 128 levels, the bit rate is 280 kbps. If we have 64 levels, the bit rate is 240 kbps.

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Example 7.8

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as shown below.

$$C = B \log_2 (1 + SNR) = B \log_2 (1 + 0) = B \log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, the data is so corrupted in this channel that it is useless when received.

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Example 7.9

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as shown below.

$$C = B \log_2 (1 + SNR) = 3000 \log_2 (1 + 3162) = 34{,}881 \text{ bps}$$

This means that the highest bit rate for a telephone line is 34.881 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to noise ratio.

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Example 7.10

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution

First, we use the Shannon formula to find the upper limit.

$$C = B \log_2 (1 + SNR) = 10^6 \log_2 (1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example. Then we use the Nyquist formula to find the number of signal levels.

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \quad \rightarrow \quad \log_2 L = 2 \quad \rightarrow \quad L = 4$$

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7.1.4 Performance

Up to now, we have discussed the tools of transmitting data (signals) over a network and how the data behave. One important issue in networking is the performance of the network—how good is it? We discuss quality of service, an overall measurement of network performance, in detail in Chapter 8.

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7.1.4 (continued)

- **Bandwidth**
  - Bandwidth in Hertz
  - Bandwidth in Bits per Seconds
  - Relationship
- **Throughput**
- **Latency (Delay)**
- **Bandwidth-Delay Product**
- **Jitter**

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Example 7.11

The bandwidth of a subscriber line is 4 kHz for voice or data. The bandwidth of this line for data transmission can be up to 56 kbps, using a sophisticated modem to change the digital signal to analog. If the telephone company improves the quality of the line and increases the bandwidth to 8 kHz, we can send 112 kbps.

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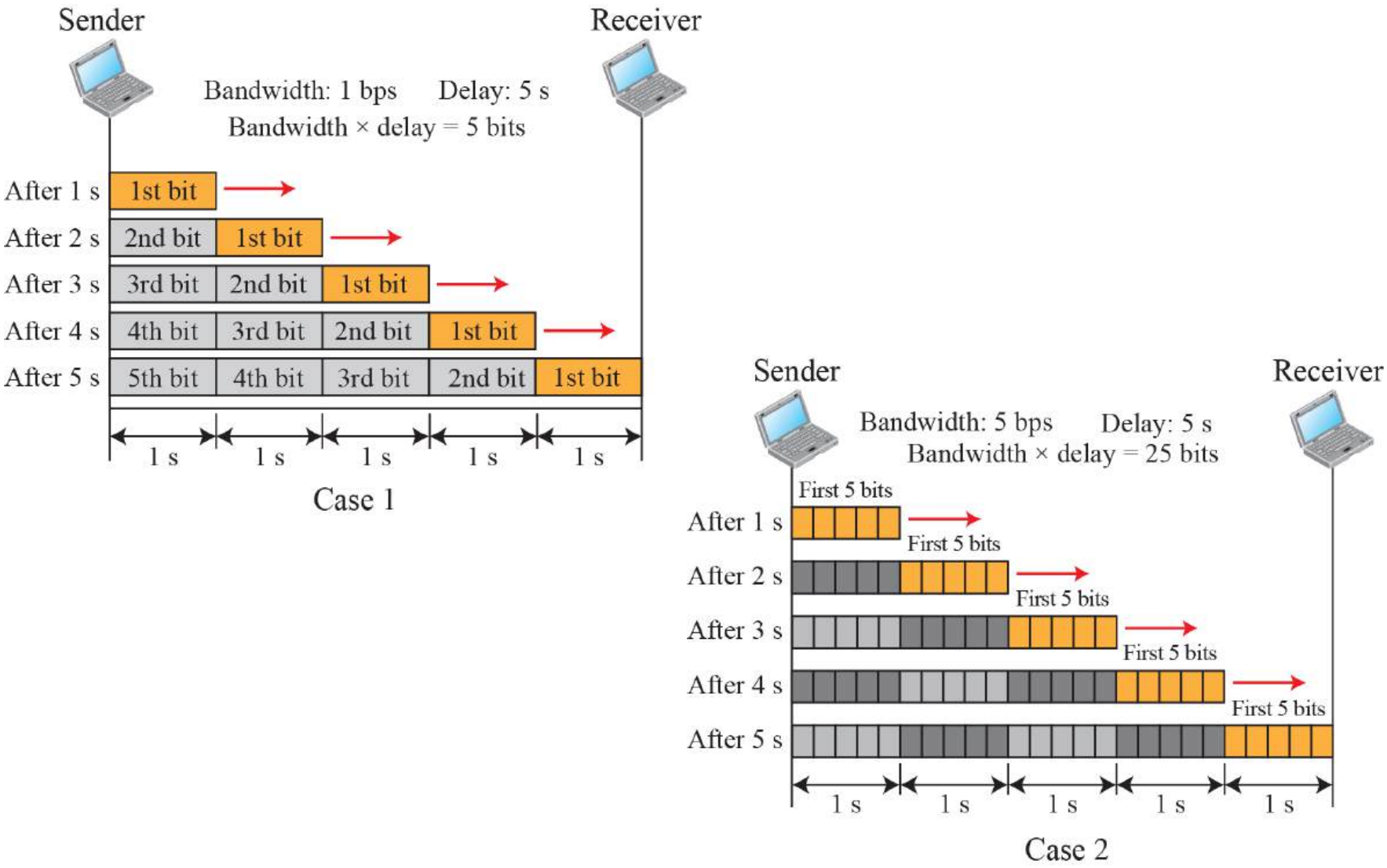
Figure 7.17: Filling the link with bits for cases 1 and 2



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Example 7.12

We can think about the link between two points as a pipe. The cross section of the pipe represents the bandwidth, and the length of the pipe represents the delay. We can say the volume of the pipe defines the bandwidth-delay product, as shown in Figure 7.18.

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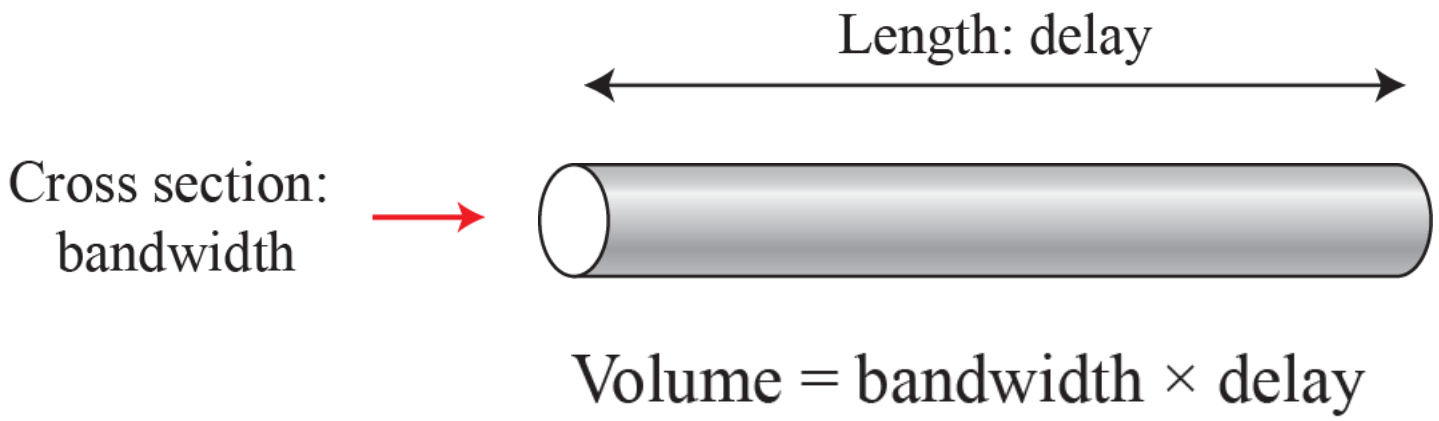
Figure 7.18: Concept of bandwidth-delay product



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7-2 DIGITAL TRANSMISSION

A computer network is designed to send information from one point to another. This information needs to be converted to either a digital signal or an analog signal for transmission. In this section, we discuss the first choice, conversion to digital signals; in the next section, we discuss the second choice, conversion to analog signals.

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7.2.1 Digital-to-Digital Conversion

In this section, we see how we can represent digital data by using digital signals. The conversion involves three techniques: line coding, block coding, and scrambling. Line coding is always needed; block coding and scrambling may or may not be needed.

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7.2.1 (continued)

- **Line Coding**
  - Polar Schemes
  - Bipolar Schemes
  - Multilevel Schemes
- **Block Coding**
  - 4B/5B Coding
  - 8B/10B Coding
- **Scrambling**
  - B8ZS Coding
  - HDB3 Coding

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Figure 7.19: Line coding and decoding

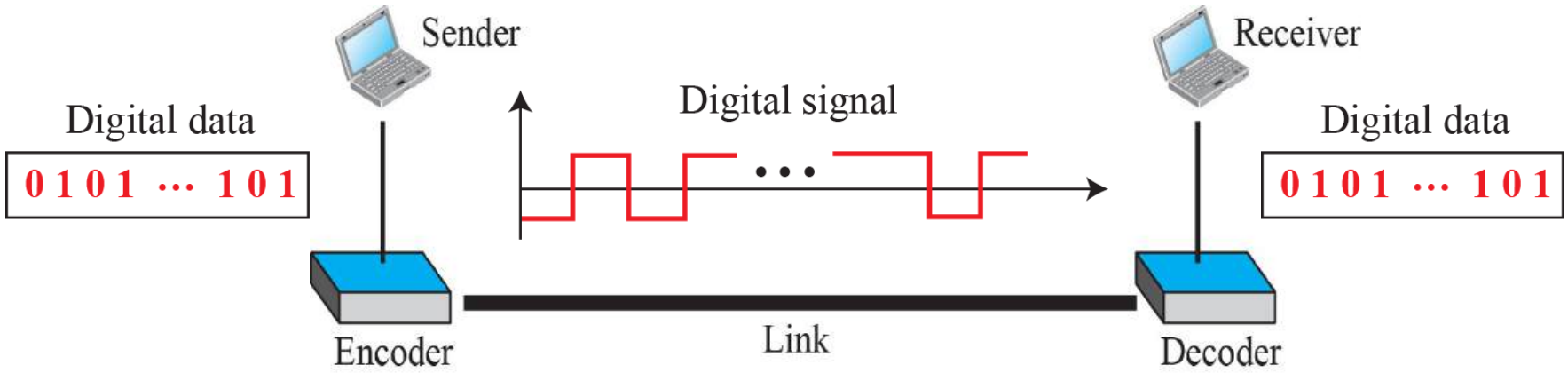

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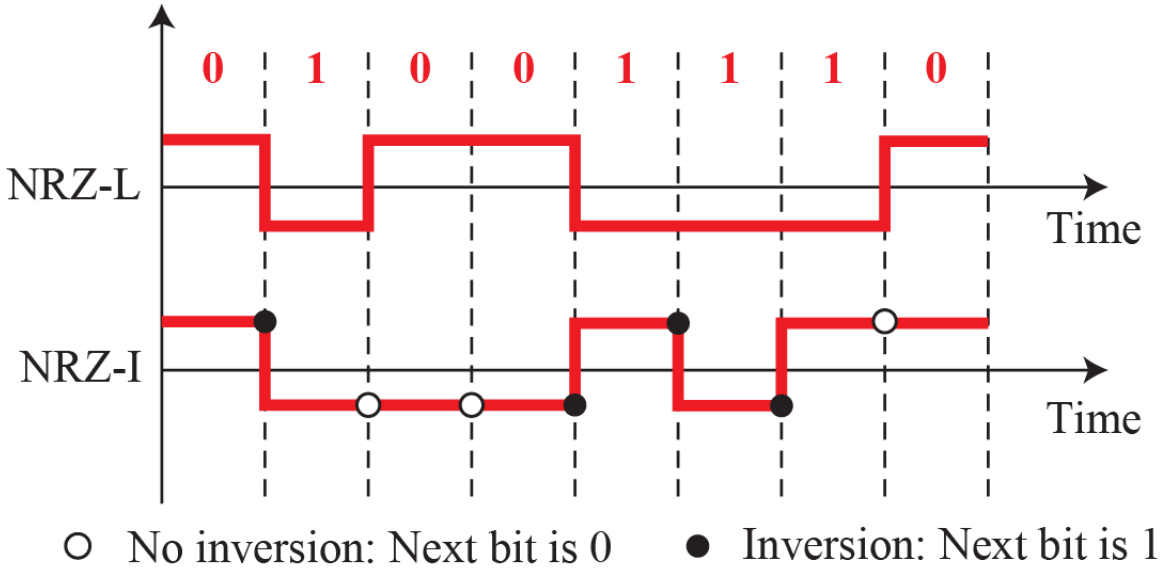
Figure 7.20: Polar schemes (Part I: NRZ)



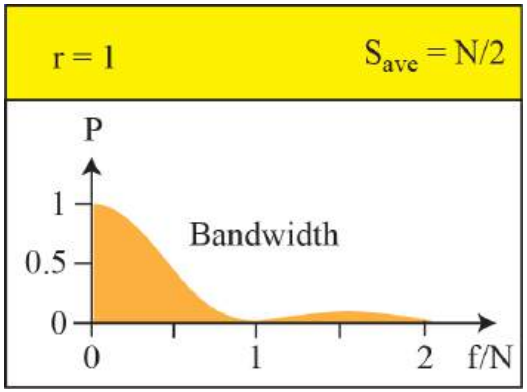


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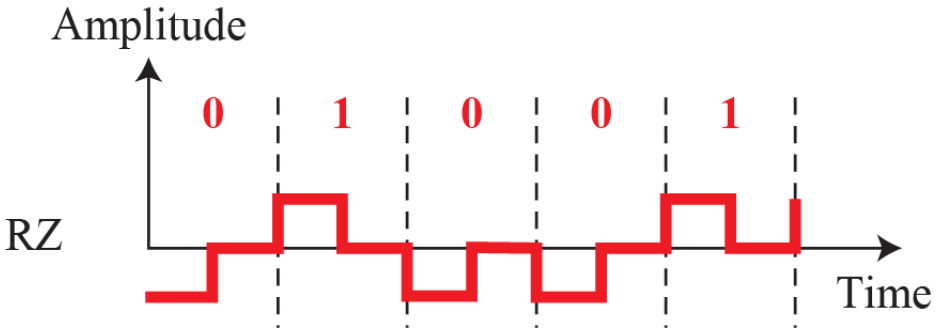
Figure 7.20: Polar schemes (Part II: RZ)



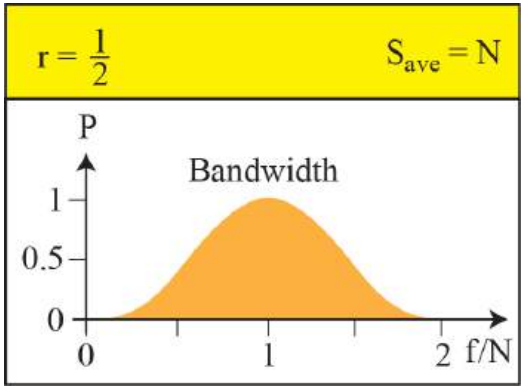


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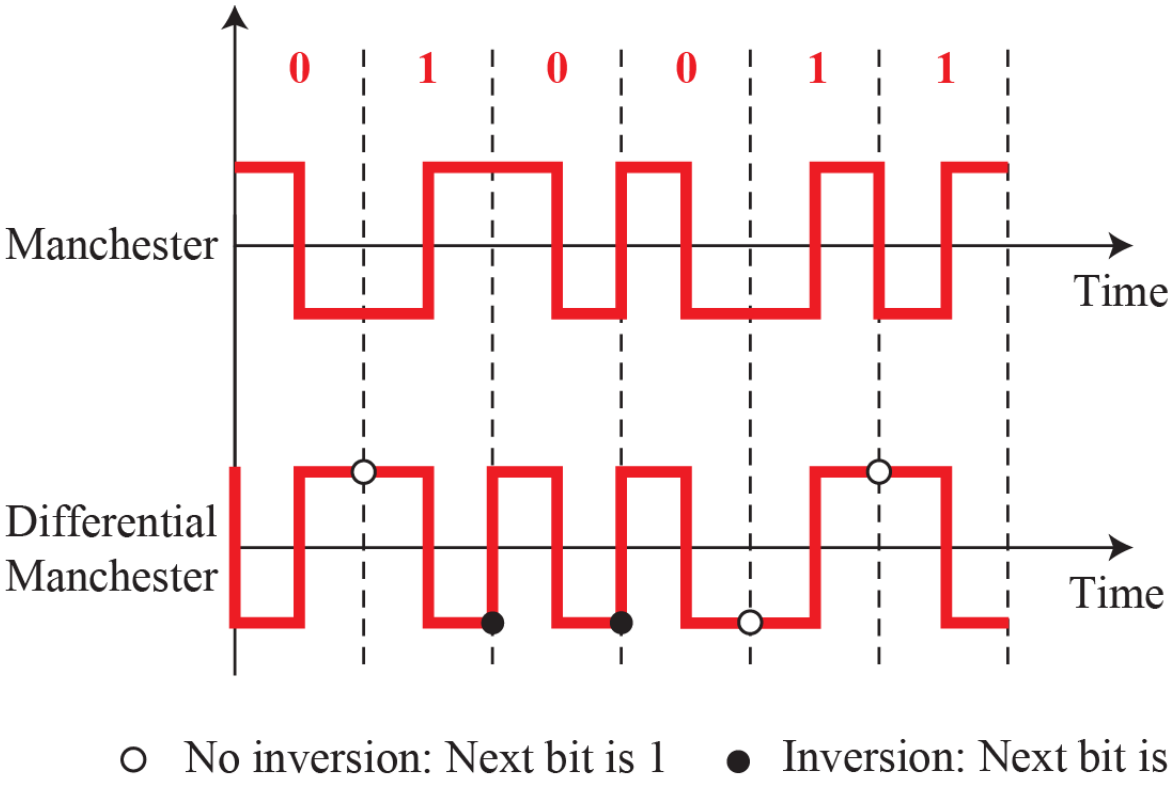
Figure 7.20: Polar schemes (Part III: Manchesters)



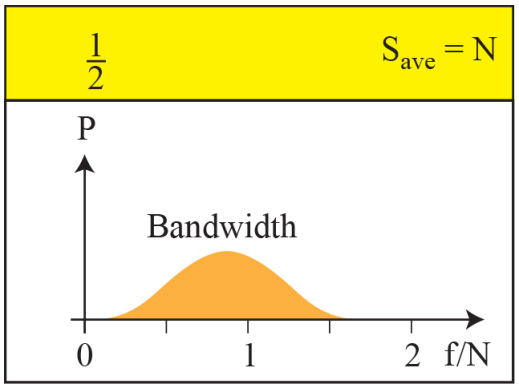


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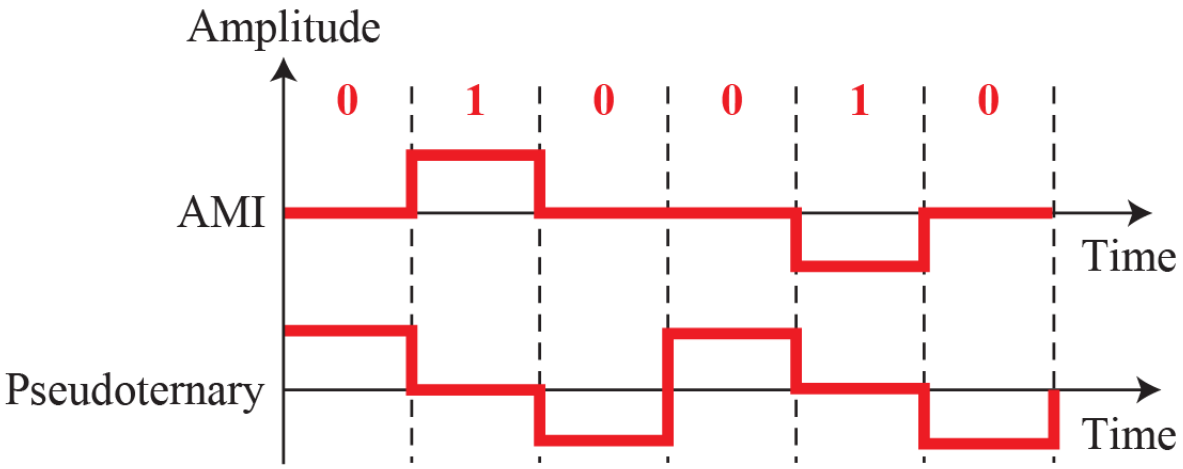
Figure 7.21: Bipolar schemes: AMI and pseudoternary



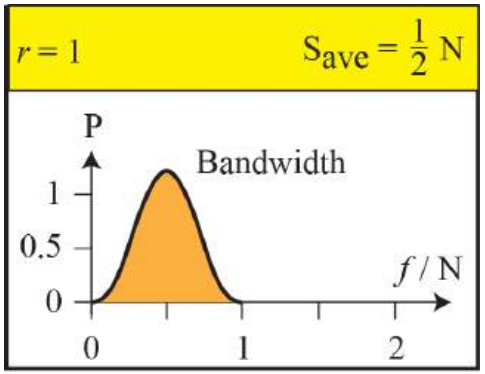


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Figure 7.22: Multilevel: 2B1Q and 8B6T

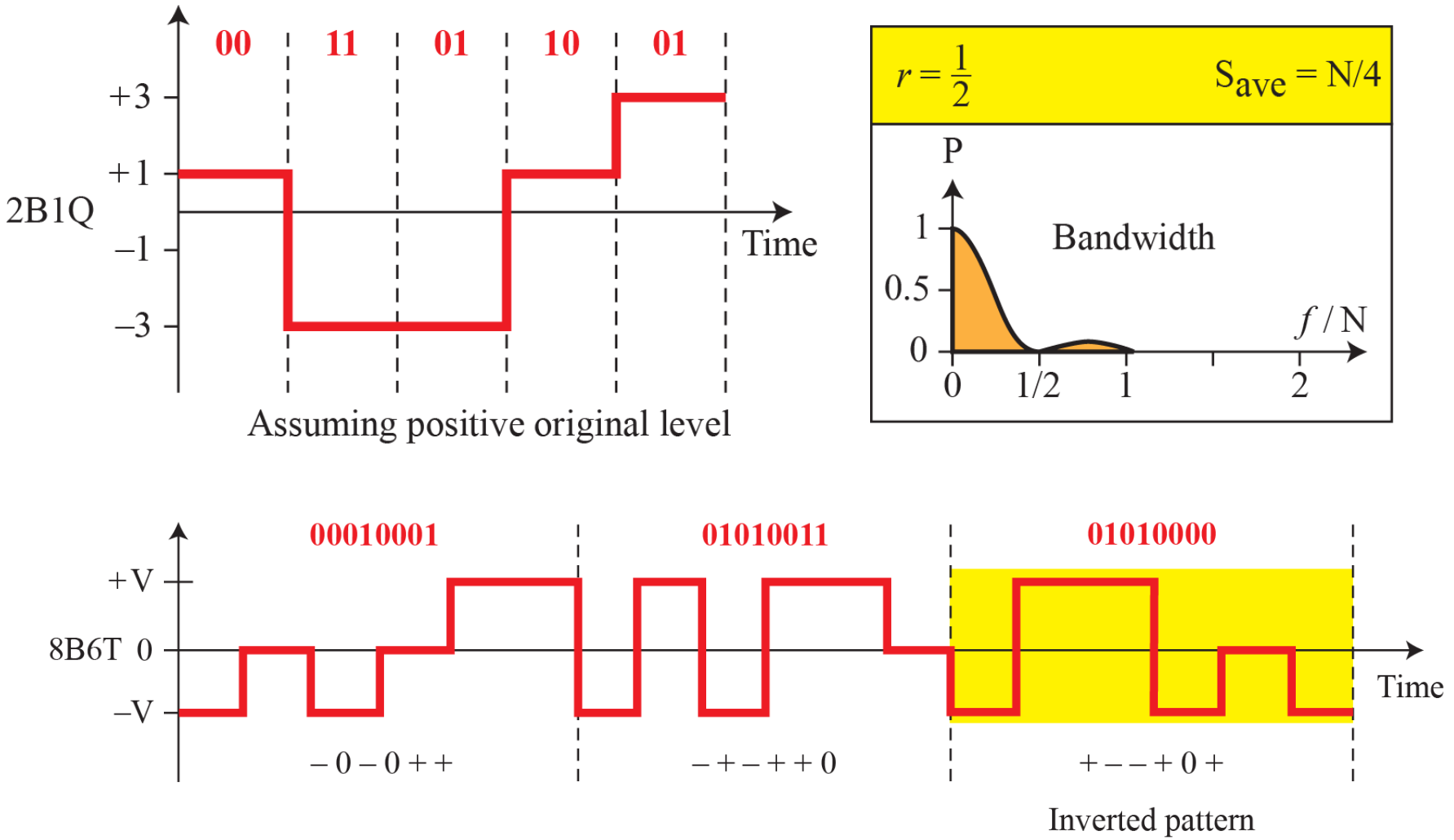

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Figure 7.23: Block coding concept

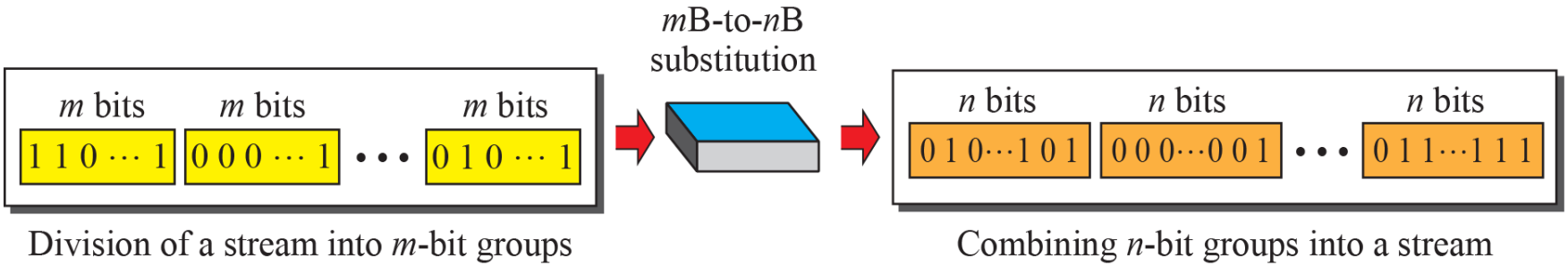

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Figure 7.24: Using block coding 4B/5B with NRZ-I line coding scheme

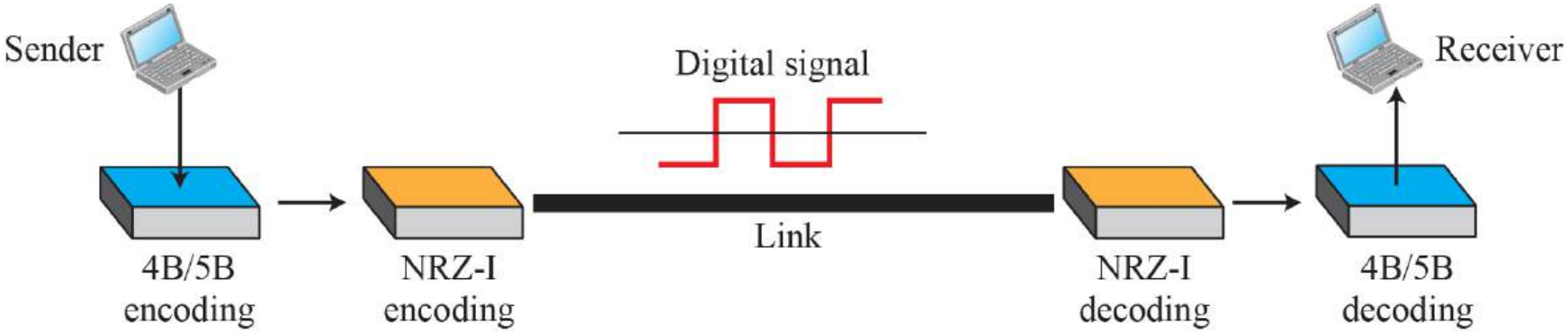

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Figure 7.25: 8B/10B block encoding

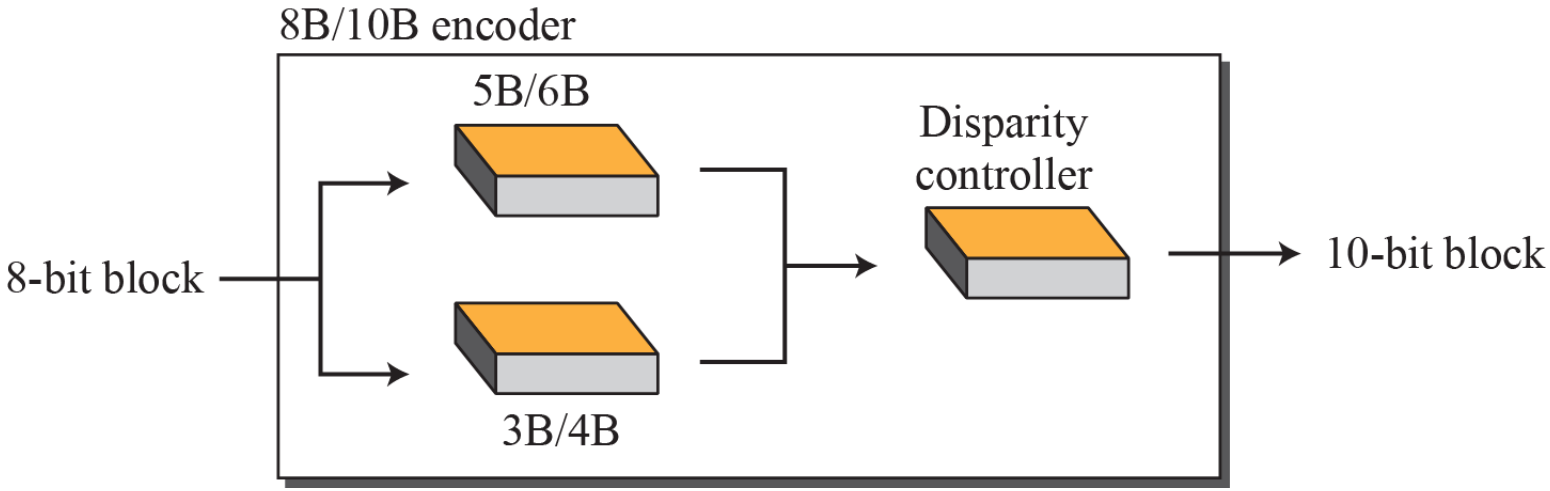

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Figure 7.26: AMI used with scrambling

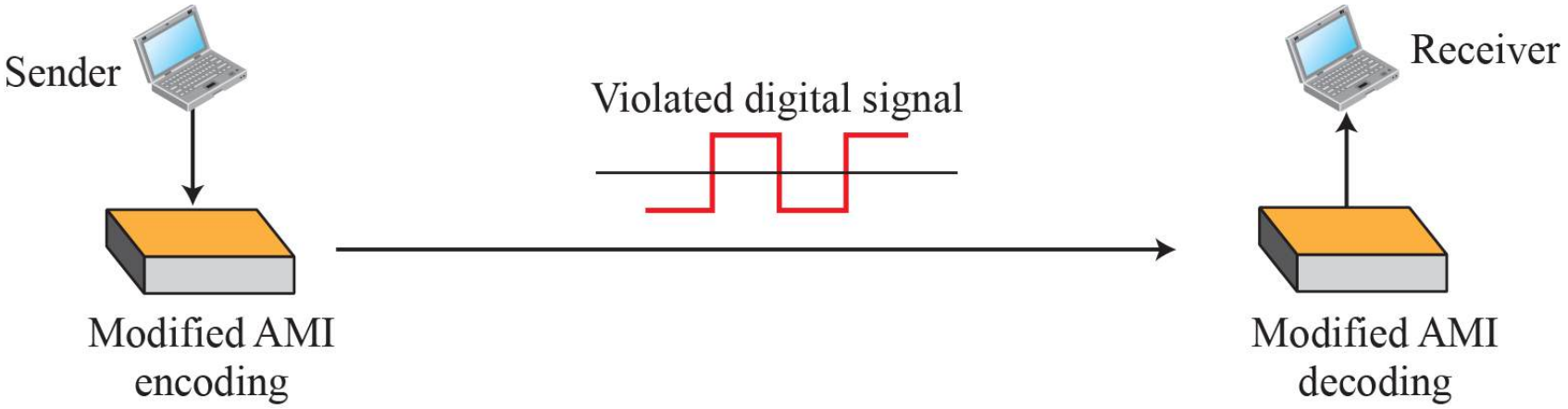

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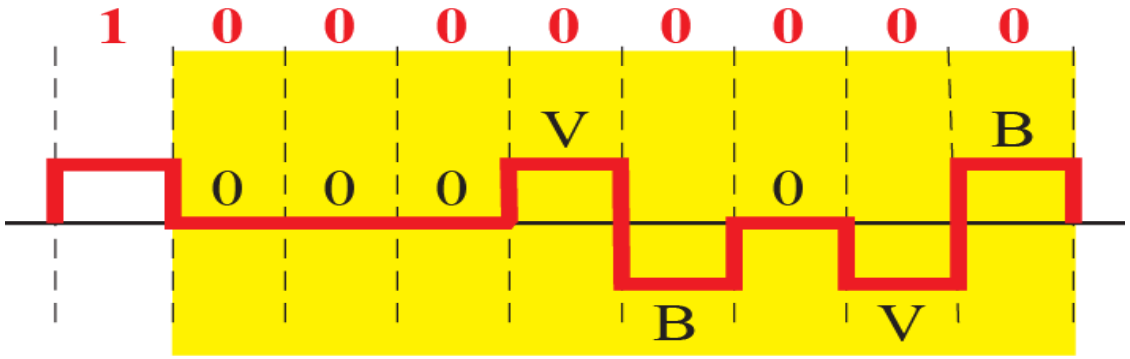
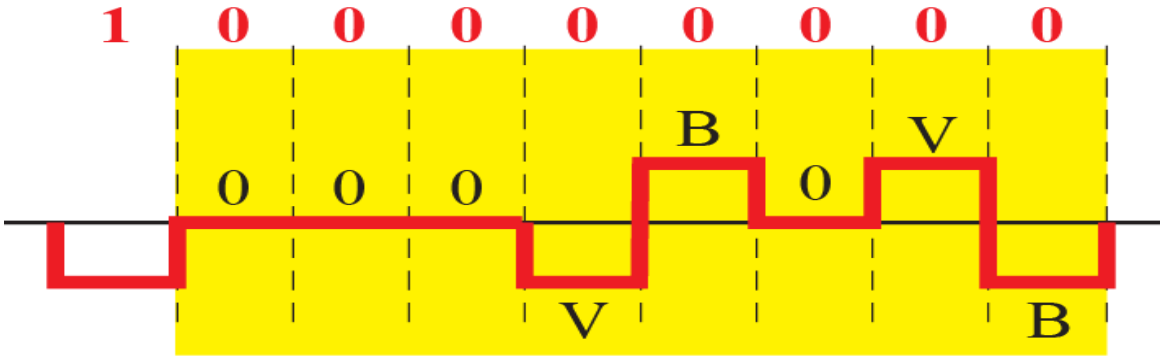
Figure 7.27: Two cases of B8ZS scrambling technique



a. Previous level is positive.



b. Previous level is negative.

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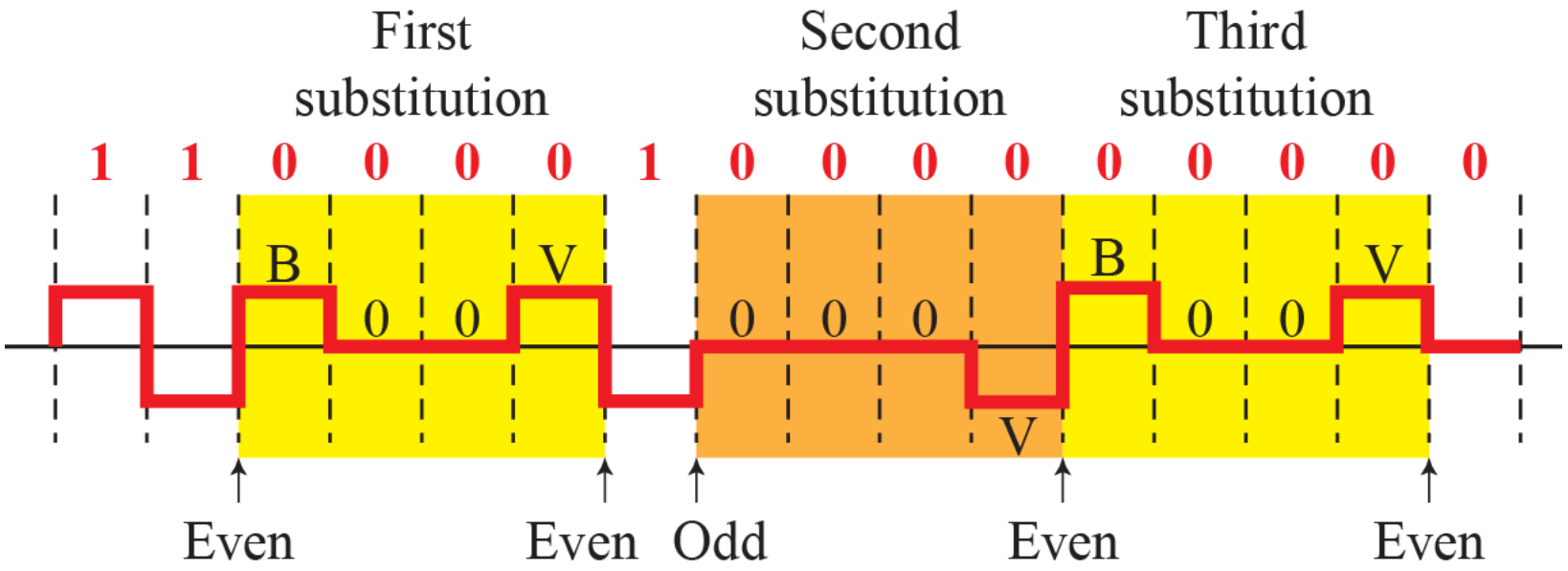
Figure 7.28: Different situations in HDB3 scrambling technique



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7.2.2 Analog-to-Digital Conversion

The techniques described in Section 7.2.1 convert digital data to digital signals. Sometimes, however, we have an analog signal such as one created by a microphone or camera. The tendency today is to change an analog signal to digital data because the digital signal is less susceptible to noise. In this section we describe two techniques, pulse code modulation and delta modulation. After the digital data are created (digitization),

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7.2.2 (continued)

- **Pulse Code Modulation (PCM)**
  - Sampling
  - Quantization
  - Encoding
  - Original Signal Recovery
  - PCM Bandwidth
- **Delta Modulation (DM)**

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Figure 7.29: Components of PCM encoder

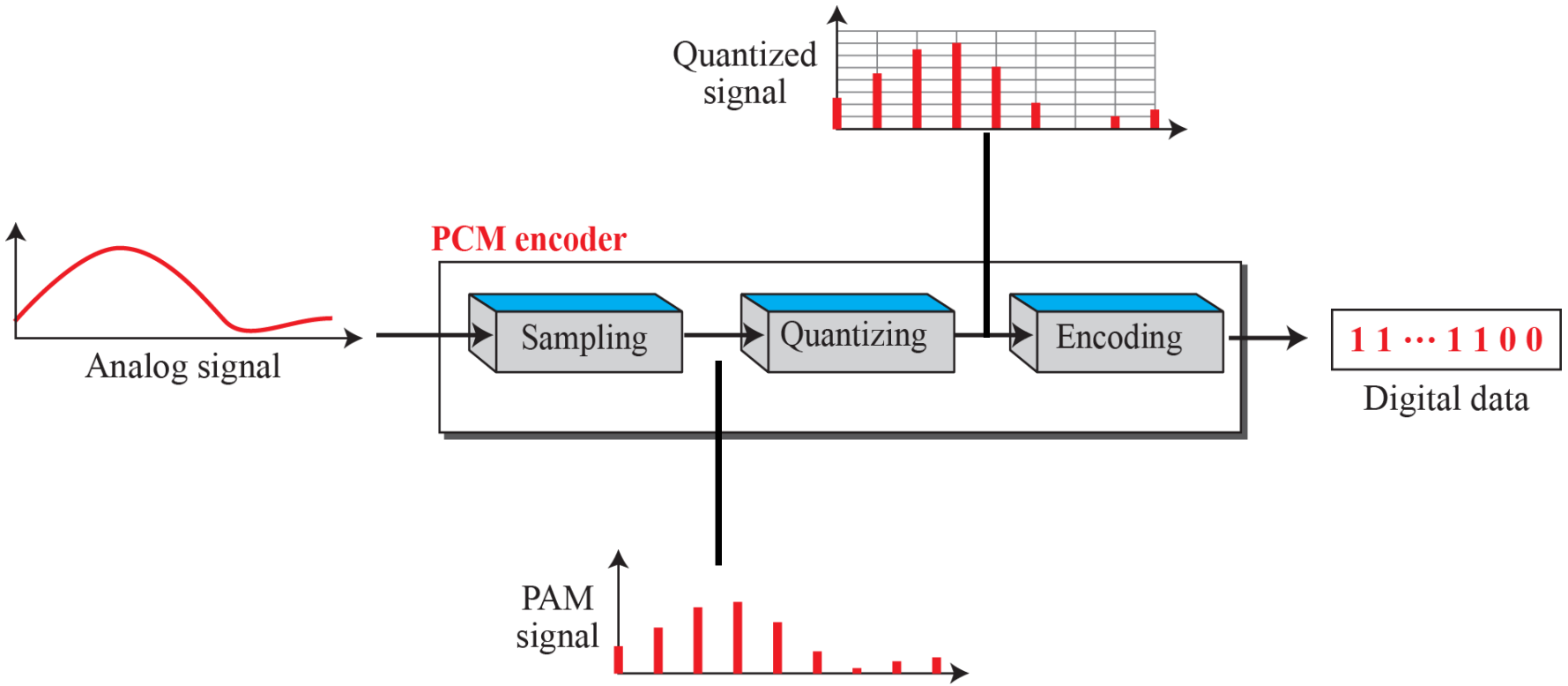

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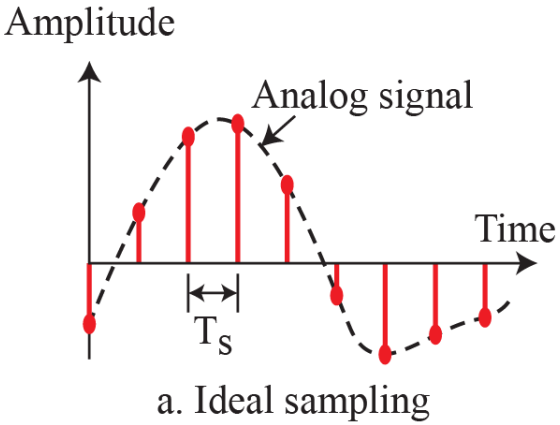
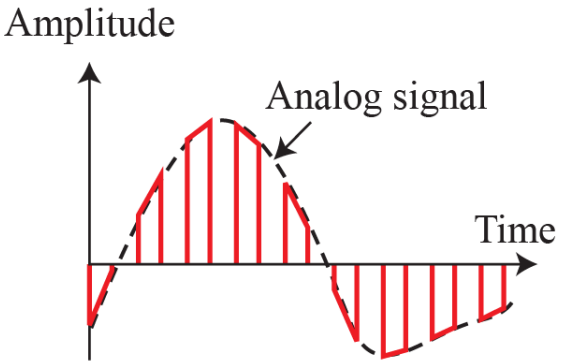
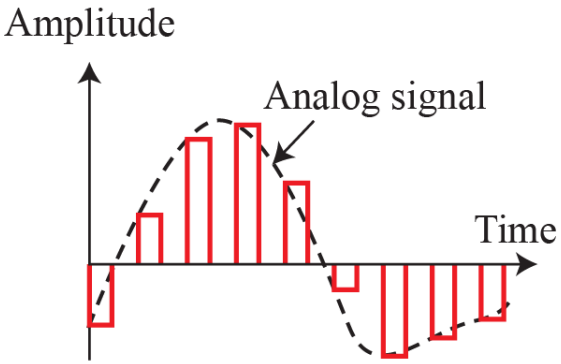
Figure 7.30: Three different sampling methods for PCM



a. Ideal sampling



b. Natural sampling



c. Flat-top sampling

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Figure 7.30: Nyquist sampling rate for low-pass and bandpass signals

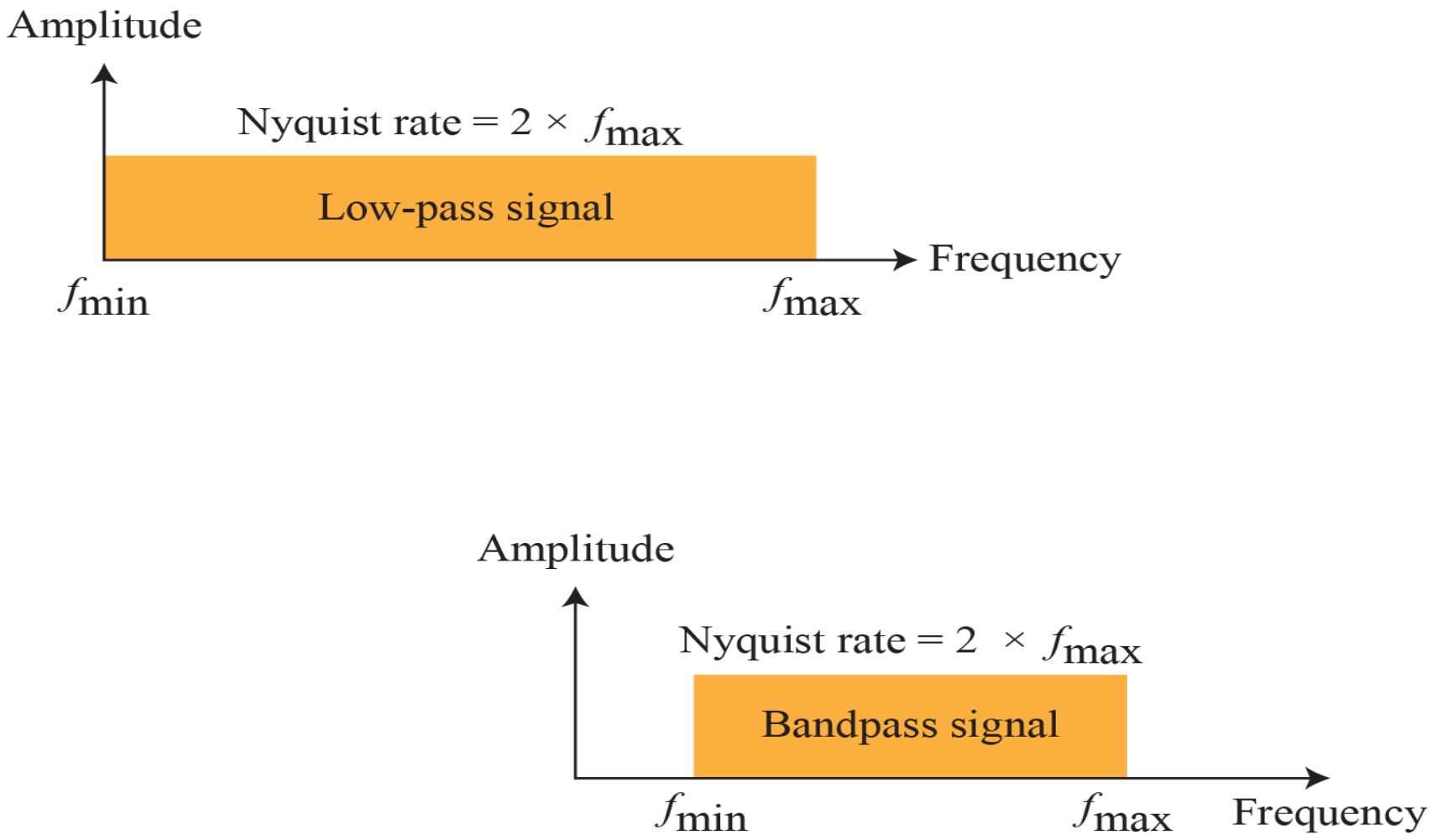

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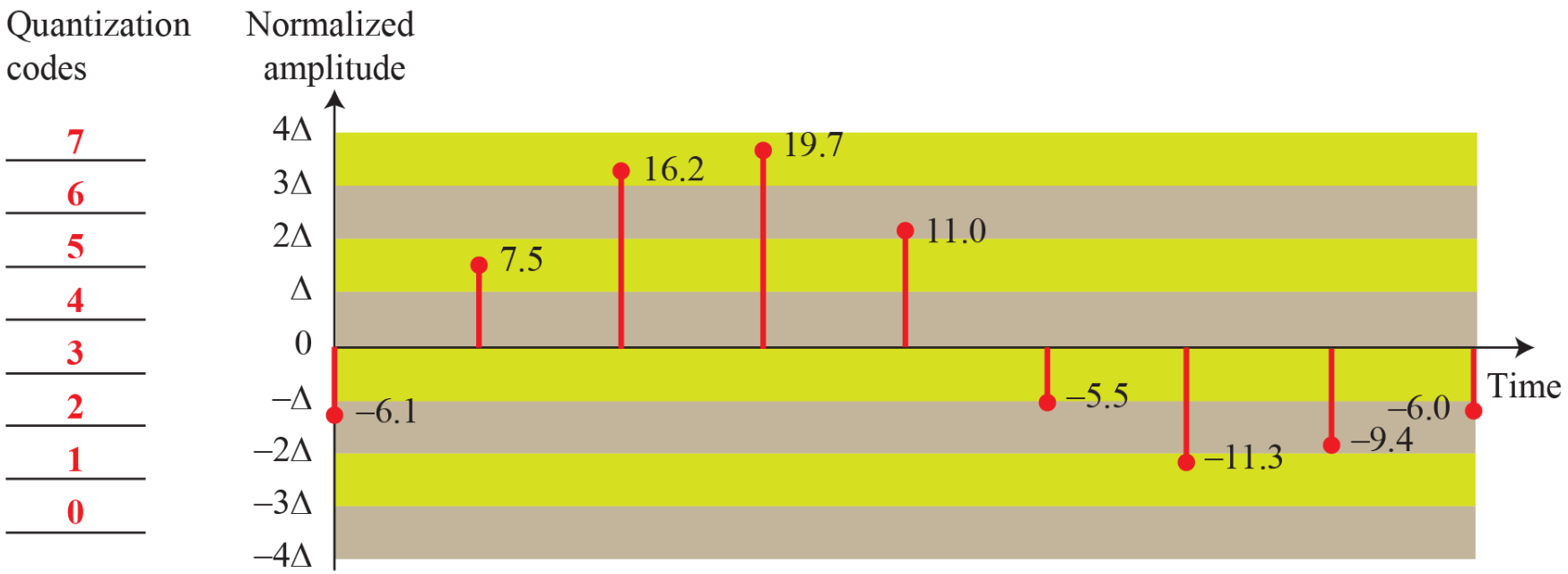
Figure 7.32: Quantization and encoding of a sampled signal



Normalized PAM values	−1.22	1.50	3.24	3.94	2.20	−1.10	−2.26	−1.88	−1.20
Normalized quantized values	−1.50	1.50	3.50	3.50	2.50	−1.50	−2.50	−1.50	−1.50
Normalized error	−0.28	0	+0.26	−0.44	+0.30	−0.40	−0.24	+0.38	−0.30
Quantization code	2	5	7	7	6	2	1	2	2
Encoded words	010	101	111	111	110	010	001	010	010

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Example 7.13

We want to digitize the human voice. What is the bit rate, assuming 8 bits per sample?

Solution

The human voice normally contains frequencies from 0 to 4000 Hz. So the sampling rate and bit rate are calculated as follows.

$$\text{Sampling rate} = 4000 \times 2 = 8000 \text{ samples/s} \rightarrow \text{Bit rate} = 8000 \times 8 = 64{,}000 \text{ bps} = 64 \text{ kbps}$$

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Figure 7.33: Components of a PCM decoder

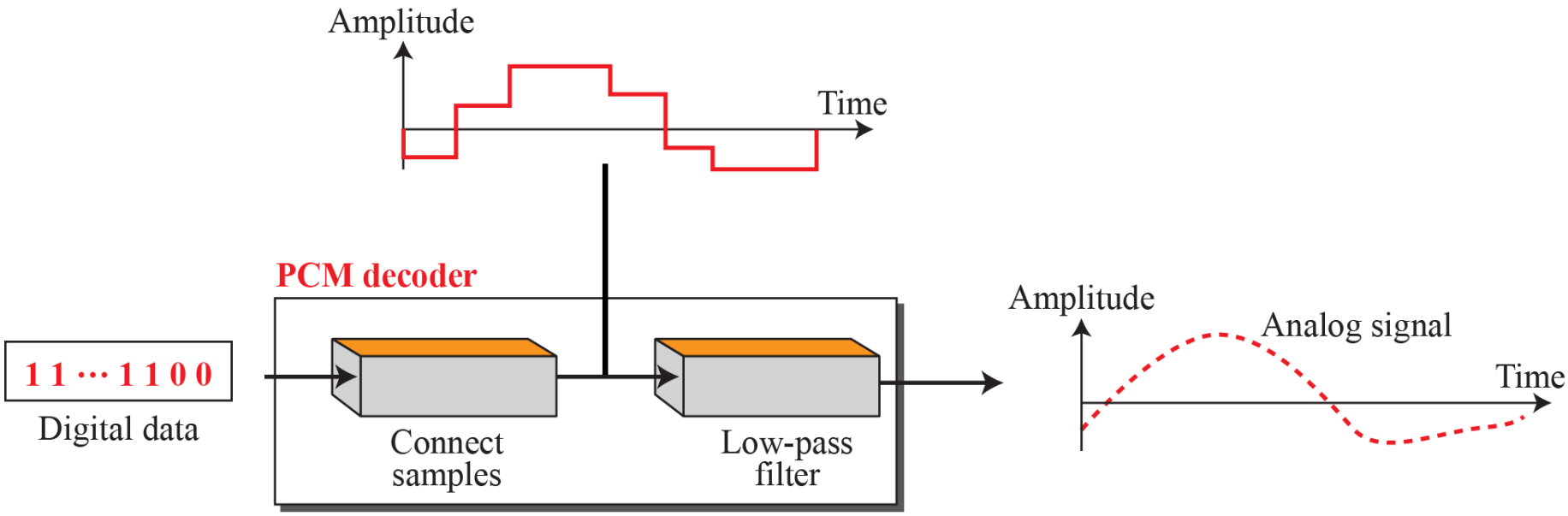

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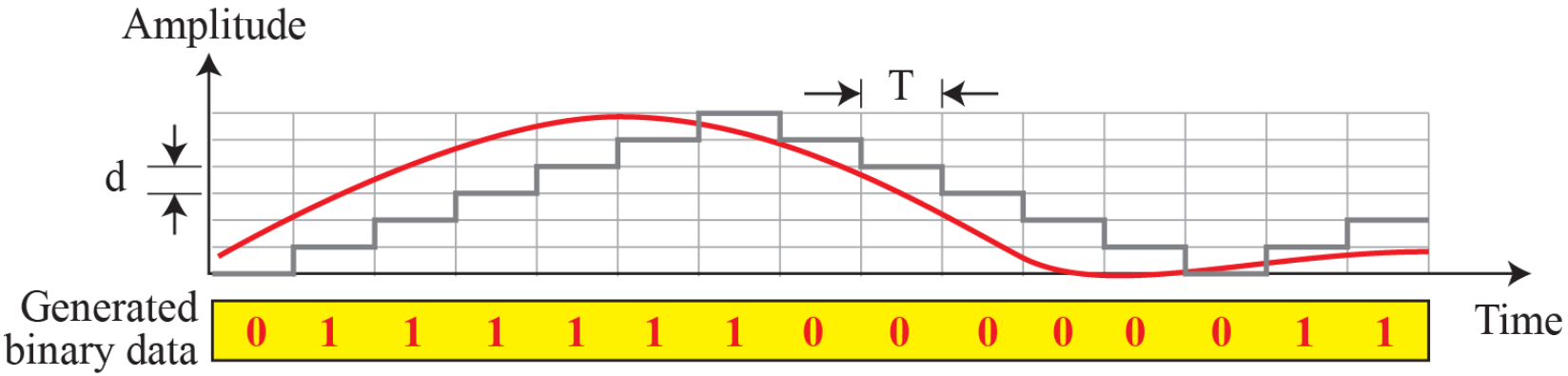
Figure 7.34: The process of delta modulation



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7-3 ANALOG TRANSMISSION

While digital transmission is desirable, it needs a low-pass channel; analog transmission is the only choice if we have a bandpass channel. Converting digital data to a bandpass analog signal is traditionally called digital-to-analog conversion. Converting a low-pass analog signal to a bandpass analog signal is traditionally called analog-to-analog conversion.

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7.3.1 Digital-to-Analog Conversion

Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data. Figure 7.35 shows the relationship between the digital information, the digital-to-analog modulating process, and the resultant analog signal.

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7.3.1 (continued)

- **Amplitude Shift Keying**
  - Binary ASK (BASK)
  - Multilevel ASK
  - Binary FSK (BFSK)
  - Multilevel FSK
- **Phase Shift Keying**
  - Binary PSK (BPSK)
  - Quadrature PSK (QPSK)
  - Constellation Diagram
- **Quadrature Amplitude Modulation**
  - Bandwidth for QAM

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Figure 7.35: Digital-to-analog conversion

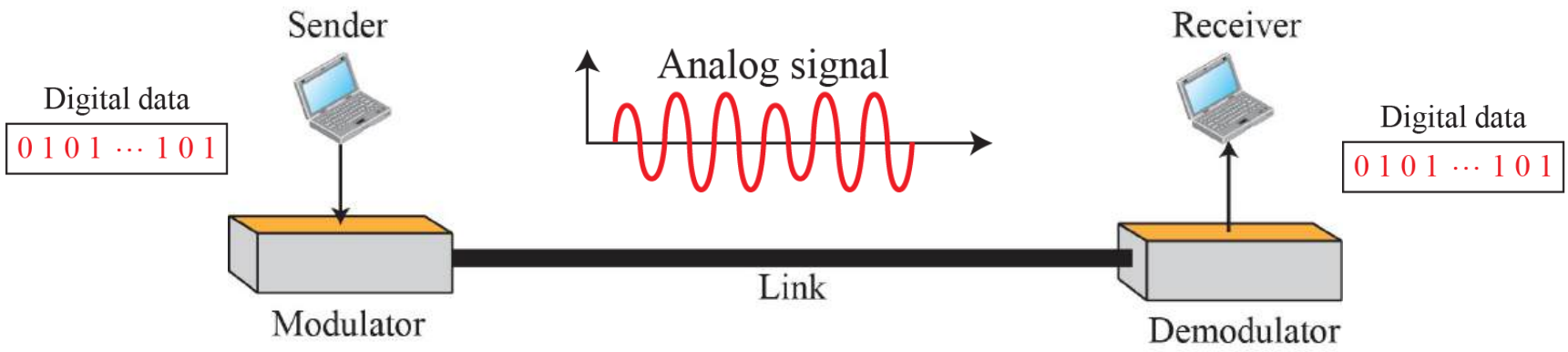

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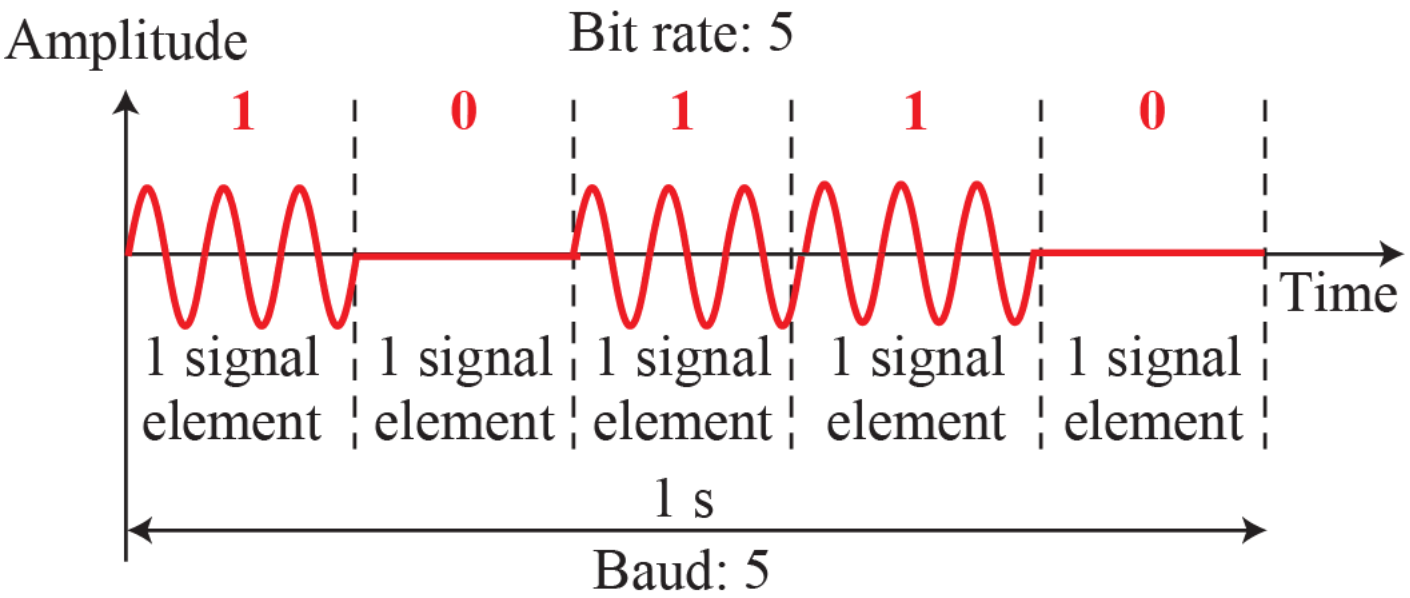
Figure 7.36: Binary amplitude shift keying



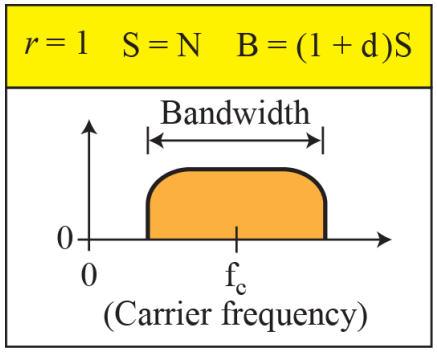


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Figure 7.37: Binary frequency shift keying

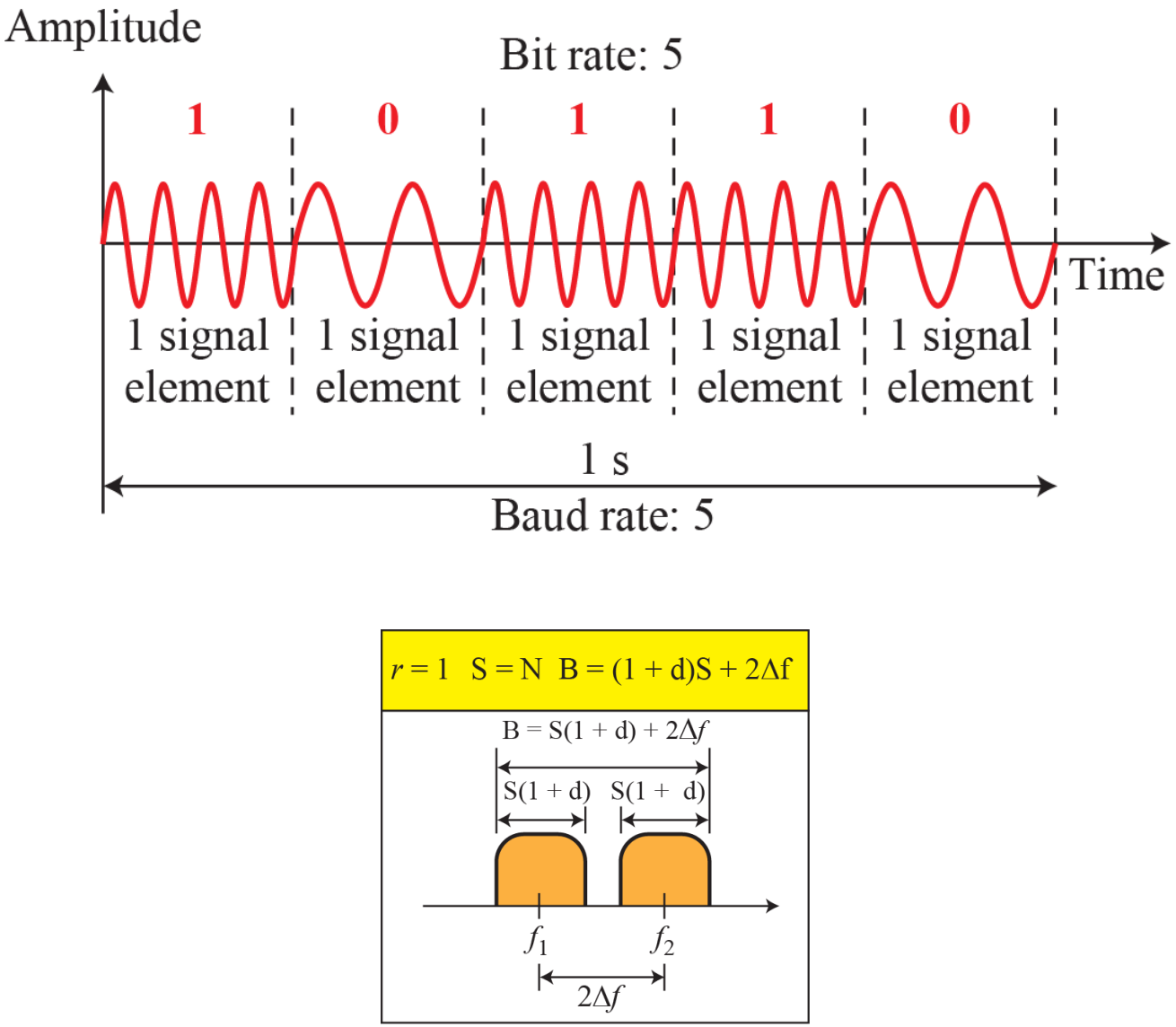

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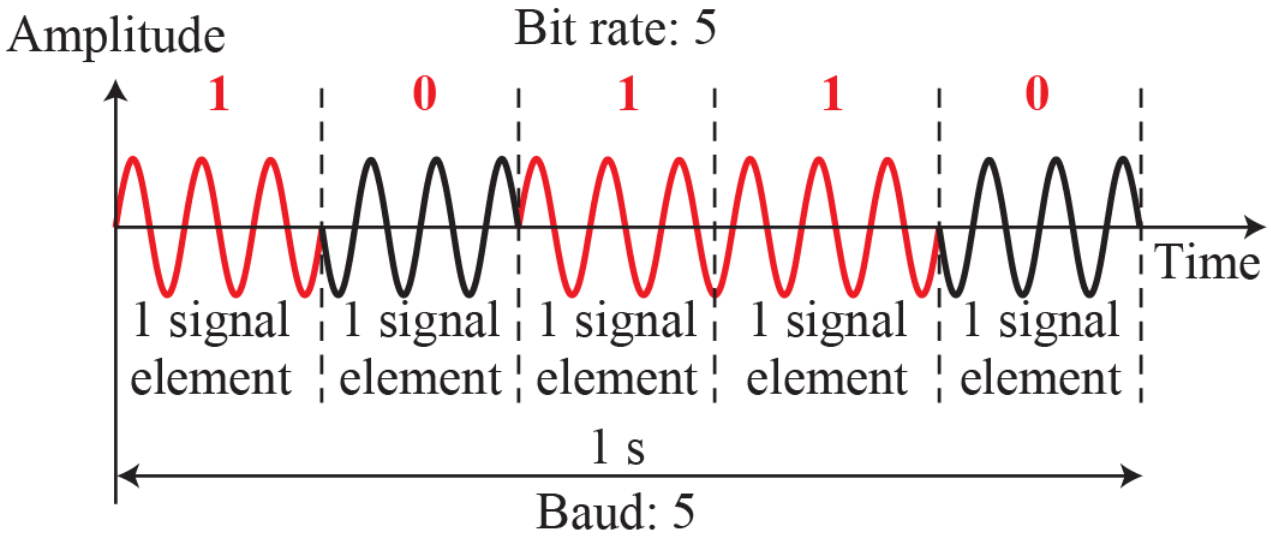
Figure 7.38: Binary phase shift keying



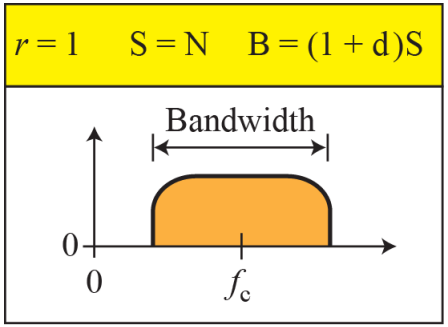


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Figure 7.39: Concept of a constellation diagram

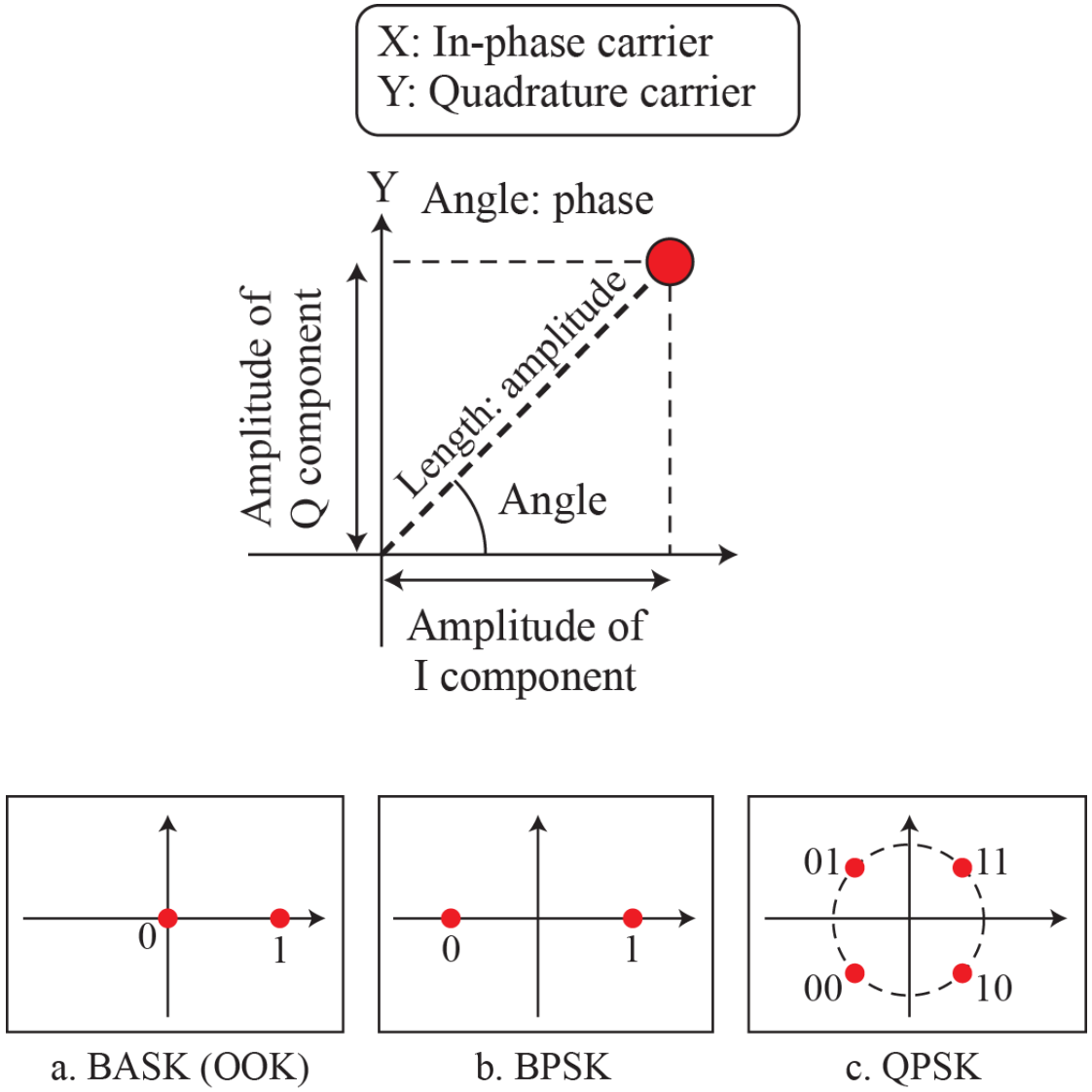

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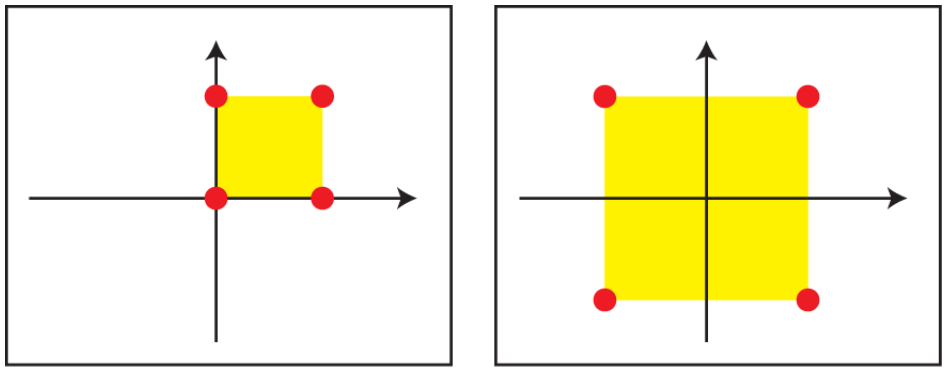
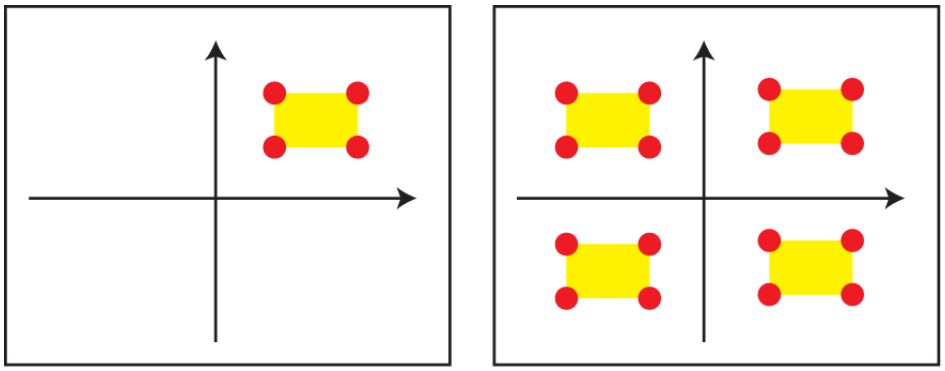
Figure 7.40: Constellation diagrams for some QAMs



a. 4-QAM b. 4-QAM



c. 4-QAM d. 16-QAM

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7.3.2 Analog-to-Analog Conversion

Analog-to-analog conversion, or analog modulation, is the representation of analog information by an analog signal. One may ask why we need to modulate an analog signal; it is already analog. Modulation is needed if the medium is bandpass in nature or if only a bandpass channel is available to us.

- ***Amplitude Modulation***
- ***Frequency Modulation***
- ***Phase Modulation***

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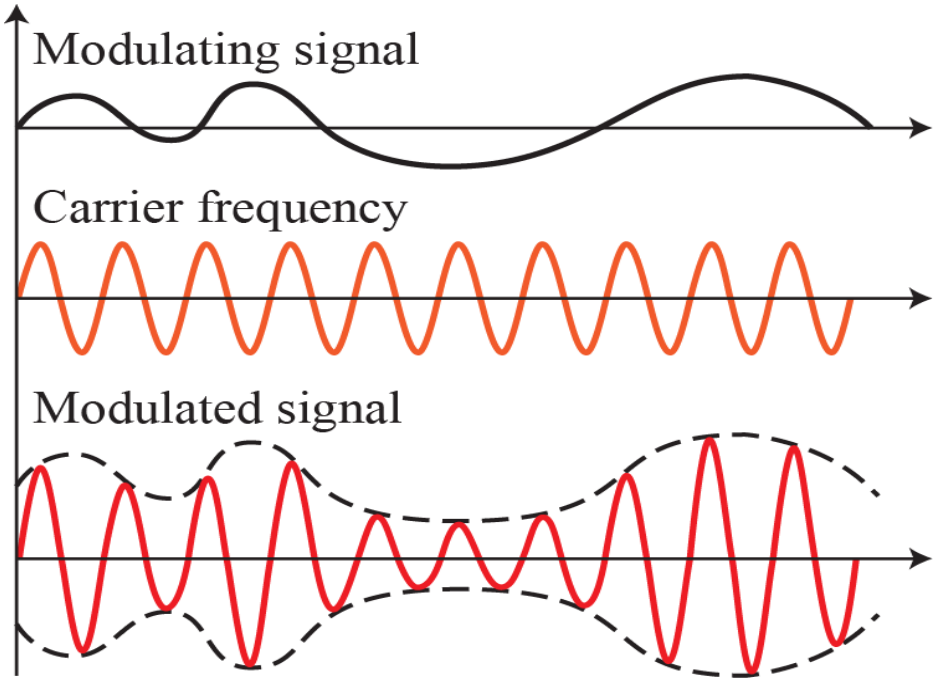
Figure 7.41: Amplitude modulation



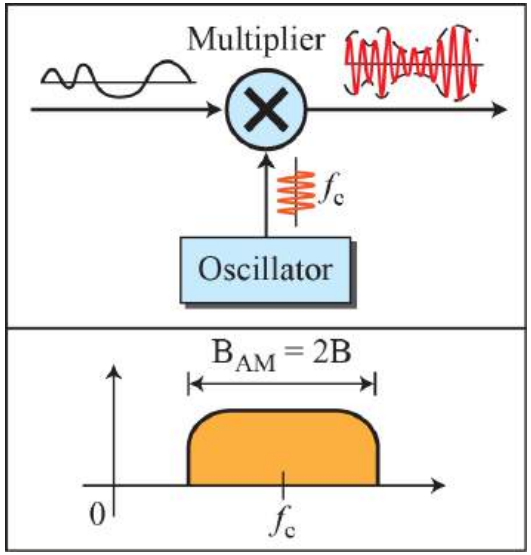


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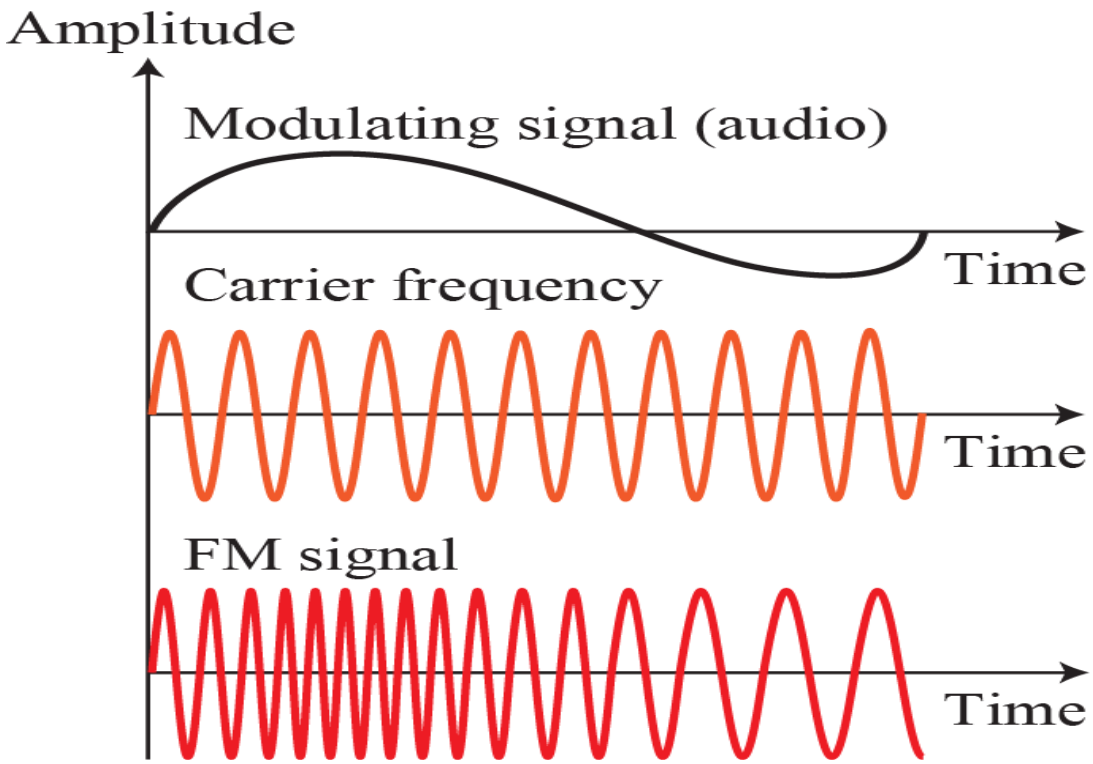
Figure 7.42: Frequency modulation



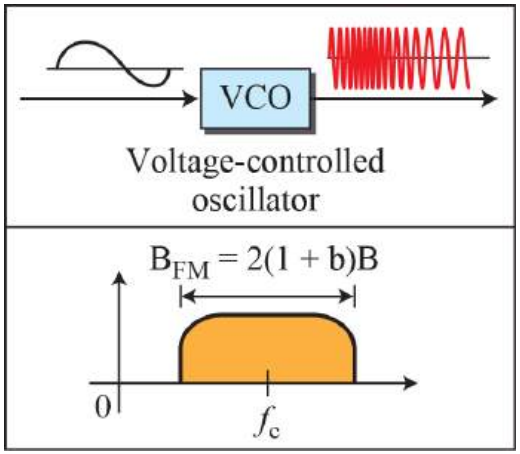


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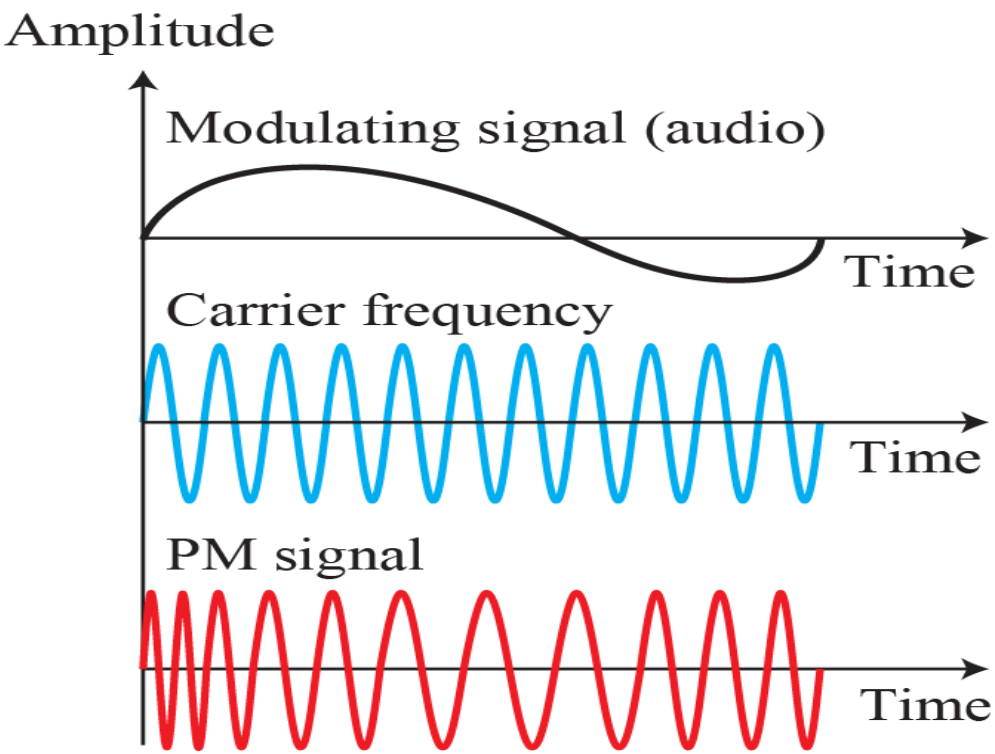
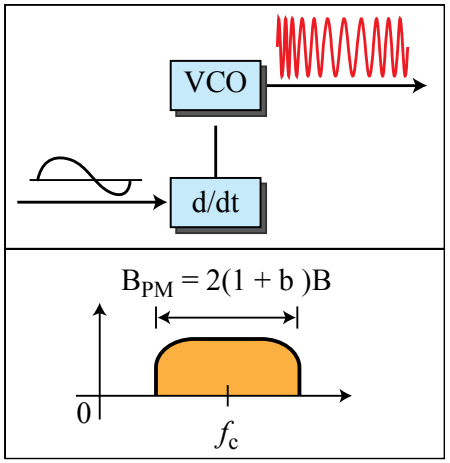
Figure 7.43: Phase modulation





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7-4 BANDWIDTH UTILIZATION

In real life, we have links with limited bandwidths. Sometimes we need to combine several low-bandwidth channels to make use of one channel with a larger bandwidth. Sometimes we need to expand the bandwidth of a channel to achieve goals such as privacy and anti-jamming.

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7.4.1 Multiplexing

Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link. As data and telecommunications use increases, so does traffic. We can accommodate this increase by continuing to add individual links each time a new channel is needed, or we can install higher-bandwidth links and use each to carry multiple signals.

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7.4.1 (continued)

- **Frequency-Division Multiplexing**
- **Wavelength-Division Multiplexing**
- **Time-Division Multiplexing**
  - Synchronous TDM
  - Statistical Time-Division Multiplexing

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Figure 7.44: Dividing a link into channels

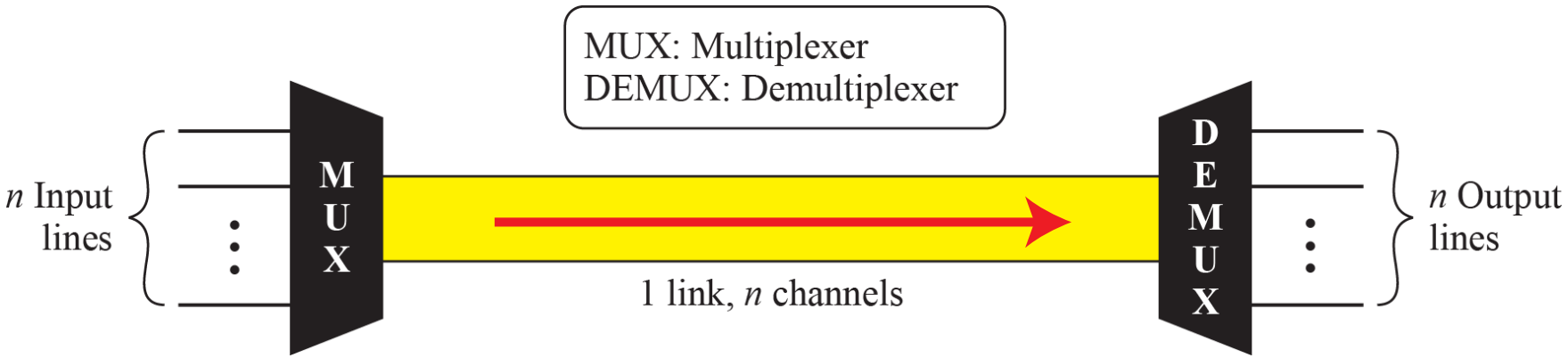

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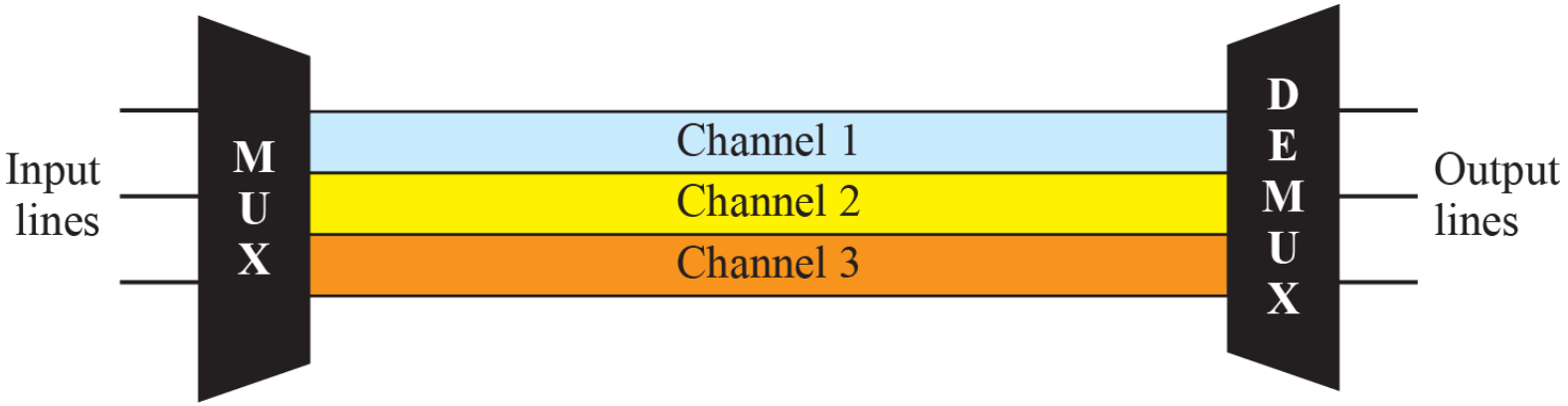
Figure 7.45: Frequency-division multiplexing



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Example 7.14

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.

Solution

We shift (modulate) each of the three voice channels to a different bandwidth, as shown in Figure 7.46. We use the 20- to 24-kHz bandwidth for the first channel, the 24- to 28-kHz bandwidth for the second channel, and the 28- to 32-kHz bandwidth for the third one. Then we combine them as shown in Figure 7.46. At the receiver, each channel receives the entire signal, using a filter to separate out its own signal. The first channel uses a filter that passes frequencies between 20 and 24 kHz and filters out (discards) any other frequencies. The second channel uses a filter that passes frequencies between 24 and 28 kHz, and the third channel uses a filter that passes frequencies between 28 and 32 kHz. Each channel then shifts the frequency to start from zero.

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Figure 7.46: Example 7.14

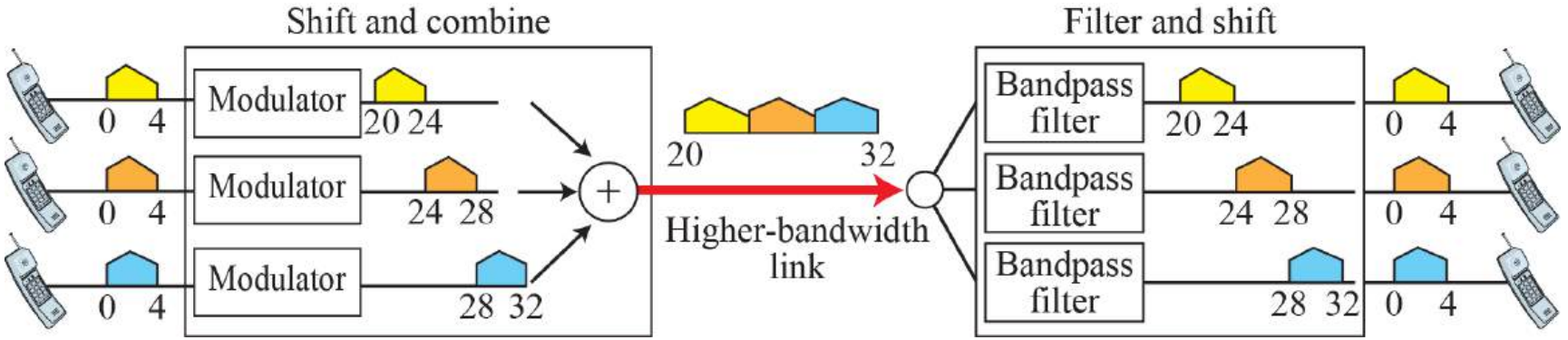

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Figure 7.47: Wavelength-division multiplexing

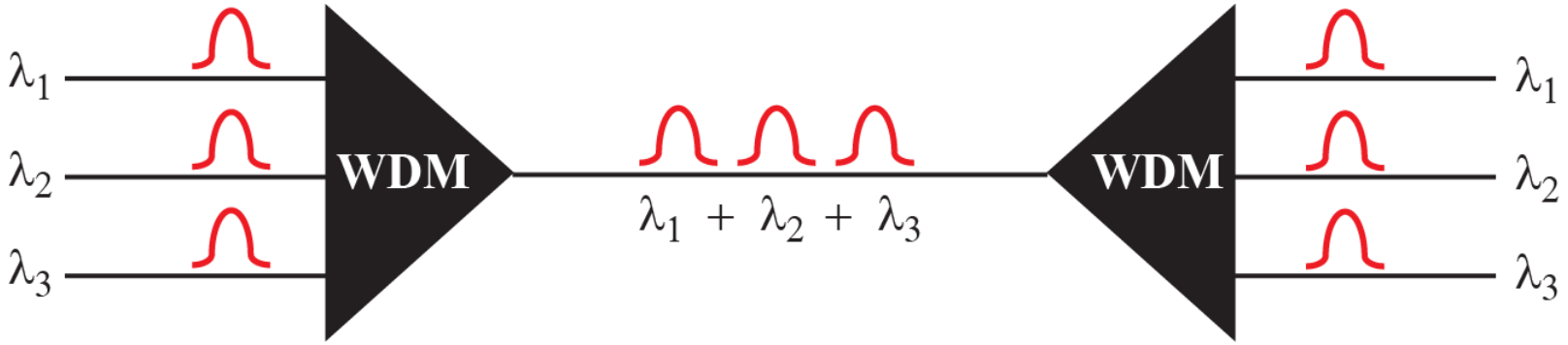

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Figure 7.48: TDM

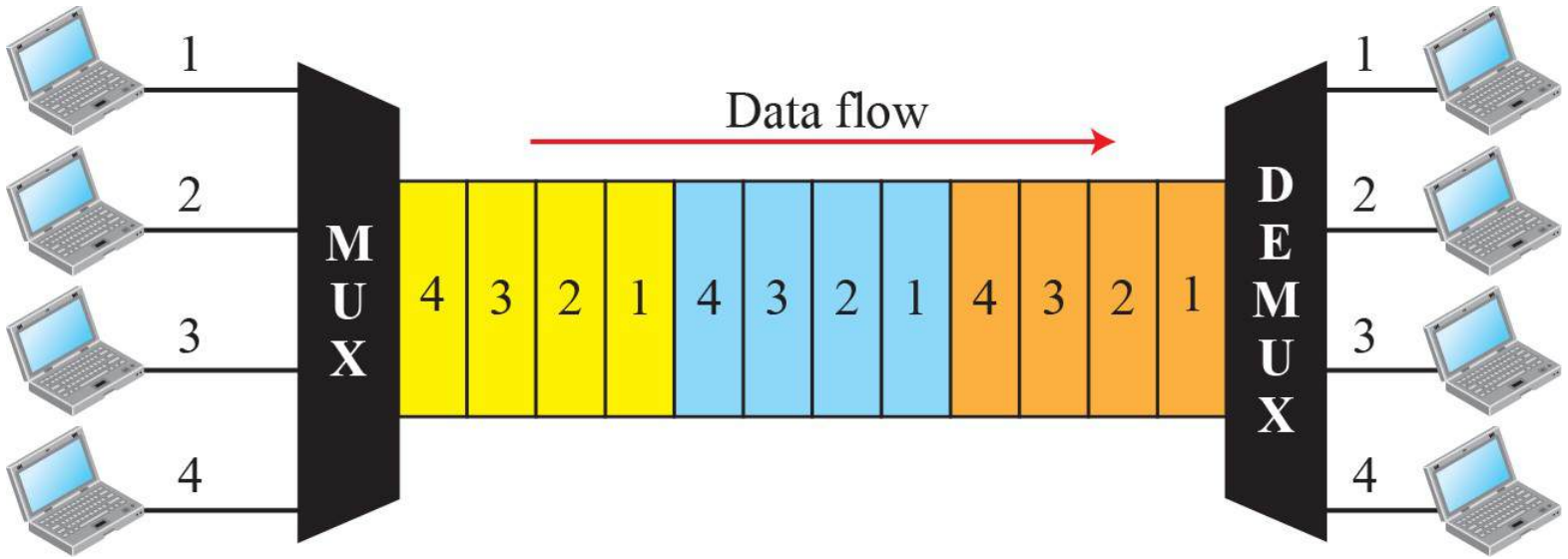

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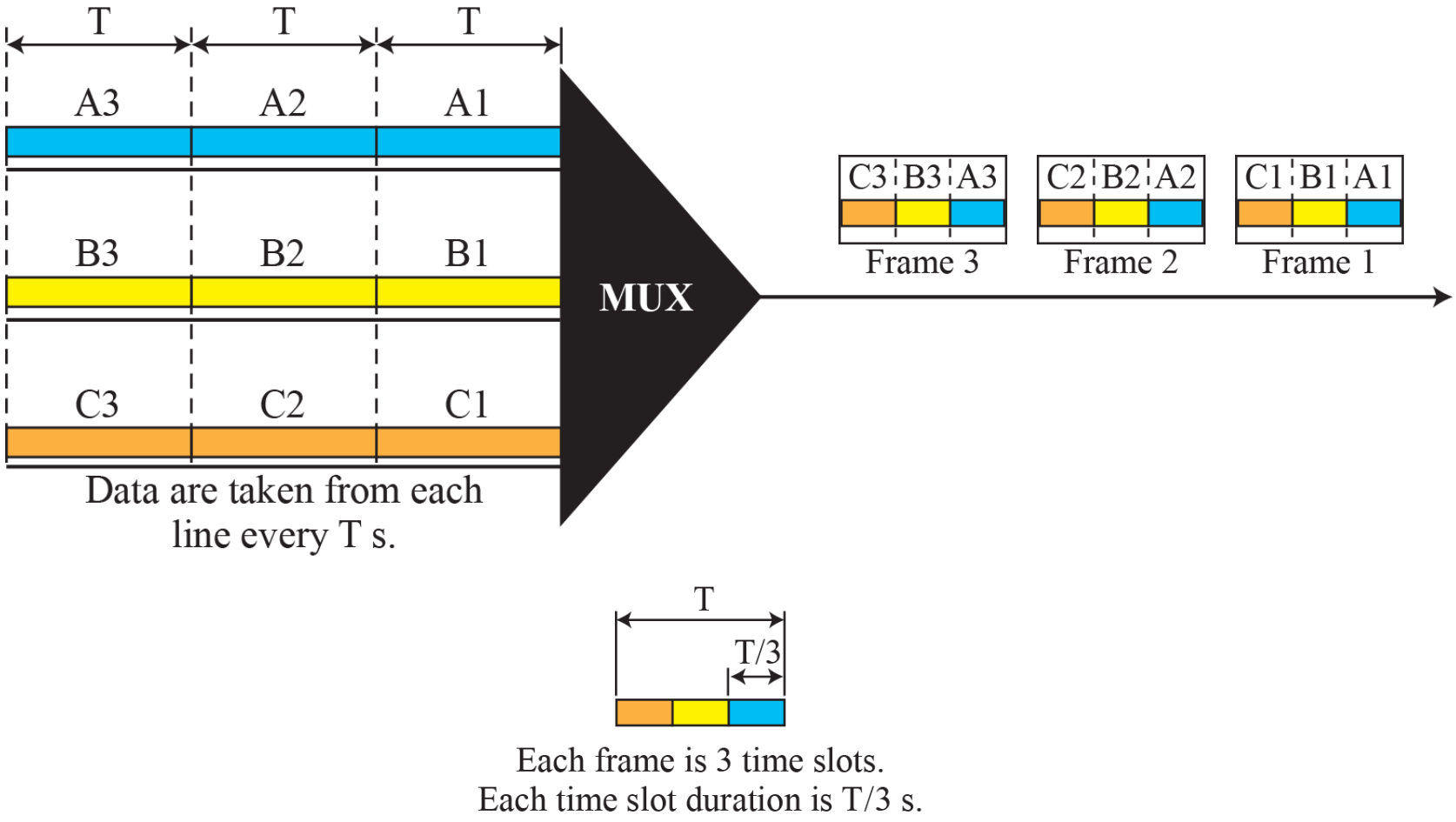
Figure 7.49: Synchronous time-division multiplexing



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Example 7.15

Figure 7.50 shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate, and (d) the output frame rate.

Solution

We can answer the questions as follows:

1. The input bit duration is the inverse of the bit rate: 1/1 Mbps = 1 μs.
2. The output bit duration is one-fourth of the input bit duration, or 1/4 μs.
3. The output bit rate is the inverse of the output bit duration, or 1/(1/4) μs or 4 Mbps. This can also be deduced from the fact that the output rate is 4 times as fast as any input rate; so the output rate = 4 × 1 Mbps = 4 Mbps.
4. The frame rate is always the same as any input rate. So the frame rate is 1,000,000 frames per second. Because we are sending 4 bits in each frame, we can verify the result of the previous question by multiplying the frame rate by the number of bits per frame.

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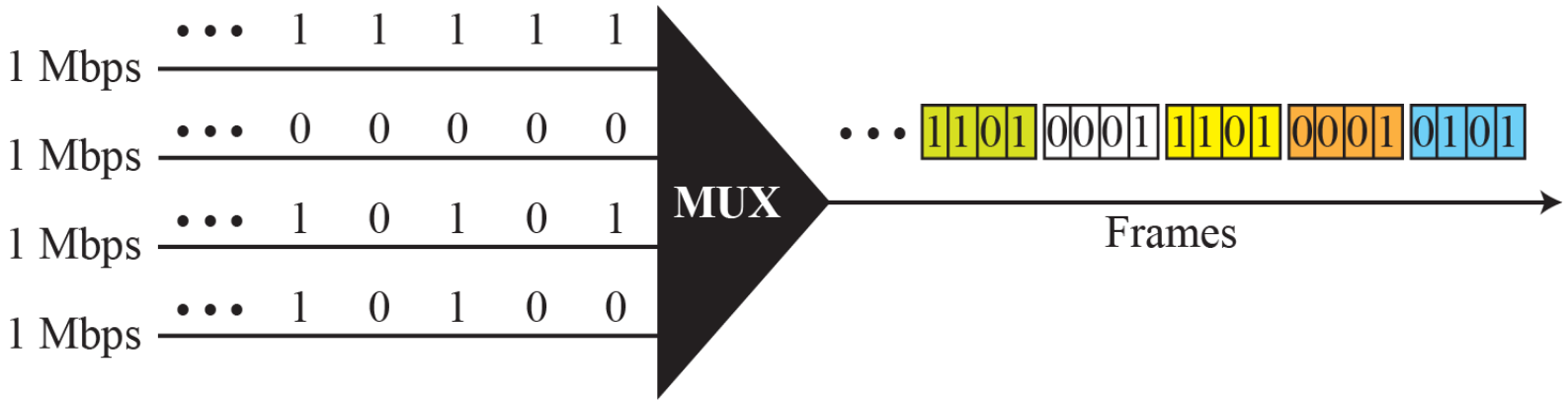
Figure 7.50: Example 7.15



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Example 7.16

Telephone companies implement TDM through a hierarchy of digital signals, called digital signal (DS) service or digital hierarchy. Figure 7.51 shows the data rates supported by each level. The commercial implementations of these services are referred to as T lines.

- **DS-0 service is a single digital channel of 64 kbps.**
- **DS-1 is a 1.544-Mbps service.**
- **DS-2 is a 6.312-Mbps service.**
- **DS-3 is a 44.376-Mbps service.**
- **DS-4 is a 274.176-Mbps service.**

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Figure 7.51: Digital hierarchy

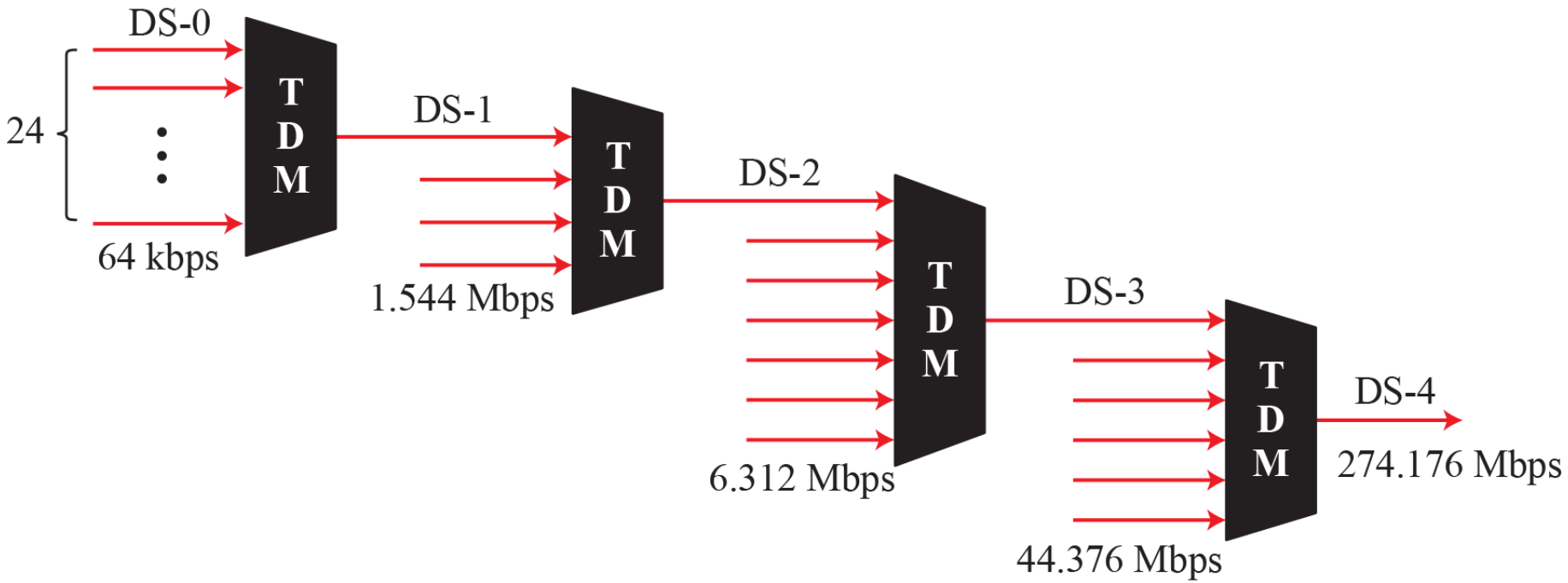

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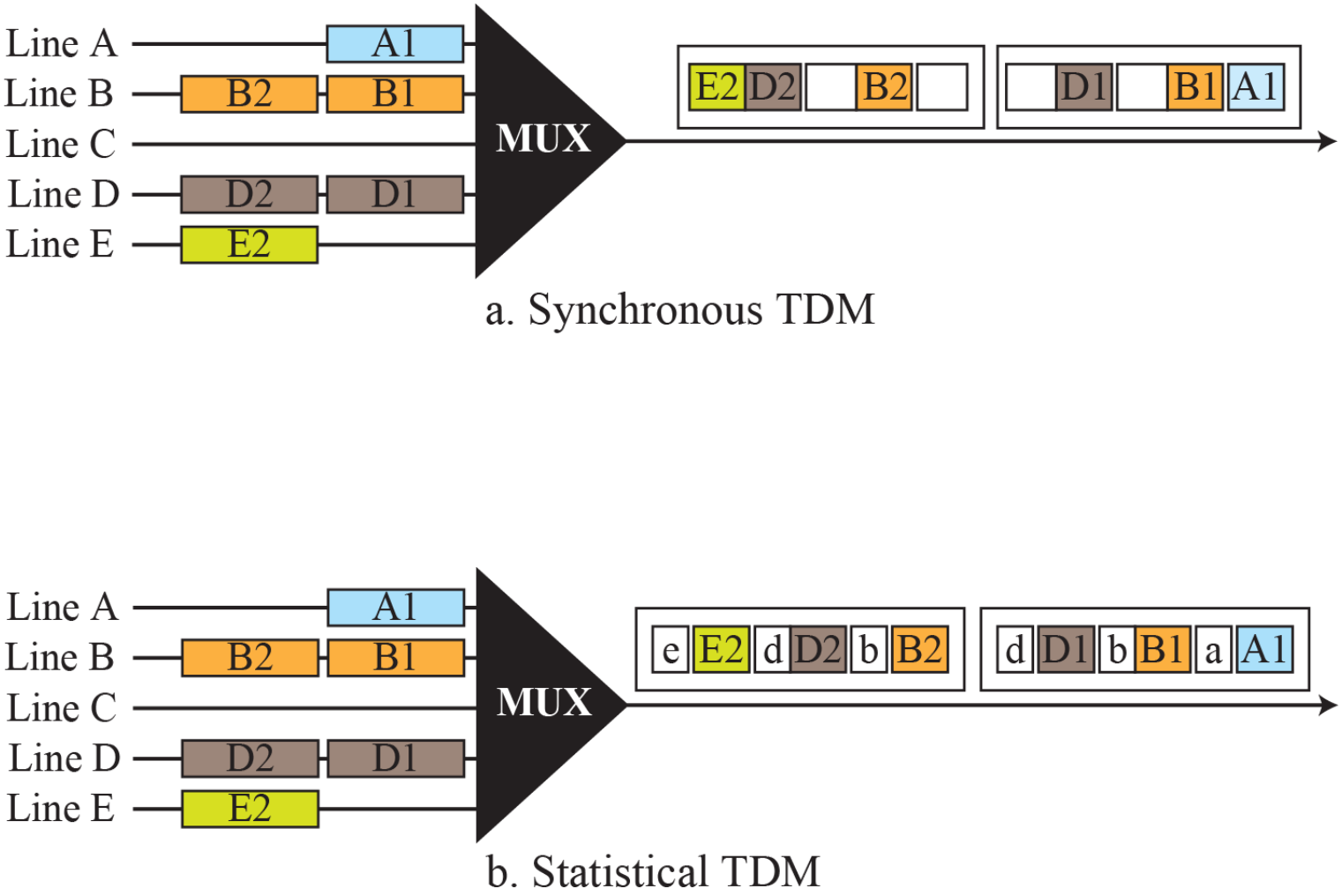
Figure 7.52: TDM slot comparison



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7.4.2 Spread Spectrum

In spread spectrum, we also combine signals from different sources to fit into a larger bandwidth, but our goals are somewhat different. In these types of applications, we have some concerns that outweigh bandwidth efficiency. In wireless applications, all stations use air (or a vacuum) as the medium for communication. Stations must be able to share this medium without interception by an eavesdropper and without being subject to jamming from a malicious intruder (in military operations, for example).

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7.4.2 (continued)

- **Frequency Hopping Spread Spectrum (FHSS)**
  - Bandwidth Sharing
- **Direct Sequence Spread Spectrum**

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Figure 7.53: Spread spectrum

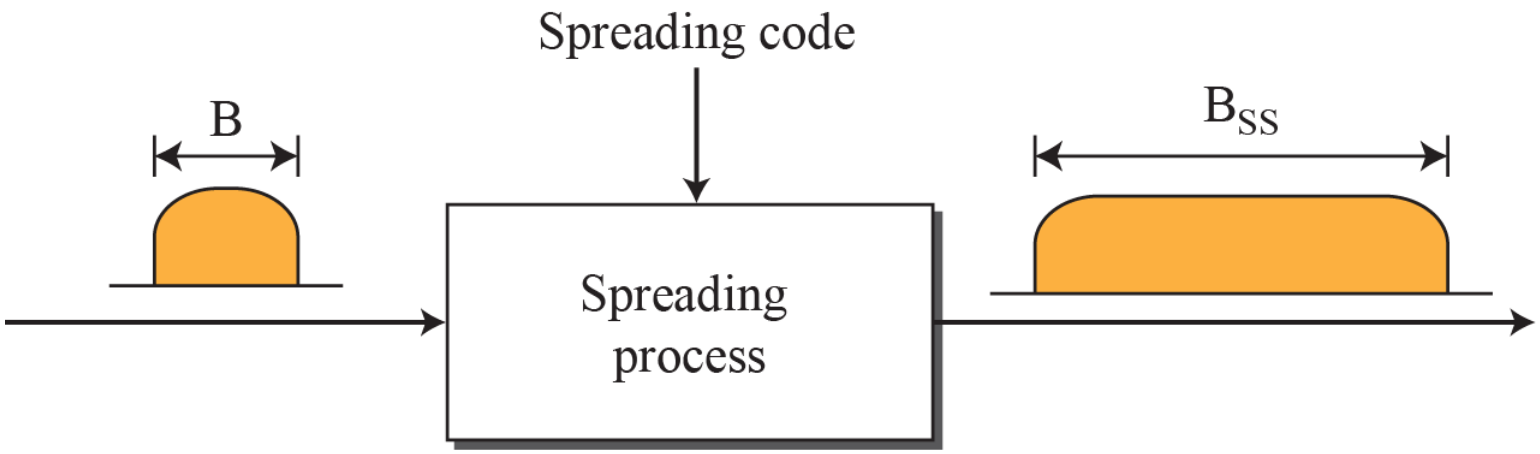

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Figure 7.54: Frequency hopping spread spectrum (FHSS)

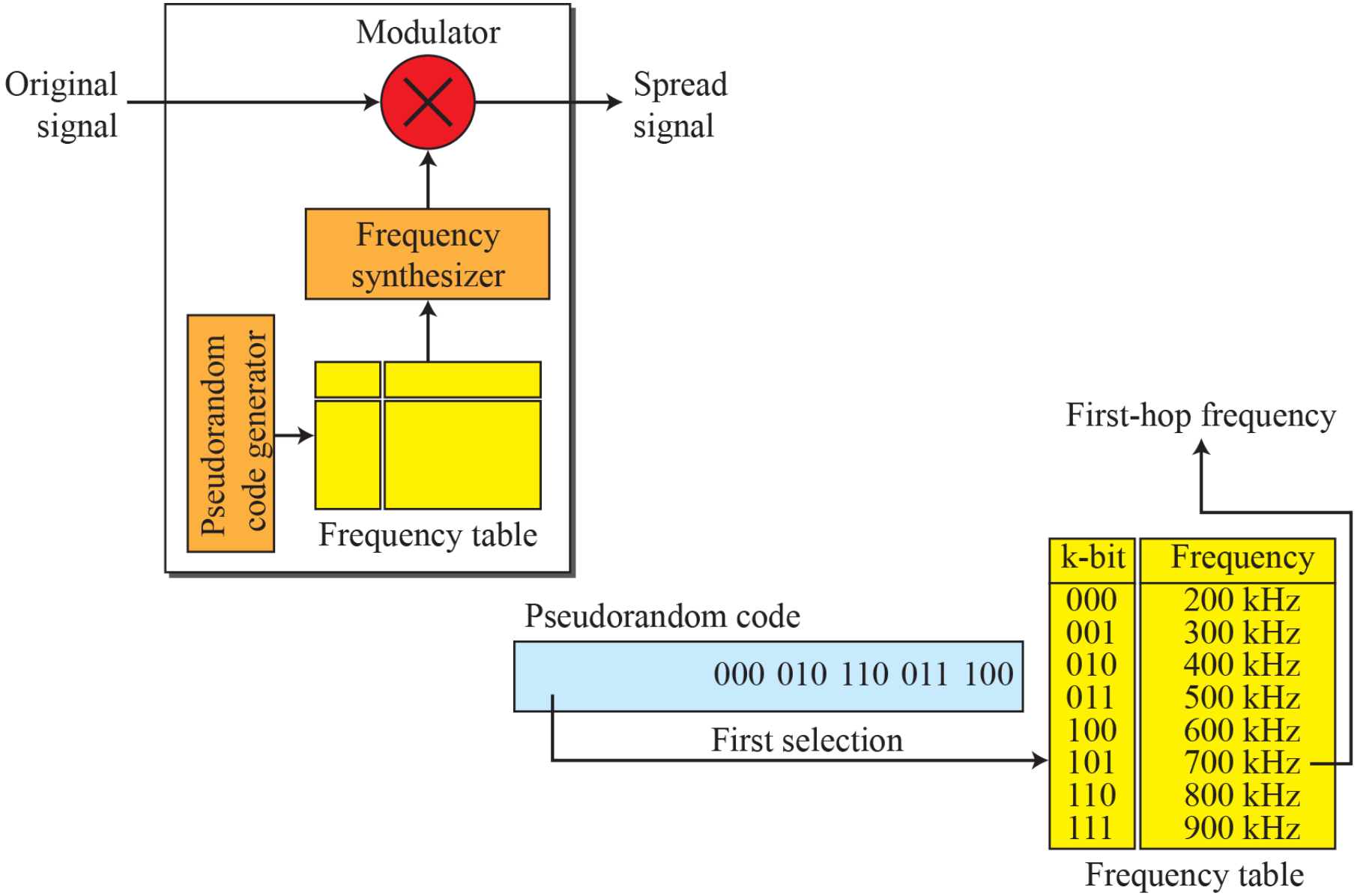

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Figure 7.55: FHSS cycles

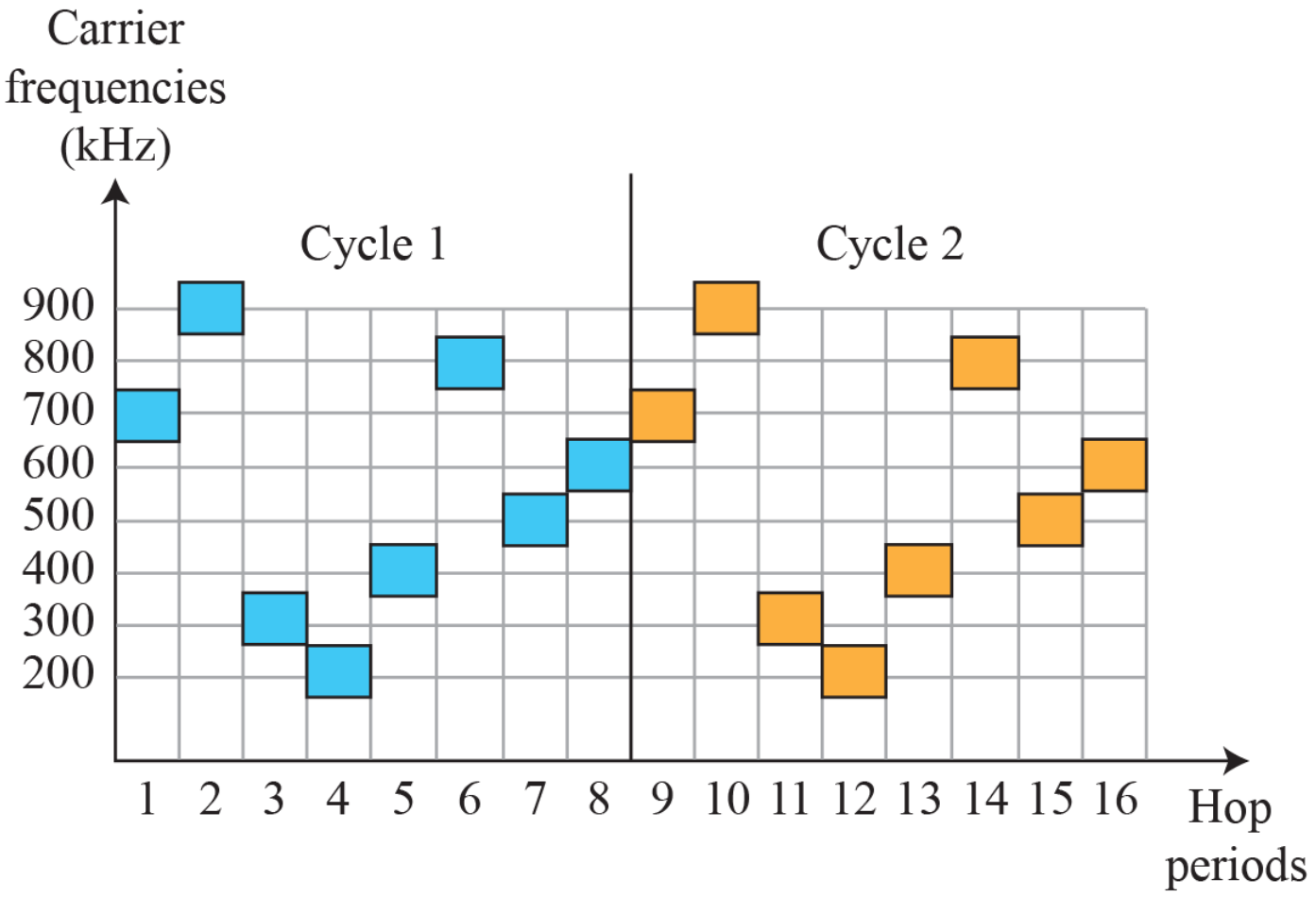

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Figure 7.56: Bandwidth sharing

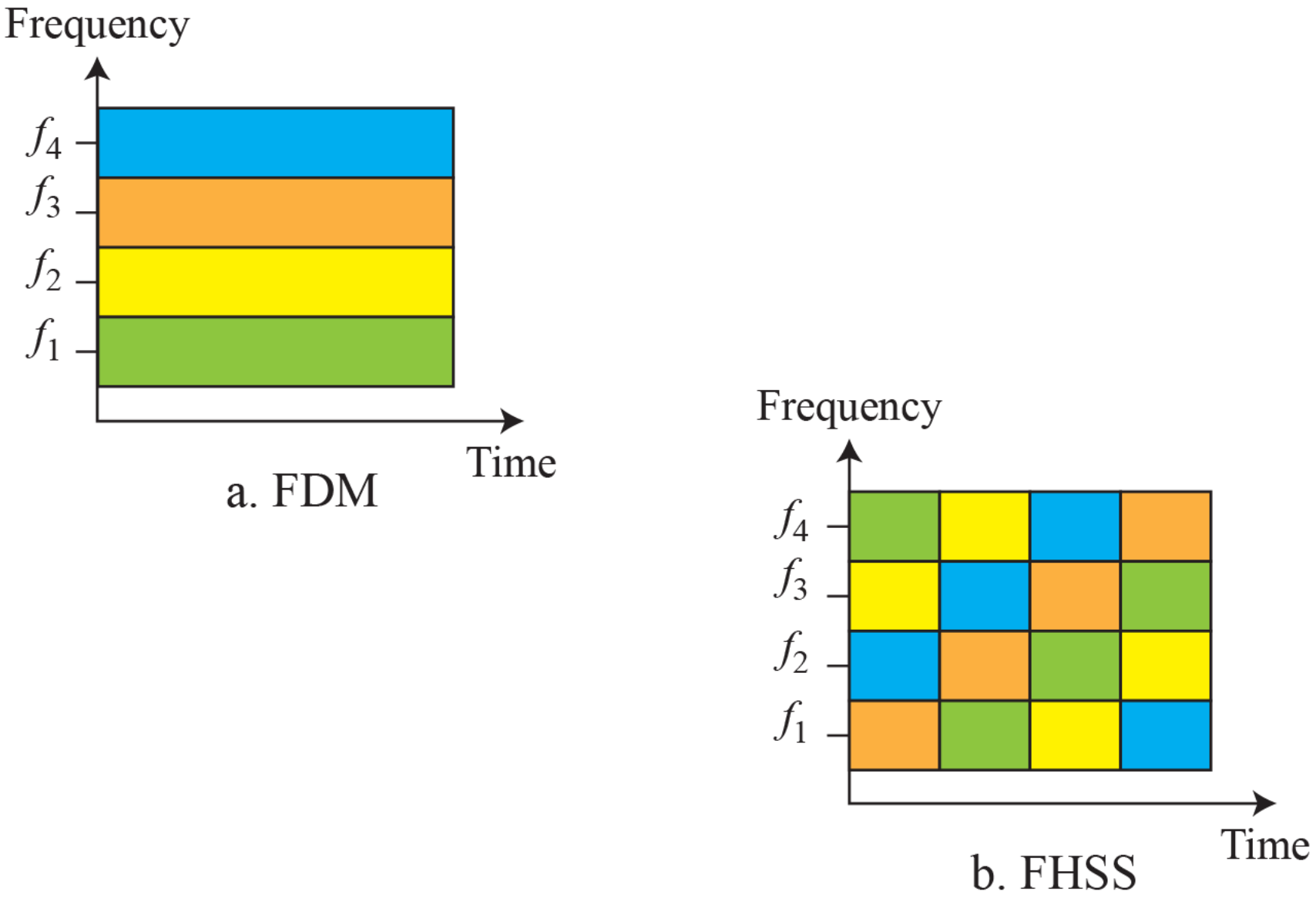

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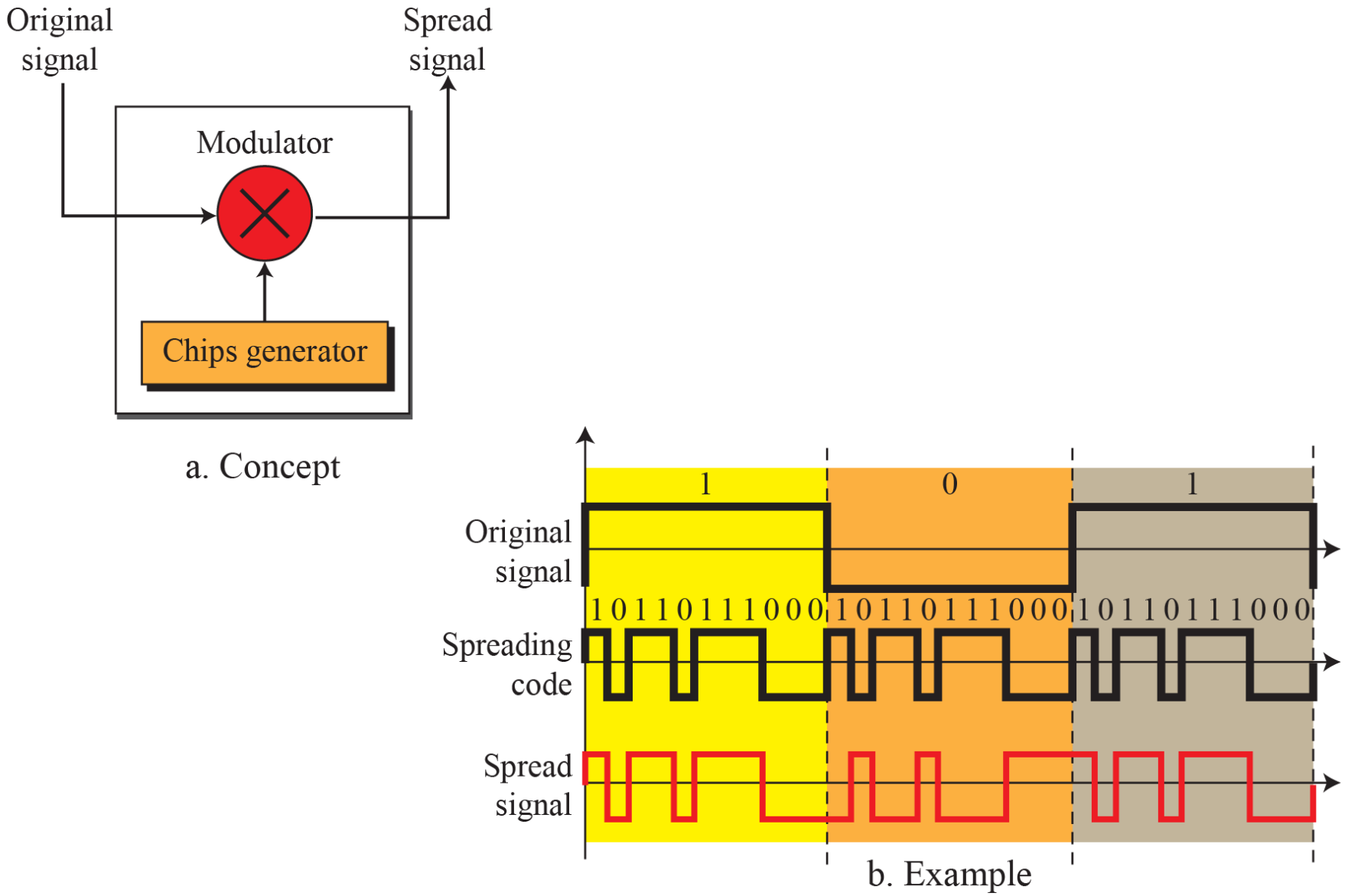
Figure 7.57: DSSS



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7-5 TRANSMISSION MEDIA

We discussed many issues related to the physical layer in this chapter. In this section, we discuss transmission media. Transmission media are actually located below the physical layer and are directly controlled by the physical layer. We could say that transmission media belong to layer zero.

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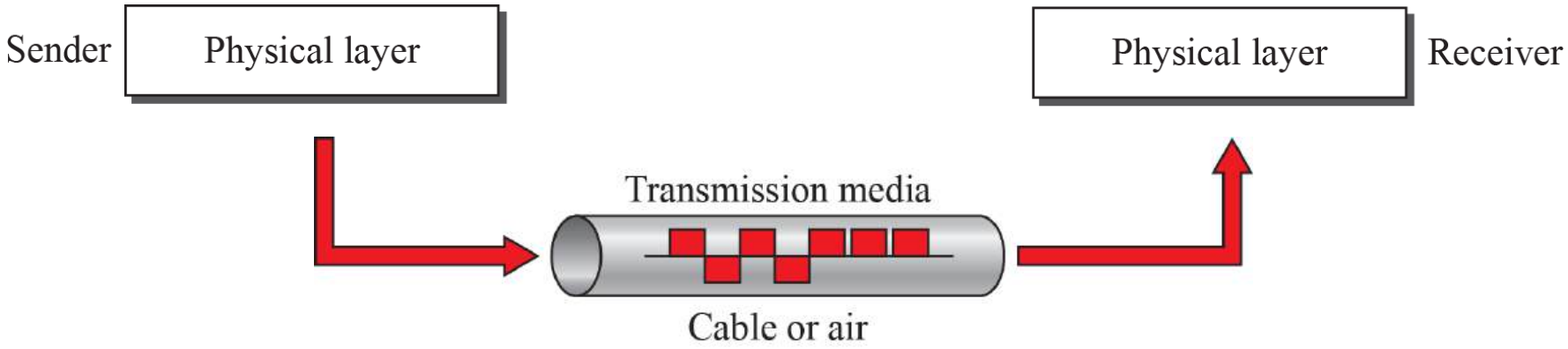
Figure 7.58: Transmission media and physical layer



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7.5.1 Guided Media

Guided media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable. A signal traveling along any of these media is directed and contained by the physical limits of the medium. Twisted-pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current. Fiber-optic cable is a cable that accepts and transports signals in the form of light.

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7.5.1 (continued)

- **Twisted-Pair Cable**
  - Performance
  - Applications
- **Coaxial Cable**
  - Performance
  - Applications
- **Fiber-Optic Cable**
  - Propagation Modes
  - Performance
  - Applications

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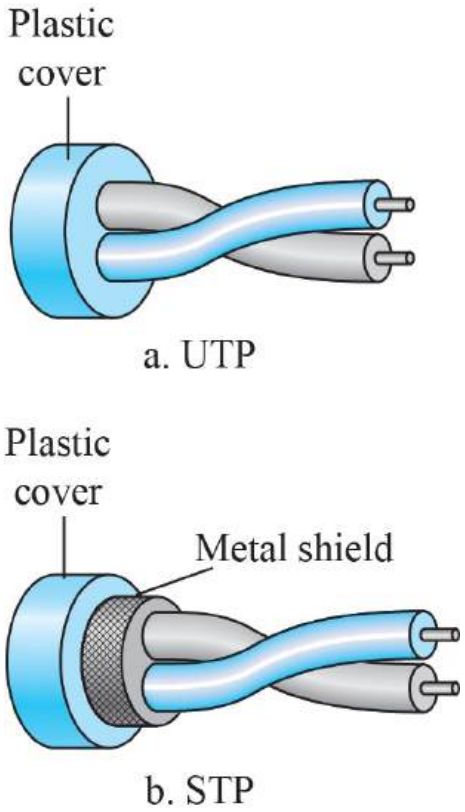
Figure 7.59: Twisted-pair cable



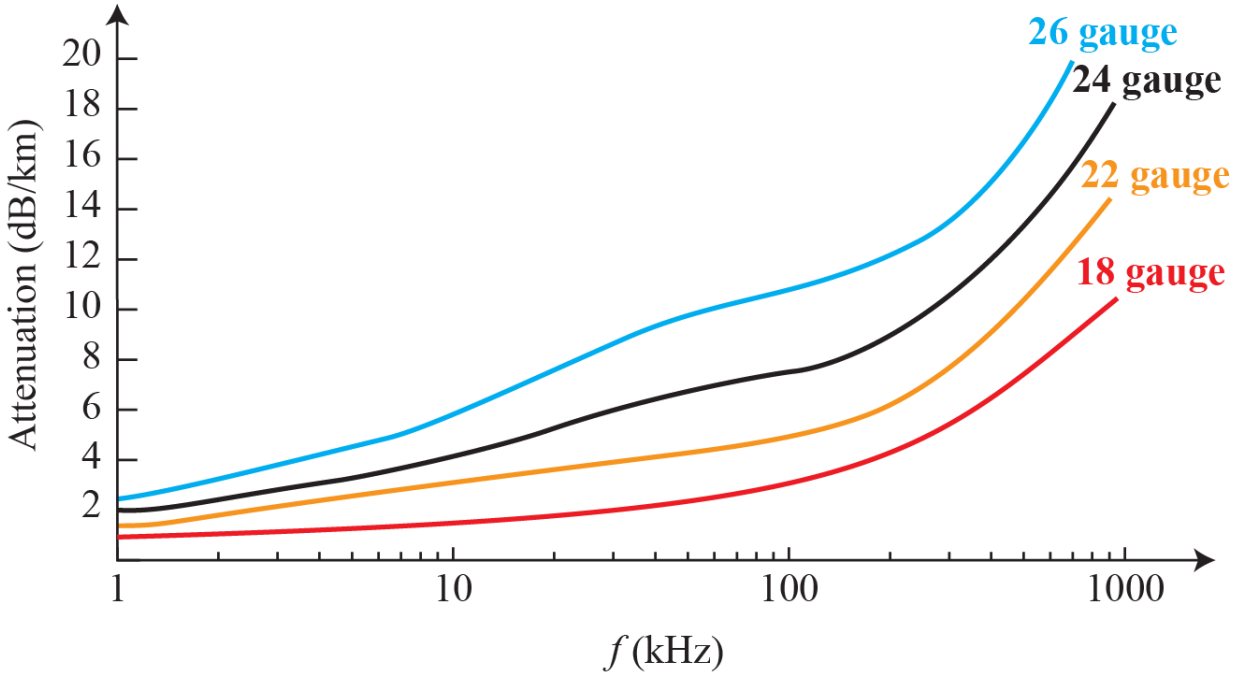


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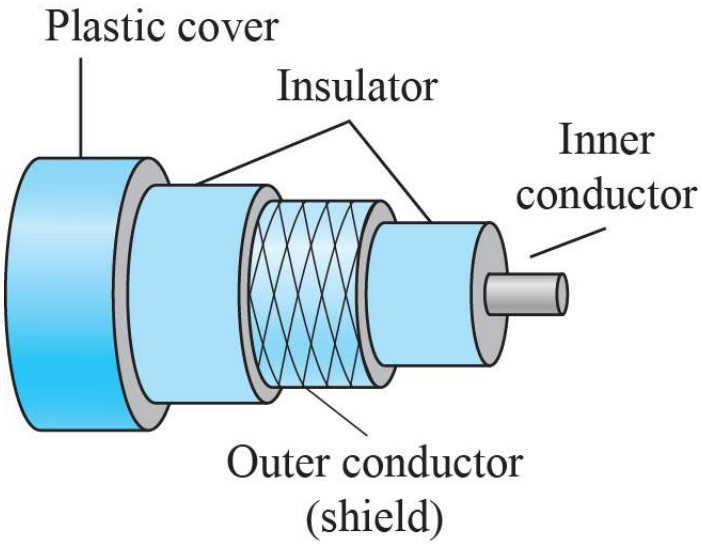
Figure 7.60: *Coaxial cable*



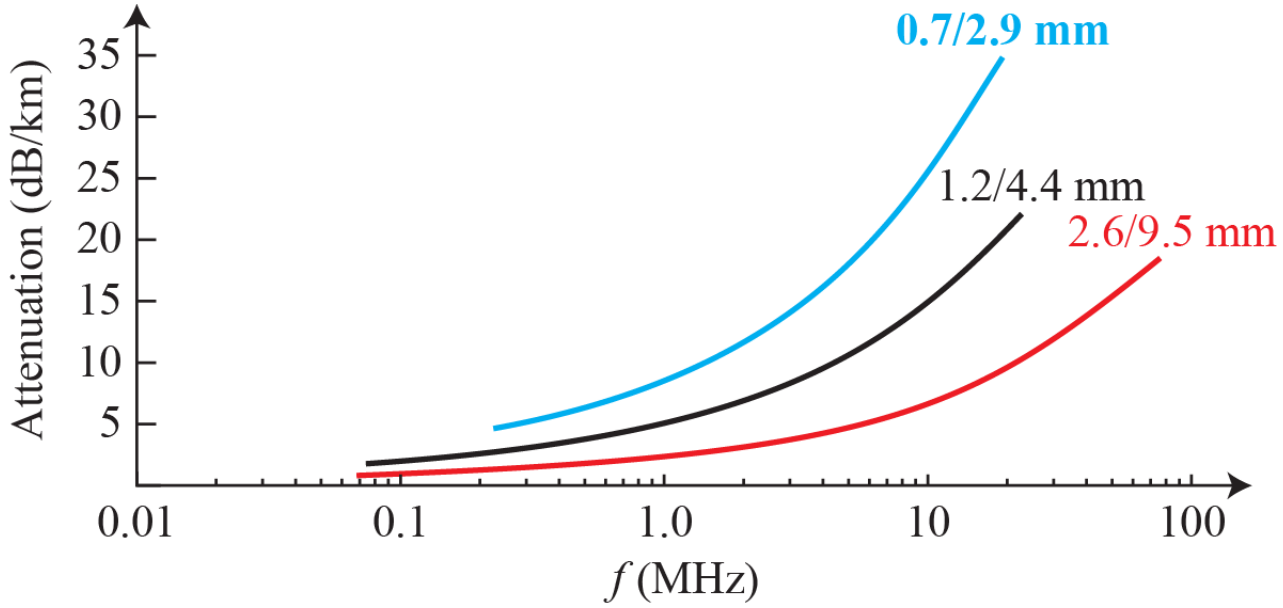


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Figure 7.61: *Bending of light ray*

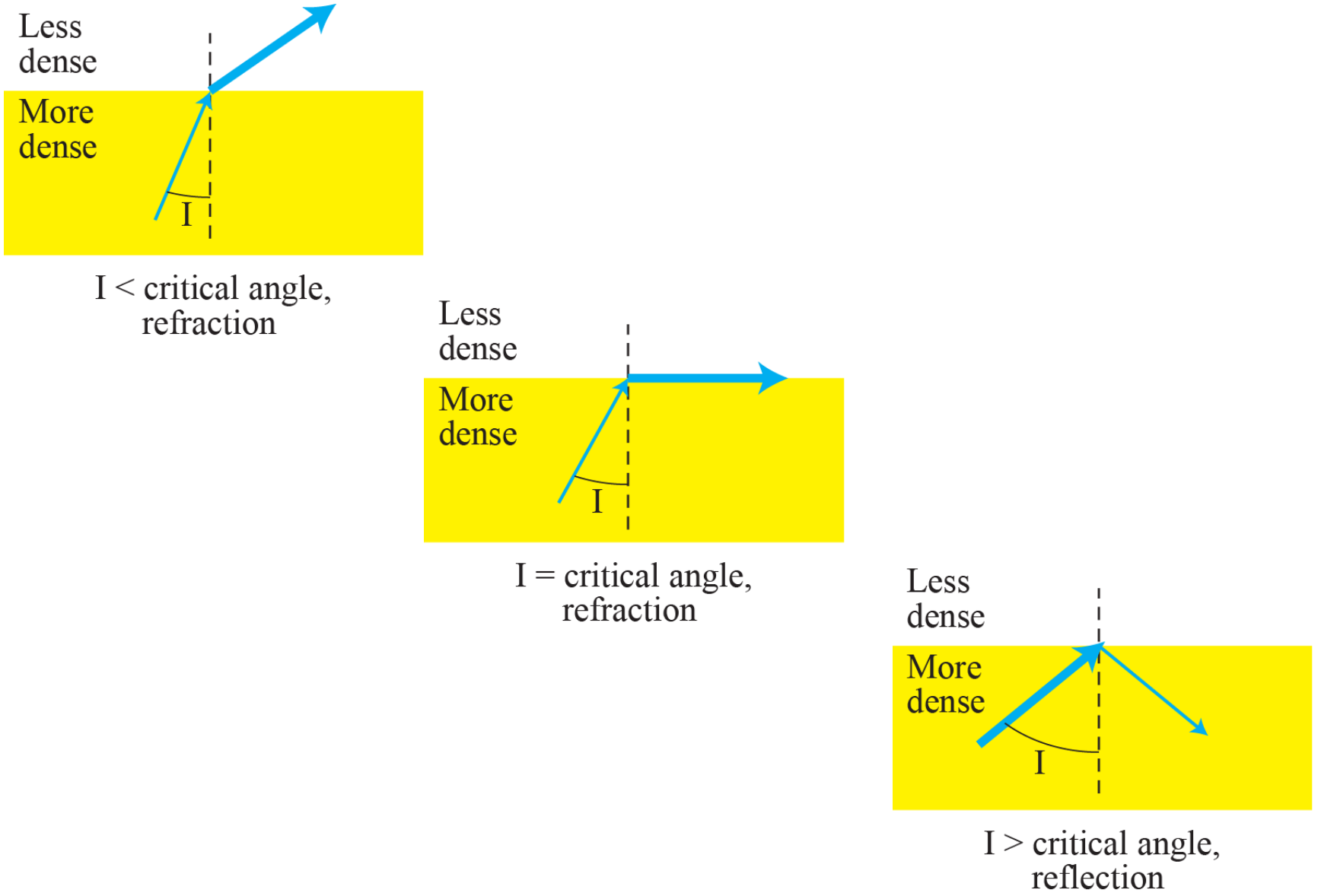

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Figure 7.62: Optical fiber

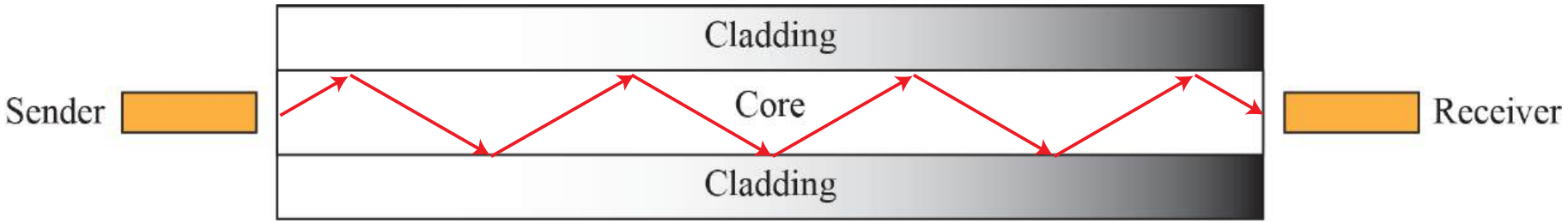

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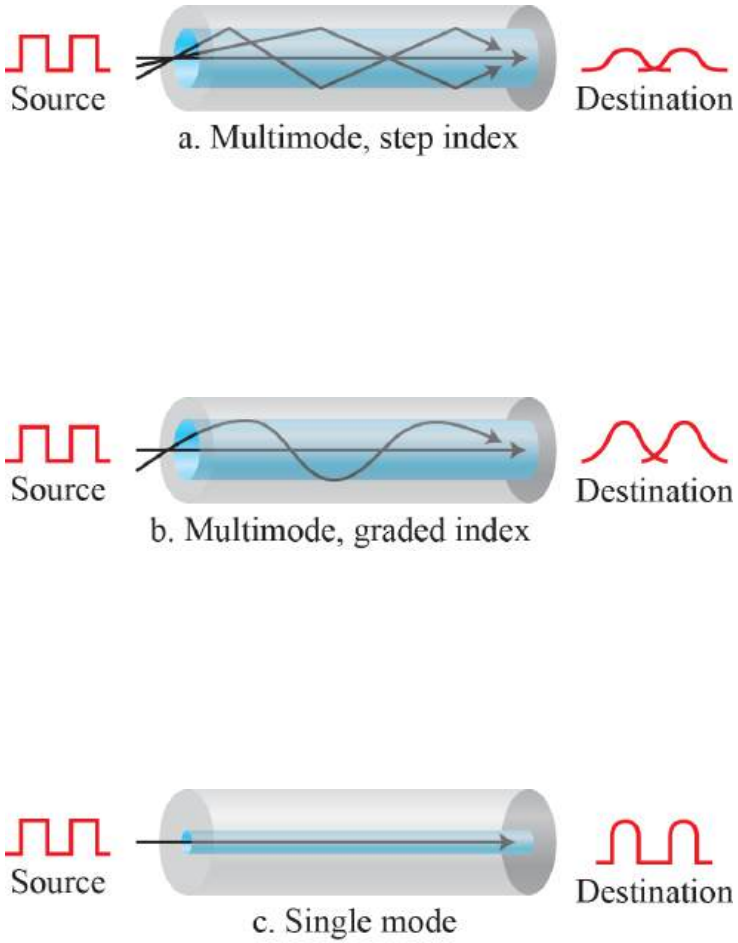
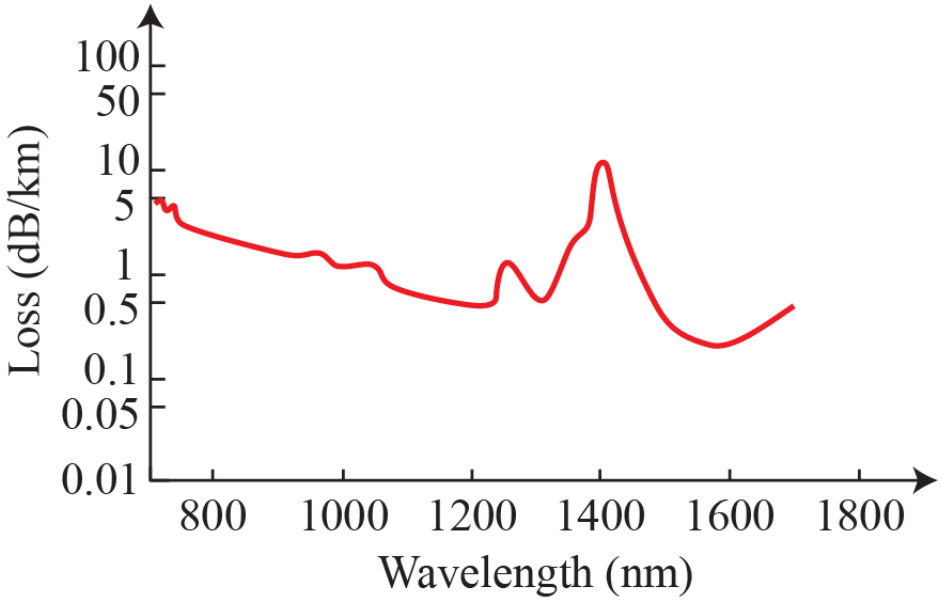
Figure 7.63: Modes





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7.5.2 Unguided Media

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication. Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

- ***Radio Waves***
- ***Microwaves***
- ***Infrared***

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Figure 7.64: Electromagnetic spectrum for wireless communication

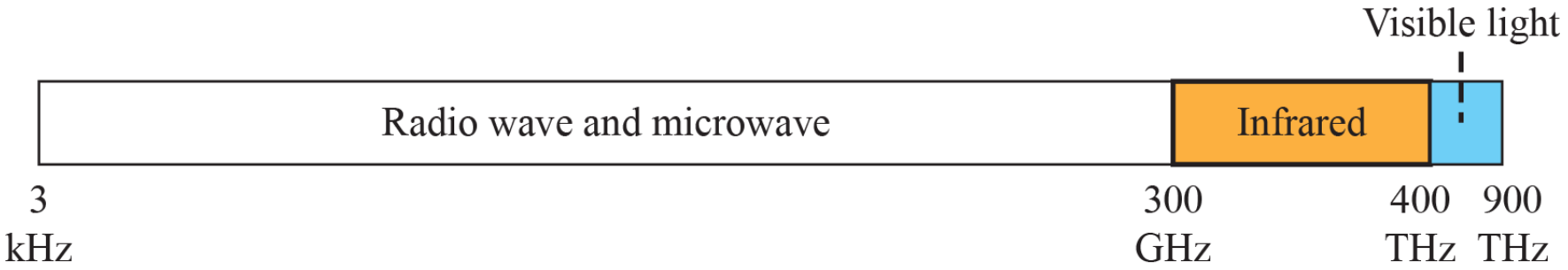

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Table 7.1: Bands

Band	*Range*	*Propagation*	*Application*
VLF (very low frequency)	3–30 kHz	Ground	Long-range radio
LF (low frequency)	30–300 kHz	Ground	Radio beacons
MF (middle frequency)	300 kHz–3 MHz	Sky	AM radio
HF (high frequency)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3–30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30–300 GHz	Line-of-sight	Radar, satellite

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Chapter 7: Summary

- ***Data must be transformed to electromagnetic signals to be transmitted. Analog data are continuous and take continuous values. Digital data have discrete states and take discrete values. Analog signals can have an infinite number of values in a range; digital signals can have only a limited number of values. In data communications, we commonly use periodic analog signals and non-periodic digital signals.***

- ***Digital-to-digital conversion involves three techniques: line coding, block coding, and scrambling. The most common technique to change an analog signal to digital data (digitization) is called pulse code modulation (PCM).***

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Chapter 7: Summary (continued)

- *Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in the digital data. Digital-to-analog can be achieved in several ways: ASK, FSK, and PSK. QAM combines ASK and PSK. Analog-to-analog conversion can be accomplished in three ways: AM, FM), and PM.*

- *Bandwidth utilization is the use of available bandwidth to achieve specific goals. Efficiency can be achieved by using multiplexing; privacy and anti-jamming can be achieved by using spreading.*

- *Transmission media lie below the physical layer. We discussed guided and unguided media.*

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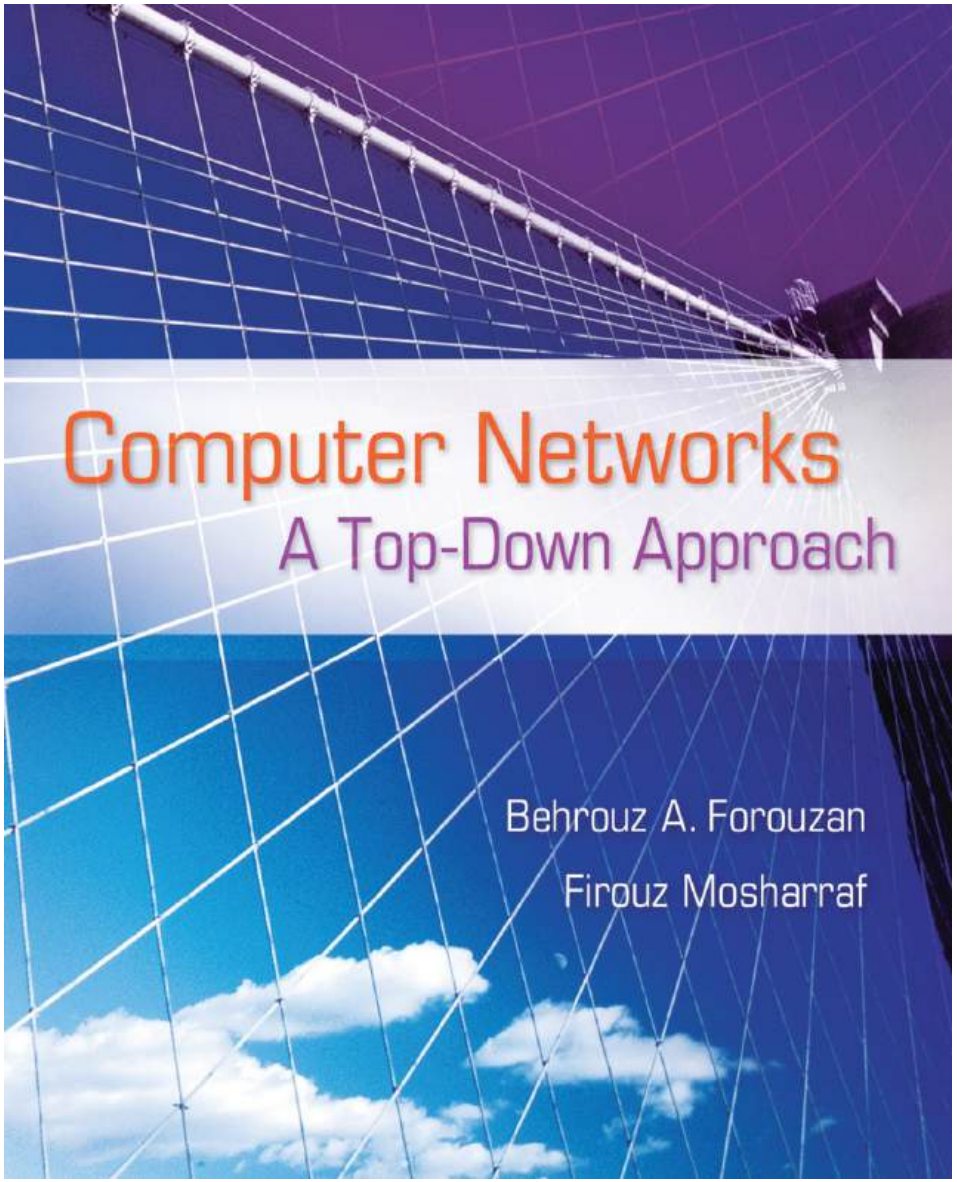
Chapter 8

Multimedia and Quality of Service



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Chapter 8: Outline

8.1 COMPRESSION

8.2 MULTIMEDIA DATA

8.3 MULTIMEDIA IN THE INTERNET

8.4 REAL-TIME INTERACTIVE PROTOCOLS

8.5 QUALITY OF SERVICE

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Chapter 8: Objective

- *We discuss the general idea behind compression. Although compression is not directly related to the subject of multimedia, multimedia transmission is not possible without first compressing the data.*

- *We discuss the elements of multimedia: text, image, video, and audio. We show how these elements are represented, encoded, and compressed using the techniques discussed in the first section.*

- *We separate multimedia in the Internet into three categories: streaming stored audio/video, streaming live audio/video, and real-time interactive audio/video. We briefly describe the features and characteristics of each and give some examples.*

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Chapter 8: Objective (continued)

- *We concentrate on the real-time interactive category. We introduce two protocols that are used in this category for signaling: SIP and H.323. These protocols are used in voice over IP (Internet telephony) and can be used for signaling protocols in future applications. We also discuss transport-layer protocols used for multimedia applications.*

- *We discuss quality of service (QoS), which is more needed for multimedia communication than for communication using only text.*

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8-1 COMPRESSION

In this section, we discuss compression, which plays a crucial role in multimedia communication due to the large volume of data exchanged. In compression, we reduce the volume of data to be exchanged. We can divide compression into two broad categories: lossless and lossy compression. We briefly discuss the common methods used in each category.

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8.1.1 Lossless Compression

In lossless compression, the integrity of the data is preserved because the compression and decompression algorithms are exact inverses of each other: no part of the data is lost in the process. Lossless compression methods are normally used when we cannot afford to lose any data. For example, we must not lose data when we compress a text file or an application program. Lossless compression is also applied as the last step in some lossy compression procedures to further reduce the size of the data.

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8.1.1 (continued)

- **Run-length Coding**
- **Dictionary Coding**
  - Encoding
  - Decoding
- **Huffman Coding**
  - Huffman Tree
  - Coding Table
  - Encoding and Decoding

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8.1.1 (continued)

- **Arithmetic Coding**
  - Encoding
  - Decoding
  - Static versus Dynamic Arithmetic Coding

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Figure 8.1 : A version of run-length coding to compress binary patterns

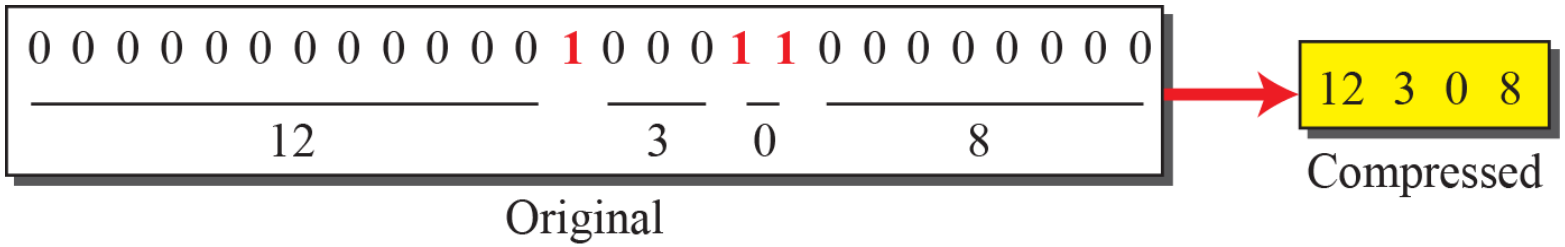

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Table 8.1: LZW encoding

```
LZWEncoding (message)
{
        Initialize (Dictionary)
        Char = Input (first character)
        S = char                                        // S is the encodable sequence
        while (more characters in message)
        {
                char = Input (next character);
                if ((S + char) is in Dictionary)        // S is not the encodable sequence
                {
                        S = S + char;
                }
                else                                    // S is the encodable sequence
                {
                        addToDictionary (S + char);
                        Output (index of S in Dictionary);
                        S = char;
                }
        }
        Output (index of S in Dictionary);
}
```


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Example 8.1

Let us show an example of LZW encoding using a text message in which the alphabet is made of two characters: A and B (Figure 8.2). The figure shows how the text "BAABABBBBAABBBBBAA" is encoded as 1002163670. Note that the buffer PreS holds the string from the previous iteration before it is updated.

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Figure 8.2 : Example 8.1

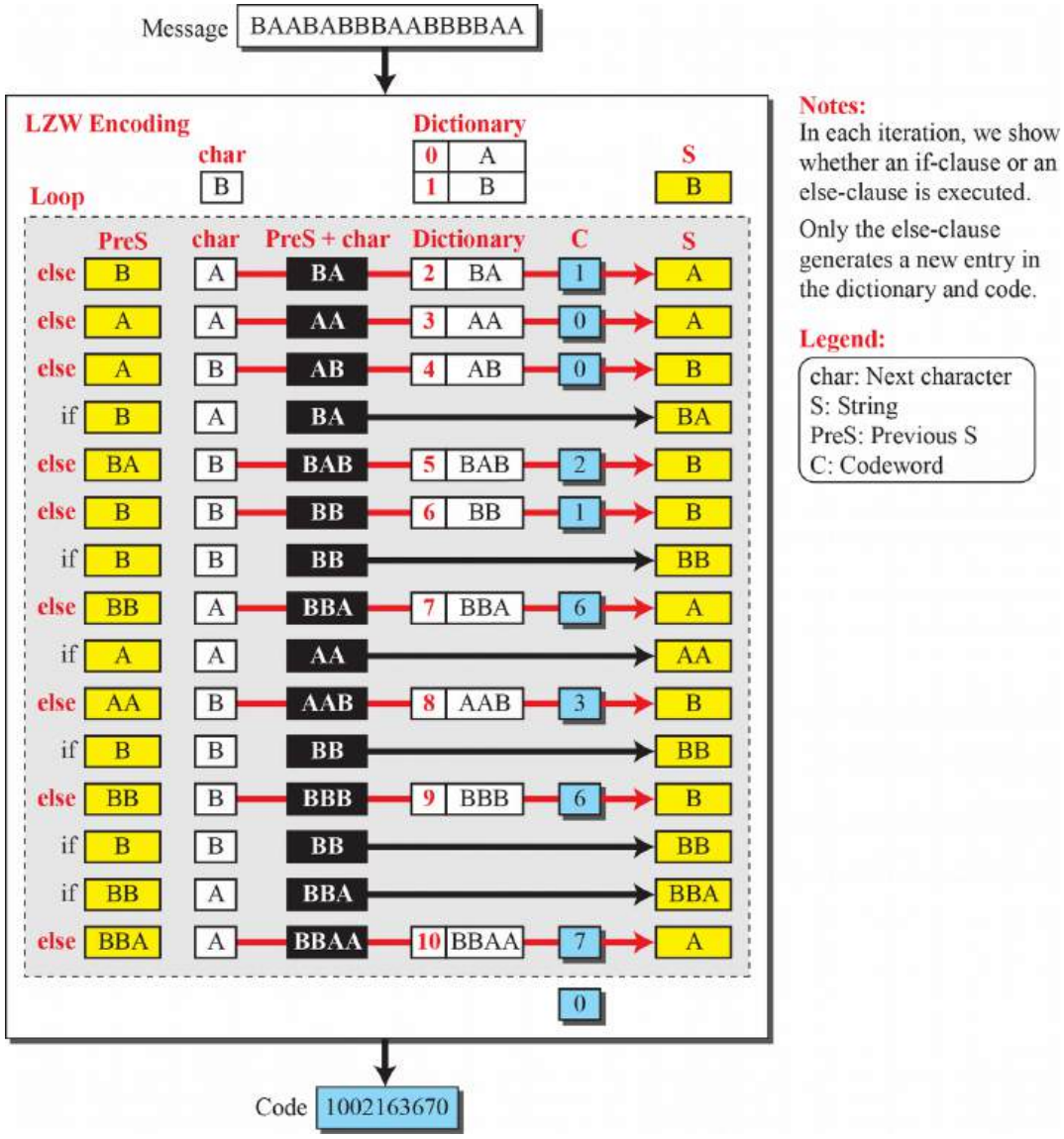

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Table 8.2: LZW decoding

```
LZWDecoding (code)
{
        Initialize (Dictionary);
        C = Input (first codeword);
        Output (Dictionary [C]);
        while (more codewords in code)
        {
                S = Dictionary[C];
                C = Input (next codeword);
                if (C is in Dictionary)                    // Normal case
                {
                        addToDictionary (S + firstSymbolOf Dictionary[C]);
                        Output (Dictionary [C]);
                }
                else                                       // Special case
                {
                        addToDictionary (S + firstSymbolOf (S));
                        Output (S + firstSymbolOf (S);

                }
        }
}
```


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Example 8.2

Let us show how the code in Example 8.1 can be decoded and the original message recovered (Figure 8.3). The box called PreC holds the codeword from the previous iteration, which is not needed in the pseudocode, but needed here to better show the process. Note that in this example there is only the special case in which the codeword is not in the dictionary. The new entry for the dictionary needs to be made from the string and the first character in the string. The output is also the same as the new entry.

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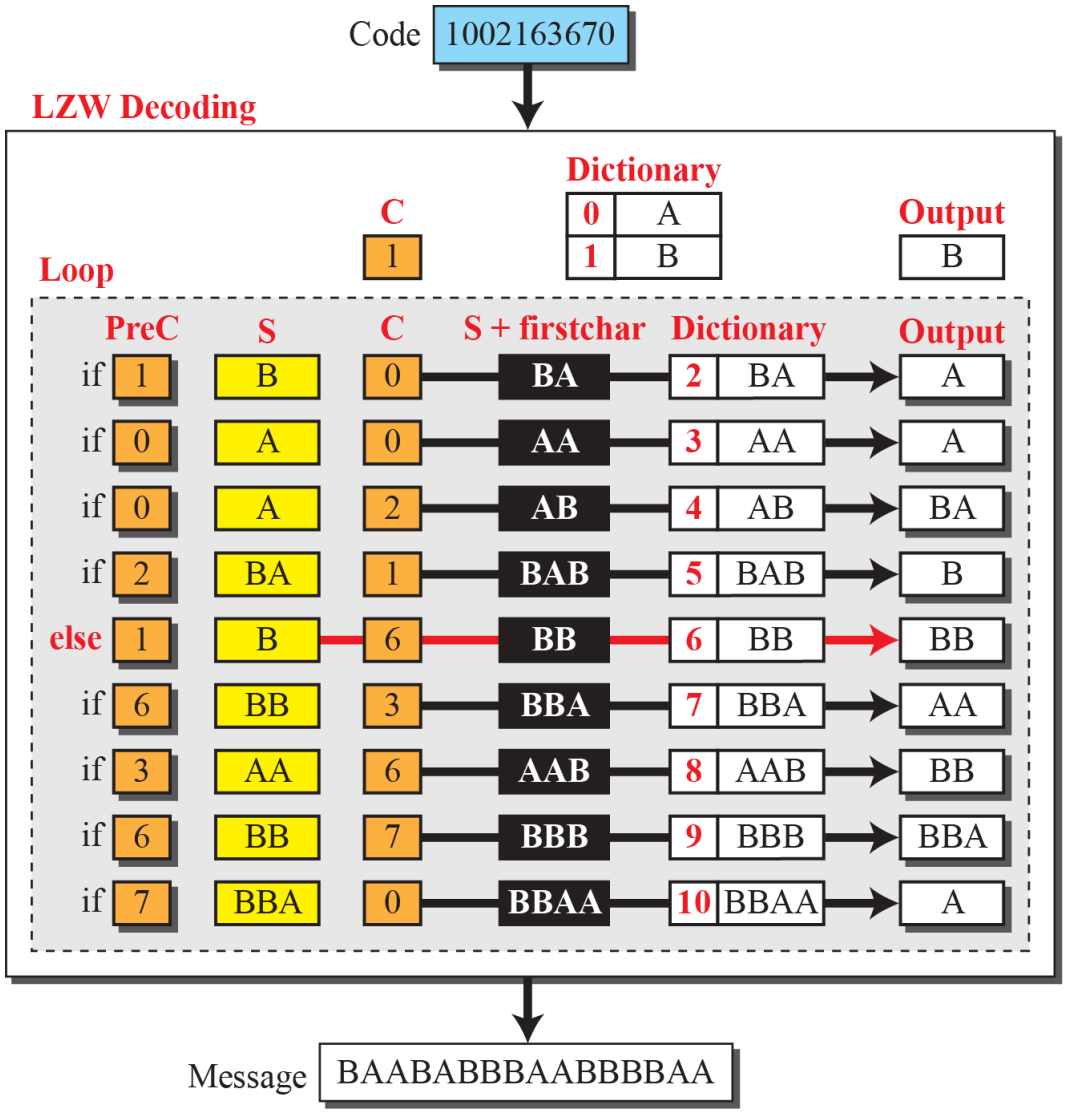
Figure 8.3 : Example 8.2



Notes:

In each iteration, we show whether an if-clause or an else-clause is executed.

An if-clause is executed when the codeword is in the dictionary; an else-clause is executed when the codeword is not in the dictionary.

S + firstChar is the string concatenated with the first character in the dictionary (if-clause) or with the first character in the string itself (else-clause).

Legend:

C: Codeword
S: String
PreC: Previous code

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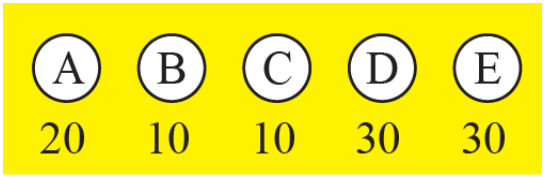
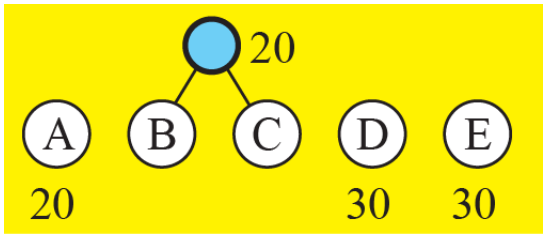
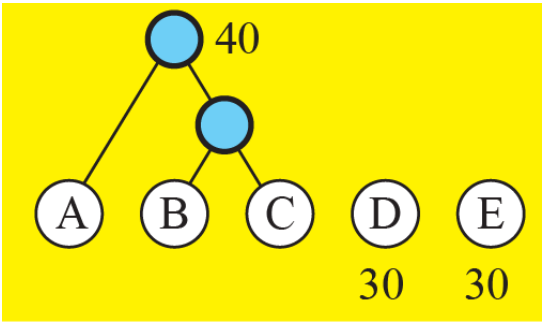
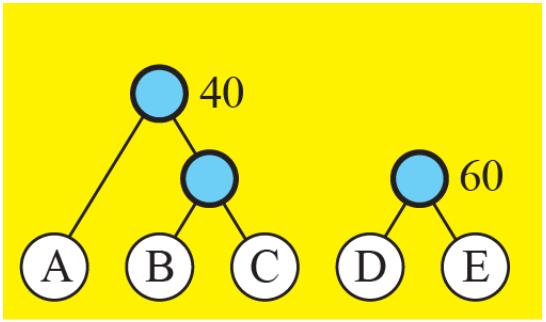
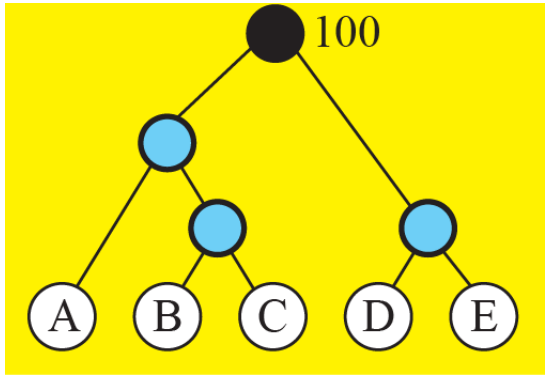
Figure 8.4 : Huffman tree











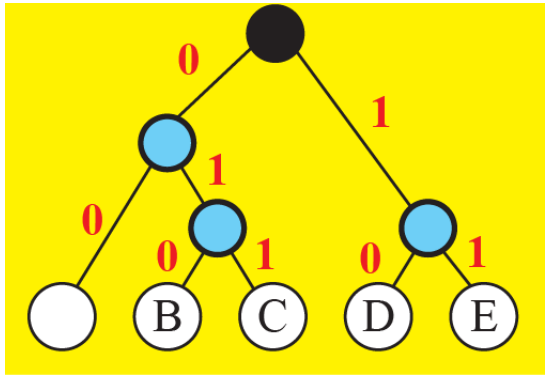


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Table 8.3: Coding Table

Symbol	Code	*Symbol*	Code	*Symbol*	Code
A	00	C	011	E	11
B	010	D	10		

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Figure 8.5 : Encoding and decoding in Huffman coding

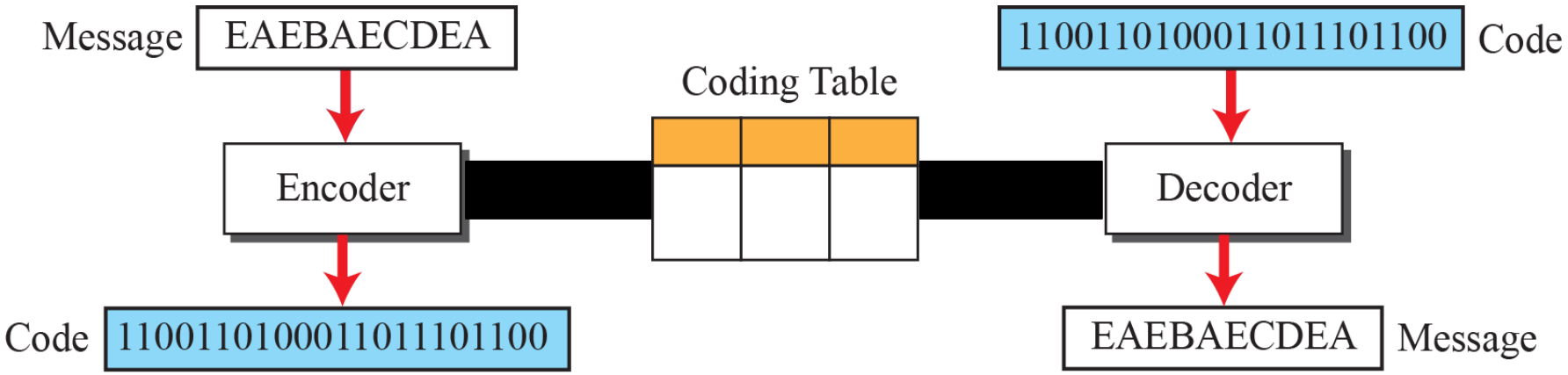

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Figure 8.6 : Arithmetic coding

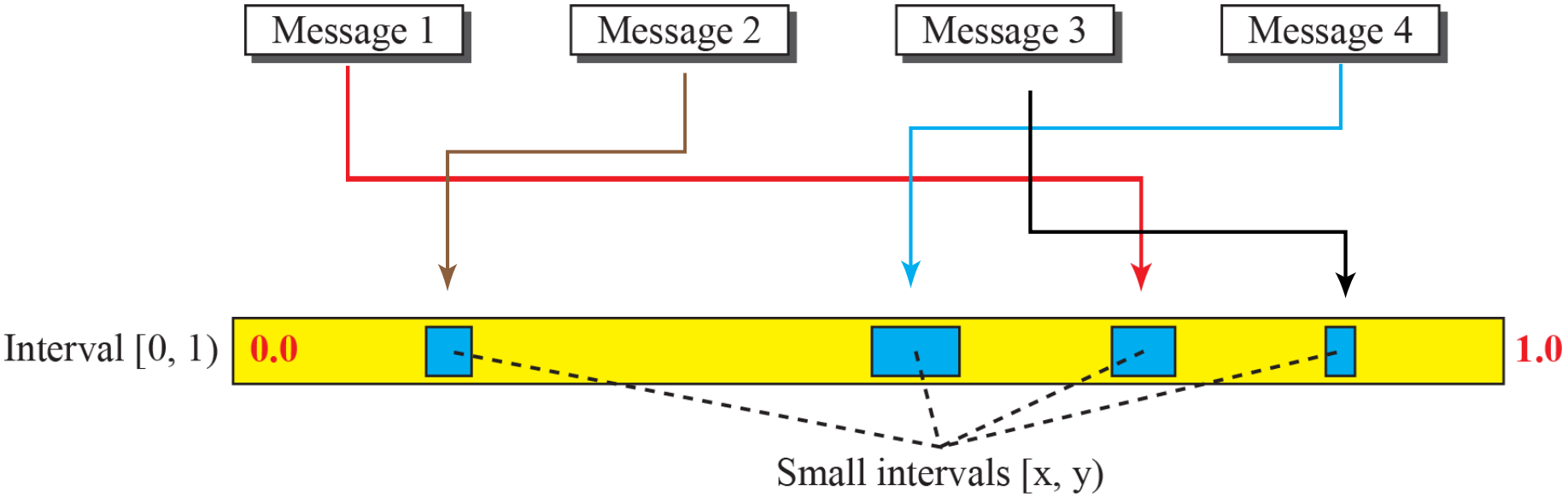

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Table 8.4: Arithmetic encoding

```
ArithmeticEncoding (message)
{
        currentInterval = [0,1);
        while (more symbols in the message)
        {
                s = Input (next symbol);
                divide currentInterval into subintervals
                subInt = subinterval related to s
                currentInterval = subInt
        }
        Output (bits related to the currentInterval)
}
```


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Example 8.3

For the sake of simplicity, let us assume that our set of symbols is S = {A, B, ∗}, in which the asterisk is the terminating symbol. We assign probability of occurrence for each symbol as

$$P_A = 0.4 \qquad P_B = 0.5 \qquad P_* = 0.1$$

Figure 8.7 shows how we find the interval and the code related to the short message "BBAB*".

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Figure 8.7 : Example 8.3

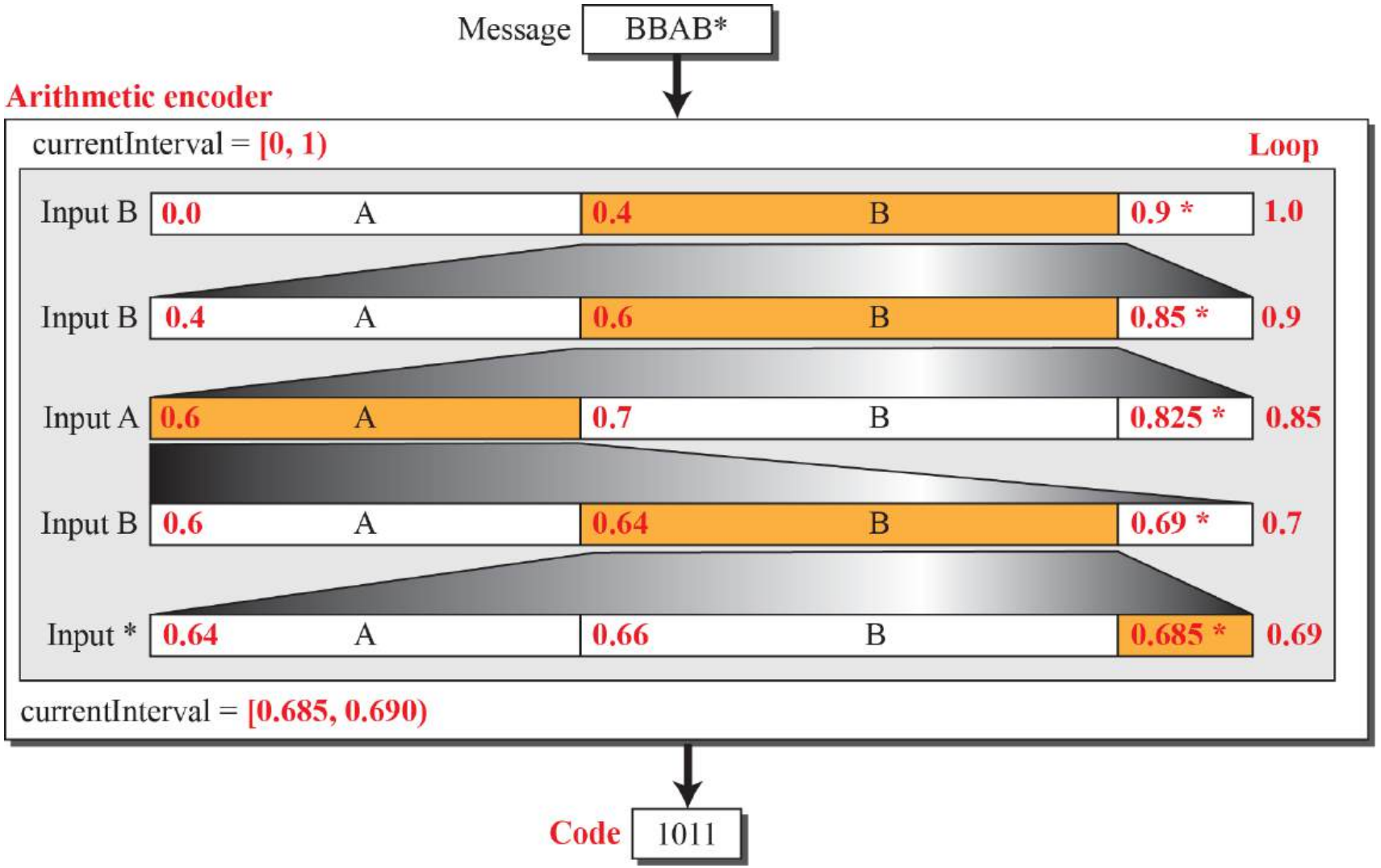

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Table 8.5: Arithmetic Decoding

```
ArithmeticDecoding (code)
{
        c = Input (code)
        num = find real number related to code
        currentInterval = [0,1);
        while (true)
        {
                divide the currentInterval into subintervals;
                subInt = subinterval related to num;
                Output (symbol related to subInt);
                if (symbol is the terminating symbol)    return;
                currentInterval = subInt;
        }
}
```


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Example 8.4

Figure 8.8 shows how we use the decoding process to decode the message in Example 8.3. Note that the hand shows the position of the number in the corresponding interval.

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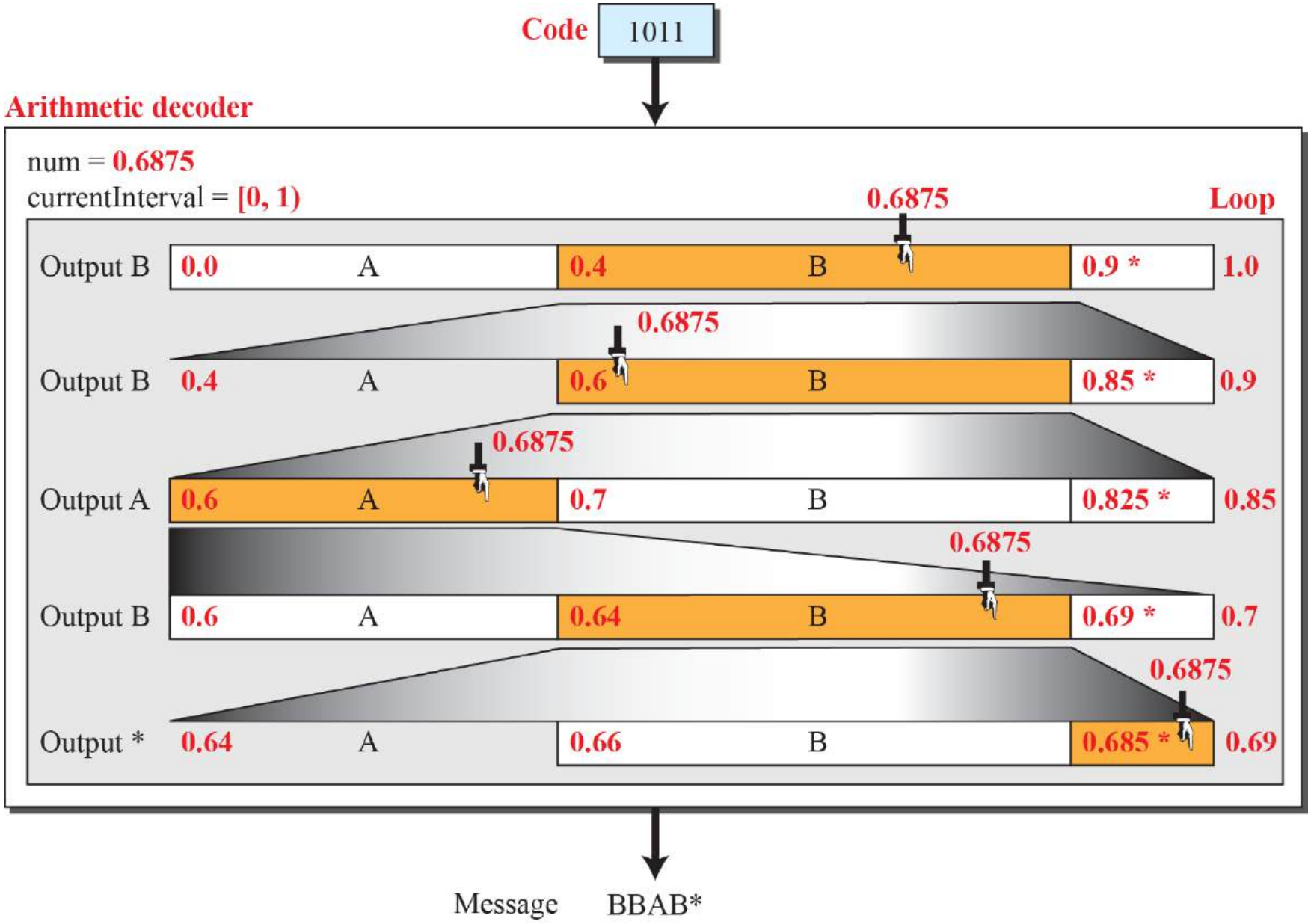
Figure 8.8 : Example 8.4



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8.1.2 Lossy Compression

Lossless compression has limits on the amount of compression. However, in some situations, we can sacrifice some accuracy to increase compression rate. Although we cannot afford to loose information in text compression, we can afford it when we are compressing images, video, and audio. For example, human vision cannot detect some small distortions that can result from lossy compression of an image. In this section, we discuss a few ideas behind lossy compression.

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8.1.2 (continued)

- **Predictive Coding**
  - Delta Modulation
  - Adaptive DM (ADM)
  - Differential PCM (DPCM)
  - Adaptive DPCM (ADPCM)
  - Linear Predictive Coding
- **Transform Coding**
  - Discrete Cosine Transform (DCT)

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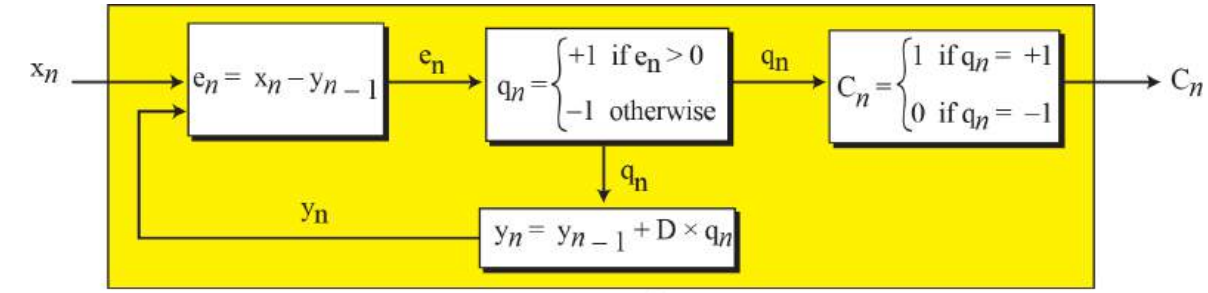
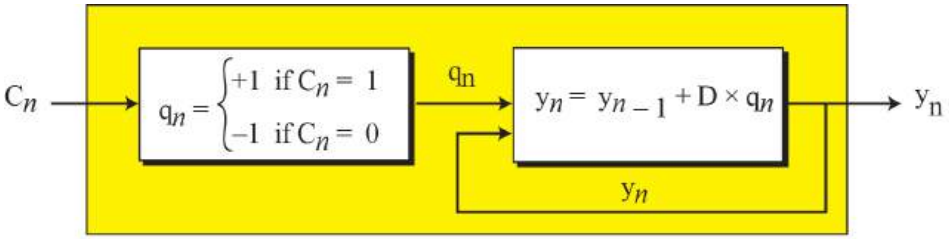
Figure 8.9 : Encoding and decoding in delta modulation



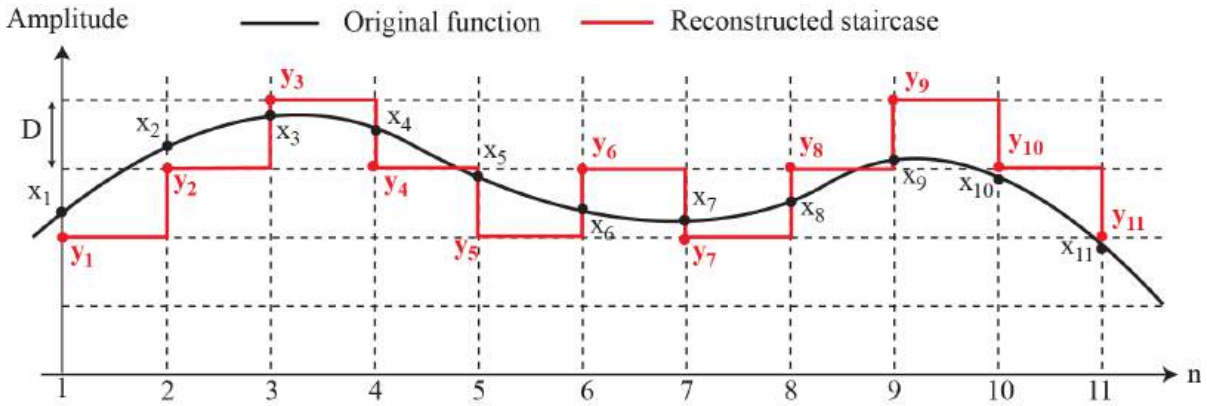
a. Encoder



b. Decoder

Notes

1. y_n becomes y_{n-1} in the next iteration.
2. y_0 and D are given.



c. Relationship between x_n and y_n

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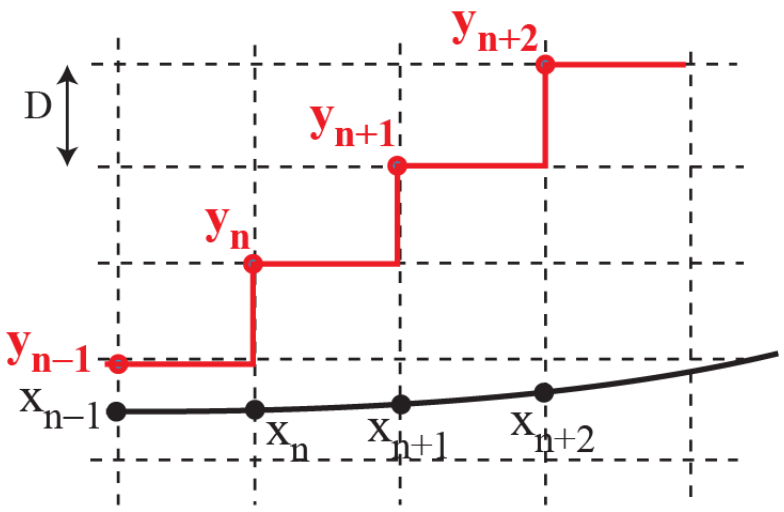
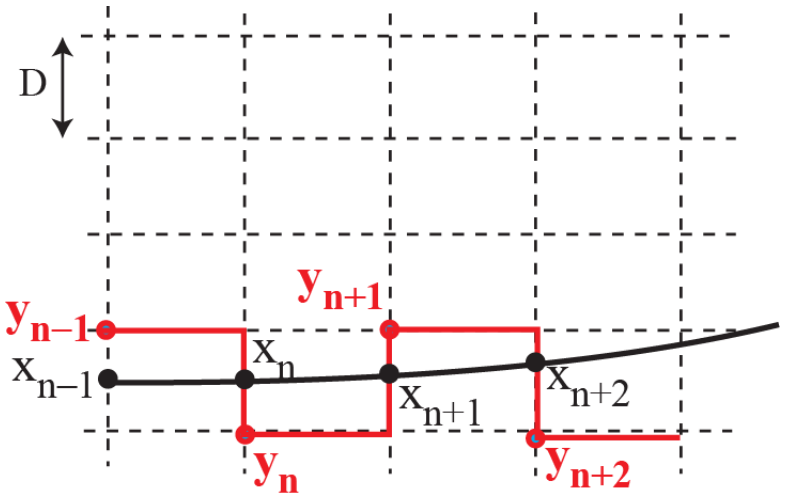
Figure 8.10 : Reconstruction of quantization of $x_n - x_{n-1}$ versus $x_n - y_{n-1}$



a. Quantizing $q_n = x_n - x_{n-1}$



b. Quantizing $q_n = x_n - y_{n-1}$

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Figure 8.11 : Slope overload and granular noise

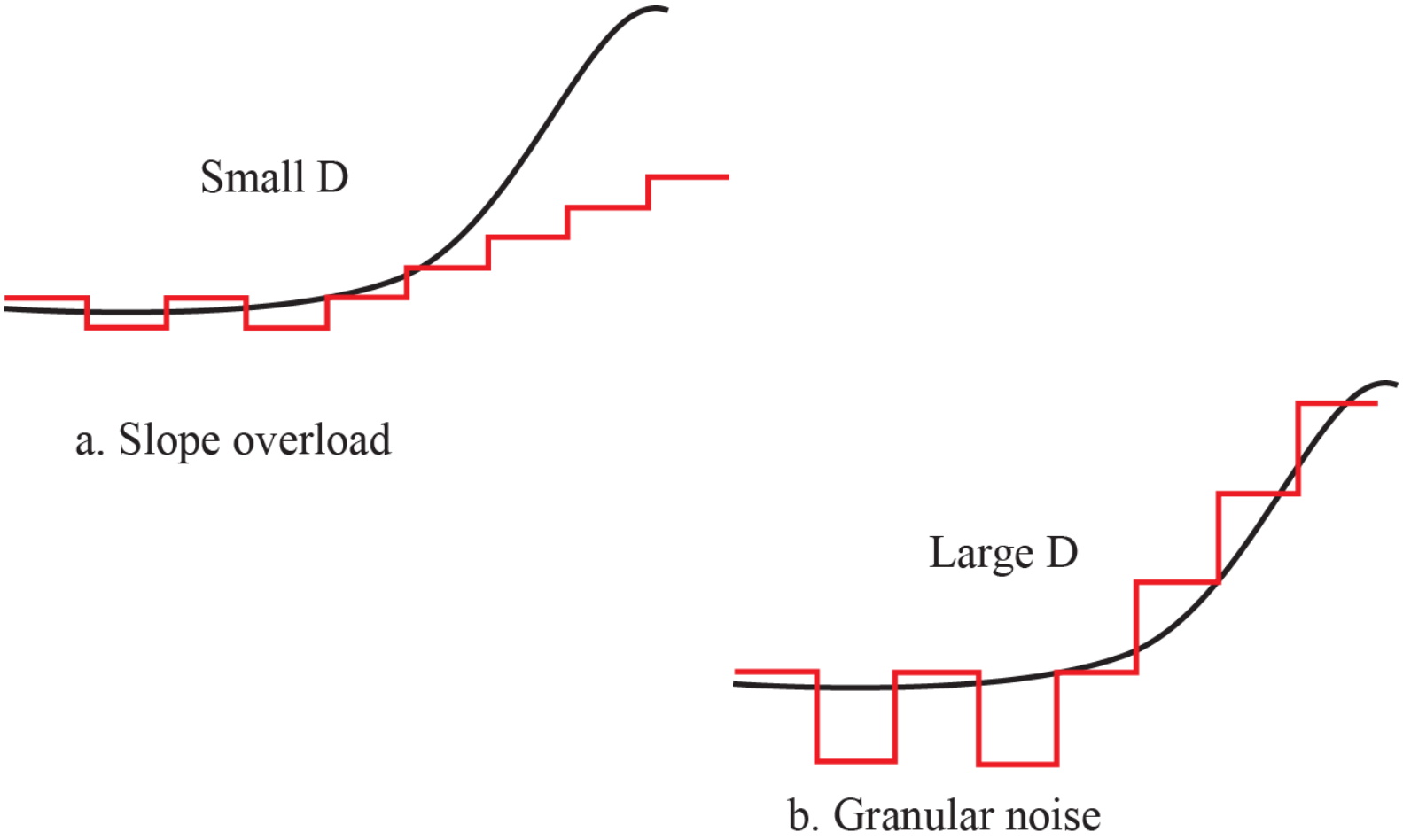

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Figure 8.12 : One-dimensional DCT

$$[\mathbf{M}] = [\quad T \quad] \times [p]$$

a. Transformation

$$T(m, n) = C(m) \cos\left[\frac{\pi n(2m+1)}{2N}\right]$$

for $m = 0$ to N–1

for $n = 0$ to N–1

$$[p] = [\quad T^T \quad] \times [\mathbf{M}]$$

b. Inverse Transformation

$$C(m) = \begin{cases} \sqrt{\frac{1}{N}} & m = 0 \\ \sqrt{\frac{2}{N}} & m > 0 \end{cases}$$

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Figure 8.13 : Formulas for one-dimensional forward and inverse transformation

$$\mathbf{M}(m) = \sum_{n=0}^{N-1} C(m) \cos\left[\pi n(2m+1)/(2N)\right] \times p(n) \qquad \text{for } m = 0, \ldots, N-1$$

$$p(n) = \sum_{m=0}^{N-1} C(n) \cos\left[\pi m(2n+1)/(2N)\right] \times \mathbf{M}(m) \qquad \text{for } n = 0, \ldots, N-1$$

$$C(i) = \begin{cases} \sqrt{\frac{1}{N}} & i = 0 \\ \sqrt{\frac{2}{N}} & i > 0 \end{cases}$$

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Example 8.5

Figure 8.14 shows the transformation matrix for N = 4. As the figure shows, the first row has four equal values, but the other rows have alternate positive and negative values. When each row is multiplied by the source data matrix, we expect that the positive and negative values result in values close to zero if the source data items are close to each other. This is what we expect from the transformation: to show that only some values in the source data are important and most values are redundant.

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Figure 8.14 : Example 8.5

$$\begin{bmatrix} \mathbf{203} \\ \mathbf{-2.22} \\ \mathbf{0.00} \\ \mathbf{-0.16} \end{bmatrix} = \begin{bmatrix} 0.50 & 0.50 & 0.50 & 0.50 \\ 0.65 & 0.27 & -0.27 & -0.65 \\ 0.50 & -0.50 & -0.50 & 0.50 \\ 0.27 & -0.65 & 0.65 & -0.27 \end{bmatrix} \times \begin{bmatrix} 100 \\ 101 \\ 102 \\ 103 \end{bmatrix}$$

$\mathbf{M}$ $\qquad$ T $\qquad$ p

a. Transformation

$$\begin{bmatrix} 100 \\ 101 \\ 102 \\ 103 \end{bmatrix} = \begin{bmatrix} 0.50 & 0.65 & 0.50 & 0.27 \\ 0.50 & 0.27 & -0.50 & -0.65 \\ 0.50 & -0.27 & -0.50 & 0.65 \\ 0.50 & -0.65 & 0.50 & -0.27 \end{bmatrix} \times \begin{bmatrix} \mathbf{203} \\ \mathbf{-2.22} \\ \mathbf{0.00} \\ \mathbf{-0.16} \end{bmatrix}$$

p $\qquad$ T^{T} $\qquad$ $\mathbf{M}$

b. Inverse Transformation

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Figure 8.15 : Two-dimensional DCT

$$[\ \mathbf{M}\ ] = [\ T\ ] \times [\ p\ ] \times [\ T^T\ ]$$

a. Transformation

$$[\ p\ ] = [\ T^T\ ] \times [\ \mathbf{M}\ ] \times [\ T\ ]$$

b. Inverse Transformation

$$T(m, n) = C(m, n)\ \mathbf{cos}\left[\frac{m\pi(2n+1)}{2N}\right]$$

for $m = 0$ to N-1
for $n = 0$ to N-1

$$C(m, n) = \begin{cases} \sqrt{\frac{1}{N}} & m = 0 \\ \sqrt{\frac{2}{N}} & m > 0 \end{cases}$$

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Figure 8.16 : Formulas for forward and inverse two-dimensional DCT

$$\mathbf{M}(m, n) = \frac{2}{\mathrm{N}} \mathrm{C}(m)\,\mathrm{C}(n) \sum_{k=0}^{\mathrm{N}-1} \sum_{l=0}^{\mathrm{N}-1} p(k, l) \cos\left[\frac{m\pi(2k+1)}{2\mathrm{N}}\right] \cos\left[\frac{n\pi(2l+1)}{2\mathrm{N}}\right] \quad \begin{array}{l} \text{for } m = 0, \ldots, \mathrm{N}-1 \\ \text{for } n = 0, \ldots, \mathrm{N}-1 \end{array}$$

$$p(k, l) = \frac{2}{\mathrm{N}} \mathrm{C}(l)\,\mathrm{C}(k) \sum_{m=0}^{\mathrm{N}-1} \sum_{n=0}^{\mathrm{N}-1} \mathbf{M}(m, n) \cos\left[\frac{k\pi(2m+1)}{2\mathrm{N}}\right] \cos\left[\frac{l\pi(2n+1)}{2\mathrm{N}}\right] \quad \begin{array}{l} \text{for } k = 0, \ldots, \mathrm{N}-1 \\ \text{for } l = 0, \ldots, \mathrm{N}-1 \end{array}$$

$$\mathrm{C}(u) = \begin{cases} \sqrt{\frac{1}{2}} & u = 0 \\ 1 & u > 0 \end{cases}$$

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8-2 MULTIMEDIA DATA

Today, multimedia data consists of text, images, video, and audio, although the definition is changing to include futuristic media types.

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8.2.1 Text

The Internet stores a large amount of text that can be downloaded and used. One often refers to plaintext, as a linear form, and hypertext, as a nonlinear form, of textual data. Text stored in the Internet uses a character set, such as Unicode, to represent symbols in the underlying language.

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8.2.2 Image

In multimedia parlance, an image (or a still image as it is often called) is the representation of a photograph, a fax page, or a frame in a moving picture.

- **Digital Image**
- **Image Compression: JPEG**
  - Transformation
  - Quantization
  - Encoding
- **Image Compression: GIF**

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Example 8.6

The following shows the time required to transmit an image of 1280 × 720 pixels using the transmission rate of 100 kbps.

a. Using a black and white image with a bit depth of 1, we need.

$$\textbf{Transmission time} = (1280 \times 720 \times 1) / 100{,}000 \approx 9 \textbf{ seconds}$$

b. Using a gray image with a bit depth of 8, we need.

$$\textbf{Transmission time} = (1280 \times 720 \times 8) / 100{,}000 \approx 74 \textbf{ seconds}$$

c. Using a color image with a bit depth of 24, we need.

$$\textbf{Transmission time} = (1280 \times 700 \times 24) / 100{,}000 \approx 215 \textbf{ seconds}$$

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Figure 8.17 : Compression in each channel of JPEG

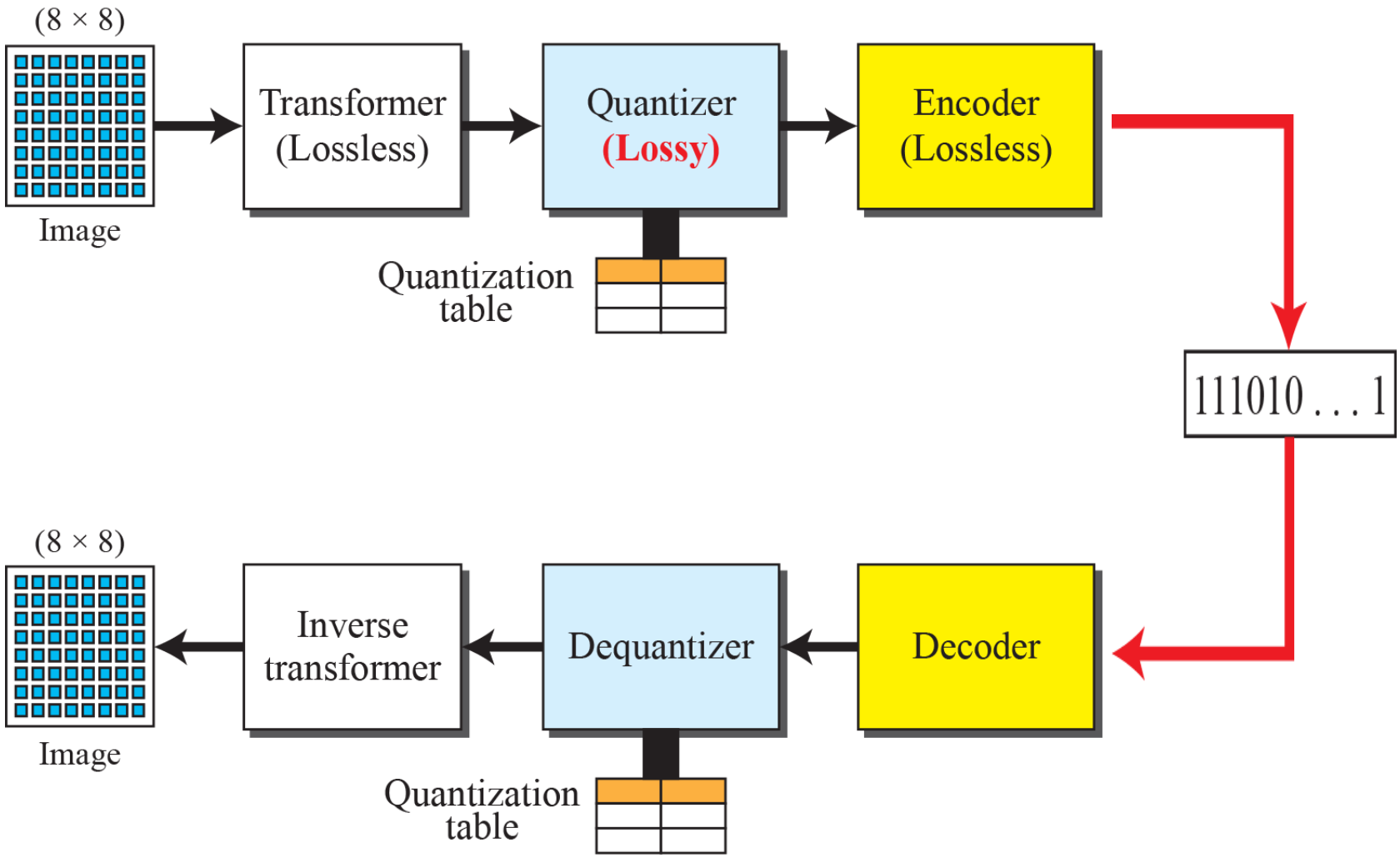

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Figure 8.18 : Three different quantization matrices

$$\begin{bmatrix} 80 & 60 & 50 & 80 & 120 & 200 & 255 & 255 \\ 55 & 60 & 70 & 95 & 130 & 255 & 255 & 255 \\ 70 & 65 & 80 & 120 & 200 & 255 & 255 & 255 \\ 70 & 85 & 110 & 145 & 255 & 255 & 255 & 255 \\ 90 & 110 & 185 & 255 & 255 & 255 & 255 & 255 \\ 120 & 175 & 255 & 255 & 255 & 255 & 255 & 255 \\ 245 & 255 & 255 & 255 & 255 & 255 & 255 & 255 \\ 255 & 255 & 255 & 255 & 255 & 255 & 255 & 255 \end{bmatrix}$$

Q10

$$\begin{bmatrix} 16 & 11 & 10 & 16 & 24 & 40 & 51 & 61 \\ 12 & 12 & 14 & 19 & 26 & 58 & 60 & 55 \\ 14 & 13 & 16 & 24 & 40 & 57 & 69 & 56 \\ 14 & 17 & 22 & 29 & 51 & 87 & 80 & 62 \\ 18 & 22 & 37 & 56 & 68 & 109 & 103 & 77 \\ 24 & 35 & 55 & 64 & 81 & 104 & 113 & 92 \\ 49 & 64 & 78 & 87 & 103 & 121 & 120 & 101 \\ 72 & 92 & 95 & 98 & 112 & 110 & 103 & 99 \end{bmatrix}$$

Q50

$$\begin{bmatrix} 3 & 2 & 2 & 3 & 5 & 8 & 10 & 12 \\ 2 & 2 & 3 & 4 & 5 & 12 & 12 & 11 \\ 3 & 3 & 3 & 5 & 8 & 11 & 14 & 11 \\ 3 & 3 & 4 & 6 & 10 & 17 & 16 & 12 \\ 4 & 4 & 7 & 11 & 14 & 22 & 21 & 15 \\ 5 & 7 & 11 & 13 & 16 & 12 & 23 & 18 \\ 10 & 13 & 16 & 17 & 21 & 24 & 24 & 21 \\ 14 & 18 & 19 & 20 & 22 & 20 & 20 & 20 \end{bmatrix}$$

Q90

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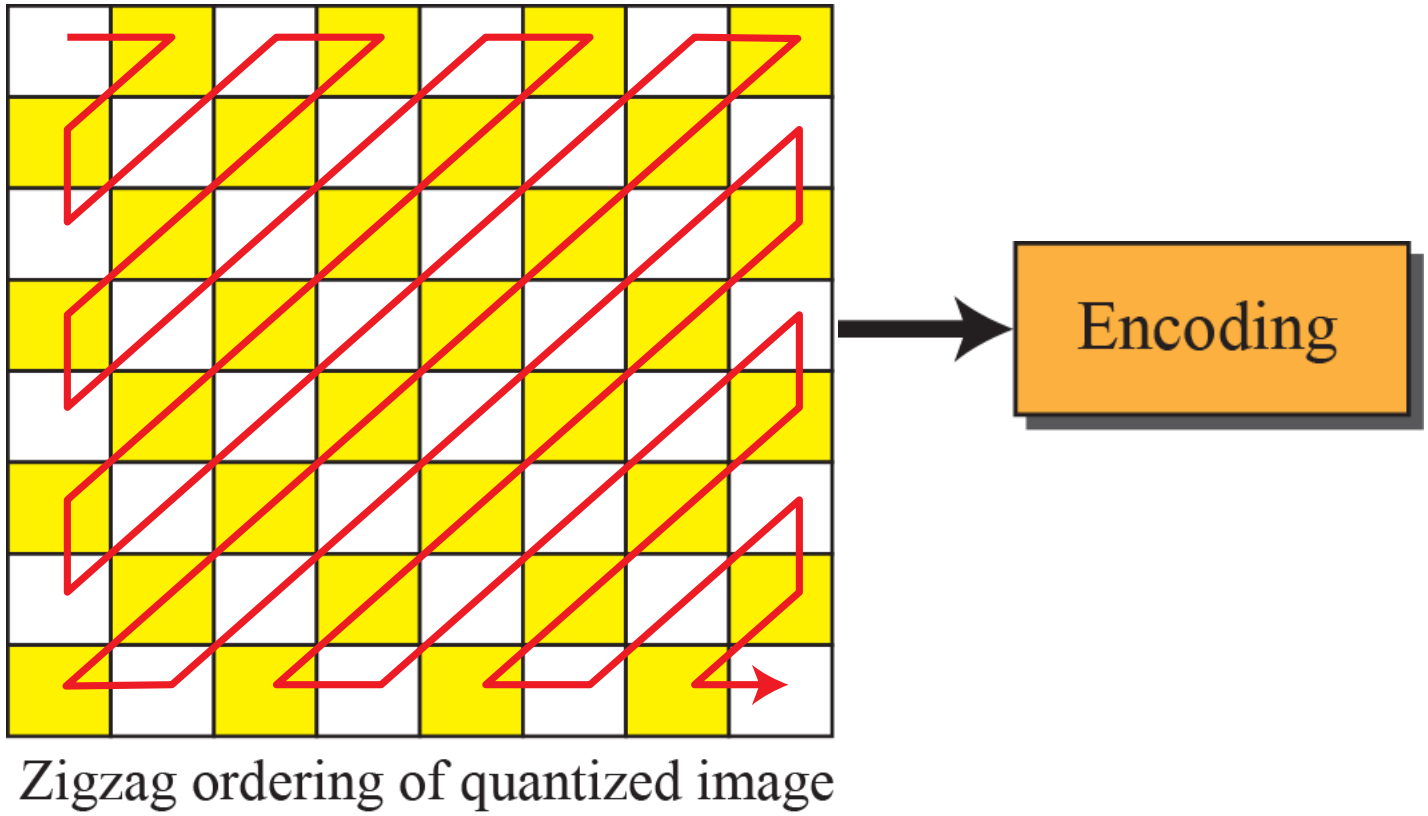
Figure 8.19 : Reading the table



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Example 8.7

To show the idea of JPEG compression, we use a block of gray image in which the bit depth for each pixel is 20. We have used a Java program to transform, quantize, and reorder the values in zigzag sequence; we have shown the encoding (Figure 8.20).

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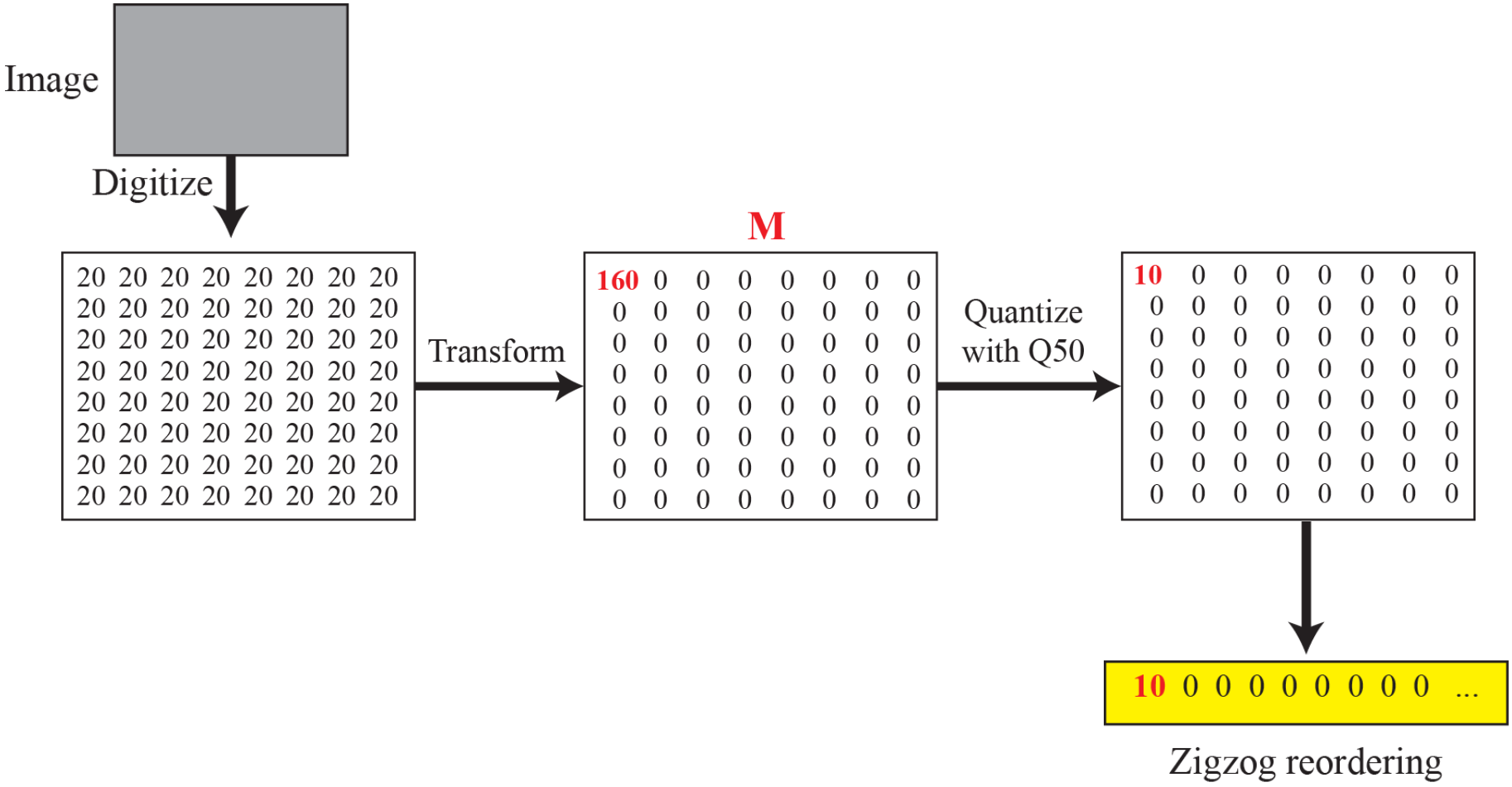
Figure 8.20 : Example 8.7: uniform gray scale



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Example 8.8

As the second example, we have a block that changes gradually; there is no sharp change between the values of neighboring pixels. We still get a lot of zero values, as shown in Figure 8.21.

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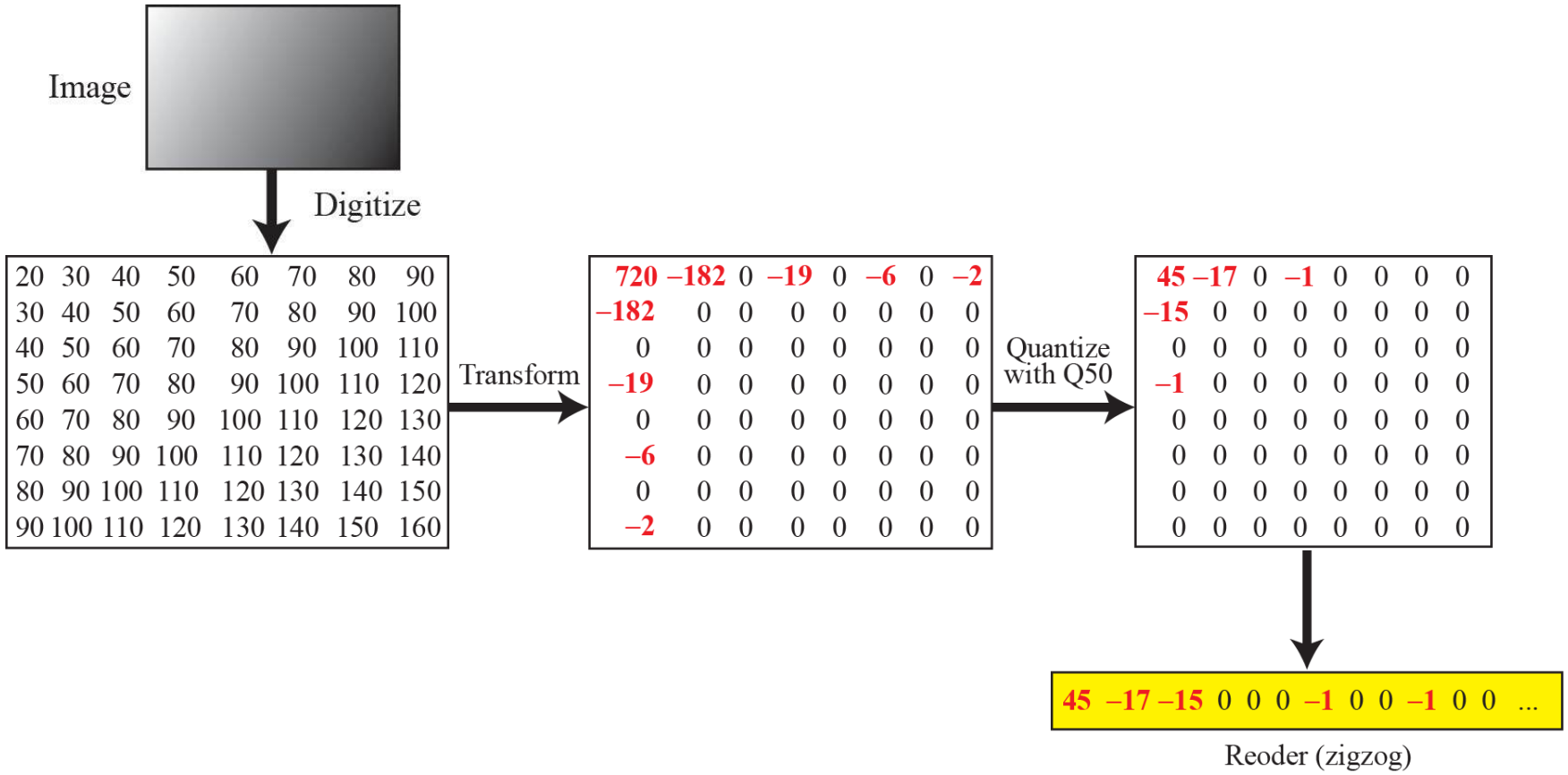
Figure 8.21 : Example 8.8: gradient gray scale



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8.2.3 Video

Video is composed of multiple frames; each frame is one image. This means that a video file requires a high transmission rate.

- **Digitizing Video**
- **Video Compression: MPEG**
  - Spatial Compression
  - Temporal Compression

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Example 8.9

Let us show the transmission rate for some video standards:

a. Color broadcast television takes 720 × 480 pixels per frame, 30 frames per second, and 24 bits per color. The transmission rate without compression is as shown below.

$$720 \times 480 \times 30 \times 24 = 248{,}832{,}000 \text{ bps} = 249 \text{ Mbps}$$

b. High definition color broadcast television takes 1920 × 1080 pixels per frame, 30 frames per second, and 24 bits per color: The transmission rate without compression is as shown below.

$$1920 \times 1080 \times 30 \times 24 = 1{,}492{,}992{,}000 \text{ bps} = 1.5 \text{ Gbps}$$

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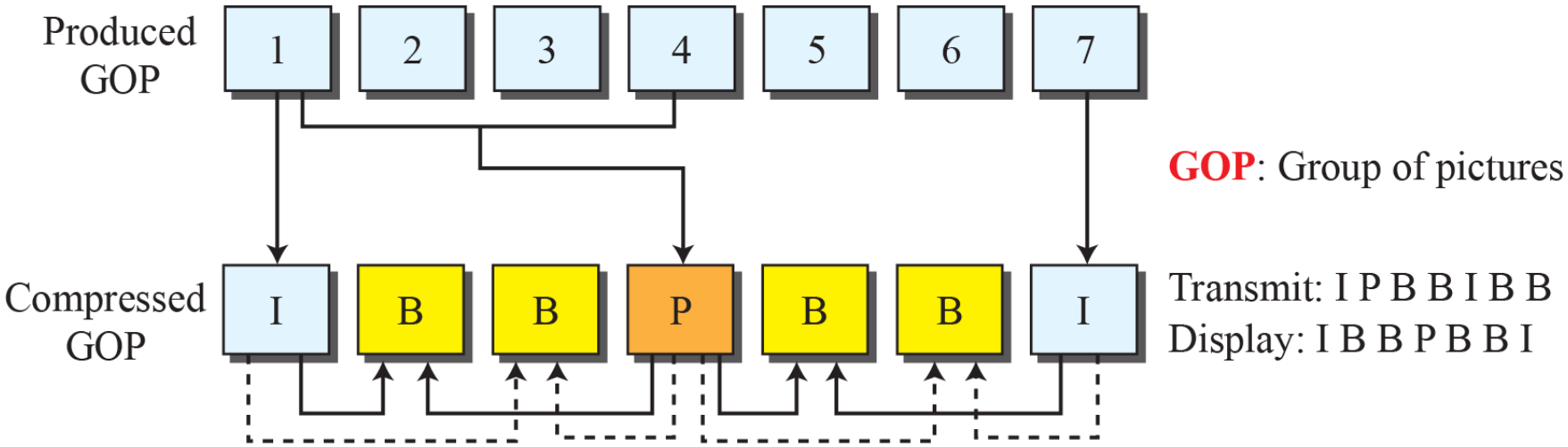
Figure 8.22 : MPEG frames



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8.2.4 Audio

Audio (sound) signals are analog signals that need a medium to travel; they cannot travel through a vacuum. The speed of the sound in the air is about 330 m/s (740 mph).

- **Digitizing Audio**
- **Audio Compression**
  - Predictive coding
  - Perceptual Coding
  - MP3

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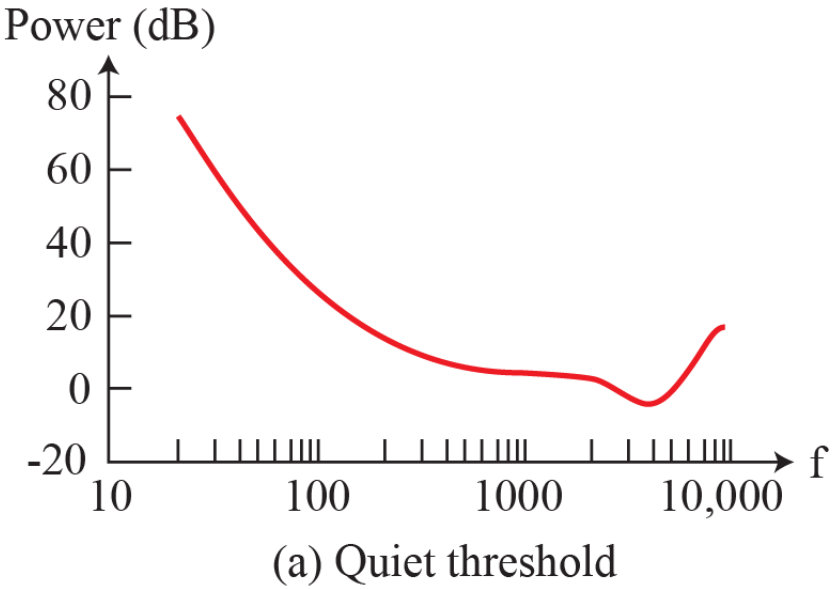
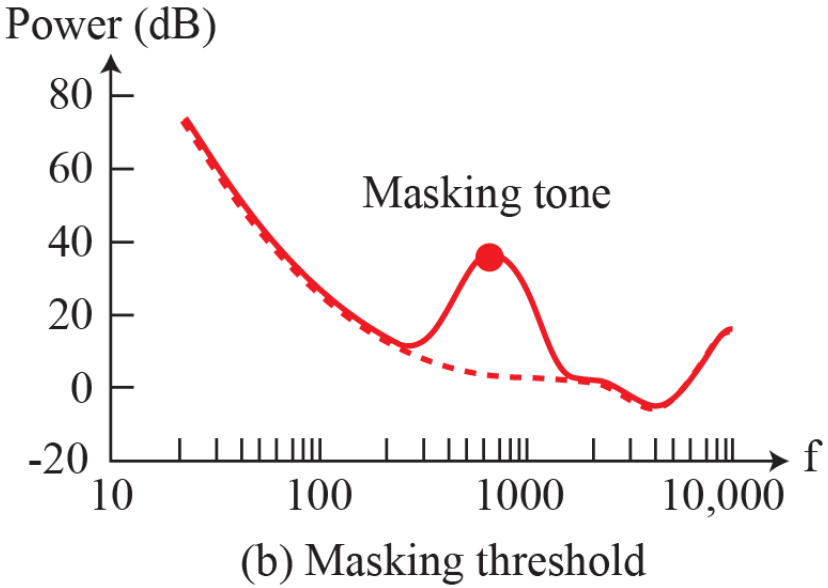
Figure 8.23 : Threshold of audibility



(a) Quiet threshold



(b) Masking threshold

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8-3 MULTIMEDIA IN THE INTERNET

We can divide audio and video services into three broad categories: streaming stored audio/video, streaming live audio/video, and interactive audio/video. Streaming means a user can listen (or watch) the file after the downloading has started.

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8.3.1 Streaming Stored Audio/Video

In the first category, streaming stored audio/video, the files are compressed and stored on a server. A client downloads the files through the Internet. This is sometimes referred to as on-demand audio/video. We can say that streaming stored audio/video refers to on-demand requests for compressed audio/video files.

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8.3.1 (continued)

- ***First Approach: Using a Web Server***
- ***Second Approach: Using a Web Server with a Metafile***
- ***Third Approach: Using a Media Server***
- ***Fourth Approach: Using a Media Server and RTSP***
- ***Example: Video on Demand (VOD)***

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Figure 8.24 : Using a Web server

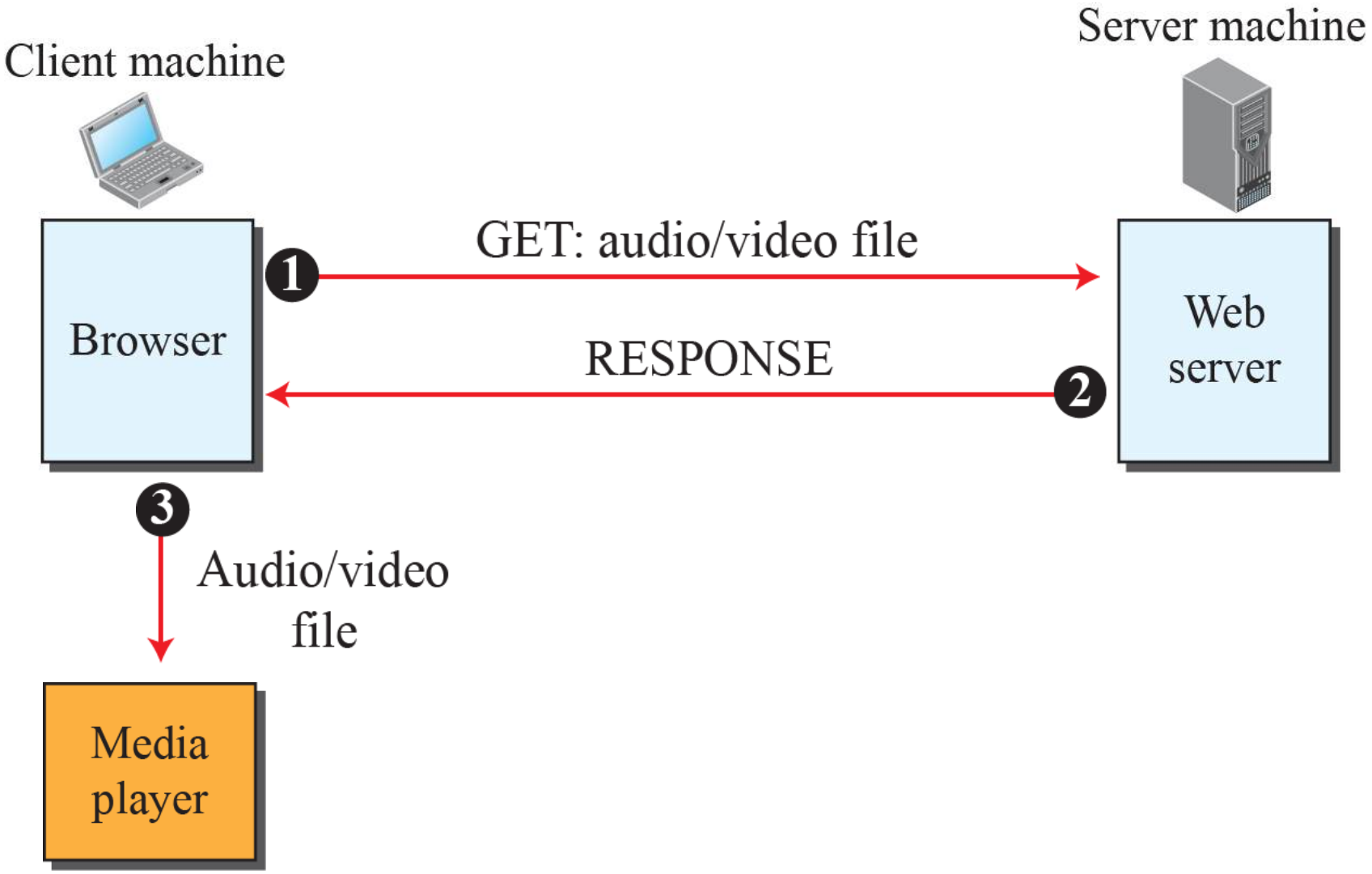

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Figure 8.25 : Using a Web server with a metafile

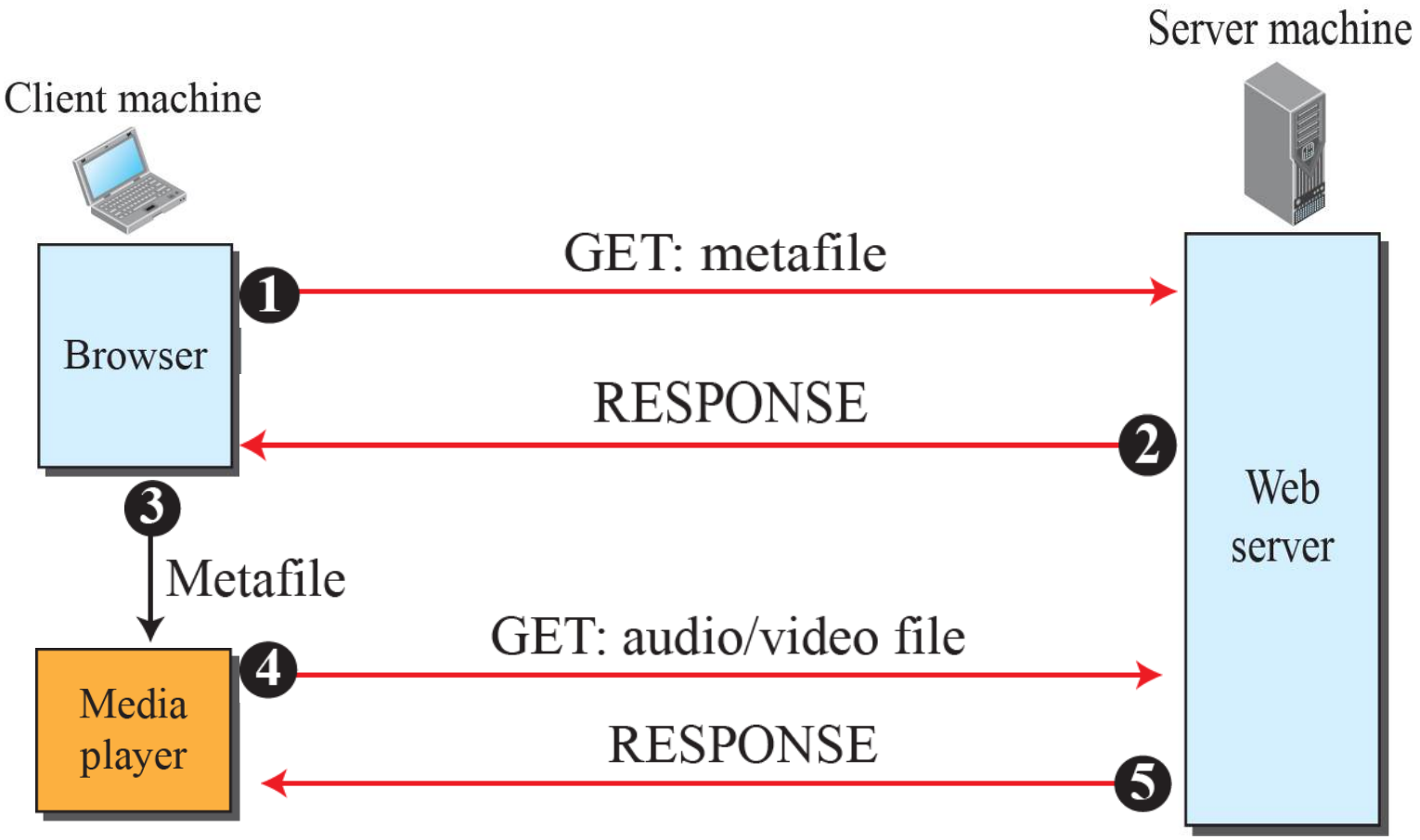

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Figure 8.26 : Using a media server

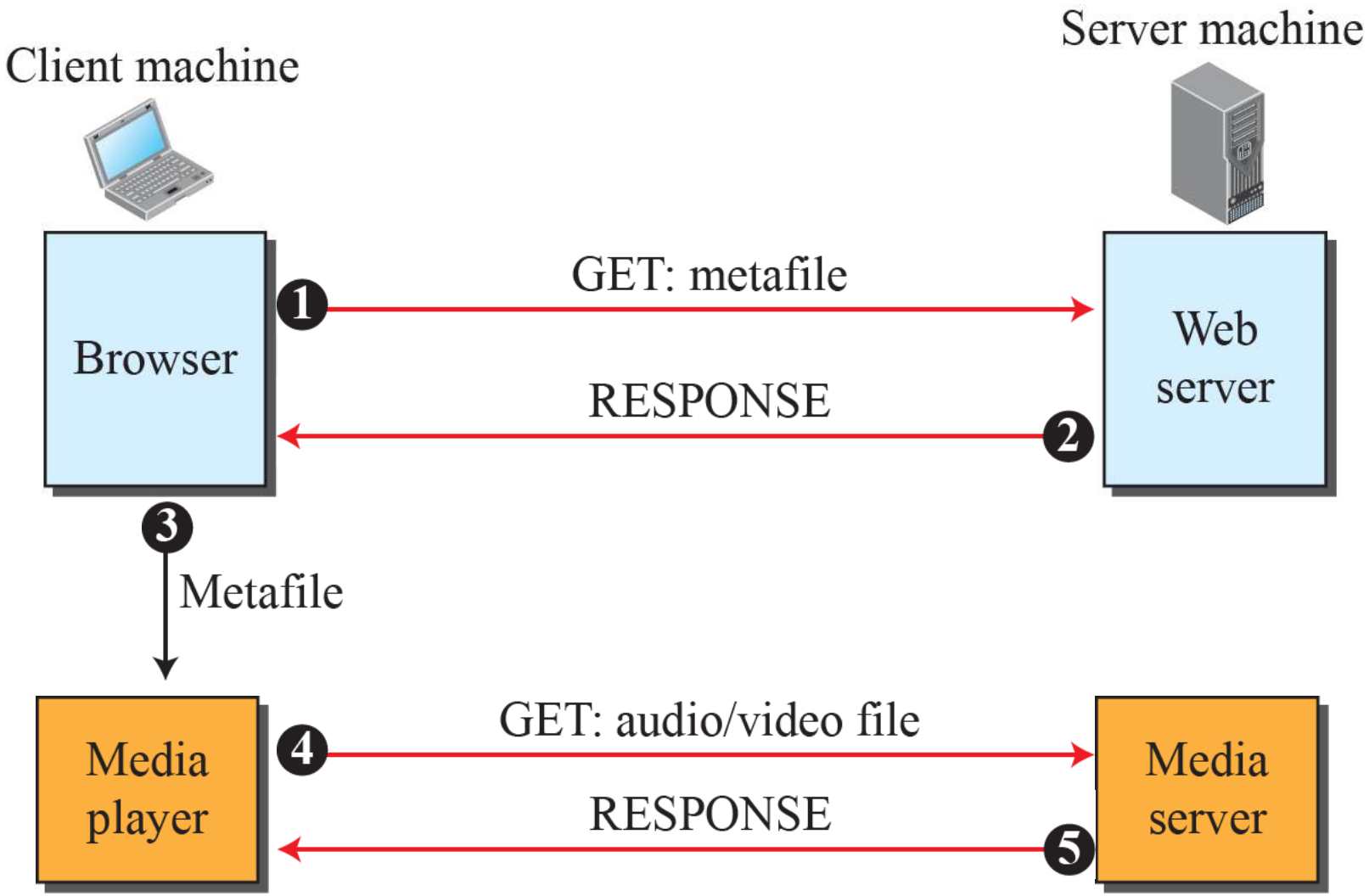

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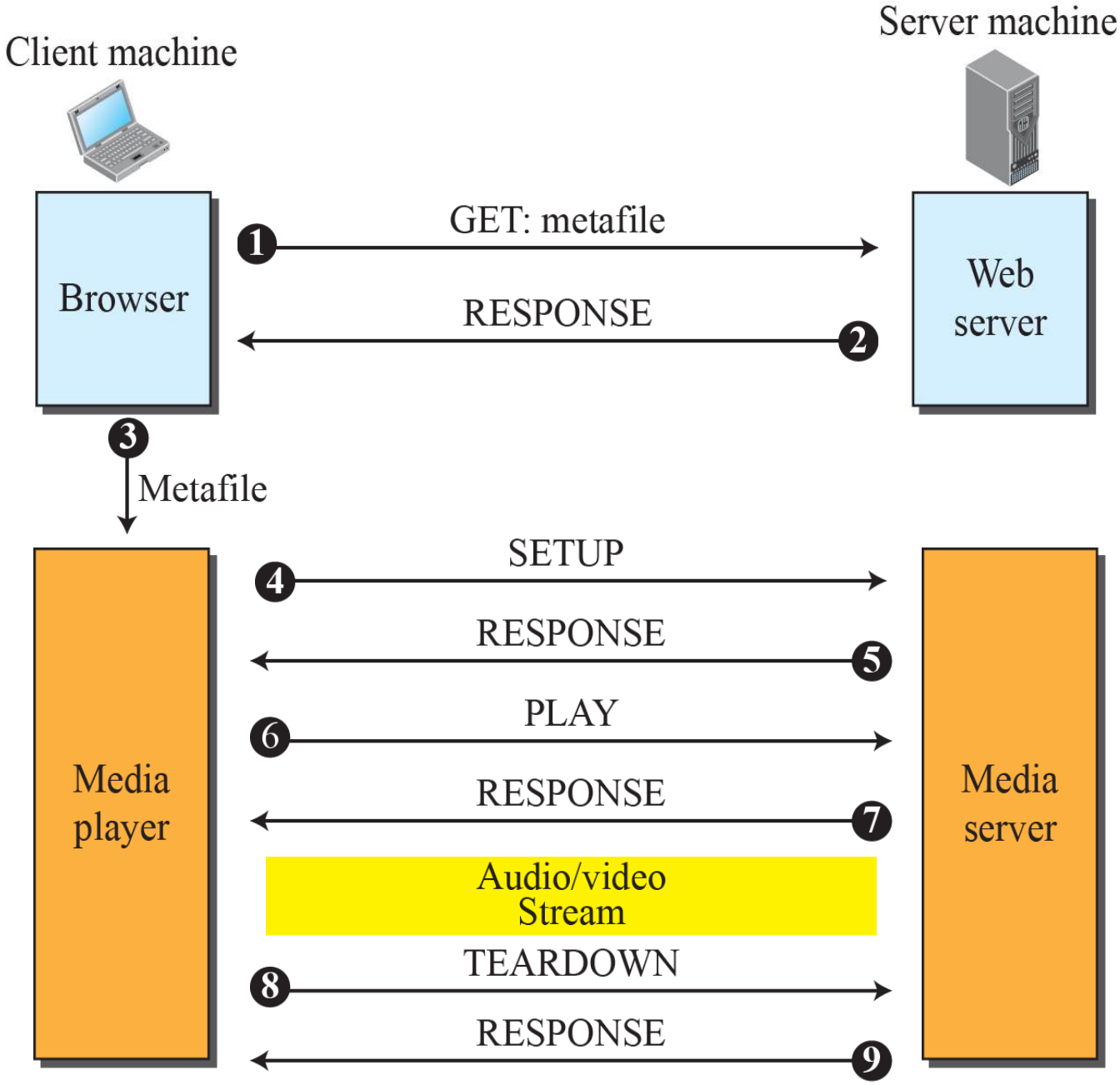
Figure 8.27 : Using a media server and RTSP



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8.3.2 Streaming Live Audio/Video

In the second category, streaming live audio/video, a user listens to broadcast audio and video through the Internet. Good examples of this type of application are Internet radio and Internet TV.

- **Example: Internet Radio**
- **Example: Internet Television (ITV)**
- **Example: IPTV**

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8.3.3 Real-Time Interactive Audio/Video

In the third category, interactive audio/video, people use the Internet to interactively communicate with one another. The Internet phone or voice over IP is an example of this type of application. Video conferencing is another example that allows people to communicate visually and orally.

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8.3.3 (continued)

- **Characteristics**
  - Time Relationship
  - Timestamp
  - Playback Buffer
  - Ordering
  - Multicasting
  - Translation
  - Mixing

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8.3.3 (continued)

- **Forward Error Correction**
  - Error Correction Using Hamming Distance
  - Error Correction Using XOR
  - Chunk Interleaving
  - Combining Hamming Distance and Interleaving
  - Compounding High- and Low-Resolution Packets
- **Example of a Real-Time Application: Skype**

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Figure 8.28 : Time relationship

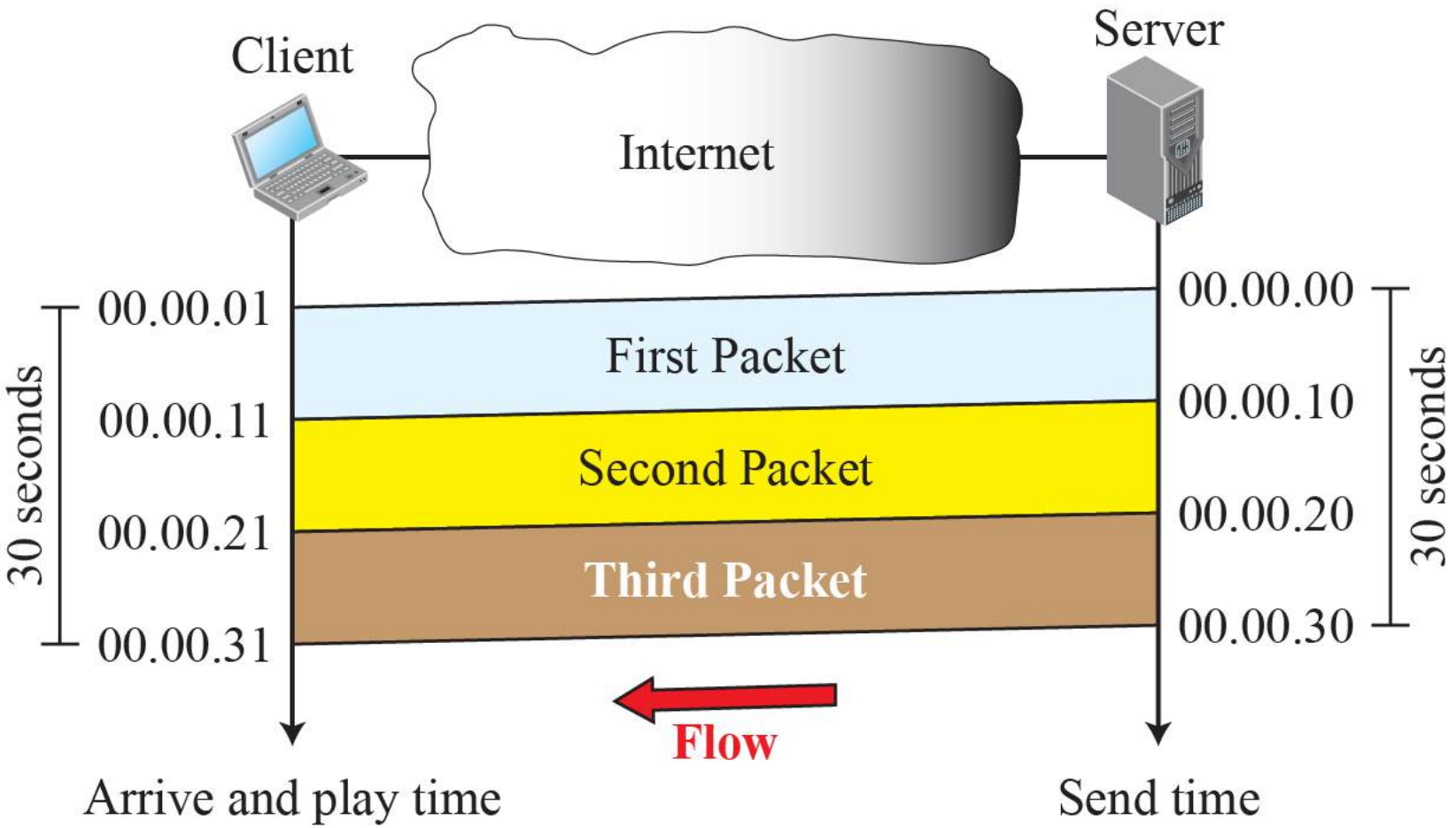

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Figure 8.29 : Jitter

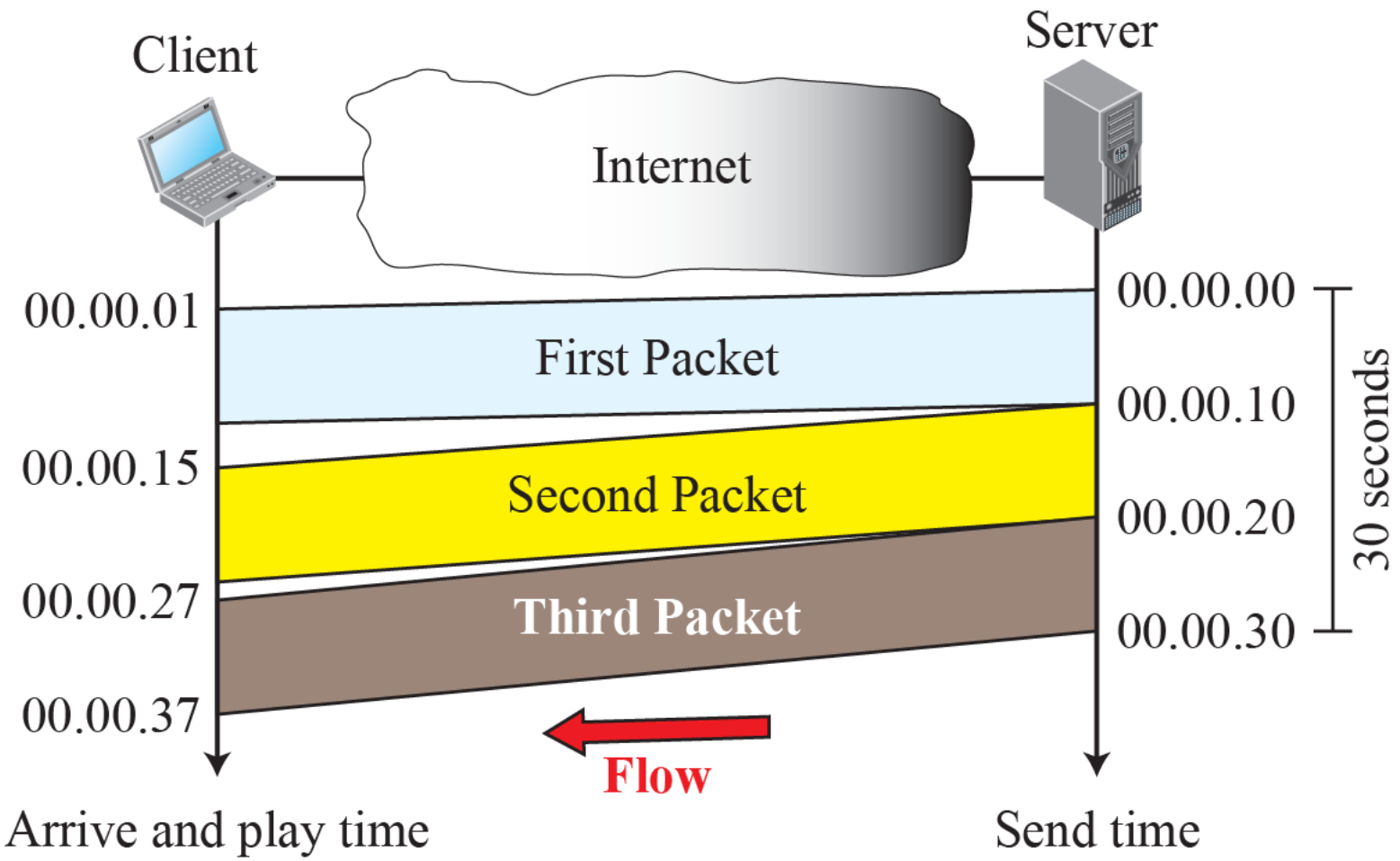

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Figure 8.30 : Timestamp

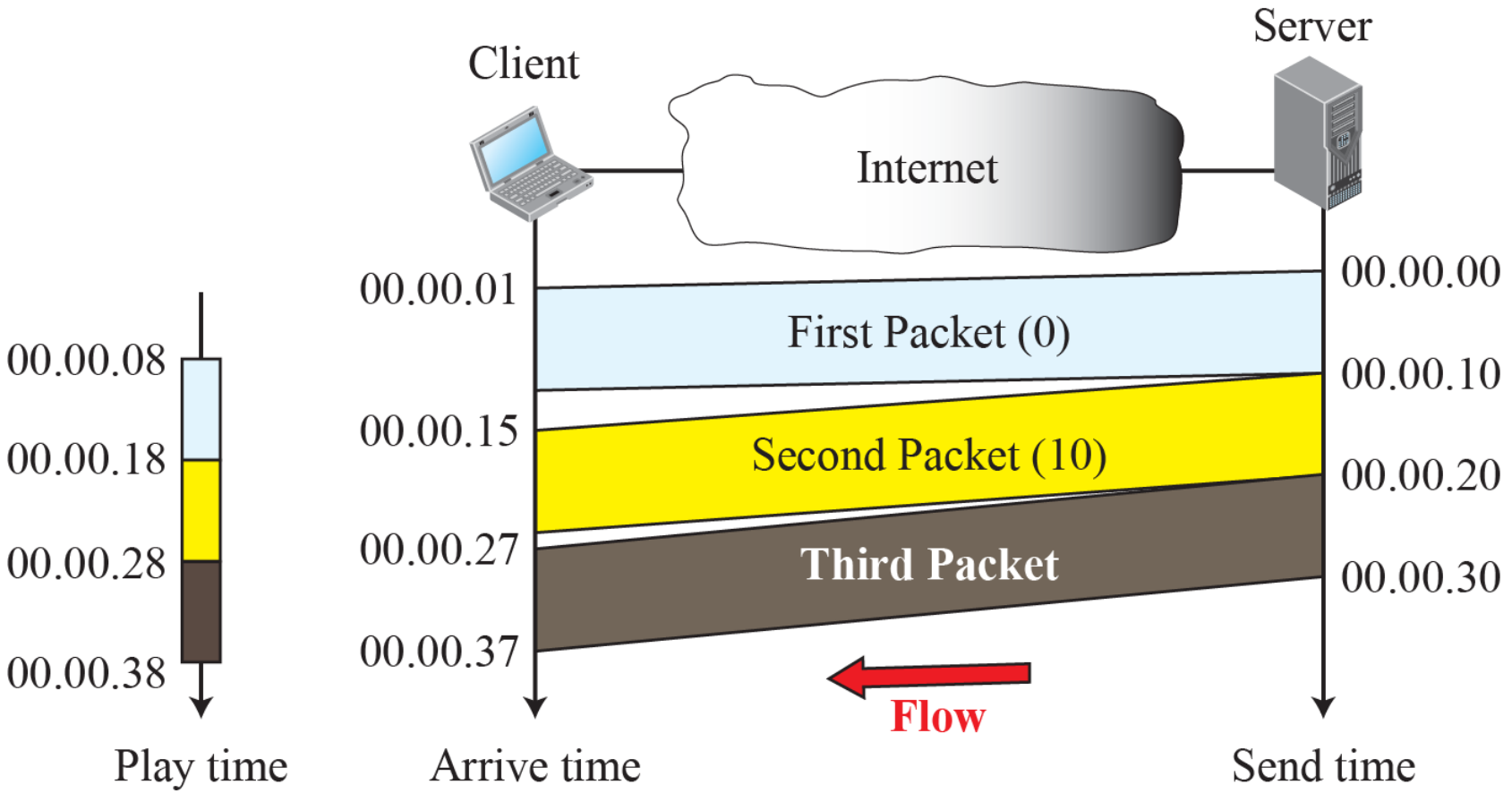

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Figure 8.31 : Playback buffer

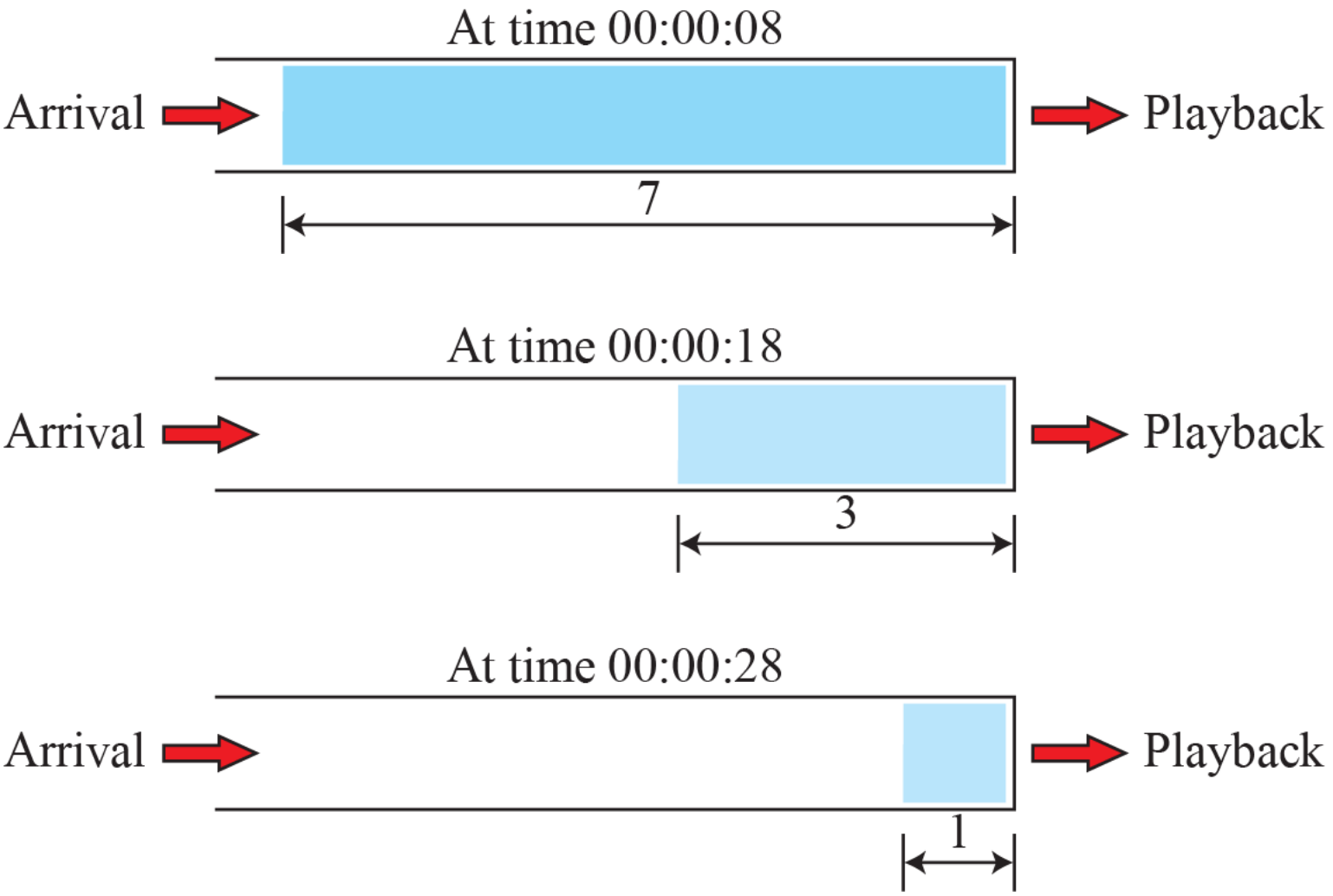

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Figure 8.32 : The time line of packets

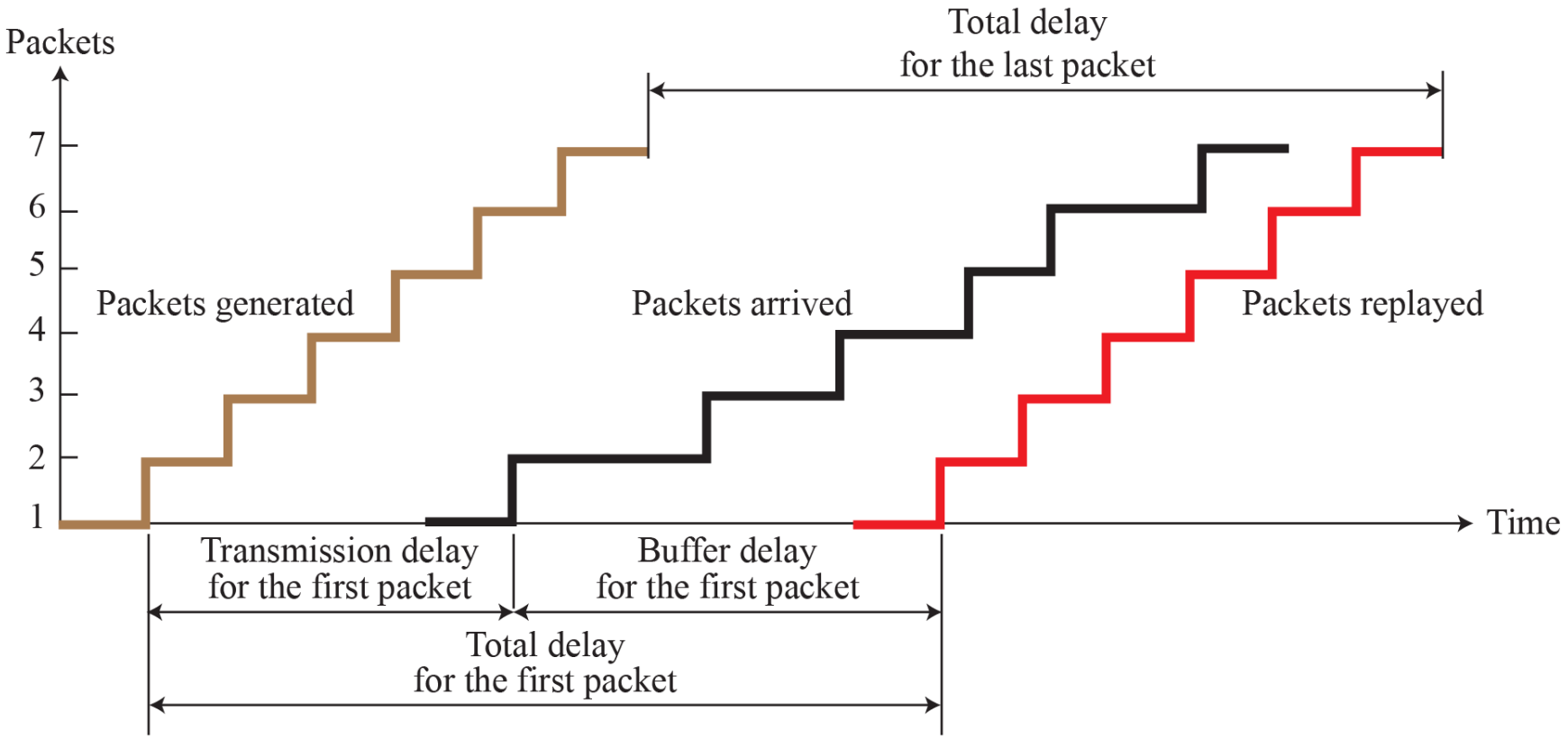

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Figure 8.33 : Interleaving

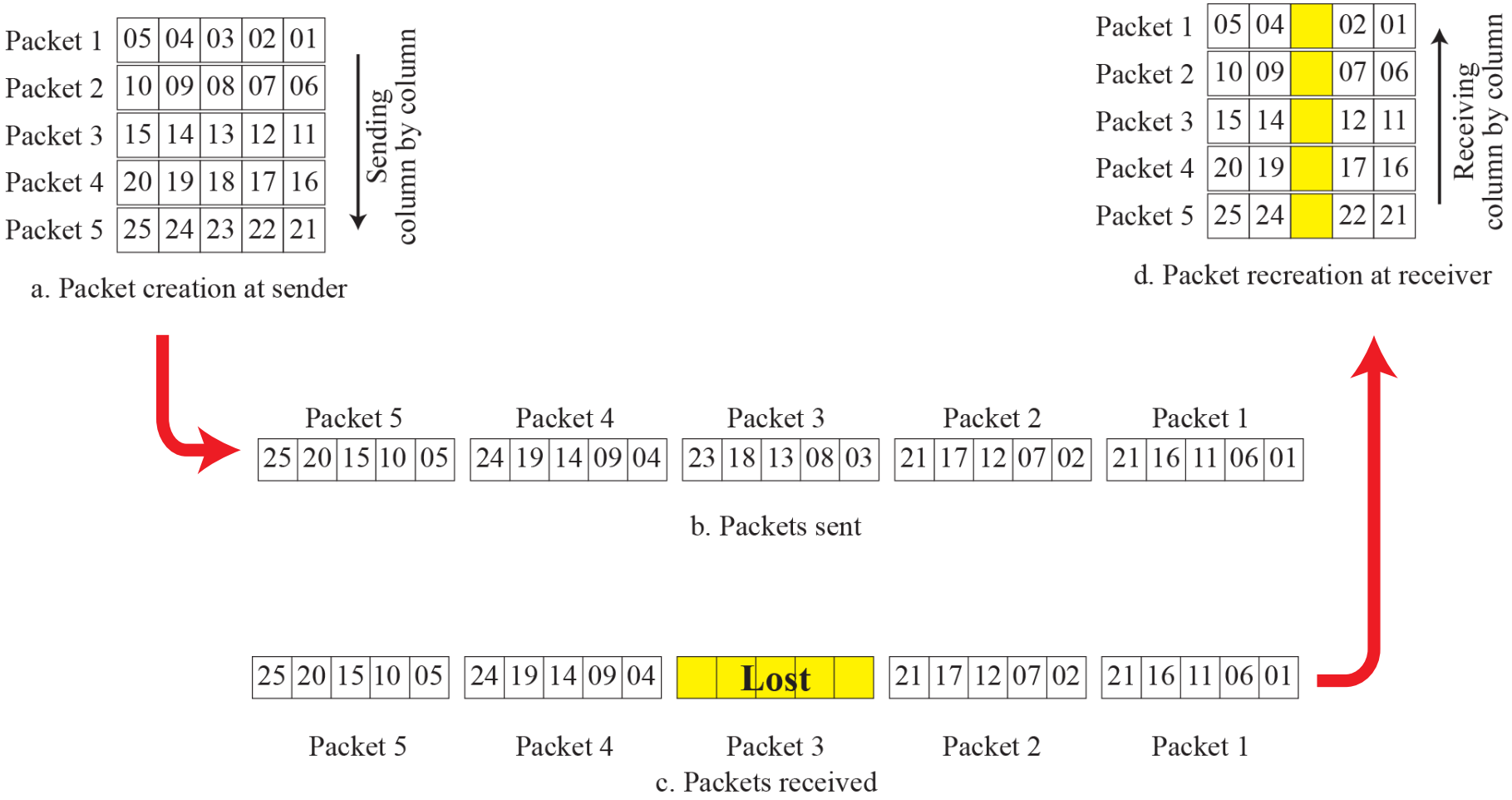

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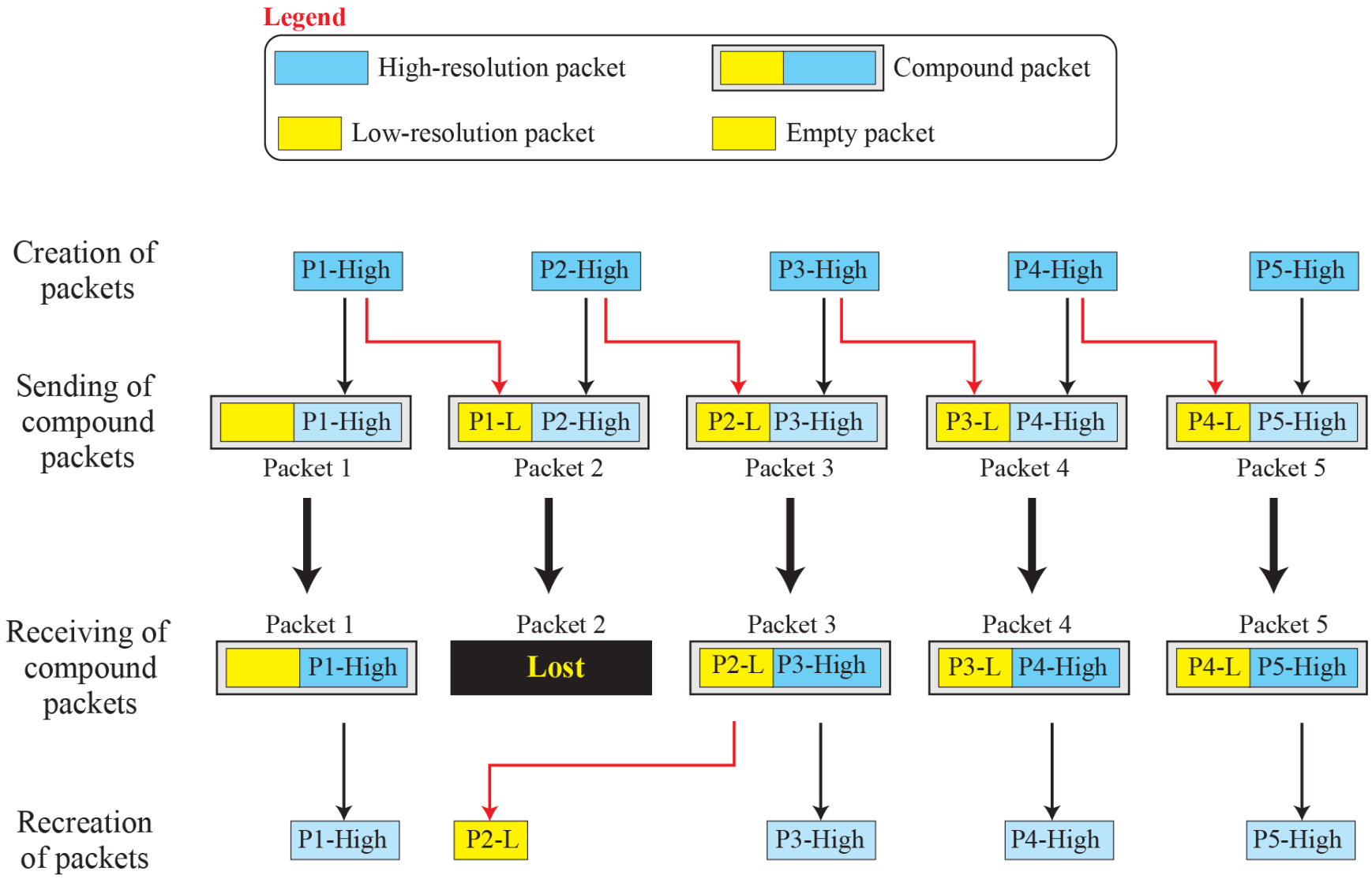
Figure 8.34 : Compounding high- and low-resolution packets



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8-4 REAL-TIME INTERACTIVE PROTOCOLS

We now concentrate on the last category, which is the most interesting and involved: real-time interactive multimedia. This application has evoked a lot of attention in the Internet society and several application-layer protocols have been designed to handle it.

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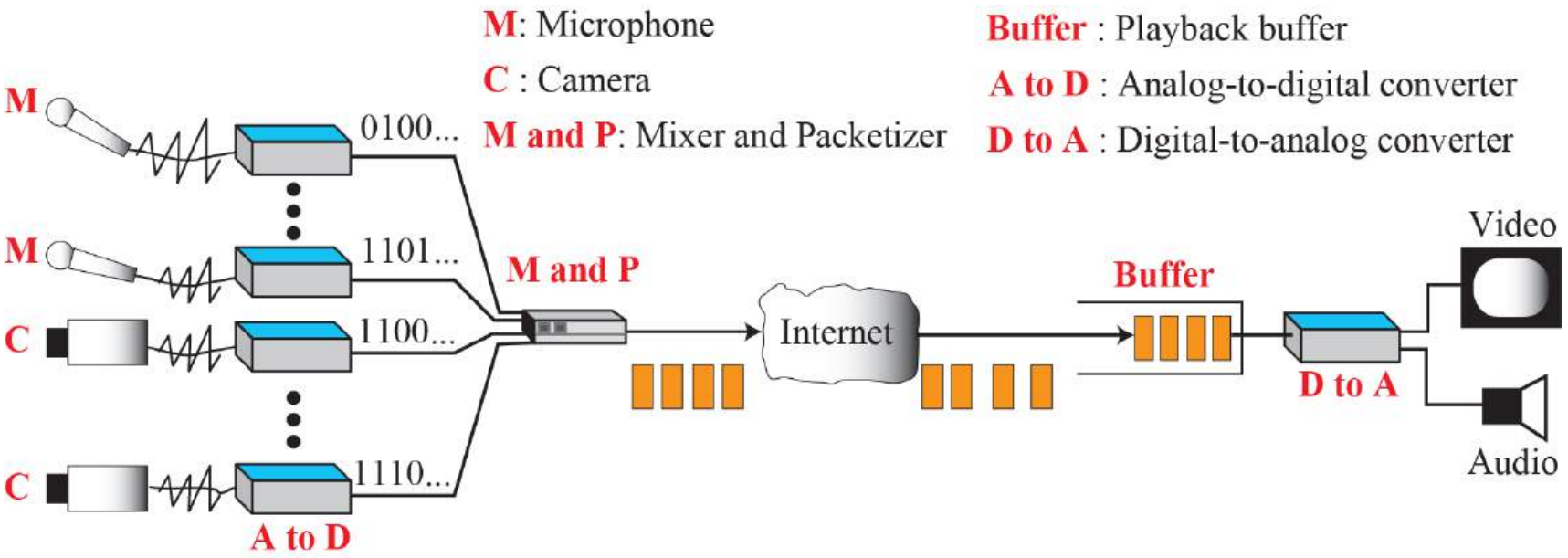
Figure 8.35 : Schematic diagram of a real-time multimedia system



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8.4.1 Rationale for New Protocols

It is clear that we do not need to change the first three layers of the TCP/IP protocol Suite because these three layers are designed to carry any type of data. It looks as if we should worry about only the application and transport layers.

- **Application Layer**
- **Transport Layer**
  - Transport-Layer Requirements
  - Capability of UDP or TCP

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Table 8.6: Capability of UDP or TCP to handle real-time data

Requirements	*UDP*	*TCP*
1. Sender-receiver negotiation for selecting the encoding type	No	No
2. Creation of packet stream	No	No
3. Source synchronization for mixing different sources	No	No
4. Error Control	No	Yes
5. Congestion Control	No	Yes
6. Jitter removal	No	No
7. Sender identification	No	No

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8.4.2 RTP

Real-time Transport Protocol (RTP) is the protocol designed to handle real-time traffic on the Internet. RTP does not have a delivery mechanism; it must be used with UDP. RTP stands between UDP and the multimedia application. The literature and standards treat RTP as the transport protocol that can be thought of as located in the application layer (see Figure 8.36).

- *UDP Port*
- *RTP Packet Format*

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Figure 8.36 : RTP location in the TCP/IP protocol suite

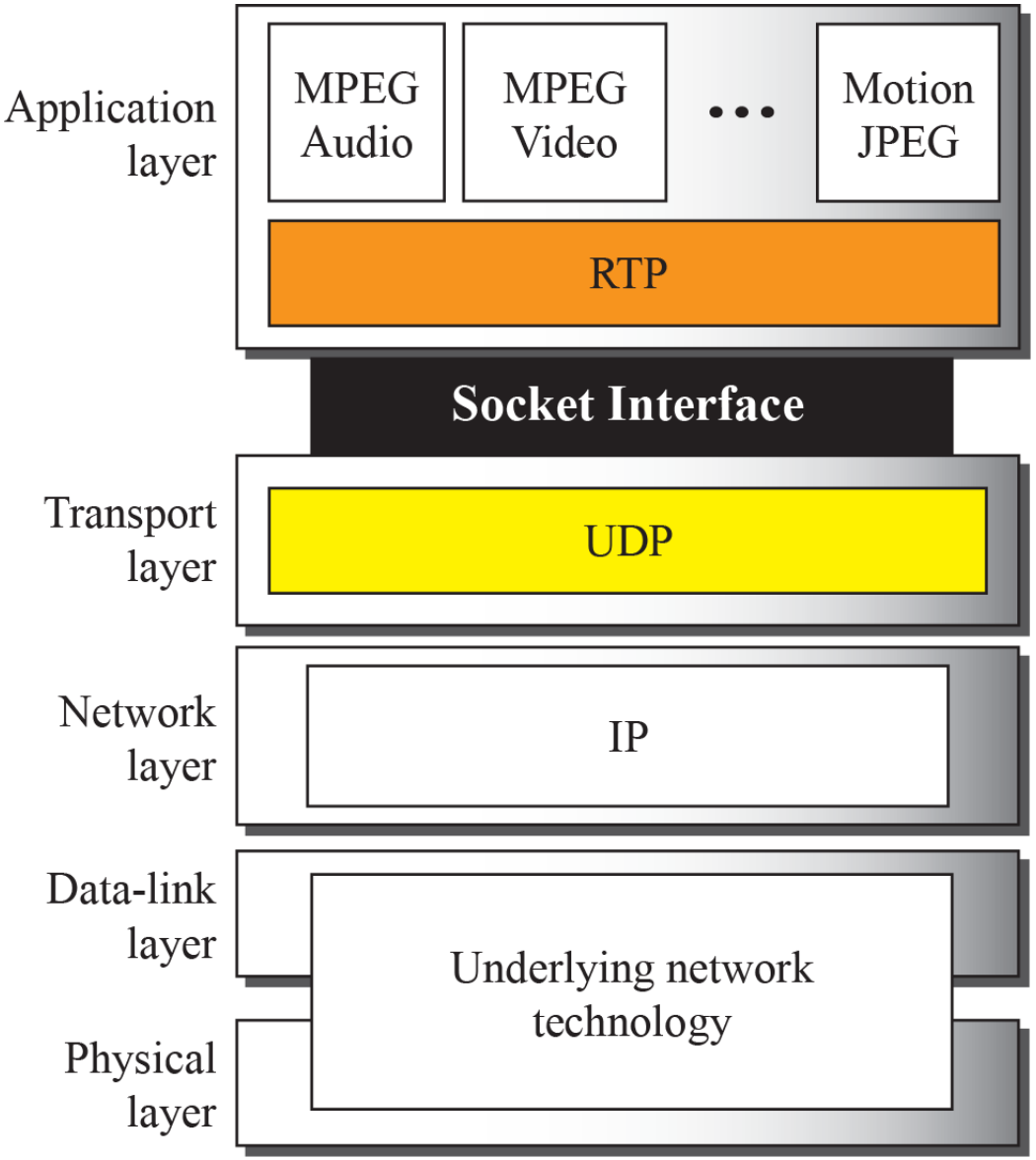

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Figure 8.37 : RTP packet header format

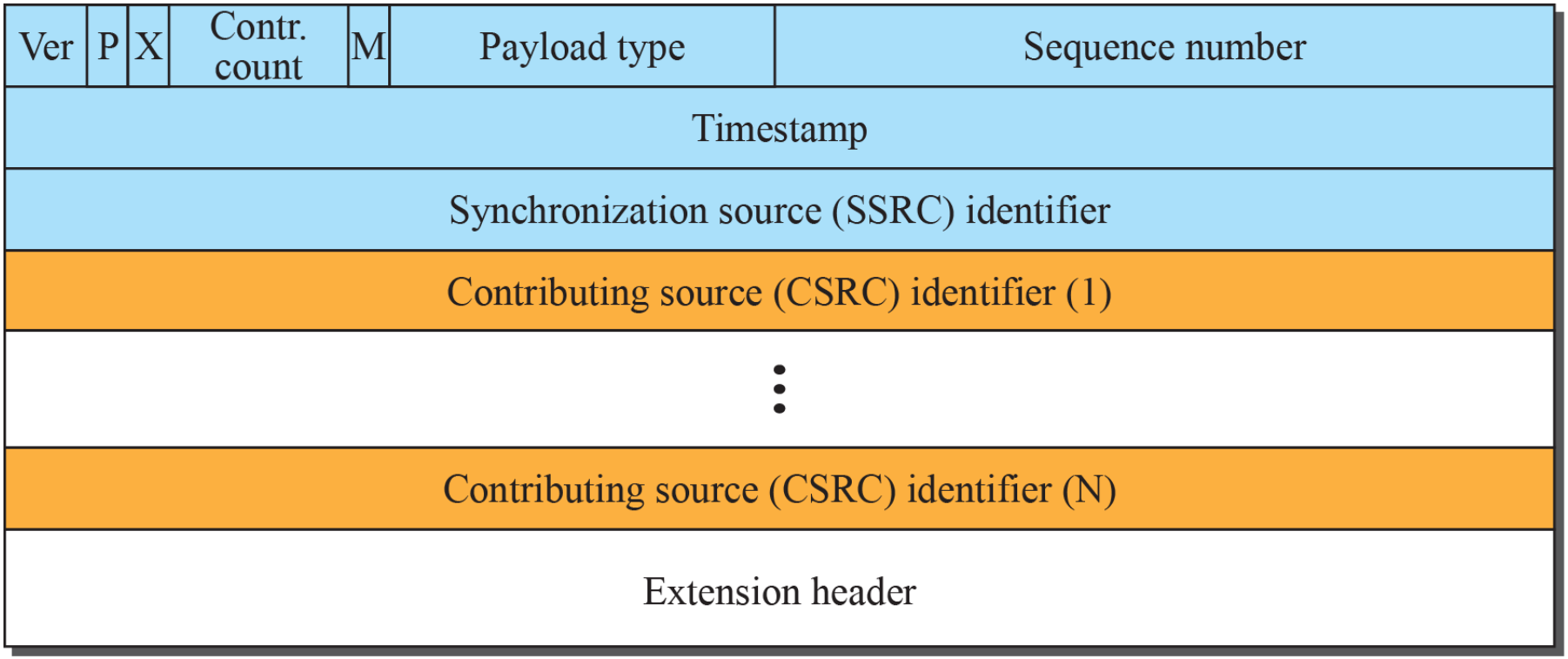

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Table 8.7: Payload types

Type	*Application*	*Type*	*Application*	*Type*	*Application*
0	PCMμ Audio	7	LPC audio	15	G728 audio
1	1016	8	PCMA audio	26	Motion JPEG
2	G721 audio	9	G722 audio	31	H.261
3	GSM audio	10–11	L16 audio	32	MPEG1 video
5–6	DV14 audio	14	MPEG audio	33	MPEG2 video

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8.4.3 RTCP

RTP allows only one type of message, one that carries data from the source to the destination. To really control the session, we need more communication between the participants in a session. Control communication in this case is assigned to a separate protocol called Real-time Transport Control Protocol (RTCP). We need to emphasize that the RTCP payloads are not carried in RTP packets; RTCP is in fact a sister protocol of RTP.

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8.4.3 (continued)

- **RTCP Packets**
  - Sender Report Packet
  - Receiver Report Packet
  - Source Description Packet
  - Bye Packet
  - Application-Specific Packet
- **UDP Port**
- **Bandwidth Utilization**
- **Requirement Fulfillment**

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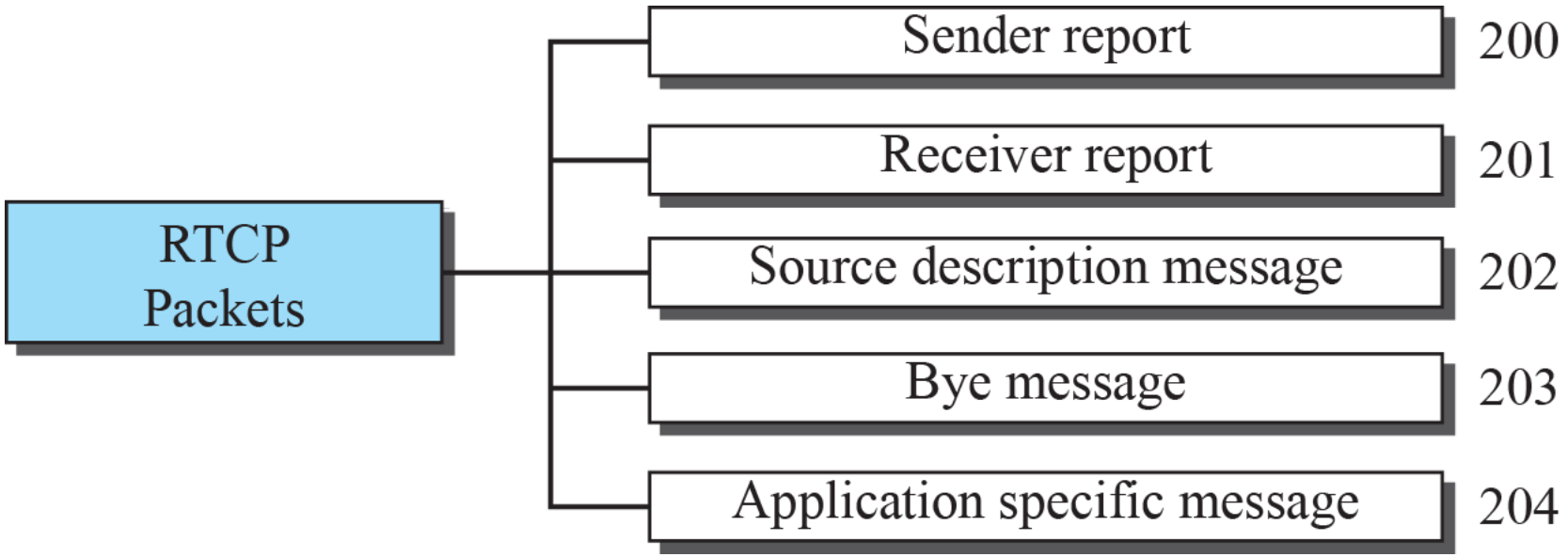
Figure 8. 38 : RTCP packet types



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Example 8.10

Let us assume that the total bandwidth allocated for a session is 1 Mbps. RTCP traffic gets only 5 percent of this bandwidth, which is 50 Kbps. If there are only 2 active senders and 8 passive receivers, it is natural that each sender or receiver gets only 5 Kbps. If the average size of the RTCP packet is 5 Kbits, then each sender or receiver can send only 1 RTCP packet per second. Note that we need to consider the packet size at the data-link layer..

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8.4.4 SIP

We discussed how to use the Internet for audio-video conferencing. Although RTP and RTCP can be used to provide these services, one component is missing: a signaling system required to call the participants.

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8.4.4 (continued)

- **Communicating Parties**
- **Addresses**
- **Messages**
  - Request Messages
  - Response Messages
- **First Scenario: Simple Session**
  - Establishing a Session
  - Communicating
  - Terminating the Session

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8.4.4 (continued)

- **Second Scenario: Tracking the Callee**
- **SIP Message Format and SDP Protocol**
  - Start Line
  - Status Line
  - Header
  - Body
  - Putting the Parts Together

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Figure 8.39 : SIP formats

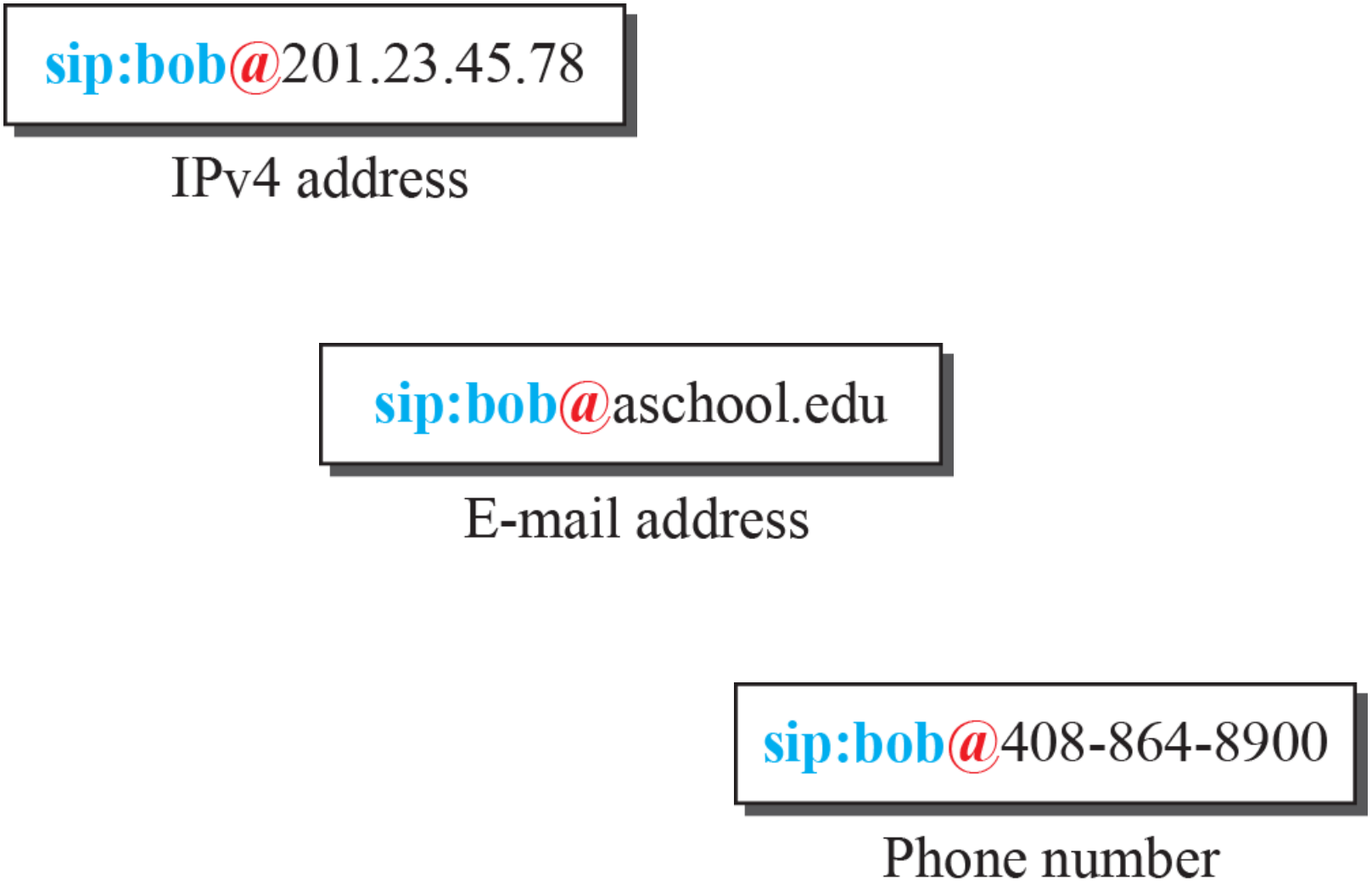

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Figure 8.40 : SIP simple session

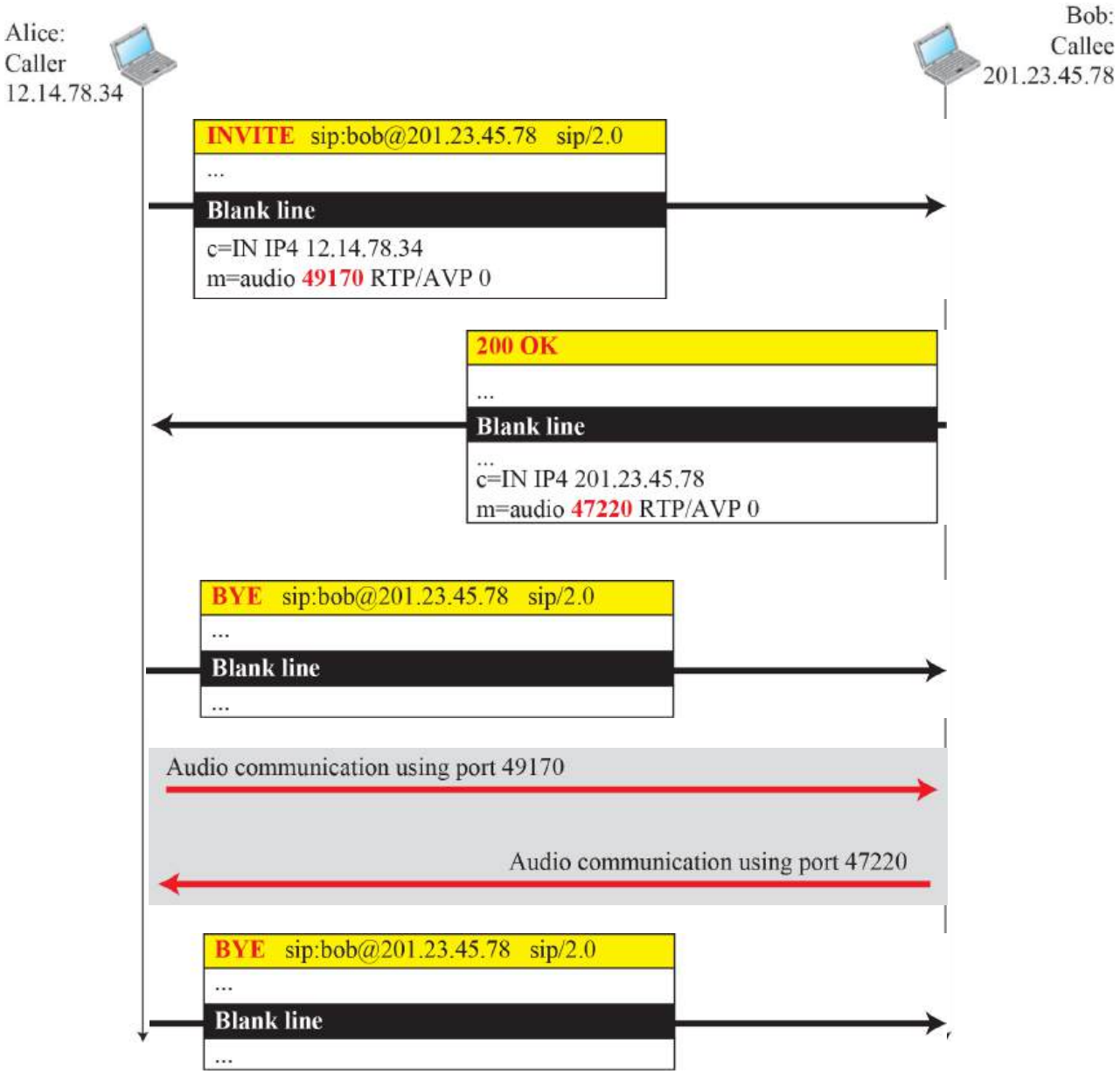

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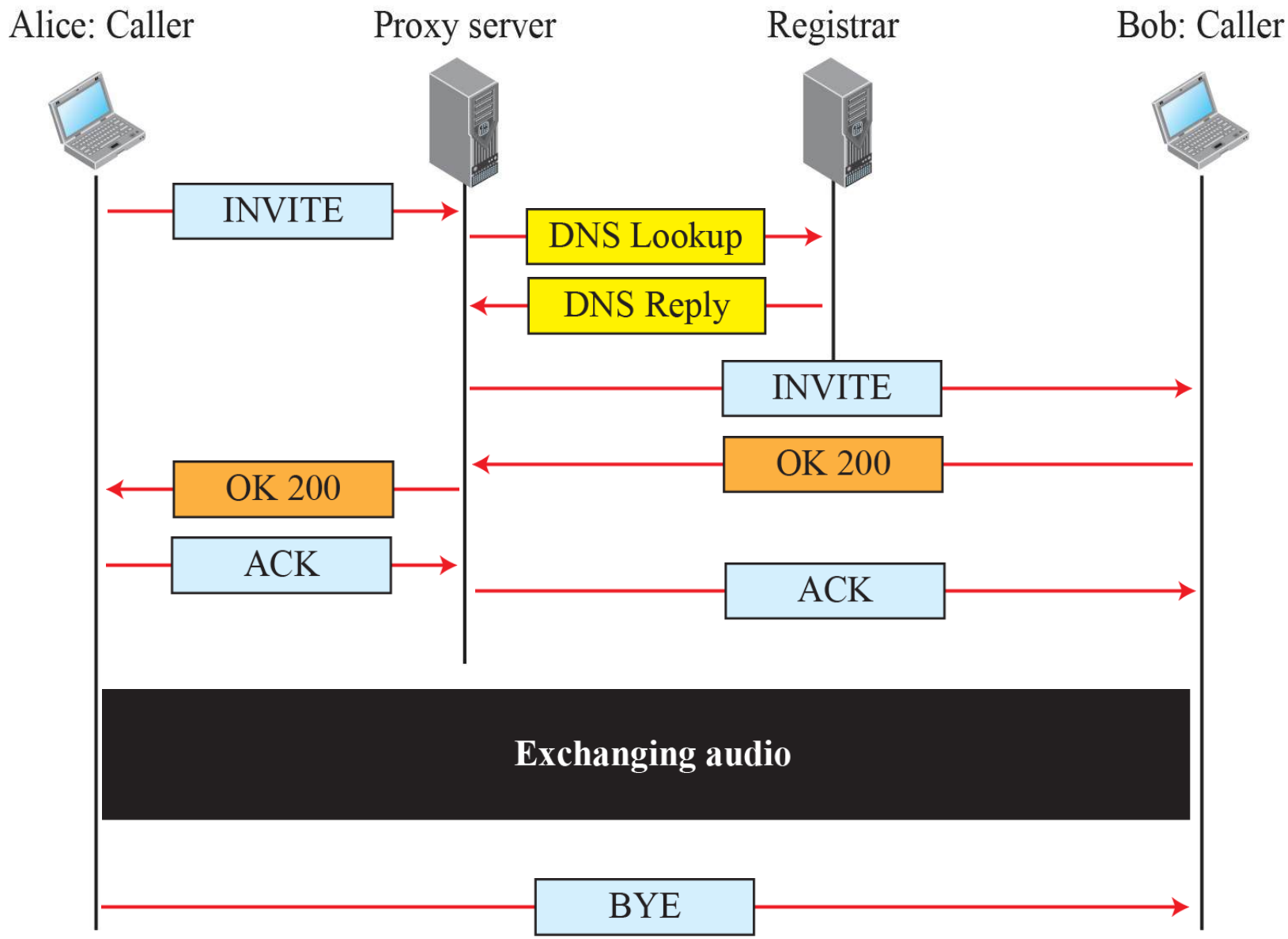
Figure 8.41 : Tracking the callee



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8.4.5 H.323

H.323 is a standard designed by ITU to allow telephones on the public telephone network to talk to computers (called terminals in H.323) connected to the Internet. Figure 8.42 shows the general architecture of H.323 for audio, but it can also be used for video.

- **Protocols**
- **Operation**

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Figure 8.42 : H.323 architecture

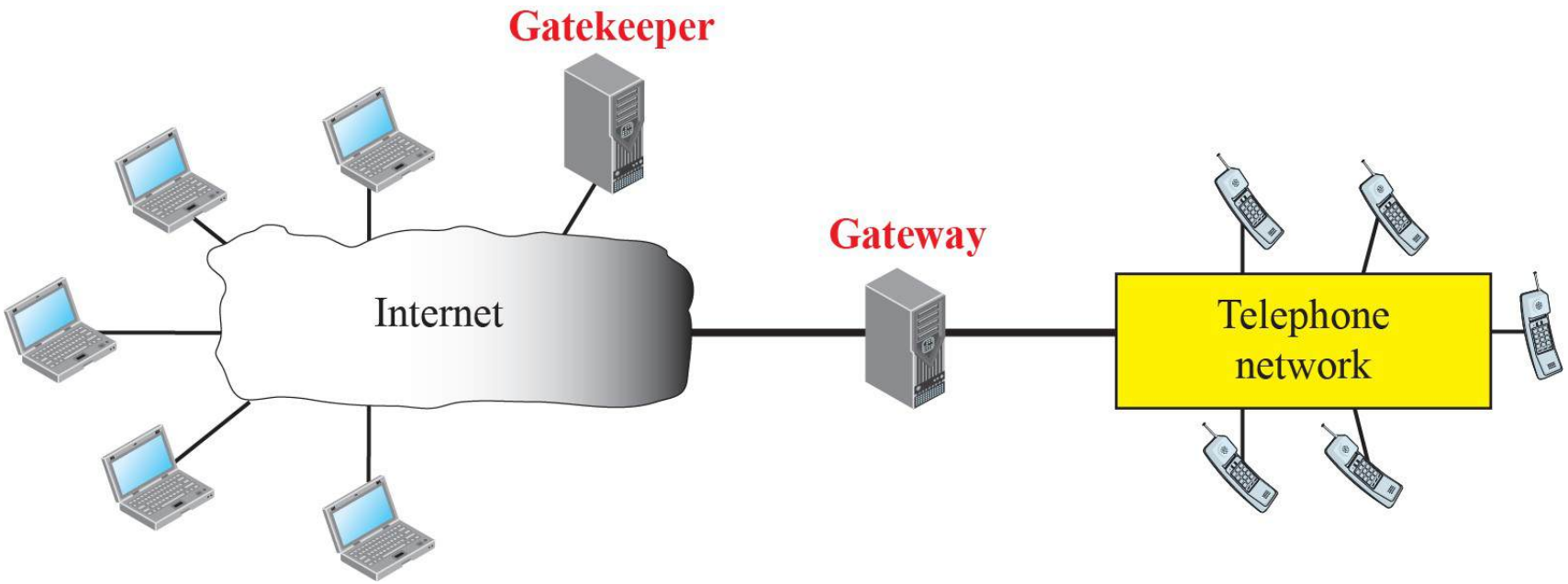

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Figure 8.43 : H.323 protocols

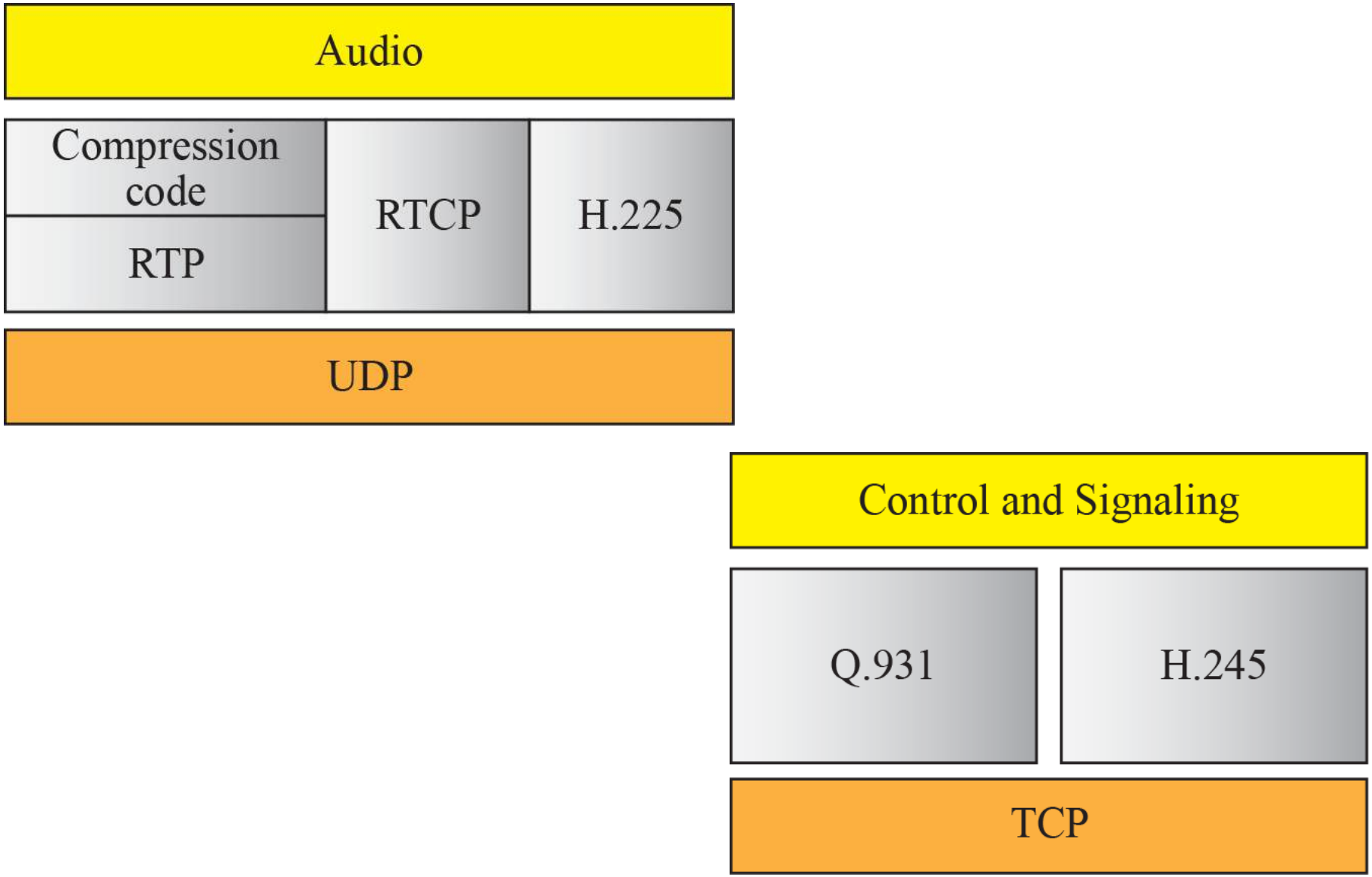

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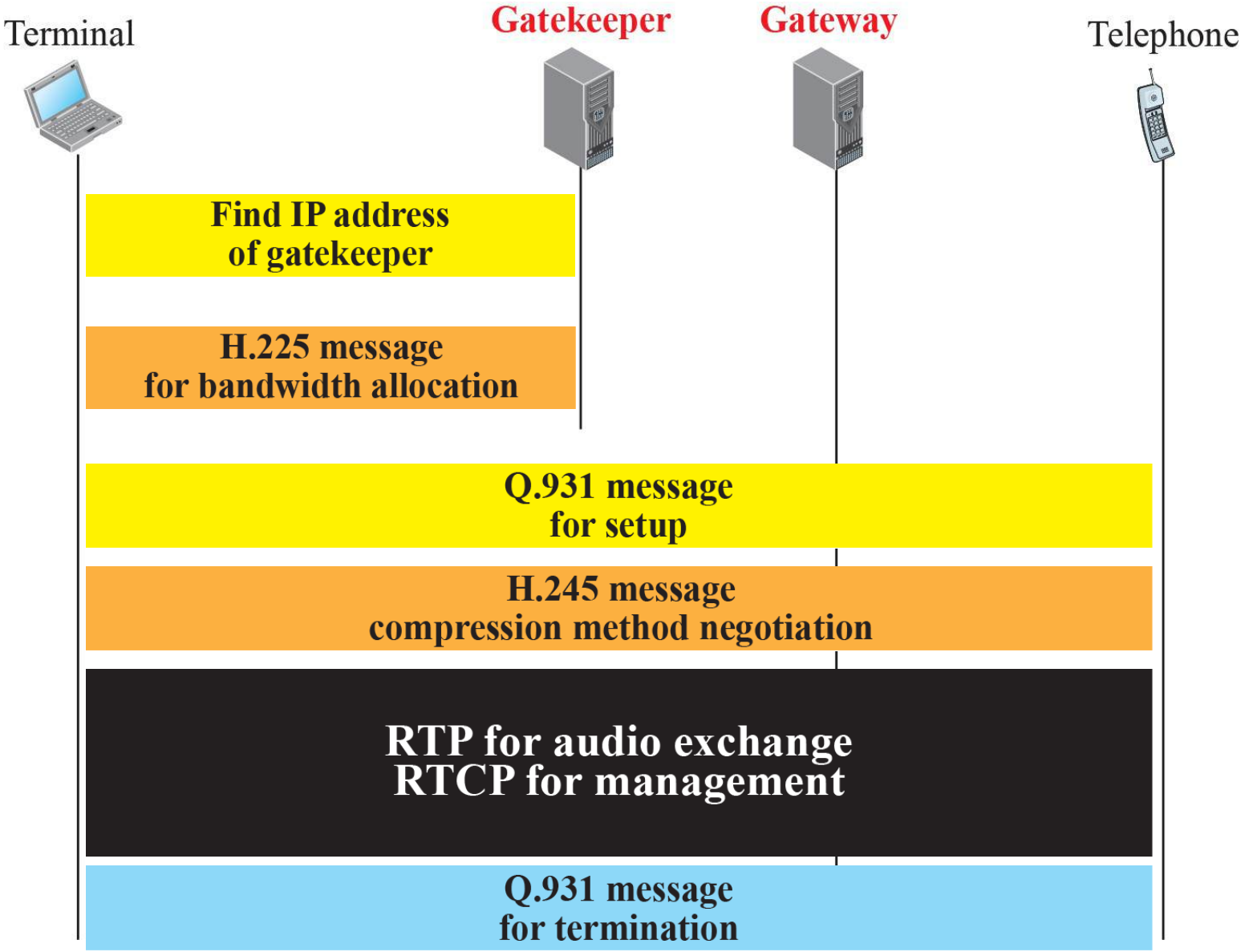
Figure 8.44 : H.323 example



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8.4.6 SCTP

Stream Control Transmission Protocol (SCTP) is a new transport-layer protocol designed to combine some features of UDP and TCP in an effort to create a better protocol for multimedia communication.

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8.4.6 (continued)

- **SCTP Services**
  - Process-to-Process Communication
  - Multiple Streams
  - Multihoming
  - Full-Duplex Communication
  - Connection-Oriented Service
  - Reliable Service
- **SCTP Features**
  - Transmission Sequence Number (TSN)
  - Stream Identifier (SI)
  - Stream Sequence Number (SSN)
  - Packets
  - Acknowledgment Number

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8.4.6 (continued)

- **Packet Format**
  - General Header
  - Chunks
- **An SCTP Association**
  - Association Establishment
  - Data Transfer
  - Association Termination
- **Flow Control**
  - Receiver Site
  - Sender Site

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8.4.6 (continued)

- **Error Control**
  - Receiver Site
  - Sender Site
  - Sending Data Chunks
  - Generating SACK Chunks
- **Congestion Control**

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Figure 8.45 : Multiple-stream concept

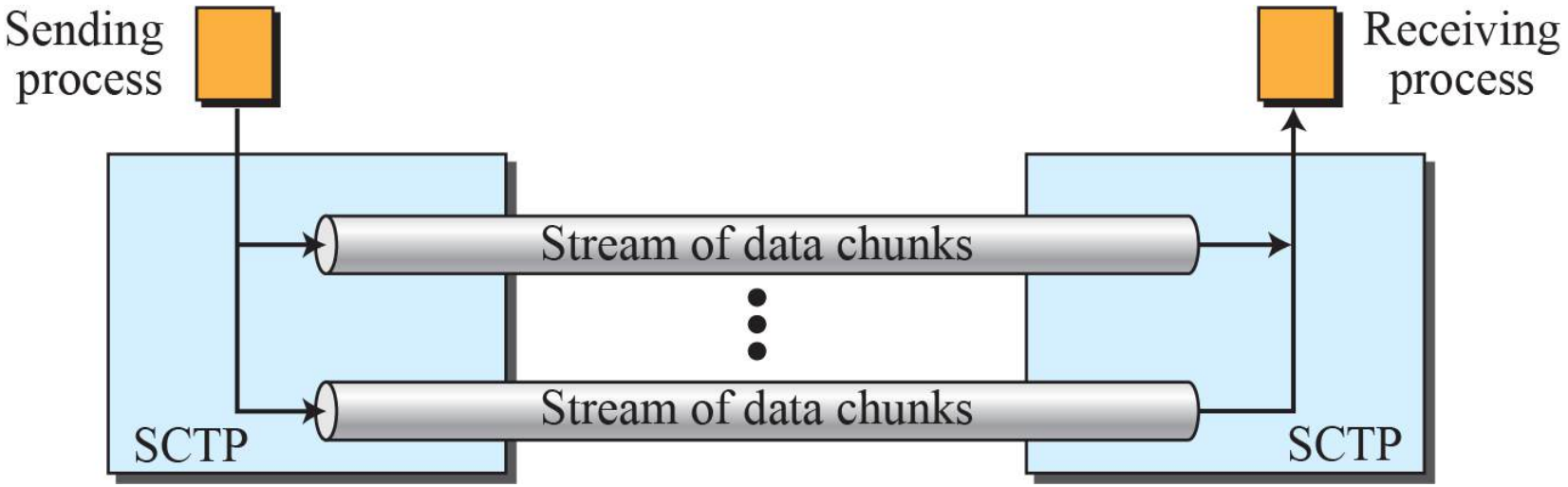

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Figure 8.46 : Multihoming concept

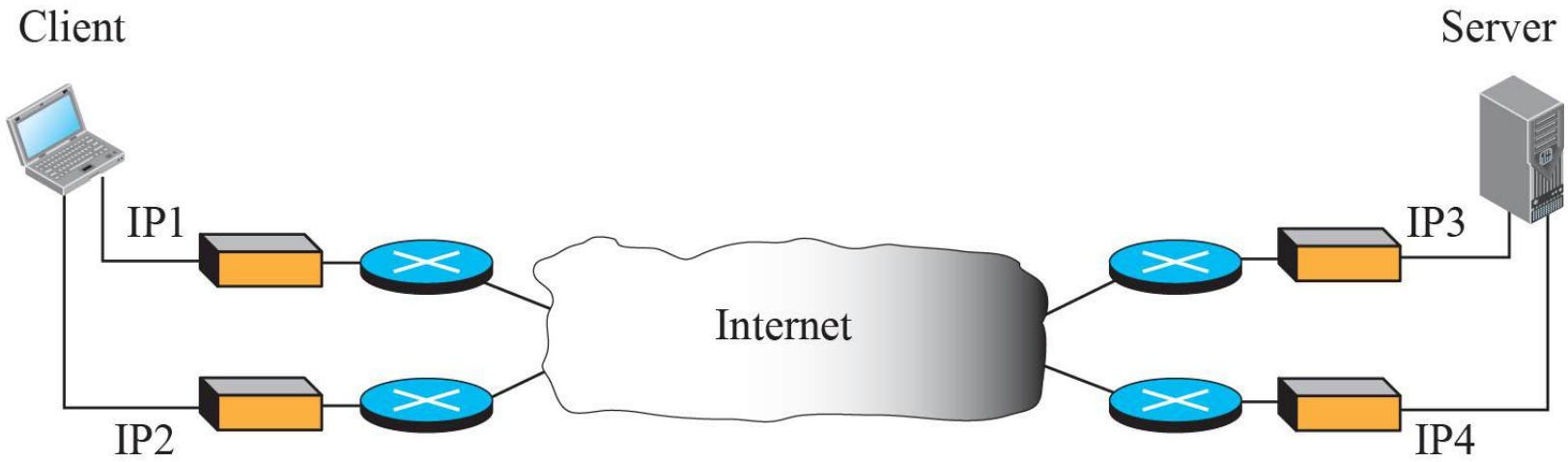

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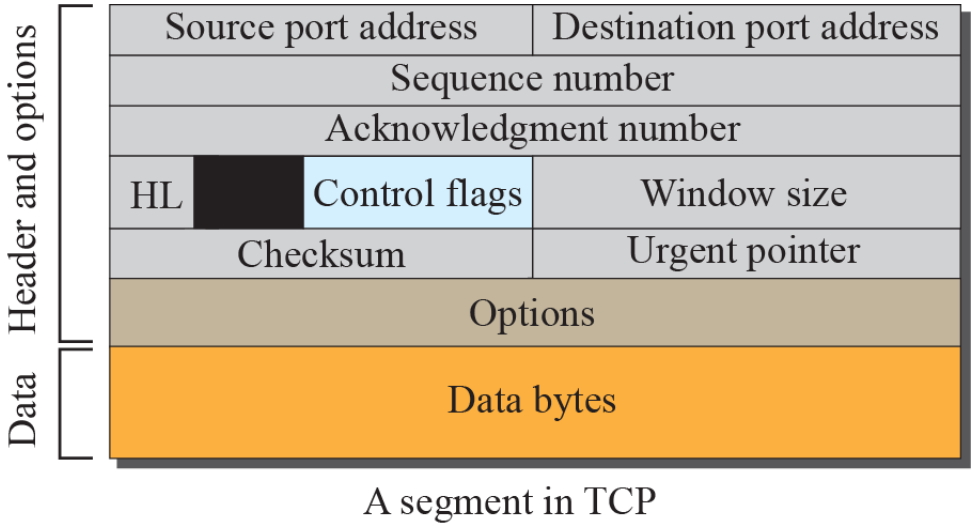
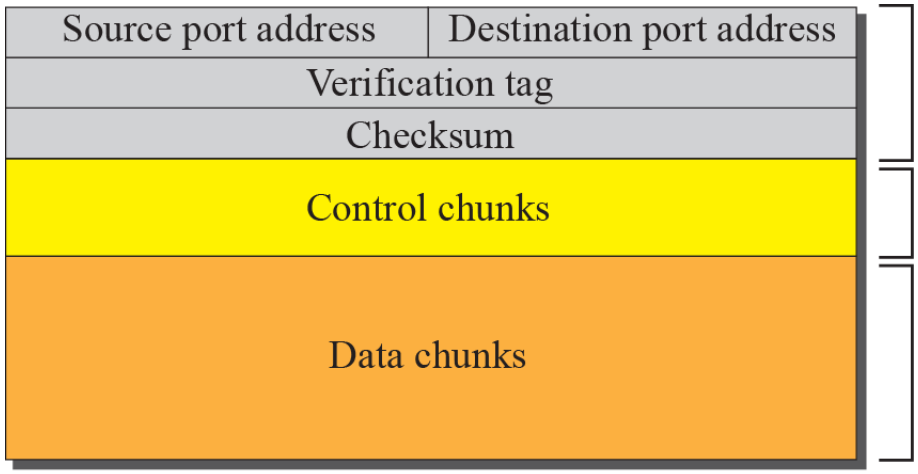
Figure 8.47 : Comparison between a TCP segment and an SCTP packet



A segment in TCP



A packet in SCTP

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Figure 8.48 : Packets, data chunks, and streams

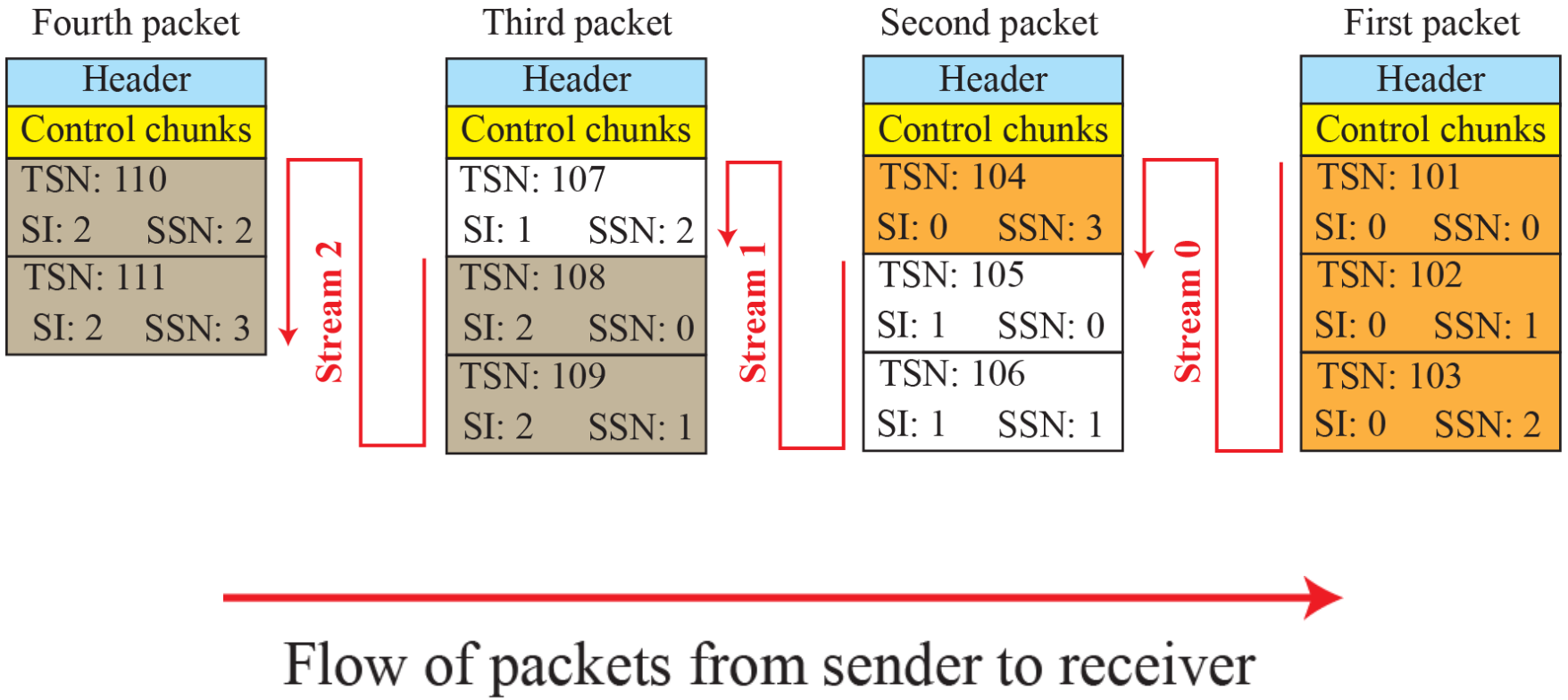

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Figure 8.49 : SCTP packet format

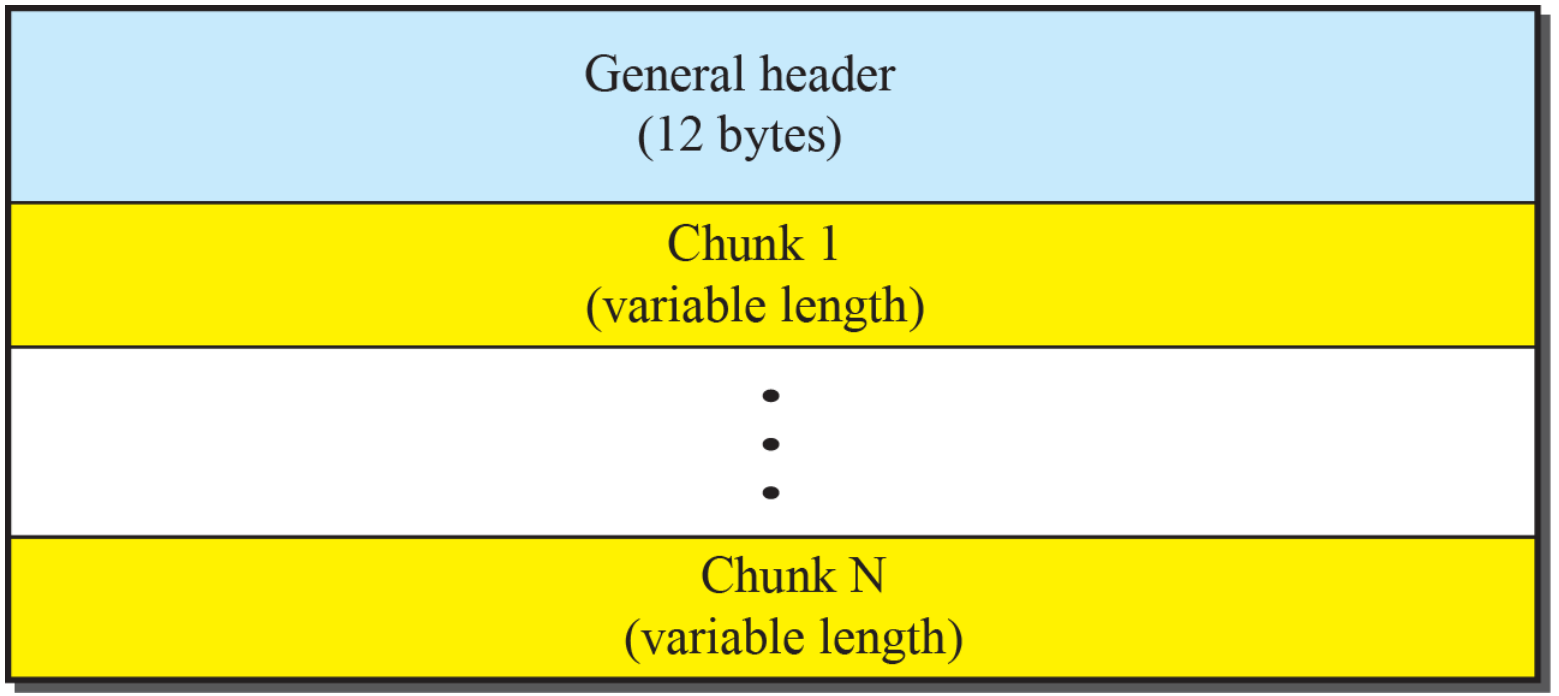

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Figure 8.50 : General header

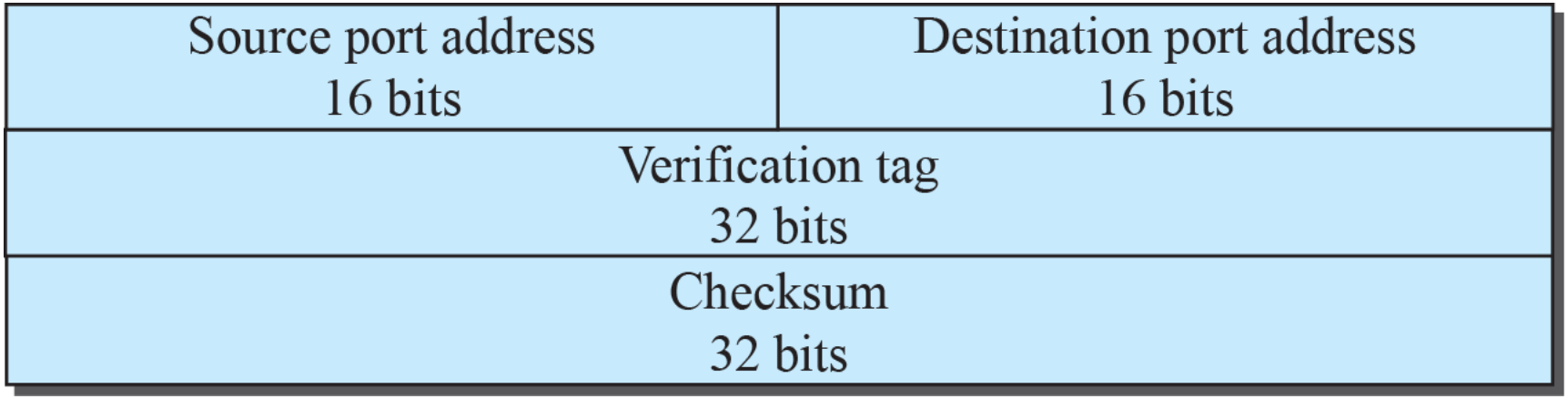

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Figure 8.51 : Common layout of a chunk

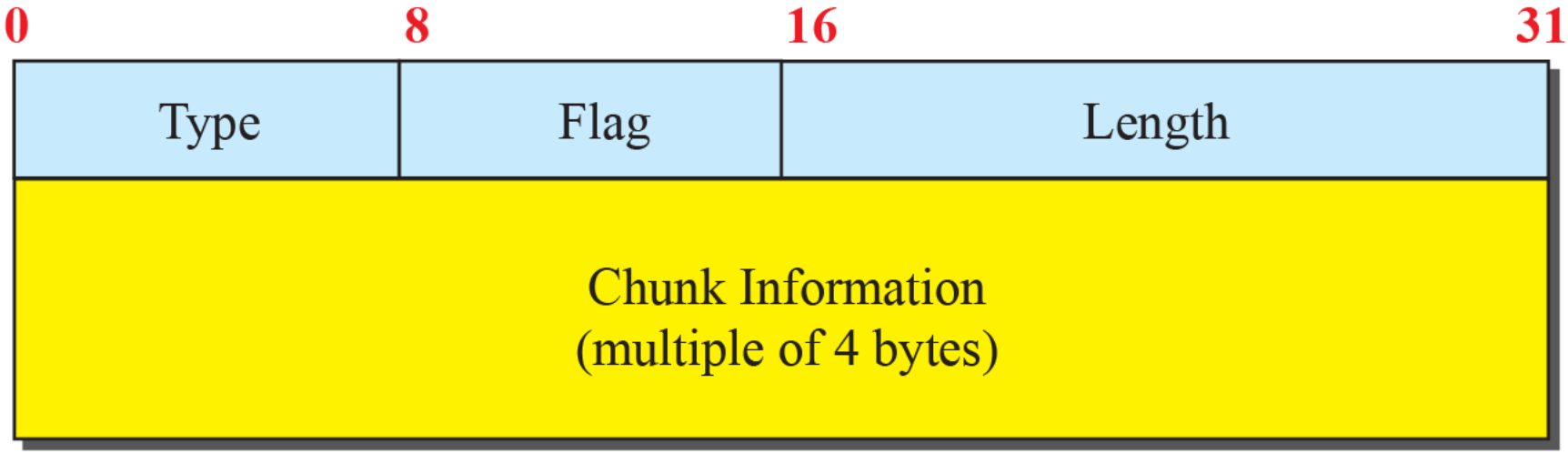

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Table 8.8: Chunks

Type	*Chunk*	*Description*
0	DATA	User data
1	INIT	Sets up an association
2	INIT ACK	Acknowledges INIT chunk
3	SACK	Selective acknowledgment
4	HEARTBEAT	Probes the peer for liveliness
5	HEARTBEAT ACK	Acknowledges HEARTBEAT chunk
6	ABORT	Aborts an association
7	SHUTDOWN	Terminates an association
8	SHUTDOWN ACK	Acknowledges SHUTDOWN chunk
9	ERROR	Reports errors without shutting down
10	COOKIE ECHO	Third packet in association establishment
11	COOKIE ACK	Acknowledges COOKIE ECHO chunk
14	SHUTDOWN COMPLETE	Third packet in association termination
192	FORWARD TSN	For adjusting cumulating TSN

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Figure 8.52 : Four-way handshaking

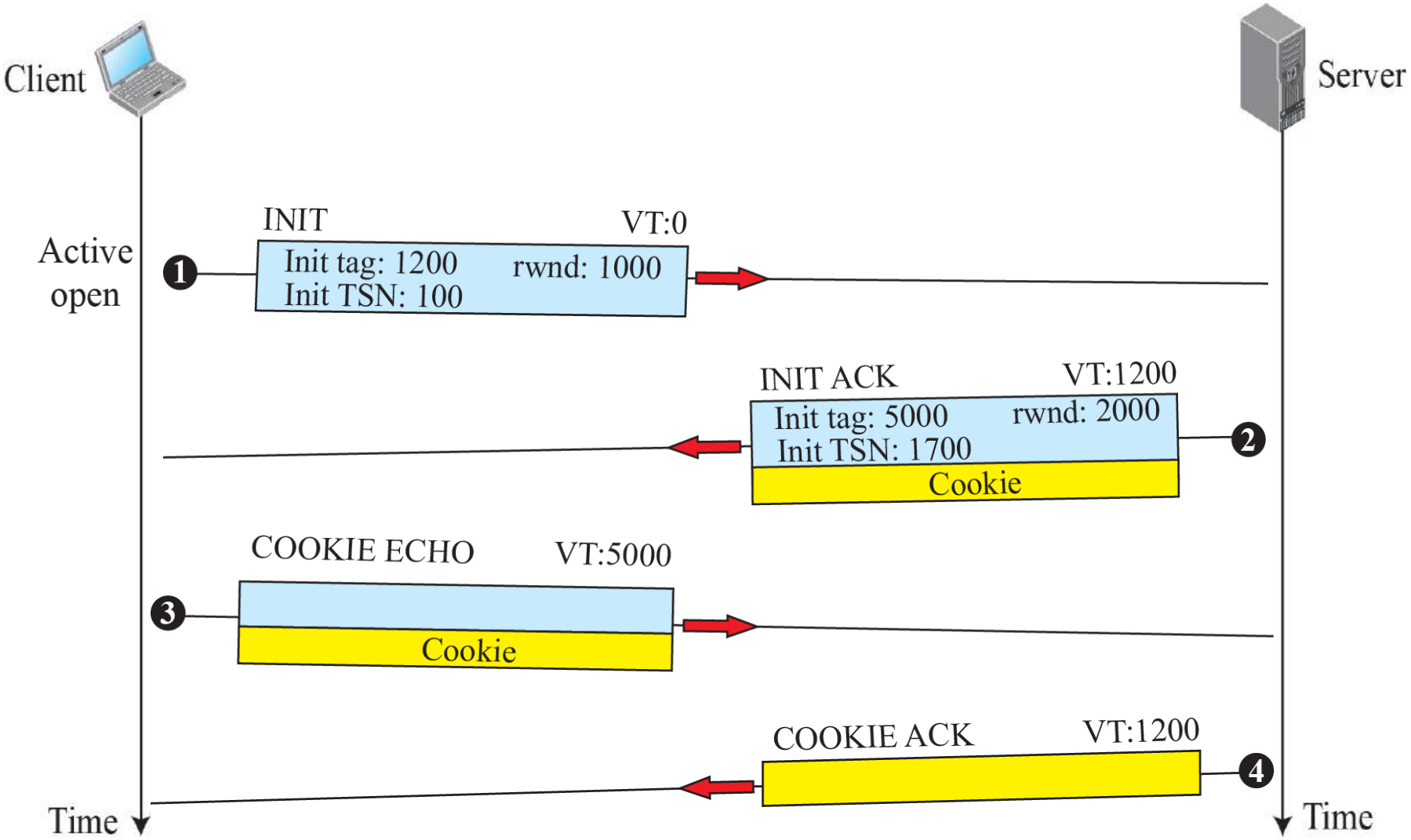

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Figure 8.53 : Association termination

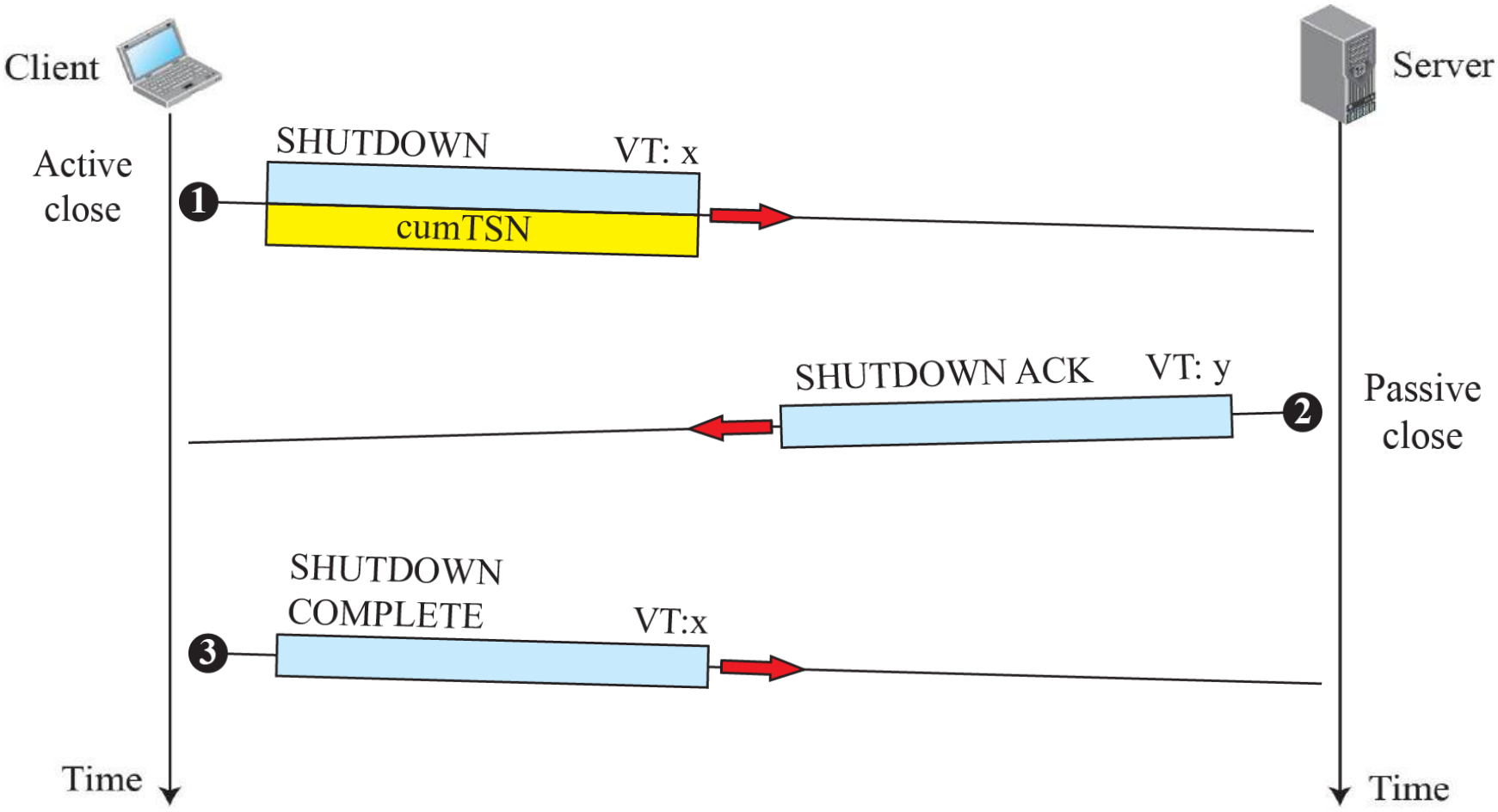

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Figure 8.54 : Flow control, receiver site

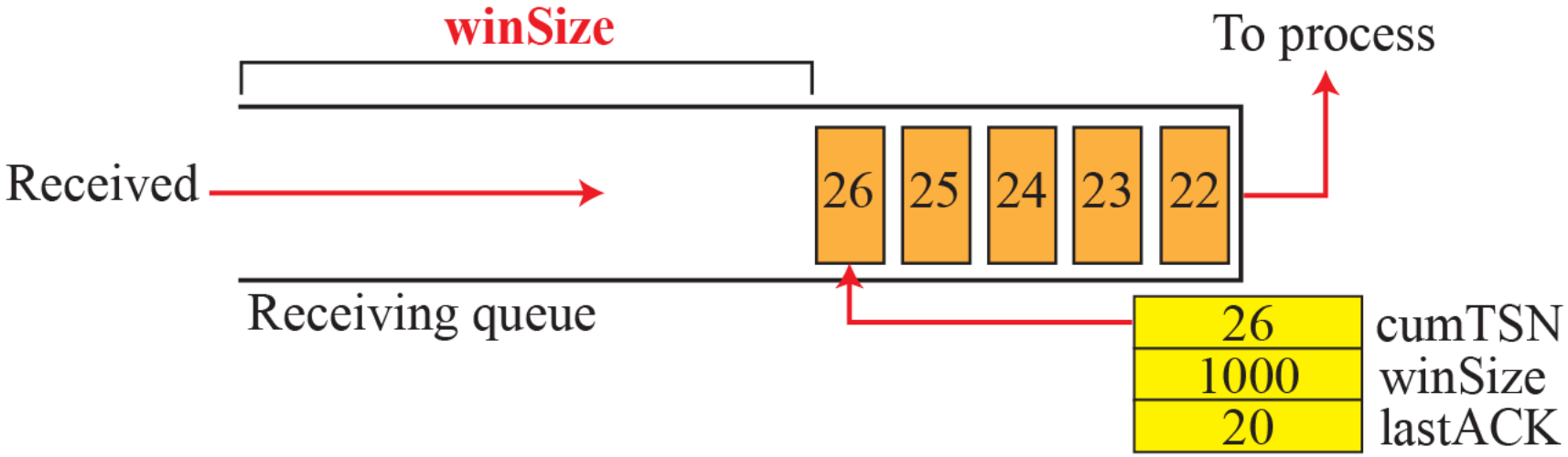

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Figure 8.55 : Flow control, sender site

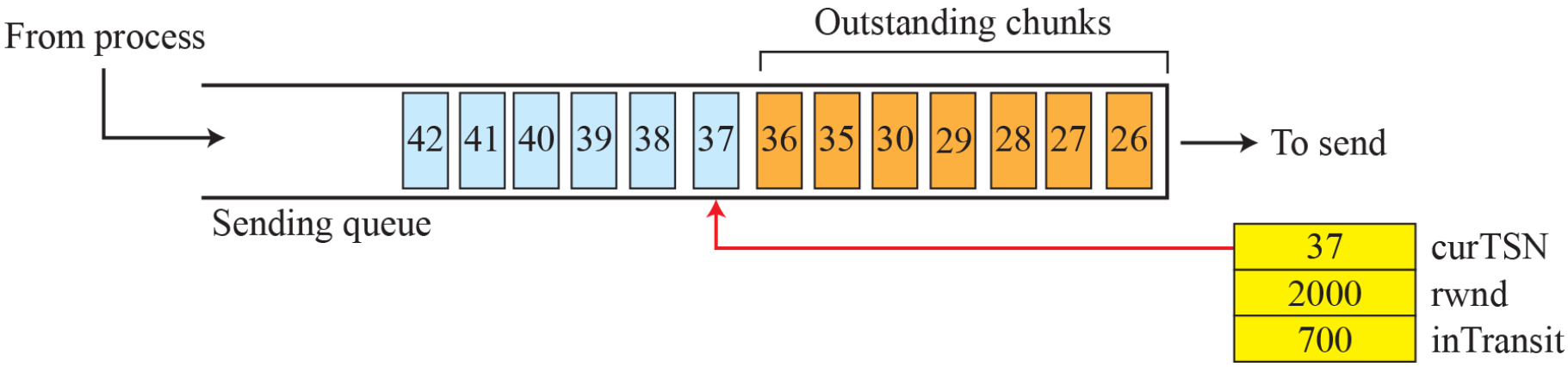

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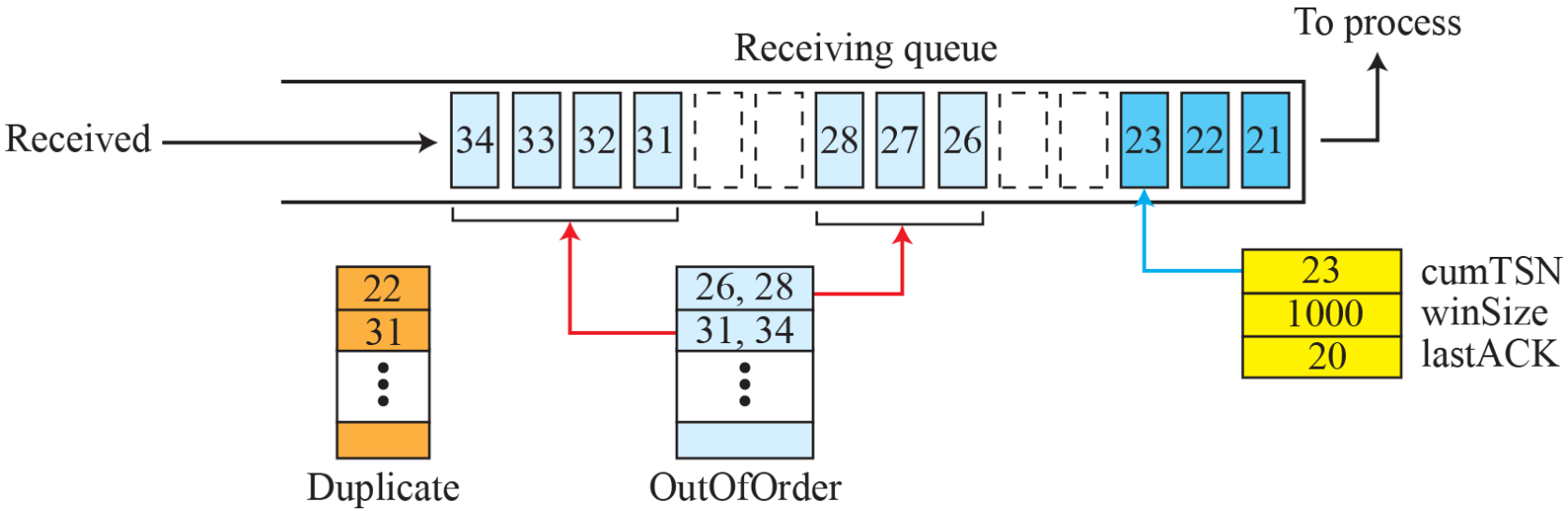
Figure 8.56 : Error control, receiver site



SACK chunk

Type: 3	Flag: 0	Length: 32
Cumulative TSN: 23		
Advertised receiver window credit: 1000		
Number of gap ACK blocks: 2	Number of duplicates: 2	
Gap ACK block #1 start: 3	Gap ACK block #1 end: 5	
Gap ACK block #2 start: 8	Gap ACK block #2 end: 11	
Duplicate TSN: 22		
Duplicate TSN: 31		

Numbers are relative to cumTSN

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Figure 8.57 : Error control, sender site

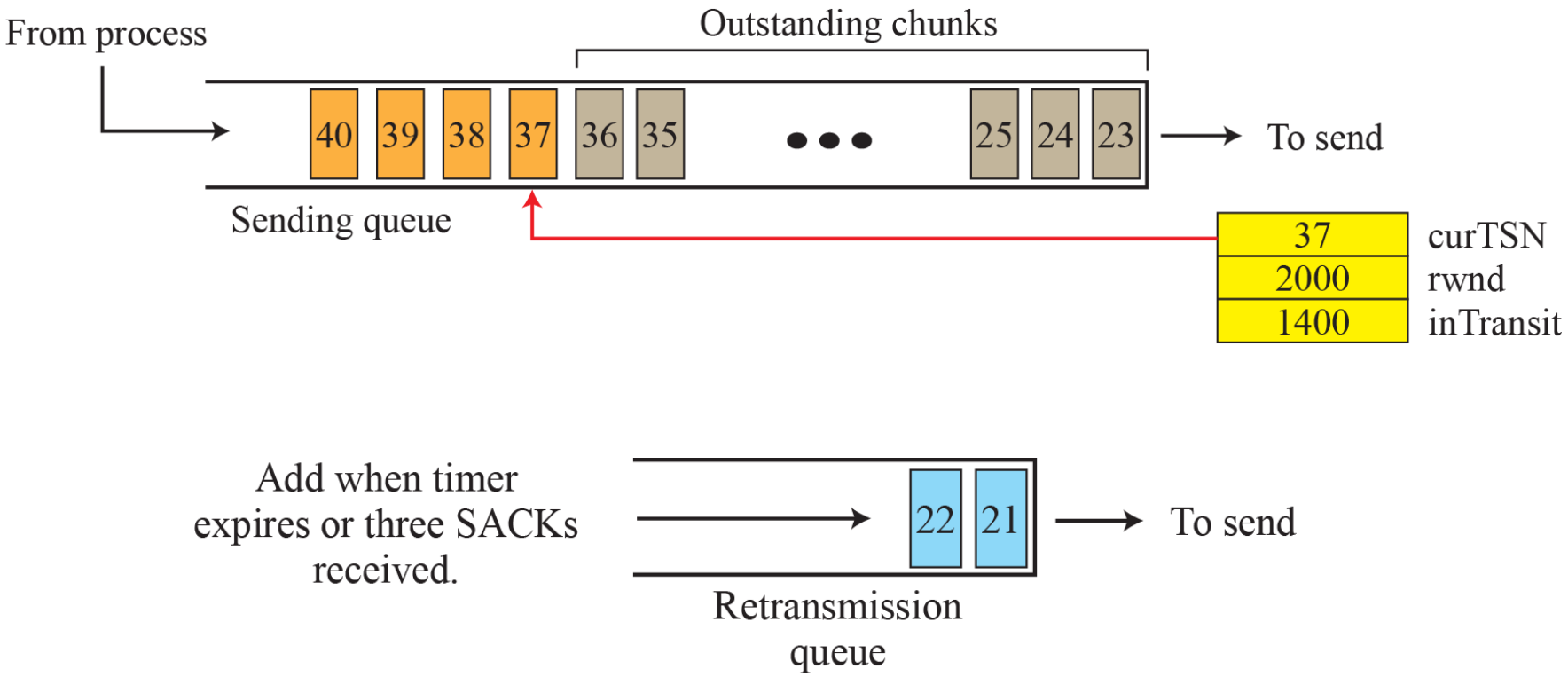

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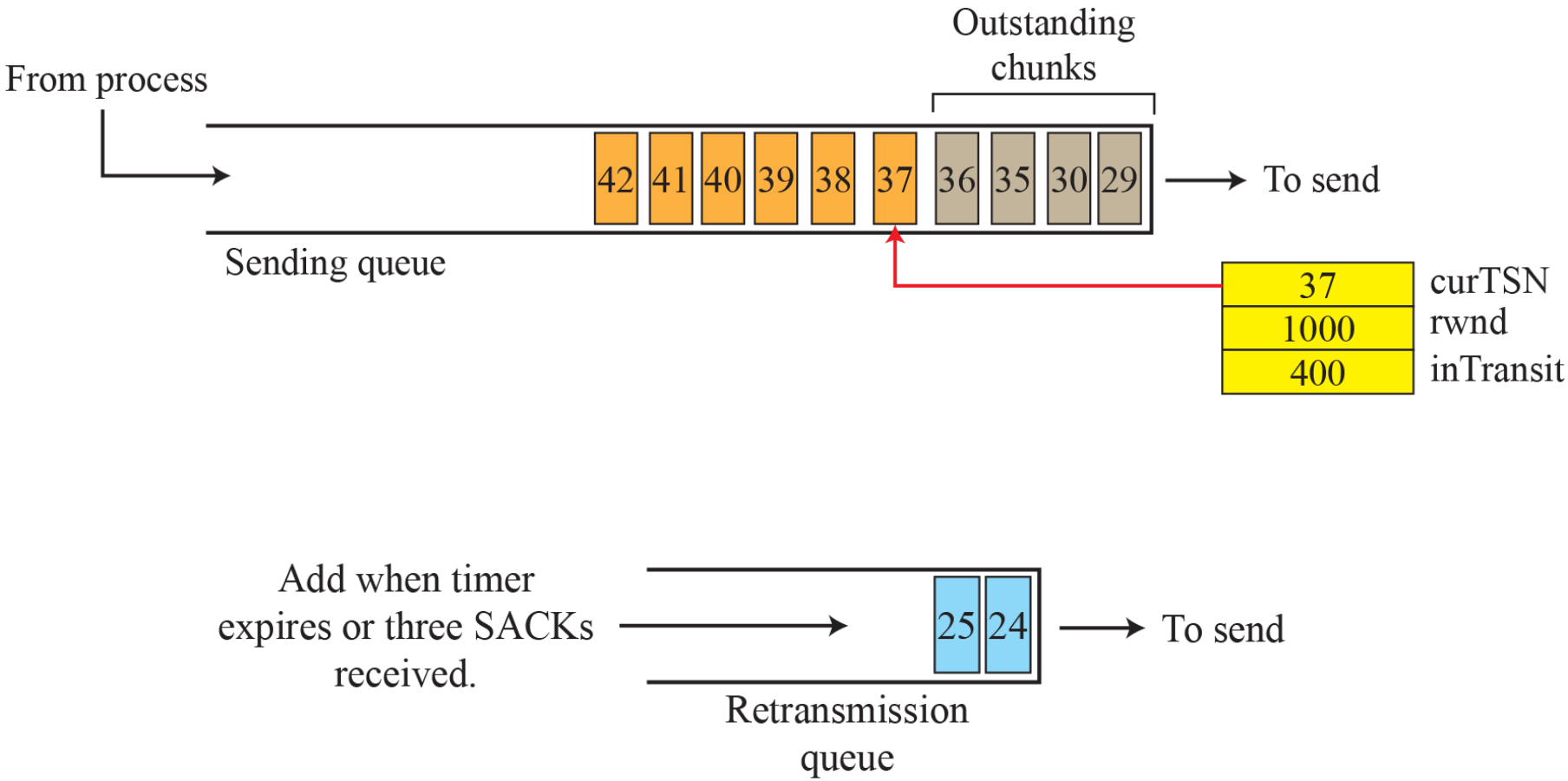
Figure 8.58 : New state at the sender site after receiving a SACK chunk



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8-5 QUALITY OF SERVICE

The Internet was originally designed for best-effort service with no guarantee of predictable performance. Quality of service is an internetworking issue that refers to a set of techniques and mechanisms that guarantees the performance of the network to deliver predictable service to an application program.

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8.5.1 Data-Flow Characterization

Traditionally, four types of characteristics are attributed to a flow: reliability, delay, jitter, and bandwidth. Let us first define these characteristics and then investigate the requirements of each application type.

- ***Definitions***
- ***Sensitivity of Applications***

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Table 8.9: Sensitivity of applications to flow characteristics

Application	*Reliability*	*Delay*	*Jitter*	*Bandwidth*
FTP	High	Low	Low	Medium
HTTP	High	Medium	Low	Medium
Audio-on-demand	Low	Low	High	Medium
Video-on-demand	Low	Low	High	High
Voice over IP	Low	High	High	Low
Video over IP	Low	High	High	High

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8.5.2 Flow Classes

Based on the flow characteristics, we can classify flows into groups, with each group having the required level of each characteristic. The Internet community has not yet defined such a classification formally. However, we know, for example, that a protocol like FTP needs a high level of reliability and probably a medium level of bandwidth, but the level of delay and jitter is not important for this protocol.

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Example 8.11

Although the Internet has not defined flow classes formally, the ATM protocol does. As per ATM specifications, there are five classes of defined service.

a. **Constant Bit Rate (CBR).**

b. **Variable Bit Rate-Non Real Time (VBR-NRT).**

c. **Variable Bit Rate-Real Time (VBR-RT).**

d. **Available Bit Rate (ABR).**

e. **Unspecified Bit Rate (UBR).**

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8.5.3 Flow Control to Improve QoS

Although formal classes of flow are not defined in the Internet, an IP datagram has a ToS field that can informally define the type of service required for a set of datagrams sent by an application. If we assign a certain type of application a single level of required service, we can then define some provisions for those levels of service. These can be done using several mechanisms.

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8.5.3 (continued)

- **Scheduling**
  - FIFO Queuing
  - Priority Queuing
  - Weighted Fair Queuing
- **Traffic Shaping or Policing**
  - Leaky Bucket
  - Token Bucket
- **Resource Reservation**
- **Admission Control**

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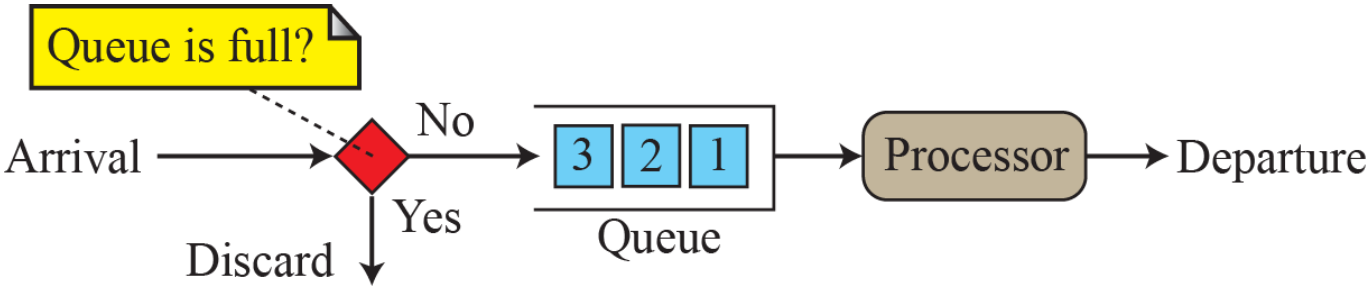
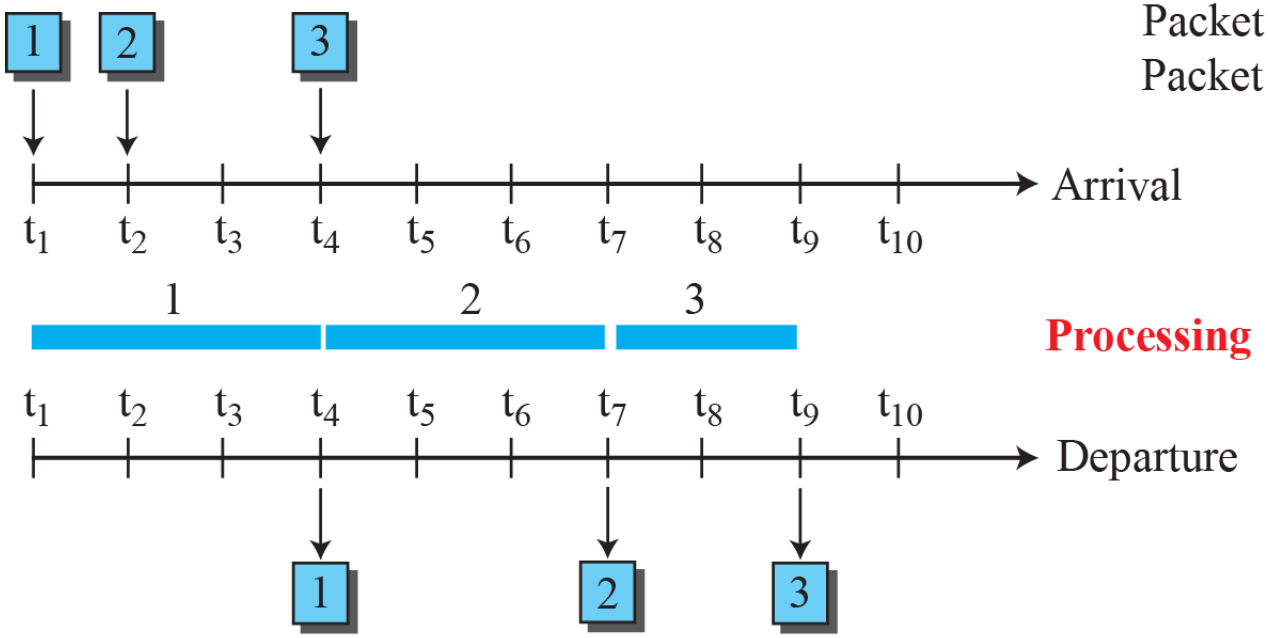
Figure 8.59 : FIFO queue



a. Processing in the router

Required processing time
Packet 1: three time units
Packet 2: three time units
Packet 3: two time units



b. Arrival and departure time

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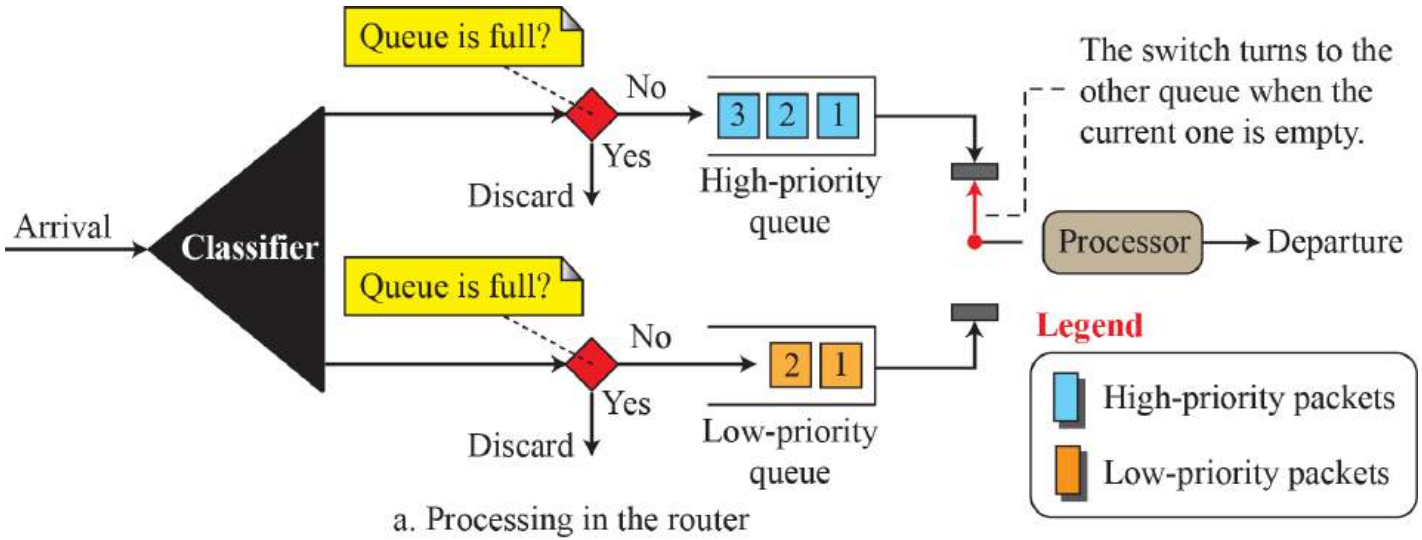
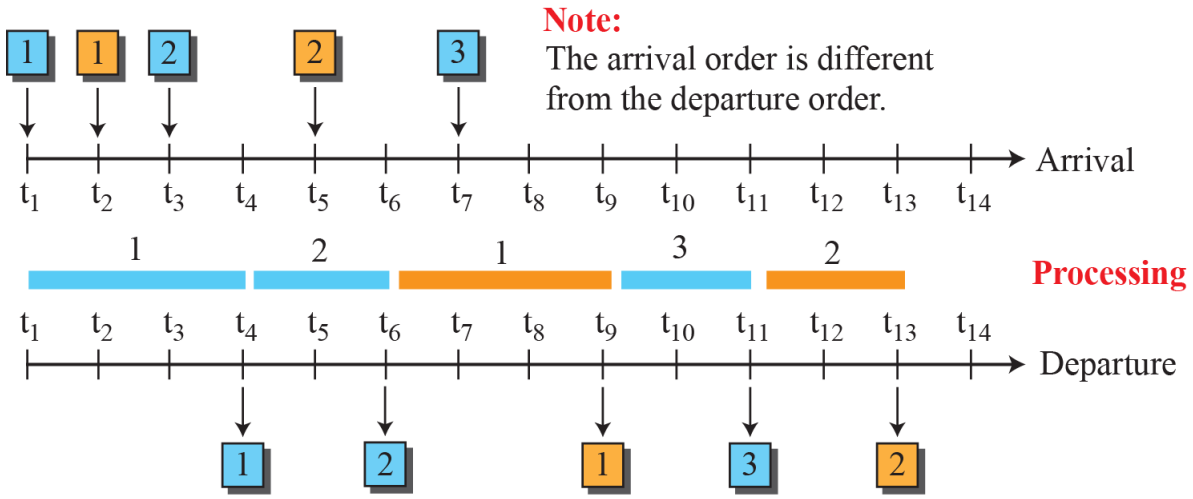
Figure 8.60 : Priority queuing



a. Processing in the router



b. Arrival and departure time

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Figure 8.61 : Weighted fair queuing

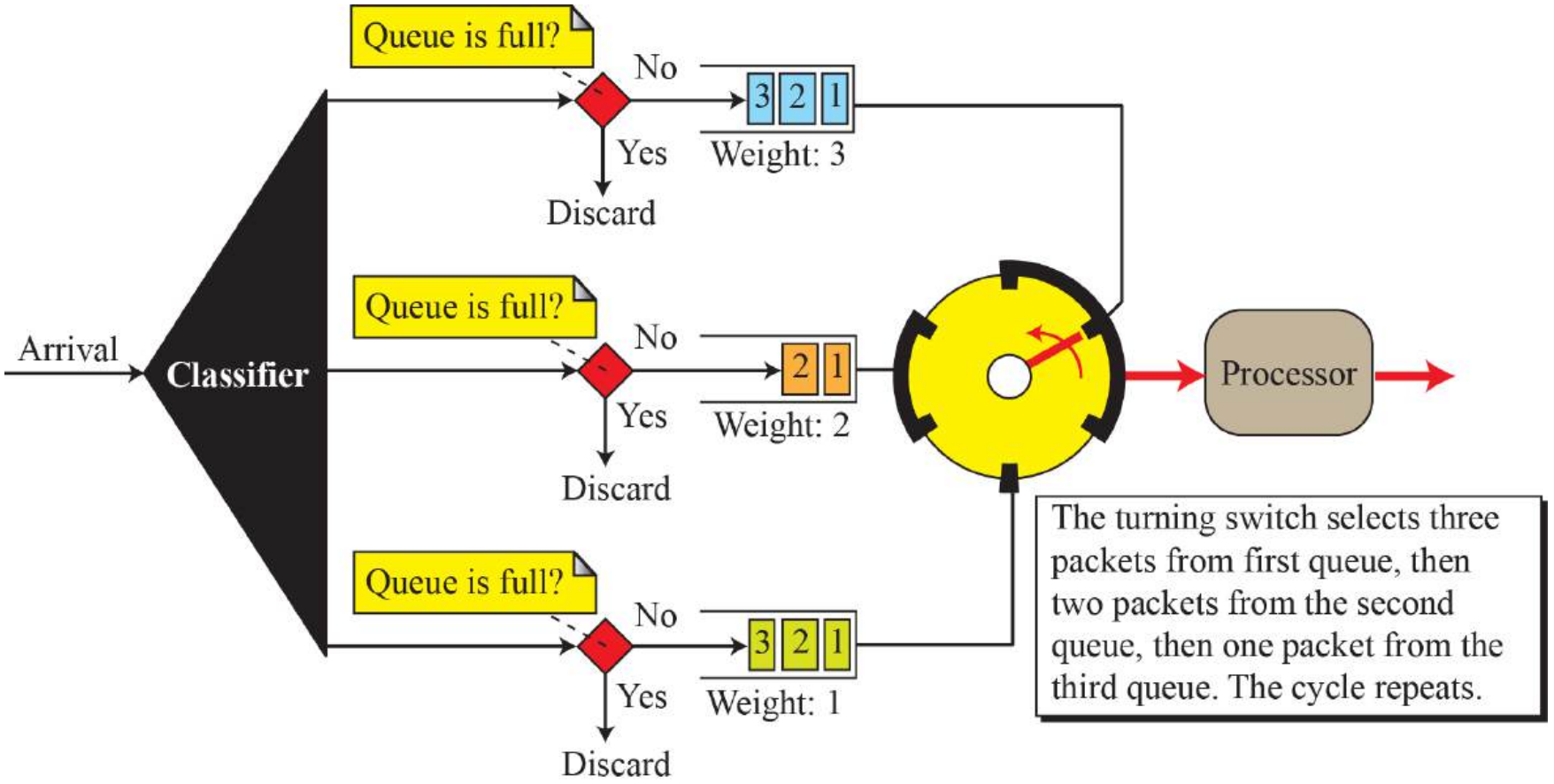

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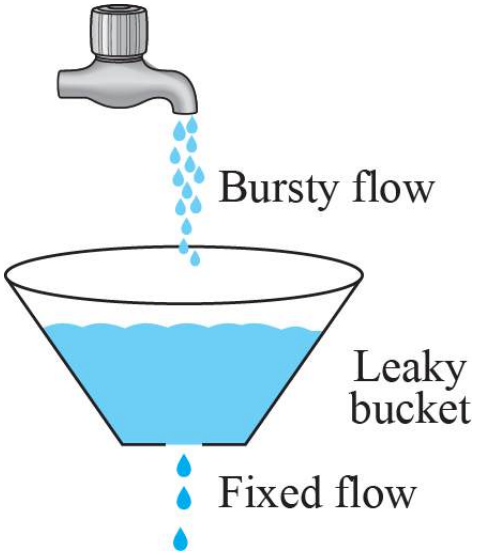
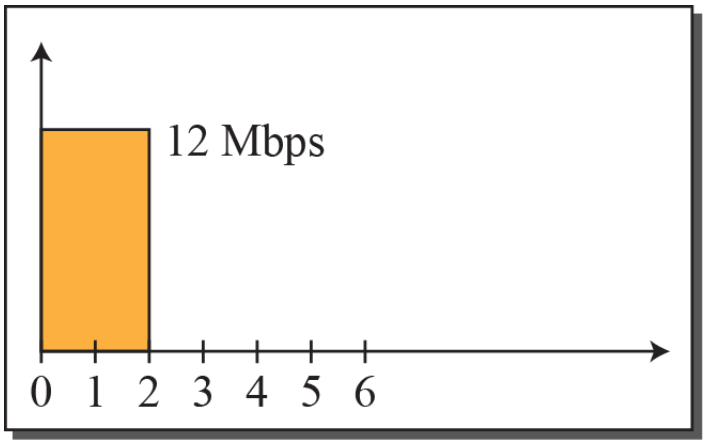
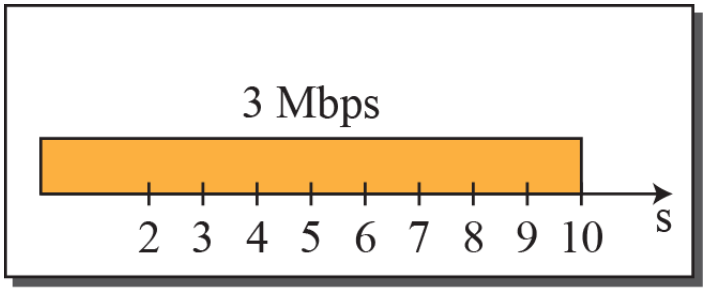
Figure 8.62 : Leaky bucket





Bursty data



Fixed-rate data

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Figure 8.63 : Leaky bucket implementation

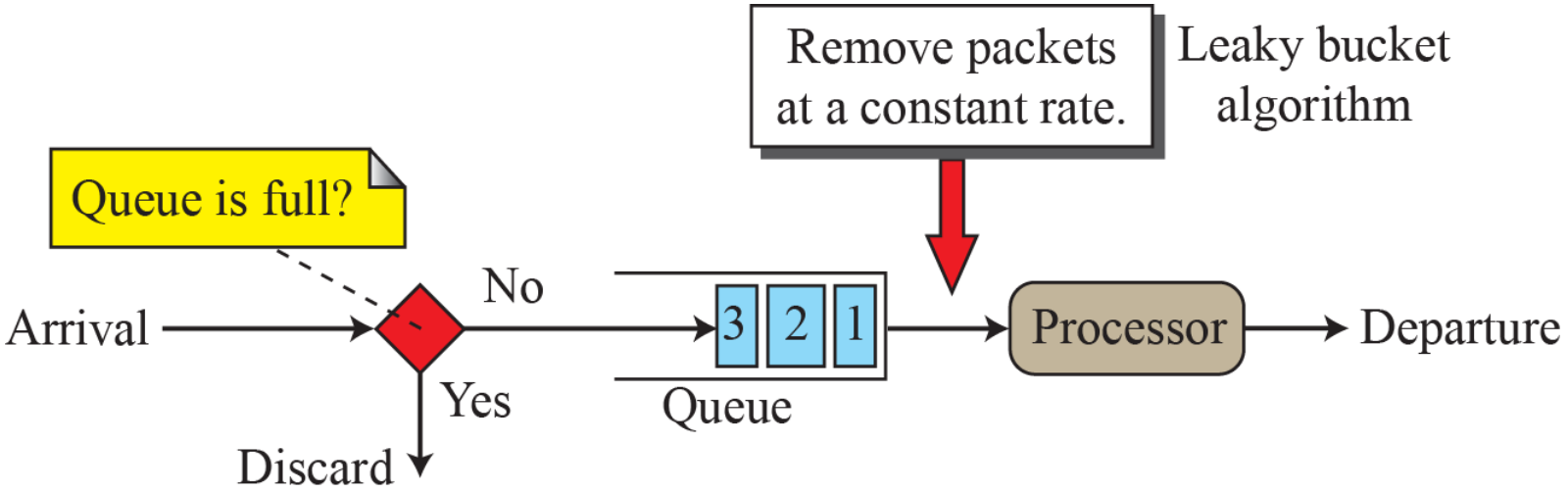

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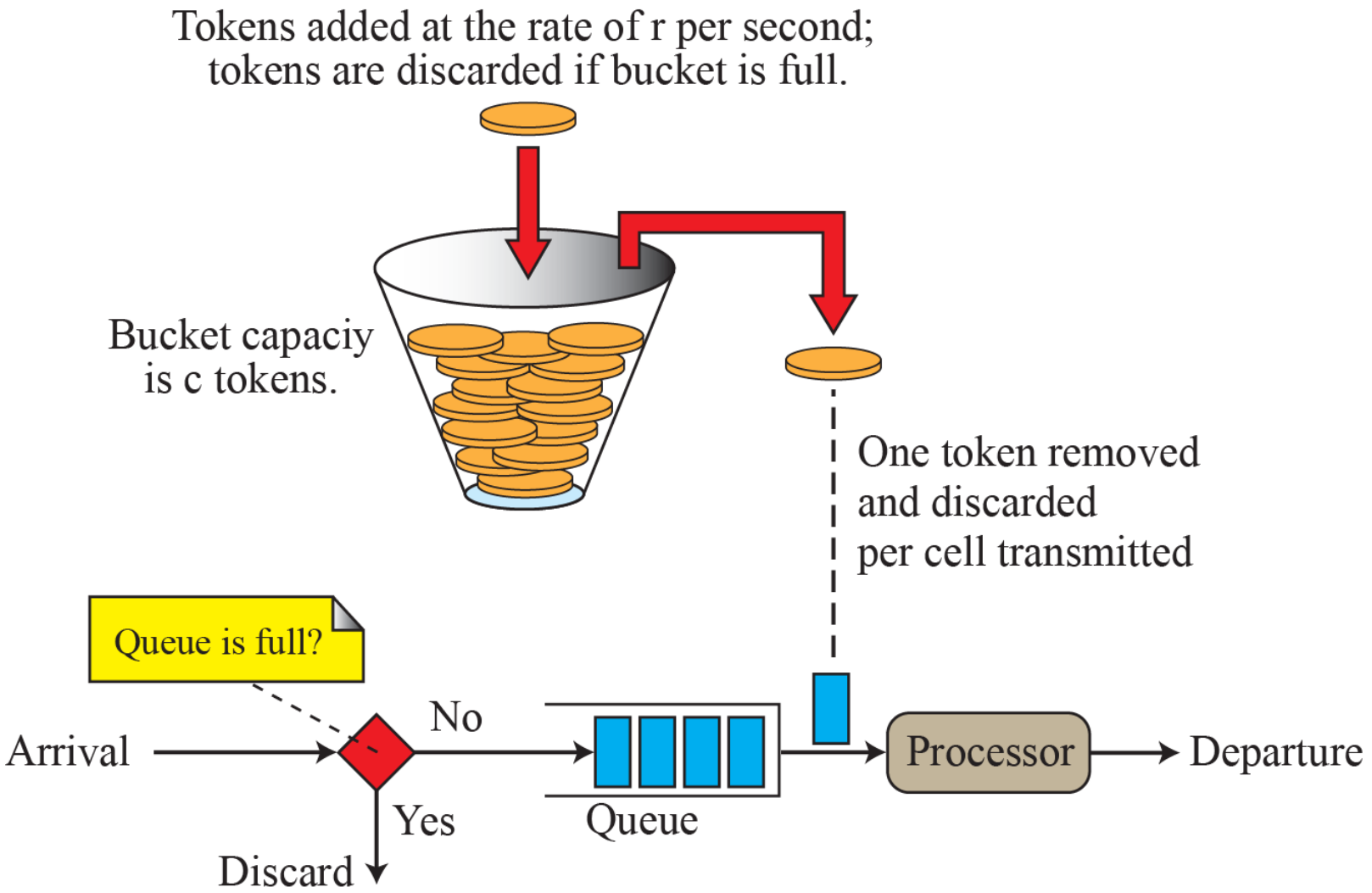
Figure 8.64 : Token bucket



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Example 8.12

Let assume that the bucket capacity is 10,000 tokens and tokens are added at the rate of 1000 tokens per second. If the system is idle for 10 seconds (or more), the bucket collects 10,000 tokens and becomes full. Any additional tokens will be discarded. The maximum average rate is shown below.

$$\text{Maximum average rate} = (1000t + 10{,}000)/t$$

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8.5.4 Integrated Services (IntServ)

Traditional Internet provided only the best-effort delivery service to all users regardless of what was needed. Some applications, however, needed a minimum amount of bandwidth to function. To provide different QoS for different applications, IETF developed the integrated services (IntServ) model. In this model, which is a flow-based architecture, resources such as bandwidth are explicitly reserved for a given data flow.

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8.5.4 (continued)

- **Flow Specification**
- **Admission**
- **Service Classes**
  - Guaranteed Service Class
  - Controlled-Load Service Class
- **Resource Reservation Protocol (RSVP)**
  - Multicast Trees
  - Receiver-Based Reservation
  - RSVP Messages

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8.5.4 (continued)

- **Problems with Integrated Services**
  - Scalability
  - Service-Type Limitation

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Figure 8.65 : Path messages

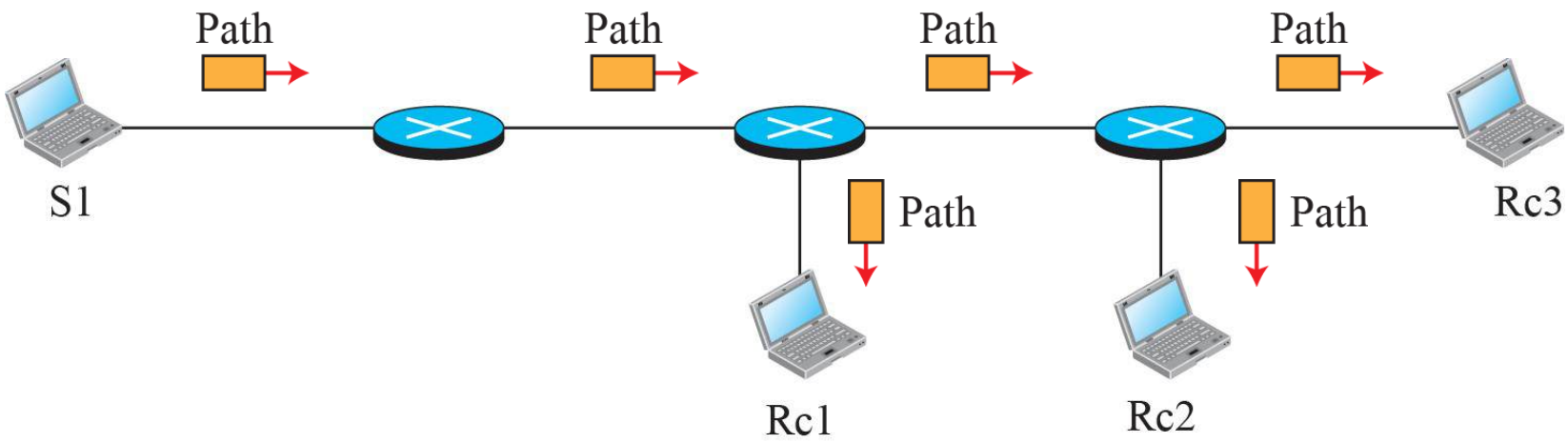

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Figure 8.66 : Resv messages

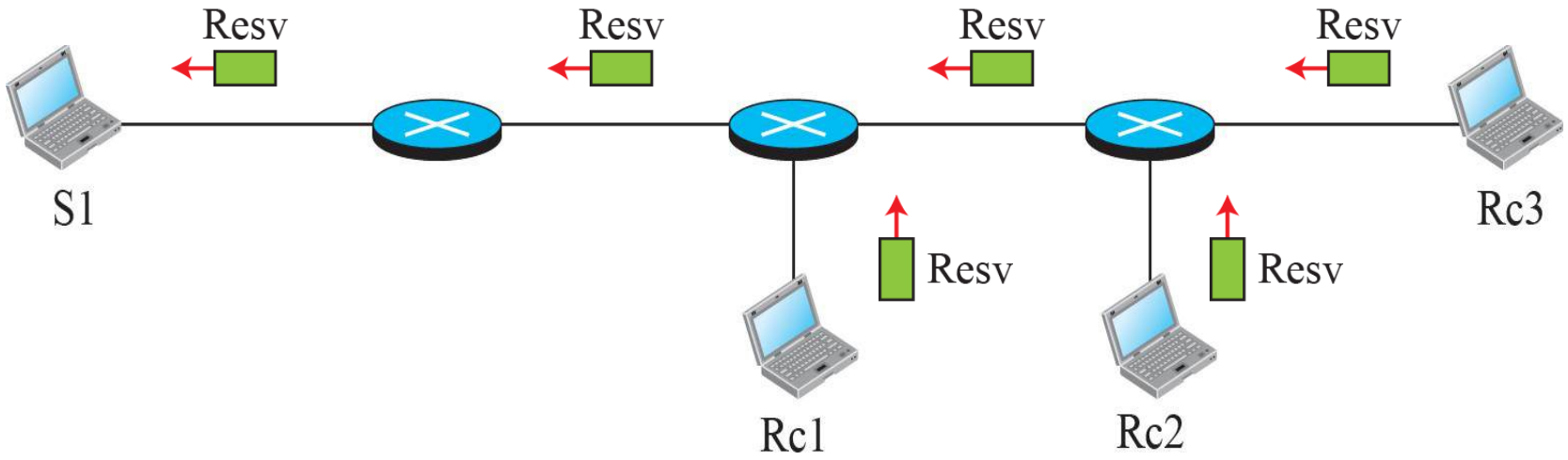

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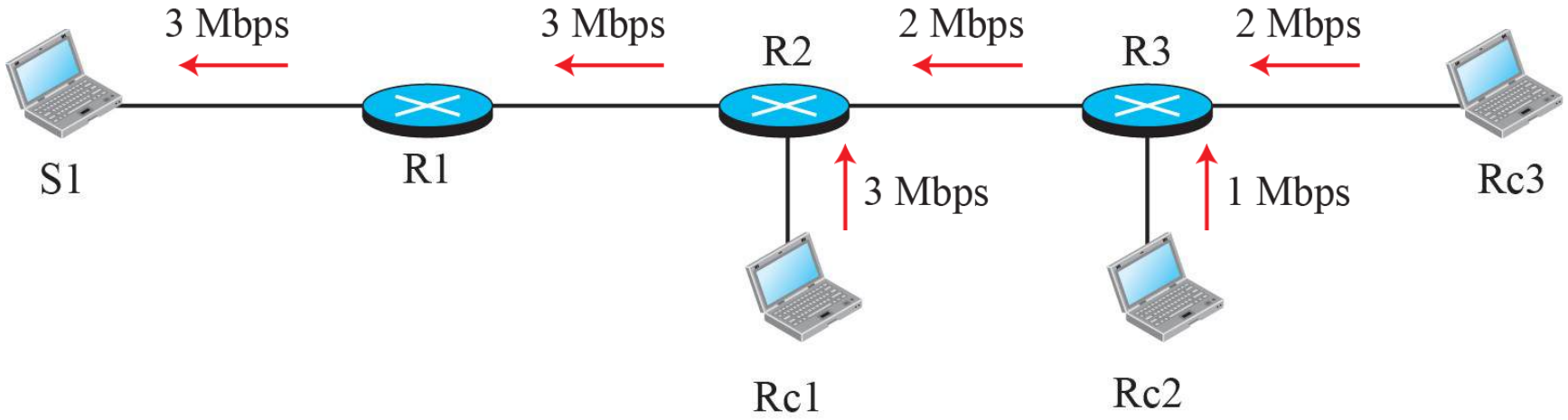
Figure 8.67 : Reservation merging



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8.5.5 Differentiated Services (DiffServ)

In this model, also called DiffServ, packets are marked by applications into classes according to their priorities. Routers and switches, using various queuing strategies, route the packets. This model was introduced by the IETF to handle the shortcomings of Integrated Services.

- **DS Field**
- **Per-Hop Behavior**
- **Traffic Conditioner**

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Figure 8.68 : DS field

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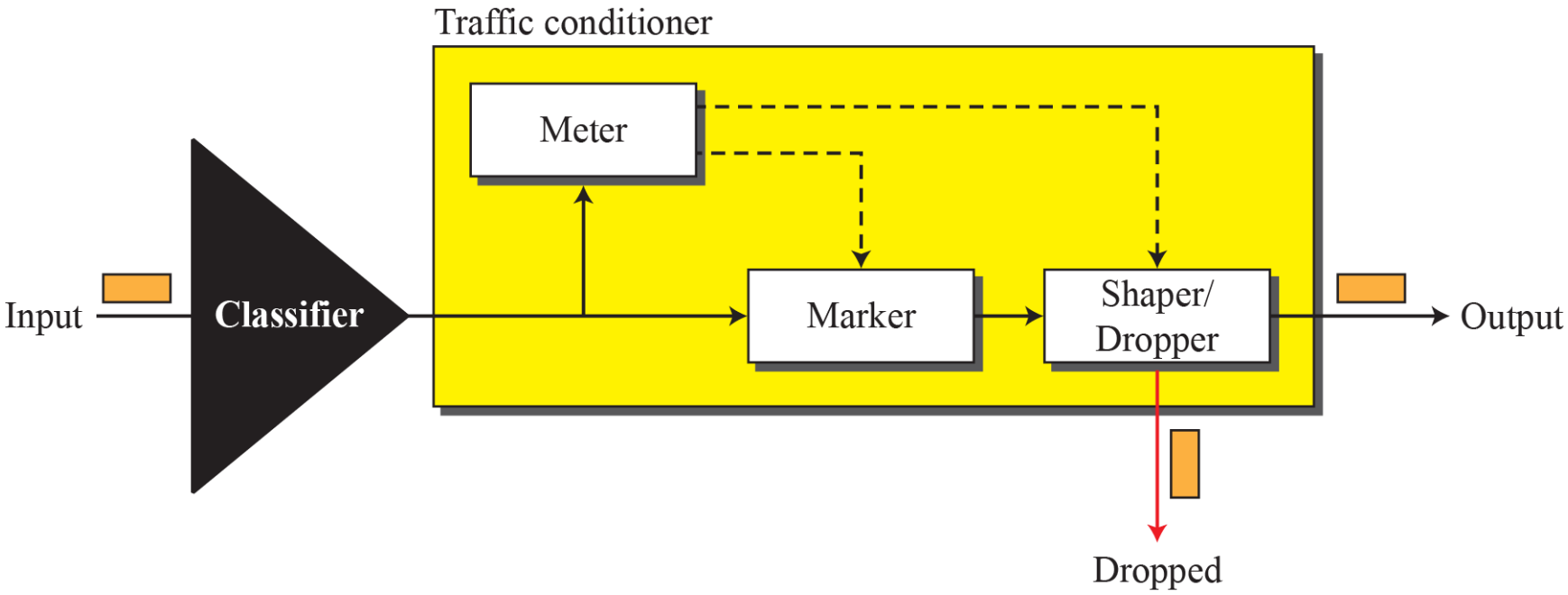
Figure 8.69 : Traffic conditioner



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Chapter 8: Summary

- *We divided compression into two broad categories: lossless and lossy compression. In lossless compression, the integrity of the data is preserved because compression and decompression algorithms are exact inverses of each other: no part of the data is lost in the process. Lossy compression cannot preserve the accuracy of data, but we gain the benefit of reducing the size of the compressed data.*

- *Audio/video files can be downloaded for future use (streaming stored audio/video) or broadcast to clients over the Internet (streaming live audio/video). The Internet can also be used for live audio/video interaction. Audio and video need to be digitized before being sent over the Internet. We can use a web server, or a web server with a metafile, or a media server, or a media server and RTSP to download a streaming audio/ video file.*

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Chapter 8: Summary (continued)

- *Real-time data on a packet-switched network requires the preservation of the time relationship between packets of a session. Gaps between consecutive packets at the receiver cause a phenomenon called jitter. Jitter can be controlled through the use of timestamps and a judicious choice of the playback time.*

- *Voice over IP is a real-time interactive audio/video application. The Session Initiation Protocol (SIP) is an application-layer protocol that establishes, manages, and terminates multimedia sessions. H.323 is an ITU standard that allows a telephone connected to a public telephone network to talk to a computer connected to the Internet.*

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Chapter 8: Summary (continued)

- *Real-time multimedia traffic requires both UDP and Real-Time Transport Protocol (RTP). RTP handles time-stamping, sequencing, and mixing. Real-Time Transport Control Protocol (RTCP) provides flow control, quality of data control, and feedback to the sources.*

- *Scheduling, traffic shaping, resource reservation, and admission control are techniques to improve quality of service (QoS). FIFO queuing, priority queuing, and weighted fair queuing are scheduling techniques. Leaky bucket and token bucket are traffic shaping techniques. Integrated Services is a flow-based QoS model designed for IP. The Resource Reservation Protocol (RSVP) is a signaling protocol that helps IP create a flow and makes a resource reservation. Differential Services is a class-based QoS model designed for IP.*

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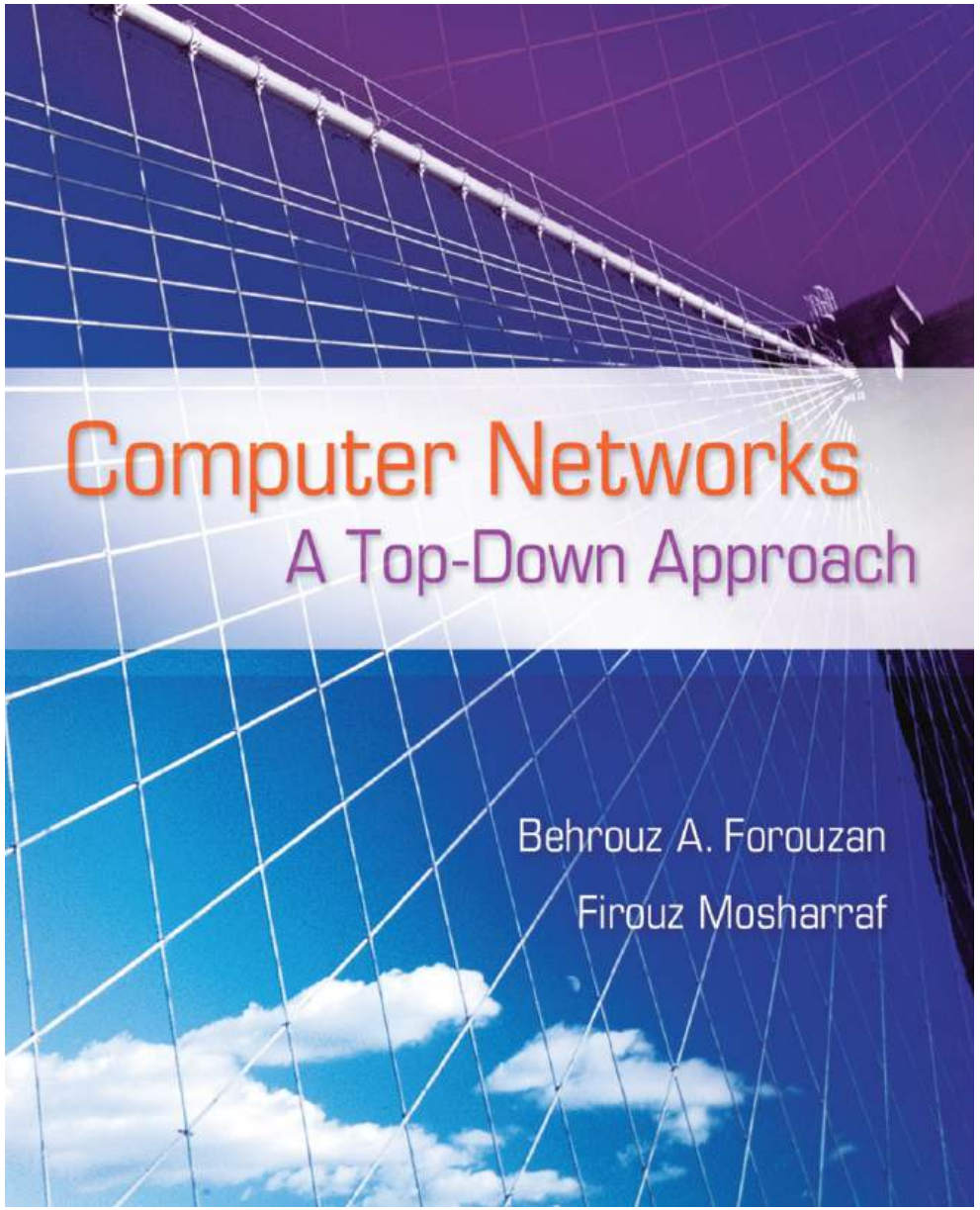
Chapter 9

Network Management



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Chapter 9: Outline

9.1 Introduction

9.2 SNMP

9.3 ASN.1

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Chapter 9: Objective

- *We introduce the concept of network management and discuss five general areas of network management: configuration, fault, performance, security, and accounting.*

- *We discuss Simple Network Management Protocol (SNMP) as a framework for managing devices in an internet using the TCP/IP protocol suite and show how a manager as a host runs an SNMP client and any agents as a router or host runs a server program.*

- *We give a brief discussion of a standard that provides the methods and rules to define data and objects. This section is very brief and only introduces the subject. Part of it is used by SMI in the second section.*

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9-1 INTRODUCTION

We can define network management as monitoring, testing, configuring, and troubleshooting network components to meet a set of requirements defined by an organization. These include the smooth, efficient operation of the network that provides the predefined quality of service for users. To accomplish this task, a network management system uses hardware, software, and humans.

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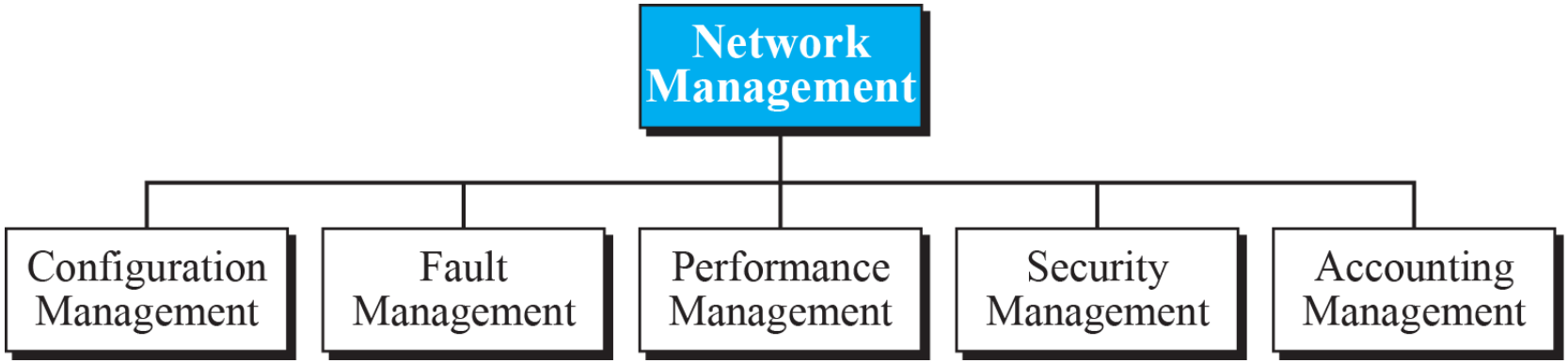
Figure 9.1: Areas of network management



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9.1.1 Configuration Management

A large network is usually made up of hundreds of entities that are physically or logically connected to each other. These entities have an initial configuration when the network is set up, but can change with time. Desktop computers may be replaced by others; application software may be updated to a newer version; and users may move from one group to another. The configuration management system must know, at any time, the status of each entity and its relation to other entities..

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9.1.1 (continued)

- **Reconfiguration**
  - Hardware Reconfiguration
  - Software Reconfiguration
  - User-Account Reconfiguration
- **Documentation**
  - Hardware Documentation
  - Software Documentation
  - User-Account Documentation
- **Arithmetic Coding**

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9.1.2 Fault Management

Complex networks today are made up of hundreds and sometimes thousands of components. Proper operation of the network depends on the proper operation of each component individually and in relation to each other. Fault management is the area of network management that handles this issue. An effective fault management system has two subsystems: reactive fault management and proactive fault management.

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9.1.2 (continued)

- **Reactive Fault Management**
  - Detecting Fault
  - Isolating Fault
  - Correcting Fault
  - Recording Fault
- **Proactive Fault Management**
- **Arithmetic Coding**

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9.1.3 Performance Management

Performance management, which is closely related to fault management, tries to monitor and control the network to ensure that it is running as efficiently as possible. Performance management tries to quantify performance using some measurable quantity, such as capacity, traffic, throughput, or response time. Some protocols, such as SNMP, which is discussed in this chapter, can be used in performance management.

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9.1.3 (continued)

- Capacity
- Traffic
- Throughput
- Response Time
- Arithmetic Coding

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9.1.4 Security Management

Security management is responsible for controlling access to the network based on predefined policy. In Chapter 10 we will discuss security tools such as encryption and authentication. Encryption allows privacy for users; authentication forces the users to identify themselves.

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9.1.5 Accounting Management

Accounting management is the controlling of users' access to network resources through charges. Under accounting management, individual users, departments, divisions, or even projects are charged for the services they receive from the network. Charging does not necessarily mean cash transfer; it may mean debiting the departments or divisions for budgeting purposes.

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9-2 SNMP

Several network management standards have been devised during the last few decades. The most important one is Simple Network Management Protocol (SNMP), used by the Internet. We discuss this standard in this section. SNMP is a framework for managing devices in an internet using the TCP/IP protocol suite.

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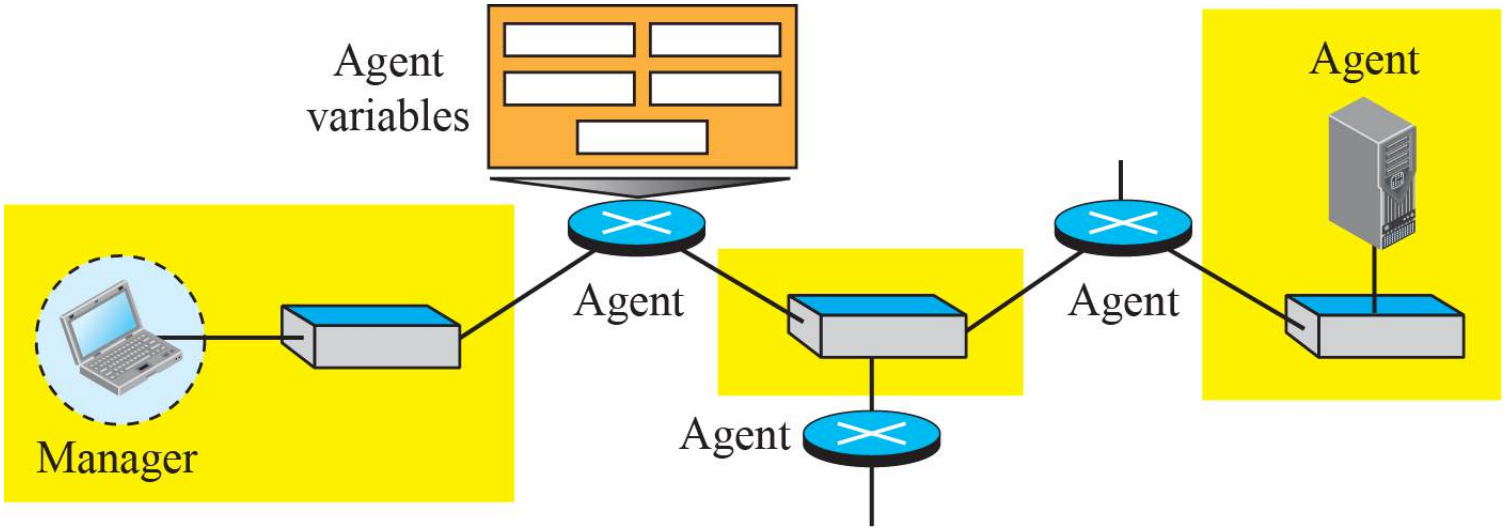
Figure 9.2: SNMP concept



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9.2.1 Managers and Agents

A management station, called a manager, is a host that runs the SNMP client program. A managed station, called an agent, is a router (or a host) that runs the SNMP server program. Management is achieved through simple interaction between a manager and an agent.

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9.2.2 Management Components

To do management tasks, SNMP uses two other protocols: Structure of Management Information (SMI) and Management Information Base (MIB). In other words, management on the Internet is done through the cooperation of three protocols: SNMP, SMI, and MIB, as shown in Figure 9.3.

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9.2.2 (continued)

- **Role of SNMP**
- **Role of SMI**
- **Role of MIB**
- **An Analogy**
  - Syntax: SMI
  - Object Declaration and Definition: MIB
  - Program Coding: SNMP
- **Arithmetic Coding**

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Figure 9.3: Components of network management on the Internet

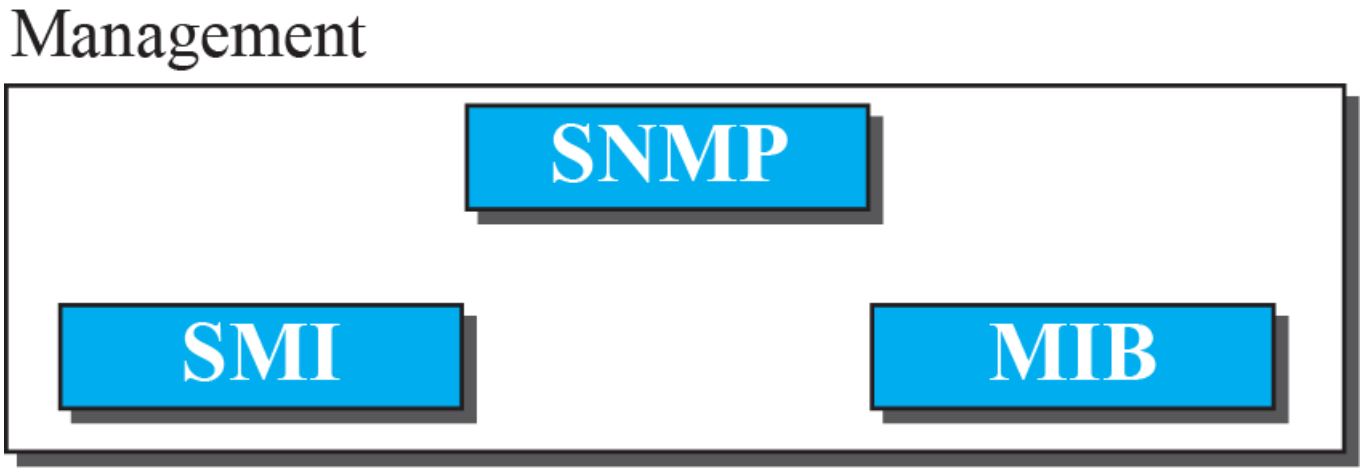

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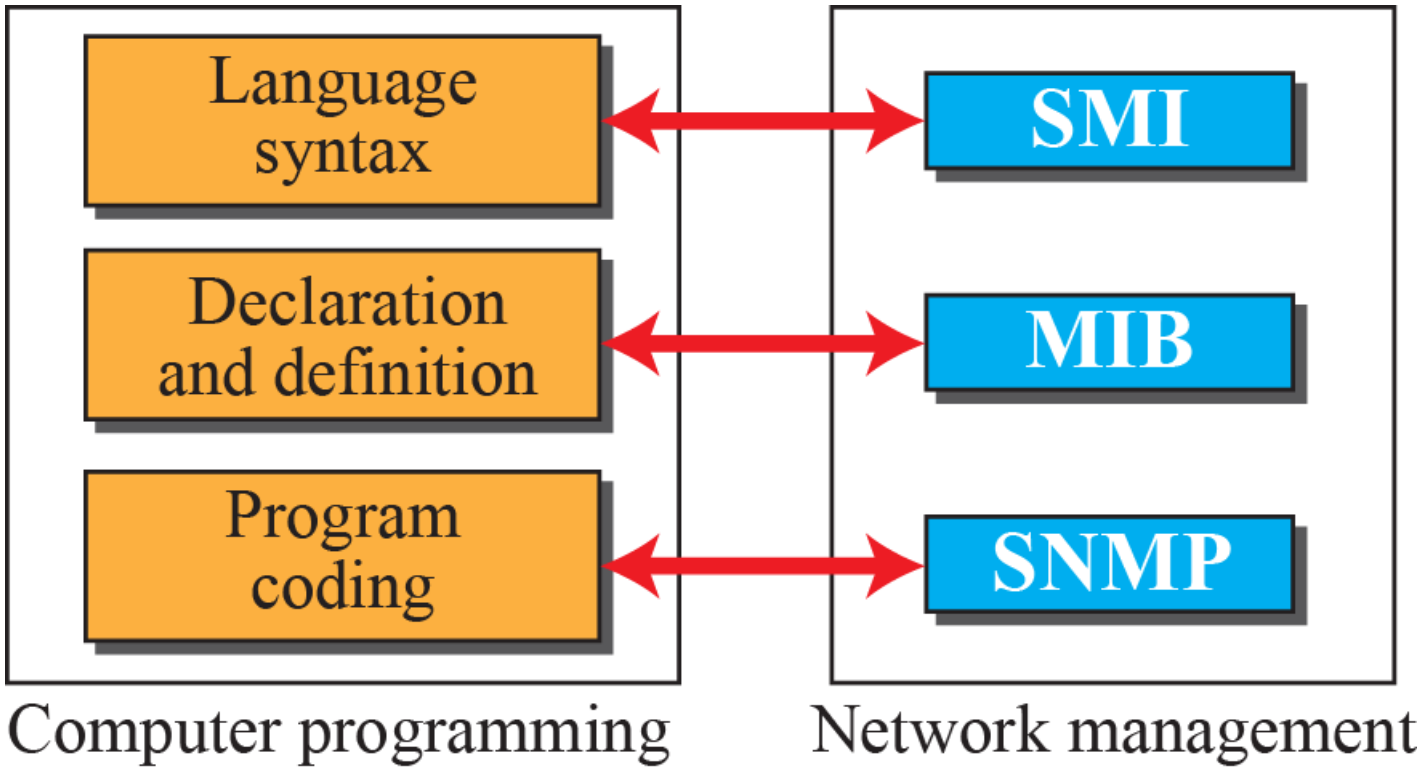
Figure 9.4: ***Comparing computer programming and network management***



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9.2.3 An Overview

Before discussing each component in more detail, let us show how each of these components is involved in a simple scenario. This is an overview that will be developed later, at the end of the chapter. A manager station (SNMP client) wants to send a message to an agent station (SNMP server) to find the number of UDP user datagrams received by the agent.

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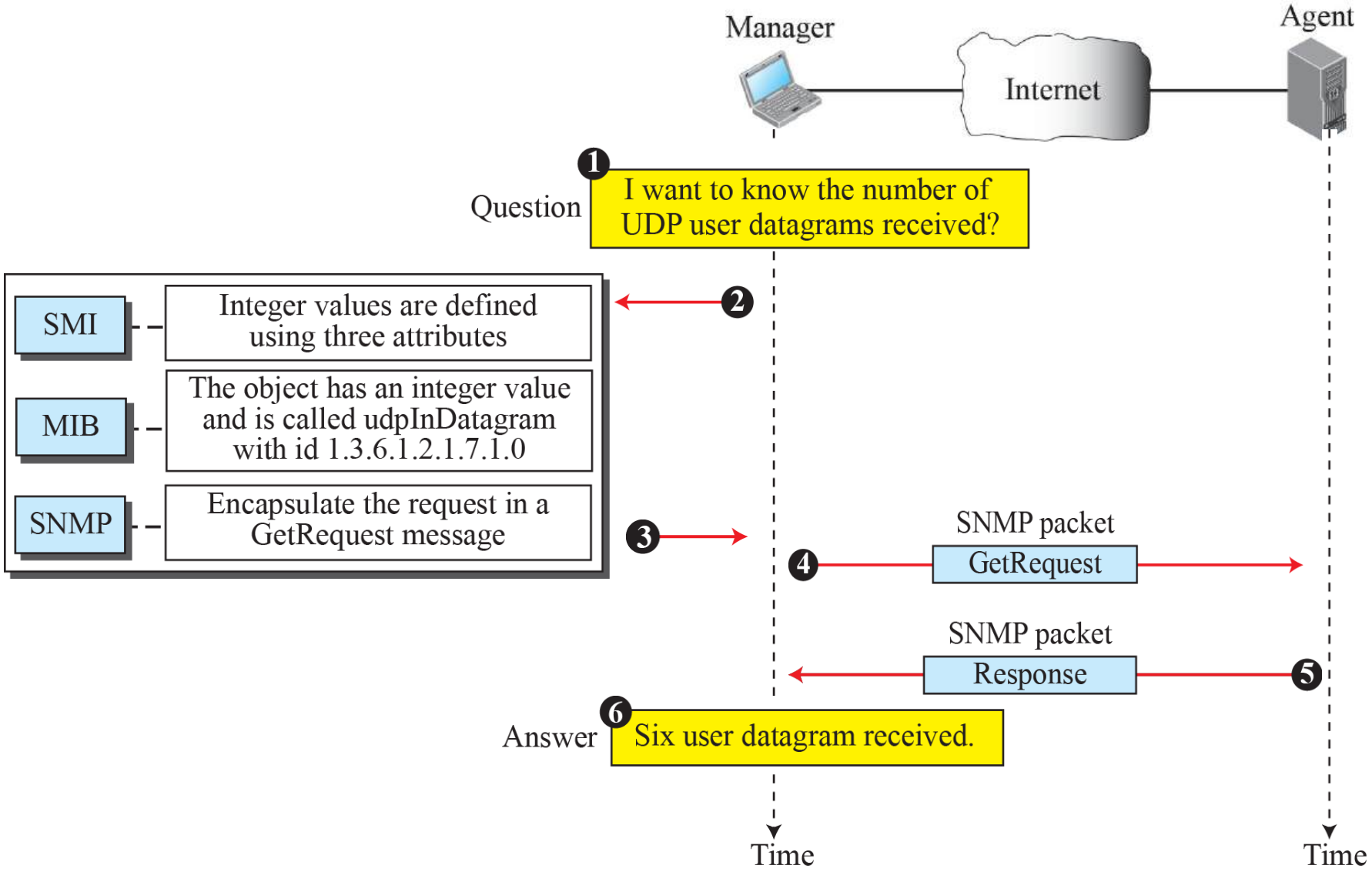
Figure 9.5: Management overview



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9.2.4 SMI

The Structure of Management Information, version 2 (SMIv2) is a component for network management. SMI is a guideline for SNMP. It emphasizes three attributes to handle an object: name, data type, and encoding method.

- **Name**
- **Type**
  - Simple Type
  - Structured Type
- **Encoding Method**

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Figure 9.6: Object identifier in SMI

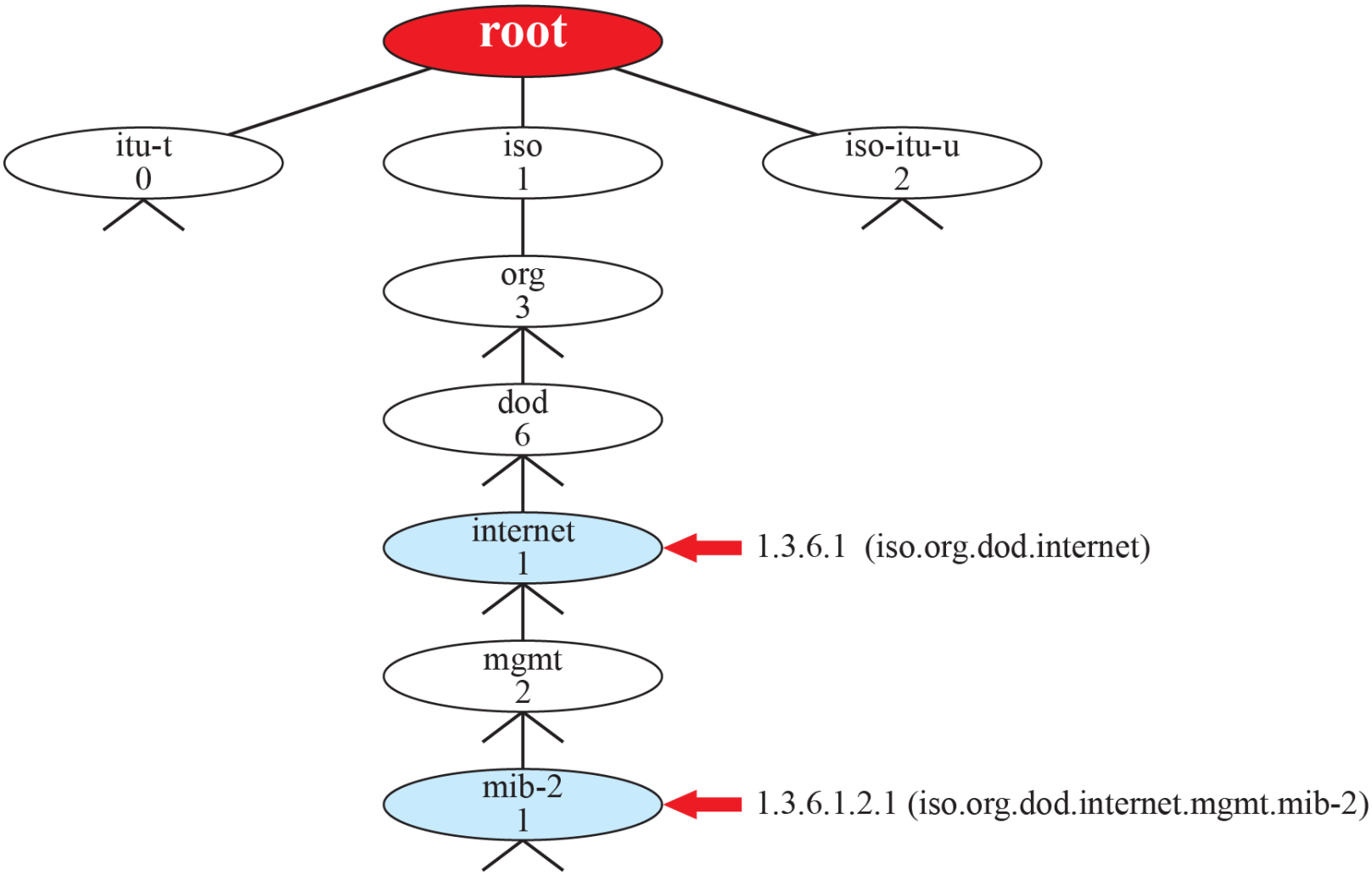

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Table 9.1: Data types

Type	*Size*	*Description*
INTEGER	4 bytes	An integer with a value between -2^{31} and $2^{31}-1$
Integer32	4 bytes	Same as INTEGER
Unsigned32	4 bytes	Unsigned with a value between 0 and $2^{32}-1$
OCTET STRING	Variable	Byte-string up to 65,535 bytes long
OBJECT IDENTIFIER	Variable	An object identifier
IPAddress	4 bytes	An IP address made of four integers
Counter32	4 bytes	An integer whose value can be incremented from zero to 2^{32}; when it reaches its maximum value it wraps back to zero
Counter64	8 bytes	64-bit counter
Gauge32	4 bytes	Same as Counter32, but when it reaches its maximum value, it does not wrap; it remains there until it is reset
TimeTicks	4 bytes	A counting value that records time in 1/100ths of a second
BITS		A string of bits
Opaque	Variable	Uninterpreted string

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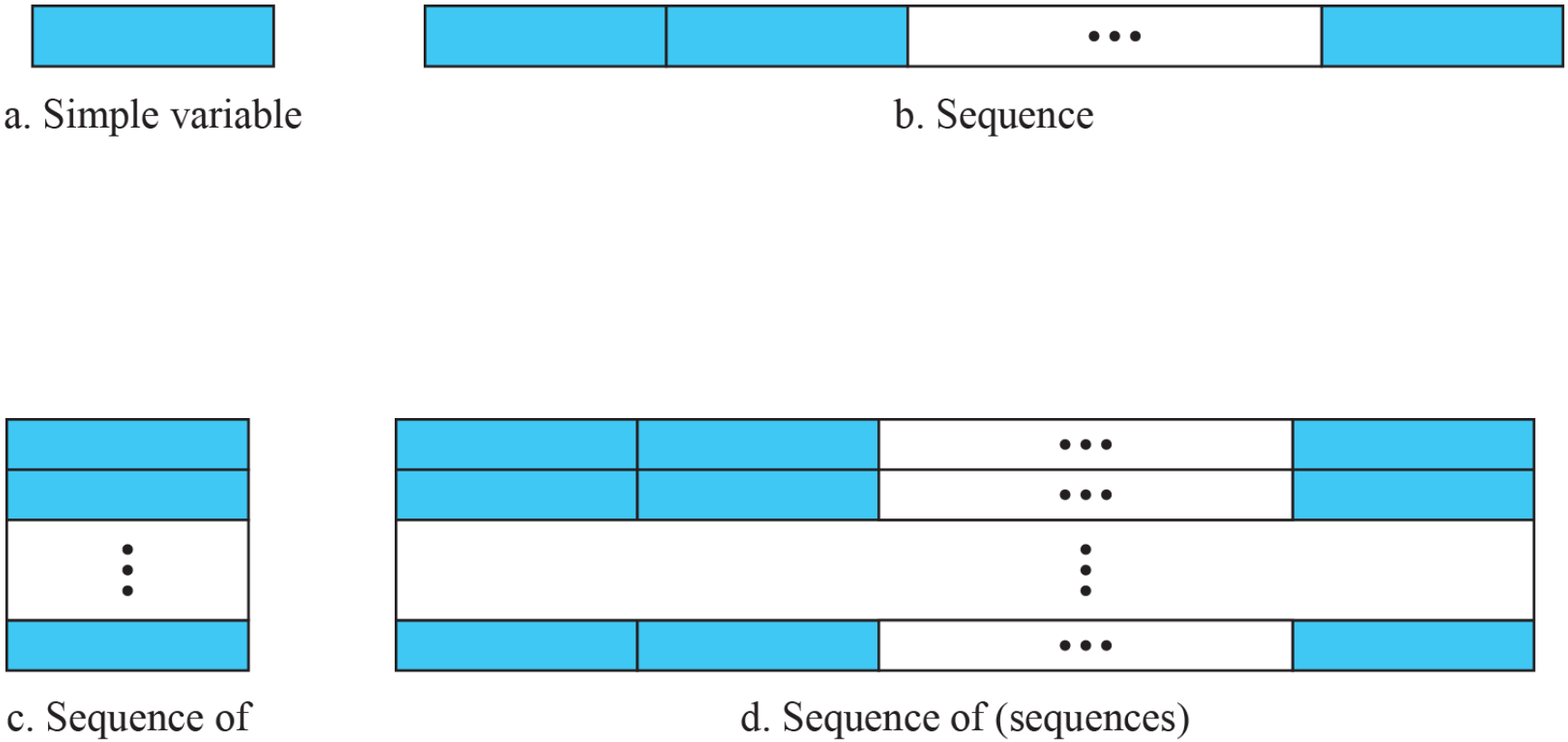
Figure 9.7: Conceptual data types



a. Simple variable

b. Sequence

c. Sequence of

d. Sequence of (sequences)

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Figure 9.8: Encoding format

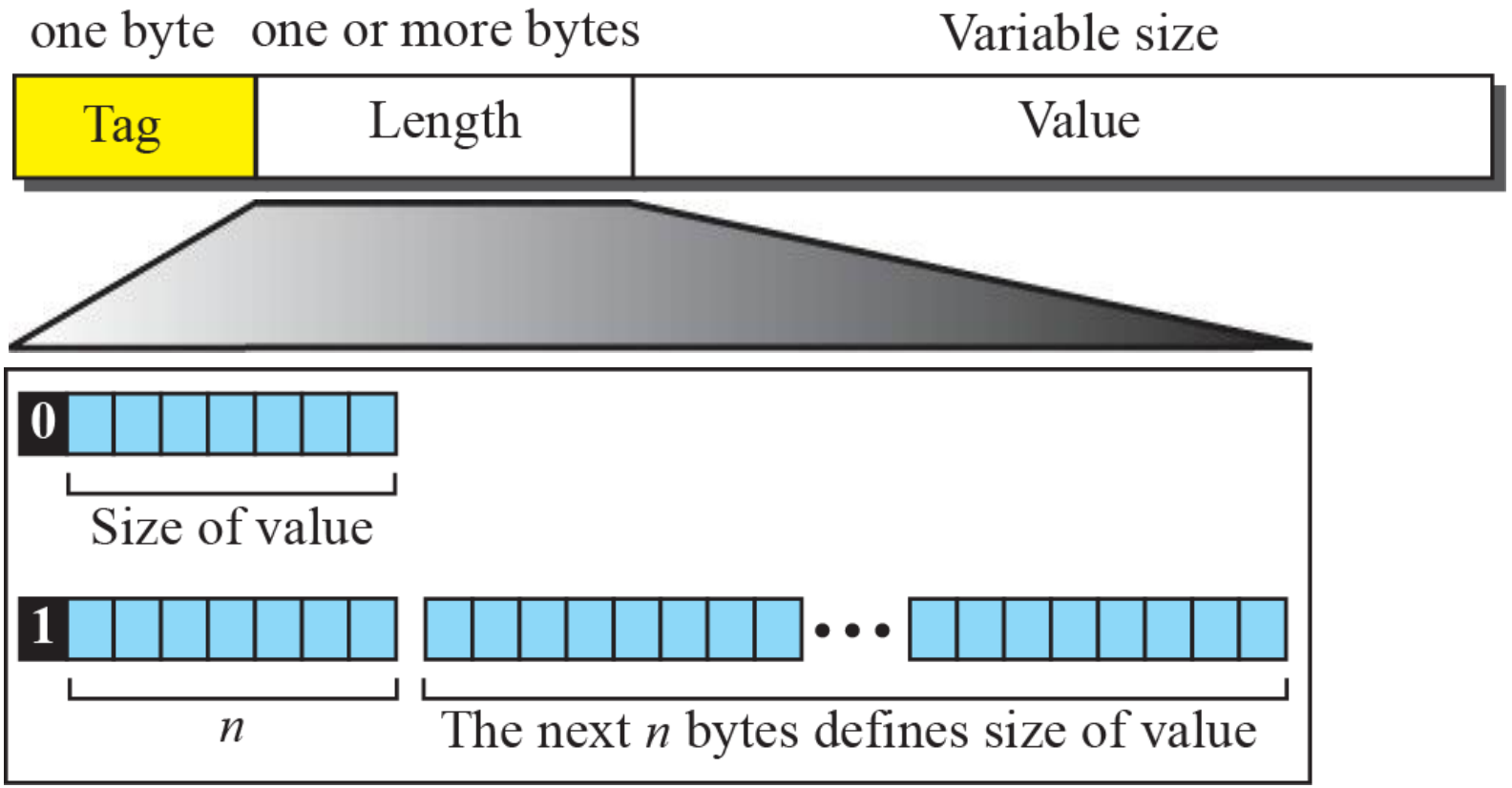

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Table 9.2: Codes for data types

Data Type	*Tag (Hex)*	*Data Type*	*Tag (Hex)*
INTEGER	**02**	IPAddress	**40**
OCTET STRING	**04**	Counter	**41**
OBJECT IDENTIFIER	**06**	Gauge	**42**
NULL	**05**	TimeTicks	**43**
SEQUENCE, SEQUENCE OF	**30**	Opaque	**44**

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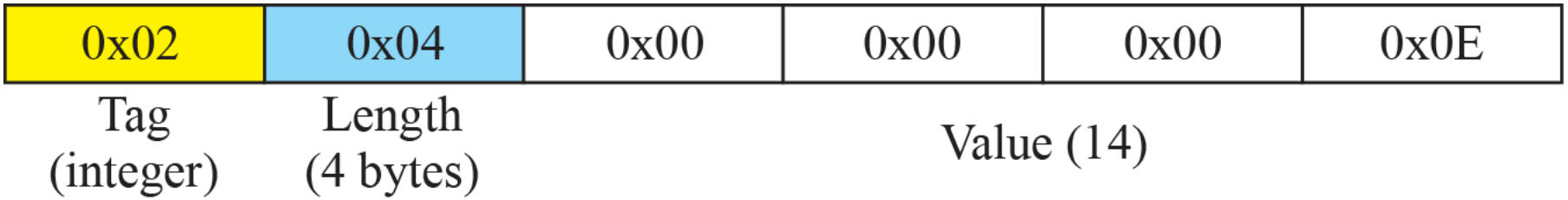
Example 9.1

Figure 9.9 shows how to define INTEGER 14. The size of the length field is from Table 9.1.

Figure 9.9: ***Example 9.1: INTEGER 14***



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Example 9.2

Figure 9.10 shows how to define the OCTET STRING "HI."

Figure 9.10: ***Example 9.2: OCTET STRING "HI"***

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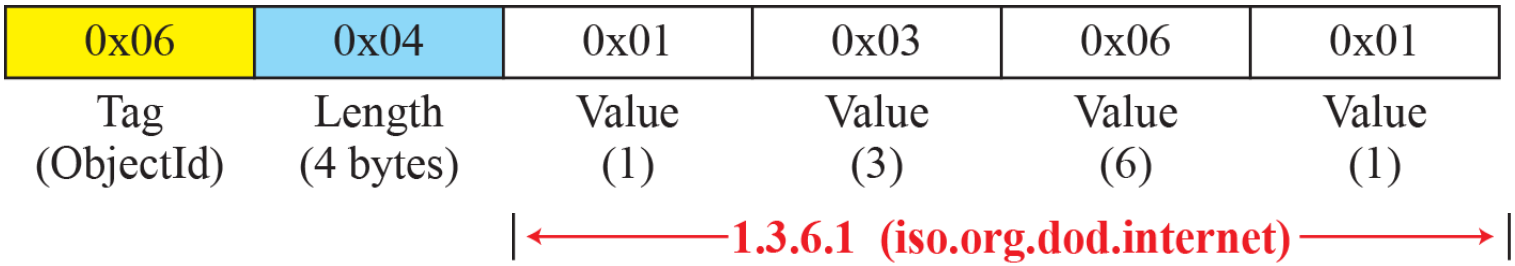
Example 9.3

Figure 9.11 shows how to define ObjectIdentifier 1.3.6.1 (iso.org.dod.internet).

Figure 9.11: ***Example 9.3: ObjectIdentifier 1.3.6.1***



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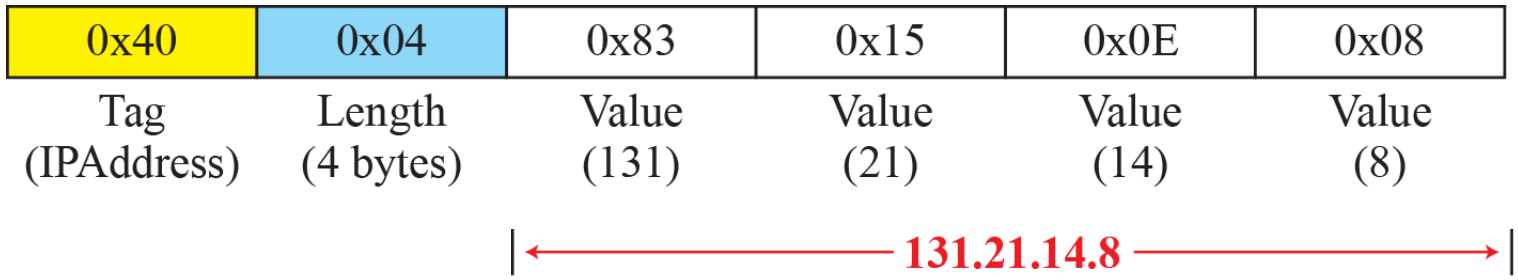
Example 9.4

Figure 9.12 shows how to define IPAddress 131.21.14.8.

Figure 9.12: ***Example 9.4: IPAddress 131.21.14.8***



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9.2.5 MIB

The Management Information Base, version 2 (MIB2) is the second component used in network management. Each agent has its own MIB2, which is a collection of all the objects that the manager can manage. (See Figure 9.13.)

- **Accessing MIB Variables**
  - Simple Variables
  - Tables

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Figure 9.13: Some mib-2 groups

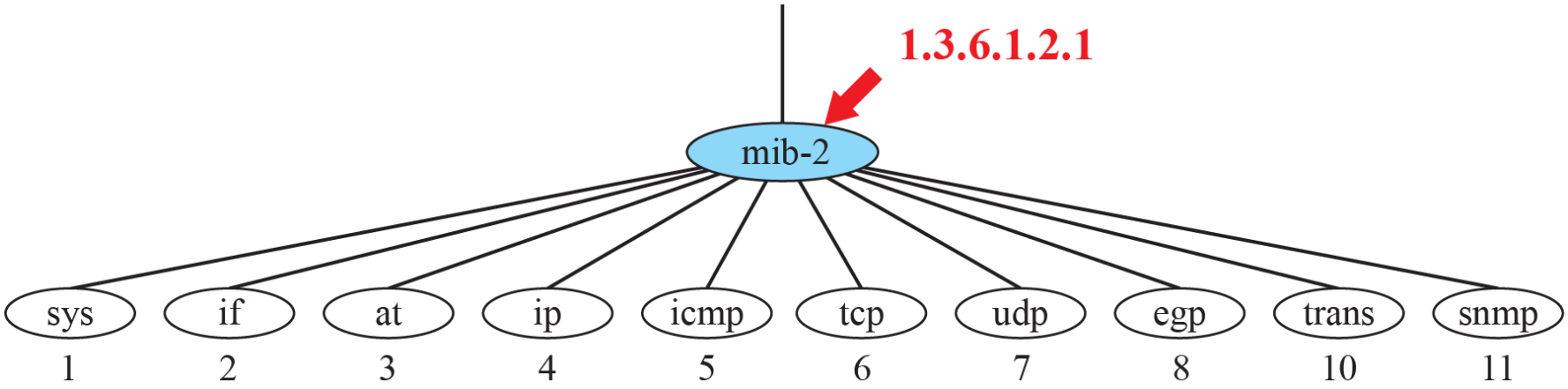

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Figure 9.14: udp group

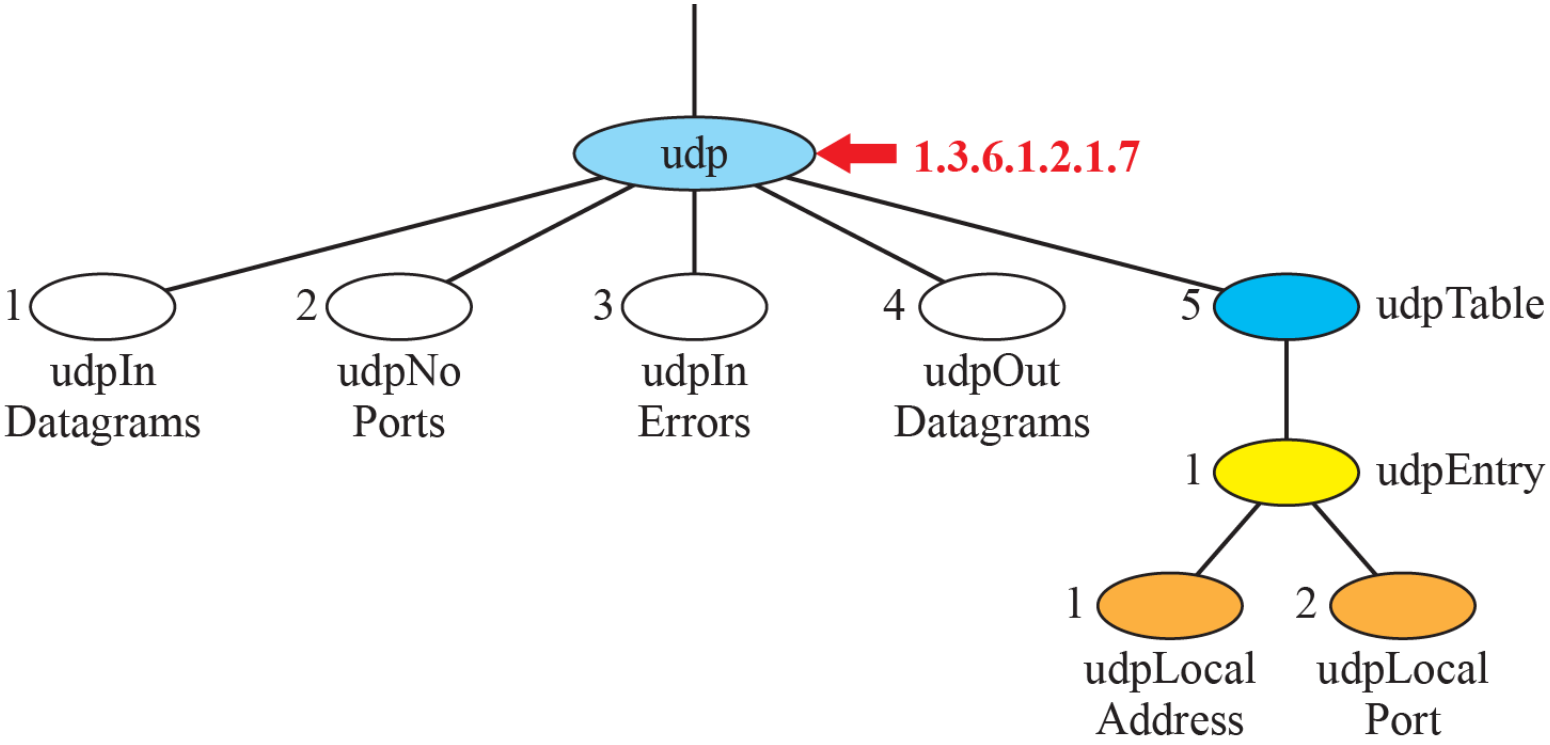

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Figure 9.15: udp variables and tables

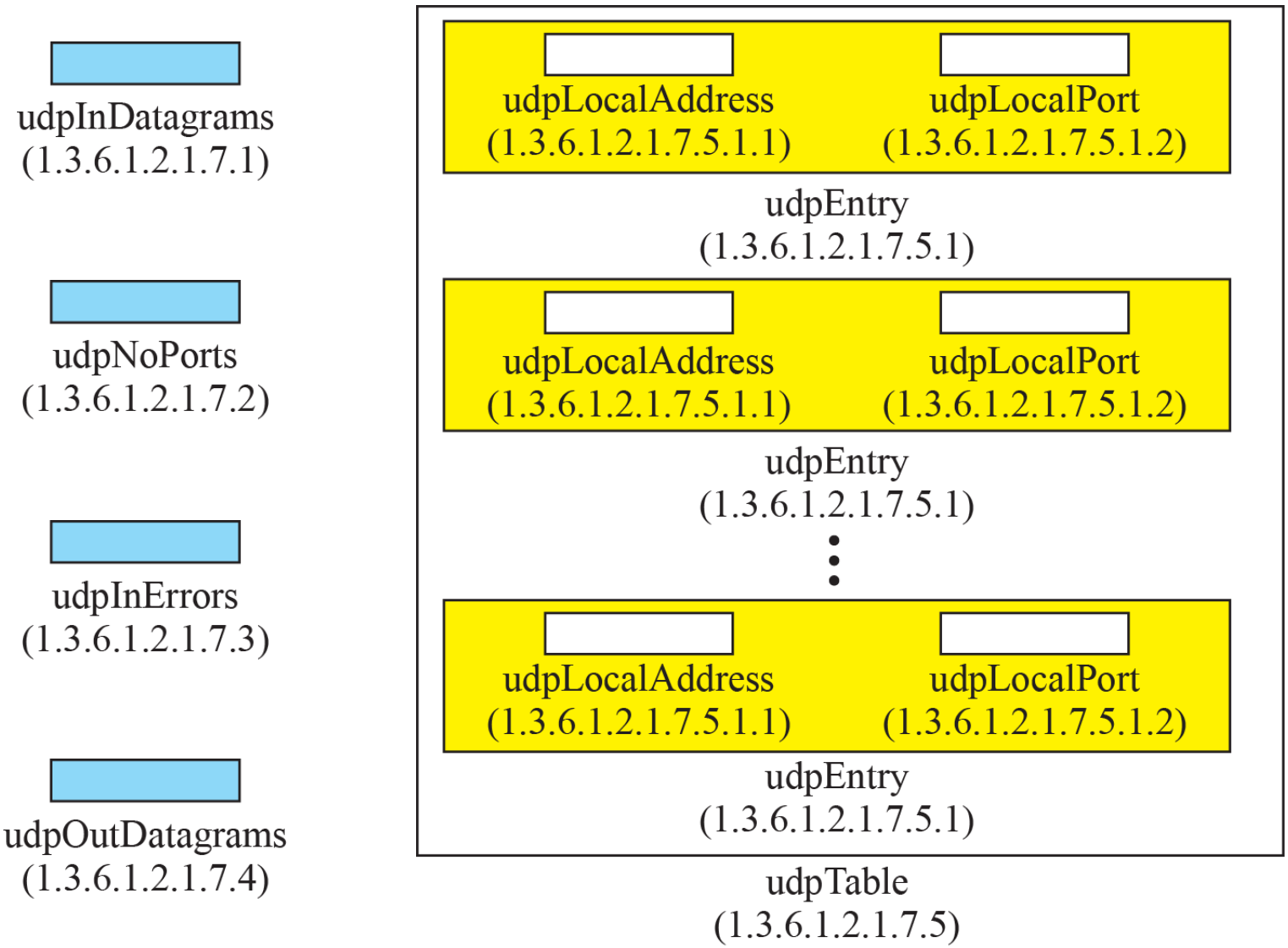

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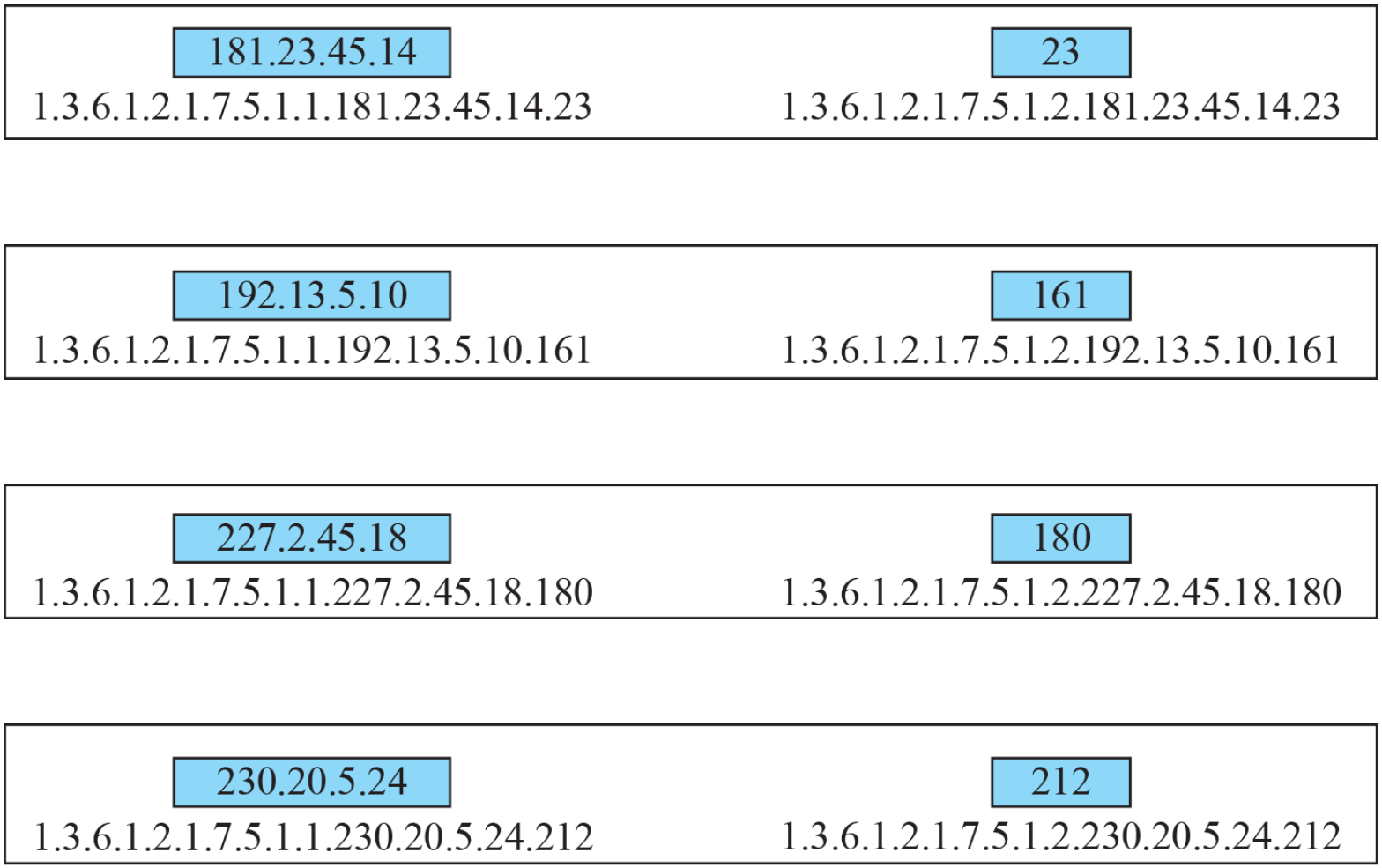
Figure 9.16: Indexes for udpTable



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9.2.6 SNMP

SNMP uses both SMI and MIB in Internet network management.

- **PDUs**
  - *GetRequest*
  - *GetNextRequest*
  - *GetBulkRequest*
  - *SetRequest*
  - *Response*
  - *Trap*
  - *InformRequest*
  - *Report*

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9.2.6 (continued)

- *Format*
- *Messages*
- *UDP Ports*
- *Security*
- *Arithmetic Coding*

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Figure 9.17: SNMP PDUs

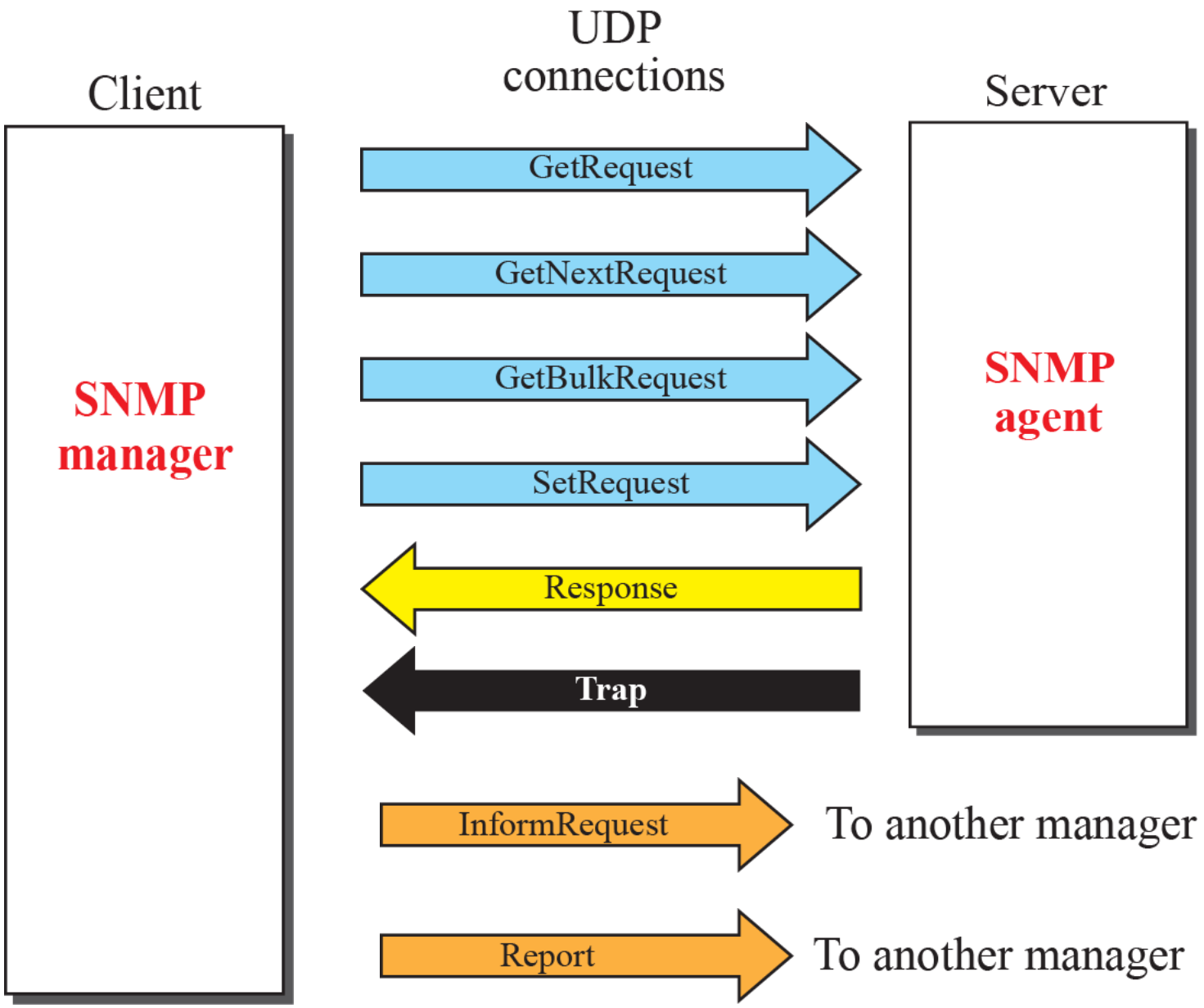

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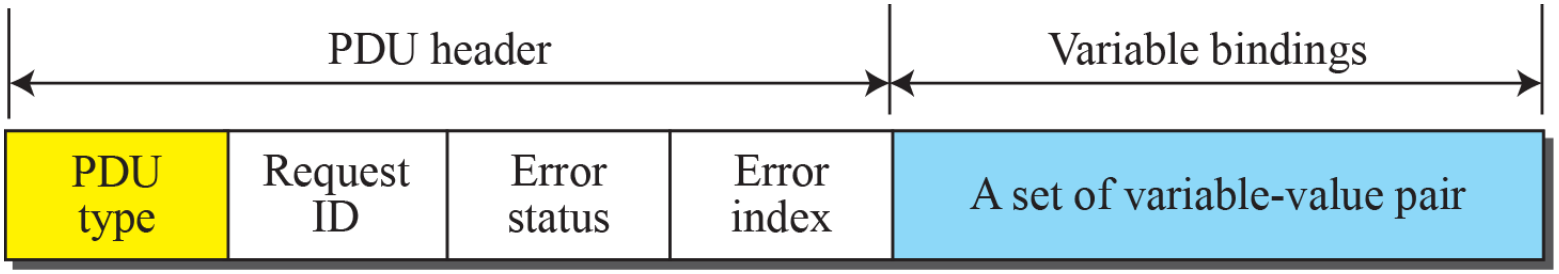
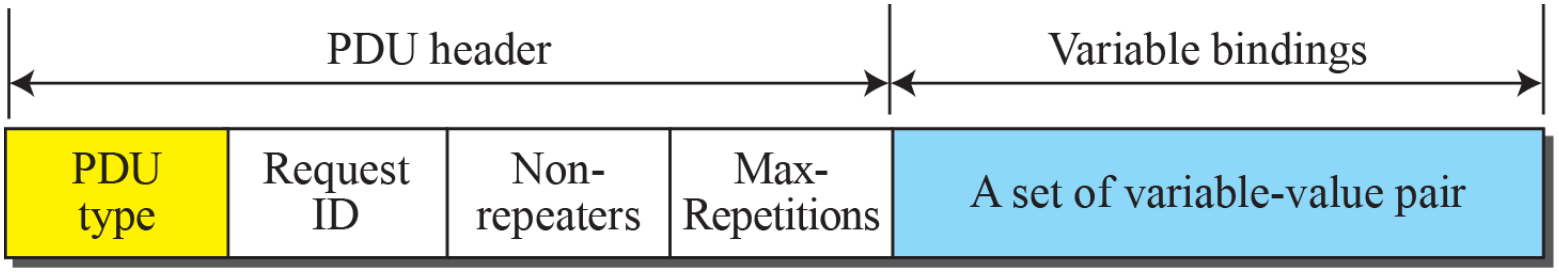
Figure 9.18: SNMP PDU format

Note:
The error status and error index values are set to 0 for all request messages.



a. All PDU types except GetBulkRequest



b. GetBulkRequest

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Table 9.3: PDU types

Type	*Tag (Hex)*	*Type*	*Tag (Hex)*
GetRequest	**A0**	GetBulkRequest	**A5**
GetNextRequest	**A1**	InformRequest	**A6**
Response	**A2**	Trap (SNMPv2)	**A7**
SetRequest	**A3**	Report	**A8**

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Table 9.4: Types of errors

Status	*Name*	*Meaning*
0	noError	No error
1	tooBig	Response too big to fit in one message
2	noSuchName	Variable does not exist
3	badValue	The value to be stored is invalid
4	readOnly	The value cannot be modified
5	genErr	Other errors

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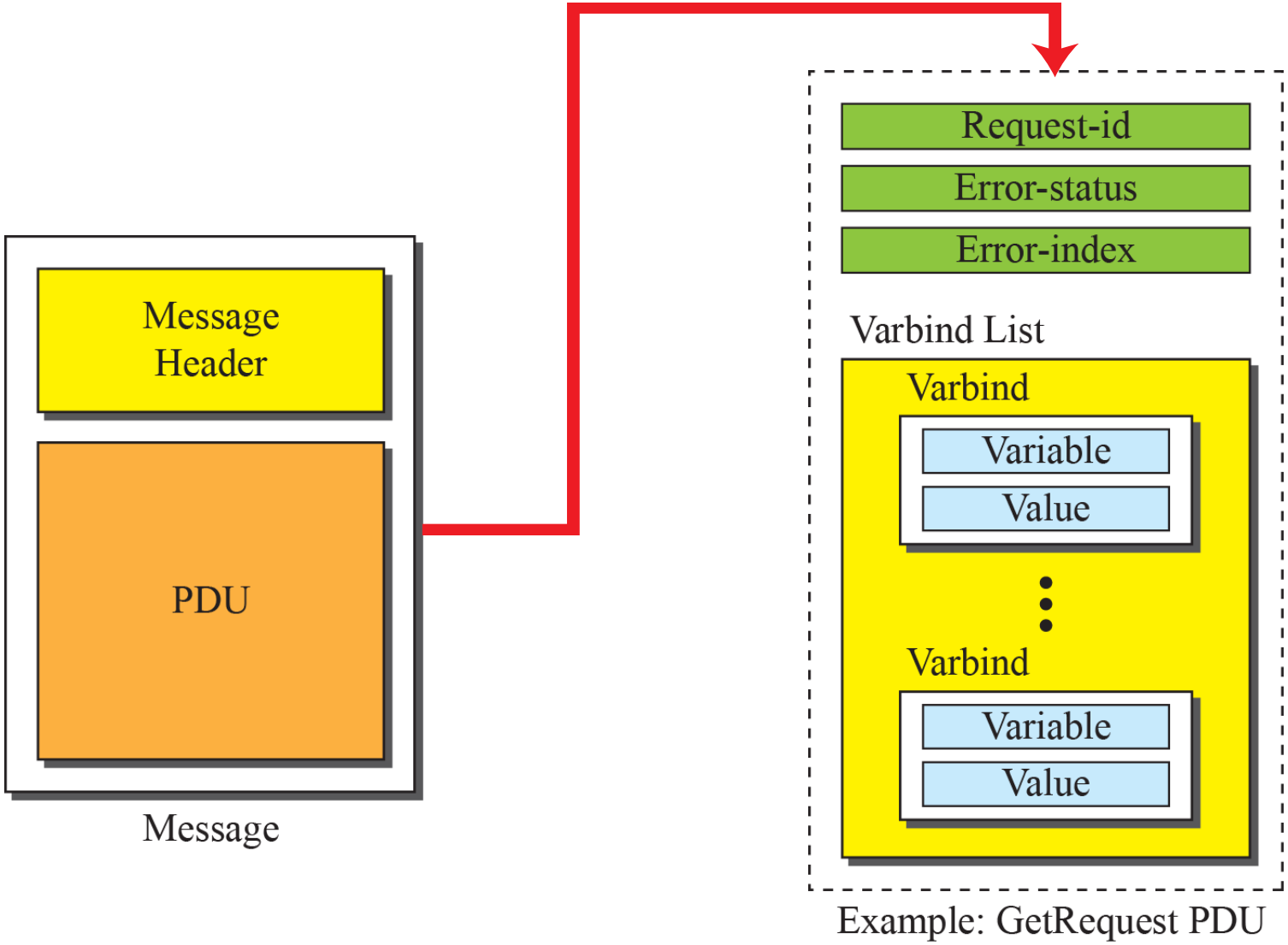
Figure 9.19: SNMP message



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Example 9.5

In this example, a manager station (SNMP client) uses a message with a GetRequest PDU to retrieve the number of UDP datagrams that a router has received (Figure 9.20).

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Figure 9.20: Example 9.5

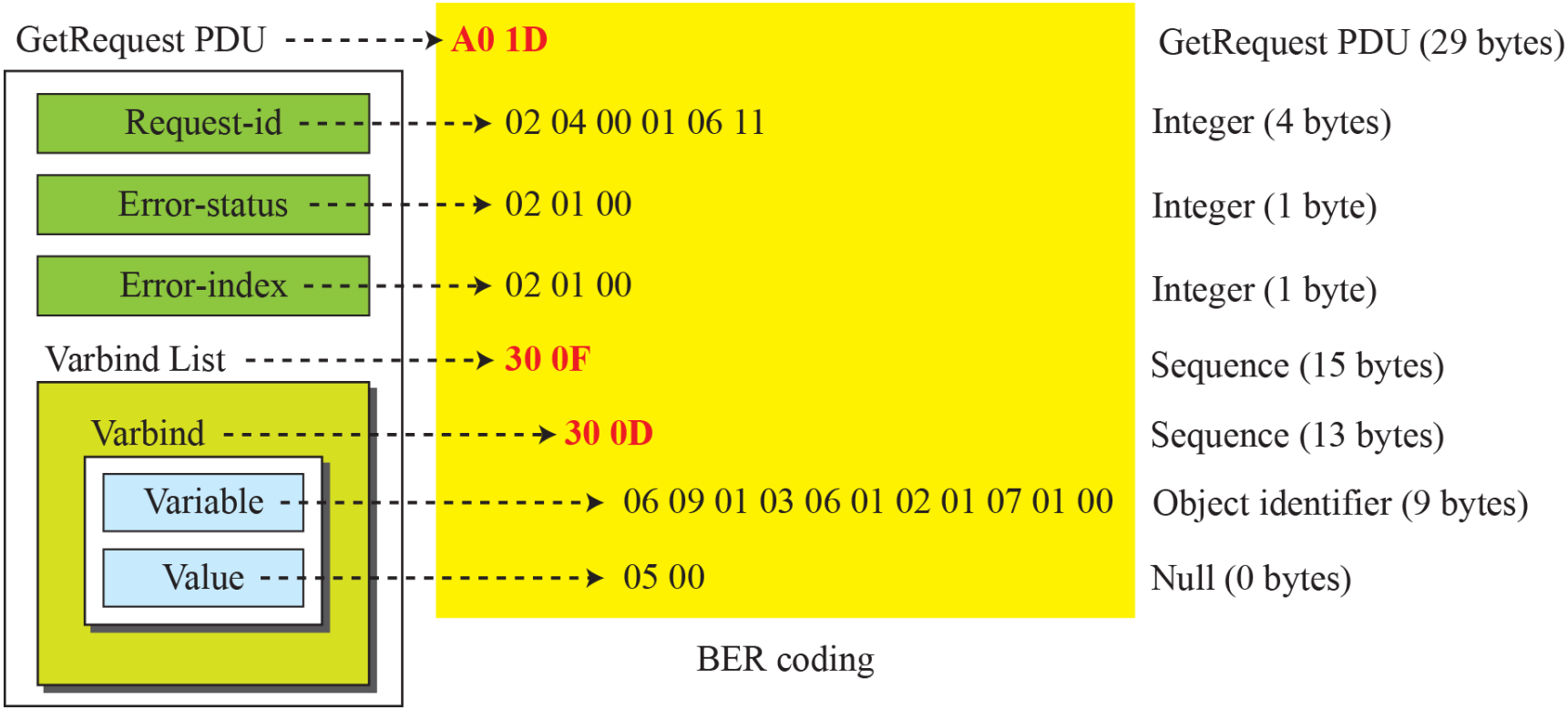

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Figure 9.21: Actual message sent for Example 9.5

30	29	--	--
--	--	--	--
--	--	--	--
A0	1D	02	04
00	01	06	11
02	01	00	02
01	00	30	0F
30	0D	06	09
01	03	06	01
02	01	07	01
00	05	00	

Message

Note:
The byte values are in hexadecimal

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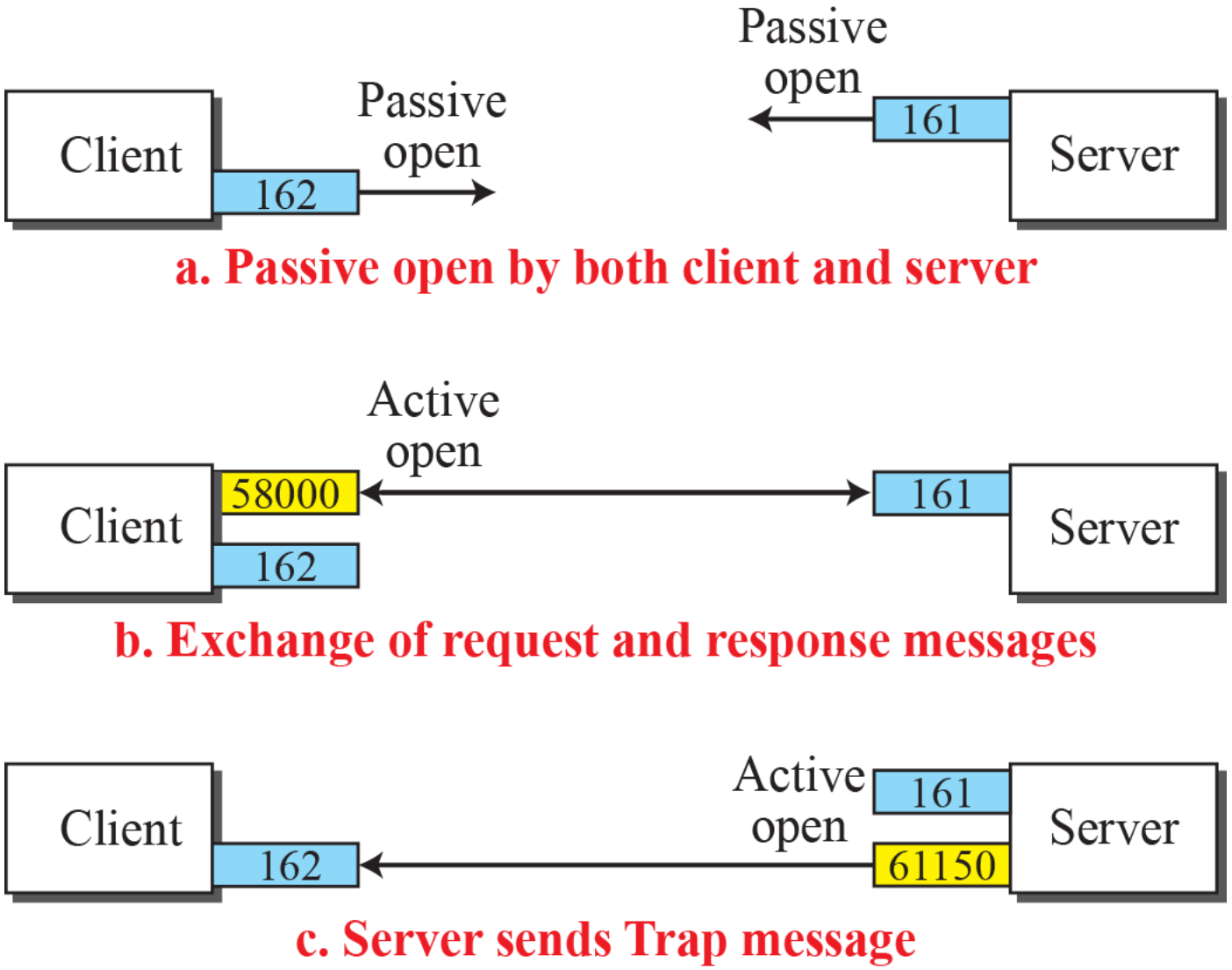
Figure 9.22: Port numbers for SNMP



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9-3 ASN.1

In data communication, when we send a continuous stream of bits to a destination, we somehow need to define the format of the data. This is done through an abstract language that uses some symbols, key words, and atomic data types and lets us make new data types out of the simple types. The language is called Abstract Syntax Notation One (ASN.1).

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9.3.1 Language Basics

Before we show how we can define objects and associated values, let us talk about the language itself. The language use some symbols and some key words and defines some primitive data types. As we said before, SMI uses a subset of these entities in its own language.

- ***Symbols***
- ***Keywords***

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Table 9.5: Symbols used in ASN.1

Symbol	*Meaning*	*Symbol*	*Meaning*
::=	Defined as or assignment	..	Range
|	Or, alternative, or option	{}	Start and end of a list
–	Negative sign	[]	Start and end of tag
––	The following is a comment	()	Start and end of a subtype

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Table 9.6: Keywords in ASN.1

Keyword	*Description*
BEGIN	Start of a module
CHOICE	List of alternatives
DEFINITIONS	Definition of a data type or an object
END	End of a module
EXPORTS	Data type that can be exported to other modules
IDENTIFIER	A sequence of non-negative numbers that identifies an object
IMPORTS	Data type defined in an external module and imported
INTEGER	Any positive, zero, or negative integer
NULL	A null value
OBJECT	Used with IDENTIFIER to uniquely define an object
OCTET	Eight-bit binary data
OF	Used with SEQUENCE or SET
SEQUENCE	An ordered list
SEQUENCE OF	An ordered array of data of the same type
SET	An unordered list
SET OF	An array of unordered lists
STRING	A string of data

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9.3.2 Data Types

After discussing the symbols and keywords used in the language, it is time to define its data types. The idea is similar to what we see in computer languages such as C, C++, or Java. In ASN.1, we have several simple data types such as integer, float, boolean, char, and so on. We can combine these data types to create a new simple data type or to define a structured data types such as array or struct.

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9.3.2 (continued)

- Simple Data Types
- New Data Types
- New Subtypes
- Simple Variables
- Structured Type
- Structure Variables
- Arithmetic Coding

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Table 9.7: Some simple ASN.1 built-in types

Tag	*Type*	*Set of values*
Universal 1	BOOLEAN	TRUE or FALSE
Universal 2	INTEGER	Integers (positive, 0, or negative)
Universal 3	BIT STRING	A string of binary digits (bits) or a null set
Universal 4	OCTET STRING	A string of octets or a null set
Universal 5	NULL	Null, single valued
Universal 6	OBJECT IDENTIFIER	A set of values that defines an object
Universal 7	ObjectDescriptor	Human readable text describing an object
Universal 8	EXTERNAL	A type which is not in the standard
Universal 9	REAL	Real numbers in scientific notation
Universal 10	ENUMERATED	A list of integers
Universal 16	SEQUENCE, SEQUENCE OF	Ordered list of types
Universal 17	SET, SET OF	Unordered list of types
Universal 18	NumericString	Digits 0-9 and space
Universal 19	PrintableString	printable characters
Universal 26	VisibleString	ISO646String
Universal 27	GeneralString	General character string
Universal 30	CHARACTER STRING	Character set

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Example 9.6

The following is an example of some new types using built-in types from Table 9.7.

```
Married ::= BOOLEAN
MaritalStatus ::= ENUMERATED {single, married, widowed, divorced}
DayOfWeek ::= ENUMERATED {sun, mon, tue, wed, thu, fri, sat}
Age ::= INTEGER
```


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Example 9.7

The following shows how we can make three new subtypes. The range of the first is the subset of INTEGER, the range of the second is the subset of REAL, and the range of the third is the subset of DayOfWeek, which we defined in Example 9.6. Note that we use the symbol (..) to define the range and the symbol (|) to define the choice.

```
NumberOfStudents ::= INTEGER (15..40)     -- An integer with the range 15 to 40
Grade ::= REAL (1.0..5.0)                 -- A real number with the range 1.0 to 5.0
Weekend ::= DayOfWeek (sun | sat)         -- A day that can be sun or sat
```


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Example 9.8

The following are a few examples of defining some variables and assigning the appropriate value from the range of those types. Note that the first and the third variables are of the built-in type, the second is of the type defined in Example 9.6, and the last is of a subtype defined in Example 9.7.

```
numberOfComputers INTEGER ::= 2
married Married ::= FALSE
herAge INTEGER ::= 35
classSize NumberOfStudents ::= 22
```


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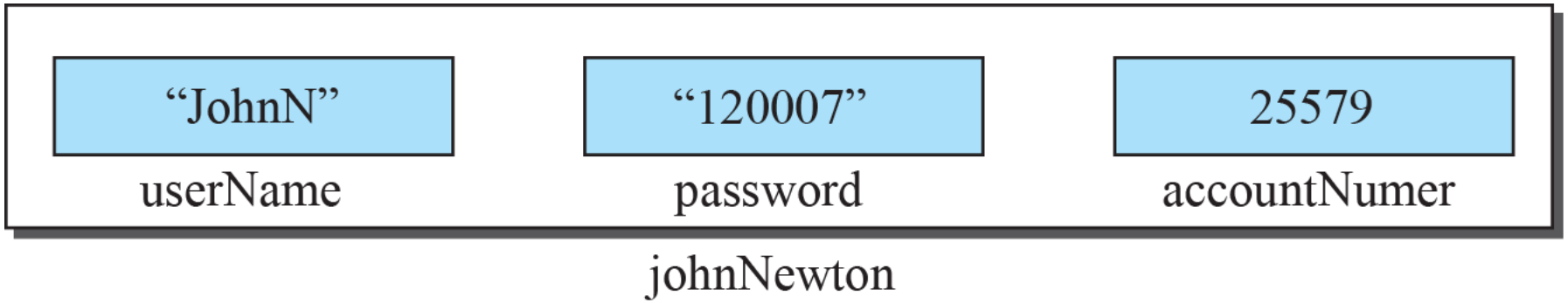
Figure 9.23: ***Record representing the type definition and variable declaration***



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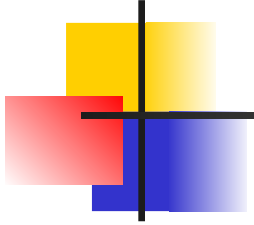

9.3.3 Encoding

After the data has been defined and values are associated with variables, ASN.1 use one of the encoding rules to encode the message to be sent. We already discussed the Basic Encoding Rule in the previous section.

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Chapter 9: Summary

- *The five areas comprising network management are configuration management, fault management, performance management, accounting management, and security management.*

- *Simple Network Management Protocol (SNMP) is a framework for managing devices in an internet using the TCP/IP protocol suite. A manager, usually a host, controls and monitors a set of agents, usually routers. SNMP uses the services of SMI and MIB. SMI names objects, defines the type of data that can be stored in an object, and encodes the data. MIB is a collection of groups of objects that can be managed by SNMP. MIB uses lexicographic ordering to manage its variables.*

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Chapter 9: Summary (continued)

- ***Abstract Syntax Notation Number One (ASN.1) is a language that defines the syntax and semantics of data. It uses some symbols, keywords, simple and structured data types. Part of ASN.1 is used by SMI to define the format of objects and values used in network management.***

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